

CHAPTER 13

INTRODUCTION TO METABOLISM



[13.1 Bioenergetics and Thermodynamics](#)

[13.2 Chemical Logic and Common Biochemical Reactions](#)

[13.3 Phosphoryl Group Transfers and ATP](#)

[13.4 Biological Oxidation-Reduction Reactions](#)

[13.5 Regulation of Metabolic Pathways](#)

Living cells and organisms must perform work to stay alive, to grow, and to reproduce. The ability to harness energy and to channel it into biological work is a fundamental property of all living organisms; it must have been acquired very early in cellular evolution. Modern organisms carry out a remarkable variety of energy transductions, conversions of one form of energy to another. They use the chemical energy in fuels to bring about the synthesis of complex, highly ordered macromolecules from simple precursors. They also convert the chemical energy of fuels into concentration gradients and electrical gradients, into motion and heat, and, in a few organisms such as fireflies and some deep-

sea fish, into light. Photosynthetic organisms transduce light energy into all these other forms of energy.

The chemical mechanisms that underlie biological energy transductions have fascinated and challenged biologists for centuries. The French chemist Antoine Lavoisier recognized that animals somehow transform chemical fuels (foods) into heat and that this process of respiration is essential to life. He observed that

in general, respiration is nothing but a slow combustion of carbon and hydrogen, which is entirely similar to that which occurs in a lighted lamp or candle, and that, from this point of view, animals that respire are true combustible bodies that burn and consume themselves.... One may say that this analogy between combustion and respiration has not escaped the notice of the poets, or rather the philosophers of antiquity, and which they had expounded and interpreted. This fire stolen from heaven, this torch of Prometheus, does not only represent an ingenious and poetic idea, it is a faithful picture of the operations of nature, at least for animals that breathe; one may therefore say, with the ancients, that the torch of life lights itself at the moment the infant breathes for the first time, and it does not extinguish itself except at death.¹



A portrait by Jacques Louis David of Antoine Lavoisier (1743–1794) in the laboratory with chemist Marie Anne Pierrette Paulze (1758–1836), his wife.

We now understand much of the chemistry underlying that “torch of life.” Biological energy transductions obey the same chemical and physical laws that govern all other natural processes, and many of the types of chemical reactions that occur in living organisms have been long known to organic chemists. One

unique feature of cellular chemistry is its exquisitely sensitive regulation by a variety of mechanisms that respond to changes in the external and internal circumstances of the cell and organism.

In this chapter we lay out the foundational principles for understanding the reactions of metabolism that follow in [Part II](#). We first review the laws of thermodynamics and the quantitative relationships among free energy, enthalpy, and entropy. We then review the common types of biochemical reactions that occur in living cells, reactions that harness, store, transfer, and release the energy taken up by organisms from their surroundings. Our focus then shifts to reactions that have special roles in biological energy exchanges, particularly those involving the cofactors ATP (for phosphoryl transfers) and NADH (for electron transfers). Finally, we look at the most common of the strategies for regulating biochemical reactions. Watch for examples of these principles as you read this chapter:

P1 The chemical changes and energy transductions in living organisms follow the laws of thermodynamics.

P2 The free-energy change is the maximum energy made available to do work when a chemical reaction occurs. If two reactions can be combined to yield a third reaction, the overall free energy change is the sum of the two. Cells accomplish energy-requiring chemical work by coupling an energy-releasing (exergonic) reaction such as the cleavage of ATP to an endergonic reaction (which requires energy input).

P3 Although thousands of different chemical reactions occur in the biosphere, most of them fall within a small set of reaction types.

P4 ATP is the universal energy currency in living organisms.

Transfer of its phosphoryl group to a water molecule or metabolic intermediates provides the energetic push for muscle contraction, the pumping of solutes against concentration gradients, and the synthesis of complex molecules.

P5 Oxidation-reduction reactions indirectly provide much of the energy needed to make ATP. Reduced substrates such as glucose are oxidized in several steps, with the energy of oxidation steps conserved in the form of a reduced cofactor, NADH. Energy stored in NADH is used to drive the synthesis of ATP.

P6 To respond to changes in external circumstances, cells must regulate enzyme activities, by changing either the number of enzyme molecules or the catalytic activity of preexisting enzyme molecules.

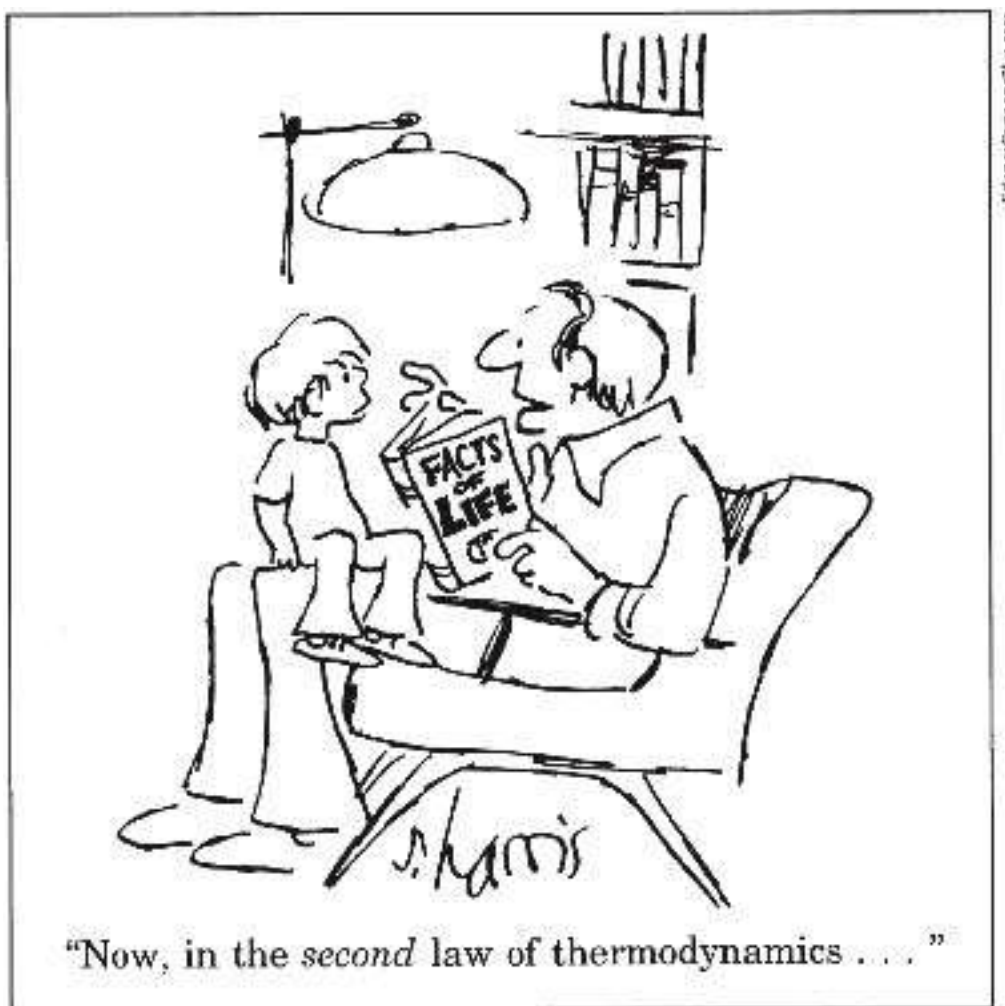
ⁱ From a memoir by Armand Seguin and Antoine Lavoisier, dated 1789, quoted in A. Lavoisier, *Oeuvres de Lavoisier*, Imprimerie Impériale, Paris, 1862.

13.1 Bioenergetics and Thermodynamics

Bioenergetics is the quantitative study of energy transductions — changes of one form of energy into another — that occur in living cells, and of the nature and function of the chemical processes underlying these transductions. Although many of the principles of thermodynamics have been introduced in earlier chapters and may be familiar to you, a review of the quantitative aspects of these principles is useful here.

Biological Energy Transformations Obey the Laws of Thermodynamics

Many quantitative observations made by physicists and chemists on the interconversion of different forms of energy led, in the nineteenth century, to the formulation of two fundamental laws of thermodynamics. The first law is the principle of the conservation of energy: *for any physical or chemical change, the total amount of energy in the universe remains constant energy may change form or it may be transported from one region to another, but it cannot be created or destroyed.* The second law of thermodynamics, which can be stated in several forms, says that the universe always tends toward increasing disorder: *in all natural processes, the entropy of the universe increases.*



Living organisms consist of collections of molecules much more highly organized than the surrounding materials from which they are constructed, and organisms maintain and produce order, seemingly immune to the second law of thermodynamics. But living organisms do not violate the second law; they operate strictly within it. To discuss the application of the second law to biological systems, we must first define those systems and their surroundings.

The *reacting system* is the collection of matter that is undergoing a particular chemical or physical process; it may be an organism, a

cell, or two reacting compounds. The reacting system and its *surroundings* together constitute the *universe*. In the laboratory, some chemical or physical processes can be carried out in isolated or closed systems, in which no material or energy is exchanged with the surroundings. Living cells and organisms, however, are open systems, exchanging both material and energy with their surroundings; living systems are never at equilibrium with their surroundings, and the constant transactions between system and surroundings explain how organisms can create order within themselves while operating within the second law of thermodynamics.

In [Chapter 1 \(p. 21\)](#) we defined three thermodynamic quantities that describe the energy changes occurring in a chemical reaction:

Free energy, G (for J. Willard Gibbs), expresses the amount of energy capable of doing work during a reaction at constant temperature and pressure. When a reaction proceeds with the release of free energy (that is, when the system changes so as to possess less free energy), the free-energy change, ΔG , has a negative value and the reaction is said to be **exergonic**. In **endergonic** reactions, the system gains free energy and ΔG is positive.

Enthalpy, H , is the heat content of the reacting system. It reflects the number and kinds of chemical bonds (covalent and noncovalent) in the reactants and products. When a chemical reaction releases heat, it is said to be **exothermic**; the heat content of the products is less than that of the reactants, and the change in enthalpy, ΔH , has, by convention, a negative value. Reacting systems that take up heat from their surroundings are **endothermic** and have positive values of ΔH .

in which ΔG is the change in Gibbs free energy of the reacting system, ΔH is the change in enthalpy of the system, T is the absolute temperature, and ΔS is the change in entropy of the system. By convention, ΔS has a positive sign when entropy increases and ΔH , as noted above, has a negative sign when heat is released by the system to its surroundings. Either of these conditions, both of which are typical of energetically favorable processes, tends to make ΔG negative. In fact, ΔG of a spontaneously reacting system is always negative.



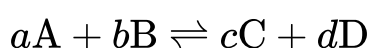
The second law of thermodynamics states that the entropy *of the universe* increases during all chemical and physical processes, but it does not require that the entropy increase take place *in the reacting system* itself. The order produced within cells as they grow and divide is more than compensated for by the disorder they create in their surroundings in the course of growth and division (see [Box 1-3](#), case 2). In short, living organisms preserve their internal order by taking from their surroundings free energy in the form of nutrients or sunlight, and returning to their surroundings an equal amount of energy as heat and entropy.

Cells are isothermal systems — they function at essentially constant temperature (and also function at constant pressure). Heat flow is not a source of energy for cells, because heat can do work only as it passes to a zone or an object at a lower

temperature. The energy that cells can and must use is free energy, described by the Gibbs free-energy function G , which allows prediction of the direction of chemical reactions, their exact equilibrium position, and the amount of work they can (in theory) perform at constant temperature and pressure. Heterotrophic cells acquire free energy from nutrient molecules, and photosynthetic cells acquire it from absorbed solar radiation. Both kinds of cells transform this free energy into ATP and other energy-rich compounds capable of providing energy for biological work at constant temperature.

Standard Free-Energy Change Is Directly Related to the Equilibrium Constant

The composition of a reacting system (a mixture of chemical reactants and products) tends to continue changing until equilibrium is reached. (In the case of an organism, equilibrium is reached only after death and complete decay.) At the equilibrium concentration of reactants and products, the rates of the forward and reverse reactions are exactly equal and no further net change occurs in the system. The concentrations of reactants and products *at equilibrium* define the equilibrium constant, K_{eq} ([p. 23](#)). In the general reaction



where a , b , c , and d are the number of molecules of A, B, C, and D participating, the equilibrium constant is given by

$$K_{\text{eq}} = \frac{[\text{C}]_{\text{eq}}^c [\text{D}]_{\text{eq}}^d}{[\text{A}]_{\text{eq}}^a [\text{B}]_{\text{eq}}^b} \quad (13-2)$$

where $[\text{A}]_{\text{eq}}$, $[\text{B}]_{\text{eq}}$, $[\text{C}]_{\text{eq}}$, and $[\text{D}]_{\text{eq}}$ are the molar concentrations of the reaction components *at the point of equilibrium*.

When a reacting system is not at equilibrium, the tendency to move toward equilibrium represents a driving force, the magnitude of which can be expressed as the free-energy change for the reaction, ΔG . Under standard conditions of temperature and pressure and when reactants and products are initially present at 1 M concentrations or, for gases, at partial pressures of 101.3 kilopascals (kPa), or 1 atm, the force driving the system toward equilibrium is defined as the standard free-energy change, ΔG° . By this definition, the standard state for reactions that involve hydrogen ions is $[\text{H}^+] = 1 \text{ M}$, or pH 0. Most biochemical reactions, however, occur in well-buffered aqueous solutions near pH 7; both the pH and the concentration of water (55.5 M) are essentially constant.

KEY CONVENTION

For convenience of calculations, biochemists define a different standard state from that used in chemistry and physics: in the biochemical standard state, $[H^+]$ is $10^{-7} M$ (pH 7) and $[H_2O]$ is 55.5 M. For reactions that involve Mg^{2+} (which include most of those with ATP as a reactant), $[Mg^{2+}]$ in solution is commonly taken to be constant at 1 mM. ■

Physical constants based on this biochemical standard state are called **standard transformed constants** and are written with a prime (such as $\Delta G'^{\circ}$ and K'_{eq}) to distinguish them from the untransformed constants used by chemists and physicists. (Note that most other textbooks use the symbol $\Delta G^{\circ'}$ rather than $\Delta G'^{\circ}$. Our use of $\Delta G'^{\circ}$, recommended by an international committee of chemists and biochemists, is intended to emphasize that the transformed free-energy change, $\Delta G'^{\circ}$, is the criterion for equilibrium.) For simplicity, we will hereafter refer to these transformed constants as **standard free-energy changes** and **standard equilibrium constants**.

KEY CONVENTION

In another simplifying convention used by biochemists, when H_2O , H^+ , and/or Mg^{2+} are reactants or products, their concentrations are not included in equations such as [Equation 13-2](#) but are instead incorporated into the constants K'_{eq} and $\Delta G'^{\circ}$. ■

Just as K'_{eq} is a physical constant characteristic for each reaction, so too is $\Delta G'^{\circ}$ a constant. As we noted in [Chapter 6](#), there is a simple relationship between K'_{eq} and $\Delta G'^{\circ}$:

$$\Delta G'^{\circ} = -RT \ln K'_{\text{eq}} \quad (13-3)$$

P1 *The standard free-energy change of a chemical reaction is simply an alternative mathematical way of expressing its equilibrium constant. [Table 13-2](#) shows the relationship between $\Delta G'^{\circ}$ and K'_{eq} . If the equilibrium constant for a given chemical reaction is 1.0, the standard free-energy change of that reaction is 0.0 (the natural logarithm of 1.0 is zero). If K'_{eq} of a reaction is greater than 1.0, its $\Delta G'^{\circ}$ is negative. If K'_{eq} is less than 1.0, $\Delta G'^{\circ}$ is positive. Because the relationship between $\Delta G'^{\circ}$ and K'_{eq} is exponential, relatively small changes in $\Delta G'^{\circ}$ correspond to large changes in K'_{eq} .*

TABLE 13-2 Relationship between Equilibrium Constants and Standard Free-Energy Changes of Chemical Reactions

K'_{eq}	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol) ^a
10^3	-17.1	-4.1
10^2	-11.4	-2.7
10^1	-5.7	-1.4

1	0.0	0.0
10^{-1}	5.7	1.4
10^{-2}	11.4	2.7
10^{-3}	17.1	4.1
10^{-4}	22.8	5.5
10^{-5}	28.5	6.8
10^{-6}	34.2	8.2

[a](#) Although joules and kilojoules are the standard units of energy and are used throughout this text, biochemists and nutritionists sometimes express $\Delta G'^{\circ}$ values in kilocalories per mole. We have therefore included values in both kilojoules and kilocalories in this table and in [Tables 13-4](#) and [13-6](#). To convert kilojoules to kilocalories, divide the number of kilojoules by 4.184.

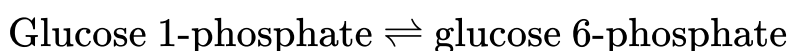
It may be helpful to think of the standard free-energy change in another way. $\Delta G'^{\circ}$ is the difference between the free-energy content of the products and the free-energy content of the reactants, under standard conditions. When $\Delta G'^{\circ}$ is negative, the products contain less free energy than the reactants and the reaction will proceed spontaneously under standard conditions; all chemical reactions tend to go in the direction that results in a decrease in the free energy of the system. A positive value of $\Delta G'^{\circ}$ means that the products of the reaction contain more free energy than the reactants, and this reaction will tend to go in the reverse direction if we start with 1.0 M concentrations of all components (standard conditions). [Table 13-3](#) summarizes these points.

TABLE 13-3 Relationships among K'_{eq} , $\Delta G'^{\circ}$, and the Direction of Chemical Reactions

When K'_{eq} is ...	$\Delta G'^{\circ}$ is ...	Starting with all components at 1 M, the reaction ...
>1.0	negative	proceeds forward
1.0	zero	is at equilibrium
<1.0	positive	proceeds in reverse

WORKED EXAMPLE 13-1 *Calculation of $\Delta G'^{\circ}$*

Calculate the standard free-energy change of the reaction catalyzed by the enzyme phosphoglucomutase,



given that, starting with 20 mM glucose 1-phosphate and no glucose 6-phosphate, the final equilibrium mixture at 25 °C and pH 7.0 contains 1.0 mM glucose 1-phosphate and 19 mM glucose 6-phosphate. Does the reaction in the direction of glucose 6-phosphate formation proceed with a loss or a gain of free energy?

SOLUTION:

First we calculate the equilibrium constant:

$$K'_{\text{eq}} = \frac{[\text{glucose 6-phosphate}]_{\text{eq}}}{[\text{glucose 1-phosphate}]_{\text{eq}}} = \frac{19 \text{ mM}}{1.0 \text{ mM}} = 19$$

We can now calculate the standard free-energy change:

$$\begin{aligned}\Delta G'^{\circ} &= -RT \ln K'_{\text{eq}} \\ &= -(8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K})(\ln 19) \\ &= -7.3 \text{ kJ/mol}\end{aligned}$$

Because the standard free-energy change is negative, the conversion of glucose 1-phosphate to glucose 6-phosphate proceeds with a loss (release) of free energy. (For the reverse reaction, $\Delta G'^{\circ}$ has the same magnitude but the *opposite* sign.)

[Table 13-4](#) gives the standard free-energy changes for some representative chemical reactions. Note that hydrolysis of simple esters, amides, peptides, and glycosides, as well as rearrangements and eliminations, proceed with relatively small standard free-energy changes, whereas hydrolysis of acid anhydrides is accompanied by relatively large decreases in standard free energy. The complete oxidation of organic compounds such as glucose or palmitate to CO_2 and H_2O , which in cells requires many steps, results in very large decreases in standard free energy. However, standard free-energy changes such as those in [Table 13-4](#) indicate how much free energy is

available from a reaction under *standard conditions*. To describe the energy released under the conditions existing in cells, an expression for the *actual* free-energy change is essential.

TABLE 13-4 Standard Free-Energy Changes of Some Chemical Reactions

Reaction type	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)
Hydrolysis reactions		
Acid anhydrides		
Acetic anhydride + H ₂ O → 2 acetate	−91.1	−21.8
ATP + H ₂ O → ADP + P _i	−30.5	−7.3
ATP + H ₂ O → AMP + PP _i	−45.6	−10.9
PP _i + H ₂ O → 2P _i	−19.2	−4.6
UDP-glucose + H ₂ O → UMP + glucose 1-phosphate	−43.0	−10.3
Esters		
Ethyl acetate + H ₂ O → ethanol + acetate	−19.6	−4.7
Glucose 6-phosphate + H ₂ O → glucose + P _i	−13.8	−3.3
Amides and peptides		
Glutamine + H ₂ O → glutamate + NH ₄ ⁺	−14.2	−3.4
Glycylglycine + H ₂ O → 2 glycine	−9.2	−2.2
Glycosides		

Maltose + H ₂ O → 2 glucose	−15.5	−3.7
Lactose + H ₂ O → glucose + galactose	−15.9	−3.8
Rearrangements		
Glucose 1-phosphate → glucose 6-phosphate	−7.3	−1.7
Fructose 6-phosphate → glucose 6-phosphate	−1.7	−0.4
Elimination of water		
Malate → fumarate + H ₂ O	3.1	0.8
Oxidations with molecular oxygen		
Glucose + 6O ₂ → 6CO ₂ + 6H ₂ O	−2,840	−686
Palmitate + 23O ₂ → 16CO ₂ + 16H ₂ O	−9,770	−2,338

Actual Free-Energy Changes Depend on Reactant and Product Concentrations

We must be careful to distinguish between two different quantities: the actual free-energy change, ΔG , and the standard free-energy change, $\Delta G'^{\circ}$. Each chemical reaction has a characteristic standard free-energy change, which may be positive, negative, or zero, depending on the equilibrium constant of the reaction.  The standard free-energy change tells us in which direction and how far a given reaction must go to reach

equilibrium when the initial concentration of each component is 1. M, the pH is 7.0, the temperature is 25 °C, and the pressure is 101.3 kPa (1 atm). Thus $\Delta G'^{\circ}$ is a constant: it has a characteristic, unchanging value for a given reaction. But the *actual* free-energy change, ΔG , is a function of reactant and product concentrations and of the temperature prevailing during the reaction, none of which will necessarily match the standard conditions as defined above. Moreover, the ΔG of any reaction proceeding spontaneously toward its equilibrium is always negative, becomes less negative as the reaction proceeds, and is zero at the point of equilibrium, indicating that no more work can be done by the reaction.

ΔG and $\Delta G'^{\circ}$ for any reaction $aA + bB \rightleftharpoons cC + dD$ are related by the equation

$$\Delta G = \Delta G'^{\circ} + RT \ln \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad (13-4)$$

in which the terms in red are those *actually prevailing* in the system under observation. The concentration terms in this equation express the effects commonly called mass action, and the term $[C]^c [D]^d / [A]^a [B]^b$ is called the mass-action ratio, Q . Thus Equation 13-4 can be expressed as $\Delta G = \Delta G'^{\circ} + RT \ln Q$. As an example, let us suppose that the reaction $A + B \rightleftharpoons C + D$ is taking place under the standard conditions of temperature (25 °C) and pressure (101.3 kPa) but that the concentrations of A, B, C,

and D are *not* equal and none of the components is present at the standard concentration of 1.0 M. To determine the actual free-energy change, ΔG , under these nonstandard conditions of concentration as the reaction proceeds from left to right, we simply enter the *actual* concentrations of A, B, C, and D in [Equation 13-4](#); the values of R , T , and $\Delta G'^{\circ}$ are the standard values. ΔG is negative and approaches zero as the reaction proceeds, because the actual concentrations of A and B decrease and the concentrations of C and D increase. Notice that when a reaction is at equilibrium — when there is no force driving the reaction in either direction and ΔG is zero — [Equation 13-4](#) reduces to

$$0 = \Delta G = \Delta G'^{\circ} + RT \ln \frac{[C]_{\text{eq}} [D]_{\text{eq}}}{[A]_{\text{eq}} [B]_{\text{eq}}}$$

or

$$\Delta G'^{\circ} = -RT \ln K'_{\text{eq}}$$

which is the equation relating the standard free-energy change and equilibrium constant ([Eqn 13-3](#)).

The criterion for spontaneity of a reaction is the value of ΔG , not $\Delta G'^{\circ}$. A reaction with a positive $\Delta G'^{\circ}$ can go in the forward

direction *if* ΔG is *negative*. This is possible if the term $RT \ln ([\text{products}]/[\text{reactants}])$ in [Equation 13-4](#) is negative and has a larger absolute value than $\Delta G'^{\circ}$. For example, the immediate removal of the products of a reaction by an enzyme that degrades the product can keep the ratio $[\text{products}]/[\text{reactants}]$ well below 1, such that the term $RT \ln ([\text{products}]/[\text{reactants}])$ has a large, negative value. This is a quantitative expression of Le Chatelier's principle. $\Delta G'^{\circ}$ and ΔG are expressions of the *maximum* amount of free energy that a given reaction can *theoretically* deliver — an amount of energy that could be realized only if a perfectly efficient device were available to trap or harness it. Given that no such device is possible (some energy is always lost to entropy during any process), the amount of work done by the reaction at constant temperature and pressure is always less than the theoretical amount.

Another important point is that some thermodynamically favorable reactions (that is, reactions for which $\Delta G'^{\circ}$ is large and negative) do not occur at measurable rates. For example, combustion of firewood to CO_2 and H_2O is very favorable thermodynamically, but firewood remains stable for years because the activation energy (see [Figs. 6-2, 6-3](#)) for the combustion reaction is higher than the energy available at room temperature. If the necessary activation energy is provided (with a lighted match, for example), combustion will begin, converting the wood to the more stable products CO_2 and H_2O and releasing energy as heat and light. The heat released by this exothermic reaction provides the activation energy for combustion of

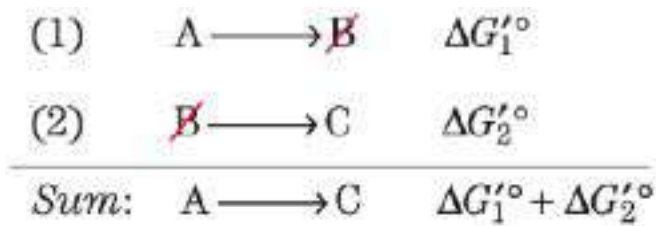
neighboring regions of the firewood; the process is self-perpetuating. Thermodynamics allows us to predict which direction a process will tend to go; how fast it will go is the subject of kinetics.

In living cells, reactions that would be extremely slow *if uncatalyzed* are caused to proceed not by supplying additional heat but by lowering the activation energy through use of an enzyme catalyst. An enzyme provides an alternative reaction pathway with a lower activation energy than the uncatalyzed reaction, so that at body temperature a large fraction of the substrate molecules have enough thermal energy to overcome the activation barrier, and the reaction rate increases dramatically. *The free-energy change for a reaction is independent of the pathway by which the reaction occurs* it depends only on the nature and concentration of the initial reactants and the final products. *Enzymes cannot, therefore, change equilibrium constants* but they can and do increase the *rate* at which a reaction proceeds in the direction dictated by thermodynamics (see [Section 6.2](#)).

Standard Free-Energy Changes Are Additive

In the case of two sequential chemical reactions, $A \rightleftharpoons B$ and $B \rightleftharpoons C$, each reaction has its own equilibrium constant and each has its characteristic standard free-energy change, $\Delta G'_1$ and $\Delta G'_2$. As the two reactions are sequential, B cancels out to give

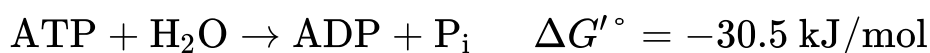
the overall reaction $A \rightleftharpoons C$, which has its own equilibrium constant and thus its own standard free-energy change, $\Delta G'_{\text{Sum}}^{\circ}$. The $\Delta G'^{\circ}$ values of sequential chemical reactions are additive. For the overall reaction $A \rightleftharpoons C$, $\Delta G'_{\text{Sum}}^{\circ}$ is the sum of the individual standard free-energy changes, $\Delta G'_1{}^{\circ}$ and $\Delta G'_2{}^{\circ}$, of the two reactions: $\Delta G'_{\text{Sum}}^{\circ} = \Delta G'_1{}^{\circ} + \Delta G'_2{}^{\circ}$.



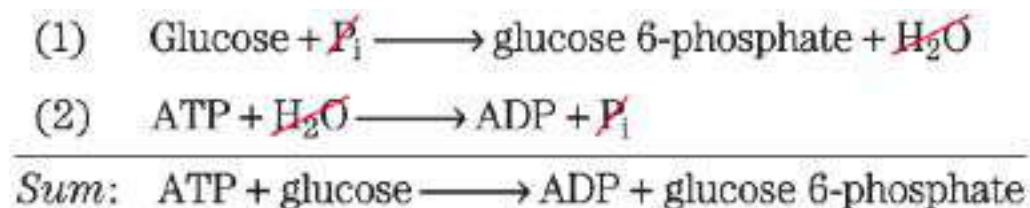
P2 This principle of bioenergetics explains how a thermodynamically unfavorable (endergonic) reaction can be driven in the forward direction by coupling it to a highly exergonic reaction. For example, in many organisms, the synthesis of glucose 6-phosphate is the first step in the utilization of glucose. In principle, the synthesis could be accomplished by this reaction:



But the positive value of $\Delta G'^{\circ}$ predicts that under standard conditions the reaction will tend not to proceed spontaneously in the direction written. Another cellular reaction, the hydrolysis of ATP to ADP and P_i , is highly exergonic:



These two reactions share the common intermediates P_i and H_2O and may be expressed as sequential reactions:



The overall standard free-energy change is obtained by adding the $\Delta G'^{\circ}$ values for individual reactions:

$$\Delta G'_{\text{Sum}} = 13.8 \text{ kJ/mol} + (-30.5 \text{ kJ/mol}) = -16.7 \text{ kJ/mol}$$

P2 The overall reaction is exergonic. In this case, energy stored in ATP is used to drive the synthesis of glucose 6-phosphate, even though its formation from glucose and inorganic phosphate (P_i) is endergonic. The *pathway* of glucose 6-phosphate formation from glucose by phosphoryl transfer from ATP is different from reactions (1) and (2), but the net result is the same as the sum of the two reactions. The standard free-energy change is a state function. In thermodynamic calculations, all that matters is the state of the system at the beginning of the process and its state at the end; the route between the initial and final states is immaterial.

We have said that $\Delta G'^{\circ}$ is a way of expressing the equilibrium constant for a reaction. For reaction (1),

$$K'_{\text{eq1}} = \frac{[\text{glucose 6-phosphate}]_{\text{eq}}}{[\text{glucose}]_{\text{eq}}[\text{P}_i]_{\text{eq}}} = 3.9 \times 10^{-3} \text{ M}^{-1}$$

Notice that H_2O is not included in this expression, as its concentration (55.5 M) is assumed to remain unchanged by the reaction. The equilibrium constant for the hydrolysis of ATP is

$$K'_{\text{eq2}} = \frac{[\text{ADP}]_{\text{eq}}[\text{P}_i]_{\text{eq}}}{[\text{ATP}]_{\text{eq}}} = 2.0 \times 10^5 \text{ M}$$

The equilibrium constant for the two coupled reactions is

$$\begin{aligned} K'_{\text{eq3}} &= \frac{[\text{glucose6-phosphate}]_{\text{eq}}[\text{ADP}]_{\text{eq}}[\text{P}_i]_{\text{eq}}}{[\text{glucose}]_{\text{eq}}[\text{P}_i]_{\text{eq}}[\text{ATP}]_{\text{eq}}} \\ &= (K'_{\text{eq1}})(K'_{\text{eq2}}) = (3.9 \times 10^{-3} \text{ M}^{-1})(2.0 \times 10^5 \text{ M}) \\ &= 7.8 \times 10^2 \end{aligned}$$

This calculation illustrates an important point about equilibrium constants: although the $\Delta G'^{\circ}$ values for two reactions that sum to a third, overall reaction are *additive*, the K'_{eq} for the overall reaction is the *product* of the individual K'_{eq} values for the two

reactions. Equilibrium constants are *multiplicative*. By coupling ATP hydrolysis to glucose 6-phosphate synthesis, the K'_{eq} for formation of glucose 6-phosphate from glucose has been raised by a factor of about 2×10^5 compared with the direct reaction between glucose and P_i .

This strategy of coupling endergonic processes to exergonic reactions that drive them is employed by all living cells in the synthesis of metabolic intermediates and cellular components. Obviously, the strategy works only if compounds such as ATP are continuously available. In the following chapters we consider several of the most important cellular pathways for producing ATP. For more practice in dealing with free-energy changes and equilibrium constants for coupled reactions, see [Worked Examples 1-1](#), [1-2](#), and [1-3](#) in [Chapter 1](#) (pp. 24–25).

SUMMARY 13.1 *Bioenergetics and Thermodynamics*

- Bioenergetics is the quantitative study of energy relationships and energy conversions in biological systems. Biological energy transformations obey the laws of thermodynamics.
- Living cells constantly perform work. They require energy for maintaining their highly organized structures, synthesizing cellular components, transporting small molecules and ions across membranes, and generating electric currents.
- All chemical reactions are influenced by two forces: the tendency to achieve the most stable bonding state (for which enthalpy, H , is a useful expression) and the tendency to achieve

the highest degree of randomness, expressed as entropy, S . The driving force in a reaction is ΔG , the free-energy change, which represents the net effect of these two factors: $\Delta G = \Delta H - T \Delta S$.

■ The standard transformed free-energy change, $\Delta G'^{\circ}$, is a physical constant that is characteristic for a given reaction and can be calculated from the equilibrium constant for the reaction: $\Delta G'^{\circ} = -RT \ln K'_{\text{eq}}$.

■ The actual free-energy change, ΔG , is a variable that depends on $\Delta G'^{\circ}$ and on the concentrations of reactants and products: $\Delta G = \Delta G'^{\circ} + RT \ln([\text{products}]/[\text{reactants}])$. When ΔG is large and negative, the reaction tends to go in the forward direction; when ΔG is large and positive, the reaction tends to go in the reverse direction; and when $\Delta G = 0$, the system is at equilibrium.

■ The free-energy change for a reaction is independent of the pathway by which the reaction occurs. Free-energy changes are additive; the net chemical reaction that results from successive reactions sharing a common intermediate has an overall free-energy change that is the sum of the ΔG values for the individual reactions.

13.2 Chemical Logic and Common Biochemical Reactions

The biological energy transductions we are concerned with in this book are chemical reactions. Cellular chemistry does not encompass every kind of reaction learned in a typical organic chemistry course. Which reactions take place in biological systems and which do not is determined by (1) their relevance to that particular metabolic system and (2) their rates. Both considerations play major roles in shaping the metabolic pathways we consider throughout the rest of the book. A relevant reaction is one that makes use of an available substrate and converts it to a useful product. However, even a potentially relevant reaction may not occur. Some chemical transformations are too slow (have activation energies that are too high) to contribute to living systems, even with the aid of powerful enzyme catalysts. The reactions that do occur in cells represent a toolbox that evolution has used to construct metabolic pathways that circumvent the “impossible” reactions. Learning to recognize the plausible reactions can be a great aid in developing a command of biochemistry.

Even so, the number of metabolic transformations taking place in a typical cell can seem overwhelming. Most cells have the capacity to carry out thousands of specific, enzyme-catalyzed reactions: for example, transformation of a simple nutrient such as glucose into amino acids, nucleotides, or lipids; extraction of

energy from fuels by oxidation; and polymerization of monomeric subunits into macromolecules.

Biochemical Reactions Occur in Repeating Patterns

To study these reactions, some organization is essential.



There are patterns within the chemistry of life; you do not need to learn every individual reaction to comprehend the molecular logic of biochemistry. Most of the reactions in living cells fall into one of five general categories: (1) reactions that make or break carbon–carbon bonds; (2) internal rearrangements, isomerizations, and eliminations; (3) free-radical reactions; (4) group transfers; and (5) oxidation-reductions. We discuss each of these in more detail below and refer to some examples of each type in later chapters. Note that the five reaction types are not mutually exclusive; for example, an isomerization reaction may involve a free-radical intermediate.

Before proceeding, however, we should review two basic chemical principles. First, a covalent bond consists of a shared pair of electrons, and the bond can be broken in two general ways ([Fig. 13-1](#)). In **homolytic cleavage**, each atom leaves the bond as a **radical**, carrying one unpaired electron. In **heterolytic cleavage**, which is more common, one atom retains both bonding electrons. The species most often generated when C—C and C—H bonds are cleaved are illustrated in [Figure 13-1](#). Carbanions,

carbocations, and hydride ions are highly unstable; this instability shapes the chemistry of these ions, as we shall see.

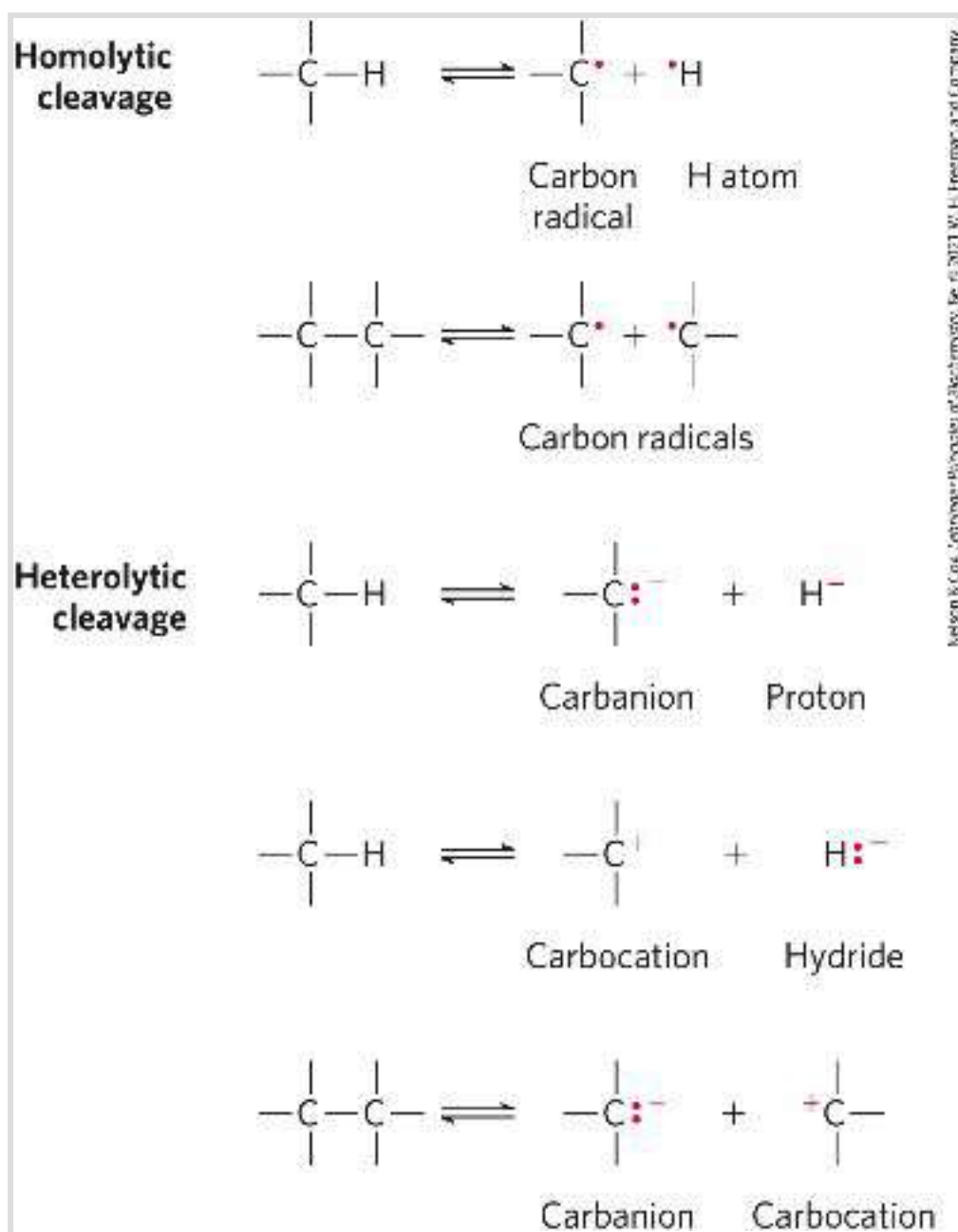


FIGURE 13-1 Two mechanisms for cleavage of a C—C or C—H bond. In a homolytic cleavage, each atom keeps one of the bonding electrons, resulting in the formation of carbon radicals (carbons having unpaired electrons) or uncharged hydrogen atoms. In a heterolytic cleavage, one of the atoms retains both bonding electrons. This can result in the formation of carbanions, carbocations, protons, or hydride ions.

The second basic principle is that many biochemical reactions involve interactions between **nucleophiles** (functional groups rich in and capable of donating electrons) and **electrophiles** (electron-deficient functional groups that seek electrons).

Nucleophiles combine with and give up electrons to electrophiles. Common biological nucleophiles and electrophiles are shown in **Figure 13-2**. Note that a carbon atom can act as either a nucleophile or an electrophile, depending on which bonds and functional groups surround it.

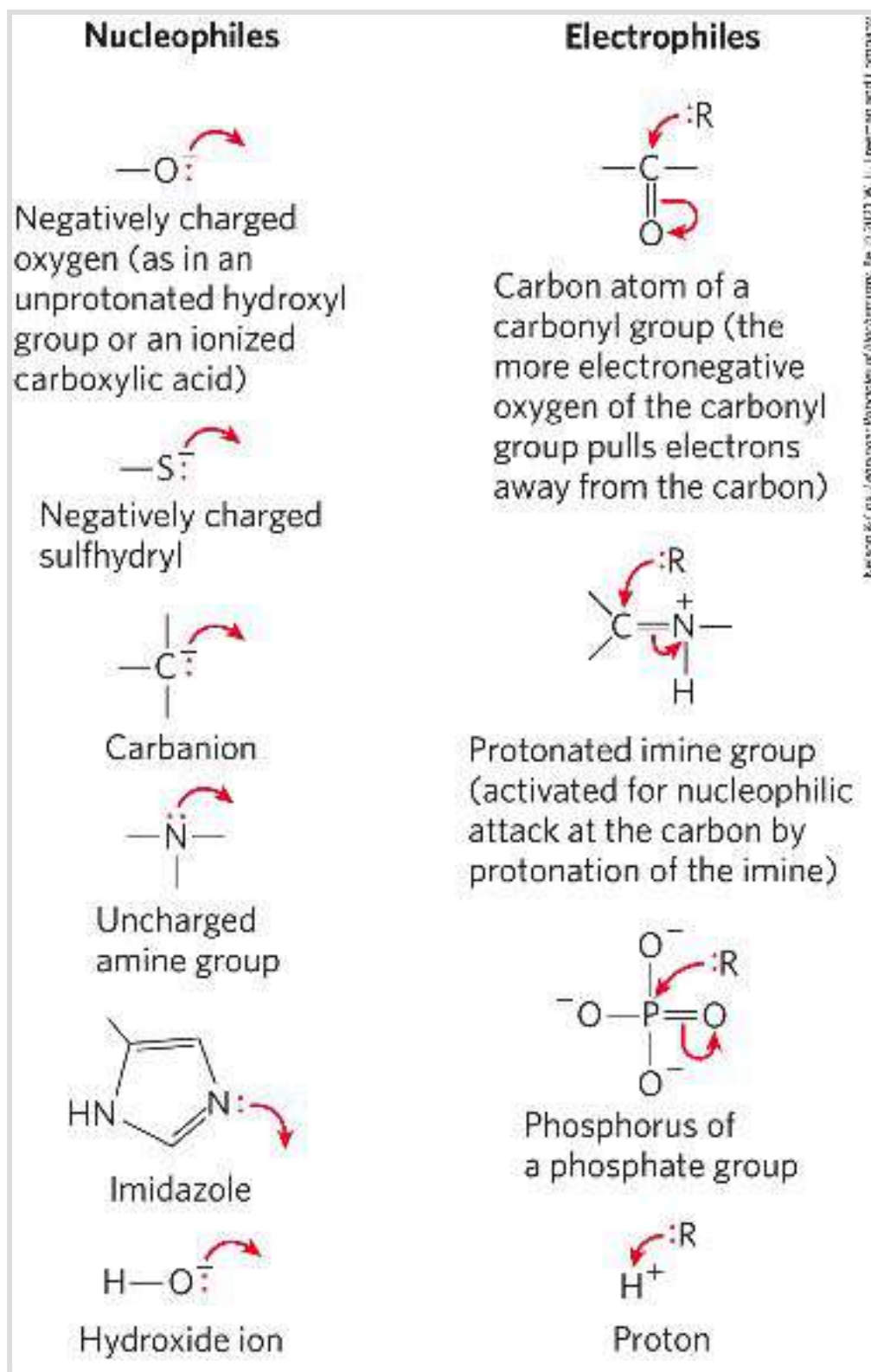


FIGURE 13-2 Common nucleophiles and electrophiles in biochemical reactions. Chemical reaction mechanisms, which trace the formation and breakage of covalent bonds, are communicated with dots and curved arrows, a convention known informally as “electron pushing.” A covalent bond consists of a shared pair of electrons. Nonbonded electrons important

to the reaction mechanism are designated by dots (:). Curved arrows () represent the movement of electron pairs. For movement of a single electron (as in a free-radical reaction), a single-headed (fishhook-type) arrow is used (). Most reaction steps involve an unshared electron pair.

Reactions That Make or Break Carbon–Carbon Bonds

Heterolytic cleavage of a C—C bond yields a [carbanion](#) and a [carbocation](#) ([Fig. 13-1](#)). Conversely, the formation of a C—C bond involves the combination of a nucleophilic carbanion and an electrophilic carbocation. Carbanions and carbocations are generally so unstable that their formation as reaction intermediates can be energetically unfeasible, even with enzyme catalysts. For the purpose of cellular biochemistry, they are impossible reactions — unless chemical assistance is provided in the form of functional groups containing electronegative atoms (O and N) that can alter the electronic structure of adjacent carbon atoms so as to stabilize and facilitate the formation of carbanion and carbocation intermediates.

Carbonyl groups are particularly important in the chemical transformations of metabolic pathways. The carbon of a carbonyl group has a partial positive charge due to the electron-withdrawing property of the carbonyl oxygen, and so is an electrophilic carbon ([Fig. 13-3a](#)). A carbonyl group can thus facilitate the formation of a carbanion on an adjoining carbon by delocalizing the carbanion's negative charge ([Fig. 13-3b](#)). An

imine group (see [Fig. 1-14](#)) can serve a similar function ([Fig. 13-3c](#)). The capacity of carbonyl and imine groups to delocalize electrons can be further enhanced by a general acid catalyst or by a metal ion (Me^{2+}) such as Mg^{2+} ([Fig. 13-3d](#)).

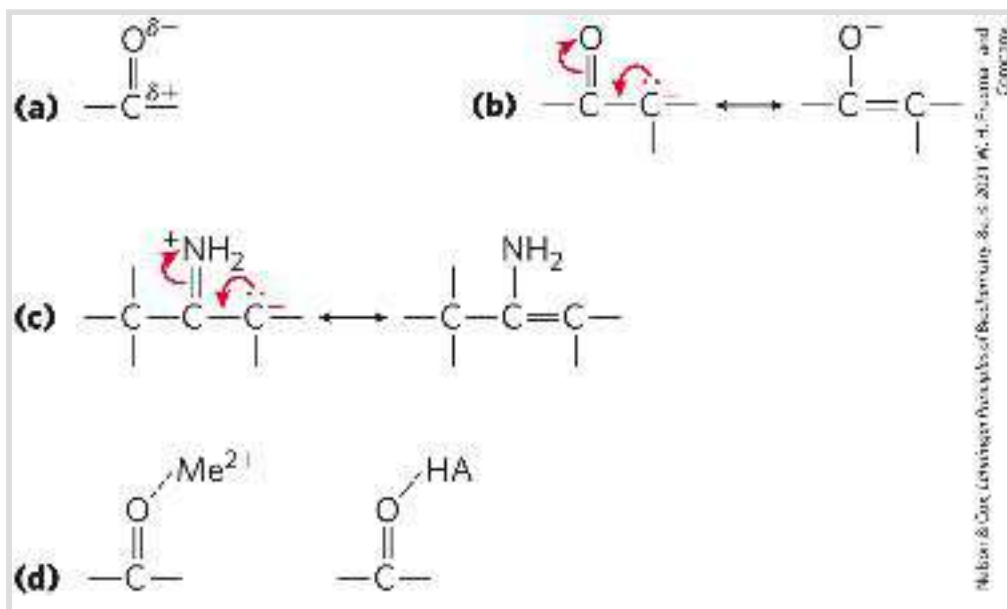


FIGURE 13-3 Chemical properties of carbonyl groups. (a) The carbon atom of a carbonyl group is an electrophile by virtue of the electron-withdrawing capacity of the electronegative oxygen atom, which results in a structure in which the carbon has a partial positive charge. (b) Within a molecule, delocalization of electrons into a carbonyl group stabilizes a carbanion on an adjacent carbon, facilitating its formation. (c) Imines function much like carbonyl groups in facilitating electron withdrawal. (d) Carbonyl groups do not always function alone; their capacity as electron sinks often is augmented by interaction with either a metal ion (Me^{2+} , such as Mg^{2+}) or a general acid (HA).

P3 The importance of a carbonyl group is evident in three major classes of reactions in which $\text{C}-\text{C}$ bonds are formed or broken ([Fig. 13-4](#)): aldol condensations, Claisen ester condensations, and decarboxylations. In each type of reaction, a

carbanion intermediate is stabilized by a carbonyl group, and in many cases another carbonyl provides the electrophile with which the nucleophilic carbanion reacts.

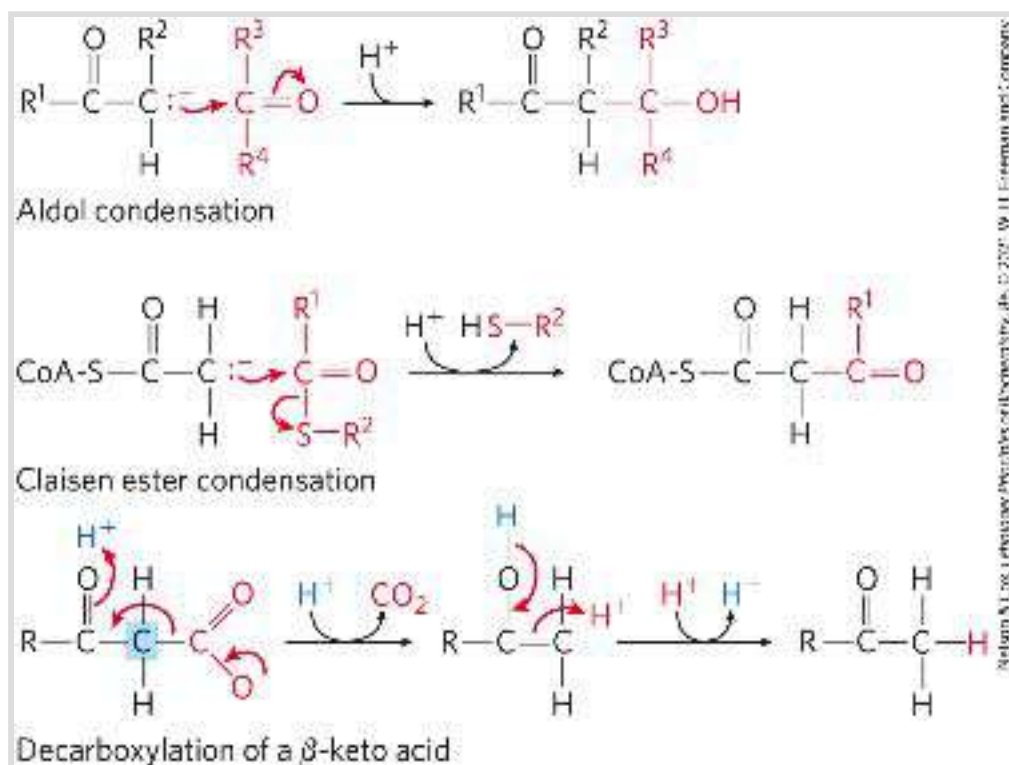


FIGURE 13-4 Some common reactions that form and break C—C bonds in biological systems. For both the aldol condensation and the Claisen condensation, a carbanion serves as nucleophile and the carbon of a carbonyl group serves as electrophile. The carbanion is stabilized in each case by another carbonyl at the adjoining carbon. In the decarboxylation reaction, a carbanion is formed on the carbon shaded blue as the CO₂ leaves. The reaction would not occur at an appreciable rate without the stabilizing effect of the carbonyl adjacent to the carbanion carbon. Wherever a carbanion is shown, a stabilizing resonance with the adjacent carbonyl, as shown in [Figure 13-3b](#), is assumed. An imine ([Fig. 13-3c](#)) or other electron-withdrawing group (including certain enzymatic cofactors such as pyridoxal) can replace the carbonyl group in the stabilization of carbanions.

An **aldol condensation** is a common route to the formation of a C—C bond; the aldolase reaction, which converts a six-carbon compound to two three-carbon compounds in glycolysis, is an aldol condensation in reverse (see [Fig. 14-5](#)). In a **Claisen condensation**, the carbanion is stabilized by the carbonyl of an adjacent thioester; an example is the synthesis of citrate in the citric acid cycle (see [Fig. 16-9](#)). Decarboxylation also commonly involves the formation of a carbanion stabilized by a carbonyl group; the acetoacetate decarboxylase reaction that occurs in the formation of ketone bodies during fatty acid catabolism provides an example (see [Fig. 17-16](#)). Entire metabolic pathways are organized around the introduction of a carbonyl group in a particular location so that a nearby carbon–carbon bond can be formed or cleaved. In some reactions, an imine or a specialized cofactor such as pyridoxal phosphate plays the electron-withdrawing role, instead of a carbonyl group.

The carbocation intermediate occurring in some reactions that form or cleave C—C bonds is generated by the elimination of an excellent leaving group, such as pyrophosphate (see “Group Transfer Reactions” below). An example is the prenyltransferase reaction ([Fig. 13-5](#)), an early step in the pathway of cholesterol biosynthesis.

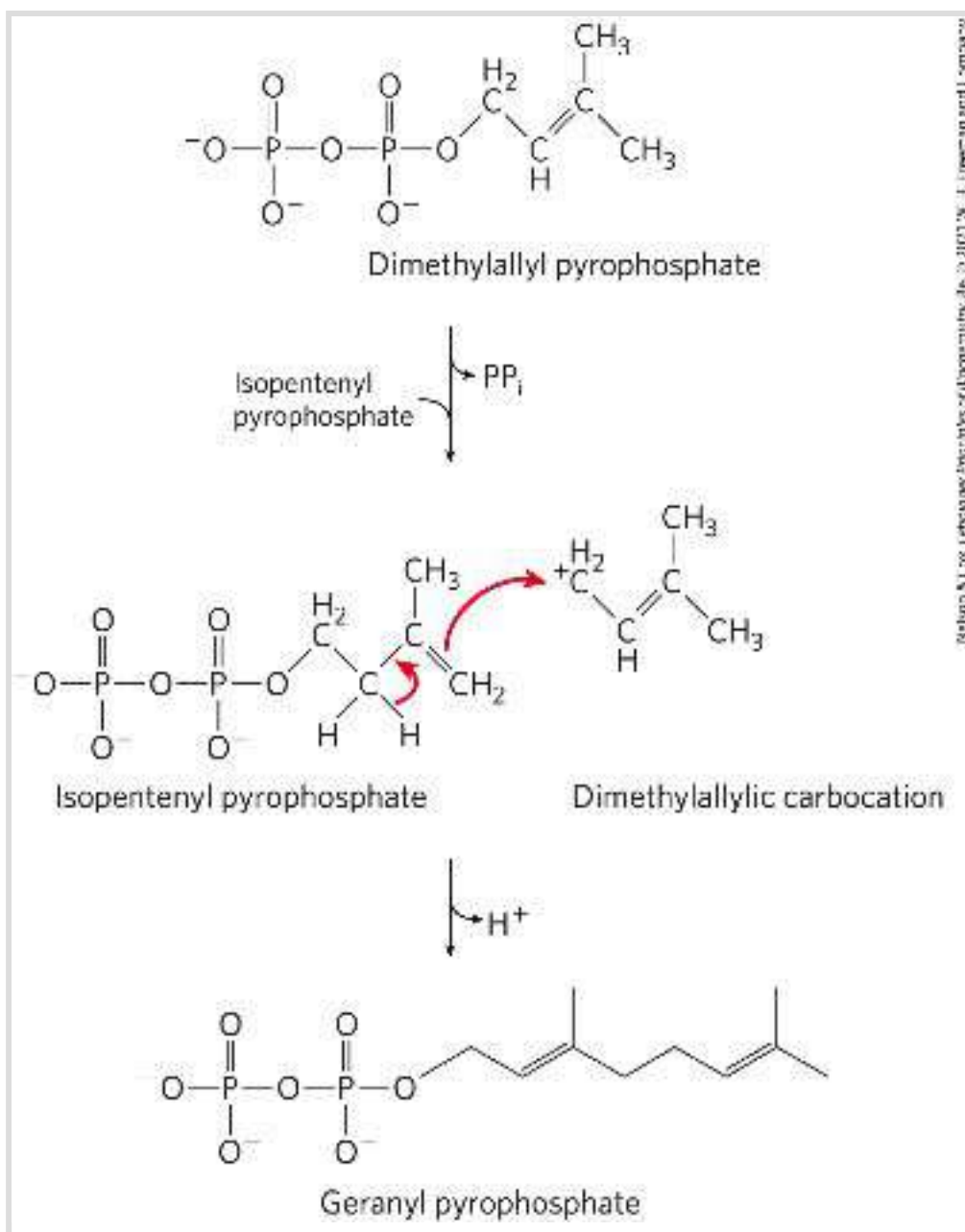


FIGURE 13-5 Carbocations in carbon–carbon bond formation. In one of the early steps in cholesterol biosynthesis, the enzyme prenyltransferase catalyzes condensation of isopentenyl pyrophosphate and dimethylallyl pyrophosphate to form geranyl pyrophosphate (see [Fig. 21-36](#)). The reaction is initiated by elimination of pyrophosphate from the dimethylallyl pyrophosphate to generate a carbocation, stabilized by resonance with the adjacent C=C bond.

Internal Rearrangements, Isomerizations, and Eliminations

Another common type of cellular reaction is an intramolecular rearrangement in which redistribution of electrons results in alterations of many different types without a change in the overall oxidation state of the molecule. For example, different groups in a molecule may undergo oxidation-reduction, with no net change in oxidation state of the molecule; groups at a double bond may undergo a cis-trans rearrangement; or the positions of double bonds may be transposed. An example of an isomerization entailing internal oxidation-reduction is the formation of fructose 6-phosphate from glucose 6-phosphate in glycolysis ([Fig. 13-6](#); this reaction is discussed in detail in [Chapter 14](#)): C-1 is reduced (aldehyde to alcohol) and C-2 is oxidized (alcohol to ketone). [Figure 13-6b](#) shows the details of the electron movements in this type of isomerization. A cis-trans rearrangement is illustrated by the prolyl cis-trans isomerase reaction in the folding of certain proteins (see p. 133). A simple transposition of a C=C bond occurs during metabolism of oleic acid, a common fatty acid (see [Fig. 17-10](#)). Some spectacular examples of double-bond repositioning occur in the biosynthesis of cholesterol (see [Fig. 21-37](#)).

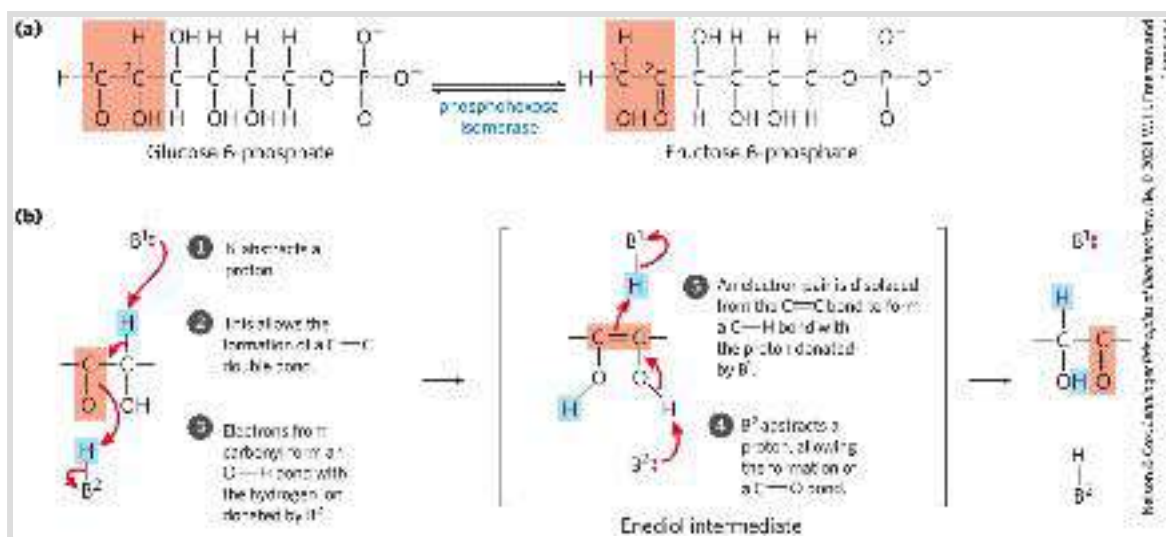
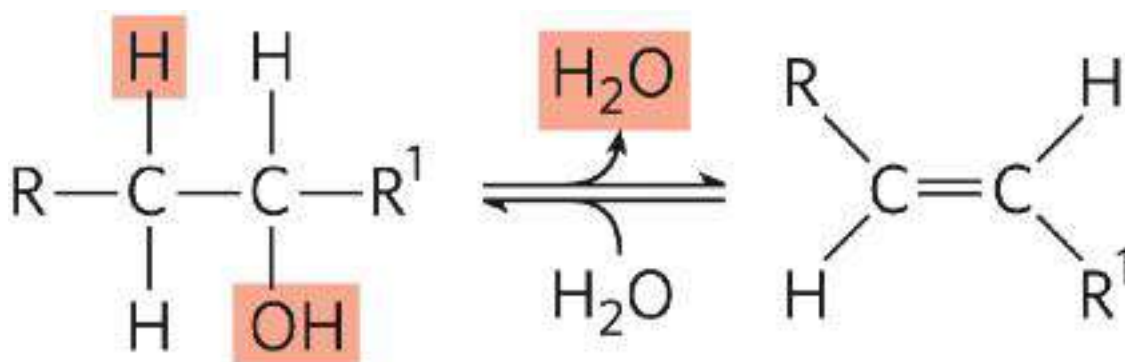


FIGURE 13-6 Isomerization and elimination reactions. (a) The conversion of glucose 6-phosphate to fructose 6-phosphate, a reaction of sugar metabolism catalyzed by phosphohexose isomerase. (b) This reaction proceeds through an enediol intermediate. Light red screens follow the path of oxidation from left to right. B^1 and B^2 are ionizable groups on the enzyme; they are capable of donating and accepting protons (acting as general acids or general bases) as the reaction proceeds.

An example of an elimination reaction that does not affect overall oxidation state is the loss of water from an alcohol, resulting in the introduction of a $C=C$ bond:



Similar reactions can result from eliminations in amines.

Free-Radical Reactions

Once thought to be rare, the homolytic cleavage of covalent bonds to generate free radicals has now been found in a wide range of biochemical processes. These include isomerizations that make use of adenosylcobalamin (vitamin B₁₂) or S-adenosylmethionine, which are initiated with a 5'-deoxyadenosyl radical (see the methylmalonyl-CoA mutase reaction in [Box 17-2](#)); certain radical-initiated decarboxylation reactions ([Fig. 13-7](#)); some reductase reactions, such as that catalyzed by ribonucleotide reductase (see [Fig. 22-43](#)); and some rearrangement reactions, such as that catalyzed by DNA photolyase (see [Fig. 25-25](#)).

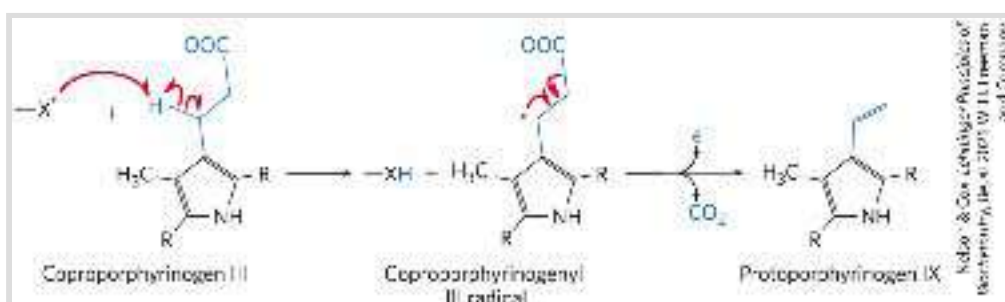
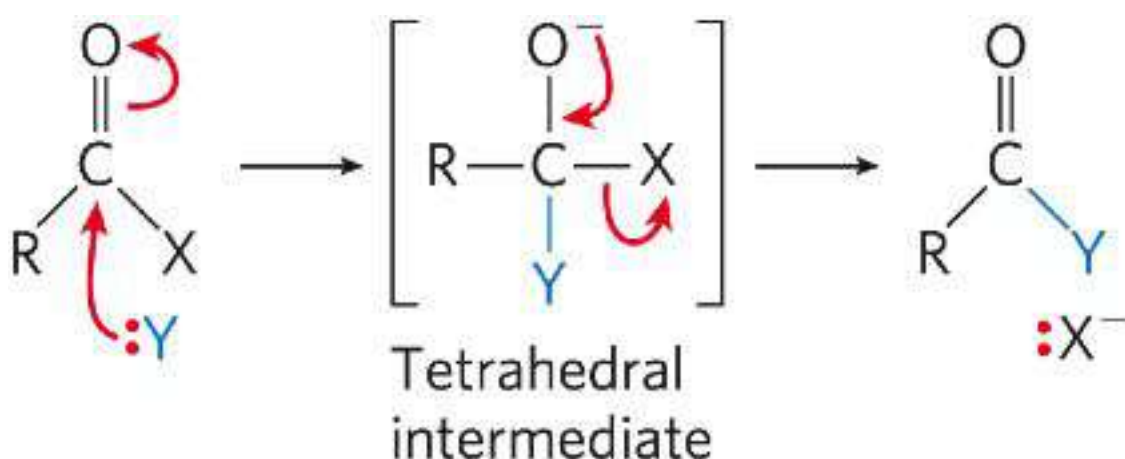


FIGURE 13-7 A free radical-initiated decarboxylation reaction. The biosynthesis of heme in *Escherichia coli* includes a decarboxylation step in which propionyl side chains on the coproporphyrinogen III intermediate are converted to the vinyl side chains of protoporphyrinogen IX. When the bacteria are grown anaerobically the enzyme oxygen-independent coproporphyrinogen III oxidase, also called HemN protein, promotes decarboxylation via the free-radical mechanism shown here. The acceptor of the released electron is not known. For simplicity, only the relevant portions of the large coproporphyrinogen III and protoporphyrinogen molecules are shown; the entire structures are given in [Figure 22-26](#). When *E. coli* is grown in the presence of oxygen, this reaction is an oxidative decarboxylation and is catalyzed by a different enzyme. [Information from G. Layer et al., *Curr. Opin. Chem. Biol.* 8:468, 2004, Fig. 4.]

Group Transfer Reactions

The transfer of acyl, glycosyl, and phosphoryl groups from one nucleophile to another is common in living cells. Acyl group transfer generally involves the addition of a nucleophile to the carbonyl carbon of an acyl group to form a tetrahedral intermediate:



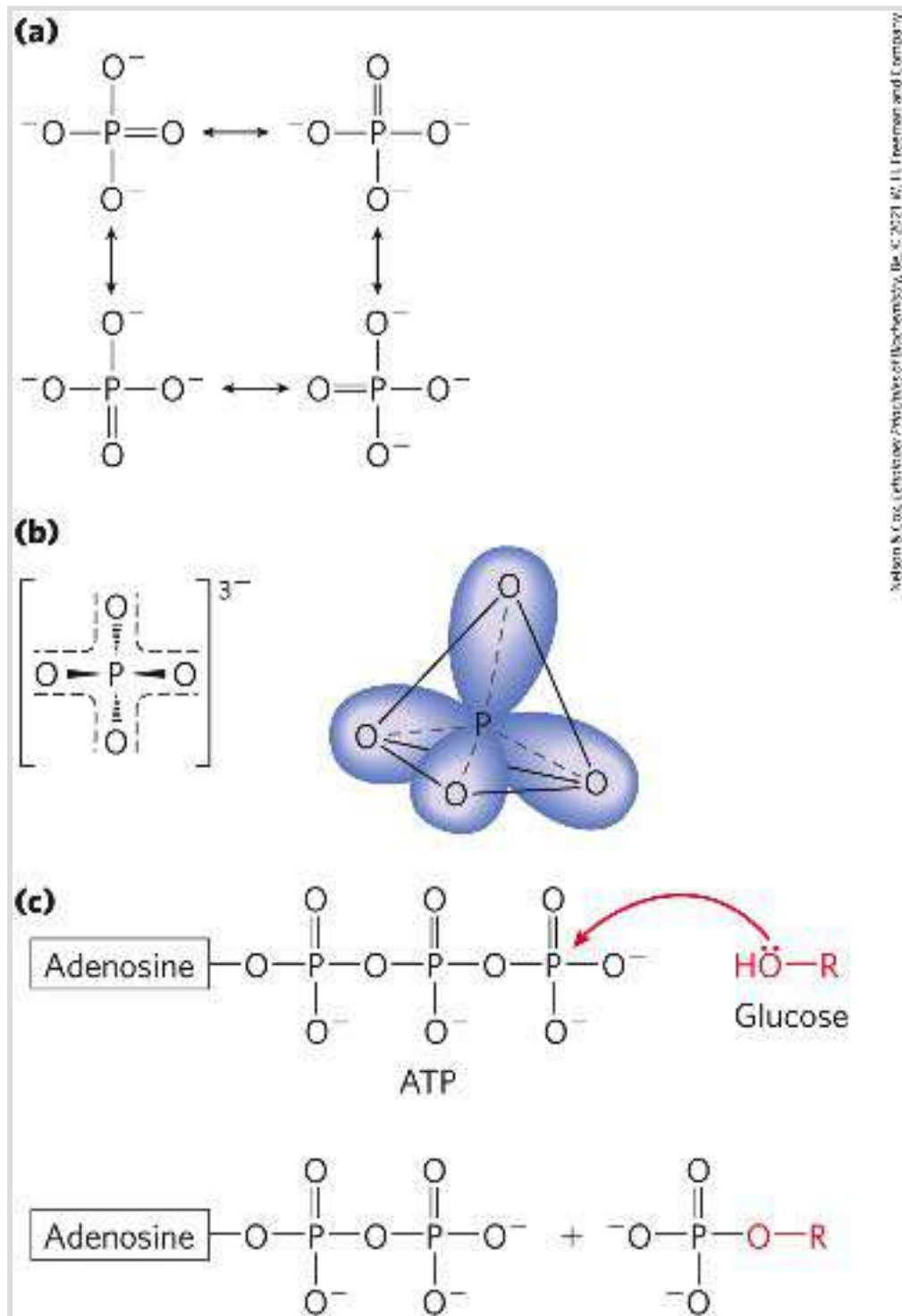
The chymotrypsin reaction is one example of acyl group transfer (see [Fig. 6-27](#)). Glycosyl group transfers involve nucleophilic substitution at C-1 of a sugar ring, which is the central atom of an acetal. In principle, the substitution could proceed by an $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ pathway.

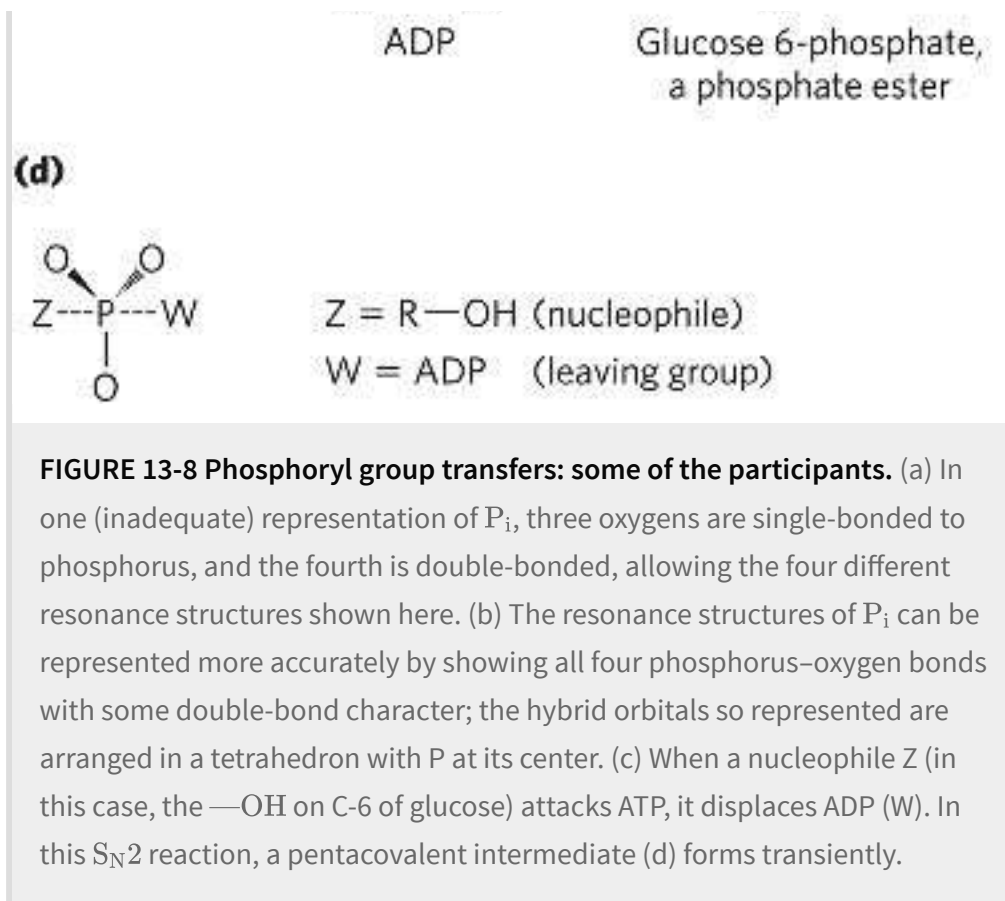
Phosphoryl group transfers play a special role in metabolic pathways, and these transfer reactions are discussed in detail in [Section 13.3](#). **P3** A general theme in metabolism is the attachment of a good leaving group to a metabolic intermediate to

“activate” the intermediate for subsequent reaction. Among the better leaving groups in nucleophilic substitution reactions are inorganic orthophosphate (the ionized form of H_3PO_4 at neutral pH, a mixture of H_2PO_4^- and HPO_4^{2-} , commonly abbreviated P_i) and inorganic pyrophosphate ($\text{P}_2\text{O}_7^{4-}$, abbreviated PP_i); esters and anhydrides of phosphoric acid are effectively activated for reaction. Nucleophilic substitution is made more favorable by the attachment of a phosphoryl group to an otherwise poor leaving group such as $-\text{OH}$. Nucleophilic substitutions in which the phosphoryl group ($-\text{PO}_3^{2-}$) serves as a leaving group occur in hundreds of metabolic reactions.

Phosphorus can form five covalent bonds. The conventional representation of P_i ([Fig. 13-8a](#)), with three $\text{P}-\text{O}$ bonds and one $\text{P}=\text{O}$ bond, is a convenient but inaccurate picture. In P_i , four equivalent phosphorus–oxygen bonds share some double-bond character, and the anion has a tetrahedral structure ([Fig. 13-8b](#)). Because oxygen is more electronegative than phosphorus, the sharing of electrons is unequal: the central phosphorus bears a partial positive charge and can therefore act as an electrophile. In a great many metabolic reactions, a phosphoryl group ($-\text{PO}_3^{2-}$) is transferred from ATP to an alcohol, forming a phosphate ester ([Fig. 13-8c](#)), or to a carboxylic acid, forming a mixed anhydride. When a nucleophile attacks the electrophilic phosphorus atom in ATP, a relatively stable pentacovalent structure forms as a reaction intermediate ([Fig. 13-8d](#)). With departure of the leaving group (ADP), the transfer of a phosphoryl group is complete. The large family of enzymes that catalyze phosphoryl group transfers

with ATP as donor are called **kinases** (Greek *kinein*, “to move”). Hexokinase, for example, “moves” a phosphoryl group from ATP to the hexose glucose. **Box 13-1** offers a primer on some of the broad classes of enzymes (including kinases) that you will encounter in your study of metabolism.





BOX 13-1

A Primer on Enzyme Names

The name **kinase** is applied to enzymes that transfer a phosphoryl group from a nucleoside triphosphate such as ATP to an acceptor molecule—a sugar (as in hexokinase and glucokinase), a protein (as in glycogen phosphorylase kinase), another nucleotide (as in nucleoside diphosphate kinase), or a metabolic intermediate such as oxaloacetate (as in PEP carboxykinase). The reaction catalyzed by a kinase is a phosphorylation. On the other hand, phosphorolysis is a displacement reaction in which phosphate is the attacking species and becomes covalently attached at the point of bond breakage. Such reactions are catalyzed by **phosphorylases**. Glycogen phosphorylase, for example, catalyzes the phosphorolysis of glycogen, producing glucose 1-phosphate.

Dephosphorylation, the removal of a phosphoryl group from a phosphate ester, is catalyzed by phosphatases, with water as the attacking species.

Fructose biphosphatase-1 converts fructose 1,6-bisphosphate to fructose 6-

phosphate in gluconeogenesis, and phosphorylase α phosphatase removes phosphoryl groups from phosphoserine in phosphorylated glycogen phosphorylase. Whew!

Citrate synthase, the first enzyme in the citric acid cycle (see [Fig. 16-7](#)), is one of many enzymes that catalyze condensation reactions, yielding a product more chemically complex than its precursors. **Synthetases** catalyze condensation reactions in which no nucleoside triphosphate (ATP, GTP, and so forth) is required as an energy source. **Synthetases** catalyze condensations that do use ATP or another nucleoside triphosphate as a source of energy for the synthetic reaction. Succinyl-CoA synthetase is such an enzyme. **Ligases** (from the Latin *ligare*, “to tie together”) are enzymes that catalyze condensation reactions in which two atoms are joined, using ATP or another energy source. (Thus, synthetases are ligases.) DNA ligase, for example, closes breaks in DNA molecules, using energy supplied by either ATP or NAD^+ ; it is widely used in joining DNA pieces for genetic engineering. Ligases are not to be confused with **lyases**, enzymes that catalyze cleavages (or, in the reverse direction, additions) in which electronic rearrangements occur. The PDH complex, which oxidatively cleaves CO_2 from pyruvate, is a member of the large class of lyases.

In some biological oxidation reactions, molecular oxygen is the electron acceptor. If the oxygen atoms *do not* appear in the oxidized product, the enzyme is an **oxidase**. If one or both of the oxygen atoms *do* appear in the oxidized product, as a new hydroxyl or carboxyl group, for example, the enzyme is an **oxygenase**. Both classes are further subdivided. **Mixed-function oxidases** oxidize two different substrates simultaneously. **Monooxygenases** and **dioxygenases** catalyze reactions in which one or two oxygen atoms, respectively, are incorporated into the organic product. These enzymes are particularly important in biosynthetic pathways of fatty acids and eicosanoids (see [Box 21-1](#)). **Dehydrogenases** catalyze oxidation-reductase reactions in which NAD^+ is electron acceptor, and molecular oxygen is generally not involved.

Unfortunately, these descriptions of enzyme types overlap, and many enzymes have two or more common names. Succinyl-CoA synthetase, for example, is also called succinate thiokinase; the enzyme is both a synthetase in the citric

acid cycle and a kinase when acting in the direction of succinyl-CoA synthesis. This raises another source of confusion in the naming of enzymes. An enzyme may have been discovered by the use of an assay in which, say, A is converted to B. The enzyme is then named for that reaction. Later work may show, however, that in the cell, the enzyme functions primarily in converting B to A. Commonly, the first name continues to be used, although the metabolic role of the enzyme would be better described by naming it for the reverse reaction. The glycolytic enzyme pyruvate kinase illustrates this situation (p. 521). To a beginner in biochemistry, this duplication in nomenclature can be bewildering. International committees have made heroic efforts to systematize the nomenclature of enzymes (see [Table 6-3](#) for a brief summary of the system), but some systematic names have proved too long and cumbersome and are not frequently used in biochemical conversation.

We have tried throughout this book to use the enzyme name most commonly employed by working biochemists and to point out cases in which an enzyme has more than one widely used name.

Phosphoryl groups are not the only groups that activate molecules for reaction. Thioalcohols (thiols), in which the oxygen atom of an alcohol is replaced with a sulfur atom, are also good leaving groups. Thiols activate carboxylic acids by forming thioesters (thiol esters). In later chapters we discuss several reactions, including those catalyzed by the fatty acyl synthases in lipid synthesis (see [Fig. 21-2](#)), in which nucleophilic substitution at the carbonyl carbon of a thioester results in transfer of the acyl group to another moiety.

Oxidation-Reduction Reactions

We will encounter carbon atoms in five oxidation states, depending on the elements with which they share electrons ([Fig. 13-9](#)), and transitions between these states are of crucial importance in metabolism (oxidation-reduction reactions are the topic of [Section 13.4](#)). In many biological oxidations, a compound loses two electrons and two hydrogen ions (that is, two hydrogen atoms); these reactions are commonly called dehydrogenations, and the enzymes that catalyze them are called dehydrogenases ([Fig. 13-10](#)). In some, but not all, biological oxidations, a carbon atom becomes covalently bonded to an oxygen atom. The enzymes that catalyze oxidations with oxygen as electron acceptor are generally called oxidases or, if the oxygen atom is derived directly from molecular oxygen (O_2) and incorporated into the product, oxygenases.

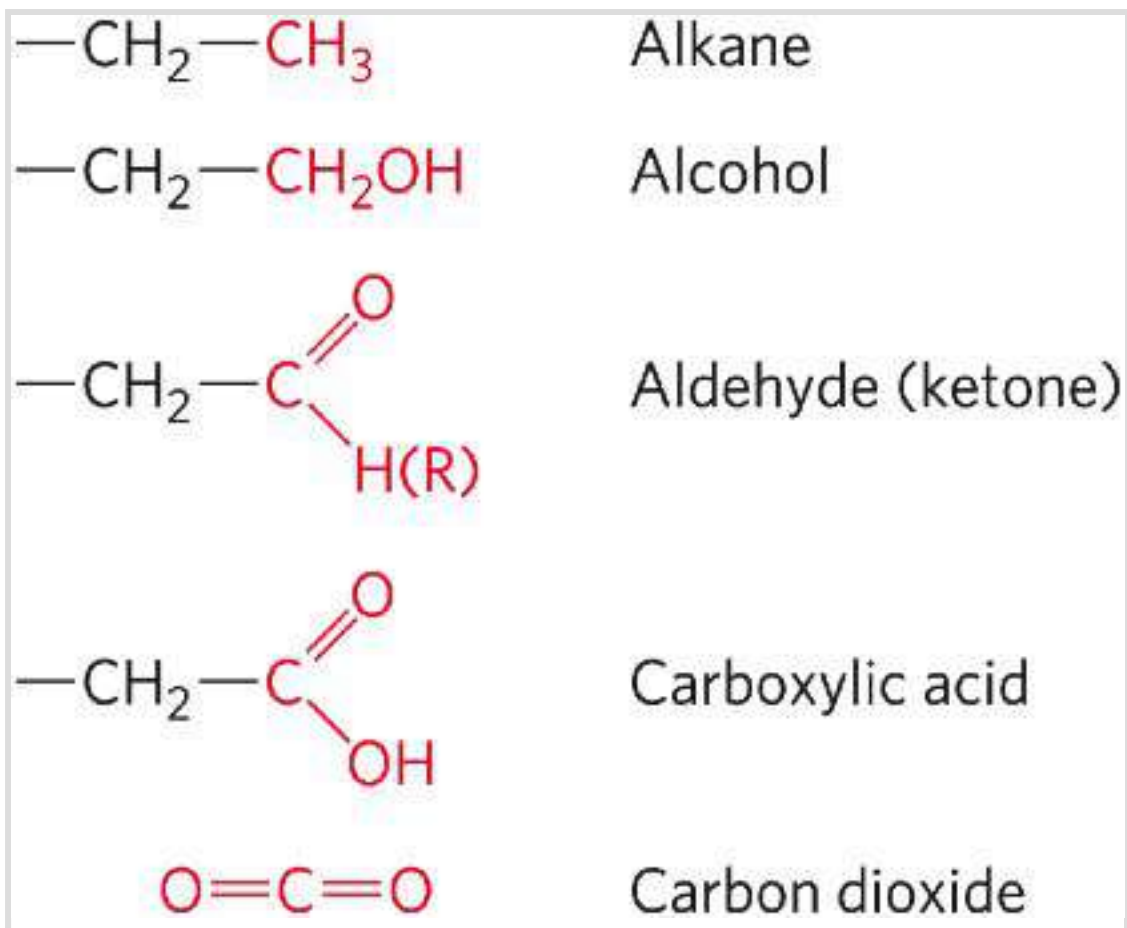


FIGURE 13-9 The oxidation levels of carbon in biomolecules. Each compound is formed by oxidation of the carbon shown in red in the compound immediately above. Carbon dioxide is the most highly oxidized form of carbon found in living systems.

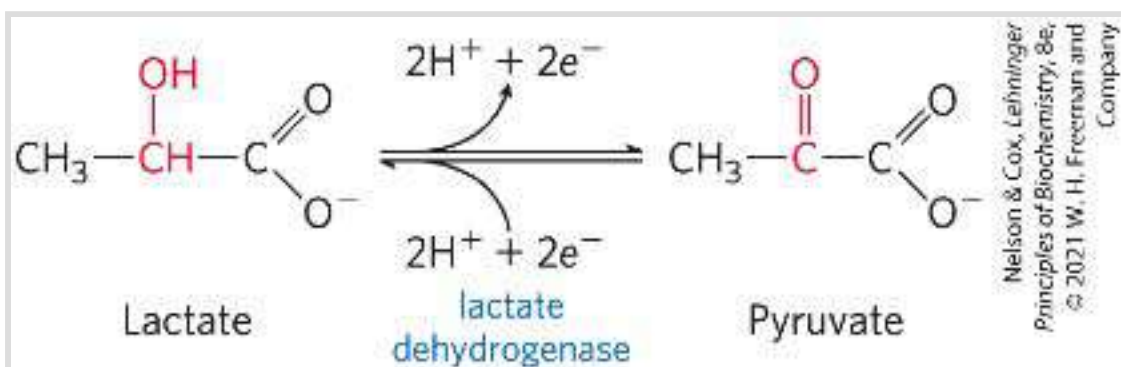


FIGURE 13-10 An oxidation-reduction reaction. Shown here is the oxidation of lactate to pyruvate. In this dehydrogenation, two electrons and two hydrogen ions (the equivalent of two hydrogen atoms) are removed from C-2 of lactate, an alcohol, to form pyruvate, a ketone. In cells the reaction is catalyzed by lactate

dehydrogenase and the electrons are transferred to the cofactor nicotinamide adenine dinucleotide (NAD^+). This reaction is fully reversible; pyruvate can be reduced by electrons transferred from the cofactor.

Every oxidation must be accompanied by a reduction, in which an electron acceptor acquires the electrons removed by oxidation. Oxidation reactions generally release energy (think of campfires: the compounds in wood are oxidized by oxygen molecules in the air).  Most living cells obtain the energy needed for cellular work by oxidizing metabolic fuels such as carbohydrates or fat (photosynthetic organisms can also trap and use the energy of sunlight). The catabolic (energy-yielding) pathways described in [Chapters 14](#) through 19 are oxidative reaction sequences that result in the transfer of electrons from fuel molecules, through a series of electron carriers, to oxygen. The high affinity of O_2 for electrons makes the overall electron-transfer process highly exergonic, providing the energy that drives ATP synthesis — the central goal of catabolism.

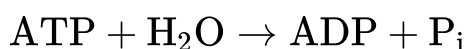
Many of the reactions within these five classes are facilitated by cofactors, in the form of coenzymes and metal ions (vitamin B_{12} , *S*-adenosylmethionine, folate, nicotinamide, and Fe^{2+} are some examples). Cofactors bind to enzymes — in some cases reversibly, in other cases almost irreversibly — and give them the capacity to promote a particular kind of chemistry ([p. 178](#)). Most cofactors participate in a narrow range of closely related reactions. In the following chapters, we introduce and discuss each important cofactor at the point where we first encounter its function. The

cofactors provide another way to organize the study of biochemical processes, given that the reactions facilitated by a given cofactor generally are mechanistically related.

Biochemical and Chemical Equations Are Not Identical

Biochemists write metabolic equations in a simplified way, and this is particularly evident for reactions involving ATP.

Phosphorylated compounds can exist in several ionization states and, as we have noted, the different species can bind Mg^{2+} . For example, at pH 7 and 2 mM Mg^{2+} , ATP exists as an equilibrium distribution of the forms ATP^{4-} , HATP^{3-} , $\text{H}_2\text{ATP}^{2-}$, MgHATP^- , and Mg_2ATP . In thinking about the biological role of ATP, however, we are not always interested in all this detail, and so we consider ATP as an entity made up of a sum of species, and we write its hydrolysis as the biochemical equation



where ATP, ADP, and P_i are sums of species. The corresponding standard transformed equilibrium constant,

$K'_{\text{eq}} = [\text{ADP}]_{\text{eq}}[\text{P}_i]_{\text{eq}}/[\text{ATP}]_{\text{eq}}$, depends on the pH and the concentration of free Mg^{2+} . Note that H^+ and Mg^{2+} do not appear in the biochemical equation, because their concentrations are not significantly changed by the reaction. Thus a biochemical

equation does not necessarily balance H, Mg, or charge, although it does balance all other elements involved in the reaction (C, N, O, and P in the equation above).

We can write a chemical equation that *does* balance for all elements and for charge. For example, when ATP is hydrolyzed at a pH above 8.5 in the absence of Mg^{2+} , the chemical reaction is represented by



The corresponding equilibrium constant,

$$K_{\text{eq}} = [\text{ADP}^{3-}]_{\text{eq}}[\text{HPO}_4^{2-}]_{\text{eq}}[\text{H}^+]_{\text{eq}}/[\text{ATP}^{4-}]_{\text{eq}},$$

depends only on temperature, pressure, and ionic strength.

Both ways of writing a metabolic reaction have value in biochemistry. Chemical equations are needed when we want to account for all atoms and charges in a reaction, as when we are considering the mechanism of a chemical reaction. Biochemical equations are used to determine in which direction a reaction will proceed spontaneously, given a specified pH and $[\text{Mg}^{2+}]$, or to calculate the equilibrium constant of such a reaction.

Throughout this book we use biochemical equations, unless the focus is on chemical mechanism, and we use values of $\Delta G'^{\circ}$ and K'_{eq} as determined at pH 7 and 1 mM Mg^{2+} .

SUMMARY 13.2 *Chemical Logic and Common Biochemical Reactions*

■ Living systems make use of a large number of chemical reactions that can be classified into five general types: reactions that make or break carbon–carbon bonds; internal rearrangements and eliminations; free-radical reactions; group transfers; and oxidation-reduction reactions. Heterolytic cleavages occur often in reactions that make or break C—C bonds.

■ Carbonyl groups play a special role in reactions that form or cleave C—C bonds. Carbanion intermediates are common and are stabilized by adjacent carbonyl groups or, less often, by imines or certain cofactors.

■ A redistribution of electrons can produce internal rearrangements, isomerizations, and eliminations. Such reactions include intramolecular oxidation-reduction, change in cis-trans arrangement at a double bond, and transposition of double bonds.

■ Homolytic cleavage of covalent bonds to generate free radicals occurs in some pathways.

■ Phosphoryl transfer reactions are an especially important type of group transfer in cells, required for the activation of molecules for reactions that would otherwise be highly unfavorable.

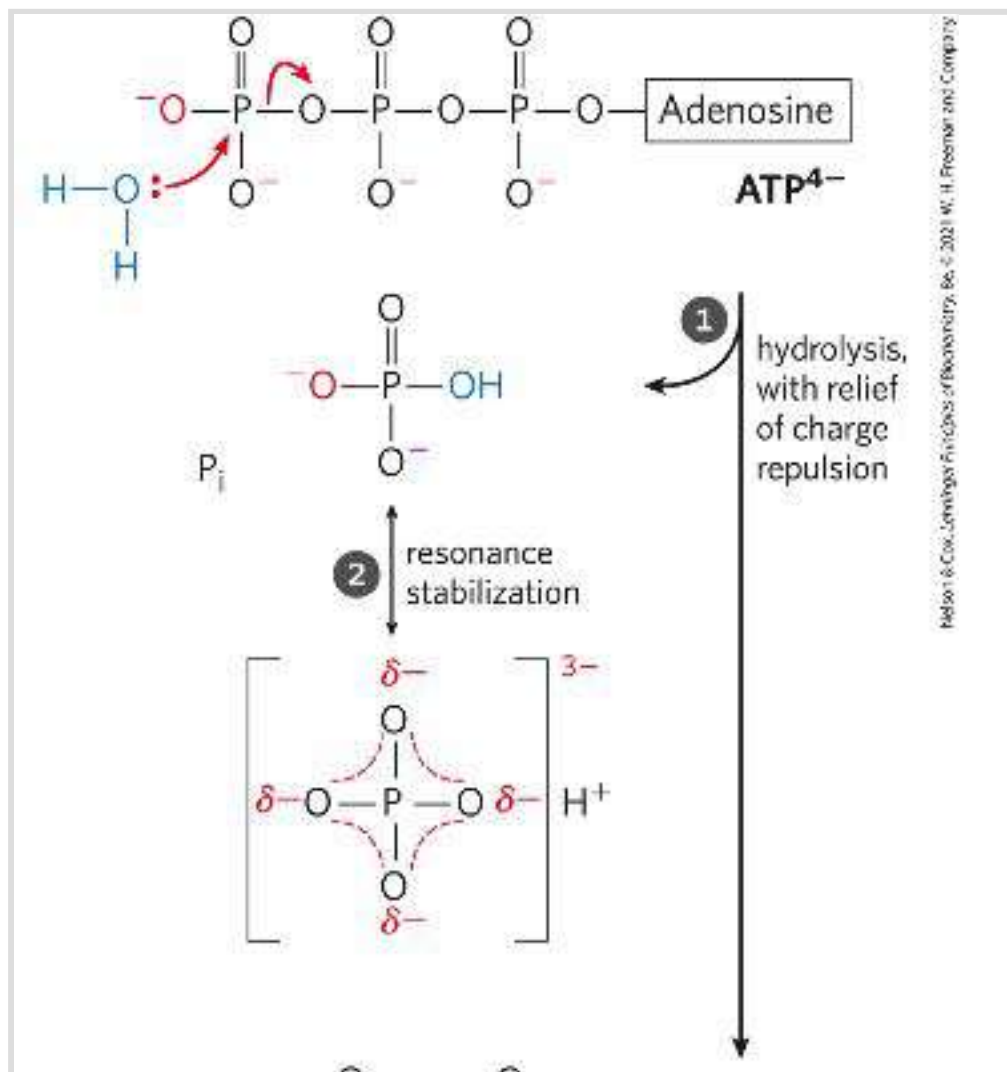
- Oxidation-reduction reactions involve the loss or gain of electrons: one reactant gains electrons and is reduced, while the other loses electrons and is oxidized. Oxidation reactions generally release energy and are important in catabolism.
- Biochemists often write reaction equations that are not balanced for H^+ and don't attempt to describe the state of phosphate ionization.

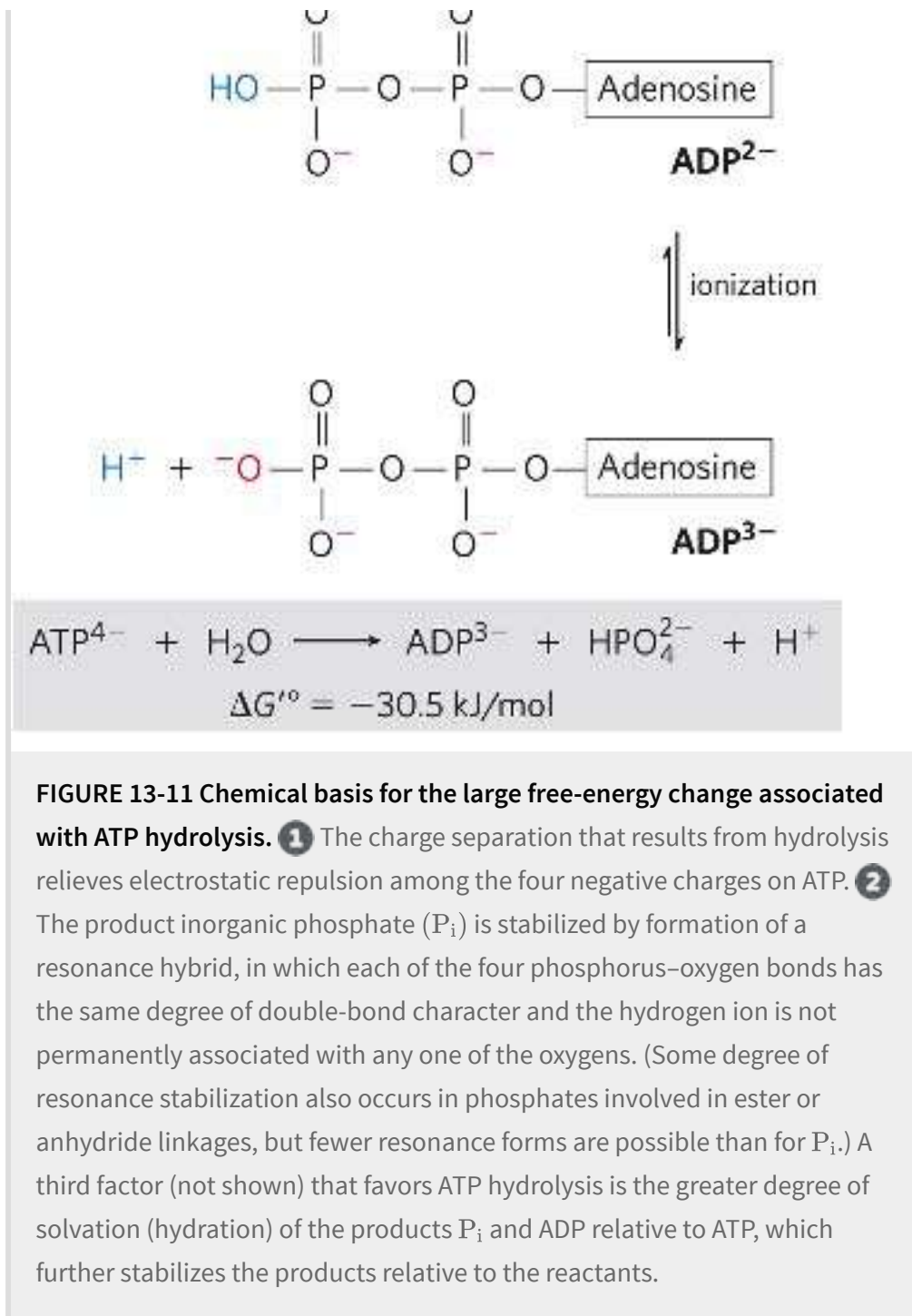
13.3 Phosphoryl Group Transfers and ATP

Having developed some fundamental principles of energy changes in chemical systems and reviewed the common classes of reactions, we can now examine the energy cycle in cells and the special role of ATP as the energy currency that links catabolism and anabolism (see [Fig. 1-28](#)). Heterotrophic cells obtain free energy in a chemical form by the catabolism of nutrient molecules, and they use that energy to make ATP from ADP and P_i . ATP then donates some of its chemical energy to endergonic processes such as the synthesis of metabolic intermediates and macromolecules from smaller precursors, the transport of substances across membranes against concentration gradients, and mechanical motion. This donation of energy from ATP generally involves the covalent participation of ATP in the reaction that is to be driven, with the eventual result that ATP is converted to ADP and P_i or, in some reactions, to AMP and $2 P_i$. We discuss here the chemical basis for the large free-energy changes that accompany hydrolysis of ATP and other high-energy phosphate compounds, and we show that most cases of energy donation by ATP involve group transfer, not simple hydrolysis of ATP. To illustrate the range of energy transductions in which ATP provides the energy, we consider the synthesis of information-rich macromolecules, the transport of solutes across membranes, and motion produced by muscle contraction.

The Free-Energy Change for ATP Hydrolysis Is Large and Negative

Figure 13-11 summarizes the chemical basis for the relatively large, negative, standard free energy of hydrolysis of ATP. The hydrolytic cleavage of the terminal phosphoric acid anhydride (phosphoanhydride) bond in ATP separates one of the three negatively charged phosphates and thus relieves some of the internal electrostatic repulsion in ATP; the P_i released is stabilized by the formation of several resonance forms not possible in ATP.





The free-energy change for ATP hydrolysis is -30.5 kJ/mol under standard conditions, but the *actual* free energy of hydrolysis (ΔG) of ATP in living cells is very different: the cellular concentrations of ATP, ADP, and P_i are not identical and are much lower than the 1.0 M of standard conditions ([Table 13-5](#)). Furthermore, Mg^{2+} in

the cytosol binds to ATP and ADP ([Fig. 13-12](#)), and for most enzymatic reactions that involve ATP as phosphoryl group donor, the true substrate is MgATP^{2-} . The relevant $\Delta G'^{\circ}$ is therefore that for MgATP^{2-} hydrolysis. We can calculate ΔG for ATP hydrolysis using data such as those in [Table 13-5](#). The actual free energy of hydrolysis of ATP under intracellular conditions is often called its **phosphorylation potential**, ΔG_p , for reasons we will explain.

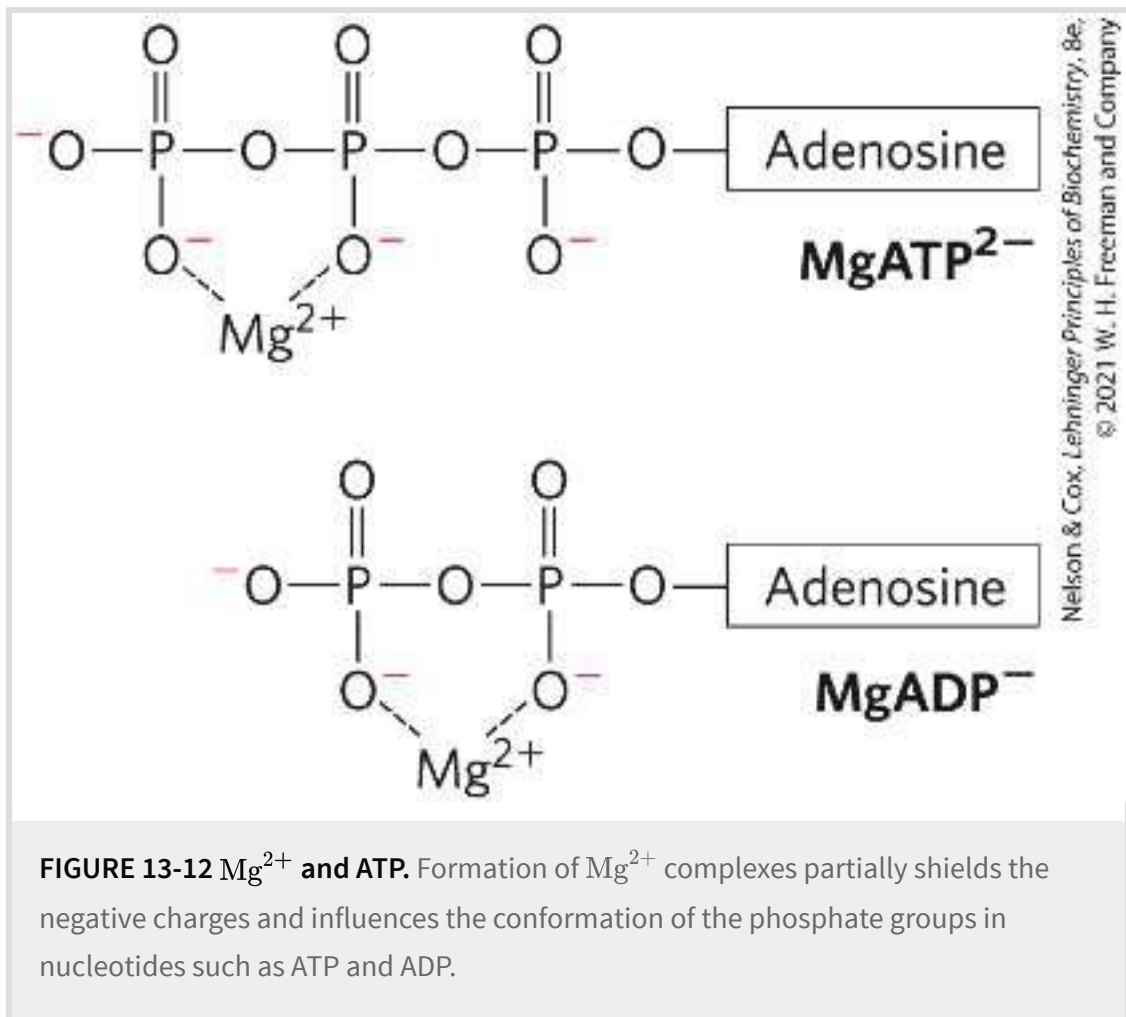
TABLE 13-5 Total Concentrations of Adenine Nucleotides, Inorganic Phosphate, and Phosphocreatine in Some Cells

Cell type	Concentration (mM) ^a				
	ATP	ADP ^b	AMP	P _i	PCr
Rat hepatocyte	3.38	1.32	0.29	4.8	0
Rat myocyte	8.05	0.93	0.04	8.05	27
Rat neuron	2.59	0.73	0.06	2.72	4.7
Human erythrocyte	2.25	0.25	0.02	1.65	0
<i>E. coli</i> cell	9.6	0.56	0.28	—	—

^a For erythrocytes, the concentrations are those of the cytosol (human erythrocytes lack a nucleus and mitochondria). In the other types of cells, the data are for the entire cell contents, although the cytosol and the mitochondria have very different concentrations of ADP. PCr is phosphocreatine, discussed on [p. 487](#).

^b This value reflects total concentration; the true value for free ADP may be much lower ([Worked Example 13-2](#)).

Mammalian data from R. L. Veech et al., *J. Biol. Chem.* 254:6538, 1979. *E. coli* data from B. D. Bennett et al., *Nat. Chem. Biol.* 5:593, 2009.



Because the concentrations of ATP, ADP, and P_i differ from one cell type to another, ΔG_p for ATP likewise differs among cells. Moreover, in any given cell, ΔG_p can vary from time to time, depending on the metabolic conditions and how they influence the concentrations of ATP, ADP, P_i , and H^+ (pH). We can calculate the actual free-energy change for any given metabolic reaction as it occurs in a cell, provided we know the concentrations of all the reactants and products and other factors (such as pH, temperature, and $[\text{Mg}^{2+}]$) that may affect the actual free-energy change.

WORKED EXAMPLE 13-2 *Calculation of ΔG_p*

Calculate the actual free energy of hydrolysis of ATP, ΔG_p , in human erythrocytes. The standard free energy of hydrolysis of ATP is -30.5 kJ/mol, and the concentrations of ATP, ADP, and P_i in erythrocytes are as shown in [Table 13-5](#). Assume that the pH is 7.0 and the temperature is 37°C (human body temperature). What does this reveal about the amount of energy required to *synthesize* ATP under the same cellular conditions?

SOLUTION:

The concentrations of ATP, ADP, and P_i in human erythrocytes are 2.25, 0.25, and 1.65 mM, respectively. The actual free energy of hydrolysis of ATP under these conditions is given by the relationship (see [Eqn 13-4](#))

$$\Delta G_p = \Delta G'^{\circ} + RT \ln \frac{[\text{ADP}][P_i]}{[\text{ATP}]}$$

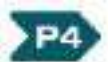
Substituting the appropriate values, we get

$$\begin{aligned}
\Delta G_p &= -30.5 \text{ kJ/mol} + \left[(8.315 \text{ J/mol} \cdot \text{K})(310 \text{ K}) \ln \frac{(0.25 \times 10^{-3})}{(2.5 \times 10^{-3})} \right] \\
&= -30.5 \text{ kJ/mol} + (2.58 \text{ kJ/mol}) \ln 1.8 \times 10^{-4} \\
&= -30.5 \text{ kJ/mol} + (2.58 \text{ kJ/mol})(-8.6) \\
&= -30.5 \text{ kJ/mol} - 22 \text{ kJ/mol} \\
&= -52 \text{ kJ/mol}
\end{aligned}$$

Thus ΔG_p , the actual free-energy change for ATP hydrolysis in the intact erythrocyte (-52 kJ/mol), is much larger than the standard free-energy change (-30.5 kJ/mol). Similarly, the free energy required to *synthesize* ATP from ADP and P_i under the conditions prevailing in the erythrocyte would be 52 kJ/mol .

To further complicate the issue, the *total* concentrations of ATP, ADP, and P_i (and H^+) in a cell — such as the values given in [Table 13-5](#) — may be substantially higher than the *free* concentrations, which are the thermodynamically relevant values. The difference is due to tight binding of ATP, ADP, and P_i to cellular proteins. For example, the free [ADP] in resting muscle has been variously estimated at between 1 and $37 \mu\text{M}$. Using the value $25 \mu\text{M}$ in [Worked Example 13-2](#), we would get a ΔG_p of -58 kJ/mol . Calculation of the exact value of ΔG_p , however, is perhaps less instructive than the generalization we can make about actual free-energy changes: *in vivo*, the energy released by ATP hydrolysis is greater than the standard free-energy change, $\Delta G'^{\circ}$.

In the following discussions we use the $\Delta G'^{\circ}$ value for ATP hydrolysis because this allows comparisons, on the same basis, with the energetics of other cellular reactions. Always keep in mind, however, that in living cells ΔG is the relevant quantity — for ATP hydrolysis and all other reactions — and may be quite different from $\Delta G'^{\circ}$.

Here we must make an important point about cellular ATP levels. We have shown (and will discuss further) how the chemical properties of ATP make it a suitable form of energy currency in cells. But it is not merely the molecule's intrinsic chemical properties that give it this ability to drive metabolic reactions and other energy-requiring processes. Even more important is that,  in the course of evolution, there has been a very strong selective pressure for regulatory mechanisms that *hold cellular ATP concentrations far above the equilibrium concentrations* for the hydrolysis reaction. When the ATP level drops, not only does the *amount* of fuel decrease, but the fuel itself *loses its potency* ΔG for its hydrolysis (that is, its phosphorylation potential, ΔG_p) is diminished. As our discussions of the metabolic pathways that produce and consume ATP will show, living cells have developed elaborate mechanisms — often at what might seem to us the expense of efficiency — to maintain high concentrations of ATP.

Other Phosphorylated Compounds and Thioesters Also Have Large,

Negative Free Energies of Hydrolysis

ATP is not the only biological compound with a large, negative free energy of hydrolysis. [Table 13-6](#) lists the standard free energies of hydrolysis for some biologically important phosphorylated compounds. In all of these phosphate-releasing reactions, a few of which we describe below, the several resonance forms available to P_i ([Fig. 13-11](#)) stabilize this product relative to the reactant, contributing to an already negative free-energy change.

TABLE 13-6 Standard Free Energies of Hydrolysis of Some Phosphorylated Compounds and Acetyl-CoA (a Thioester)

Compounds and Acetyl-CoA	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)
Phosphoenolpyruvate	−61.9	−14.8
1,3-Bisphosphoglycerate ($\rightarrow$ 3-phosphoglycerate + P_i)	−49.3	−11.8
Phosphocreatine	−43.0	−10.3
ADP ($\rightarrow$ AMP + P_i)	−32.8	−7.8
ATP ($\rightarrow$ ADP + P_i)	−30.5	−7.3
ATP ($\rightarrow$ AMP + PP_i)	−45.6	−10.9
AMP ($\rightarrow$ adenosine + P_i)	−14.2	−3.4
PP_i ($\rightarrow$ 2 P_i)	−19.2	−4.0
Glucose 3-phosphate	−20.9	−5.0

Fructose 6-phosphate	−15.9	−3.8
Glucose 6-phosphate	−13.8	−3.3
Glycerol 3-phosphate	−9.2	−2.2
Acetyl-CoA	−31.4	−7.5

Data mostly from W. P. Jencks, in *Handbook of Biochemistry and Molecular Biology*, 3rd edn (G. D. Fasman, ed.), *Physical and Chemical Data*, Vol. 1, p. 296, CRC Press, 1976. Value for the free energy of hydrolysis of PP_i from P. A. Frey and A. Arabshahi, *Biochemistry* 34:11,307, 1995.

Phosphoenolpyruvate (PEP; [Fig. 13-13](#)), a central intermediate in the energy-conserving process of glycolysis ([Chapter 14](#)), contains a phosphate ester bond that undergoes hydrolysis to yield the enol form of pyruvate, and this direct product can tautomerize to the more stable keto form. Because the reactant (PEP) has only one form (enol) and the product (pyruvate) has two possible forms, the reaction occurs with a gain in entropy, and the product is therefore stabilized relative to the reactant. This is a major contributor to the high standard free energy of hydrolysis of phosphoenolpyruvate: $\Delta G'^{\circ} = -61.9 \text{ kJ/mol}$. A second contributor is the greater resonance stabilization in the P_i released by cleavage of PEP.

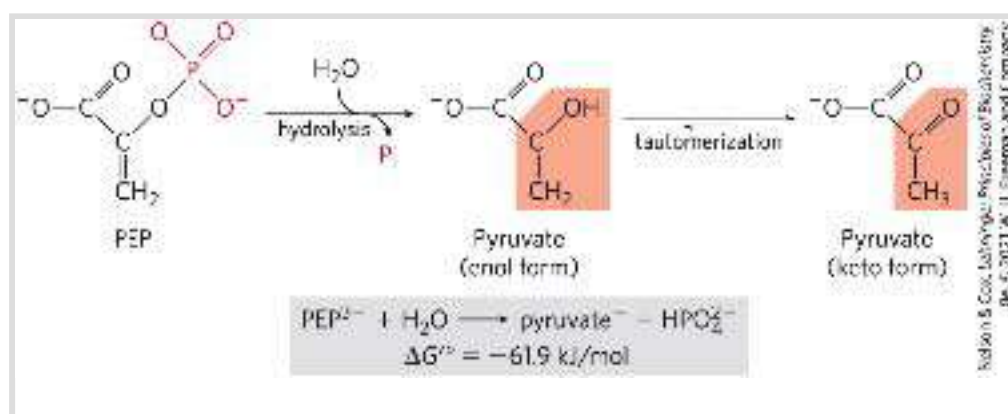


FIGURE 13-13 Hydrolysis of phosphoenolpyruvate (PEP). Catalyzed by pyruvate kinase, this reaction is followed by spontaneous tautomerization of the product, pyruvate. Tautomerization is not possible in PEP, and thus the products of hydrolysis are stabilized relative to the reactants. Resonance stabilization of P_i also occurs, as shown in [Figure 13-11](#).

Another intermediate in glycolysis, the three-carbon compound 1,3-bisphosphoglycerate ([Fig. 13-14](#)), contains an anhydride bond between the C-1 carboxyl group and phosphoric acid. Hydrolysis of this acyl phosphate is accompanied by a large, negative, standard free-energy change ($\Delta G'^{\circ} = -49.3 \text{ kJ/mol}$), which can, again, be explained in terms of the structure of reactant and products. When H_2O is added across the anhydride bond of 1,3-bisphosphoglycerate, one of the direct products, 3-phosphoglyceric acid, can lose a proton to give the carboxylate ion, 3-phosphoglycerate, which has two equally probable resonance forms. Removal of the direct product (3-phosphoglyceric acid) by its further metabolism and formation of the resonance-stabilized ion both favor the forward reaction.

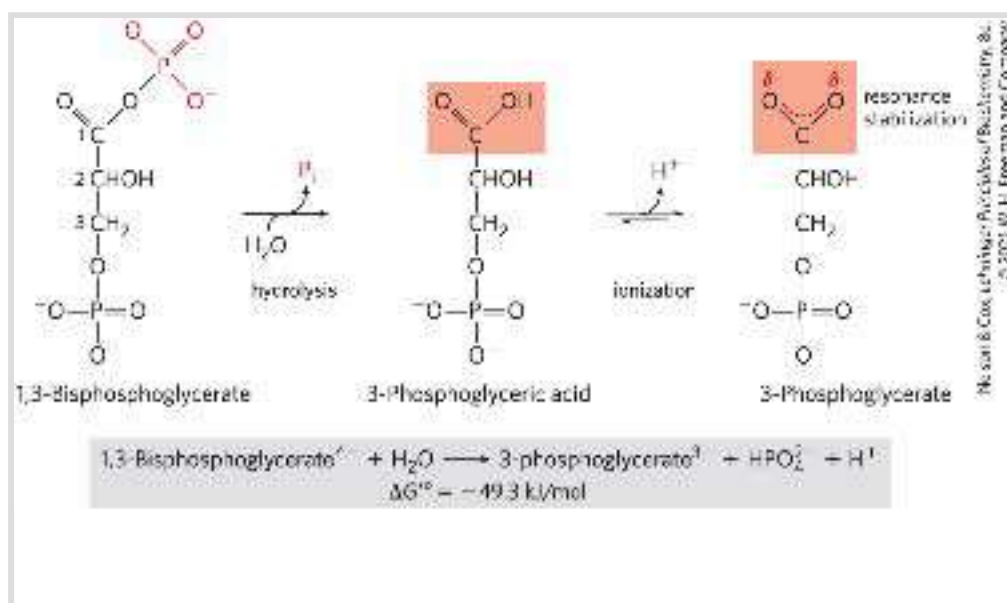


FIGURE 13-14 Hydrolysis of 1,3-bisphosphoglycerate. The direct product of hydrolysis is 3-phosphoglyceric acid, with an undissociated carboxylic acid. Its dissociation allows resonance structures that stabilize the product relative to the reactants. Resonance stabilization of P_i further contributes to the negative free-energy change.

In phosphocreatine ([Fig. 13-15](#)), which is used in muscle tissue to replenish ATP after its use in contraction, the P—N bond can be hydrolyzed to generate free creatine and P_i . The release of P_i and the resonance stabilization of creatine favor the forward reaction. The standard free-energy change of phosphocreatine hydrolysis is therefore large: -43.0 kJ/mol .

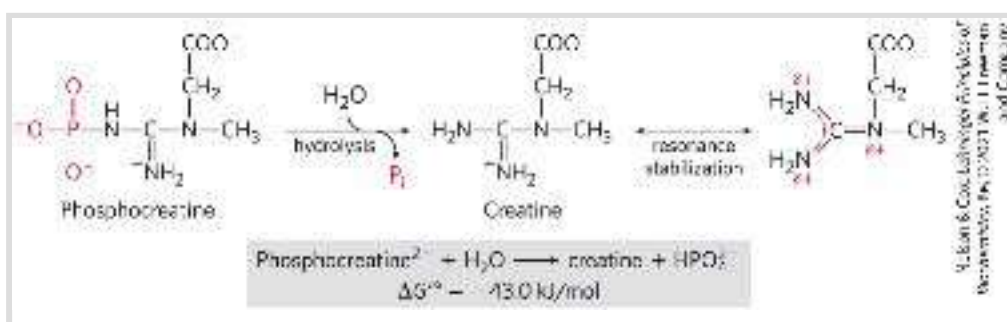


FIGURE 13-15 Hydrolysis of phosphocreatine. Breakage of the P—N bond in phosphocreatine produces creatine, which is stabilized by formation of a resonance hybrid. The other product, P_i , is also resonance stabilized.

Thioesters, in which a sulfur atom replaces the usual oxygen in the ester bond, also have large, negative, standard free energies of hydrolysis. Acetyl-coenzyme A, or acetyl-CoA ([Fig. 13-16](#)), is one of many thioesters important in metabolism. The acyl group in these compounds is activated for transacylation and condensation. Thioesters undergo much less resonance stabilization than do oxygen esters; consequently, the difference

in free energy between the reactant and its hydrolysis products, which *are* resonance-stabilized, is greater for thioesters than for comparable oxygen esters ([Fig. 13-17](#)). In both cases, hydrolysis of the ester generates a carboxylic acid, which can ionize and assume several resonance forms. Together, these factors result in the large, negative $\Delta G'^{\circ}$ (-31.4 kJ/mol) for acetyl-CoA hydrolysis.

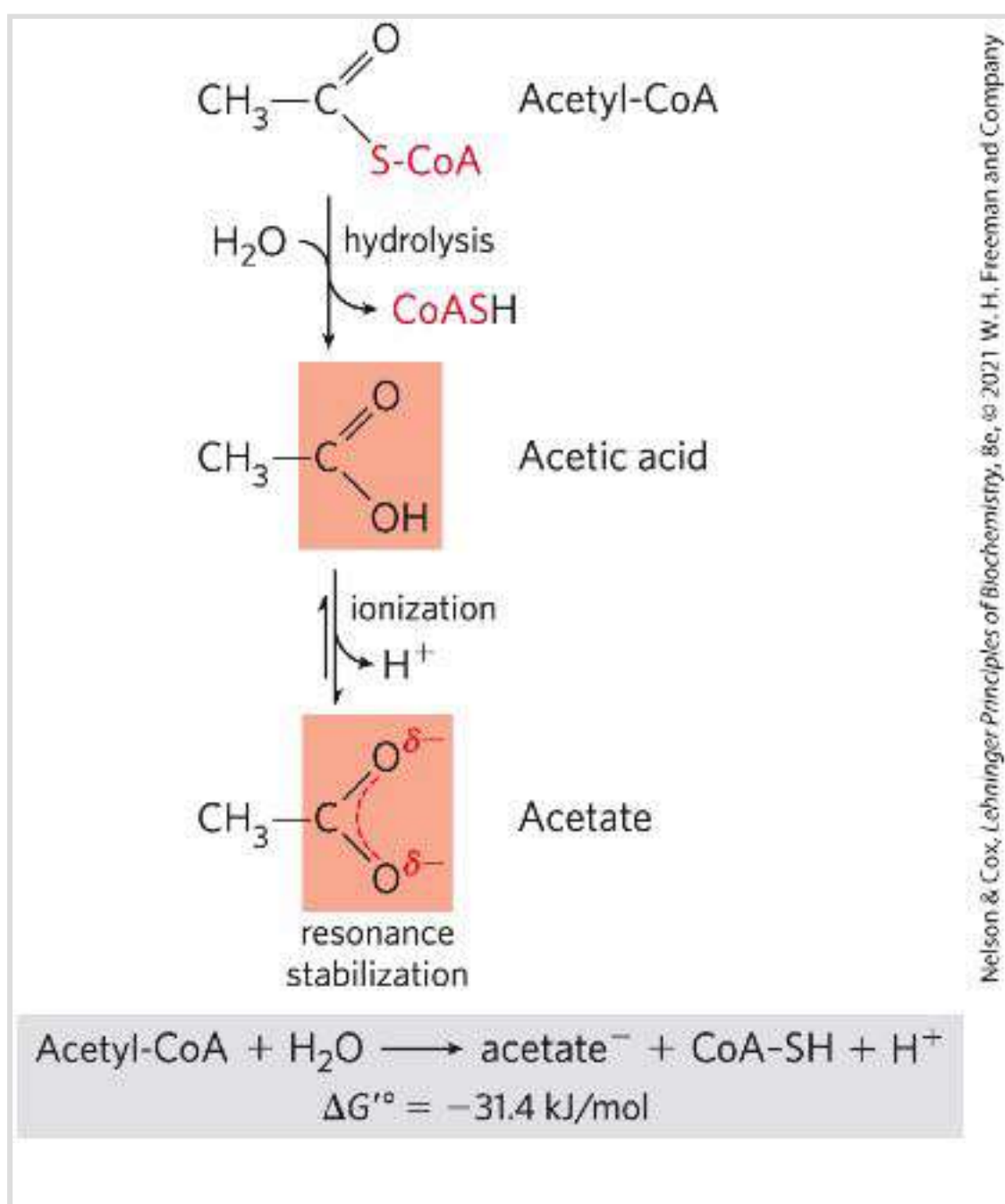


FIGURE 13-16 Hydrolysis of acetyl-coenzyme A. Acetyl-CoA is a thioester with a large, negative, standard free energy of hydrolysis. Thioesters contain a sulfur atom in the position occupied by an oxygen atom in oxygen esters. The complete structure of coenzyme A (CoA, or CoASH) is shown in [Figure 8-41](#).

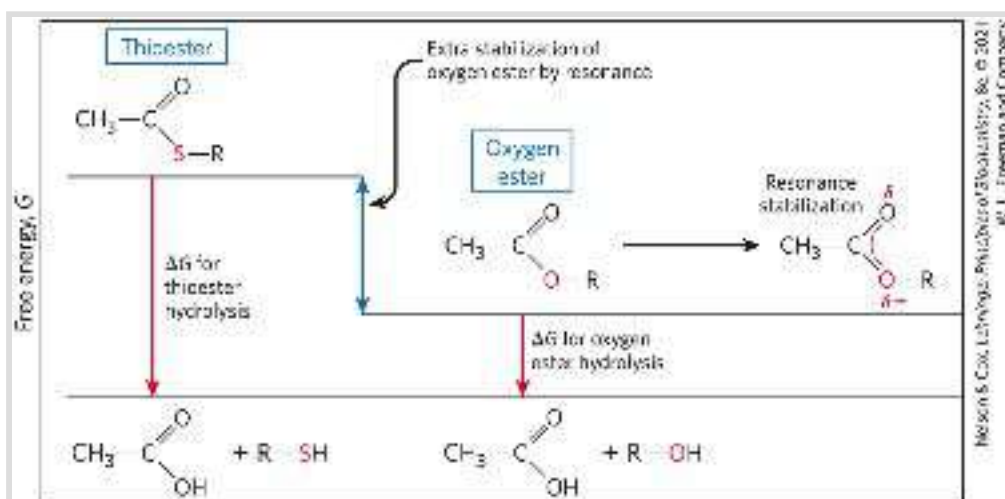


FIGURE 13-17 Free energy of hydrolysis for thioesters and oxygen esters.

The *products* of both types of hydrolysis reaction have about the same free-energy content (G), but the thioester has a higher free-energy content than the oxygen ester. Orbital overlap between the O and C atoms allows resonance stabilization in oxygen esters; orbital overlap between S and C atoms is poorer and provides little resonance stabilization.

To summarize, for hydrolysis reactions with large, negative, standard free-energy changes, the products are more stable than the reactants for one or more of the following reasons: (1) the bond strain in reactants due to electrostatic repulsion is relieved by charge separation, as for ATP; (2) the products are stabilized by ionization, as for ATP, acyl phosphates, and thioesters; (3) the products are stabilized by isomerization (tautomerization), as for PEP; and/or (4) the products are stabilized by resonance, as for creatine released from phosphocreatine, carboxylate ion released

from acyl phosphates and thioesters, and phosphate (P_i) released from anhydride or ester linkages.

ATP Provides Energy by Group Transfers, Not by Simple Hydrolysis

Throughout this book you will encounter reactions or processes for which ATP supplies energy, and the contribution of ATP to these reactions is commonly indicated as in [Figure 13-18a](#), with a single arrow showing the conversion of ATP to ADP and P_i (or, in some cases, of ATP to AMP and pyrophosphate, PP_i). When written this way, these reactions of ATP seem to be simple hydrolysis reactions in which water displaces P_i (or PP_i), and one is tempted to say that an ATP-dependent reaction is “driven by the hydrolysis of ATP.” This is *not* the case. ATP hydrolysis per se usually accomplishes nothing but the liberation of heat, which cannot drive a chemical process in an isothermal system. A single reaction arrow such as that in [Figure 13-18a](#) almost invariably represents a two-step process ([Fig. 13-18b](#)) in which part of the ATP molecule, a phosphoryl or pyrophosphoryl group or the adenylate moiety (AMP), is first transferred to a substrate molecule or to an amino acid residue in an enzyme, becoming covalently attached to the substrate or the enzyme and raising its free-energy content (activating it). Then, in a second step, the phosphate-containing moiety transferred in the first step is displaced, generating P_i , PP_i , or AMP as the leaving group. Thus,

ATP participates *covalently* in the enzyme-catalyzed reaction to which it contributes free energy.

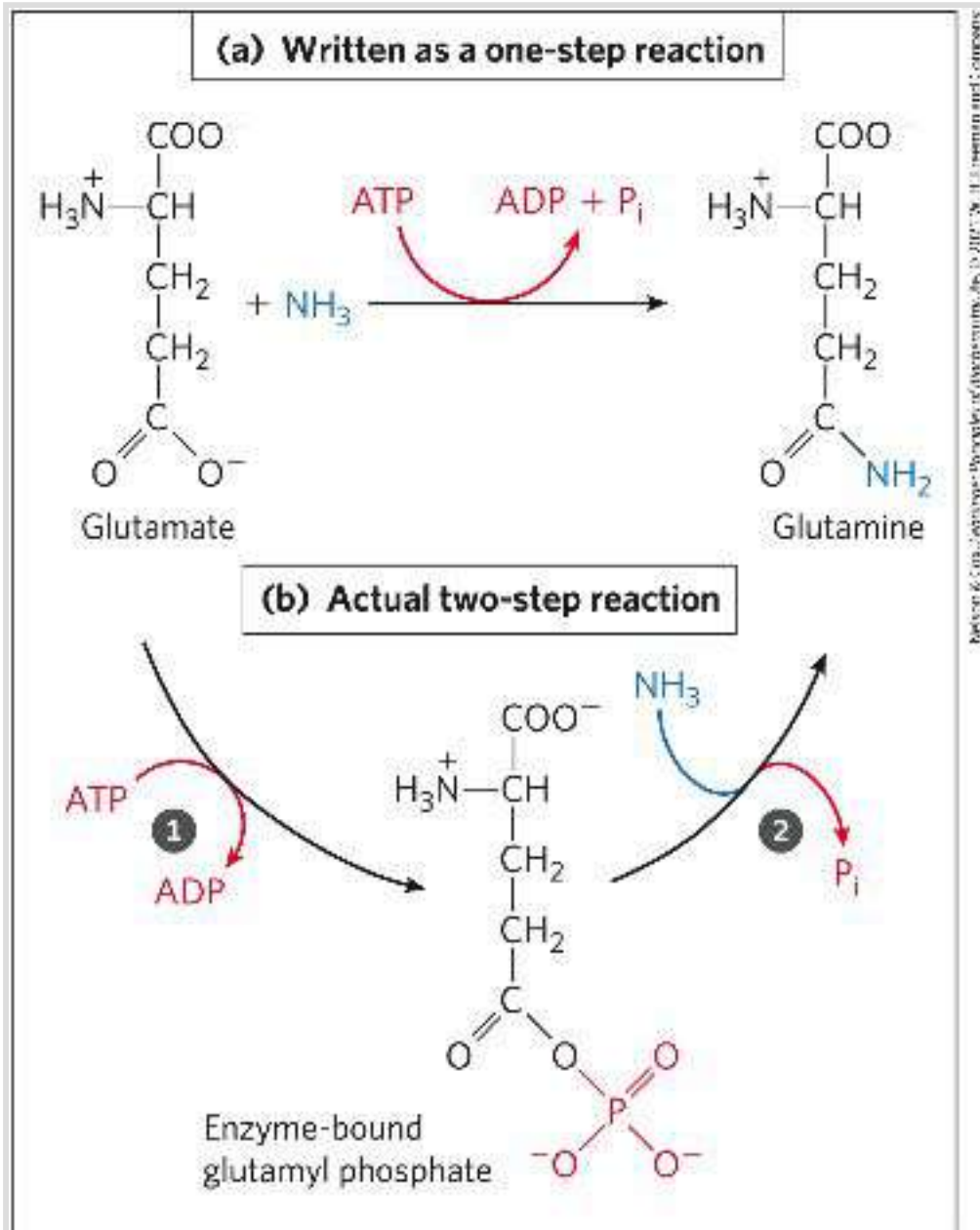


FIGURE 13-18 ATP hydrolysis in two steps. (a) The contribution of ATP to a reaction is often shown as a single step but is almost always a two-step process. (b) Shown here is the reaction catalyzed by ATP-dependent glutamine synthetase. **1** A phosphoryl group is transferred from ATP to glutamate, then **2** the phosphoryl group is displaced by NH_3 and released as P_i .

Some processes *do* involve direct hydrolysis of ATP (or the related GTP, guanosine triphosphate), however. For example, noncovalent binding of ATP (or GTP), followed by its hydrolysis to ADP (or GDP, guanosine diphosphate) and P_i , can provide the energy to cycle some proteins between two conformations, producing mechanical motion. This occurs in muscle contraction (see [Fig. 5-29](#)) and in the movement of enzymes along DNA (see [Fig. 25-30](#)) or of ribosomes along messenger RNA (see [Fig. 27-30](#)). The energy-dependent reactions catalyzed by helicases, RecA protein, and some topoisomerases ([Chapter 25](#)) also involve direct hydrolysis of phosphoanhydride bonds. The AAA+ ATPases involved in DNA replication and other processes described in [Chapter 25](#) use ATP hydrolysis to cycle associated proteins between active and inactive forms. GTP-binding proteins that act in signaling pathways directly hydrolyze GTP to drive conformational changes that terminate signals triggered by hormones or by other extracellular factors ([Chapter 12](#)).

The phosphate compounds found in living organisms can be divided, somewhat arbitrarily, into two groups, based on their standard free energies of hydrolysis ([Fig. 13-19](#)).  “High-energy” compounds have a $\Delta G'^{\circ}$ of hydrolysis more negative than -25 kJ/mol ; “low-energy” compounds have a less negative $\Delta G'^{\circ}$. Based on this criterion, ATP, with a $\Delta G'^{\circ}$ of hydrolysis of -30.5 kJ/mol (-7.3 kcal/mol), is a high-energy compound; glucose 6-phosphate, with a $\Delta G'^{\circ}$ of hydrolysis of -13.8 kJ/mol (-3.3 kcal/mol), is a low-energy compound.

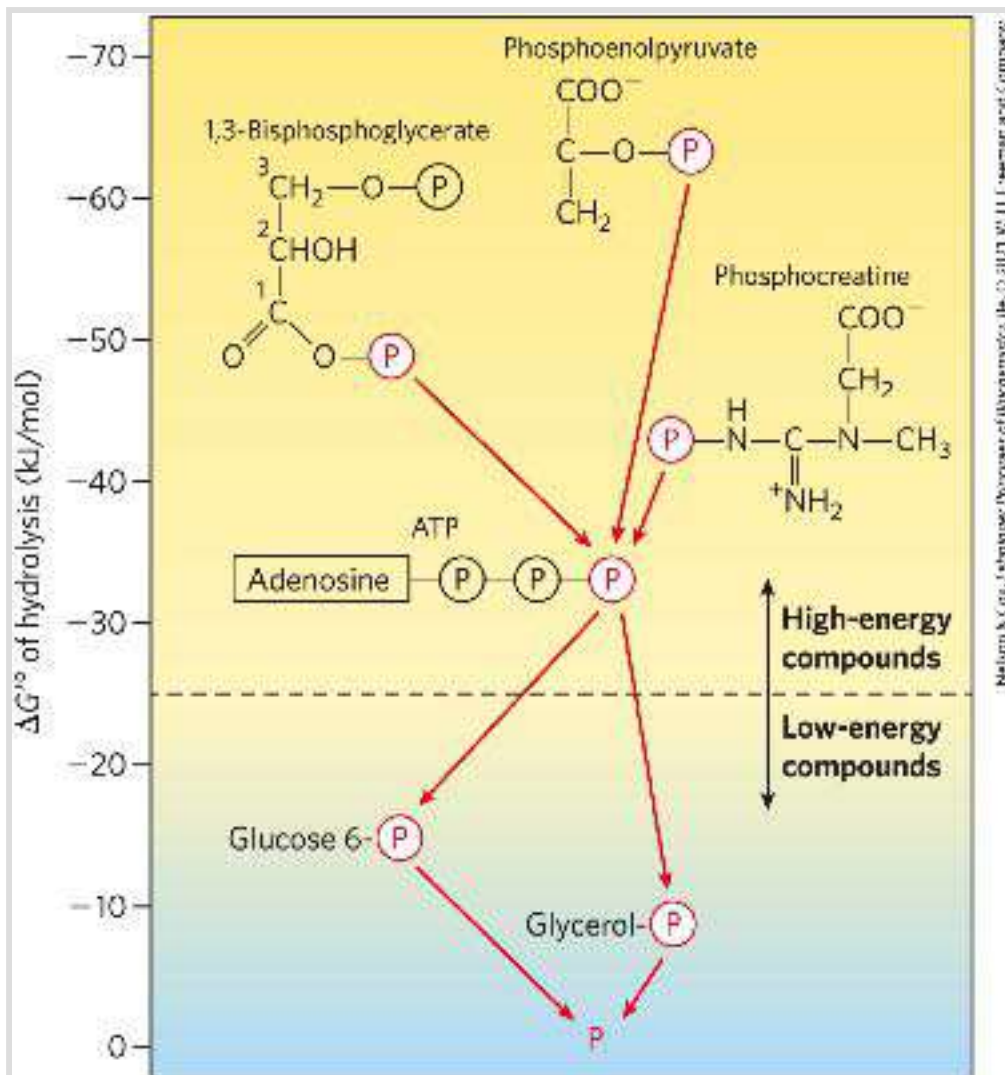
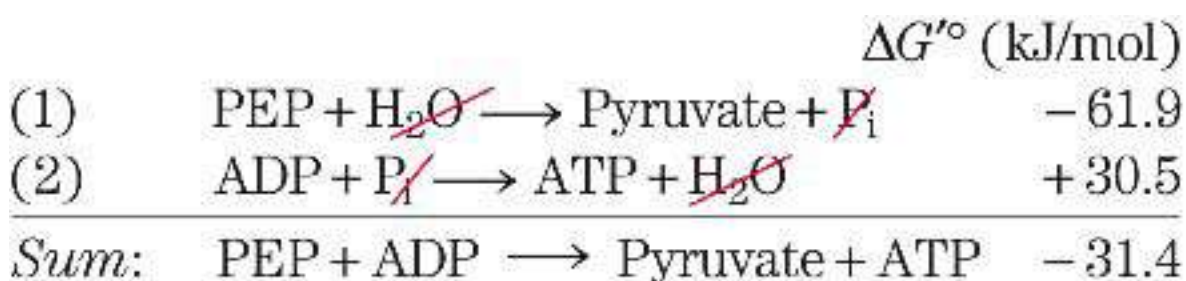


FIGURE 13-19 Ranking of biological phosphate compounds by standard free energies of hydrolysis. Phosphoryl groups, represented by (P), flow from high-energy phosphoryl group donors via ATP to acceptor molecules (such as glucose and glycerol) to form their low-energy phosphate derivatives. (The location of each compound's donor phosphoryl group along the scale is an approximate indication of the compound's $\Delta G'^{\circ}$ of hydrolysis.) This flow of phosphoryl groups, catalyzed by kinases, proceeds with an overall loss of free energy under intracellular conditions. Hydrolysis of low-energy phosphate compounds releases P_i , which has an even lower phosphoryl group transfer potential.

The term “high-energy phosphate bond,” long used by biochemists to describe the P—O bond broken in hydrolysis

reactions, is incorrect and misleading, as it wrongly suggests that the bond itself contains the energy. In fact, the breaking of all chemical bonds requires an *input* of energy. The free energy released by hydrolysis of phosphate compounds does not come from the specific bond that is broken; it results from the products of the reaction having a lower free-energy content than the reactants. For simplicity, we sometimes use the term “high-energy phosphate compound” when referring to ATP or other phosphate compounds with a large, negative, standard free energy of hydrolysis.

P2 As is evident from the additivity of free-energy changes of sequential reactions (see [Section 13.1](#)), any phosphorylated compound can be synthesized by coupling the synthesis to the breakdown of another phosphorylated compound with a more negative free energy of hydrolysis. For example, because cleavage of P_i from phosphoenolpyruvate releases more energy than is needed to drive the condensation of P_i with ADP, the direct donation of a phosphoryl group from PEP to ADP is thermodynamically feasible:



Notice that although the actual reaction is represented as the algebraic sum of the first two reactions, the actual reaction is a

third, distinct reaction that does not involve P_i ; PEP donates a *phosphoryl group directly* to ADP. We can describe phosphorylated compounds as having a high or low *phosphoryl group transfer potential*, on the basis of their standard free energies of hydrolysis (as listed in [Table 13-6](#)). The phosphoryl group transfer potential of PEP is very high, that of ATP is high, and that of glucose 6-phosphate is low ([Fig. 13-19](#)).

Much of catabolism is directed toward the synthesis of high-energy phosphate compounds, but their formation is not an end in itself; they are the means of activating a wide variety of compounds for further chemical transformation. The transfer of a phosphoryl group to a compound effectively puts free energy into that compound, so that it has more free energy to give up during subsequent metabolic transformations. We described above how the synthesis of glucose 6-phosphate is accomplished by phosphoryl group transfer from ATP. In the next chapter we see how this phosphorylation of glucose activates, or “primes,” the glucose for catabolic reactions that occur in nearly every living cell. Because of its intermediate position on the scale of group transfer potential, ATP can carry energy from high-energy phosphate compounds produced by catabolism (phosphoenolpyruvate, for example) to compounds such as glucose, converting them into more reactive species with better leaving groups. ATP thus serves as the universal energy currency in all living cells.

One more chemical feature of ATP is crucial to its role in metabolism: although, in aqueous solution, ATP is thermodynamically unstable and is therefore a good phosphoryl group donor, it is *kinetically* stable. Because of the huge activation energies (200 to 400 kJ/mol) required for uncatalyzed cleavage of its phosphoanhydride bonds, ATP does not spontaneously donate phosphoryl groups to water or to the hundreds of other potential acceptors in the cell. Only when specific enzymes are present to lower the energy of activation does phosphoryl group transfer from ATP proceed. The cell is therefore able to regulate the disposition of the energy carried by ATP by regulating the various enzymes that act on it.

ATP Donates Phosphoryl, Pyrophosphoryl, and Adenylyl Groups

The reactions of ATP are generally S_N2 nucleophilic displacements (see [Section 13.2](#)) in which the nucleophile may be, for example, the oxygen of an alcohol or carboxylate, or a nitrogen of creatine or of the side chain of arginine or histidine. Each of the three phosphates of ATP is susceptible to nucleophilic attack ([Fig. 13-20](#)), and each position of attack yields a different type of product.

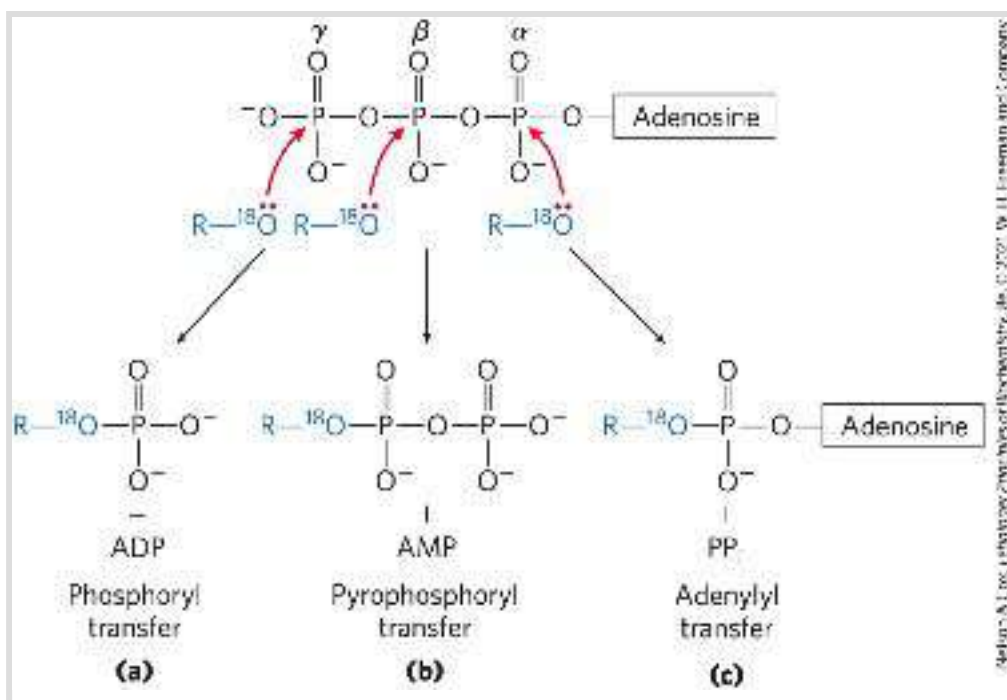


FIGURE 13-20 Three positions on ATP for attack by the nucleophile

$R-^{18}O$. Any of the three P atoms (α , β , or γ) may serve as the electrophilic target for nucleophilic attack, in this case by the labeled nucleophile $R-^{18}O$. The nucleophile may be an alcohol (ROH), a carboxyl group ($RCOO^-$), or a phosphoanhydride (a nucleoside mono- or diphosphate, for example). (a) When the oxygen of the nucleophile attacks the γ position, the bridge oxygen of the product is labeled, indicating that the group transferred from ATP is a phosphoryl ($-PO_3^{2-}$), not a phosphate ($-OPO_3^{2-}$). (b) Attack on the β position displaces AMP and leads to the transfer of a pyrophosphoryl (not pyrophosphate) group to the nucleophile. (c) Attack on the α position displaces PP_i and transfers the adenylyl group to the nucleophile.

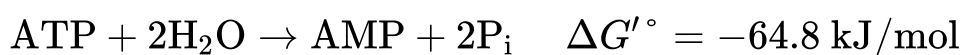
Nucleophilic attack by an alcohol on the γ phosphate ([Fig. 13-20a](#)) displaces ADP and produces a new phosphate ester. Studies with ^{18}O -labeled reactants have shown that the bridge oxygen in the new compound is derived from the alcohol, not from ATP; the group transferred from ATP is therefore a phosphoryl ($-PO_3^{2-}$), not a phosphate ($-OPO_3^{2-}$). Phosphoryl group transfer from ATP

to glutamate ([Fig. 13-18](#)) or to glucose ([p. 209](#)) involves attack at the γ position of the ATP molecule.

Attack at the β phosphate of ATP displaces AMP and transfers a pyrophosphoryl (not pyrophosphate) group to the attacking nucleophile ([Fig. 13-20b](#)). For example, the formation of 5-phosphoribosyl-1-pyrophosphate ([Chapter 22](#)), a key intermediate in nucleotide synthesis, results from attack of an —OH of the ribose on the β phosphate.

Nucleophilic attack at the α position of ATP displaces PP_i and transfers adenylate (5'-AMP) as an adenylyl group ([Fig. 13-20c](#)); the reaction is an **adenylylation** (a-den'-i-li-la'-shun, one of the most ungainly words in the biochemical language). Notice that hydrolysis of the $\alpha - \beta$ phosphoanhydride bond releases considerably more energy (~ 46 kJ/mol) than hydrolysis of the $\beta - \gamma$ bond (~ 31 kJ/mol) ([Table 13-6](#)). Furthermore, the PP_i formed as a byproduct of the adenylylation is hydrolyzed to two P_i by the ubiquitous enzyme [inorganic pyrophosphatase](#), releasing 19 kJ/mol and thereby providing a further energy “push” for the adenylylation reaction. In effect, both phosphoanhydride bonds of ATP are split in the overall reaction. Adenylylation reactions are therefore thermodynamically very favorable.  When the energy of ATP is used to drive a particularly unfavorable metabolic reaction, adenylylation is often the mechanism of energy coupling. Fatty acid activation is a good example of this energy-coupling strategy.

The first step in the activation of a fatty acid — either for energy-yielding oxidation or for use in the synthesis of more complex lipids — is the formation of its thiol ester (see [Fig. 17-5](#)). The direct condensation of a fatty acid with coenzyme A is endergonic, but the formation of a fatty acyl-CoA is made exergonic by stepwise removal of *two* phosphoryl groups from ATP. First, adenylate (AMP) is transferred from ATP to the carboxyl group of the fatty acid, forming a mixed anhydride (fatty acyl adenylate) and liberating PP_i. The thiol group of coenzyme A then displaces the adenylyl group and forms a thioester with the fatty acid. The sum of these two reactions is energetically equivalent to the exergonic hydrolysis of ATP to AMP and PP_i ($\Delta G'^{\circ} = -45.6 \text{ kJ/mol}$) and the endergonic formation of fatty acyl-CoA. The formation of fatty acyl-CoA ($\Delta G'^{\circ} = -31.4 \text{ kJ/mol}$) is made energetically favorable by hydrolysis of the PP_i by inorganic pyrophosphatase. Thus, in the activation of a fatty acid, both phosphoanhydride bonds of ATP are broken. The resulting $\Delta G'^{\circ}$ is the sum of the $\Delta G'^{\circ}$ values for the breakage of these bonds, or $-45.6 \text{ kJ/mol} + (-19.2) \text{ kJ/mol}$:



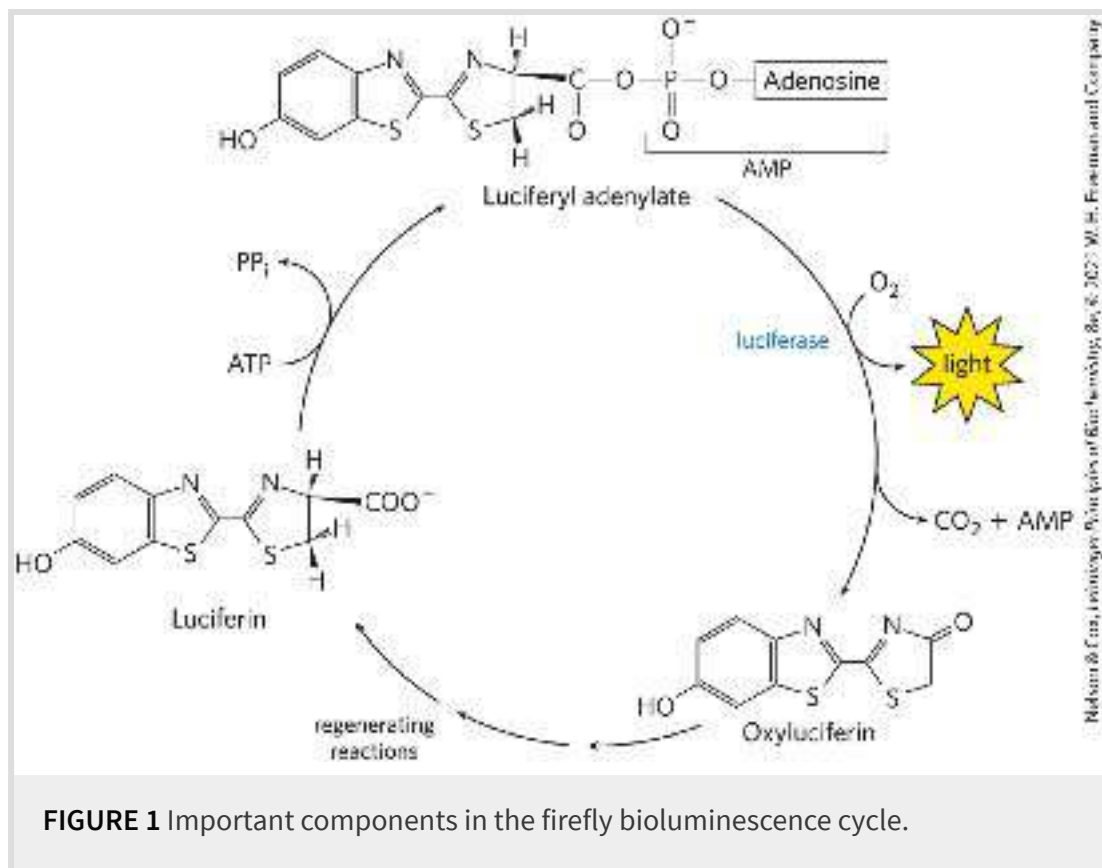
The activation of amino acids before their polymerization into proteins (see [Fig. 27-19](#)) is accomplished by an analogous set of reactions in which a transfer RNA molecule takes the place of coenzyme A. An interesting use of the cleavage of ATP to AMP

and PP_i occurs in the firefly, which uses ATP as an energy source to produce flashes of light ([Box 13-2](#)).

BOX 13-2

Firefly Flashes: Glowing Reports of ATP

Bioluminescence requires considerable amounts of energy. The firefly uses ATP to convert chemical energy into light energy. Males emit a flash of light to attract females, who flash in return to signal their interest. In the 1950s, from many thousands of fireflies collected by children in and around Baltimore, William McElroy and his colleagues at Johns Hopkins University isolated the principal biochemical components: luciferin, a complex carboxylic acid, and luciferase, an enzyme. Activation of luciferin by an enzymatic reaction involving pyrophosphate cleavage of ATP to form luciferyl adenylate generates the light flash ([Fig. 1](#)). In the presence of molecular oxygen and luciferase, the luciferin undergoes a multistep oxidative decarboxylation to oxyluciferin. This process is accompanied by emission of light. The color of the light flash differs from one firefly species to another and seems to be determined by differences in the structure of the luciferase. Luciferin is regenerated from oxyluciferin in a subsequent series of reactions.





Cathy Keifer/Fotolia

The firefly, a beetle of the *Lampyridae* family.

In the laboratory, pure firefly luciferin and luciferase are used to measure minute quantities of ATP by the intensity of the light flash produced. As little as a few picomoles (10^{-12} mol) of ATP can be measured in this way.

Assembly of Informational Macromolecules Requires Energy

When simple precursors are assembled into high molecular weight polymers with defined sequences (DNA, RNA, proteins), as described in detail in Part III, energy is required both for the condensation of monomeric units and for the creation of *ordered* sequences. The precursors for DNA and RNA synthesis are nucleoside triphosphates, and polymerization is accompanied by cleavage of the phosphoanhydride linkage between the α and β phosphates, with the release of PP_i ([Fig. 13-20](#)). The moieties transferred to the growing polymer in these reactions are adenylate (AMP), guanylate (GMP), cytidylate (CMP), or uridylate (UMP) for RNA synthesis, and their deoxy analogs (with TMP in place of UMP) for DNA synthesis. As noted above, the activation of amino acids for protein synthesis involves the donation of adenylyl groups from ATP, and we shall see in [Chapter 27](#) that several steps of protein synthesis on the ribosome are also accompanied by GTP hydrolysis. In all these cases, the exergonic breakdown of a nucleoside triphosphate is coupled to the endergonic process of synthesizing a polymer of a specific sequence.

ATP can supply the energy for transporting an ion or a molecule across a membrane into another aqueous compartment where its concentration is higher (see [Fig. 11-39](#)). Transport processes are major consumers of energy; in human kidney and brain, for example, as much as two-thirds of the energy consumed at rest is

used to pump Na^+ and K^+ across plasma membranes via the Na^+K^+ ATPase. The transport of Na^+ and K^+ is driven by cyclic phosphorylation and dephosphorylation of the transporter protein, with ATP as the phosphoryl group donor. Na^+ -dependent phosphorylation of the Na^+K^+ ATPase forces a change in the protein's conformation, and K^+ -dependent dephosphorylation favors return to the original conformation. Each cycle in the transport process results in the conversion of ATP to ADP and P_i , and it is the free-energy change of ATP hydrolysis that drives the cyclic changes in protein conformation that result in the electrogenic pumping of Na^+ and K^+ . Note that in this case, ATP interacts covalently by phosphoryl group transfer to the *enzyme*, not to the *substrate*.

In the contractile system of skeletal muscle cells, myosin and actin are specialized to transduce the chemical energy of ATP into motion (see [Fig. 5-29](#)). ATP binds tightly but noncovalently to one conformation of myosin, holding the protein in that conformation. When myosin catalyzes the hydrolysis of its bound ATP, the ADP and P_i dissociate from the protein, allowing it to relax into a second conformation until another molecule of ATP binds. The binding and subsequent hydrolysis of ATP (by myosin ATPase) provide the energy that forces cyclic changes in the conformation of the myosin head. The change in conformation of many individual myosin molecules results in the sliding of myosin fibrils along actin filaments (see [Fig. 5-28](#)), which translates into macroscopic contraction of the muscle fiber. As we noted earlier, this production of mechanical motion at the

expense of ATP is one of the few cases in which ATP hydrolysis per se, rather than group transfer from ATP, is the source of the chemical energy in a coupled process.

Transphosphorylations between Nucleotides Occur in All Cell Types

 Although we have focused on ATP as the cell's energy currency and donor of phosphoryl groups, all other nucleoside triphosphates (GTP, UTP, and CTP) and all deoxynucleoside triphosphates (dATP, dGTP, dTTP, and dCTP) are energetically equivalent to ATP. The standard free-energy changes associated with hydrolysis of their phosphoanhydride linkages are very nearly identical with those shown in [Table 13-6](#) for ATP. In preparation for their various biological roles, these other nucleotides are generated and maintained as the nucleoside triphosphate (NTP) forms by phosphoryl group transfer to the corresponding nucleoside diphosphates (NDPs) and monophosphates (NMPs).

ATP is the primary high-energy phosphate compound produced by catabolism, in the processes of glycolysis, oxidative phosphorylation, and, in photosynthetic cells, photophosphorylation. Several enzymes then carry phosphoryl groups from ATP to the other nucleotides. [Nucleoside diphosphate kinase](#), found in all cells, catalyzes the reaction



Although this reaction is fully reversible, the relatively high [ATP]/[ADP] ratio in cells normally drives the reaction to the right, with the net formation of NTPs and dNTPs. The enzyme actually catalyzes a two-step phosphoryl group transfer, which is a classic case of a double-displacement (Ping-Pong) mechanism ([Fig. 13-21](#); see also [Fig. 6-15b](#)). First, phosphoryl group transfer from ATP to an active-site His residue produces a phosphoenzyme intermediate; then the phosphoryl group is transferred from the P-His residue to an NDP acceptor. Because the enzyme is nonspecific for the base in the NDP and works equally well on dNDPs and NDPs, it can synthesize all NTPs and dNTPs, given the corresponding NDPs and a supply of ATP.

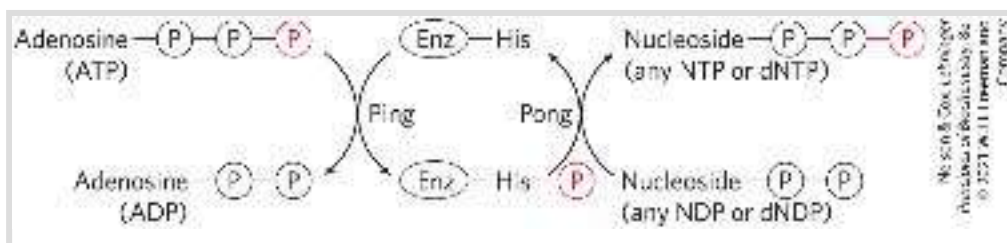


FIGURE 13-21 Ping-Pong mechanism of nucleoside diphosphate kinase.

The enzyme binds its first substrate (ATP in our example), and a phosphoryl group is transferred to the side chain of a His residue. ADP departs, another nucleoside (or deoxynucleoside) diphosphate replaces it, and this is converted to the corresponding triphosphate by transfer of the phosphoryl group from the phosphohistidine residue.

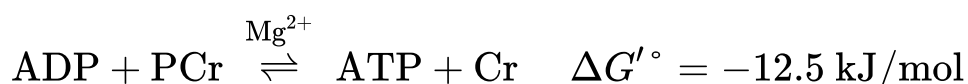
Phosphoryl group transfers from ATP result in an accumulation of ADP; for example, when muscle is contracting vigorously, ADP accumulates and interferes with ATP-dependent contraction.

During periods of intense demand for ATP, the cell lowers the ADP concentration, and at the same time replenishes ATP, by the action of **adenylate kinase**:



This reaction is fully reversible, so, after the intense demand for ATP ends, the enzyme can recycle AMP by converting it to ADP, which can then be phosphorylated to ATP in mitochondria. A similar enzyme, guanylate kinase, converts GMP to GDP at the expense of ATP. By pathways such as these, energy conserved in the catabolic production of ATP is used to supply the cell with all required NTPs and dNTPs.

Phosphocreatine (PCr; [Fig. 13-15](#)), also called creatine phosphate, serves as a ready source of phosphoryl groups for the quick synthesis of ATP from ADP. The PCr concentration in skeletal muscle is approximately 30 mM, nearly 10 times the concentration of ATP, and in other tissues such as smooth muscle, brain, and kidney, [PCr] is 5 to 10 mM. The enzyme **creatine kinase** catalyzes the reversible reaction



When a sudden demand for energy depletes ATP, the PCr reservoir is used to replenish ATP at a rate considerably faster than ATP can be synthesized by catabolic pathways. When the demand for energy slackens, ATP produced by catabolism is used to replenish the PCr reservoir by reversal of the creatine kinase reaction (see [Box 23-1](#)). Organisms in the lower phyla employ other PCr-like molecules (collectively called **phosphagens**) as phosphoryl reservoirs.

SUMMARY 13.3 *Phosphoryl Group Transfers and ATP*

- ATP is the chemical link between catabolism and anabolism. It is the energy currency of the living cell. The exergonic conversion of ATP to ADP and P_i , or to AMP and PP_i , is coupled to many endergonic reactions and processes.
- The free-energy change for ATP hydrolysis under cellular conditions is its phosphorylation potential, ΔG_p .
- Direct hydrolysis of ATP is the source of energy in some processes driven by conformational changes. In general, however, it is not ATP hydrolysis but the transfer of a phosphoryl group from ATP to a substrate or an enzyme that couples the energy of ATP breakdown to endergonic transformations of substrates.
- Phosphate compounds with high free energies of hydrolysis can donate their phosphoryl group to form another phosphate compound with a smaller free energy of hydrolysis.
- ATP can also donate a pyrophosphoryl (PP_i) or adenylyl (AMP) group to a variety of metabolic intermediates, activating them for

nucleophilic displacement reactions.

■ Through these group transfer reactions, ATP provides the energy for a large number of anabolic reactions, including the synthesis of informational macromolecules, and for the transport of molecules and ions across membranes against concentration gradients and electrical potential gradients. Muscle contraction is also powered by ATP.

■ To maintain its high group transfer potential, ATP concentration must be held far above the equilibrium concentration by energy-yielding reactions of catabolism.

■ ATP can donate a phosphoryl group to nucleoside diphosphates by transphosphorylation to keep the levels of GTP, UTP, CTP, and the deoxynucleotides far above their equilibrium concentrations.

13.4 Biological Oxidation-Reduction Reactions

The transfer of phosphoryl groups is a central feature of metabolism. Equally important is another kind of transfer: electron transfer in oxidation-reduction reactions, sometimes referred to as redox reactions. These reactions involve the loss of electrons by one chemical species, which is thereby oxidized, and the gain of electrons by another, which is reduced.  The flow of electrons in oxidation-reduction reactions is responsible, directly or indirectly, for all work done by living organisms. In nonphotosynthetic organisms, the sources of electrons are reduced compounds (foods); in photosynthetic organisms, the initial electron donor is a chemical species excited by the absorption of light. The path of electron flow in metabolism is complex. Electrons move from various metabolic intermediates to specialized electron carriers in enzyme-catalyzed reactions. The carriers, in turn, donate electrons to acceptors with higher electron affinities, with the release of energy. Cells possess a variety of molecular energy transducers, which convert the energy of electron flow into useful work.

We begin by discussing how work can be accomplished by an electromotive force (emf), then consider the theoretical and experimental basis for measuring energy changes in oxidation reactions in terms of emf and the relationship between this force, expressed in volts, and the free-energy change, expressed in joules. We also describe the structures and oxidation-reduction

chemistry of the most common of the specialized electron carriers, which you will encounter repeatedly in later chapters.

The Flow of Electrons Can Do Biological Work

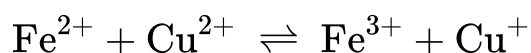
Every time we use a motor, an electric light or heater, or a spark to ignite gasoline in a car engine, we use the flow of electrons to accomplish work. In the circuit that powers a motor, the source of electrons can be a battery containing two chemical species that differ in affinity for electrons. Electrical wires provide a pathway for electron flow from the chemical species at one pole of the battery, through the motor, to the chemical species at the other pole of the battery. Because the two chemical species differ in their affinity for electrons, electrons flow spontaneously through the circuit, driven by a force proportional to the difference in electron affinity, the **electromotive force, emf**. The emf (typically a few volts) can accomplish work if an appropriate energy transducer — in this case a motor — is placed in the circuit. The motor can be coupled to a variety of mechanical devices to do useful work.

 Living cells have an analogous biological “circuit,” with a relatively reduced compound such as glucose as the source of electrons. As glucose is enzymatically oxidized, the released electrons flow spontaneously through a series of electron-carrier intermediates to another chemical species, such as O_2 . This

electron flow is exergonic, because O_2 has a higher affinity for electrons than do the electron-carrier intermediates. The resulting emf provides energy to a variety of molecular energy transducers (enzymes and other proteins) that do biological work. In the mitochondrion, for example, membrane-bound enzymes couple electron flow to the production of a transmembrane pH difference and a transmembrane electrical potential, accomplishing chemiosmotic and electrical work. The proton gradient thus formed has potential energy, sometimes called the proton-motive force by analogy with electromotive force. Another enzyme, ATP synthase in the inner mitochondrial membrane, uses the proton-motive force to do chemical work: synthesis of ATP from ADP and P_i as protons flow spontaneously across the membrane. Similarly, membrane-localized enzymes in *Escherichia coli* convert emf to proton-motive force, which is then used to power flagellar motion. The principles of electrochemistry that govern energy changes in the macroscopic circuit with a motor and battery apply with equal validity to the molecular processes accompanying electron flow in living cells.

Oxidation-Reductions Can Be Described as Half-Reactions

Although oxidation and reduction must occur together, it is convenient when describing electron transfers to consider the two halves of an oxidation-reduction reaction separately. For example, the oxidation of ferrous ion by cupric ion,



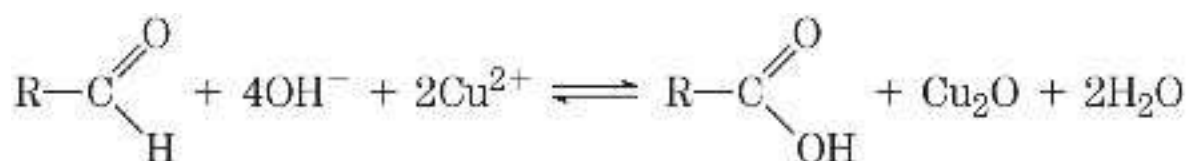
can be described in terms of two half-reactions:

1. $\text{Fe}^{2+} \rightleftharpoons \text{Fe}^{3+} + e^{-}$
2. $\text{Cu}^{2+} + e^{-} \rightleftharpoons \text{Cu}^{+}$

The electron-donating molecule in an oxidation-reduction reaction is called the reducing agent or reductant; the electron-accepting molecule is the oxidizing agent or oxidant. A given agent, such as an iron cation existing in the ferrous (Fe^{2+}) or ferric (Fe^{3+}) state, functions as a conjugate reductant-oxidant pair (redox pair), just as an acid and corresponding base function as a conjugate acid-base pair. Recall from [Chapter 2](#) that in acid-base reactions we can write a general equation: proton donor $\rightleftharpoons \text{H}^{+} +$ proton acceptor. In redox reactions we can write a similar general equation: electron donor (reductant) $\rightleftharpoons e^{-} +$ electron acceptor (oxidant). In the reversible half-reaction (1) above, Fe^{2+} is the electron donor and Fe^{3+} is the electron acceptor; together, Fe^{2+} and Fe^{3+} constitute a [conjugate redox pair](#). The mnemonic OIL RIG — *oxidation is losing, reduction is gaining* — may be helpful in remembering what happens to electrons in redox reactions.

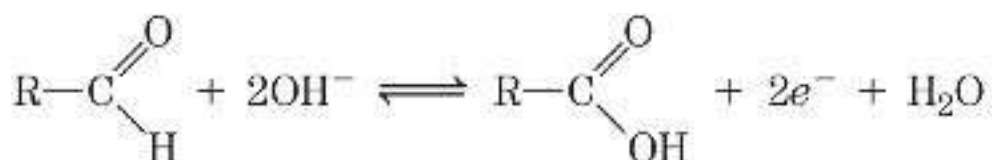
The electron transfers in the oxidation-reduction reactions of organic compounds are not fundamentally different from those of

inorganic species. Consider the oxidation of a reducing sugar (an aldehyde or a ketone) by cupric ion:



This overall reaction can be expressed as two half-reactions:

1.

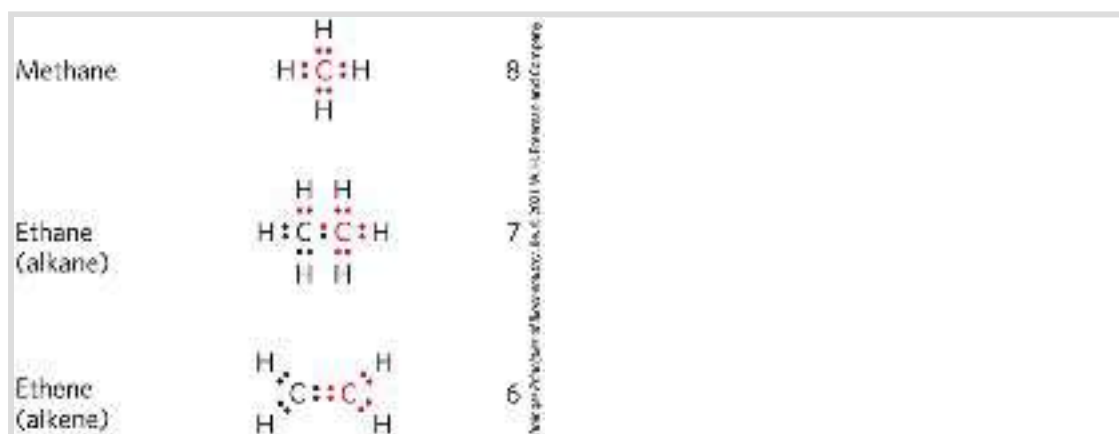


Notice that because two electrons are removed from the aldehyde carbon, the second half-reaction (the one-electron reduction of cupric to cuprous ion) must be doubled to balance the overall equation.

Biological Oxidations Often Involve Dehydrogenation

The carbon in living cells exists in a range of oxidation states ([Fig. 13-22](#)). When a carbon atom shares an electron pair with another atom (typically H, C, S, N, or O), the sharing is unequal, in favor of the more electronegative atom. The order of increasing

electronegativity is $H < C < S < N < O$. In oversimplified but useful terms, the more electronegative atom “owns” the bonding electrons it shares with another atom. For example, in methane (CH_4), carbon is more electronegative than the four hydrogens bonded to it, and the C atom therefore owns all eight bonding electrons ([Fig. 13-22](#)). In ethane, the electrons in the C—C bond are shared equally, so each C atom owns only seven of its eight bonding electrons. In ethanol, C-1 is less electronegative than the oxygen to which it is bonded, and the O atom therefore owns both electrons of the C—O bond, leaving C-1 with only five bonding electrons. With each formal loss of “owned” electrons, the carbon atom has undergone oxidation — even when no oxygen is involved, as in the conversion of an alkane ($-CH_2-CH_2-$) to an alkene ($-CH=CH-$). In this case, oxidation (loss of electrons) is coincident with the loss of hydrogen. In biological systems, as we noted earlier in the chapter, oxidation is often synonymous with [dehydrogenation](#), and many enzymes that catalyze oxidation reactions are **dehydrogenases**. Notice that the more reduced compounds in [Figure 13-22](#) (top) are richer in hydrogen than in oxygen, whereas the more oxidized compounds (bottom) have more oxygen and less hydrogen.



Ethanol (alcohol)		5
Acetylene (alkyne)		5
Formaldehyde		4
Acetaldehyde (aldehyde)		3
Acetone (ketone)		2
Formic acid (carboxylic acid)		2
Carbon monoxide		2
Acetic acid (carboxylic acid)		1
Carbon dioxide		0

FIGURE 13-22 Different levels of oxidation of carbon compounds in the biosphere.

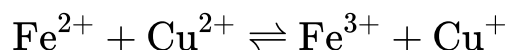
To approximate the level of oxidation of these compounds, focus on the red carbon atom and its bonding electrons. When this carbon is bonded to the less electronegative H atom, both bonding electrons (red) are assigned to the carbon. When carbon is bonded to another carbon, bonding electrons are shared equally, so one of the two electrons is assigned to the red carbon. When the red carbon is bonded to the more electronegative O atom, the bonding electrons are assigned to

the oxygen. The number to the right of each compound is the number of electrons “owned” by the red carbon, a rough expression of the degree of oxidation of that compound. As the red carbon undergoes oxidation (loses electrons), the number gets smaller.

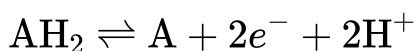
Not all biological oxidation-reduction reactions involve carbon. For example, in the conversion of molecular nitrogen to ammonia, $6\text{H}^+ + 6e^- + \text{N}_2 \rightarrow 2\text{NH}_3$, the nitrogen atoms are reduced.

Electrons are transferred from one molecule (electron donor) to another (electron acceptor) in one of four ways:

1. Directly as *electrons*. For example, the $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox pair can transfer an electron to the $\text{Cu}^+/\text{Cu}^{2+}$ redox pair:

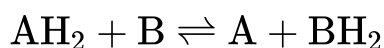


2. As *hydrogen atoms*. Recall that a hydrogen atom consists of a proton (H^+) and a single electron (e^-). In this case we can write the general equation

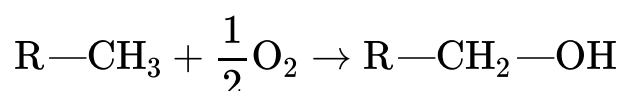


where AH_2 is the hydrogen/electron donor. (Do not mistake the above reaction for an acid dissociation, which involves a proton and no electron.) AH_2 and A together constitute a conjugate redox pair (A/AH_2), which can reduce another

compound B (or redox pair, B/BH₂) by transfer of hydrogen atoms:



3. As a *hydride ion* (:H⁻), which has two electrons. This occurs in the case of NAD-linked dehydrogenases, described below.
4. Through direct *combination with oxygen*. In this case, oxygen combines with an organic reductant and is covalently incorporated in the product, as in the oxidation of a hydrocarbon to an alcohol:

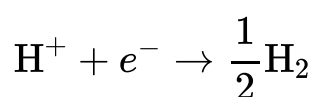


The hydrocarbon is the electron donor, and the oxygen atom is the electron acceptor.

All four types of electron transfer occur in cells. The neutral term **reducing equivalent** is commonly used to designate a single electron equivalent participating in an oxidation-reduction reaction, no matter whether this equivalent is an electron per se or is part of a hydrogen atom or a hydride ion, or whether the electron transfer takes place in a reaction with oxygen to yield an oxygenated product.

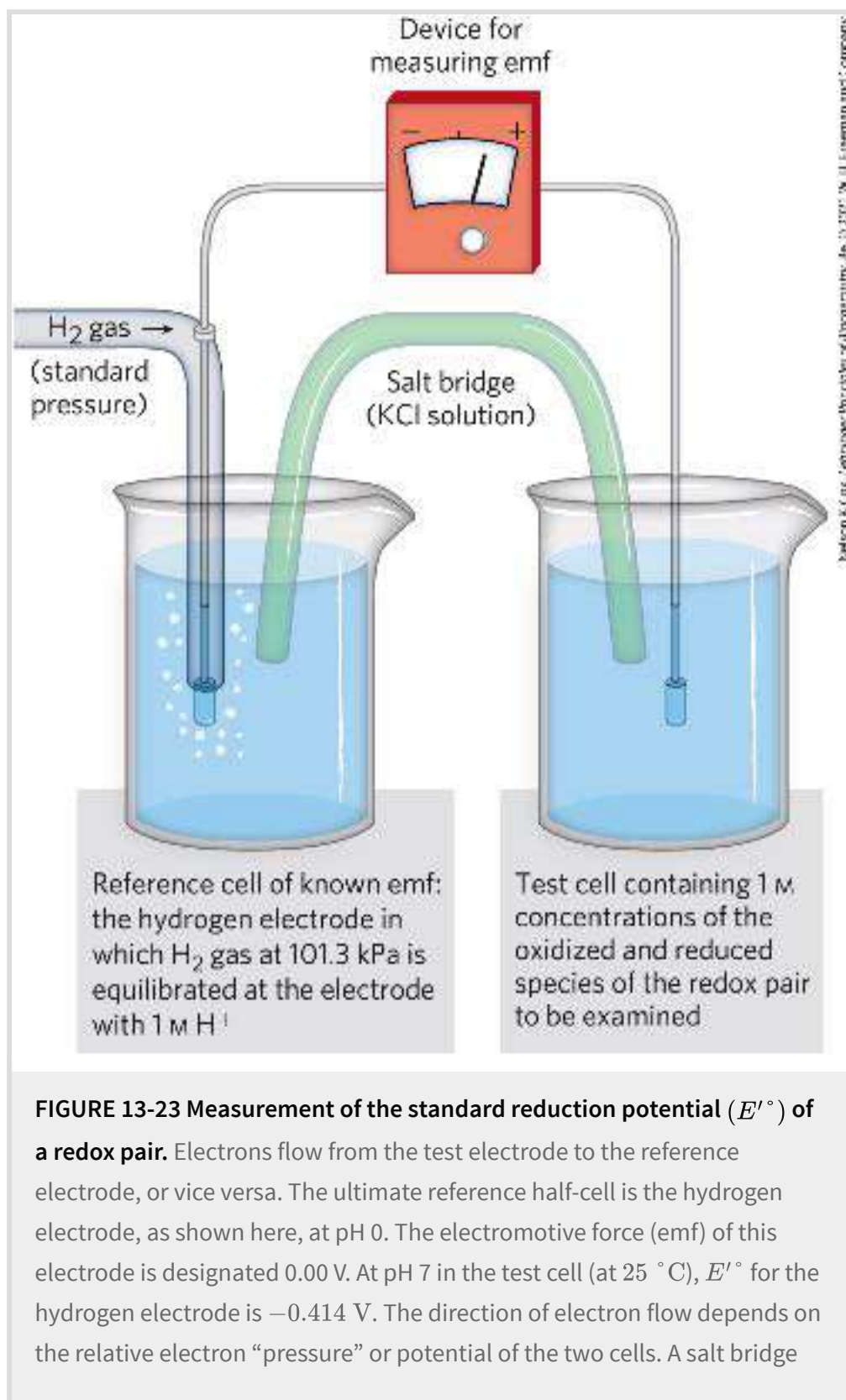
Reduction Potentials Measure Affinity for Electrons

When two conjugate redox pairs are together in solution, electron transfer from the electron donor of one pair to the electron acceptor of the other may proceed spontaneously. The tendency for such a reaction depends on the relative affinity of the electron acceptor of each redox pair for electrons. The standard reduction potential, E° , a measure (in volts) of this affinity, can be determined in an experiment such as that described in [Figure 13-23](#). Electrochemists have chosen as a standard of reference the half-reaction



The electrode at which this half-reaction occurs (called a half-cell) is arbitrarily assigned an E° of 0.00 V. When this hydrogen electrode is connected through an external circuit to another half-cell in which an oxidized species and its corresponding reduced species are present at standard concentrations (at 25 °C, each solute at 1 M, each gas at 101.3 kPa), electrons tend to flow through the external circuit from the half-cell of lower E° to the half-cell of higher E° . By convention, a half-cell that takes electrons from the standard hydrogen cell is assigned a positive value of E° , and one that donates electrons to the hydrogen cell, a negative value. When any two half-cells are connected, that with

the larger (more positive) E° will be reduced; it has the greater reduction potential.



containing a saturated KCl solution provides a path for counter-ion movement between the test cell and the reference cell. From the observed emf and the known emf of the reference cell, the experimenter can find the emf of the test cell containing the redox pair. The cell that gains electrons has, by convention, the more positive reduction potential.

The reduction potential of a half-cell depends not only on the chemical species present but also on their activities, approximated by their concentrations. The Nernst equation relates standard reduction potential (E°) to the actual reduction potential (E) at any concentration of oxidized and reduced species in a living cell:

$$E = E^\circ + \frac{RT}{nF} \ln \frac{[\text{electron acceptor}]}{[\text{electron donor}]}$$

(13-5)

where R and T have their usual meanings, n is the number of electrons transferred per molecule, and F is the Faraday constant, a proportionality constant that converts volts to joules ([Table 13-1](#)). At 298 K (25 °C), this expression reduces to

$$E = E^\circ + \frac{0.026 \text{ V}}{n} \ln \frac{[\text{electron acceptor}]}{[\text{electron donor}]}$$

(13-6)

KEY CONVENTION

Many half-reactions of interest to biochemists involve protons. As in the definition of $\Delta G'^{\circ}$, biochemists define the standard state for oxidation-reduction reactions as pH 7 and express a standard transformed reduction potential, E'° , the standard reduction potential at pH 7 and 25 °C. By convention, $\Delta E'^{\circ}$ for any redox reaction is given as E'° of the electron acceptor minus E'° of the electron donor. ■

The standard reduction potentials given in [Table 13-7](#) and used throughout this book are values for E'° and are therefore valid only for systems at neutral pH. Each value represents the potential difference when the conjugate redox pair, at 1 M concentrations, 25 °C, and pH 7, is connected with the standard (pH 0) hydrogen electrode. Notice in [Table 13-7](#) that when the conjugate pair $2\text{H}^{+}/\text{H}_2$ at pH 7 is connected with the standard hydrogen electrode (pH 0), electrons tend to flow from the pH 7 cell to the standard (pH 0) cell; the measured E'° for the $2\text{H}^{+}/\text{H}_2$ pair is -0.414 V .

TABLE 13-7 Standard Reduction Potentials of Some Biologically Important Half-Reactions

Half-reaction	E'° (V)
$\frac{1}{2}\text{O}_2 + 2\text{H}^{+} + 2e^{-} \rightarrow \text{H}_2\text{O}$	0.816

$\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$	0.771
$\text{NO}_3^- + 2\text{H}^+ + 2e^- \rightarrow \text{NO}_2^- + \text{H}_2\text{O}$	0.421
Cytochrome <i>f</i> (Fe^{3+}) + $e^- \rightarrow$ cytochrome <i>f</i> (Fe^{2+})	0.365
$\text{Fe}(\text{CN})_6^{3-}$ (ferricyanide) + $e^- \rightarrow \text{Fe}(\text{CN})_6^{4-}$	0.36
Cytochrome <i>a</i> ₃ (Fe^{3+}) + $e^- \rightarrow$ cytochrome <i>a</i> ₃ (Fe^{2+})	0.35
$\text{O}_2 + 2\text{H}^+ + 2e^- \rightarrow \text{H}_2\text{O}_2$	0.295
Cytochrome <i>a</i> (Fe^{3+}) + $e^- \rightarrow$ cytochrome <i>a</i> (Fe^{2+})	0.29
Cytochrome <i>c</i> (Fe^{3+}) + $e^- \rightarrow$ cytochrome <i>c</i> (Fe^{2+})	0.254
Cytochrome <i>c</i> ₁ (Fe^{3+}) + $e^- \rightarrow$ cytochrome <i>c</i> ₁ (Fe^{2+})	0.22
Cytochrome <i>b</i> (Fe^{3+}) + $e^- \rightarrow$ cytochrome <i>b</i> (Fe^{2+})	0.077
Ubiquinone + $2\text{H}^+ + 2e^- \rightarrow$ ubiquinol	0.045
$\text{Fumarate}^{2-} + 2\text{H}^+ + 2e^- \rightarrow \text{succinate}^{2-}$	0.031
$2\text{H}^+ + 2e^- \rightarrow \text{H}_2$ (at standard conditions, pH 0)	0.000
Crotonyl-CoA + $2\text{H}^+ + 2e^- \rightarrow$ butyryl-CoA	−0.015
$\text{Oxaloacetate}^{2-} + 2\text{H}^+ + 2e^- \rightarrow \text{malate}^{2-}$	−0.166
$\text{Pyruvate}^- + 2\text{H}^+ + 2e^- \rightarrow \text{lactate}^-$	−0.185
Acetaldehyde + $2\text{H}^+ + 2e^- \rightarrow$ ethanol	−0.197
$\text{FAD} + 2\text{H}^+ + 2e^- \rightarrow \text{FADH}_2$	−0.219 ^a
Glutathione + $2\text{H}^+ + 2e^- \rightarrow$ 2 reduced glutathione	−0.23
$\text{S} + 2\text{H}^+ + 2e^- \rightarrow \text{H}_2\text{S}$	−0.243

Lipoic acid + 2H ⁺ + 2e ⁻ → dihydrolipoic acid	-0.29
NAD ⁺ + H ⁺ + 2e ⁻ → NADH	-0.320
NADP ⁺ + H ⁺ + 2e ⁻ → NADPH	-0.324
Acetoacetate + 2H ⁺ + 2e ⁻ → β-hydroxybutyrate	-0.346
α-Ketoglutarate + CO ₂ + 2H ⁺ + 2e ⁻ → isocitrate	-0.38
2H ⁺ + 2e ⁻ → H ₂ (at pH 7)	-0.414
Ferredoxin (Fe ³⁺) + e ⁻ → ferredoxin (Fe ²⁺)	-0.432
Data mostly from R. A. Loach, in <i>Handbook of Biochemistry and Molecular Biology</i>, 3rd edn (G. D. Fasman, ed.), <i>Physical and Chemical Data</i>, Vol. 1, p. 122, CRC Press, 1976. aThis is the value for free FAD; FAD bound to a specific flavoprotein (e.g., succinate dehydrogenase) has a different E'° that depends on its protein environment.	

Standard Reduction Potentials Can Be Used to Calculate Free-Energy Change

Why are reduction potentials so useful to the biochemist? When E values have been determined for any two half-cells, relative to the standard hydrogen electrode, we also know their reduction potentials relative to each other. We can then predict the direction in which electrons will tend to flow when the two half-cells are connected through an external circuit or when components of both half-cells are present in the same solution. Electrons tend to flow to the half-cell with the more positive E , and the strength of

that tendency is proportional to ΔE , the difference in reduction potential. The energy made available by this spontaneous electron flow (the free-energy change, ΔG , for the oxidation-reduction reaction) is proportional to ΔE :

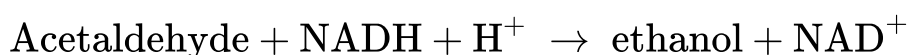
$$\Delta G = -nF\Delta E \text{ or } \Delta G'^{\circ} = -nF\Delta E'^{\circ} \quad (13-7)$$

where n is the number of electrons transferred in the reaction.

 With this equation we can calculate the actual free-energy change for any oxidation-reduction reaction from the values of $\Delta E'^{\circ}$ in a table of reduction potentials ([Table 13-7](#)) and the concentrations of reacting species.

WORKED EXAMPLE 13-3 *Calculation of $\Delta G'^{\circ}$ and ΔG of a Redox Reaction*

Calculate the standard free-energy change, $\Delta G'^{\circ}$, for the reaction in which acetaldehyde is reduced by the biological electron carrier NADH:



Then calculate the *actual* free-energy change, ΔG , when [acetaldehyde] and [NADH] are 1.00 M, and [ethanol] and [NAD⁺] are 0.100 M. The relevant half-reactions and their E'° values are

1. Acetaldehyde + 2H⁺ + 2e⁻ → ethanol $E'^{\circ} = -0.197 \text{ V}$
2. NAD⁺ + 2H⁺ + 2e⁻ → NADH + H⁺ $E'^{\circ} = -0.320 \text{ V}$

Remember that, by convention, $\Delta E'^{\circ}$ is E'° of the electron acceptor minus E'° of the electron donor. It represents the difference between the electron affinities of the two half-reactions in the table of reduction potentials ([Table 13-7](#)). Note that the more widely separated the two half-reactions in the table, the more energetic the electron-transfer reaction when the two half-reactions occur together. By convention, in tables of reduction potentials, all half-reactions are represented as reductions, but when two half-reactions occur together, one of them must be an oxidation. Although that half-reaction will go in the opposite direction from that shown in [Table 13-7](#), we *do not change the sign* of that half-reaction before calculating $\Delta E'^{\circ}$, because $\Delta E'^{\circ}$ is *defined* as a difference of reduction potentials.

SOLUTION:

Because acetaldehyde is accepting electrons ($n = 2$) from NADH, $\Delta E'^{\circ} = -0.197 \text{ V} - (-0.320 \text{ V}) = 0.123 \text{ V}$. Therefore,

$$\Delta G'^{\circ} = -nF\Delta E'^{\circ} = -2(96.5 \text{ kJ/V} \cdot \text{mol})(0.123 \text{ V}) = -23.7 \text{ kJ/mol}$$



This is the free-energy change for the oxidation-reduction reaction at 25 °C and pH 7, when acetaldehyde, ethanol, NAD⁺, and NADH are all present at 1.00 M concentrations.

To calculate ΔG when [acetaldehyde] and [NADH] are 1.00 M, and [ethanol] and [NAD⁺] are 0.100 M, we can use [Equation 13-4](#) and the standard free-energy change calculated above:

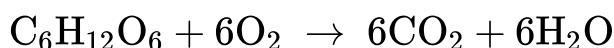
$$\begin{aligned}\Delta G &= \Delta G'^{\circ} + RT \ln \frac{[\text{ethanol}][\text{NAD}^+]}{[\text{acetaldehyde}][\text{NADH}]} \\&= -23.7 \text{ kJ/mol} + (8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K}) \ln \frac{(0.100 \text{ M})(}{(1.00 \text{ M})(} \\&= -23.7 \text{ kJ/mol} + (2.48 \text{ J/mol}) \ln 0.01 \\&= -35.1 \text{ kJ/mol}\end{aligned}$$

This is the actual free-energy change at the specified concentrations of the redox pairs.

A Few Types of Coenzymes and Proteins Serve as Universal Electron Carriers

The principles of oxidation-reduction energetics described above apply to the many metabolic reactions that involve electron

transfers. For example, in many organisms, the oxidation of glucose supplies energy for the production of ATP. The complete oxidation of glucose

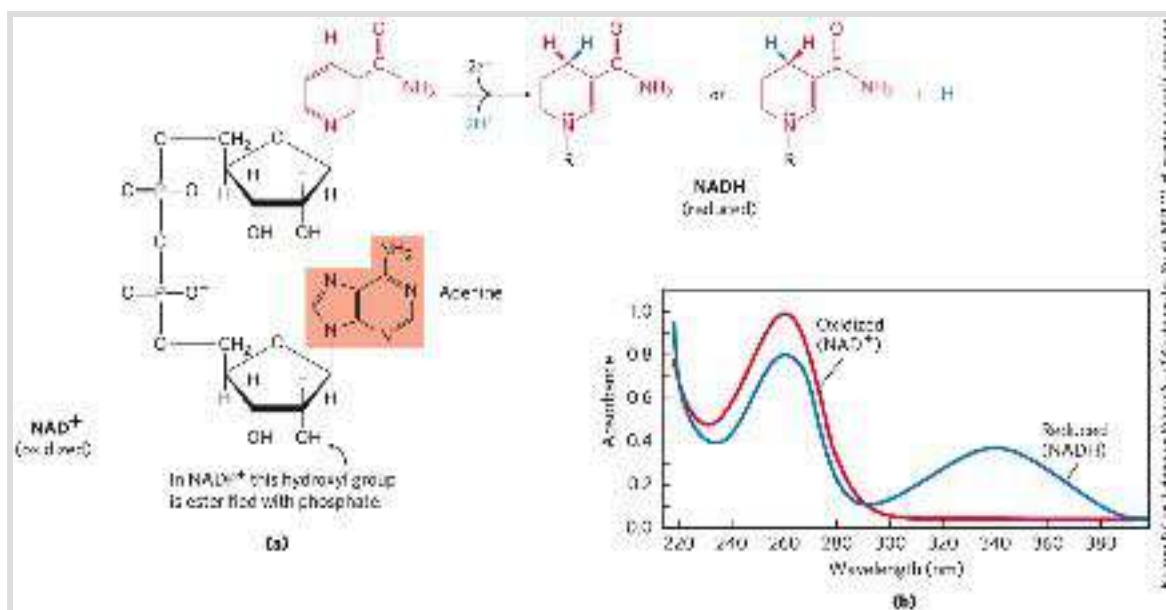


has a $\Delta G'^{\circ}$ of $-2,840$ kJ/mol. This is a much larger release of free energy than is required for ATP synthesis in cells (50 to 60 kJ/mol; see [Worked Example 13-2](#)). Cells convert glucose to CO_2 not in a single, high-energy-releasing reaction but rather in a series of controlled reactions, some of which are oxidations. The free energy released in these oxidation steps is of the same order of magnitude as that required for ATP synthesis from ADP, with some energy to spare. Electrons removed in these oxidation steps are transferred to coenzymes specialized for carrying electrons, such as NAD^+ and FAD (described below).

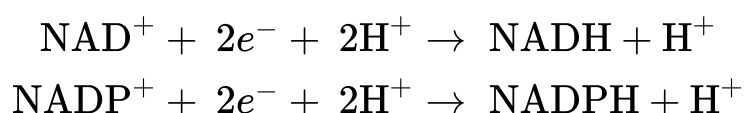
The multitude of enzymes that catalyze cellular oxidations channel electrons from their hundreds of different substrates into just a few types of universal electron carriers. The reduction of these carriers in catabolic processes results in the conservation of free energy released by substrate oxidation. NAD, NADP, FMN, and FAD are water-soluble coenzymes that undergo reversible oxidation and reduction in many of the electron-transfer reactions of metabolism. The nucleotides NAD and NADP move readily from one enzyme to another; the flavin nucleotides FMN and FAD are usually very tightly bound to the enzymes, called

flavoproteins, for which they serve as prosthetic groups. Lipid-soluble quinones such as ubiquinone and plastoquinone act as electron carriers and proton donors in the nonaqueous environment of membranes. Iron-sulfur proteins and cytochromes, which have tightly bound prosthetic groups that undergo reversible oxidation and reduction, also serve as electron carriers in many oxidation-reduction reactions. Some of these proteins are water-soluble, but others are peripheral or integral membrane proteins. The oxidation-reduction chemistry of quinones, iron-sulfur proteins, and cytochromes is discussed in [Chapters 19](#) and [20](#).

Nicotinamide adenine dinucleotide (NAD; NAD^+ in its oxidized form) and its close analog nicotinamide adenine dinucleotide phosphate (NADP; NADP^+ when oxidized) are composed of two nucleotides joined through their phosphate groups by a phosphoanhydride bond ([Fig. 13-24a](#)). Because the nicotinamide ring resembles pyridine, these compounds are sometimes called [pyridine nucleotides](#). The vitamin niacin is the source of the nicotinamide moiety in nicotinamide nucleotides.



Both coenzymes undergo reversible reduction of the nicotinamide ring ([Fig. 13-24](#)). As a substrate molecule undergoes oxidation (dehydrogenation), giving up two hydrogen atoms, the oxidized form of the nucleotide (NAD^+ or NADP^+) accepts a hydride ion ($:\text{H}^-$, the equivalent of a proton and two electrons) and is reduced (to NADH or NADPH). The second proton removed from the substrate is released to the aqueous solvent. The half-reactions for these nucleotide cofactors are

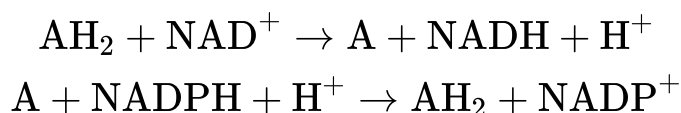


Reduction of NAD^+ or NADP^+ converts the benzenoid ring of the nicotinamide moiety (with a fixed positive charge on the ring nitrogen) to the quinonoid form (with no charge on the nitrogen). The reduced nucleotides absorb light at 340 nm; the oxidized forms do not ([Fig. 13-24b](#)). Biochemists use this difference in absorption to assay reactions involving these coenzymes. Note that the plus sign in the abbreviations NAD^+ and NADP^+ does *not* indicate the net charge on these molecules (in fact, both are negatively charged); rather, it indicates that the nicotinamide ring is in its oxidized form, with a positive charge on the nitrogen atom. In the abbreviations NADH and NADPH, the “H” denotes the added hydride ion. To refer to these nucleotides without specifying their oxidation state, we use NAD and NADP.

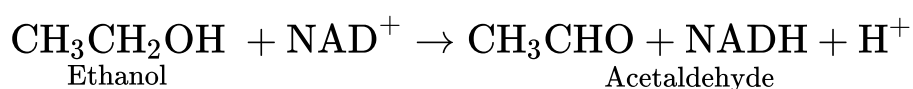
The total concentration of $\text{NAD}^+ + \text{NADH}$ in most tissues is about 10^{-5}M ; that of $\text{NADP}^+ + \text{NADPH}$ is about 10^{-6}M . In many cells and tissues, the ratio of NAD^+ (oxidized) to NADH (reduced) is high, favoring hydride transfer from a substrate *to* NAD^+ to form NADH. By contrast, NADPH is generally present at a higher concentration than NADP^+ , favoring hydride transfer *from* NADPH to a substrate. This reflects the specialized metabolic roles of the two coenzymes: NAD^+ generally functions in oxidations — usually as part of a catabolic reaction; NADPH is the usual coenzyme in reductions — nearly always as part of an anabolic reaction. A few enzymes can use either coenzyme, but

most show a strong preference for one over the other. Also, the processes in which these two cofactors function are segregated in eukaryotic cells: for example, oxidations of fuels such as pyruvate, fatty acids, and α -keto acids derived from amino acids occur in the mitochondrial matrix, whereas reductive biosynthetic processes such as fatty acid synthesis take place in the cytosol. This functional and spatial specialization allows a cell to maintain two distinct pools of electron carriers, with two distinct functions.

More than 200 enzymes are known to catalyze reactions in which NAD^+ (or NADP^+) accepts a hydride ion from a reduced substrate, or NADPH (or NADH) donates a hydride ion to an oxidized substrate. The general reactions are

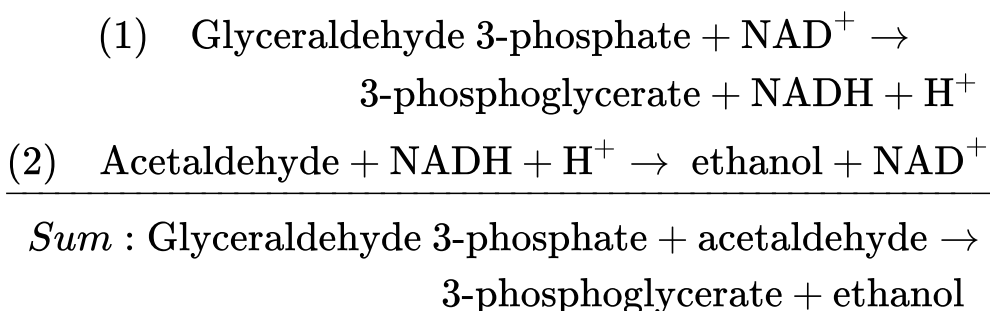


where AH_2 is the reduced substrate and A is the oxidized substrate. The general name for an enzyme of this type is **oxidoreductase**; they are also commonly called dehydrogenases. For example, alcohol dehydrogenase catalyzes the first step in the catabolism of ethanol, in which ethanol is oxidized to acetaldehyde:



Notice that one of the carbon atoms in ethanol has lost a hydrogen; the compound has been oxidized from an alcohol to an aldehyde (refer again to [Fig. 13-22](#) for the oxidation states of carbon).

The association between a dehydrogenase and NAD or NADP is relatively loose; the coenzyme readily diffuses from one enzyme to another, acting as a water-soluble carrier of electrons from one metabolite to another. For example, in the production of alcohol during fermentation of glucose by yeast cells, a hydride ion is removed from glyceraldehyde 3-phosphate by one enzyme (glyceraldehyde 3-phosphate dehydrogenase) and transferred to NAD^+ . The NADH produced then leaves the enzyme surface and diffuses to another enzyme (alcohol dehydrogenase), which transfers a hydride ion to acetaldehyde, producing ethanol:



Notice that in the overall reaction there is no net production or consumption of NAD^+ or NADH; the coenzymes function catalytically and are recycled repeatedly without a net change in the total amount of $\text{NAD}^+ + \text{NADH}$.

Both reduced and oxidized forms of NAD and NADP serve as allosteric effectors of proteins in catabolic pathways. As we describe in later chapters, the ratios NAD^+/NADH and $\text{NADP}^+/\text{NADPH}$ serve as sensitive gauges of a cell's fuel supply, allowing rapid, appropriate changes in energy-yielding and energy-dependent metabolism.

NAD Has Important Functions in Addition to Electron Transfer

Some key cellular functions are regulated by enzymes that use NAD^+ not as a redox cofactor but as a substrate in a coupled reaction in which the availability of NAD^+ can be an indicator of the cell's energy status. In DNA replication and repair, the enzyme DNA ligase is adenylylated and then transfers the AMP to a 5' phosphate in a nicked DNA (see [Fig. 25-15](#)); in bacteria, NAD^+ serves as the source of the activating AMP group. A family of proteins called sirtuins regulate the activity of proteins in diverse cellular pathways by deacetylating the ϵ -amino group of an acetylated Lys residue. The deacetylation is coupled to NAD^+ hydrolysis, yielding O-acetyl-ADP-ribose and nicotinamide. Among the cellular processes regulated by sirtuins are inflammation, apoptosis, aging, and DNA transcription; deacetylation by a sirtuin alters the charge on histones, influencing which genes are expressed. The availability of NAD^+ for these types of reactions may indicate that the cell is

undergoing stress and that pathways designed to respond to stress should be activated.



NAD^+ also plays an important role in cholera pathology (see [Section 12.2](#)). Cholera toxin has an enzymatic activity that transfers ADP-ribose from NAD^+ to a G protein involved in regulating ion fluxes in the cells lining the gut. This ADP-ribosylation blocks water retention, causing the diarrhea and dehydration characteristic of cholera.

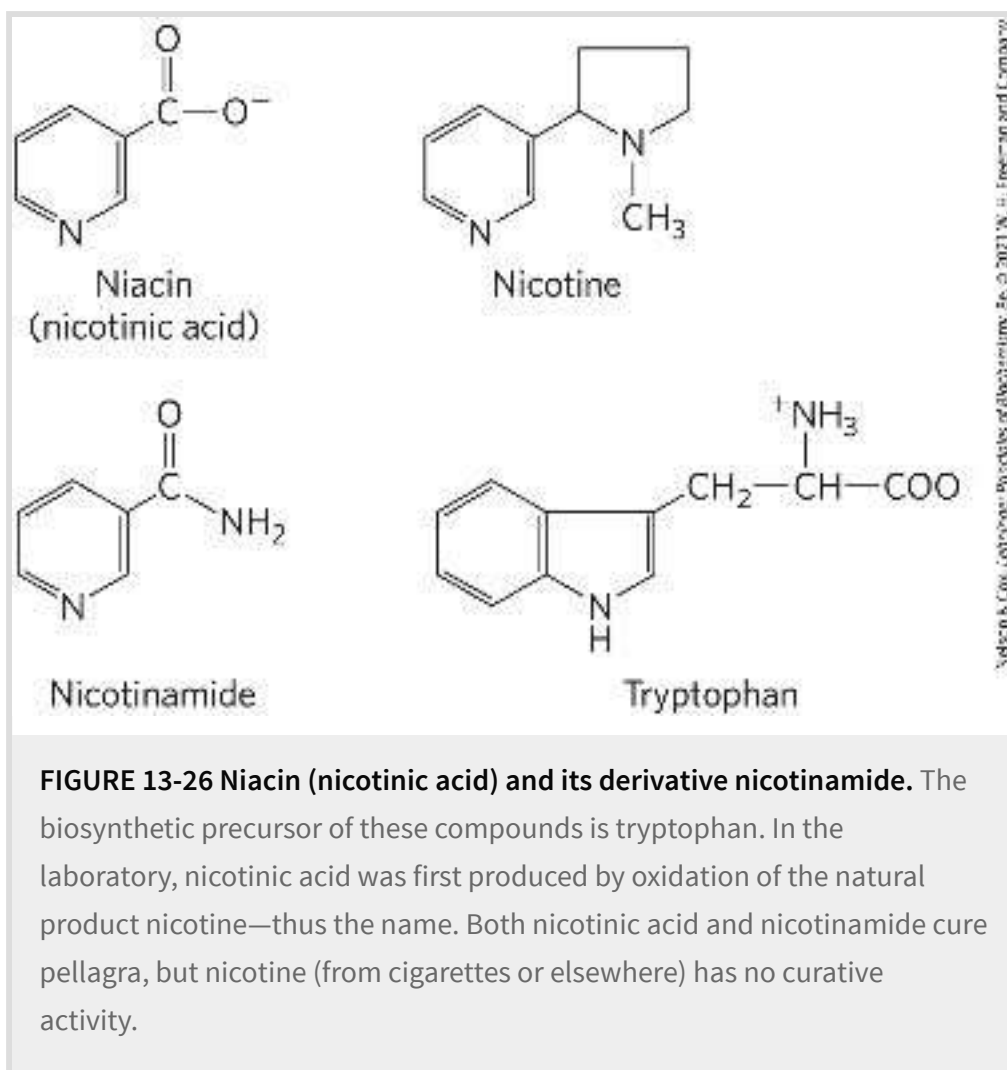
Dietary deficiency of niacin, the vitamin form of NAD and NADP, causes pellagra ([Fig. 13-25](#)). The pyridine-like rings of NAD and NADP are derived from the vitamin niacin (nicotinic acid; [Fig. 13-26](#)), which is synthesized from tryptophan. Humans generally cannot synthesize sufficient quantities of niacin, and this is especially so for individuals with diets low in tryptophan (maize, for example, has a low tryptophan content). Niacin deficiency, which affects all the NAD(P)-dependent dehydrogenases, causes the serious human disease pellagra (Italian for “rough skin”) and a related disease in dogs, called black tongue. Pellagra is characterized by the “three Ds”: dermatitis, diarrhea, and dementia, followed in many cases by death. A century ago, pellagra was a common human disease; in the southern United States, where maize was a dietary staple, about 100,000 people were afflicted and about 10,000 died as a result of this disease between 1912 and 1916. In 1920, Joseph Goldberger showed pellagra to be caused by a dietary insufficiency, and in 1937, Frank Strong, D. Wayne Woolley, and Conrad Elvehjem identified

niacin as the curative agent for the dog version of pellagra, black tongue. Supplementation of the human diet with this inexpensive compound has nearly eradicated pellagra in the populations of the developed world, with one significant exception: people who drink excessive amounts of alcohol. In these individuals, intestinal absorption of niacin is much reduced, and caloric needs are often met with distilled spirits that are virtually devoid of vitamins, including niacin. ■



Double Vision/Science Source

FIGURE 13-25 Dermatitis associated with pellagra. Dermatitis involving the face, hands, and feet is an early sign of pellagra, a serious human disease that results from insufficient niacin in the diet. Untreated, pellagra leads to dementia and ultimately is fatal.



Flavin Nucleotides Are Tightly Bound in Flavoproteins

Flavoproteins are enzymes that catalyze oxidation-reduction reactions using either flavin mononucleotide (FMN) or flavin adenine dinucleotide (FAD) as coenzyme ([Fig. 13-27](#)). These coenzymes, the **flavin nucleotides**, are derived from the vitamin riboflavin. The fused ring structure of flavin nucleotides (the isoalloxazine ring) undergoes reversible reduction, accepting

either one or two electrons in the form of one or two hydrogen atoms (each atom an electron plus a proton) from a reduced substrate. The fully reduced forms are abbreviated FADH_2 and FMNH_2 . When a fully oxidized flavin nucleotide accepts only one electron (one hydrogen atom), the semiquinone form of the isoalloxazine ring is produced, abbreviated $\text{FADH}^\bullet$ and $\text{FMNH}^\bullet$. Because flavin nucleotides have a chemical specialty that is slightly different from that of the nicotinamide coenzymes — the ability to participate in either one- or two-electron transfers — flavoproteins are involved in a greater diversity of reactions than the NAD(P)-linked dehydrogenases.

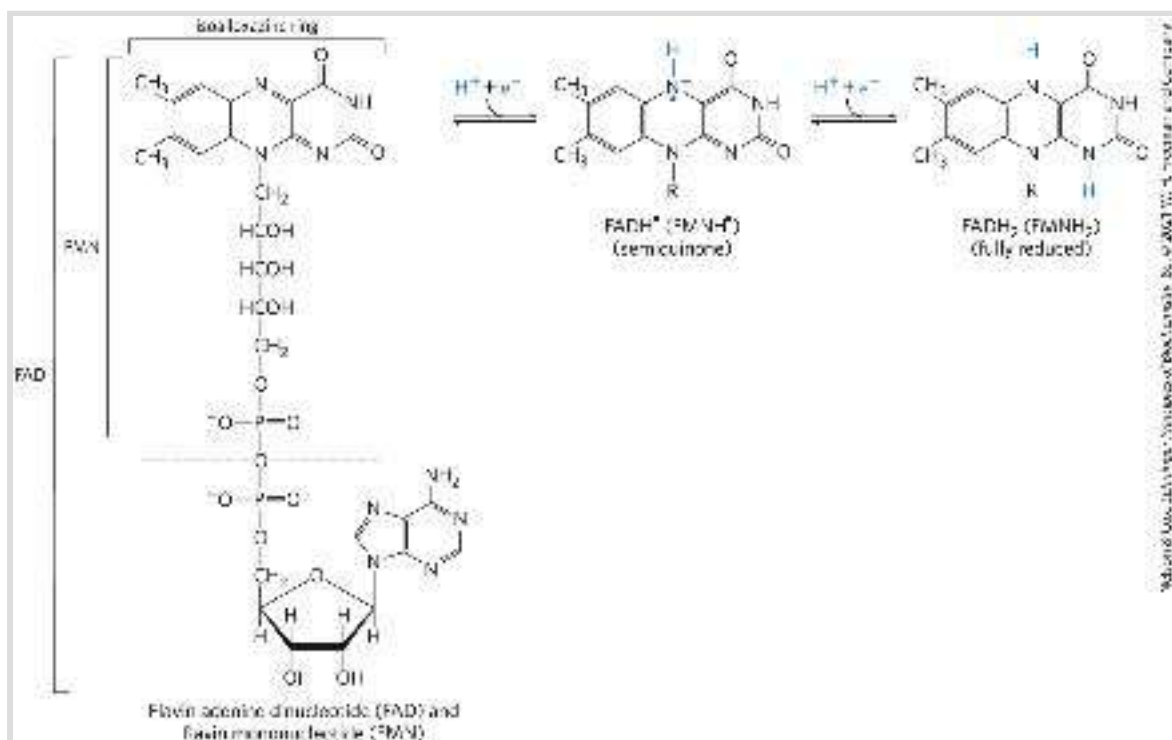


FIGURE 13-27 Oxidized and reduced FAD and FMN. FMN consists of the structure above the dashed red line across the FAD molecule (oxidized form). The flavin nucleotides accept two hydrogen atoms (two electrons and two protons), both of which appear in the flavin ring system (isoalloxazine ring). When FAD or FMN accepts only one hydrogen atom, the semiquinone, a stable free radical, forms.

Like the nicotinamide coenzymes ([Fig. 13-24](#)), the flavin nucleotides undergo a shift in a major absorption band on reduction (again, useful to biochemists who want to monitor reactions involving these coenzymes). Flavoproteins that are fully reduced (two electrons accepted) generally have an absorption maximum near 360 nm. When partially reduced (one electron), they acquire another absorption maximum at about 450 nm; when fully oxidized, the flavin has maxima at 370 and 440 nm.

The flavin nucleotide in most flavoproteins is bound rather tightly to the protein, and in some enzymes, such as succinate dehydrogenase, it is bound covalently. Such tightly bound coenzymes are properly called prosthetic groups. They do not transfer electrons by diffusing from one enzyme to another; rather, they provide a means by which the flavoprotein can temporarily hold electrons while it catalyzes electron transfer from a reduced substrate to an electron acceptor. One important feature of the flavoproteins is the variability in the standard reduction potential (E'°) of the bound flavin nucleotide. Tight association between the enzyme and prosthetic group confers on the flavin ring a reduction potential typical of that particular flavoprotein, sometimes quite different from the reduction potential of the free flavin nucleotide. FAD bound to succinate dehydrogenase, for example, has an E'° close to 0.0 V, compared with -0.219 V for free FAD; E'° for other flavoproteins ranges from -0.40 V to $+0.06$ V. Flavoproteins are often very complex; some have, in addition to a flavin nucleotide, tightly bound

inorganic ions (iron or molybdenum, for example) capable of participating in electron transfers.

We examine the function of flavoproteins as electron carriers in [Chapters 19](#) and [20](#), when we consider their roles in oxidative phosphorylation (in mitochondria) and photophosphorylation (in chloroplasts).

SUMMARY 13.4 *Biological Oxidation-Reduction Reactions*

■ In many organisms, a central energy-conserving process is the stepwise oxidation of glucose to CO_2 , in which some of the energy of oxidation is conserved in ATP as electrons are passed to O_2 .

■ Biological oxidation-reduction reactions can be described in terms of two half-reactions, each with a characteristic standard reduction potential, E'° .

■ Many biological oxidation reactions are dehydrogenations in which one or two hydrogen atoms ($\text{H}^+ + e^-$) are transferred from a substrate to a hydrogen acceptor. In some biological redox reactions, the substrate loses both electrons and protons, the equivalent of losing hydrogen. The many enzymes that catalyze such reactions are called dehydrogenases.

■ When two electrochemical half-cells, each containing the components of a half-reaction, are connected, electrons tend to flow to the half-cell with the higher reduction potential. The strength of this tendency is proportional to the difference

between the two reduction potentials (ΔE) and is a function of the concentrations of oxidized and reduced species.

■ The standard free-energy change for an oxidation-reduction reaction is directly proportional to the difference in standard reduction potentials of the two half-cells: $\Delta G'^{\circ} = -nF\Delta E'^{\circ}$.

■ Oxidation-reduction reactions in living cells involve specialized electron carriers. NAD and NADP are the freely diffusible coenzymes of many dehydrogenases. Both NAD^{+} and NADP^{+} accept two electrons and one proton. In addition to its role in oxidation-reduction reactions, NAD^{+} is the source of AMP in the bacterial DNA ligase reaction and of ADP-ribose in the cholera toxin reaction, and it is hydrolyzed in the deacetylation of proteins by some sirtuins.

■ Lack of the vitamin niacin prevents NAD synthesis and leads to pellagra.

■ FAD and FMN, the flavin nucleotides, serve as tightly bound prosthetic groups of flavoproteins. They can accept either one or two electrons and one or two protons. Their reduction potentials depend on the flavoprotein with which they are associated.

13.5 Regulation of Metabolic Pathways

Metabolic regulation is one of the most remarkable features of living organisms. Of the thousands of enzyme-catalyzed reactions that can take place in a cell, there is probably not one that escapes some form of regulation. This need to regulate every aspect of cellular metabolism becomes clear as one examines the complexity of metabolic reaction sequences. Although it is convenient for the student of biochemistry to divide metabolic processes into “pathways” that play discrete roles in the cell’s economy, no such separation exists in the living cell. Rather, every pathway we discuss in this book is inextricably intertwined with all the other cellular pathways in a multidimensional network of reactions ([Fig. 13-28](#)).

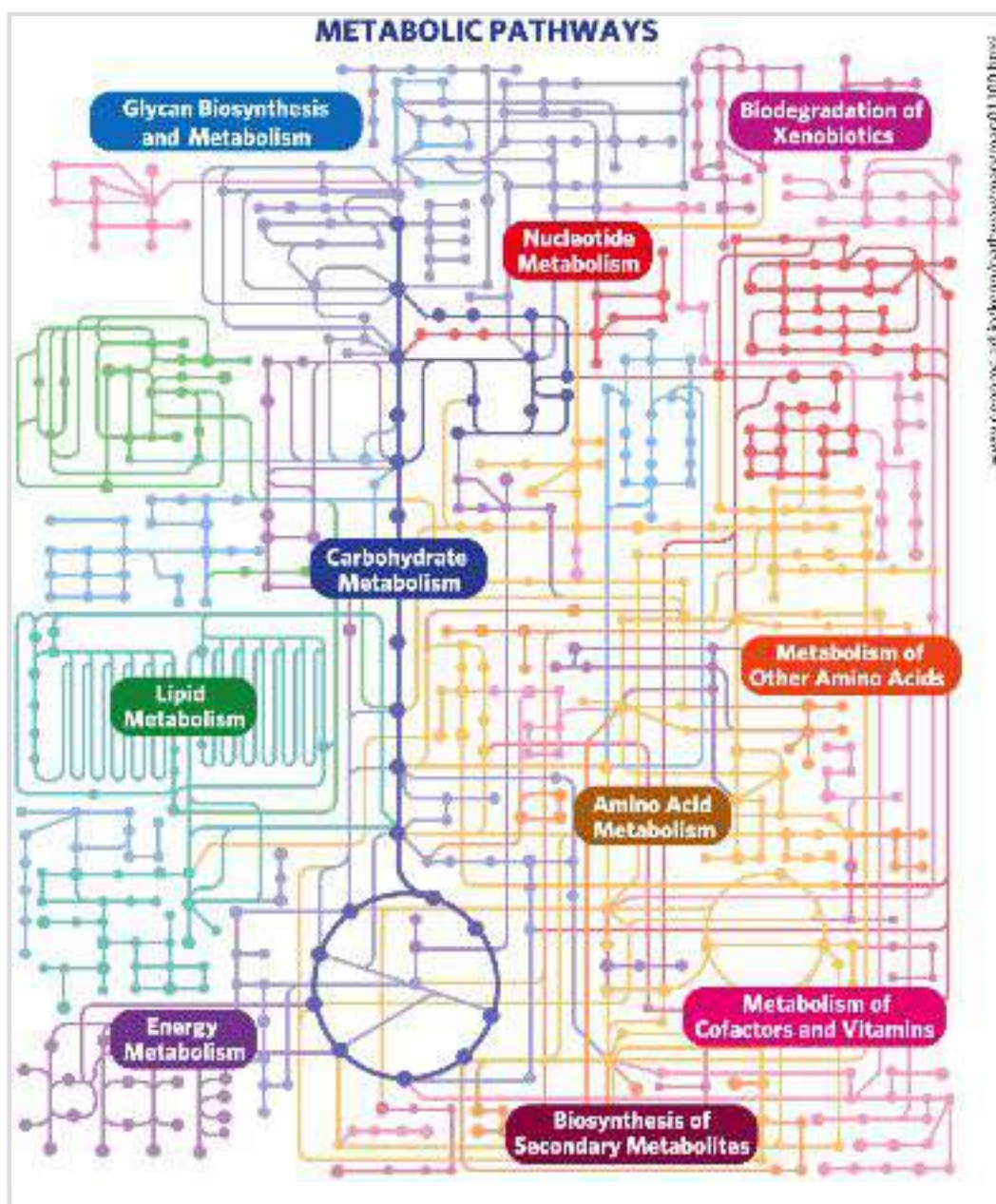


FIGURE 13-28 Metabolism as a three-dimensional meshwork. A typical eukaryotic cell has the capacity to make about 30,000 different proteins, which catalyze thousands of different reactions involving many hundreds of metabolites, most shared by more than one “pathway.” In this much-simplified overview of metabolic pathways, each dot represents an intermediate compound and each connecting line represents an enzymatic reaction. For a more realistic and far more complex diagram of metabolism, see the online KEGG PATHWAY database (www.genome.ad.jp/kegg/pathway/map/map01100.html); in this interactive map, you can click on each dot to obtain extensive data about the compound and the enzymes for which it is a substrate.

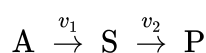
For example, in [Chapter 14](#) we will discuss four possible fates for **glucose 6-phosphate** in a hepatocyte: breakdown by glycolysis for the production of ATP, breakdown in the pentose phosphate pathway for the production of NADPH and pentose phosphates, use in the synthesis of complex polysaccharides of the extracellular matrix, or hydrolysis to glucose and phosphate to replenish blood glucose. In fact, glucose 6-phosphate has other possible fates in hepatocytes, too; it may, for example, be used to synthesize other sugars, such as glucosamine, galactose, galactosamine, fucose, and neuraminic acid, for use in protein glycosylation, or it may be partially degraded to provide acetyl-CoA for fatty acid and sterol synthesis. And *E. coli* can use glucose to produce the carbon skeleton of every one of its several thousand types of molecules. When any cell uses glucose 6-phosphate for one purpose, that “decision” affects all the other pathways for which glucose 6-phosphate is a precursor or intermediate: any change in the allocation of glucose 6-phosphate to one pathway affects, directly or indirectly, the flow of metabolites through all the others.

Cells and Organisms Maintain a Dynamic Steady State

The pathways of glucose metabolism provide, in the catabolic direction, the energy essential to oppose the forces of entropy and, in the anabolic direction, biosynthetic precursors and a storage form of metabolic energy. These reactions are so important to survival that very complex regulatory mechanisms have evolved to ensure that metabolites move through each pathway in the correct direction and at the correct rate to match exactly the cell's or the organism's changing circumstances. By a variety of mechanisms operating on different time scales, adjustments are made in the rate of metabolite flow through an entire pathway when external circumstances change.

Circumstances do change, sometimes dramatically. The availability of oxygen may decrease due to hypoxia (diminished delivery of oxygen to tissues) or ischemia (diminished flow of blood to tissues). Wound healing requires huge amounts of energy and biosynthetic precursors. The relative proportions of carbohydrate, fat, and protein in the diet vary from meal to meal, and the supply of fuels obtained in the diet is intermittent, requiring metabolic adjustments between meals and during periods of starvation.

Fuels such as glucose enter a cell, and waste products such as CO₂ leave, but the mass and the gross composition of a typical cell, organ, or adult animal do not change appreciably over time; cells and organisms exist in a dynamic steady state. For each metabolic reaction in a pathway, the substrate is provided by the preceding reaction at the same rate at which it is converted to product. Thus, although the rate (v) of metabolite flow, or **flux**, through this step of the pathway may be high and variable, the concentration of substrate, S , remains constant. So, for the two-step reaction



when $v_1 = v_2$, $[S]$ is constant.  For example, changes in v_1 for the entry of glucose from various sources into the blood are balanced by changes in v_2 for the uptake of glucose from the blood into various tissues, so the concentration of glucose in the blood ($[S]$) is held nearly constant at 5 mM. This is [homeostasis](#) for blood glucose. The failure of homeostatic mechanisms is often at the root of human disease. In diabetes mellitus, for example, the regulation of blood glucose concentration is defective as a result of the lack of or insensitivity to insulin, with profound medical consequences.

In the course of evolution, organisms have acquired a remarkable collection of regulatory mechanisms for maintaining homeostasis at the molecular, cellular, and organismal levels, as reflected in the proportion of genes that encode regulatory machinery. In humans, about 2,500 genes (~12% of all genes) encode regulatory proteins, including a variety of receptors, regulators of gene expression, and more than 500 different protein kinases! In many cases, the regulatory mechanisms overlap: one enzyme is subject to regulation by several different mechanisms.

Both the Amount and the Catalytic Activity of an Enzyme Can Be Regulated

The flux through an enzyme-catalyzed reaction can be modulated by changes in the *number* of enzyme molecules or by changes in the *catalytic activity* of each enzyme molecule already present. Such changes occur on time scales from milliseconds to many hours, in response to signals from within or outside the cell. Very rapid allosteric changes in enzyme activity are generally triggered locally, by changes in the local concentration of a small molecule — a substrate of the pathway in which that reaction is a step (say, glucose for glycolysis), a product of the pathway (ATP from glycolysis), or a key metabolite or cofactor (such as NADH) that indicates the cell's metabolic state. Second messengers (such as cyclic AMP and Ca^{2+}) generated intracellularly in response to extracellular signals (hormones, cytokines, and so forth) also mediate allosteric regulation, on a slightly slower time scale set by the rate of the signal-transduction mechanism (see [Chapter 12](#)).

Extracellular signals ([Fig. 13-29](#), ①) may be hormonal (insulin or epinephrine, for example) or neuronal (acetylcholine), or may be growth factors or cytokines. The number of molecules of a given enzyme in a cell is a function of the relative rates of synthesis and degradation of that enzyme. The rate of synthesis can be adjusted by the activation (in response to some outside signal) of a transcription factor ([Fig. 13-29](#), ②; described in more detail in [Chapter 28](#)). **Transcription factors** are nuclear proteins that, when activated, bind specific DNA regions (**response elements**) near a gene's promoter (its transcriptional starting point) and activate or repress the transcription of that gene, leading to increased or decreased synthesis of the encoded protein. Activation of a transcription factor is sometimes the result of its binding of a specific ligand and sometimes the result of its phosphorylation or dephosphorylation. Each gene is controlled by one or more response elements that are recognized by specific transcription factors. Genes that have several response elements are therefore controlled by several different transcription factors responding to several different signals. Groups of genes encoding proteins that act together, such as the enzymes of glycolysis, often share common response element sequences, so that a single signal, acting through a particular transcription factor, turns all of these genes on and off together.

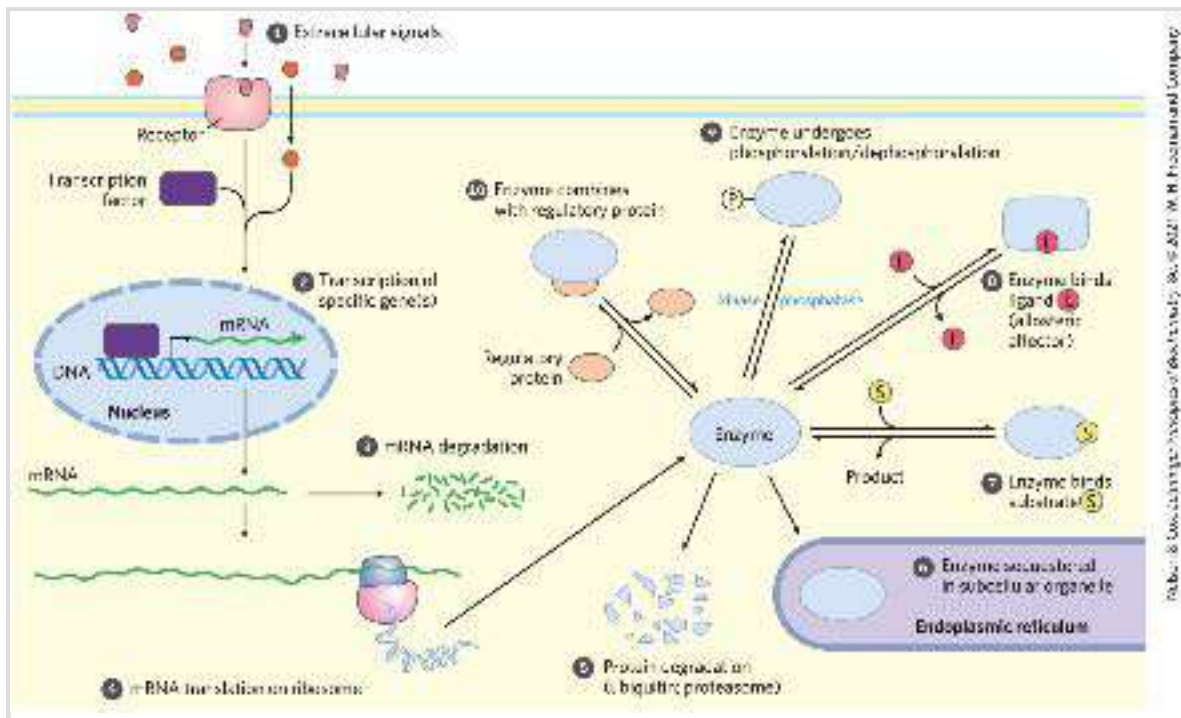


FIGURE 13-29 Factors affecting the activity of enzymes. The total activity of an enzyme can be changed by altering the *number* of its molecules in the cell, or its *effective* activity in a subcellular compartment (1 through 6), or by modulating the *activity* of existing molecules (7 through 10), as detailed in the text. An enzyme may be influenced by a combination of such factors.

The stability of messenger RNAs — their resistance to degradation by cellular ribonucleases ([Fig. 13-29](#), 3) — varies, and the amount of a given mRNA in the cell is a function of its rates of synthesis and degradation ([Chapter 26](#)). The rate at which an mRNA is translated into a protein by ribosomes ([Fig. 13-29](#), 4) is also regulated, and depends on several factors described in detail in [Chapter 27](#). Note that an n -fold increase in an mRNA does not always mean an n -fold increase in its protein product.

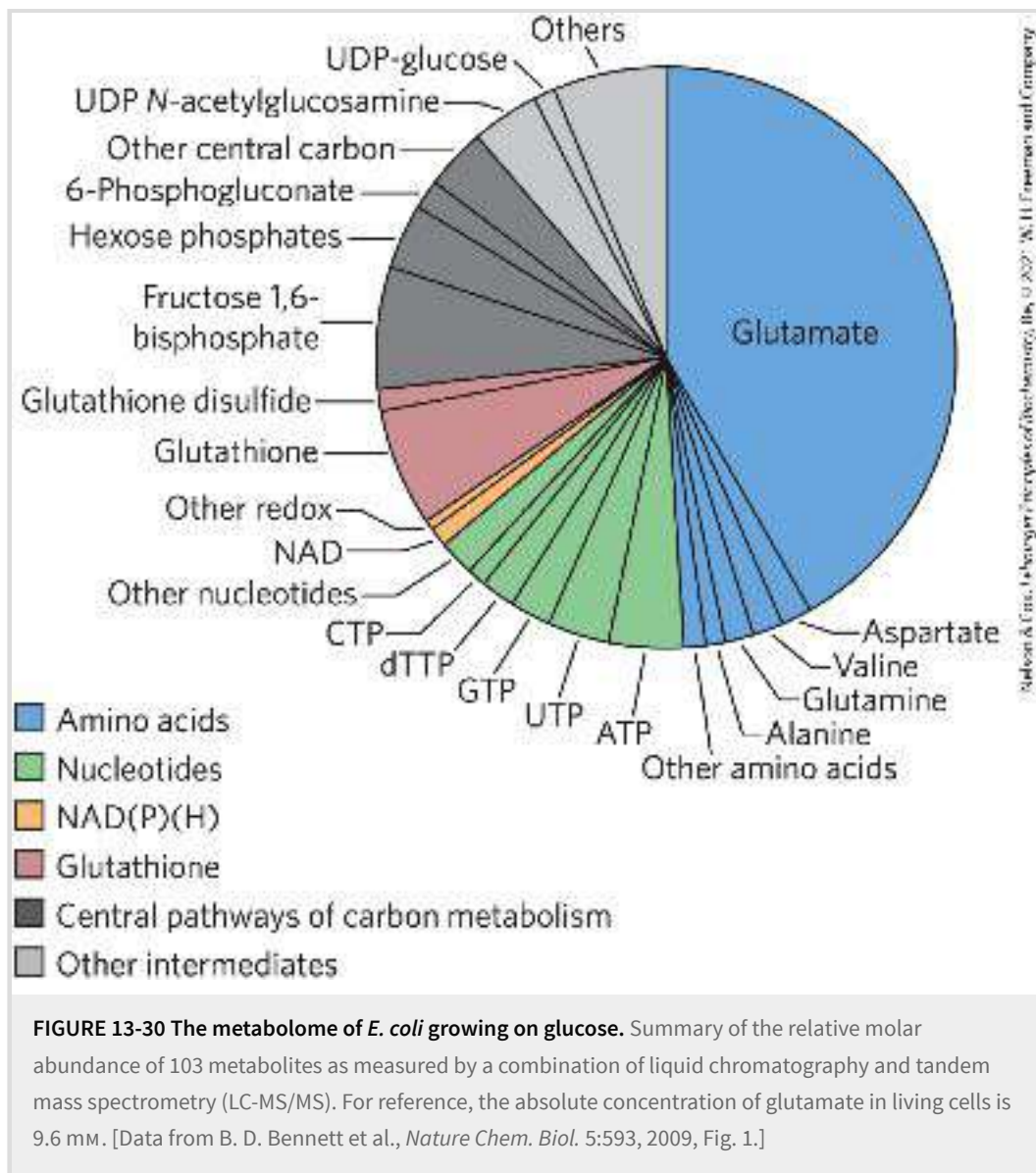
Once synthesized, protein molecules have a finite lifetime, which may range from minutes to many days ([Table 13-8](#)). The rate of protein degradation ([Fig. 13-29](#), 5) differs from one protein to another and depends on the conditions in the cell. Some proteins are tagged by the covalent attachment of ubiquitin for degradation in proteasomes, as discussed in [Chapter 27](#) (see, for example, the case of cyclin, in [Fig. 12-38](#)). **P6** Rapid **turnover** (synthesis followed by degradation) is energetically expensive, but proteins with a short half-life can reach new steady-state levels much faster than those with a long half-life, and the benefit of this quick responsiveness must balance or outweigh the cost to the cell.

TABLE 13-8 Average Half-Life of Proteins in Mammalian Tissues

Tissue	Average half-life (days)
Liver	0.9
Kidney	1.7
Heart	4.1
Brain	4.6
Muscle	10.7

Yet another way to alter the *effective* activity of an enzyme is to sequester the enzyme and its substrate in different compartments ([Fig. 13-29](#), 6). In muscle, for example, hexokinase cannot act on glucose until the sugar enters the myocyte from the blood, and the rate at which it enters depends on the activity of glucose transporters (see [Table 11-1](#)) in the plasma membrane. Within cells, membrane-bounded compartments segregate certain enzymes and enzyme systems, and the transport of substrate across these intracellular membranes may be the limiting factor in enzyme action.

By these several mechanisms for regulating enzyme level, cells can dramatically change their complement of enzymes in response to changes in metabolic circumstances. In vertebrates, liver is the most adaptable tissue; a change from a high-carbohydrate diet to a high-lipid diet, for example, affects the transcription of hundreds of genes and thus the levels of hundreds of proteins. These global changes in gene expression can be quantified in the entire complement of mRNAs (the [transcriptome](#)) or protein ([proteome](#)) of a cell type or organ, offering great insights into metabolic regulation. The effect of changes in the proteome is often a change in the total ensemble of low molecular weight metabolites, the [metabolome](#) ([Fig. 13-30](#)). The metabolome of *E. coli* growing on glucose is dominated by a few classes of metabolites: glutamate (49%); nucleotides (mainly ribonucleoside triphosphates) (15%); intermediates of glycolysis, the citric acid cycle, and the pentose phosphate pathway (central pathways of carbon metabolism) (15%); and redox cofactors and glutathione (9%).



Once the regulatory mechanisms that involve protein synthesis and degradation have produced a certain number of molecules of each enzyme in a cell, the activity of those enzymes can be further regulated in several other ways: by the concentration of substrate, the presence of allosteric effectors, covalent modifications, or binding of regulatory proteins — all of which can change the activity of an individual enzyme molecule ([Fig. 13-29](#), 7 to 10).

All enzymes are sensitive to the concentration of their substrate(s) ([Fig. 13-29](#), 7). Recall that in the simplest case (an enzyme that follows Michaelis-Menten kinetics), the initial rate of the reaction is half-maximal when the substrate is present at a concentration equal to K_m (that is, when the enzyme is half-saturated with

substrate). Activity drops off at lower $[S]$, and when $[S] \ll K_m$, the reaction rate is linearly dependent on $[S]$.

The relationship between $[S]$ and K_m is important because intracellular concentrations of substrate are often in the same range as, or lower than, K_m . The activity of hexokinase, for example, changes with $[glucose]$, and intracellular $[glucose]$ varies with the concentration of glucose in the blood. As we will see, the different forms (isozymes) of hexokinase have different K_m values and are therefore affected differently by changes in intracellular $[glucose]$, in ways that make sense physiologically. For a number of phosphoryl transfers from ATP, and for redox reactions using NADPH or NAD^+ , the metabolite concentration is well above the K_m (Fig. 13-31); these cofactors are not likely to be limiting in such reactions.

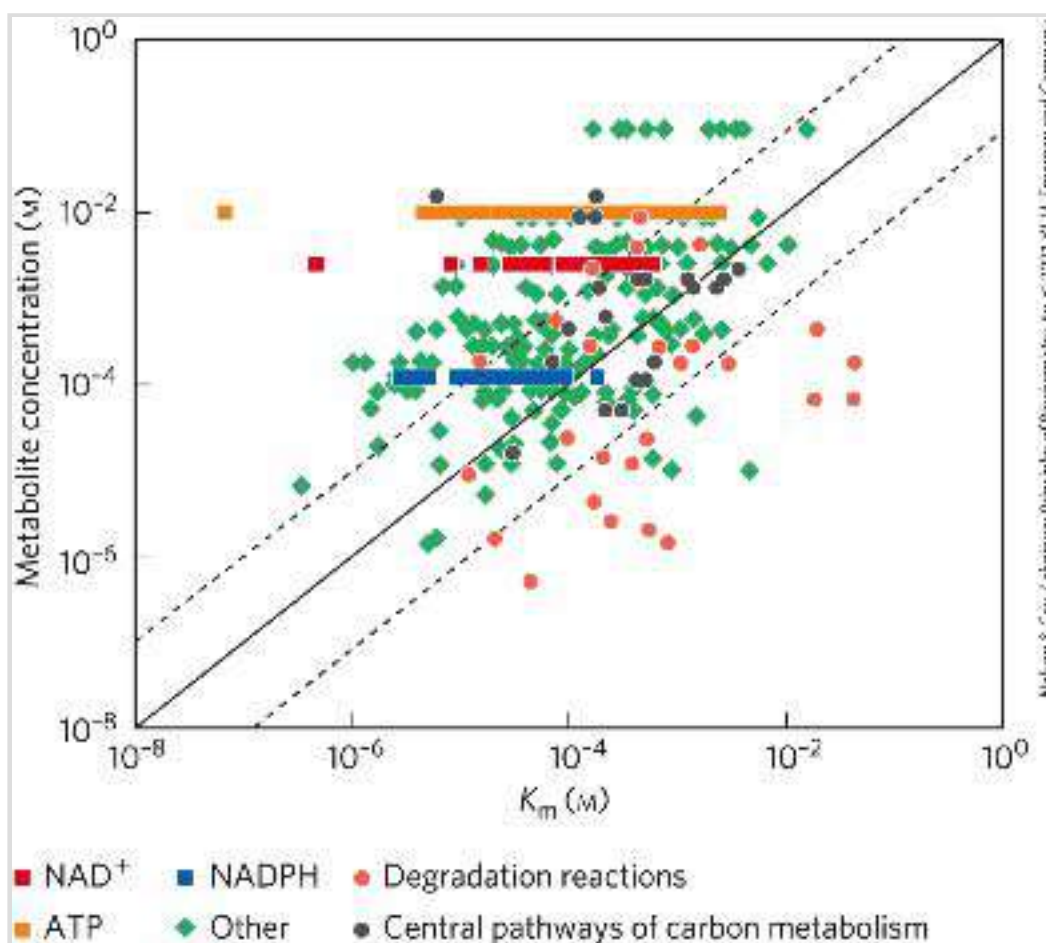


FIGURE 13-31 Comparison of K_m and substrate concentration for some metabolic enzymes.

Measured metabolite concentrations for *E. coli* growing on glucose are plotted against the known K_m for enzymes that consume that metabolite. The solid line is the line of unity (where metabolite concentration = K_m), and the dashed lines each denote a 10-fold deviation from the line of unity.

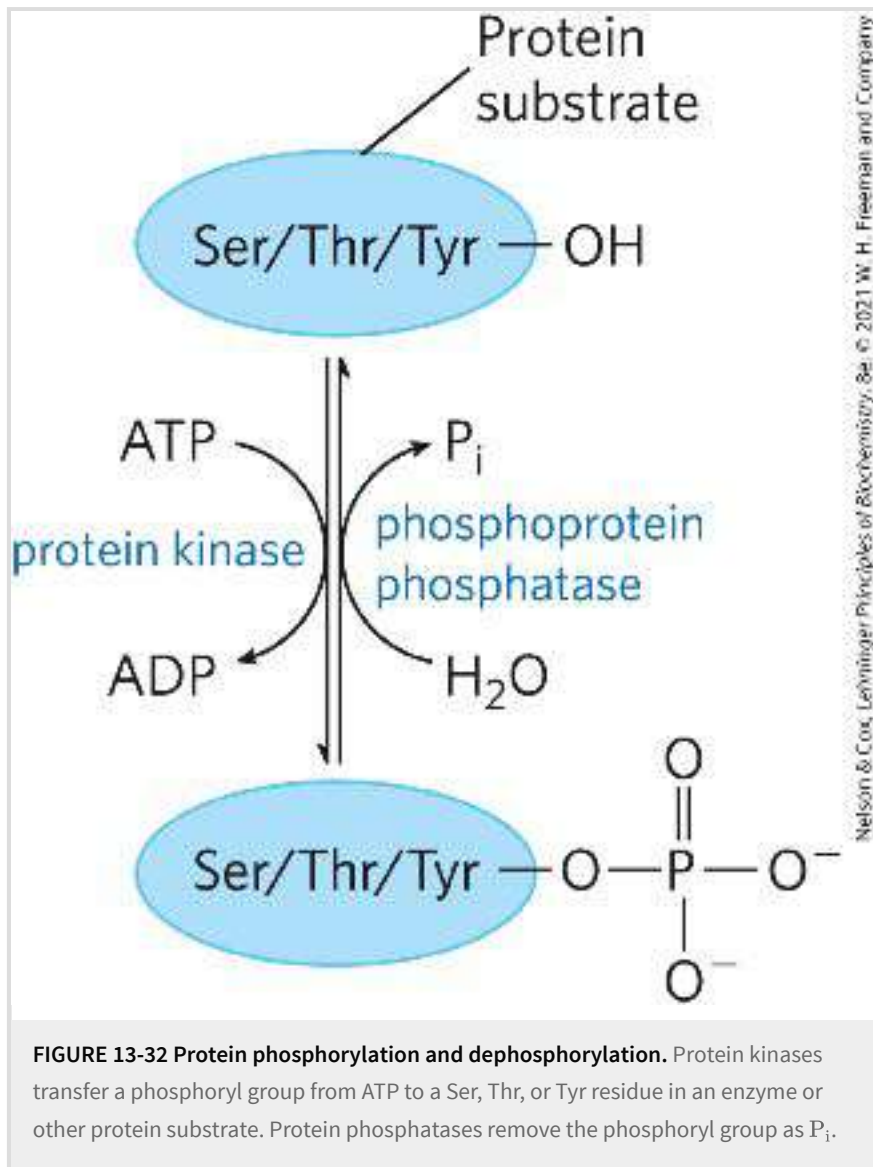
[Data from B. D. Bennett et al., *Nature Chem. Biol.* 5:593, 2009, Fig. 2.]

Enzyme activity can be either increased or decreased by an allosteric effector (Fig. 13-29, 8; see Fig. 6-37). Allosteric effectors typically convert hyperbolic kinetics to sigmoid kinetics, or vice versa (see Fig. 14-24b, for example). In the steepest part of the sigmoid curve, a small change in the concentration of substrate, or of allosteric effector, can have a large impact on reaction rate. Recall from Chapter 5 (p. 157) that the cooperativity of an allosteric protein can be expressed as a Hill coefficient, with higher coefficients meaning greater cooperativity. For an allosteric enzyme with a Hill coefficient of 4, activity increases from 10% $V_{\max}$ to 90% $V_{\max}$ with only a 3-fold increase in [S], compared with the 81-fold rise in [S] needed by an enzyme with no cooperative effects (Hill coefficient of 1; Table 13-9).

TABLE 13-9 Relationship between Hill Coefficient and the Effect of Substrate Concentration on Reaction Rate for Allosteric Enzymes

Hill coefficient (n_H)	Required change in [S] to increase V_0 from 10% to 90% $V_{\max}$
0.5	$\times 6,600$
1.0	$\times 81$
2.0	$\times 9$
3.0	$\times 4.3$
4.0	$\times 3$

Covalent modifications of enzymes or other proteins (Fig. 13-29, 9) occur within seconds or minutes of a regulatory signal, typically an extracellular signal. By far the most common modifications are phosphorylation and dephosphorylation (Fig. 13-32); up to half the proteins in a eukaryotic cell are phosphorylated under some circumstances. Phosphorylation by a specific protein kinase may alter the electrostatic features of an enzyme's active site, cause movement of an inhibitory region of the enzyme out of the active site, alter the enzyme's interaction with other proteins, or force conformational changes that translate into changes in $V_{\max}$ or K_m . For covalent modification to be useful in regulation, the cell must be able to restore the altered enzyme to its original activity state. A family of phosphoprotein phosphatases, at least some of which are themselves under regulation, catalyzes the dephosphorylation of proteins.



Finally, many enzymes are regulated by association with and dissociation from another, regulatory protein ([Fig. 13-29](#), ) . For example, the cyclic AMP-dependent protein kinase (PKA; see [Fig. 12-6](#)) is inactive until cAMP binding separates catalytic from regulatory (inhibitory) subunits of the enzyme.

These several mechanisms for altering the flux through a step in a metabolic pathway are not mutually exclusive.  It is very common for a single enzyme to be regulated at the level of transcription and by both allosteric and covalent mechanisms. The combination provides fast, smooth, effective regulation in response to a very wide array of perturbations and signals.

In the discussions that follow, it is useful to think of changes in enzymatic activity as serving two distinct though complementary roles. We use the term **metabolic regulation** to refer to processes that serve to maintain homeostasis at the molecular level — to hold some cellular parameter (concentration of a metabolite, for example) at a steady level over time, even as the flow of metabolites through the pathway changes. The term **metabolic control** refers to a process that leads to a change in the output of a metabolic pathway over time, in response to some outside signal or change in circumstances. The distinction, although useful, is not always easy to make.

Reactions Far from Equilibrium in Cells Are Common Points of Regulation

For some steps in a metabolic pathway the reaction is close to equilibrium, with the cell in its dynamic steady state ([Fig. 13-33](#)). The net flow of metabolites through these steps is the small difference between the rates of the forward and reverse reactions, rates that are very similar when a reaction is near equilibrium. Small changes in substrate or product concentration can produce large changes in the net rate, and they can even change the direction of the net flow. We can identify these near-equilibrium reactions in a cell by comparing the **mass-action ratio**, Q , with the equilibrium constant for the reaction, K'_{eq} . Recall that for the reaction $A + B \rightarrow C + D$, $Q = [C][D]/[A][B]$. In practice, when Q and K'_{eq} are within 1 to 2 orders of magnitude of each other, the reaction is near equilibrium. This is the case for more than half of the enzymes in the glycolytic pathway, for example ([Table 13-10](#)).

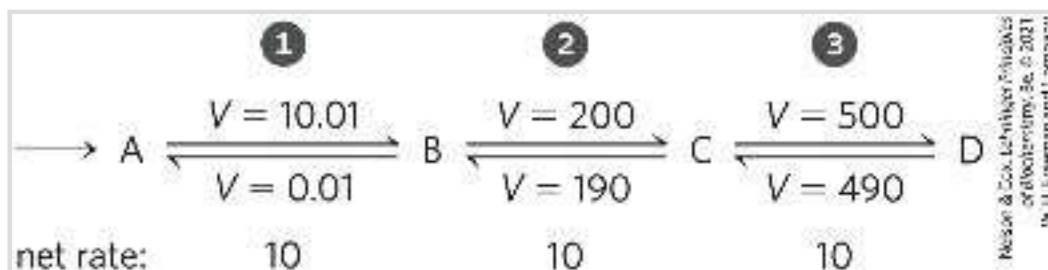


FIGURE 13-33 Near-equilibrium and nonequilibrium steps in a metabolic pathway. Steps **2** and **3** of this pathway are near equilibrium in the cell; for each step, the rate (V) of the forward reaction is only slightly greater than the reverse rate, so the net forward rate (10) is relatively low and the free-energy change, ΔG , is close to zero. An increase in $[C]$ or $[D]$ can reverse the direction of these steps. Step **1** is maintained in the cell far from equilibrium; its forward rate greatly exceeds its reverse rate. The net rate

of step ① (10) is much larger than the reverse rate (0.01) and is identical to the net rates of steps ② and ③ when the pathway is operating in the steady state. Step ① has a large, negative ΔG .

TABLE 13-10 Equilibrium Constants, Mass-Action Ratios, and Free-Energy Changes for Enzymes of Carbohydrate Metabolism

Enzyme	K'_{eq}	Mass-action ratio, Q		Reaction near equilibrium in vivo? ^a	$\Delta G'^{\circ}$ (kJ/mol)	ΔG (kJ/mol) in heart
		Liver	Heart			
Hexokinase	1×10^3	2×10^{-2}	8×10^{-2}	No	-17	-27
PFK-1	1.0×10^3	9×10^{-2}	3×10^{-2}	No	-14	-23
Aldolase	1.0×10^{-4}	1.2×10^{-6}	9×10^{-6}	Yes	+24	-6.0
Triose phosphate isomerase	4×10^{-2}	— ^b	2.4×10^{-1}	Yes	+7.5	+3.8
Glyceraldehyde 3-phosphate dehydrogenase + phosphoglycerate kinase	2×10^3	6×10^2	9.0	Yes	-13	+3.5
Phosphoglycerate mutase	1×10^{-1}	1×10^{-1}	1.2×10^{-1}	Yes	+4.4	+0.6
Enolase	3	2.9	1.4	Yes	-3.2	-0.5
Pyruvate kinase	2×10^4	7×10^{-1}	40	No	-31	-17
Phosphohexose isomerase	4×10^{-1}	3.1×10^{-1}	2.4×10^{-1}	Yes	+2.2	-1.4
Pyruvate carboxylase + PEP carboxykinase	7	1×10^{-3}	— ^b	No	-5.0	-23
Glucose 6-phosphatase	8.5×10^2	1.2×10^2	— ^b	Yes	-17	-5.0

Data for K'_{eq} and Q from E. A. Newsholme and C. Start, *Regulation in Metabolism*, pp. 97, 263, Wiley Press, 1973. ΔG and $\Delta G'^{\circ}$ were calculated from these data.

^aFor simplicity, any reaction for which the absolute value of the calculated ΔG is less than 6 is considered near equilibrium.

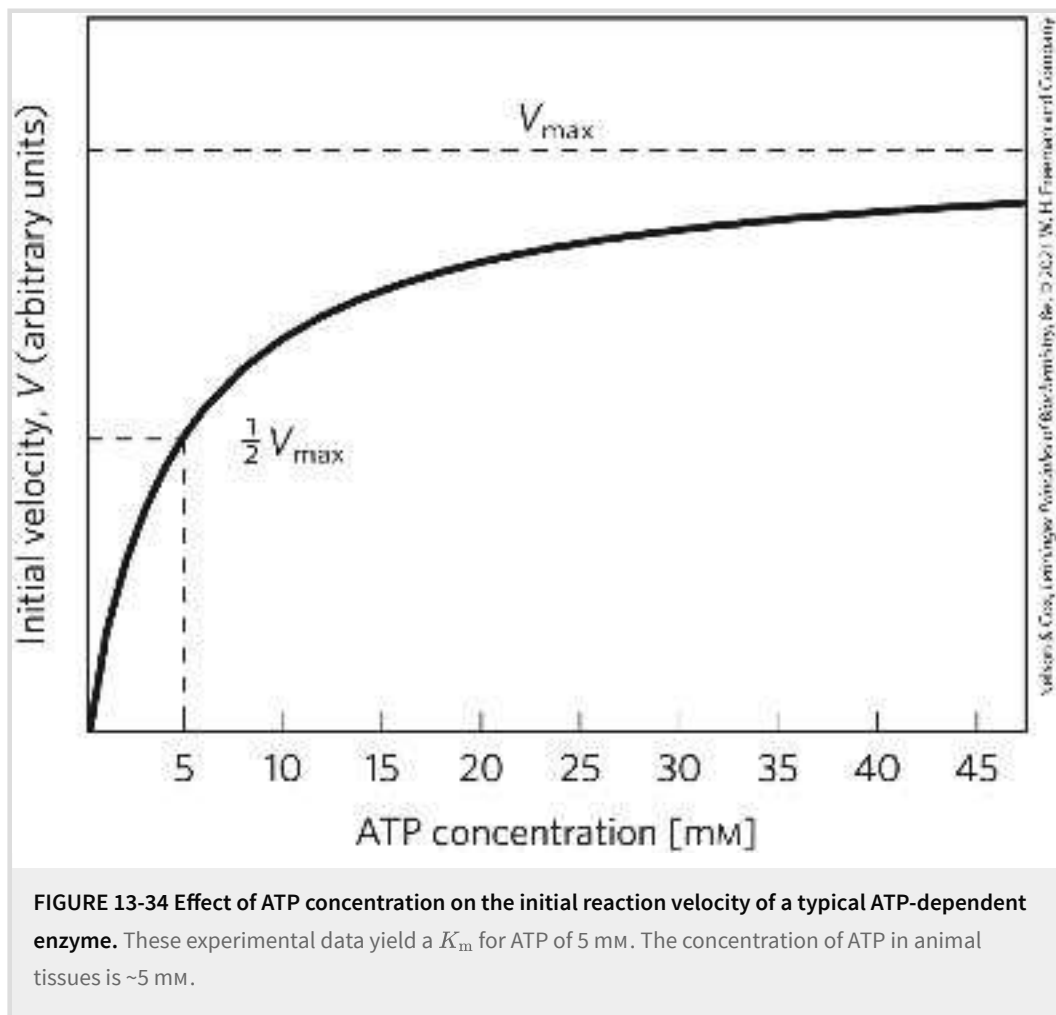
^bData not available.

Other reactions are far from equilibrium in the cell. For example, K'_{eq} for the phosphofructokinase-1 (PFK-1) reaction is about 1,000, but Q ($[\text{fructose 1,6-bisphosphate}][\text{ADP}]/[\text{fructose 6-phosphate}][\text{ATP}]$) in a hepatocyte in the steady state is about 0.1 ([Table 13-10](#)). It is *because* the reaction is so far from equilibrium in the cell that the process is exergonic under cellular conditions and tends to go in the forward direction. The reaction is held far from equilibrium because, under prevailing cellular conditions of substrate, product, and effector concentrations, the rate of conversion of fructose 6-phosphate to fructose 1,6-bisphosphate is limited by the activity of PFK-1. PFK-1 activity itself is limited by the number of PFK-1 molecules present and by the actions of allosteric effectors. Thus, the net forward rate of the enzyme-catalyzed reaction is equal to the net flow of glycolytic intermediates through other steps in the pathway, and the reverse flow through PFK-1 remains near zero.

The cell *cannot* allow reactions with large equilibrium constants to reach equilibrium. If $[\text{fructose 6-phosphate}]$, $[\text{ATP}]$, and $[\text{ADP}]$ in the cell were held at typical levels (low millimolar concentrations) and the PFK-1 reaction were allowed to reach equilibrium by an increase in $[\text{fructose 1,6-bisphosphate}]$, the concentration of fructose 1,6-bisphosphate would rise into the molar range, wreaking osmotic havoc on the cell. Consider another case: if the reaction $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ were allowed to approach equilibrium in the cell, the actual free-energy change (ΔG) for that reaction (ΔG_p ; see [Worked Example 13-2, p. 480](#)) would approach zero, and ATP would lose the high phosphoryl group transfer potential that makes it valuable to the cell. It is therefore essential that enzymes catalyzing ATP breakdown and other highly exergonic reactions in a cell be sensitive to regulation, so that when metabolic changes are forced by external circumstances, the flow through these enzymes will be adjusted to ensure that $[\text{ATP}]$ remains far above its equilibrium level. When such metabolic changes occur, the activities of enzymes in all interconnected pathways adjust to keep these critical steps away from equilibrium. Thus, not surprisingly, many enzymes that catalyze highly exergonic reactions are subject to a variety of subtle regulatory mechanisms. The multiplicity of these adjustments is so great that we cannot predict by examining the properties of any one enzyme in a pathway whether that enzyme has a strong influence on net flow through the entire pathway.

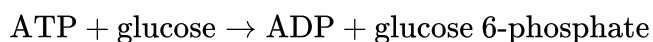
Adenine Nucleotides Play Special Roles in Metabolic Regulation

After the protection of its DNA from damage, perhaps nothing is more important to a cell than maintaining a constant supply and concentration of ATP. Many ATP-using enzymes have K_m values between 0.1 and 1 mM, and the ATP concentration in a typical cell is about 5 to 10 mM (Fig. 13-31). If [ATP] were to drop significantly, these enzymes would be less than fully saturated by their substrate (ATP), and the rates of hundreds of reactions that involve ATP would decrease (Fig. 13-34); the cell would probably not survive this kinetic effect on so many reactions.



There is also an important *thermodynamic* effect of lowered [ATP]. Because ATP is converted to ADP or AMP when “spent” to accomplish cellular work, the [ATP]/[ADP] ratio profoundly affects all reactions that employ these cofactors. The same is true

for other important cofactors, such as NADH/NAD⁺ and NADPH/NADP⁺. For example, consider the reaction catalyzed by hexokinase:



$$K'_{\text{eq}} = \frac{[\text{ADP}]_{\text{eq}}[\text{glucose 6-phosphate}]_{\text{eq}}}{[\text{ATP}]_{\text{eq}}[\text{glucose}]_{\text{eq}}} = 2 \times 10^3$$

Note that this expression holds true *only* when reactants and products are at their *equilibrium* concentrations, where $\Delta G = 0$. At any other set of concentrations, ΔG is not zero. Recall (from [Section 13.1](#)) that the ratio of products to substrates (the mass-action ratio, Q) determines the magnitude and sign of ΔG and therefore the driving force, ΔG , of the reaction:

$$\Delta G = \Delta G'^{\circ} + RT \ln \frac{[\text{ADP}][\text{glucose 6-phosphate}]}{[\text{ATP}][\text{glucose}]}$$

Because an alteration of this driving force profoundly influences every reaction that involves ATP, organisms have evolved under strong pressure to develop regulatory mechanisms responsive to the [ATP]/[ADP] ratio.

AMP concentration is an even more sensitive indicator of a cell's energetic state than is [ATP]. Normally, cells have a far higher concentration of ATP (5 to 10 mM) than of AMP (<0.1 mM). When some process (say, muscle contraction) consumes ATP, AMP is produced in two steps. First, hydrolysis of ATP produces ADP, then the reaction catalyzed by [adenylate kinase](#) produces AMP:



If ATP is consumed such that its concentration drops 10%, the relative increase in [AMP] is much greater than that of [ADP] ([Table 13-11](#)). It is not surprising, therefore,

that many regulatory processes are keyed to changes in [AMP].  Probably the most important mediator of regulation by AMP is [AMP-activated protein kinase \(AMPK\)](#), which responds to an increase in [AMP] by phosphorylating key proteins and thus regulating their activities. (AMPK is not to be confused with the *cyclic* AMP-dependent protein kinase PKA; see [Section 12.2](#).) The rise in [AMP] may be caused by a reduced nutrient supply or by increased physical exercise. AMP activates AMPK allosterically, which increases glucose transport and activates glycolysis and fatty acid oxidation, while suppressing energy-requiring processes such as the synthesis of glycogen, fatty acids, cholesterol, and protein. All of the changes effected by AMPK serve to raise [ATP] and lower [AMP]. In [Chapter 23](#), we discuss the role of AMPK in balancing anabolism and catabolism in the whole organism.

TABLE 13-11 Relative Changes in [ATP] and [AMP] When ATP Is Consumed

Adenine nucleotide	Concentration before ATP depletion (mM)	Concentration after ATP depletion (mM)	Relative change
ATP	5.0	4.5	10%
ADP	1.0	1.0	0
AMP	0.1	0.6	600%

SUMMARY 13.5 Regulation of Metabolic Pathways

- In a metabolically active cell in a steady state, intermediates are formed and consumed at equal rates. When a transient perturbation alters the rate of formation or consumption of a metabolite, compensating changes in enzyme activities return the system to the steady state.
- Cells regulate their metabolism by a variety of mechanisms over a time scale ranging from less than a millisecond to days, either by changing the activity of existing enzyme molecules or by changing the number of molecules of a specific enzyme.
- Various signals activate or inactivate transcription factors, which act in the nucleus to regulate gene expression. Changes in the transcriptome lead to changes in the proteome, and ultimately in the metabolome of a cell or tissue.

- In multistep processes such as glycolysis, certain reactions are essentially at equilibrium in the steady state; the rates of these reactions rise and fall with substrate concentration. Other reactions are far from equilibrium; these steps are typically the points of regulation of the overall pathway.
- Regulatory mechanisms maintain nearly constant levels of key metabolites such as ATP and NADH in cells and glucose in the blood, while matching the use or production of glucose to the organism's changing needs.
- The levels of ATP and AMP are a sensitive reflection of a cell's energy status, and when the $[ATP]/[AMP]$ ratio decreases, the AMP-activated protein kinase (AMPK) triggers a variety of cellular responses to raise $[ATP]$ and lower $[AMP]$.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

autotroph

heterotroph

metabolite

intermediary metabolism

catabolism

anabolism

energy transduction

free energy, G

exergonic

endergonic

enthalpy, H

exothermic

endothermic

entropy, S

standard transformed constants

mass-action ratio, Q

homolytic cleavage

radical

heterolytic cleavage

nucleophile

electrophile

carbanion

carbocation

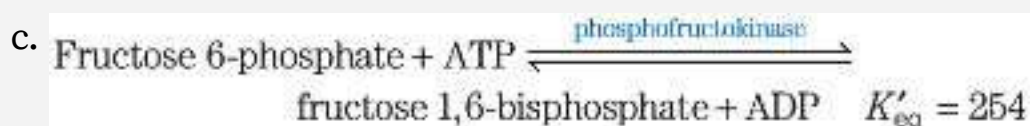
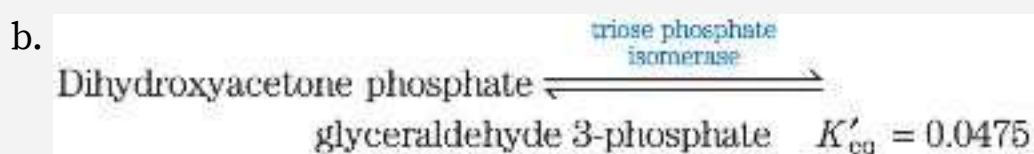
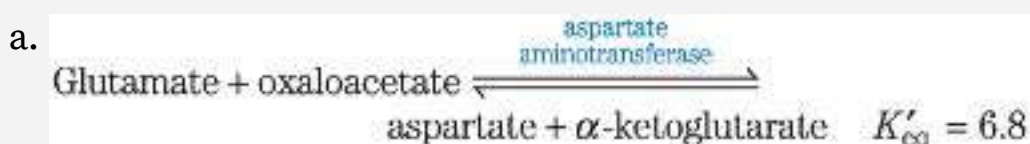
aldol condensation
Claisen condensation
kinases
phosphorylation potential (ΔG_p)
thioester
adenylylation
inorganic pyrophosphatase
nucleoside diphosphate kinase
adenylate kinase
creatine kinase
phosphagens
electromotive force (emf)
conjugate redox pair
dehydrogenases
reducing equivalent
standard reduction potential (E°)
pyridine nucleotide
oxidoreductase
flavoprotein
flavin nucleotides
glucose 6-phosphate
homeostasis
transcription factor
response element
turnover
transcriptome
proteome
metabolome
metabolic regulation
metabolic control
adenylate kinase
AMP-activated protein kinase (AMPK)

PROBLEMS

1. Entropy Changes during Egg Development Consider a system consisting of an egg in an incubator. The white and yolk of the egg contain proteins, carbohydrates, and lipids. If fertilized, the egg transforms from a single cell to a complex organism. Discuss this irreversible process in terms of the entropy changes in the system and surroundings. Be sure that you first clearly define the system and surroundings.

2. Calculation of $\Delta G'^{\circ}$ from an Equilibrium Constant

Calculate the standard free-energy change for each of the three metabolically important enzyme-catalyzed reactions, using the equilibrium constants given for the reactions at 25 °C and pH 7.0.

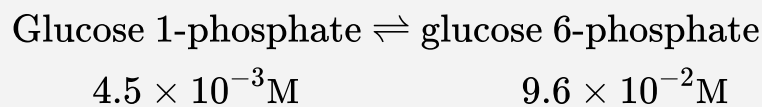


3. Calculation of the Equilibrium Constant from $\Delta G'^{\circ}$

Calculate the equilibrium constant K'_{eq} for each of the three reactions at pH 7.0 and 25 °C, using the $\Delta G'^{\circ}$ values in [Table 13-4](#).

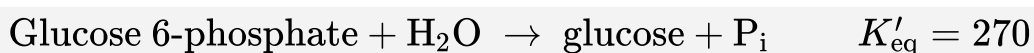
- a.
$$\text{Glucose 6-phosphate} + \text{H}_2\text{O} \xrightleftharpoons{\text{glucose 6-phosphatase}} \text{glucose} + \text{P}_i$$
- b.
$$\text{Lactose} + \text{H}_2\text{O} \xrightleftharpoons{\beta\text{-galactosidase}} \text{glucose} + \text{galactose}$$
- c.
$$\text{Malate} \xrightleftharpoons{\text{fumarase}} \text{fumarate} + \text{H}_2\text{O}$$

4. Experimental Determination of K'_{eq} and $\Delta G'^{\circ}$ Incubating a 0.1 M solution of glucose 1-phosphate at 25 °C with a catalytic amount of phosphoglucomutase transforms some of the glucose 1-phosphate to glucose 6-phosphate. At equilibrium, the concentrations of the reaction components are



Calculate K'_{eq} and $\Delta G'^{\circ}$ for this reaction.

5. Experimental Determination of $\Delta G'^{\circ}$ for ATP Hydrolysis A direct measurement of the standard free-energy change associated with the hydrolysis of ATP is technically demanding because the minute amount of ATP remaining at equilibrium is difficult to measure accurately. The value of $\Delta G'^{\circ}$ can be calculated indirectly, however, from the equilibrium constants of two other enzymatic reactions having less favorable equilibrium constants:



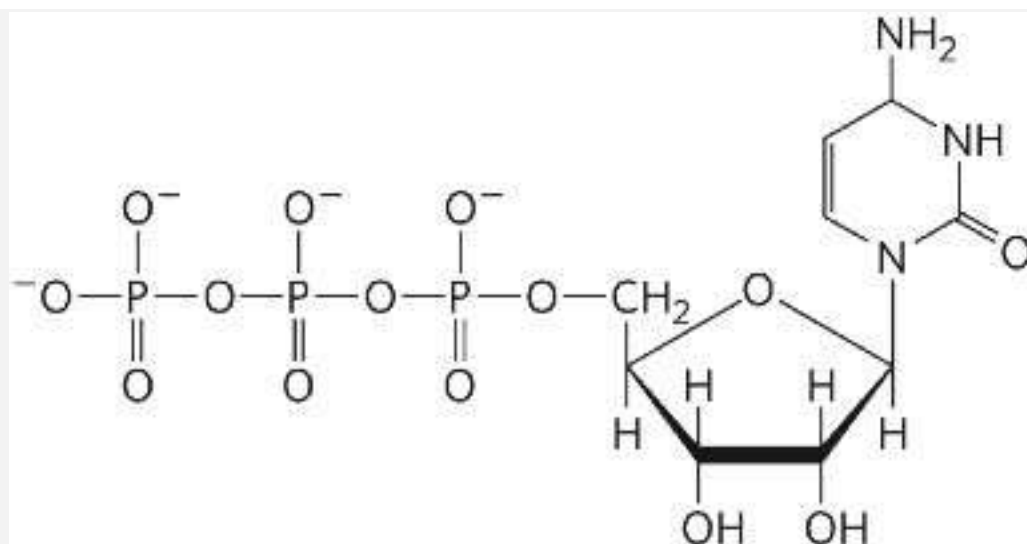
Using this information for equilibrium constants determined at 25 °C, calculate the standard free energy of hydrolysis of ATP.

6. Difference between $\Delta G'^{\circ}$ and ΔG Consider the interconversion shown, which occurs in glycolysis ([Chapter 14](#)):

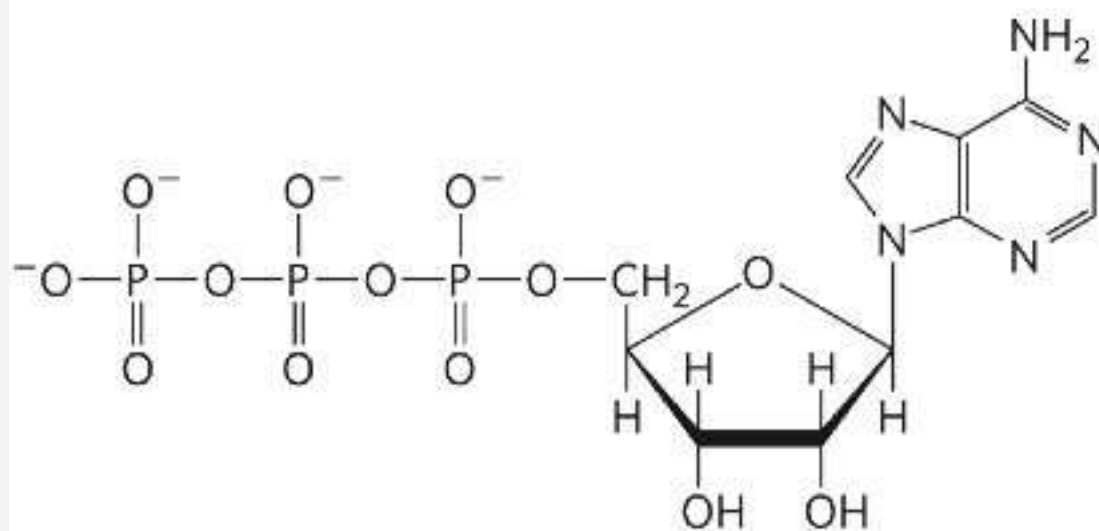


- What is $\Delta G'^{\circ}$ for the reaction (K'_{eq} measured at 25 °C)?
- If the concentration of fructose 6-phosphate is adjusted to 1.5 M and that of glucose 6-phosphate is adjusted to 0.50 M, what is ΔG ?
- Why are $\Delta G'^{\circ}$ and ΔG different?

7. Free Energy of Hydrolysis of CTP Compare the structure of the nucleoside triphosphate CTP with the structure of ATP.

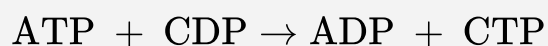


Cytidine triphosphate (CTP)



Adenosine triphosphate (ATP)

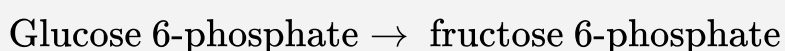
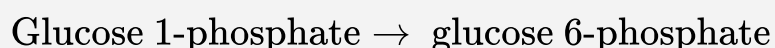
Now predict the K'_{eq} and $\Delta G'^{\circ}$ for the reaction:



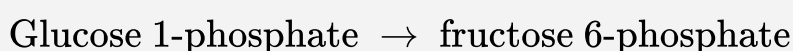
8. Dependence of ΔG on pH The free energy released by the hydrolysis of ATP under standard conditions is -30.5 kJ/mol . If

ATP is hydrolyzed under standard conditions except at pH 5.0, is more or less free energy released? Explain.

9. The $\Delta G'^{\circ}$ for Coupled Reactions Glucose 1-phosphate is converted into fructose 6-phosphate in two successive reactions:



Using the $\Delta G'^{\circ}$ values in [Table 13-4](#), calculate the equilibrium constant, K'_{eq} , for the sum of the two reactions:



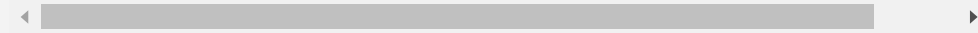
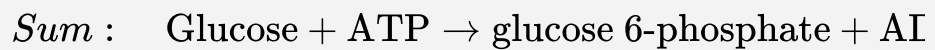
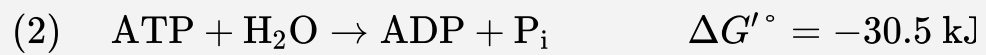
10. Effect of [ATP]/[ADP] Ratio on Free Energy of Hydrolysis of ATP Using [Equation 13-4](#), plot ΔG against $\ln Q$ (mass-action ratio) at 25 °C for the concentrations of ATP, ADP, and P_i in the table shown. $\Delta G'^{\circ}$ for the reaction is -30.5 kJ/mol . Use the resulting plot to explain why metabolism is regulated to keep the ratio [ATP]/[ADP] high.

Concentration (mM)					
ATP	5	3	1	0.2	5
ADP	0.2	2.2	4.2	5.0	25
P_i	10	12.1	14.1	14.9	10

11. Strategy for Overcoming an Unfavorable Reaction: ATP-Dependent Chemical Coupling The phosphorylation of glucose to glucose 6-phosphate is the initial step in the catabolism of glucose. The direct phosphorylation of glucose by P_i is described by the equation



- ◀ ▶
- Calculate the equilibrium constant for this reaction at 37 °C. In the rat hepatocyte, the physiological concentrations of glucose and P_i are maintained at approximately 4.8 mM. What is the equilibrium concentration of glucose 6-phosphate obtained by the direct phosphorylation of glucose by P_i ? Does this reaction represent a reasonable metabolic step for the catabolism of glucose? Explain.
 - In principle at least, one way to increase the concentration of glucose 6-phosphate is to drive the equilibrium reaction to the right by increasing the intracellular concentrations of glucose and P_i . Assuming a fixed concentration of P_i at 4.8 mM, how high would the intracellular concentration of glucose have to be to give an equilibrium concentration of glucose 6-phosphate of 250 μM (the normal physiological concentration)? Would this route be physiologically reasonable, given that the maximum solubility of glucose is less than 1 M?
 - The phosphorylation of glucose in the cell is coupled to the hydrolysis of ATP; that is, part of the free energy of ATP hydrolysis is used to phosphorylate glucose:



Calculate K'_{eq} at 37 °C for the overall reaction. For the ATP-dependent phosphorylation of glucose, what concentration of glucose is needed to achieve a 250 μM intracellular concentration of glucose 6-phosphate when the concentrations of ATP and ADP are 3.38 mM and 1.32 mM, respectively? Does this coupling process provide a feasible route, at least in principle, for the phosphorylation of glucose in the cell? Explain.

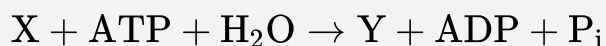
- d. Although coupling ATP hydrolysis to glucose phosphorylation makes thermodynamic sense, we have not yet specified how this coupling is to take place. Given that coupling requires a common intermediate, one conceivable route is to use ATP hydrolysis to raise the intracellular concentration of P_i and thus drive the unfavorable phosphorylation of glucose by P_i . Is this a reasonable route? (Think about the solubility product, K_{sp} , of metabolic intermediates.)
- e. The ATP-coupled phosphorylation of glucose is catalyzed in hepatocytes by the enzyme glucokinase. This enzyme binds ATP and glucose to form a glucose-ATP-enzyme complex, and the phosphoryl group is transferred directly from ATP to glucose. Explain the advantages of this route.

12. Calculations of $\Delta G'^{\circ}$ for ATP-Coupled Reactions From data in [Table 13-6](#), calculate the $\Delta G'^{\circ}$ value for each reaction:

- Phosphocreatine + ADP $\rightarrow$ creatine + ATP
- ATP + fructose $\rightarrow$ ADP + fructose 6-phosphate

13. Coupling ATP Cleavage to an Unfavorable Reaction To explore the consequences of coupling ATP hydrolysis under physiological conditions to a thermodynamically unfavorable biochemical reaction, consider the hypothetical transformation $X \rightarrow Y$, for which $\Delta G'^{\circ} = 20.0 \text{ kJ/mol}$.

- What is the ratio $[Y]/[X]$ at equilibrium?
- Suppose X and Y participate in a sequence of reactions during which ATP is hydrolyzed to ADP and P_i . The overall reaction is



Calculate $[Y]/[X]$ for this reaction at equilibrium. Assume that the temperature is 25.0°C and the equilibrium concentrations of ATP, ADP, and P_i are 1 M .

- We know that $[\text{ATP}]$, $[\text{ADP}]$, and $[P_i]$ are *not* 1 M under physiological conditions. Calculate $[Y]/[X]$ for the ATP-coupled reaction when the values of $[\text{ATP}]$, $[\text{ADP}]$, and $[P_i]$ are those found in rat myocytes ([Table 13-5](#)).

14. Calculations of ΔG at Physiological Concentrations

Calculate the actual, physiological ΔG for the reaction



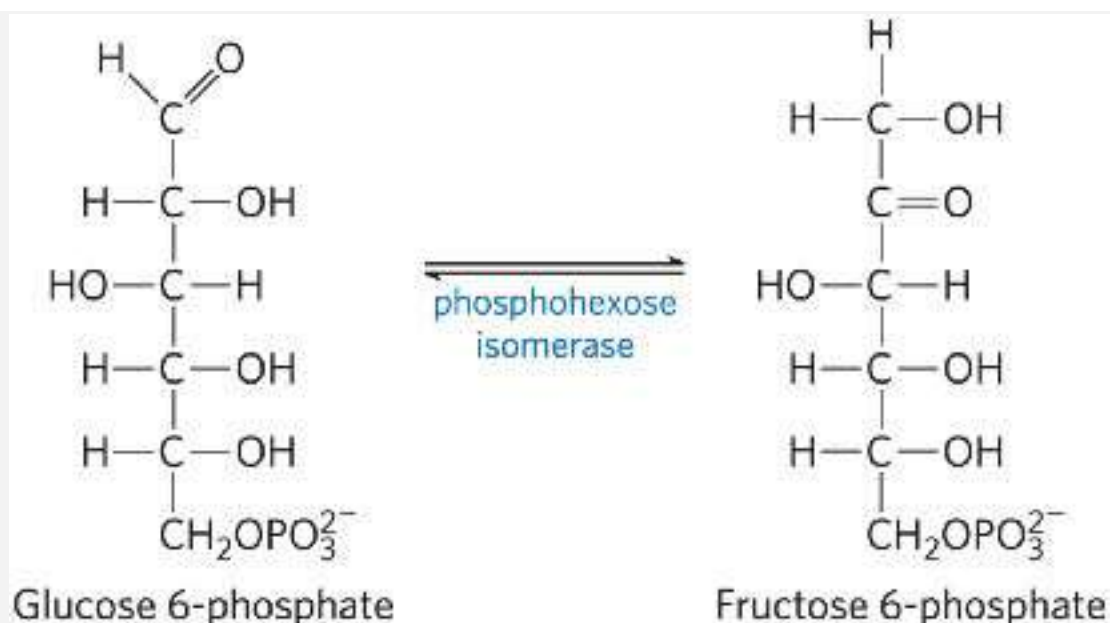
at 37 °C, as it occurs in the cytosol of neurons, with phosphocreatine at 4.7 mM, creatine at 1.0 mM, ADP at 0.73 mM, and ATP at 2.6 mM.

15. Free Energy Required for ATP Synthesis under Physiological Conditions In the cytosol of rat hepatocytes, the temperature is 37 °C and the mass-action ratio, Q , is

$$\frac{[\text{ATP}]}{[\text{ADP}][\text{P}_i]} = 5.33 \times 10^2 \text{ M}^{-1}$$

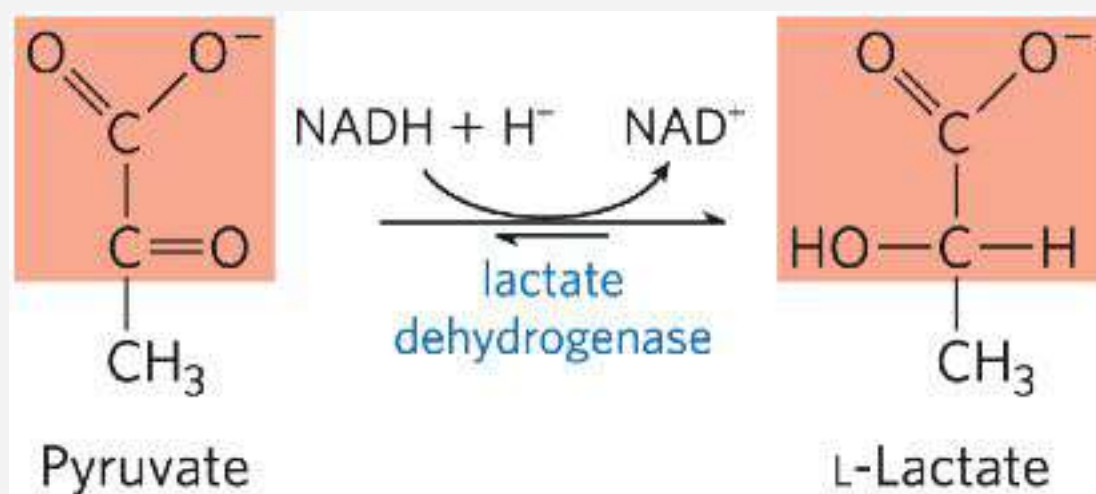
Calculate the free energy required to synthesize ATP in a rat hepatocyte.

16. Chemical Logic In the glycolytic pathway, a six-carbon sugar (fructose 1,6-bisphosphate) is cleaved to form two three-carbon sugars, which undergo further metabolism. In this pathway, an isomerization of glucose 6-phosphate to fructose 6-phosphate (as shown in the diagram) occurs two steps before the cleavage reaction. The intervening step is phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate (p. 516).



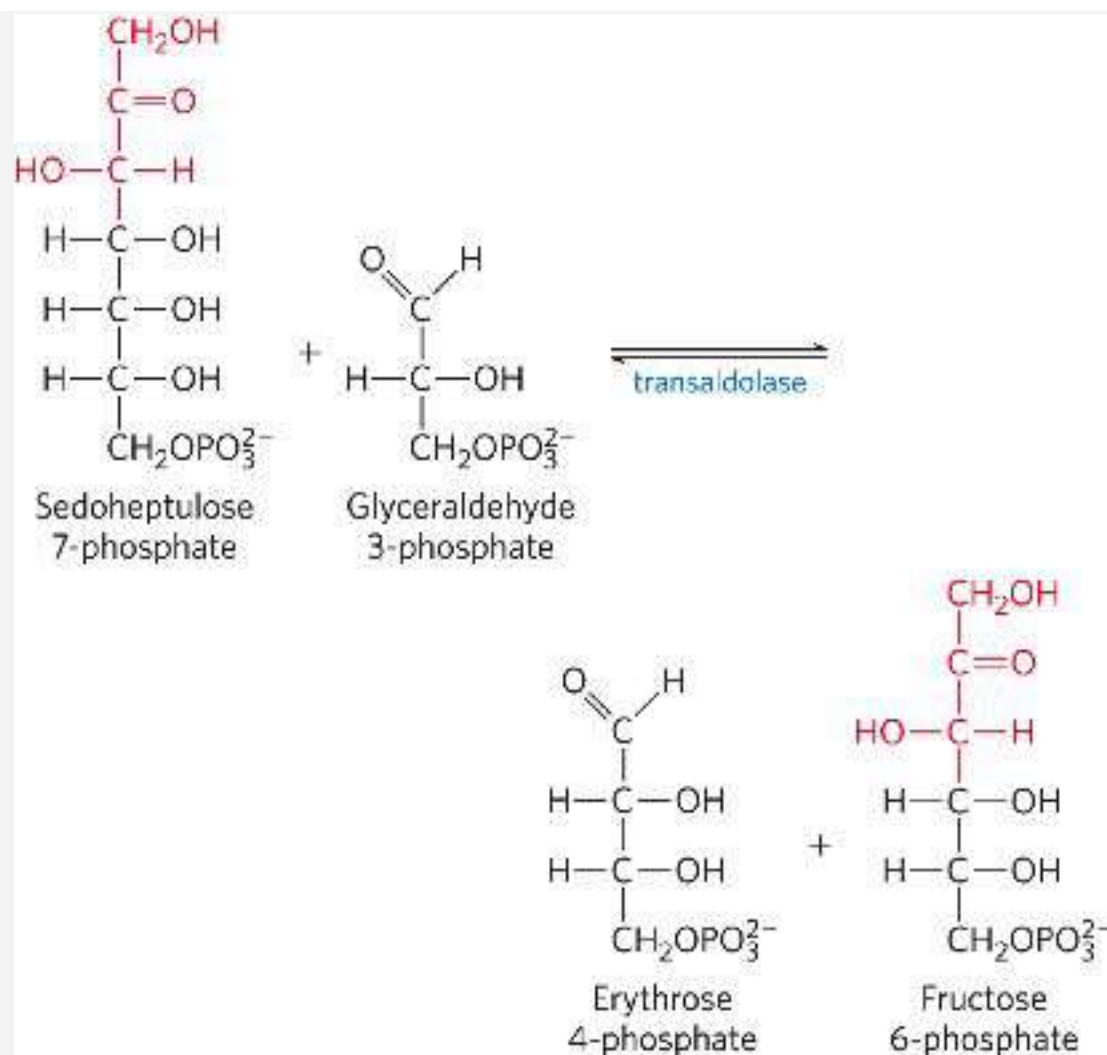
What does the isomerization step accomplish from a chemical perspective? (Hint: Consider what might happen if the C—C bond cleavage were to proceed without the preceding isomerization.)

17. Enzymatic Reaction Mechanisms I Lactate dehydrogenase is one of the many enzymes that require NADH as coenzyme. It catalyzes the conversion of pyruvate to lactate:



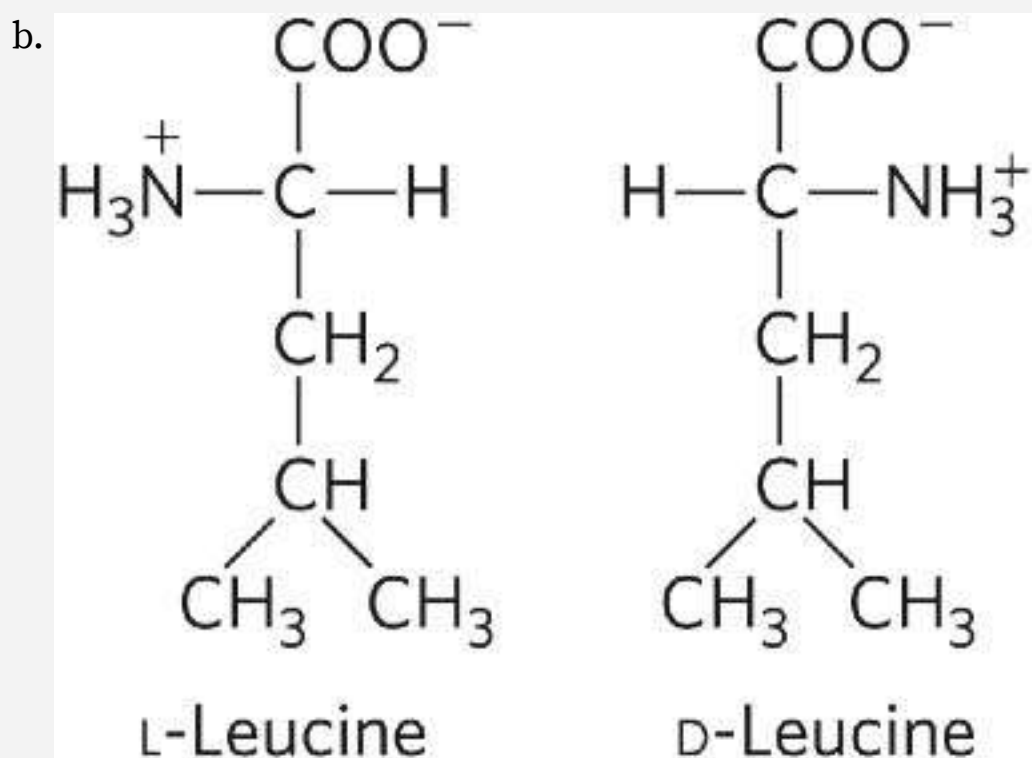
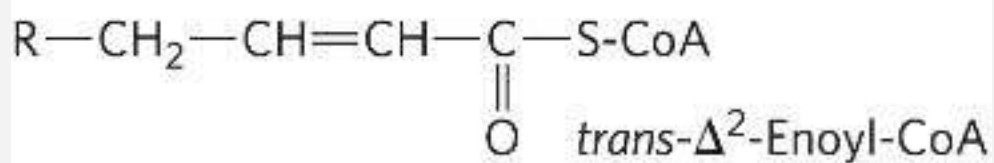
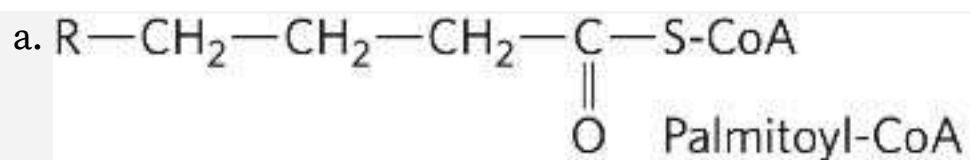
Draw the mechanism of this reaction (show electron-pushing arrows). (Hint: This is a common reaction throughout metabolism; the mechanism is similar to that catalyzed by other dehydrogenases that use NADH, such as alcohol dehydrogenase.)

18. Enzymatic Reaction Mechanisms II Biochemical reactions often look more complex than they really are. In the pentose phosphate pathway ([Chapter 14](#)), sedoheptulose 7-phosphate and glyceraldehyde 3-phosphate react to form erythrose 4-phosphate and fructose 6-phosphate in a reaction catalyzed by transaldolase.

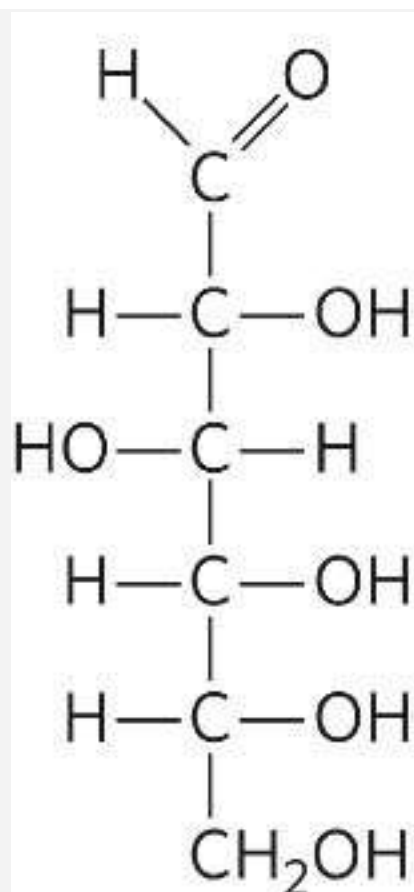


Draw a mechanism for this reaction (show electron-pushing arrows). (Hint: Take another look at aldol condensations, then consider the name of this enzyme.)

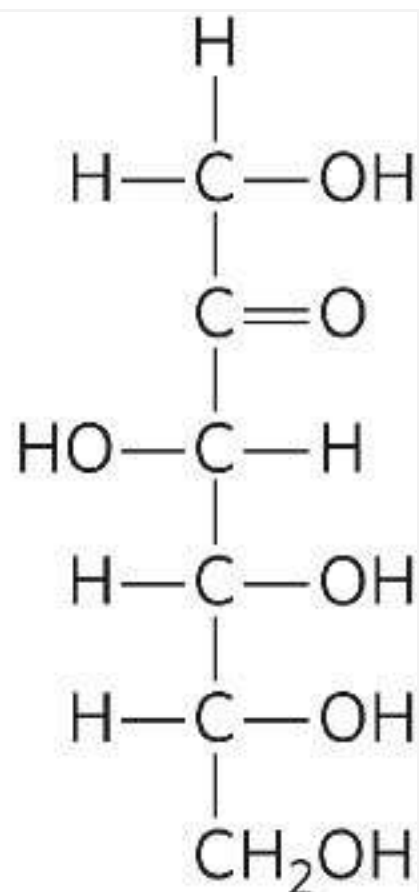
19. Recognizing Reaction Types For each pair of biomolecules, identify the type of reaction (oxidation-reduction, hydrolysis, isomerization, group transfer, or internal rearrangement) required to convert the first molecule to the second. In each case, indicate the general type of enzyme and cofactor(s) or reactants that would be required, and any other products that would result.



c.

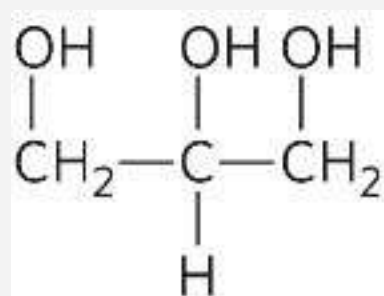


Glucose

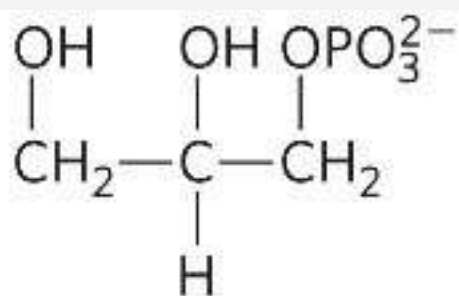


Fructose

d.

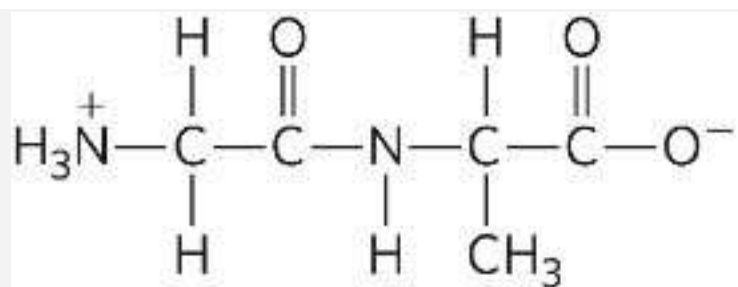


Glycerol

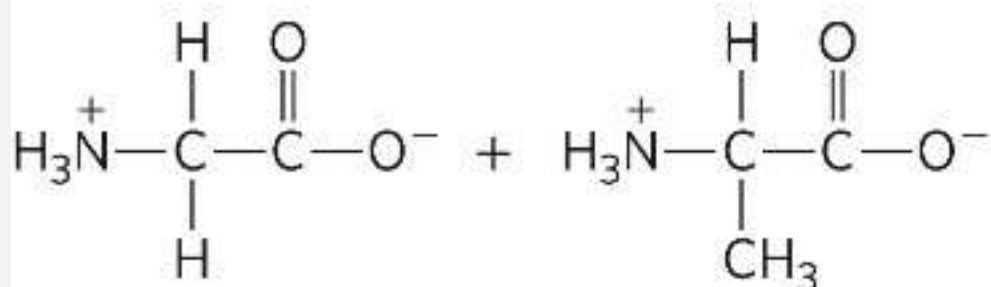


Glycerol 3-phosphate

e.



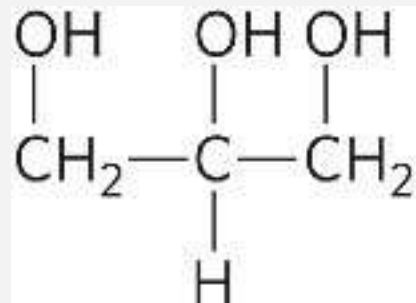
Glycylalanine



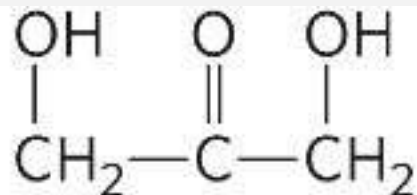
Glycine

Alanine

f.

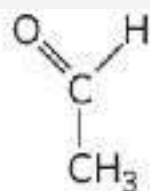


Glycerol

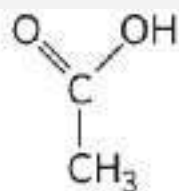


Dihydroxyacetone

g.

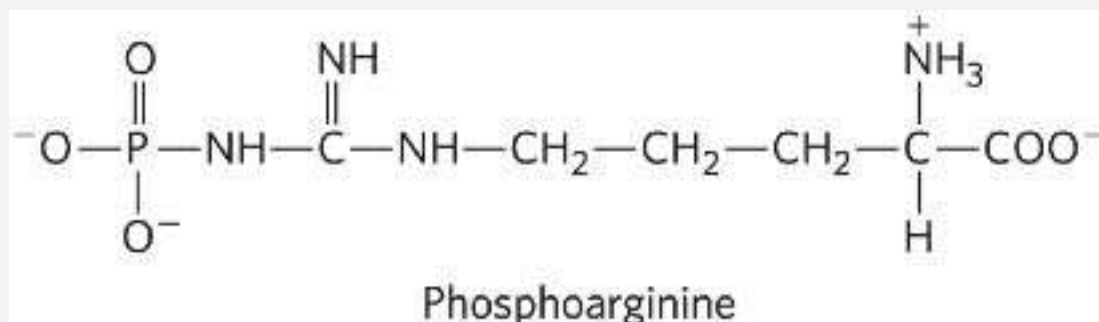


Acetaldehyde

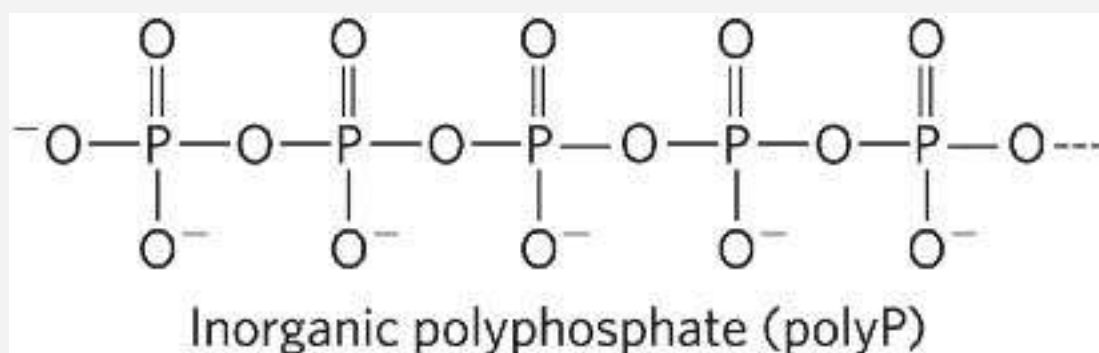


Acetic acid

20. Effect of Structure on Group Transfer Potential Some invertebrates contain phosphoarginine. Is the standard free energy of hydrolysis of this molecule more similar to that of glucose 6-phosphate or of ATP? Explain your answer.



21. Polyphosphate as a Possible Energy Source The standard free energy of hydrolysis of inorganic polyphosphate (polyP) is about -20 kJ/mol for each P_i released. We calculated in [Worked Example 13-2](#) that, in a cell, it takes about 50 kJ/mol of energy to synthesize ATP from ADP and P_i .



Is it feasible for a cell to use polyphosphate to synthesize ATP from ADP? Explain your answer.

22. Daily ATP Utilization by Human Adults

- a. The synthesis of ATP from ADP and P_i requires a total of 30.5 kJ/mol of free energy when the reactants and

products are at 1 M concentrations and the temperature is 25 °C (standard state). However, the actual physiological concentrations of ATP, ADP, and P_i are not 1 M, and the physiological temperature is 37 °C. Thus, the free energy required to synthesize ATP under physiological conditions is different from $\Delta G'^\circ$.

Calculate the free energy required to synthesize ATP in the human hepatocyte when the physiological concentrations of ATP, ADP, and P_i are 3.5, 1.50, and 5.0 mM, respectively.

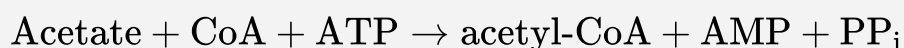
- b. A 68 kg (150 lb) adult requires a caloric intake of 2,000 kcal (8,360 kJ) of food per day (24 hours). The body metabolizes the food and uses the free energy to synthesize ATP, which then provides energy for the body's daily chemical and mechanical work. Assuming that the efficiency of converting food energy into ATP is 50%, calculate the weight of ATP used by a human adult in 24 hours. What percentage of the body weight does this represent?
- c. Although adults synthesize large amounts of ATP daily, their body weight, structure, and composition do not change significantly during this period. Explain this apparent contradiction.

23. Rates of Turnover of γ and β Phosphates of ATP After adding a small amount of ATP labeled with radioactive phosphorus in the terminal position, $[\gamma\text{-}^{32}\text{P}]\text{ATP}$, to a yeast extract, a researcher finds about half of the ^{32}P activity in P_i within a few minutes, but the concentration of ATP remains unchanged. Explain. She then carries out the same experiment

using ATP labeled with ^{32}P in the central position, $[\beta\text{-}^{32}\text{P}]\text{ATP}$, but the ^{32}P does not appear in P_i within such a short time. Why?

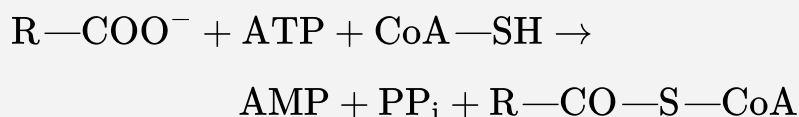
24. Cleavage of ATP to AMP and PP_i during Metabolism

Synthesis of the activated form of acetate (acetyl-CoA) is carried out in an ATP-dependent process:



- a. The $\Delta G'^{\circ}$ for hydrolysis of acetyl-CoA to acetate and CoA is -32.2 kJ/mol . The $\Delta G'^{\circ}$ for hydrolysis of ATP to AMP and PP_i is -30.5 kJ/mol . Calculate $\Delta G'^{\circ}$ for the ATP-dependent synthesis of acetyl-CoA.
- b. Almost all cells contain the enzyme inorganic pyrophosphatase, which catalyzes the hydrolysis of PP_i to P_i . What effect does the presence of this enzyme have on the synthesis of acetyl-CoA? Explain.

25. Activation of a Fatty Acid by Reaction with Coenzyme A In the reaction sequence for fatty acid breakdown, coenzyme A (CoA), with its thiol (—SH) group, joins to the fatty acid as a thiol ester, as ATP is converted into AMP and PP_i :



The oxidation of fatty acids as fuels requires two steps. The first step transfers an activating group from ATP to the carboxyl group of the fatty acid. In the second step, CoA—SH displaces

the activating group to form fatty acyl-S—CoA. Given the known products of the reaction, what is the activating group?

26. Energy for H^+ Pumping The parietal cells of the stomach lining contain membrane “pumps” that transport hydrogen ions from the cytosol (pH 7.0) into the stomach, contributing to the acidity of gastric juice (pH 1.0). Calculate the free energy required to transport 1 mol of hydrogen ions through these pumps. (Hint: See [Chapter 11](#).) Assume a temperature of 37 °C.

27. Most-Reduced Carbon Compounds Arrange the four structures in order from most reduced to most oxidized.

- a. $R-CH_2-CH_2-OH$
- b. $R-CH_2-COO^-$
- c. $R-CH_2-CHO$
- d. $R-CH_2-CH_3$

28. Standard Reduction Potentials The standard reduction potential, E'° , of any redox pair is defined for the half-cell reaction



The E'° values for the $NAD^+/NADH$ and pyruvate/lactate conjugate redox pairs are -0.32 V and -0.19 V , respectively.

- a. Which redox pair has the greater tendency to lose electrons? Explain.
- b. Which pair is the stronger oxidizing agent? Explain.
- c. Beginning with 1 M concentrations of each reactant and product at pH 7 and 25 °C, in which direction will the

following reaction proceed?

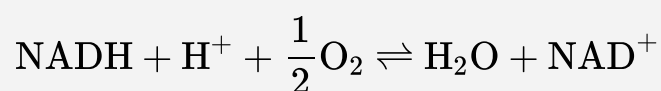


- d. What is the standard free-energy change ($\Delta G'^{\circ}$) for the conversion of pyruvate to lactate?
- e. What is the equilibrium constant (K'_{eq}) for this reaction?

29. Simple Biobattery Suppose you set up a simple battery using half-reactions as pictured in [Figure 13-23](#). One electrode contains pyruvate and lactate at 1 mM, and the other electrode contains fumarate and succinate at 1 mM (see [Table 13-7](#)).

- a. In which direction will electrons initially flow?
- b. Calculate the standard reduction potential and standard free-energy change for your biological battery.
- c. When a flashlight battery “runs out,” net electron movement has essentially ended. What is the equivalent situation for your biobattery?

30. Energy Span of the Respiratory Chain Electron transfer in the mitochondrial respiratory chain may be represented by the net reaction equation



- a. Calculate $\Delta E'^{\circ}$ for the net reaction of mitochondrial electron transfer. Use E'° values in [Table 13-7](#).
- b. Calculate $\Delta G'^{\circ}$ for this reaction.
- c. How many ATP molecules can *theoretically* be generated by this reaction if the free energy of ATP synthesis under

cellular conditions is 52 kJ/mol?

31. Dependence of Electromotive Force on Concentrations

Suppose that you place an electrode into solutions of various concentrations of NAD^+ and NADH at pH 7.0 and 25 °C.

Calculate the electromotive force (in volts) registered by the electrode when immersed in each solution, with reference to a half-cell of E'° 0.00 V.

- a. 1.0 mM NAD^+ and 10 mM NADH
- b. 1.0 mM NAD^+ and 1.0 mM NADH
- c. 10 mM NAD^+ and 1.0 mM NADH

32. Electron Affinity of Compounds List the four compounds or reactions in order of increasing tendency to accept electrons:

- a. α -ketoglutarate + CO_2 (yielding isocitrate)
- b. oxaloacetate
- c. O_2
- d. NADP^+

33. Direction of Oxidation-Reduction Reactions Which of the reactions listed would you expect to proceed in the direction shown, under standard conditions, in the presence of the appropriate enzymes?

- a. $\text{Malate} + \text{NAD}^+ \rightarrow \text{oxaloacetate} + \text{NADH} + \text{H}^+$
- b. $\text{Acetoacetate} + \text{NADH} + \text{H}^+ \rightarrow \beta\text{-hydroxybutyrate} + \text{NAD}^+$
- c. $\text{Pyruvate} + \text{NADH} + \text{H}^+ \rightarrow \text{lactate} + \text{NAD}^+$
- d. $\text{Pyruvate} + \beta\text{-hydroxybutyrate} \rightarrow \text{lactate} + \text{acetoacetate}$
- e. $\text{Malate} + \text{pyruvate} \rightarrow \text{oxaloacetate} + \text{lactate}$
- f. $\text{Acetaldehyde} + \text{succinate} \rightarrow \text{ethanol} + \text{fumarate}$

34. Measurement of Intracellular Metabolite Concentrations

Measuring the concentrations of metabolic intermediates in a

living cell presents great experimental difficulties — usually, a cell must be destroyed before metabolite concentrations can be measured. Yet enzymes catalyze metabolic interconversions very rapidly, so a common problem associated with these types of measurements is that the findings reflect not the physiological concentrations of metabolites but the equilibrium concentrations. To prevent changes in metabolite concentrations during sample preparation, cells were quick-frozen in liquid nitrogen, then extracted under conditions that prevented enzymatic activity.

The table gives the intracellular concentrations of the substrates and products of the phosphofructokinase-1 reaction in isolated rat heart tissue.

Metabolite	Concentration (μM) ^a
Fructose 6-phosphate	87.0
Fructose 1,6-bisphosphate	22.0
ATP	11,400
ADP	1,320

Data from J. R. Williamson, *J. Biol. Chem.* 240:2308, 1965.

^aCalculated as $\mu\text{mol/mL}$ of intracellular water.

- Calculate Q , $[\text{fructose 1,6-bisphosphate}][\text{ADP}]/[\text{fructose 6-phosphate}][\text{ATP}]$, for the PFK-1 reaction under physiological conditions.
- Given a $\Delta G'^{\circ}$ for the PFK-1 reaction of -14.2 kJ/mol , calculate the equilibrium constant for this reaction.

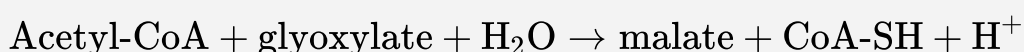
- c. Compare the values of Q and K'_{eq} . Is the physiological reaction near or far from equilibrium? Explain. What does this experiment suggest about the role of PFK-1 as a regulatory enzyme?

35. Are All Metabolic Reactions at Equilibrium?

- a. Phosphoenolpyruvate (PEP) is one of the two phosphoryl group donors in the synthesis of ATP during glycolysis. In human erythrocytes, the steady-state concentration of ATP is 2.24 mM, that of ADP is 0.25 mM, and that of pyruvate is 0.051 mM. Calculate the concentration of PEP at 25 °C, assuming that the pyruvate kinase reaction (see [Fig. 13-13](#)) is at equilibrium in the cell.
- b. The physiological concentration of PEP in human erythrocytes is 0.023 mM. Compare this with the value obtained in (a). Explain the significance of this difference.

36. Michaelis Constant K_m Compared with Substrate

Concentration Malate synthase in *E. coli* catalyzes the reaction

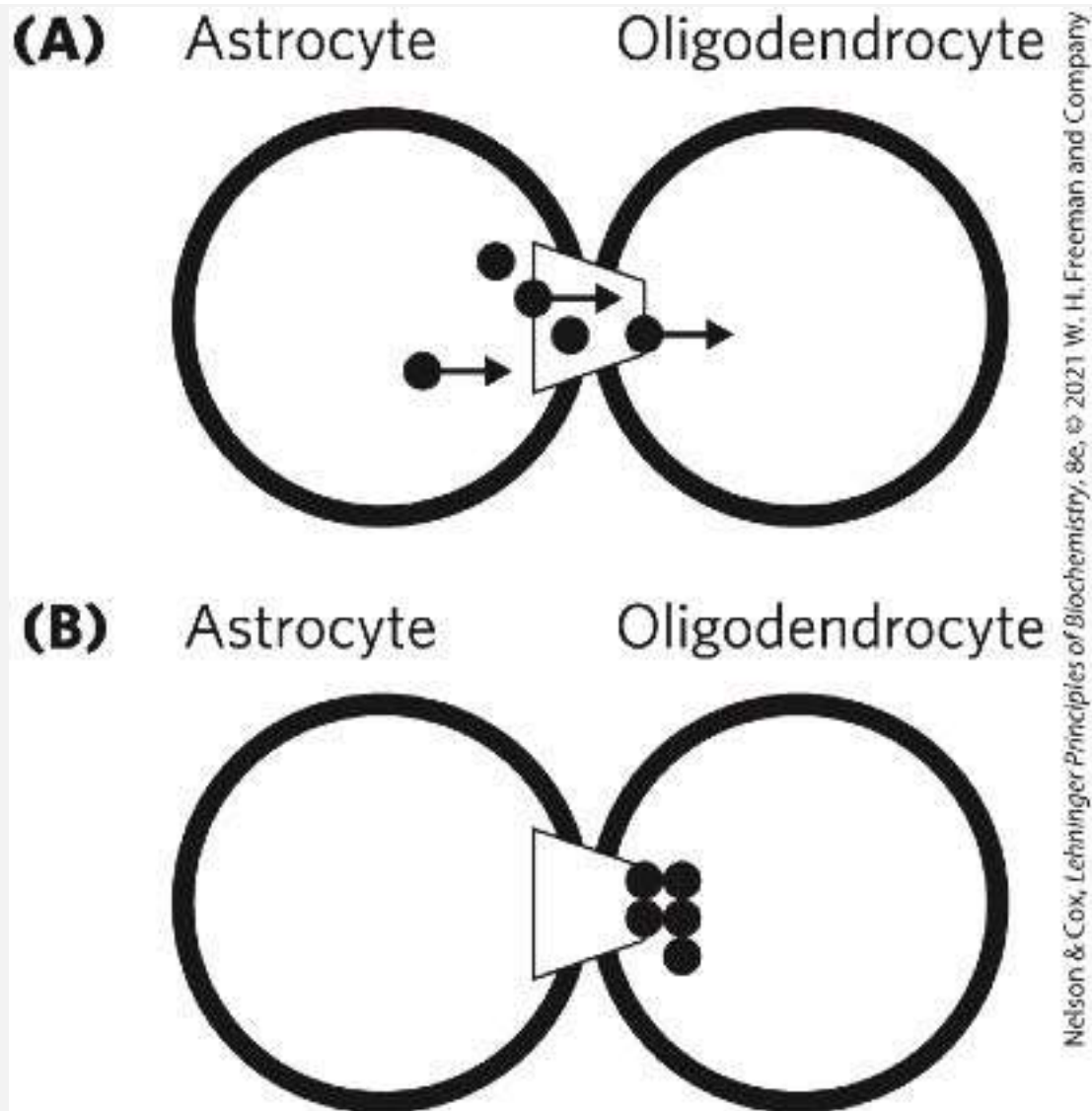


The experimentally measured K_m for acetyl-CoA is 9×10^{-6} M. In a growing culture of *E. coli*, the measured concentration of acetyl-CoA is 6×10^{-4} M. Is malate synthase operating at its V_{max} under these conditions?

DATA ANALYSIS PROBLEM

37. Thermodynamics Can Be Tricky Thermodynamics is a challenging area of study and one with many opportunities for confusion. An interesting example is found in an article by Robinson, Hampson, Munro, and Vaney, published in *Science* in 1993. Robinson and colleagues studied the movement of small molecules between neighboring cells of the nervous system through cell-to-cell channels (gap junctions). They found that the dyes Lucifer yellow (a small, negatively charged molecule) and biocytin (a small zwitterionic molecule) moved in only one direction between two particular types of glia (nonneuronal cells of the nervous system). Dye injected into astrocytes would rapidly pass into adjacent astrocytes, oligodendrocytes, or Müller cells, but dye injected into oligodendrocytes or Müller cells passed slowly if at all into astrocytes. All of these cell types are connected by gap junctions.

Although it was not a central point of their article, the authors presented a molecular model for how this unidirectional transport might occur, as shown in their Figure 3:



The figure legend reads: “Model of the unidirectional diffusion of dye between coupled oligodendrocytes and astrocytes, based on differences in connection pore diameter. Like a fish in a fish trap, dye molecules (black circles) can pass from an astrocyte to an oligodendrocyte (A) but not back in the other direction (B).”

Although this article clearly passed review at a well-respected journal, several letters to the editor (1994) followed, showing that Robinson and coauthors’ model violated the second law of thermodynamics.

- a. Explain how the model violates the second law. Hint: Consider what would happen to the entropy of the system if one started with equal concentrations of dye in the astrocyte and oligodendrocyte connected by the “fish trap” type of gap junctions.
- b. Explain why this model cannot work for small molecules, although it may allow one to catch fish.
- c. Explain why a fish trap *does* work for fish.
- d. Provide two plausible mechanisms for the unidirectional transport of dye molecules between the cells that do not violate the second law of thermodynamics.

References

Letters to the editor. 1994. *Science* 265:1017–1019.

Robinson, S.R., E.C.G.M. Hampson, M.N. Munro, and D.I. Vaney. 1993. Unidirectional coupling of gap junctions between neuroglia. *Science* 262:1072–1074.

CHAPTER 14

GLYCOLYSIS, GLUCONEOGENESIS, AND THE PENTOSE PHOSPHATE PATHWAY



14.1 Glycolysis

14.2 Feeder Pathways for Glycolysis

14.3 Fates of Pyruvate

14.4 Gluconeogenesis

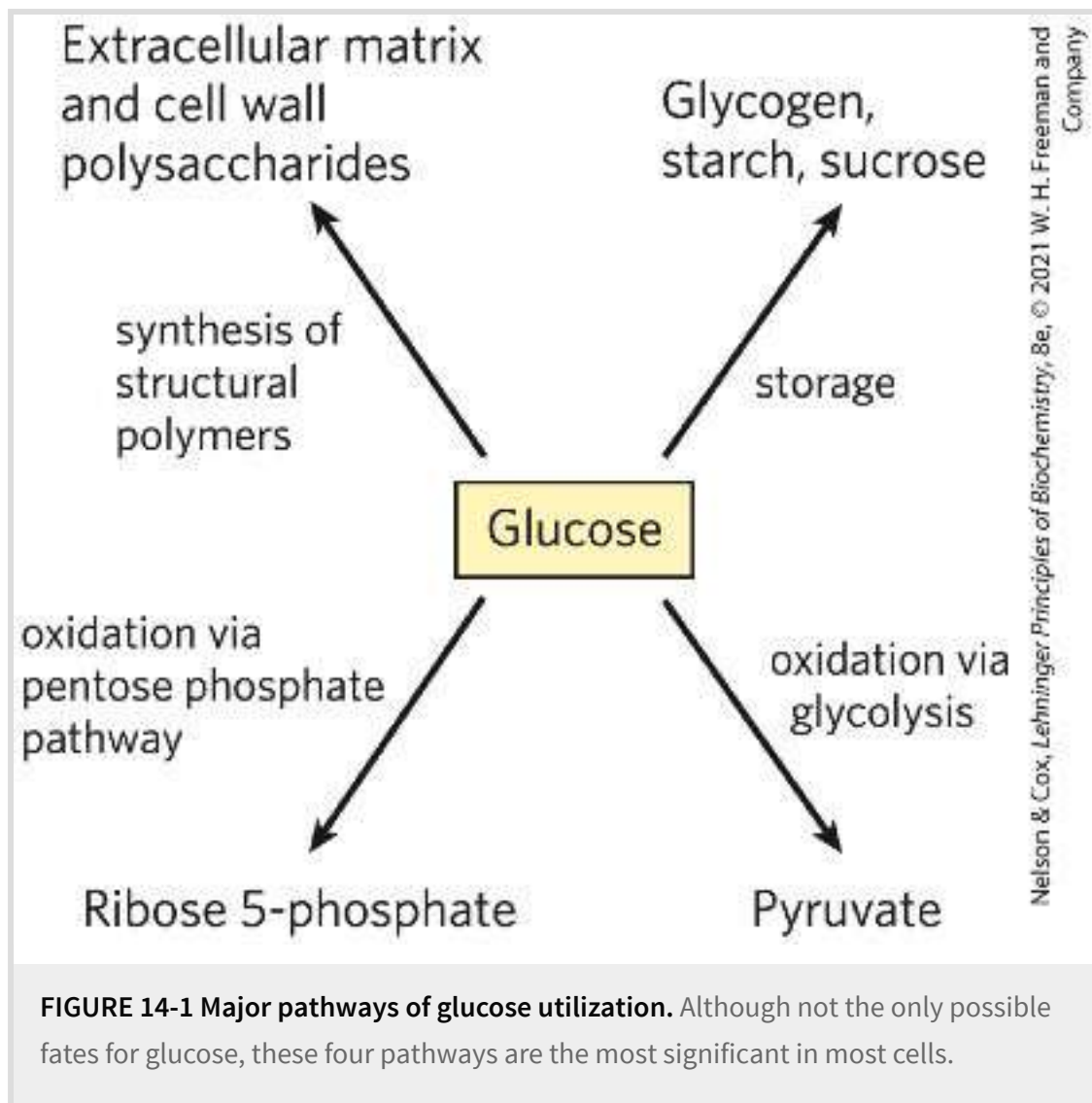
14.5 Coordinated Regulation of Glycolysis and Gluconeogenesis

14.6 Pentose Phosphate Pathway of Glucose Oxidation

We begin with the major pathways of glucose metabolism: glycolysis and fermentation, gluconeogenesis, and the pentose phosphate pathway. Glucose occupies a central position in the metabolism of plants, animals, and many microorganisms. It is relatively rich in potential energy, and thus a good fuel; the complete oxidation of glucose to carbon dioxide and water proceeds with a standard free-energy change of $-2,840 \text{ kJ/mol}$. By storing glucose as a high molecular weight polymer such as

starch or glycogen, a cell can stockpile large quantities of hexose units while maintaining a relatively low cytosolic osmolarity. When energy demands increase, glucose can be released from these intracellular storage polymers and used to produce ATP either aerobically or anaerobically.

Glucose is not only an excellent fuel, but also a remarkably versatile precursor, capable of supplying a huge array of metabolic intermediates for biosynthetic reactions. A bacterium such as *Escherichia coli* can obtain from glucose the carbon skeletons for every amino acid, nucleotide, coenzyme, fatty acid, or other metabolic intermediate it needs for growth. A comprehensive study of the metabolic fates of glucose would encompass hundreds or thousands of transformations. In animals and vascular plants, glucose has four major fates: it may be (1) used in the synthesis of complex polysaccharides destined for the extracellular space; (2) stored in cells (as a polysaccharide or as sucrose); (3) oxidized to a three-carbon compound (pyruvate) via glycolysis to provide ATP and metabolic intermediates; or (4) oxidized via the pentose phosphate (phosphogluconate) pathway to yield ribose 5-phosphate for nucleic acid synthesis and NADPH for reductive biosynthetic processes ([Fig. 14-1](#)).



Organisms that do not have access to glucose from other sources must make it. Photosynthetic organisms make glucose by first reducing atmospheric CO_2 to trioses, then converting the trioses to glucose. Nonphotosynthetic cells make glucose from simpler three- and four-carbon precursors by the process of gluconeogenesis, effectively reversing glycolysis in a pathway that uses many of the glycolytic enzymes.

These principles are central to understanding glucose metabolism, but many apply to all metabolic pathways:

P1 Metabolites like glucose are often activated with a high-energy group before their catabolism. Glycolysis is a nearly universal 10-step metabolic pathway for producing ATP by the oxidation of glucose. In this process, two molecules of ATP are invested to activate glucose, but the products of the pathway include four ATP, as well as NADH (a form of reducing power) and the triose pyruvate, which can be metabolized further in other pathways.

P2 Glucose and other hexoses and hexose phosphates obtained from stored polysaccharides or dietary carbohydrates feed into the glycolytic pathway. By using a common pathway for a number of starting materials, the cell economizes on the number of enzymes that must be synthesized and simplifies the regulation of the common pathway.

P3 Pyruvate formed under anaerobic conditions is reduced to lactate with electrons from NADH, recycling NADH to NAD^+ and allowing continued glycolysis in the processes of lactate or alcohol fermentation. Manipulation of the fermentable material and the microorganisms present allows the synthesis of a variety of industrial products and foods.

P4 Gluconeogenesis is the synthesis of glucose from simpler precursors like pyruvate and lactate. Although it uses seven of the ten enzymes that also act in glycolysis, gluconeogenesis must bypass three of the most exergonic steps in glycolysis

with energetically favorable reactions unique to gluconeogenesis.

P5 Glycolysis and gluconeogenesis are reciprocally regulated so that both processes don't occur simultaneously in a futile cycle. Most regulatory mechanisms act on reactions that are unique to each pathway.

P6 The pentose phosphate pathway is an alternative pathway for glucose oxidation. It yields pentoses for nucleotide synthesis and reduced cofactors for biosynthesis of fatty acids, sterols, and many other compounds.

14.1 Glycolysis

 In **glycolysis** (from the Greek *glykys*, “sweet” or “sugar,” and **lysis**, “splitting”), a molecule of glucose is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon compound pyruvate. During the sequential reactions of glycolysis, some of the free energy released from glucose is conserved in the form of ATP and NADH. Glycolysis was the first metabolic pathway to be elucidated and is probably the best understood. From Eduard Buchner’s discovery in 1897 of fermentation in cell-free extracts of yeast until the elucidation of the whole pathway in yeast and in muscle in the 1930s, the reactions of glycolysis were a major focus of biochemical research. These discoveries showed that the reactions of life could be explained chemically, without reliance on a mystical life force. This philosophical shift led physiologist Jacques Loeb to observe in 1906, “The history of this problem is instructive, as it warns us against considering problems as beyond our reach because they have not yet found their solution.”¹

The development of methods of enzyme purification, the discovery and recognition of the importance of coenzymes such as NAD, and the discovery of the pivotal metabolic role of ATP and other phosphorylated compounds all came out of studies of glycolysis. The glycolytic enzymes of many species have long since been purified and thoroughly studied.

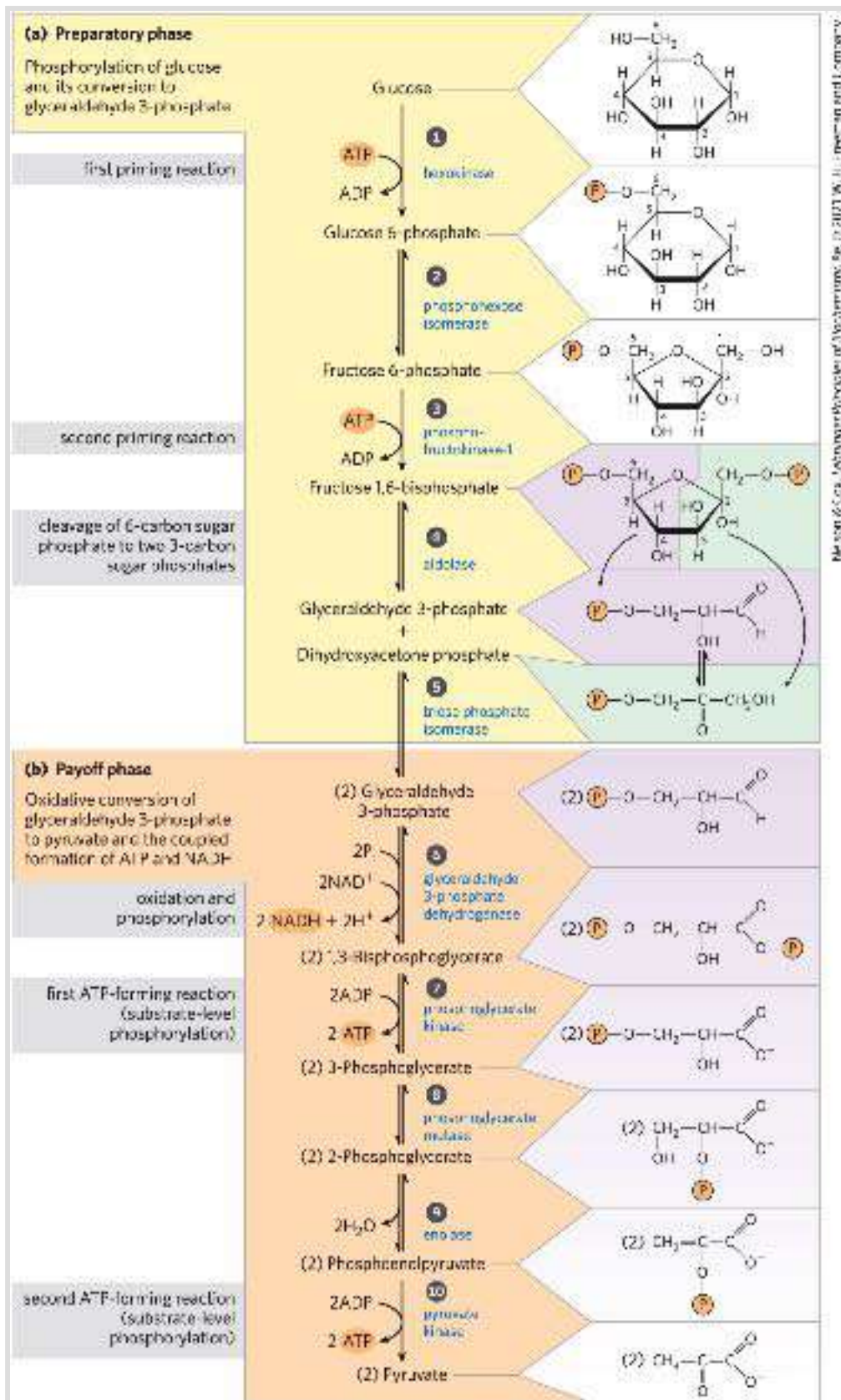
Glycolysis is an almost universal central pathway of glucose catabolism, the pathway with the largest flux of carbon in most cells. The glycolytic breakdown of glucose is the sole source of metabolic energy in some mammalian tissues and cell types (erythrocytes, renal medulla, brain, and sperm, for example). Some plant tissues that are modified to store starch (such as potato tubers) and some aquatic plants (watercress, for example) derive most of their energy from glycolysis; many anaerobic microorganisms are entirely dependent on glycolysis.

In the course of evolution, the chemistry of the reactions of glycolysis has been completely conserved. Genome sequencing and structural studies have shown that the glycolytic enzymes of vertebrates are closely similar in amino acid sequence and three-dimensional structure to their homologs in yeast and spinach. Although some archaea and parasitic microorganisms lack one or more of the enzymes of glycolysis, they retain the core of the pathway. The glycolytic pathway, of central importance in itself, is governed by thermodynamic principles and regulatory mechanisms that are common to all pathways of cell metabolism. It serves as a model of principles we will revisit throughout [Part II](#) of this book.

An Overview: Glycolysis Has Two Phases

Before examining each step of the pathway in some detail, we take a look at glycolysis as a whole. As all sugar derivatives in

glycolysis are the D isomers, we will usually omit the D designation except when emphasizing stereochemistry. The breakdown of the six-carbon glucose into two molecules of the three-carbon pyruvate occurs in 10 steps, the first 5 of which constitute the *preparatory phase* ([Fig. 14-2a](#)). In these reactions, glucose is first phosphorylated at the hydroxyl group on C-6 (step ①). The glucose 6-phosphate thus formed is converted to fructose 6-phosphate (step ②), which is again phosphorylated, this time at C-1, to yield fructose 1,6-bisphosphate (step ③). For both phosphorylations, ATP is the phosphoryl group donor.



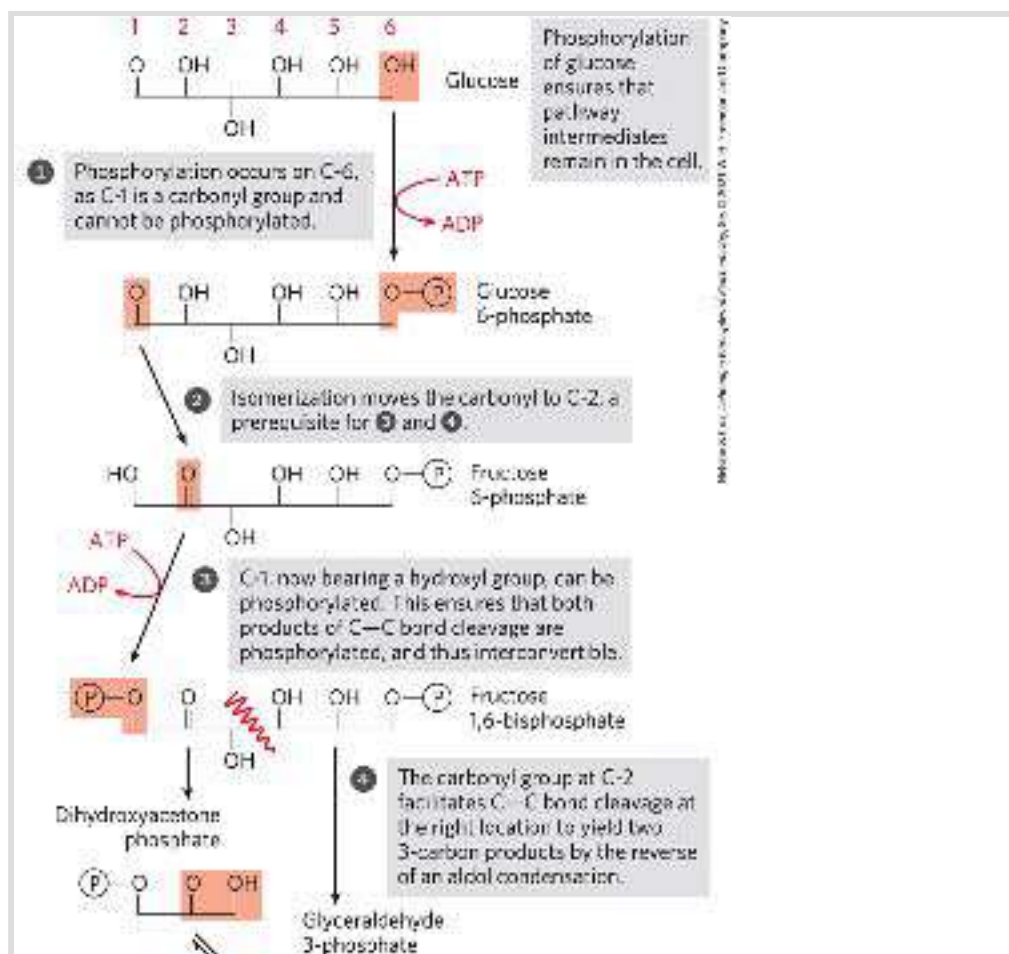
glyceraldehyde 3-phosphate are formed; both pass through the payoff phase (b). Pyruvate is the end product of the second phase of glycolysis. For each glucose molecule, two ATP are consumed in the preparatory phase and four ATP are produced in the payoff phase, giving a net yield of two ATP and two NADH per molecule of glucose converted to pyruvate. The numbered reaction steps correspond to the numbered headings in the text discussion. Keep in mind that each phosphoryl group, represented here as , has two negative charges ($-\text{PO}_3^{2-}$).

Fructose 1,6-bisphosphate is split to yield two different three-carbon molecules, dihydroxyacetone phosphate and glyceraldehyde 3-phosphate (step ); this is the “lysis” step that gives the pathway its name. The dihydroxyacetone phosphate is isomerized to form a second molecule of glyceraldehyde 3-phosphate (step ), ending the first phase of glycolysis. Note that two molecules of ATP are invested before the cleavage of glucose into two three-carbon pieces; there will be a good return on this investment. To summarize: in the preparatory phase of glycolysis the energy of ATP is invested, raising the free-energy content of the intermediates, and the carbon chains of all the metabolized hexoses are converted to a common product, glyceraldehyde 3-phosphate.

The energy gain comes in the *payoff phase* of glycolysis ([Fig. 14-2b](#)). Each molecule of glyceraldehyde 3-phosphate is oxidized and phosphorylated by inorganic phosphate (*not* by ATP) to form 1,3-bisphosphoglycerate (step ). Energy is then released as the two molecules of 1,3-bisphosphoglycerate are converted to two molecules of pyruvate (steps  through ). Much of this energy is conserved by the coupled phosphorylation of four molecules of

ADP to ATP. The net yield is two molecules of ATP per molecule of glucose used, because two molecules of ATP were invested in the preparatory phase. Energy is also conserved in the payoff phase in the formation of two molecules of the electron carrier NADH per molecule of glucose.

P1 In the sequential reactions of glycolysis, three types of chemical transformations are particularly noteworthy: (1) degradation of the carbon skeleton of glucose to yield pyruvate; (2) phosphorylation of ADP to ATP by compounds with high phosphoryl group transfer potential, formed during glycolysis; and (3) transfer of a hydride ion to NAD^+ , forming NADH. The overall chemical logic of the pathway is described in [Figure 14-3](#).



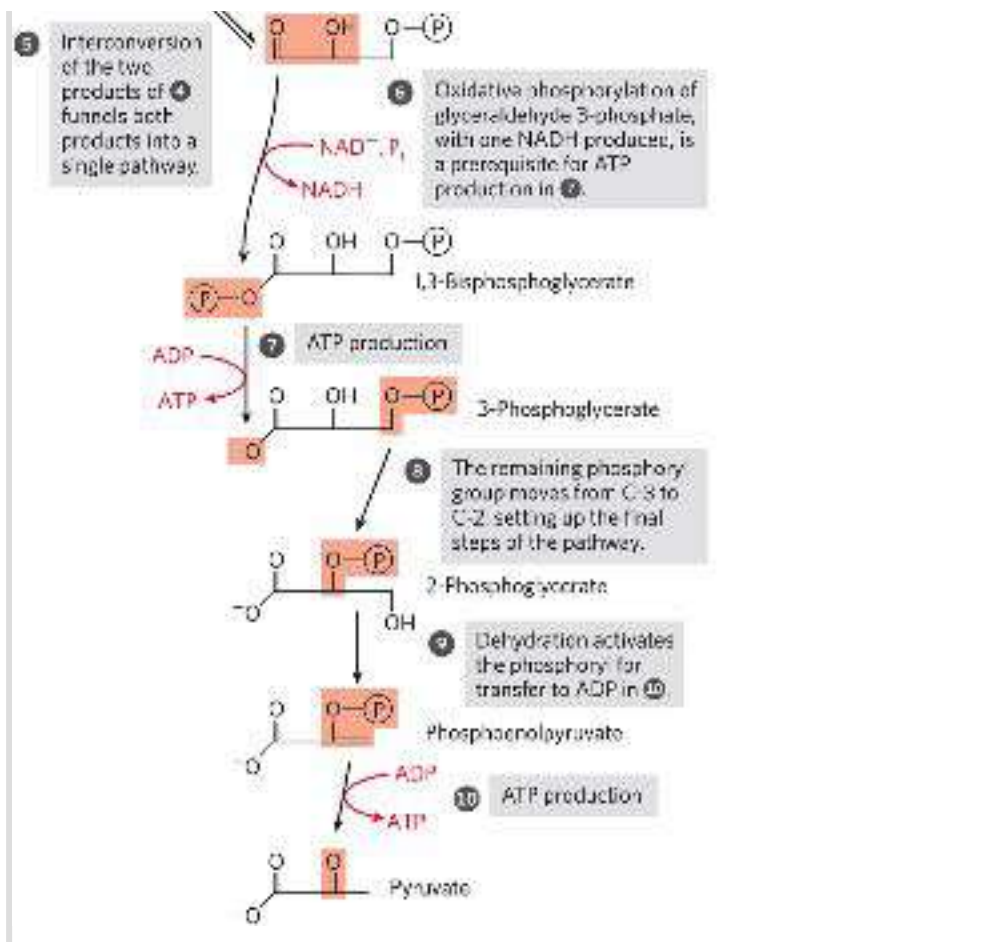
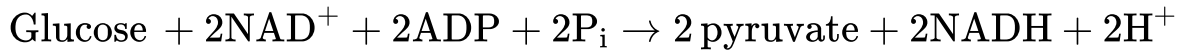


FIGURE 14-3 The chemical logic of the glycolytic pathway. In this simplified version of the pathway, each molecule is shown in a linear form, with carbon and hydrogen atoms not depicted, in order to highlight chemical transformations. Remember that glucose and fructose are present mostly in their cyclized forms in solution, although they are transiently present in linear form at the active sites of some of the enzymes in this pathway.

The preparatory phase, steps **1** to **5**, converts the six-carbon glucose into two three-carbon units, each of them phosphorylated. Oxidation of the three-carbon units is initiated in the payoff phase, steps **6** to **10**. To produce pyruvate, the chemical steps must occur in the order shown.

ATP and NADH Formation Coupled to Glycolysis

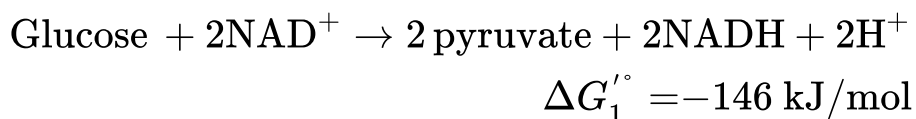
During glycolysis some of the energy of the glucose molecule is conserved in ATP, while much remains in the product, pyruvate. The overall equation for glycolysis is



(14-1)

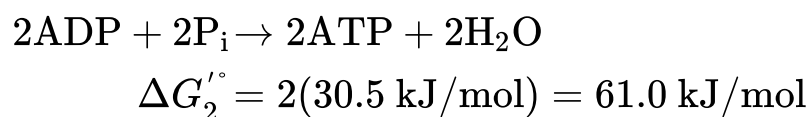
For each molecule of glucose degraded to pyruvate, two molecules of ATP are generated from ADP and P_i , and two molecules of NADH are produced by the reduction of NAD^+ . The reduction of NAD^+ (see [Fig. 13-24](#)) proceeds by the enzymatic transfer of a hydride ion ($:\text{H}^-$) from the aldehyde group of glyceraldehyde 3-phosphate to the nicotinamide ring of NAD^+ , yielding the reduced coenzyme NADH. The other hydrogen atom of the substrate molecule is released to the solution as H^+ .

We can now resolve the equation of glycolysis into two processes — the conversion of glucose to pyruvate, which is exergonic:



(14-2)

and the formation of ATP from ADP and P_i , which is endergonic:



(14-3)

The sum of [Equations 14-2](#) and [14-3](#) gives the overall standard free-energy change of glycolysis, $\Delta G_{\text{Sum}}'^{\circ}$:

$$\begin{aligned}\Delta G_{\text{Sum}}'^{\circ} &= \Delta G_1'^{\circ} + \Delta G_2'^{\circ} = -146 \text{ kJ/mol} + 61.0 \text{ kJ/mol} \\ &= -85 \text{ kJ/mol}\end{aligned}$$

Under standard conditions, and under the (nonstandard) conditions that prevail in a cell, glycolysis is an essentially irreversible process, driven to completion by a large net decrease in free energy.

Energy Remaining in Pyruvate

Glycolysis releases only a small fraction of the total available energy of the glucose molecule. The two molecules of pyruvate formed by glycolysis still contain most of the chemical potential energy of glucose, energy that can be extracted by oxidative reactions in the citric acid cycle ([Chapter 16](#)) and oxidative phosphorylation ([Chapter 19](#)) — aerobic processes. Under anaerobic conditions, pyruvate can be reduced to lactate or ethanol ([Section 14.3](#)). The oxidation of pyruvate is an important catabolic process, but pyruvate has anabolic fates as well. It can,

for example, provide the carbon skeleton for the synthesis of the amino acid alanine or for the synthesis of fatty acids. We return to these anabolic reactions of pyruvate in later chapters.

Importance of Phosphorylated Intermediates



Each of the nine glycolytic intermediates between glucose and pyruvate is phosphorylated ([Fig. 14-2](#)). The phosphoryl groups have three functions.

1. Because the plasma membrane lacks transporters for phosphorylated sugars, the phosphorylated glycolytic intermediates cannot leave the cell. After the initial phosphorylation, no further energy is necessary to retain phosphorylated intermediates in the cell, despite the large difference in their intracellular and extracellular concentrations.
2. Phosphoryl groups are essential components in the enzymatic conservation of metabolic energy. Energy made available with the transfer of phosphoryl groups from compounds with phosphoanhydride bonds (such as those in ATP) is partially conserved in the formation of phosphate esters such as glucose 6-phosphate. Compounds with higher group transfer potential than ATP, which are formed in glycolysis (1,3-bisphosphoglycerate and phosphoenolpyruvate), donate phosphoryl groups to ADP to form ATP.

3. Binding energy resulting from the binding of phosphate groups to the active sites of enzymes lowers the activation energy and increases the specificity of the enzymatic reactions ([Chapter 6](#)). The phosphate groups of ADP, ATP, and the glycolytic intermediates form complexes with Mg^{2+} , and the substrate binding sites of many glycolytic enzymes are specific for these Mg^{2+} complexes. Most glycolytic enzymes require Mg^{2+} for activity.

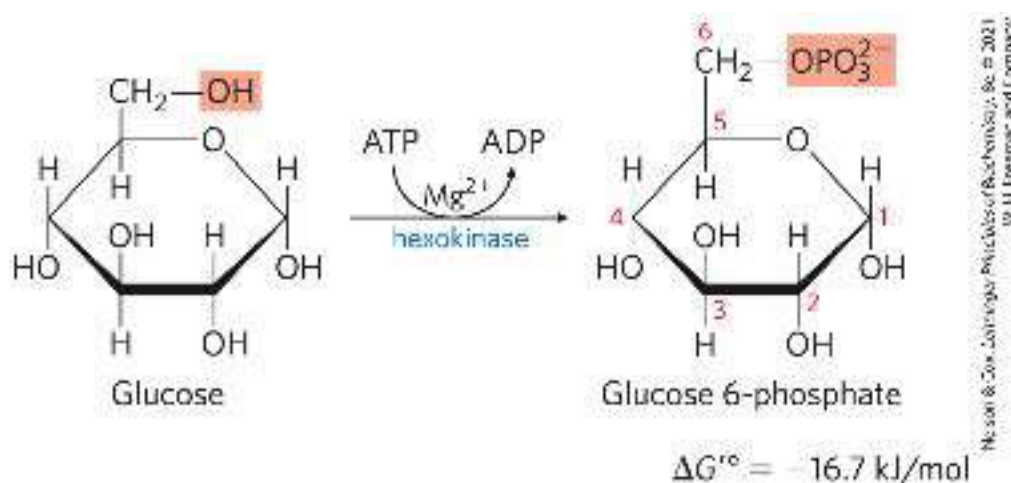
The Preparatory Phase of Glycolysis Requires ATP

In the preparatory phase of glycolysis, two molecules of ATP are invested and the hexose chain is cleaved into two triose phosphates. The realization that *phosphorylated* hexoses were intermediates in glycolysis came slowly and serendipitously. In 1906, Arthur Harden and William Young tested their hypothesis that inhibitors of proteolytic enzymes would stabilize the glucose-fermenting enzymes in yeast extract. They added blood serum (known to contain inhibitors of proteolytic enzymes) to yeast extracts and observed the predicted stimulation of glucose metabolism. However, in a control experiment intended to show that boiling the serum destroyed the stimulatory activity, they discovered that boiled serum was just as effective at stimulating glycolysis! Careful examination and testing of the contents of the boiled serum revealed that inorganic phosphate was responsible for the stimulation. Harden and Young soon discovered that glucose added to their yeast extract was converted to a hexose

bisphosphate (the “Harden-Young ester,” eventually identified as fructose 1,6-bisphosphate). This was the beginning of a long series of investigations on the role of organic esters and anhydrides of phosphate in biochemistry, which has led to our current understanding of the central role of phosphoryl group transfer in biology.

1 Phosphorylation of Glucose

In the first step of glycolysis ([Fig. 14-2](#)), glucose is activated for subsequent reactions by its phosphorylation at C-6 to yield **glucose 6-phosphate**, with ATP as the phosphoryl donor:



This reaction, which is irreversible under intracellular conditions, is catalyzed by **hexokinase**. Recall that kinases are enzymes that catalyze the transfer of the terminal phosphoryl group from ATP to an acceptor nucleophile (see [Fig. 13-8c](#)). Kinases are a subclass of transferases (see [Table 6-3](#)). The acceptor in the case of hexokinase is a hexose, normally glucose, although hexokinase

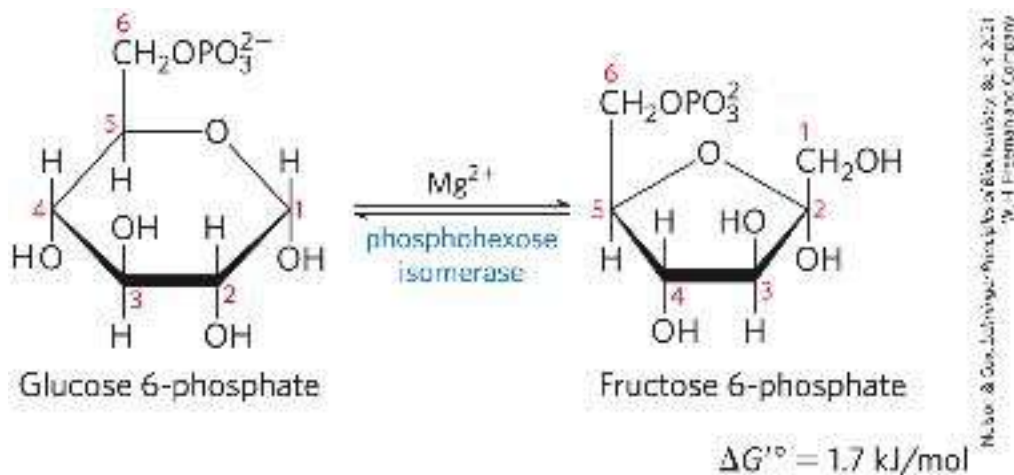
also catalyzes the phosphorylation of other common hexoses, such as fructose and mannose, in some tissues.

Hexokinase, like many other kinases, requires Mg^{2+} for its activity, because the true substrate of the enzyme is not ATP^{4-} but the MgATP^{2-} complex (see [Fig. 13-12](#)). Mg^{2+} shields the negative charges of the phosphoryl groups in ATP, making the terminal phosphorus atom an easier target for nucleophilic attack by an —OH of glucose. Hexokinase undergoes a profound change in shape, an induced fit, when it binds glucose; two domains of the protein move about 8 Å closer to each other when ATP binds (see [Fig. 6-30](#)). This movement brings bound ATP closer to a molecule of glucose also bound to the enzyme and blocks the access of water (from the solvent), which might otherwise enter the active site and attack (hydrolyze) the phosphoanhydride bonds of ATP. Like the other nine enzymes of glycolysis, hexokinase is a soluble, cytosolic protein.

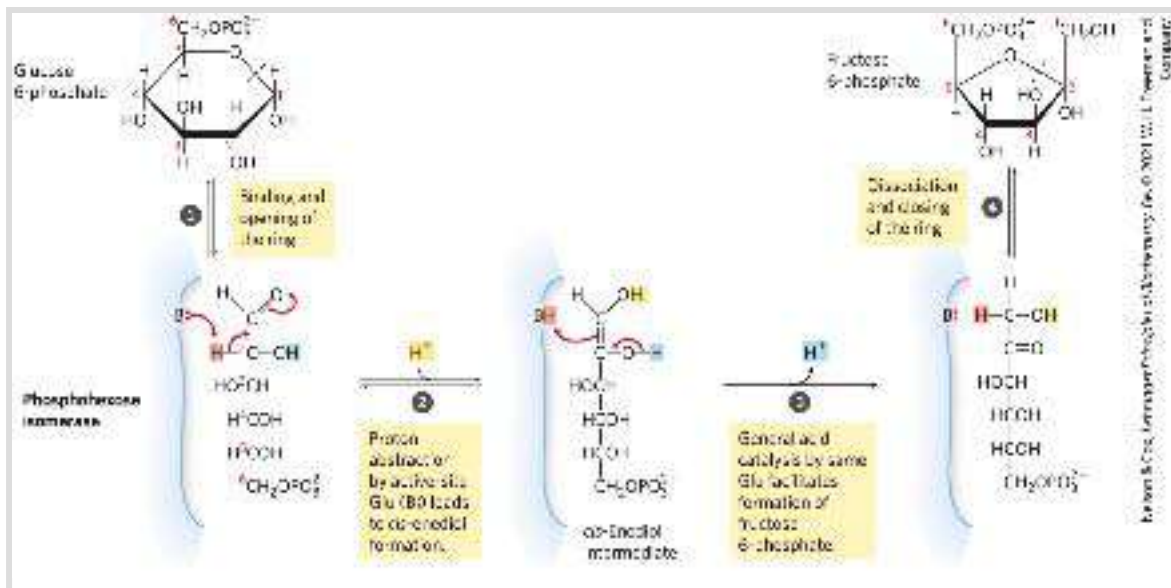
Hexokinase is present in nearly all organisms. The human genome encodes four different hexokinases (I to IV), all of which catalyze the same reaction, but differ in kinetics, regulation, and location. Two or more enzymes that catalyze the same reaction but are encoded by different genes are called **isozymes** (see [Box 14-3](#)). One of the isozymes present in hepatocytes, hexokinase IV (also called glucokinase), differs from other forms of hexokinase in kinetic and regulatory properties, with important physiological consequences that are described in [Section 14.5](#).

② Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate

The enzyme **phosphohexose isomerase** (**phosphoglucose isomerase**) catalyzes the reversible isomerization of glucose 6-phosphate, an aldose, to **fructose 6-phosphate**, a ketose:



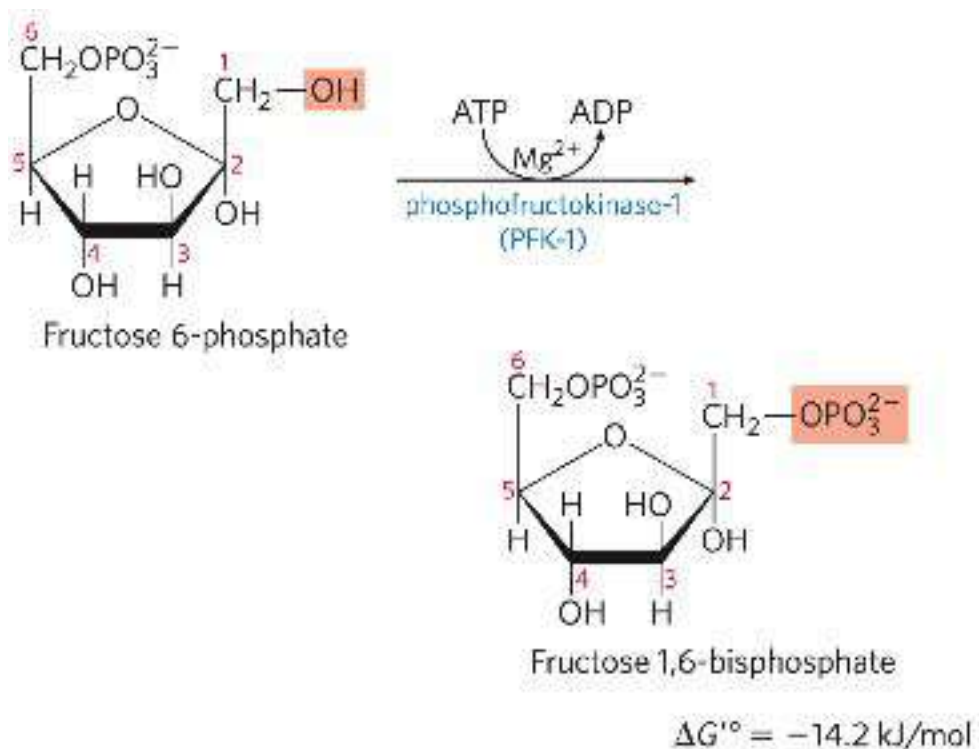
The mechanism for this reaction involves an enediol intermediate ([Fig. 14-4](#)). The reaction proceeds readily in either direction, as might be expected from the relatively small change in standard free energy.



MECHANISM FIGURE 14-4 The phosphohexose isomerase reaction. The ring opening and closing reactions (steps ① and ④) are catalyzed by an active-site His residue, by mechanisms omitted here for simplicity. The proton (light red) initially at C-2 is made more easily abstractable by electron withdrawal by the adjacent carbonyl and nearby hydroxyl groups. After its transfer from C-2 to the active-site Glu residue (a weak acid), the proton is freely exchanged with the surrounding solution; that is, the proton abstracted from C-2 in step ② is not necessarily the same one that is added to C-1 in step ③.

③ Phosphorylation of Fructose 6-Phosphate to Fructose 1,6-Bisphosphate

In the second of the two priming reactions of glycolysis, **phosphofructokinase-1 (PFK-1)** catalyzes the transfer of a phosphoryl group from ATP to fructose 6-phosphate to yield **fructose 1,6-bisphosphate**:



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KEY CONVENTION

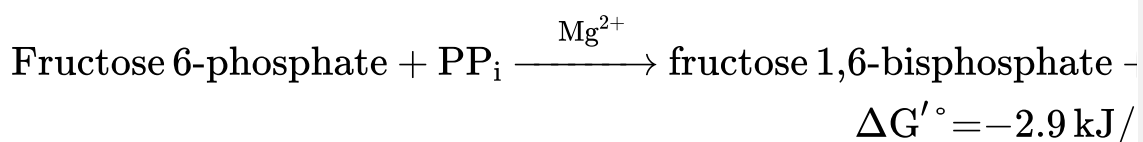
Compounds that contain two phosphate or phosphoryl groups attached at different positions in the molecule are named *bisphosphates* (or *bisphospho* compounds); for example, fructose 1,6-bisphosphate and 1,3-bisphosphoglycerate. Compounds with two phosphates linked together as a pyrophosphoryl group are named *diphosphates*; for example, adenosine diphosphate (ADP). Similar rules apply for the naming of *trisphosphates* (such as inositol 1,4,5-trisphosphate; see [p. 425](#)) and *triphosphates* (such as adenosine triphosphate, ATP). ■

The enzyme that forms fructose 1,6-bisphosphate is called PFK-1 to distinguish it from a second enzyme (PFK-2) that catalyzes the

formation of fructose 2,6-bisphosphate from fructose 6-phosphate in a separate pathway (the roles of PFK-2 and fructose 2,6-bisphosphate are discussed in [Section 14.5](#)). The PFK-1 reaction is essentially irreversible under cellular conditions, and it is the first “committed” step in the glycolytic pathway; glucose 6-phosphate and fructose 6-phosphate have other possible fates, but fructose 1,6-bisphosphate is targeted for glycolysis.

Phosphofructokinase-1 is subject to complex allosteric regulation; its activity is increased whenever the cell’s ATP supply is depleted or when the ATP breakdown products, ADP and AMP (particularly the latter), accumulate. The enzyme is inhibited whenever the cell has ample ATP and is well supplied by other fuels such as fatty acids. In some organisms, fructose 2,6-bisphosphate (not to be confused with the PFK-1 reaction product, fructose 1,6-bisphosphate) is a potent allosteric activator of PFK-1. Ribulose 5-phosphate, an intermediate in the pentose phosphate pathway discussed in [Section 14.6](#), also activates phosphofructokinase indirectly. The multiple layers of regulation of this step in glycolysis are discussed in greater detail in [Section 14.5](#).

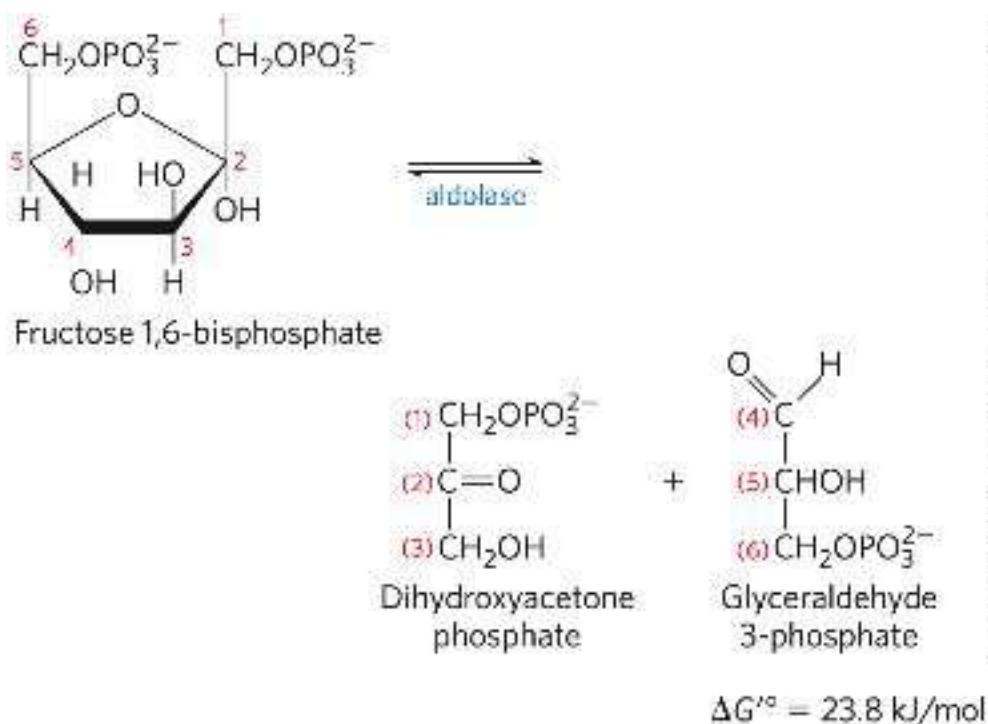
Some bacteria and protists and perhaps all plants have a different phosphofructokinase (PP-PFK-1) that uses pyrophosphate (PP_i), not ATP, as the phosphoryl group donor in the synthesis of fructose 1,6-bisphosphate:

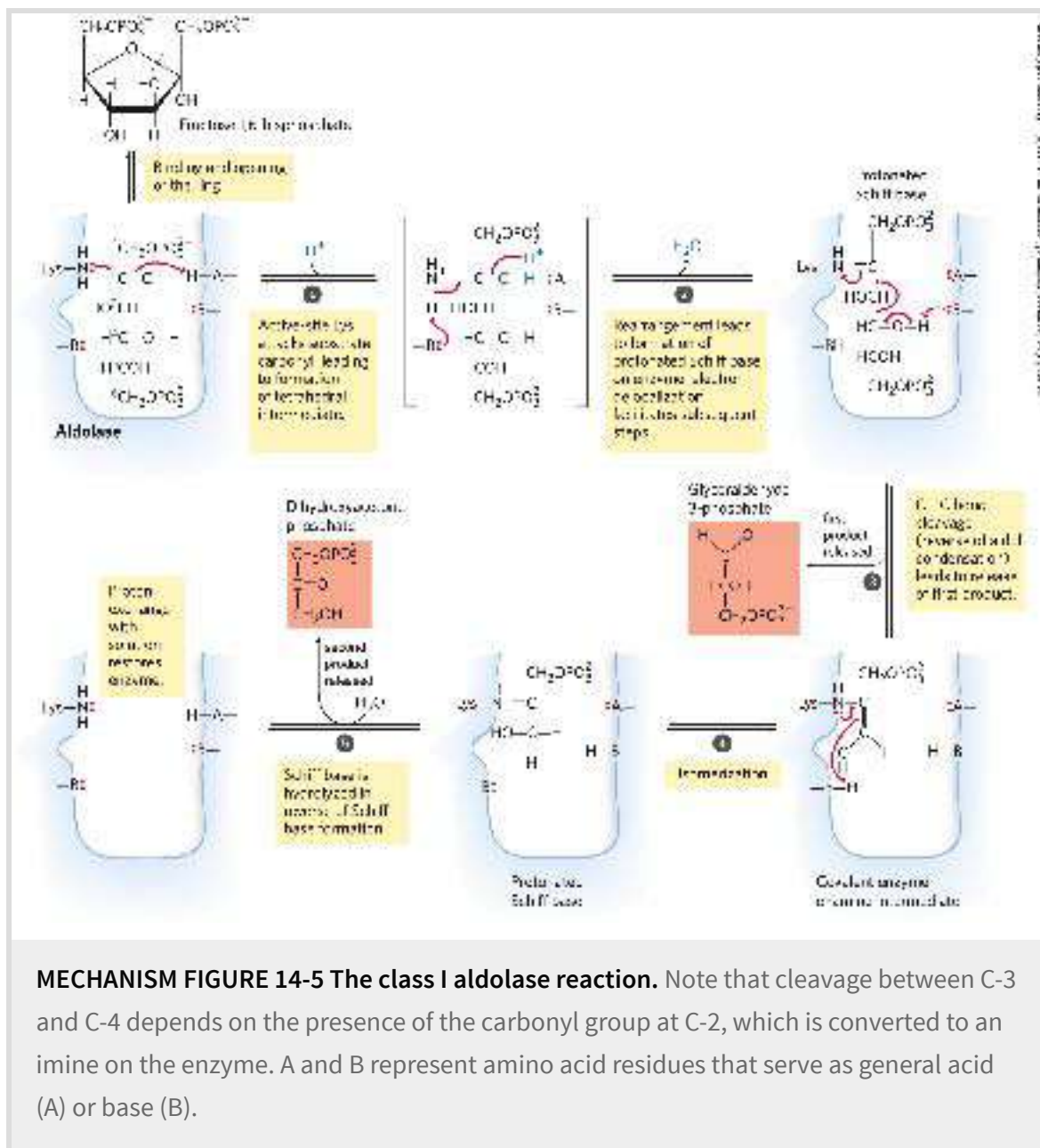


We will discuss this enzyme in [Chapter 20](#).

4 Cleavage of Fructose 1,6-Bisphosphate

The enzyme **fructose 1,6-bisphosphate aldolase**, often called simply **aldolase**, catalyzes a reverse aldol condensation ([Fig. 14-5](#); see [Fig. 13-4](#)). Fructose 1,6-bisphosphate is cleaved to yield two different triose phosphates, namely, **glyceraldehyde 3-phosphate** (an aldose) and **dihydroxyacetone phosphate** (a ketose):



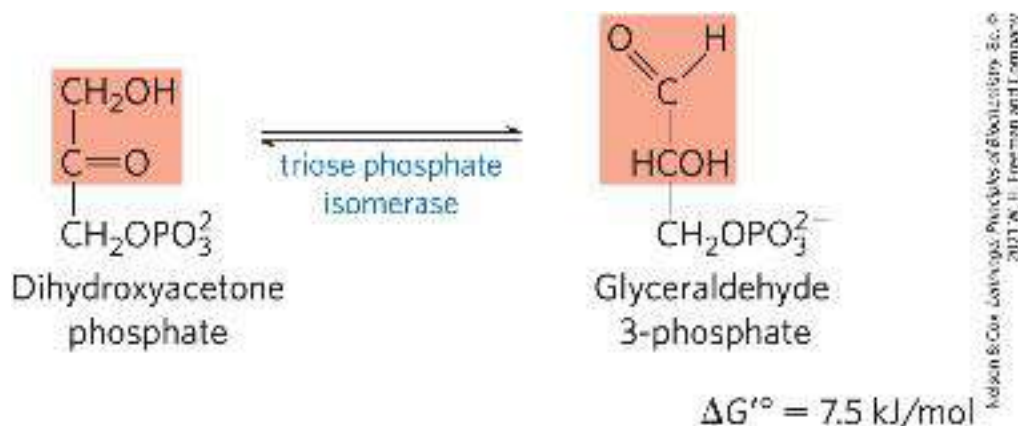


There are two classes of aldolases. Class I aldolases, found in animals and plants, use the mechanism shown in [Figure 14-5](#). Class II enzymes, in fungi and bacteria, do not form the Schiff base intermediate. Instead, a zinc ion at the active site is coordinated with the carbonyl oxygen at C-2; the Zn^{2+} polarizes the carbonyl group and stabilizes the enolate intermediate created in the C—C bond cleavage step (see [Fig. 6-23](#)).

Although the aldolase reaction has a strongly positive standard free-energy change in the direction of fructose 1,6-bisphosphate cleavage, at the lower concentrations of reactants present in cells the actual free-energy change is small and the aldolase reaction is readily reversible. We shall see later that aldolase acts in the reverse direction during the process of gluconeogenesis.

5 Interconversion of the Triose Phosphates

Only one of the two triose phosphates formed by aldolase, glyceraldehyde 3-phosphate, can be directly degraded in the subsequent steps of glycolysis. The other product, dihydroxyacetone phosphate, is immediately and reversibly converted to glyceraldehyde 3-phosphate by the fifth enzyme of the glycolytic sequence, **triose phosphate isomerase**:



The reaction mechanism is similar to the reaction promoted by phosphohexose isomerase in step 2 of glycolysis ([Fig. 14-4](#)). After the triose phosphate isomerase reaction, the carbon atoms

derived from C-1, C-2, and C-3 of the starting glucose are chemically indistinguishable from C-6, C-5, and C-4, respectively ([Fig. 14-6](#)); both “halves” of glucose have yielded glyceraldehyde 3-phosphate.

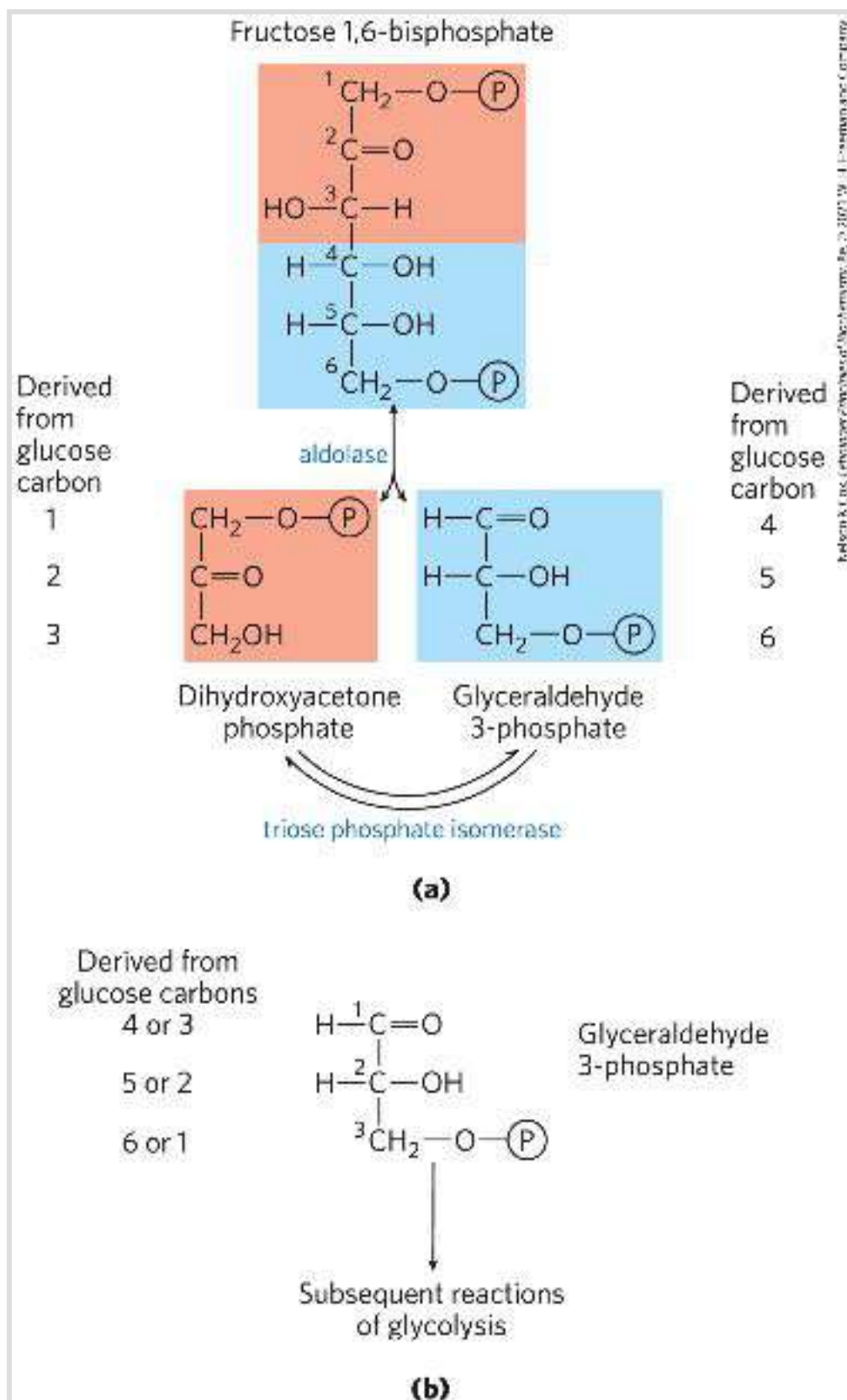


FIGURE 14-6 Fate of the glucose carbons in the formation of glyceraldehyde 3-phosphate. (a) The origin of the carbons in the two three-carbon products of the aldolase and triose phosphate isomerase reactions.

The end product of the two reactions is glyceraldehyde 3-phosphate (two molecules). (b) Each carbon of glyceraldehyde 3-phosphate is derived from either of two specific carbons of glucose. Note that the numbering of the carbon atoms of glyceraldehyde 3-phosphate differs from that of the glucose from which it is derived. In glyceraldehyde 3-phosphate, the most complex functional group (the carbonyl) is specified as C-1. This numbering change is important for interpreting experiments with glucose in which a single carbon is labeled with a radioisotope. (See Problems 5 and 22 at the end of this chapter.)

This reaction completes the preparatory phase of glycolysis. The hexose molecule has been phosphorylated at C-1 and C-6 and then cleaved to form two molecules of glyceraldehyde 3-phosphate.

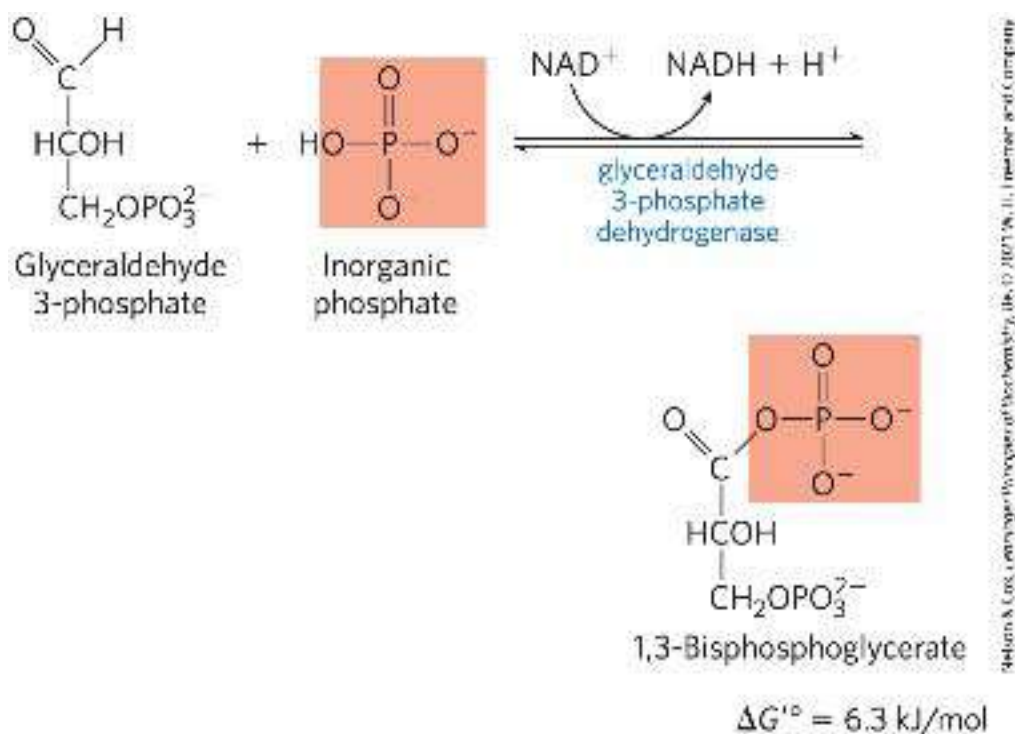
The Payoff Phase of Glycolysis Yields ATP and NADH

The payoff phase of glycolysis ([Fig. 14-2b](#)) includes the energy-conserving phosphorylation steps in which some of the chemical energy of the glucose molecule is conserved in the form of ATP and NADH. Remember that one molecule of glucose yields two molecules of glyceraldehyde 3-phosphate, and both halves of the glucose molecule follow the same pathway in the second phase of glycolysis. The conversion of two molecules of glyceraldehyde 3-phosphate to two molecules of pyruvate is accompanied by the formation of four molecules of ATP from ADP. However, the net yield of ATP per molecule of glucose degraded is only two,

because two ATP were invested in the preparatory phase of glycolysis to phosphorylate the two ends of the hexose molecule.

6 Oxidation of Glyceraldehyde 3-Phosphate to 1,3-Bisphosphoglycerate

P1 The first step in the payoff phase is the oxidation of glyceraldehyde 3-phosphate to **1,3-bisphosphoglycerate**, catalyzed by **glyceraldehyde 3-phosphate dehydrogenase**:

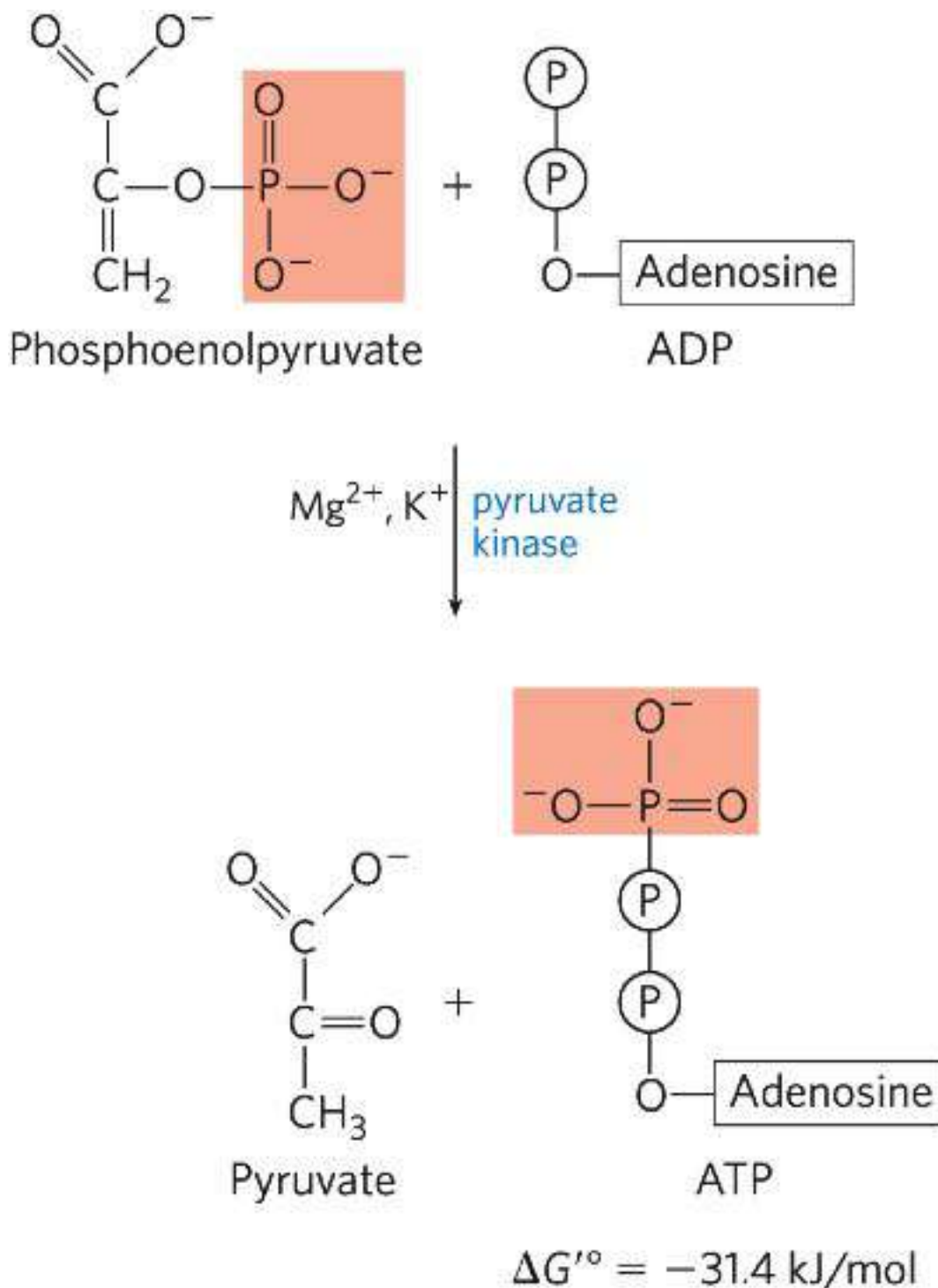


This is the first of the two energy-conserving reactions of glycolysis that eventually lead to the formation of ATP. The aldehyde group of glyceraldehyde 3-phosphate is oxidized, not to a free carboxyl group but to a carboxylic acid anhydride with phosphoric acid. This type of anhydride, called an **acyl**

The amount of NAD^+ in a cell ($\leq 10^{-5} \text{ M}$) is far smaller than the amount of glucose metabolized in a few minutes. Glycolysis would soon come to a halt if the NADH formed in this step of glycolysis were not continuously reoxidized and recycled. We return to a discussion of this recycling of NAD^+ later in the chapter.

7 Phosphoryl Transfer from 1,3-Bisphosphoglycerate to ADP

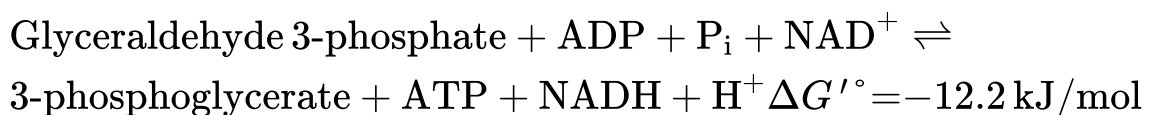
The enzyme **phosphoglycerate kinase** transfers the high-energy phosphoryl group from the carboxyl group of 1,3-bisphosphoglycerate to ADP, forming ATP and 3-phosphoglycerate:



Notice that phosphoglycerate kinase is named for the reverse reaction, in which it transfers a phosphoryl group from ATP to 3-phosphoglycerate. Like all enzymes, it catalyzes the reaction in both directions. This enzyme acts in the direction suggested by its name during gluconeogenesis (see [Fig. 14-16](#)) and during

photosynthetic CO₂ assimilation (see [Fig. 20-26](#)). In glycolysis, the reaction it catalyzes proceeds as shown above, in the direction of ATP synthesis.

Steps 6 and 7 of glycolysis together constitute an energy-coupling process in which 1,3-bisphosphoglycerate is the common intermediate; it is formed in the first reaction (which would be endergonic in isolation), and its acyl phosphate group is transferred to ADP in the second reaction (which is strongly exergonic). The sum of these two reactions is



Thus the overall reaction is exergonic.

Recall from [Chapter 13](#) that the actual free-energy change, ΔG , is determined by the standard free-energy change, $\Delta G'^{\circ}$, and the mass-action ratio, Q , which is the ratio [products]/[reactants] (see [Eqn 13-4](#)). For step 6,

$$\begin{aligned} \Delta G &= \Delta G'^{\circ} + RT \ln Q \\ &= \Delta G'^{\circ} + RT \ln \frac{[\text{1,3-bisphosphoglycerate}][\text{NADH}]}{[\text{glyceraldehyde 3-phosphate}][\text{P}_i][\text{NAD}^+]} \end{aligned}$$

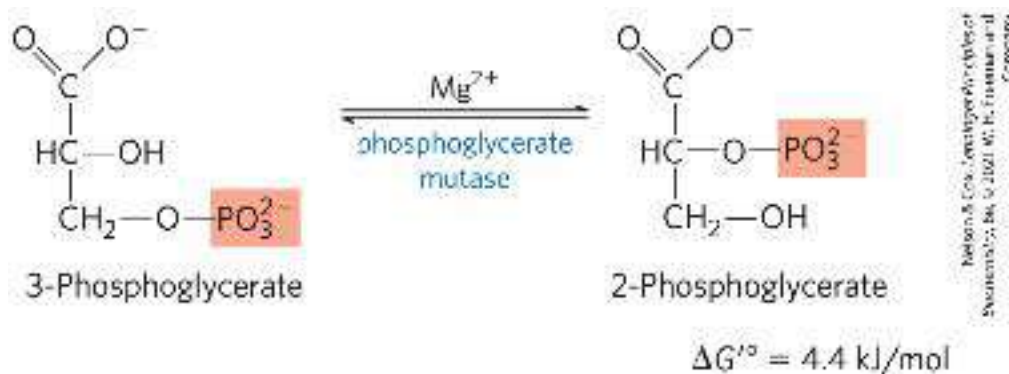
Notice that $[H^+]$ is not included in Q . In biochemical calculations, $[H^+]$ is assumed to be a constant (10^{-7} M), and this constant is included in the definition of $\Delta G'^{\circ}$ ([p. 468](#)).

When the mass-action ratio is less than 1.0, its natural logarithm has a negative sign. In the cytosol, where these reactions are taking place, the ratio $[NADH]/[NAD^+]$ is a small fraction, contributing to a low Q . Step 7, by consuming the product of step 6 (1,3-bisphosphoglycerate), keeps [1,3-bisphosphoglycerate] relatively low in the steady state and thereby keeps Q for the overall energy-coupling process small. When Q is small, the contribution of $\ln Q$ can make ΔG strongly negative. This is simply another way of showing how the two reactions, steps 6 and 7, are coupled through a common intermediate.

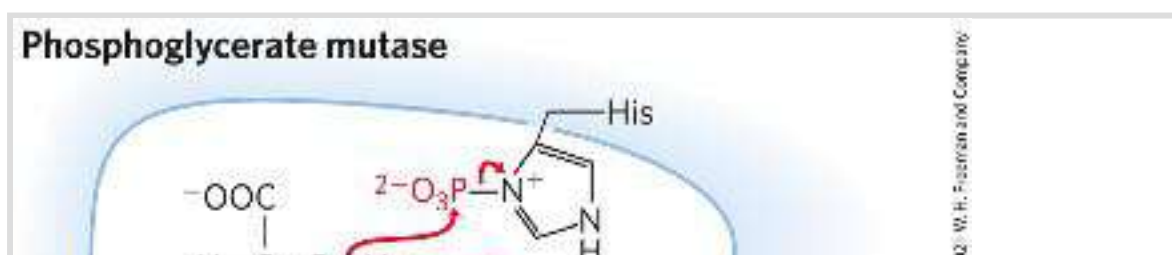
The outcome of these coupled reactions, both reversible under cellular conditions, is that the energy released on oxidation of an aldehyde to a carboxylate group is conserved by the coupled formation of ATP from ADP and P_i . The formation of ATP by phosphoryl group transfer from a substrate such as 1,3-bisphosphoglycerate is referred to as a [substrate-level phosphorylation](#), to distinguish this mechanism from [respiration-linked phosphorylation](#). Substrate-level phosphorylations involve soluble enzymes and chemical intermediates (1,3-bisphosphoglycerate in this case). Respiration-linked phosphorylations, on the other hand, involve membrane-bound enzymes and transmembrane gradients of protons ([Chapter 19](#)).

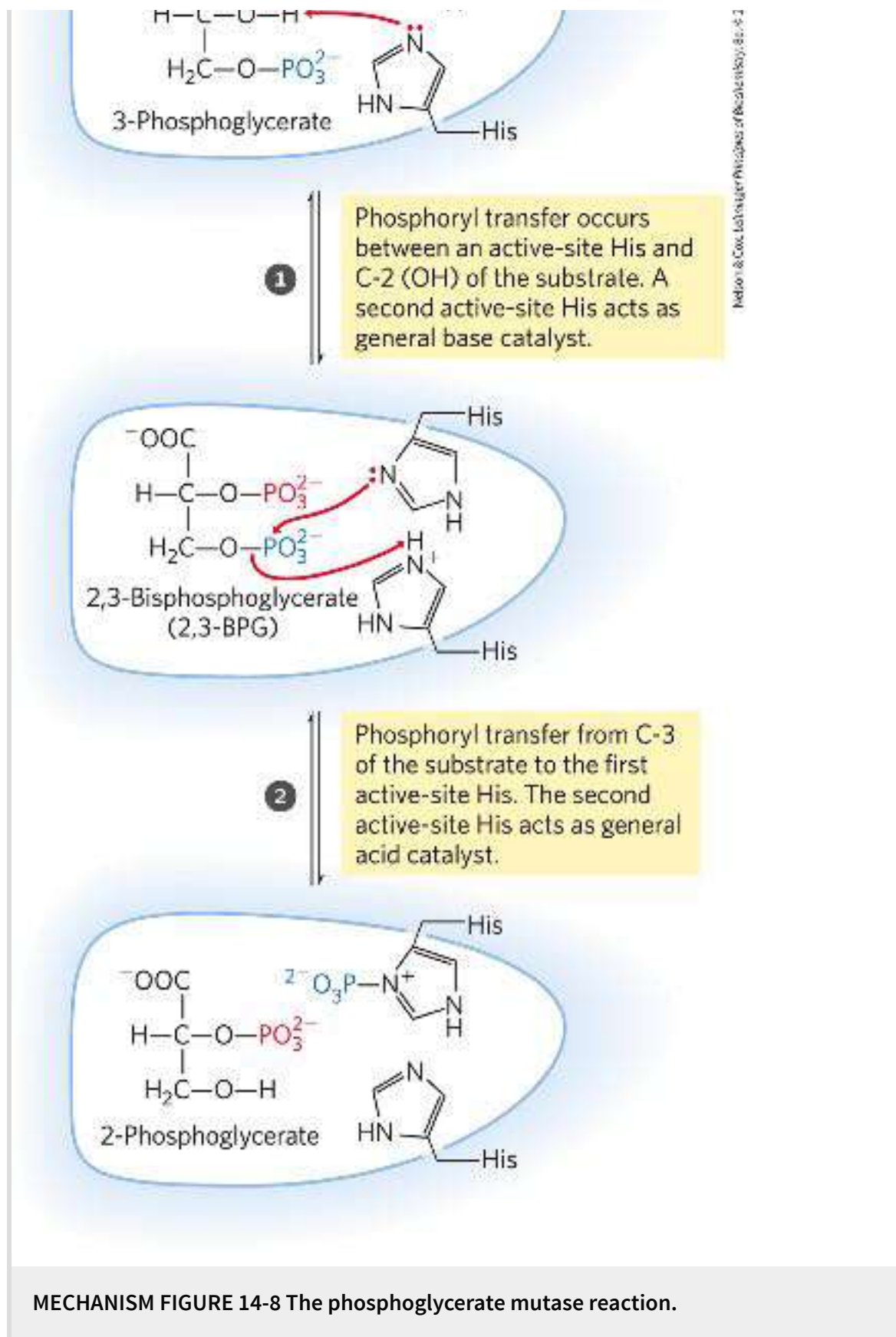
8 Conversion of 3-Phosphoglycerate to 2-Phosphoglycerate

The enzyme **phosphoglycerate mutase** catalyzes a reversible shift of the phosphoryl group between C-2 and C-3 of glycerate; Mg^{2+} is essential for this reaction:



The reaction occurs in two steps ([Fig. 14-8](#)). A phosphoryl group initially attached to a His residue of the mutase is transferred to the hydroxyl group at C-2 of 3-phosphoglycerate, forming 2,3-bisphosphoglycerate (2,3-BPG). The phosphoryl group at C-3 of 2,3-BPG is then transferred to the same His residue, producing 2-phosphoglycerate and regenerating the phosphorylated enzyme. Phosphoglycerate mutase is initially phosphorylated by phosphoryl transfer from 2,3-BPG, which is required in small quantities to initiate the catalytic cycle and is continuously regenerated by that cycle.

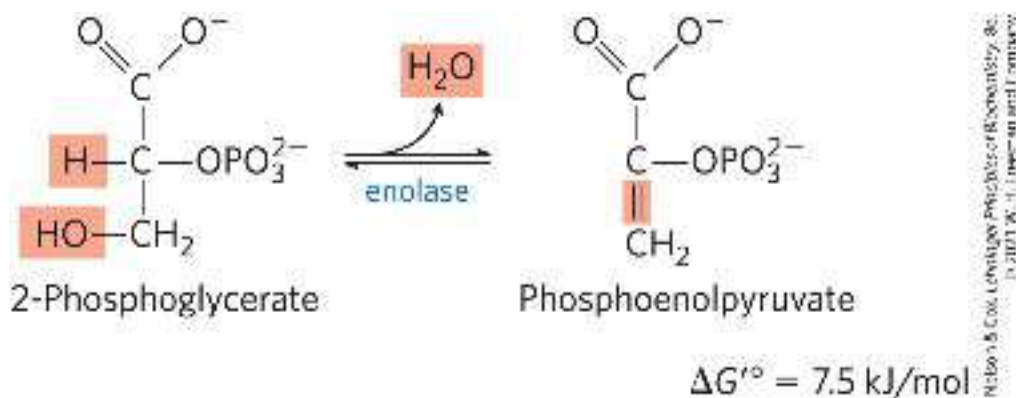




MECHANISM FIGURE 14-8 The phosphoglycerate mutase reaction.

9 Dehydration of 2-Phosphoglycerate to Phosphoenolpyruvate

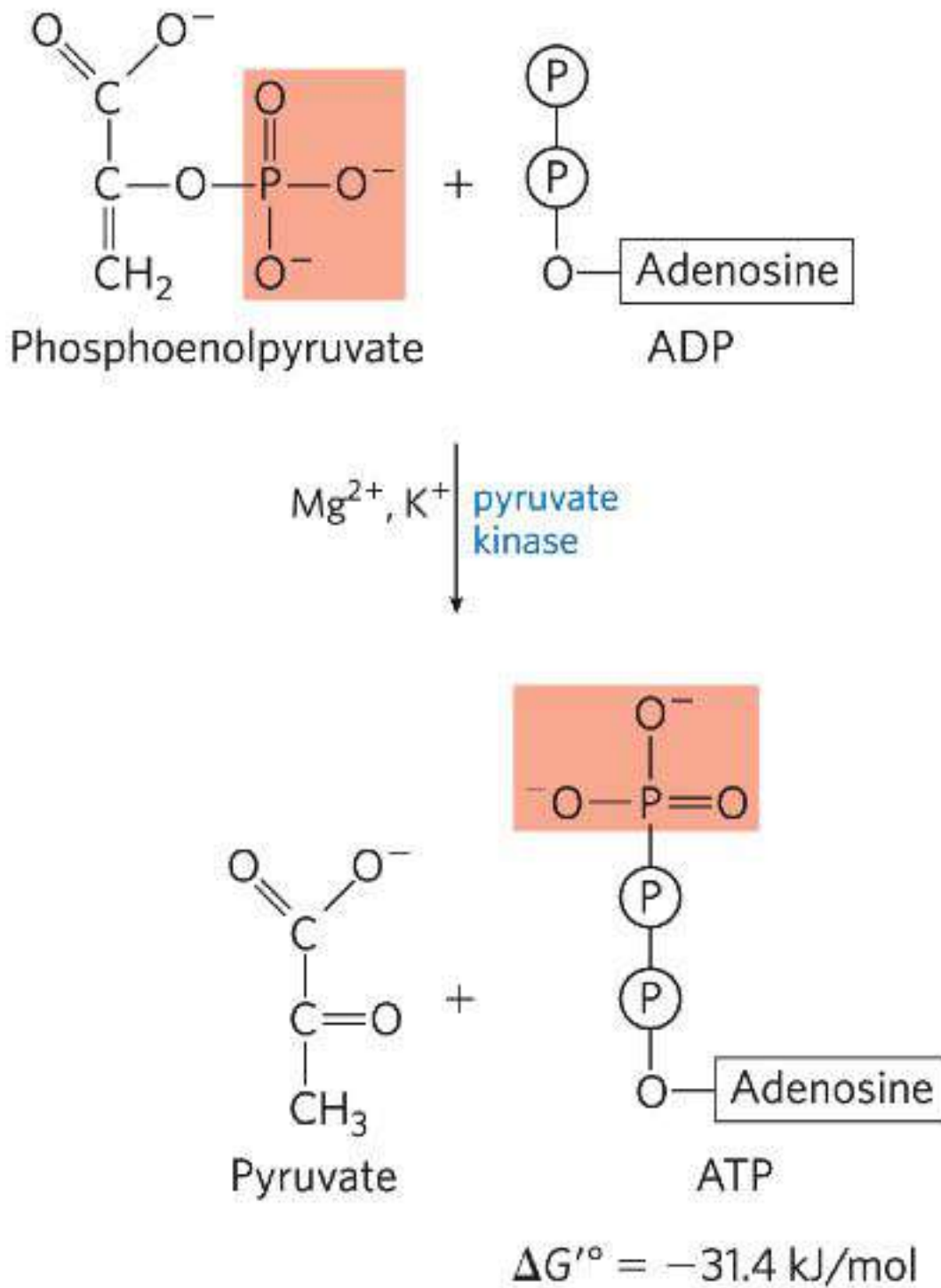
P1 In the second glycolytic reaction that generates a compound with high phosphoryl group transfer potential (the first was step 6), **enolase** promotes reversible removal of a molecule of water from 2-phosphoglycerate to yield **phosphoenolpyruvate (PEP)**:



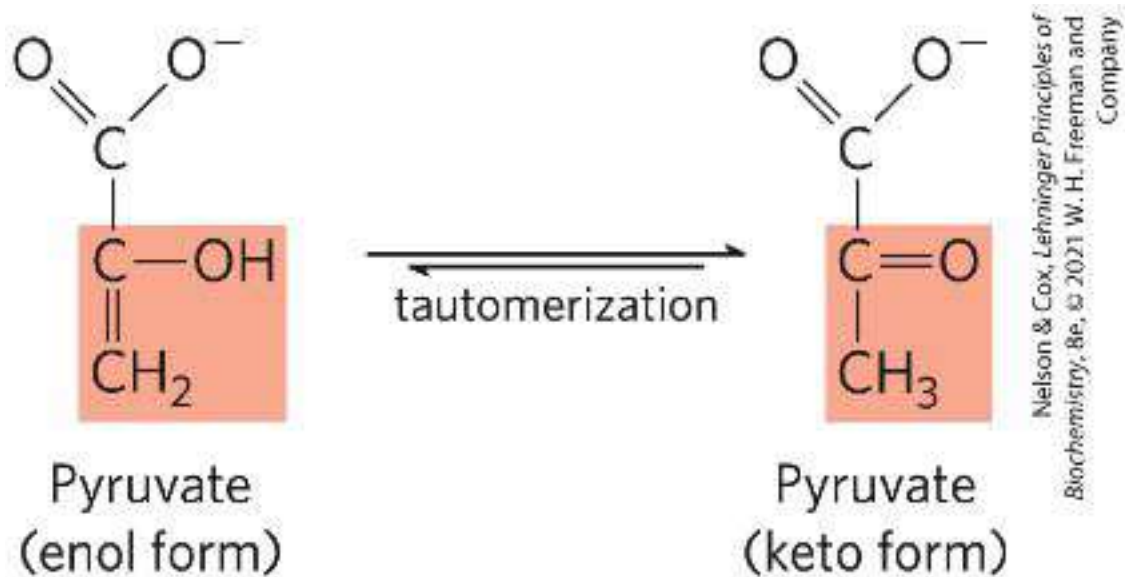
The mechanism of the enolase reaction involves an enolic intermediate stabilized by Mg^{2+} (see [Fig. 6-31](#)). The reaction converts a compound with a relatively low phosphoryl group transfer potential ($\Delta G'^{\circ}$ for hydrolysis of 2-phosphoglycerate is -17.6 kJ/mol) to one with high phosphoryl group transfer potential ($\Delta G'^{\circ}$ for PEP hydrolysis is -61.9 kJ/mol) (see [Fig. 13-13](#)).

10 Transfer of the Phosphoryl Group from Phosphoenolpyruvate to ADP

The last step in glycolysis is the transfer of the phosphoryl group from phosphoenolpyruvate to ADP, catalyzed by **pyruvate kinase**, which requires K^+ and either Mg^{2+} or Mn^{2+} :



In this substrate-level phosphorylation, the product **pyruvate** first appears in its enol form, then tautomerizes nonenzymatically to its keto form, which predominates at pH 7:

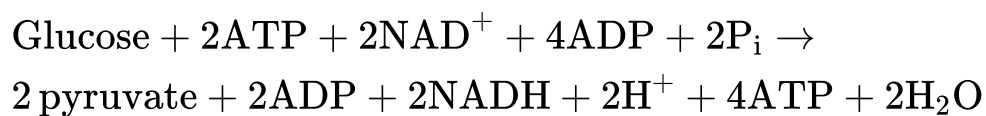


The overall reaction has a large, negative standard free-energy change, due in large part to the spontaneous conversion of the enol form of pyruvate to the keto form (see [Fig. 13-13](#)). About half of the energy released by PEP hydrolysis ($\Delta G'^{\circ} = -61.9 \text{ kJ/mol}$) is conserved in the formation of the phosphoanhydride bond of ATP ($\Delta G'^{\circ} = -30.5 \text{ kJ/mol}$), and the rest (-31.4 kJ/mol) constitutes a large driving force pushing the reaction toward ATP synthesis. We discuss the regulation of pyruvate kinase in [Section 14.5](#).

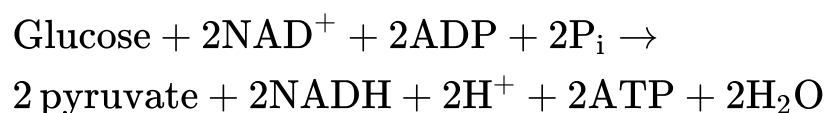
The Overall Balance Sheet Shows a Net Gain of Two ATP and Two NADH Per Glucose



We can now construct a balance sheet for glycolysis to account for (1) the fate of the carbon skeleton of glucose, (2) the input of P_i and ADP and output of ATP, and (3) the pathway of electrons in the oxidation-reduction reactions. The left side of the following equation shows all the inputs of ATP, NAD^+ , ADP, and P_i (consult [Fig. 14-2](#)), and the right side shows all the outputs (keep in mind that each molecule of glucose yields two molecules of pyruvate):



Canceling out common terms on both sides of the equation gives the overall equation for glycolysis:



In the overall glycolytic process, one molecule of glucose is converted to two molecules of pyruvate (the pathway of carbon). Two molecules of ADP and two of P_i are converted to two molecules of ATP (the pathway of phosphoryl groups). Four electrons, as two hydride ions, are transferred from two molecules of glyceraldehyde 3-phosphate to two of NAD^+ (the pathway of electrons).

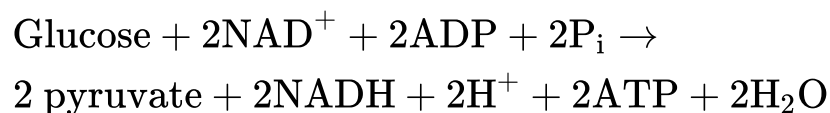
SUMMARY 14.1 *Glycolysis*

■ Glycolysis is a near-universal pathway by which a glucose molecule is oxidized, in two phases, to two molecules of pyruvate, with energy conserved as ATP and NADH. Ten cytosolic enzymes act sequentially in glycolysis. The overall reaction converts glucose to two molecules of pyruvate, and energy is conserved in the synthesis of two molecules of ATP and two molecules of NADH.

■ In the preparatory phase of glycolysis, two molecules of ATP are invested to activate glucose to fructose 1,6-bisphosphate. The bond between C-3 and C-4 is then broken to yield two molecules of triose phosphate.

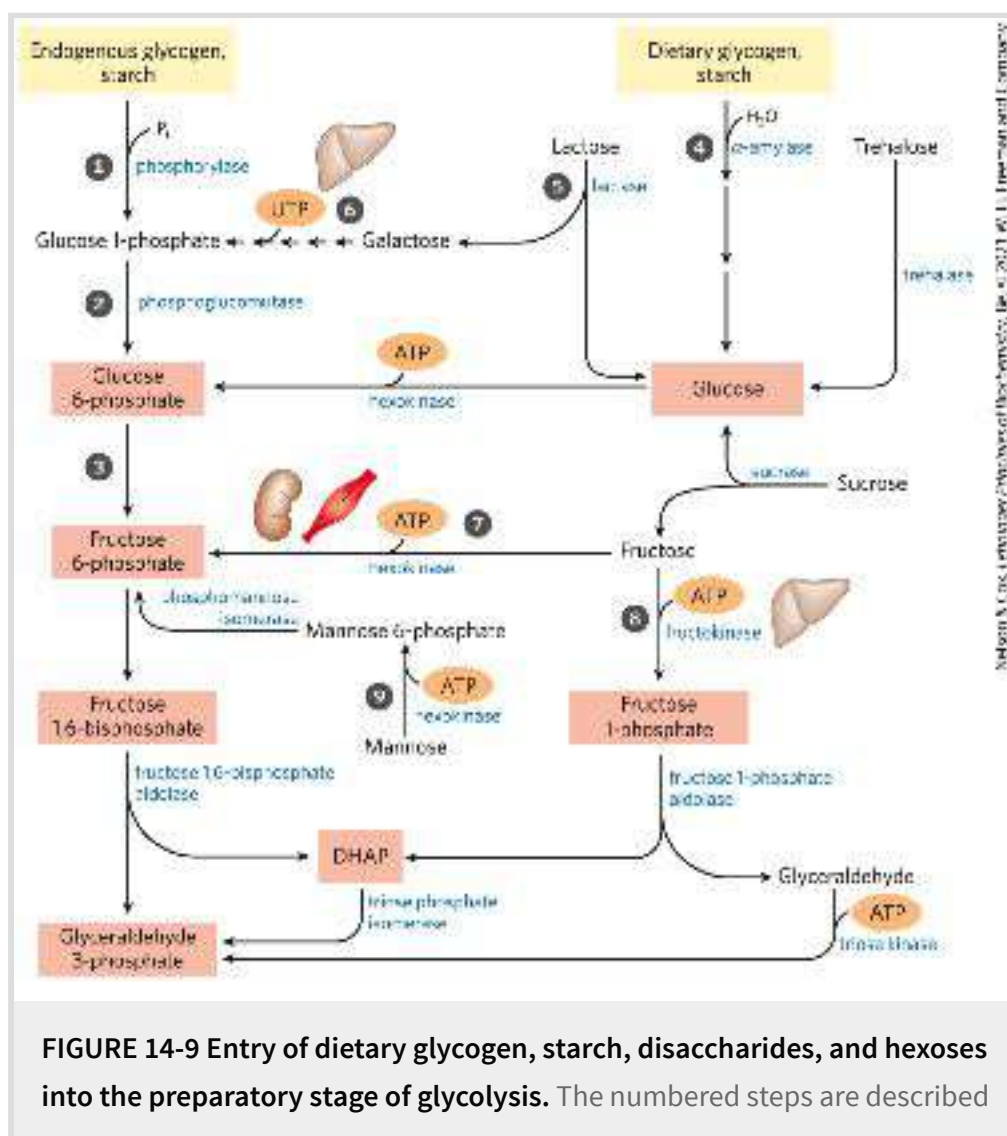
■ In the payoff phase, each of the two molecules of glyceraldehyde 3-phosphate derived from glucose undergoes oxidation at C-1; some of the energy of this oxidation reaction is conserved in the form of one NADH and two ATP per triose phosphate oxidized.

■ Subtracting the two ATP spent in the preparatory phase, the net equation for the overall process is



14.2 Feeder Pathways for Glycolysis

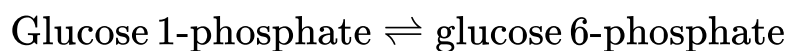
Many carbohydrates besides glucose meet their catabolic fate in glycolysis, after being transformed into one of the glycolytic intermediates. The most significant are the storage polysaccharides glycogen and starch, either within cells (endogenous) or obtained in the diet; the disaccharides maltose, lactose, and sucrose; and the monosaccharides fructose, mannose, and galactose ([Fig. 14-9](#)).



Endogenous Glycogen and Starch Are Degraded by Phosphorolysis

P2 Glycogen stored in animal tissues (primarily liver and skeletal muscle) and in microorganisms is mobilized for use within the same cell by a *phosphorolytic* reaction (❶ in [Fig. 14-9](#)) catalyzed by **glycogen phosphorylase**. The product of this reaction is not free glucose, but glucose 1-phosphate. We discuss glycogen metabolism in more detail in [Chapter 15](#). In plant tissues, starch is mobilized by a similar *phosphorolytic* reaction catalyzed by **starch phosphorylase**.

Glucose 1-phosphate produced by glycogen phosphorylase is converted to glucose 6-phosphate by **phosphoglucomutase** (❷), which catalyzes the reversible reaction



Phosphoglucomutase employs essentially the same mechanism as phosphoglycerate mutase ([Fig. 14-8](#)): both entail a bisphosphate intermediate, and the enzyme is transiently phosphorylated in each catalytic cycle. The general name **mutase** is given to enzymes that catalyze the transfer of a functional group from one position to another in the same molecule.

Mutases are a subclass of [isomerases](#), enzymes that interconvert stereoisomers or structural or positional isomers (see [Table 6-3](#)). The glucose 6-phosphate formed in the phosphoglucomutase reaction can continue through glycolysis [3](#) or enter another pathway such as the pentose phosphate pathway, described in [Section 14.6](#).

WORKED EXAMPLE 14-1 *Energy Savings for Glycogen Breakdown by Phosphorolysis*

Calculate the energy savings (in ATP molecules per glucose monomer) achieved by breaking down glycogen by [phosphorolysis](#) rather than *hydrolysis* to begin the process of glycolysis.

SOLUTION:

Phosphorolysis produces a phosphorylated glucose (glucose 1-phosphate), which is then converted to glucose 6-phosphate — without expenditure of the cellular energy (1 ATP) needed for formation of glucose 6-phosphate from free glucose. Thus only 1 ATP is consumed per glucose monomer in the preparatory phase, compared with 2 ATP when glycolysis starts with free glucose. The cell therefore gains 3 ATP per glucose monomer (4 ATP produced in the payoff phase minus 1 ATP used in the

preparatory phase), rather than 2— a savings of 1 ATP per glucose monomer.

Dietary Polysaccharides and Disaccharides Undergo Hydrolysis to Monosaccharides

For most humans, starch is the major source of carbohydrates in the diet (4, [Fig. 14-9](#)). Dietary starch has essentially the same structure as glycogen, and its digestion proceeds by the same pathway. Digestion begins in the mouth, where salivary α -**amylase** hydrolyzes the internal ($\alpha 1 \rightarrow 4$) glycosidic linkages of starch and glycogen, producing di- and trisaccharides. These are produced by *hydrolysis* reactions, in which water, not P_i , is the attacking species. In the stomach, salivary α -amylase is inactivated by the low pH, but a second form of α -amylase, secreted by the pancreas into the small intestine, continues the digestion process.

Pancreatic α -amylase release into the small intestine yields mainly maltose and maltotriose (the di- and trisaccharides of glucose) and oligosaccharides called limit **dextrins**, fragments of amylopectin containing ($\alpha 1 \rightarrow 6$) branch points, which are removed by limit **dextrinases**. Disaccharides are hydrolyzed by a

family of membrane-bound hydrolases in the intestinal brush border:



Only monosaccharides are taken up from the intestine. They are actively transported into the intestinal epithelial cells (see [Fig. 11-42](#)), then passed into the blood to be carried to various tissues, where they are catabolized via glycolysis.

As we noted in [Chapter 7](#), most animals cannot digest cellulose for lack of the enzyme **cellulase**, which attacks the ($\beta 1 \rightarrow 4$) glycosidic bonds of cellulose. In ruminant animals, the extended stomach includes a chamber in which symbiotic microorganisms that produce cellulase break down cellulose into glucose molecules. These microorganisms use the resulting glucose in an anaerobic fermentation that produces large quantities of propionate. This propionate, after conversion to succinate (see [Fig. 17-12](#)), serves as the starting material for gluconeogenesis, which produces much of the lactose in milk.



Lactose Digestion and Lactose Intolerance

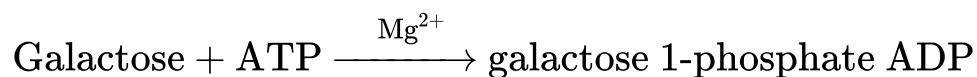
The defining feature of mammals is, of course, mammary glands, which produce the disaccharide lactose for the nourishment of infants. The enzyme lactase converts lactose to glucose and galactose (5, [Fig. 14-9](#)), both of which are taken up from the small intestine and metabolized in the tissues by glycolysis. As infants are weaned, their lactase levels diminish, and lactase is absent in most adults — except in certain populations. About one in three adults in northern Europe and in some parts of Africa shows the **lactase persistence** phenotype. They continue to produce lactase and thus are able to digest milk into adulthood. The other two-thirds experience **lactose intolerance** due to the disappearance after childhood of most or all of the lactase activity of the intestinal epithelial cells. Without intestinal lactase, lactose cannot be completely digested and absorbed in the small intestine, and it passes into the large intestine, where bacteria convert it to toxic products that cause abdominal cramps and diarrhea. The problem is further complicated because undigested lactose and its metabolites increase the osmolarity of the intestinal contents, favoring retention of water in the intestine, and causing diarrhea. In most parts of the world where lactose intolerance is prevalent, milk is not used as a food by adults, although milk products predigested with lactase are commercially available. In certain human disorders, several or all of the intestinal disaccharidases are missing. In these cases, the digestive disturbances triggered by dietary disaccharides can

sometimes be minimized by a controlled diet lacking the undigestible carbohydrates.

One way to determine whether lactase is present and active in the intestine is to compare the rise in blood glucose after the ingestion of a quantity of either glucose or lactose. When glucose is ingested, the blood glucose level increases rapidly and transiently. When lactose is ingested, lactase, if present in the intestine, will hydrolyze the lactose into glucose and galactose and the blood glucose level will rise. If lactase is absent or less active, ingesting lactose will lead to little or no transient increase in blood glucose. ■

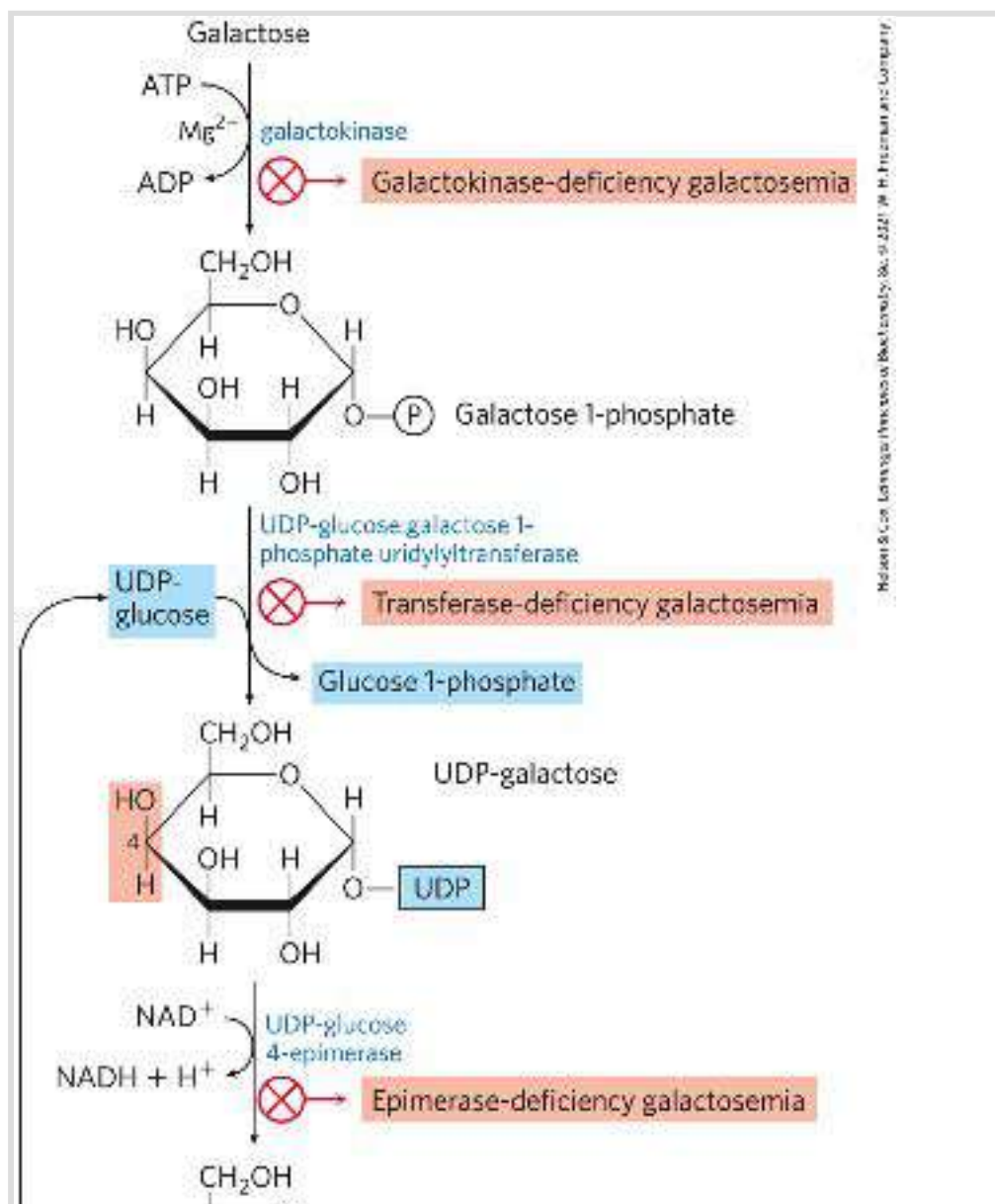
Galactose Metabolism and Disease

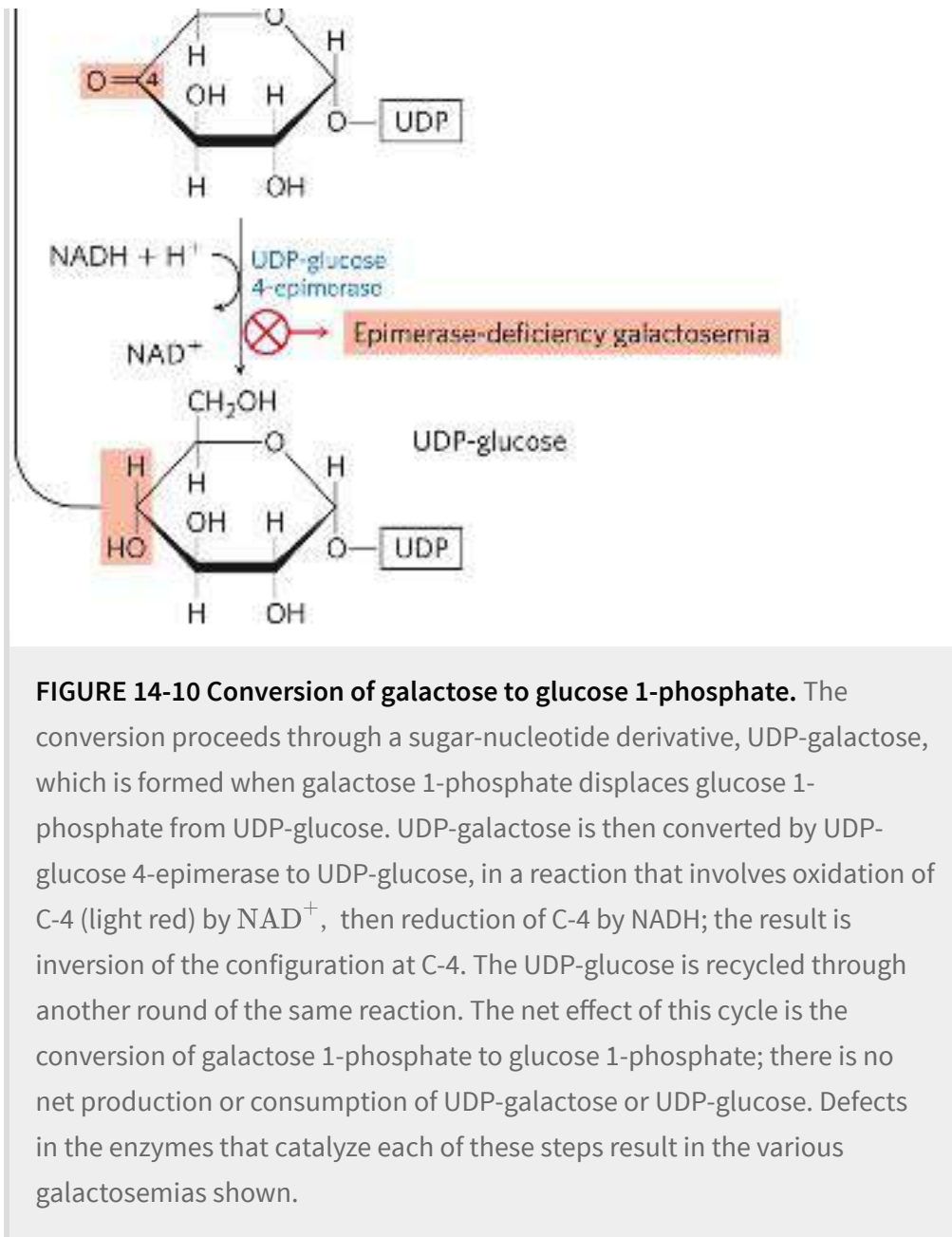
Galactose (6, [Fig. 14-9](#)), a product of the hydrolysis of lactose and therefore an important component in the diet of infants, passes in the blood from the intestine to the liver, where it is first phosphorylated at C-1, at the expense of ATP, by the enzyme **galactokinase**:



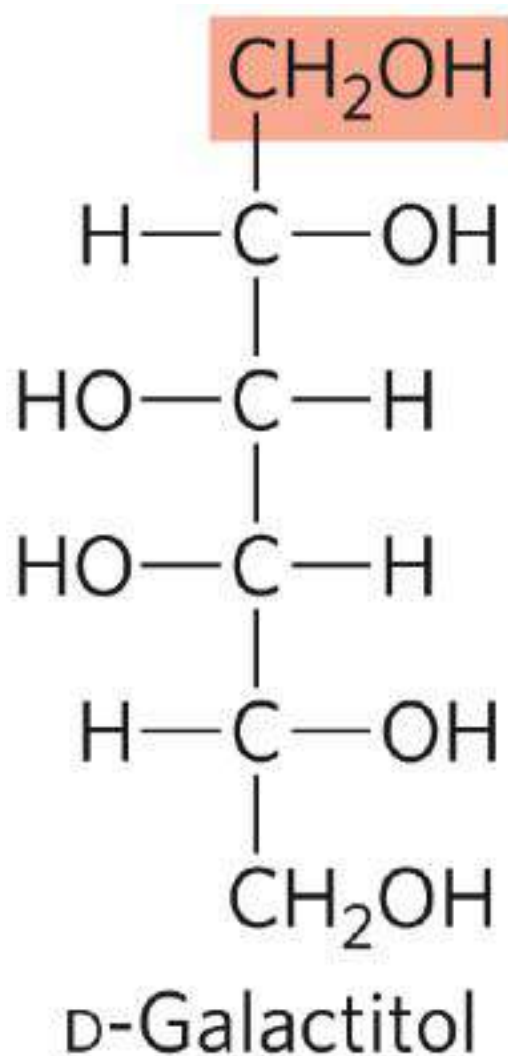
The galactose 1-phosphate is then transferred to a uridine nucleotide by a transferase. The resulting UDP-galactose is epimerized at C-4, forming UDP-glucose by a set of reactions in

which UDP functions as an activator of hexose groups ([Fig. 14-10](#)) and a “tag” that these hexoses are in a separate pool from those destined for another process such as glycolysis. The epimerization, catalyzed by UDP-glucose 4-epimerase, involves first the oxidation of the C-4 —OH group to a ketone, then reduction of the ketone to an —OH, with inversion of the configuration at C-4. NAD is the cofactor for both the oxidation and the reduction. The glucose 1-phosphate made this way is converted to glucose 6-phosphate by phosphoglucomutase.





A defect in any of the enzymes in this pathway has serious medical consequences. In galactokinase-deficiency **galactosemia**, caused by a defect in the *GALK* gene, high galactose concentrations are found in blood and urine. Affected individuals develop cataracts in infancy, caused by deposition of the galactose metabolite galactitol in the lens.



The other symptoms in this disorder are relatively mild, and strict limitation of galactose in the diet greatly diminishes their severity. Transferase-deficiency galactosemia, caused by a defect in the *GALT* gene, is more serious; it is characterized by poor growth in childhood, speech abnormality, mental deficiency, and liver damage that may be fatal, even when galactose is withheld from the diet. Epimerase-deficiency galactosemia, caused by a defect in the *GALE* gene, leads to similar symptoms, but they are less severe when dietary galactose is carefully controlled. ■

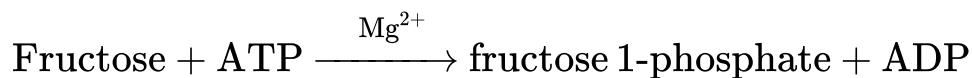
Fructose and Mannose



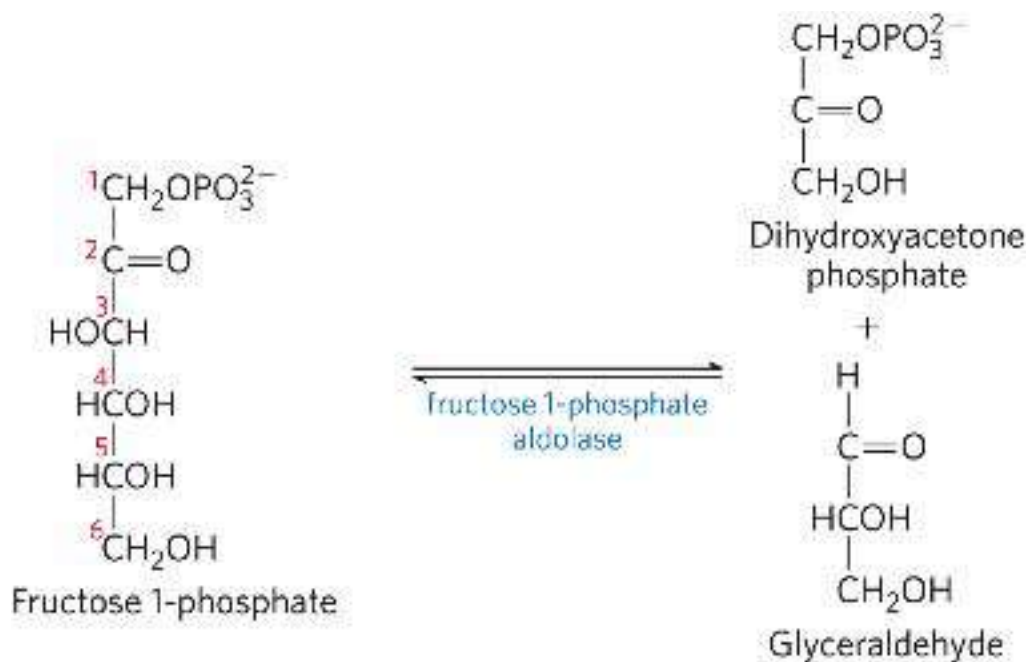
In most organisms, hexoses other than glucose can undergo glycolysis after conversion to a phosphorylated derivative. Fructose, present in free form in many fruits and formed by hydrolysis of sucrose in the small intestine of vertebrates, is phosphorylated by hexokinase:



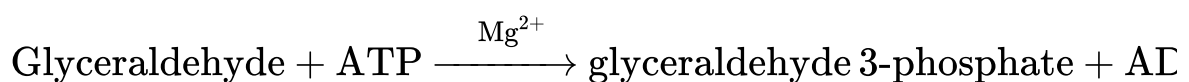
This is a major pathway of fructose entry into glycolysis in the muscles and kidney (7, [Fig. 14-9](#)). In the liver, fructose enters by a different pathway. The liver enzyme **fructokinase** catalyzes the phosphorylation of fructose at C-1 rather than C-6 (8, [Fig. 14-9](#)):



The fructose 1-phosphate is then cleaved to glyceraldehyde and dihydroxyacetone phosphate by **fructose 1-phosphate aldolase**:



Dihydroxyacetone phosphate is converted to glyceraldehyde 3-phosphate by the glycolytic enzyme triose phosphate isomerase. Glyceraldehyde is phosphorylated by ATP and **triose kinase** to glyceraldehyde 3-phosphate:



Thus, both products of fructose 1-phosphate hydrolysis enter the glycolytic pathway as glyceraldehyde 3-phosphate.

Mannose, released in the digestion of various polysaccharides and glycoproteins of foods, is phosphorylated at C-6 by hexokinase (9, [Fig. 14-9](#)):



Phosphohexose isomerase converts mannose 6-phosphate to fructose 6-phosphate, which enters glycolysis.

SUMMARY 14.2 *Feeder Pathways for Glycolysis*

■ Endogenous glycogen and starch, polymeric storage forms of glucose, undergo sequential phosphorolysis of glucose residues, forming glucose 1-phosphate. Phosphoglucomutase converts the glucose 1-phosphate to glucose 6-phosphate, which can enter glycolysis at a point in the preparatory phase that requires the investment of only one more ATP.

■ Ingested polysaccharides and disaccharides are converted to monosaccharides by hydrolytic enzymes in saliva and in the small intestine. The monosaccharides pass through intestinal cells to the bloodstream, which transports them to the liver or other tissues.

■ Lactase is present in infants but often absent in adults, producing lactose intolerance. D-Hexoses, including galactose, fructose, and mannose, can be phosphorylated and funneled into glycolysis. Galactose is converted to glucose 1-phosphate through UDP-galactose and UDP-glucose intermediates. A genetic defect in enzymes of this pathway results in one of several galactosemias of varying severity.

14.3 Fates of Pyruvate

With the exception of some interesting variations in the bacterial realm, the pyruvate formed by glycolysis is further metabolized via one of three catabolic routes ([Fig. 14-11](#)).  Under aerobic conditions, glycolysis is only the first stage in the *complete* degradation of glucose. The pyruvate formed in the final step of glycolysis is oxidized to acetate (acetyl-CoA), which enters the citric acid cycle and is oxidized completely to CO₂ and H₂O ([Chapter 16](#)). The electrons from these oxidations are carried by NADH and FADH₂, which are ultimately reoxidized to NAD⁺ and FAD by passage of electrons to O₂ through a chain of carriers in mitochondrial respiration, to form H₂O. The energy from the electron-transfer reactions drives the synthesis of ATP ([Chapter 19](#)).

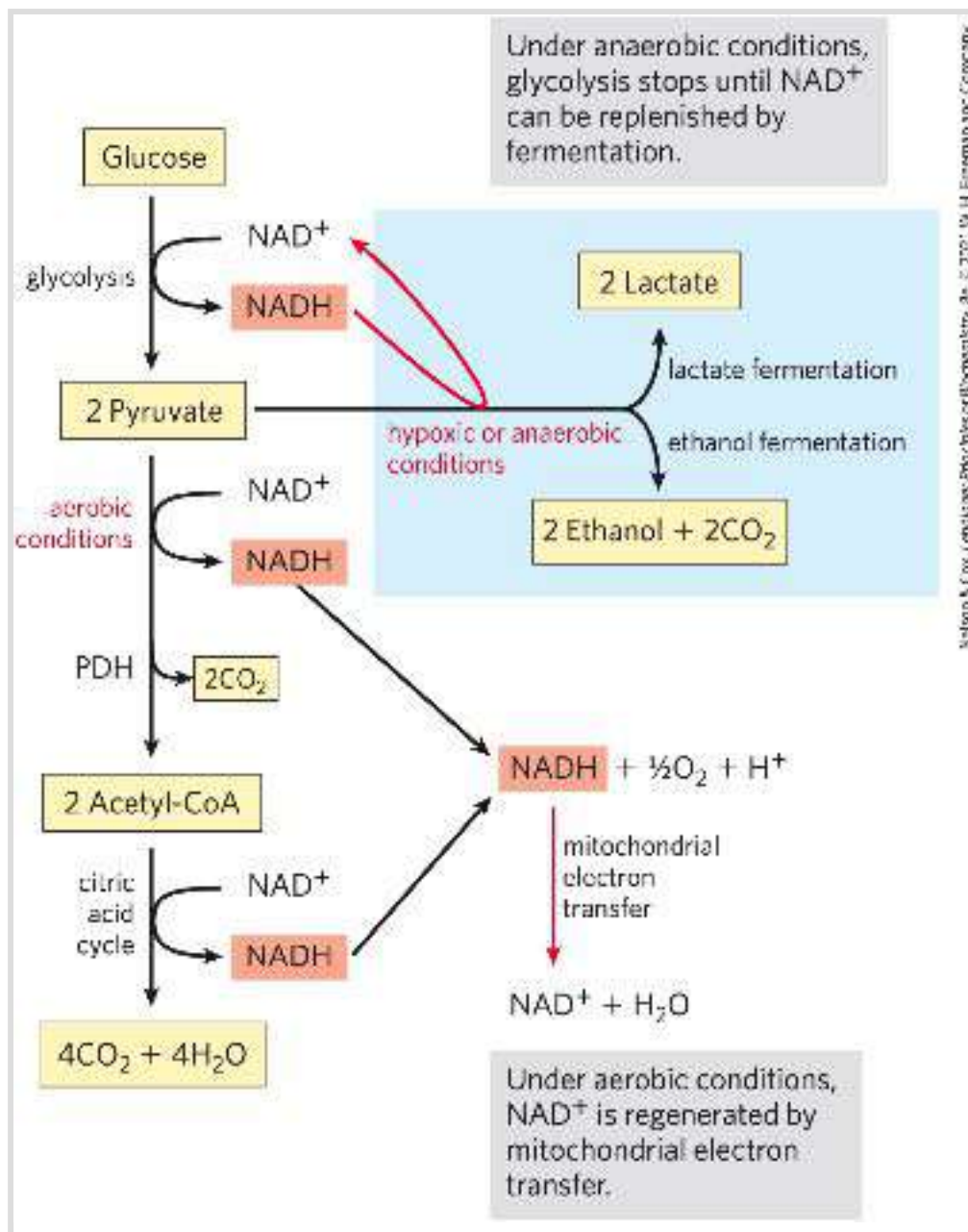


FIGURE 14-11 Three possible catabolic fates of the pyruvate formed in glycolysis and the recycling of NADH. Red arrows follow the regeneration of NAD⁺ from NADH. Under *aerobic* conditions, pyruvate is activated to acetyl-CoA and is completely oxidized to CO₂ and water through the citric acid cycle and mitochondrial oxidative phosphorylation. NADH produced in this pathway is oxidized to NAD⁺ through mitochondrial electron transfer. Under *anaerobic* conditions, pyruvate reduction to lactate or to ethanol is required to produce the NAD⁺ needed for glycolysis to continue. Pyruvate also serves as a precursor in many anabolic reactions, not shown here.

The earliest cells lived in an atmosphere almost devoid of oxygen and evolved deriving energy from fuel molecules under anaerobic conditions. Under anaerobic or low-oxygen conditions ([hypoxia](#)), NADH cannot be reoxidized to NAD^+ by passing its electrons to O_2 . But for glycolysis to continue, NAD^+ must be regenerated. Under these conditions, glucose is degraded by [fermentation](#) (defined below), which leads to one of two different fates for the pyruvate formed by glycolysis. In **lactic acid fermentation**, pyruvate accepts electrons from NADH and is reduced to lactate while regenerating the NAD^+ necessary for glycolysis. In [ethanol \(alcohol\) fermentation](#), pyruvate is further catabolized to ethanol ([Fig. 14-11](#)).

The Pasteur and Warburg Effects Are Due to Dependence on Glycolysis Alone for ATP Production

During his studies on the fermentation of glucose by yeast, Louis Pasteur discovered that both the rate and the total amount of glucose consumption under anaerobic conditions were many times greater than under aerobic conditions. Later studies of muscle showed the same large difference in the rates of glycolysis under anaerobic and aerobic conditions. The biochemical basis of this “Pasteur effect” is now clear. The ATP yield from glycolysis alone (2 ATP per molecule of glucose) is much smaller than that from the complete oxidation of glucose to CO_2 under aerobic conditions (30 or 32 ATP per glucose; see [Table 19-5](#)). About 15

times as much glucose must therefore be consumed anaerobically as aerobically to yield the same amount of ATP.

The flux of glucose through the glycolytic pathway is regulated to maintain nearly constant ATP levels (as well as adequate supplies of glycolytic intermediates that serve biosynthetic roles). The required adjustment in the rate of glycolysis is achieved by a complex interplay among ATP consumption, NAD^+ regeneration from NADH formed in glycolysis, and allosteric regulation of several glycolytic enzymes — including hexokinase, PFK-1, and pyruvate kinase — and by second-to-second fluctuations in the concentration of key metabolites that reflect the cellular balance between ATP production and consumption. On a slightly longer time scale, glycolysis is regulated by the hormones glucagon, epinephrine, and insulin, and by changes in the expression of the genes for several glycolytic enzymes. An especially interesting case is glycolysis in tumors. The German biochemist Otto Warburg first observed in 1928 that tumors of nearly all types carry out glycolysis at a much higher rate than normal tissue, *even when oxygen is available*. This “Warburg effect” is the basis for several methods of detecting and treating cancer ([Box 14-1](#)).

BOX 14-1 **MEDICINE**

High Rate of Glycolysis in Tumors Suggests Targets for Chemotherapy and Facilitates Diagnosis

In many types of tumors found in humans and other animals, glucose uptake and glycolysis proceed about 10 times faster than in normal, noncancerous tissues. Most tumor cells grow under hypoxic conditions (i.e., with limited oxygen supply) because, at least initially, they lack the capillary network to supply sufficient oxygen. Cancer cells located more than 100 to 200 μM from the nearest capillaries must depend on glycolysis alone (without further oxidation of pyruvate) for much of their ATP production. The energy yield (2 ATP per glucose) is far lower than can be obtained by the complete oxidation of pyruvate to CO_2 in mitochondria (about 30 ATP per glucose; [Chapter 19](#)). So, to make the same amount of ATP, tumor cells must take up much more glucose than do normal cells, converting it to pyruvate and then to lactate as they recycle NADH. It is likely that two early steps in the transformation of a normal cell into a tumor cell are (1) the change to dependence on glycolysis alone for ATP production and (2) the development of tolerance to a low pH in the extracellular fluid (caused by release of lactic acid). In general, the more aggressive the tumor, the greater its rate of glycolysis.

This increase in glycolysis is achieved, at least in part, by increased synthesis of the glycolytic enzymes and of the plasma membrane transporters GLUT1 and GLUT3 (see [Table 11-1](#)) that carry glucose into cells. (GLUT1 and GLUT3 are not dependent on insulin.) The **hypoxia-inducible transcription factor (HIF-1)** is a protein that acts at the level of mRNA synthesis to stimulate the production of at least eight glycolytic enzymes and the glucose transporters when oxygen supply is limited ([Fig. 1](#)). With the resulting high rate of glycolysis, the tumor cell can survive anaerobic conditions until the supply of blood vessels has caught up with tumor growth. Another protein induced by HIF-1 is the peptide hormone VEGF (vascular endothelial growth factor), which stimulates the outgrowth of blood vessels (angiogenesis) toward the tumor.

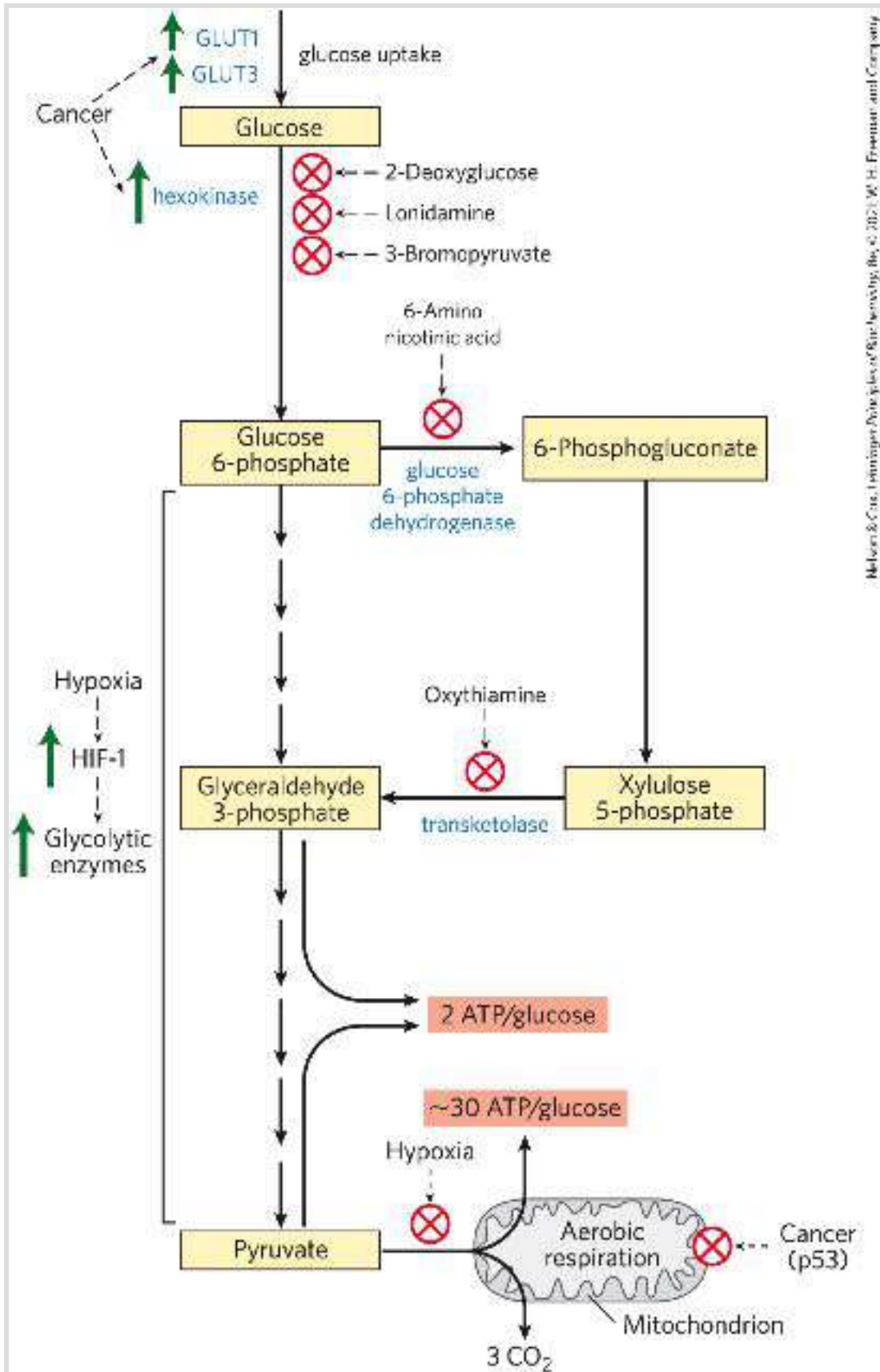


FIGURE 1 The anaerobic metabolism of glucose in tumor cells yields far less ATP (2 per glucose) than the complete oxidation to CO₂ that takes place in healthy cells

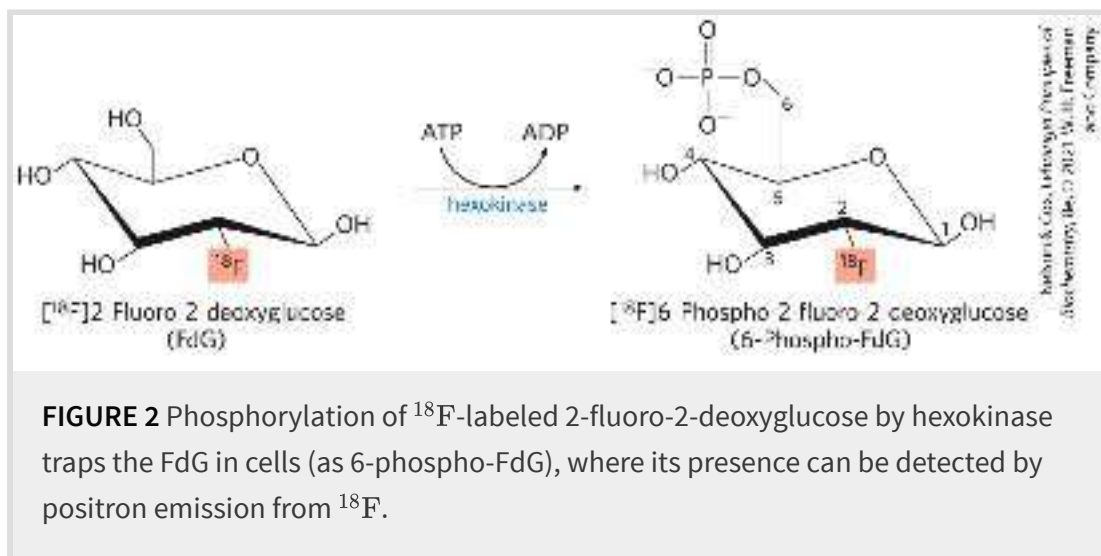
under aerobic conditions (~30 ATP per glucose), so a tumor cell must consume much more glucose to produce the same amount of ATP. Glucose transporters and most of the glycolytic enzymes are overproduced in tumors. Compounds that inhibit hexokinase, glucose 6-phosphate dehydrogenase, or transketolase block ATP production by glycolysis, thus depriving the cancer cell of energy and killing it.

There is also evidence that the tumor suppressor protein p53, which is mutated in most types of cancer (see [Section 12.9](#)), controls the synthesis and assembly of mitochondrial proteins essential to the passage of electrons to O₂. Cells with mutant p53 are defective in mitochondrial electron transfer and are forced to rely more heavily on glycolysis for ATP production ([Fig. 1](#)).

This heavier reliance of tumors than of normal tissue on glycolysis suggests a possibility for anticancer therapy: inhibitors of glycolysis might target and kill tumors by depleting their supply of ATP. Three inhibitors of hexokinase have shown promise as chemotherapeutic agents: 2-deoxyglucose, lonidamine, and 3-bromopyruvate. By preventing the formation of glucose 6-phosphate, these compounds not only deprive tumor cells of glycolytically produced ATP but also prevent the formation of pentose phosphates via the pentose phosphate pathway, which also begins with glucose 6-phosphate. Without pentose phosphates, a cell cannot synthesize the nucleotides essential to DNA and RNA synthesis and thus cannot grow or divide. Another anticancer drug already approved for clinical use is imatinib (Gleevec), described in [Box 12-4](#). It inhibits a specific tyrosine kinase, preventing the increased synthesis of hexokinase normally triggered by that kinase. The thiamine analog oxythiamine, which blocks the action of a transketolase-like enzyme that converts xylulose 5-phosphate to glyceraldehyde 3-phosphate ([Fig. 1](#)), is in preclinical trials as an antitumor drug.

The high glycolytic rate in tumor cells also has diagnostic usefulness. The relative rates at which tissues take up glucose can be used in some cases to pinpoint the location of tumors. In positron emission tomography (PET), individuals are injected with a harmless, isotopically labeled glucose analog that is taken up but not metabolized by tissues. The labeled compound is 2-fluoro-2-deoxyglucose (FdG), in which the hydroxyl group at the C-2 of glucose is replaced with ¹⁸F ([Fig. 2](#)). This compound is taken up via GLUT transporters

and is a good substrate for hexokinase, but it cannot be converted to the enediol intermediate in the phosphohexose isomerase reaction (see [Fig. 14-4](#)) and therefore accumulates as 6-phospho-FdG. The extent of its accumulation depends on its rate of uptake and phosphorylation, which, as noted above, is typically 10 or more times higher in tumors than in normal tissue. Decay of ^{18}F yields positrons (two per ^{18}F atom) that can be detected by a series of sensitive detectors positioned around the body, which allows accurate localization of accumulated 6-phospho-FdG ([Fig. 3](#)).



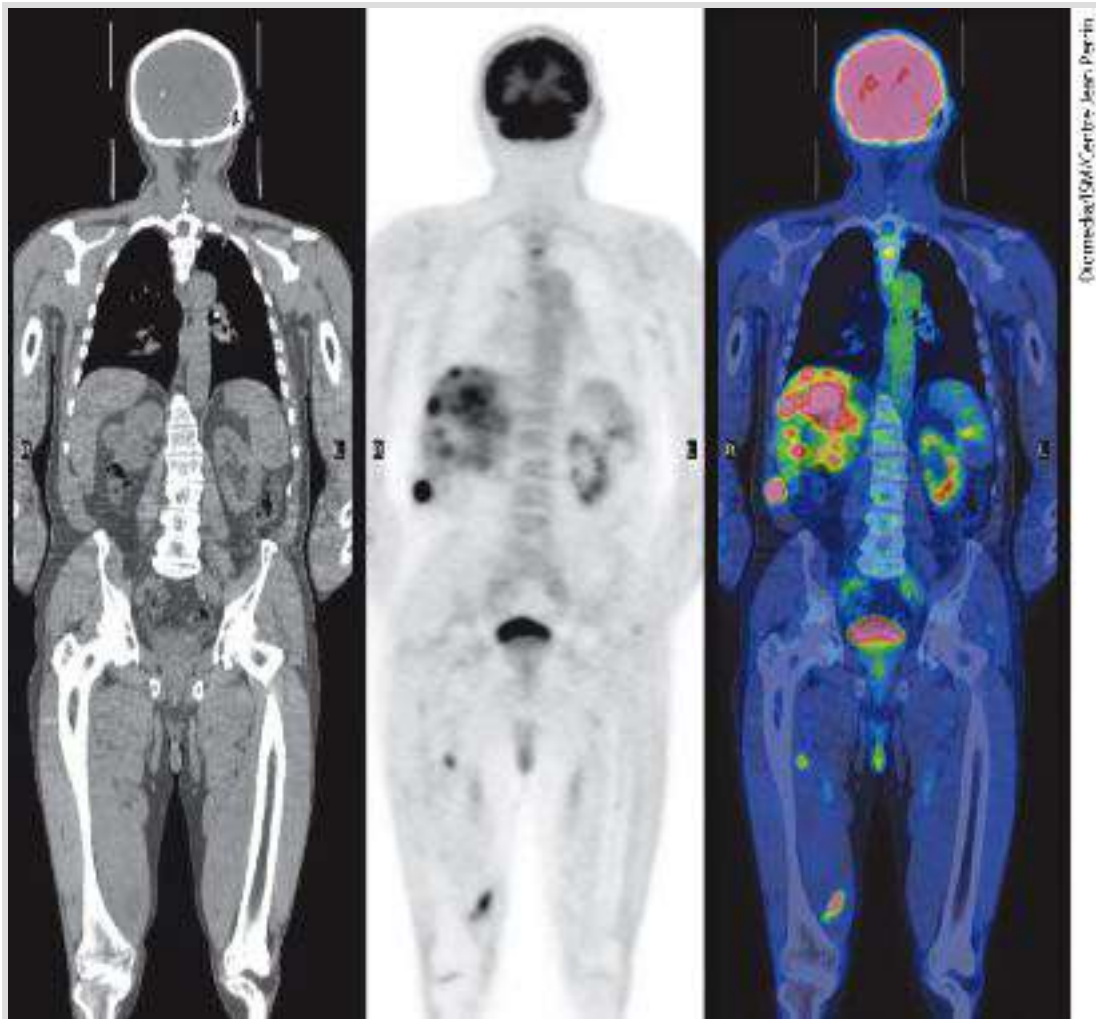


FIGURE 3 Detection of cancerous tissue by positron emission tomography (PET). The adult male patient had undergone surgical removal of a primary skin cancer (malignant melanoma). The image on the left, obtained by whole-body computed tomography (CT scan), shows the location of the soft tissues and bones. The central panel is a PET scan after the patient had ingested ^{18}F -labeled 2-fluoro-2-deoxyglucose (FdG). Dark spots indicate regions of high glucose utilization. As expected, the brain and bladder are heavily labeled — the brain because it uses most of the glucose consumed in the body, and the bladder because the ^{18}F -labeled 6-phospho-FdG is excreted in the urine. When the intensity of the label in the PET scan is translated into false color (the intensity increases from green to yellow to red) and the image is superimposed on the CT scan, the fused image (right) reveals cancer in the bones of the upper spine, in the liver, and in some regions of muscle, all the result of cancer spreading from the primary malignant melanoma.



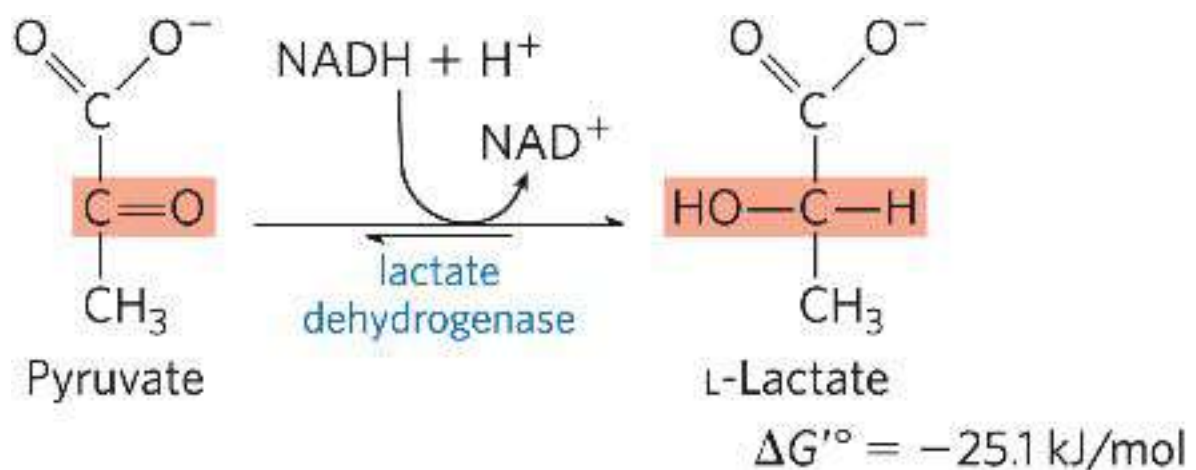
Science Photo Library/Science Source

Otto Warburg, 1883–1970

Warburg is generally considered the preeminent biochemist of the first half of the twentieth century. He made seminal contributions to many other areas of biochemistry, including respiration, photosynthesis, and the enzymology of intermediary metabolism. Beginning in 1930, Warburg and his associates purified and crystallized seven of the enzymes of glycolysis. They developed an experimental tool that revolutionized biochemical studies of oxidative metabolism: the Warburg manometer, which directly measured the oxygen consumption of tissues by monitoring changes in gas volume, and thus allowed quantitative measurement of any enzyme with oxidase activity.

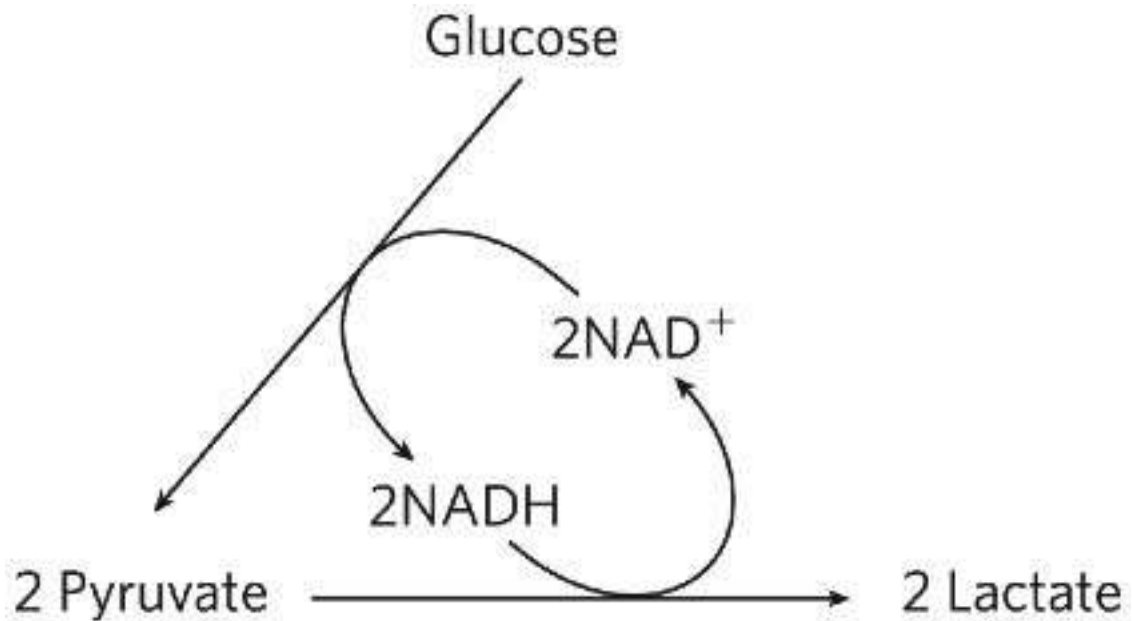
Pyruvate Is the Terminal Electron Acceptor in Lactic Acid Fermentation

P3 When animal tissues cannot be supplied with sufficient oxygen to support aerobic oxidation of the pyruvate and NADH produced in glycolysis, NAD^+ is regenerated from NADH by the reduction of pyruvate to **lactate**. Some tissues and cell types (such as erythrocytes, which have no mitochondria and thus cannot oxidize pyruvate to CO_2) produce lactate from glucose even under aerobic conditions. The reduction of pyruvate in this pathway is catalyzed by **lactate dehydrogenase**, which forms the L isomer of lactate at pH 7:



The overall equilibrium of the reaction strongly favors lactate formation, as shown by the large negative standard free-energy change.

In glycolysis, dehydrogenation of the two molecules of glyceraldehyde 3-phosphate derived from each molecule of glucose converts two molecules of NAD^+ to two of NADH. Because the reduction of two molecules of pyruvate to two of lactate regenerates two molecules of NAD^+ , there is no net change in NAD^+ or NADH:



The lactate formed by active skeletal muscles (or by erythrocytes or retinal cells) can be recycled; it is carried in the blood to the liver, where it is converted to glucose during the recovery from strenuous muscular activity. When lactate is produced in large quantities during vigorous muscle contraction (during a sprint, for example), the acidification that results from ionization of lactic acid in muscle and blood limits the period of vigorous activity. The best-conditioned athletes can sprint at top speed for no more than a minute ([Box 14-2](#)).

BOX 14-2

Glucose Catabolism at Limiting Concentrations of Oxygen

Most vertebrates are essentially aerobic organisms; they convert glucose to pyruvate by glycolysis, then use molecular oxygen to oxidize the pyruvate completely to CO_2 and H_2O . Anaerobic catabolism of glucose to lactate occurs during short bursts of extreme muscular activity — for example, in a

sprint — during which oxygen cannot be carried to the muscles fast enough to oxidize pyruvate. Instead, the muscles use their stored glucose (glycogen) as fuel to generate ATP by fermentation, with lactate as the end product. In a sprint, lactate in the blood builds up to high concentrations. It is slowly converted back to glucose by gluconeogenesis in the liver in the subsequent rest or recovery period, during which oxygen is consumed at a gradually diminishing rate until the breathing rate returns to normal. The excess oxygen consumed in the recovery period represents a repayment of the oxygen debt. This is the amount of oxygen required to supply ATP for gluconeogenesis during recovery respiration, in order to regenerate the glycogen “borrowed” from liver and muscle to carry out intense muscular activity in the sprint. The cycle of reactions that includes glucose conversion to lactate in muscle and lactate conversion to glucose in liver is called the Cori cycle, for Carl and Gerty Cori, whose studies in the 1930s and 1940s clarified the pathway and its role (see [Box 15-1](#)).

The circulatory systems of most small vertebrates can carry oxygen to their muscles fast enough to avoid having to use muscle glycogen anaerobically. For example, migrating birds often fly great distances at high speeds without rest and without incurring an oxygen debt. Many running animals of moderate size also maintain an essentially aerobic metabolism in their skeletal muscle. However, the circulatory systems of larger animals, including humans, cannot completely sustain aerobic metabolism in skeletal muscles over long periods of intense muscular activity. These animals generally are slow-moving under normal circumstances and engage in intense muscular activity only in the gravest emergencies, because such bursts of activity require long recovery periods to repay the oxygen debt.

Alligators and crocodiles, for example, are normally sluggish animals. Yet, when provoked, they are capable of lightning-fast charges and dangerous lashings of their powerful tails. Such intense bursts of activity are short and must be followed by long periods of recovery. The fast emergency movements require lactic acid fermentation to generate ATP in skeletal muscles. The stores of muscle glycogen are rapidly expended in intense muscular activity, and lactate reaches very high concentrations in myocytes and extracellular fluid. Whereas a trained athlete can recover from a 100 m sprint in 30 min or less, an alligator

may require many hours of rest and extra oxygen consumption to clear the excess lactate from its blood and regenerate muscle glycogen after a burst of activity.

Other large animals, such as the elephant and the rhinoceros, have similar metabolic characteristics, as do diving mammals such as whales and seals. Dinosaurs and other huge, now-extinct animals probably had to depend on lactic acid fermentation to supply energy for muscular activity, followed by very long recovery periods during which they were vulnerable to attack by smaller predators that were better able to use oxygen and thus better adapted to continuous, sustained muscular activity.

Deep-sea explorations have revealed many species of marine life at great ocean depths, where the oxygen concentration is near zero. For example, the primitive coelacanth, a large fish recovered from depths of 4,000 m or more off the coast of South Africa, has an essentially anaerobic metabolism in virtually all its tissues. It converts carbohydrates to lactate and other products, most of which must be excreted. Some marine vertebrates ferment glucose to ethanol and CO_2 in order to generate ATP.



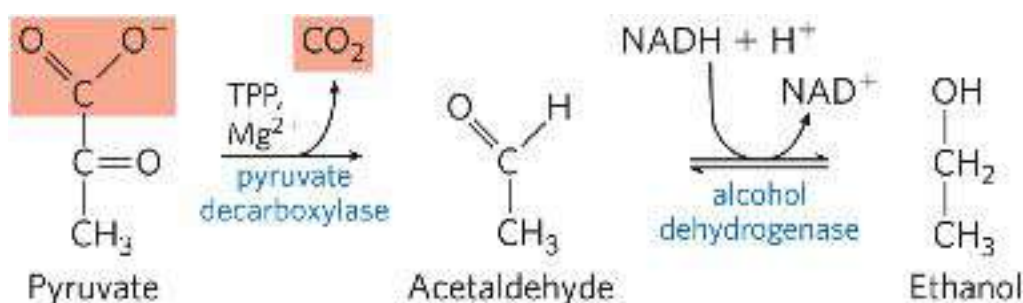
Ezra Shaw/Getty Images.

Francena McCorory, Olympic sprinter

Although conversion of glucose to lactate includes two oxidation-reduction steps, there is no net change in the oxidation state of carbon; in glucose ($C_6H_{12}O_6$) and lactic acid ($C_3H_6O_3$), the H:C ratio is the same. Nevertheless, some of the energy of the glucose molecule has been extracted by its conversion to lactate — enough to give a net yield of two molecules of ATP for every glucose molecule consumed. **Fermentation** is the general term for such processes, which extract energy (as ATP) but do not consume oxygen or change the concentrations of NAD^+ or NADH.

Ethanol Is the Reduced Product in Ethanol Fermentation

P3 Yeast and other microorganisms ferment glucose to ethanol and CO_2 , rather than to lactate. Glucose is metabolized to pyruvate by glycolysis, and the pyruvate is converted to ethanol and CO_2 in a two-step process:

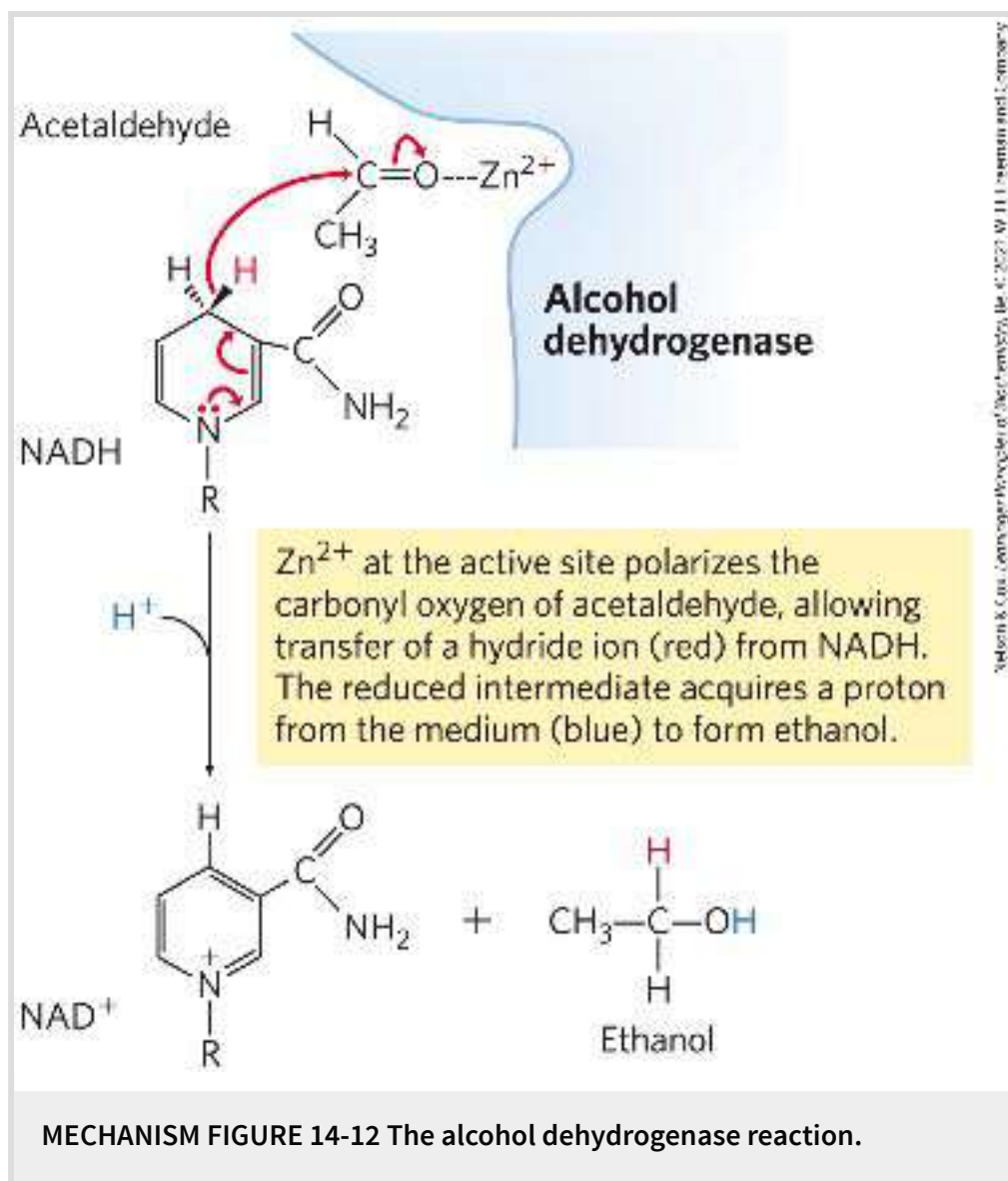


In the first step, pyruvate is decarboxylated to form acetaldehyde in an irreversible reaction catalyzed by **pyruvate decarboxylase**.

This reaction is a simple decarboxylation and does not involve the net oxidation of pyruvate. Pyruvate decarboxylase requires Mg^{2+} and has a tightly bound coenzyme, thiamine pyrophosphate, which is discussed below. In the second step, acetaldehyde is reduced to ethanol through the action of **alcohol dehydrogenase**, with the reducing power furnished by NADH derived from the dehydrogenation of glyceraldehyde 3-phosphate. This reaction is a well-studied case of hydride transfer from NADH ([Fig. 14-12](#)). Ethanol and CO_2 are thus the end products of ethanol fermentation, and the overall equation is



As in lactic acid fermentation, there is no net change in the ratio of hydrogen to carbon atoms when glucose (H:C ratio = $12/6 = 2$) is fermented to two ethanol and two CO_2 (combined H:C ratio = $12/6 = 2$). In all fermentations, the H:C ratio of the reactants and products remains the same.

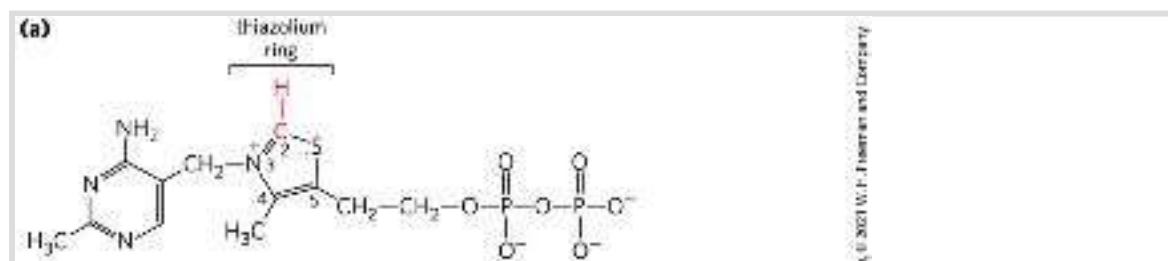


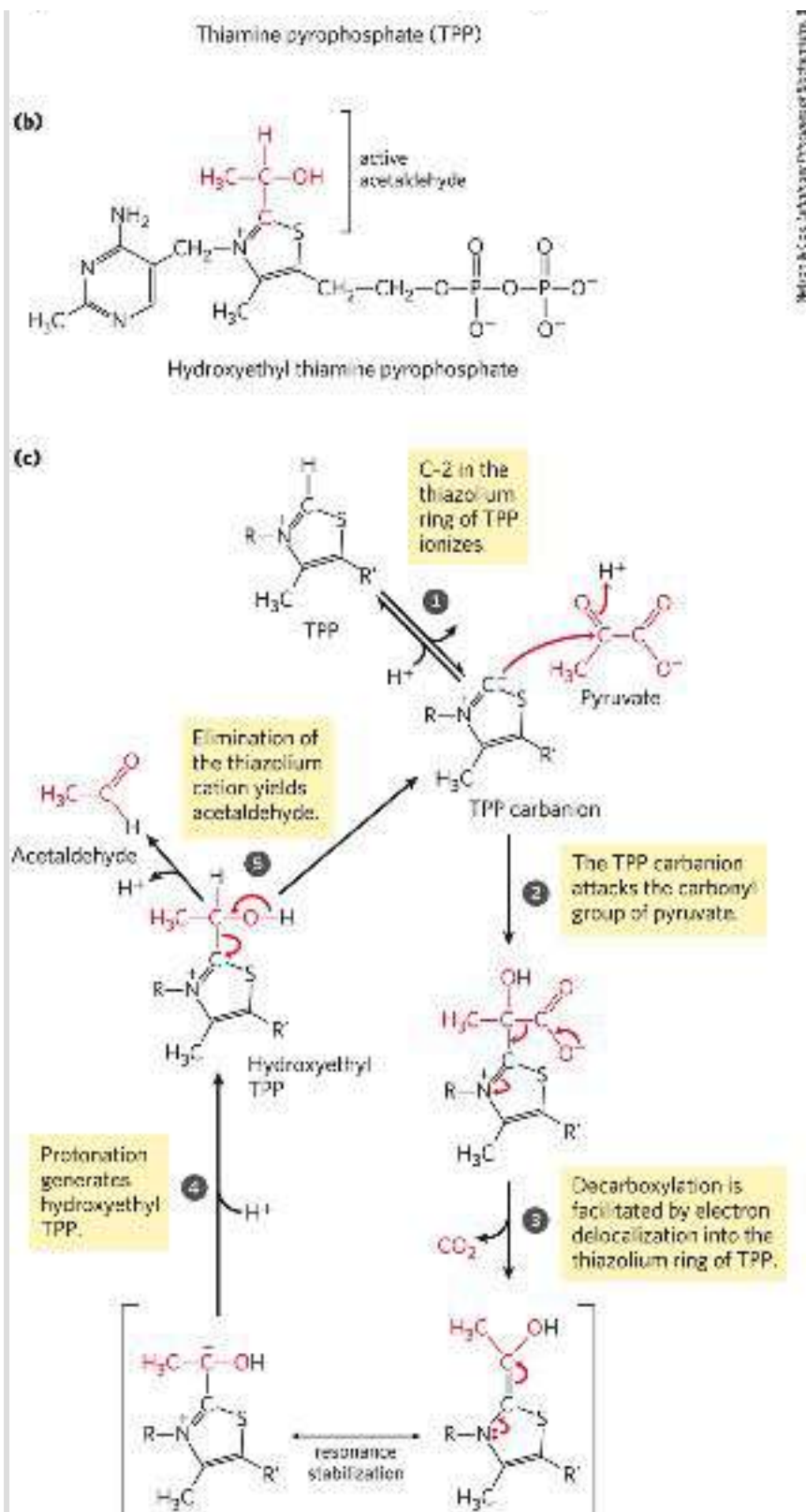
Pyruvate decarboxylase is present in brewer's and baker's yeast (different strains of the species *Saccharomyces cerevisiae*) and in all other organisms that ferment glucose to ethanol, including some plants. The CO_2 produced by pyruvate decarboxylation in brewer's yeast is responsible for the characteristic carbonation of champagne. The ancient art of brewing beer involves several enzymatic processes in addition to the reactions of ethanol fermentation. In baking, CO_2 released by pyruvate decarboxylase when yeast is mixed with a fermentable sugar causes dough to

rise. The enzyme is absent in vertebrate tissues and in other organisms that carry out lactic acid fermentation.

Alcohol dehydrogenase is present in many organisms that metabolize ethanol, including humans. In the liver it catalyzes the oxidation of ethanol, either ingested or produced by intestinal microorganisms, with the concomitant reduction of NAD^+ to NADH. In this case, the reaction proceeds in the direction opposite to that involved in the production of ethanol by fermentation.

The pyruvate decarboxylase reaction provides our first encounter with **thiamine pyrophosphate (TPP)** (**Fig. 14-13**), a coenzyme derived from vitamin B_1 . TPP plays an important role in the cleavage of bonds adjacent to a carbonyl group, such as the decarboxylation of α -keto acids, and in chemical rearrangements in which an activated acetaldehyde group is transferred from one carbon atom to another (**Table 14-1**). The functional part of TPP, the thiazolium ring, has a relatively acidic proton at C-2. Loss of this proton produces a carbanion that is the active species in TPP-dependent reactions. The carbanion readily adds to carbonyl groups, and the thiazolium ring is thereby positioned to act as an “electron sink” that greatly facilitates reactions such as the decarboxylation catalyzed by pyruvate decarboxylase.





MECHANISM FIGURE 14-13 Thiamine pyrophosphate (TPP) and its role in pyruvate decarboxylation. (a) TPP is the coenzyme form of vitamin B₁ (thiamine). The reactive carbon atom in the thiazolium ring of TPP is shown in red. In the reaction catalyzed by pyruvate decarboxylase, two of the three carbons of pyruvate are carried transiently on

TPP in the form of a hydroxyethyl, or “active acetaldehyde,” group (b), which is subsequently released as acetaldehyde. (c) The thiazolium ring of TPP stabilizes carbanion intermediates by providing an electrophilic (electron-deficient) structure into which the carbanion electrons can be delocalized by resonance. Structures with this property, often called “electron sinks,” play a role in many biochemical reactions — here, facilitating carbon–carbon bond cleavage. Dietary insufficiency of thiamine causes the serious disease beriberi and the Wernicke-Korsakoff syndrome.

TABLE 14-1 Some TPP-Dependent Reactions

Enzyme	Pathway(s)	Bond cleaved	Bond formed
Pyruvate decarboxylase	Ethanol fermentation		
Pyruvate dehydrogenase α -Ketoglutarate dehydrogenase	Synthesis of acetyl-CoA Citric acid cycle		
Transketolase	Carbon-assimilation reactions Pentose phosphate pathway		

Fermentations Produce Some Common Foods and Industrial

Chemicals

Our progenitors learned millennia ago to use fermentation in the production and preservation of foods and beverages. 

Certain microorganisms present in raw food products ferment the carbohydrates and yield metabolic products that give the foods their characteristic forms, textures, and tastes. In modern times, industrial fermentation produces organic chemicals and fuels.

Fermented Foods

Yogurt, already known in biblical times, is produced when the bacterium *Lactobacillus bulgaricus* ferments the carbohydrate in milk, producing lactic acid; the resulting drop in pH causes the milk proteins to precipitate, producing the thick texture and sour taste of unsweetened yogurt. Another bacterium,

Propionibacterium freudenreichii, ferments milk to produce propionic acid and CO₂; the propionic acid precipitates milk proteins, and bubbles of CO₂ cause the holes characteristic of Swiss cheese. Many other food products are the result of fermentations: pickles, sauerkraut, sausage, soy sauce, kimchi, kefir, dahi, and kombucha. The drop in pH associated with fermentation also helps to preserve foods, because most of the microorganisms that cause food spoilage cannot grow at low pH. In agriculture, plant byproducts such as corn stalks are preserved for use as animal feed by packing them into a large container (a silo) with limited access to air; microbial fermentation produces

acids that lower the pH. The silage that results from this fermentation process can be kept as animal feed for long periods without spoilage.

Fermented Beverages

 Beer brewing was a science learned early in human history, and later refined for larger-scale production ([Fig. 14-14](#)). Brewers prepare beer by ethanol fermentation of the carbohydrates in cereal grains (seeds) such as barley, carried out by yeast glycolytic enzymes. The carbohydrates, largely polysaccharides, must first be degraded to disaccharides and monosaccharides. In the malting process, the barley seeds are allowed to germinate until they form the hydrolytic enzymes required to break down their polysaccharides, at which point germination is stopped by controlled heating. The product is malt, which contains enzymes that catalyze the hydrolysis of the β linkages of cellulose and other cell wall polysaccharides of the barley husks, and enzymes such as α -amylase and maltase. The malt is mixed with water, mashed, and boiled with hops to add flavor. Yeast cells added to this mixture grow and reproduce rapidly, using energy obtained from available sugars. No ethanol forms during this stage, because the yeast, amply supplied with oxygen, oxidizes the pyruvate formed by glycolysis to CO_2 and H_2O via the citric acid cycle.



FIGURE 14-14 Beer brewing. Large breweries and microbreweries produce beers with a wide variety of flavors, the result of differences in materials and fermentation conditions.

When all the dissolved oxygen in the vat has been consumed, the yeast cells switch to anaerobic metabolism and ferment the sugars into ethanol and CO_2 . The fermentation process is controlled in part by the concentration of the ethanol formed, by the pH, and by the amount of remaining sugar. After fermentation has been stopped, the cells are removed and the “raw” beer is ready for final processing.

Chemical Production by Fermentation

In 1910, Chaim Weizmann (later to become the first president of Israel) discovered that the bacterium *Clostridium acetobutyricum* ferments starch to butanol and acetone.  This discovery

opened the field of industrial fermentations, in which some readily available material rich in carbohydrate is supplied to a pure culture of a specific microorganism, which ferments it into a product of greater commercial value. Microbial fermentations produce formic, acetic, propionic, butyric, and succinic acids, and ethanol, glycerol, methanol, isopropanol, butanol, and butanediol. Industrial fermentations are also used to produce certain antibiotics, including penicillin, streptomycin, and chloramphenicol. These fermentations are generally carried out in huge closed vats in which temperature and access to air are controlled to favor the multiplication of the desired microorganism and to exclude contaminating organisms. The beauty of industrial fermentations is that complicated, multistep chemical transformations are carried out in high yields and with few side products by chemical factories that reproduce themselves — microbial cells.

Fuel Production by Fermentation

Much of the technology developed for large-scale production of alcoholic beverages can be applied to the production of ethanol as a renewable fuel. The principal advantage of ethanol as a fuel is that it can be produced from relatively *inexpensive* and *renewable* resources rich in sucrose, starch, or cellulose — starch from corn or wheat; sucrose from beets or cane; and cellulose from straw, forest industry waste, or municipal solid waste. Typically, the raw material (feedstock) is converted chemically to monosaccharides, then fed to a hardy strain of yeast in an industrial-scale

fermenter. The fermentation can yield not only ethanol for fuel but also side products such as proteins that can be used as animal feed.

SUMMARY 14.3 *Fates of Pyruvate*

■ The NADH formed in glycolysis must be recycled to regenerate NAD^+ , which is required as an electron acceptor in the first step of the payoff phase. Under aerobic conditions, electrons pass from NADH to O_2 in mitochondrial respiration. The Warburg effect is the observation that tumor cells have high rates of glycolysis, with fermentation of glucose to lactate, even in the presence of oxygen. It is the basis of PET scanning used to diagnose tumors.

■ Under anaerobic or hypoxic conditions, many organisms regenerate NAD^+ by transferring electrons from NADH to pyruvate, forming lactate. Other organisms, such as yeast, regenerate NAD^+ by reducing pyruvate to ethanol and CO_2 .

■ A variety of microorganisms can ferment sugar in fresh foods, resulting in changes in pH, taste, and texture, and preserving food from spoilage. Fermentations are used in industry to produce many commercially valuable organic compounds from inexpensive starting materials.

14.4 Gluconeogenesis

The central role of glucose in metabolism arose early in evolution, and this sugar remains the nearly universal fuel and building block in modern organisms, from microbes to humans. In mammals, some tissues depend almost completely on glucose for their metabolic energy. For the human brain and nervous system, as well as the erythrocytes, testes, renal medulla, and embryonic tissues, glucose from the blood is the sole or major fuel source. The brain alone requires about 120 g of glucose each day — more than half of all the glucose stored as glycogen in muscle and liver. However, the supply of glucose from these stores is not always sufficient; between meals and during longer fasts, or after vigorous exercise, glycogen is depleted. For these times, organisms need a method for synthesizing glucose from noncarbohydrate precursors. This is accomplished by a pathway called **gluconeogenesis** (“new formation of sugar”), which converts pyruvate and related three- and four-carbon compounds to glucose.



Gluconeogenesis occurs in all animals, plants, fungi, and microorganisms. The reactions are essentially the same in all tissues and all species. The important precursors of glucose in animals are three-carbon compounds such as lactate, pyruvate, and glycerol, as well as certain amino acids ([Fig. 14-15](#)). In mammals, gluconeogenesis takes place mainly in the liver, and to a lesser extent in the renal cortex and in the epithelial cells that line the small intestine. The glucose produced passes into the blood to supply other tissues. After vigorous exercise, lactate produced by anaerobic glycolysis in skeletal muscle returns to the liver and is converted to glucose, which moves back to muscle and is converted to glycogen — a circuit called the Cori cycle (see [Fig. 23-17](#)). In plant seedlings, stored fats and proteins are converted, via paths that include gluconeogenesis, to the disaccharide sucrose for transport throughout the developing plant. Glucose and its derivatives are precursors for the synthesis of plant cell walls, nucleotides and coenzymes, and a variety of other essential metabolites. In many microorganisms, gluconeogenesis starts from simple organic compounds of two or three carbons, such as acetate, lactate, and propionate, in their growth medium.

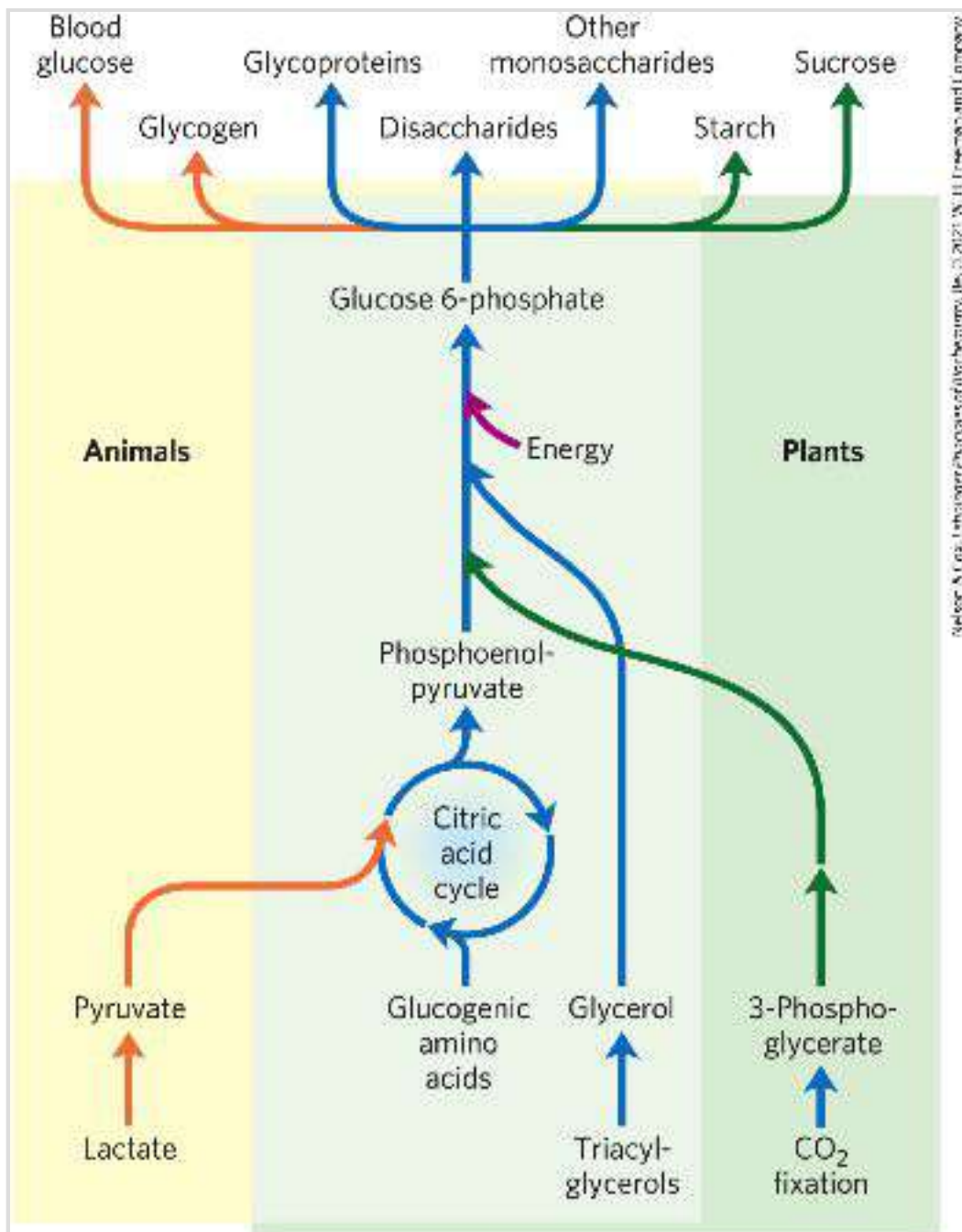


FIGURE 14-15 Carbohydrate synthesis from simple precursors. The pathway from phosphoenolpyruvate to glucose 6-phosphate is common to the biosynthetic conversion of many different precursors of carbohydrates in animals and plants. The path from pyruvate to phosphoenolpyruvate leads through oxaloacetate, an intermediate of the citric acid cycle, which we discuss in [Chapter 16](#). Any compound that can be converted to either pyruvate or oxaloacetate can therefore serve as starting material for gluconeogenesis. This includes alanine and aspartate, which are convertible to pyruvate and oxaloacetate, respectively, and other amino acids that can also yield three- or four-carbon fragments, the so-called glucogenic amino acids. Plants and photosynthetic bacteria are uniquely able to convert CO₂ to carbohydrates, using the Calvin cycle, as we shall see in [Section 20.4](#).

Although the reactions of gluconeogenesis are the same in all organisms, the metabolic context and the regulation of the pathway differ from one species to another and from tissue to tissue. In this section we focus on gluconeogenesis as it occurs in the mammalian liver. In [Chapter 20](#) we show how photosynthetic organisms use this pathway to convert the primary products of photosynthesis into glucose, to be stored as sucrose or starch.

 Gluconeogenesis and glycolysis are not identical pathways running in opposite directions, although they do share several steps ([Fig. 14-16](#)); 7 of the 10 enzymatic reactions of gluconeogenesis are the reverse of glycolytic reactions. However, three reactions of glycolysis are essentially irreversible in vivo and cannot be used in gluconeogenesis: the conversion of glucose to glucose 6-phosphate by hexokinase, the phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate by phosphofructokinase-1, and the conversion of phosphoenolpyruvate to pyruvate by pyruvate kinase. In cells, these three reactions are characterized by a large negative free-energy change, whereas other glycolytic reactions have a ΔG near 0 ([Table 14-2](#)). In gluconeogenesis, the three irreversible steps are bypassed by a separate set of enzymes, catalyzing reactions that are sufficiently exergonic to be effectively irreversible in the direction of glucose synthesis. Thus, both glycolysis and gluconeogenesis are irreversible processes in cells. In animals, both pathways occur largely in the cytosol, necessitating their reciprocal and coordinated regulation, described in [Section 14.5](#).

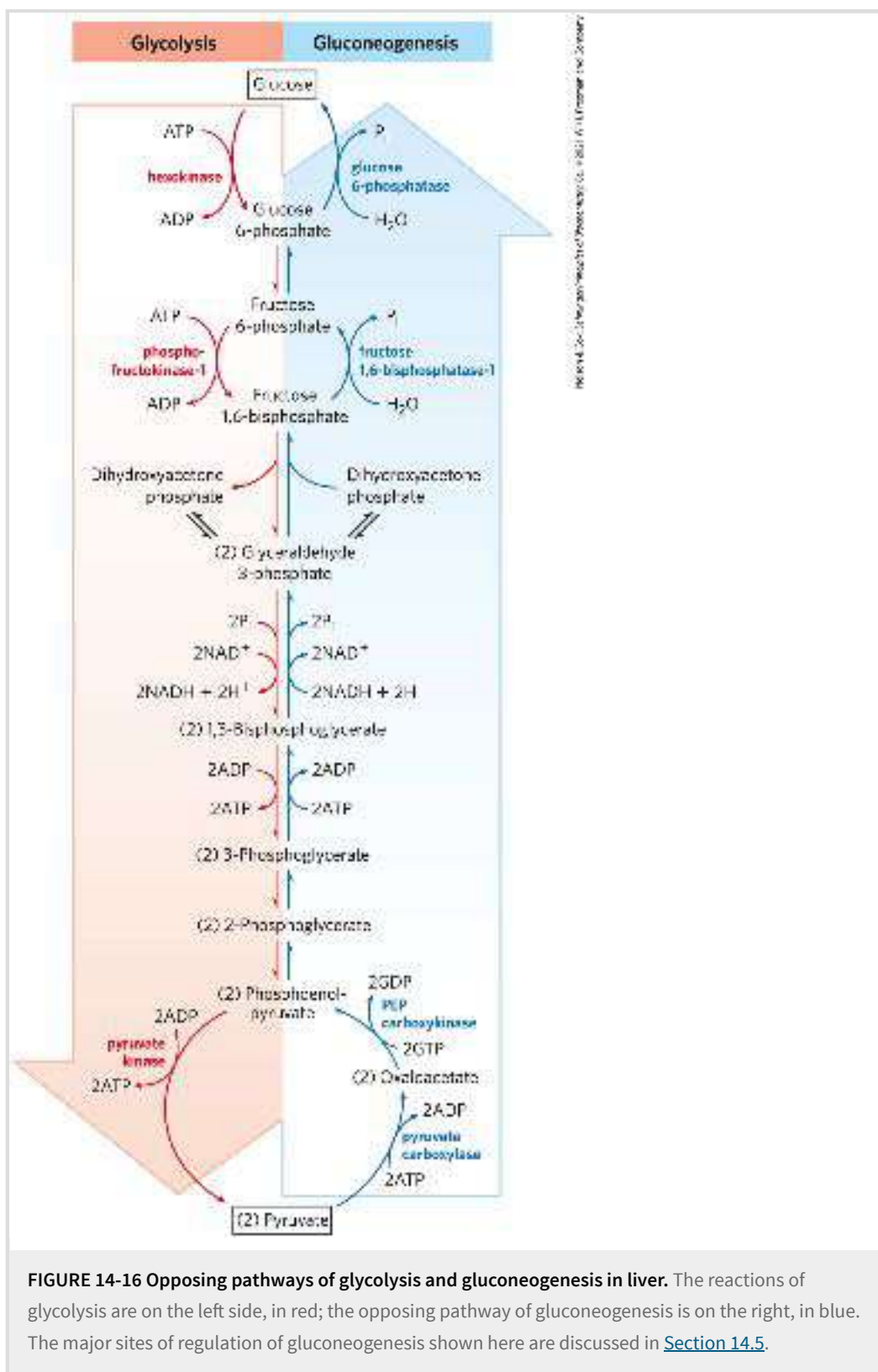


TABLE 14-2 Free-Energy Changes of Glycolytic Reactions in Erythrocytes

Glycolytic reaction step	$\Delta G'^{\circ}$ (kJ/mol)	ΔG (kJ/mol)

1 Glucose + ATP → glucose 6-phosphate + ADP	−16.7	−33.4
2 Glucose 6-phosphate ⇌ fructose 6-phosphate	1.7	0 to 25
3 Fructose 6-phosphate + ATP → fructose1,6-bisphosphate + ADP	−14.2	−22.2
4 Fructose1,6-bisphosphate ⇌ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate	23.8	−6 to 0
5 Dihydroxyacetone phosphate ⇌ glyceraldehyde 3-phosphate	7.5	0 to 4
6 Glyceraldehyde 3-phosphate + P _i + NAD ⁺ ⇌ 1,3-bisphosphoglycerate + NADH + H ⁺	6.3	−2 to 2
7 1,3-Bisphosphoglycerate + ADP ⇌ 3-phosphoglycerate + ATP	−18.8	0 to 2
8 3-Phosphoglycerate ⇌ 2-phosphoglycerate	4.4	0 to 0.8
9 3-Phosphoglycerate ⇌ phosphoenolpyruvate + H ₂ O	7.5	0 to 3.3
10 Phosphoenolpyruvate + ADP → pyruvate + ATP	−31.4	−16.7

Note: $\Delta G'^{\circ}$ is the standard free-energy change, as defined in [Chapter 13 \(p. 468\)](#). ΔG is the free-energy change calculated from the actual concentrations of glycolytic intermediates present under physiological conditions in erythrocytes, at pH 7. The glycolytic reactions bypassed in gluconeogenesis are shown in red. Biochemical equations are not necessarily balanced for H or charge ([p. 478](#)).

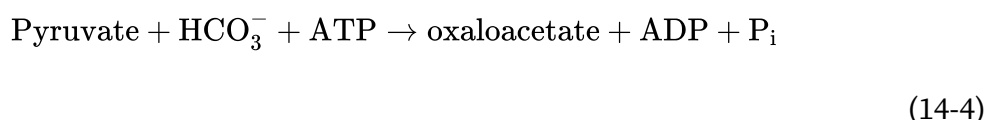
We begin by considering the three bypass reactions of gluconeogenesis. (Keep in mind that “bypass” refers throughout to the bypass of irreversible glycolytic reactions.)

The First Bypass: Conversion of Pyruvate to Phosphoenolpyruvate Requires Two Exergonic Reactions

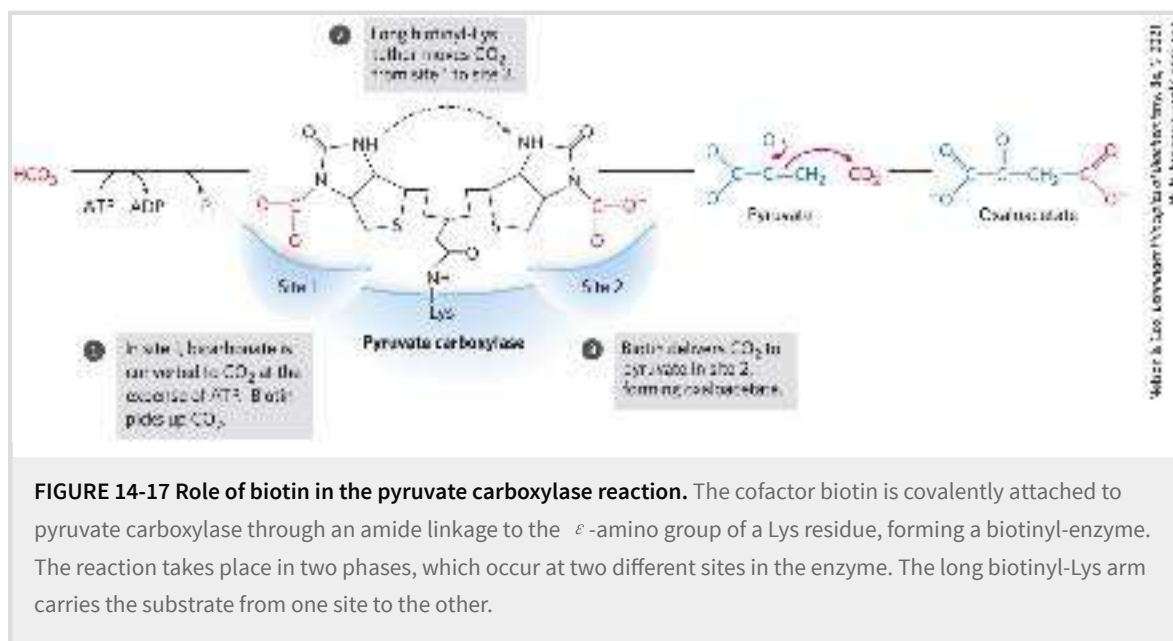
P4 The first of the bypass reactions in gluconeogenesis is the conversion of pyruvate to phosphoenolpyruvate (PEP). This reaction cannot occur by simple reversal of the pyruvate kinase reaction of glycolysis ([p. 521](#)), which has a large, negative free-energy change and is therefore irreversible under the conditions prevailing in intact cells ([Table 14-2](#), step 10). Instead, the phosphorylation of pyruvate is achieved by a roundabout sequence of reactions that in eukaryotes

requires enzymes in both the cytosol and mitochondria. As we shall see, the pathway shown in [Figure 14-16](#) and described in detail here is one of two routes from pyruvate to PEP; it is the predominant path when pyruvate or alanine is the glucogenic precursor. A second pathway, described later, predominates when lactate is the glucogenic precursor.

Pyruvate is first transported from the cytosol into mitochondria or is generated from alanine within mitochondria by transamination, in which the α -amino group is transferred from alanine (leaving pyruvate) to an α -keto carboxylic acid (transamination reactions are discussed in detail in [Chapter 18](#)). Then **pyruvate carboxylase**, a mitochondrial enzyme that requires the coenzyme [biotin](#), converts the pyruvate to oxaloacetate:

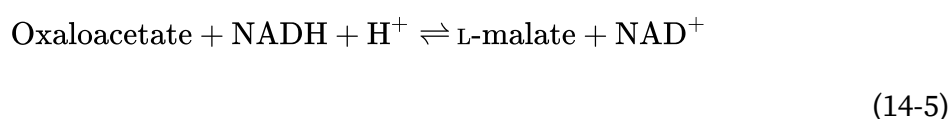


The carboxylation reaction involves biotin as a carrier of activated bicarbonate, as shown in [Figure 14-17](#); the reaction mechanism is shown in [Figure 16-16](#). (Note that HCO_3^- is formed by ionization of carbonic acid formed from $\text{CO}_2 + \text{H}_2\text{O}$.) HCO_3^- is phosphorylated by ATP to form a mixed anhydride (a carboxyphosphate); then biotin displaces the phosphate in the formation of carboxybiotin.



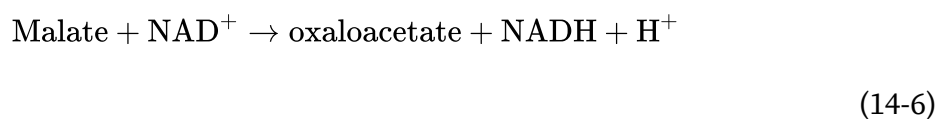
Pyruvate carboxylase is the first regulatory enzyme in the gluconeogenic pathway, requiring acetyl-CoA as a positive effector. Acetyl-CoA is produced by fatty acid oxidation ([Chapter 17](#)), and its accumulation signals the availability of fatty acids as fuel. As we shall see in [Chapter 16](#), the pyruvate carboxylase reaction can replenish intermediates in another central metabolic pathway, the citric acid cycle.

Because the mitochondrial membrane has no transporter for oxaloacetate, before export to the cytosol the oxaloacetate formed from pyruvate must be reduced to malate by mitochondrial **malate dehydrogenase**, at the expense of NADH:

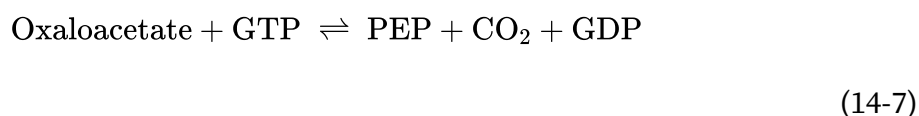


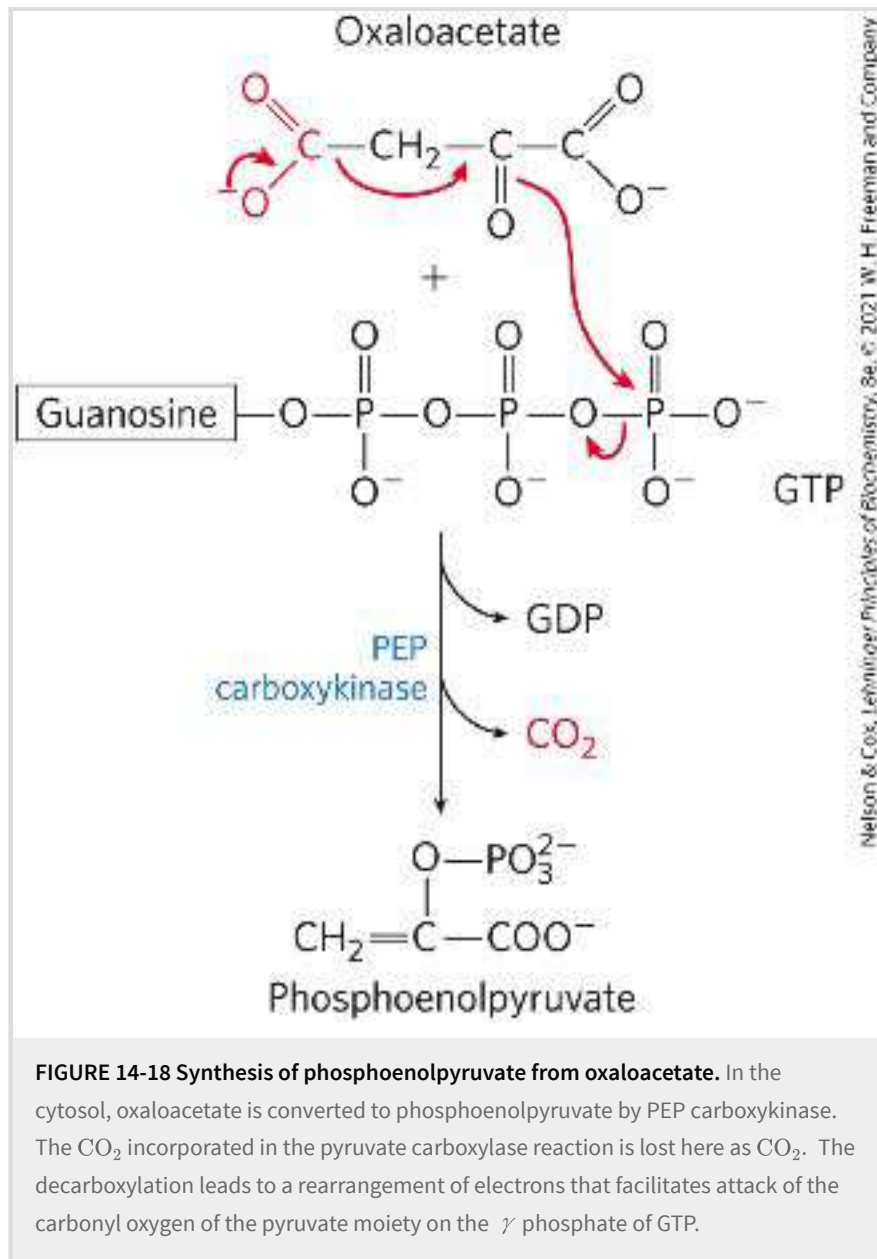
The standard free-energy change for this reaction is quite high, but under physiological conditions (including a very low concentration of oxaloacetate) $\Delta G \approx 0$ and the reaction is readily reversible. Mitochondrial malate dehydrogenase functions in both gluconeogenesis and the citric acid cycle, but the overall flow of metabolites in the two processes is in opposite directions.

Malate leaves the mitochondrion through a specific transporter in the inner mitochondrial membrane (see [Fig. 19-31](#)), and in the cytosol it is reoxidized to oxaloacetate, with the production of cytosolic NADH:

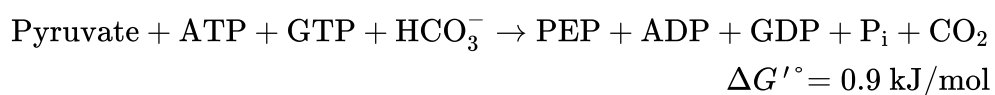


The oxaloacetate is then converted to PEP by **phosphoenolpyruvate carboxykinase** ([Fig. 14-18](#)). This Mg^{2+} -dependent reaction requires GTP as the phosphoryl group donor:





The reaction is reversible under intracellular conditions; the formation of one high-energy phosphate compound (PEP) is balanced by the hydrolysis of another (GTP). The overall equation for this set of bypass reactions, the sum of [Equations 14-4](#) through [14-7](#), is



(14-8)

Two high-energy phosphate equivalents (one from ATP and one from GTP), each yielding about 50 kJ/mol under cellular conditions, must be expended to phosphorylate one molecule of pyruvate to PEP. In contrast, when PEP is converted to pyruvate during glycolysis, only one ATP is generated from ADP. Although the standard free-energy change ($\Delta G'^{\circ}$) of the two-step path from pyruvate to PEP is 0.9 kJ/mol, the actual free-energy change (ΔG), calculated from measured cellular concentrations of intermediates, is very strongly negative (-25 kJ/mol); this results from the ready consumption of PEP in other reactions such that its concentration remains relatively low. The reaction is thus effectively irreversible in the cell.

Note that the CO_2 added to pyruvate in the pyruvate carboxylase step ([Fig. 14-17](#)) is the same molecule that is lost in the PEP carboxykinase reaction ([Fig. 14-18](#)). This carboxylation-decarboxylation sequence represents a way of “activating” pyruvate, in that the decarboxylation of oxaloacetate facilitates PEP formation. In [Chapter 21](#) we shall see how a similar carboxylation-decarboxylation sequence is used to activate acetyl-CoA for fatty acid biosynthesis (see [Fig. 21-1](#)).

There is a logic to the route of these reactions through the mitochondrion. The $[\text{NADH}]/[\text{NAD}^+]$ ratio in the cytosol is several orders of magnitude lower than in mitochondria. Because cytosolic NADH is consumed in gluconeogenesis (in the conversion of 1,3-bisphosphoglycerate to glyceraldehyde 3-phosphate; [Fig. 14-16](#)), glucose biosynthesis cannot proceed unless NADH is available. The transport of malate from the mitochondrion to the cytosol and its reconversion there to oxaloacetate effectively moves reducing equivalents to the cytosol, where they are scarce. This path from pyruvate to PEP therefore provides an important balance between NADH produced and consumed in the cytosol during gluconeogenesis.

 A second pyruvate $\rightarrow$ PEP bypass predominates when lactate is the glucogenic precursor ([Fig. 14-19](#)). This pathway makes use of lactate produced by glycolysis in erythrocytes or anaerobic muscle, for example, and it is particularly important in large vertebrates after vigorous exercise ([Box 14-2](#)). The conversion of lactate to pyruvate in the cytosol of hepatocytes yields NADH, and the export of reducing equivalents (as malate) from mitochondria is therefore unnecessary. After the pyruvate produced by the lactate dehydrogenase reaction is transported into the mitochondrion (by a transporter in the inner mitochondrial membrane specific for pyruvate), it is converted to oxaloacetate by pyruvate carboxylase, as described

above. This oxaloacetate, however, is converted directly to PEP by a mitochondrial isozyme of PEP carboxykinase, and the PEP is transported out of the mitochondrion to continue on the gluconeogenic path. The mitochondrial and cytosolic isozymes of PEP carboxykinase are encoded by separate genes in the nuclear chromosomes, providing another example of two distinct enzymes catalyzing the same reaction but having different cellular locations or metabolic roles (recall the isozymes of hexokinase).

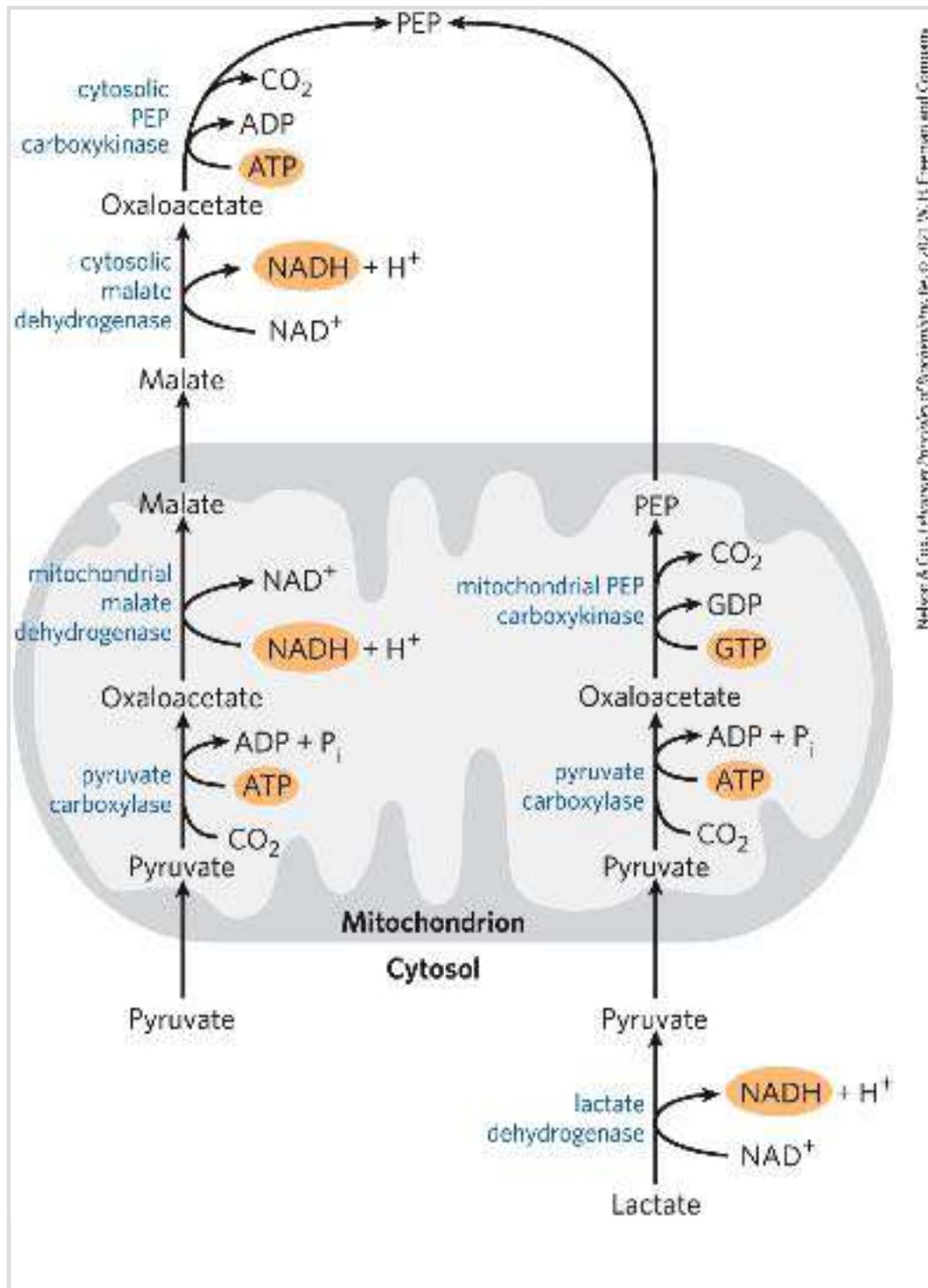
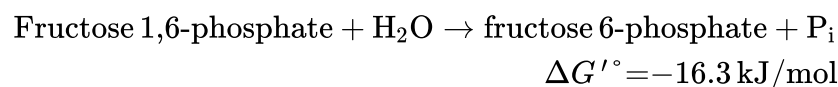


FIGURE 14-19 Alternative paths from pyruvate to phosphoenolpyruvate. The relative importance of the two pathways depends on the availability of lactate or pyruvate and the cytosolic requirements for NADH for gluconeogenesis. The path on the right predominates when lactate is the precursor, because cytosolic NADH is generated in the lactate dehydrogenase reaction and does not have to be shuttled out of the mitochondrion (see text).

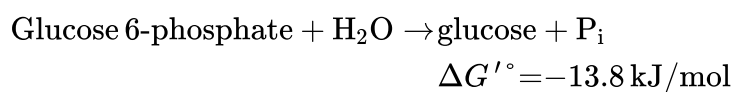
The Second and Third Bypasses Are Simple Dephosphorylations by Phosphatases

The second glycolytic reaction that cannot participate in gluconeogenesis is the phosphorylation of fructose 6-phosphate by PFK-1 ([Table 14-2](#), step ③). Because this reaction is highly exergonic and therefore irreversible in intact cells, the generation of fructose 6-phosphate from fructose 1,6-bisphosphate ([Fig. 14-16](#)) is catalyzed by a different enzyme, Mg^{2+} -dependent **fructose 1,6-bisphosphatase (FBPase-1)**, which promotes the essentially irreversible *hydrolysis* of the C-1 phosphate (*not* phosphoryl group transfer to ADP):



FBPase-1 is so named to distinguish it from another, similar enzyme (FBPase-2) with a regulatory role, which we discuss in [Section 14.5](#).

The third bypass is the final reaction of gluconeogenesis, the dephosphorylation of glucose 6-phosphate to yield glucose ([Fig. 14-16](#)). Reversal of the hexokinase reaction ([p. 514](#)) would require phosphoryl group transfer from glucose 6-phosphate to ADP, forming ATP, an energetically unfavorable reaction ([Table 14-2](#), step ①). The reaction catalyzed by **glucose 6-phosphatase** does not require synthesis of ATP; it is a simple hydrolysis of a phosphate ester:



This Mg^{2+} -activated enzyme is a membrane protein in the lumen of the endoplasmic reticulum of hepatocytes, renal cells, and epithelial cells of the small intestine (see [Fig. 15-6](#)), but not in other tissues, which are therefore unable to supply glucose to the blood. If other tissues had glucose 6-phosphatase, this enzyme's activity would hydrolyze the glucose 6-phosphate needed within those tissues for glycolysis. Glucose produced by gluconeogenesis in the liver or kidney or ingested in the diet is delivered to these other tissues, including brain and muscle, through the bloodstream.

Gluconeogenesis Is Energetically Expensive, But Essential

 The sum of the biosynthetic reactions leading from pyruvate to free blood glucose ([Table 14-3](#)) is

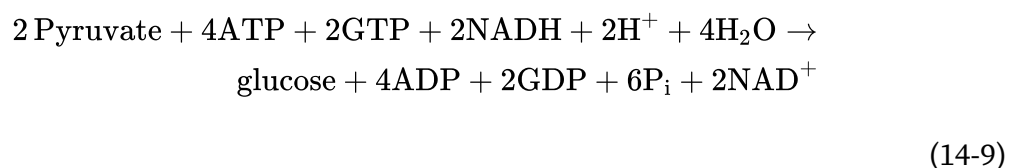


TABLE 14-3 Sequential Reactions in Gluconeogenesis Starting from Pyruvate

Pyruvate + HCO_3^- + ATP $\rightarrow$ oxaloacetate + ADP + P_i	$\times 2$
Oxaloacetate + GTP $\rightleftharpoons$ phosphoenolpyruvate + CO_2 + GDP	$\times 2$
Phosphoenolpyruvate + H_2O $\rightleftharpoons$ 2-phosphoglycerate	$\times 2$
2-Phosphoglycerate $\rightleftharpoons$ 3-phosphoglycerate	$\times 2$
3-Phosphoglycerate + ATP $\rightleftharpoons$ 1,3-bisphosphoglycerate + ADP	$\times 2$
1,3-Bisphosphoglycerate + NADH + H^+ $\rightleftharpoons$ glyceraldehyde 3-phosphate + NAD^+ + P_i	$\times 2$
Glyceraldehyde 3-phosphate $\rightleftharpoons$ dihydroxyacetone phosphate	
Glyceraldehyde 3-phosphate + dihydroxyacetone phosphate $\rightleftharpoons$ fructose 1,6-bisphosphate	
Fructose 1, 6-bisphosphate $\rightarrow$ fructose 6-phosphate + P_i	

Fructose 6-phosphate $\rightleftharpoons$ glucose 6-phosphate	
Glucose 6-phosphate + H ₂ O $\rightarrow$ glucose + P _i	
<i>Sum:</i> 2 Pyruvate + 4ATP + 2GTP + 2NADH + 2H ⁺ + 4H ₂ O $\rightarrow$ glucose + 4ADP + 2GDP + 6P _i + 2NAD ⁺	

Note: The bypass reactions are in red; all other reactions are reversible steps of glycolysis. The figures at the right indicate that the reaction is to be counted twice, because two three-carbon precursors are required to make a molecule of glucose. The reactions required to replace the cytosolic NADH consumed in the glyceraldehyde 3-phosphate dehydrogenase reaction (the conversion of lactate to pyruvate in the cytosol or the transport of reducing equivalents from mitochondria to the cytosol in the form of malate) are not considered in this summary. Biochemical equations are not necessarily balanced for H and charge ([p. 478](#)).

For each molecule of glucose formed from pyruvate, six high-energy phosphate groups are required: four from ATP and two from GTP. In addition, two molecules of NADH are required for the reduction of two molecules of 1,3-bisphosphoglycerate. Clearly, [Equation 14-9](#) is not simply the reverse of the equation for conversion of glucose to pyruvate by glycolysis, which would require only two molecules of ATP:



This makes the synthesis of glucose from pyruvate a relatively expensive process. Much of this high energy cost is necessary to ensure the irreversibility of gluconeogenesis. Under intracellular conditions, the overall free-energy change of glycolysis is at least -63 kJ/mol . Under the same conditions the overall ΔG of gluconeogenesis is -16 kJ/mol . Thus both glycolysis and gluconeogenesis are essentially irreversible processes in cells. A second advantage to investing energy to convert pyruvate to glucose is that if pyruvate were instead excreted, its considerable potential for ATP production by complete, aerobic oxidation would be lost (more than 10 ATP are produced per pyruvate, as we shall see in [Chapter 16](#)).

The biosynthetic pathway to glucose described above allows the net synthesis of glucose not only from pyruvate but also from the four-, five-, and six-carbon intermediates of the citric acid cycle ([Chapter 16](#)). The citric acid cycle intermediates can undergo oxidation to oxaloacetate (see [Fig. 16-7](#)). Some or all of the carbon atoms of most amino acids derived from proteins are ultimately catabolized to pyruvate or to intermediates of the citric acid cycle. Such amino acids can therefore undergo net conversion to glucose and are said to be **glucogenic** ([Table 14-4](#)).

Alanine and glutamine, the principal molecules that transport amino groups from extrahepatic tissues to the liver (see [Fig. 18-9](#)), are particularly important glucogenic amino acids in mammals. After removal of their amino groups in liver mitochondria, the carbon skeletons remaining (pyruvate and α -ketoglutarate, respectively) are readily funneled into gluconeogenesis.

TABLE 14-4 Glucogenic Amino Acids, Grouped by Site of Entry

Pyruvate	Succinyl-CoA
Alanine	Isoleucine ^a
Cysteine	Methionine
Glycine	Threonine
Serine	Valine
Threonine	Fumarate
Tryptophan ^a	Phenylalanine ^a
α-Ketoglutarate	Tyrosine ^a
Arginine	Oxaloacetate
Glutamate	Asparagine
Glutamine	Aspartate
Histidine	
Proline	

Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis.

^aThese amino acids are also ketogenic (see [Fig. 18-15](#)).

Mammals Cannot Convert Fatty Acids to Glucose; Plants and Microorganisms Can

No net conversion of fatty acids to glucose occurs in mammals. As we shall see in [Chapter 17](#), the catabolism of most fatty acids yields only acetyl-CoA. Mammals

cannot use acetyl-CoA as a precursor of glucose, because the pyruvate dehydrogenase reaction is irreversible and cells have no other pathway to convert acetyl-CoA to pyruvate. Plants, yeast, and many bacteria do have a pathway (the glyoxylate cycle; see [Fig. 20-45](#)) for converting acetyl-CoA to oxaloacetate, so these organisms can use fatty acids as the starting material for gluconeogenesis. This is important during the germination of seedlings, for example; before leaves develop and photosynthesis can provide energy and carbohydrates, the seedling relies on stored seed oils for energy production and cell wall biosynthesis.

Although mammals cannot convert fatty acids to carbohydrate, they can use the small amount of glycerol produced from the breakdown of fats (triacylglycerols) for gluconeogenesis. Phosphorylation of glycerol by glycerol kinase, followed by oxidation of the central carbon, yields dihydroxyacetone phosphate, an intermediate in gluconeogenesis in liver.

As we shall see in [Chapter 21](#), glycerol phosphate is an essential intermediate in triacylglycerol synthesis in adipocytes, but these cells lack glycerol kinase and so cannot simply phosphorylate glycerol. Instead, adipocytes carry out a truncated version of gluconeogenesis, known as [glyceroneogenesis](#): the conversion of pyruvate to dihydroxyacetone phosphate via the early reactions of gluconeogenesis, followed by reduction of the dihydroxyacetone phosphate to glycerol 3-phosphate (see [Fig. 21-21](#)).

SUMMARY 14.4 *Gluconeogenesis*

■ Gluconeogenesis is a multistep process in which glucose is produced from lactate, pyruvate, or oxaloacetate, or any compound (including citric acid cycle intermediates) that can be converted to one of these intermediates. Seven steps are the reversal of glycolytic reactions; three differ and must be bypassed with exergonic reactions.

■ In the first bypass, pyruvate is converted to PEP via oxaloacetate in two steps catalyzed by pyruvate carboxylase (which uses ATP) and PEP carboxykinase (which uses GTP).

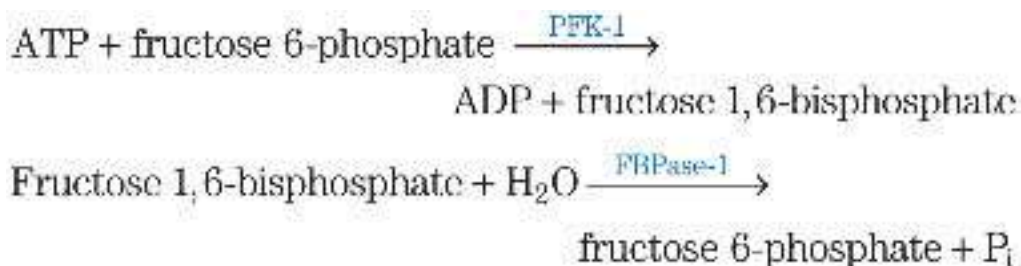
■ In the second bypass, FBPase-1 removes a phosphate group from fructose 1,6-bisphosphate, producing fructose 6-phosphate. In the third bypass, glucose 6-phosphatase converts glucose 6-phosphate to glucose.

- In mammals, gluconeogenesis in the liver, kidney, and small intestine provides glucose for use by the brain, muscles, and erythrocytes. Formation of one molecule of glucose from pyruvate requires four ATP, two GTP, and two NADH; it is expensive.
- Animals cannot convert acetyl-CoA derived from fatty acids into glucose; they lack the enzymatic machinery to convert acetyl-CoA to pyruvate. Plants and microorganisms have the glyoxylate pathway, which allows them to make glucose from fatty acids.

14.5 Coordinated Regulation of Glycolysis and Gluconeogenesis

 Glycolysis (the conversion of glucose to pyruvate) and gluconeogenesis (the conversion of pyruvate to glucose) generally do not both occur at the same time in the same tissues. In mammals, gluconeogenesis occurs primarily in the liver, where its role is to provide glucose for export to other tissues when glycogen stores are exhausted and when no dietary glucose is available. Glycolysis occurs in most tissues, including, brain, kidney, muscle and liver. Glycolysis provides ATP to support all of the energy-requiring activities of cells: active transport of ions; synthesis of macromolecules and of their precursors; synthesis of lipids and storage compounds like glycogen; and muscle contraction.

At each of the three points where glycolytic reactions are bypassed by alternative, gluconeogenic reactions ([Fig. 14-16](#)), simultaneous operation of both pathways would consume ATP without accomplishing any chemical or biological work. For example, PFK-1 and FBPase-1 catalyze opposing reactions:



The sum of these two reactions is



that is, hydrolysis of ATP without any useful metabolic work being done. Clearly, if these two reactions were allowed to proceed simultaneously at a high rate in the same cell, a large amount of chemical energy would be dissipated as heat.

We look now in some detail at the mechanisms that regulate glycolysis and gluconeogenesis at the three points where these pathways diverge.

Hexokinase Isozymes Are Affected Differently by Their Product, Glucose 6-Phosphate

Hexokinase, which catalyzes the entry of glucose into the glycolytic pathway, is a regulatory enzyme. As noted in [Section 14.1](#), humans have four isozymes of hexokinase (designated I to IV), encoded by four different genes ([Box 14-3](#)).  The different hexokinase isozymes of liver and muscle reflect the different roles of these organs in carbohydrate metabolism: muscle consumes glucose, using it for energy production; liver maintains blood glucose homeostasis by consuming or producing glucose, depending on the prevailing blood glucose concentration.

BOX 14-3

Isozymes: Different Proteins That Catalyze the Same Reaction

The four forms of hexokinase found in mammalian tissues are but one example of a common biological situation: the same reaction catalyzed by two or more different molecular forms of an enzyme. These multiple forms, called isozymes or isoenzymes, may occur in the same species, in the same tissue, even in the same cell. The different forms (isoforms) of the enzyme generally differ in kinetic or regulatory properties, in the cofactor they use (NADH or NADPH for dehydrogenase isozymes, for example), or in their subcellular distribution (soluble or membrane-bound). Isozymes may have similar, but not identical, amino acid sequences, and in many cases they clearly share a common evolutionary origin.

One of the first enzymes found to have isozymes was lactate dehydrogenase (LDH; [p. 526](#)), which in vertebrate tissues exists as at least five different isozymes separable by electrophoresis. All LDH isozymes contain four polypeptide chains (each of M_r 33,500), each type containing a different ratio of two kinds of polypeptides. The M (for muscle) chain and the H (for heart) chain are encoded by two different genes.

In skeletal muscle the predominant isozyme contains four M chains, and in heart the predominant isozyme contains four H chains. Other tissues have some combination of the five possible types of LDH isozymes:

Type	Composition	Location
LDH ₁	HHHH	Heart and erythrocyte
LDH ₂	HHHM	Heart and erythrocyte
LDH ₃	HHMM	Brain and kidney
LDH ₄	HMMM	Skeletal muscle and liver



Differences in the isozyme content of tissues can be used to assess the timing and extent of heart damage due to myocardial infarction (heart attack). Damage to heart tissue results in the release of heart LDH into the blood. Shortly after a heart attack, the blood level of total LDH increases, and there is more LDH₂ than LDH₁. After 12 hours the amounts of LDH₁ and LDH₂ are very similar, and after 24 hours there is more LDH₁ than LDH₂. This switch in the $[\text{LDH}_1]/[\text{LDH}_2]$ ratio, combined with increased concentrations in the blood of another heart enzyme, creatine kinase, is very strong evidence of a recent myocardial infarction. ■

The different LDH isozymes have significantly different values of V_{max} and K_m , particularly for pyruvate. The properties of LDH₄ favor rapid reduction of very low concentrations of pyruvate to lactate in skeletal muscle, whereas those of isozyme LDH₁ favor rapid oxidation of lactate to pyruvate in the heart.

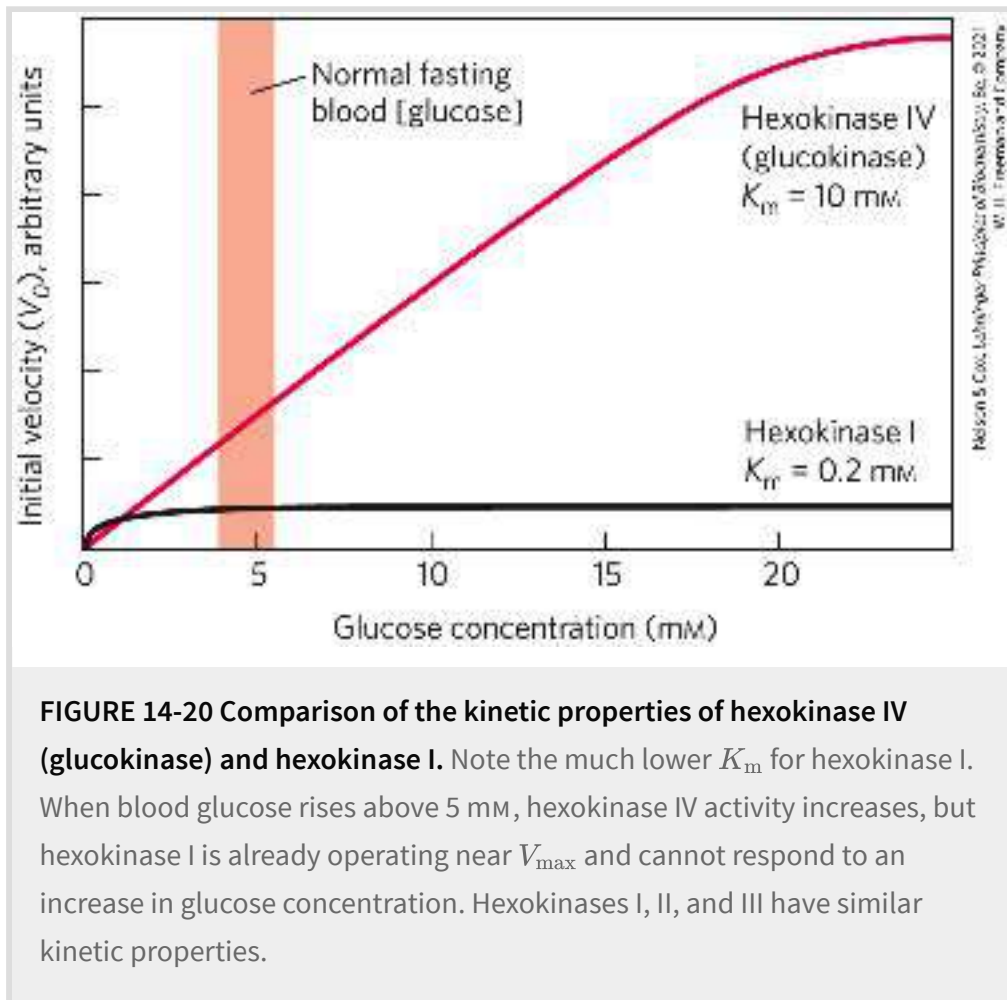
In general, the distribution of different isozymes of a given enzyme reflects at least four factors:

1. *Different metabolic patterns in different organs.* For glycogen phosphorylase, the isozymes in skeletal muscle and liver have different regulatory properties, reflecting the different roles of glycogen breakdown in these two tissues.
2. *Different locations and metabolic roles for isozymes in the same cell.* The isocitrate dehydrogenase isozymes of the cytosol and the mitochondrion are an example ([Chapter 16](#)).
3. *Different stages of development in embryonic or fetal tissues and in adult tissues.* For example, the fetal liver has a characteristic isozyme distribution of LDH, which changes as the organ develops into its adult form. Some enzymes of glucose catabolism in malignant (cancer) cells occur as their fetal, not adult, isozymes.
4. *Different responses of isozymes to allosteric modulators.* This difference is useful in fine-tuning metabolic rates. Hexokinase IV (glucokinase) of liver and the hexokinase isozymes of other tissues differ in their sensitivity to inhibition by glucose 6-phosphate.

The predominant hexokinase isozyme of myocytes (**hexokinase II**) has a high affinity for glucose — it is half-saturated at about 0.1 mM. Because glucose entering myocytes from the blood (where the glucose concentration is 4 to 5 mM) produces an intracellular glucose concentration high enough to saturate hexokinase II, the muscle enzyme normally acts at or near its $V_{\max}$. Muscle **hexokinase I** and hexokinase II are allosterically inhibited by their product, glucose 6-phosphate, so whenever the cellular concentration of glucose 6-phosphate rises above its normal level, these isozymes are temporarily and reversibly inhibited, bringing the rate of glucose 6-phosphate formation into balance with the rate of its utilization and reestablishing the steady state.

The predominant hexokinase isozyme of liver is hexokinase IV (also called glucokinase), which differs in three important respects from hexokinases I, II, and III of muscle. First, the glucose concentration at which hexokinase IV is half-saturated (about 10 mM) is higher than the usual concentration of glucose in the blood. Because an efficient glucose transporter in hepatocytes (GLUT2) rapidly equilibrates the glucose concentrations in cytosol and blood, the high K_m of hexokinase IV allows its direct regulation by the blood glucose concentration ([Fig. 14-20](#)). When blood glucose is high, as it is after a meal rich in carbohydrates, excess glucose is transported into hepatocytes, where hexokinase IV converts it to glucose 6-phosphate. Because hexokinase IV is not saturated at 10 mM glucose, its activity continues to increase as the glucose concentration rises to 10 mM or more. Under conditions of low blood glucose, the glucose concentration in a

hepatocyte is low relative to the K_m of hexokinase IV, and the glucose generated by gluconeogenesis leaves the cell before being trapped by phosphorylation.



Second, hexokinase IV is not inhibited by glucose 6-phosphate, and it can therefore continue to operate when the accumulation of glucose 6-phosphate completely inhibits hexokinases I, II, and III. Third, hexokinase IV is subject to inhibition by the reversible binding of a regulatory protein specific to liver ([Fig. 14-21](#)). The binding is much tighter in the presence of the allosteric effector fructose 6-phosphate. Glucose competes with fructose 6-phosphate for binding and causes dissociation of the regulatory

protein from the hexokinase, relieving the inhibition. Immediately after a carbohydrate-rich meal, when blood glucose is high, glucose enters the hepatocyte via GLUT2 and activates hexokinase IV by this mechanism. During a fast, when blood glucose drops below 5 mM, fructose 6-phosphate triggers the inhibition of hexokinase IV by the regulatory protein, so the liver does not compete with other organs for the scarce glucose. The mechanism of inhibition by the regulatory protein is interesting: the protein anchors hexokinase IV inside the nucleus, where it is segregated from the other enzymes of glycolysis in the cytosol. When the glucose concentration in the cytosol rises, it equilibrates with glucose in the nucleus by transport through the nuclear pores. Glucose causes dissociation of the regulatory protein, and hexokinase IV enters the cytosol and begins to phosphorylate glucose.

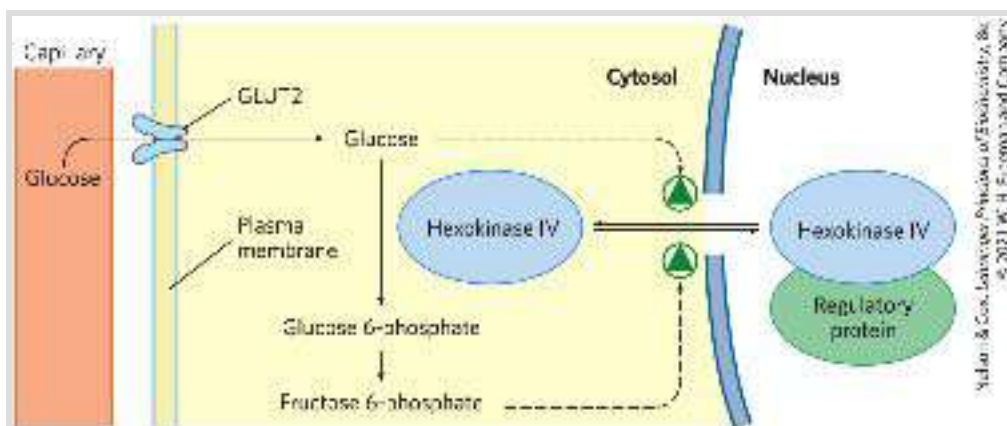


FIGURE 14-21 Regulation of hexokinase IV (glucokinase) by sequestration in the nucleus. The protein inhibitor of hexokinase IV is a nuclear binding protein that draws hexokinase IV into the nucleus when the fructose 6-phosphate concentration in liver is high and releases it to the cytosol when the glucose concentration is high.

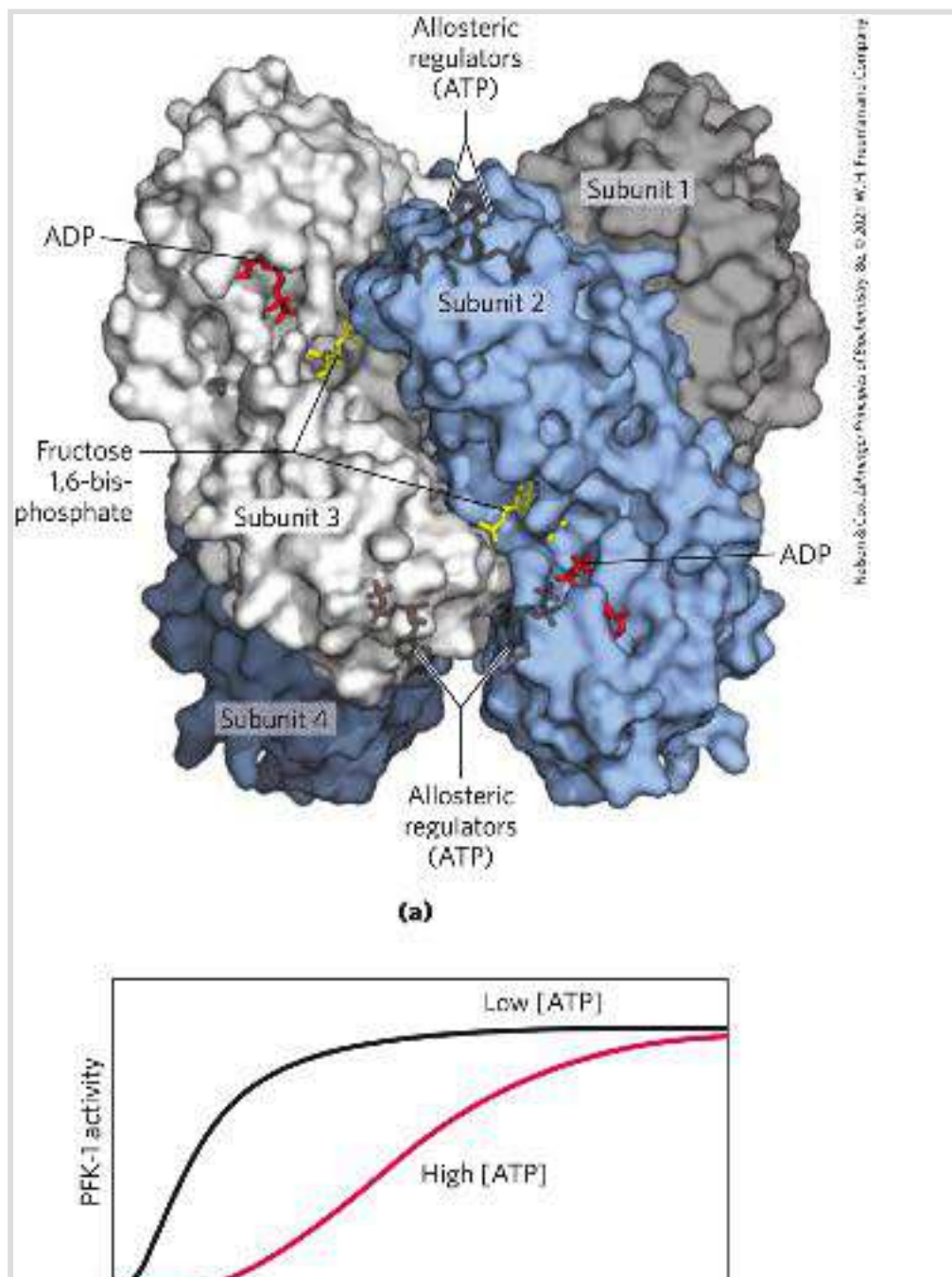
Hexokinase IV is also regulated at the level of protein synthesis. Circumstances that call for greater energy production (low [ATP], high [AMP], vigorous muscle contraction) or for greater glucose consumption (high blood [glucose], for example) cause increased transcription of the hexokinase IV gene. Glucose 6-phosphatase, the gluconeogenic enzyme that bypasses the hexokinase step of glycolysis, is transcriptionally regulated by factors that call for increased production of glucose (low blood glucose, glucagon signaling). The transcriptional regulation of these two enzymes (along with other enzymes of glycolysis and gluconeogenesis) is described below.

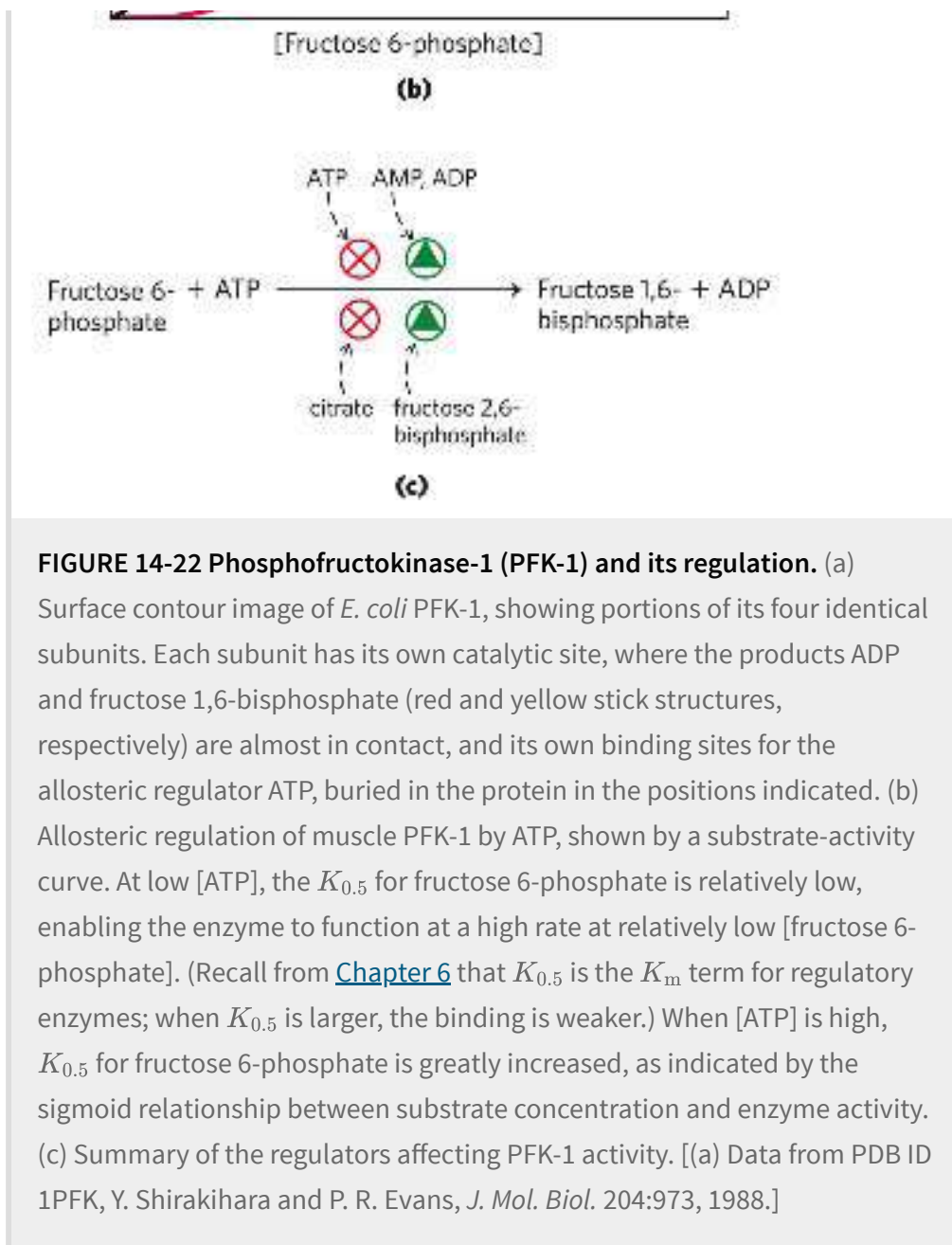
Phosphofructokinase-1 and Fructose 1,6-Bisphosphatase Are Reciprocally Regulated

Glucose 6-phosphate can flow either into glycolysis or through any of several other pathways, including glycogen synthesis and the pentose phosphate pathway. The metabolically irreversible reaction catalyzed by PFK-1 is the step that commits glucose to glycolysis. In addition to its substrate-binding sites, this complex enzyme has several regulatory sites at which allosteric activators or inhibitors bind.

ATP is not only a substrate for PFK-1 but also an end product of the glycolytic pathway. When high cellular [ATP] signals that ATP is being produced faster than it is being consumed, ATP inhibits

PFK-1 by binding to an allosteric site and lowering the affinity of the enzyme for its substrate fructose 6-phosphate ([Fig. 14-22](#)). ADP and AMP, which increase in concentration as consumption of ATP outpaces production, act allosterically to relieve this inhibition by ATP. These effects combine to produce higher enzyme activity when ADP or AMP accumulates and lower activity when ATP accumulates.

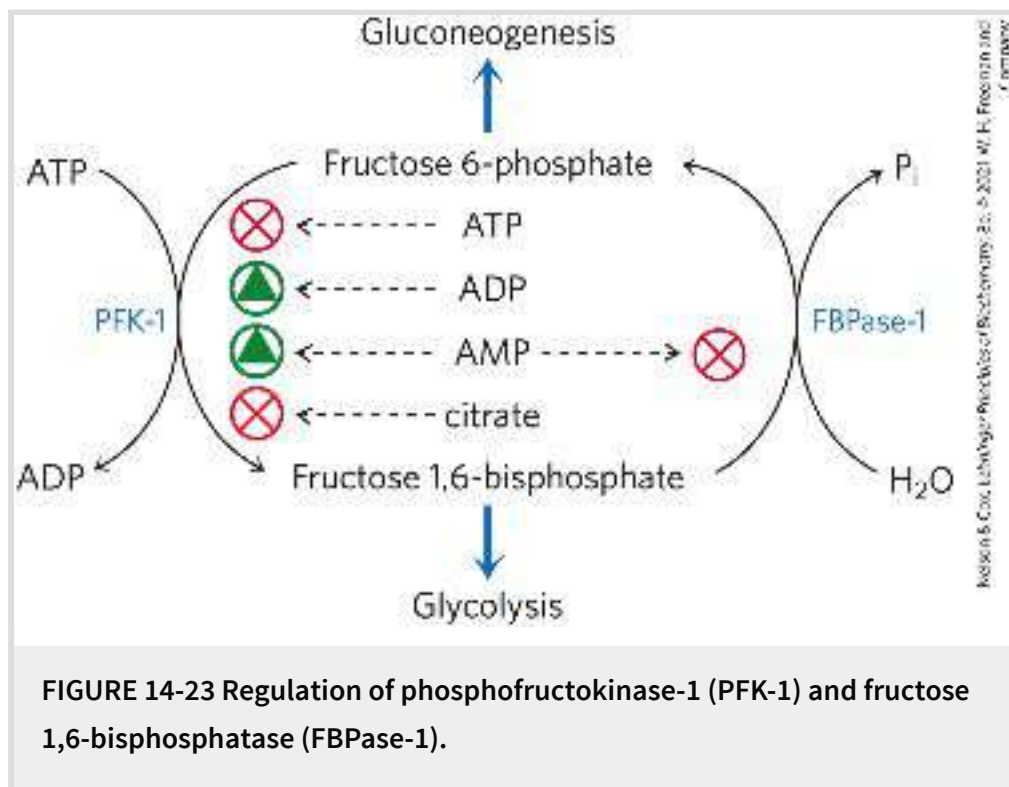




Citrate (the ionized form of citric acid), a key intermediate in the aerobic oxidation of pyruvate, fatty acids, and amino acids, is also an allosteric regulator of PFK-1. High citrate concentration increases the inhibitory effect of ATP, further reducing the flow of glucose through glycolysis. In this case, as in several others encountered later, citrate serves as an intracellular signal that the

cell is meeting its current needs for energy-yielding metabolism by the oxidation of fats and proteins.

The corresponding step in gluconeogenesis is the conversion of fructose 1,6-bisphosphate to fructose 6-phosphate ([Fig. 14-23](#)). The enzyme that catalyzes this reaction, FBPase-1, is strongly inhibited (allosterically) by AMP; when the cell's supply of ATP is low (corresponding to high [AMP]), the ATP-requiring synthesis of glucose slows.



Thus, these opposing steps in the glycolytic and gluconeogenic pathways — those catalyzed by PFK-1 and FBPase-1 — are regulated in a coordinated and reciprocal manner. In general, when sufficient concentrations of acetyl-CoA or citrate (the product of acetyl-CoA condensation with oxaloacetate) are

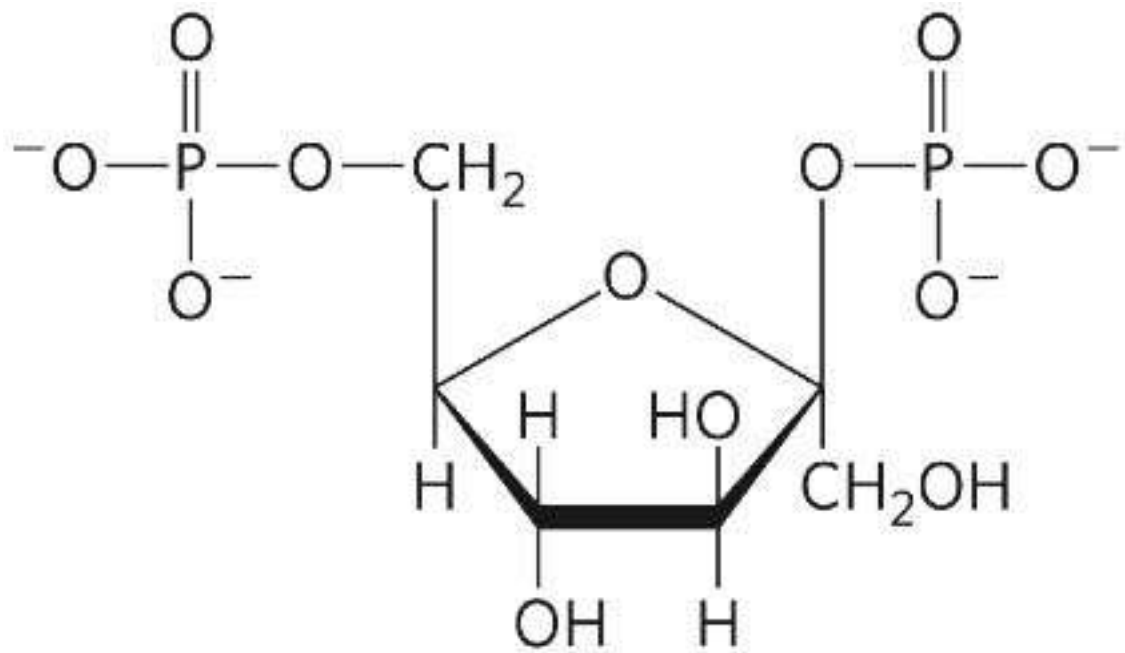
present, or when a high proportion of the cell's adenylate is in the form of ATP, gluconeogenesis is favored. When the concentration of AMP increases, it promotes glycolysis by stimulating PFK-1 (and, as we shall see in [Section 15.3](#), promotes glycogen degradation by activating glycogen phosphorylase).

Fructose 2,6-Bisphosphate Is a Potent Allosteric Regulator of PFK-1 and FBPase-1

The special role of the liver in maintaining a constant blood glucose level requires additional regulatory mechanisms to coordinate glucose production and consumption. When the blood glucose level decreases, the hormone **glucagon** signals the liver to produce and release more glucose and to stop consuming it for its own needs. One source of glucose is glycogen stored in the liver; another source is gluconeogenesis, using pyruvate, lactate, glycerol, or certain amino acids as starting material. When blood glucose is high, insulin signals the liver to use glucose as a fuel and as a precursor for the synthesis and storage of glycogen and triacylglycerol.



The rapid hormonal regulation of glycolysis and gluconeogenesis is mediated by **fructose 2,6-bisphosphate**, an allosteric effector for the enzymes PFK-1 and FBPase-1:



Fructose 2,6-bisphosphate

When fructose 2,6-bisphosphate binds to its allosteric site on PFK-1, it increases the enzyme's affinity for its substrate fructose 6-phosphate ([Fig. 14-24a](#)) and reduces its affinity for the allosteric inhibitors ATP and citrate. At the physiological concentrations of its substrates, ATP and fructose 6-phosphate, and of its other positive and negative effectors (ATP, AMP, citrate), PFK-1 is virtually inactive in the absence of fructose 2,6-bisphosphate. Fructose 2,6-bisphosphate has the opposite effect on FBPase-1: it reduces its affinity for its substrate ([Fig. 14-24b](#)), thereby slowing gluconeogenesis.

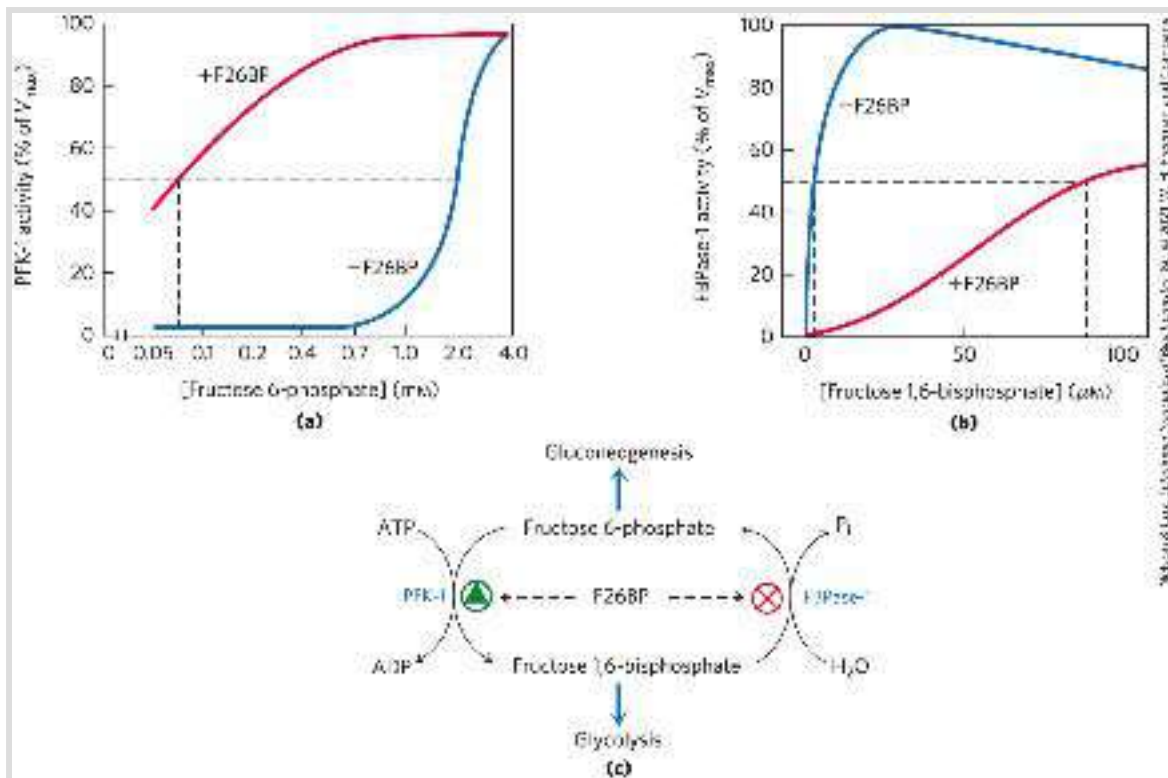


FIGURE 14-24 Role of fructose 2,6-bisphosphate in regulation of glycolysis and gluconeogenesis. Fructose 2,6-bisphosphate (F26BP) has opposite effects on the enzymatic activities of phosphofructokinase-1 (PFK-1, a glycolytic enzyme) and fructose 1,6-bisphosphatase (FBPase-1, a gluconeogenic enzyme). (a) PFK-1 activity in the absence of F26BP (blue curve) is half-maximal when the concentration of fructose 6-phosphate is 2 mM (that is, $K_{0.5} = 2$ mM). When $0.13 \mu\text{M}$ F26BP is present (red curve), the $K_{0.5}$ for fructose 6-phosphate is only 0.08 mM. Thus F26BP activates PFK-1 by increasing its apparent affinity for fructose 6-phosphate (see [Fig. 14-23b](#)). (b) FBPase-1 activity is inhibited by as little as $1 \mu\text{M}$ F26BP and is strongly inhibited by $25 \mu\text{M}$. In the absence of this inhibitor (blue curve), the $K_{0.5}$ for fructose 1,6-bisphosphate is $5 \mu\text{M}$, but in the presence of $25 \mu\text{M}$ F26BP (red curve), the $K_{0.5}$ is $>70 \mu\text{M}$. Fructose 2,6-bisphosphate also makes FBPase-1 more sensitive to inhibition by another allosteric regulator, AMP. (c) Summary of regulation by F26BP.

The cellular concentration of the allosteric regulator fructose 2,6-bisphosphate is set by the relative rates of its formation and breakdown ([Fig. 14-25a](#)). It is formed by phosphorylation of fructose 6-phosphate, catalyzed by **phosphofructokinase-2 (PFK-2)**, and broken down by **fructose 2,6-bisphosphatase (FBPase-2)**.

(Note that these enzymes are distinct from PFK-1 and FBPase-1, which catalyze the formation and breakdown, respectively, of fructose 1,6-bisphosphate.) PFK-2 and FBPase-2 are two separate enzymatic activities of a single, bifunctional protein. The balance of these two activities in the liver, which determines the cellular level of fructose 2,6-bisphosphate, is set by glucagon and insulin ([Fig. 14-25b](#)).

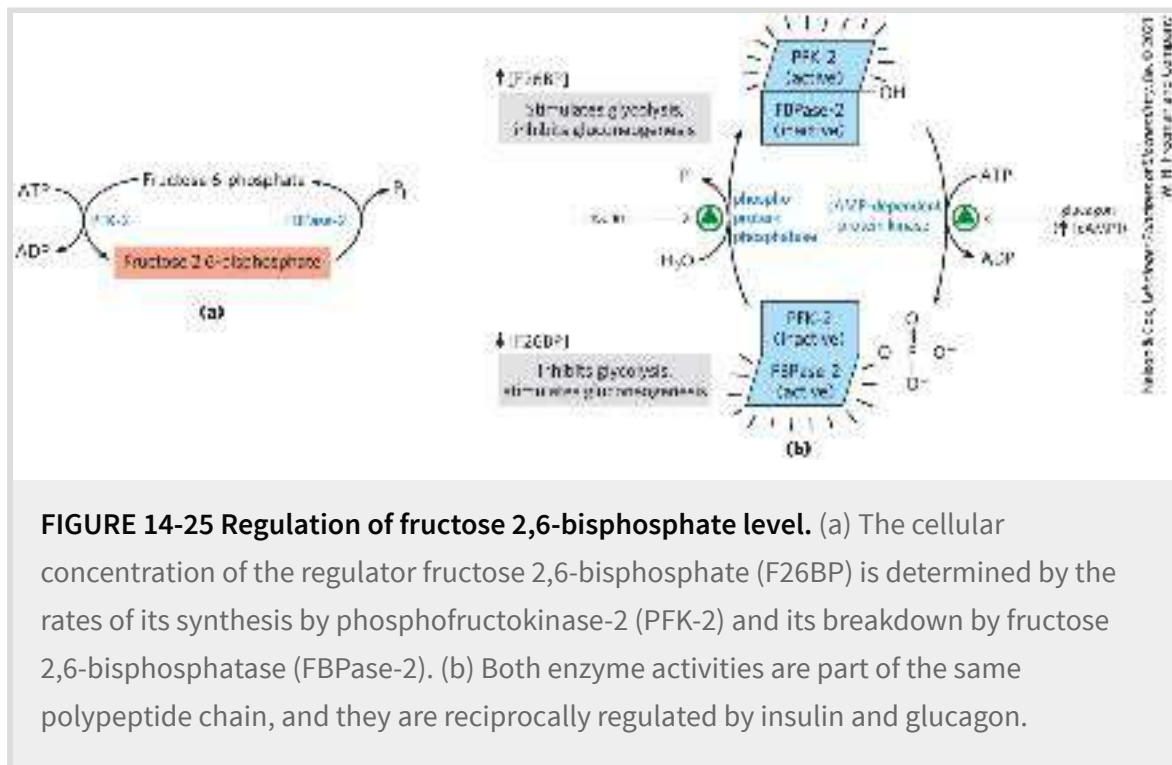


FIGURE 14-25 Regulation of fructose 2,6-bisphosphate level. (a) The cellular concentration of the regulator fructose 2,6-bisphosphate (F26BP) is determined by the rates of its synthesis by phosphofructokinase-2 (PFK-2) and its breakdown by fructose 2,6-bisphosphatase (FBPase-2). (b) Both enzyme activities are part of the same polypeptide chain, and they are reciprocally regulated by insulin and glucagon.

As we saw in [Chapter 12](#), glucagon stimulates the adenylyl cyclase of liver to synthesize 3', 5'-cyclic AMP (cAMP) from ATP. Cyclic AMP then activates cAMP-dependent protein kinase, which transfers a phosphoryl group from ATP to the bifunctional protein PFK-2/FBPase-2. Phosphorylation of this protein enhances its FBPase-2 activity and inhibits its PFK-2 activity. Glucagon thereby lowers the cellular level of fructose 2,6-bisphosphate, inhibiting glycolysis and stimulating

gluconeogenesis. The resulting production of more glucose enables the liver to replenish blood glucose in response to glucagon. Insulin has the opposite effect, stimulating the activity of a phosphoprotein phosphatase that catalyzes removal of the phosphoryl group from the bifunctional protein PFK-2/FBPase-2, activating its PFK-2 activity, increasing the level of fructose 2,6-bisphosphate, stimulating glycolysis, and inhibiting gluconeogenesis.

Xylulose 5-Phosphate Is a Key Regulator of Carbohydrate and Fat Metabolism

Another regulatory mechanism also acts by controlling the level of fructose 2,6-bisphosphate. In the mammalian liver, xylulose 5-phosphate, a product of the pentose phosphate pathway, mediates the increase in glycolysis that follows ingestion of a high-carbohydrate meal. The xylulose 5-phosphate concentration rises as glucose entering the liver is converted to glucose 6-phosphate and enters both the glycolytic and pentose phosphate pathways. Xylulose 5-phosphate activates phosphoprotein phosphatase 2A, which dephosphorylates the bifunctional PFK-2/FBPase-2 enzyme ([Fig. 14-25](#)). Dephosphorylation activates PFK-2 and inhibits FBPase-2, and the resulting rise in fructose 2,6-bisphosphate concentration stimulates glycolysis and inhibits gluconeogenesis. The increased glycolysis boosts the production of acetyl-CoA, while the increased flow of hexose through the pentose

phosphate pathway generates NADPH. Acetyl-CoA and NADPH are the starting materials for fatty acid synthesis, which increases dramatically in response to intake of a high-carbohydrate meal. Xylulose 5-phosphate also increases the synthesis of *all* the enzymes required for fatty acid synthesis, as we shall see ([Fig. 14-28](#)).

The Glycolytic Enzyme Pyruvate Kinase Is Allosterically Inhibited by ATP

At least three isozymes of pyruvate kinase are found in vertebrates, differing in their tissue distribution and their response to modulators. High concentrations of ATP, acetyl-CoA, and long-chain fatty acids (signs of abundant energy supply) allosterically inhibit all isozymes of pyruvate kinase ([Fig. 14-26](#)). The liver isozyme (L form), but not the muscle isozyme (M form), is subject to further regulation by phosphorylation. When low blood glucose causes glucagon release, cAMP-dependent protein kinase phosphorylates the L isozyme of pyruvate kinase, inactivating it. This slows the use of glucose as a fuel in liver, sparing it for export to the brain and other organs. In muscle, the effect of increased [cAMP] is quite different. In response to epinephrine, cAMP activates glycogen breakdown and glycolysis and provides the fuel needed for the fight-or-flight response.

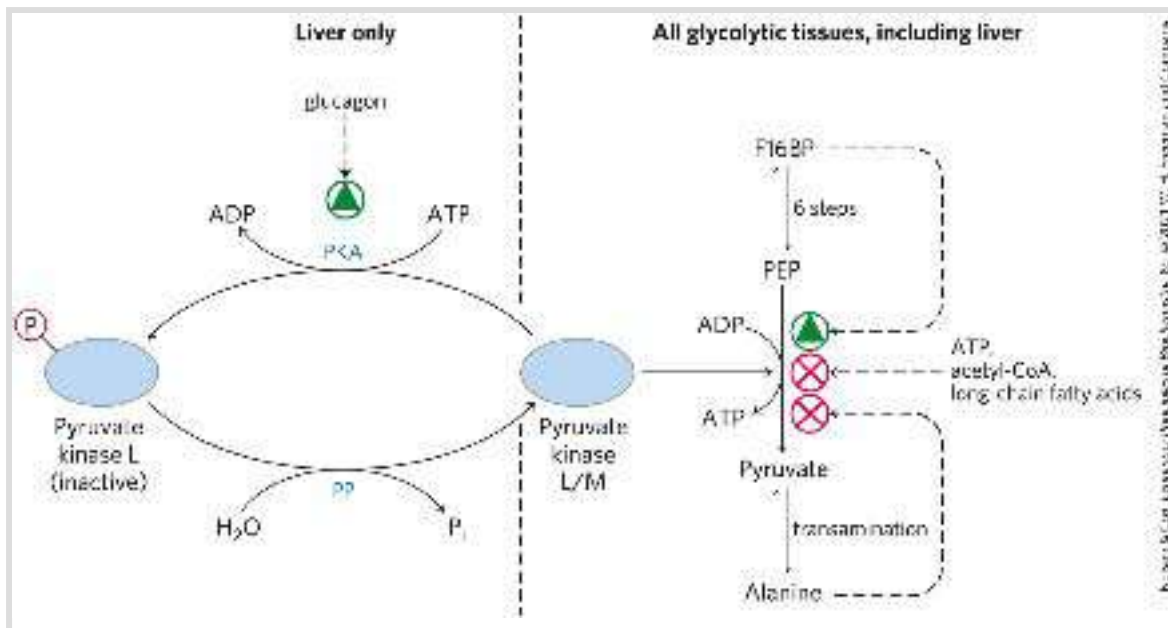
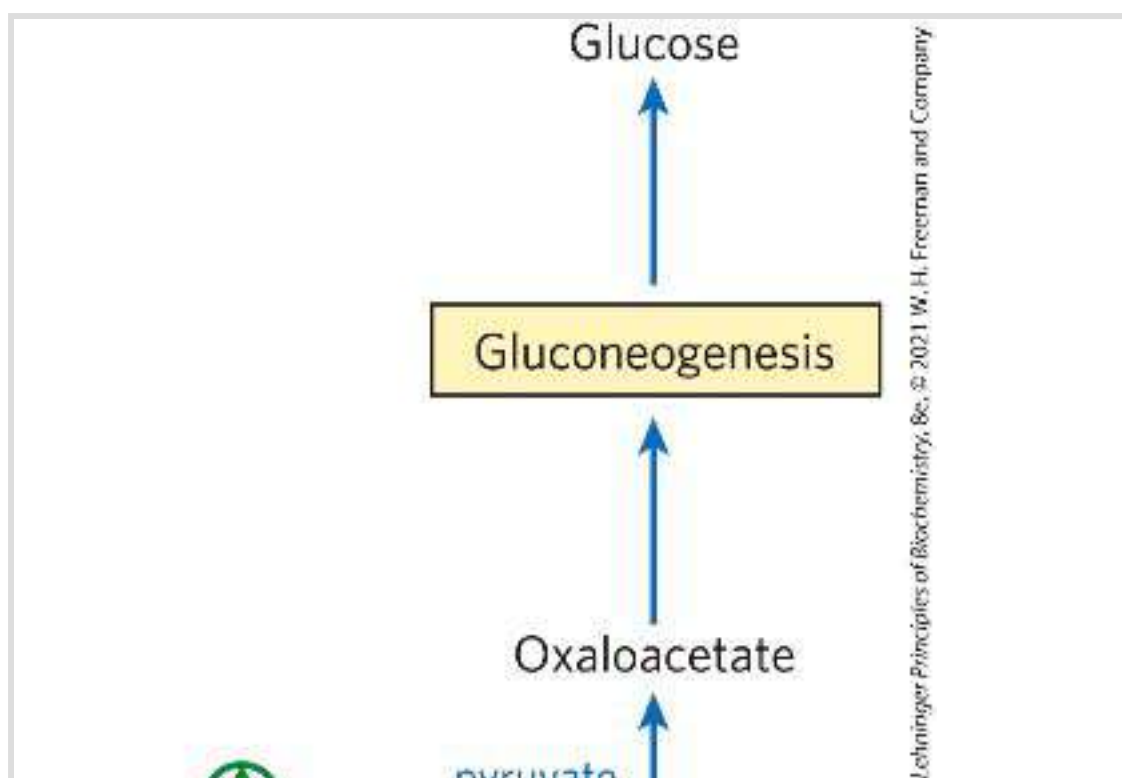


FIGURE 14-26 Regulation of pyruvate kinase. The enzyme is allosterically inhibited by ATP, acetyl-CoA, and long-chain fatty acids (all signs of an abundant energy supply), and the accumulation of fructose 1,6-bisphosphate triggers its activation. Accumulation of alanine, which can be synthesized from pyruvate in one step, allosterically inhibits pyruvate kinase, slowing the production of pyruvate by glycolysis. The liver isozyme (L form) is also regulated hormonally. Glucagon activates cAMP-dependent protein kinase (PKA; see [Fig. 15-12](#)), which phosphorylates the pyruvate kinase L isozyme, inactivating it. When the glucagon level drops, a protein phosphatase (PP) dephosphorylates pyruvate kinase, activating it. This mechanism prevents the liver from consuming glucose by glycolysis when blood glucose is low; instead, the liver exports glucose. The muscle isozyme (M form) is not affected by this phosphorylation mechanism.

Conversion of Pyruvate to Phosphoenolpyruvate Is Stimulated When Fatty Acids Are Available

P5 In the pathway leading from pyruvate to glucose, the first control point determines the fate of pyruvate in the mitochondrion: its conversion either to acetyl-CoA (by the

pyruvate dehydrogenase complex) to fuel the citric acid cycle ([Chapter 16](#)) or to oxaloacetate (by pyruvate carboxylase) to start the process of gluconeogenesis ([Fig. 14-27](#)). When fatty acids are readily available as fuels, their breakdown in liver mitochondria yields acetyl-CoA, a signal that further oxidation of glucose for fuel is not necessary. Acetyl-CoA is a positive allosteric modulator of pyruvate carboxylase and a negative modulator of pyruvate dehydrogenase, through stimulation of a protein kinase that inactivates the dehydrogenase. When the cell's energy needs are being met, oxidative phosphorylation slows, [NADH] rises relative to $[NAD^+]$ and inhibits the citric acid cycle, and acetyl-CoA accumulates. The increased concentration of acetyl-CoA inhibits the pyruvate dehydrogenase complex, slowing the formation of acetyl-CoA from pyruvate, and stimulates gluconeogenesis by activating pyruvate carboxylase, allowing conversion of excess pyruvate to oxaloacetate (and, eventually, glucose).



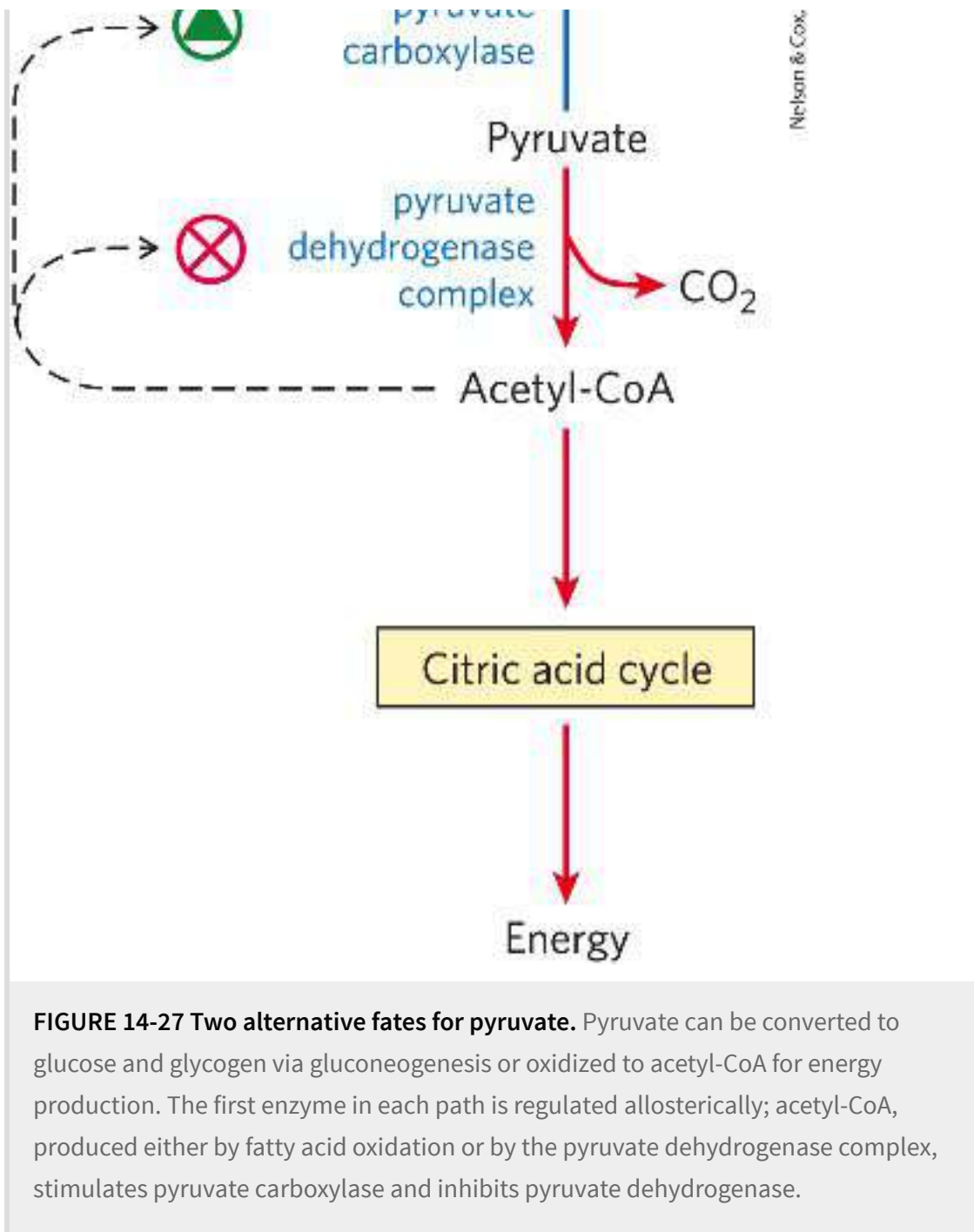


FIGURE 14-27 Two alternative fates for pyruvate. Pyruvate can be converted to glucose and glycogen via gluconeogenesis or oxidized to acetyl-CoA for energy production. The first enzyme in each path is regulated allosterically; acetyl-CoA, produced either by fatty acid oxidation or by the pyruvate dehydrogenase complex, stimulates pyruvate carboxylase and inhibits pyruvate dehydrogenase.

Oxaloacetate formed in this way is converted to phosphoenolpyruvate (PEP) in the reaction catalyzed by PEP carboxykinase ([Fig. 14-16](#)). In mammals, the regulation of this key enzyme occurs primarily at the level of its synthesis and breakdown, in response to dietary and hormonal signals. Fasting or high glucagon levels act through cAMP to increase the rate of

transcription and to stabilize the mRNA. Insulin, or high blood [glucose], has the opposite effects. We discuss this transcriptional regulation in more detail below. Generally triggered by a signal from outside the cell, these changes take place on a time scale of minutes to days.

Transcriptional Regulation Changes the Number of Enzyme Molecules

Most of the regulatory actions discussed thus far are mediated by fast, reversible mechanisms to change the activity of existing enzyme molecules: allosteric effects, covalent alteration (phosphorylation) of the enzyme, or binding of a regulatory protein. Another set of regulatory processes involves changes in the number of molecules of an enzyme in the cell, through changes in the balance of enzyme synthesis and breakdown. Our discussion now turns briefly to regulation of transcription through signal-activated transcription factors. Transcriptional control is discussed in more detail in [Chapter 28](#).

In [Chapter 12](#) we encountered nuclear receptors and transcription factors in the context of insulin signaling. Insulin acts through its receptor in the plasma membrane to turn on at least two distinct signaling pathways, each involving activation of a protein kinase (MAP kinase and protein kinase B). The kinases phosphorylate transcription factors, which then act in the nucleus to stimulate the synthesis of enzymes needed for cell growth and division. More than 150 genes are transcriptionally

regulated by insulin, many of which encode proteins we have described here ([Table 14-5](#)).

TABLE 14-5 Some of the Many Genes Regulated by Insulin

Change in gene expression	Role in glucose metabolism
Increased expression Hexokinase II Hexokinase IV Phosphofructokinase-1 (PFK-1) PFK-2/FBPase-2 Pyruvate kinase	Essential for glycolysis, which consumes glucose for energy
Glucose 6-phosphate dehydrogenase 6-Phosphogluconate dehydrogenase Malic enzyme	Produce NADPH, which is essential for conversion of glucose to lipids
ATP-citrate lyase Pyruvate dehydrogenase	Produce acetyl-CoA, which is essential for conversion of glucose to lipids
Acetyl-CoA carboxylase Fatty acid synthase complex Stearyl-CoA dehydrogenase Acyl-CoA-glycerol transferases	Essential for conversion of glucose to lipids
Decreased expression PEP carboxykinase Glucose 6-phosphatase (catalytic subunit)	Essential for glucose production by gluconeogenesis



One transcription factor important to carbohydrate metabolism is **ChREBP (carbohydrate response element binding protein; Fig. 14-28)**, which is expressed primarily in liver, adipose tissue, and kidney. It coordinates the synthesis of enzymes needed for carbohydrate and fat synthesis. ChREBP in its

phosphorylated form is inactive and is located in the cytosol. When the phosphoprotein phosphatase PP2A removes a phosphoryl group from ChREBP, the transcription factor can enter the nucleus. Here, nuclear PP2A removes another phosphoryl group, and ChREBP now joins with a partner protein, Mlx, and turns on the synthesis of several enzymes: pyruvate kinase; fatty acid synthase; and acetyl-CoA carboxylase, the first enzyme in the path to fatty acid synthesis.

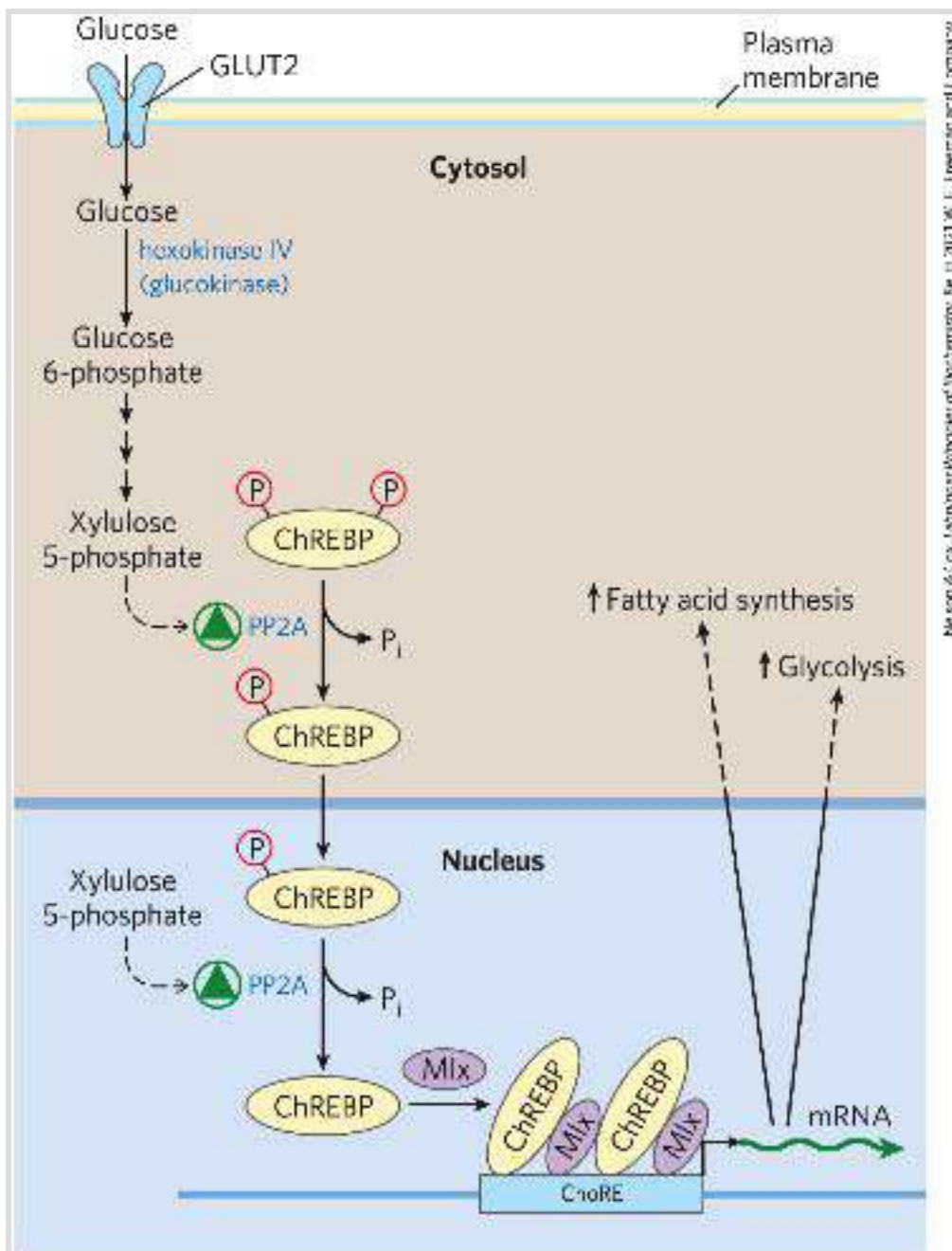


FIGURE 14-28 Mechanism of gene regulation by the transcription factor ChREBP. When ChREBP in the cytosol of a hepatocyte is phosphorylated on a Ser residue and a Thr residue, it cannot enter the nucleus. Dephosphorylation of (P)—Ser by protein phosphatase PP2A allows ChREBP to enter the nucleus, where a second dephosphorylation, of (P)—Thr, activates ChREBP so that it can associate with its partner protein, Mlx. ChREBP-Mlx now binds to the carbohydrate response element (ChoRE) in the promoter and stimulates transcription. PP2A is allosterically activated by xylulose 5-phosphate, an intermediate in the pentose phosphate pathway.

Controlling the activity of PP2A — and thus, ultimately, the synthesis of this group of metabolic enzymes — is xylulose 5-phosphate, an intermediate of the pentose phosphate pathway (see [Fig. 14-31](#)). When blood glucose concentration is high, glucose enters the liver and is phosphorylated by hexokinase IV. The glucose 6-phosphate thus formed can enter either the glycolytic pathway or the pentose phosphate pathway. If the latter, two initial oxidations produce xylulose 5-phosphate, which serves as a signal that the glucose-utilizing pathways are well-supplied with substrate. It accomplishes this by allosterically activating PP2A, which then dephosphorylates ChREBP, allowing the transcription factor to turn on the expression of genes for enzymes of glycolysis and fat synthesis ([Fig. 14-28](#)).

SUMMARY 14.5 *Coordinated Regulation of Glycolysis and Gluconeogenesis*

- Glycolysis and gluconeogenesis are reciprocally regulated to prevent wasteful operation of both pathways at the same time.
- Hexokinase IV (glucokinase) has kinetic properties related to its special role in the liver: releasing glucose to the blood when blood [glucose] is low; taking up and metabolizing glucose when blood [glucose] is high. Hexokinases I, II, and III are all inhibited by their product, glucose 6-phosphate.

- PFK-1 is allosterically inhibited by high [ATP]; low [AMP] inhibits FBPase-1. High [ATP] therefore slows glycolysis and speeds gluconeogenesis.
- Reciprocal allosteric control of glycolysis and gluconeogenesis is mainly achieved by the opposing effects of fructose 2,6-bisphosphate on PFK-1 and FBPase-1. Fructose 2,6-bisphosphate formation is stimulated, indirectly, by insulin, and inhibited by epinephrine.
- Xylulose 5-phosphate, an intermediate of the pentose phosphate pathway, activates phosphoprotein phosphatase PP2A. Activated PP2A tips the balance toward glucose uptake, glycogen synthesis, and lipid synthesis in the liver.
- Pyruvate kinase is allosterically inhibited by ATP, and the liver isozyme also is inhibited by cAMP-dependent phosphorylation. When [ATP] is high, glycolysis is slowed.
- When fatty acids are readily available as fuels, their breakdown in liver mitochondria yields acetyl-CoA, a signal that further oxidation of glucose for fuel is not necessary. Acetyl-CoA activates pyruvate carboxylase, thus favoring gluconeogenesis.
- Transcription factors such as ChREBP act in the nucleus to regulate the expression of specific genes coding for enzymes of the glycolytic and gluconeogenic pathways.

14.6 Pentose Phosphate Pathway of Glucose Oxidation



In most animal tissues, the major catabolic fate of glucose 6-phosphate is glycolytic breakdown to pyruvate, much of which is then oxidized via the citric acid cycle, ultimately leading to the formation of ATP. Glucose 6-phosphate does have other catabolic fates, however, which lead to specialized products needed by the cell. Of particular importance in some tissues is the oxidation of glucose 6-phosphate to pentose phosphates by the pentose phosphate pathway (also called the phosphogluconate pathway or the hexose monophosphate pathway; Fig. 14-29). In this oxidative pathway, NADP^+ is the electron acceptor, yielding NADPH. Rapidly dividing cells, such as those of bone marrow, skin, and intestinal mucosa, and those of tumors, use the pentose ribose 5-phosphate to make RNA, DNA, and such coenzymes as ATP, NADH, FADH_2 , and coenzyme A.

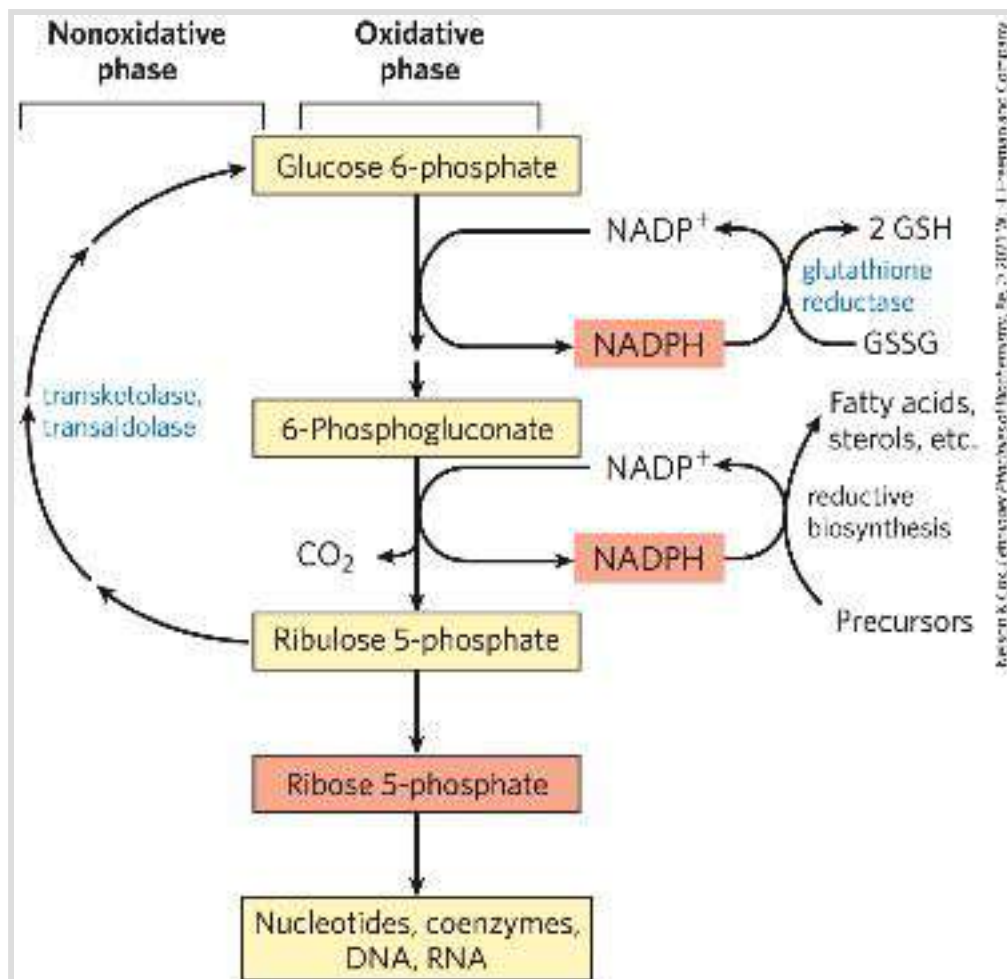


FIGURE 14-29 General scheme of the pentose phosphate pathway. NADPH formed in the oxidative phase is used to reduce glutathione, GSSG (see [Box 14-4](#)), and to support reductive biosynthesis. The other product of the oxidative phase is ribose 5-phosphate, which serves as a precursor for nucleotides, coenzymes, and nucleic acids. In cells that are not using ribose 5-phosphate for biosynthesis, the nonoxidative phase recycles six molecules of the pentose into five molecules of the hexose glucose 6-phosphate, allowing continued production of NADPH and converting glucose 6-phosphate (in six cycles) to CO_2 .

In other tissues, the essential product of the pentose phosphate pathway is not the pentoses but the electron donor NADPH, needed for reductive biosynthesis or to counter the damaging effects of oxygen radicals. Tissues that carry out extensive fatty

acid synthesis (liver, adipose, lactating mammary gland) or very active synthesis of cholesterol and steroid hormones (liver, adrenal glands, gonads) require the NADPH provided by this pathway. Erythrocytes and the cells of the lens and cornea are directly exposed to oxygen and thus to the damaging free radicals generated by oxygen. By maintaining a reducing environment (a high ratio of NADPH to NADP^+ and a high ratio of reduced glutathione to oxidized glutathione), such cells can prevent or undo oxidative damage to proteins, lipids, and other sensitive molecules. In erythrocytes, the NADPH produced by the pentose phosphate pathway is so important in preventing oxidative damage that a genetic defect in glucose 6-phosphate dehydrogenase, the first enzyme of the pathway, can have serious medical consequences ([Box 14-4](#)).

BOX 14-4 MEDICINE

Why Pythagoras Wouldn't Eat Falafel: Glucose 6-Phosphate Dehydrogenase Deficiency

Fava beans, an ingredient of falafel, have been an important food source in the Mediterranean and the Middle East since antiquity. The Greek philosopher and mathematician Pythagoras prohibited his followers from dining on fava beans, perhaps because they make many people sick with a condition called favism, which can be fatal. In favism, erythrocytes begin to lyse 24 to 48 hours after ingestion of the beans, releasing free hemoglobin into the blood. Jaundice and sometimes kidney failure can result. Similar symptoms can occur with ingestion of the antimalarial drug primaquine or of sulfa antibiotics, or following exposure to certain herbicides. These symptoms have a genetic basis: glucose 6-phosphate dehydrogenase (G6PD) deficiency, which affects about 400 million people worldwide. Most G6PD-deficient individuals are

asymptomatic; only the combination of G6PD deficiency and certain environmental factors produces the clinical manifestations.

Glucose 6-phosphate dehydrogenase catalyzes the first step in the pentose phosphate pathway (see [Fig. 14-30](#)), which produces NADPH. This reductant, essential in many biosynthetic pathways, also protects cells from oxidative damage by hydrogen peroxide (H_2O_2) and superoxide free radicals, highly reactive oxidants generated as metabolic byproducts and through the actions of drugs such as primaquine and natural products such as divicine — the toxic ingredient of fava beans. During normal detoxification, H_2O_2 is converted to H_2O by reduced glutathione and glutathione peroxidase, and the oxidized glutathione is converted back to the reduced form by glutathione reductase and NADPH ([Fig. 1](#)). H_2O_2 is also broken down to H_2O and O_2 by catalase, which also requires NADPH. In G6PD-deficient individuals, the NADPH production is diminished and detoxification of H_2O_2 is inhibited. Cellular damage results: lipid peroxidation leading to breakdown of erythrocyte membranes and oxidation of proteins and DNA.

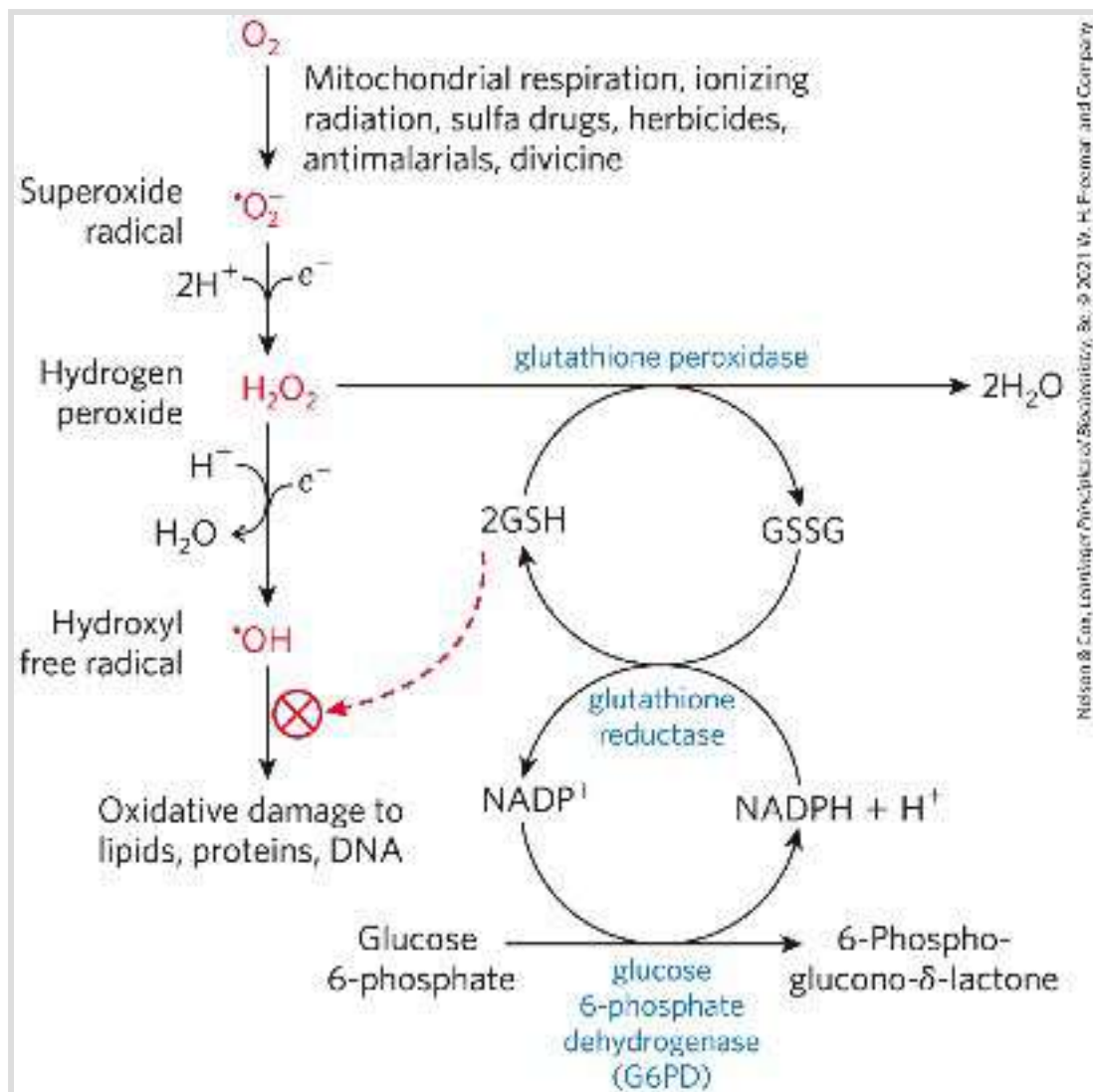


FIGURE 1 Role of NADPH and glutathione in protecting cells against highly reactive oxygen derivatives. Reduced glutathione (GSH) protects the cell by destroying hydrogen peroxide and hydroxyl free radicals. Regeneration of GSH from its oxidized form (GSSG) requires the NADPH produced in the glucose 6-phosphate dehydrogenase reaction.

The geographic distribution of G6PD deficiency is instructive. Frequencies as high as 25% occur in tropical Africa, parts of the Middle East, and Southeast Asia, areas where malaria is most prevalent. In addition to such epidemiological observations, in vitro studies show that growth of one malaria parasite, *Plasmodium falciparum*, is inhibited in G6PD-deficient erythrocytes. The parasite is very sensitive to oxidative damage and is killed by a level of oxidative stress that is tolerable to a G6PD-deficient human host. Because the advantage of resistance to malaria balances the disadvantage of lowered

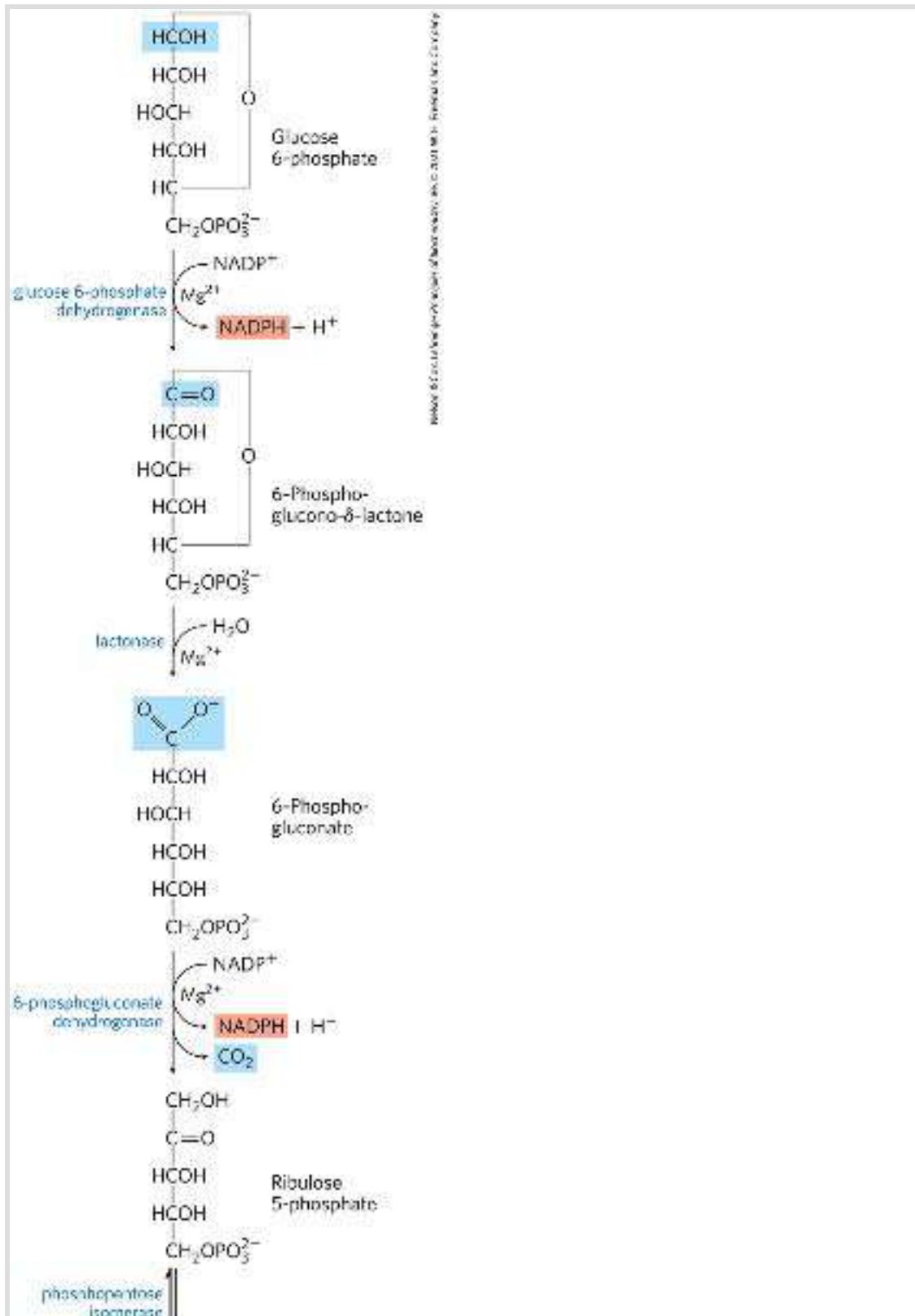
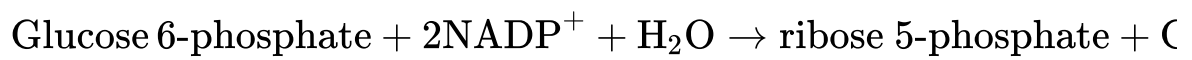
resistance to oxidative damage, natural selection sustains the G6PD-deficient genotype in human populations where malaria is prevalent. Only under overwhelming oxidative stress, caused by drugs, herbicides, or divicine, does G6PD deficiency cause serious medical problems.

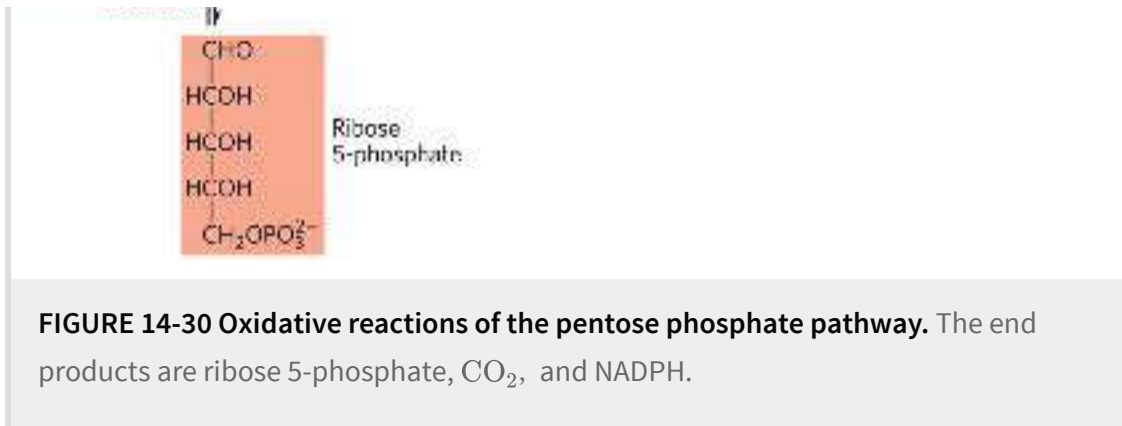
An antimalarial drug such as primaquine is believed to act by causing oxidative stress to the parasite. It is ironic that antimalarial drugs can cause human illness through the same biochemical mechanism that provides resistance to malaria. Divicine also acts as an antimalarial drug, and ingestion of fava beans may protect against malaria. By refusing to eat falafel, many Pythagoreans with normal G6PD activity may have unwittingly increased their risk of malaria!

The Oxidative Phase Produces NADPH and Pentose Phosphates



The first reaction of the pentose phosphate pathway ([Fig. 14-30](#)) is the oxidation of glucose 6-phosphate by **glucose 6-phosphate dehydrogenase (G6PD)** to form 6-phosphoglucono- δ -lactone, an intramolecular ester. NADP^+ is the electron acceptor, and the overall equilibrium lies far in the direction of NADPH formation. The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific **lactonase**, then 6-phosphogluconate undergoes oxidation and decarboxylation by **6-phosphogluconate dehydrogenase** to form the ketopentose ribulose 5-phosphate; the reaction also generates a second molecule of NADPH. **Phosphopentose isomerase** converts ribulose 5-phosphate to its aldose isomer, ribose 5-phosphate. In some tissues, the pentose phosphate pathway ends at this point, and its overall equation is

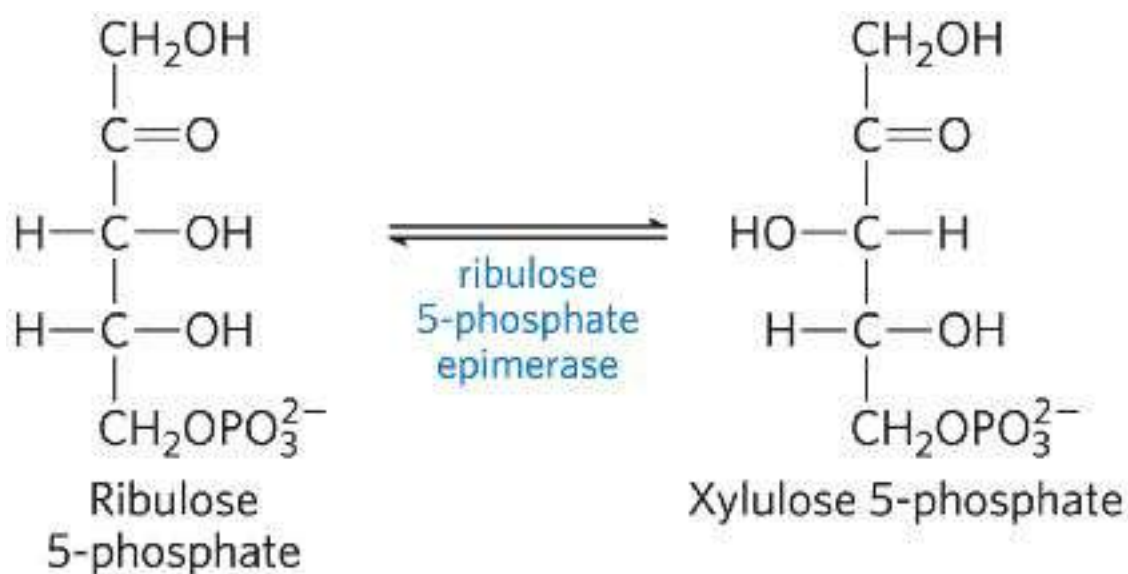




The net result is the production of NADPH, a reductant for biosynthetic reactions, and ribose 5-phosphate, a precursor for nucleotide synthesis.

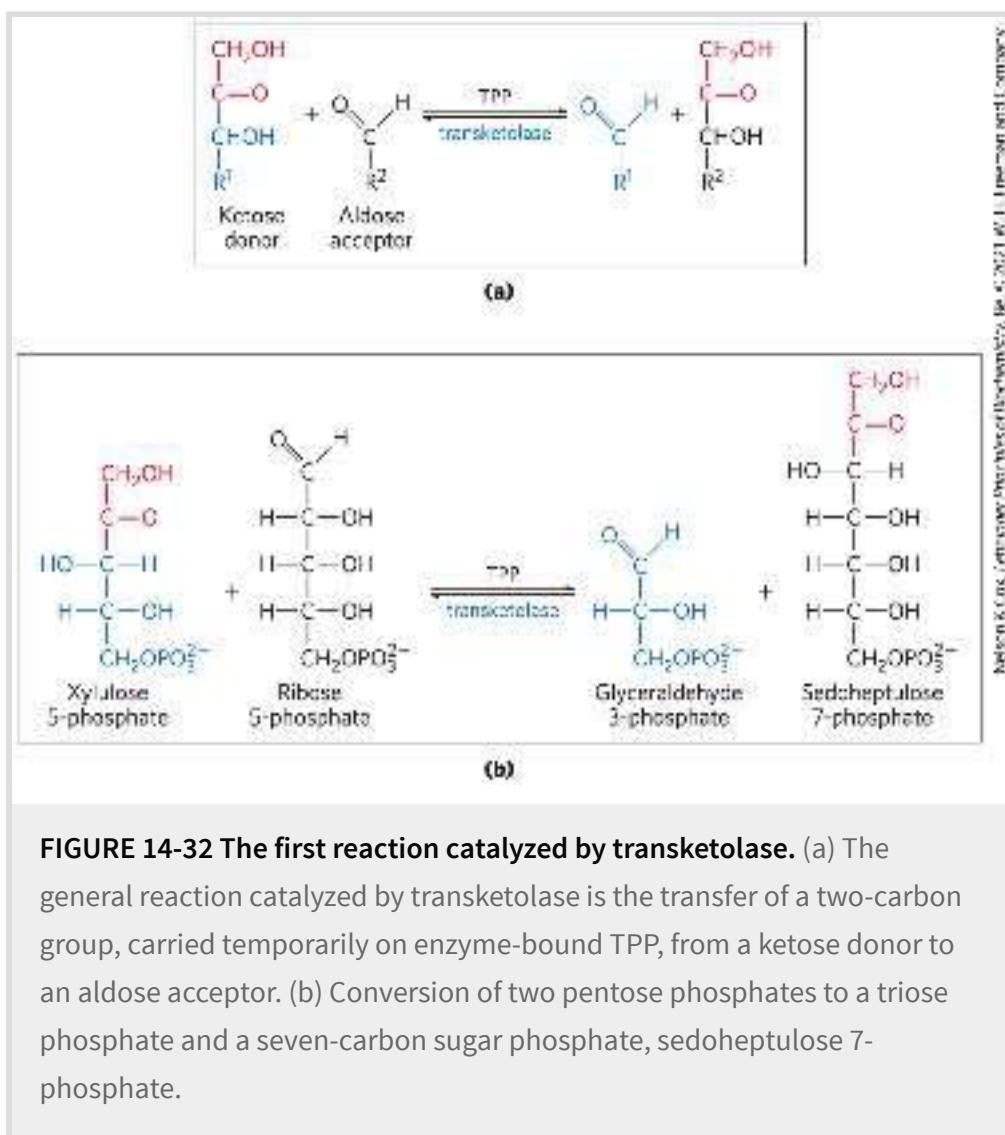
The Nonoxidative Phase Recycles Pentose Phosphates to Glucose 6-Phosphate

P6 In tissues that require primarily NADPH, the pentose phosphates produced in the oxidative phase of the pathway are recycled into glucose 6-phosphate. In this nonoxidative phase, ribulose 5-phosphate is first epimerized to xylulose 5-phosphate:



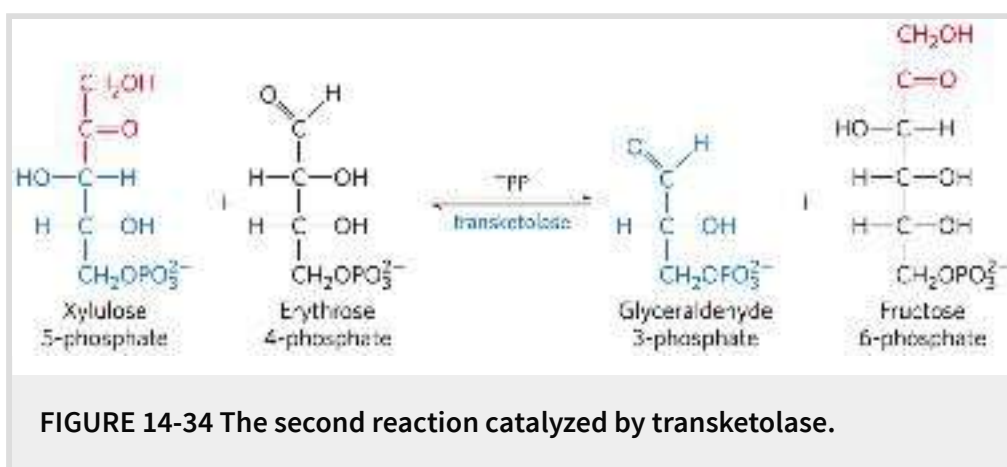
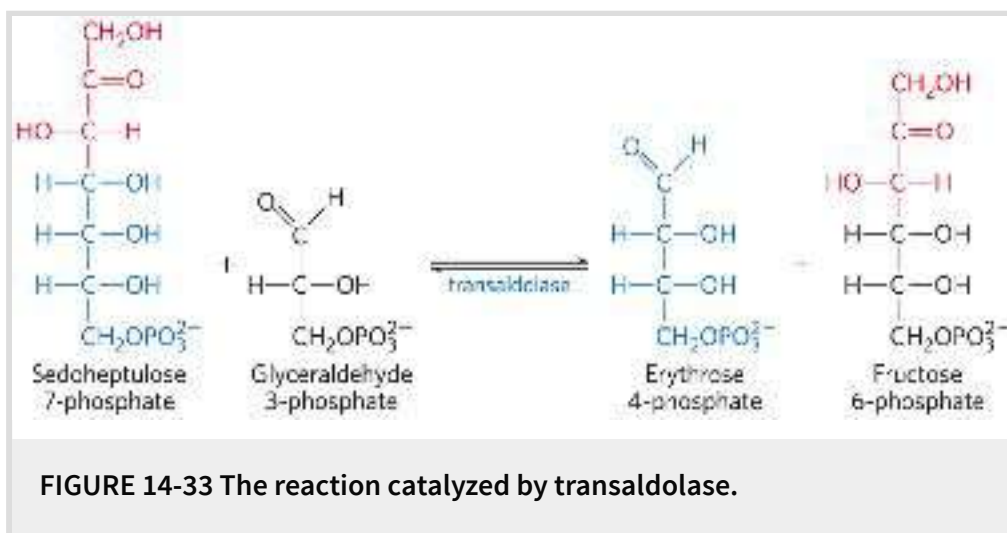
Then, in a series of rearrangements of the carbon skeletons ([Fig. 14-31](#)), six five-carbon sugar phosphates are converted to five six-carbon sugar phosphates, completing the cycle and allowing continued oxidation of glucose 6-phosphate with production of NADPH. Continued recycling leads ultimately to the conversion of glucose 6-phosphate to six CO_2 . Two enzymes unique to the pentose phosphate pathway act in these interconversions of sugars: transketolase and transaldolase. **Transketolase** catalyzes the transfer of a two-carbon fragment from a ketose donor to an aldose acceptor ([Fig. 14-32a](#)). In its first appearance in the pentose phosphate pathway, transketolase transfers C-1 and C-2 of xylulose 5-phosphate to ribose 5-phosphate, forming the seven-carbon product sedoheptulose 7-phosphate ([Fig. 14-32b](#)). The remaining three-carbon fragment from xylulose is glyceraldehyde 3-phosphate.

other enzymes also serve in the glycolytic or gluconeogenic pathways. (b) A schematic diagram showing the pathway from six pentoses (5C) to five hexoses (6C). Note that this involves two sets of the interconversions shown in (a). Every reaction shown here is reversible; unidirectional arrows are used only to make clear the direction of the reactions during continuous oxidation of glucose 6-phosphate. In the light-independent reactions of photosynthesis, the direction of these reactions is reversed.



Next, **transaldolase** catalyzes a reaction similar to the aldolase reaction of glycolysis: a three-carbon fragment is removed from sedoheptulose 7-phosphate and condensed with glyceraldehyde 3-phosphate, forming fructose 6-phosphate and the tetrose

erythrose 4-phosphate ([Fig. 14-33](#)). Now transketolase acts again, forming fructose 6-phosphate and glyceraldehyde 3-phosphate from erythrose 4-phosphate and xylulose 5-phosphate ([Fig. 14-34](#)). Two molecules of glyceraldehyde 3-phosphate formed by two iterations of these reactions can be converted to a molecule of fructose 1,6-bisphosphate as in gluconeogenesis ([Fig. 14-16](#)), and finally FBPase-1 and phosphohexose isomerase convert fructose 1,6-bisphosphate to glucose 6-phosphate. Overall, six pentose phosphates have been converted to five hexose phosphates ([Fig. 14-32b](#)) — the cycle is now complete.



Transketolase requires the cofactor thiamine pyrophosphate (TPP), which stabilizes a two-carbon carbanion in this reaction ([Fig. 14-35a](#)), just as it does in the pyruvate decarboxylase reaction ([Fig. 14-13](#)). Transaldolase uses a Lys side chain to form a Schiff base with the carbonyl group of its substrate, a ketose, thereby stabilizing a carbanion ([Fig. 14-35b](#)) that is central to the reaction mechanism.

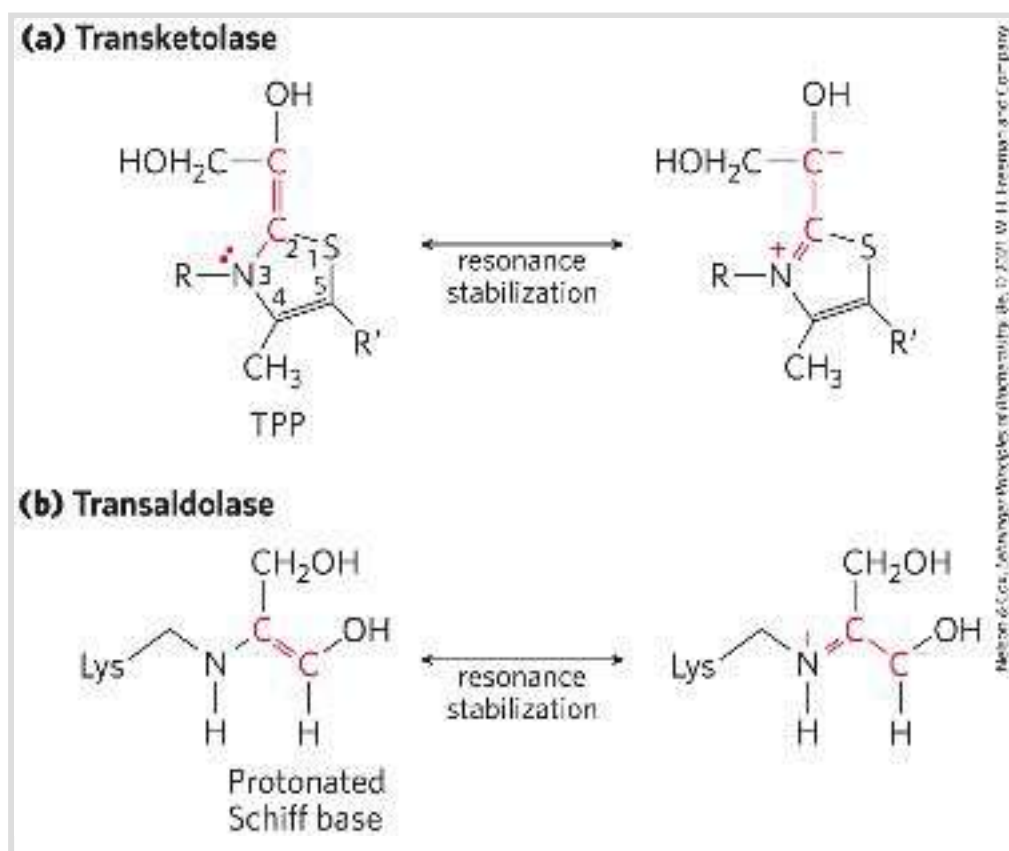


FIGURE 14-35 Carbanion intermediates stabilized by covalent interactions with transketolase and transaldolase. (a) The ring of TPP stabilizes the carbanion in the dihydroxyethyl group carried by transketolase. (b) In the transaldolase reaction, the protonated Schiff base formed between the ϵ -amino group of a Lys side chain and the substrate stabilizes the C-3 carbanion formed after aldol cleavage.

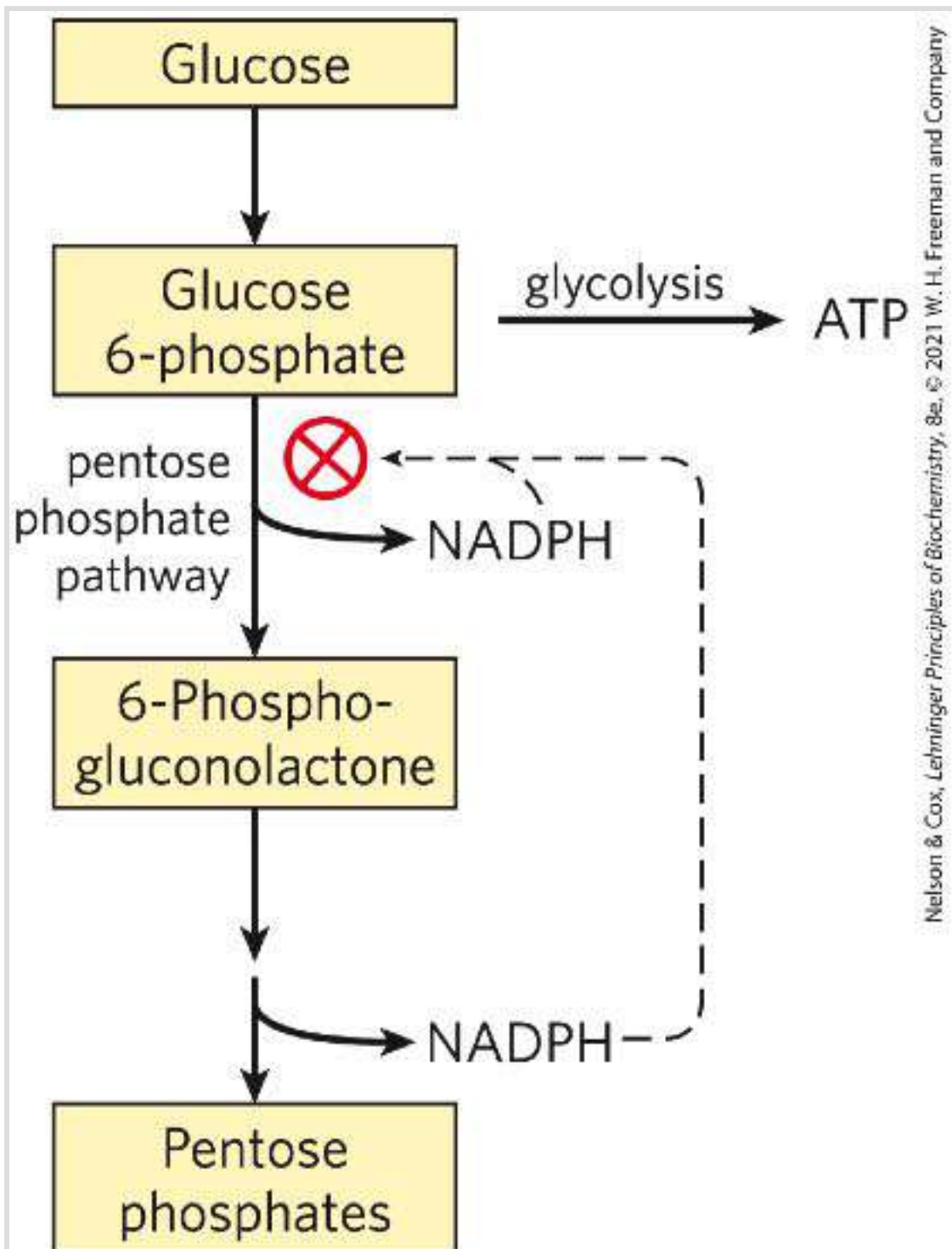
The first and third steps of the **oxidative pentose phosphate pathway** shown in [Figure 14-30](#) are oxidations with large, negative standard free-energy changes and are essentially irreversible in the cell. The reactions of the nonoxidative part of the pentose phosphate pathway ([Fig. 14-31](#)) are readily reversible and thus also provide a means of converting hexose phosphates to pentose phosphates. As we shall see in [Chapter 20](#), a process that converts hexose phosphates to pentose phosphates is central to the photosynthetic assimilation of CO₂ by plants. That pathway, the **reductive pentose phosphate pathway**, is essentially the reversal of the reactions shown in [Figure 14-31](#) and employs many of the same enzymes.

All the enzymes of the pentose phosphate pathway are located in the cytosol, like those of glycolysis and most of those of gluconeogenesis. In fact, these three pathways are connected through several shared intermediates and enzymes. The glyceraldehyde 3-phosphate formed by the action of transketolase is readily converted to dihydroxyacetone phosphate by the glycolytic enzyme triose phosphate isomerase, and these two trioses can be joined by the aldolase as in gluconeogenesis, forming fructose 1,6-bisphosphate. Alternatively, the triose phosphates can be oxidized to pyruvate by the glycolytic reactions. The fate of the trioses is determined by the cell's relative needs for pentose phosphates, NADPH, and ATP.

Glucose 6-Phosphate Is Partitioned between Glycolysis and the Pentose Phosphate Pathway



Whether glucose 6-phosphate enters glycolysis or the pentose phosphate pathway depends on the current needs of the cell and on the concentration of NADP^+ in the cytosol. Without this electron acceptor, the first reaction of the pentose phosphate pathway (catalyzed by glucose 6-phosphate dehydrogenase) cannot proceed. When a cell is rapidly converting NADPH to NADP^+ in biosynthetic reductions, $[\text{NADP}^+]$ rises, allosterically stimulating glucose 6-phosphate dehydrogenase and thereby increasing the flux of glucose 6-phosphate through the pentose phosphate pathway ([Fig. 14-36](#)). When the demand for NADPH slows, the level of NADP^+ drops, the pentose phosphate pathway slows, and glucose 6-phosphate is instead used to fuel glycolysis.



Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2011 W. H. Freeman and Company

FIGURE 14-36 Role of NADPH in regulating the partitioning of glucose 6-phosphate between glycolysis and the pentose phosphate pathway. When NADPH is forming faster than it is being used for biosynthesis and glutathione reduction, [NADPH] rises and inhibits the first enzyme in the pentose phosphate pathway. As a result, more glucose 6-phosphate is available for glycolysis.

Thiamine Deficiency Causes Beriberi and Wernicke-Korsakoff Syndrome



Thiamine, precursor to the cofactor thiamine pyrophosphate (TPP), is one of the B vitamins, essential in humans. Lack of vitamin B₁ in the diet leads to a range of medical problems. The condition known as beriberi is characterized by an accumulation of body fluids (swelling), pain, paralysis, and ultimately, without treatment, death. Wernicke-Korsakoff syndrome, also caused by a severe deficiency of thiamine, typically includes problems with voluntary movements, reflected in abnormal eye movements and gait, and neurological defects. The syndrome is more common among heavy drinkers than in the general population because chronic, heavy alcohol consumption interferes with the intestinal absorption of thiamine. The syndrome can be exacerbated by a mutation in the gene for transketolase that results in an enzyme with a lowered affinity for TPP — an affinity one-tenth that of the normal enzyme. This defect makes individuals much more sensitive to a thiamine deficiency: even a moderate thiamine deficiency (tolerable in individuals with an unmutated transketolase) can result in a transketolase that is not saturated with TPP at its normal concentration. The result is a slowing down of the whole pentose phosphate pathway. In people with Wernicke-Korsakoff syndrome, this mutation results in a worsening of symptoms, which can include severe memory loss, mental confusion, and partial paralysis. ■

SUMMARY 14.6 *Pentose Phosphate Pathway of Glucose Oxidation*

- The *oxidative* pentose phosphate pathway produces NADPH and pentose phosphates. Tissues that carry out extensive fatty acid synthesis (liver, adipose, lactating mammary gland) or very active synthesis of cholesterol and steroid hormones (liver, adrenal glands, gonads) require the NADPH provided by this pathway.
- Ribose 5-phosphate is a precursor for nucleotide and nucleic acid synthesis.
- The first, oxidative phase of the pentose phosphate pathway consists of two oxidations that convert glucose 6-phosphate to ribulose 5-phosphate and reduce NADP^+ to NADPH.
- The second, nonoxidative phase of the pentose phosphate pathway comprises steps that convert pentose phosphates to glucose 6-phosphate, which begins the oxidative cycle again.
- Entry of glucose 6-phosphate either into glycolysis or into the pentose phosphate pathway is largely determined by the relative concentrations of NADP^+ and NADPH.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

glycolysis

hexokinase

isozymes

phosphofructokinase-1 (PFK-1)

fructose 1,6-bisphosphate aldolase

aldolase

triose phosphate isomerase

glyceraldehyde 3-phosphate dehydrogenase

acyl phosphate

phosphoglycerate kinase

substrate-level phosphorylation

respiration-linked phosphorylation

phosphoglycerate mutase

enolase

phosphoenolpyruvate (PEP)

pyruvate kinase

glycogen phosphorylase

mutases

isomerases

lactose intolerance

galactosemia

hypoxia

fermentation

lactic acid fermentation

ethanol (alcohol) fermentation

lactate dehydrogenase

pyruvate decarboxylase

alcohol dehydrogenase

thiamine pyrophosphate (TPP)

gluconeogenesis

biotin

fructose 1,6-bisphosphatase (FBPase-1)

glucogenic

glyceroneogenesis

glucagon

fructose 2,6-bisphosphate

phosphofructokinase-2 (PFK-2)

fructose 2,6-bisphosphatase (FBPase-2)

carbohydrate response element binding protein (ChREBP)

pentose phosphate pathway

phosphogluconate pathway

hexose monophosphate pathway

glucose 6-phosphate dehydrogenase (G6PD)

6-phosphogluconate dehydrogenase

PROBLEMS

1. Is the Hexokinase Reaction at Equilibrium in Cells? For the reaction catalyzed by the enzyme hexokinase





the equilibrium constant, K_{eq} , is 7.8×10^2 . In living *E. coli* cells, $[\text{ATP}] = 5 \text{ mM}$, $[\text{ADP}] = 0.5 \text{ mM}$, $[\text{glucose}] = 2 \text{ mM}$, and $[\text{glucose 6-phosphate}] = 1 \text{ mM}$. Is the reaction at equilibrium in *E. coli*?

2. Equation for the Preparatory Phase of Glycolysis Write balanced biochemical equations for all the reactions in the catabolism of glucose to two molecules of glyceraldehyde 3-phosphate (the preparatory phase of glycolysis), including the standard free-energy change for each reaction. Then write the overall or net equation for the preparatory phase of glycolysis, with the net standard free-energy change.

3. Payoff Phase of Glycolysis: Fate of Pyruvate in Active Skeletal Muscle In working skeletal muscle under anaerobic conditions, glyceraldehyde 3-phosphate is converted to pyruvate (the payoff phase of glycolysis), and the pyruvate is reduced to lactate. Write balanced biochemical equations for all the reactions in this process, with the standard free-energy change for each reaction. Then write the overall or net equation for the payoff phase of glycolysis with fermentation to lactate, including the net standard free-energy change.

4. Energetics of the Aldolase Reaction Aldolase catalyzes the glycolytic reaction

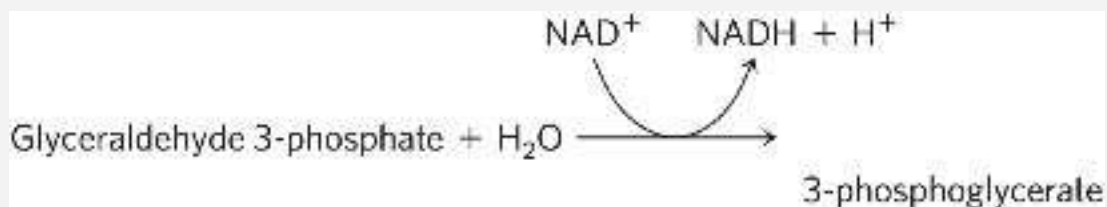




The standard free-energy change for this reaction in the direction written is +23.8 kJ/mol. The concentrations of the three intermediates in the hepatocyte of a mammal are fructose 1,6-bisphosphate, 1.4×10^{-5} M; glyceraldehyde 3-phosphate, 3×10^{-6} M; and dihydroxyacetone phosphate, 1.6×10^{-5} M. At body temperature (37 °C), what is the actual free-energy change for the reaction?

5. Equivalence of Triose Phosphates A researcher adds ^{14}C -labeled glyceraldehyde 3-phosphate to a yeast extract. After a short time, she isolates fructose 1,6-bisphosphate labeled with ^{14}C at C-3 and C-4. What was the location of the ^{14}C label in the starting glyceraldehyde 3-phosphate? Where did the second ^{14}C label in fructose 1,6-bisphosphate come from? Explain.

6. Glycolysis Shortcut Suppose you discovered a mutant yeast whose glycolytic pathway was shorter because of the presence of a new enzyme catalyzing the reaction



Would shortening the glycolytic pathway in this way benefit the cell? Explain.

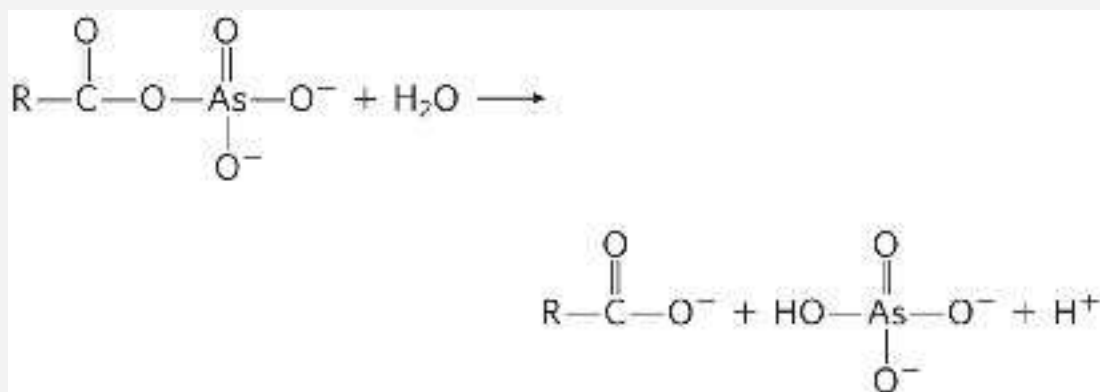
7. Role of Lactate Dehydrogenase During strenuous activity, the demand for ATP in muscle tissue vastly increases. In rabbit leg muscle or turkey flight muscle, ATP production is almost exclusively a product of lactic acid fermentation. Phosphoglycerate kinase and pyruvate kinase catalyze the two reactions that form ATP in the payoff phase of glycolysis. Suppose skeletal muscle were devoid of lactate dehydrogenase. Could it carry out strenuous physical activity; that is, could it generate ATP at a high rate by glycolysis? Explain.

8. Efficiency of ATP Production in Muscle The transformation of glucose to lactate in myocytes releases only about 7% of the free energy released when glucose is completely oxidized to CO_2 and H_2O . Does this mean that glycolysis with lactate fermentation under anaerobic conditions in muscle is a wasteful use of glucose? Explain.

9. Free-Energy Change for Triose Phosphate Oxidation The oxidation of glyceraldehyde 3-phosphate to 1,3-bisphosphoglycerate, catalyzed by glyceraldehyde 3-phosphate dehydrogenase, proceeds with an unfavorable equilibrium constant ($K'_{\text{eq}} = 0.08$; $\Delta G'^{\circ} = 6.3 \text{ kJ/mol}$), yet the flow through this point in the glycolytic pathway proceeds smoothly. How does the cell overcome the unfavorable equilibrium?

10. Arsenate Poisoning Arsenate is structurally and chemically similar to inorganic phosphate (P_i), and many enzymes that require phosphate will also use arsenate.

Organic compounds of arsenate are less stable than analogous phosphate compounds, however. For example, acyl *arsenates* decompose rapidly by hydrolysis, as shown.



On the other hand, acyl *phosphates*, such as 1,3-bisphosphoglycerate, are more stable and undergo further enzyme-catalyzed transformation in cells.

- a. Predict the effect on the net reaction catalyzed by glyceraldehyde 3-phosphate dehydrogenase if phosphate were replaced by arsenate.
- b. What would be the consequence to an organism if arsenate were substituted for phosphate? Arsenate is very toxic to most organisms. Explain why.

11. Requirement for Phosphate in Ethanol Fermentation In 1906 Harden and Young, in a series of classic studies on the fermentation of glucose to ethanol and CO₂ by extracts of brewer's yeast, made the following observations: (1) Inorganic phosphate was essential to fermentation; when the supply of phosphate was exhausted, fermentation ceased before all the glucose was used. (2) During fermentation under these conditions (with no phosphate), ethanol, CO₂,

and a hexose biphosphate accumulated. (3) When arsenate was substituted for phosphate, no hexose biphosphate accumulated, but the fermentation proceeded until all the glucose was converted to ethanol and CO_2 .

- a. Why did fermentation cease when the supply of phosphate was exhausted?
- b. Why did ethanol and CO_2 accumulate? Was the conversion of pyruvate to ethanol and CO_2 essential? Why? Identify the hexose biphosphate that accumulated. Why did it accumulate?
- c. Why did the substitution of arsenate for phosphate prevent the accumulation of the hexose biphosphate yet allow fermentation to ethanol and CO_2 to go to completion? (See Problem 10.)

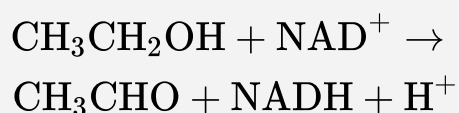
12. Role of the Vitamin Niacin Adults engaged in strenuous physical activity require an intake of about 160 g of carbohydrate daily but only about 20 mg of niacin for optimal nutrition. Given the role of niacin in glycolysis, how do you explain the observation?

13. Synthesis of Glycerol Phosphate The glycerol 3-phosphate required for the synthesis of glycerophospholipids can be synthesized from a glycolytic intermediate. Propose a reaction sequence for this conversion.

14.  Severity of Clinical Symptoms Due to Enzyme Deficiency The clinical symptoms of two forms of galactosemia — galactokinase-deficiency galactosemia and transferase-deficiency galactosemia — show radically

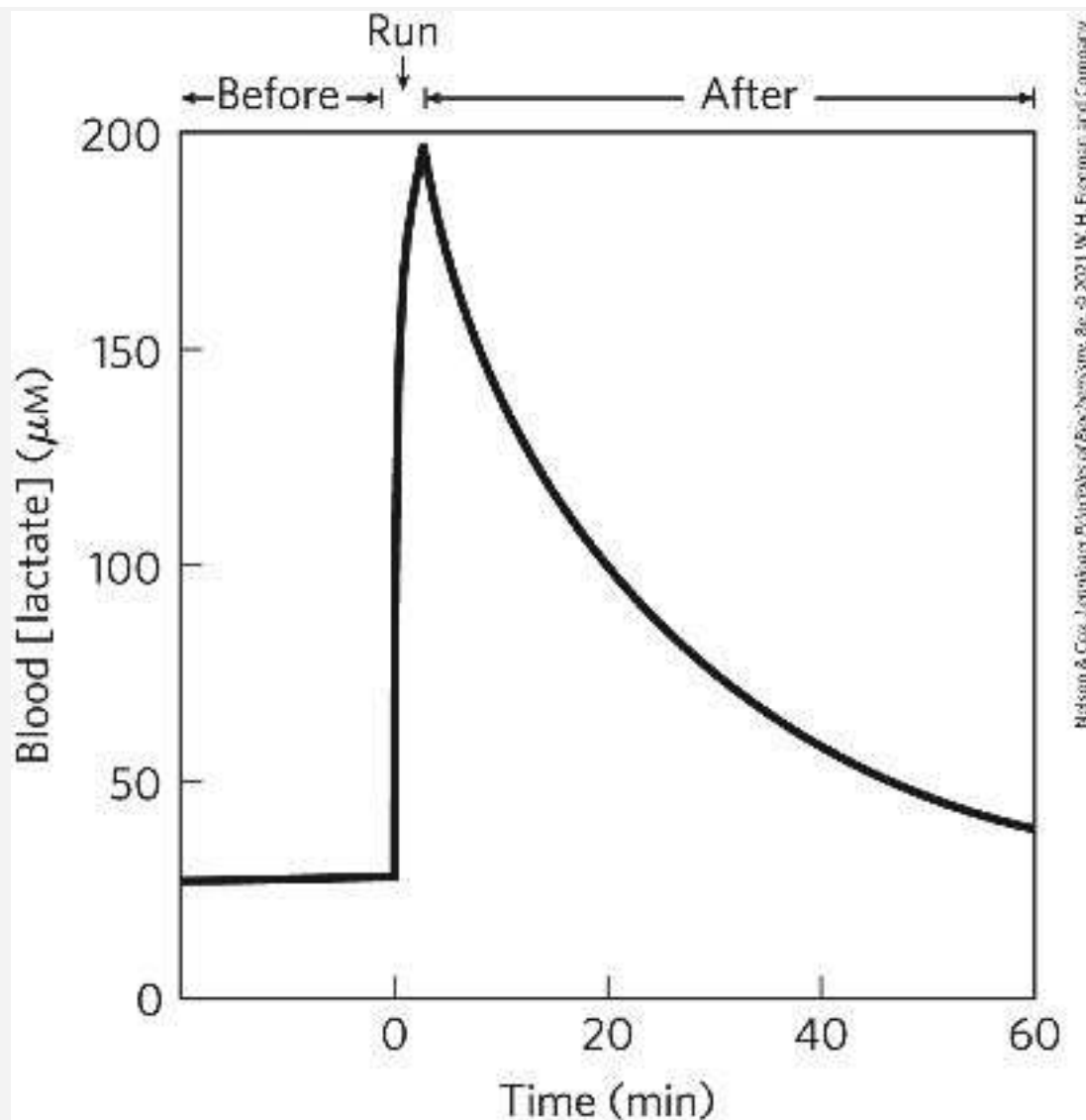
different severity. Although both types produce gastric discomfort after milk ingestion, deficiency of the transferase also leads to liver, kidney, spleen, and brain dysfunction and eventual death. What products accumulate in the blood and tissues with each type of enzyme deficiency? Estimate the relative toxicities of these products from the above information.

15.  Ethanol Affects Blood Glucose Levels The consumption of alcohol (ethanol), especially after periods of strenuous activity or after not eating for several hours, results in a deficiency of glucose in the blood, a condition known as hypoglycemia. The first step in the metabolism of ethanol by the liver is oxidation to acetaldehyde, catalyzed by liver alcohol dehydrogenase:



Explain how this reaction inhibits the transformation of lactate to pyruvate. Why does this lead to hypoglycemia?

16. Blood Lactate Levels during Vigorous Exercise The graph shows the concentrations of lactate in blood plasma before, during, and after a 400 m sprint.

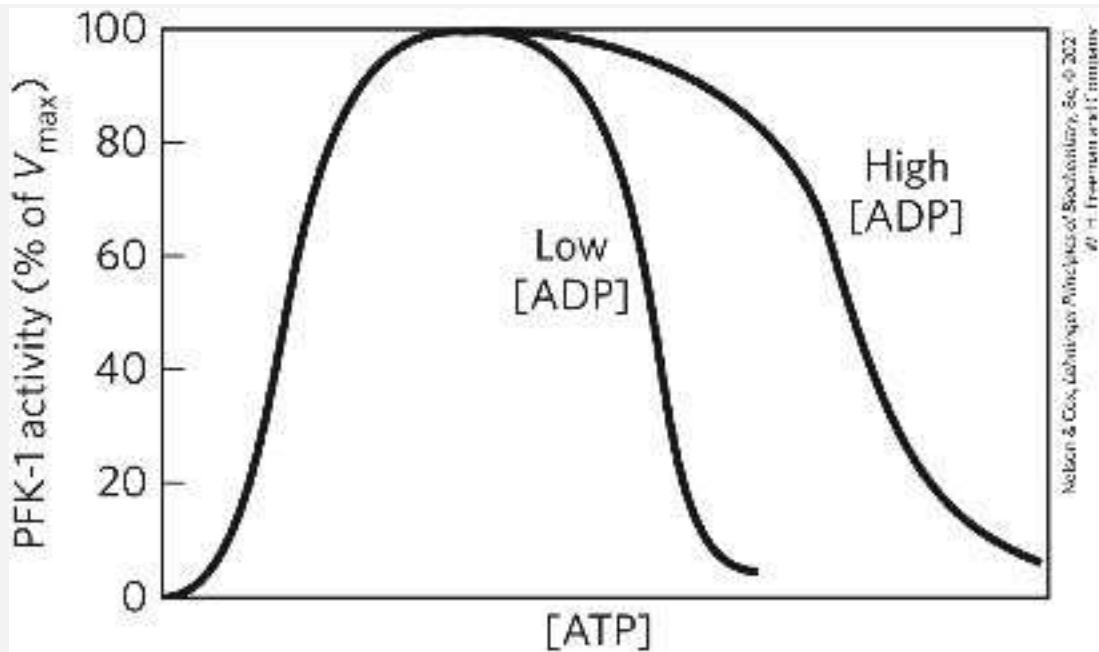


- What causes the rapid rise in lactate concentration?
- What causes the decline in lactate concentration after completion of the sprint? Why does the decline occur more slowly than the increase?
- Why is the concentration of lactate not zero during the resting state?

17. 🏥 Relationship between Fructose 1,6-Bisphosphatase and Blood Lactate Levels A congenital defect in the liver enzyme fructose 1,6-bisphosphatase results in abnormally high levels of lactate in the blood plasma. Explain.

18. Effect of O₂ Supply on Glycolytic Rates The regulated steps of glycolysis in intact cells can be identified by studying the catabolism of glucose in whole tissues or organs. For example, the glucose consumption by heart muscle can be measured by artificially circulating blood through an isolated intact heart and measuring the concentration of glucose before and after the blood passes through the heart. If the circulating blood is deoxygenated, heart muscle consumes glucose at a steady rate. When oxygen is added to the blood, the rate of glucose consumption drops dramatically, then is maintained at the new, lower rate. Explain.

19. Regulation of PFK-1 The graph shows the effect of ATP on the allosteric enzyme PFK-1. For a given concentration of fructose 6-phosphate, the PFK-1 activity increases with increasing concentrations of ATP, but there is a point beyond which increasing the concentration of ATP inhibits the enzyme.



- Explain how ATP can be both a substrate and an inhibitor of PFK-1. How is the enzyme regulated by ATP?
- How do ATP levels regulate glycolysis?
- The inhibition of PFK-1 by ATP diminishes when the ADP concentration is high, as shown in the graph. What explains this observation?

20. Cellular Glucose Concentration Homeostatic mechanisms maintain the concentration of glucose in human blood at about 5 mM. The concentration of free glucose inside a myocyte is much lower. Why is the concentration so low in the cell? What happens to glucose after entry into the cell? Physicians administer glucose intravenously as a food source in certain clinical situations. Given that the transformation of glucose to glucose 6-phosphate consumes ATP, why not administer intravenous glucose 6-phosphate instead?

21. Ethanol Production in Yeast When grown anaerobically on glucose, yeast (*S. cerevisiae*) converts pyruvate to acetaldehyde, then reduces acetaldehyde to ethanol using electrons from NADH. Write the equation for the second reaction, and calculate its equilibrium constant at 25 °C, given the standard reduction potentials in [Table 13-7](#).

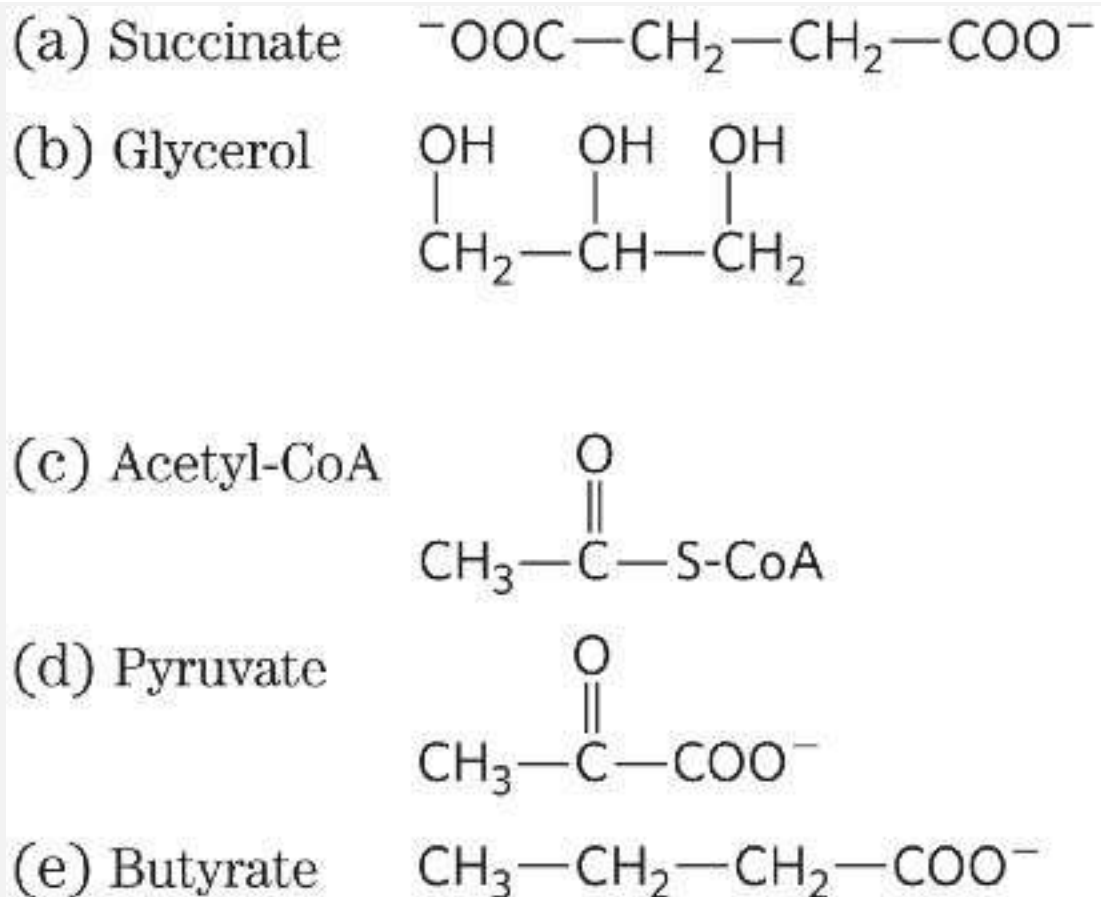
22. Pathway of Atoms in Fermentation An investigator carries out a “pulse-chase” experiment using ^{14}C -labeled carbon sources on a yeast extract maintained under strictly anaerobic conditions to produce ethanol. The experiment consists of incubating a small amount of ^{14}C -labeled substrate (the pulse) with the yeast extract just long enough for each intermediate in the fermentation pathway to become labeled. The addition of excess unlabeled glucose then “chases” the label through the pathway. The chase effectively prevents any further entry of labeled glucose into the pathway.

- a. If the investigator uses $[1\text{-}^{14}\text{C}]\text{glucose}$ (glucose labeled at C-1 with ^{14}C) as a substrate, what is the location of ^{14}C in the product ethanol? Explain.
- b. Where would ^{14}C have to be located in the starting glucose to ensure that all the ^{14}C activity is liberated as $^{14}\text{CO}_2$ during fermentation to ethanol? Explain.

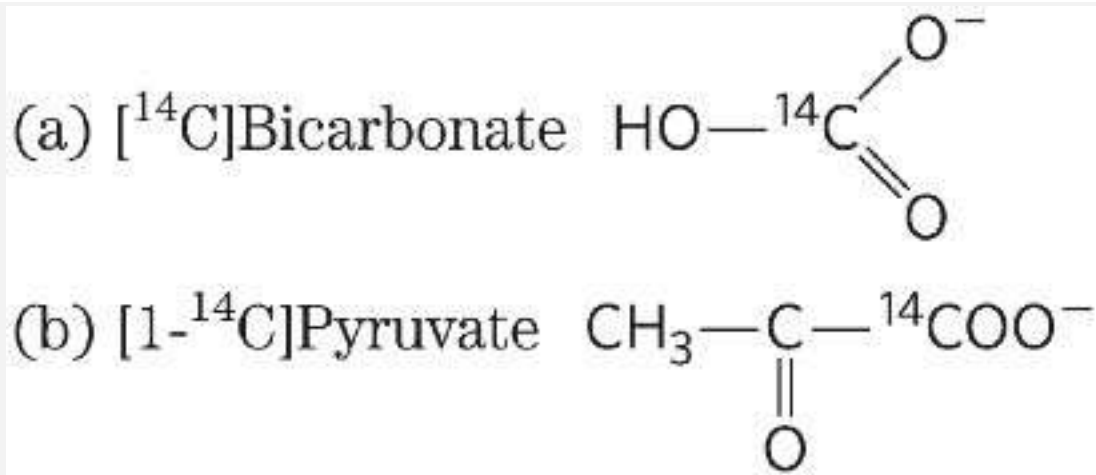
23. Heat from Fermentations Large-scale industrial fermenters generally require constant, vigorous cooling. Why?

24. Fermentation to Produce Soy Sauce Soy sauce preparation involves fermenting a salted mixture of soybeans and wheat with several microorganisms, including yeast, over a period of 8 to 12 months. The resulting sauce (after solids are removed) is rich in lactate and ethanol. How are these two compounds produced? To prevent the soy sauce from having a strong vinegary taste (vinegar is dilute acetic acid), oxygen must be kept out of the fermentation tank. Why?

25. Glucogenic Substrates A common procedure for determining the effectiveness of compounds as precursors of glucose in mammals is to starve the animal until the liver glycogen stores are depleted and then administer the compound in question. A substrate that leads to a *net* increase in liver glycogen is termed glucogenic, because it must first be converted to glucose 6-phosphate. Show by means of known enzymatic reactions which of these substances are glucogenic:



26. Pathway of Atoms in Gluconeogenesis An investigator briefly incubates a liver extract capable of carrying out all the normal metabolic reactions of the liver in separate experiments with two different ^{14}C -labeled precursors: $[^{14}\text{C}]$ bicarbonate and $[^{14}\text{C}]$ pyruvate.



Trace the pathway of each precursor through gluconeogenesis. Indicate the location of ^{14}C in all intermediates and in the product, glucose.

27. Energy Cost of a Cycle of Glycolysis and

Gluconeogenesis What is the cost (in ATP equivalents) of transforming glucose to pyruvate via glycolysis and back again to glucose via gluconeogenesis?

28. Relationship between Gluconeogenesis and Glycolysis

Why is it important that gluconeogenesis is not the exact reversal of glycolysis?

29. Energetics of the Pyruvate Kinase Reaction Explain in bioenergetic terms how the conversion of pyruvate to phosphoenolpyruvate in gluconeogenesis overcomes the large, negative, standard free-energy change of the pyruvate kinase reaction in glycolysis.

30. Muscle Wasting in Starvation One consequence of starvation is a reduction in muscle mass. What happens to

the muscle proteins?

31. Effect of Phloridzin on Carbohydrate Metabolism

Phloridzin, a toxic glycoside from the bark of the pear tree, blocks the normal reabsorption of glucose from the kidney tubule, thus causing blood glucose to be almost completely excreted in the urine. In an experiment, rats fed phloridzin and sodium succinate excreted about 0.5 mol of glucose (made by gluconeogenesis) for every 1 mol of sodium succinate ingested. How do rats transform the succinate to glucose? Explain the stoichiometry.

32. Excess O₂ Uptake during Gluconeogenesis

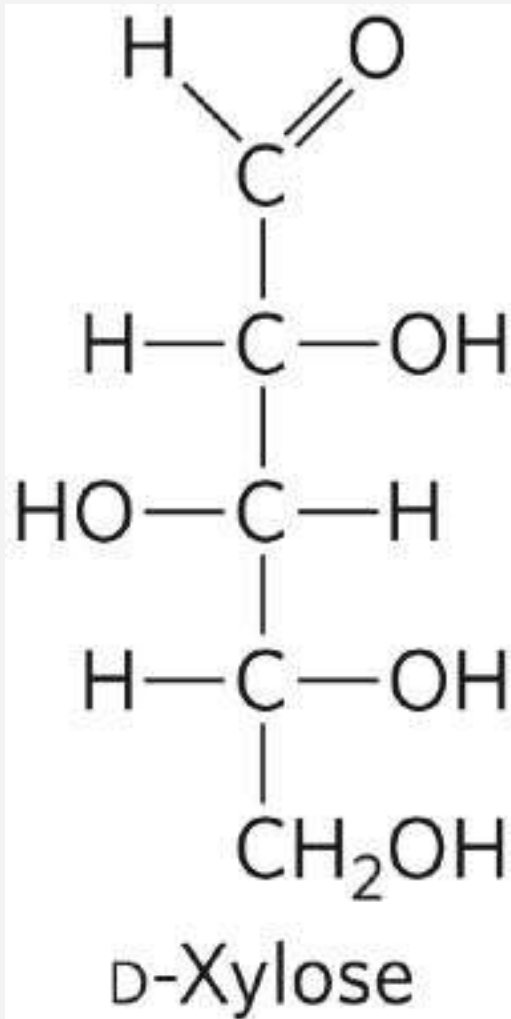
The conversion of lactate to glucose in the liver requires the input of 6 mol of ATP for every mol of glucose produced. Investigators can monitor the extent of this process in a rat liver preparation by administering [¹⁴C]lactate and measuring the amount of [¹⁴C]glucose produced. Because the stoichiometry of O₂ consumption and ATP production is known (about 5 ATP per O₂), investigators can predict the extra O₂ consumption above the normal rate after administering a given amount of lactate. However, when they actually measure extra O₂ used in the synthesis of glucose from lactate, it is always higher than what the stoichiometric relationships predict. Suggest a possible explanation for this observation.

33. Role of the Pentose Phosphate Pathway If the oxidation of glucose 6-phosphate via the pentose phosphate pathway

were being used primarily to generate NADPH for biosynthesis, the other product, ribose 5-phosphate, would accumulate. What problems might this cause?

DATA ANALYSIS PROBLEM

34. Engineering a Fermentation System Fermentation of plant matter to produce ethanol for fuel is one potential method for reducing the use of fossil fuels and thus the CO₂ emissions that lead to global warming. Many microorganisms can break down cellulose, then ferment the glucose to ethanol. However, many potential cellulose sources, including agricultural residues and switchgrass, also contain substantial amounts of arabinose, which is not as easily fermented.



Escherichia coli is capable of fermenting arabinose to ethanol, but it is not naturally tolerant of high ethanol levels, thus limiting its utility for commercial ethanol production.

Another bacterium, *Zymomonas mobilis*, is naturally tolerant of high levels of ethanol but cannot ferment arabinose.

[Deanda, Zhang, Eddy, and Picataggio \(1996\)](#) described their efforts to combine the most useful features of these two organisms by introducing the *E. coli* genes for the arabinose-metabolizing enzymes into *Z. mobilis*.

- a. Why is this a simpler strategy than the reverse: engineering *E. coli* to be more ethanol-tolerant?

Deanda and colleagues inserted five *E. coli* genes into the *Z. mobilis* genome: *araA*, coding for L-arabinose isomerase, which interconverts L-arabinose and L-ribulose; *araB*, L-ribulokinase, which uses ATP to phosphorylate L-ribulose at C-5; *araD*, L-ribulose 5-phosphate epimerase, which interconverts L-ribulose 5-phosphate and L-xylulose 5-phosphate; *talB*, transaldolase; and *tktA*, transketolase.

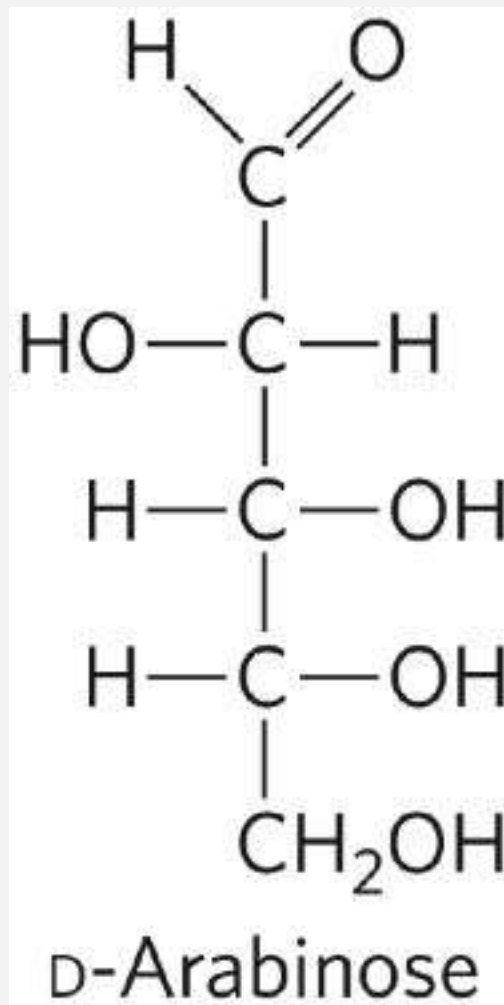
- b. For each of the three *ara* enzymes, briefly describe the chemical transformation it catalyzes and, where possible, name an enzyme discussed in this chapter that carries out an analogous reaction.

The five *E. coli* genes inserted in *Z. mobilis* allowed the entry of arabinose into the nonoxidative phase of the pentose phosphate pathway ([Fig. 14-31](#)), where it was converted to glucose 6-phosphate and fermented to ethanol.

- c. The three *ara* enzymes eventually converted arabinose into which sugar?
- d. The product from part (c) feeds into the pathway shown in [Figure 14-31a](#). Combining the five *E. coli* enzymes listed above with the enzymes of this pathway, describe the overall pathway for the fermentation of six molecules of arabinose to ethanol.
- e. What is the stoichiometry of the fermentation of six molecules of arabinose to ethanol and CO₂? How many ATP molecules would you expect this reaction to generate?

f. *Z. mobilis* uses a pathway for ethanol fermentation that is slightly different from the one described in this chapter. As a result, the expected ATP yield is only 1 ATP per molecule of arabinose. Although this is less beneficial for the bacterium, it is better for ethanol production. Why?

Another sugar commonly found in plant matter is xylose.



g. What additional enzymes would you need to introduce into the modified *Z. mobilis* strain described above to enable it to use xylose as well as arabinose to produce

ethanol? You don't need to name the enzymes (they may not even exist in the real world); just give the reactions they would need to catalyze.

Reference

Deanda, K., M. Zhang, C. Eddy, and S. Picataggio. 1996.

Development of an arabinose-fermenting *Zymomonas mobilis* strain by metabolic pathway engineering. *Appl. Environ. Microbiol.* :4465–4470.

CHAPTER 15

THE METABOLISM OF GLYCOGEN IN ANIMALS



[15.1 The Structure and Function of Glycogen](#)

[15.2 Breakdown and Synthesis of Glycogen](#)

[15.3 Coordinated Regulation of Glycogen Breakdown and Synthesis](#)

In [Chapter 14](#) we examined universal pathways by which hexoses are metabolized through glycolysis, fermentation, gluconeogenesis, and the pentose phosphate pathways to provide energy and components for the biosynthesis of amino acids, fats, and nucleotides. In this chapter, we focus more narrowly on the metabolism of glycogen, the polymeric storage form of glucose employed by animals. These are the principles that guide our discussion:



Glycogen provides vertebrate animals with a ready source of glucose to supply the brain and skeletal muscles with energy. Although animals store about 100 times more energy as fat than as glycogen, they cannot metabolize fat into glucose. The highly branched polymeric structure of glycogen

granules allows cells in the liver and muscle to make large numbers of glucose and glucose phosphate monomers available quickly, without raising the osmolarity of the cytosol by storing them in monomeric form.

P2 **Monomers are released from glycogen granules by a phosphorolysis reaction that creates phosphorylated glucose molecules that can enter glycolysis to supply energy to the cell.** Skeletal muscle cells especially require stores of glycogen to supply energy for bursts of activity. In the liver, the phosphate can be removed, allowing free glucose to be transported out of the cell to the blood for use in the brain and other tissues when dietary glucose is not sufficient.

P3 **Glycogen synthesis requires a protein primer and an activated glucose precursor.** Individual glucose molecules activated as sugar nucleotides are added to the nonreducing end of the growing linear chains in the outer tiers of the glycogen β -granules, and a branching enzyme adds branches periodically.

P4 **Regulation of the balance between the formation of glycogen from excess glucose and the release of glucose from glycogen polymers when it is needed in metabolism is a critical function of cellular and organismal homeostasis.** This balance, ultimately controlled by the hormones epinephrine, glucagon, and insulin, is achieved through allosteric regulation and phosphorylation of the synthetic and degradative enzymes. These enzymes, and the regulatory

proteins that act on them, are integral parts of the glycogen granule.

Glycogen was discovered in the mid-1800s by Claude Bernard. The French physiologist also found that a liver “ferment” (enzyme) released a reducing sugar from liver tissue. He named this reducing sugar *mati re glycog ne* — sugar-forming substance. In the first half of the twentieth century, scientists in laboratories around the world followed up this early work, purifying the “ferments” that synthesize and degrade glycogen and characterizing the regulation of these enzymes by insulin and epinephrine. These studies characterized the enzymes and also uncovered multiple regulatory mechanisms that proved to be universal: second messengers responsive to extracellular signals, protein kinase cascades, and protein phosphorylation, for example. In this chapter, we begin by exploring the structure and function of glycogen particles, describe the pathways of glycogen breakdown and synthesis, and finally, dig into the complex web of regulatory controls that exquisitely deliver the necessary amount of energy from glucose that each organ system requires to function in the moment.

15.1 The Structure and Function of Glycogen

Structure–function relationships are key to understanding biomolecules and biochemical systems, and glycogen is no exception. The compact structure of glycogen granules allows cells to store glucose when it is available in excess, and to make it available on short notice when needed. Subtle tissue-specific differences in the enzymes (isozymes) that act on glycogen determine the dynamics of glycogen metabolism in each tissue.

Vertebrate Animals Require a Ready Fuel Source for Brain and Muscle

 For all vertebrate animals, maintaining a ready supply of glucose for the brain and muscles is a top metabolic priority. The challenge for cells is to be able to store glucose in a form that rapidly sequesters it when the glucose concentration in the blood is high (say, after a meal) but allows it to be accessed quickly for use particularly by the brain and skeletal muscle. Recall from our discussion in [Chapter 7 \(p. 242\)](#) that a typical hepatocyte in the fed state stores an amount of glucose, polymerized as glycogen, that in monomeric form would be equivalent to about 0.4 M. At this concentration, the osmolarity of the cell would be so high relative to the surrounding fluid that water would enter the cell and likely rupture it.

When the diet temporarily provides more carbohydrate than is needed immediately as fuel, excess glucose is polymerized into glycogen. Small amounts of glycogen are present in all animal cells, but it is stored primarily in liver and muscle, where it is a significant portion of the wet weight of the organ (5% to 10% of liver and 1% to 2% of muscle). A 70 kg human stores about 100 g of glycogen in the liver and up to 400 g in skeletal muscle. The total amount of energy stored in the body as glycogen is far less (about 1%) than the amount stored as fat (triacylglycerol), but fats cannot be converted to glucose in vertebrates and cannot be catabolized anaerobically through glycolysis, as is often required in skeletal muscle.



When a sudden burst of physical activity demands a quick source of energy in muscle, the rapid breakdown of glycogen stored there provides glucose for glycolysis within seconds. Between meals or during a fast, the release of glucose from glycogen stored in the liver provides a steady supply of glucose in the blood. This is especially important for the brain, a major consumer of metabolic energy, which, unlike muscle, cannot use fatty acids as fuel; long-chain fatty acids do not cross the blood-brain barrier. The brain therefore depends on a constant supply of glucose, from the diet or from the liver.

Glycogen Granules Have Many Tiers of Branched Chains of D-Glucose

Glycogen is stored as cytosolic granules, called β -granules, that vary in size, structure, and subcellular location depending on the tissue or cell type. (In this chapter we focus on liver and muscle.) The size of β -granules also varies with the state of activity and feeding of the animal. In muscle, β -granules are 20–30 nm in diameter and have an M_r of $10^6 - 10^7$. They consist of up to 55,000 glucose residues with about 2,000 nonreducing ends available for degradative enzymes to work on. In liver, 20 to 40 β -granules cluster together to form protein-rich α -granules as large as 300 nm in diameter and of M_r greater than 10^8 . They are visible with the electron microscope in tissue samples from well-fed animals ([Fig. 15-1](#)), but they are essentially absent after a 24-hour fast. The β -granules of muscle release glucose more quickly than the α -granules of liver, consistent with the different needs of these tissues for glucose.

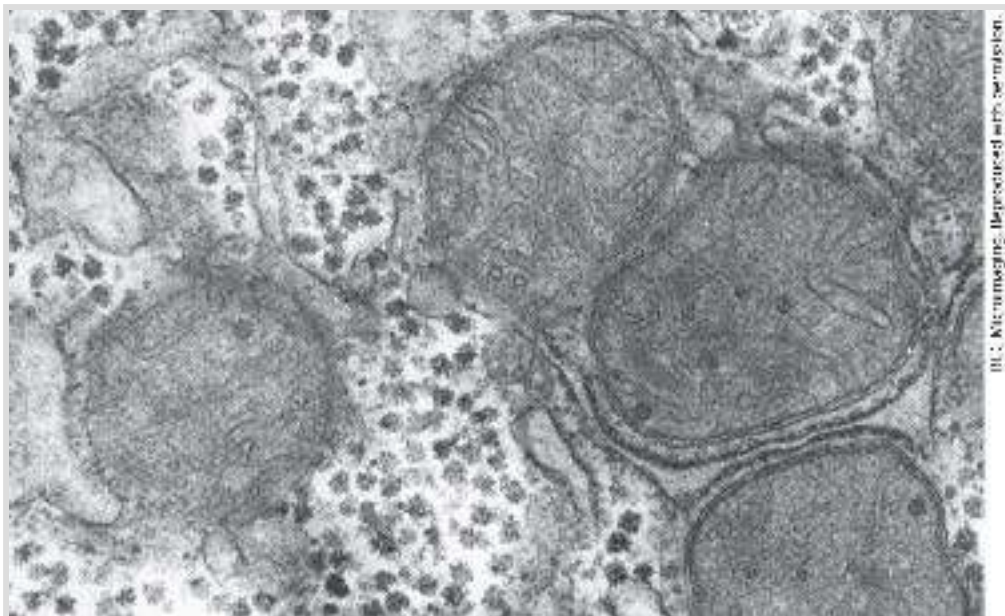
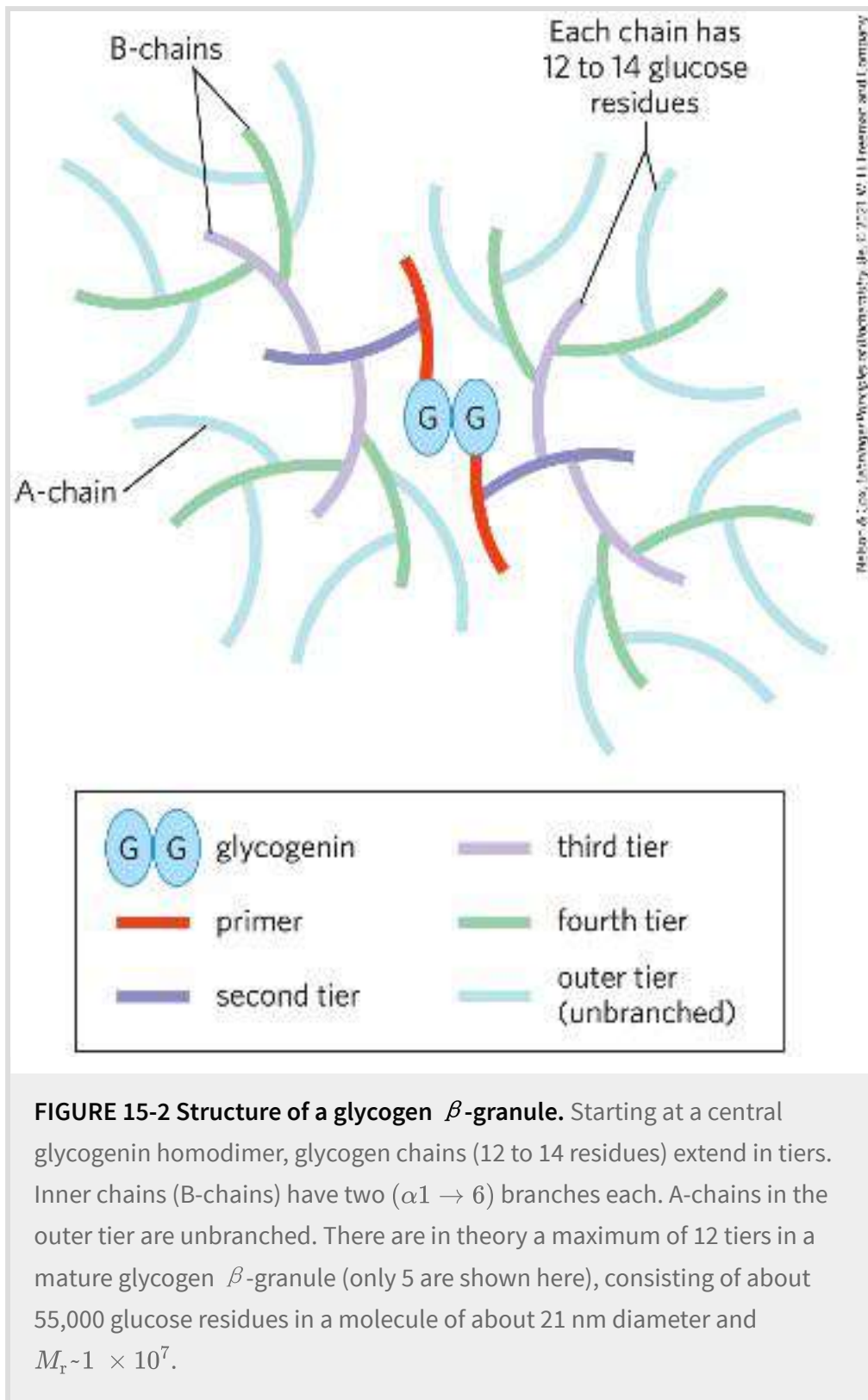


FIGURE 15-1 Glycogen granules in a hepatocyte. Glycogen β -granules appear as electron-dense particles. In liver they form larger clusters called α -granules and are often associated with tubules of the smooth

endoplasmic reticulum. Four mitochondria are also evident in this micrograph.

All glycogen granules have at their core a dimer of the protein glycogenin, which serves as a primer for the synthesis of polymers of D-glucose. In the tiered β -granule model, the central glycogenin dimer is surrounded by tier upon tier of chains of about 13 glucose residues in ($\alpha 1 \rightarrow 4$) linkage, with ($\alpha 1 \rightarrow 6$)-linked branches. Inner B-chains contain two branch points, and outer A-chains are unbranched ([Fig. 15-2](#)). Granules typically have 6 or 7 tiers, with the outermost tier of unbranched A-chains making up the majority of the granule. Associated with each β -granule are patches of electron-dense, protein-rich material, called γ -particles. Among the associated proteins are the enzymes that synthesize and break down glycogen.



The general mechanisms for storing and mobilizing glycogen are the same in muscle and liver, but the enzymes involved differ in

subtle yet important ways that reflect the different roles of glycogen in the two tissues. In the next two sections we will look at the enzymatic basis for glycogen synthesis and breakdown, and at the regulation of these processes.

SUMMARY 15.1 *The Structure and Function of Glycogen*

■ All cells need ready access to glucose, either from the diet or from supplies stored in cells. Glycogen is a polymeric storage form of glucose in animals that is found primarily in muscle and liver. Glycogen breakdown in muscle delivers glucose needed for muscle contraction. Glycogen stored in the liver provides a reservoir that maintains homeostasis of blood glucose throughout the body.

■ Glycogen β -granules have tiers of glucose residues in ($\alpha 1 \rightarrow 4$) linkage, with ($\alpha 1 \rightarrow 6$)-linked branches, providing many free nonreducing ends for synthetic and degradative enzymes to access. In liver, β -granules cluster into larger α -granules, which release glucose more slowly.

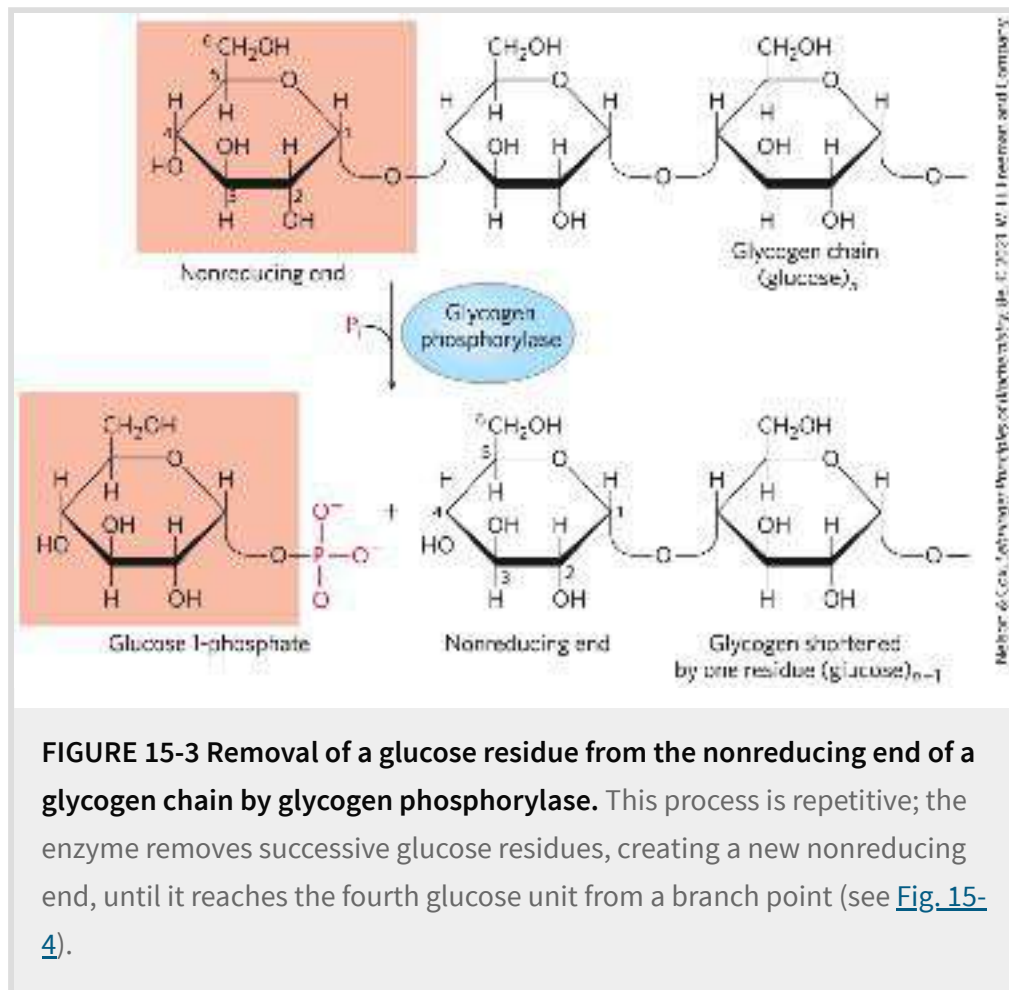
15.2 Breakdown and Synthesis of Glycogen

As we saw in [Chapter 14 \(Fig. 14-9\)](#), glycogen obtained in the diet is broken down by α -amylases, hydrolytic enzymes that act in the mouth and gut to convert glycogen to free glucose. (Dietary starch is hydrolyzed in a similar way.) But glycogen stored in cells (endogenous glycogen) is degraded by a different pathway. We begin with the breakdown of cellular glycogen to glucose 1-phosphate ([glycogenolysis](#)), then turn to synthesis of glycogen ([glycogenesis](#)).

Glycogen Breakdown Is Catalyzed by Glycogen Phosphorylase

 In skeletal muscle and liver, the glucose units of the outer branches of glycogen enter the glycolytic pathway through the action of three enzymes: glycogen phosphorylase, glycogen debranching enzyme, and phosphoglucomutase. Glycogen phosphorylase catalyzes the reaction in which an ($\alpha 1 \rightarrow 4$) glycosidic linkage between two glucose residues at a nonreducing end of glycogen undergoes attack by inorganic phosphate (P_i), removing the terminal glucose residue as α -D-glucose 1-phosphate ([Fig. 15-3](#)). This *phosphorolysis* reaction is different from the *hydrolysis* of glycosidic bonds by amylase during intestinal degradation of dietary glycogen and starch. In phosphorolysis, some of the energy of the glycosidic bond is

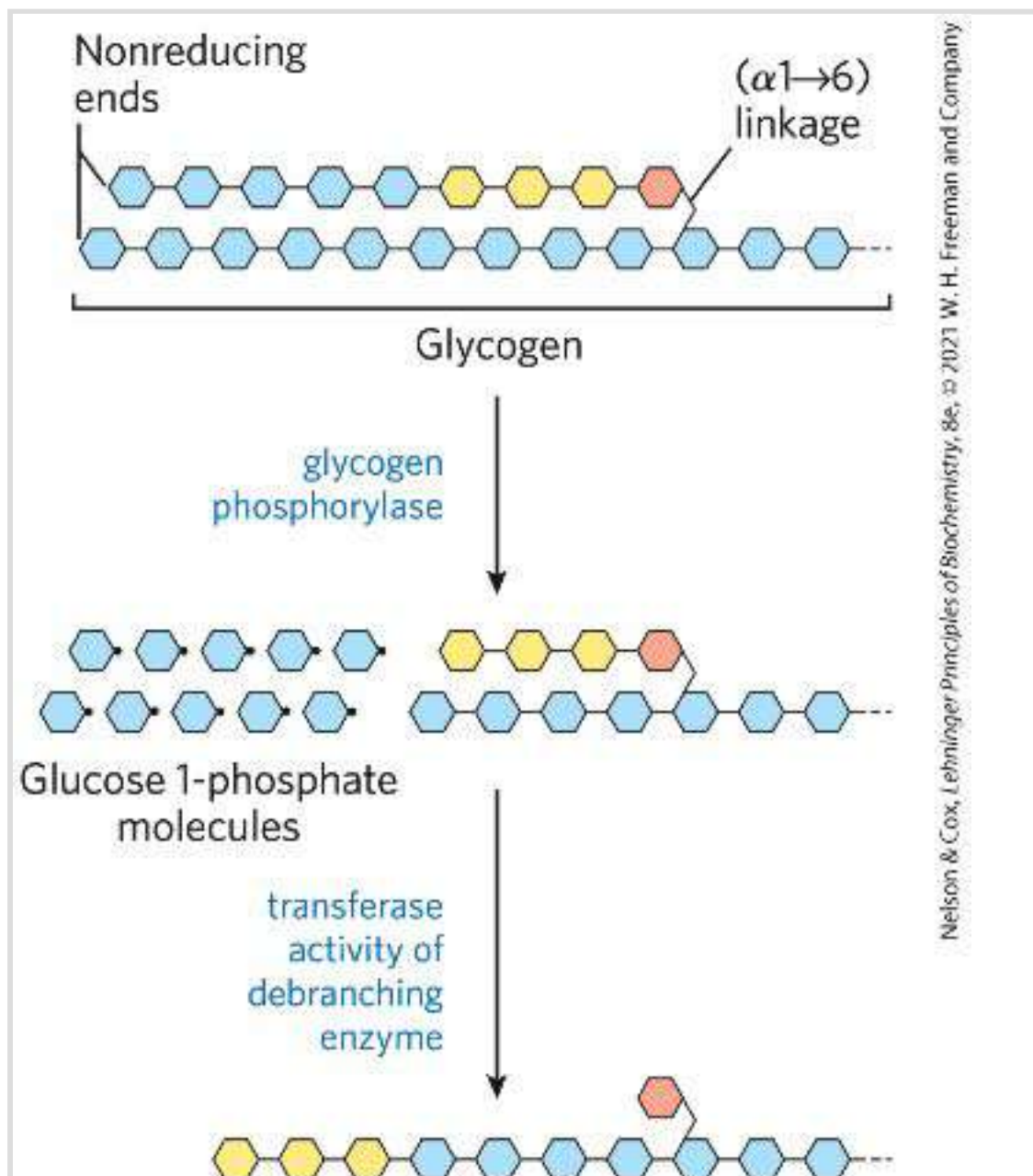
conserved in the formation of the phosphate ester glucose 1-phosphate.



Pyridoxal phosphate is an essential cofactor in the glycogen phosphorylase reaction. It is covalently attached near the enzyme active site, where its phosphate group acts as a general acid catalyst, promoting attack by P_i on the glycosidic bond. (This is an unusual role for pyridoxal phosphate; its more typical role is as a cofactor in amino acid metabolism; see [Fig. 18-6](#).)

Glycogen phosphorylase acts repetitively on the nonreducing ends of glycogen branches until it reaches a point four glucose

residues away from an ($\alpha 1 \rightarrow 6$) branch point, where its action stops. Further degradation by glycogen phosphorylase can occur only after the **debranching enzyme**, formally known as oligo ($\alpha 1 \rightarrow 6$) to ($\alpha 1 \rightarrow 4$) glucantransferase, catalyzes two successive reactions that transfer branches ([Fig. 15-4](#)) to form straight chains. Once these branches are transferred and the glucosyl residue at C-6 is hydrolyzed, glycogen phosphorylase activity can continue.



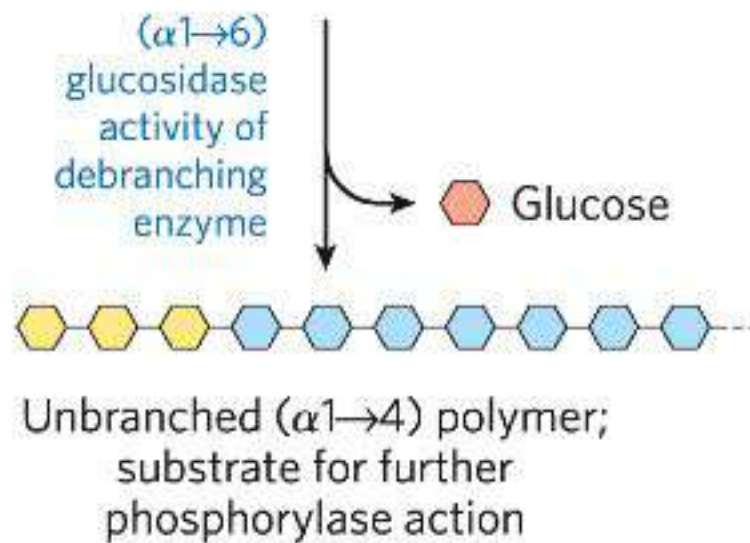
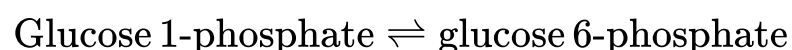


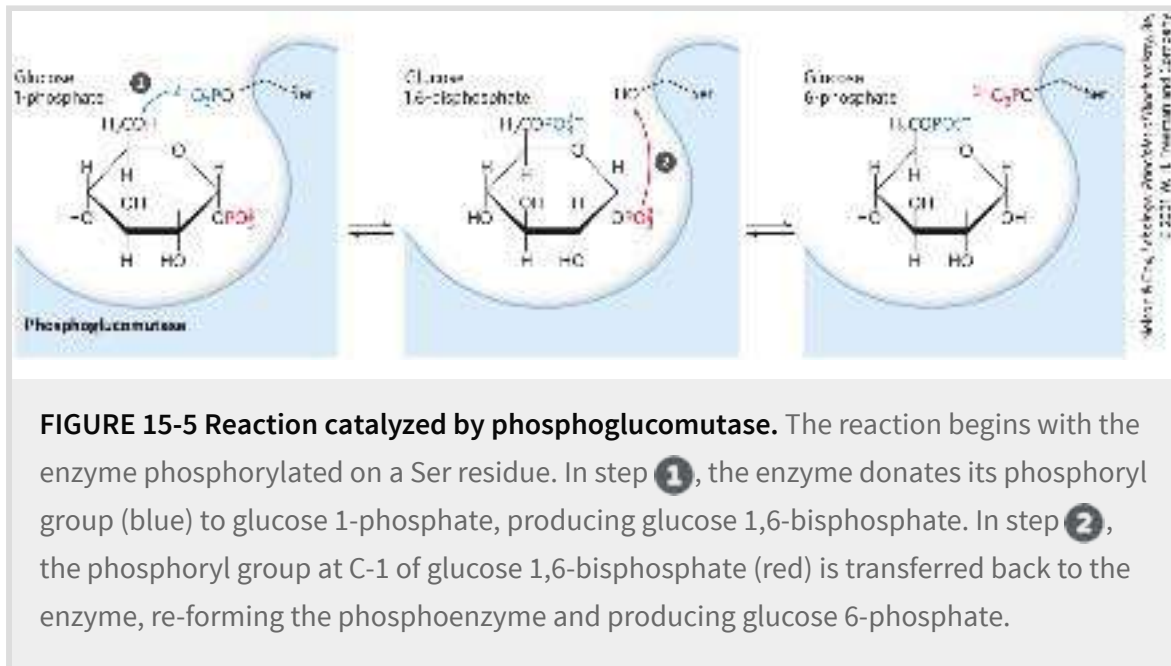
FIGURE 15-4 Glycogen breakdown near an ($\alpha 1 \rightarrow 6$) branch point. Following sequential removal of terminal glucose residues by glycogen phosphorylase (see [Fig. 15-3](#)), glucose residues near a branch are removed in a two-step process that requires a bifunctional debranching enzyme. First, the transferase activity of the enzyme shifts a block of three glucose residues from the branch to a nearby nonreducing end, to which the segment is reattached in ($\alpha 1 \rightarrow 4$) linkage. The single glucose residue remaining at the branch point, in ($\alpha 1 \rightarrow 6$) linkage, is then released as free glucose by the ($\alpha 1 \rightarrow 6$) glucosidase activity of the debranching enzyme. The glucose residues are shown in shorthand form.

Glucose 1-Phosphate Can Enter Glycolysis or, in Liver, Replenish Blood Glucose

Glucose 1-phosphate, the end product of the glycogen phosphorylase reaction, is converted to glucose 6-phosphate by **phosphoglucomutase**, which catalyzes the reversible reaction:

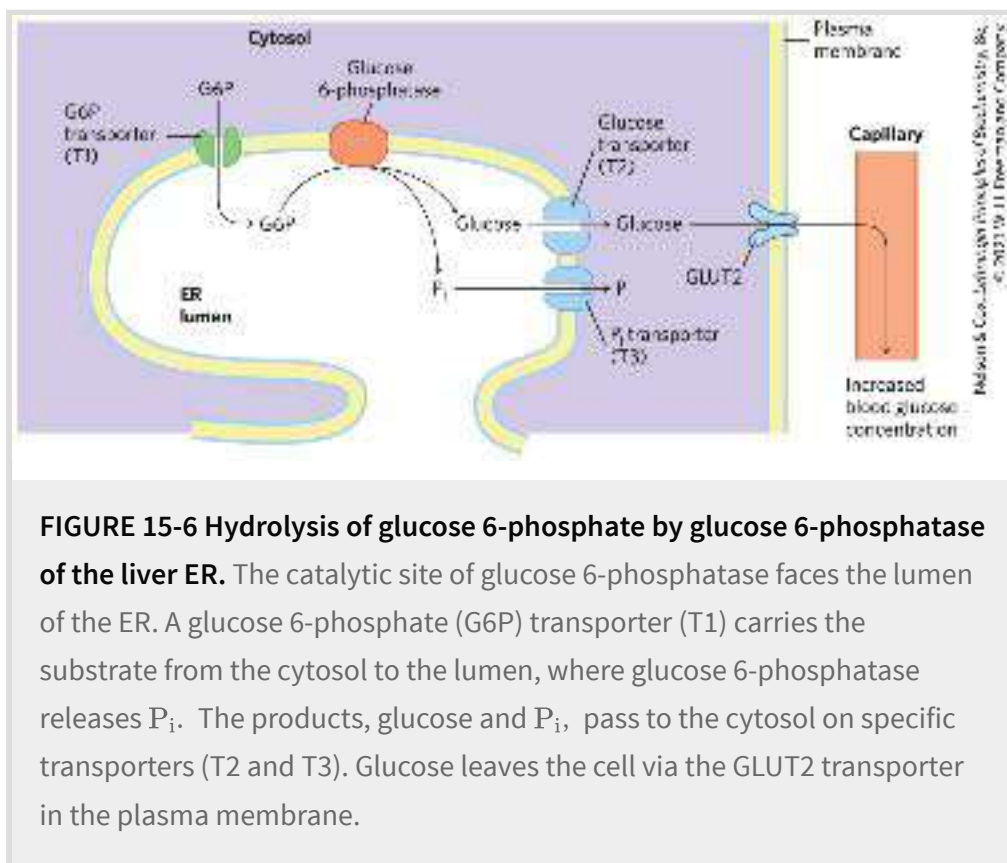


Initially phosphorylated at a Ser residue, the enzyme donates its phosphoryl group to C-6 of the substrate, then accepts a phosphoryl group from C-1 ([Fig. 15-5](#)).



P2 The glucose 6-phosphate formed from glycogen in skeletal muscle can enter glycolysis and serve as an energy source to support muscle contraction. In liver, glycogen breakdown serves a different purpose: to release glucose into the blood when the blood glucose level drops, as it does between meals. This requires the enzyme glucose 6-phosphatase, present in liver and kidney but not in other tissues. The enzyme is an integral protein of the endoplasmic reticulum, with its active site on the luminal side of the ER. Glucose 6-phosphate formed in the cytosol is transported into the ER lumen by a specific transporter (T1) ([Fig. 15-6](#)) and hydrolyzed at the luminal surface by glucose 6-phosphatase. The

resulting P_i and glucose are carried back into the cytosol by two different transporters (T2 and T3), and the glucose enters the blood via the plasma membrane transporter, GLUT2. Notice that by having the active site of glucose 6-phosphatase in the ER lumen, the cell separates this reaction from the process of glycolysis, which takes place in the cytosol and would be aborted by the action of glucose 6-phosphatase. Genetic defects in either glucose 6-phosphatase or T1 lead to serious derangement of glycogen metabolism, resulting in type Ia glycogen storage disease ([Box 15-1](#)).



BOX 15-1 MEDICINE

Carl and Gerty Cori: Pioneers in Glycogen Metabolism and Disease

Much of what is written in present-day biochemistry textbooks about the metabolism of glycogen was discovered between about 1925 and 1950 by the remarkable husband and wife team of Carl F. Cori and Gerty T. Cori. Both trained in medicine in Europe at the end of World War I (she completed premedical studies and medical school in one year!). They left Europe together in 1922 to establish research laboratories in the United States, first for nine years in Buffalo, New York, at what is now the Roswell Park Comprehensive Cancer Center, then from 1931 until the end of their lives at Washington University in St. Louis.

In their early physiological studies of the origin and fate of glycogen in animal muscle, the Coris demonstrated the conversion of glycogen to lactate in tissues; movement of lactate in the blood to the liver; and, in the liver, reconversion of lactate to glycogen — a pathway that came to be known as the Cori cycle (see [Fig. 23-17](#)). Pursuing these observations at the biochemical level, they showed that glycogen was mobilized in a phosphorolysis reaction catalyzed by the enzyme they discovered, glycogen phosphorylase. They identified the product of this reaction (the “Cori ester”) as glucose 1-phosphate and showed that it could be reincorporated into glycogen in the reverse reaction. Although this did not prove to be the reaction by which glycogen is synthesized in cells, it was the first in vitro demonstration of the synthesis of a macromolecule from simple monomeric subunits, and it inspired others to search for polymerizing enzymes. Arthur Kornberg, discoverer of the first DNA polymerase, said of his experience in the Coris’s lab, “Glycogen phosphorylase, not base pairing, was what led me to DNA polymerase.”



FIGURE 1 The Coris in Gerty Cori’s laboratory, around 1947.

Gerty Cori became interested in human genetic diseases in which too much glycogen is stored in the liver. She was able to identify the biochemical defect in several of these diseases and to show that the diseases could be diagnosed by assays of the enzymes of glycogen metabolism in small samples of tissue obtained by biopsy. [Table 1](#) summarizes what we now know about 13 genetic diseases of this sort.

TABLE 1 Glycogen Storage Diseases of Humans

Type (name)	Enzyme affected	Primary organ/cells affected	Symptoms

Type 0	Glycogen synthase	Liver	Low blood glucose, high ketone bodies, early death
Type Ia (von Gierke)	Glucose 6-phosphatase	Liver	Enlarged liver, kidney failure
Type Ib	Microsomal glucose 6-phosphate translocase	Liver	As in type Ia; also high susceptibility to bacterial infections
Type Ic	Microsomal P_i transporter	Liver	As in type Ia
Type II (Pompe)	Lysosomal glucosidase	Skeletal and cardiac muscle	Infantile form: death by age 2; juvenile form: muscle defects (myopathy); adult form: as in muscular dystrophy
Type IIIa (Cori or Forbes)	Debranching enzyme	Liver, skeletal and cardiac muscle	Enlarged liver in infants; myopathy
Type IIIb	Liver debranching enzyme (muscle enzyme normal)	Liver	Enlarged liver in infants
Type IV (Andersen)	Branching enzyme	Liver, skeletal muscle	Enlarged liver and spleen,

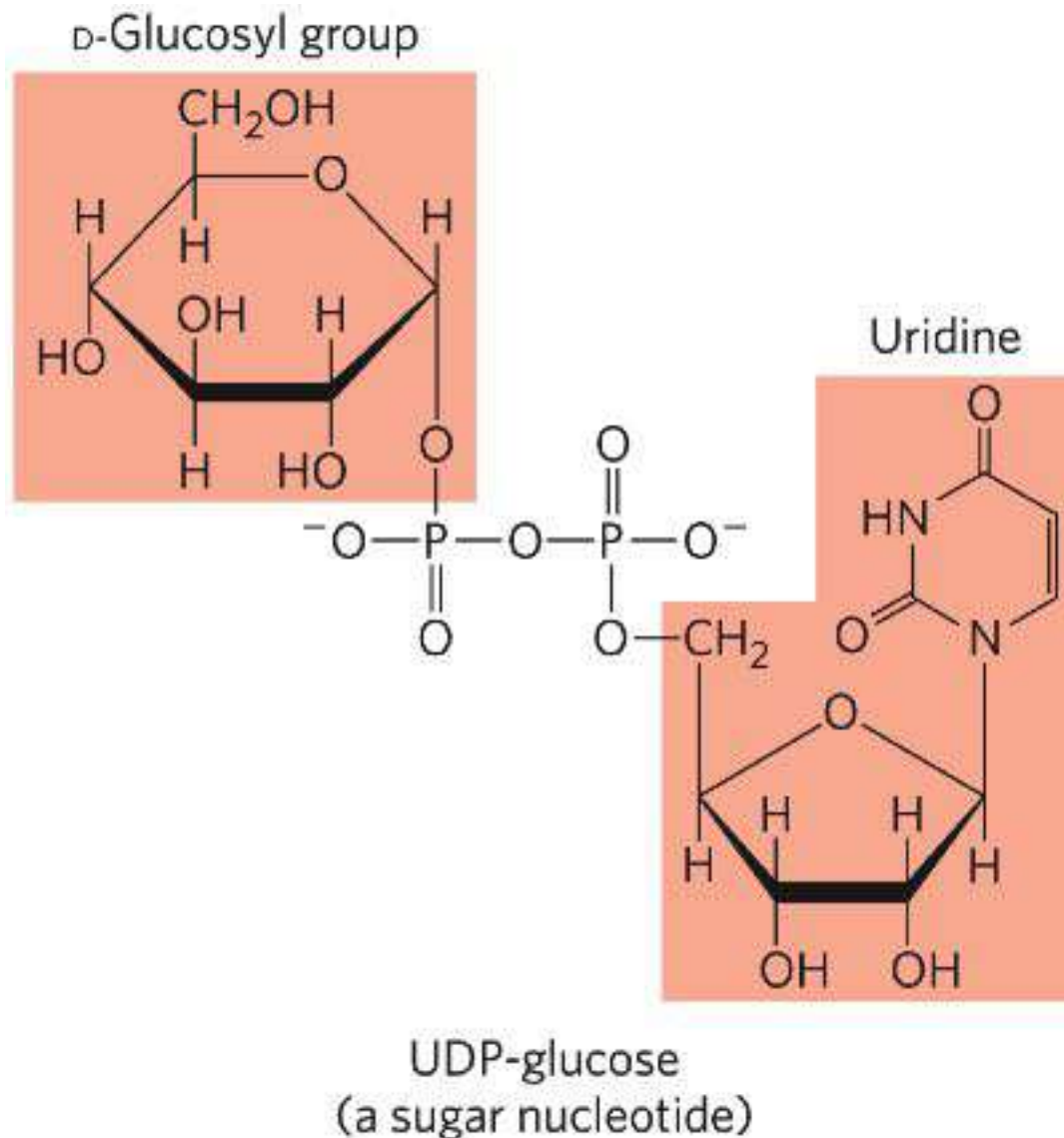
			myoglobin in urine
Type V (McArdle)	Muscle phosphorylase	Skeletal muscle	Exercise-induced cramps and pain; myoglobin in urine
Type VI (Hers)	Liver phosphorylase	Liver	Enlarged liver
Type VII (Tarui)	Muscle PFK-1	Muscle, erythrocytes	As in type V; also hemolytic anemia
Type VIb, VIII, or IX	Phosphorylase kinase	Liver, leukocytes, muscle	Enlarged liver
Type XI (Fanconi-Bickel)	Glucose transporter (GLUT2)	Liver	Failure to thrive, enlarged liver, rickets, kidney dysfunction

Carl and Gerty Cori shared the Nobel Prize in Physiology or Medicine in 1947 with Bernardo Houssay of Argentina, who was cited for his studies of hormonal regulation of carbohydrate metabolism. The Cori laboratories in St. Louis became an international center of biochemical research in the 1940s and 1950s, and at least six scientists who trained with the Coris became Nobel laureates: Arthur Kornberg (for DNA synthesis, 1959), Severo Ochoa (for RNA synthesis, 1959), Luis Leloir (for the role of sugar nucleotides in polysaccharide synthesis, 1970), Earl Sutherland (for the discovery of cAMP in the regulation of carbohydrate metabolism, 1971), Christian de Duve (for subcellular fractionation, 1974), and Edwin Krebs (for the discovery of phosphorylase kinase, 1991).

Because muscle and adipose tissue lack glucose 6-phosphatase, they cannot convert the glucose 6-phosphate formed by glycogen breakdown to glucose, and these tissues therefore do not contribute glucose to the blood.

The Sugar Nucleotide UDP-Glucose Donates Glucose for Glycogen Synthesis

Many of the reactions in which hexoses are transformed or polymerized involve **sugar nucleotides**, compounds in which the anomeric carbon of a sugar is activated by attachment to a nucleotide through a phosphate ester linkage. Sugar nucleotides are the substrates for polymerization of monosaccharides into disaccharides, glycogen, starch, cellulose, and more complex extracellular polysaccharides. They are also key intermediates in the production of the aminohexoses and deoxyhexoses found in some of these polysaccharides, and in the synthesis of vitamin C (L-ascorbic acid). The role of sugar nucleotides in the biosynthesis of glycogen and many other carbohydrate derivatives was discovered in 1953 by the Argentine biochemist Luis Leloir.



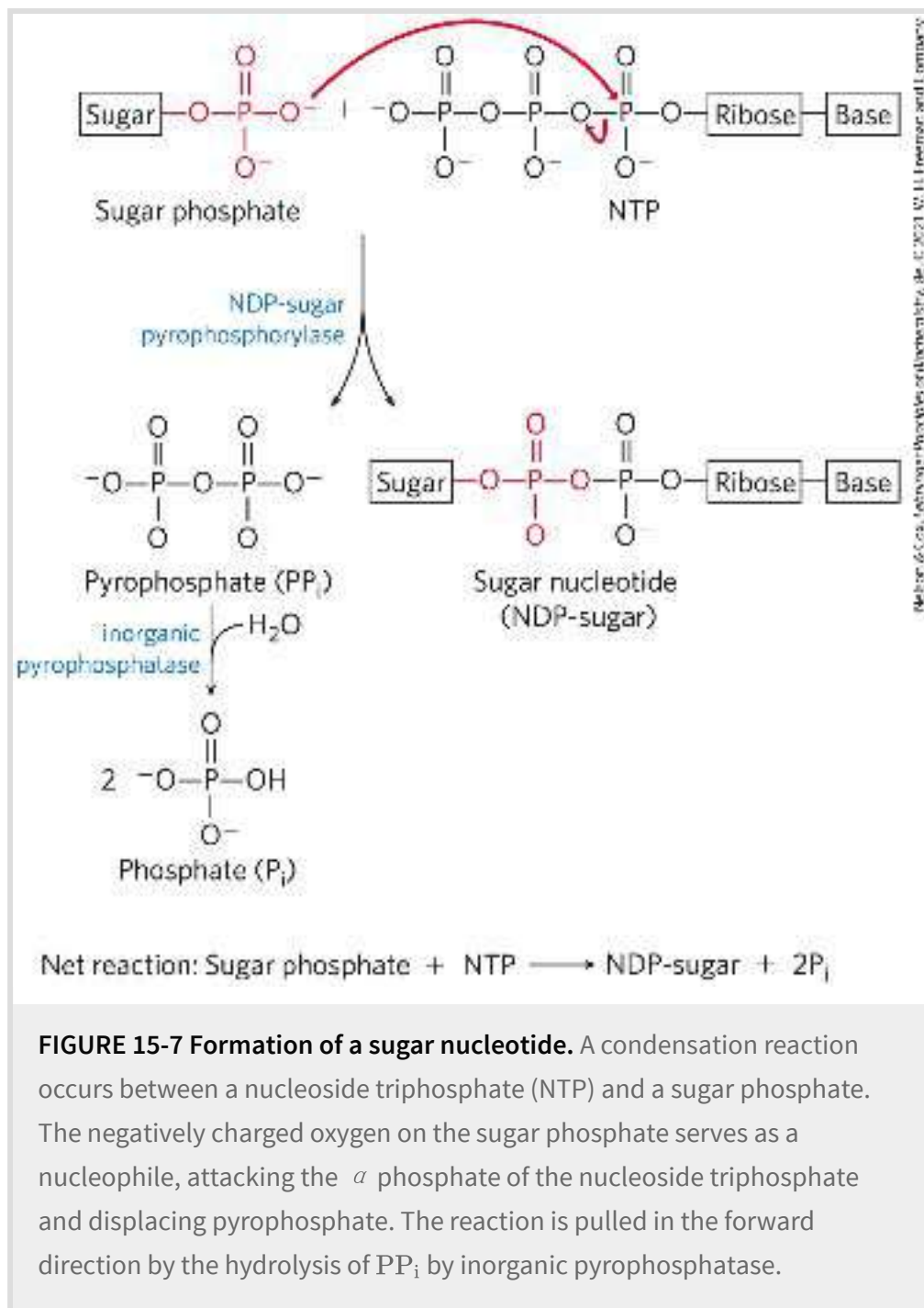
Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021 W. H. Freeman and Company

P3 The suitability of sugar nucleotides for biosynthetic reactions stems from several properties:

1. Their formation is metabolically irreversible, contributing to the irreversibility of the synthetic pathways in which they are intermediates. The condensation of a nucleoside triphosphate with a hexose 1-phosphate to form a sugar nucleotide has a small positive free-energy change, but the reaction releases PP_i , which is immediately hydrolyzed by inorganic pyrophosphatase ([Fig. 15-7](#)), in a reaction that is

strongly exergonic ($\Delta G'^{\circ} = -19.2 \text{ kJ/mol}$). This keeps the cellular concentration of PP_i low, ensuring that the actual free-energy change in the cell is favorable. In effect, rapid removal of the product, driven by the large, negative free-energy change of PP_i hydrolysis, pulls the synthetic reaction forward. This is a common strategy in biological polymerization reactions.

2. Although the chemical transformations of sugar nucleotides do not involve the atoms of the nucleotide itself, the nucleotide moiety has many groups that can undergo noncovalent interactions with enzymes; the additional free energy of binding can contribute significantly to catalytic activity ([Chapter 6](#); see also [p. 294](#)).
3. Like phosphate, the nucleotidyl group (UDP or ADP, for example) is an excellent leaving group, facilitating nucleophilic attack by activating the sugar carbon to which it is attached.
4. By “tagging” some hexoses with nucleotidyl groups, cells can set them aside in a pool for a particular purpose (glycogen synthesis, for example), separate from hexose phosphates destined for another purpose (such as glycolysis).



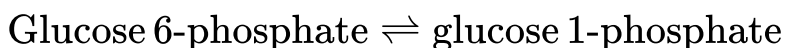
Glycogen synthesis takes place in virtually all animal tissues but is especially prominent in the liver and skeletal muscles. The starting point for synthesis of glycogen is glucose 6-phosphate. This can be derived from free glucose in a reaction catalyzed by

the isozymes hexokinase I and II in muscle and hexokinase IV (glucokinase) in liver:



However, some ingested glucose takes a more roundabout path to glycogen. It is first taken up by erythrocytes and converted to lactate via glycolysis; the lactate is then taken up by the liver and converted to glucose 6-phosphate by gluconeogenesis.

To start glycogen synthesis, the glucose 6-phosphate is converted to glucose 1-phosphate in the phosphogluco-mutase reaction:



The product is then converted to UDP-glucose by the action of **UDP-glucose pyrophosphorylase**, in a key step of glycogen biosynthesis:



Notice that this enzyme is named for the reverse reaction; in the cell, the reaction proceeds in the direction of UDP-glucose formation, because the pyrophosphate concentration is kept low

by its immediate hydrolysis by inorganic pyrophosphatase ([Fig. 15-7](#)).

P3 UDP-glucose is the immediate donor of glucose residues in the reaction catalyzed by **glycogen synthase**, which promotes the transfer of the glucose residue from UDP-glucose to a nonreducing end of a branched glycogen molecule, forming an ($\alpha 1 \rightarrow 4$) linkage ([Fig. 15-8](#)). The overall equilibrium of the path from glucose 6-phosphate to glycogen lengthened by one glucose unit greatly favors synthesis of glycogen.

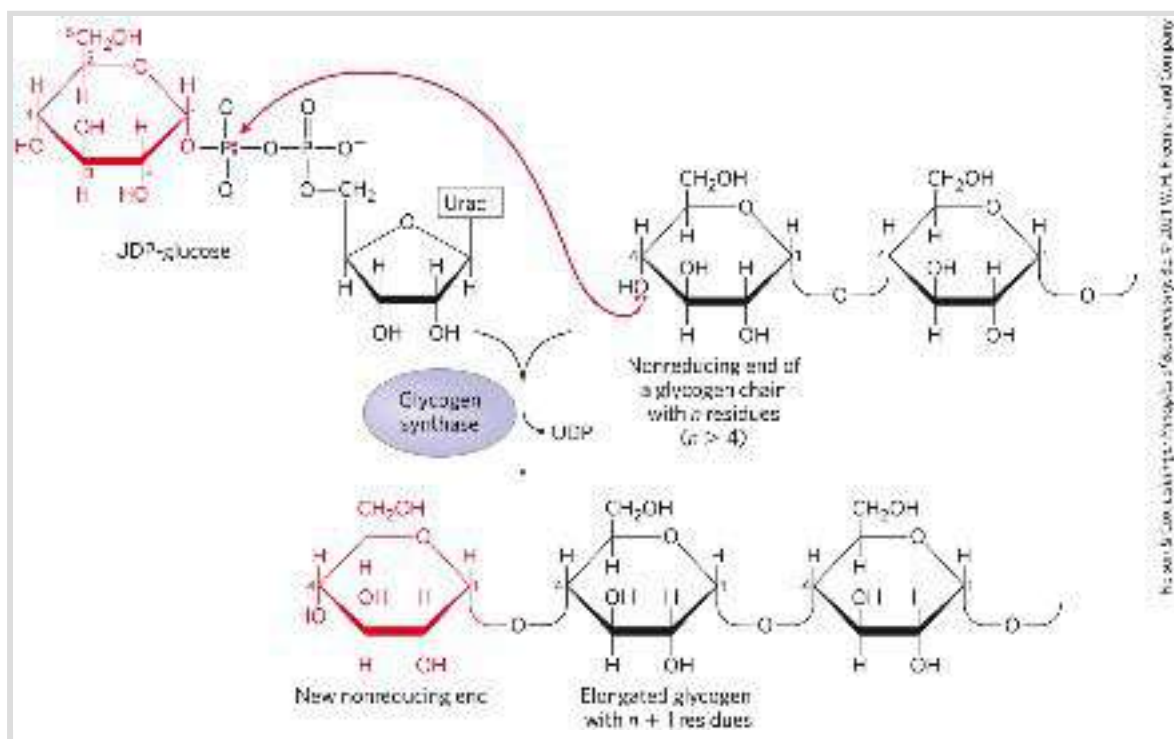
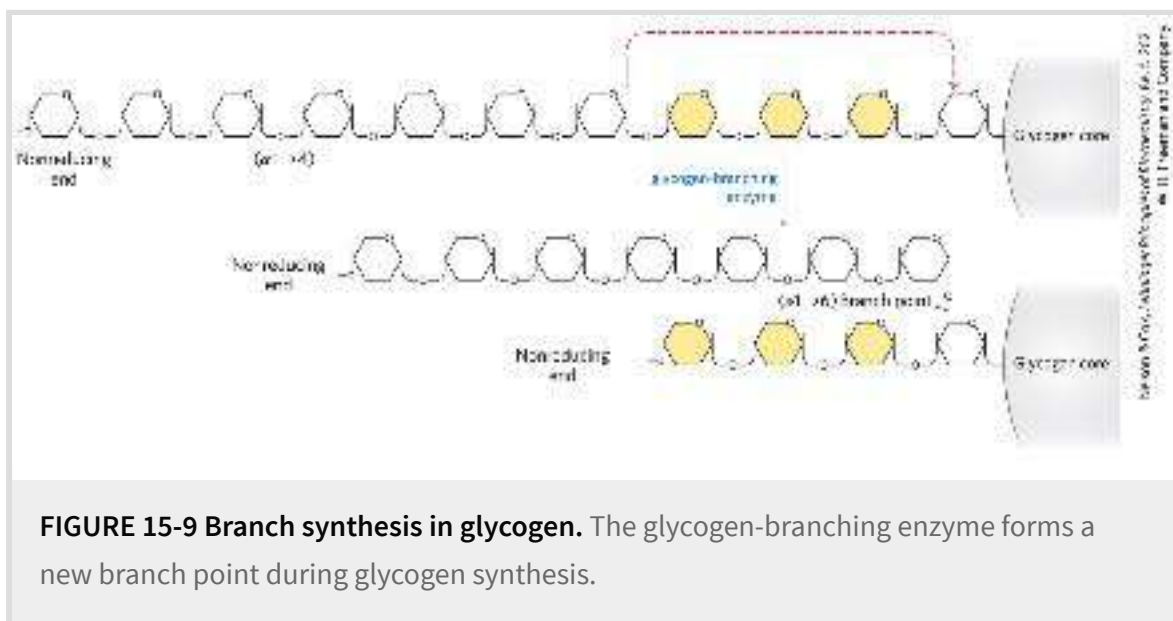


FIGURE 15-8 Glycogen synthesis. A glycogen chain is elongated by glycogen synthase. The enzyme transfers the glucose residue of UDP-glucose to the nonreducing end of a glycogen branch to make a new ($\alpha 1 \rightarrow 4$) linkage.

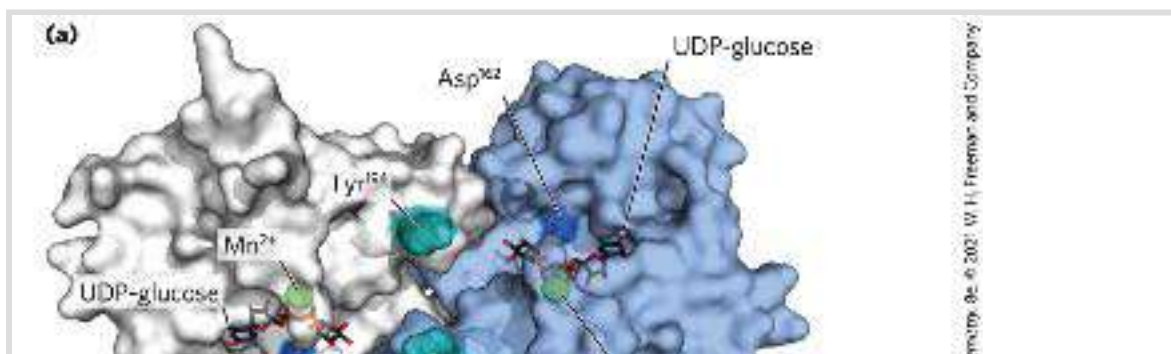
Glycogen synthase cannot make the ($\alpha 1 \rightarrow 6$) bonds found at the branch points of glycogen; these are formed by the **glycogen-**

branching enzyme, also called amylo (1 → 4) to (1 → 6) transglycosylase, or glycosyl (4 → 6) transferase. The glycogen-branching enzyme catalyzes transfer of a terminal fragment of 6 or 7 glucose residues from the nonreducing end of a glycogen branch having at least 11 residues to the C-6 hydroxyl group of a glucose residue at a more interior position of the same or another glycogen chain, thus creating a new branch ([Fig. 15-9](#)). Further glucose residues may be added to the new branch by glycogen synthase. The biological effect of branching is to increase the number of nonreducing ends. This increases the number of sites accessible to glycogen phosphorylase and glycogen synthase, both of which act only at nonreducing ends.



Glycogenin Primes the Initial Sugar Residues in Glycogen

Glycogen synthase cannot initiate a new glycogen chain de novo. It requires a primer, usually a preformed ($\alpha 1 \rightarrow 4$) polyglucose chain. **P3** So, how is a *new* glycogen molecule initiated? The intriguing protein **glycogenin** ([Fig. 15-10](#)) is both the primer on which new chains are assembled and the enzyme that catalyzes their assembly. The first step in the synthesis of a new glycogen molecule is the transfer of a glucose residue from UDP-glucose to the hydroxyl group of Tyr¹⁹⁴ of glycogenin. Each subunit of the glycogenin homodimer glycosylates Tyr¹⁹⁴ of the other subunit. The glycosidic bond in the product has the same configuration about the C-1 of glucose as the substrate UDP-glucose, suggesting that the transfer of glucose from UDP to Tyr¹⁹⁴ occurs in two steps. The first step is probably a nucleophilic attack by Asp¹⁶², forming a temporary intermediate with inverted configuration. A second nucleophilic attack by Tyr¹⁹⁴ then restores the starting configuration. The nascent chains are extended by the sequential addition of seven more glucose residues, each derived from UDP-glucose; the reactions are catalyzed by the chain-extending activity of glycogenin. At this point, glycogen synthase takes over, further extending the glycogen chain. Glycogenin remains buried within the β -particle, covalently attached to the two reducing ends of the glycogen molecule.



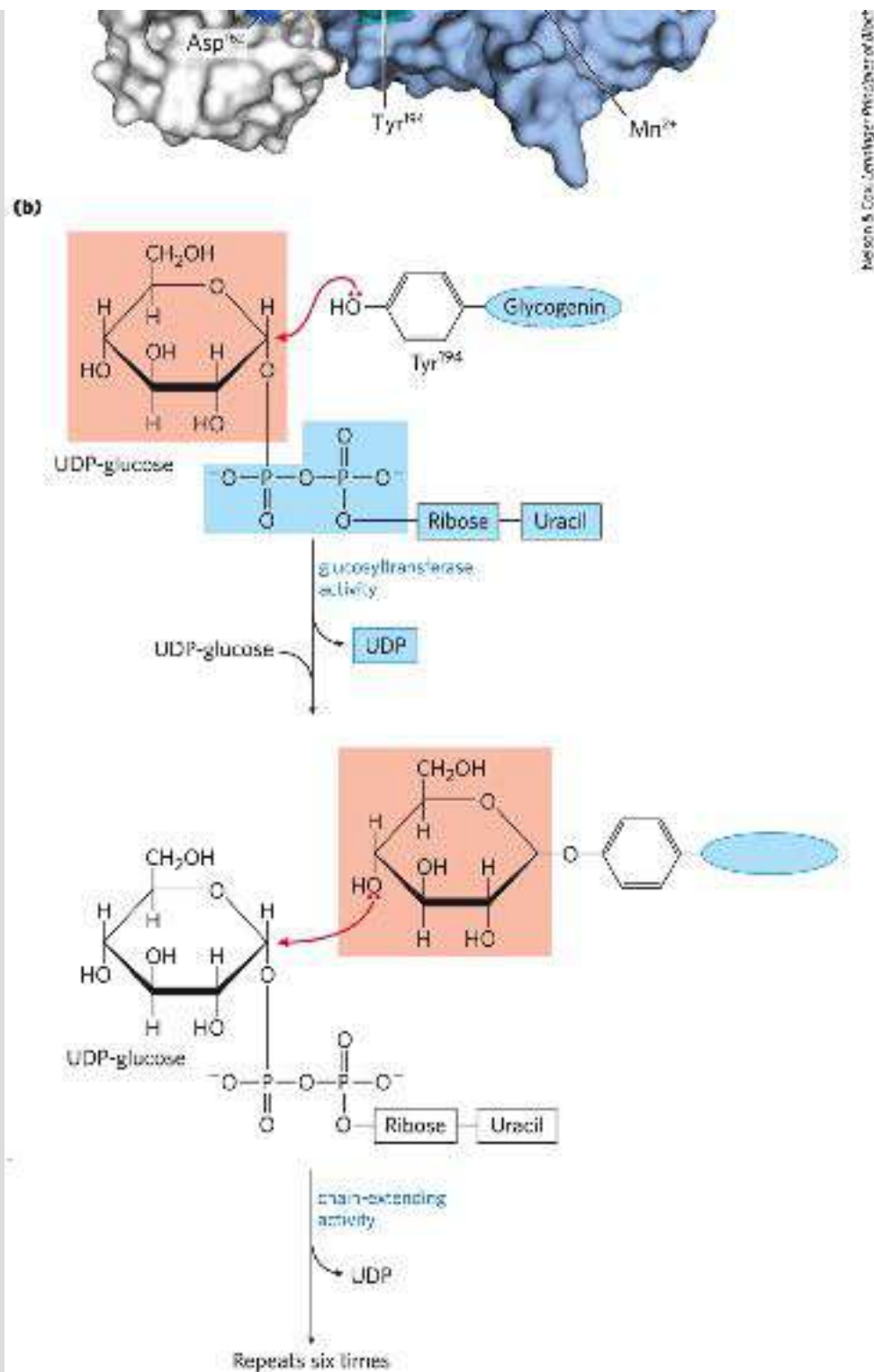


FIGURE 15-10 Glycogenin. (a) The protein is a homodimer. The substrate, UDP-glucose, is bound in a region near the amino terminus and is some distance from the Tyr¹⁹⁴ residues — 15 Å from the Tyr in the same monomer, 12 Å from the Tyr in the dimeric partner. Each UDP-glucose is bound through its phosphates to a Mn²⁺ ion, which is essential to catalysis. Mn²⁺ is believed to function as an electron-pair acceptor (Lewis acid) to stabilize the leaving group, UDP. (b) Glycogenin catalyzes two distinct reactions.

Initial attack by the hydroxyl group of Tyr¹⁹⁴ on C-1 of the glucosyl moiety of UDP-glucose results in a glucosylated Tyr residue. The C-1 of another UDP-glucose molecule is now attacked by the C-4 hydroxyl group of the terminal glucose, and this sequence repeats to form a nascent glycogen molecule of eight glucose residues attached by ($\alpha 1 \rightarrow 4$) glycosidic linkages. [(a) Data from PDB ID 1LL2, B. J. Gibbons et al., *J. Mol. Biol.* 319:463, 2002.]

SUMMARY 15.2 *Breakdown and Synthesis of Glycogen*

- Glycogen phosphorylase catalyzes phosphorolytic cleavage at the nonreducing ends of glycogen chains, producing glucose 1-phosphate. The debranching enzyme transfers branches onto main chains and releases the residue at the ($\alpha 1 \rightarrow 6$) branch as free glucose.
- Phosphoglucomutase interconverts glucose 1-phosphate and glucose 6-phosphate. Glucose 6-phosphate can enter glycolysis or, in liver, it can be converted to free glucose by glucose 6-phosphatase in the endoplasmic reticulum, then exported to replenish blood glucose.
- The sugar nucleotide UDP-glucose donates glucose residues to the nonreducing end of glycogen in the reaction catalyzed by glycogen synthase, producing short ($\alpha 1 \rightarrow 4$)-linked segments. A separate branching enzyme produces the ($\alpha 1 \rightarrow 6$) linkages at branch points.
- New glycogen particles begin with the autocatalytic formation of a glycosidic bond between the glucose of UDP-glucose and a Tyr residue of the protein glycogenin, followed by addition of several glucose residues to form a primer that can be acted on by glycogen synthase.

15.3 Coordinated Regulation of Glycogen Breakdown and Synthesis

As we have seen, the mobilization of stored glycogen is brought about by glycogen phosphorylase, which degrades glycogen to glucose 1-phosphate ([Fig. 15-3](#)). Glycogen phosphorylase provides an especially instructive case of enzyme regulation. It was one of the first known examples of an allosterically regulated enzyme and the first enzyme shown to be controlled by reversible phosphorylation. It was also one of the first allosteric enzymes for which the detailed three-dimensional structures of the active and inactive forms were revealed by x-ray crystallographic studies. Glycogen phosphorylase also illustrates how isozymes play their tissue-specific roles.

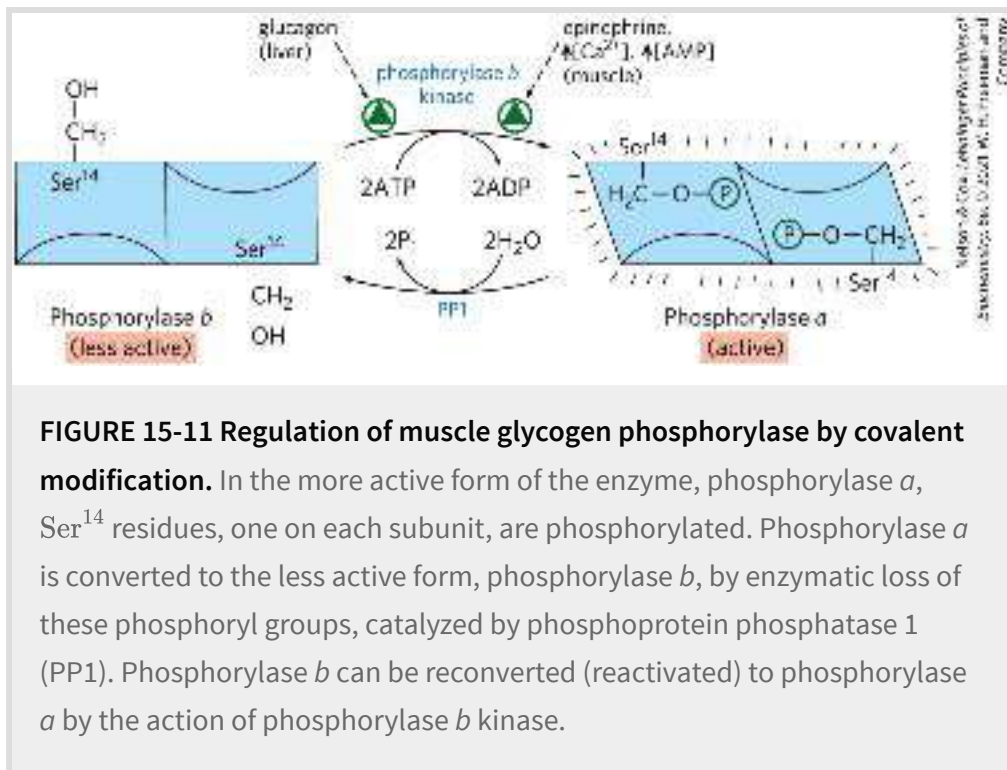
Glycogen Phosphorylase Is Regulated by Hormone-Stimulated Phosphorylation and by Allosteric Effectors



NIH/Science Source

Earl W. Sutherland, Jr., 1915–1974

P4 In the late 1930s, Carl and Gerty Cori ([Box 15-1](#)) discovered that the glycogen phosphorylase of skeletal muscle exists in two interconvertible forms: **glycogen phosphorylase *a***, which is catalytically active, and **glycogen phosphorylase *b***, which is much less active. (Note that glycogen phosphorylase is often referred to simply as phosphorylase — so honored because it was the first phosphorylase to be **discovered**; the shortened name has persisted in common usage and in the literature.) Subsequent studies by Earl Sutherland showed that phosphorylase *b* predominates in resting muscle, but during vigorous muscular activity, epinephrine triggers phosphorylation of phosphorylase *b*, converting it to its more active form, phosphorylase *a* ([Fig. 15-11](#)). In the liver, glucagon triggers phosphorylation of phosphorylase *b*, converting it to its active form.



Sutherland discovered the second messenger cAMP, which increases in concentration in response to stimulation by epinephrine (in muscle; see [Fig. 12-4](#)) or glucagon (in liver). Elevated [cAMP] initiates an **enzyme cascade** ([Fig. 15-12](#)), in which a catalyst activates a catalyst, which activates a catalyst. As we saw in [Chapter 12](#), such cascades allow exponential amplification of the initial signal. The rise in [cAMP] activates cAMP-dependent protein kinase, also called protein kinase A (PKA). PKA then phosphorylates and activates **phosphorylase *b* kinase**, which catalyzes the phosphorylation of glycogen phosphorylase *b*, activating it and thus stimulating glycogen breakdown.

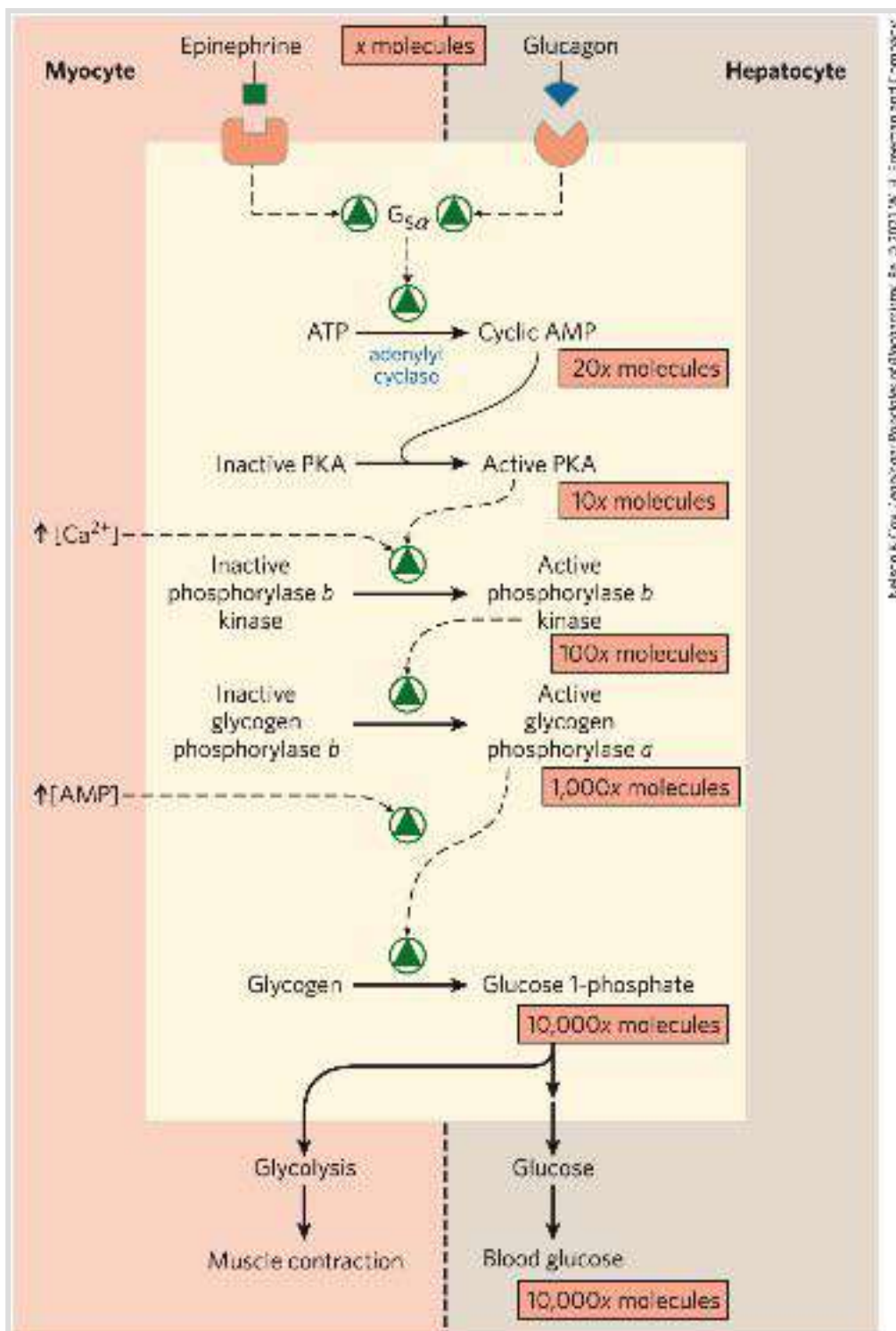


FIGURE 15-12 Cascade mechanism of epinephrine and glucagon action.

By binding to specific surface receptors, either epinephrine acting on a myocyte (left) or glucagon acting on a hepatocyte (right) activates a GTP-binding protein, $G_{s\alpha}$. Active $G_{s\alpha}$ triggers a rise in [cAMP], activating PKA. This sets off a cascade of phosphorylations; PKA activates phosphorylase b kinase, which then activates glycogen phosphorylase. Such cascades effect

a large amplification of the initial signal; the figures in pink boxes are certainly low estimates of the actual increase in number of molecules at each stage of the cascade. The resulting breakdown of glycogen provides glucose, which in the myocyte can supply ATP (via glycolysis) for muscle contraction and in the hepatocyte is released into the blood to counter the low blood glucose.

In muscle, this provides fuel for glycolysis to sustain muscle contraction for the fight-or-flight response signaled by epinephrine. In liver, glycogen breakdown counters the low blood glucose signaled by glucagon, by releasing glucose into the blood. These different roles are reflected in subtle differences in the regulatory mechanisms in muscle and liver. The glycogen phosphorylases of liver and muscle are isozymes, encoded by different genes and differing in their regulatory properties.

[Figure 15-12](#) shows the pathway for hormonal regulation of glycogen phosphorylase activity: epinephrine or glucagon initiates the cascade that activates phosphorylase *b* kinase. Phosphorylase *b* kinase activates phosphorylase by transferring a phosphoryl group to Ser¹⁴ on each of the two identical subunits of phosphorylase *b*, triggering a conformation change from the T state (phosphorylase *b*) to the R state (phosphorylase *a*), shown in detail in [Figure 6-40](#).  Superimposed on the hormonal activation of phosphorylase *b* kinase are allosteric control mechanisms ([Fig. 15-12](#)). In muscle, Ca²⁺, the signal for muscle contraction, binds to and activates phosphorylase *b* kinase. Ca²⁺ binds to phosphorylase *b* kinase through its δ subunit, which is calmodulin (see [Fig. 12-17](#)). AMP, which accumulates in

vigorously contracting muscle as a result of ATP breakdown, binds to and activates phosphorylase, speeding the release of glucose 1-phosphate from glycogen. When ATP levels are adequate, ATP blocks the allosteric site to which AMP binds (see [Fig. 6-40](#)), inactivating phosphorylase.

When the muscle returns to rest, a second enzyme, **phosphoprotein phosphatase 1 (PP1)**, removes the phosphoryl groups from phosphorylase *a*, converting it to the less active form, phosphorylase *b* (see [Fig. 15-11](#)).

Like the enzyme of muscle, the glycogen phosphorylase of liver is regulated hormonally (by phosphorylation/dephosphorylation) and allosterically. The dephosphorylated form is essentially inactive. When the blood glucose level is too low, glucagon (acting through the cascade mechanism shown in [Fig. 15-12](#)) activates phosphorylase *b* kinase, which in turn converts phosphorylase *b* to its active *a* form, initiating the release of glucose into the blood. When blood glucose levels return to normal, glucose enters hepatocytes and binds to an inhibitory allosteric site on phosphorylase *a*. This binding also produces a conformational change that exposes the phosphorylated Ser residues to PP1, which catalyzes their dephosphorylation and inactivates the phosphorylase ([Fig. 15-13](#)). The allosteric site for glucose allows liver glycogen phosphorylase to act as its own glucose sensor and to respond appropriately to changes in blood glucose.

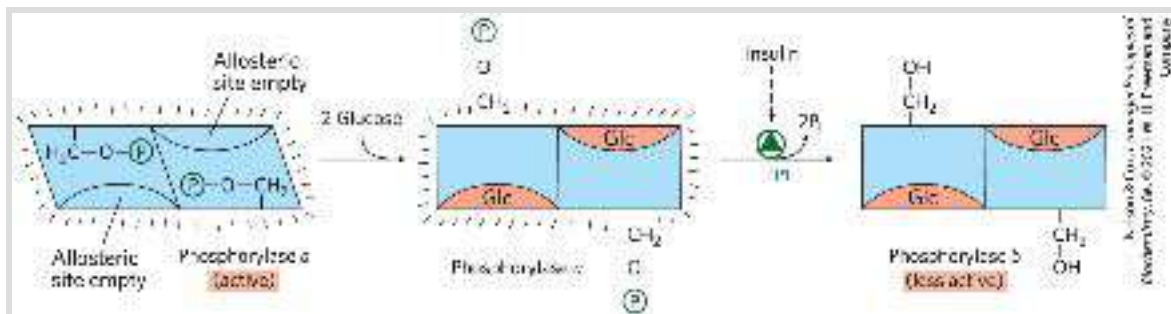


FIGURE 15-13 Glycogen phosphorylase of liver as a glucose sensor. Glucose binding to an allosteric site of the phosphorylase *a* isozyme of liver induces a conformational change that exposes its phosphorylated Ser residues to the action of phosphoprotein phosphatase 1 (PP1). This phosphatase converts phosphorylase *a* to phosphorylase *b*, sharply reducing the activity of phosphorylase and slowing glycogen breakdown in response to high blood glucose. Insulin also acts indirectly to stimulate PP1 and slow glycogen breakdown.

Glycogen Synthase Also Is Subject to Multiple Levels of Regulation

Like glycogen phosphorylase, glycogen synthase can exist in phosphorylated and unphosphorylated forms ([Fig. 15-14](#)). Its active form, **glycogen synthase *a***, is unphosphorylated. P4

Glycogen synthase kinase 3 (GSK3) adds phosphoryl groups to three Ser residues near the carboxyl terminus of glycogen synthase *a* converting it to **glycogen synthase *b***, which is inactive unless its allosteric activator, glucose 6-phosphate, is present. The action of GSK3 is hierarchical; it cannot phosphorylate glycogen synthase until another protein kinase, **casein kinase II (CKII)**, has first phosphorylated the glycogen synthase on a nearby residue, an event called **priming** ([Fig. 15-15a](#)). AMP-activated protein kinase (AMPK), which associates with glycogen granules

through its carbohydrate-binding domain, also phosphorylates glycogen synthase, inhibiting glycogen synthesis during periods of metabolic stress, signaled by high [AMP] and low [ATP].

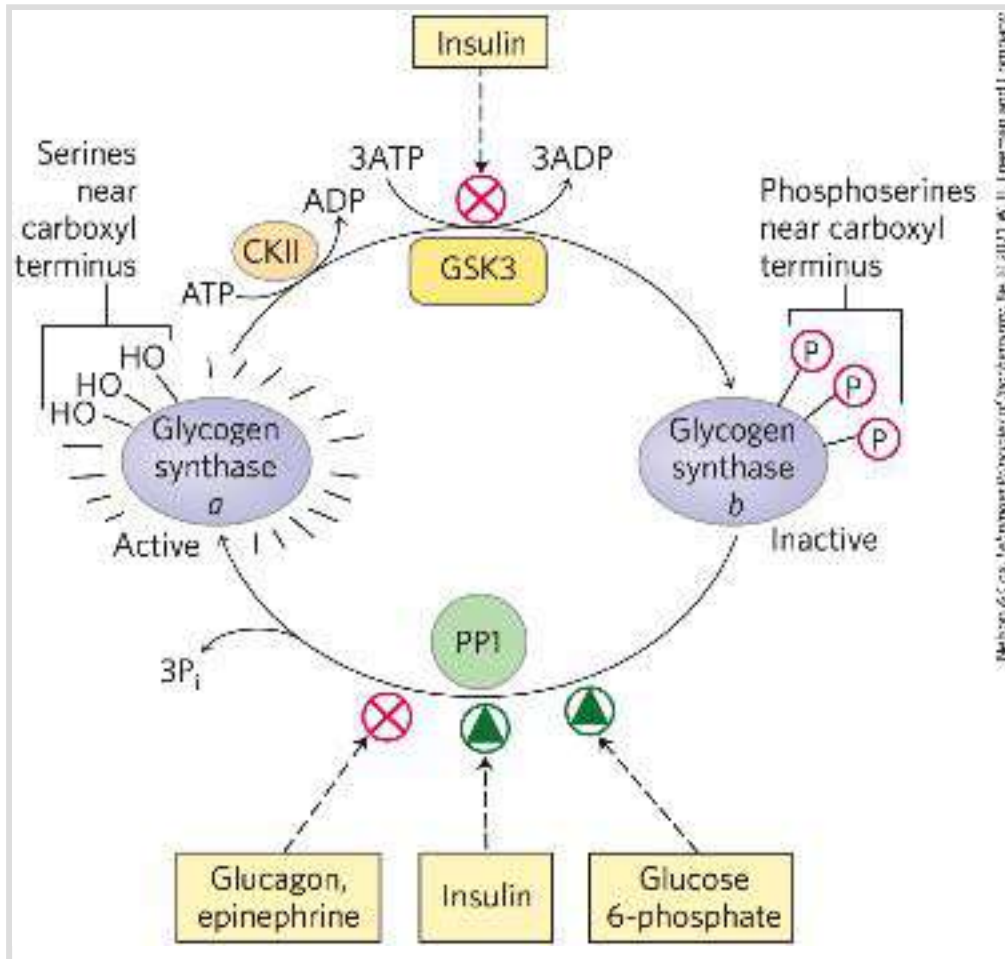


FIGURE 15-14 Effects of GSK3 on glycogen synthase activity. Glycogen synthase *a*, the active form, has three Ser residues near its carboxyl terminus. Their phosphorylation by glycogen synthase kinase 3 (GSK3) converts glycogen synthase to its inactive *b* form. Insulin favors the active *a* form of glycogen synthase by blocking the activity of GSK3 and activating phosphoprotein phosphatase 1 (PP1). In muscle, epinephrine activates PKA, which phosphorylates the glycogen-targeting protein Gm on a site that causes dissociation of PP1 from glycogen. Glucose 6-phosphate favors dephosphorylation of glycogen synthase by binding to it and promoting a conformation that is a good substrate for PP1.

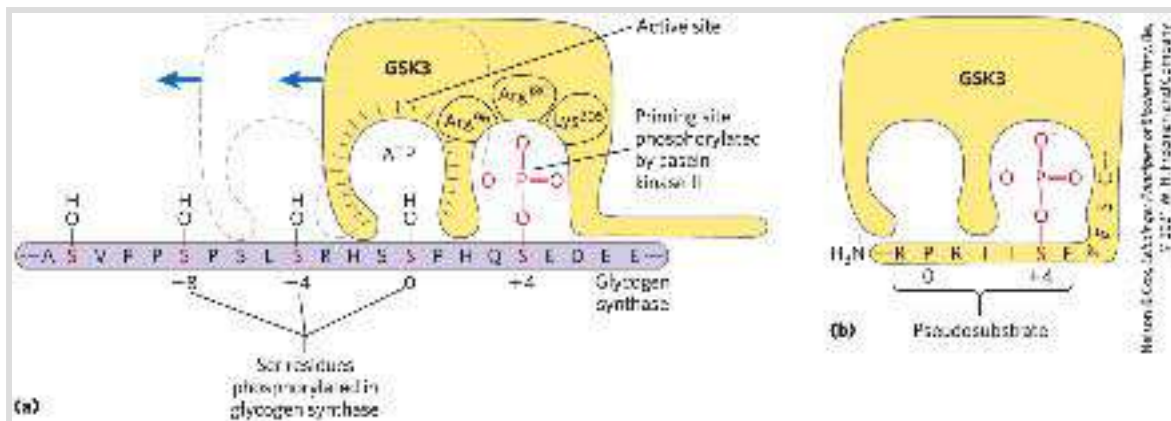


FIGURE 15-15 Priming of GSK3 phosphorylation of glycogen synthase. (a) Glycogen synthase kinase 3 first associates with its substrate (glycogen synthase) by interaction between three positively charged residues (Arg⁹⁶, Arg¹⁸⁰, Lys²⁰⁵) and a phosphoserine residue at position +4 in the substrate. (For orientation, the Ser or Thr residue to be phosphorylated in the substrate is assigned the index 0. Residues on the amino-terminal side of this residue are numbered -1, -2, and so forth; residues on the carboxyl-terminal side are numbered +1, +2, and so forth.) This association aligns the active site of the enzyme with a Ser residue at position 0, which it phosphorylates. This creates a new priming site, and the enzyme moves down the protein to phosphorylate the Ser residue at position -4, and then the Ser at -8. (b) GSK3 has a Ser residue near its amino terminus that can be phosphorylated by PKA or PKB. This produces a “pseudosubstrate” region in GSK3 that folds into the priming site and makes the active site inaccessible to another protein substrate, inhibiting GSK3 until the priming phosphoryl group of its pseudosubstrate region is removed by PP1. Other proteins that are substrates for GSK3 also have a priming site at position +4, which must be phosphorylated by another protein kinase before GSK3 can act on them.

Insulin favors activation of glycogen synthase by blocking the activity of GSK3 and activating PP1. In muscle, epinephrine activates PKA, which phosphorylates the glycogen-targeting protein G_M (see [Fig. 15-16](#)), a regulatory subunit of PP1 in muscle, on a site that causes dissociation of PP1 from glycogen, effectively blocking its action on glycogen synthase.

In liver, conversion of glycogen synthase *b* to the active form is promoted by PP1, which is bound to the glycogen particle by its regulatory subunit in liver, GL. PP1 removes the phosphoryl groups from the three Ser residues phosphorylated by GSK3. Glucose 6-phosphate binds to an allosteric site on glycogen synthase, making the enzyme a better substrate for dephosphorylation by PP1 and causing its activation. By analogy with glycogen phosphorylase, which acts as a glucose sensor, glycogen synthase can be regarded as a glucose 6-phosphate sensor. In muscle, a different phosphatase may have the role played by PP1 in liver, activating glycogen synthase by dephosphorylating it.

As we saw in [Chapter 12](#), one way in which insulin triggers intracellular changes is by activating a protein kinase (PKB) that, in turn, phosphorylates and inactivates GSK3 (see [Fig. 12-23](#)). Phosphorylation of a Ser residue near the amino terminus of GSK3 converts that region of the protein to a pseudosubstrate, which folds into the site at which the priming phosphorylated Ser residue normally binds ([Fig. 15-15b](#)). This prevents GSK3 from binding the priming site of a real substrate, thereby inactivating the enzyme and tipping the balance in favor of dephosphorylation of glycogen synthase by PP1. Glycogen phosphorylase can also affect the phosphorylation of glycogen synthase: active glycogen phosphorylase directly inhibits PP1, preventing it from activating glycogen synthase ([Fig. 15-14](#)).

P4 Insulin stimulates glycogen synthesis by activating PP1 and by inactivating GSK3. This single enzyme, PP1, can remove phosphoryl groups from all three of the enzymes phosphorylated in response to glucagon (liver) and epinephrine (liver and muscle): phosphorylase kinase, glycogen phosphorylase, and glycogen synthase. The catalytic subunit of PP1 (PP1c) does not exist free in the cytosol, but is tightly bound to its target proteins through a tissue-specific regulatory subunit, one of a family of **glycogen-targeting proteins** that bind glycogen and each of the three enzymes (**Fig. 15-16**). PP1 is itself subject to covalent and allosteric regulation: it is inactivated when phosphorylated by PKA and is allosterically activated by glucose 6-phosphate.

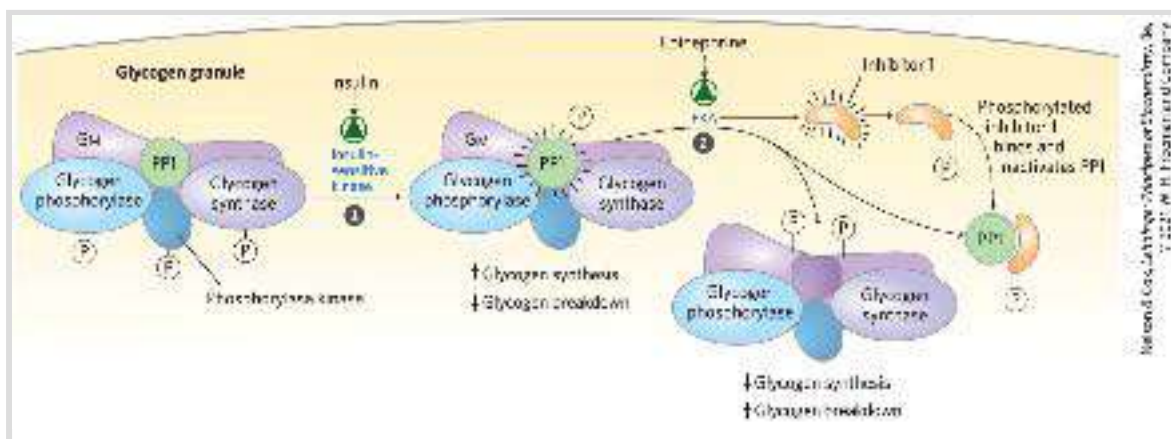


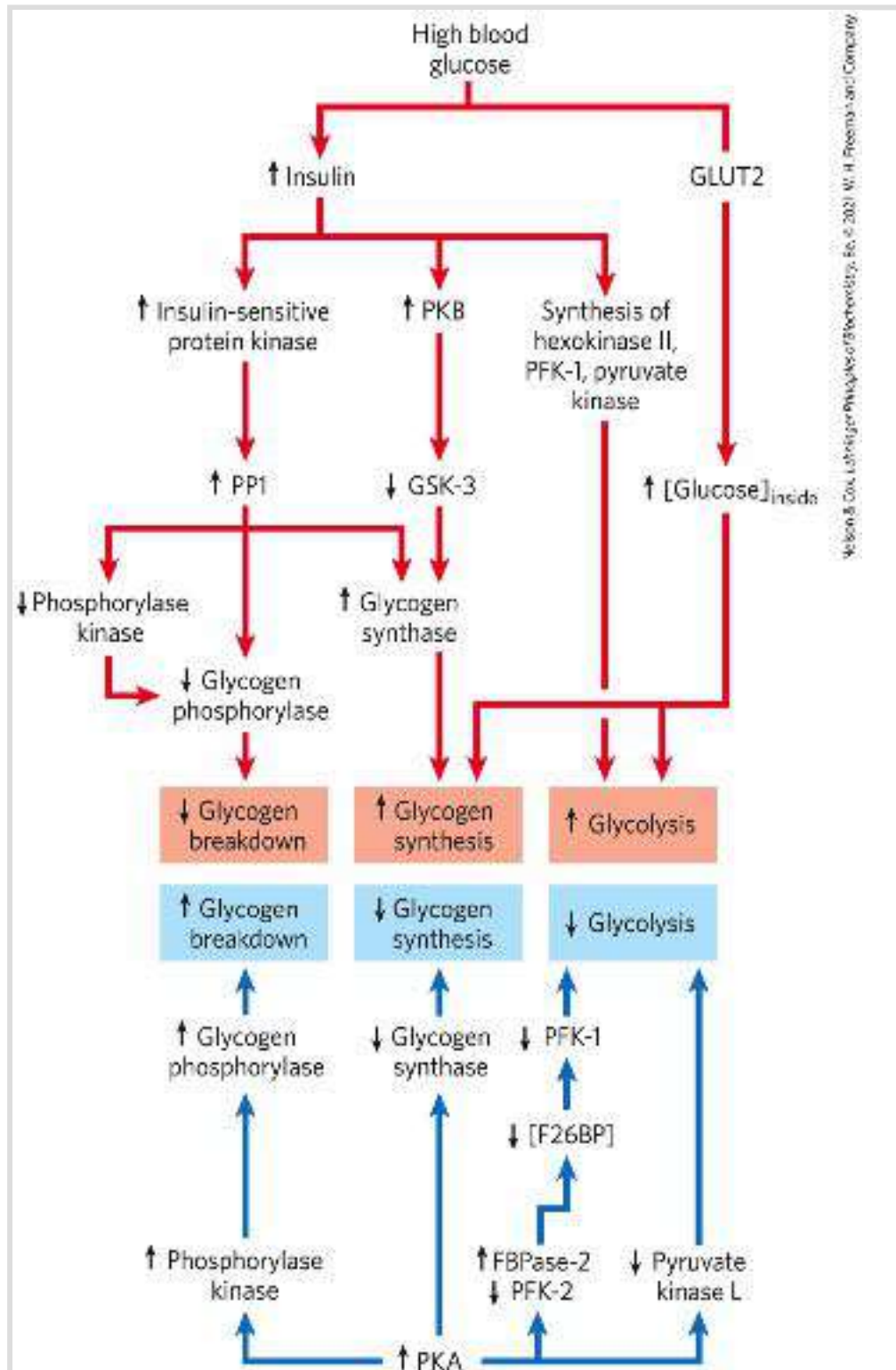
FIGURE 15-16 Glycogen-targeting protein GM. GM is a regulatory subunit of PP1 in muscle, and one of a family of glycogen-targeting proteins that serve as a scaffold, binding other proteins (including PP1) to glycogen particles. GM can be phosphorylated at two different sites in response to insulin or epinephrine. **1** Insulin-stimulated phosphorylation of GM site 1 activates PP1, which dephosphorylates phosphorylase kinase, glycogen phosphorylase, and glycogen synthase. **2** Epinephrine-stimulated phosphorylation of GM site 2 by PKA causes dissociation of PP1 from the glycogen particle, preventing its access to glycogen phosphorylase and glycogen synthase. PKA also phosphorylates a protein (inhibitor 1) that, when phosphorylated, inhibits PP1. By these means, insulin stimulates glycogen synthesis and inhibits glycogen breakdown, whereas epinephrine (or glucagon in the liver) has the opposite effects.

Allosteric and Hormonal Signals Coordinate Carbohydrate Metabolism Globally

Having looked at the mechanisms that regulate individual enzymes, we can now consider the overall shifts in carbohydrate metabolism that occur in the well-fed state, during fasting, and in the fight-or-flight response — signaled by insulin, glucagon, and epinephrine, respectively. We need to contrast two cases in which regulation serves different ends: (1) the role of hepatocytes in supplying glucose to the blood; and (2) the selfish use of carbohydrate fuels by extrahepatic tissues, typified by skeletal muscle (myocytes), to support their own activities.

After ingestion of a carbohydrate-rich meal, the elevation of blood glucose triggers insulin release ([Fig. 15-17](#), top). In a hepatocyte, insulin has two immediate effects: it inactivates GSK3; and it activates a protein phosphatase, probably PP1. These two actions fully activate glycogen synthase. PP1 also inactivates glycogen phosphorylase *a* and phosphorylase kinase by dephosphorylating both, effectively stopping glycogen breakdown. Glucose enters the hepatocyte through the high-capacity transporter GLUT2, always present in the plasma membrane, and the elevated intracellular glucose leads to dissociation of hexokinase IV (glucokinase) from its nuclear regulatory protein ([Fig. 14-21](#)). Hexokinase IV enters the cytosol and phosphorylates glucose,

stimulating glycolysis and supplying the precursor for glycogen synthesis. Under these conditions, hepatocytes use the excess glucose in the blood to synthesize glycogen, up to the limit of about 10% of the total weight of the liver.



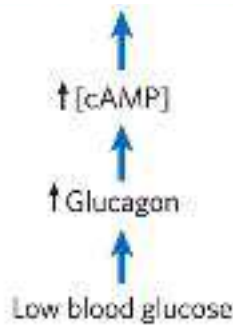
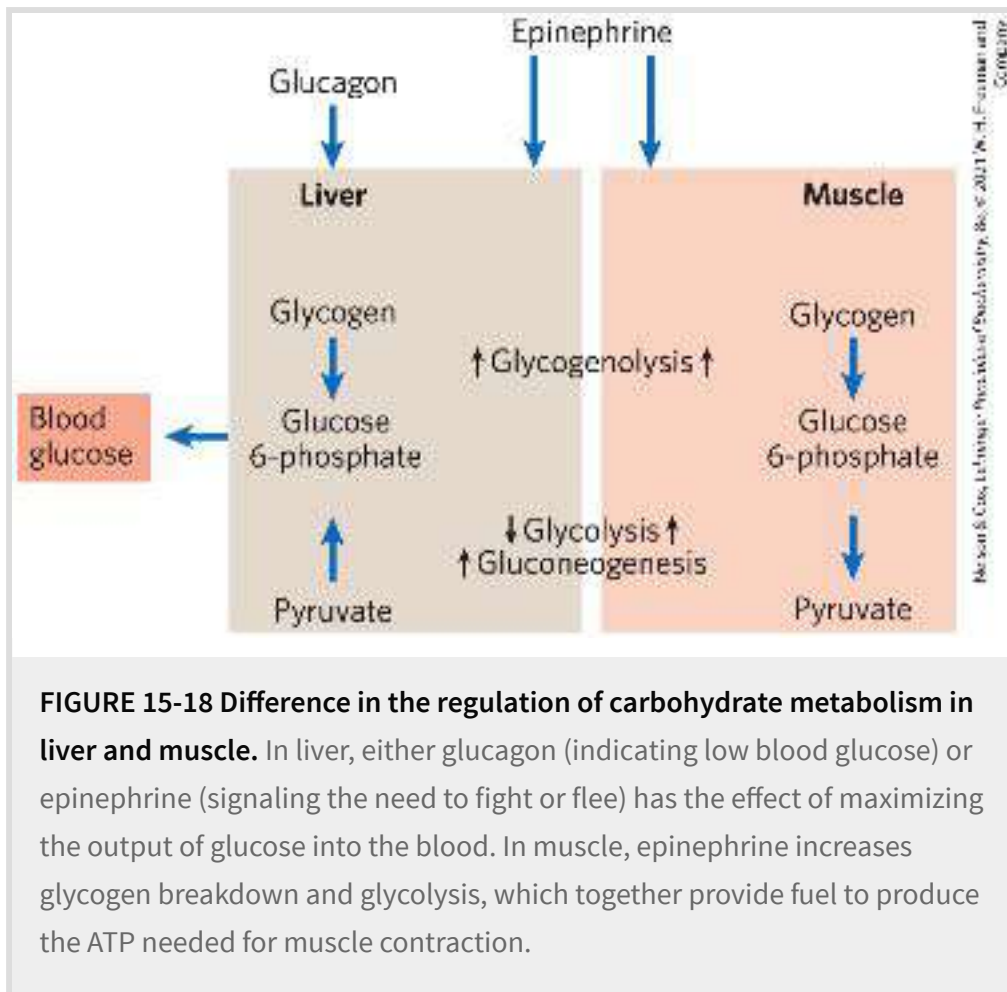


FIGURE 15-17 Regulation of carbohydrate metabolism in the liver. Colored arrows indicate causal relationships between the changes they connect. For example, an arrow from $\downarrow$ A to $\uparrow$ B means that a decrease in A causes an increase in B. Red arrows connect events that result from high blood glucose; blue arrows connect events that result from low blood glucose.

Between meals, or during an extended fast, the blood glucose level drops, triggering the release of glucagon, which, acting through the cascade shown in [Figure 15-12](#), activates PKA. PKA mediates all the effects of glucagon ([Fig. 15-17](#), bottom). It phosphorylates phosphorylase kinase, activating it and leading to the activation of glycogen phosphorylase. It phosphorylates glycogen synthase, inactivating it and blocking glycogen synthesis. It phosphorylates PFK-2/FBPase-2, leading to a drop in the concentration of the regulator fructose 2,6-bisphosphate, which has the effect of inactivating the glycolytic enzyme PFK-1 and activating the gluconeogenic enzyme FBPase-1 (see [Fig. 14-24](#)). And it phosphorylates and inactivates the glycolytic enzyme pyruvate kinase. Under these conditions, the liver produces glucose 6-phosphate by glycogen breakdown and by gluconeogenesis, and it stops using glucose to fuel glycolysis or make glycogen, maximizing the amount of glucose it can release to the blood. This release of glucose is possible only in liver and

kidney, because other tissues lack glucose 6-phosphatase ([Fig. 15-6](#)).

 The physiology of skeletal muscle differs from that of liver in three ways important to our discussion of metabolic regulation ([Fig. 15-18](#)): (1) muscle uses its stored glycogen only for its own needs; (2) as it goes from rest to vigorous contraction, muscle undergoes very large changes in its demand for ATP, which is provided by glycolysis; (3) muscle lacks the enzymatic machinery for gluconeogenesis. The regulation of carbohydrate metabolism in muscle reflects these differences from liver. First, myocytes lack receptors for glucagon, which are present on hepatocytes. Second, the muscle isozyme of pyruvate kinase is not phosphorylated by PKA, so glycolysis is not turned off when [cAMP] is high. In fact, cAMP *increases* the rate of glycolysis in muscle, probably by activating glycogen phosphorylase. When epinephrine is released into the blood in a fight-or-flight situation, PKA is activated by the rise in [cAMP] and phosphorylates and activates glycogen phosphorylase kinase. The resulting phosphorylation and activation of glycogen phosphorylase results in faster glycogen breakdown. Epinephrine is not released under low-stress conditions, but with each neuronal stimulation of muscle contraction, cytosolic $[Ca^{2+}]$ rises briefly and activates phosphorylase kinase through its calmodulin subunit.



Elevated insulin triggers increased glycogen synthesis in myocytes by activating PP1 and inactivating GSK3. Unlike hepatocytes, myocytes have a reserve of the glucose transporter GLUT4 sequestered in intracellular vesicles. Insulin triggers their movement to the plasma membrane (see [Fig. 12-23](#)), where they allow increased glucose uptake. In response to insulin, therefore, myocytes help to lower blood glucose by increasing their rates of glucose uptake, glycogen synthesis, and glycolysis.

Carbohydrate and Lipid Metabolism Are Integrated by Hormonal and

Allosteric Mechanisms

As complex as the regulation of carbohydrate metabolism is, it is far from the whole story of fuel metabolism.  The metabolism of fats and fatty acids is very closely tied to that of carbohydrates. Hormonal signals such as insulin and changes in diet or exercise are equally important in regulating fat metabolism and integrating it with that of carbohydrates. We return to this overall metabolic integration in mammals in [Chapter 23](#), after first considering the metabolic pathways for fats and amino acids ([Chapters 17](#) and [18](#)). The message we wish to convey here is that metabolic pathways are overlaid with complex regulatory controls that are exquisitely sensitive to changes in metabolic circumstances. These mechanisms act to adjust the flow of metabolites through various metabolic pathways, as needed by the cell and organism, without causing major changes in the concentrations of intermediates shared with other pathways.

SUMMARY 15.3 *Coordinated Regulation of Glycogen Breakdown and Synthesis*

■ Glycogen phosphorylase is activated in response to epinephrine (in muscle) or glucagon (in liver), which raise [cAMP] and thereby activate PKA. PKA phosphorylates and activates phosphorylase kinase, which converts glycogen phosphorylase *b* to its active *a* form. In muscle, Ca^{2+} and AMP act allosterically to

amplify the activity of the enzymes in this cascade. In liver, glucose acts allosterically to make phosphorylase *a* more susceptible to dephosphorylation/inactivation by phosphoprotein phosphatase 1 (PP1).

■ Glycogen synthase *a* is inactivated by phosphorylation, ultimately catalyzed by GSK3, but primed by phosphorylation by other kinases. Insulin, by causing inactivation of GSK3 and activating PP1, favors glycogen synthase *a* and stimulates glycogen synthesis. Glucose 6-phosphate acts allosterically to make glycogen synthase *b* a better substrate for PP1, which dephosphorylates it to its active *a* form.

■ In liver, glucagon stimulates glycogen breakdown and gluconeogenesis while blocking glycolysis, thereby sparing glucose for export to the brain and other tissues. In muscle, epinephrine stimulates glycogen breakdown and glycolysis, providing ATP to support contraction.

■ Further hormonal and allosteric mechanisms regulate the use of carbohydrates and lipids as metabolic fuels.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

glycogenolysis

glycogenesis

debranching enzyme

phosphoglucomutase

sugar nucleotides

UDP-glucose pyrophosphorylase

glycogen-branching enzyme

glycogenin

glycogen phosphorylase *a*

glycogen phosphorylase *b*

enzyme cascade

phosphorylase *b* kinase

phosphoprotein phosphatase 1 (PP1)

glycogen synthase *a*

glycogen synthase kinase 3 (GSK3)

glycogen synthase *b*

casein kinase II (CKII)

priming

glycogen-targeting proteins

PROBLEMS

1. Glycogen as Energy Storage: How Long Can a Game Bird Fly? Since ancient times, people have observed that certain game birds, such as grouse, quail, and pheasants, fatigue easily. The Greek historian Xenophon wrote: “The bustards ... can be caught if one is quick in starting them up, for they will fly only a short distance, like partridges, and soon tire; and their flesh is delicious.” The flight muscles of game birds rely almost entirely on the use of glucose 1-phosphate to drive ATP synthesis ([Chapter 14](#)). The glucose 1-phosphate derives from the breakdown of stored muscle glycogen, catalyzed by the enzyme glycogen phosphorylase. The rate of ATP production is limited by the rate at which glycogen can be broken down. During a “panic flight,” the game bird’s rate of glycogen breakdown is quite high, approximately $120\ \mu\text{mol}/\text{min}$ of glucose 1-phosphate produced per gram of fresh tissue. Given that the flight muscles usually contain about 0.35% glycogen by weight, calculate how long a game bird can fly. (Assume the average molecular weight of a glucose residue in glycogen is $162\ \text{g}/\text{mol}$.)

2. Enzyme Activity and Physiological Function The V_{max} of the glycogen phosphorylase from skeletal muscle is much greater than the V_{max} of the same enzyme from liver tissue.

- a. What is the physiological function of glycogen phosphorylase in skeletal muscle? In liver tissue?

- b. Why does the $V_{\max}$ of the muscle enzyme need to be greater than that of the liver enzyme?

3. Glycogen Phosphorylase Equilibrium

Glycogen phosphorylase catalyzes the removal of glucose from glycogen. The $\Delta G'^{\circ}$ for this reaction is 3.1 kJ/mol.

- Calculate the ratio of $[P_i]$ to [glucose 1-phosphate] when the reaction is at equilibrium. (Hint: The removal of glucose units from glycogen does not change the glycogen concentration.)
- The measured ratio $[P_i]/[\text{glucose 1-phosphate}]$ in myocytes under physiological conditions is more than 100:1. What does this indicate about the direction of metabolite flow through the glycogen phosphorylase reaction in muscle?
- Why are the equilibrium and physiological ratios different? What is the possible significance of this difference?

4. Regulation of Glycogen Phosphorylase In muscle tissue, the rate of conversion of glycogen to glucose 6-phosphate is determined by the ratio of phosphorylase *a* (active) to phosphorylase *b* (less active). Determine what happens to the rate of glycogen breakdown if a broken cell extract of muscle containing glycogen phosphorylase is treated with **(a)** phosphorylase kinase and ATP **(b)** PP1 **(c)** epinephrine.

5. Glycogen Breakdown in Rabbit Muscle The intracellular use of glucose and glycogen is tightly regulated at four points. To compare the regulation of glycolysis when oxygen is

plentiful and when it is depleted, consider the utilization of glucose and glycogen by rabbit leg muscle in two physiological settings: a resting rabbit, with low ATP demands, and a rabbit that sights its mortal enemy, the coyote, and dashes into its burrow. For each setting, determine the relative levels (high, intermediate, or low) of AMP, ATP, citrate, and acetyl-CoA and describe how these levels affect the flow of metabolites through glycolysis by regulating specific enzymes. (Hint: In periods of stress, rabbit leg muscle produces much of its ATP by anaerobic glycolysis (lactate fermentation) and very little by oxidation of acetyl-CoA derived from fat breakdown.)

6. Glycogen Breakdown in Migrating Birds Unlike a rabbit, running all-out for a few moments to escape a predator, migratory birds require energy for extended periods of time. For example, ducks generally fly several thousand miles during their annual migration. The flight muscles of migratory birds have a high oxidative capacity and obtain the necessary ATP through the oxidation of acetyl-CoA (obtained from fats) via the citric acid cycle. Compare the regulation of muscle glycolysis during short-term intense activity, as in a fleeing rabbit, and during extended activity, as in a migrating duck. Why must the regulation in these two settings be different?

7. Enzyme Defects in Carbohydrate Metabolism

Consider the four clinical case studies, A through D. For each case, determine which enzyme is defective and designate the

appropriate treatment from the lists provided at the end of the problem. Justify your choices. Answer the questions contained in each case study. (You may need to refer to information in [Chapter 14](#).)

Case A The patient develops vomiting and diarrhea shortly after milk ingestion. The physician administers a lactose tolerance test. (The patient ingests a standard amount of lactose, and the physician measures the glucose and galactose concentrations in his blood plasma at intervals. In individuals with normal carbohydrate metabolism, the levels increase to a maximum in about 1 hour, then decline.) The patient's blood glucose and galactose concentrations do not increase during the test. Why do blood glucose and galactose increase and then decrease during the test in healthy individuals? Why do they fail to rise in the patient?

Case B The patient develops vomiting and diarrhea after ingestion of milk. Blood tests show a low concentration of glucose but a much higher than normal concentration of reducing sugars. The urine tests positive for galactose. Why is the concentration of reducing sugar in the blood high? Why does galactose appear in the urine?

Case C The patient complains of painful muscle cramps when performing strenuous physical exercise but has no other symptoms. A muscle biopsy indicates a muscle glycogen concentration much higher than normal. Why does glycogen accumulate?

Case D The patient is lethargic, her liver is enlarged, and a biopsy of the liver shows large amounts of excess glycogen. She also has a lower than normal blood glucose level. What is the reason for the low blood glucose in this patient?

Defective Enzyme

- a. Muscle PFK-1
- b. Phosphomannose isomerase
- c. Galactose 1-phosphate uridylyltransferase
- d. Liver glycogen phosphorylase
- e. Triose kinase
- f. Lactase in intestinal mucosa
- g. Maltase in intestinal mucosa
- h. Muscle debranching enzyme

Treatment

- 1. Jogging 5 km each day
- 2. Fat-free diet
- 3. Low-lactose diet
- 4. Avoiding strenuous exercise
- 5. Large doses of niacin (the precursor of NAD)
- 6. Frequent feedings (smaller portions) of a normal diet

8. Effects of Insufficient Insulin in a Person with

Diabetes A man with insulin-dependent diabetes is brought to the hospital emergency department in a near-comatose state. While vacationing in an isolated place, he lost his insulin medication and has not taken any insulin for two days.

- a. For each tissue listed at the end of the problem, is each pathway faster, slower, or unchanged in this patient, compared with the normal level when he is getting appropriate amounts of insulin?
- b. For each pathway, describe at least one control mechanism responsible for the change you predict.

Tissue and Pathways

1. Adipose: fatty acid synthesis
2. Muscle: glycolysis; fatty acid synthesis; glycogen synthesis
3. Liver: glycolysis; gluconeogenesis; glycogen synthesis; fatty acid synthesis; pentose phosphate pathway

9.  **Blood Metabolites in Insulin Insufficiency** For the patient described in Problem 8, predict the levels of each listed metabolite in his blood *before* treatment in the emergency room, relative to levels maintained during adequate insulin treatment: **(a)** glucose **(b)** ketone bodies **(c)** free fatty acids.

10. Metabolic Effects of Mutant Enzymes Predict and explain the effect on glycogen metabolism of each of the listed defects caused by mutation: **(a)** Loss of the cAMP-binding site on the regulatory subunit of protein kinase A (PKA) **(b)** Loss of the protein phosphatase inhibitor (inhibitor 1 in [Fig. 15-16](#)) **(c)** Overexpression of phosphorylase *b* kinase in liver **(d)** Defective glucagon receptors in liver.

11. Hormonal Control of Metabolic Fuel Between your evening meal and breakfast, your blood glucose drops and your liver becomes a net producer rather than consumer of glucose. Describe the hormonal basis for this switch, and explain how the hormonal change triggers glucose production by the liver.

12. Altered Metabolism in Genetically Manipulated Mice Researchers can manipulate the genes of a mouse so that a single gene in a single tissue either produces an inactive protein (a “knockout” mouse) or produces a protein that is always (constitutively) active. What effects on metabolism would you predict for mice with the listed genetic changes? **(a)** Knockout of glycogen debranching enzyme in the liver **(b)** Knockout of hexokinase IV in liver **(c)** Knockout of FBPase-2 in liver **(d)** Constitutively active FBPase-2 in liver **(e)** Constitutively active AMPK in muscle **(f)** Constitutively active ChREBP in liver (see [Fig. 14-28](#))

DATA ANALYSIS PROBLEM

13. Optimal Glycogen Structure Muscle cells need rapid access to large amounts of glucose during heavy exercise. This glucose is stored in liver and skeletal muscle in polymeric form as particles of glycogen. The typical glycogen β -particle contains about 55,000 glucose residues (see [Fig. 15-2](#)). [Meléndez-Hevia, Waddell, and Shelton \(1993\)](#) explored

some theoretical aspects of the structure of glycogen, as described in this problem.

- a. The cellular concentration of glycogen in liver is about $0.01\ \mu\text{M}$. What cellular concentration of free glucose would be required to store an equivalent amount of glucose? Why would this concentration of free glucose present a problem for the cell?

Glucose is released from glycogen by glycogen phosphorylase, an enzyme that can remove glucose molecules, one at a time, from one end of a glycogen chain (see [Fig. 15-3](#)). Glycogen chains are branched (see [Fig. 15-2](#)), and the degree of branching — the number of branches per chain — has a powerful influence on the rate at which glycogen phosphorylase can release glucose.

- b. Why would a degree of branching that was too low (i.e., below an optimum level) reduce the rate of glucose release? (Hint: Consider the extreme case of no branches in a chain of 55,000 glucose residues.)
- c. Why would a degree of branching that was too high also reduce the rate of glucose release? (Hint: Think of the physical constraints.)

Meléndez-Hevia and colleagues did a series of calculations and found that two branches per chain (see [Fig. 15-2](#)) was optimal for the constraints described above. This is what is found in glycogen stored in muscle and liver.

To determine the optimum number of glucose residues per chain, Meléndez-Hevia and coauthors considered two key parameters that define the structure of a glycogen particle:

t = the number of tiers of glucose chains in a particle (the molecule in [Fig. 15-2](#) has five tiers);

g_c = the number of glucose residues in each chain. They set out to find the values of t and g_c that would maximize three quantities: (1) the amount of glucose stored in the particle (G_T) per unit volume; (2) the number of unbranched glucose chains (C_A) per unit volume (i.e., number of A chains in the outermost tier, readily accessible to glycogen phosphorylase); and (3) the amount of glucose available to phosphorylase in these unbranched chains (G_{PT}).

- d. Show that $C_A = 2^{t-1}$. This is the number of chains available to glycogen phosphorylase before the action of the debranching enzyme.
- e. Show that C_T , the total number of chains in the particle, is given by $C_T = 2^t - 1$. For purposes of this calculation, consider the primers to be a single chain. Thus $G_T = g_c(C_T) = g_c(2^t - 1)$, the total number of glucose residues in the particle.
- f. Glycogen phosphorylase cannot remove glucose from glycogen chains that are shorter than five glucose residues. Show that $G_{PT} = (g_c - 4)(2^{t-1})$. This is the amount of glucose readily available to glycogen phosphorylase.

- g. Based on the size of a glucose residue and the location of branches, the thickness of one tier of glycogen is $0.12 g_c \text{ nm} + 0.35 \text{ nm}$. Show that the volume of a particle, V_s , is given by the equation

$$V_s = \frac{4}{3} \pi t^3 (0.12 g_c + 0.35)^3 \text{ nm}^3.$$

Meléndez-Hevia and coauthors then determined the optimum values of t and g_c — those that gave the maximum value of a quality function, f , that maximizes G_T , C_A , and G_{PT} , while minimizing V_s : $f = \frac{G_T C_A G_{PT}}{V_s}$. They found that the optimum value of g_c is independent of t .

- h. Choose a value of t between 5 and 15 and find the optimum value of g_c . How does this compare with the g_c found in liver glycogen (see [Fig. 15-2](#))? (Hint: You may find it useful to use a spreadsheet program.)

Reference

Meléndez-Hevia, E., T.G. Waddell, and E.D. Shelton. 1993.
Optimization of molecular design in the evolution of
metabolism: the glycogen molecule. *Biochem. J.* 295:477–483.

CHAPTER 16

THE CITRIC ACID CYCLE



[16.1 Production of Acetyl-CoA \(Activated Acetate\)](#)

[16.2 Reactions of the Citric Acid Cycle](#)

[16.3 The Hub of Intermediary Metabolism](#)

[16.4 Regulation of the Citric Acid Cycle](#)

As we saw in [Chapter 14](#), some cells obtain energy (ATP) by fermentation, breaking down glucose in the absence of oxygen. For most eukaryotic cells and many bacteria, which live under aerobic conditions and oxidize their organic fuels to carbon dioxide and water, glycolysis is but the first stage in the complete oxidation of glucose. Rather than being reduced to lactate, ethanol, or some other fermentation product, the pyruvate produced by glycolysis is further oxidized to H_2O and CO_2 through the process of **cellular respiration**.

Cellular respiration occurs in three major stages ([Fig. 16-1](#)), the first two of which we discuss in this chapter. In the first, organic fuel molecules — glucose, fatty acids, and some amino acids — are oxidized to yield two-carbon fragments in the form of the acetyl group of acetyl-coenzyme A (acetyl-CoA). In the second stage, the

acetyl groups are oxidized to CO_2 in the [citric acid cycle](#), also called the [tricarboxylic acid \(TCA\) cycle](#) or the [Krebs cycle](#) (after its discoverer, Hans Krebs). Much of the energy of these oxidations is conserved in the reduced electron carriers NADH and FADH_2 . In the third stage of respiration, these reduced coenzymes are themselves oxidized, giving up protons (H^+) and electrons. The electrons are transferred to O_2 via a series of electron-carrying molecules known as the respiratory chain, resulting in the formation of water. In the course of electron transfer, much of the energy available from redox reactions is conserved in the form of ATP, by a process called oxidative phosphorylation. We discuss this third stage in [Chapter 19](#).

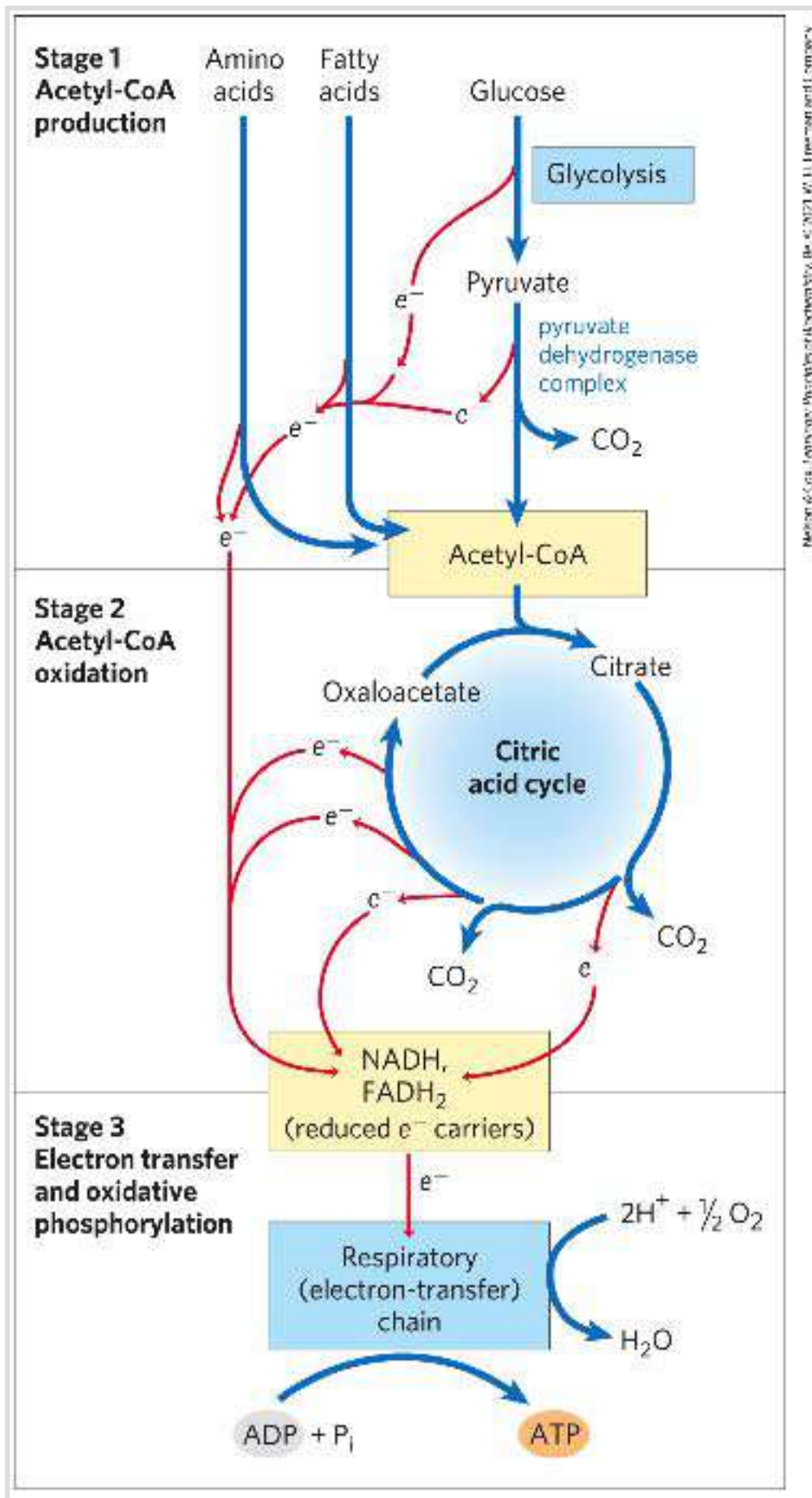


FIGURE 16-1 Catabolism of proteins, fats, and carbohydrates in the three stages of cellular respiration. Stage 1: oxidation of fatty acids, glucose, and some amino acids yields acetyl-CoA. Stage 2: oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted. Stage 3: electrons carried by NADH and FADH_2 are funneled into a chain of mitochondrial (or, in bacteria, plasma membrane-bound) electron carriers — the respiratory chain — ultimately reducing O_2 to H_2O . This electron flow drives the production of ATP.



Hans Krebs, 1900–1981

We use these principles as our guide to this chapter:



Pyruvate is the metabolite that links two central catabolic pathways, glycolysis and the citric acid cycle. It is therefore a logical point for regulation that determines the rate of catabolic activity and the partitioning of pyruvate among its possible uses.

P2 The reactions of the citric acid cycle follow a chemical logic. In its catabolic role the citric acid cycle oxidizes acetyl-CoA to CO_2 and H_2O . Energy from the oxidations in the cycle drives the synthesis of ATP. The chemical strategies for activating groups for oxidation and for conserving energy in the form of reducing power and high-energy compounds are used in many other biochemical pathways.

P3 The citric acid cycle is a hub of metabolism, with catabolic pathways leading in and anabolic pathways leading out. Acetate groups (acetyl-CoA) from the catabolism of various fuels are used in the synthesis of such metabolites as amino acids, fatty acids, and sterols. The breakdown products of many amino acids and nucleotides are intermediates of the cycle, and they can be fed in or siphoned off as needed by the cell.

P4 The central role of the citric acid cycle in metabolism requires that it be regulated in coordination with many other pathways. Regulation occurs by both allosteric and covalent mechanisms that overlap and interact to achieve homeostasis. Some mutations that affect the reactions of the citric acid cycle lead to tumor formation.

P5 Enzymes have evolved to form complexes to efficiently achieve a series of chemical transformations without releasing the intermediates into the bulk solvent. This strategy, seen in the pyruvate dehydrogenase complex and the metabolons of the citric acid cycle, is ubiquitous in other

pathways of metabolism, in respiration, and in the many “ —
somes” that assemble and disassemble informational
macromolecules.

16.1 Production of Acetyl-CoA (Activated Acetate)

Coenzyme A, or CoA ([Fig. 16-2](#)), has a reactive thiol (—SH) group that is critical to its role as an acyl carrier in many metabolic processes, including the citric acid cycle. Acyl groups are covalently linked to the thiol group, forming **thioesters**. Because of their relatively high standard free energies of hydrolysis (see [Figs. 13-16](#), [13-17](#)), thioesters have a high acyl group transfer potential — that is, donation of their acyl groups to a variety of acceptor molecules is a favorable reaction. The acyl group attached to coenzyme A may thus be thought of as “activated” for group transfer.

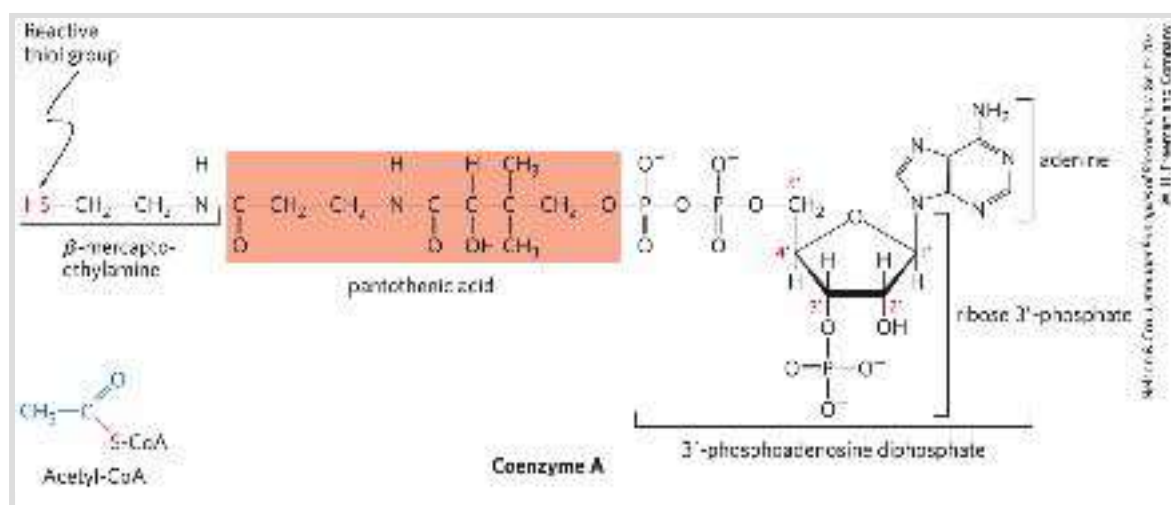


FIGURE 16-2 Coenzyme A (CoA-SH). A hydroxyl group of pantothenic acid is joined to a modified ADP moiety by a phosphate ester bond, and its carboxyl group is attached to β -mercaptoethylamine in amide linkage. The hydroxyl group at the 3' position of the ADP moiety has a phosphoryl group not present in free ADP. The —SH group of the mercaptoethylamine moiety forms a thioester with acetate in acetyl-coenzyme A (acetyl-CoA) (lower left).

In aerobic organisms, glucose and other sugars, fatty acids, and most amino acids are ultimately oxidized to CO_2 and H_2O via the citric acid cycle and the respiratory chain. Before entering the citric acid cycle, the carbon skeletons of monosaccharides and fatty acids are degraded to the acetyl group of acetyl-CoA, the form in which the cycle accepts much of its fuel input. Many amino acid carbons also enter the cycle as acetate, although several amino acids are degraded to other cycle intermediates such as succinate and malate, which then enter the cycle.

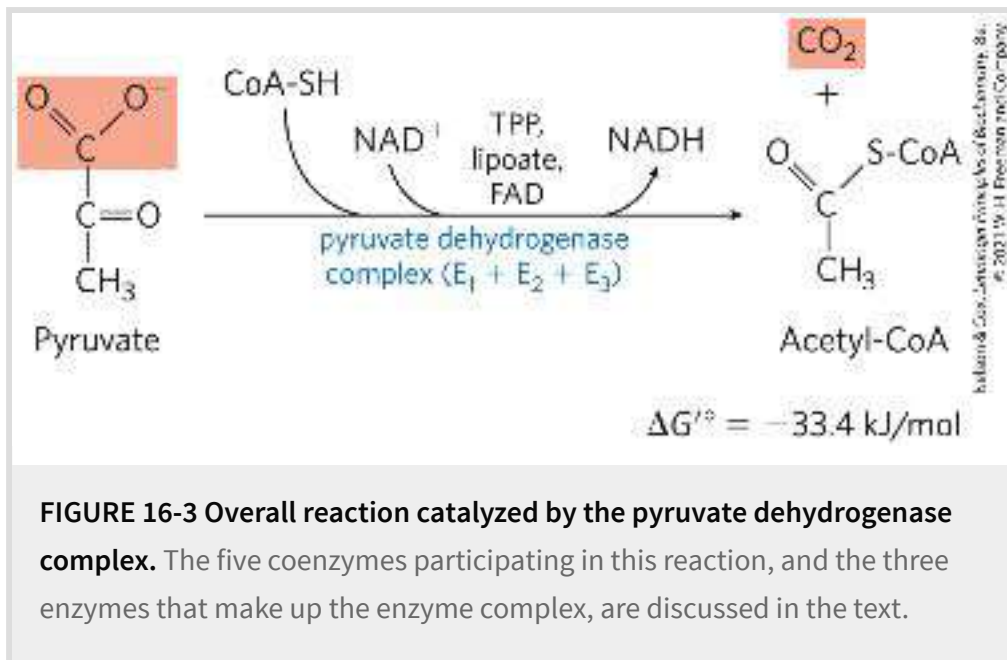
Pyruvate Is Oxidized to Acetyl-CoA and CO_2



Pyruvate generated in the cytosol by glycolysis represents a node in the metabolism of carbohydrates, proteins, and fats. Under anaerobic conditions, pyruvate may simply be reduced to lactate in the cytosol, regenerating NAD^+ for continued ATP production by glycolysis. Pyruvate may serve as a precursor for the synthesis of amino acids ([Chapter 22](#)). In eukaryotes, pyruvate may diffuse into mitochondria, first through large openings in the outer mitochondrial membrane and then into the matrix via an H^+ -coupled pyruvate-specific symporter in the inner mitochondrial membrane, the **mitochondrial pyruvate carrier (MPC)**. Pyruvate that enters mitochondria may be oxidized by the citric acid cycle to generate energy or, after conversion to acetyl-CoA, may be used as the starting material for synthesis of fatty acids and sterols.

Pyruvate in the mitochondrial matrix is oxidized to acetyl-CoA and CO₂ by the **pyruvate dehydrogenase (PDH) complex**, a highly ordered cluster of enzymes and cofactors.  In the PDH complex, a series of chemical intermediates remain bound to the enzyme subunits as a substrate (pyruvate) is transformed into the final product (acetyl-CoA). Five cofactors, four of which are derived from vitamins, participate in the reaction mechanism.  The regulation of this enzyme complex illustrates how a combination of covalent modification and allosteric mechanism results in precisely regulated flux through a metabolic step.

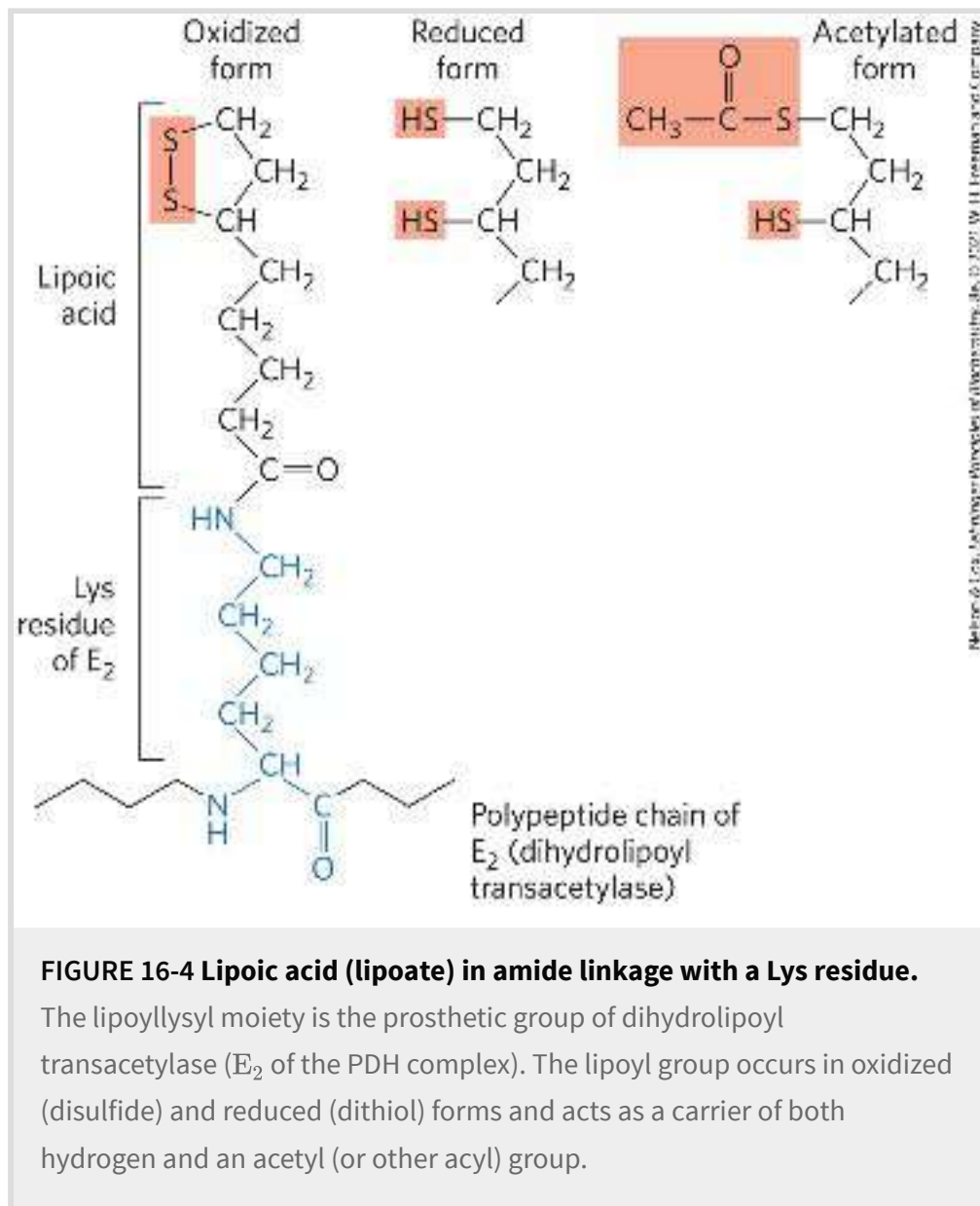
The overall reaction catalyzed by the pyruvate dehydrogenase complex is an **oxidative decarboxylation**, an irreversible oxidation process in which the carboxyl group is removed from pyruvate as a molecule of CO₂ and the two remaining carbons become the acetyl group of acetyl-CoA ([Fig. 16-3](#)). The NADH formed in this reaction gives up a hydride ion (:H⁻) to the respiratory chain ([Fig. 16-1](#)), which carries the two electrons to oxygen or, in anaerobic microorganisms, to an alternative electron acceptor such as nitrate or sulfate. The transfer of electrons from NADH to oxygen ultimately generates 2.5 molecules of ATP per pair of electrons. The irreversibility of the PDH complex reaction has been demonstrated by isotopic labeling experiments: the complex cannot reattach radioactively labeled CO₂ to acetyl-CoA to yield carboxyl-labeled pyruvate.



The PDH Complex Employs Three Enzymes and Five Coenzymes to Oxidize Pyruvate

The combined dehydrogenation and decarboxylation of pyruvate to the acetyl group of acetyl-CoA ([Fig. 16-3](#)) requires the sequential action of three different enzymes and five different coenzymes or prosthetic groups — thiamine pyrophosphate (TPP), lipoate, coenzyme A (CoA, sometimes denoted CoA-SH, to emphasize the role of the —SH group), flavin adenine dinucleotide (FAD), and nicotinamide adenine dinucleotide (NAD). We encountered TPP as the coenzyme of pyruvate decarboxylase (see [Fig. 14-13](#)). **Lipoate** ([Fig. 16-4](#)), has two thiol groups that can undergo reversible oxidation to a disulfide bond (—S—S—), similar to that between two Cys residues in a protein. Because of its capacity to undergo oxidation-reduction reactions,

lipoate can serve both as an electron (hydrogen) carrier and as an acyl carrier, as we shall see. CoA serves as the carrier of the activated acyl group. We described the roles of FAD and NAD as electron carriers in [Chapter 13](#).



The PDH complex contains multiple copies of three enzymes — **pyruvate dehydrogenase (E₁)**, **dihydrolipoyl transacetylase (E₂)**, and **dihydrolipoyl dehydrogenase (E₃)**—that catalyze the

oxidation of pyruvate. The number of copies of each enzyme and therefore the size of the complex varies among species. A central core is formed of many copies (24 to 60) of E_2 surrounded by multiple and variable numbers of copies of E_1 and E_3 ([Fig. 16-5](#)). Two regulatory proteins are also part of the complex: a protein kinase and a phosphoprotein phosphatase, discussed below.

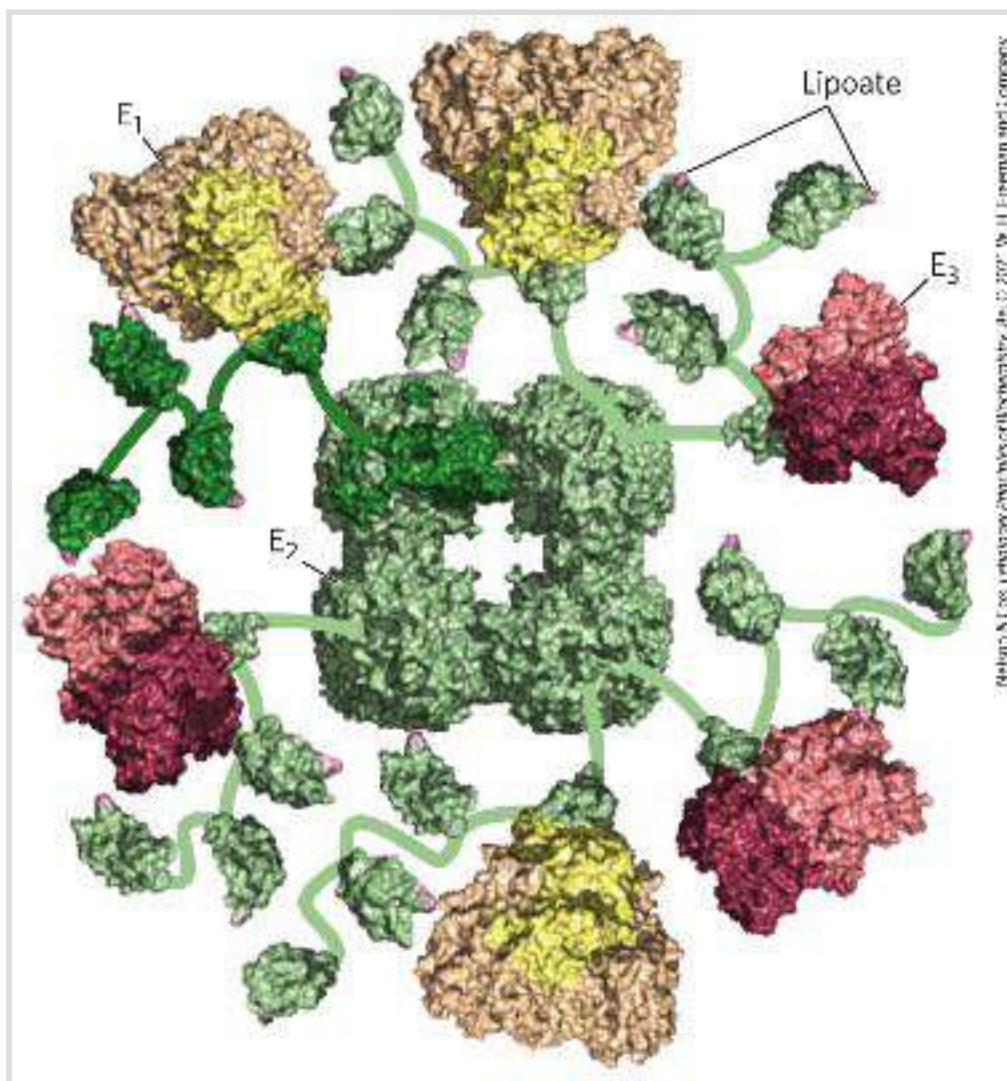


FIGURE 16-5 Structure of the pyruvate dehydrogenase complex. The complex is so big and so flexible that solving its structure required a combination of methods, including x-ray crystallography and NMR spectroscopy; once the individual pieces were solved, cryo-EM of the whole structure was used to assemble the pieces from several organisms to get this view. The core (E_2) is from the gram-negative bacterium *Azotobacter*

vinelandii. E₁ and E₃ are from the thermophilic gram-positive bacterium *Geobacillus stearothermophilus*. E₁, pyruvate dehydrogenase (yellow); E₂, dihydrolipoyl transacetylase (green); and E₃, dihydrolipoyl dehydrogenase (red). The central core of the *Azotobacter* PDH complex consists of 24 copies of E₂, but to simplify the structure, only six are shown here. Multiple copies of E₁ and E₃ surround the central core, and flexible arms (shown schematically) reach out from E₂ to E₁ and E₃, carrying the lipoyl moiety (pink) from the active site of one enzyme to that of the next. The amino acid sequences and three-dimensional structures of individual domains show that both catalytic mechanism and structure have been conserved in evolution. [Information from D. Goodsell, doi:10.2210/rcsb_pdb/mom_2012_9. Data from PDB ID 1LAC F. Dardel et al., *J. Mol. Biol.* 229:1037, 1993; PDB ID 1EAA A. Mattevi et al., *Biochemistry* 32:3887, 1993; PDB ID 1W85 R. A. Frank et al., *Science* 306:872, 2004; PDB ID 1EBD S. S. Mande et al., *Structure* 4:277, 1996.]

The active site of E₁ has noncovalently bound TPP. E₂ has the prosthetic group lipoate, attached through an amide bond to the ϵ -amino group of a Lys residue ([Fig. 16-4](#)). E₂ has three functionally distinct domains: one or more (depending on the species) amino-terminal *lipoyl domains*, containing the lipoyl-Lys residue(s); the central E₁- and E₃-*binding domain*; and the inner-core *acyltransferase domain*, which contains the acyltransferase active site. The domains of E₂ are separated by linkers, sequences of 20 to 30 amino acid residues, rich in Ala and Pro and interspersed with charged residues; these linkers tend to assume their extended forms, holding the three domains apart. The attachment of lipoate to the end of a Lys side chain in E₂ produces a long, flexible arm that can move from the active site of E₁ to the active sites of E₂ and E₃, a distance of perhaps 5 nm or

more, while holding the intermediate captive throughout the reaction sequence. E_3 has tightly bound FAD.

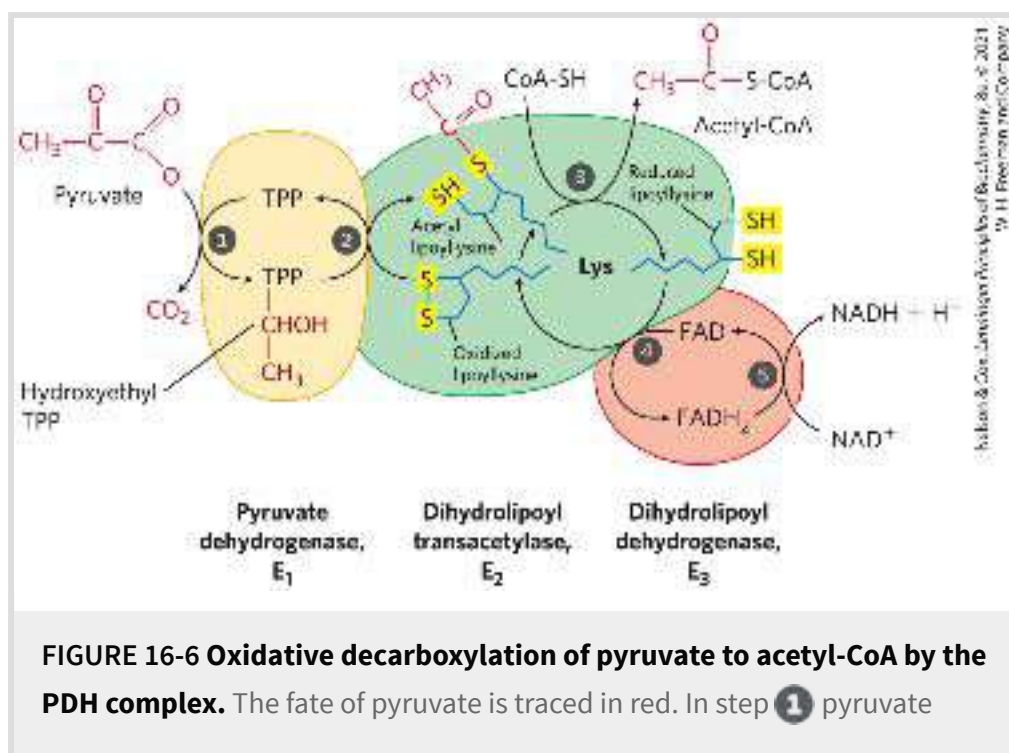
 This basic E_1 — E_2 — E_3 structure of the PDH complex is seen in two other enzyme complexes that catalyze similar reactions: α -ketoglutarate dehydrogenase, which oxidizes α -ketoglutarate in the citric acid cycle (described below), and the branched-chain α -keto acid dehydrogenase, which oxidizes α -keto acids derived from the breakdown of the branched-chain amino acids valine, isoleucine, and leucine (see [Fig. 18-28](#)).

Within a given species, E_3 is identical in all three complexes. The remarkable similarity in protein structure, cofactor requirements, and reaction mechanisms of these three complexes doubtless reflects a common evolutionary origin; they are paralogs.

The PDH Complex Channels Its Intermediates through Five Reactions

[Figure 16-6](#) shows schematically how  the pyruvate dehydrogenase complex carries out the five consecutive reactions in the decarboxylation and dehydrogenation of pyruvate without allowing the intermediates to leave the surface of the complex. Step ❶ is essentially identical to the reaction catalyzed by pyruvate decarboxylase (see [Fig. 14-13c](#)); C-1 of pyruvate is released as CO_2 , and C-2, which in pyruvate has the oxidation

state of an aldehyde, is attached to TPP as a hydroxyethyl group. This first step is the slowest and therefore limits the rate of the overall reaction. In step ② the hydroxyethyl group is oxidized to the level of a carboxylic acid (acetate). The two electrons removed in this reaction reduce the —S—S— of a lipoyl group on E₂ to two thiol (—SH) groups. The acetyl moiety produced in this oxidation-reduction reaction is first esterified to one of the lipoyl —SH groups, then transesterified to CoA to form acetyl-CoA (step ③). Thus the energy of oxidation drives the formation of a high-energy thioester of acetate. The remaining reactions catalyzed by the PDH complex (by E₃, in steps ④ and ⑤) are electron transfers necessary to regenerate the oxidized (disulfide) form of the lipoyl group of E₂, to prepare the enzyme complex for another round of oxidation. The electrons removed from the hydroxyethyl group derived from pyruvate pass through FAD to NAD⁺, forming NADH, which can enter the respiratory chain.



reacts with the bound thiamine pyrophosphate (TPP) of pyruvate dehydrogenase (E_1) and is decarboxylated to the hydroxyethyl derivative. Pyruvate dehydrogenase also carries out step **2**, the transfer of two electrons and the acetyl group from TPP to the oxidized form of the lipoyllysyl group of the core enzyme, dihydrolipoyl transacetylase (E_2), to form the acetyl thioester of the reduced lipoyl group. Step **3** is a transesterification in which the —SH group of CoA replaces the —SH group of E_2 to yield acetyl-CoA and the fully reduced (dithiol) form of the lipoyl group. In step **4** dihydrolipoyl dehydrogenase (E_3) promotes transfer of two hydrogen atoms from the reduced lipoyl groups of E_2 to the FAD prosthetic group of E_3 , restoring the oxidized form of the lipoyllysyl group of E_2 . In step **5** the reduced $FADH_2$ of E_3 transfers a hydride ion to NAD^+ , forming NADH. The enzyme complex is now ready for another catalytic cycle. (Subunit colors correspond to those in [Fig. 16-5](#).)



The five-reaction sequence shown in [Figure 16-6](#) is an example of **substrate channeling**. The swinging lipoyllysyl arms of E_2 accept from E_1 the two electrons and the acetyl group that it derived from pyruvate, and pass them to E_3 . The intermediates of the multistep sequence never leave the complex, and the local concentration of the substrate of E_2 is kept very high. This substrate channeling also prevents theft of the activated acetyl group by other enzymes that use this group as substrate. We will encounter other enzymes that use a similar tethering mechanism to channel substrate between active sites, with lipoate, biotin, or a CoA-like moiety serving as cofactor.



Four different vitamins required in human nutrition are vital components of the pyruvate dehydrogenase complex: thiamine (TPP), pantothenate (CoA), riboflavin (FAD), and niacin (NAD). As one might predict, mutations in the genes for the subunits of the

PDH complex, or a dietary vitamin deficiency, can have severe consequences. Thiamine-deficient animals are unable to oxidize pyruvate normally. This is of particular importance to the brain, which usually obtains all its energy from the aerobic oxidation of glucose in a pathway that necessarily includes the oxidation of pyruvate. Beriberi, a disease that results from thiamine deficiency, is characterized by loss of neural function. This disease occurs primarily in populations that rely on a diet consisting mainly of white (polished) rice, which lacks the hulls that contain most of the thiamine found in rice. People who habitually consume large amounts of alcohol can also develop thiamine deficiency, because much of their dietary intake consists of the vitamin-free “empty calories” of distilled spirits. An elevated level of pyruvate in the blood is often an indicator of defects in pyruvate oxidation due to one of these causes. ■

SUMMARY 16.1 *Production of Acetyl-CoA (Activated Acetate)*

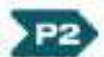
■ Pyruvate, the end product of glycolysis, enters the mitochondrial matrix, where the pyruvate dehydrogenase (PDH) complex oxidizes it to CO_2 , acetyl-CoA — the starting material for the citric acid cycle — and NADH.

■ The supramolecular PDH complex includes multiple copies of three enzymes. Pyruvate dehydrogenase, E_1 (with bound TPP), decarboxylates pyruvate, producing hydroxyethyl-TPP, which is then oxidized to an acetyl group. The electrons from this oxidation reduce the disulfide of lipoate bound to E_2 , the dihydrolipoyl transacetylase, and the acetyl group is transferred

to one of the —SH groups of the reduced lipoate through a thioester bond. E₂ catalyzes the transfer of the acetyl group to coenzyme A, forming acetyl-CoA. Dihydrolipoyl dehydrogenase, E₃, catalyzes the regeneration of the disulfide (oxidized) form of lipoate, passing electrons first to FAD, then to NAD⁺.

■ In the PDH complex we see examples of two strategies that we will see repeated in other metabolic enzyme systems. Its organization is very similar to that of the enzyme complexes that catalyze the oxidation of α -ketoglutarate and the branched-chain α -keto acids. And the long lipoyllysyl arm of E₂ is used to channel the substrate from the active site of E₁ to E₂ to E₃, tethering the intermediates to the enzyme complex and increasing the efficiency of the overall reaction.

16.2 Reactions of the Citric Acid Cycle

We are now ready to trace the process by which acetyl-CoA undergoes oxidation. This chemical transformation is carried out by the citric acid cycle, the first *cyclic* pathway we have encountered ([Fig. 16-7](#)). To begin a turn of the cycle, acetyl-CoA donates its acetyl group to the four-carbon compound oxaloacetate to form the six-carbon citrate. Citrate is then transformed into isocitrate, also a six-carbon molecule, which is dehydrogenated with loss of CO₂ to yield the five-carbon compound α -ketoglutarate (also called 2-oxoglutarate). α -Ketoglutarate undergoes loss of a second molecule of CO₂ and ultimately yields the four-carbon compound succinate. Succinate is enzymatically converted in three steps into the four-carbon oxaloacetate — which is then ready to react with another molecule of acetyl-CoA. In each turn of the cycle, one acetyl group (two carbons) enters as acetyl-CoA and two molecules of CO₂ leave; one molecule of oxaloacetate is used to form citrate and one molecule of oxaloacetate is regenerated. No net removal of oxaloacetate occurs;  one molecule of oxaloacetate can theoretically bring about oxidation of an infinite number of acetyl groups, effectively acting catalytically; the steady-state concentration of oxaloacetate is very low (micromolar). Four of the eight steps in this process are oxidations, in which the energy of oxidation is very efficiently conserved in the form of the reduced coenzymes NADH and FADH₂. These two carriers donate their electrons to the respiratory chain where electron flow drives ATP synthesis.

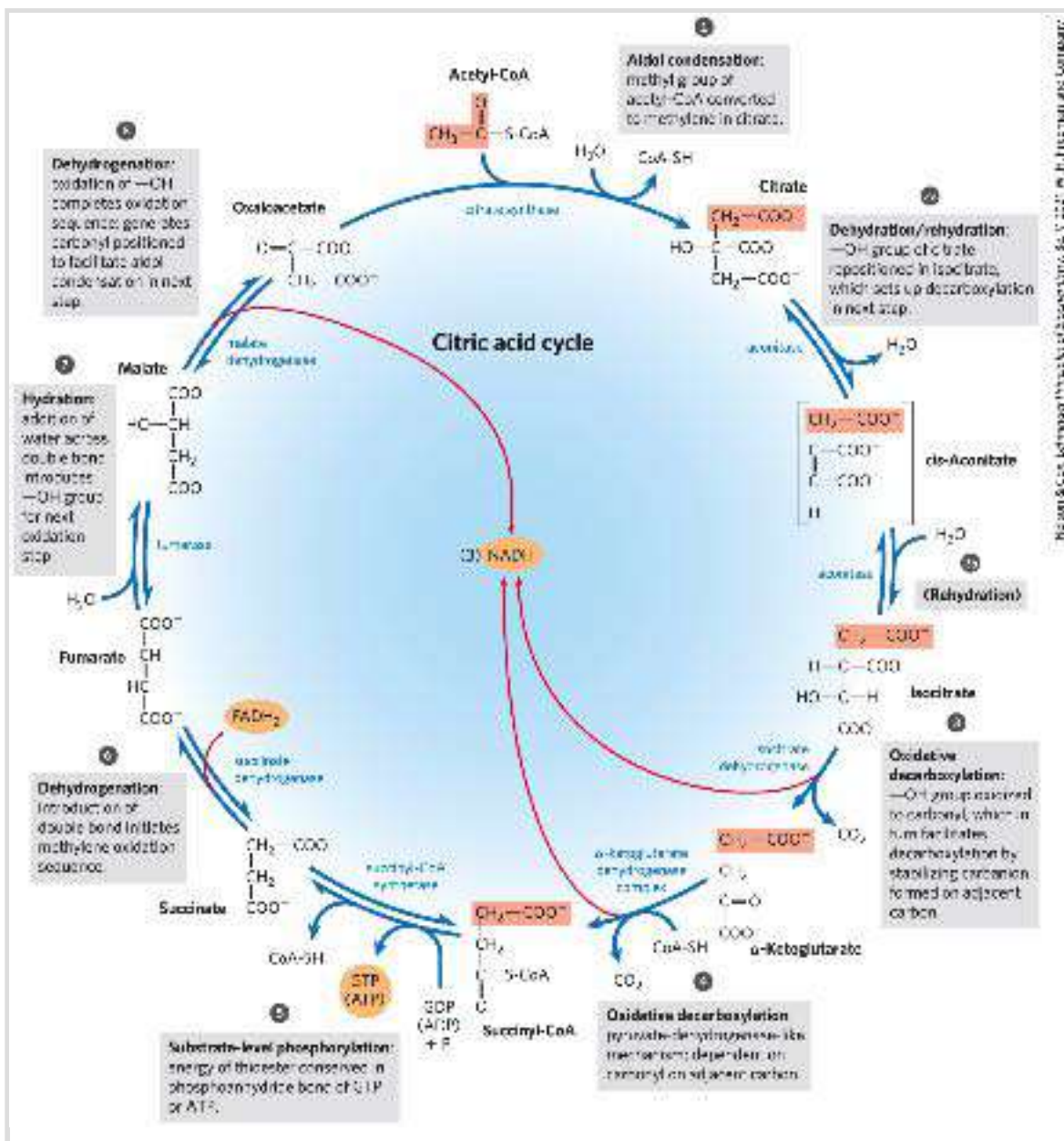


FIGURE 16-7 Reactions of the citric acid cycle. The carbon atoms shaded in pink are those derived from the acetate of acetyl-CoA in the first turn of the cycle; these are *not* the carbons released as CO₂ in the first turn. Note that in succinate and fumarate, the two-carbon group derived from acetate can no longer be specifically denoted; because succinate and fumarate are symmetric molecules, C-1 and C-2 are indistinguishable from C-4 and C-3. The red arrows show where energy is conserved by electron transfer to FAD or NAD⁺, forming FADH₂ or NADH + H⁺. Steps **1**, **3**, and **4** are essentially irreversible in the cell; all other steps are reversible. The nucleoside triphosphate product of step **5** may be either ATP or GTP, depending on which succinyl-CoA synthetase isozyme is the catalyst.

Although the citric acid cycle is central to energy-yielding metabolism, its role is not limited to energy conservation. Four- and five-carbon intermediates of the cycle serve as precursors for a wide variety of products. To replace intermediates removed for this purpose, cells employ anaplerotic (replenishing) reactions, which are described below.

Eugene Kennedy and Albert Lehninger showed in 1948 that, in eukaryotes, the entire set of reactions of the citric acid cycle takes place in mitochondria. Isolated mitochondria were found to contain not only all the enzymes and coenzymes required for the citric acid cycle, but also all the enzymes and proteins necessary for the last stage of respiration — electron transfer and ATP synthesis by oxidative phosphorylation. As we shall see in later chapters, mitochondria also contain the enzymes for the oxidation of fatty acids and some amino acids to acetyl-CoA, and the oxidative degradation of other amino acids to α -ketoglutarate, succinyl-CoA, or oxaloacetate. Thus, in nonphotosynthetic eukaryotes, the mitochondrion is the site of most energy-yielding oxidative reactions and of the coupled synthesis of ATP. In photosynthetic eukaryotes, mitochondria are the major site of ATP production in the dark, but in daylight, chloroplasts produce most of the organism's ATP. In most bacteria, the enzymes of the citric acid cycle are in the cytosol, and the plasma membrane plays a role analogous to that of the inner mitochondrial membrane in ATP synthesis ([Chapter 19](#)).

The Sequence of Reactions in the Citric Acid Cycle Makes Chemical Sense

Acetyl-CoA produced in the breakdown of carbohydrates, fats, and proteins must be completely oxidized to CO_2 if the maximum potential energy is to be extracted from these fuels.  However, it is not biochemically feasible to directly oxidize acetate (or acetyl-CoA) to CO_2 . Decarboxylation of this two-carbon acid would yield CO_2 and methane (CH_4). Methane is chemically rather stable, and except for certain methanotrophic bacteria that grow in methane-rich niches, organisms do not have the cofactors and enzymes needed to oxidize methane. Methylene groups ($-\text{CH}_2-$), however, are readily metabolized by enzyme systems present in most organisms. In typical oxidation sequences, two adjacent methylene groups ($-\text{CH}_2-\text{CH}_2-$) are involved, at least one of which is adjacent to a carbonyl group. As we noted in [Chapter 13 \(p. 473\)](#), carbonyl groups are particularly important in the chemical transformations of metabolic pathways. The carbon of the carbonyl group has a partial positive charge due to the electron-withdrawing property of the carbonyl oxygen and is therefore an electrophilic center. A carbonyl group can facilitate the formation of a carbanion on an adjoining carbon by delocalizing the carbanion's negative charge. We see an example of the oxidation of a methylene group in the citric acid cycle, as succinate is oxidized (steps 6 to 8 in [Fig. 16-7](#)) to form a carbonyl (in oxaloacetate) that is more chemically reactive than either a methylene group or methane.

In short, if acetyl-CoA is to be oxidized efficiently, the methyl group of the acetyl-CoA must be attached to something. The first step of the citric acid cycle — the condensation of acetyl-CoA with oxaloacetate — neatly solves the problem of the unreactive methyl group. The carbonyl of oxaloacetate acts as an electrophilic center, which is attacked by the methyl carbon of acetyl-CoA in an aldol condensation

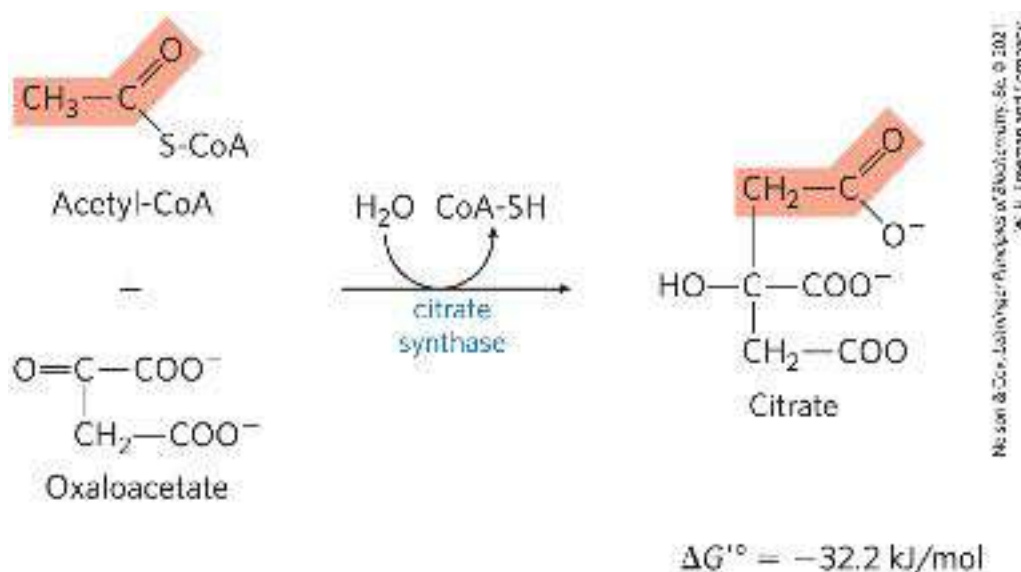
(p. 473) to form citrate (step ① in Fig. 16-7). The methyl group of acetate has been converted into a methylene in citric acid. This tricarboxylic acid then readily undergoes a series of oxidations that eliminate two carbons as CO_2 .  Note that all steps featuring the breakage or formation of carbon-carbon bonds (steps ①, ③, and ④) rely on properly positioned carbonyl groups. As in all metabolic pathways, there is a chemical logic to the sequence of steps in the citric acid cycle: each step either involves an energy-conserving oxidation or is a necessary prelude to the oxidation, placing functional groups in position to facilitate oxidation or oxidative decarboxylation. As you learn the steps of the cycle, keep in mind the chemical rationale for each; it will make the process easier to understand and remember.

The Citric Acid Cycle Has Eight Steps

In examining the eight successive reaction steps of the citric acid cycle, we place special emphasis on the chemical transformations taking place as citrate formed from acetyl-CoA and oxaloacetate is oxidized to yield CO_2 and the energy of this oxidation is conserved in the reduced coenzymes NADH and FADH_2 .

① Formation of Citrate

The first reaction of the cycle is the condensation of acetyl-CoA with **oxaloacetate** to form **citrate**, catalyzed by **citrate synthase**.



In this reaction, the methyl carbon of the acetyl group is joined to the carbonyl group (C-2) of oxaloacetate. Citroyl-CoA is a transient intermediate formed on the active site of the enzyme (see [Fig. 16-9](#)). It undergoes hydrolysis to free CoA and citrate, which are released from the active site. The hydrolysis of this high-energy thioester intermediate makes the forward reaction highly exergonic. The large, negative standard free-energy change of the forward citrate synthase reaction is essential to the operation of the cycle because the concentration of oxaloacetate is normally very low (micromolar). The CoA liberated in this reaction is recycled to participate in the oxidative decarboxylation of another molecule of pyruvate by the PDH complex.

Citrate synthase from mitochondria is a homodimer ([Fig. 16-8](#)). Each subunit is a single polypeptide with two domains, one large and rigid, the other smaller and more flexible, with the active site between them. Oxaloacetate, the first substrate to bind to the enzyme, induces a large conformational change in the flexible domain, creating a binding site for the second substrate, acetyl-CoA. When citroyl-CoA has formed in the enzyme active site, another conformational change

brings about thioester hydrolysis, releasing CoA-SH. This induced fit of the enzyme first to its substrate and then to its reaction intermediate decreases the likelihood of premature and unproductive cleavage of the thioester bond of acetyl-CoA. Kinetic studies of the enzyme are consistent with this ordered bisubstrate mechanism (see [Fig. 6-15](#)). The reaction catalyzed by citrate synthase is essentially an aldol condensation ([p. 473](#)), involving a thioester (acetyl-CoA) and a ketone (oxaloacetate) ([Fig. 16-9](#)).

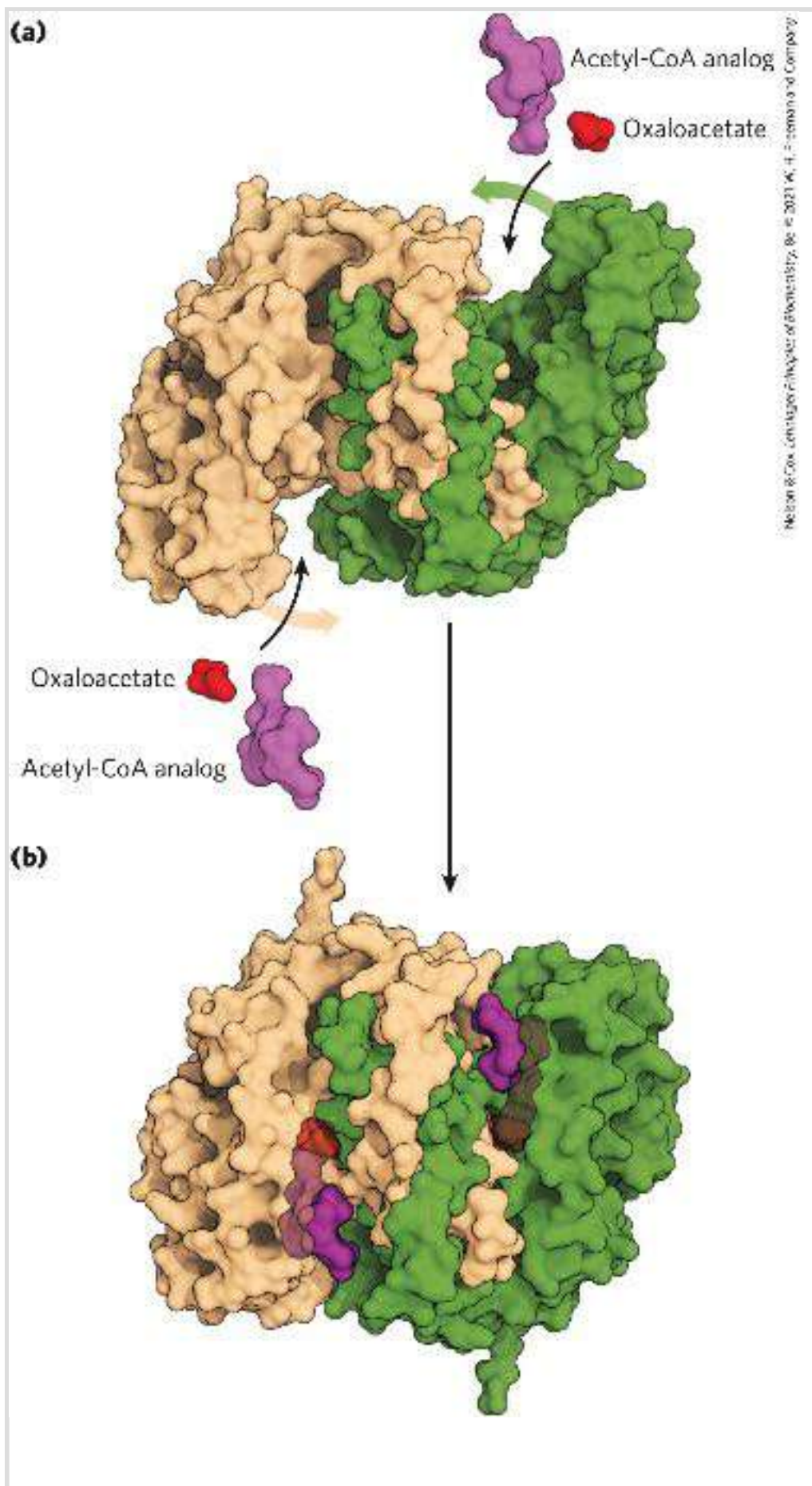
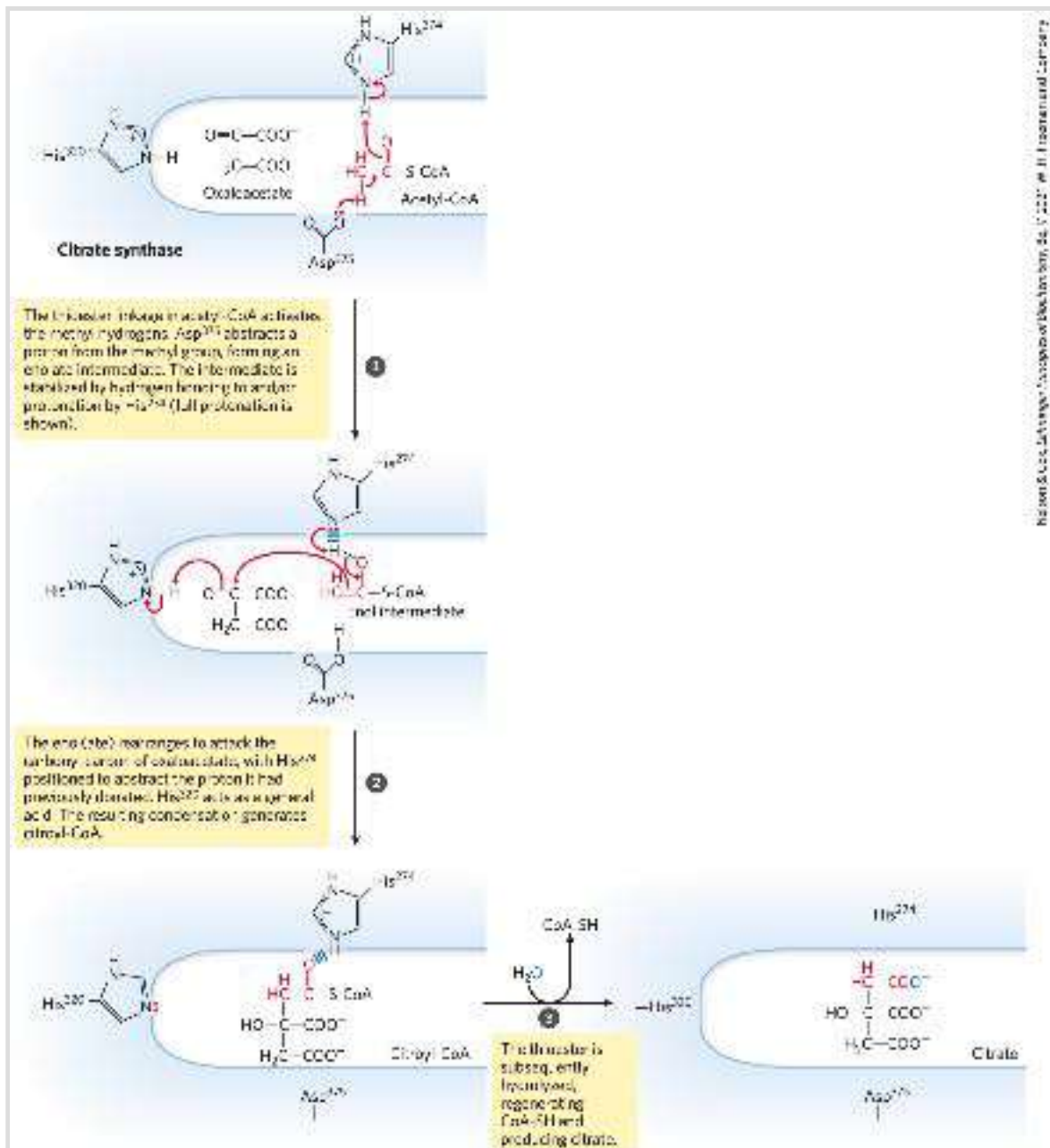


FIGURE 16-8 Structure of citrate synthase. The flexible domain of each subunit undergoes a large conformational change on binding oxaloacetate, creating a binding site for acetyl-CoA. (a) Open form of the enzyme alone; (b) closed form with bound oxaloacetate and a stable analog of acetyl-CoA (carboxymethyl-CoA). In these representations one subunit is colored tan and one green. [Data from (a) PDB ID 5CSC, D.-I. Liao et al., *Biochemistry* 30:6031, 1991;]

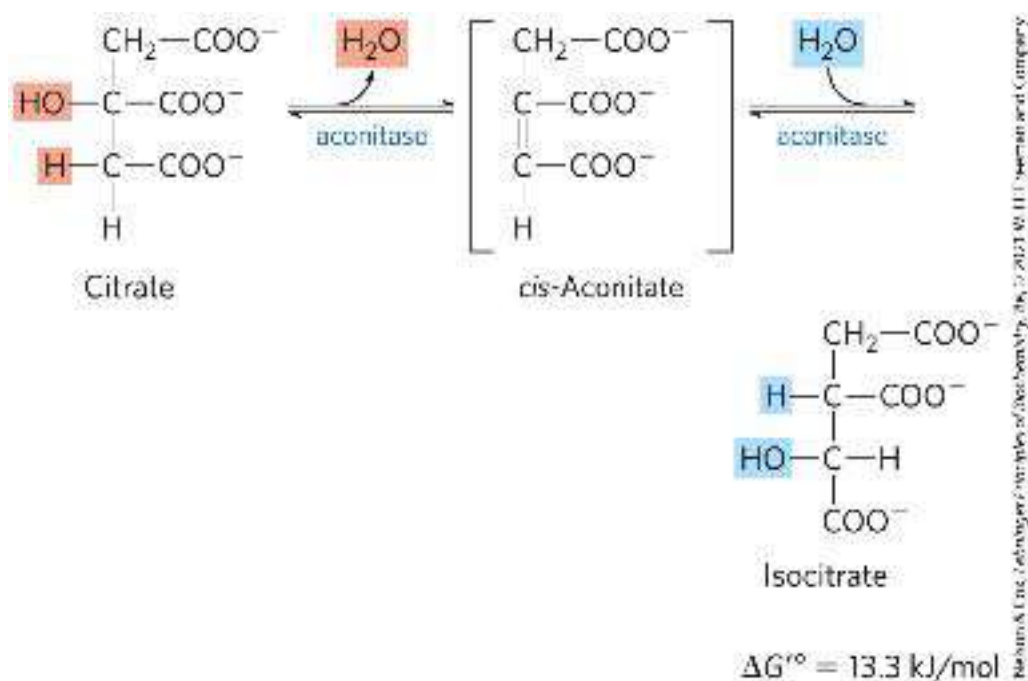


MECHANISM FIGURE 16-9 Citrate synthase. In the citrate synthase reaction in mammals, oxaloacetate binds first, in a strictly ordered reaction sequence. This binding triggers a conformational change that opens up the binding site for acetyl-CoA. Oxaloacetate is

specifically oriented in the active site of citrate synthase by interaction of its two carboxylates with two positively charged Arg residues (not shown here). [Information from S. J. Remington, *Curr. Opin. Struct. Biol.* 2:730, 1992.]

2 Formation of Isocitrate via *cis*-Aconitate

The enzyme **aconitase** (more formally, **aconitate hydratase**) catalyzes the reversible transformation of citrate to **isocitrate**, through the intermediary formation of the tricarboxylic acid ***cis*-aconitate**, which normally does not dissociate from the active site. Aconitase can promote the reversible addition of H₂O to the double bond of enzyme-bound *cis*-aconitate in two different ways, one leading to citrate and the other to isocitrate:



Although the equilibrium mixture at pH 7.4 and 25 °C contains less than 10% isocitrate, in the cell the reaction is pulled to the right because isocitrate is immediately consumed in the next step of the

cycle, lowering its steady-state concentration. Aconitase contains an **iron-sulfur center** ([Fig. 16-10](#)), which acts both in the binding of the substrate at the active site and in the catalytic addition or removal of H_2O . In iron-depleted cells, aconitase loses its iron-sulfur center and acquires a new role in the regulation of iron homeostasis. Aconitase is one of many enzymes known to “moonlight” in a second role ([Box 16-1](#)).

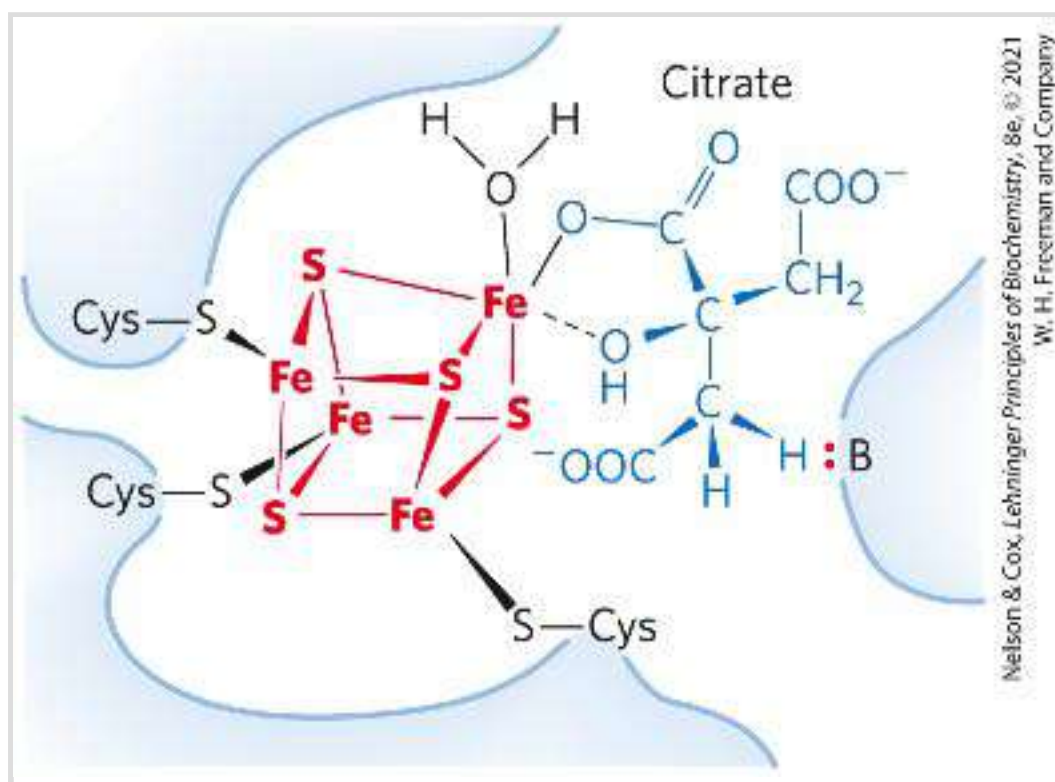


FIGURE 16-10 Iron-sulfur center in aconitase. The iron-sulfur center is in red, the citrate molecule in blue. Three Cys residues of the enzyme bind three iron atoms; the fourth iron is bound to one of the carboxyl groups of citrate and also interacts noncovalently with a hydroxyl group of citrate (dashed bond). A basic residue (:B) in the enzyme helps to position the citrate in the active site. The iron-sulfur center acts in both substrate binding and catalysis. The general properties of iron-sulfur proteins are discussed in [Chapter 19](#).

BOX 16-1

Moonlighting Enzymes: Proteins with More Than One Job

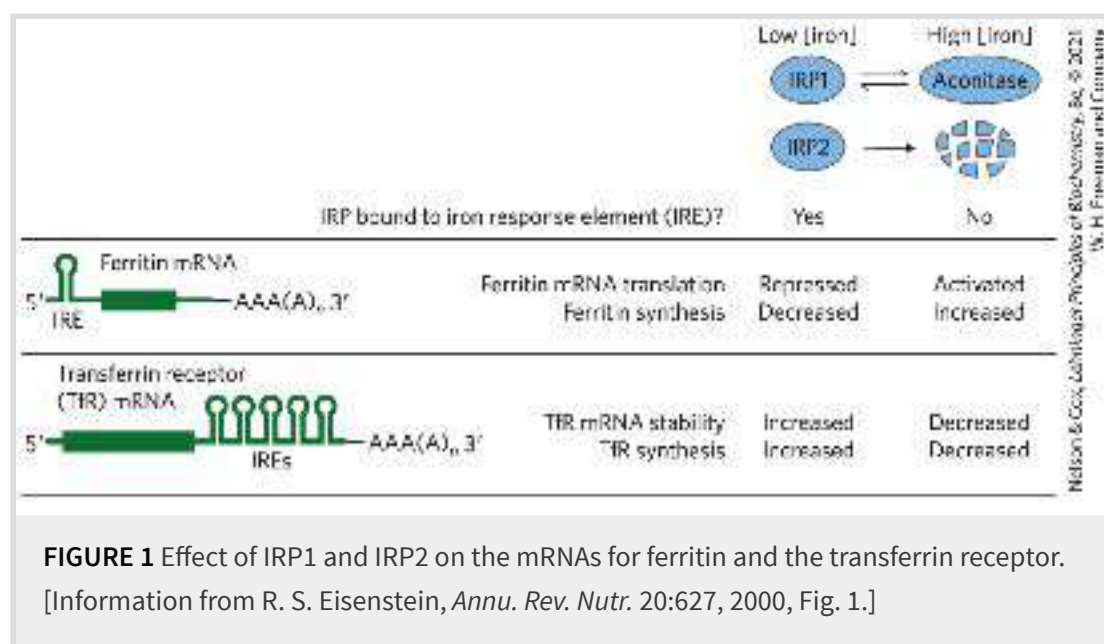
Before protein and DNA sequencing became routine, two groups of investigators studying two unrelated questions sometimes discovered that “their” proteins had similar properties. Further comparison sometimes showed that they were studying the *same* protein by following *different* functions. We now know that many proteins **moonlight** in a second role, a phenomenon sometimes called **gene sharing**. Today, researchers using annotated protein and DNA sequence databases can more easily find proteins of the same sequence that have been identified as having different functions, and a protein sequence annotated as having a given function doesn’t necessarily have *only* that function. When a protein with a known function is made inactive by a mutation, and the resulting mutant organisms show a phenotype with no obvious relation to that function, protein moonlighting can sometimes explain the puzzling result.

One citric acid cycle enzyme is a known moonlighter. Eukaryotic cells have two isozymes of aconitase. The mitochondrial isozyme converts citrate to isocitrate in the citric acid cycle. The cytosolic isozyme has two distinct functions. It catalyzes the conversion of citrate to isocitrate, providing the substrate for a cytosolic isocitrate dehydrogenase that generates NADPH as reducing power for fatty acid synthesis and other anabolic processes in the cytosol. But it also has a role in cellular iron homeostasis.

All cells must obtain iron for the proteins that require it as a cofactor. In humans, severe iron deficiency results in anemia, an insufficient supply of erythrocytes and a reduced oxygen-carrying capacity that can be life-threatening. Too much iron is also harmful: it accumulates in and damages the liver in hemochromatosis and other diseases. Iron obtained in the diet is carried in the blood by the protein **transferrin** and enters cells via endocytosis mediated by the **transferrin receptor**. Once inside cells, iron is used in the synthesis of hemes, cytochromes, Fe-S proteins, and other Fe-dependent proteins; excess iron is stored bound to the protein **ferritin**. The levels of transferrin, transferrin receptor, and ferritin are therefore crucial to cellular iron homeostasis. The synthesis of these three proteins is regulated in response to iron availability — and aconitase, in its moonlighting job, plays a key regulatory role.

Aconitase has an essential Fe-S cluster at its active site (see [Fig. 16-10](#)). When a cell is depleted of iron, this Fe-S cluster is disassembled and the enzyme loses its

aconitase activity. But the apoenzyme (apoaconitase, lacking its Fe-S cluster) so formed has now acquired its second activity: regulating iron metabolism. Cytosolic apoaconitase is identical to **iron regulatory protein 1 (IRP1)** and closely related to **IRP2**. Both IRP1 and IRP2 bind to regions in the mRNAs encoding ferritin and the transferrin receptor, with effects on iron mobilization and iron uptake. These mRNA sequences are part of hairpin structures (see [Fig. 8-23](#)) called **iron response elements (IREs)**, located at the 5' and 3' ends of the mRNAs ([Fig. 1](#)). When bound to the 5'-untranslated IRE sequence in the ferritin mRNA, IRPs block ferritin synthesis; when bound to the 3'-untranslated IRE sequence in the transferrin receptor mRNA, they stabilize the mRNA, preventing its degradation and thus allowing the synthesis of more copies of the receptor protein per mRNA molecule. So, in iron-deficient cells, iron uptake becomes more efficient and iron storage (bound to ferritin) is reduced. When cellular iron concentrations return to normal levels, IRP1 regains its Fe-S cluster, converting to aconitase, and IRP2 undergoes proteolytic degradation, ending the low-iron response.



The enzymatically active aconitase and the moonlighting, regulatory apoaconitase have different structures. As the active aconitase, the protein has two lobes that close around the Fe-S cluster; as IRP1, the two lobes open, exposing the mRNA-binding site ([Fig. 2](#)).

It makes evolutionary sense that the enzymes of central metabolism have had a long time to evolve additional functions. The glycolytic enzyme pyruvate kinase acts in the nucleus to regulate the transcription of genes that respond to thyroid

hormone. Glyceraldehyde 3-phosphate dehydrogenase moonlights both as uracil DNA glycosylase, effecting the repair of damaged DNA, and as a regulator of histone H2B transcription. Several glycolytic enzymes, including phosphoglycerate kinase, triose phosphate isomerase, and lactate dehydrogenase, moonlight as crystallins in the lens of the vertebrate eye.

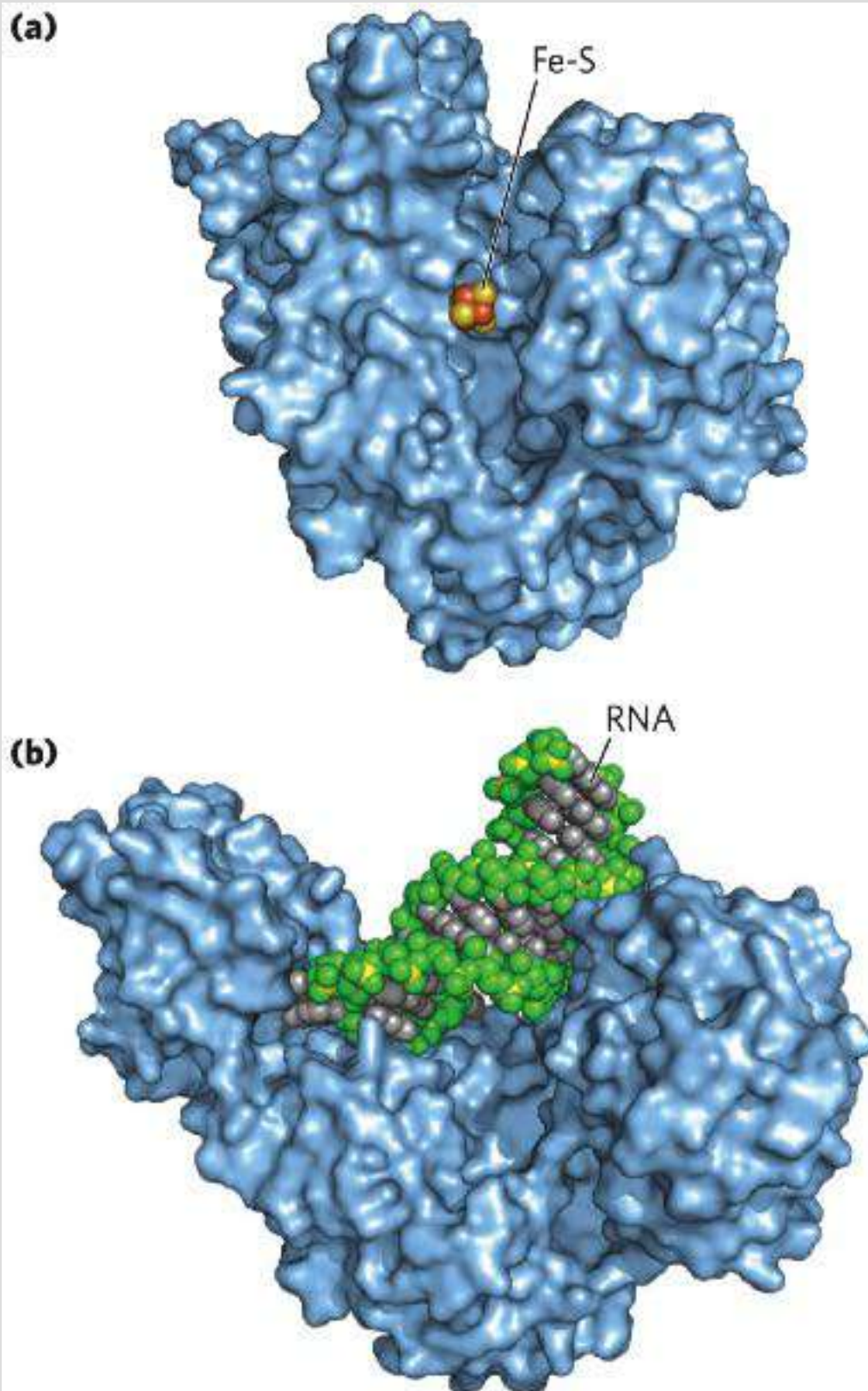


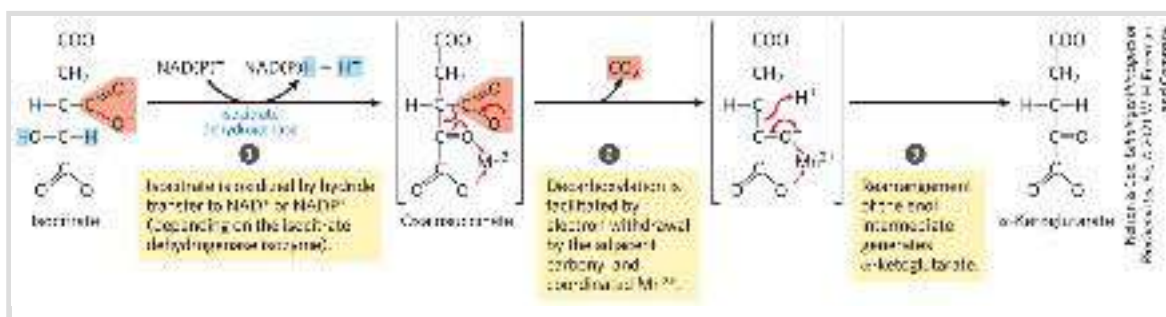
FIGURE 2 Two forms of cytosolic aconitase/IRP1 with two distinct functions. (a) In aconitase, the two major lobes are closed and the Fe-S cluster is buried; the protein has been made transparent here to show the Fe-S cluster. (b) In IRP1, the lobes open,

exposing a binding site for the mRNA hairpin. [Data from (a) PDB ID 2B3Y, J. Dupuy et al., *Structure* 14:129, 2006; (b) PDB ID 2IPY, W. E. Walden et al., *Science* 314:1903, 2006.]

3 Oxidation of Isocitrate to α -Ketoglutarate and CO_2

In the next step, **isocitrate dehydrogenase** catalyzes oxidative decarboxylation of isocitrate to form α -ketoglutarate ([Fig. 16-11](#)).

Mn^{2+} in the active site interacts with the carbonyl group of the intermediate oxalosuccinate, which is formed transiently but does not leave the binding site until decarboxylation converts it to α -ketoglutarate. Mn^{2+} also stabilizes the enol formed transiently by decarboxylation.



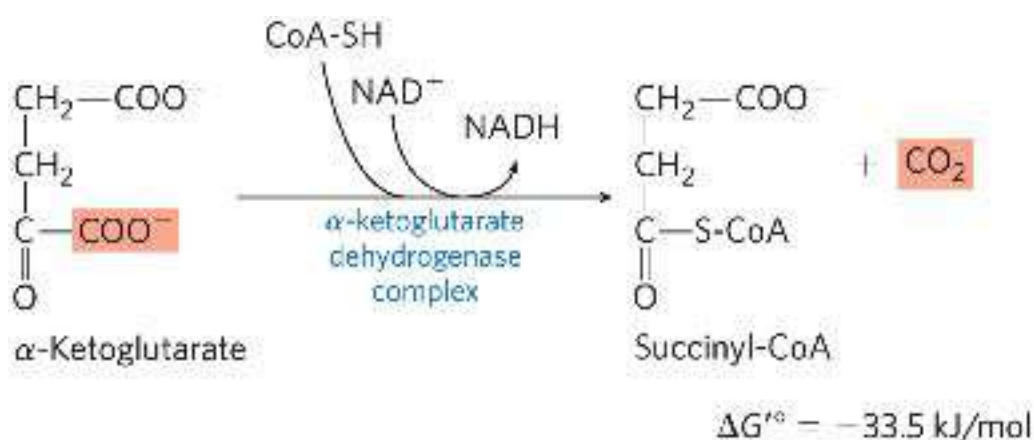
MECHANISM FIGURE 16-11 Isocitrate dehydrogenase. In this reaction, the substrate, isocitrate, loses one carbon by oxidative decarboxylation.

There are two different forms of isocitrate dehydrogenase in all cells, one requiring NAD^+ as electron acceptor and the other requiring NADP^+ . The overall reactions are otherwise identical. In eukaryotic cells, the NAD-dependent enzyme occurs in the mitochondrial matrix and serves in the citric acid cycle. The main function of the NADP-

dependent enzyme, found in both the mitochondrial matrix and the cytosol, is the generation of NADPH, which is essential for reductive anabolic pathways such as fatty acid and sterol synthesis.

4 Oxidation of α -Ketoglutarate to Succinyl-CoA and CO_2

The next step is another oxidative decarboxylation, in which α -ketoglutarate is converted to **succinyl-CoA** and CO_2 by the action of the **α -ketoglutarate dehydrogenase complex**; NAD^+ serves as electron acceptor and CoA as the carrier of the succinyl group. The energy of oxidation of α -ketoglutarate is conserved in the formation of the thioester bond of succinyl-CoA:



P2 This reaction is virtually identical to the pyruvate dehydrogenase reaction discussed above and to the reaction sequence responsible for the breakdown of branched-chain amino acids ([Fig. 16-12](#)). The α -ketoglutarate dehydrogenase complex closely resembles the PDH complex and the complex that degrades branched-chain α -keto acids in both structure and function. All three have homologous E_1 and E_2 components, identical E_3 components,

enzyme-bound TPP and lipoate, coenzyme A, FAD, and NAD. These related enzymes can employ the same E_3 subunit because the substrate for E_3 — a reduced lipoate — is the same for both complexes. They are certainly derived from a common evolutionary ancestor by gene duplication and subsequent **divergent evolution**, as described in [Figure 1-32](#).

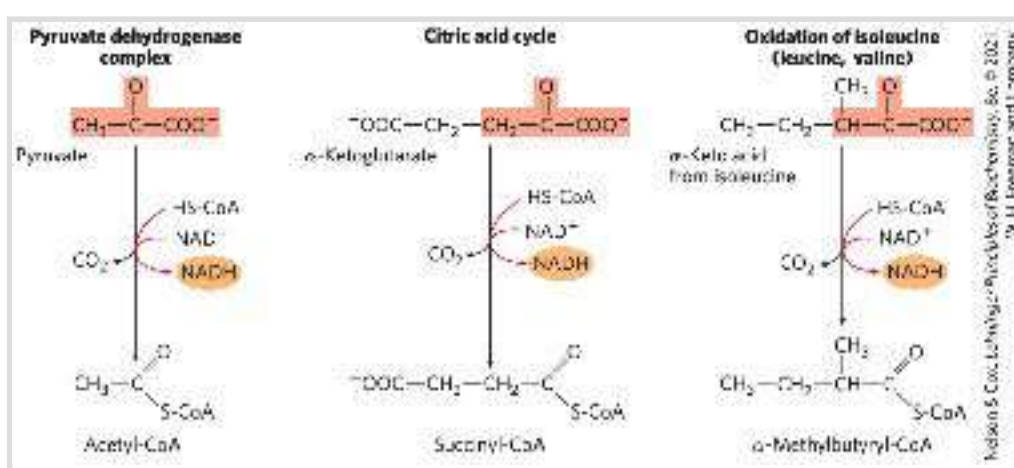
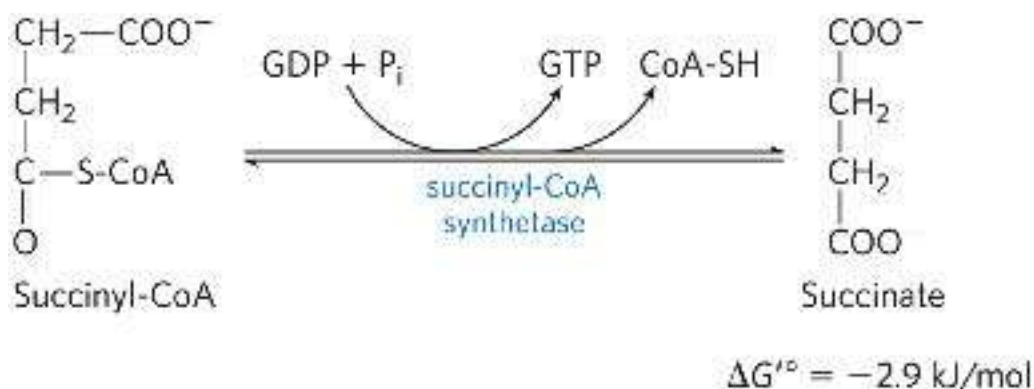


FIGURE 16-12 A conserved mechanism for oxidative decarboxylation. The pathways shown employ the same five cofactors (thiamine pyrophosphate, coenzyme A, lipoate, FAD, and NAD⁺), closely similar multienzyme complexes, and the same enzymatic mechanism to carry out oxidative decarboxylations of pyruvate (by the pyruvate dehydrogenase complex), α -ketoglutarate (in the citric acid cycle), and the carbon skeletons of the three branched-chain amino acids, isoleucine (shown here), leucine, and valine.

5 Conversion of Succinyl-CoA to Succinate

Succinyl-CoA, like acetyl-CoA, has a thioester bond with a strongly negative standard free energy of hydrolysis ($\Delta G'^{\circ} \approx -36 \text{ kJ/mol}$). In the next step of the citric acid cycle, energy released in the breakage of this bond is used to drive the synthesis of a phosphoanhydride bond in either GTP or ATP, with a net $\Delta G'^{\circ}$ of only -2.9 kJ/mol .

Succinate is formed in the process:



The enzyme that catalyzes this reversible reaction is called **succinyl-CoA synthetase** or **succinic thiokinase**; both names indicate the participation of a nucleoside triphosphate in the reaction.

This energy-conserving reaction involves an intermediate step in which the enzyme molecule itself becomes phosphorylated at a His residue in the active site ([Fig. 16-13a](#)). This phosphoryl group, which has a high group transfer potential, is transferred to ADP (or GDP) to form ATP (or GTP). Animal cells have two isozymes of succinyl-CoA synthetase, one specific for ADP and the other for GDP. The enzyme has two subunits, α (M_r 32,000), which has the P-His residue (His^{246}) and the binding site for CoA, and β (M_r 42,000), which confers specificity for either ADP or GDP. The active site is at the interface between subunits. The crystal structure of succinyl-CoA synthetase reveals two “power helices” (one from each subunit), oriented so that their electric dipoles situate partial positive charges close to the negatively charged P-His ([Fig. 16-13b](#)), stabilizing the phosphoenzyme intermediate. (Recall the similar role of helix dipoles in stabilizing K^+ ions in the K^+ channel; see [Fig. 11-45](#).)

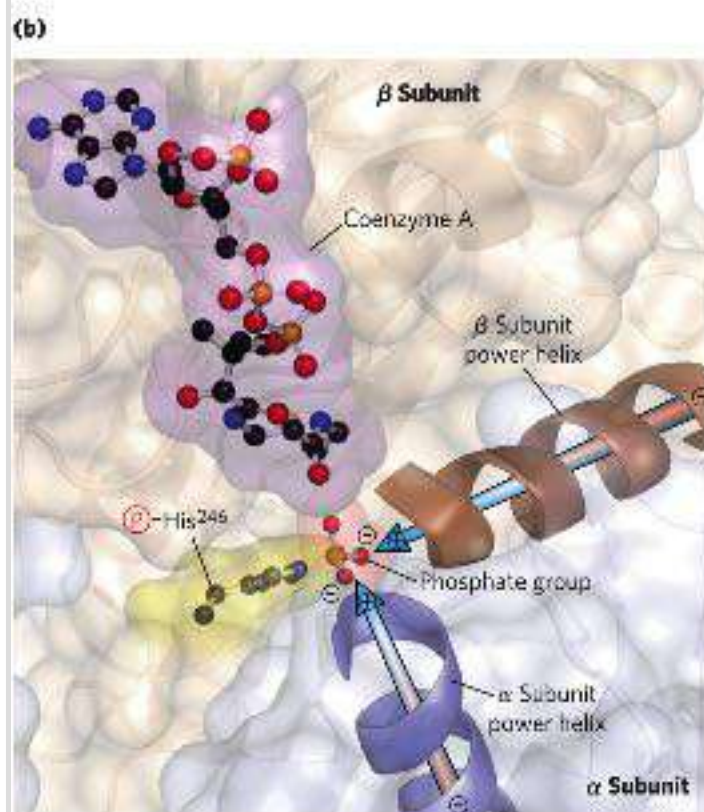
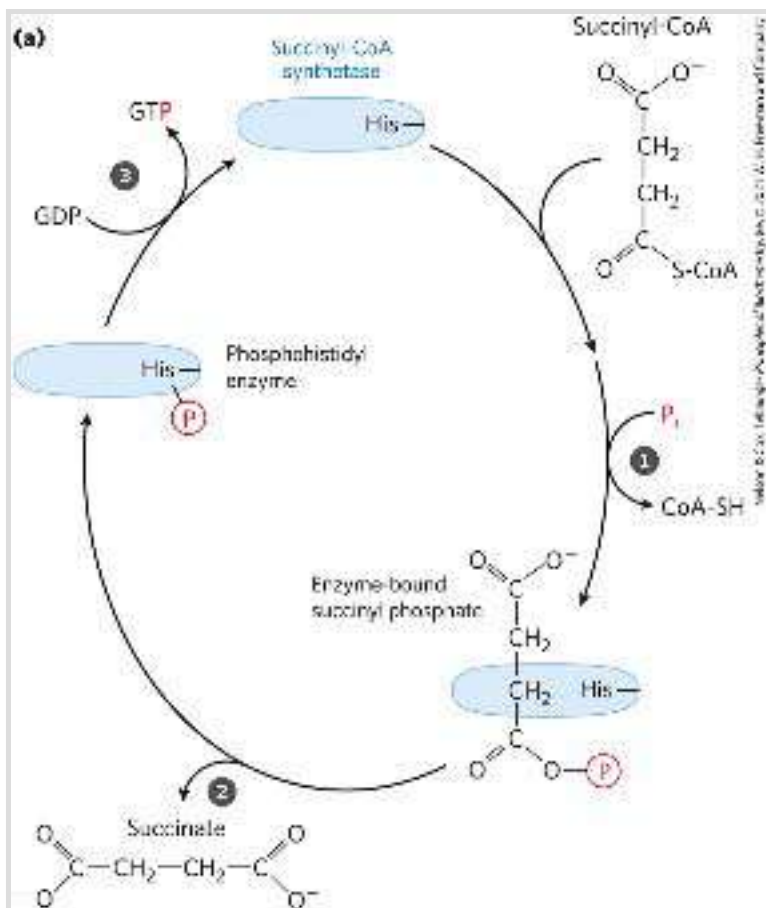
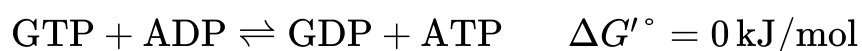


FIGURE 16-13 The succinyl-CoA synthetase reaction. (a) In step **1** a phosphoryl group replaces the CoA of succinyl-CoA bound to the enzyme, forming a high-energy acyl phosphate. In step **2** the succinyl phosphate donates its phosphoryl group to a His residue of the enzyme, forming a high-energy phosphohistidyl enzyme. In step **3** the phosphoryl group is transferred from the His residue to the terminal phosphate of GDP (or ADP), forming GTP (or ATP). (b) Active site of succinyl-CoA synthetase of *Escherichia coli*. The active site includes part of both the α (blue) and the β (brown) subunits. The power helices (blue, brown) place the partial positive charges of the helix dipole near the phosphate group of -His²⁴⁶ in the α chain, stabilizing the phosphohistidyl enzyme. The bacterial and mammalian enzymes have similar amino acid sequences and three-dimensional structures. [Data from PDB ID 1SCU, W. T. Wolodko et al., J. Biol. Chem. 269:10,883, 1994.]

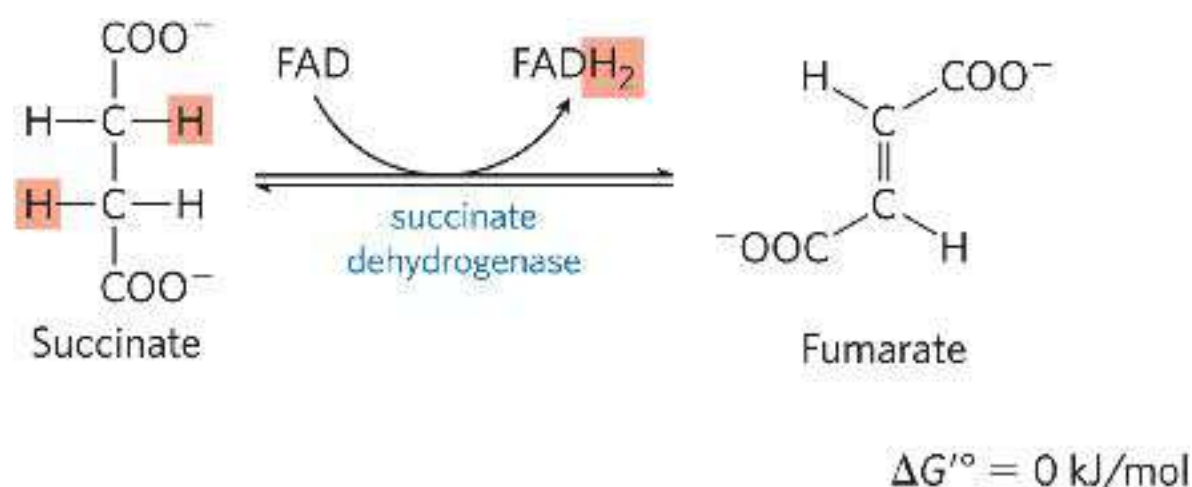
The formation of ATP (or GTP) at the expense of the energy released by the oxidative decarboxylation of α -ketoglutarate is a substrate-level phosphorylation, like the synthesis of ATP in the glycolytic reactions catalyzed by phosphoglycerate kinase and pyruvate kinase (see [Fig. 14-2](#)). The GTP formed by succinyl-CoA synthetase can donate its terminal phosphoryl group to ADP to form ATP, in a reversible reaction catalyzed by **nucleoside diphosphate kinase** ([p. 487](#)):



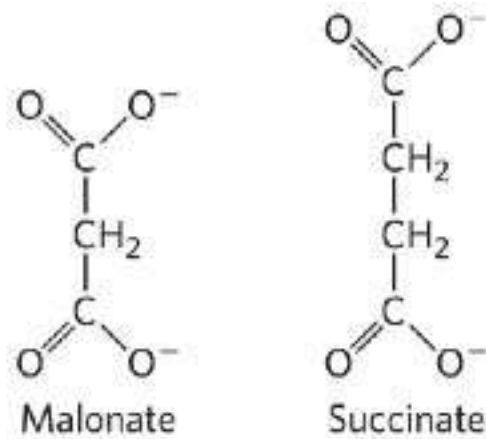
Thus the net result of the activity of either isozyme of succinyl-CoA synthetase is the conservation of energy as ATP. There is no change in free energy for the nucleoside diphosphate kinase reaction; ATP and GTP are energetically equivalent.

6 Oxidation of Succinate to Fumarate

The succinate formed from succinyl-CoA is oxidized to **fumarate** by the flavoprotein **succinate dehydrogenase**:

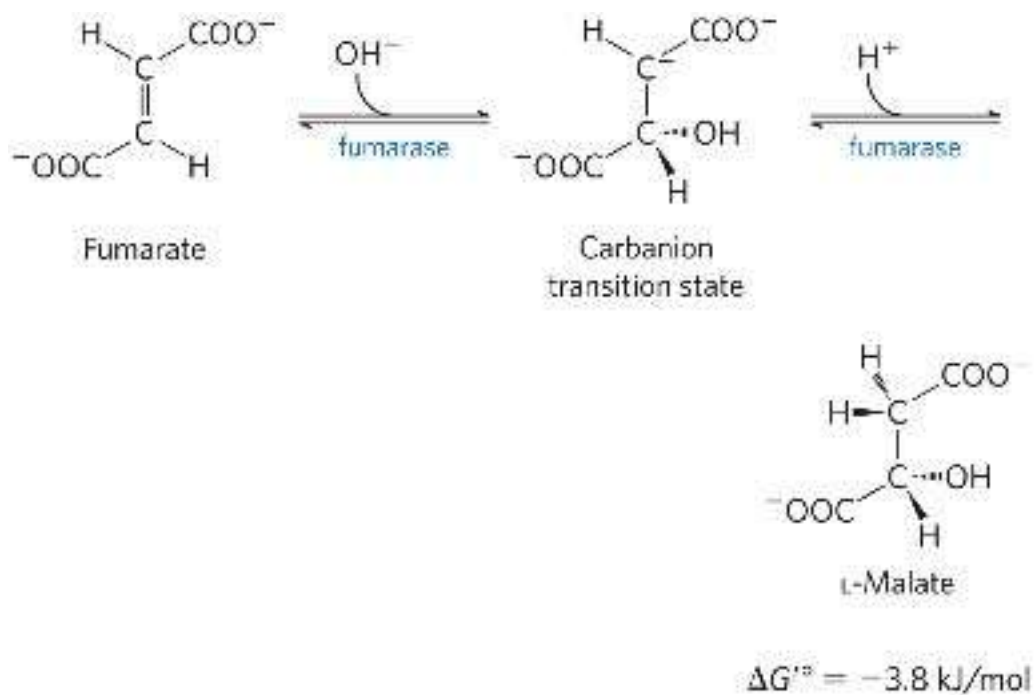


In eukaryotes, succinate dehydrogenase is an integral protein of the mitochondrial inner membrane; in bacteria, of the plasma membrane. The enzyme contains three different iron-sulfur clusters and one molecule of covalently bound FAD (see [Fig. 19-9](#)). Electrons pass from succinate through the FAD and iron-sulfur centers before entering the chain of electron carriers in the mitochondrial inner membrane (the plasma membrane in bacteria). Electron flow from succinate through these carriers to the final electron acceptor, O_2 , is coupled to the synthesis of about 1.5 ATP molecules per pair of electrons (respiration-linked phosphorylation). Malonate, an analog of succinate not normally present in cells, is a strong competitive inhibitor of succinate dehydrogenase, and its addition to mitochondria in the laboratory blocks the activity of the citric acid cycle.



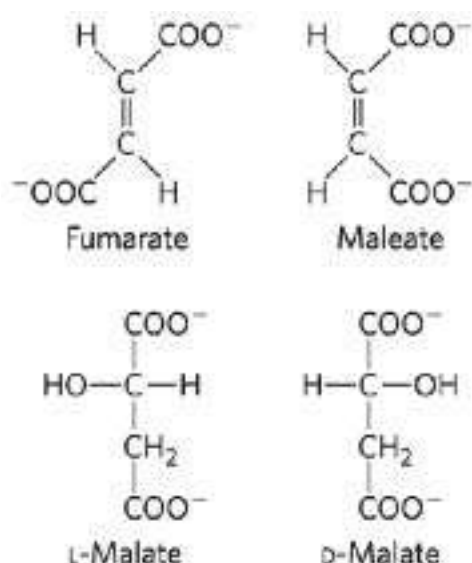
7 Hydration of Fumarate to Malate

The reversible hydration of fumarate to **L-malate** is catalyzed by **fumarase** (formally, **fumarate hydratase**). The transition state in this reaction is a carbanion:



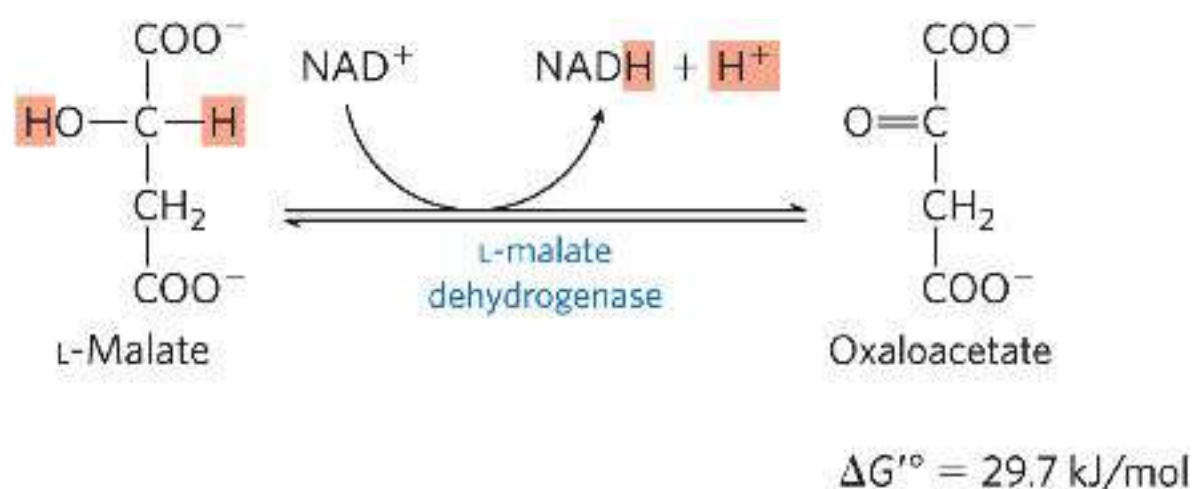
This enzyme is highly stereospecific; it catalyzes hydration of the trans double bond of fumarate but not the cis double bond of

maleate. In the reverse direction (from L-malate to fumarate), fumarase is equally stereospecific: D-malate is not a substrate.



8 Oxidation of Malate to Oxaloacetate

In the last reaction of the citric acid cycle, **L-malate dehydrogenase** catalyzes the oxidation of L-malate to oxaloacetate, coupled to the reduction of NAD^+ to NADH :



The equilibrium of this reaction lies far to the left under standard thermodynamic conditions, but in intact cells oxaloacetate is continually removed by the highly exergonic citrate synthase reaction (step 2 of [Fig. 16-7](#)). This keeps the concentration of oxaloacetate in the cell extremely low ($<10^{-6}$ M), pulling the malate dehydrogenase reaction toward the formation of oxaloacetate.

Although the individual reactions of the citric acid cycle were initially worked out in vitro, using minced muscle tissue, the pathway and its regulation have also been studied extensively in vivo. By using precursors isotopically labeled with ^{14}C , researchers have traced the fate of individual carbon atoms through the citric acid cycle. Some of the earliest experiments with ^{14}C produced an unexpected result, however, which aroused considerable controversy about the pathway and mechanism of the citric acid cycle. In fact, these experiments at first seemed to show that citrate was not the first tricarboxylic acid to be formed. [Box 16-2](#) gives some details of this episode in the history of citric acid cycle research. Today biochemists use ^{13}C -labeled precursors and whole-tissue NMR spectroscopy to monitor metabolic flux through the cycle in living tissue. Because the NMR signal is unique to the compound containing the ^{13}C , precursor carbons can be traced into each cycle intermediate and into compounds derived from the intermediates. This technique makes it possible to study regulation of the citric acid cycle and its interconnections with other metabolic pathways.

BOX 16-2

Citrate: A Symmetric Molecule That Reacts Asymmetrically

When compounds enriched in the heavy-carbon isotope ^{13}C and the radioactive carbon isotopes ^{11}C and ^{14}C became available in the mid-1940s, they were soon put to use in tracing the pathway of carbon atoms through the citric acid cycle. One such experiment initiated the controversy over the role of citrate. Acetate labeled in the carboxyl group (designated $[1-^{14}\text{C}]$ acetate) was incubated aerobically with an animal tissue preparation. Acetate is enzymatically converted to acetyl-CoA in animal tissues, and the pathway of the labeled carboxyl carbon of the acetyl group in the cycle reactions could thus be traced. α -Ketoglutarate was isolated from the tissue after incubation, then degraded by known chemical reactions to establish the position(s) of the isotopic carbon.

Condensation of unlabeled oxaloacetate with carboxyl-labeled acetate would be expected to produce citrate labeled in one of the two primary carboxyl groups. Citrate is a symmetric molecule, its two terminal carboxyl groups being chemically indistinguishable. Therefore, half the labeled citrate molecules were expected to yield α -ketoglutarate labeled in the α -carboxyl group and the other half to yield α -ketoglutarate labeled in the γ -carboxyl group; that is, the α -ketoglutarate isolated was expected to be a mixture of the two types of labeled molecules (Fig. 1, pathways ① and ②). Contrary to this expectation, the labeled α -ketoglutarate isolated from the tissue suspension contained ^{14}C only in the γ -carboxyl group (Fig. 1, pathway ①). The investigators concluded that citrate (or any other symmetric molecule) could not be an intermediate in the pathway from acetate to α -ketoglutarate. Rather, an asymmetric tricarboxylic acid, presumably *cis*-aconitate or isocitrate, must be the first product formed from condensation of acetate and oxaloacetate.

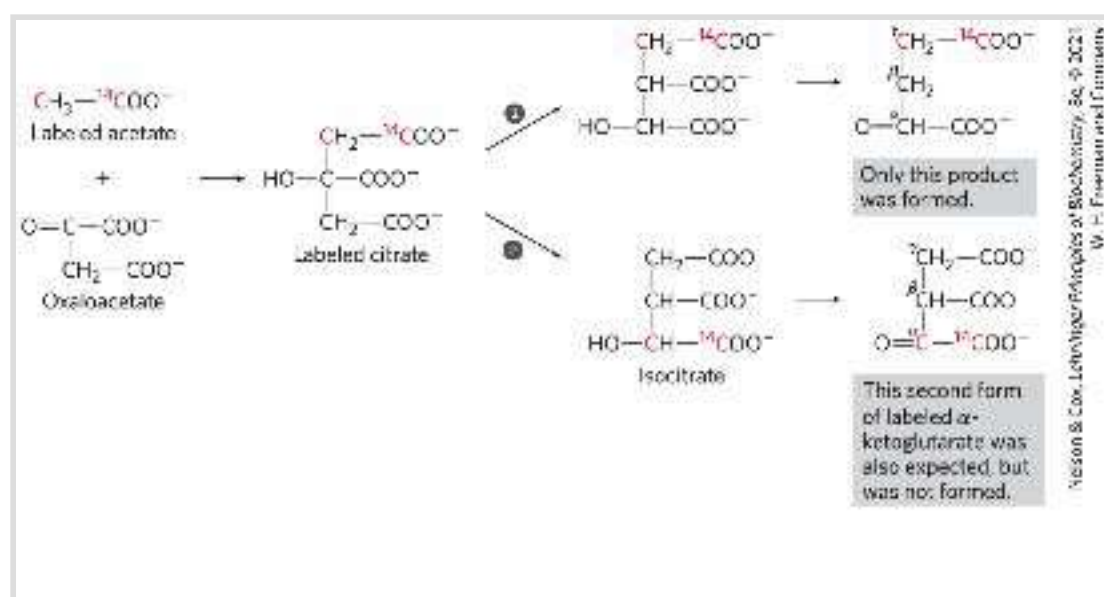


FIGURE 1 Incorporation of the isotopic carbon (^{14}C) of the labeled acetyl group into α -ketoglutarate by the citric acid cycle. The carbon atoms of the entering acetyl group are shown in red.

In 1948, however, Alexander Ogston pointed out that although citrate has no chiral center (see [Fig. 1-19](#)), it has the *potential* to react asymmetrically if an enzyme with which it interacts has an active site that is asymmetric. He suggested that the active site of aconitase may have three points to which the citrate must be bound and that the citrate must undergo a specific three-point attachment to these binding points. As seen in [Figure 2](#), the binding of citrate to three such points could happen in only one way, and this would account for the formation of only one type of labeled α -ketoglutarate. Organic molecules such as citrate that have no chiral center but are potentially capable of reacting asymmetrically with an asymmetric active site are now called **prochiral molecules**.

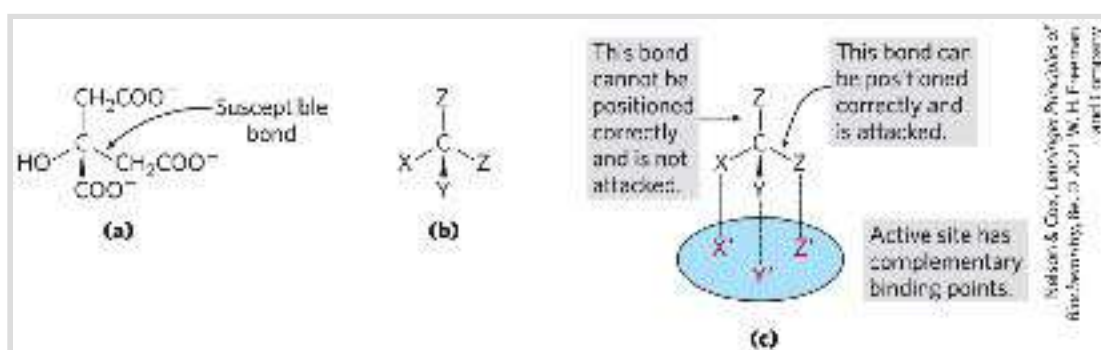
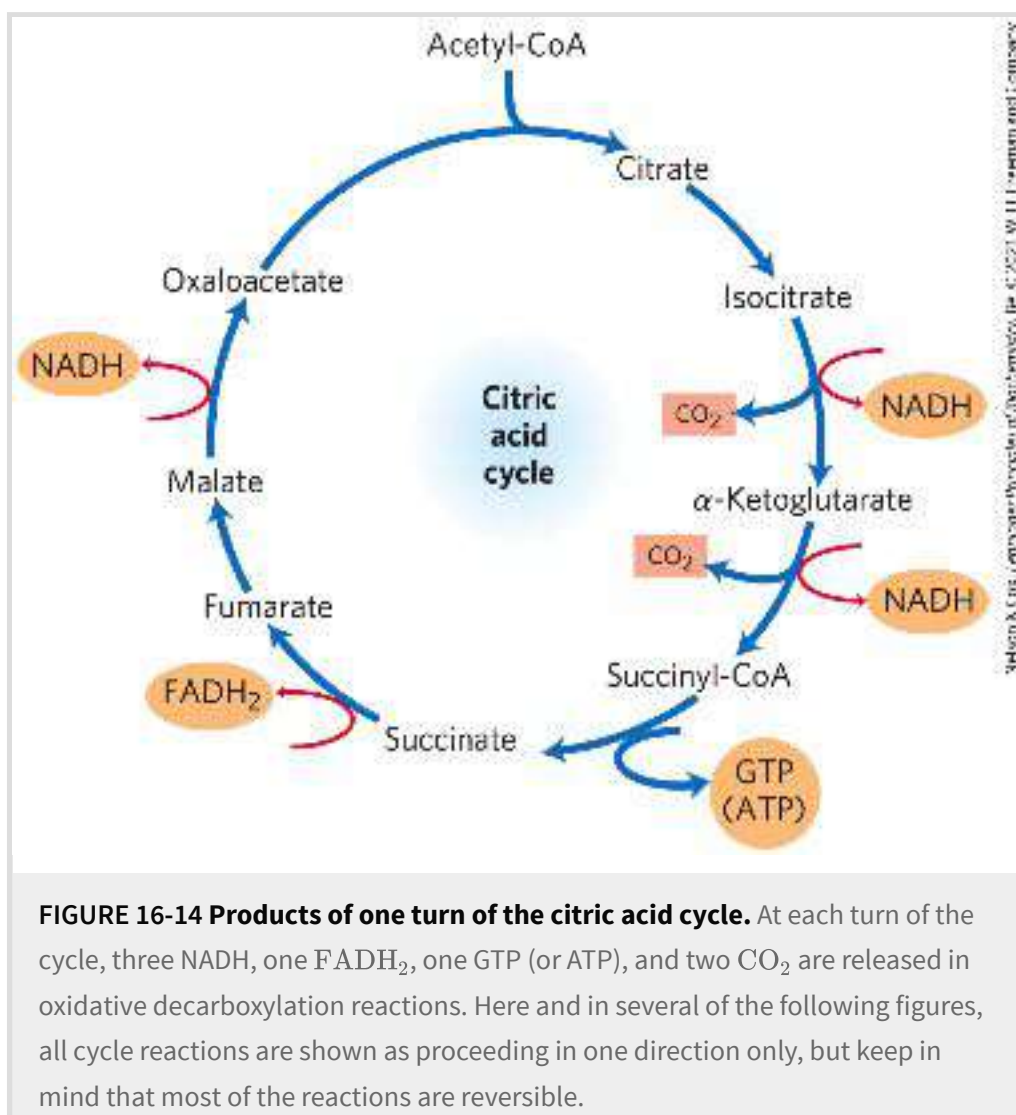


FIGURE 2 The prochiral nature of citrate. (a) Structure of citrate; (b) schematic representation of citrate: $\text{X} = \text{—OH}$; $\text{Y} = \text{—COO}^-$; $\text{Z} = \text{—CH}_2\text{COO}^-$. (c) Correct complementary fit of citrate to the binding site of aconitase. There is only one way in which the three specified groups of citrate can fit on the three points of the binding site. Thus only one of the two $\text{—CH}_2\text{COO}^-$ groups is bound by aconitase.

The Energy of Oxidations in the Cycle Is Efficiently Conserved

We have now covered one complete turn of the citric acid cycle ([Fig. 16-14](#)). A two-carbon acetyl group entered the cycle by combining with oxaloacetate. Two carbon atoms emerged from the cycle as CO_2 from the oxidation of isocitrate and α -ketoglutarate. **P2** The energy released by these oxidations was conserved in the reduction of three NAD^+ and one FAD and the production of one ATP or GTP. At the end of the cycle a molecule of oxaloacetate was regenerated. Note that the two carbon atoms appearing as CO_2 are not the same two carbons that entered in the form of the acetyl group; additional turns around the cycle are required to release these carbons as CO_2 ([Fig. 16-7](#)).





Although the citric acid cycle directly generates only one ATP per turn (in the conversion of succinyl-CoA to succinate), the four oxidation steps in the cycle provide a large flow of electrons into the respiratory chain via NADH and FADH₂ and thus lead to formation of almost 10 times more ATP during oxidative phosphorylation.

We saw in [Chapter 14](#) that the production of two molecules of pyruvate from one molecule of glucose in glycolysis yields 2 ATP and 2 NADH. In oxidative phosphorylation ([Chapter 19](#)), passage of two electrons from NADH to O₂ drives the formation of about 2.5 ATP, and passage of two electrons from FADH₂ to O₂ yields about 1.5 ATP. This stoichiometry allows us to calculate the overall yield of ATP from the complete oxidation of glucose. When both pyruvate molecules are oxidized to 6 CO₂ via the pyruvate dehydrogenase complex and the citric acid cycle, and the electrons are transferred to O₂ via oxidative phosphorylation, as many as 32 ATP are obtained per glucose ([Table 16-1](#)). In round numbers, this represents the conservation of $32 \times 30.5 \text{ kJ/mol} = 976 \text{ kJ/mol}$, or 34% of the theoretical maximum of about 2,840 kJ/mol available from the complete oxidation of glucose. These calculations employ the standard free-energy changes; when corrected for the actual free energy required to form ATP within cells (see [Worked Example 13-2, p. 480](#)), the calculated efficiency of the process is closer to 65%. When cells under anaerobic conditions depend on glycolysis for ATP, one glucose yields just 2 ATP. Aerobic metabolism is far more effective in capturing the energy in glucose as a fuel.

TABLE 16-1 Stoichiometry of Coenzyme Reduction and ATP

Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

Reaction	Number of ATP or reduced coenzyme directly formed	Number of ATP ultimately formed ^a
Glucose → glucose 6-phosphate	−1 ATP	−1
Fructose 6-phosphate → fructose 1,6-bisphosphate	−1 ATP	−1
2 Glyceraldehyde 3-phosphate → 2 1,3-bisphosphoglycerate	2 NADH	3 or 5 ^b
2 1,3-Bisphosphoglycerate → 2 3-phosphoglycerate	2 ATP	2
2 Phosphoenolpyruvate → 2 pyruvate	2 ATP	2
2 Pyruvate → 2 acetyl-CoA	2 NADH	5
2 Isocitrate → 2 α-ketoglutarate	2 NADH	5
2 α-Ketoglutarate → 2 succinyl-CoA	2 NADH	5
2 Succinyl-CoA → 2 succinate	2 ATP (or 2 GTP)	2
2 Succinate → 2 fumarate	2 FADH ₂	3
2 Malate → 2 oxaloacetate	2 NADH	5
Total		30–32

^aThis is calculated as 2.5 ATP per NADH and 1.5 ATP per FADH₂. A negative value indicates consumption.

^bThis number is either 3 or 5, depending on the mechanism used to shuttle NADH equivalents from the cytosol to the mitochondrial matrix; see [Figures 19-31](#) and [19-32](#).

SUMMARY 16.2 *Reactions of the Citric Acid Cycle*

■ The citric acid cycle (Krebs cycle, TCA cycle) is a nearly universal central catabolic pathway in which compounds derived from the breakdown of carbohydrates, fats, and proteins are oxidized to CO_2 , with most of the energy of oxidation temporarily held in the electron carriers FADH_2 and NADH . During aerobic metabolism, these electrons are transferred to O_2 and the energy of electron flow is conserved as ATP.

■ Acetyl-CoA enters the citric acid cycle as citrate synthase catalyzes its condensation with oxaloacetate to form citrate. In seven sequential reactions, including two decarboxylations, the citric acid cycle converts citrate to oxaloacetate and releases two CO_2 . The pathway is cyclic in that the intermediates of the cycle are not used up; for each oxaloacetate consumed in the path, one is produced.

■ For each acetyl-CoA oxidized by the citric acid cycle, the energy gain consists of three molecules of NADH , one FADH_2 , and one nucleoside triphosphate (either ATP or GTP).

16.3 The Hub of Intermediary Metabolism

The eight-step cyclic process for oxidation of simple two-carbon acetyl groups to CO_2 may seem unnecessarily complex and not in keeping with the biological principle of maximum economy. Remember, though, that the role of the citric acid cycle is not confined to the oxidation of acetate from carbohydrates, fatty acids, or amino acids.  The cycle also accepts 3-, 4-, and 5-carbon skeletons, especially from the **breakdown of amino acids**, at other points in the pathway. For example, when deaminated, the amino acids aspartate and glutamate become the cycle intermediates oxaloacetate and α -ketoglutarate, respectively.

The Citric Acid Cycle Serves in Both Catabolic and Anabolic Processes

In aerobic organisms, the citric acid cycle is an [amphibolic pathway](#), one that serves in both catabolic and anabolic processes. In the cycle's anabolic role, oxaloacetate and α -ketoglutarate can be withdrawn from the cycle to serve as precursors of aspartate and glutamate by simple transamination ([Chapter 22](#)). Through aspartate and glutamate, the carbons of oxaloacetate and α -ketoglutarate are then used to build other amino acids, as well as purine and pyrimidine nucleotides. Succinyl-CoA is a central intermediate in the synthesis of the porphyrin ring of heme groups, which serve as oxygen carriers

(in hemoglobin and myoglobin) and electron carriers (in cytochromes) (see [Figs. 22-25](#), [22-26](#)). Finally, oxaloacetate can be converted to glucose via gluconeogenesis (see [Fig. 14-16](#)).

 One biosynthetic process is not possible for animals: the conversion of acetate or acetyl-CoA to glucose. Given that the carbon atoms of acetate molecules entering the citric acid cycle appear eight steps later in oxaloacetate, it might seem that the cycle could generate oxaloacetate from acetate, then the oxaloacetate could be used to synthesize glucose by gluconeogenesis. However, there is no *net* conversion of acetate to oxaloacetate; for every two carbons that enter the cycle as acetate (acetyl-CoA), two leave as CO₂. In bacteria, plants, fungi, and protists, another reaction sequence, the [glyoxylate cycle](#), serves as a mechanism for converting acetate to carbohydrate. The glyoxylate cycle, which shares some reactions with the citric acid cycle, converts *two* molecules of acetate to one of oxaloacetate in a variant of the citric acid cycle in which the two decarboxylation steps are bypassed (see [Fig. 20-45](#)). Thus, plants and many simpler organisms can synthesize glucose from fatty acids, but humans and other animals cannot.

Anaplerotic Reactions Replenish Citric Acid Cycle Intermediates

Under most circumstances, there is a dynamic steady state between reactions that siphon intermediates away from the citric

acid cycle and those that supply additional carbon skeletons.

P3 When the withdrawal of cycle intermediates for use in biosynthesis lowers the concentrations of citric acid cycle intermediates enough to slow the cycle, the intermediates are replenished by **anaplerotic reactions** (Greek, “to refill”) (**Fig. 16-15**). **Table 16-2** shows the most common anaplerotic reactions, all of which, in various tissues and organisms, convert either pyruvate or phosphoenolpyruvate to oxaloacetate or malate. The most important anaplerotic reaction in mammalian liver, kidney, and brown adipose tissue is the reversible carboxylation of pyruvate by HCO_3^- to form oxaloacetate, catalyzed by **pyruvate carboxylase**. The enzymatic addition of a carboxyl group to pyruvate requires energy, which is supplied by ATP.

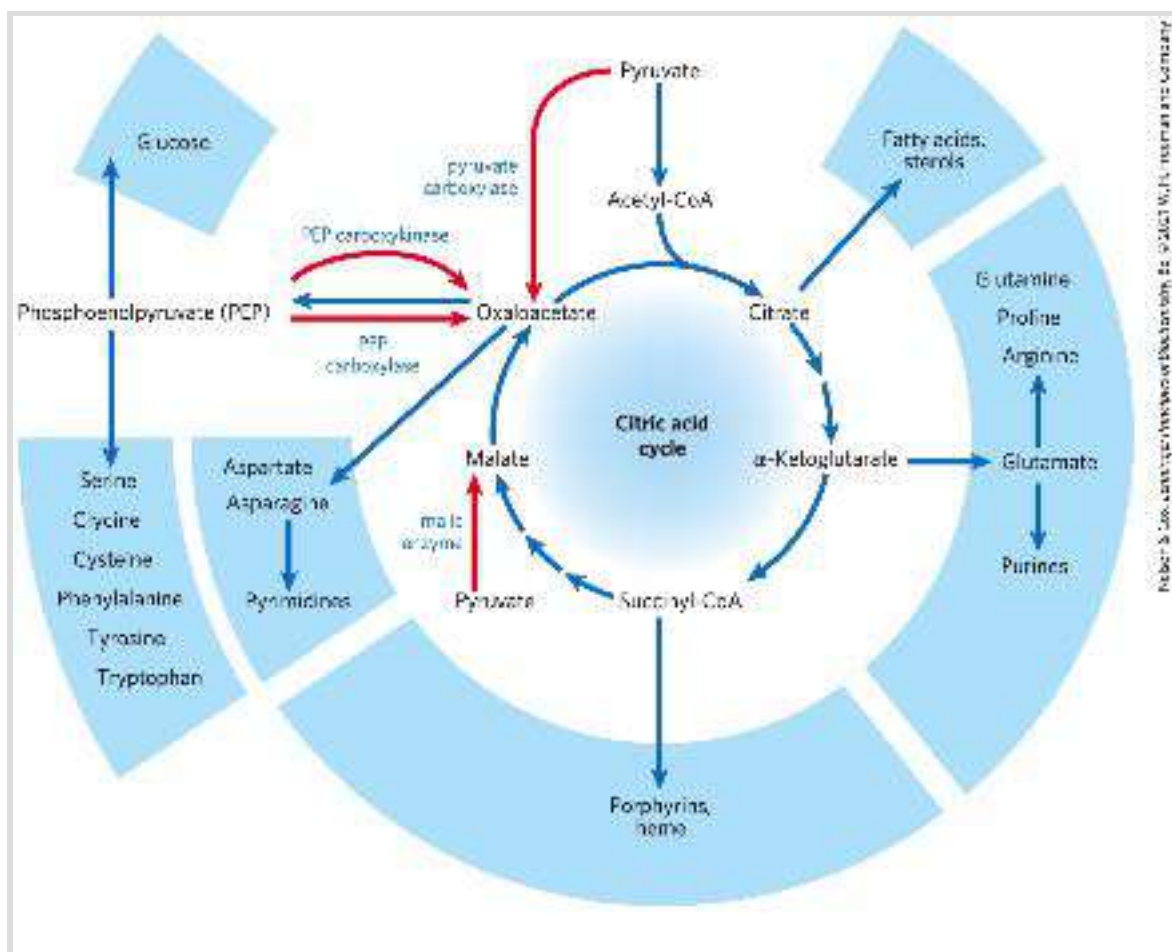


FIGURE 16-15 Role of the citric acid cycle in anabolism. Intermediates of the citric acid cycle are drawn off as precursors in many biosynthetic pathways. Shown in red are four anaplerotic reactions that replenish depleted cycle intermediates (see [Table 16-2](#)).

TABLE 16-2 Anaplerotic Reactions

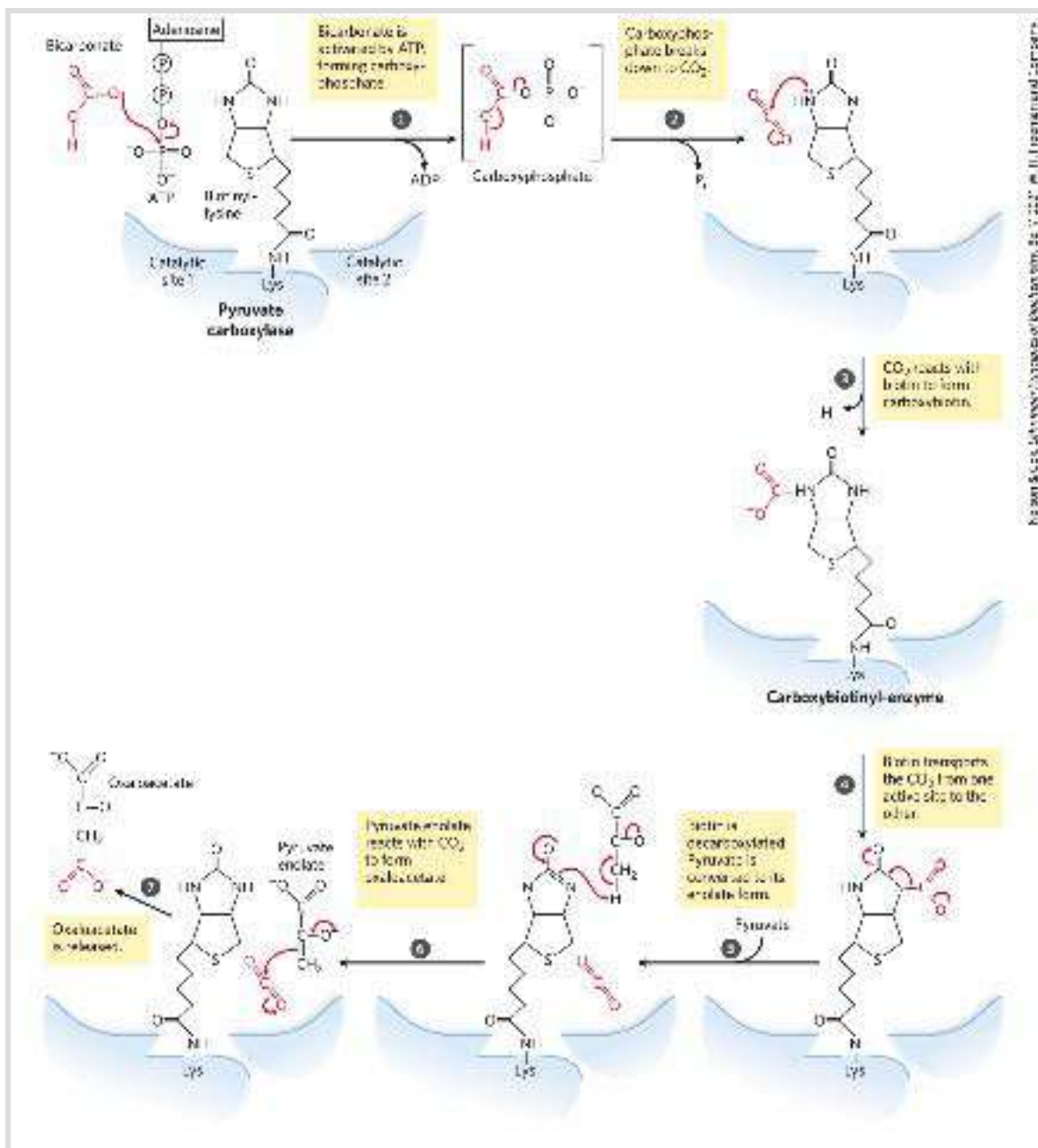
Reaction	Tissue(s)/organism(s)
$\text{Pyruvate} + \text{HCO}_3^- + \text{ATP} \xrightleftharpoons{\text{pyruvate carboxylase}} \text{oxaloacetate} + \text{ADP} + \text{P}_i$	Liver, kidney
$\text{Phosphoenolpyruvate} + \text{CO}_2 + \text{GDP} \xrightleftharpoons{\text{PEP carboxylase}} \text{oxaloacetate} + \text{GTP}$	Heart, skeletal muscle
$\text{Phosphoenolpyruvate} + \text{HCO}_3^- \xrightleftharpoons{\text{PEP carboxykinase}} \text{oxaloacetate} + \text{P}_i$	Higher plants, yeast, bacteria
$\text{Pyruvate} + \text{HCO}_3^- + \text{NAD(P)H} \xrightleftharpoons{\text{malic enzyme}} \text{malate} + \text{NAD(P)}^+$	Widely distributed in eukaryotes and bacteria

Pyruvate carboxylase is a regulatory enzyme and is virtually inactive in the absence of acetyl-CoA, its positive allosteric modulator. Whenever acetyl-CoA, the fuel for the citric acid cycle, is present in excess, it stimulates the pyruvate carboxylase reaction to produce more oxaloacetate, enabling the cycle to use more acetyl-CoA in the citrate synthase reaction.

Biotin in Pyruvate Carboxylase Carries One-Carbon (CO_2) Groups

The pyruvate carboxylase reaction requires the vitamin [biotin](#) ([Fig. 16-16](#)), which is the prosthetic group of the enzyme. Biotin,

which plays a key role in many carboxylation reactions, is a specialized carrier of one-carbon groups in their most oxidized form: CO_2 . (The transfer of one-carbon groups in more reduced forms is mediated by other cofactors, notably tetrahydrofolate and *S*-adenosylmethionine, as described in [Chapter 18](#).) Carboxyl groups are activated in a reaction that consumes ATP and joins CO_2 to enzyme-bound biotin. This “activated” CO_2 is then passed to an acceptor (pyruvate in this case) in a carboxylation reaction.



MECHANISM FIGURE 16-16 The role of biotin in the reaction catalyzed by pyruvate carboxylase. Biotin is attached to the enzyme through an amide bond with the ϵ -amino group of a Lys residue, forming biotinyl-enzyme. Biotin-mediated carboxylation reactions occur in two phases, generally catalyzed in separate active sites on the enzyme, as exemplified by the pyruvate carboxylase reaction. In the first phase (steps 1 to 3), bicarbonate is converted to the more activated CO_2 , and then used to carboxylate biotin. The biotin acts as a carrier to transport the CO_2 from one active site to another on an adjacent monomer of the tetrameric enzyme (step 4). In the second phase (steps 5 to 7), catalyzed in this second active site, the CO_2 reacts with pyruvate to form oxaloacetate.

Pyruvate carboxylase has four identical subunits, each containing a molecule of biotin covalently attached through an amide linkage to the ϵ -amino group of a specific Lys residue in the enzyme active site. Carboxylation of pyruvate proceeds in two steps ([Fig. 16-16](#)): first, a carboxyl group derived from HCO_3^- is attached to biotin, then the carboxyl group is transferred to pyruvate to form oxaloacetate.  These two steps occur at separate active sites; the long flexible arm of biotin transfers activated carboxyl groups from the first active site (on one monomer of the tetramer) to the second (on the adjacent monomer), functioning much like the long lipoyllysyl arm of E_2 in the PDH complex ([Fig. 16-6](#)) and the long arm of the CoA-like moiety in the acyl carrier protein involved in fatty acid synthesis (see [Fig. 21-5](#)); these are compared in [Figure 16-17](#). Lipoate, biotin, and pantothenate all enter cells on the same transporter; all become covalently attached to proteins by similar reactions; and all provide a flexible tether that allows bound reaction intermediates to move from one active site to another in an enzyme complex, without dissociating from it. That is, all participate in substrate channeling.

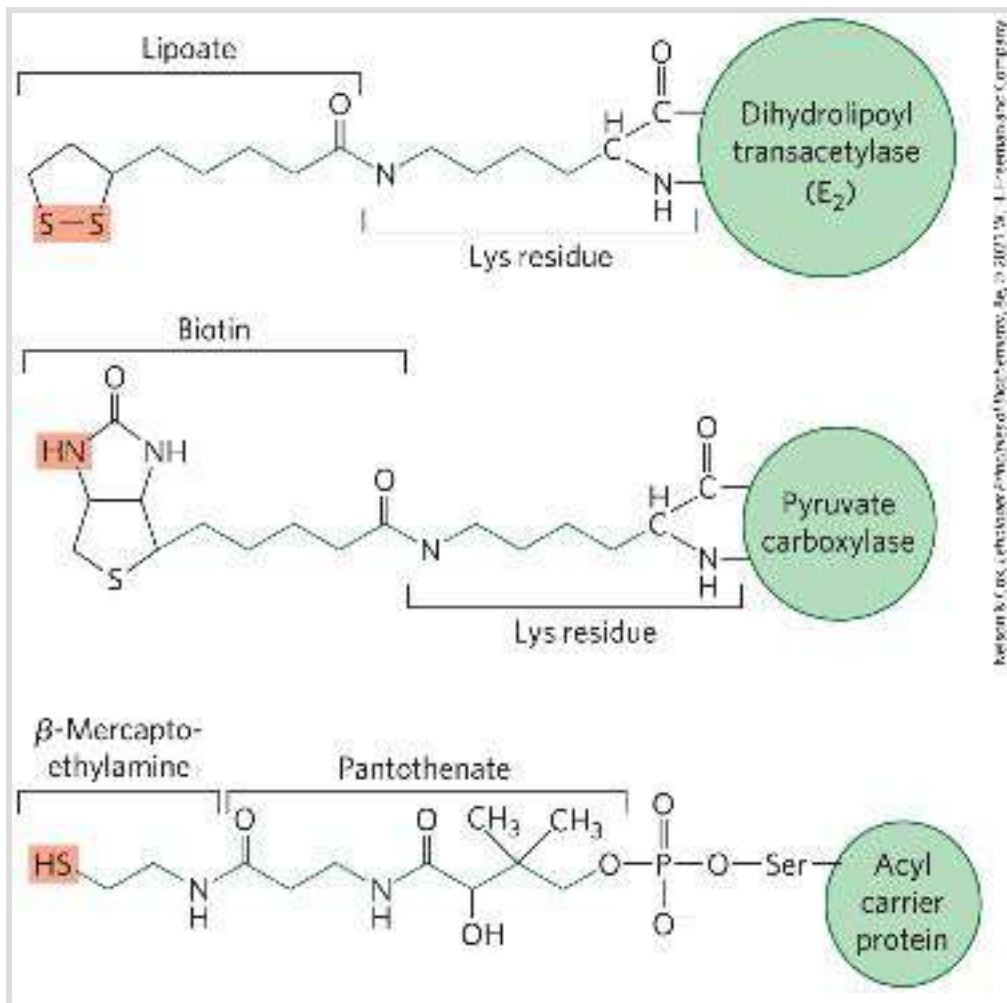


FIGURE 16-17 Biological tethers. The cofactors lipoate, biotin, and the combination of β -mercaptoethylamine and pantothenate form long, flexible arms (green) on the enzymes to which they are covalently bound, acting as tethers that move intermediates from one active site to the next. The group shaded light red is, in each case, the point of attachment of the activated intermediate to the tether.

SUMMARY 16.3 *The Hub of Intermediary Metabolism*

■ The citric acid cycle is amphibolic, serving in both catabolism and anabolism. Besides acetyl-CoA, any compound that gives rise to a four- or five-carbon intermediate of the citric acid cycle — for

example, the breakdown products of many amino acids — can be oxidized by the cycle. Cycle intermediates can be drawn off and used as the starting material for a variety of biosynthetic products.

■ When intermediates are shunted from the citric acid cycle to other pathways, they are replenished by several anaplerotic reactions, which produce four-carbon intermediates by carboxylation of three-carbon compounds; pyruvate carboxylase is a major anaplerotic enzyme.

■ Enzymes that catalyze carboxylations commonly employ biotin to activate CO_2 and to carry it to acceptors such as pyruvate or phosphoenolpyruvate.

16.4 Regulation of the Citric Acid Cycle

 The central role of the citric acid cycle in metabolism requires stringent regulation to balance the supply of key intermediates with the demands of energy production and biosynthetic processes. Regulation occurs at several levels, including the oxidation of pyruvate to acetyl-CoA (catalyzed by the PDH complex) and the entry of acetyl-CoA into the cycle (the citrate synthase reaction). The transport of pyruvate into mitochondria by the mitochondrial pyruvate carrier (MPC) determines to some degree the fate of pyruvate produced by glycolysis. Most cells also produce acetyl-CoA from the oxidation of fatty acids ([Chapter 17](#)) and certain amino acids ([Chapter 18](#)), and the availability of intermediates from those other pathways is important in the regulation of pyruvate oxidation and of the citric acid cycle. There are therefore multiple points at which the metabolism of pyruvate and the citric acid cycle may be regulated.

Production of Acetyl-CoA by the PDH Complex Is Regulated by Allosteric and Covalent Mechanisms

The PDH complex of mammals is strongly inhibited allosterically by ATP and by acetyl-CoA and NADH, the products of the reaction catalyzed by the complex ([Fig. 16-18](#)). Long-chain fatty acids,

which can be broken down to acetyl-CoA, are also inhibitory. AMP, CoA, and NAD^+ , all of which accumulate when too little acetate flows into the citric acid cycle, allosterically activate the PDH complex.  Thus, PDH complex activity is turned off when ample fuel is available in the form of fatty acids and acetyl-CoA and when the cell's $[\text{ATP}]/[\text{ADP}]$ and $[\text{NADH}]/[\text{NAD}^+]$ ratios are high; it is turned on again when energy demands are high and the cell requires greater flux of acetyl-CoA into the citric acid cycle.

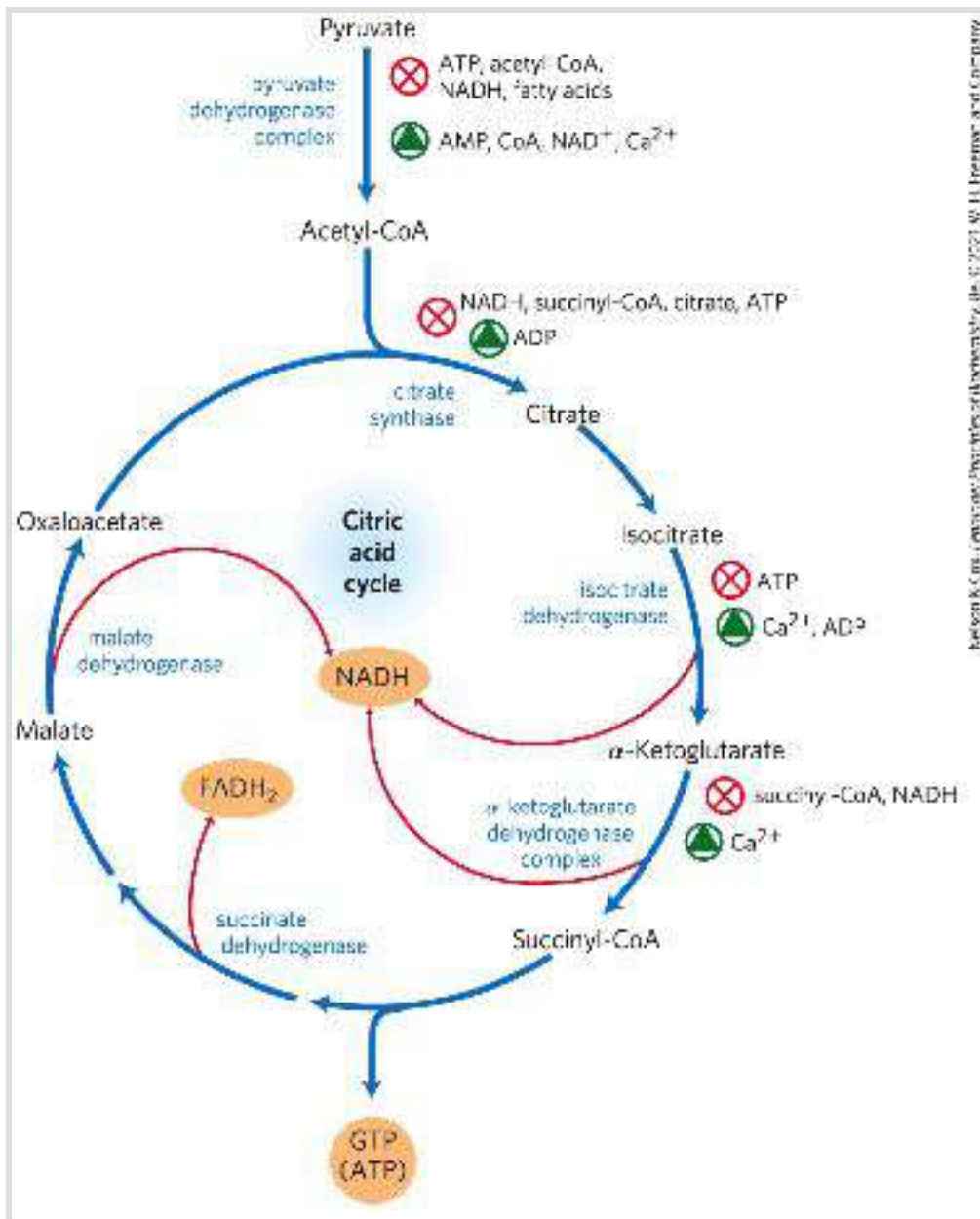
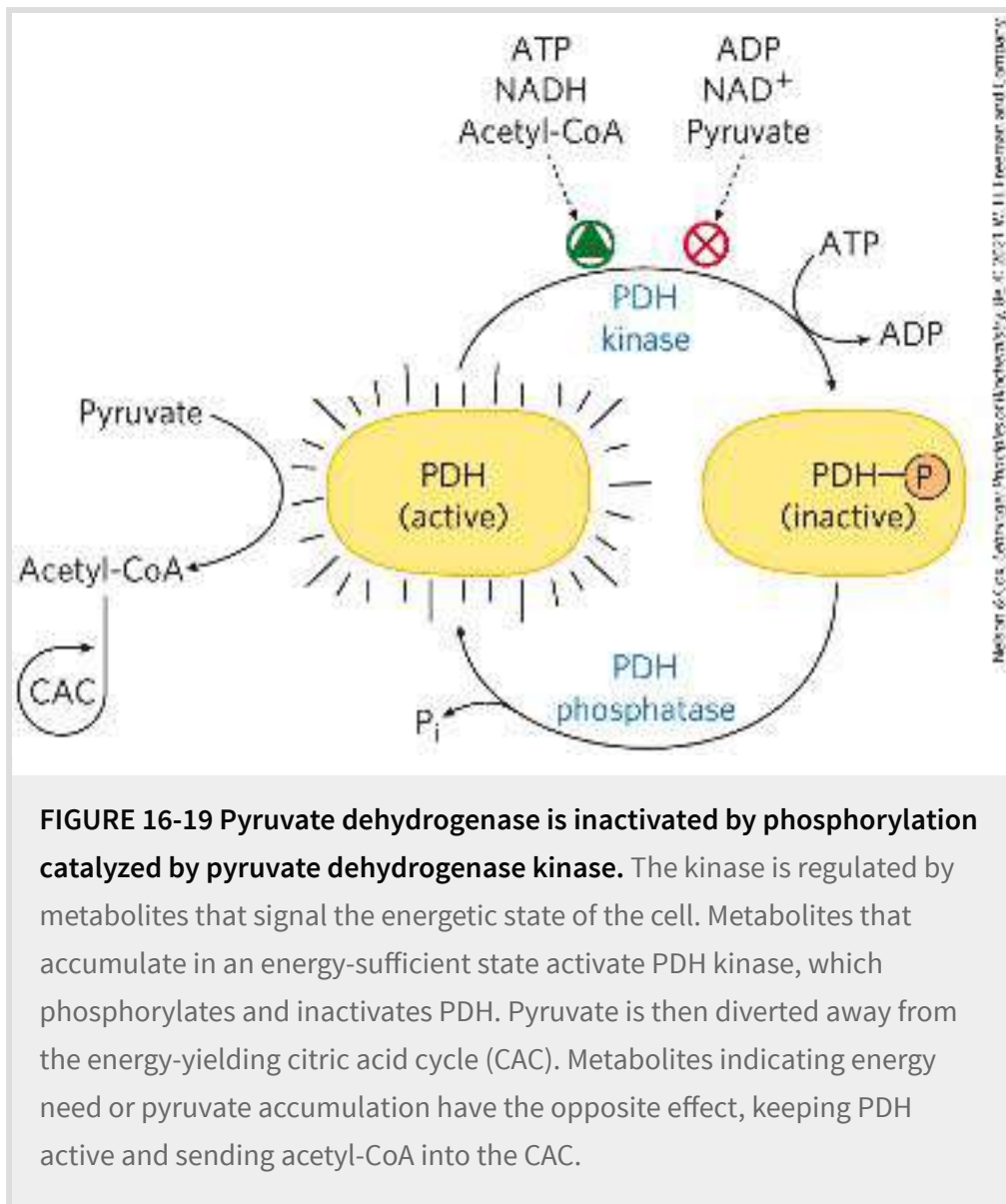


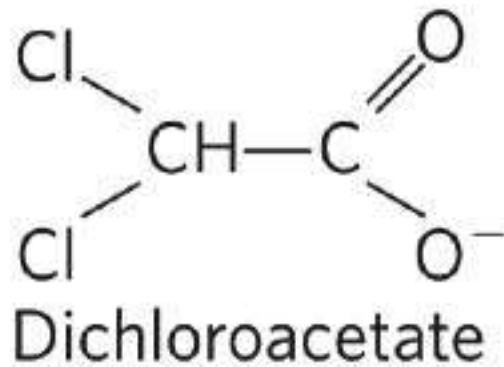
FIGURE 16-18 Regulation of metabolite flow from the PDH complex through the citric acid cycle in mammals. The PDH complex is allosterically inhibited when $[ATP]/[ADP]$, $[NADH]/[NAD^+]$, and $[acetyl-CoA]/[CoA]$ ratios are high, all of which indicate an energy-sufficient metabolic state. When these ratios decrease, allosteric activation of pyruvate oxidation results. The rate of flow through the citric acid cycle can be limited by the availability of the citrate synthase substrates, oxaloacetate and acetyl-CoA, or of NAD^+ , which is depleted by its conversion to NADH, slowing the three NAD^+ -dependent oxidation steps. Feedback inhibition by succinyl-CoA, citrate, and ATP also slows the cycle by inhibiting early steps. In muscle tissue, Ca^{2+} stimulates contraction and,

as shown here, stimulates energy-yielding metabolism to replace the ATP consumed by contraction.

In mammals, these allosteric regulatory mechanisms are complemented by a second level of regulation: phosphorylation/dephosphorylation. The PDH complex is inhibited by reversible phosphorylation of Ser residues on E₁ by **PDH kinase**, which is an intrinsic part of the PDH complex. PDH kinase is allosterically activated by the products of the PDH complex (ATP, NADH, and acetyl-CoA), and it is inhibited by the substrates for the PDH complex (ADP, NAD⁺, and pyruvate) ([Fig. 16-19](#)). The complex also contains **PDH phosphatase**, which reverses the inhibition by PDH kinase.  Together, the kinase and phosphatase exert strong control over the entry of acetyl-CoA from pyruvate into the citric acid cycle. Higher concentrations of ATP, NADH, or acetyl-CoA lead to inactivation of the PDH complex by phosphorylation of PDH. When the concentrations of ADP, NAD⁺, or pyruvate rise, kinase activity decreases and pyruvate dehydrogenase phosphatase removes the phosphoryl group, reactivating the PDH complex and thereby stimulating the citric acid cycle.



 The simple compound dichloroacetate (DCA), a structural analog of acetate, inhibits PDH kinase in the laboratory and so relieves the inhibition of the PDH complex. This stimulates pyruvate oxidation via the citric acid cycle ([Fig. 16-19](#)) and thus may have a use in directing metabolism in tumor cells away from aerobic glycolysis (see [Box 14-1](#)). The increase in mitochondrial oxidation also stimulates apoptosis ([Chapter 19](#)), thereby suppressing tumor growth. Phase III clinical trials of DCA began in 2019.



The Citric Acid Cycle Is Also Regulated at Three Exergonic Steps

Each of the three strongly exergonic steps in the cycle — those catalyzed by citrate synthase, isocitrate dehydrogenase, and α -ketoglutarate dehydrogenase ([Fig. 16-18](#)) — can become the rate-limiting step under some circumstances. The availability of the substrates for citrate synthase (acetyl-CoA and oxaloacetate) varies with the metabolic state of the cell and sometimes limits the rate of citrate formation. NADH, a product of isocitrate and α -ketoglutarate oxidation, accumulates under some conditions, and at high $[\text{NADH}]/[\text{NAD}^+]$ both dehydrogenase reactions are severely inhibited by mass action. Similarly, the malate dehydrogenase reaction is essentially at equilibrium in the cell (that is, it is substrate-limited), and when $[\text{NADH}]/[\text{NAD}^+]$ is high the concentration of oxaloacetate is low, slowing the first step in the cycle. Product accumulation inhibits all three limiting steps of the cycle: succinyl-CoA inhibits α -ketoglutarate dehydrogenase

(and also citrate synthase); citrate blocks citrate synthase; and the end product, ATP, inhibits both citrate synthase and isocitrate dehydrogenase. The inhibition of citrate synthase by ATP is relieved by ADP, an allosteric activator of this enzyme. In vertebrate muscle, Ca^{2+} , the signal for contraction and for a concomitant increase in demand for ATP, activates both isocitrate dehydrogenase and α -ketoglutarate dehydrogenase, as well as the PDH complex.  In short, the concentrations of substrates and intermediates in the citric acid cycle set the flux through this pathway at a rate that provides optimal concentrations of ATP and NADH.

Under normal conditions, the rates of glycolysis and of the citric acid cycle are integrated so that only as much glucose is metabolized to pyruvate as is needed to supply the citric acid cycle with fuel (acetyl-CoA). Both pathways are inhibited by high levels of ATP and NADH, but also by the concentration of citrate, the product of the first step of the citric acid cycle, and an important allosteric inhibitor of phosphofructokinase-1 of the glycolytic pathway (see [Fig. 14-23](#)).

Citric Acid Cycle Activity Changes in Tumors



The mitochondrial pyruvate carrier (MPC) is down-regulated in tumor cells, leading to pyruvate accumulation in the cytosol. Several other mitochondrial enzymes are also inactivated in

tumor cells, including the PDH complex and succinate dehydrogenase. As a result, tumor cells accumulate lactate (from the pyruvate produced by glycolysis) and succinate. Both of these intermediates are **oncometabolites**; they stimulate tumor growth, acting through specific G protein–coupled receptors (GPCRs; see [Chapter 12](#)) in the plasma membrane. The membrane receptor for lactate is upregulated in most cancers. L-Lactate acting through its membrane receptor, lowers [cAMP] and raises $[Ca^{2+}]$, with downstream effects currently under investigation.

 Mutations in citric acid cycle enzymes are very rare in humans and other mammals, but those that do occur are devastating. Genetic defects in succinate dehydrogenase lead to tumors of the adrenal gland (pheochromocytomas), and mutations in the fumarase gene lead to tumors of smooth muscle (leiomyomas) and kidney. Their activities thus define both enzymes as tumor suppressors ([p. 451](#)).

Another remarkable connection between citric acid cycle intermediates and cancer is the finding that in many glial cell tumors (gliomas), the NADPH-dependent isocitrate dehydrogenase has an unusual genetic defect. The mutant enzyme loses its normal activity (converting isocitrate to α -ketoglutarate) but *gains* a new activity: it converts α -ketoglutarate to 2-hydroxyglutarate ([Fig. 16-20](#)), which accumulates in the tumor cells. α -Ketoglutarate and Fe^{3+} are essential cofactors for a family of histone demethylases that alter gene expression. They

do so by removing methyl groups from Arg and Lys residues in the histones that organize nuclear DNA. By competing with α -ketoglutarate for binding to the histone demethylases, 2-hydroxyglutarate inhibits their activity. Inhibition of the histone demethylases interferes with normal gene regulation, leading to unrestricted glial cell growth. The family of more than 60 dioxygenases that use α -ketoglutarate and Fe^{3+} as cofactors are also competitively inhibited by 2-hydroxyglutarate. Inhibition of one or more of these enzymes could interfere with normal regulation of cell division and thus produce a tumor. ■

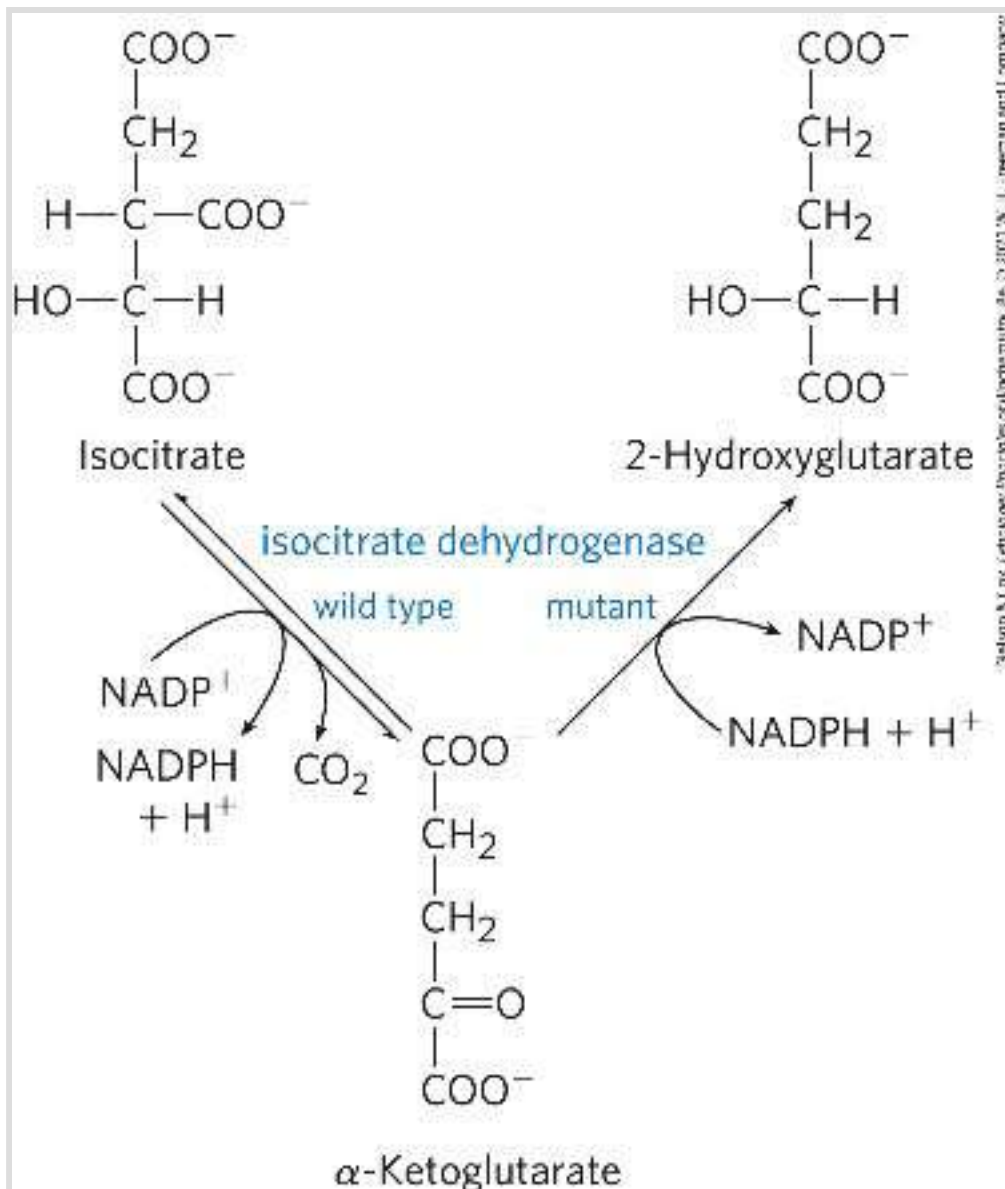


FIGURE 16-20 A mutant isocitrate dehydrogenase acquires a new activity. Wild-type isocitrate dehydrogenase catalyzes the conversion of isocitrate to α -ketoglutarate, but mutations that alter the binding site for isocitrate cause loss of the normal enzymatic activity and gain of a new activity: conversion of α -ketoglutarate to 2-hydroxyglutarate. Accumulation of this product inhibits histone demethylase, altering gene regulation and leading to glial cell tumors in the brain.

Certain Intermediates Are Channeled through Metabolons

We saw an example of substrate channeling in the five-reaction sequence of the PDH complex. Many other reactions occur in similar multienzyme complexes that ensure efficient passage of the product of one enzyme reaction to the next enzyme in the pathway. Such integrated multienzyme complexes, **metabolons**, are held together by noncovalent interactions and are not easily extracted intact from the cell. In the classic approach of enzymology — purification of individual proteins from extracts of broken cells — once cells are broken open, their contents, including enzymes, are diluted 100- or 1,000-fold (**Fig. 16-21**), which favors dissociation of noncovalent complexes such as metabolons.

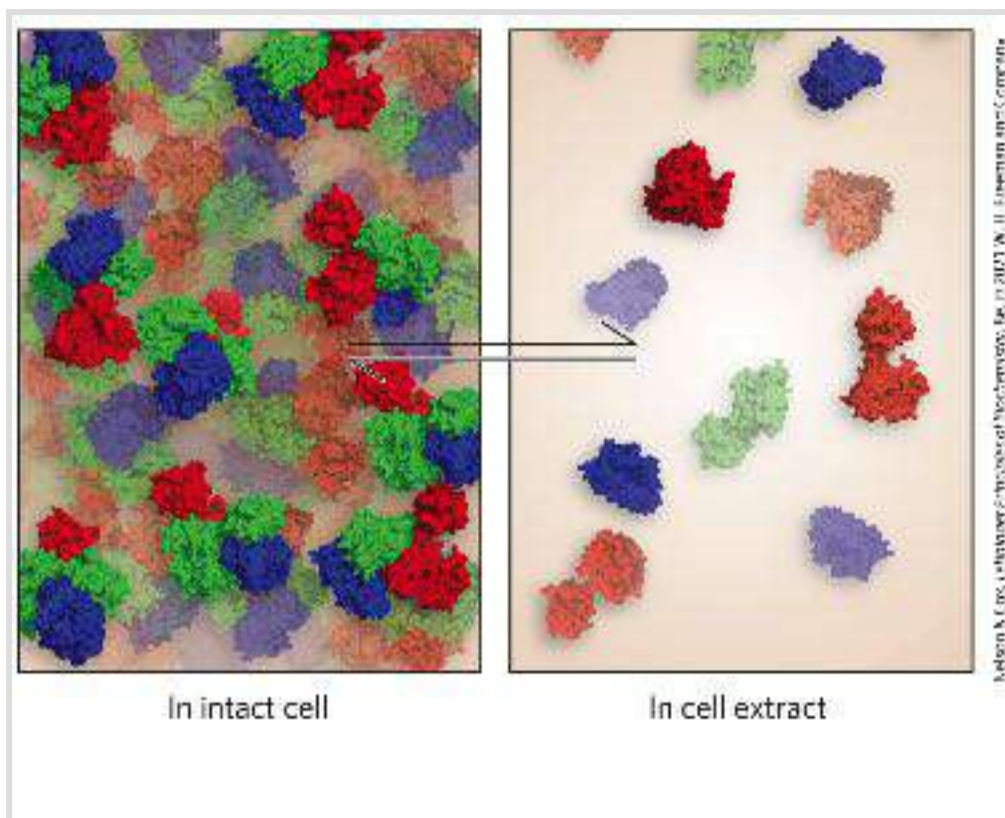


FIGURE 16-21 Effect of protein concentration on complex stability.

Dilution of a solution containing a noncovalently bound protein complex — such as a metabolon consisting of three enzymes (illustrated here in red, blue, and green) — favors dissociation of the complex into its constituents.

The enzymes of the citric acid cycle are usually described as *soluble* components of the mitochondrial matrix (except for succinate dehydrogenase, which is membrane-bound), but there is evidence that at least three sequential enzymes of the citric acid cycle (malate dehydrogenase, citrate synthase, and aconitase) constitute a metabolon ([Fig. 16-22](#)). We will see more examples of substrate channeling in some amino acid biosynthesis pathways in [Chapter 22](#). As cryo-EM increases our understanding of larger complexes and “—somes” (apoptosomes, respirasomes, and replisomes, for example), it seems likely that many more enzymes once thought to function as individual soluble proteins will be found to act in multienzyme complexes that facilitate the channeling of substrates.

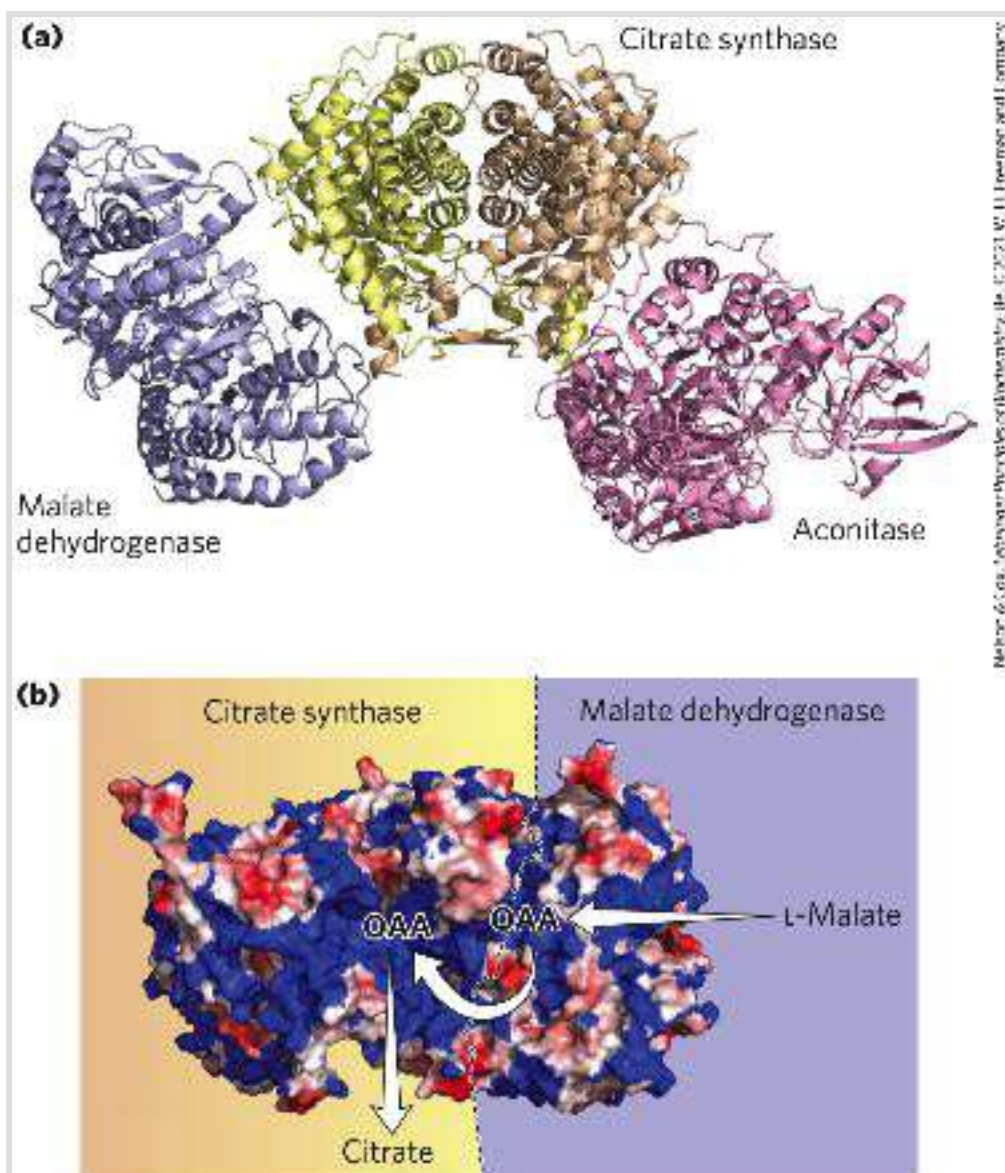


FIGURE 16-22 A three-enzyme metabolon of the citric acid cycle. (a)

Purified porcine enzymes malate dehydrogenase (MDH), citrate synthase

(CS), and aconitase form a metabolon when combined in vitro. (b)

Electrostatic modeling shows that a broad path of positive potential along the surface of a MDH-CS complex connects the active sites of MDH and CS.

This path provides a channel for the passage of the negatively charged oxaloacetate (OAA) from the active site of MDH, where it is formed from L-malate, to the active site of CS, where it condenses with acetyl-CoA to form citrate. Engineered mutations that replaced a positively charged Arg residue along this path with a negatively charged Asp residue greatly reduced the rate of substrate channeling through the complex, providing evidence that the functional unit is a metabolon. [Information from B. Bulutoglu et al., *ACS Chem. Biol.* 11:2847, 2016. Data from PDB ID 1MLD, W.

B. Gleason et al., *Biochemistry* 33:2078, 1994; PDB ID 1CTS, S. Remington et al., *J. Mol. Biol.* 158:111, 1982; PDB ID 7ACN, H. Lauble et al., *Biochemistry* 31:2735, 1992.]

SUMMARY 16.4 *Regulation of the Citric Acid Cycle*

■ The production of acetyl-CoA for the citric acid cycle by the PDH complex is inhibited allosterically by metabolites that signal a sufficiency of metabolic energy (ATP, acetyl-CoA, NADH, and fatty acids) and is stimulated by metabolites that indicate a reduced energy supply (AMP, CoA-SH, NAD^+).

■ The PDH complex is regulated by allosteric mechanisms and covalent modification (phosphorylation). The overall rate of the citric acid cycle is controlled by the rate of conversion of pyruvate to acetyl-CoA and by the flux through citrate synthase, isocitrate dehydrogenase, and α -ketoglutarate dehydrogenase. These fluxes are affected by the concentrations of substrates and products: the end products ATP and NADH are inhibitory, and the substrates NAD^+ and ADP are stimulatory. Long-chain fatty acids, which can break down to acetyl-CoA, are also inhibitory.

■ Some mutations that affect the PDH complex or citric acid cycle enzymes are oncogenic: they occur very commonly in certain types of cancer.

■ Complexes of consecutive enzymes in a pathway (metabolons) allow substrate channeling and more efficient passage of substrates through the reaction sequences.

Chapter Review

Key Terms

Terms in bold are defined in the glossary.

cellular respiration

citric acid cycle

tricarboxylic acid (TCA) cycle

Krebs cycle

thioester

mitochondrial pyruvate carrier (MPC)

pyruvate dehydrogenase (PDH) complex

oxidative decarboxylation

lipoate

substrate channeling

iron-sulfur center

α -ketoglutarate dehydrogenase complex

moonlighting enzymes

nucleoside diphosphate kinase

prochiral molecule

amphibolic pathway

glyoxylate cycle

anaplerotic reaction

biotin

oncometabolite

metabolon

PROBLEMS

1. Balance Sheet for the Citric Acid Cycle The citric acid cycle has eight enzymes: citrate synthase, aconitase, isocitrate dehydrogenase, α -ketoglutarate dehydrogenase, succinyl-CoA synthetase, succinate dehydrogenase, fumarase, and malate dehydrogenase.

- Write a balanced equation for the reaction catalyzed by each enzyme.
- Name the cofactor(s) required by each enzyme reaction.
- For each enzyme, determine which of the following describes the type of reaction(s) catalyzed: condensation (carbon-carbon bond formation); dehydration (loss of water); hydration (addition of water); decarboxylation (loss of CO_2); oxidation-reduction; substrate-level phosphorylation; isomerization.
- Write a balanced net equation for the catabolism of acetyl-CoA to CO_2 .

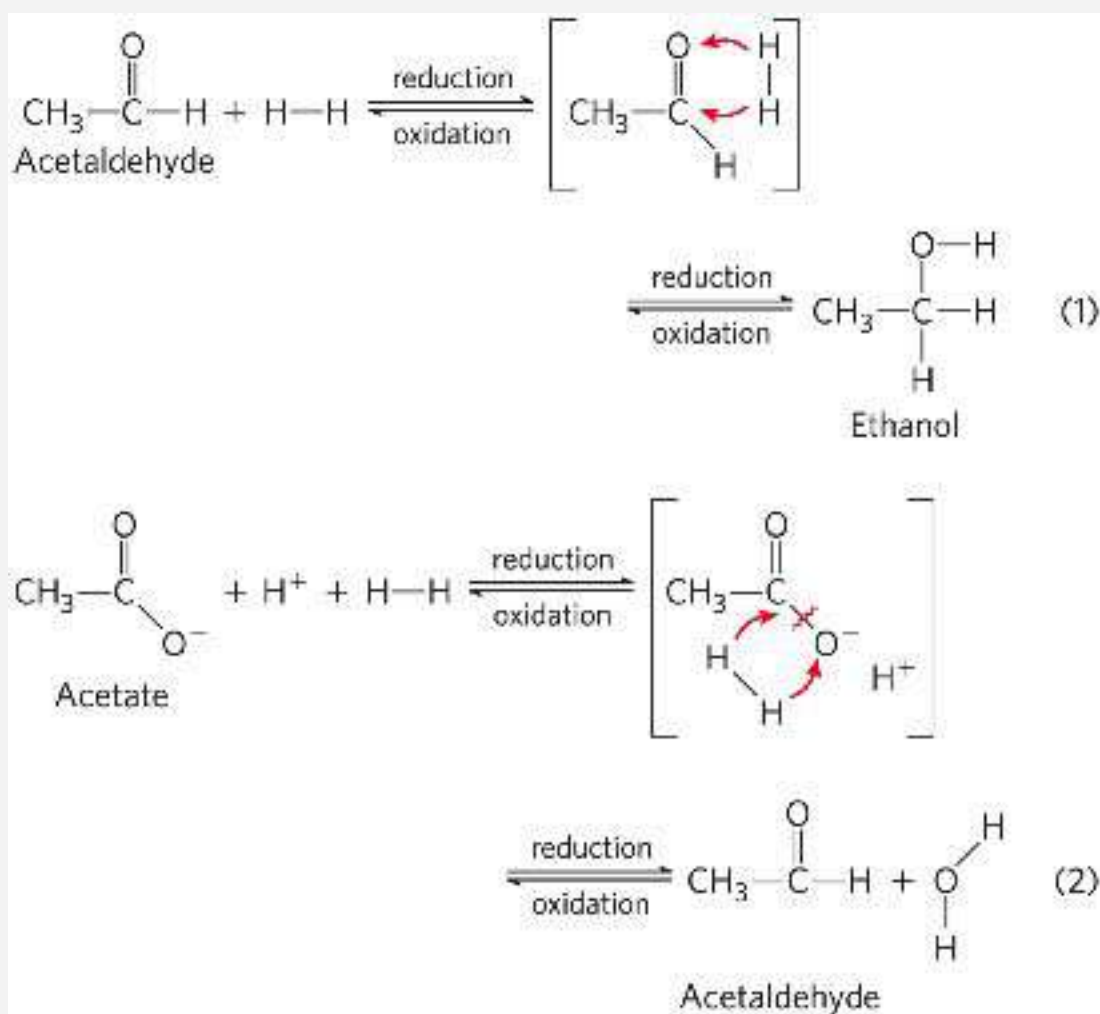
2. Net Equation for Glycolysis and the Citric Acid Cycle

Write the net biochemical equation for the metabolism of a molecule of glucose by glycolysis and the citric acid cycle, including all cofactors.

3. Recognizing Oxidation and Reduction Reactions

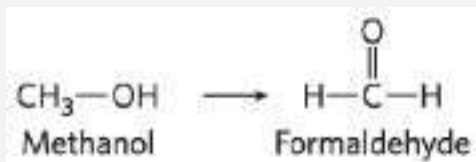
One biochemical strategy of many living organisms is the stepwise oxidation of organic compounds to CO_2 and H_2O

and the conservation of a major part of the energy thus produced in the form of ATP. It is important to be able to recognize oxidation-reduction processes in metabolism. Reduction of an organic molecule results from the hydrogenation of a double bond (Eqn 1, below) or of a single bond with accompanying cleavage (Eqn 2). Conversely, oxidation results from dehydrogenation. In biochemical redox reactions, the coenzymes NAD and FAD dehydrogenate/hydrogenate organic molecules in the presence of the proper enzymes.

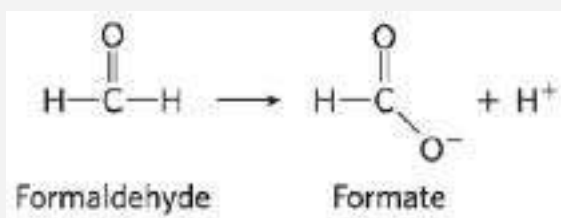


For each of the metabolic transformations (a) through (h), determine whether the compound on the left has undergone oxidation or reduction. Balance each transformation by inserting H—H and, where necessary, H₂O.

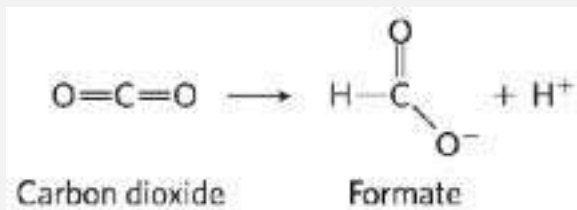
a.



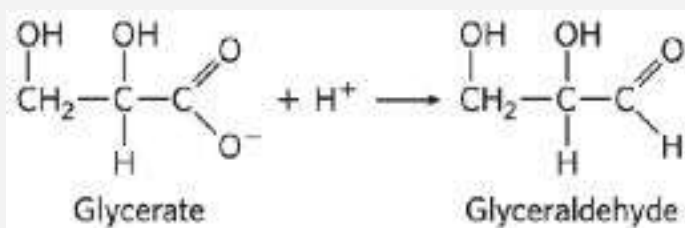
b.



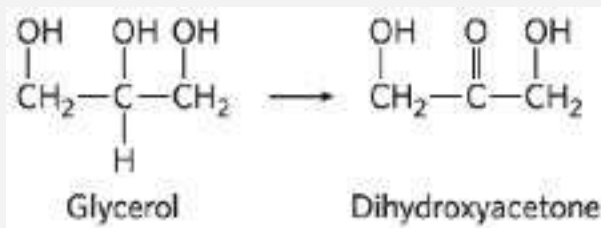
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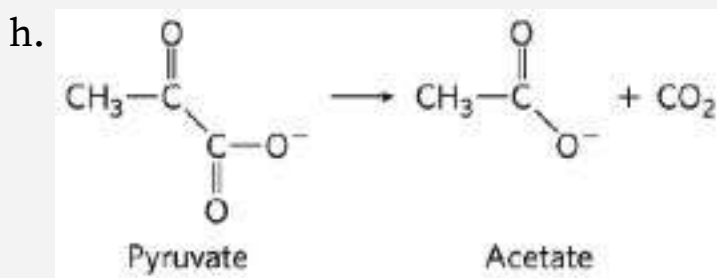
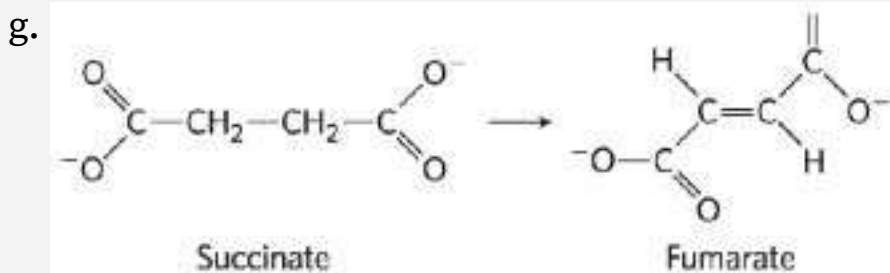
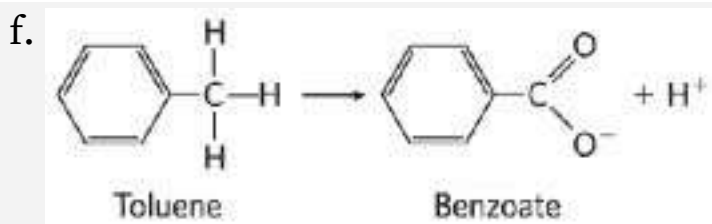


d.



e.



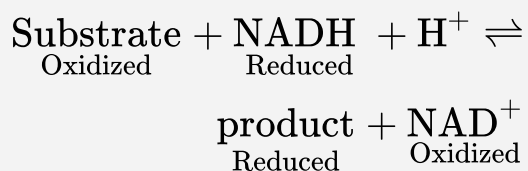


4. Relationship between Energy Release and the Oxidation State of Carbon A eukaryotic cell can use glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and hexanoate ($\text{C}_6\text{H}_{11}\text{O}_2$) as fuels for cellular respiration. On the basis of their structural formulas, which substance releases more energy per gram on complete combustion to CO_2 and H_2O ?

5. Nicotinamide Coenzymes as Reversible Redox Carriers

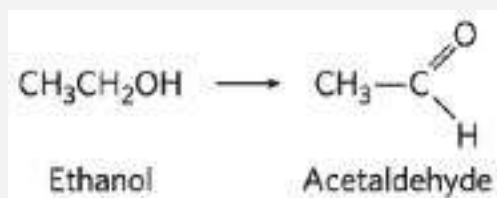
The nicotinamide coenzymes (see [Fig. 13-24](#)) can undergo reversible oxidation-reduction reactions with specific substrates in the presence of the appropriate dehydrogenase. In these reactions, $\text{NADH} + \text{H}^+$ serves as the hydrogen

source, as described in Problem 3. Whenever the coenzyme is oxidized, a substrate must be simultaneously reduced:

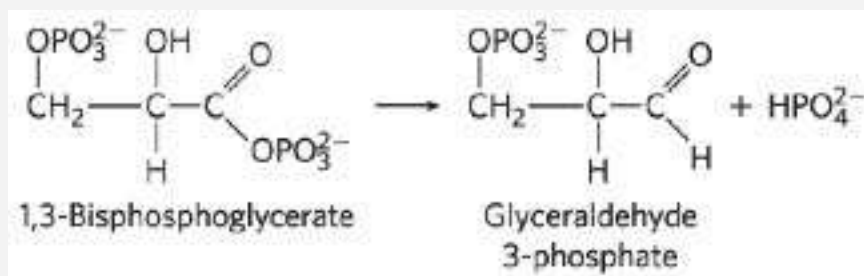


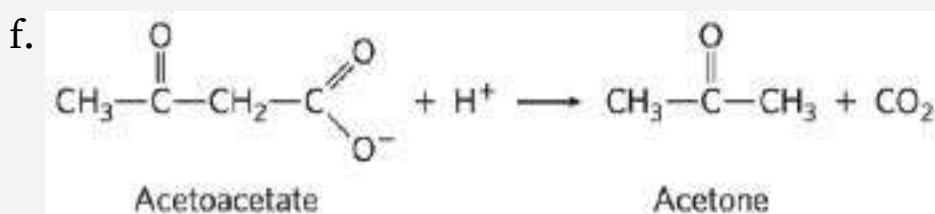
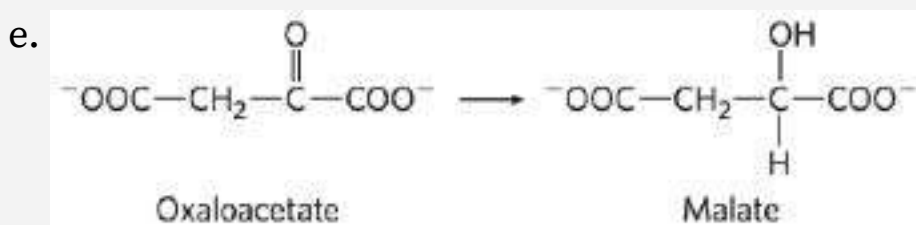
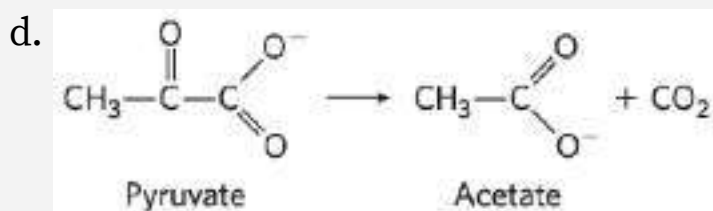
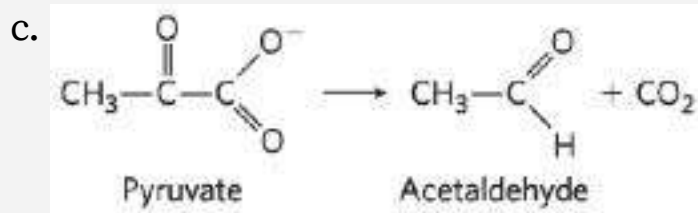
For each of the reactions in (a) through (f) shown below, determine whether the substrate has been oxidized or reduced or is unchanged in oxidation state (see [Problem 3](#)). If a redox change has occurred, balance the reaction with the necessary amount of NAD^+ , NADH , H^+ , and H_2O . The objective is to recognize when a redox coenzyme is necessary in a metabolic reaction.

a.



b.





6. Pyruvate Dehydrogenase Cofactors and Mechanism

Describe the role of each cofactor involved in the reaction catalyzed by the pyruvate dehydrogenase complex.

7. Thiamine Deficiency Individuals with a thiamine-deficient diet have relatively high levels of pyruvate in their blood. Explain this in biochemical terms.

8. Isocitrate Dehydrogenase Reaction What type of chemical reaction is involved in the conversion of isocitrate to α -

ketoglutarate? Name and describe the role of any cofactors. What other reaction(s) of the citric acid cycle are of this same type?

9. Stimulation of Oxygen Consumption by Oxaloacetate and Malate

In the early 1930s, Albert Szent-Györgyi reported the interesting observation that the addition of small amounts of oxaloacetate or malate to suspensions of minced pigeon breast muscle stimulated the oxygen consumption of the preparation. Surprisingly, the amount of oxygen consumed was about seven times more than the amount necessary for complete oxidation (to CO₂ and H₂O) of the added oxaloacetate or malate. Why did the addition of oxaloacetate or malate stimulate oxygen consumption? Why was the amount of oxygen consumed so much greater than the amount necessary to completely oxidize the added oxaloacetate or malate?

10. Formation of Oxaloacetate in a Mitochondrion In the last reaction of the citric acid cycle, malate is dehydrogenated to regenerate the oxaloacetate necessary for the entry of acetyl-CoA into the cycle:



- a. Calculate the equilibrium constant for this reaction at 25 °C.

- b. Because $\Delta G'^{\circ}$ assumes a standard pH of 7, the equilibrium constant calculated in (a) corresponds to

$$K'_{\text{eq}} = \frac{[\text{oxaloacetate}][\text{NADH}]}{[\text{L-malate}][\text{NAD}^+]}$$

The measured concentration of L-malate in rat liver mitochondria is about 0.20 mM when $[\text{NAD}^+]/[\text{NADH}]$ is 10. Calculate the concentration of oxaloacetate at pH 7 in these mitochondria.

- c. To appreciate the magnitude of the mitochondrial oxaloacetate concentration, calculate the number of oxaloacetate molecules in a single rat liver mitochondrion. Assume the mitochondrion is a sphere of diameter 2.0 μm .

11. Cofactors for the Citric Acid Cycle Suppose you have prepared a mitochondrial extract that contains all the soluble enzymes of the matrix but has lost (by dialysis) all the low molecular weight cofactors. What must you add to the extract so that the preparation will oxidize acetyl-CoA to CO_2 ?

12. Riboflavin Deficiency How would a riboflavin deficiency affect the functioning of the citric acid cycle? Explain your answer.

13. Oxaloacetate Pool What factors might decrease the pool of oxaloacetate available for the activity of the citric acid cycle? How can the pool of oxaloacetate be replenished?

14. Energy Yield from the Citric Acid Cycle The reaction catalyzed by succinyl-CoA synthetase produces the high-energy compound GTP. How is the free energy contained in GTP incorporated into the cellular ATP pool?

15. Respiration Studies in Isolated Mitochondria Cellular respiration can be studied in isolated mitochondria by measuring oxygen consumption under different conditions. If 0.01 M sodium malonate is added to actively respiring mitochondria that are using pyruvate as fuel, respiration soon stops and a metabolic intermediate accumulates.

- What is the structure of this intermediate?
- Explain why it accumulates.
- Explain why oxygen consumption stops.
- Aside from removal of the malonate, what can overcome this inhibition of respiration? Explain.

16. Labeling Studies in Isolated Mitochondria Biochemists have often delineated the metabolic pathways of organic compounds by using a radioactively labeled substrate and following the fate of the label.

- How can you determine whether a suspension of isolated mitochondria metabolizes added glucose to CO_2 and H_2O ?
- Suppose you add a brief pulse of $[3\text{-}^{14}\text{C}]$ pyruvate (labeled in the methyl position) to the mitochondria. After one turn of the citric acid cycle, what is the location of the ^{14}C in the oxaloacetate? Explain by tracing the ^{14}C label through the pathway. How many

turns of the cycle are required to release all the $[3-^{14}\text{C}]$ pyruvate as CO_2 ?

17. $[1-^{14}\text{C}]$ Glucose Catabolism An investigator briefly incubates an actively respiring bacterial culture with $[1-^{14}\text{C}]$ glucose and isolates the glycolytic and citric acid cycle intermediates. Where is the ^{14}C located in each of the intermediates listed? Consider only the initial incorporation of ^{14}C in the first pass of labeled glucose through the pathways.

- a. Fructose 1,6-bisphosphate
- b. Glyceraldehyde 3-phosphate
- c. Phosphoenolpyruvate
- d. Acetyl-CoA
- e. Citrate
- f. α -Ketoglutarate
- g. Oxaloacetate

18.  Role of the Vitamin Thiamine People with beriberi, a disease caused by thiamine deficiency, have elevated levels of blood pyruvate and α -ketoglutarate, especially after consuming a meal rich in glucose. How are these effects related to a deficiency of thiamine?

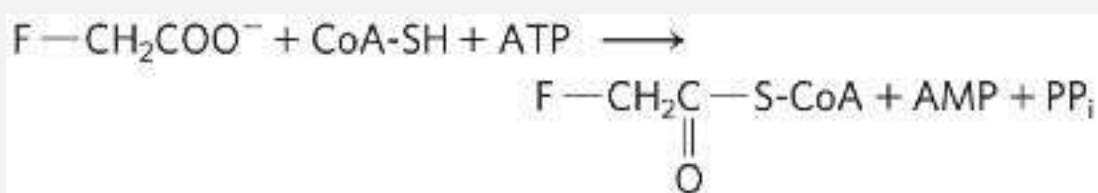
19. Synthesis of Oxaloacetate by the Citric Acid Cycle In the last step of the citric acid cycle, NAD^+ -dependent oxidation of L-malate forms oxaloacetate. Can a net synthesis of oxaloacetate from acetyl-CoA occur using only the enzymes and cofactors of the citric acid cycle, without depleting the intermediates of the cycle? Explain. How do cells replenish

the oxaloacetate that is lost from the cycle to biosynthetic reactions?

20. Oxaloacetate Depletion Mammalian liver can carry out gluconeogenesis using oxaloacetate as the starting material ([Chapter 14](#)). Would the extensive use of oxaloacetate for gluconeogenesis affect the operation of the citric acid cycle? Explain your answer.

21. Mode of Action of the Rodenticide Fluoroacetate

Fluoroacetate, prepared commercially for rodent control, is also produced by a South African plant. After entering a cell, fluoroacetate is converted to fluoroacetyl-CoA in a reaction catalyzed by the enzyme acetate thiokinase:



You perform a perfusion experiment to study the toxic effect of fluoroacetate using intact isolated rat heart. After perfusing the heart with 0.22 mM fluoroacetate, you see a decrease in the measured rate of glucose uptake and glycolysis as well as an accumulation of glucose 6-phosphate and fructose 6-phosphate. Examination of the citric acid cycle intermediates reveals that their concentrations are below normal, except for citrate, which has a concentration 10 times higher than normal.

- a. Where did the block in the citric acid cycle occur?
What caused citrate to accumulate and the other cycle intermediates to be depleted?
- b. Fluoroacetyl-CoA is enzymatically transformed in the citric acid cycle. What is the structure of the end product of fluoroacetate metabolism? Why does it block the citric acid cycle? How might the inhibition be overcome?
- c. In the heart perfusion experiments, why did glucose uptake and glycolysis decrease? Why did hexose monophosphates accumulate?
- d. Why is fluoroacetate poisoning fatal?

22. Synthesis of L-Malate in Wine Making The tartness of some wines is due to high concentrations of L-malate. Write a sequence of reactions showing how yeast cells synthesize L-malate from glucose under anaerobic conditions in the presence of dissolved CO₂ (HCO₃⁻). Note that the overall reaction for this fermentation cannot involve the consumption of nicotinamide coenzymes or citric acid cycle intermediates.

23. Net Synthesis of α -Ketoglutarate α -Ketoglutarate plays a central role in the biosynthesis of several amino acids. Write a sequence of enzymatic reactions that could result in the net synthesis of α -ketoglutarate from pyruvate. Your proposed sequence must not involve the net consumption of other citric acid cycle intermediates. Write an equation for the overall reaction.

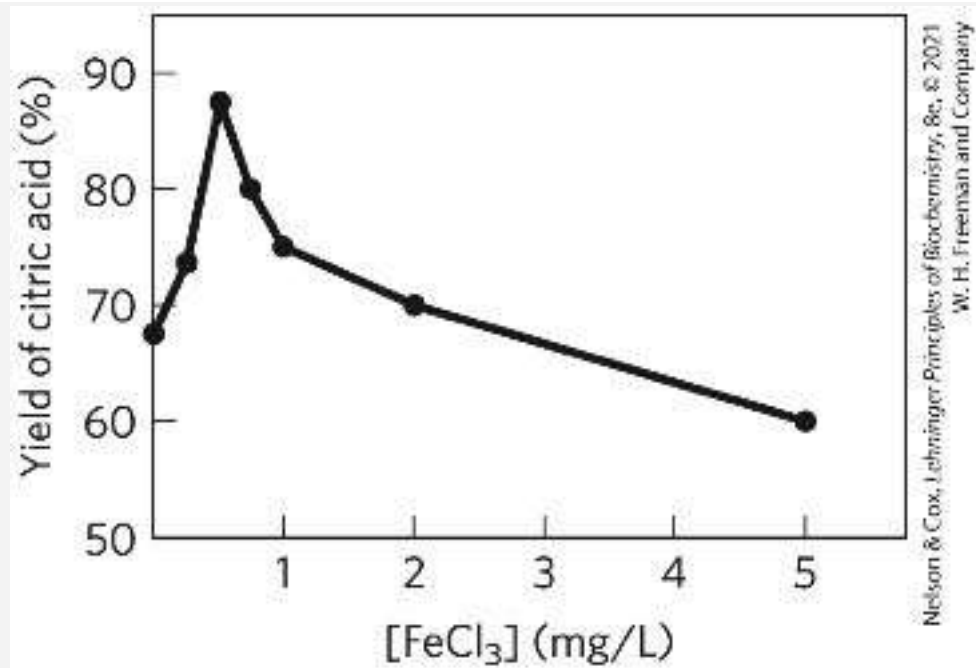
24. Amphibolic Pathways Explain, giving examples, what is meant by the statement that the citric acid cycle is amphibolic.

25. Regulation of the Pyruvate Dehydrogenase Complex In animal tissues, the ratio of active, unphosphorylated to inactive, phosphorylated PDH complex regulates the rate of conversion of pyruvate to acetyl-CoA. Determine what happens to the rate of this reaction when a preparation of rabbit muscle mitochondria containing the PDH complex is treated with **(a)** pyruvate dehydrogenase kinase, ATP, and NADH; **(b)** pyruvate dehydrogenase phosphatase and Ca^{2+} ; **(c)** malonate.

26. Commercial Synthesis of Citric Acid Manufacturers use citric acid as a flavoring agent in soft drinks, fruit juices, and many other foods. Worldwide, the market for citric acid is valued at hundreds of millions of dollars per year.

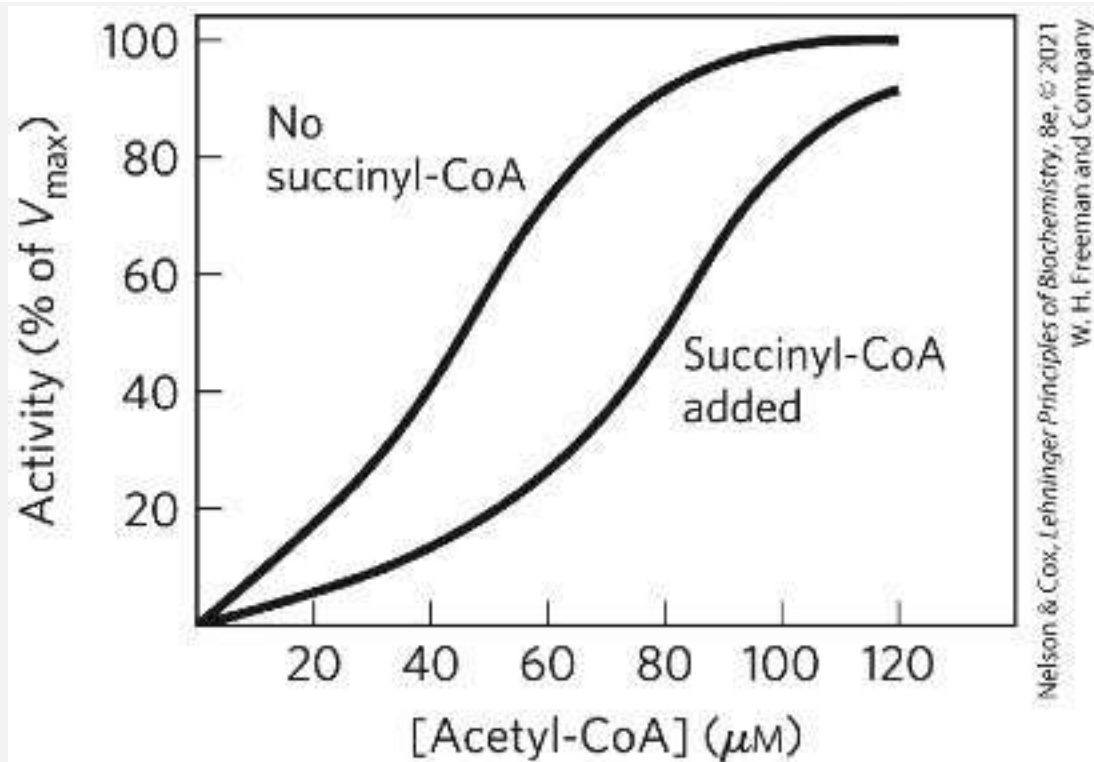
Commercial production uses the mold *Aspergillus niger*, which metabolizes sucrose under carefully controlled conditions.

- a. The yield of citric acid strongly depends on the concentration of FeCl_3 in the culture medium, as indicated in the graph. Why does the yield decrease when the concentration of Fe^{3+} is above or below the optimal value of 0.5 mg/L?



- b. Write the sequence of reactions by which *A. niger* synthesizes citric acid from sucrose. Write an equation for the overall reaction.
- c. Does the commercial process require the culture medium to be aerated — that is, is this a fermentation or an aerobic process? Explain.

27. Regulation of Citrate Synthase In the presence of saturating amounts of oxaloacetate, the activity of citrate synthase from pig heart tissue shows a sigmoid dependence on the concentration of acetyl-CoA, as shown in the graph. Adding succinyl-CoA shifts the curve to the right and makes the sigmoid dependence more pronounced.



On the basis of these observations, suggest how succinyl-CoA regulates the activity of citrate synthase. (Hint: See [Fig. 6-37](#).) Why is succinyl-CoA an appropriate signal for regulation of the citric acid cycle? How does the regulation of citrate synthase control the rate of cellular respiration in pig heart tissue?

28. Regulation of Pyruvate Carboxylase The carboxylation of pyruvate by pyruvate carboxylase occurs at a very low rate unless acetyl-CoA, a positive allosteric modulator, is present. If you have just eaten a meal rich in fatty acids (triacylglycerols) but low in carbohydrates (glucose), how does this regulatory property shut down the oxidation of glucose to CO_2 and H_2O but increase the oxidation of acetyl-CoA derived from fatty acids?

29. Relationship between Respiration and the Citric Acid Cycle

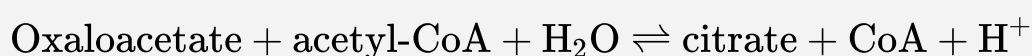
Although oxygen does not participate directly in the citric acid cycle, the cycle operates only when O₂ is present. Why?

30. Effect of [NADH]/[NAD⁺] on the Citric Acid Cycle

How would you expect the operation of the citric acid cycle to respond to a rapid increase in the [NADH]/[NAD⁺] ratio in the mitochondrial matrix? Why?

31. Thermodynamics of Citrate Synthase Reaction in Cells

Citrate is formed by the condensation of acetyl-CoA with oxaloacetate, catalyzed by citrate synthase:



In rat heart mitochondria at pH 7.0 and 25 °C, the concentrations of reactants and products are oxaloacetate, 1 μM; acetyl-CoA, 1 μM; citrate, 220 μM; and CoA, 65 μM. The standard free-energy change for the citrate synthase reaction is −32.2 kJ/mol. What is the direction of metabolite flow through the citrate synthase reaction in rat heart cells? Explain.

32. Reactions of the Pyruvate Dehydrogenase Complex

Two of the steps in the oxidative decarboxylation of pyruvate (steps ④ and ⑤ in [Fig. 16-6](#)) do not involve any of the three carbons of pyruvate, yet are essential to the operation of the PDH complex. Explain.

33.  Pyruvate Transport into Mitochondria The mitochondrial pyruvate carrier (MPC) is a heterodimer of the proteins MPC1 and MPC2. In a high proportion (80%) of certain cancers, including gliomas (tumors of the glial cells of the brain), the gene for one of these proteins is mutated such that pyruvate cannot enter the mitochondrial matrix. Name three metabolic effects that you would expect to see if cytosolic pyruvate could not gain access to the machinery of the citric acid cycle. (Hint: [Box 14-1](#) may be helpful.)

34. Citric Acid Cycle Mutants There are many cases of human disease in which one or another enzyme activity is lacking due to genetic mutation. Why are cases in which individuals lack one of the enzymes of the citric acid cycle extremely rare?

DATA ANALYSIS PROBLEM

35. How the Citric Acid Cycle Was Discovered The detailed biochemistry of the citric acid cycle was determined by several researchers over a period of decades. In a 1937 article, Krebs and Johnson summarized their work and the work of others in the first published description of this pathway.

The methods used by these researchers were very different from those of modern biochemistry. Radioactive tracers were not commonly available until the 1940s, so Krebs and other

researchers had to use nontracer techniques to work out the pathway. Using freshly prepared samples of pigeon breast muscle, they determined oxygen consumption by suspending minced muscle in buffer in a sealed flask and measuring the volume (in μL) of oxygen consumed under different conditions. They measured levels of substrates (intermediates) by treating samples with acid to remove contaminating proteins, then assaying the quantities of various small organic molecules. The two key observations that led Krebs and colleagues to propose a citric acid *cycle* as opposed to a *linear pathway* (like that of glycolysis) were made in the following experiments.

Experiment I They incubated 460 mg of minced muscle in 3 mL of buffer at 40 °C for 150 minutes. Addition of *citrate* increased O_2 consumption by 893 μL compared with samples without added citrate. They calculated, based on the O_2 consumed during respiration of other carbon-containing compounds, that the expected O_2 consumption for complete respiration of this quantity of citrate was only 302 μL .

Experiment II They measured O_2 consumption by 460 mg of minced muscle in 3 mL of buffer when incubated with *citrate* and/or with *1-phosphoglycerol* (glycerol 1-phosphate; this was known to be readily oxidized by cellular respiration) at 40 °C for 140 minutes. The results are shown in the table.

Sample	Substrate(s) added	$\mu\text{L O}_2$
--------	--------------------	-------------------

		absorbed
1	No extra	342
2	0.3 mL 0.2 M 1-phosphoglycerol	757
3	0.15 mL 0.02 M citrate	431
4	0.3 mL 0.2 M 1-phosphoglycerol and 0.15 mL 0.02 M citrate	1,385

- Why is O₂ consumption a good measure of cellular respiration?
- Why does sample 1 (unsupplemented muscle tissue) consume some oxygen?
- Based on the results for samples 2 and 3, can you conclude that 1-phosphoglycerol and citrate serve as substrates for cellular respiration in this system? Explain your reasoning.
- Krebs and colleagues used the results from these experiments to argue that citrate was “catalytic” — that it helped the muscle tissue samples metabolize 1-phosphoglycerol more completely. How would you use their data to make this argument?
- Krebs and colleagues further argued that citrate was not simply consumed by these reactions, but had to be *regenerated*. Therefore, the reactions had to be a *cycle* rather than a linear pathway. How would you make this argument?

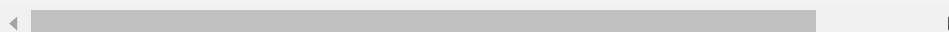
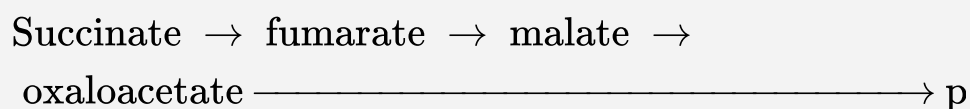
Other researchers had found that *arsenate* (AsO₄³⁻) inhibits α -ketoglutarate dehydrogenase and that

malonate inhibits succinate dehydrogenase.

- f. Krebs and coworkers found that muscle tissue samples treated with arsenate and citrate would consume citrate only in the presence of oxygen; under these conditions, oxygen was consumed. Based on the pathway in [Figure 16-7](#), what was the citrate converted to in this experiment, and why did the samples consume oxygen?

In their article, Krebs and Johnson further reported the following: (1) In the presence of arsenate, 5.48 mmol of citrate was converted to 5.07 mmol of α -ketoglutarate. (2) In the presence of malonate, citrate was quantitatively converted to large amounts of succinate and small amounts of α -ketoglutarate. (3) Addition of oxaloacetate in the absence of oxygen led to production of a large amount of citrate; the amount was increased if glucose was also added.

Other workers had found the following pathway in similar muscle tissue preparations:



- g. Based only on the data presented in this problem, what is the order of the intermediates in the citric acid cycle? How does this compare with [Figure 16-7](#)? Explain your reasoning.

h. Why was it important to show the *quantitative* conversion of citrate to α -ketoglutarate?

The Krebs and Johnson article also contains other data that filled in most of the missing components of the cycle. The only component left unresolved was the molecule that reacted with oxaloacetate to form citrate.

Reference

Krebs, H.A., and W.A. Johnson. 1937. The role of citric acid in intermediate metabolism in animal tissues. *Enzymologia* 4:148–156. Reprinted in *FEBS Lett.* 117(Suppl.):K2–K10, 1980.

CHAPTER 17

FATTY ACID CATABOLISM



17.1 Digestion, Mobilization, and Transport of Fats

17.2 Oxidation of Fatty Acids

17.3 Ketone Bodies

The oxidation of long-chain fatty acids to acetyl-CoA is a central energy-yielding pathway in many organisms and tissues. In mammalian heart and liver, for example, it provides as much as 80% of the energetic needs under all physiological circumstances. The electrons removed from fatty acids during oxidation pass through the respiratory chain, driving ATP synthesis; the acetyl-CoA produced from the fatty acids may be completely oxidized to CO_2 in the citric acid cycle, resulting in further energy conservation.

In [Chapter 10](#) we described the properties of triacylglycerols (also called triglycerides or neutral fats) that make them especially suitable as storage fuels. The long alkyl chains of their constituent fatty acids are essentially hydrocarbons, highly reduced structures with an energy of complete oxidation ($\sim 38 \text{ kJ/g}$) more than twice that for the same weight of carbohydrate or protein.

This advantage is compounded by the extreme insolubility of lipids in water; cellular triacylglycerols aggregate in lipid droplets, which do not raise the osmolarity of the cytosol, and they are unsolvated. (In storage polysaccharides, by contrast, water of solvation can account for two-thirds of the overall weight of the stored molecules.) And because of their relative chemical inertness, triacylglycerols can be stored in large quantity in cells without the risk of undesired chemical reactions with other cellular constituents.

The properties that make triacylglycerols good storage compounds, however, present problems in their role as fuels. Because they are insoluble in water, ingested triacylglycerols must be emulsified before they can be digested by water-soluble enzymes in the intestine, and triacylglycerols absorbed in the intestine or mobilized from storage tissues must be carried in the blood bound to proteins that counteract their insolubility. Also, to overcome the relative stability of the C—C bonds in a fatty acid, the carboxyl group at C-1 is activated by attachment to coenzyme A, which allows stepwise oxidation of the fatty acyl group at the C-3, or β , position — hence the name [β oxidation](#).

The principles we emphasize in this chapter are not new. They apply to the catabolic pathways of carbohydrates that we just studied.



Metabolites of diverse origin funnel into a few central pathways. Fatty acid catabolism and glycolysis convert quite different starting materials into the same product (acetyl-CoA).

The electrons from the oxidative reactions of these pathways and of the citric acid cycle are carried by common cofactors (NAD and FAD) to the mitochondrial respiratory chain leading to oxygen, providing the energy for ATP synthesis by oxidative phosphorylation.

P2 Evolution selects for chemical mechanisms that make useful reactions more energetically favorable, and those same mechanisms are used in different pathways. In the breakdown of fatty acids we see the activation of a carboxylic acid by its conversion to a thioester, as we saw with acetyl-CoA in the citric acid cycle. To break the C—C bonds in the long chain of relatively inert —CH₂—CH₂— groups in fatty acids, a carbonyl group is created adjacent to the —CH₂— group, as we saw in the reactions of the citric acid cycle.

P3 Allosteric mechanisms and posttranslational regulation (protein phosphorylation) coordinate metabolic processes within a cell. Hormones and growth factors coordinate metabolic activities among tissues and organs. Reciprocal regulation of catabolic and anabolic pathways prevents the inefficiency of futile cycling.

P4 When a process lacks a critical component — an enzyme, a cofactor, or a regulatory agent — the resulting loss of homeostasis may cause disease across a spectrum of severity. Defects in fatty acid breakdown are no exception.



The liver plays a unique role in whole-body metabolism.

When glucose is unavailable, the liver makes glucose by gluconeogenesis and releases it to the blood for distribution to other tissues, including the brain. During starvation, the liver processes fatty acids into ketone bodies, which, unlike fatty acids, can cross the blood brain barrier and fuel the brain.

17.1 Digestion, Mobilization, and Transport of Fats

Cells can obtain fatty acid fuels from four sources: fats consumed in the diet, fats stored in cells as lipid droplets, fats synthesized in one organ for export to another, and fats obtained by autophagy (which degrades the cell's own organelles). Some species use all four sources under various circumstances; others use one or two. Vertebrates, for example, obtain fats in the diet; mobilize fats stored in specialized tissue (adipose tissue, consisting of cells called adipocytes); and, in the liver, convert excess dietary carbohydrates to fats for export to other tissues. During starvation, they can recycle lipids by autophagy. On average, 40% or more of the daily energy requirement of humans in highly industrialized countries is supplied by dietary triacylglycerols. Triacylglycerols provide more than half the energy requirements of some organs, particularly the liver, heart, and resting skeletal muscle. Stored triacylglycerols are virtually the sole source of energy in hibernating animals and migrating birds. Vascular plants mobilize fats stored in seeds during germination, but do not otherwise depend on fats for energy.

Dietary Fats Are Absorbed in the Small Intestine

In vertebrates, before ingested triacylglycerols can be absorbed through the intestinal wall they must be converted from insoluble

macroscopic fat particles to finely dispersed microscopic micelles. This solubilization is carried out by bile salts, such as taurocholic acid ([p. 352](#)), which are synthesized from cholesterol in the liver, stored in the gallbladder, and released into the small intestine after ingestion of a fatty meal. Bile salts are amphipathic compounds that act as biological detergents, emulsifying dietary fats into mixed micelles of bile salts and triacylglycerols ([Fig. 17-1](#), step ❶). Micelle formation enormously increases the fraction of lipid molecules accessible to the action of water-soluble lipases in the intestine, and lipase action converts triacylglycerols to monoacylglycerols (monoglycerides) and diacylglycerols (diglycerides) and free fatty acids (step ❷). These products of lipase action diffuse or are transported into the epithelial cells lining the intestinal surface (the intestinal mucosa) (step ❸), where they are reconverted to triacylglycerols and packaged with dietary cholesterol and specific [apolipoproteins](#) (from the Greek *apo*, meaning “detached” or “separate,” designating the protein in its lipid-free form) into lipoprotein aggregates called [chylomicrons](#) (step ❹). It is the protein moieties that target triacylglycerols, phospholipids, cholesterol, and cholesteryl esters for transport between organs.

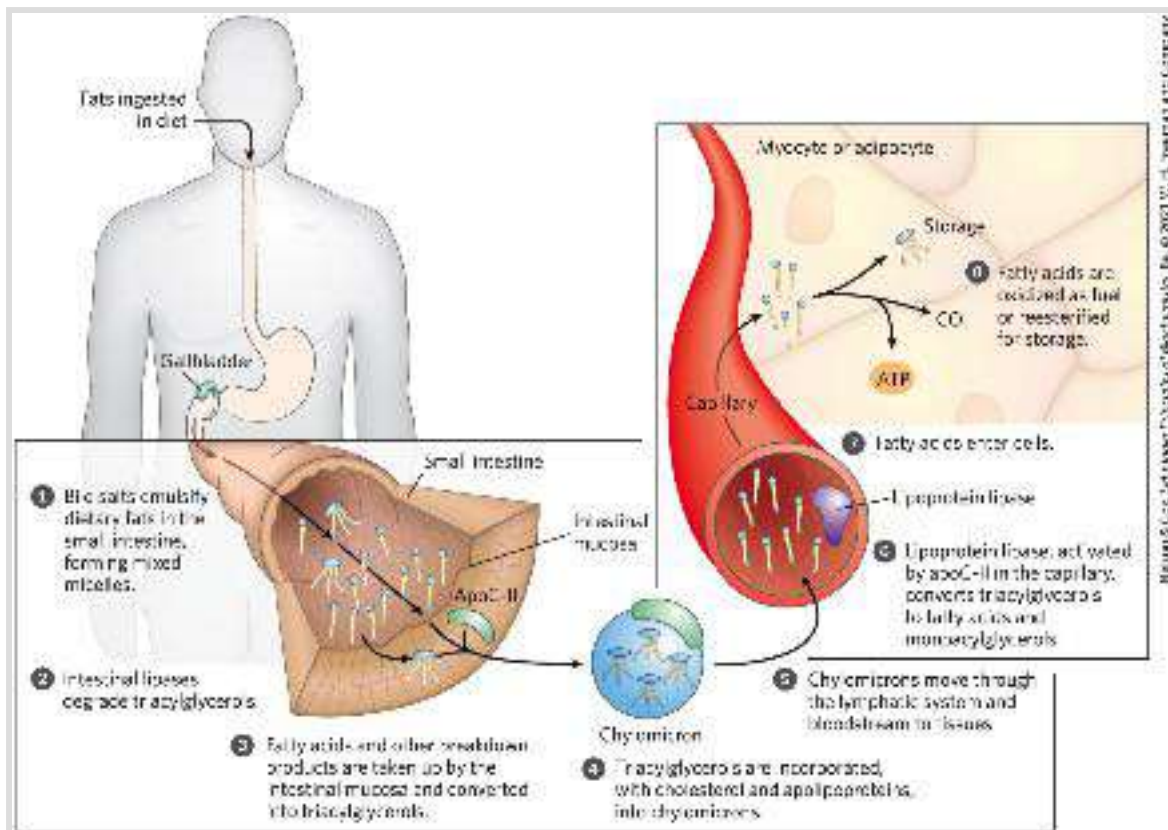


FIGURE 17-1 Processing of dietary lipids in vertebrates. Digestion and absorption of dietary lipids occur in the small intestine, and the fatty acids released from triacylglycerols are packaged and delivered to muscle and adipose tissues. The eight steps are discussed in the text.

Apolipoproteins combine with lipids to form several classes of **lipoprotein particles**, spherical aggregates with hydrophobic lipids at the core and hydrophilic protein side chains and lipid head groups at the surface. Various combinations of lipid and protein produce particles of different densities, ranging from chylomicrons and very-low-density lipoproteins (VLDL) to very-high-density lipoproteins (VHDL). These particles can be separated by ultracentrifugation. The structures of these lipoprotein particles and their roles in lipid transport are detailed in [Chapter 21](#) ([Fig. 21-39](#)).

Apolipoprotein B-48 (apoB-48) is the primary protein component of chylomicrons. In lipid uptake from the intestine, chylomicrons move from the intestinal mucosa into the lymphatic system, and then enter the blood, where they can exchange apolipoproteins with other types of circulating lipoprotein. In the blood, chylomicrons pick up **apolipoprotein C-II (apoC-II)** from **high-density lipoprotein (HDL)** particles and are carried to muscle and adipose tissue ([Fig. 17-1](#), step ⑤). In the capillaries of these tissues, the extracellular enzyme **lipoprotein lipase**, activated by apoC-II, hydrolyzes triacylglycerols to free fatty acids and monoacylglycerols (step ⑥), which are taken up by specific transporters in the plasma membranes of cells in the target tissues (step ⑦). In muscle, the fatty acids are oxidized for energy; in adipose tissue, they are reesterified for storage as triacylglycerols (step ⑧).

The remnants of chylomicrons, depleted of most of their triacylglycerols but still containing cholesterol and apolipoproteins, travel in the blood to the liver, where they are taken up by endocytosis mediated by receptors for their apolipoproteins. Triacylglycerols that enter the liver by this route may be oxidized to provide energy or to provide precursors for the synthesis of ketone bodies, as described in [Section 17.3](#).

When the diet contains more fatty acids than are needed immediately for fuel or as precursors, the liver converts them to triacylglycerols, which are packaged with specific apolipoproteins into VLDLs. These VLDLs are secreted by hepatocytes and transported in the blood to adipose tissue, where the

triacylglycerols are removed and stored in lipid droplets within adipocytes.

Hormones Trigger Mobilization of Stored Triacylglycerols

Neutral lipids (triacylglycerols, sterols, and sterol esters) are stored in adipocytes (and in steroid-synthesizing cells of the adrenal cortex, ovary, and testis) in **lipid droplets**, which have a core of triacylglycerols and sterol esters surrounded by a monolayer of phospholipids. The surface of these droplets is coated with **perilipins**, a family of proteins that restrict access to lipid droplets, preventing untimely lipid mobilization. 

When hormones signal the need for metabolic energy, triacylglycerols stored in adipose tissue are mobilized (brought out of storage) and transported to tissues (skeletal muscle, heart, and renal cortex) in which fatty acids can be oxidized for energy production. The hormones epinephrine and glucagon, secreted in response to low blood glucose levels or a fight-or-flight situation, stimulate the enzyme adenylyl cyclase in the adipocyte plasma membrane ([Fig. 17-2](#)), which produces the intracellular second messenger cyclic AMP (cAMP; see [Fig. 12-4](#)). Cyclic AMP-dependent protein kinase (PKA) triggers changes that open the lipid droplet to the action of three cytosolic lipases, which act on tri-, di-, and monoacylglycerols, releasing fatty acids and glycerol.

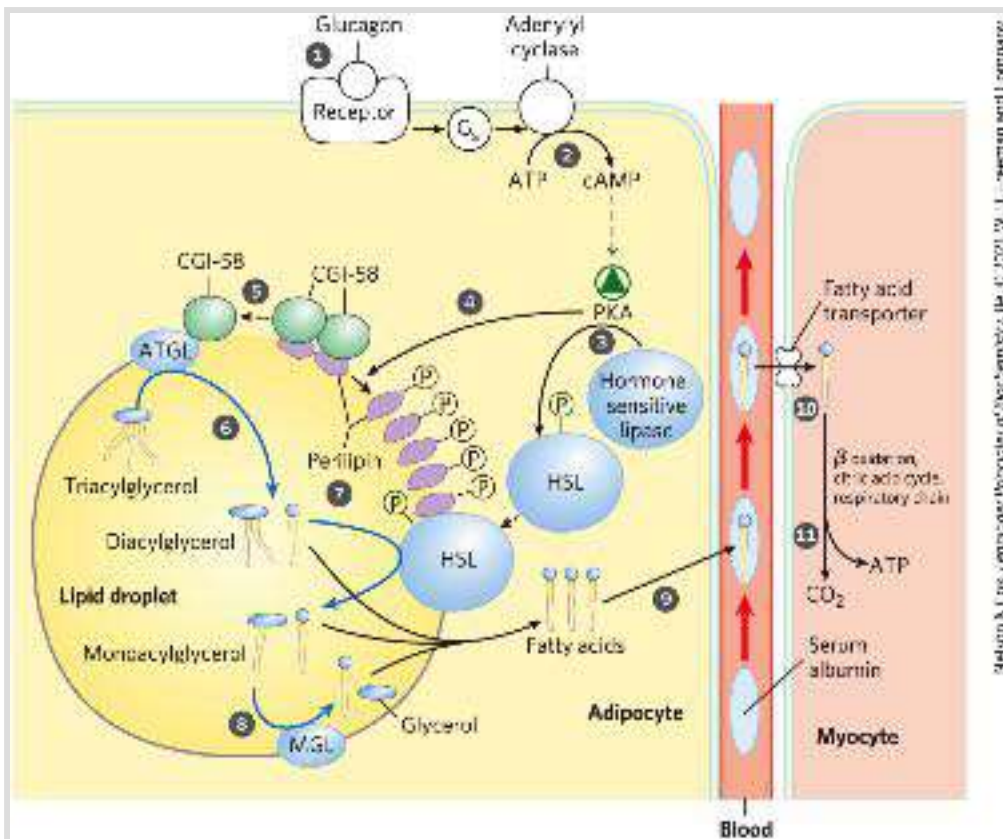
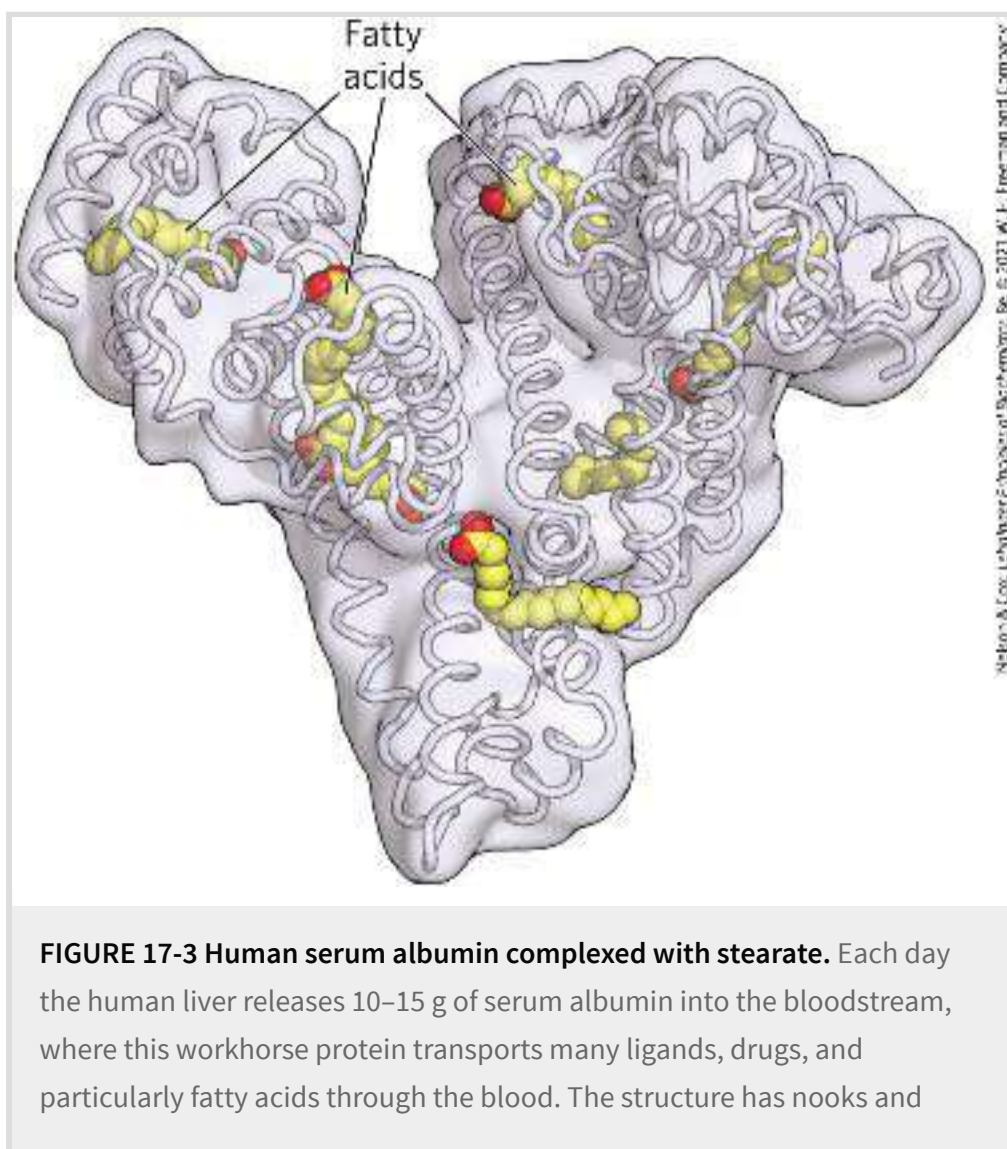


FIGURE 17-2 Mobilization of triacylglycerols stored in adipose tissue.

When low levels of glucose in the blood trigger the release of glucagon, **1** the hormone binds its receptor in the adipocyte membrane and thus **2** stimulates adenyl cyclase, via a G protein, to produce cAMP. This activates PKA, which phosphorylates **3** the hormone-sensitive lipase (HSL) and **4** perilipin molecules on the surface of the lipid droplet. Phosphorylation of perilipin causes **5** dissociation of the protein CGI-58 from perilipin. CGI-58 (comparative gene identification-58), a protein closely associated with lipid droplets, then recruits adipose triacylglycerol lipase (ATGL) to the droplet surface and stimulates its lipase activity. Active ATGL **6** converts triacylglycerols to diacylglycerols. The phosphorylated perilipin associates with phosphorylated HSL, allowing it access to the surface of the lipid droplet, where **7** it converts diacylglycerols to monoacylglycerols. **8** A third lipase, monoacylglycerol lipase (MGL), hydrolyzes monoacylglycerols. **9** Fatty acids leave the adipocyte, and are transported in the blood bound to serum albumin. They are released from the albumin and **10** enter a myocyte via a specific fatty acid transporter. **11** In the myocyte, fatty acids are oxidized to CO_2 , and the energy of oxidation is conserved in ATP, which fuels muscle contraction and other energy-requiring metabolism in the myocyte.

The fatty acids thus released (**free fatty acids, FFAs**) pass from the adipocyte into the blood, where they bind to the blood protein **serum albumin** ([Fig. 17-3](#)). This protein (M_r 66,000), which makes up about half of the total serum protein, noncovalently binds as many as seven fatty acids. Bound to this soluble protein, the otherwise insoluble fatty acids are carried to tissues such as skeletal muscle, heart, and renal cortex. In these target tissues, fatty acids dissociate from albumin and are moved by plasma membrane transporters into cells to serve as fuel.



crannies that can carry up to seven fatty acids. Its passengers are both hydrophobic and hydrophilic, and include steroid hormones, the blood thinner warfarin, the antibiotic penicillin, the anti-inflammatory drug ibuprofen, and the anxiolytic diazepam.

About 95% of the biologically available energy of triacylglycerols resides in their three long-chain fatty acids; only 5% is contributed by the glycerol moiety. The glycerol released by lipase action is phosphorylated by **glycerol kinase** ([Fig. 17-4](#)), and the resulting glycerol 3-phosphate is oxidized to dihydroxyacetone phosphate. The glycolytic enzyme triose phosphate isomerase converts this compound to glyceraldehyde 3-phosphate, which is oxidized via glycolysis.

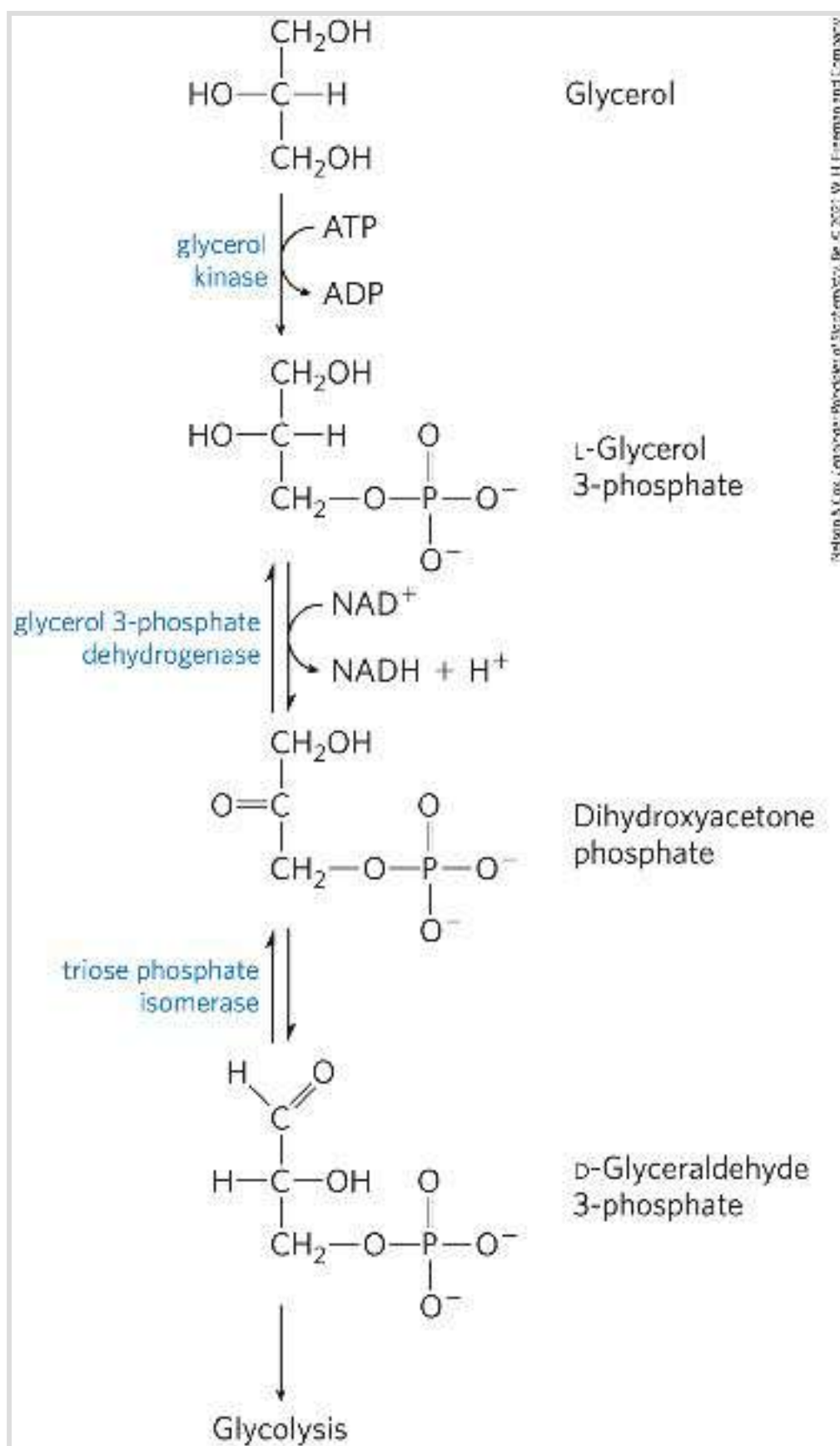
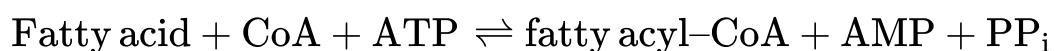


FIGURE 17-4 Entry of glycerol into the glycolytic pathway.

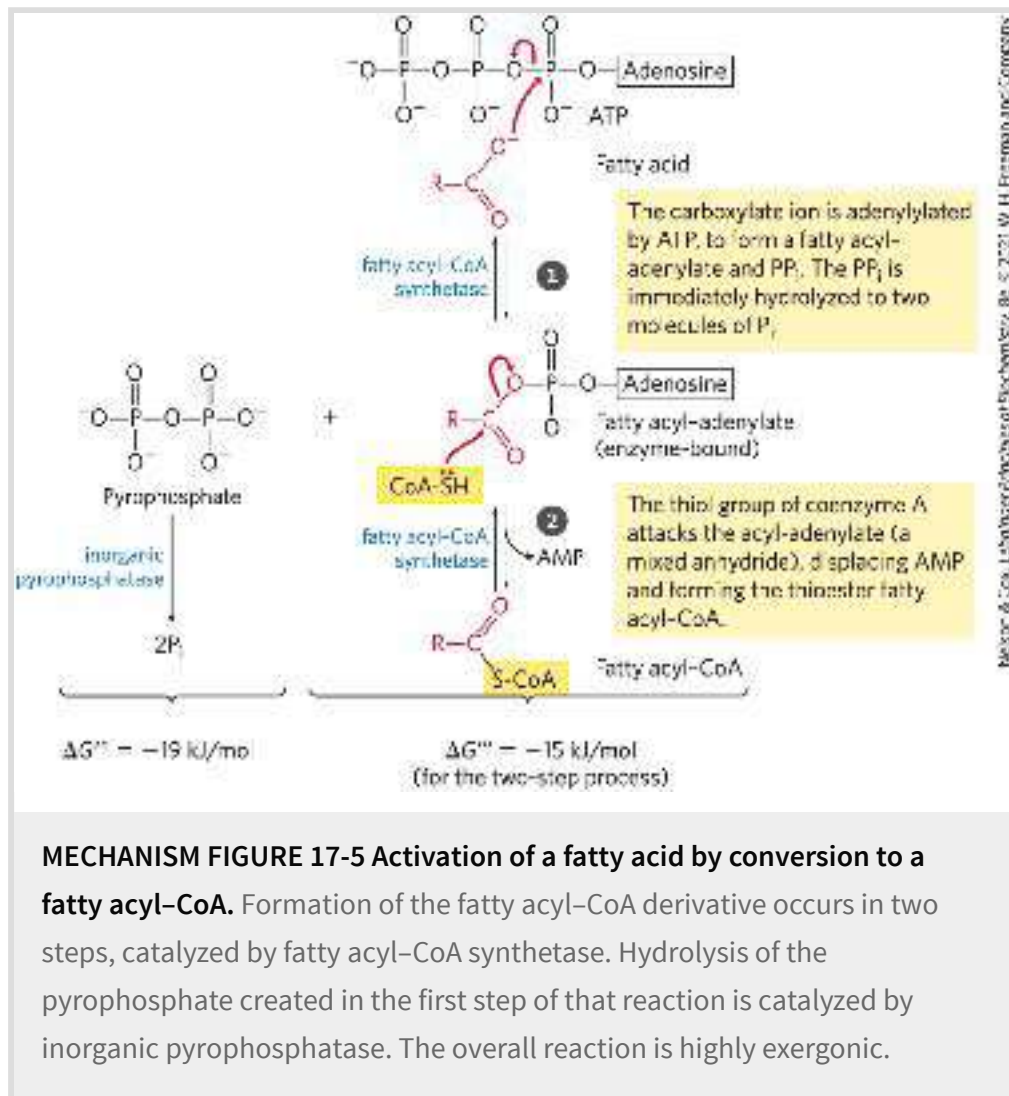
Fatty Acids Are Activated and Transported into Mitochondria

The enzymes of fatty acid oxidation in animal cells are located in the mitochondrial matrix, as demonstrated in 1948 by Eugene P. Kennedy and Albert Lehninger. Short- and medium-chain fatty acids, those with chain lengths of 12 or fewer carbons, enter mitochondria without the help of membrane transporters. Long-chain fatty acids, those with 14 or more carbons, which constitute the majority of the FFAs obtained in the diet or released from adipose tissue, cannot pass directly through the mitochondrial membranes: they must be transported through the [carnitine shuttle](#). First, the fatty acid must be activated by a **fatty acyl-CoA synthetase** isozyme specific for long-chain fatty acids. The isozymes are present in the outer mitochondrial membrane, where they promote the general reaction



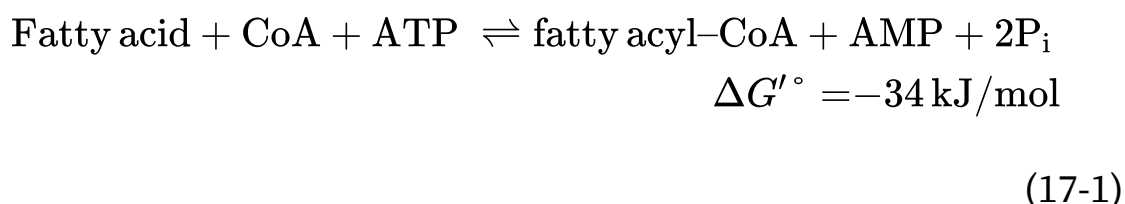
 Fatty acyl-CoA synthetases catalyze the formation of a thioester linkage between the fatty acid carboxyl group and the thiol group of coenzyme A to yield a **fatty acyl-CoA**, coupled to the cleavage of ATP to AMP and PP_i. (Recall the description of this reaction in [Chapter 13](#), which illustrates how the free energy released by cleavage of phosphoanhydride bonds in ATP could be coupled to the formation of a high-energy compound; [p. 485](#).) The

reaction catalyzed by a fatty acyl-CoA synthetase occurs in two steps and involves a fatty acyl-adenylate intermediate ([Fig. 17-5](#)).



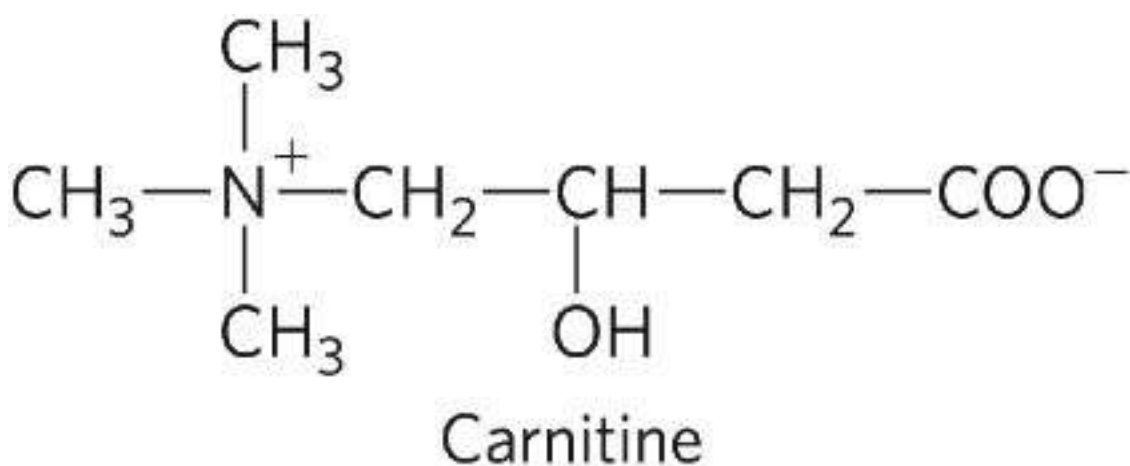
Fatty acyl-CoAs, like acetyl-CoA, are high-energy compounds; their hydrolysis to FFAs and CoA has a large, negative standard free-energy change. The formation of a fatty acyl-CoA is made more favorable by the hydrolysis of *two* high-energy bonds in ATP; the pyrophosphate formed in the activation reaction is immediately hydrolyzed by inorganic pyrophosphatase (left side of [Fig. 17-5](#)), which pulls the preceding activation reaction in the

direction of fatty acyl-CoA formation. The overall reaction (three steps) is



Fatty acyl-CoA esters formed on the cytosolic side of the outer mitochondrial membrane can be transported into the mitochondrion and oxidized to produce ATP, or they can be used in the cytosol to synthesize membrane lipids.

Fatty acyl-CoAs destined for mitochondrial oxidation must be attached to **carnitine** to be shuttled across the inner mitochondrial membrane.



In a transesterification catalyzed by **carnitine acyltransferase 1**, **CAT1** (also called **carnitine palmitoyltransferase 1**, **CPT1**) in the outer mitochondrial membrane, the fatty acyl-CoA is transiently

attached to the hydroxyl group of carnitine to form fatty acyl–carnitine ([Fig. 17-6](#)). The fatty acyl–carnitine ester then diffuses across the intermembrane space and enters the matrix by passive transport through the **acyl-carnitine/carnitine cotransporter** of the inner mitochondrial membrane. This cotransporter moves one molecule of carnitine from the matrix to the intermembrane space as one molecule of fatty acyl–carnitine moves into the matrix. Once inside the matrix, the fatty acyl group is transferred from carnitine back to coenzyme A from the intramitochondrial pool by **carnitine acyltransferase 2 (CAT2, or CPT2)**. This isozyme, located on the inner face of the inner mitochondrial membrane, regenerates fatty acyl–CoA and releases it, along with free carnitine, into the matrix. The carnitine is then available to be transferred back through the acyl-carnitine/carnitine cotransporter to be used to shuttle the next fatty acid across. Once inside the mitochondrion, the fatty acyl–CoA is acted upon by a set of enzymes in the matrix.

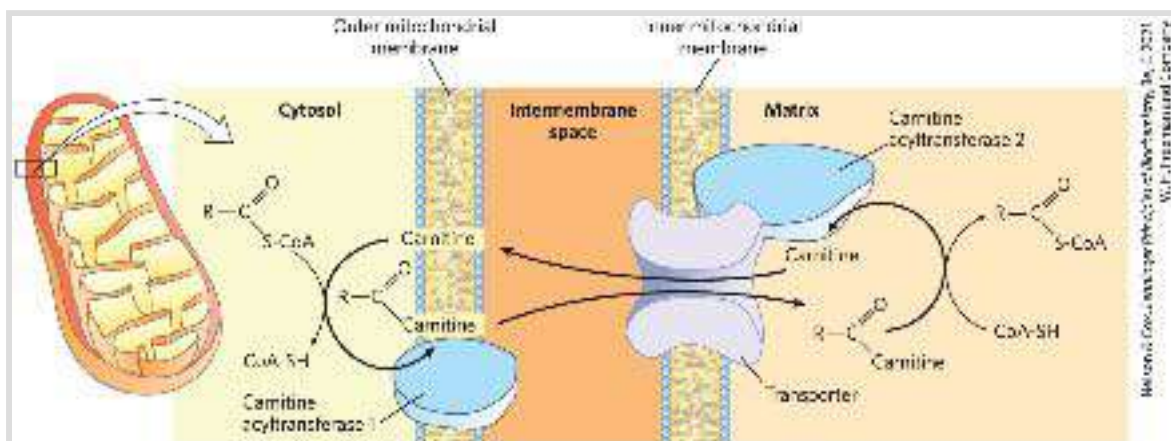


FIGURE 17-6 Fatty acid entry into mitochondria via the acyl-carnitine/carnitine transporter. Fatty acyl–carnitine formed on the outer mitochondrial membrane moves into the matrix by passive cotransport through the inner membrane. In the matrix, the

acyl group is transferred to mitochondrial coenzyme A, freeing carnitine to leave the matrix through the same transporter.

This process for transferring fatty acids into the mitochondrion — esterification to CoA, transesterification to carnitine, translocation across the inner membrane, and transesterification back to CoA — links two separate pools of coenzyme A and of fatty acyl-CoA, one in the cytosol, the other in mitochondria. These pools have different functions. Coenzyme A in the mitochondrial matrix is largely used in oxidative degradation of pyruvate, fatty acids, and some amino acids, whereas cytosolic coenzyme A is used in the biosynthesis of fatty acids (see [Fig. 21-10](#)). Fatty acyl-CoA in the cytosolic pool can be used there for membrane lipid synthesis or can be moved into the mitochondrial matrix for oxidation and ATP production. Conversion to the carnitine ester commits the fatty acyl moiety to the oxidative fate.  The carnitine-mediated entry process is the rate-limiting step for oxidation of fatty acids in mitochondria and, as discussed later, is a control point. Carnitine acyltransferase 1 is inhibited by malonyl-CoA, the first intermediate in fatty acid synthesis (see [Fig. 21-1](#)). This inhibition prevents the simultaneous synthesis and degradation of fatty acids, a wasteful futile cycle.

SUMMARY 17.1 *Digestion, Mobilization, and Transport of Fats*

■ Dietary triacylglycerols are emulsified in the small intestine by bile salts, hydrolyzed by intestinal lipases, absorbed by intestinal epithelial cells, and reconverted into triacylglycerols. They are combined with specific apolipoproteins for passage through the blood and lymph to adipose tissue, where they are stored as lipid droplets.

■ Triacylglycerols stored in adipose tissue are mobilized by a hormone-sensitive triacylglycerol lipase. The released fatty acids bind to serum albumin and are carried in the blood to the heart, skeletal muscle, and other tissues that use fatty acids for fuel.

■ Once inside cells, fatty acids are activated at the outer mitochondrial membrane by conversion to fatty acyl-CoA thioesters. Fatty acyl-CoA that is to be oxidized enters mitochondria via the carnitine shuttle, which is a major control point. Malonyl-CoA, the first intermediate in fatty acid synthesis, inhibits carnitine acyltransferase 1, assuring that fatty acid oxidation and fatty acid synthesis do not occur simultaneously.

17.2 Oxidation of Fatty Acids

As noted earlier, mitochondrial oxidation of fatty acids takes place in three stages ([Fig. 17-7](#)). In the first stage — β oxidation — fatty acids undergo oxidative removal of successive two-carbon units in the form of acetyl-CoA, starting from the carboxyl end of the fatty acyl chain. For example, the 16-carbon palmitic acid (palmitate at pH 7) undergoes seven passes through the oxidative sequence, in each pass losing two carbons as acetyl-CoA. At the end of seven cycles, the last two carbons of palmitate (originally C-15 and C-16) remain as acetyl-CoA. The overall result is the conversion of the 16-carbon chain of palmitate to eight two-carbon acetyl groups of acetyl-CoA molecules. Formation of each acetyl-CoA requires removal of four hydrogen atoms (two pairs of electrons and four H^+) from the fatty acyl moiety by dehydrogenases.

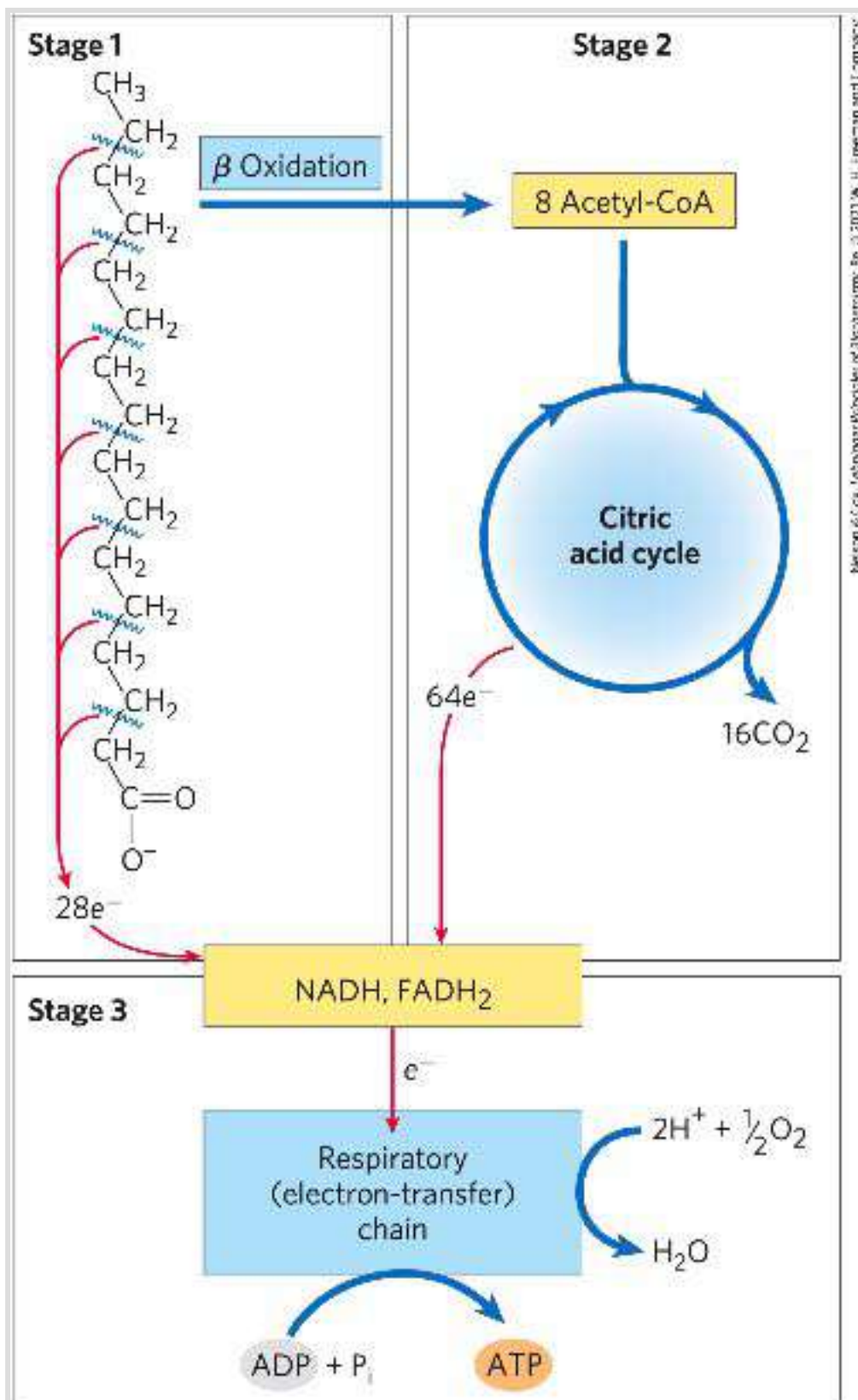


FIGURE 17-7 Stages of fatty acid oxidation. Stage 1: A long-chain fatty acid is oxidized to yield acetyl residues in the form of acetyl-CoA. This process is called β oxidation. Stage 2: The acetyl groups are oxidized to CO₂ via the

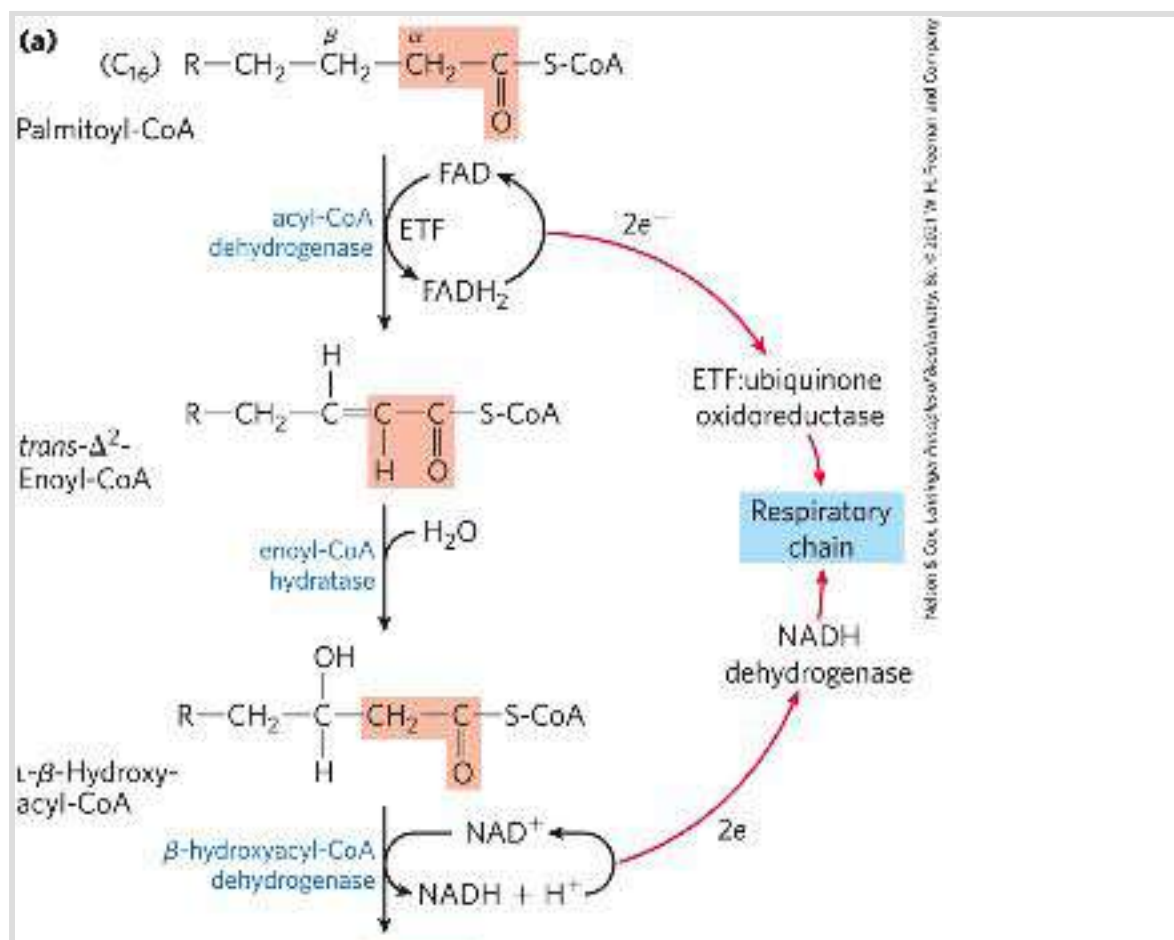
citric acid cycle. Stage 3: Electrons derived from the oxidations of stages 1 and 2 pass to O_2 via the mitochondrial respiratory chain, providing the energy for ATP synthesis by oxidative phosphorylation.

In the second stage of fatty acid oxidation, the acetyl groups of acetyl-CoA are oxidized to CO_2 in the citric acid cycle, which also takes place in the mitochondrial matrix. Acetyl-CoA derived from fatty acids thus enters a final common pathway of oxidation with the acetyl-CoA derived from glucose via glycolysis and pyruvate oxidation (see [Fig. 16-1](#)). The first two stages of fatty acid oxidation produce the reduced electron carriers NADH and $FADH_2$, which in the third stage donate electrons to the mitochondrial respiratory chain, through which the electrons pass to oxygen with the concomitant phosphorylation of ADP to ATP ([Fig. 17-7](#)). The energy released by fatty acid oxidation is thus conserved as ATP.

We now take a closer look at the first stage of fatty acid oxidation, beginning with the simple case of a saturated fatty acyl chain with an even number of carbons, then turning to the slightly more complicated cases of unsaturated and odd-number chains. We also consider the regulation of fatty acid oxidation, the β -oxidative processes as they occur in organelles other than mitochondria, and, finally, a less-common mode of fatty acid catabolism — α oxidation.

The β Oxidation of Saturated Fatty Acids Has Four Basic Steps

Four enzyme-catalyzed reactions make up the first stage of fatty acid oxidation ([Fig. 17-8a](#)). First, dehydrogenation of fatty acyl-CoA produces a double bond between the α and β carbon atoms (C-2 and C-3), yielding a ***trans*- Δ^2 -enoyl-CoA** (the symbol Δ^2 designates the position of the double bond; you may want to review fatty acid nomenclature, [p. 342](#)). Note that the new double bond has the *trans* configuration, whereas the double bonds in naturally occurring unsaturated fatty acids are normally in the *cis* configuration. We consider the significance of this difference later.



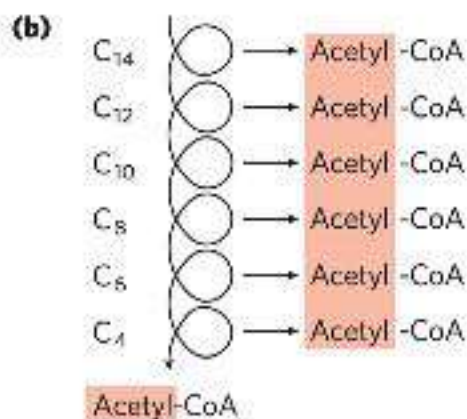
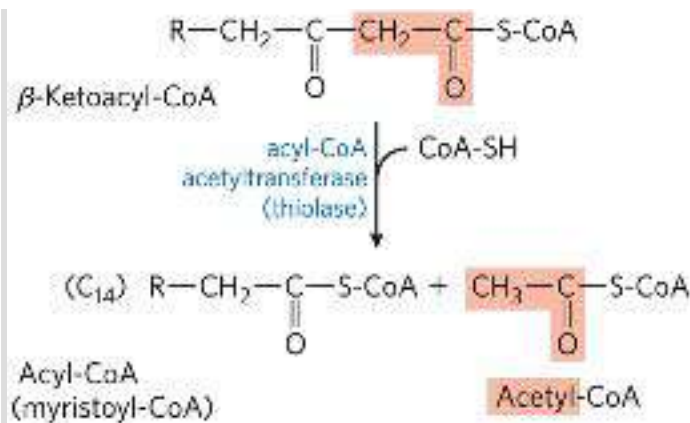
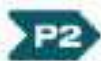


FIGURE 17-8 The β -oxidation pathway. (a) In each pass through this four-step sequence, one acetyl residue (shaded in light red) is removed in the form of acetyl-CoA from the carboxyl end of the fatty acyl chain — in this example palmitate (C_{16}), which enters as palmitoyl-CoA. Electrons from the first oxidation pass through electron transfer flavoprotein (ETF), and then through a second flavoprotein (ETF:ubiquinone oxidoreductase), into the respiratory chain. Electrons from the second oxidation enter the respiratory chain through NADH dehydrogenase. (b) Six more passes through the β -oxidation pathway yield seven more molecules of acetyl-CoA, the seventh arising from the last two carbon atoms of the 16-carbon chain. Eight molecules of acetyl-CoA are formed in all. The acetyl-CoA may be oxidized in the citric acid cycle, donating more electrons to the respiratory chain.

This first step is catalyzed by three isozymes of **acyl-CoA dehydrogenase**, each specific for a range of fatty-acyl chain lengths: very-long-chain acyl-CoA dehydrogenase (VLCAD), acting on fatty acids of 12 to 18 carbons; medium-chain (MCAD), acting

on fatty acids of 4 to 14 carbons; and short-chain (SCAD), acting on fatty acids of 4 to 8 carbons. VLCAD is in the inner mitochondrial membrane; MCAD and SCAD are in the matrix. All three isozymes are flavoproteins with tightly bound FAD (see [Fig. 13-27](#)) as a prosthetic group. The electrons removed from the fatty acyl-CoA are transferred to FAD, and the reduced form of the dehydrogenase immediately donates its electrons to an electron carrier, the **electron transfer flavoprotein (ETF)** (see [Fig. 19-15](#)). Electrons move from ETF to a second flavoprotein, **ETF:ubiquinone oxidoreductase**, and through ubiquinone into the mitochondrial respiratory chain.  The oxidation catalyzed by an acyl-CoA dehydrogenase is analogous to succinate dehydrogenation in the citric acid cycle ([p. 586](#)); in both reactions the enzyme is bound to the inner membrane, a double bond is introduced into a carboxylic acid between the α and β carbons, FAD is the electron acceptor, and electrons from the reaction ultimately enter the respiratory chain and pass to O_2 , with the concomitant synthesis of about 1.5 ATP molecules per electron pair.

In the second step of the β -oxidation cycle ([Fig. 17-8a](#)), water is added to the double bond of the *trans*- Δ^2 -enoyl-CoA to form the L stereoisomer of **β -hydroxyacyl-CoA (3-hydroxyacyl-CoA)**. This reaction, catalyzed by **enoyl-CoA hydratase**, is formally analogous to the fumarase reaction in the citric acid cycle, in which H_2O adds across an α - β double bond ([p. 587](#)).

In the third step, L- β -hydroxyacyl-CoA is dehydrogenated to form β -ketoacyl-CoA, by the action of β -hydroxyacyl-CoA **dehydrogenase**; NAD^+ is the electron acceptor. This enzyme is absolutely specific for the L stereoisomer of hydroxyacyl-CoA. The NADH formed in the reaction donates its electrons to [NADH dehydrogenase \(Complex I\)](#), an electron carrier of the respiratory chain (see [Fig. 19-15](#)), and ATP is formed from ADP as the electrons pass to O_2 .  The reaction catalyzed by β -hydroxyacyl-CoA dehydrogenase is closely analogous to the malate dehydrogenase reaction of the citric acid cycle ([p. 587](#)).

The fourth and last step of the β -oxidation cycle is catalyzed by **acyl-CoA acetyltransferase**, more commonly called **thiolase**, which promotes reaction of β -ketoacyl-CoA with a molecule of free coenzyme A to split off the carboxyl-terminal two-carbon fragment of the original fatty acid as acetyl-CoA. The other product is the coenzyme A thioester of the fatty acid, now shortened by two carbon atoms ([Fig. 17-8a](#)). This reaction is called thiolysis, by analogy with the process of hydrolysis, because the β -ketoacyl-CoA is cleaved by reaction with the thiol group of coenzyme A. The thiolase reaction is a reverse Claisen condensation (see [Fig. 13-4](#)).

The last three steps of this four-step sequence are catalyzed by either of two sets of enzymes, with the enzymes employed depending on the length of the fatty acyl chain. For fatty acyl chains of 12 or more carbons, the reactions are catalyzed by a multienzyme complex associated with the inner mitochondrial

membrane, the **trifunctional protein (TFP)**. TFP is a heterooctamer of $\alpha_4\beta_4$ subunits. Each α subunit contains two activities, the enoyl-CoA hydratase and the β -hydroxyacyl-CoA dehydrogenase; the β subunits contain the thiolase activity. This tight association of three enzymes allows efficient substrate channeling from one active site to the next, without diffusion of the intermediates away from the enzyme surface. When TFP has shortened the fatty acyl chain to 12 or fewer carbons, further oxidations are catalyzed by a set of four soluble enzymes in the matrix.

 As noted earlier, the single bond between methylene ($-\text{CH}_2-$) groups in fatty acids is relatively stable. The β -oxidation sequence is an elegant mechanism for destabilizing and breaking these bonds. The first three reactions of β oxidation create a much less stable C—C bond, in which the α carbon (C-2) is bonded to *two* carbonyl carbons (the β -ketoacyl-CoA intermediate). The ketone function on the β carbon (C-3) makes it a good target for nucleophilic attack by the $-\text{SH}$ of coenzyme A, catalyzed by thiolase. The acidity of the α hydrogen and the resonance stabilization of the carbanion generated by the departure of this hydrogen make the terminal $-\text{CH}_2-\text{CO}-\text{S-CoA}$ a good leaving group, facilitating breakage of the α - β bond.

We have already seen a reaction sequence nearly identical with these four steps of fatty acid oxidation, in the citric acid cycle reaction steps between succinate and oxaloacetate (see [Fig. 16-7](#)).

A nearly identical reaction sequence occurs again in the pathways by which the branched-chain amino acids (isoleucine, leucine, and valine) are oxidized as fuels (see [Fig. 18-28](#)). **Figure 17-9** shows the common features of these three sequences, almost certainly an example of the conservation of a mechanism by gene duplication and evolution of a new specificity in the enzyme products of the duplicated genes.

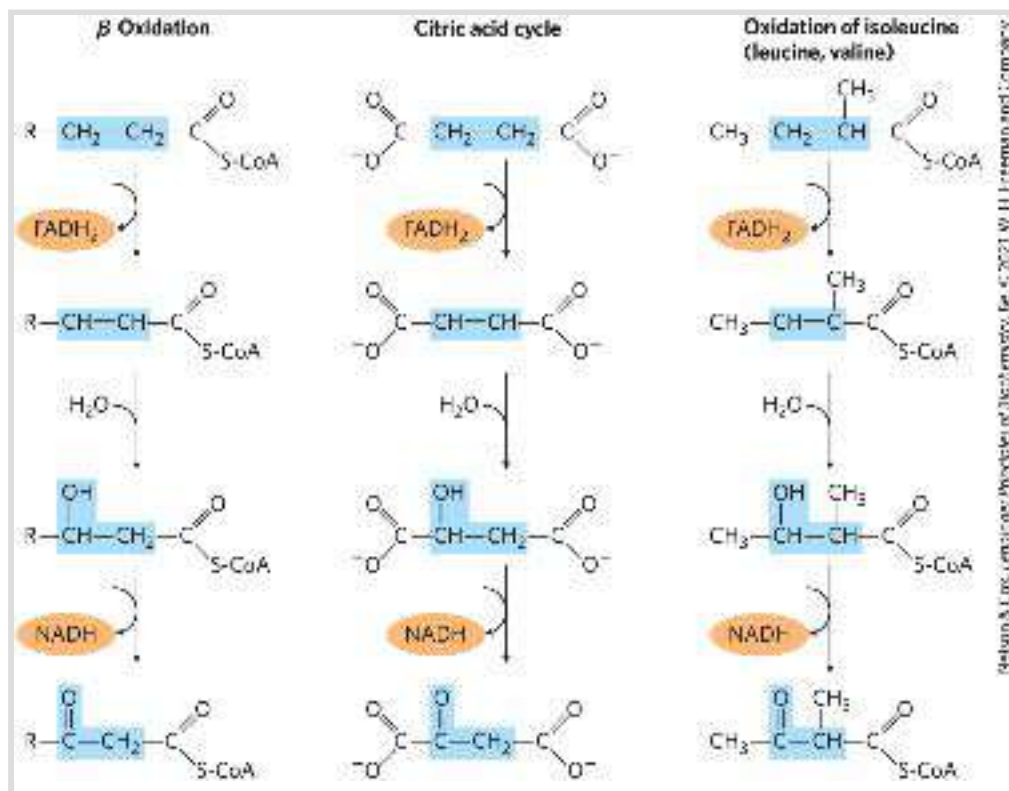
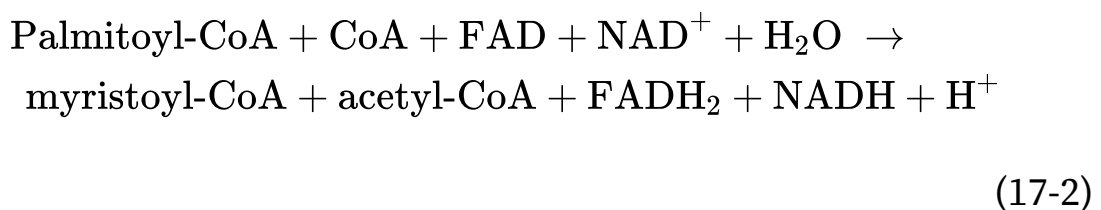


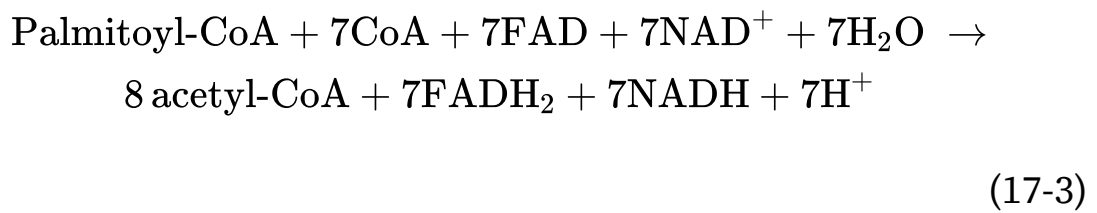
FIGURE 17-9 A conserved reaction sequence to introduce a carbonyl function on the carbon β to a carboxyl group. The β -oxidation pathway for fatty acyl-CoAs, the pathway from succinate to oxaloacetate in the citric acid cycle, and the pathway by which the deaminated carbon skeletons from isoleucine, leucine, and valine are oxidized as fuels — all use the same reaction sequence.

The Four β -Oxidation Steps Are Repeated to Yield Acetyl-CoA and ATP

In one pass through the β -oxidation sequence, one molecule of acetyl-CoA, two pairs of electrons, and four protons (H^+) are removed from the long-chain fatty acyl-CoA, shortening it by two carbon atoms. The equation for one pass, beginning with the coenzyme A ester of our example, palmitate, is



Following removal of one acetyl-CoA unit from palmitoyl-CoA, the coenzyme A thioester of the shortened fatty acid (now the 14-carbon myristate) remains. The myristoyl-CoA can now go through another set of four β -oxidation reactions, exactly analogous to the first, to yield a second molecule of acetyl-CoA and lauroyl-CoA, the coenzyme A thioester of the 12-carbon laurate. Altogether, seven passes through the β -oxidation sequence are required to oxidize one molecule of palmitoyl-CoA to eight molecules of acetyl-CoA ([Fig. 17-8b](#)). The overall equation is



 Each molecule of FADH₂ formed during oxidation of the fatty acid donates a pair of electrons to ETF of the respiratory chain, and about 1.5 molecules of ATP are generated during the ensuing transfer of each electron pair to O₂. Similarly, each molecule of NADH formed delivers a pair of electrons to the mitochondrial NADH dehydrogenase, and the subsequent transfer of each pair of electrons to O₂ results in formation of about 2.5 molecules of ATP. Thus four molecules of ATP are formed for each two-carbon unit removed in one pass through the sequence.

Note that water is also produced in this process. Each pair of electrons transferred from NADH or FADH₂ to O₂ yields one H₂O, referred to as “metabolic water.” Reduction of O₂ by NADH also consumes one H⁺ per NADH molecule:

$\text{NADH} + \text{H}^+ + \frac{1}{2}\text{O}_2 \rightarrow \text{NAD}^+ + \text{H}_2\text{O}$. In hibernating animals, fatty acid oxidation provides metabolic energy, heat, and water — all essential for survival of an animal that neither eats nor drinks for long periods ([Box 17-1](#)). Camels obtain water to supplement the meager supply available in their natural environment by oxidation of fats stored in their hump.

BOX 17-1

A Long Winter's Nap: Oxidizing Fats during Hibernation

Many animals depend on fat stores for energy during hibernation, during migratory periods, and in other situations involving radical metabolic adjustments. One of the most pronounced adjustments of fat metabolism occurs in hibernating grizzly bears. These animals remain in a continuous state of dormancy for as long as seven months. Unlike most hibernating species, the bear maintains a body temperature of about 31 °C, close to its normal (nonhibernating) temperature near 40 °C. Although expending about 25,000 kJ/day (6,000 kcal/day) while hibernating, the bear does not eat, drink, urinate, or defecate for months at a time. Its heart rate drops from 90 to 8 beats per minute, and its respiration rate drops from 6 to 10 breaths to approximately 1 breath per minute.

As we shall see in [Chapter 19](#), mitochondrial electron transfer can be uncoupled from ATP production so that all of the energy of fuel oxidation is dissipated as heat, to maintain a body temperature near normal in the face of much lower ambient temperatures. One form of fat tissue, **brown adipose tissue**, is especially important in thermogenesis; we discuss it in more detail in [Chapter 23](#).



Stockier Productions/Animals Animals

A grizzly bear prepares its hibernation nest near the McNeil River in Canada.

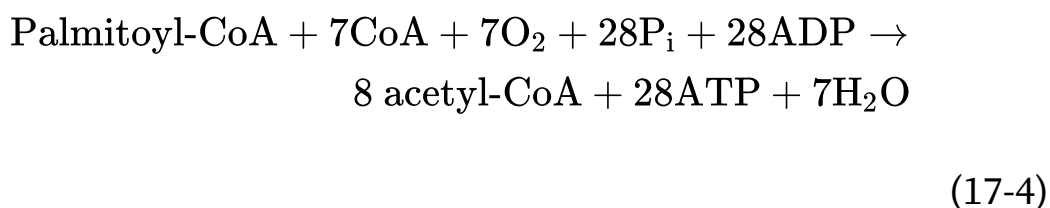
Experimental studies have shown that hibernating grizzly bears use body fat as their sole fuel. Fat oxidation yields sufficient energy to maintain body temperature, synthesize amino acids and proteins, and carry out other energy-requiring activities, such as membrane transport. Fat oxidation also releases large amounts of water, as described in the text, which replenishes water lost in breathing. The glycerol released by degradation of triacylglycerols is converted into blood glucose by gluconeogenesis. Urea formed during breakdown of amino acids is reabsorbed in the kidneys and recycled, with the amino groups reused to make new amino acids for maintaining body proteins.

Bears store an enormous amount of body fat in preparation for their long sleep. An adult grizzly consumes about 38,000 kJ/day during the late spring and summer, but as winter approaches it feeds for 20 hours a day, consuming up to 84,000 kJ daily. This increase in feeding is a response to a seasonal change in hormone secretion. Large amounts of triacylglycerols are formed from the huge intake of carbohydrates during the fattening-up period. The bear will emerge from hibernation having lost 15% to 40% of its maximum body weight.

The winter sleep of bears is sometimes called torpor, and it differs in important ways from the hibernation behavior of a group of smaller animals that undergo alternating periods of high and low body temperature. In these animals, body temperature approaches ambient temperature, close to 0 °C, for much of the time during hibernation, but rises to almost prehibernation level during brief periods of wakefulness. During these periods, the animals eat, drink, and defecate. In the Arctic ground squirrel (*Urocitellus parryii*), for example, body temperature (37 °C prehibernation) drops to 0 °C during hibernation, and respiration drops to less than 10% of its prehibernation rate.

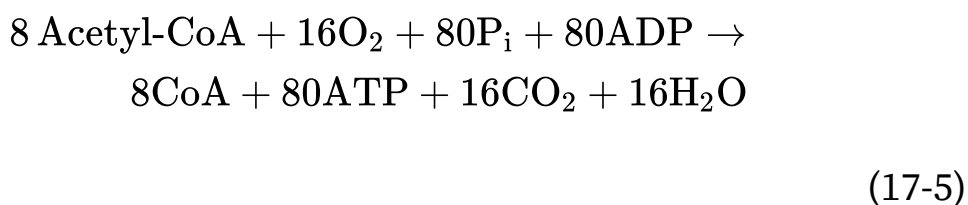
Studies of hibernation mechanisms may yield insight into several problems in human medicine; for example, slowing the metabolism of organs donated for transplantation might extend the period of their viability. And if humans are to make long trips into space, inducing a torporlike state might relieve the monotony of long missions and conserve onboard resources such as food and oxygen.

The overall equation for the oxidation of palmitoyl-CoA to eight molecules of acetyl-CoA, including the electron transfers and oxidative phosphorylations, is



Acetyl-CoA Can Be Further Oxidized in the Citric Acid Cycle

 The acetyl-CoA produced from the oxidation of fatty acids can be oxidized to CO₂ and H₂O by the citric acid cycle. The following equation represents the balance sheet for the second stage in the oxidation of palmitoyl-CoA, together with the coupled phosphorylations of the third stage:



Combining [Equations 17-4](#) and [17-5](#), we obtain the overall equation for the complete oxidation of palmitoyl-CoA to carbon dioxide and water:

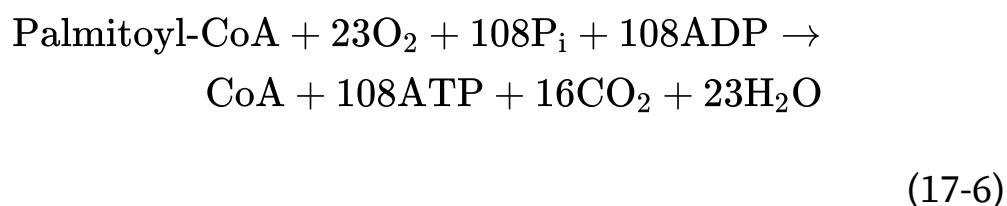


Table 17-1 summarizes the yields of NADH, FADH₂, and ATP in the successive steps of palmitoyl-CoA oxidation. Note that because the activation of palmitate to palmitoyl-CoA breaks both phosphoanhydride bonds in ATP ([Fig. 17-5](#)), the energetic cost of activating a fatty acid is equivalent to two ATP, and the net gain per molecule of palmitate is 106 ATP. The standard free-energy change for the oxidation of palmitate to CO₂ and H₂O is about 9,800 kJ/mol. Under standard conditions, the energy recovered as the phosphate bond energy of ATP is $106 \times 30.5 \text{ kJ/mol} = 3,230 \text{ kJ/mol}$, about 33% of the theoretical maximum. However, when the free-energy changes are calculated from actual concentrations of reactants and products under intracellular conditions, the free-energy recovery is more than 60%; the energy conservation is remarkably efficient (see [Worked Example 13-2, p. 480](#)).

TABLE 17.1 Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO₂ and H₂O

Enzyme catalyzing the oxidation step	Number of NADH or FADH ₂ formed	Number of ATP ultimately formed ^a
<i>β</i> Oxidation		
Acyl-CoA dehydrogenase	7 FADH ₂	10.5
<i>β</i> -Hydroxyacyl-CoA	7 NADH	17.5

dehydrogenase

Citric acid cycle

Isocitrate dehydrogenase	8 NADH	20
α -Ketoglutarate dehydrogenase	8 NADH	20
Succinyl-CoA synthetase		8 ^{b}
Succinate dehydrogenase	8 FADH ₂	12
Malate dehydrogenase	8 NADH	20
Total		108

^{[a](#)}These calculations assume that mitochondrial oxidative phosphorylation produces 1.5 ATP per FADH₂ oxidized and 2.5 ATP per NADH oxidized.

^{[b](#)}GTP produced directly in this step yields ATP in the reaction catalyzed by nucleoside diphosphate kinase ([p. 487](#)).

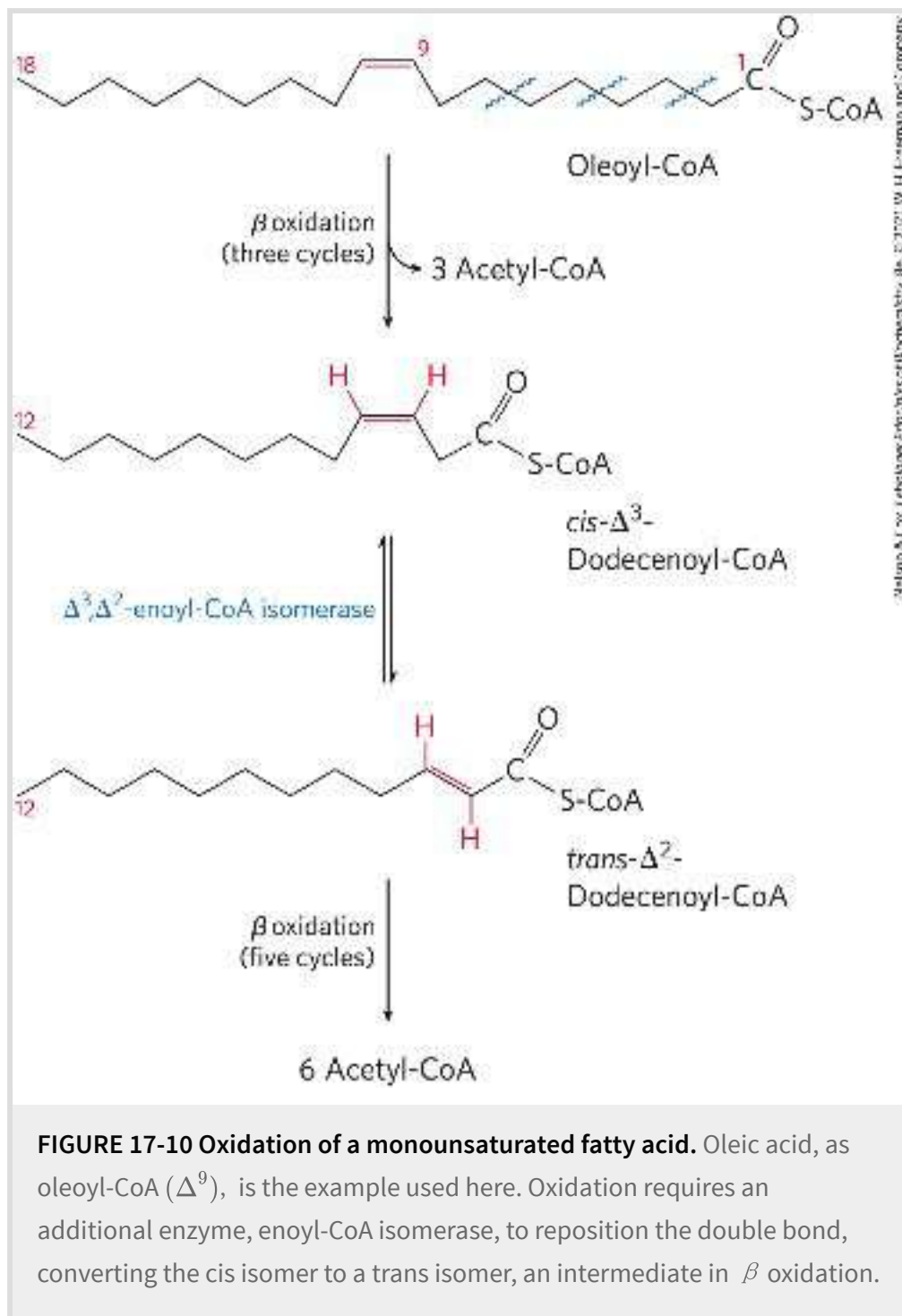
Oxidation of Unsaturated Fatty Acids Requires Two Additional Reactions

The fatty acid oxidation sequence just described is typical when the incoming fatty acid is saturated (that is, has only single bonds in its carbon chain). However, most of the fatty acids in the triacylglycerols and phospholipids of animals and plants are unsaturated, having one or more double bonds. These bonds are in the *cis* configuration and cannot be acted upon by enoyl-CoA hydratase, the enzyme catalyzing the addition of H₂O to the *trans* double bond of the Δ^2 -enoyl-CoA generated during β oxidation.



Two auxiliary enzymes — an isomerase and a reductase — must act on the common unsaturated fatty acids to transform them into substrates for the β -oxidation pathway. We illustrate these auxiliary reactions with two examples.

Oleate is an abundant 18-carbon monounsaturated fatty acid with a *cis* double bond between C-9 and C-10 (denoted Δ^9). In the first step of oxidation, oleate is converted to oleoyl-CoA and, like the saturated fatty acids, enters the mitochondrial matrix via the carnitine shuttle ([Fig. 17-6](#)). Oleoyl-CoA then undergoes three passes through the fatty acid oxidation cycle to yield three molecules of acetyl-CoA and the coenzyme A ester of a Δ^3 , 12-carbon unsaturated fatty acid, *cis*- Δ^3 -dodecenoyl-CoA ([Fig. 17-10](#)). This product cannot serve as a substrate for enoyl-CoA hydratase, which acts only on *trans* double bonds. The auxiliary enzyme **Δ^3, Δ^2 -enoyl-CoA isomerase** isomerizes the *cis*- Δ^3 -enoyl-CoA to the *trans*- Δ^2 -enoyl-CoA, which is converted by enoyl-CoA hydratase into the corresponding L- β hydroxyacyl-CoA (*trans*- Δ^2 -dodecenoyl-CoA). This intermediate is now acted upon by the remaining enzymes of β oxidation to yield acetyl-CoA and the coenzyme A ester of a 10-carbon saturated fatty acid, decanoyl-CoA. The latter undergoes four more passes through the β -oxidation pathway to yield five more molecules of acetyl-CoA. Altogether, nine acetyl-CoAs are produced from one molecule of the 18-carbon oleate.



The other auxiliary enzyme (a reductase) is required for oxidation of polyunsaturated fatty acids — for example, the 18-carbon linoleate, which has a *cis*- Δ^9 , *cis*- Δ^{12} configuration ([Fig. 17-11](#)). Linoleoyl-CoA undergoes three passes through the β -oxidation

sequence to yield three molecules of acetyl-CoA and the coenzyme A ester of a 12-carbon unsaturated fatty acid with a *cis*- Δ^3 , *cis*- Δ^6 configuration. This intermediate cannot be used by the enzymes of the β -oxidation pathway: its double bonds are in the wrong position and have the wrong configuration (cis, not trans). However, the combined action of enoyl-CoA isomerase and **2,4-dienoyl-CoA reductase**, as shown in [Figure 17-11](#), transforms this intermediate into one that can enter the β -oxidation pathway and be degraded to six acetyl-CoAs. The overall result is conversion of linoleate to nine molecules of acetyl-CoA.

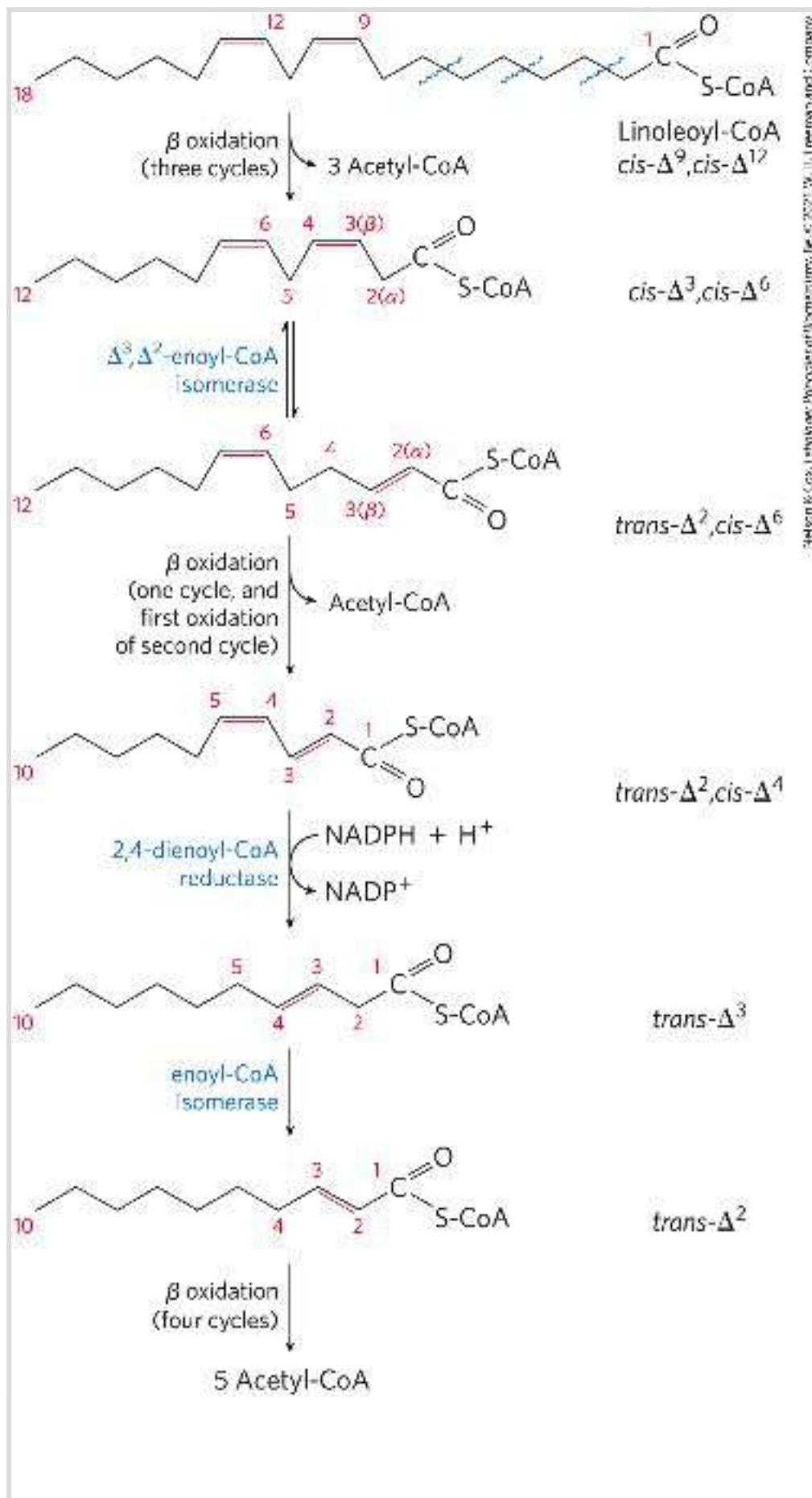


FIGURE 17-11 Oxidation of a polyunsaturated fatty acid. The example here is linoleic acid, as linoleoyl-CoA ($\Delta^{9,12}$). Oxidation requires a second auxiliary enzyme in addition to enoyl-CoA isomerase: NADPH-dependent 2,4-dienoyl-CoA reductase. The combined action of these two enzymes converts a *trans*- Δ^2 , *cis*- Δ^4 -dienoyl-CoA intermediate to the *trans*- Δ^2 -enoyl-CoA substrate necessary for β oxidation.

Complete Oxidation of Odd-Number Fatty Acids Requires Three Extra Reactions

Although most naturally occurring lipids contain fatty acids with an even number of carbon atoms, fatty acids with an odd number of carbons are common in the lipids of many plants and some marine organisms. Cattle and other ruminant animals form large amounts of the three-carbon **propionate** ($\text{CH}_3\text{—CH}_2\text{—COO}^-$) during fermentation of carbohydrates in the rumen. The propionate is absorbed into the blood and oxidized by the liver and other tissues.

Long-chain odd-number fatty acids are oxidized in the same pathway as the even-number acids, beginning at the carboxyl end of the chain. However, the substrate for the last pass through the β -oxidation sequence is a fatty acyl-CoA with a five-carbon fatty acid. When this is oxidized and cleaved, the products are acetyl-CoA and **propionyl-CoA**. The acetyl-CoA can be oxidized in the citric acid cycle, of course, but propionyl-CoA enters a different pathway, having three enzymes.

Propionyl-CoA is first carboxylated to form the D stereoisomer of **methyalmalonyl-CoA** by **propionyl-CoA carboxylase**, which contains the cofactor biotin ([Fig. 17-12](#)). In this enzymatic reaction, as in the pyruvate carboxylase reaction (see [Fig. 16-16](#)), bicarbonate ion (HCO_3^-) is activated by attachment to biotin before its transfer to the substrate, in this case the propionate moiety. Formation of the carboxybiotin intermediate requires energy, which is provided by ATP. The D-methyalmalonyl-CoA thus formed is enzymatically epimerized to its L stereoisomer by **methyalmalonyl-CoA epimerase**. The L-methyalmalonyl-CoA then undergoes an intramolecular rearrangement to form succinyl-CoA, which can enter the citric acid cycle. This rearrangement is catalyzed by **methyalmalonyl-CoA mutase**, which requires as its coenzyme 5'-deoxyadenosylcobalamin, or **coenzyme B₁₂**, which is derived from vitamin B₁₂ (cobalamin). [Box 17-2](#) describes the role of coenzyme B₁₂ in this remarkable exchange reaction.

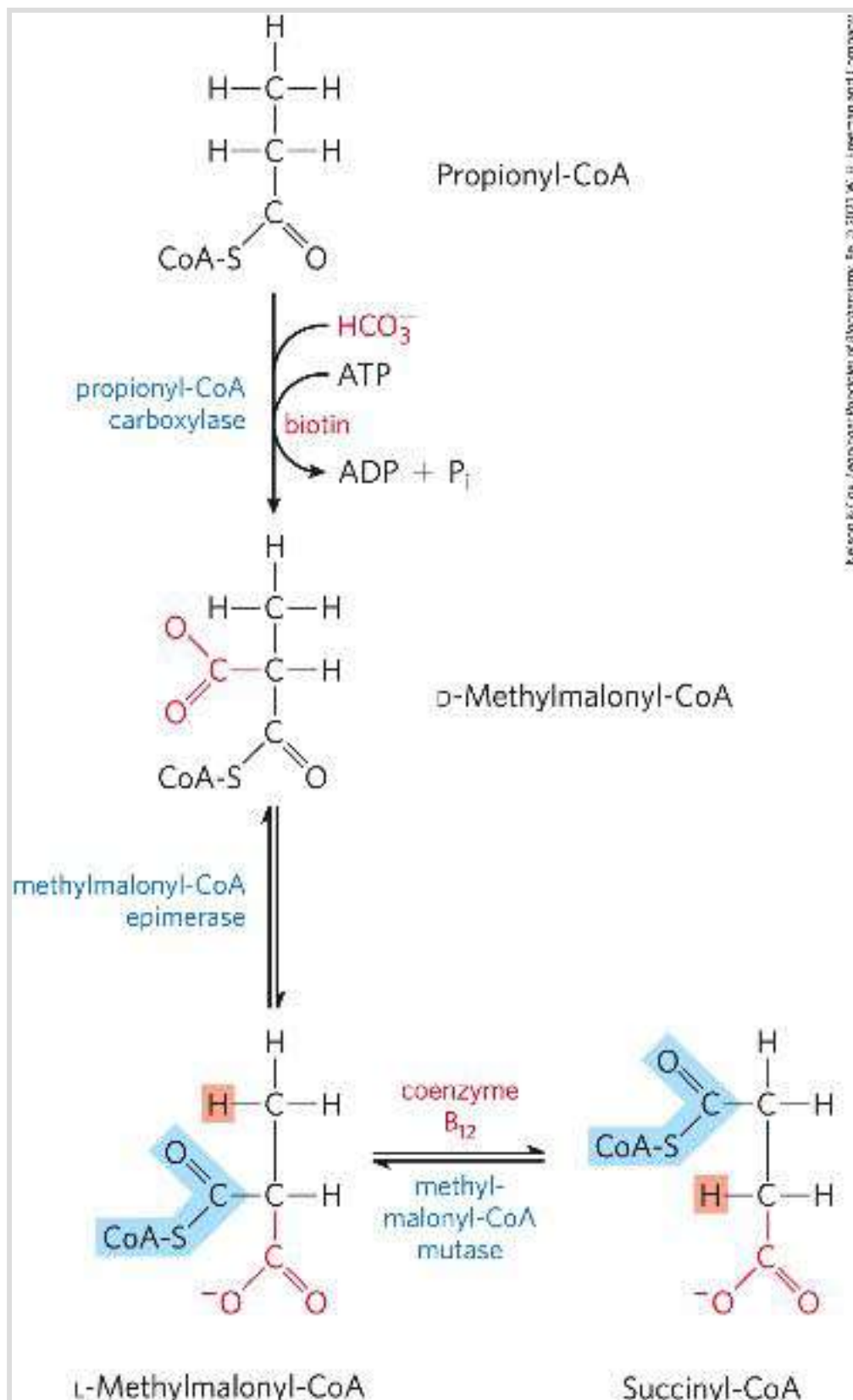


FIGURE 17-12 Oxidation of propionyl-CoA produced by β oxidation of odd-number fatty acids. The sequence involves the carboxylation of propionyl-CoA to D-methylmalonyl-CoA and conversion of the latter to

succinyl-CoA. This conversion requires epimerization of D- to L-methylmalonyl-CoA, followed by a remarkable reaction in which substituents on adjacent carbon atoms exchange positions (see [Box 17-2](#)).

BOX 17-2

Coenzyme B₁₂: A Radical Solution to a Perplexing Problem

In the methylmalonyl-CoA mutase reaction (see [Fig. 17-12](#)), the group —CO—S-CoA at C-2 of the original propionate exchanges position with a hydrogen atom at C-3 of the original propionate ([Fig. 1a](#)). Coenzyme B₁₂ is the cofactor for this general type of reaction ([Fig. 1b](#)). These coenzyme B₁₂-dependent processes are among the very few enzymatic reactions in biology in which there is an exchange of an alkyl or substituted alkyl group (X) with a hydrogen atom on an adjacent carbon, *with no mixing of the transferred hydrogen atom with the hydrogen of the solvent, H₂O*. How can the hydrogen atom move between two carbons without mixing with the enormous excess of hydrogen atoms in the solvent?

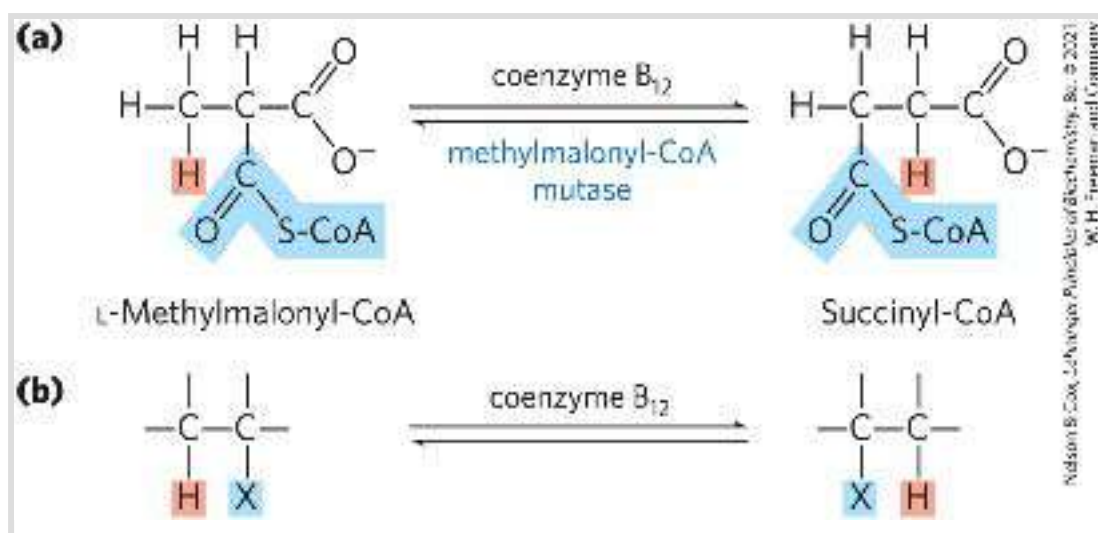


FIGURE 1

Coenzyme B₁₂ is the cofactor form of vitamin B₁₂, which is unique among all the vitamins in that it contains an essential trace element, cobalt. The complex **corrin ring system** of vitamin B₁₂ (colored blue in [Fig. 2](#)), to which cobalt (as Co³⁺) is coordinated, is chemically related to the porphyrin ring system of heme (see [Fig. 5-1](#)). A fifth coordination position of cobalt is filled by dimethylbenzimidazole ribonucleotide (shaded yellow), bound covalently by its 3'-phosphate group to a side chain of the corrin ring, through aminoisopropanol. The formation of this complex cofactor occurs in one of only two known reactions in which triphosphate is cleaved from ATP ([Fig. 3](#)); the other reaction is the formation of S-adenosylmethionine from ATP and methionine (see [Fig. 18-18](#)).

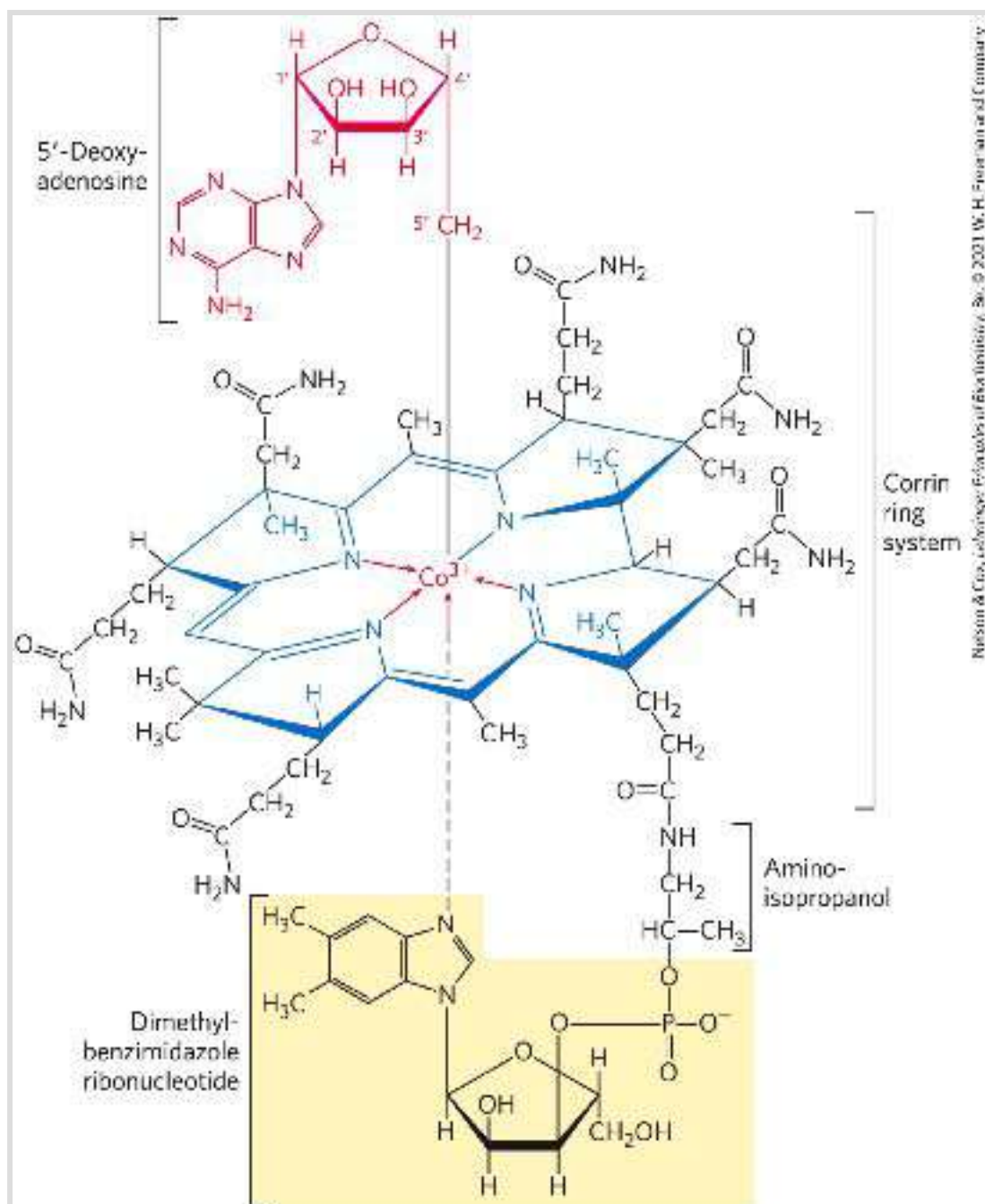
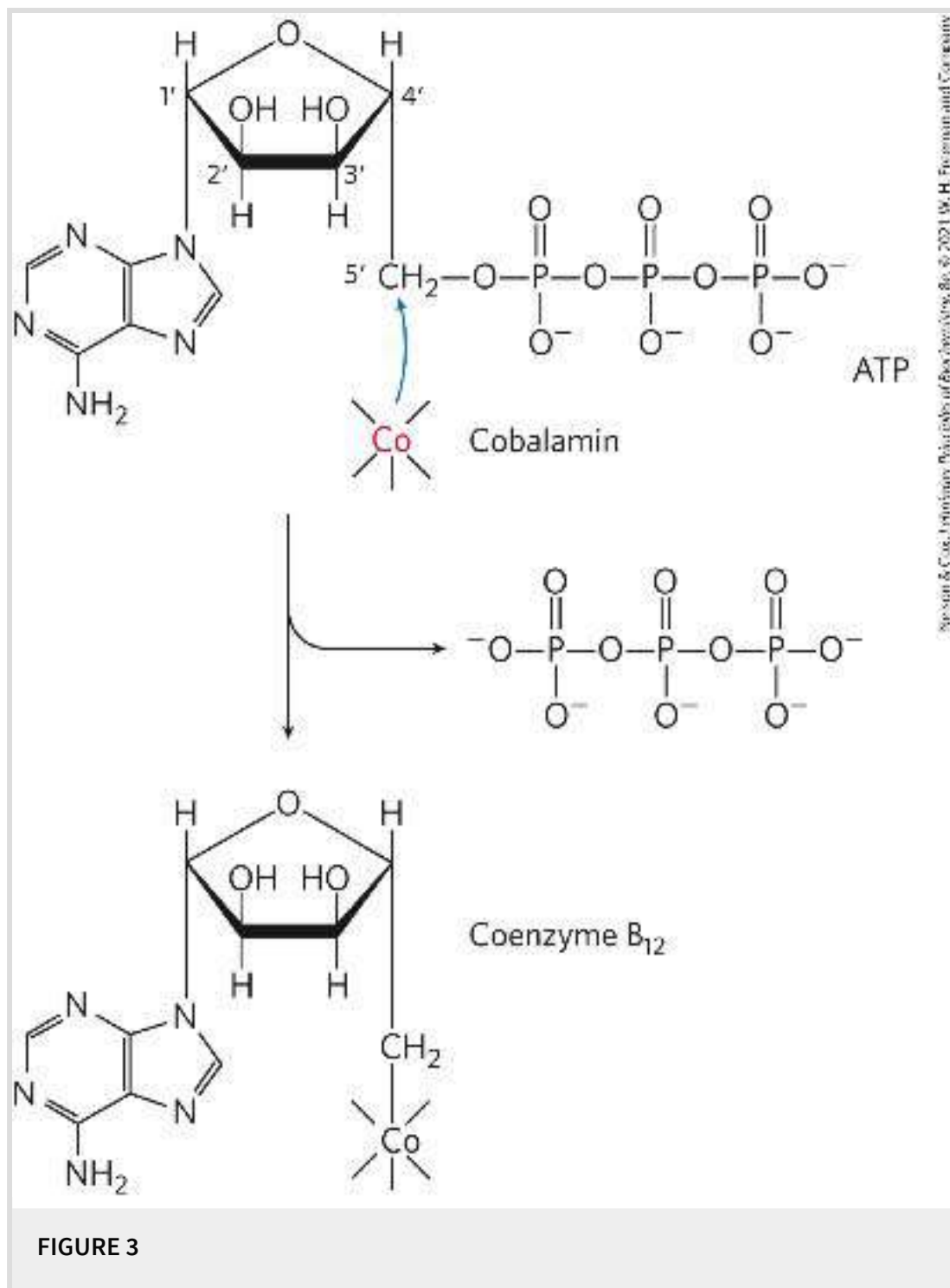


FIGURE 2



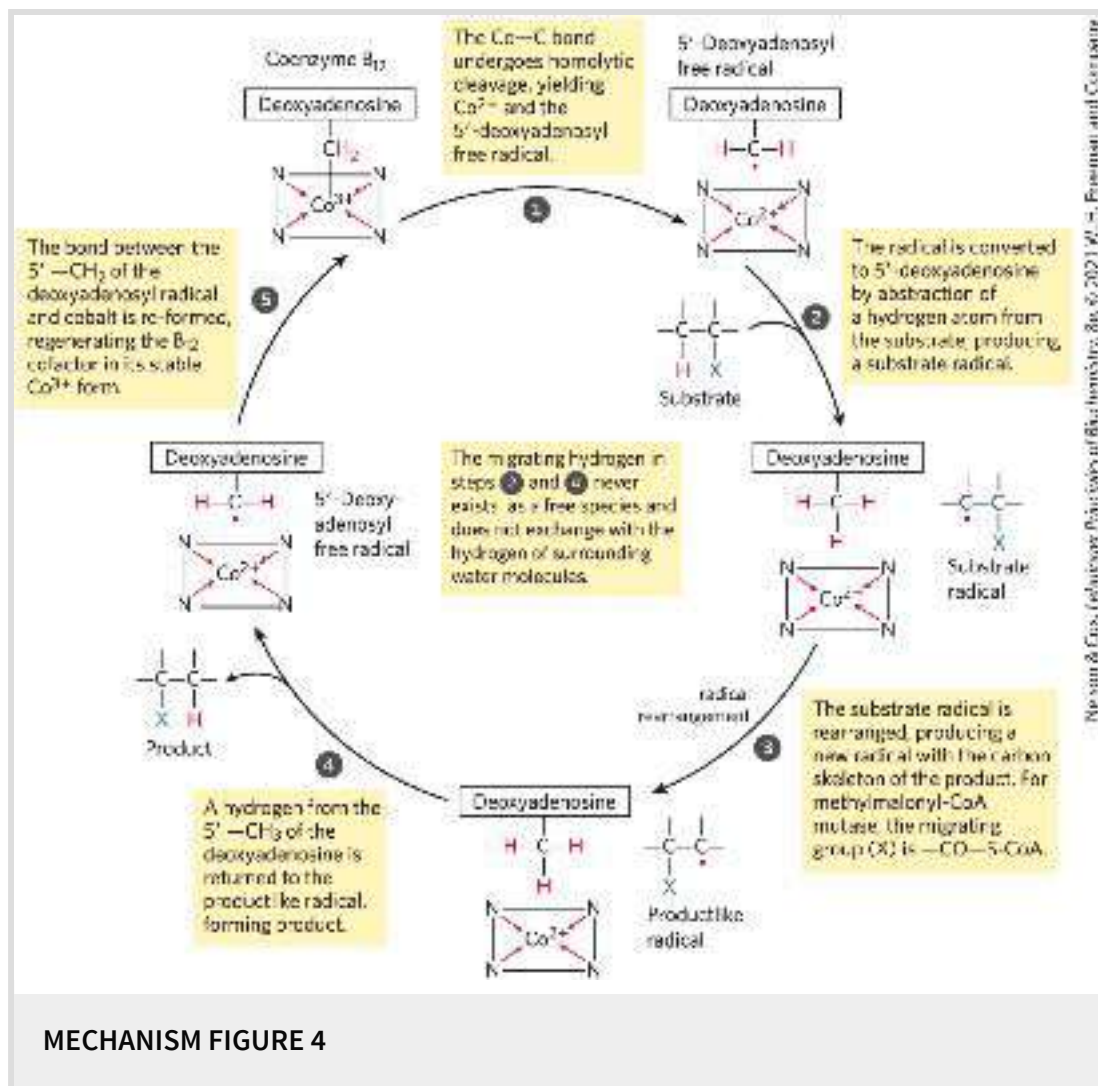
Vitamin B₁₂ is usually isolated as **cyanocobalamin**, because it contains a cyano group (picked up during purification) attached to cobalt in the sixth coordination position. In **5'-deoxyadenosylcobalamin**, the cofactor for methylmalonyl-CoA mutase, the cyano group is replaced by the **5'-deoxyadenosyl** group (red in [Fig. 2](#)), covalently bound through C-5' to the

cobalt. The three-dimensional structure of the cofactor was determined by Dorothy Crowfoot Hodgkin in 1956, using x-ray crystallography.



Dorothy Crowfoot Hodgkin, 1910–1994

The key to understanding how coenzyme B₁₂ catalyzes hydrogen exchange lies in the properties of the covalent bond between cobalt and C-5' of the deoxyadenosyl group ([Fig. 2](#)). This is a relatively weak bond; merely illuminating the compound with visible light is enough to break this Co—C bond. (This extreme photolability probably accounts for the absence of vitamin B₁₂ in plants.) Dissociation produces a 5'-deoxyadenosyl radical and the Co²⁺ form of the vitamin. By generating free radicals in this way, 5'-deoxyadenosylcobalamin initiates a series of transformations such as that illustrated in [Figure 4](#) — a postulated mechanism for the reaction catalyzed by methylmalonyl-CoA mutase and several other coenzyme B₁₂-dependent transformations. In this postulated mechanism, the migrating hydrogen atom never exists as a free species and is thus never free to exchange with the hydrogen of surrounding water molecules.



Vitamin B₁₂ is not made by plants or animals and can be synthesized only by a few species of microorganisms. It is required by healthy people in only minute amounts, about 3 μg/day. Vitamin B₁₂ deficiency results in **pernicious anemia**, caused by a failure to absorb vitamin B₁₂ efficiently from the intestine, where it is synthesized by intestinal bacteria or obtained from digestion of meat. Individuals with this disease do not produce sufficient amounts of **intrinsic factor**, a glycoprotein essential to vitamin B₁₂ absorption. The pathology in pernicious anemia includes reduced production of erythrocytes, reduced levels of hemoglobin, and severe, progressive impairment of the central nervous system. Administration of large doses of vitamin B₁₂ alleviates these symptoms in at least some cases. ■



About 1 in 100,000 babies are born with a genetic deficiency in propionyl-CoA carboxylase, making them unable to metabolize the propionyl-CoA that results from catabolism of odd-number fatty acids, fatty acids with methyl branches, and the amino acids isoleucine, valine, threonine, and methionine.  The

absence of functional propionyl-CoA carboxylase leads to an accumulation of propionyl-CoA in mitochondria, depleting the available supply of coenzyme A for continuing β -oxidation and other metabolism. The propionyl-CoA is esterified to carnitine, transported out of the mitochondria through the carnitine shuttle, and eventually released to the blood as propionate, which severely acidifies the blood and urine, a condition called propionic acidemia. Symptoms generally manifest in the first few days of life, with vomiting, low blood glucose, seizures, and neurologic abnormalities. The condition is diagnosed by detection of propionate and its metabolites in the blood and urine. Treatments include severe restriction of dietary protein, supplying carnitine, and the use of antibiotics targeted at gut bacteria that produce odd-chain and branched-chain fatty acids. Some individuals who are defective in the incorporation of biotin into the enzyme benefit from treatment with high doses of biotin.

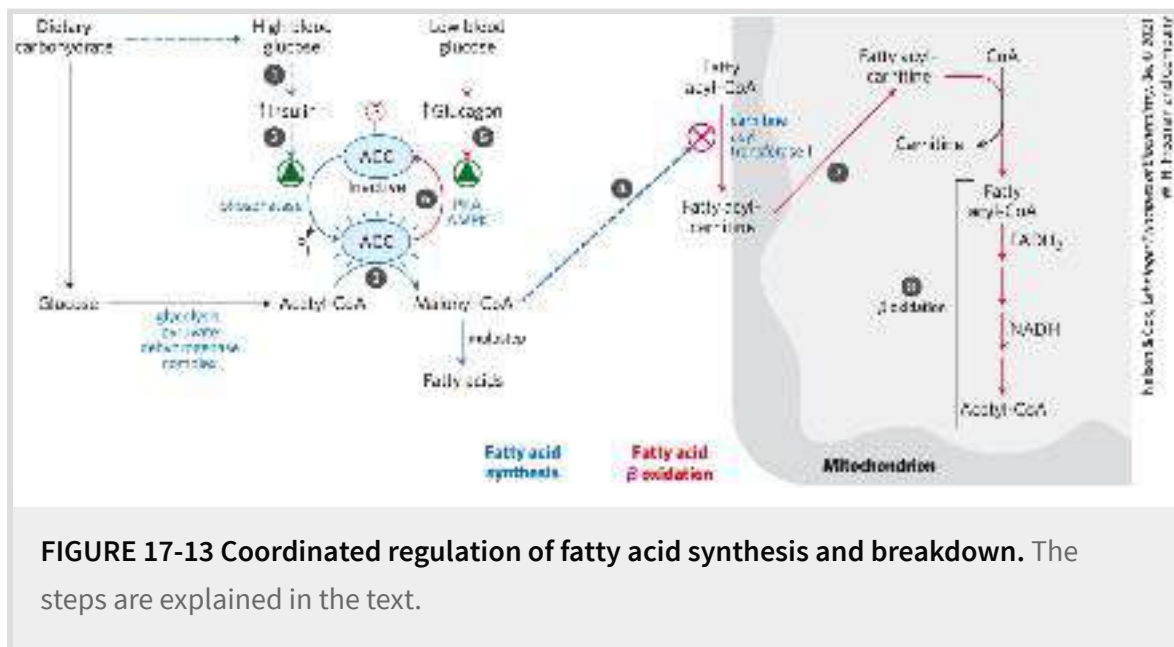
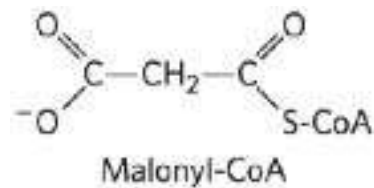


Fatty Acid Oxidation Is Tightly Regulated

P3 Oxidation of fatty acids consumes a precious fuel, and it is regulated so as to occur only when the organism's need for energy requires it. In the liver, fatty acyl-CoA formed in the cytosol has two major pathways open to it: (1) β oxidation by enzymes in mitochondria or (2) conversion into triacylglycerols and phospholipids by enzymes in the cytosol. The pathway taken depends on the rate of transfer of long-chain fatty acyl-CoA into mitochondria. The three-step carnitine shuttle by which fatty acyl groups are carried into the mitochondrial matrix as fatty acyl-carnitine ([Fig. 17-6](#)) is rate-limiting for fatty acid oxidation and is an important point of regulation. Once fatty acyl groups have entered the mitochondrion, they are committed to oxidation to acetyl-CoA.

[Figure 17-13](#) illustrates, with numbered steps described below, the coordinated down-regulation of β oxidation when carbohydrate levels are sufficient. Ingestion of a high-carbohydrate meal raises the blood glucose level and thus ❶ triggers the release of insulin. ❷ Insulin-dependent protein phosphatase dephosphorylates the enzyme acetyl-CoA carboxylase (ACC), activating it. ❸ ACC catalyzes the formation of **malonyl-CoA** (the first intermediate of cytosolic fatty acid synthesis; see [Figs 21-1](#) and [21-2](#)), and ❹ malonyl-CoA inhibits carnitine acyltransferase 1, thereby preventing fatty acid entry into the mitochondrial matrix through the carnitine shuttle. The inhibition of carnitine acyltransferase 1 by malonyl-CoA ensures that the oxidation of fatty acids is inhibited whenever the liver is amply supplied with glucose as fuel and is actively making

triacylglycerols from excess glucose. Two of the enzymes of β oxidation (8) are also regulated by metabolites that signal energy sufficiency. When the $[\text{NADH}]/[\text{NAD}^+]$ ratio is high, β -hydroxyacyl-CoA dehydrogenase is inhibited; in addition, high concentrations of acetyl-CoA inhibit thiolase.



[Figure 17-13](#) also shows that conversely, when blood glucose levels drop between meals, (5) glucagon release activates protein kinase A (PKA), which (6) phosphorylates and inactivates ACC. Recall from [Chapter 14](#) that in a low-energy state, [AMP] rises relative to [ATP], activating AMP kinase (AMPK). Activated AMPK also phosphorylates and inactivates ACC. The concentration of malonyl-CoA falls, and the inhibition of fatty acid entry into

mitochondria is relieved. Fatty acids ⑦ enter the mitochondrial matrix, ⑧ allowing β oxidation to replenish the supply of ATP. Because glucagon also triggers the mobilization of fatty acids in adipose tissue, a supply of fatty acids begins arriving in the blood.

Transcription Factors Turn on the Synthesis of Proteins for Lipid Catabolism



In addition to the various short-term regulatory mechanisms that modulate the activity of existing enzymes, transcriptional regulation can change the number of molecules of the enzymes of fatty acid oxidation on a longer time scale — minutes to hours. The [PPAR](#) family of nuclear receptors are transcription factors that affect many metabolic processes in response to a variety of fatty acid-like ligands. (They were originally recognized as *peroxisome proliferator-activated receptors*, then were found to function more broadly.) PPAR α acts in muscle, adipose tissue, and liver to turn on a set of genes essential for fatty acid oxidation, including the fatty acid transporter, carnitine acyltransferases 1 and 2; the fatty acyl-CoA dehydrogenases for short, medium, long, and very long acyl chains; and related enzymes. This response is triggered when a cell or an organism has an increased demand for energy from fat catabolism, such as during a fast between meals or under conditions of longer-term starvation. Glucagon, released in response to low blood glucose, can act through cAMP and the

transcription factor CREB to turn on certain genes for lipid catabolism.

Another situation that is accompanied by major changes in the expression of the enzymes of fatty acid oxidation is the transition from fetal to neonatal metabolism in the heart. In the fetus, the principal fuels in heart muscle are glucose and lactate, but in the neonatal heart, fatty acids are the main fuel. At the time of this transition, PPAR α is activated and in turn activates the genes essential for fatty acid metabolism. As we will see in [Chapter 23](#), two other transcription factors in the PPAR family also play crucial roles in determining the enzyme complements — and therefore the metabolic activities — of specific tissues at particular times (see [Fig. 23-38](#)).

The major sites of fatty acid oxidation, at rest and during exercise, are skeletal and heart muscle. Endurance training increases PPAR α expression in muscle, leading to increased levels of fatty acid-oxidizing enzymes and increased oxidative capacity of the muscle.

Genetic Defects in Fatty Acyl–CoA Dehydrogenases Cause Serious Disease



Stored triacylglycerols are typically the chief source of energy for muscle contraction, and an inability to oxidize fatty

acids from triacylglycerols has serious consequences for health.

 The most common genetic defect in fatty acid catabolism in U.S. and northern European populations is due to a mutation in the gene encoding the **medium-chain acyl-CoA dehydrogenase (MCAD)**, mentioned earlier in the chapter. Among northern Europeans, the frequency of carriers (individuals with this recessive mutation on one of the two homologous chromosomes) is about 1 in 40, and about 1 individual in 10,000 has the disease — that is, has two copies of the mutant MCAD allele and is unable to oxidize fatty acids of 6 to 12 carbons. The disease is characterized by recurring episodes of a syndrome that includes fat accumulation in the liver, high blood levels of octanoic acid (8:0), low blood glucose (hypoglycemia), sleepiness, vomiting, and coma. Although individuals may have no symptoms between episodes, the episodes are very serious; mortality due to this disease is 25% to 60% in early childhood. If the genetic defect is detected shortly after birth, the infant can be started on a low-fat, high-carbohydrate diet. With early detection and careful management of the diet — including avoiding long intervals between meals, to prevent the body from turning to its fat reserves for energy — the prognosis for these individuals is good.

 More than 20 human genetic defects in fatty acid transport or oxidation have been documented, most much less common than the defect in MCAD. One of the most severe disorders results from loss of the long-chain β -hydroxyacyl-CoA dehydrogenase activity of the trifunctional protein, TFP. Other disorders include defects in the α or β subunits that affect all three activities of

TFP and cause serious heart disease and abnormal skeletal muscle. ■

Peroxisomes Also Carry Out β Oxidation

The mitochondrial matrix is the major site of fatty acid oxidation in animal cells, but other compartments also contain enzymes capable of oxidizing fatty acids to acetyl-CoA, by a pathway similar but not identical to that in mitochondria. In plant cells, the major site of β oxidation is not mitochondria but peroxisomes. In peroxisomes, organelles also found in animal cells, the intermediates for β oxidation of fatty acids are coenzyme A derivatives, and the process consists of four steps, as in mitochondrial β oxidation ([Fig. 17-14](#)): (1) dehydrogenation, (2) addition of water to the resulting double bond, (3) oxidation of the β -hydroxyacyl-CoA to a ketone, and (4) thiolytic cleavage by coenzyme A. The identical reactions also occur in glyoxysomes, organelles found only in germinating seeds.

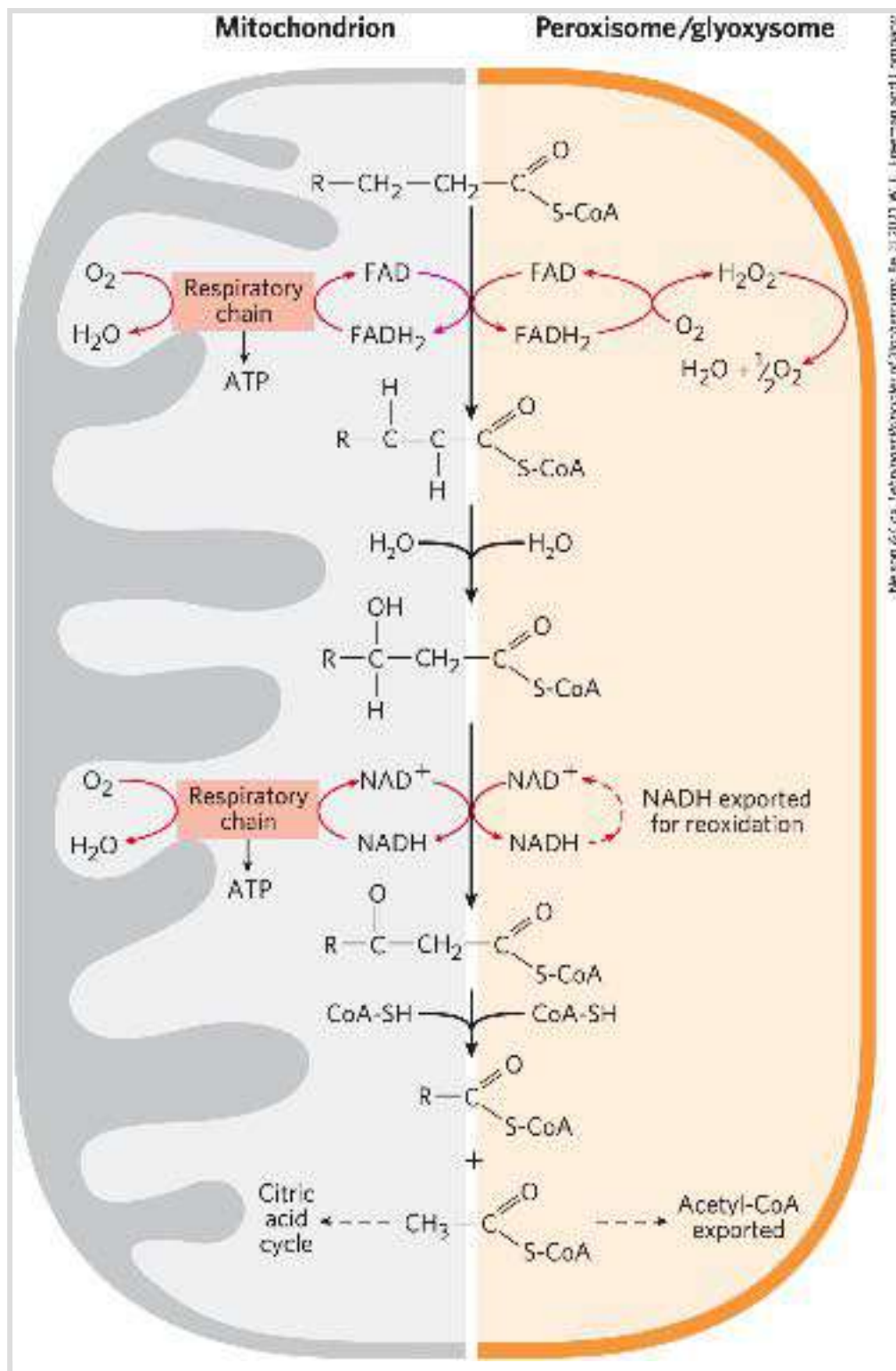


FIGURE 17-14 Comparison of β oxidation in mitochondria and in peroxisomes and glyoxysomes. The peroxisomal/glyoxysomal system differs from the mitochondrial system in three respects: (1) the peroxisomal system prefers very-long-chain fatty acids; (2) in the first oxidative step, electrons pass directly to O_2 , generating H_2O_2 ; and (3) the $NADH$ formed in the second oxidative step cannot be reoxidized in the peroxisome or

glyoxysome, so reducing equivalents are exported to the cytosol, eventually entering mitochondria. The acetyl-CoA produced by peroxisomes and glyoxysomes is also exported; the acetate from glyoxysomes serves as a biosynthetic precursor. Acetyl-CoA produced in mitochondria is further oxidized in the citric acid cycle.

One difference between the peroxisomal and mitochondrial pathways is in the chemistry of the first step ([Fig. 17-14](#)). In peroxisomes, the flavoprotein acyl-CoA oxidase that introduces the double bond passes electrons directly to O_2 , producing H_2O_2 (thus the name “peroxisomes”). This strong and potentially damaging oxidant is immediately cleaved to H_2O and O_2 by **catalase**. Recall that in mitochondria, the electrons removed in the first oxidation step pass through the respiratory chain to O_2 to produce H_2O , and this process is accompanied by ATP synthesis. In peroxisomes, the energy released in the first oxidative step of fatty acid breakdown is not conserved as ATP, but is dissipated as heat.



A second important difference between mitochondrial and peroxisomal β oxidation in mammals is in the specificity for fatty acyl-CoAs; the peroxisomal system is much more active on very-long-chain fatty acids such as hexacosanoic acid (26:0) and on branched-chain fatty acids such as phytanic acid and pristanic acid (see [Fig. 17-15](#)). These less-common fatty acids are obtained from dietary intake of dairy products, the fat of ruminant animals, meat, and fish. Their catabolism in the peroxisome involves several auxiliary enzymes unique to this organelle.



The inability to oxidize these compounds is responsible for

several serious human genetic diseases. Individuals with **Zellweger syndrome** are unable to make peroxisomes and therefore lack all the metabolism unique to that organelle. In **X-linked adrenoleukodystrophy (XALD)**, peroxisomes fail to oxidize very-long-chain fatty acids, apparently due to lack of a functional transporter for these fatty acids in the peroxisomal membrane. Both defects lead to accumulation in the blood of very-long-chain fatty acids, especially 26:0. XALD affects young boys before the age of 10 years, causing loss of vision, behavioral disturbances, and death within a few years. ■

In mammals, high concentrations of fats in the diet result in increased synthesis of the enzymes of peroxisomal β oxidation in the liver. Liver peroxisomes do not contain the enzymes of the citric acid cycle and cannot catalyze the oxidation of acetyl-CoA to CO_2 . Instead, long-chain or branched fatty acids are catabolized in peroxisomes to shorter-chain products, such as hexanoyl-CoA, which are exported to mitochondria and completely oxidized there. As we will see in [Chapter 20](#), the germinating seeds of plants can synthesize carbohydrates and many other metabolites from acetyl-CoA produced in peroxisomes, using a pathway (the glyoxylate cycle) not present in vertebrates.

Phytanic Acid Undergoes α Oxidation in Peroxisomes

Phytanic acid, a long-chain fatty acid with methyl branches, is derived from the phytol side chain of chlorophyll (see [Fig. 20-5](#)). The presence of a methyl group on the β carbon of this fatty acid prevents the formation of a β -keto intermediate, making its β oxidation impossible. Humans obtain phytanic acid in the diet, primarily from dairy products and from the fats of ruminant animals; microorganisms in the rumen of these animals produce phytanic acid as they digest plant chlorophyll. The typical western diet includes 50 to 100 mg of phytanic acid per day.



Phytanic acid is metabolized in peroxisomes by [\$\alpha\$ oxidation](#), in which a single carbon is removed from the carboxyl end of the fatty acid ([Fig. 17-15](#)). Phytanoyl-CoA is first hydroxylated on its α carbon in a reaction that involves molecular oxygen. The product is decarboxylated to form an aldehyde one carbon shorter, and then oxidized to the corresponding carboxylic acid, which now has no substituent on the β carbon. Further β oxidation produces acetyl-CoA and then propionyl-CoA in successive oxidation cycles. **Refsum disease**, resulting from a genetic defect in phytanoyl-CoA hydroxylase, leads to the accumulation of very high blood levels of phytanic acid, causing (by unknown mechanisms) severe neurological deficits, including blindness and deafness. ■

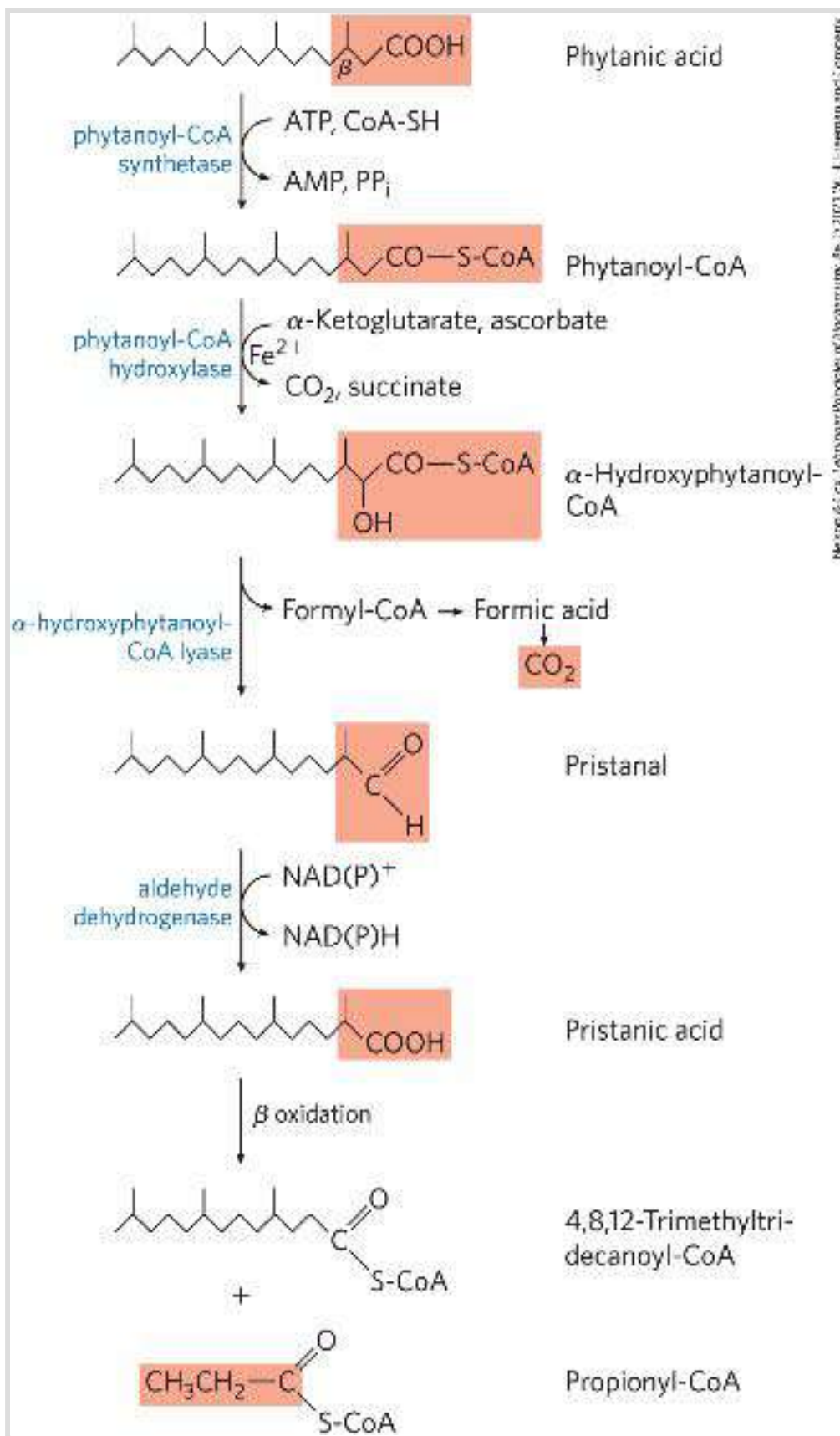


FIGURE 17-15 The α oxidation of a branched-chain fatty acid (phytanic acid) in peroxisomes. Phytanic acid has a methyl-substituted β carbon

and therefore cannot undergo β oxidation. The combined action of the enzymes shown here removes the carboxyl carbon of phytanic acid to produce pristanic acid, in which the β carbon is unsubstituted, allowing β oxidation. Notice that β oxidation of pristanic acid releases propionyl-CoA, not acetyl-CoA. This is further catabolized as in [Figure 17-12](#). (The details of the reaction that produces pristanal remain controversial.)

SUMMARY 17.2 *Oxidation of Fatty Acids*

■ In the first stage of β oxidation, four sequential reactions remove each acetyl-CoA unit, in turn, from the carboxyl end of a saturated fatty acyl-CoA: (1) dehydrogenation of the α and β carbons by acyl-CoA dehydrogenases; (2) hydration of the resulting *trans*- Δ^2 double bond by enoyl-CoA hydratase; (3) dehydrogenation of the resulting L- β -hydroxyacyl-CoA; and (4) cleavage of the resulting β -ketoacyl-CoA by thiolase, to form acetyl-CoA and a fatty acyl-CoA shortened by two carbons.

■ The shortened fatty acyl-CoA then reenters the β oxidation sequence for removal of another, and then another, acetyl-CoA.

■ In the second stage of fatty acid oxidation, the acetyl-CoA is oxidized to CO₂ in the citric acid cycle. Much of the free energy from fatty acid oxidation is recovered as ATP by oxidative phosphorylation, the final stage of the oxidative pathway.

■ Oxidation of unsaturated fatty acids requires two additional enzymes: enoyl-CoA isomerase and 2,4-dienoyl-CoA reductase.

■ Odd-number fatty acids are oxidized by the β -oxidation pathway to yield acetyl-CoA and a molecule of propionyl-CoA. The propionyl-CoA is carboxylated to methylmalonyl-CoA, which is

isomerized to succinyl-CoA in a reaction catalyzed by methylmalonyl-CoA mutase, an enzyme requiring coenzyme B₁₂.

■ To prevent futile cycling, entry of fatty acids into mitochondria is blocked by the first intermediate in fatty acid synthesis, malonyl-CoA.

■ The transcription factor PPAR α stimulates the synthesis of several enzymes required in β oxidation when glucose is not available as an energy source.

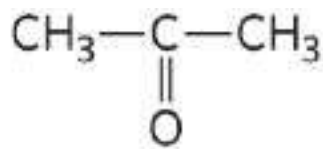
■ Individuals who lack medium-chain acyl-CoA dehydrogenase experience fatty liver, high blood levels of octanoic acid, coma, and sometimes death.

■ Peroxisomes, of plants and animals, and glyoxysomes, of plants, carry out β oxidation in four steps similar to those of the mitochondrial pathway. The first oxidation step, however, transfers electrons directly to O₂, generating H₂O₂. Peroxisomes of animal tissues specialize in the oxidation of very-long-chain fatty acids and branched fatty acids.

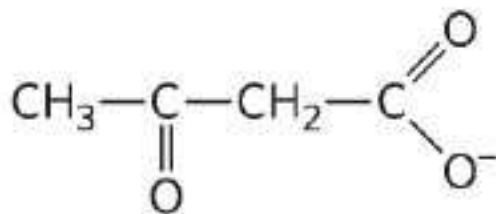
■ The reactions of α oxidation convert branched fatty acids such as phytanic acid to products that can undergo β oxidation, yielding acetyl-CoA and propionyl-CoA.

17.3 Ketone Bodies

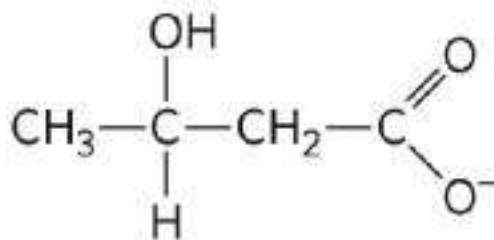
In humans and most other mammals, acetyl-CoA formed in the liver during oxidation of fatty acids can either enter the citric acid cycle (stage 2 of [Fig. 17-7](#)) or undergo conversion to the “[ketone bodies](#),” **acetone**, **acetoacetate**, and **D- β -hydroxybutyrate**, for export to other tissues. (The term “bodies” is an historical artifact; the compounds are soluble in blood and urine, not particulate, and not all of them are ketones.)



Acetone



Acetoacetate



D- β -Hydroxybutyrate

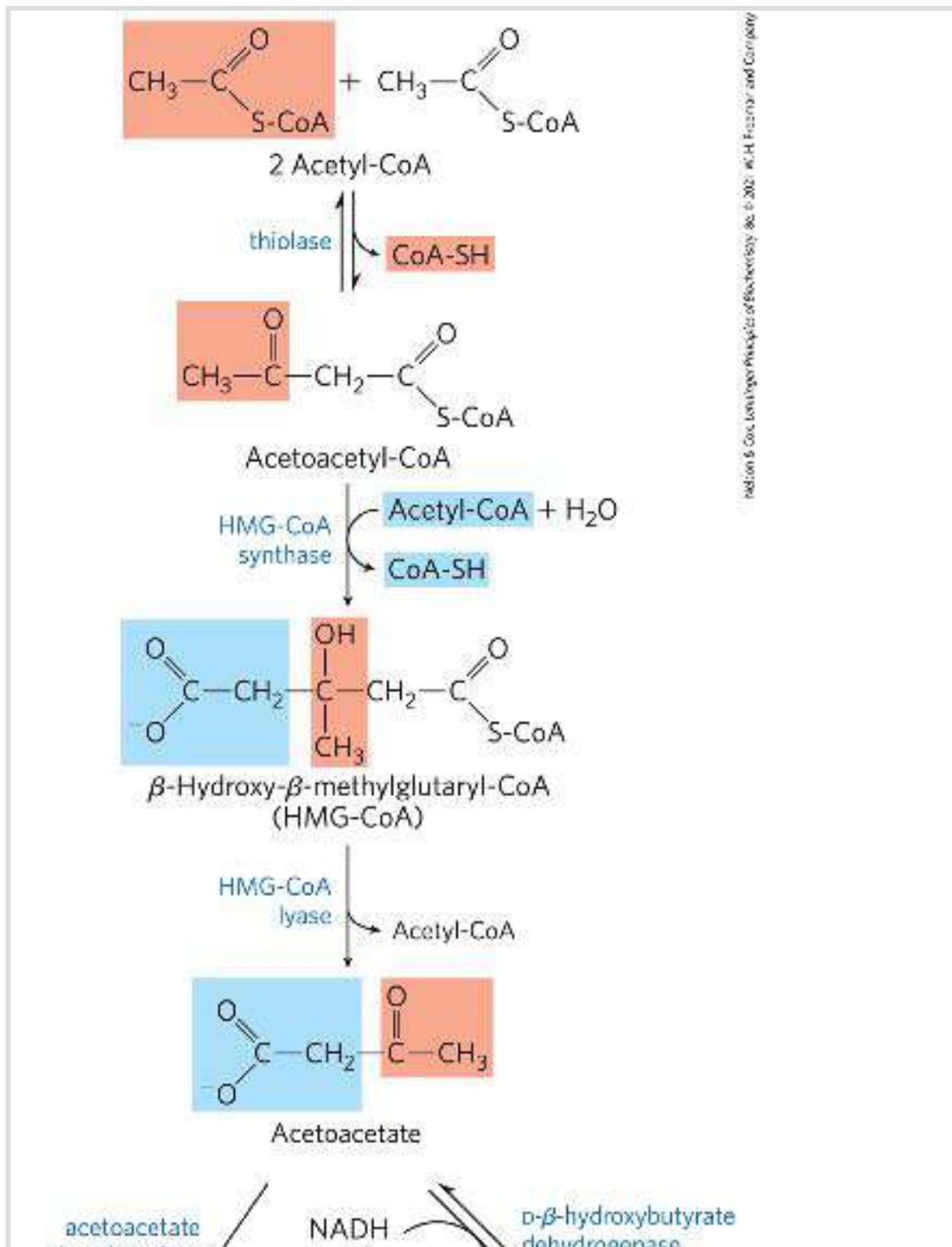
Acetone, produced in smaller quantities than the other ketone bodies, is exhaled. **P5** Acetoacetate and D- β -hydroxybutyrate are transported by the blood to tissues other than the liver

(extrahepatic tissues), where they are converted to acetyl-CoA and oxidized in the citric acid cycle, providing much of the energy required by tissues such as skeletal and heart muscle and the renal cortex. The brain, which preferentially uses glucose as fuel, can adapt to the use of acetoacetate and D- β -hydroxybutyrate under starvation conditions, when glucose is unavailable. The brain cannot use fatty acids as fuel, because they do not cross the blood-brain barrier. The production and export of ketone bodies from the liver to extrahepatic tissues allows continued oxidation of fatty acids in the liver when acetyl-CoA is not being oxidized in the citric acid cycle.

Ketone Bodies, Formed in the Liver, Are Exported to Other Organs as Fuel

The first step in the formation of acetoacetate, occurring in the liver ([Fig. 17-16](#)), is the enzymatic condensation of two molecules of acetyl-CoA, catalyzed by thiolase; this is simply the reversal of the last step of β oxidation. The acetoacetyl-CoA then condenses with another molecule of acetyl-CoA to form β -**hydroxy- β -methylglutaryl-CoA (HMG-CoA)**, which is cleaved to free acetoacetate and acetyl-CoA. The acetoacetate is reversibly reduced by D- β -hydroxybutyrate dehydrogenase, a mitochondrial enzyme, to D- β -hydroxybutyrate. This enzyme is specific for the D stereoisomer; it does not act on L- β -hydroxyacyl-CoAs and is distinct from L- β -hydroxyacyl-CoA dehydrogenase of the β -oxidation pathway. This difference in stereospecificity of the two

enzymes that use β -hydroxyacyl-CoAs as substrates in fatty acid breakdown and fatty acid synthesis means that the cell can maintain separate pools of β -hydroxyacyl-CoAs, earmarked for either breakdown or synthesis.



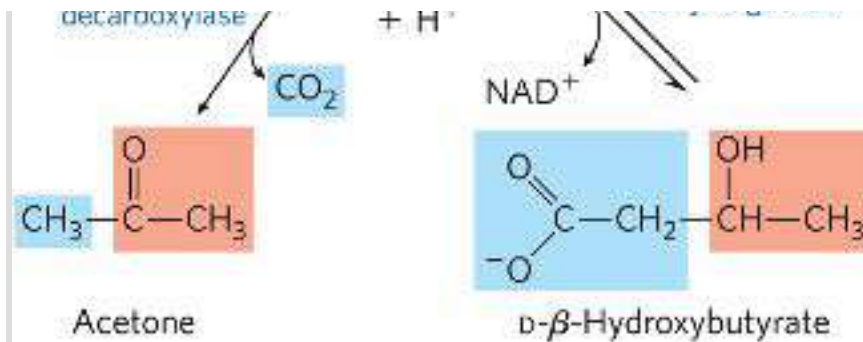


FIGURE 17-16 Formation of ketone bodies from acetyl-CoA. Healthy, well-nourished individuals produce ketone bodies at a relatively low rate. When acetyl-CoA accumulates (as in starvation or untreated diabetes, for example), thiolase catalyzes the condensation of two acetyl-CoA molecules to acetoacetyl-CoA, the parent compound of the three ketone bodies. The reactions of ketone body formation occur in the matrix of liver mitochondria. The six-carbon compound β -hydroxy- β -methylglutaryl-CoA (HMG-CoA) is also an intermediate of sterol biosynthesis, but the enzyme that forms HMG-CoA in that pathway is cytosolic. HMG-CoA lyase is present only in the mitochondrial matrix.

In extrahepatic tissues, D- β -hydroxybutyrate is oxidized to acetoacetate by D- β -hydroxybutyrate dehydrogenase ([Fig. 17-17](#)). The acetoacetate is activated to its coenzyme A ester by transfer of CoA from succinyl-CoA, an intermediate of the citric acid cycle (see [Fig. 16-7](#)), in a reaction catalyzed by β -**ketoacyl-CoA transferase**, also called thiophorase. The acetoacetyl-CoA is then cleaved by thiolase to yield two molecules of acetyl-CoA, which enter the citric acid cycle. Thus the ketone bodies are used as fuels in all tissues except liver, which lacks β -ketoacyl-CoA transferase. **P5** The liver is therefore a producer of ketone bodies for other tissues, but not a consumer.

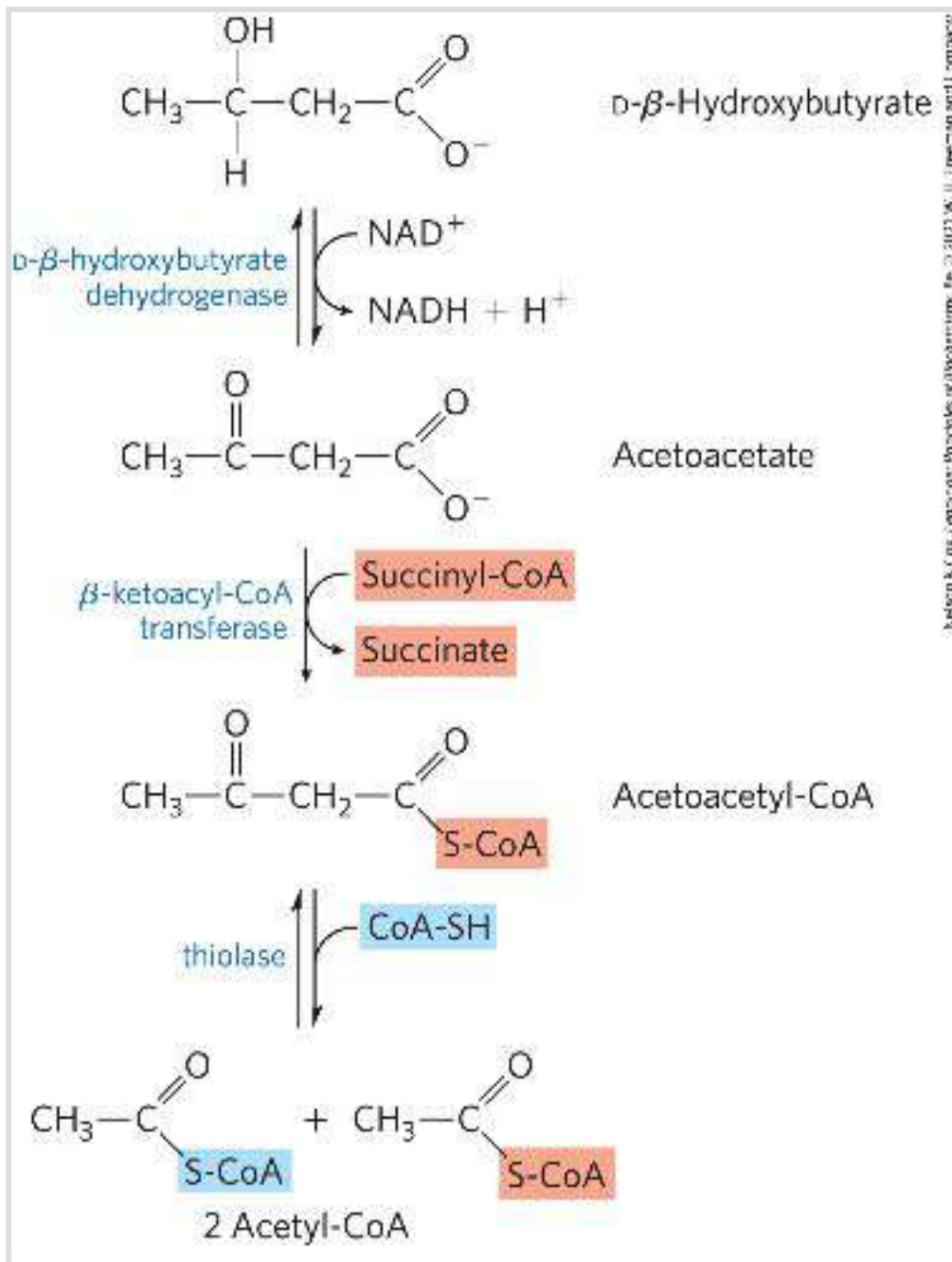


FIGURE 17-17 D- β -Hydroxybutyrate as a fuel. D- β -Hydroxybutyrate, synthesized in the liver, passes into the blood and thus to other tissues, where it is converted in three steps to acetyl-CoA. It is first oxidized to acetoacetate, which is activated with coenzyme A donated from succinyl-CoA, then split by thiolase. The acetyl-CoA thus formed enters the citric acid cycle.

The production and export of ketone bodies by the liver allows continued oxidation of fatty acids with only minimal oxidation of acetyl-CoA. When intermediates of the citric acid cycle are being siphoned off for glucose synthesis by gluconeogenesis, for example, oxidation of cycle intermediates slows — and so does acetyl-CoA oxidation. Moreover, the liver contains only a limited amount of coenzyme A, and when most of it is tied up in acetyl-CoA, β oxidation slows for want of the free coenzyme. The production and export of ketone bodies frees coenzyme A, allowing continued fatty acid oxidation.

Ketone Bodies Are Overproduced in Diabetes and during Starvation



Starvation (including very-low-calorie diets) and untreated diabetes mellitus lead to overproduction of ketone bodies in the liver, with several adverse effects on health. During starvation, gluconeogenesis depletes citric acid cycle intermediates, diverting acetyl-CoA from oxidation of mobilized stored fats to ketone body production ([Fig. 17-18](#)). In untreated diabetes, when the insulin level is insufficient, extrahepatic tissues cannot take up glucose efficiently from the blood, either for fuel or for conversion to fat. Under these conditions, levels of malonyl-CoA fall, inhibition of carnitine acyltransferase 1 is relieved, and fatty acids enter mitochondria to be degraded to acetyl-CoA (see [Fig. 17-13](#)). But this acetyl-CoA cannot pass through the citric acid cycle because cycle intermediates have been drawn off for use as

substrates in gluconeogenesis.   The resulting accumulation of acetyl-CoA accelerates the formation of ketone bodies and their release into the blood beyond the capacity of extrahepatic tissues to oxidize them. The increased blood levels of acetoacetate and D- β -hydroxybutyrate lower the blood pH, causing the condition known as acidosis. Extreme acidosis can lead to coma and in some cases death. Ketone bodies in the blood and urine of individuals with untreated diabetes can reach extraordinary levels — a blood concentration of 90 mg/100 mL (compared with a normal level of <3 mg/100 mL) and urinary excretion of 5,000 mg/24 h (compared with a normal rate of $\leq$ 125 mg/24 h). This condition is called ketosis or, when combined with acidosis, ketoacidosis.

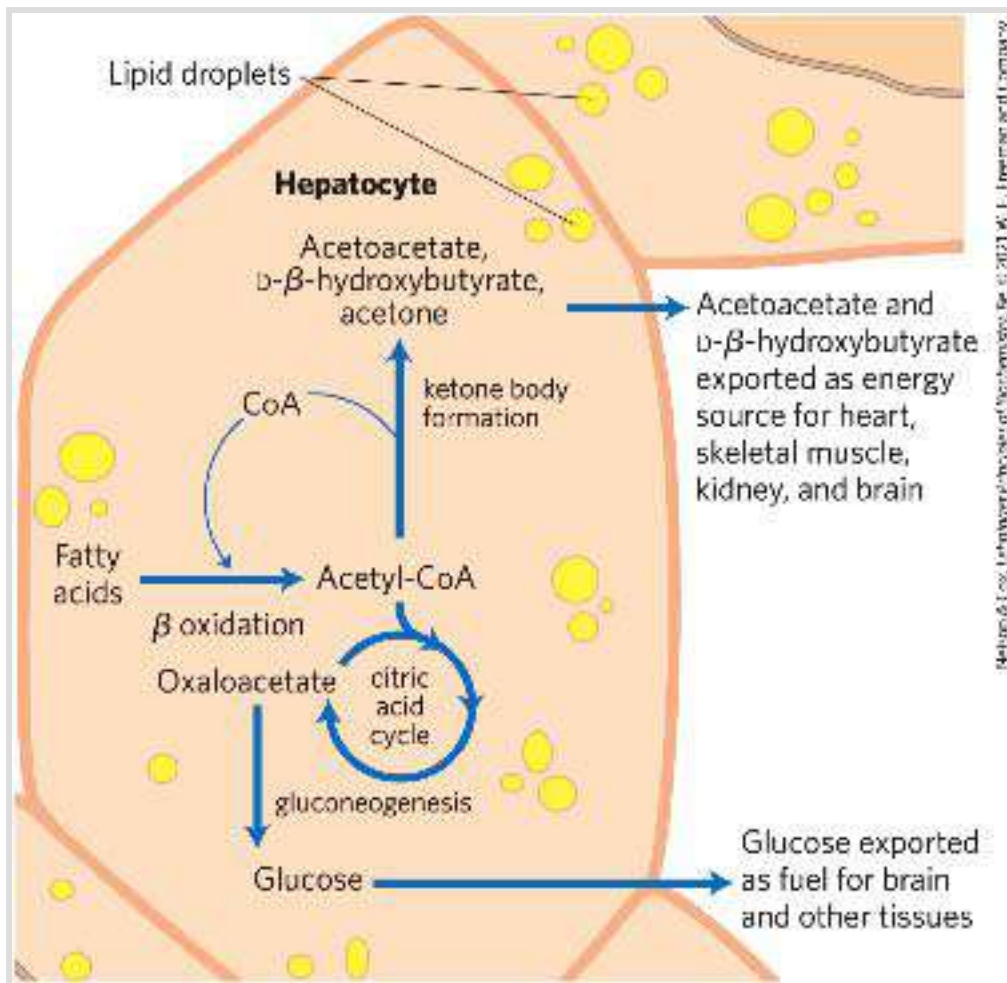


FIGURE 17-18 Ketone body formation and export from the liver.

Conditions that promote gluconeogenesis (untreated diabetes, severely reduced food intake) slow the citric acid cycle (by drawing off oxaloacetate) and enhance the conversion of acetyl-CoA to acetoacetate. The released coenzyme A allows continued β oxidation of fatty acids.

In healthy people, acetone is formed in very small amounts from acetoacetate, which is easily decarboxylated, either spontaneously or by the action of **acetoacetate decarboxylase** ([Fig. 17-16](#)). Because individuals with untreated diabetes produce large quantities of acetoacetate, their blood contains significant amounts of acetone, which is toxic. Acetone is volatile and

imparts a characteristic odor to the breath, which is sometimes useful in diagnosing diabetes. ■

SUMMARY 17.3 *Ketone Bodies*

■ The ketone bodies — acetone, acetoacetate, and D- β -hydroxybutyrate — are formed in the liver when fatty acids are the principal fuel supporting whole-body metabolism.

Acetoacetate and β -hydroxybutyrate serve as fuel molecules in extrahepatic tissues, including the brain, through oxidation to acetyl-CoA and entry into the citric acid cycle.

■ Overproduction of ketone bodies in uncontrolled diabetes or severely reduced calorie intake can lead to life-threatening ketoacidosis, characterized by high concentrations of ketones in blood and urine and lower blood pH.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

β oxidation

apolipoprotein

chylomicron

lipoprotein

perilipin

free fatty acids

serum albumin

carnitine shuttle

fatty acyl-CoA synthetase

fatty acyl-CoA

carnitine acyltransferase 1 (CAT1)

acyl-carnitine/carnitine cotransporter

carnitine acyltransferase 2 (CAT2)

electron-transfer flavoprotein (ETF)

NADH dehydrogenase (Complex I)

trifunctional protein (TFP)

methylmalonyl-CoA mutase

coenzyme B₁₂

malonyl-CoA

pernicious anemia

intrinsic factor

PPAR (peroxisome proliferator-activated receptor)

medium-chain acyl-CoA dehydrogenase (MCAD)

peroxisome

α oxidation

ketone bodies

acetone

acetoacetate

D- β -hydroxybutyrate

acidosis

ketosis

ketoacidosis

PROBLEMS

1. Energy in Triacylglycerols On a per-carbon basis, where does the largest amount of biologically available energy in triacylglycerols reside: in the fatty acid portions or in the glycerol portion? Indicate how knowledge of the chemical structure of triacylglycerols provides the answer.

2. Effect of PDE Inhibitor on Adipocytes How would the addition of a cAMP phosphodiesterase (PDE) inhibitor affect the response of an adipocyte to epinephrine? (Hint: See [Fig. 12-4](#).)

3. Compartmentation in β Oxidation The activation of free palmitate to its coenzyme A derivative (palmitoyl-CoA) in the cytosol occurs before it can be oxidized in the mitochondrion. After adding palmitate and [^{14}C]coenzyme A to a liver homogenate, you find palmitoyl-CoA isolated from the cytosolic fraction is radioactive, but that isolated from the mitochondrial fraction is not. Explain.

4. Mutant Carnitine Acyltransferase What changes in metabolic pattern would result from a mutation in the muscle carnitine acyltransferase 1 in which the mutant protein has lost its affinity for malonyl-CoA but not its catalytic activity?

5.  Effect of Carnitine Deficiency An individual developed a condition characterized by progressive muscular weakness and aching muscle cramps. The symptoms were aggravated by fasting, exercise, and a high-fat diet. An homogenate of a skeletal muscle specimen from the patient oxidized added oleate more slowly than did control homogenates consisting of muscle specimens from healthy individuals. When the pathologist added carnitine to the patient's muscle homogenate, the rate of oleate oxidation equaled that in the control homogenates. Based on these results, the attending physician diagnosed the patient as having a carnitine deficiency.

- Why did added carnitine increase the rate of oleate oxidation in the patient's muscle homogenate?
- Why did fasting, exercise, and a high-fat diet aggravate the patient's symptoms?
- Suggest two possible reasons for the deficiency of muscle carnitine in this individual.

6. Fuel Reserves in Adipose Tissue Triacylglycerols, with their hydrocarbon-like fatty acids, have the highest energy content of the major nutrients.

- If 15% of the body mass of a 70.0 kg adult consists of triacylglycerols, what is the total available fuel reserve, in both kilojoules and kilocalories, in the form of triacylglycerols? Recall that $1.00 \text{ kcal} = 4.18 \text{ kJ}$.

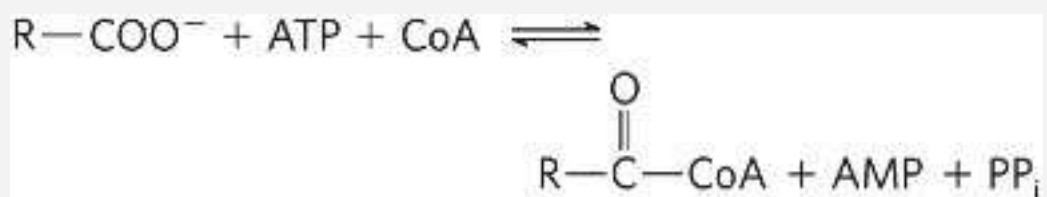
- b. If the basal energy requirement is approximately 8,400 kJ/day (2,000 kcal/day), how long could this person survive if the oxidation of fatty acids stored as triacylglycerols were the only source of energy?
- c. What would be the weight loss in pounds per day under such starvation conditions (1 lb = 0.454 kg)?

7. Common Reaction Steps in the Fatty Acid Oxidation Cycle and Citric Acid Cycle

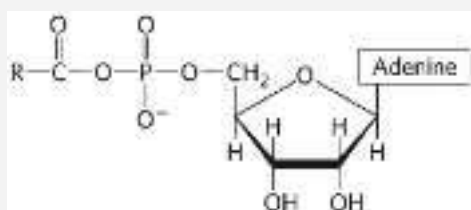
Cells often use the same enzyme reaction pattern for analogous metabolic conversions. For example, the steps in the oxidation of pyruvate to acetyl-CoA and of α -ketoglutarate to succinyl-CoA, although catalyzed by different enzymes, are very similar. The first stage of fatty acid oxidation follows a reaction sequence closely resembling a sequence in the citric acid cycle. Use equations to show the analogous reaction sequences in the two pathways.

8. **β Oxidation: How Many Cycles?** How many cycles of β oxidation are required for the complete oxidation of activated oleic acid, 18:1(Δ^9)?

9. **Chemistry of the Acyl-CoA Synthetase Reaction** Fatty acids are converted to their coenzyme A esters in a reversible reaction catalyzed by acyl-CoA synthetase:



- a. The enzyme-bound intermediate in this reaction has been identified as the mixed anhydride of the fatty acid and adenosine monophosphate (AMP), acyl-AMP:



Write two equations corresponding to the two steps of the reaction catalyzed by acyl-CoA synthetase.

- b. The acyl-CoA synthetase reaction is readily reversible, with an equilibrium constant near 1. How can this reaction be made to favor formation of fatty acyl-CoA?

10. Intermediates in Oleic Acid Oxidation What is the structure of the partially oxidized fatty acyl group that is formed when oleic acid, 18:1(Δ^9), has undergone three cycles of β oxidation? What are the next two steps in the continued oxidation of this intermediate?

11. β Oxidation of an Odd-Number Fatty Acid What are the direct products of β oxidation of a fully saturated, straight-chain fatty acid of 11 carbons?

12. Oxidation of Tritiated Palmitate An investigator adds palmitate uniformly labeled with tritium (^3H) to a specific activity of 2.48×10^8 counts per minute (cpm) per micromole of palmitate to a mitochondrial preparation that oxidizes it to acetyl-CoA. She then isolates the acetyl-CoA and hydrolyzes it to acetate. The specific activity of the isolated acetate is 1.00×10^7 cpm/ μmol . Is this result consistent with the β -oxidation pathway? Explain. What is the final fate of the removed tritium? (Note: Specific activity is a measure of the degree of labeling with a radioactive tracer expressed as radioactivity per unit mass. In a uniformly labeled compound, all atoms of a given type are labeled.)

13. Comparative Biochemistry: Energy-Generating Pathways in Birds One indication of the relative importance of various ATP-producing pathways is the V_{max} of certain enzymes of these pathways. The values of V_{max} of several enzymes from the pectoral muscles (chest muscles used for flying) of pigeon and pheasant are listed below.

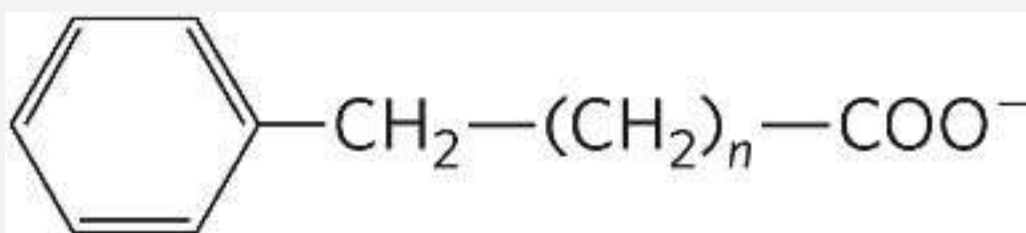
Enzyme	V_{max} ($\mu\text{mol substrate/min/g tissue}$)	
	Pigeon	Pheasant
Hexokinase	3.0	2.3
Glycogen phosphorylase	18.0	120.0
Phosphofructokinase-1	24.0	143.0
Citrate synthase	100.0	15.0
Triacylglycerol lipase	0.07	0.01

- Discuss the relative importance of glycogen metabolism and fat metabolism in generating ATP in the pectoral muscles of these birds.
- Compare oxygen consumption in the two birds.
- Judging from the data in the table, which bird is the long-distance flyer? Justify your answer.
- Why were these particular enzymes selected for comparison? Would the activities of triose phosphate isomerase and malate dehydrogenase be equally

good bases for comparison? Explain.

14. Fatty Acids as a Source of Water Contrary to legend, camels do not store water in their humps, which actually consist of large fat deposits. How can these fat deposits serve as a source of water? Calculate the amount of water (in liters) that a camel can produce from 1.0 kg of fat. Assume for simplicity that the fat consists entirely of tripalmitoylglycerol.

15. Metabolism of a Straight-Chain Phenylated Fatty Acid Investigators isolate a crystalline metabolite from the urine of a rabbit that has been fed a straight-chain fatty acid containing a terminal phenyl group:



The addition of 22.2 mL of 0.100 M NaOH completely neutralized a 302 mg sample of the metabolite in aqueous solution.

- What is the probable molecular weight and structure of the metabolite?
- Did the straight-chain fatty acid contain an even or an odd number of methylene ($-\text{CH}_2-$) groups (i.e., is n even or odd)? Explain.

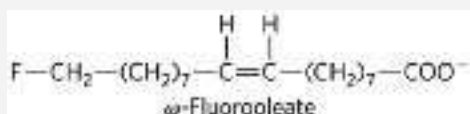
16.  Fatty Acid Oxidation in Uncontrolled Diabetes When the acetyl-CoA produced during β oxidation in the liver exceeds the capacity of the citric acid cycle, the excess acetyl-CoA forms ketone bodies — acetone, acetoacetate, and D- β -hydroxybutyrate. This occurs in people with severe, uncontrolled diabetes; because their tissues cannot use glucose, they oxidize large amounts of fatty acids instead. Although acetyl-CoA is not toxic, the mitochondrion must divert the acetyl-CoA to ketone bodies. What problem would arise if acetyl-CoA were not converted to ketone bodies? How does the diversion to ketone bodies solve the problem?

17. Consequences of a High-Fat Diet with No Carbohydrates Suppose you had to subsist on a diet of whale blubber and seal blubber, with little or no carbohydrate.

- What would be the effect of carbohydrate deprivation on the utilization of fats for energy?
- If your diet were totally devoid of carbohydrate, would it be better to consume odd- or even-number fatty acids? Explain.

18. Even- and Odd-Number Fatty Acids in the Diet In a laboratory experiment, investigators feed two groups of rats two different fatty acids as their sole source of carbon for a month. The first group gets heptanoic acid (7:0), and the second gets octanoic acid (8:0). After the experiment, those in the first group are healthy and have gained weight, whereas those in the second group are weak and have lost weight as a result of losing muscle mass. What is the biochemical basis for this difference?

19. Metabolic Consequences of Ingesting ω -Fluorooleate The shrub *Dichapetalum toxicarium*, native to Sierra Leone, produces ω -fluorooleate, which is highly toxic to warm-blooded animals.



This substance has been used as an arrow poison, and powdered fruit from the plant is sometimes used as a rat poison (hence the plant's common name, ratsbane). Why is this substance so toxic? (Hint: Review [Chapter 16, Problem 21](#).)

20. Mutant Acetyl-CoA Carboxylase What would be the consequences for fat metabolism of a mutation in acetyl-CoA carboxylase that replaced the Ser residue normally phosphorylated by AMPK with an Ala residue? What might happen if the same Ser were replaced by Asp? (Hint: Compare the structures of phosphoserine, alanine, aspartate; see [Fig. 17-13](#).)

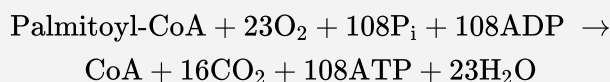
21. Role of FAD as Electron Acceptor Acyl-CoA dehydrogenase uses enzyme-bound FAD as a prosthetic group to dehydrogenate the α and β carbons of fatty acyl-CoA. What is the advantage of using FAD as an electron acceptor rather than NAD^+ ? Explain in terms of the standard reduction potentials for the Enz-FAD/FADH_2 ($E'^{\circ} = -0.219 \text{ V}$) and NAD^+/NADH ($E'^{\circ} = -0.320 \text{ V}$) half-reactions.

22. β Oxidation of Arachidic Acid How many turns of the fatty acid oxidation cycle are required for complete oxidation of arachidic acid (20:0) to acetyl-CoA?

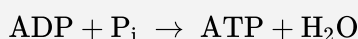
23. Fate of Labeled Propionate Adding $[3-^{14}\text{C}]$ propionate (^{14}C in the methyl group) to a liver homogenate leads to the rapid production of ^{14}C -labeled oxaloacetate. Draw a flowchart for the pathway by which propionate is transformed to oxaloacetate, and indicate the location of the ^{14}C in oxaloacetate.

24. Phytanic Acid Metabolism A mouse fed phytanic acid uniformly labeled with ^{14}C produces detectable levels of radioactive malate, a citric acid cycle intermediate, within minutes. Draw a metabolic pathway that could account for this. Which of the carbon atoms in malate would contain ^{14}C label?

25. Sources of H_2O Produced in β Oxidation The complete oxidation of palmitoyl-CoA to carbon dioxide and water is represented by the overall equation



Water also forms in the reaction



but is not included as a product in the overall equation. Why?

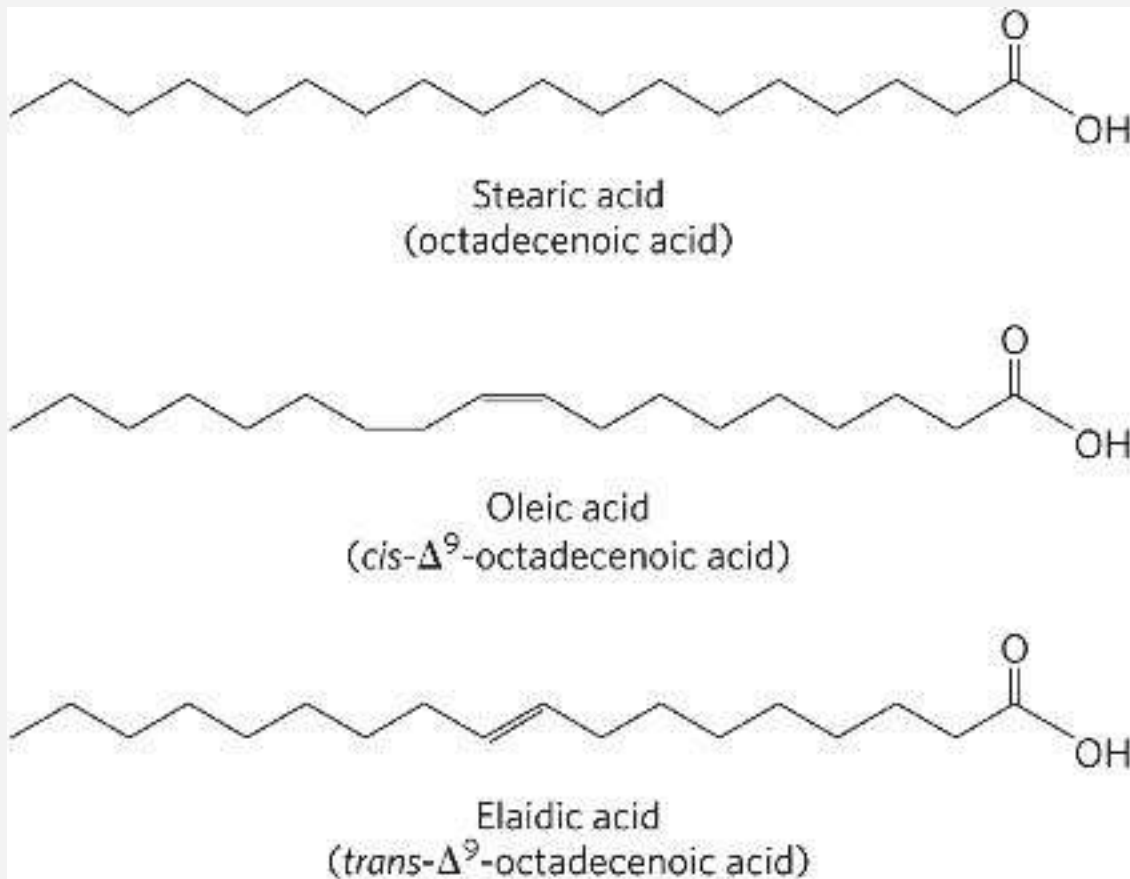
26. Biological Importance of Cobalt Cattle, deer, sheep, and other ruminant animals produce large amounts of propionate in the rumen through the bacterial fermentation of ingested plant matter. Propionate is the principal source of glucose for these animals, via the route propionate $\rightarrow$ oxaloacetate $\rightarrow$ glucose. In some areas of the world, notably Australia, ruminant animals sometimes show symptoms of anemia with concomitant loss of appetite and retarded growth, resulting from an inability to transform propionate to oxaloacetate. This condition is due to a cobalt deficiency caused by very low cobalt levels in the soil and thus in plant matter. Explain.

27. Fat Loss during Hibernation Bears expend about $25 \times 10^6 \text{ J/day}$ during periods of hibernation, which may last as long as seven months. The energy required to sustain life is obtained from fatty acid oxidation. How much weight (in kilograms) do bears lose after 7 months of hibernation? How could a bear's body minimize ketosis during hibernation? (Assume the oxidation of fat yields 38 kJ/g .)

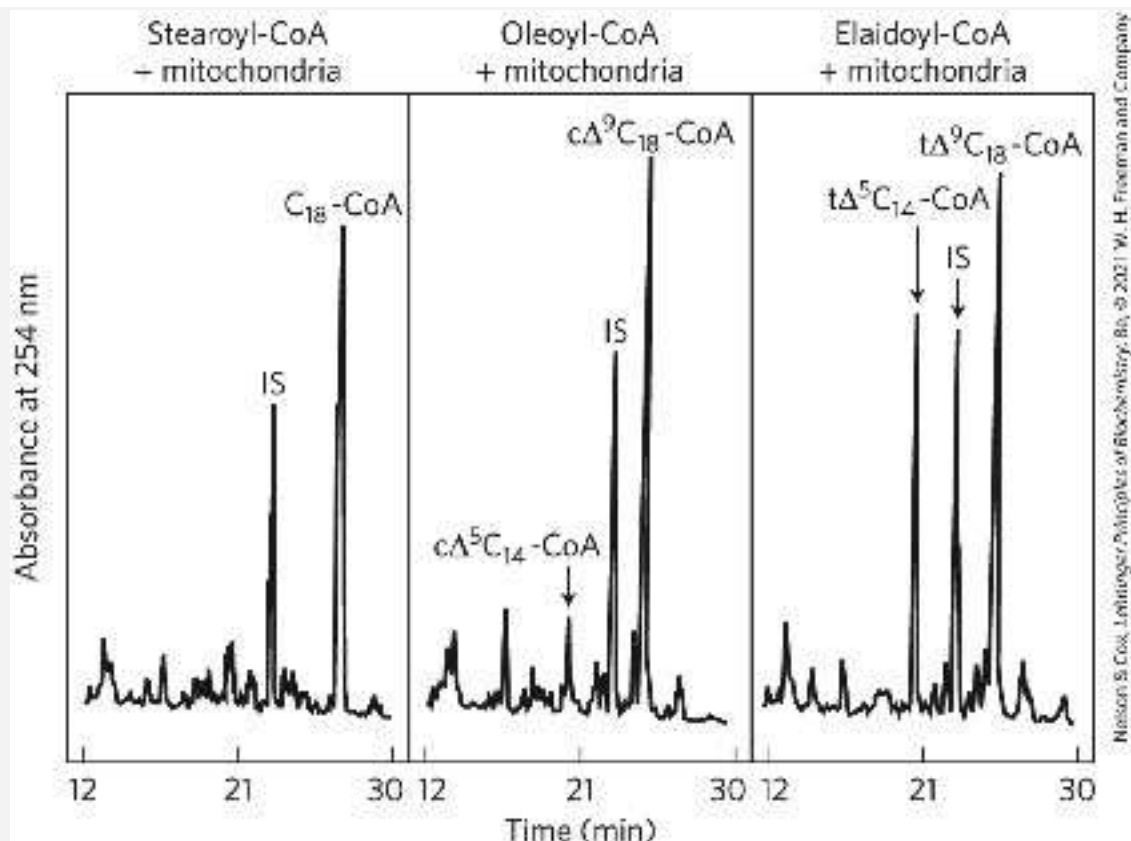
DATA ANALYSIS PROBLEM

28. β Oxidation of Trans Fats Unsaturated fats with trans double bonds are commonly referred to as "trans fats." In their investigations of the effects of trans fatty

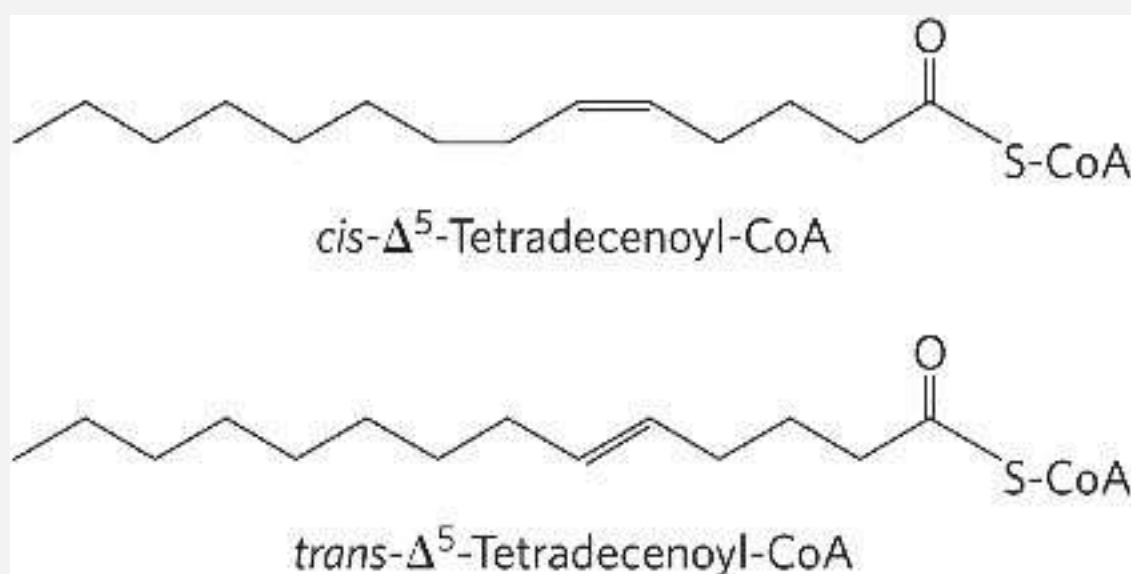
acid metabolism on health, Yu and colleagues (2004) showed that a model trans fatty acid was processed differently from its cis isomer. They used three related 18-carbon fatty acids to explore the difference in β oxidation between cis and trans isomers of the same-size fatty acid.



The researchers incubated the coenzyme A derivative of each acid with rat liver mitochondria for 5 minutes, then separated the remaining CoA derivatives in each mixture by HPLC (high-performance liquid chromatography). The results are shown below, with separate panels for the three experiments.



In the figure, IS indicates an internal standard (pentadecanoyl-CoA) added to the mixture after the reaction as a molecular marker. The researchers abbreviated the CoA derivatives as follows: stearoyl-CoA, C₁₈-CoA; *cis*-Δ⁵-tetradecenoyl-CoA, cΔ⁵C₁₄-CoA; oleoyl-CoA, cΔ⁹C₁₈-CoA; *trans*-Δ⁵-tetradecenoyl-CoA, tΔ⁵C₁₄-CoA; and elaidoyl-CoA, tΔ⁹C₁₈-CoA.



- Why did Yu and colleagues need to use CoA derivatives rather than the free fatty acids in these experiments?
- Why were no lower molecular weight CoA derivatives found in the reaction with stearoyl-CoA?
- How many rounds of β oxidation would be required to convert the oleoyl-CoA and the elaidoyl-CoA to *cis*- Δ^5 -tetradecenoyl-CoA and *trans*- Δ^5 -tetradecenoyl-CoA, respectively?

Yu and coworkers measured the kinetic parameters of two forms of the enzyme acyl-CoA dehydrogenase: long-chain acyl-CoA dehydrogenase (LCAD) and very-long-chain acyl-CoA dehydrogenase (VLCAD). They used the CoA derivatives of three fatty acids: tetradecanoyl-CoA (C_{14} -CoA), *cis*- Δ^5 -tetradecenoyl-CoA ($c\Delta^5 C_{14}$ -CoA), and *trans*- Δ^5 -tetradecenoyl-CoA ($t\Delta^5 C_{14}$ -CoA). The results are shown below. (See [Chapter 6](#) for definitions of the kinetic parameters.)

	LCAD			VLCAD		
	C_{14} -CoA	$c\Delta^5 C_{14}$ -CoA	$t\Delta^5 C_{14}$ -CoA	C_{14} -CoA	$c\Delta^5 C_{14}$ -CoA	$t\Delta^5 C_{14}$ -CoA
$V_{\max}$	3.3	3.0	2.9	1.4	0.32	0.88
K_m	0.41	0.40	1.6	0.57	0.44	0.97
k_{cat}	9.9	8.9	8.5	2.0	0.42	1.12
k_{cat}/K_m	24	22	5	4	1	1

- For LCAD, the K_m differs dramatically for the *cis* and *trans* substrates. Provide a plausible explanation for this observation in terms of the structures of the substrate molecules. (Hint: You may want to refer to [Fig. 10-1](#).)
- The kinetic parameters of the two enzymes are relevant to the differential processing of these fatty acids *only* if the LCAD or VLCAD reaction (or both) is the rate-limiting step in the pathway. What evidence is there to support this assumption?
- How do these different kinetic parameters explain the different levels of the CoA derivatives found after incubation of rat liver mitochondria with stearoyl-CoA, oleoyl-CoA, and elaidoyl-CoA (shown in the three-panel figure)?

Yu and coworkers measured the substrate specificity of rat liver mitochondrial thioesterase, which hydrolyzes acyl-CoA to CoA and free fatty acid. This enzyme was approximately twice as active with C_{14} -CoA thioesters as with C_{18} -CoA thioesters.

- g. Other research has suggested that free fatty acids can pass through membranes. In their experiments, Yu and colleagues found *trans*- Δ^5 -tetradecenoic acid outside (i.e., in the medium surrounding) mitochondria that had been incubated with elaidoyl-CoA. Describe the pathway that led to this extramitochondrial *trans*- Δ^5 -tetradecenoic acid. Be sure to indicate where in the cell the various transformations take place, as well as the enzymes that catalyze the transformations.
- h. It is sometimes said in the popular press that “trans fats are not broken down by your cells and instead accumulate in your body.” In what sense is this statement correct and in what sense is it an oversimplification?

Reference

Yu, W., X. Liang, R. Ensenauer, J. Vockley, L. Sweetman, and H. Schultz. 2004. Leaky β -oxidation of a *trans*-fatty acid. *J. Biol. Chem.* 279:52,160–52,167.

CHAPTER 18

AMINO ACID OXIDATION AND THE PRODUCTION OF UREA



18.1 Metabolic Fates of Amino Groups

18.2 Nitrogen Excretion and the Urea Cycle

18.3 Pathways of Amino Acid Degradation

We now turn our attention to the amino acids, the final class of biomolecules that, through their oxidative degradation, make a significant contribution to the generation of metabolic energy. The fraction of metabolic energy obtained from amino acids, whether they are derived from dietary protein or from tissue protein, varies greatly with the type of organism and with metabolic conditions. Carnivores consume primarily protein and thus must obtain most of their energy from amino acids, whereas herbivores may fill only a small fraction of their energy needs by this route. Most microorganisms can scavenge amino acids from their environment and use them as fuel when required by metabolic conditions. Plants, however, rarely if ever oxidize amino acids to provide energy; the carbohydrate produced from

CO₂ and H₂O in photosynthesis is generally their sole energy source. Amino acid concentrations in plant tissues are carefully regulated to just meet the requirements for biosynthesis of proteins, nucleic acids, and other molecules needed to support growth. Amino acid catabolism does occur in plants, but only to produce metabolites for other biosynthetic pathways.

Amino acid oxidation pathways can seem complex, and they are best understood in the context of five principles:

P1 The many paths for amino acid catabolism have two broad parts, one involving the amino groups and the other involving the carbon skeletons. All of the pathways for amino acid degradation include a key step, always involving a pyridoxal phosphate cofactor, in which the α -amino group is separated from the carbon skeleton and shunted into the pathways of amino group metabolism. The carbon skeletons are broken down to citric acid cycle intermediates ([Fig. 18-1](#)).

P2 Four amino acids — alanine, glutamate, glutamine, and aspartate — play key roles in the transport and distribution of amino groups. All are present in relatively high concentrations in one or many mammalian tissues. All are readily converted to key citric acid cycle intermediates.

P3 Metabolic pathways are not distinct. The various pathways for amino acid catabolism are elaborately intertwined with other catabolic and anabolic pathways.

P4 **Free ammonia is toxic.** Excess amino groups must be safely excreted. In mammals, the urea cycle serves this purpose.

P5 **Each amino acid has a different catabolic fate.** The varied carbon skeletons of amino acids are broken down via equally varied pathways. All can be oxidized to generate ATP. All but leucine and lysine can contribute to gluconeogenesis when needed.

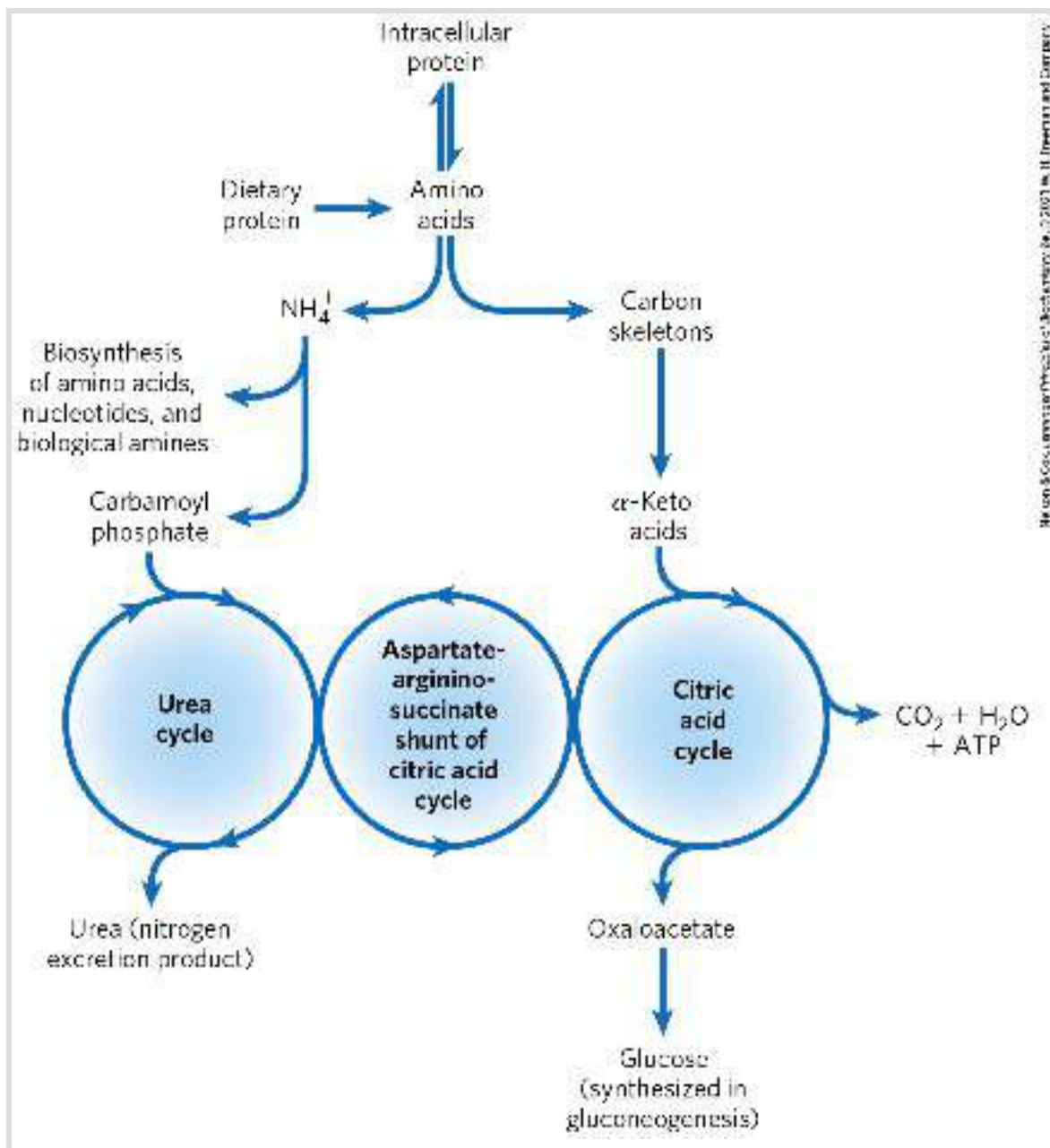


FIGURE 18-1 Overview of amino acid catabolism in mammals. The amino groups and the carbon skeleton take separate but interconnected pathways.

In mammals, blood glucose must be supplemented by gluconeogenesis, often just a few hours after a meal. Amino acids, particularly alanine and glutamine, can make a significant contribution to the fuel for gluconeogenesis. Amino acids

undergo oxidative degradation in three different metabolic circumstances:

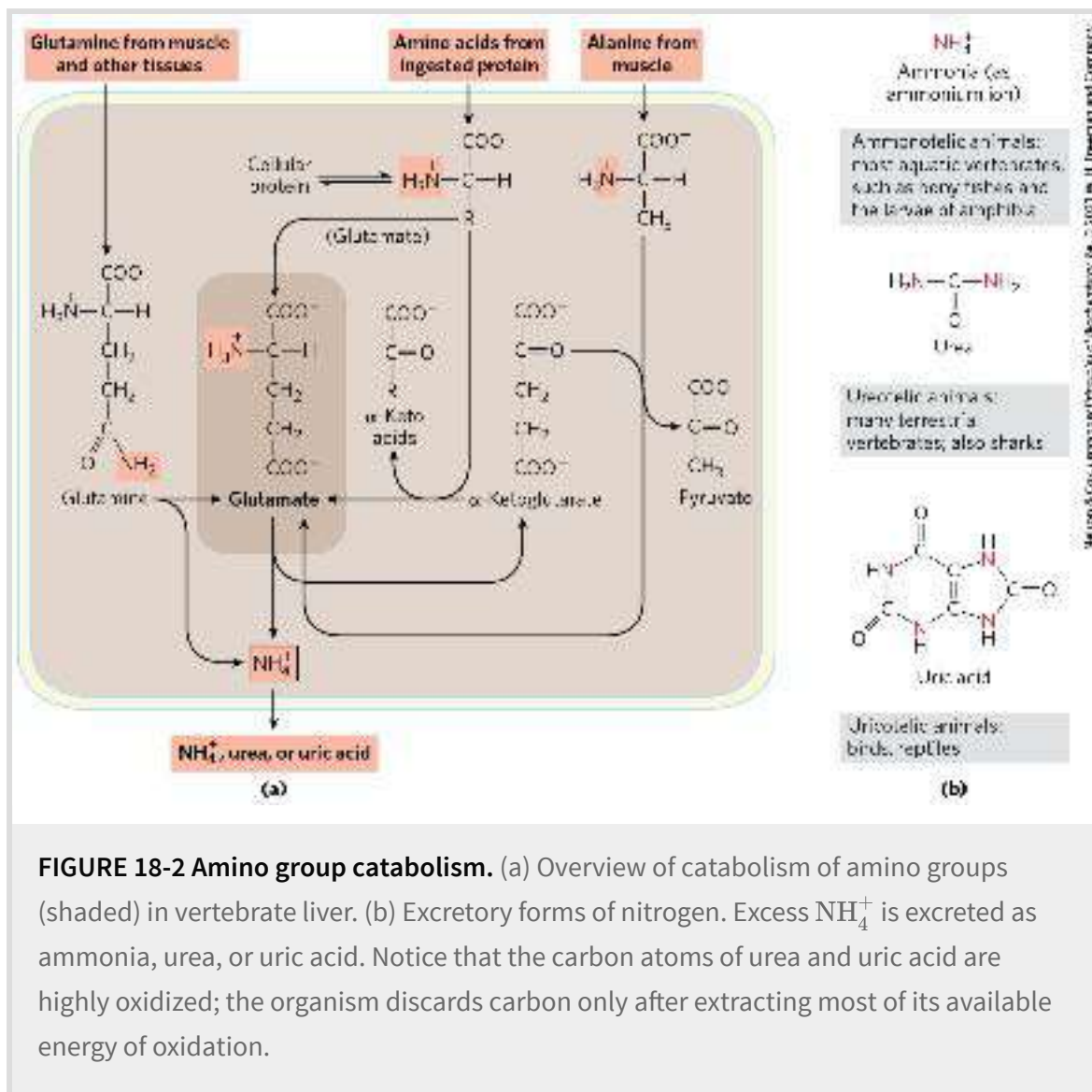
1. Amino acids released during normal protein turnover are not needed for new protein synthesis.
2. Ingested amino acids exceed the body's needs for protein synthesis.
3. Cellular proteins are used as fuel because carbohydrates are either unavailable or not properly utilized due to starvation or uncontrolled diabetes mellitus.

The pathways of amino acid catabolism are quite similar in most organisms. The focus of this chapter is on the pathways in vertebrates, because these have received the most research attention. As in carbohydrate and fatty acid catabolism, the processes of amino acid degradation converge on the central catabolic pathways, with the carbon skeletons of most amino acids finding their way to the citric acid cycle. In some cases, the reaction pathways of amino acid breakdown closely parallel steps in the catabolism of fatty acids (see [Fig. 17-9](#)). We begin with amino group metabolism and nitrogen excretion, before turning to the fate of the carbon skeletons derived from the amino acids; along the way we see how the pathways are interconnected.

18.1 Metabolic Fates of Amino Groups

Nitrogen, N_2 , is abundant in the atmosphere but is too unreactive for use in most biochemical processes. Reduced nitrogen is essential for life but bioenergetically expensive. Only a few microorganisms can convert N_2 to biologically useful forms such as NH_3 ([Chapter 22](#)). Amino groups are thus used efficiently in biological systems, and the lack of reactive nitrogen can limit growth.

 If not reused for the synthesis of new amino acids or other nitrogenous products, amino groups are channeled into a single excretory end product ([Fig. 18-2](#)). Most aquatic species, such as the bony fishes, are [ammonotelic](#), excreting amino nitrogen as ammonia. The toxic ammonia is simply diluted in the surrounding water. Terrestrial animals require pathways for nitrogen excretion that minimize toxicity and water loss. Most terrestrial animals are [ureotelic](#), excreting amino nitrogen in the form of urea; birds and reptiles are [uricotelic](#), excreting amino nitrogen as uric acid. (The pathway of uric acid synthesis is described in [Fig. 22-48](#).) Plants recycle virtually all amino groups — they excrete nitrogen only under rare circumstances. Reactive nitrogen excreted in any form is rapidly assimilated into the global nitrogen web (see [Fig. 22-1](#)), metabolized by microorganisms that are ubiquitous in aqueous and soil environments.



[Figure 18-2a](#) provides an overview of the catabolic pathways of ammonia and amino groups in vertebrates. Glutamine, glutamate, and alanine are prominent in this scheme. Amino acids derived from dietary protein are the source of most amino groups. Most amino acids are metabolized in the liver. Some of the ammonia generated in this process is recycled and used in biosynthetic pathways where glutamine, glutamate, and aspartate play major roles ([Chapter 22](#)). Excess amino groups are either excreted directly or converted to urea or uric acid for excretion,

depending on the organism ([Fig. 18-2b](#)). In mammals, including marsupials, most excess ammonia generated in other (extrahepatic) tissues travels to the liver for conversion to urea.

 The special place of glutamate, glutamine, alanine, and aspartate in nitrogen metabolism is not an evolutionary accident. These particular amino acids are the ones most easily converted into citric acid cycle intermediates: glutamate and glutamine to α -ketoglutarate, alanine to pyruvate, and aspartate to oxaloacetate. Glutamate and glutamine are especially important, acting as general collection points for amino groups. In the cytosol of liver cells (hepatocytes), amino groups from most amino acids are transferred to α -ketoglutarate to form glutamate, which enters mitochondria and gives up its amino group to form NH_4^+ . Excess ammonia generated in most other tissues is converted to the amide nitrogen of glutamine, which passes to the liver, then into liver mitochondria. Glutamine or glutamate or both are present in higher concentrations than other amino acids in most tissues. In skeletal muscle, excess amino groups are generally transferred to pyruvate to form alanine, another important molecule in the transport of amino groups to the liver. We begin with a discussion of the breakdown of dietary proteins, then proceed to a general description of the metabolic fates of amino groups.

Dietary Protein Is Enzymatically Degraded to Amino Acids

In humans, the degradation of ingested proteins to their constituent amino acids occurs in the gastrointestinal tract. Entry of dietary protein into the stomach stimulates the gastric mucosa to secrete the hormone **gastrin**, which in turn stimulates the secretion of hydrochloric acid by the parietal cells and pepsinogen by the chief cells of the gastric glands ([Fig. 18-3a](#)). The acidic gastric juice (pH 1.0 to 2.5) is both an antiseptic, killing most bacteria and other foreign cells, and a denaturing agent, unfolding globular proteins and rendering their internal peptide bonds more accessible to enzymatic hydrolysis. **Pepsinogen** (M_r 40,554), an inactive precursor, or zymogen ([p. 220](#)), is converted to active pepsin (M_r 34,614) by an autocatalytic cleavage (a cleavage mediated by the pepsinogen itself) that occurs only at low pH. In the stomach, pepsin cleaves long polypeptide chains into a mixture of smaller peptides.

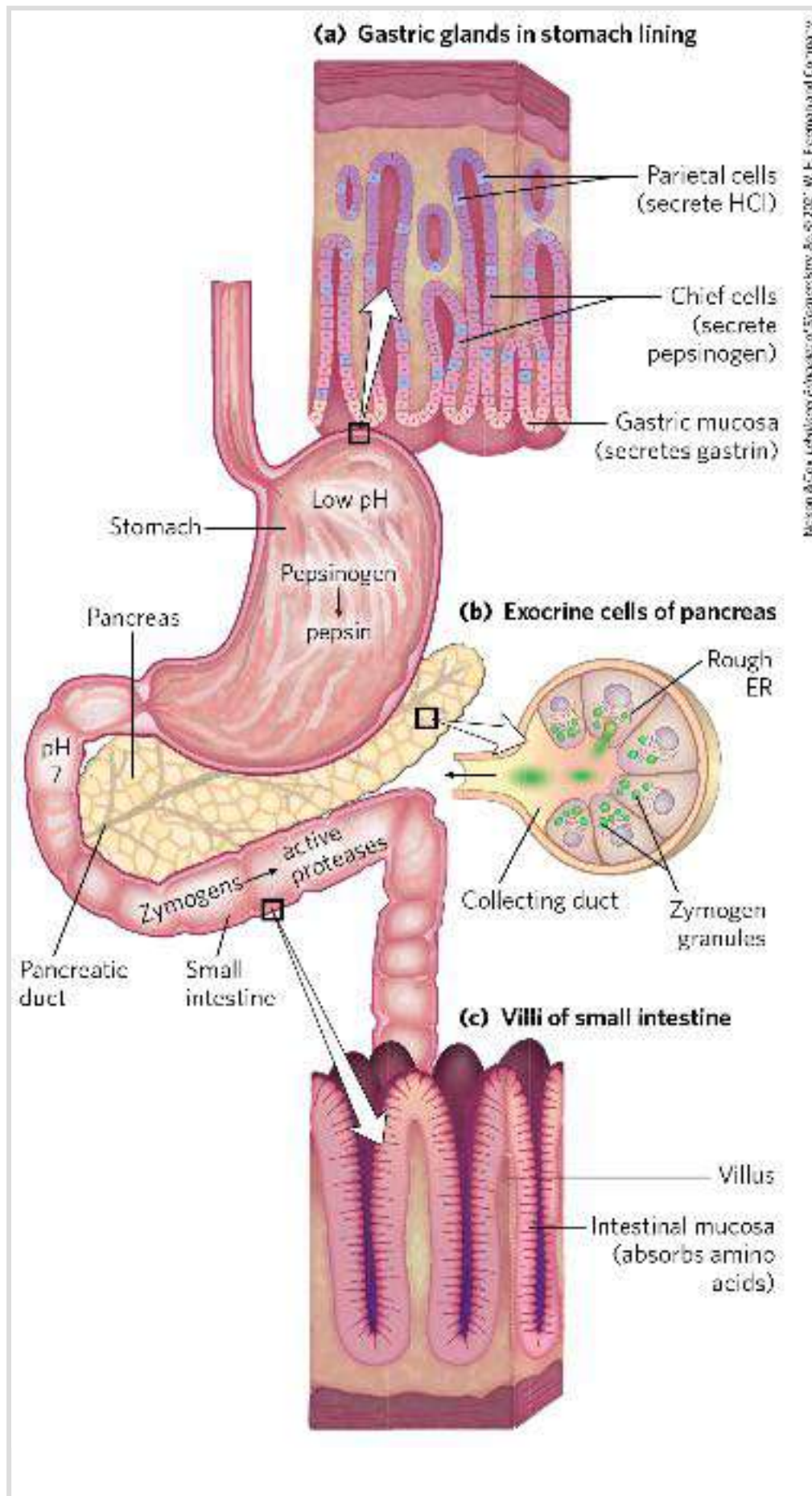


FIGURE 18-3 Part of the human digestive (gastrointestinal) tract. (a) The parietal cells and chief cells of the gastric glands secrete their products in response to the hormone gastrin. Pepsin begins the process of protein degradation in the stomach. (b) The cytoplasm of exocrine cells of the pancreas is completely filled with rough endoplasmic reticulum, the site of synthesis of the zymogens of many digestive enzymes. The zymogens are concentrated in membrane-enclosed transport particles called zymogen granules. When an exocrine cell is stimulated, its plasma membrane fuses with the zymogen granule membrane and zymogens are released into the lumen of the collecting duct by exocytosis. The collecting ducts ultimately lead to the pancreatic duct and from there to the small intestine. (c) In the small intestine, amino acids are absorbed through the epithelial cell layer (intestinal mucosa) of the villi and enter the capillaries.

As the acidic stomach contents pass into the small intestine, the low pH triggers secretion of the hormone **secretin** into the blood. Secretin stimulates the pancreas to secrete bicarbonate into the small intestine to neutralize the gastric HCl, abruptly increasing the pH to about 7. Arrival of peptides in the upper part of the intestine (duodenum) causes release into the blood of the hormone **cholecystokinin**, which stimulates secretion of several pancreatic proteases with activity optima at pH 7 to 8. Trypsinogen, chymotrypsinogen, and procarboxypeptidases A and B—the zymogens of **trypsin**, **chymotrypsin**, and **carboxypeptidases A and B**—are synthesized and secreted by the exocrine cells of the pancreas ([Fig. 18-3b](#)). Trypsinogen is converted to its active form, trypsin, by **enteropeptidase**, a proteolytic enzyme secreted by intestinal cells. Free trypsin then catalyzes the conversion of additional trypsinogen to trypsin (see [Fig. 6-42](#)). Trypsin also activates chymotrypsinogen, the procarboxypeptidases, and proelastase.

Why this elaborate mechanism for getting active digestive enzymes into the gastrointestinal tract? Synthesis of the enzymes as inactive precursors protects the exocrine cells from destructive proteolytic attack. The pancreas further protects itself against self-digestion by making a specific inhibitor, a protein called **pancreatic trypsin inhibitor** ([p. 220](#)). Given the key role of trypsin in proteolytic activation pathways, inhibition of trypsin effectively prevents premature production of active proteolytic enzymes within pancreatic cells.

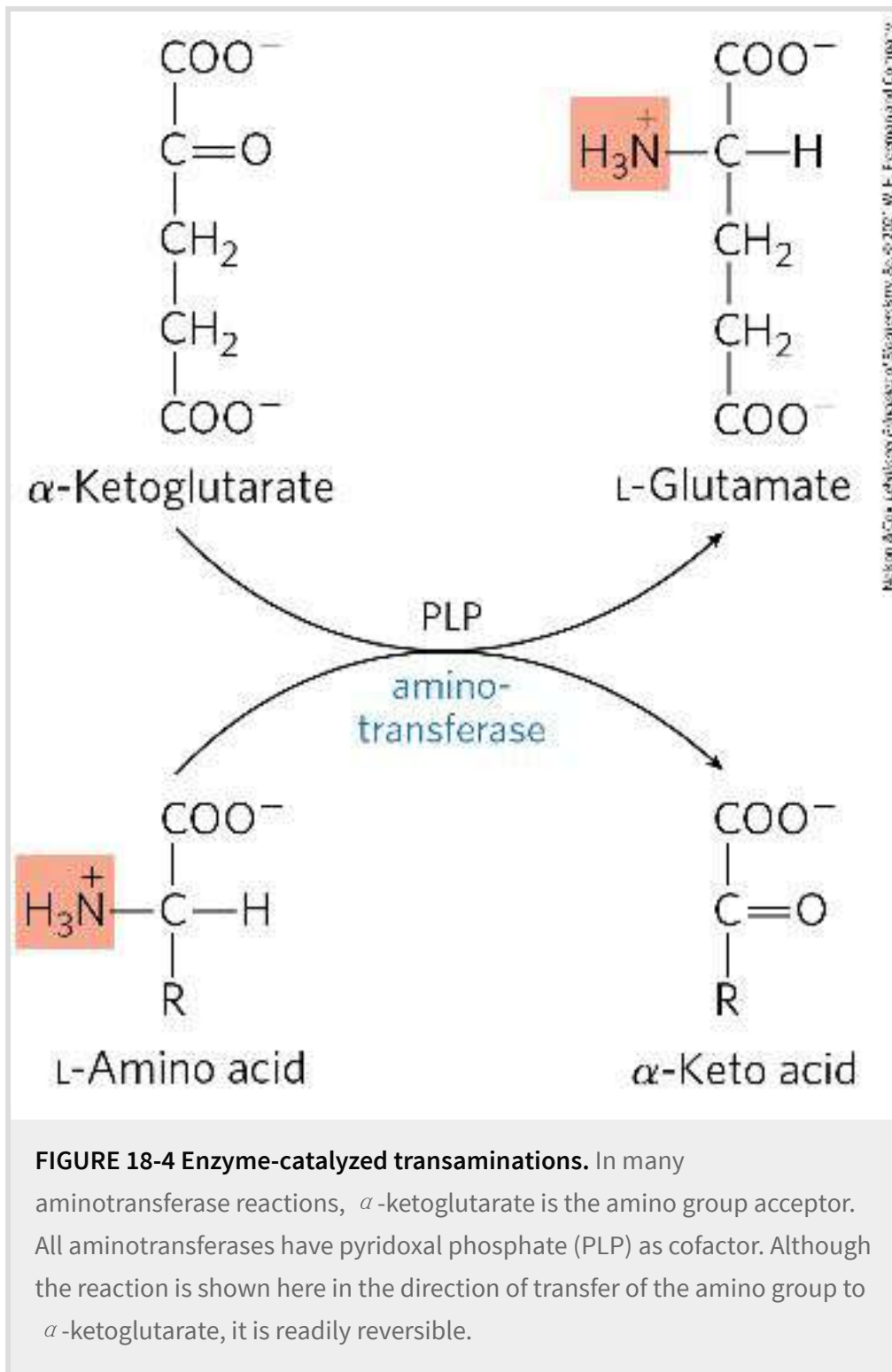
Trypsin and chymotrypsin further hydrolyze the peptides that were produced by pepsin in the stomach. This stage of protein digestion is accomplished very efficiently, because pepsin, trypsin, chymotrypsin, and the carboxypeptidases have different catalytic specificities and cleave different sets of peptide bonds (see [Table 3-6](#)). The resulting mixture of free amino acids is transported into the epithelial cells lining the small intestine ([Fig. 18-3c](#)), through which the amino acids enter the blood capillaries in the villi and travel to the liver.



Acute pancreatitis is a disease caused by obstruction of the normal pathway by which pancreatic secretions enter the intestine. The zymogens of the proteolytic enzymes are converted to their catalytically active forms prematurely, *inside* the pancreatic cells, and attack the pancreatic tissue itself. This causes excruciating pain and damage to the organ that can prove fatal. ■

Pyridoxal Phosphate Participates in the Transfer of α -Amino Groups to α -Ketoglutarate

 The first step in the catabolism of most L-amino acids, once they have reached the liver, is removal of the α -amino groups, promoted by enzymes called aminotransferases or transaminases. In these transamination reactions, the α -amino group is transferred to the α -carbon atom of α -ketoglutarate, leaving behind the corresponding α -keto acid analog of the amino acid (Fig. 18-4). There is no net deamination (loss of amino groups) in these reactions, because the α -ketoglutarate becomes aminated as the α -amino acid is deaminated.  These are the reactions that effectively collect the amino groups from many different amino acids in the form of L-glutamate.

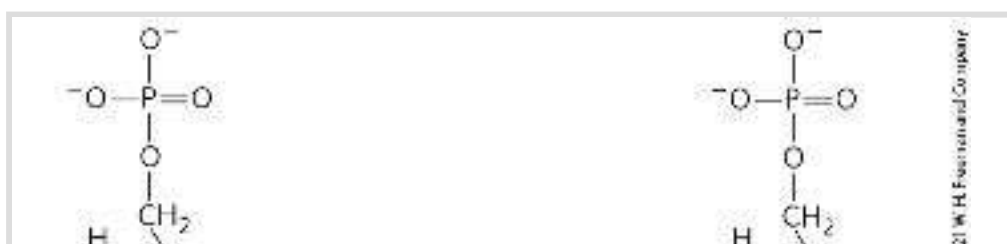


Cells contain different types of aminotransferases. Many are specific for α -ketoglutarate as the amino group acceptor but differ in their specificity for the L-amino acid. The enzymes are

named for the amino group donor (alanine aminotransferase and aspartate aminotransferase, for example). The reactions catalyzed by aminotransferases are freely reversible, having an equilibrium constant of about 1.0 ($\Delta G' \approx 0$ kJ/mol).

P1 All aminotransferases have **pyridoxal phosphate (PLP)**, the coenzyme form of pyridoxine, or vitamin B₆, as a prosthetic group. We encountered pyridoxal phosphate in [Chapter 15](#), as a coenzyme in the glycogen phosphorylase reaction, but its role in that reaction is not representative of its usual coenzyme function. Its primary role in cells is in the metabolism of molecules with amino groups.

Pyridoxal phosphate functions as an intermediate carrier of amino groups at the active site of aminotransferases. It undergoes reversible transformations between its aldehyde form, pyridoxal phosphate, which can accept an amino group, and its aminated form, pyridoxamine phosphate, which can donate its amino group to an α -keto acid ([Fig. 18-5a](#)). Pyridoxal phosphate is generally covalently bound to the enzyme's active site through an aldimine (Schiff base) linkage to the ϵ -amino group of a Lys residue ([Fig. 18-5b, c](#)). This linkage is replaced by the amino group of the amino acid as the first step in most PLP-catalyzed reactions ([Fig. 18-5d](#)).



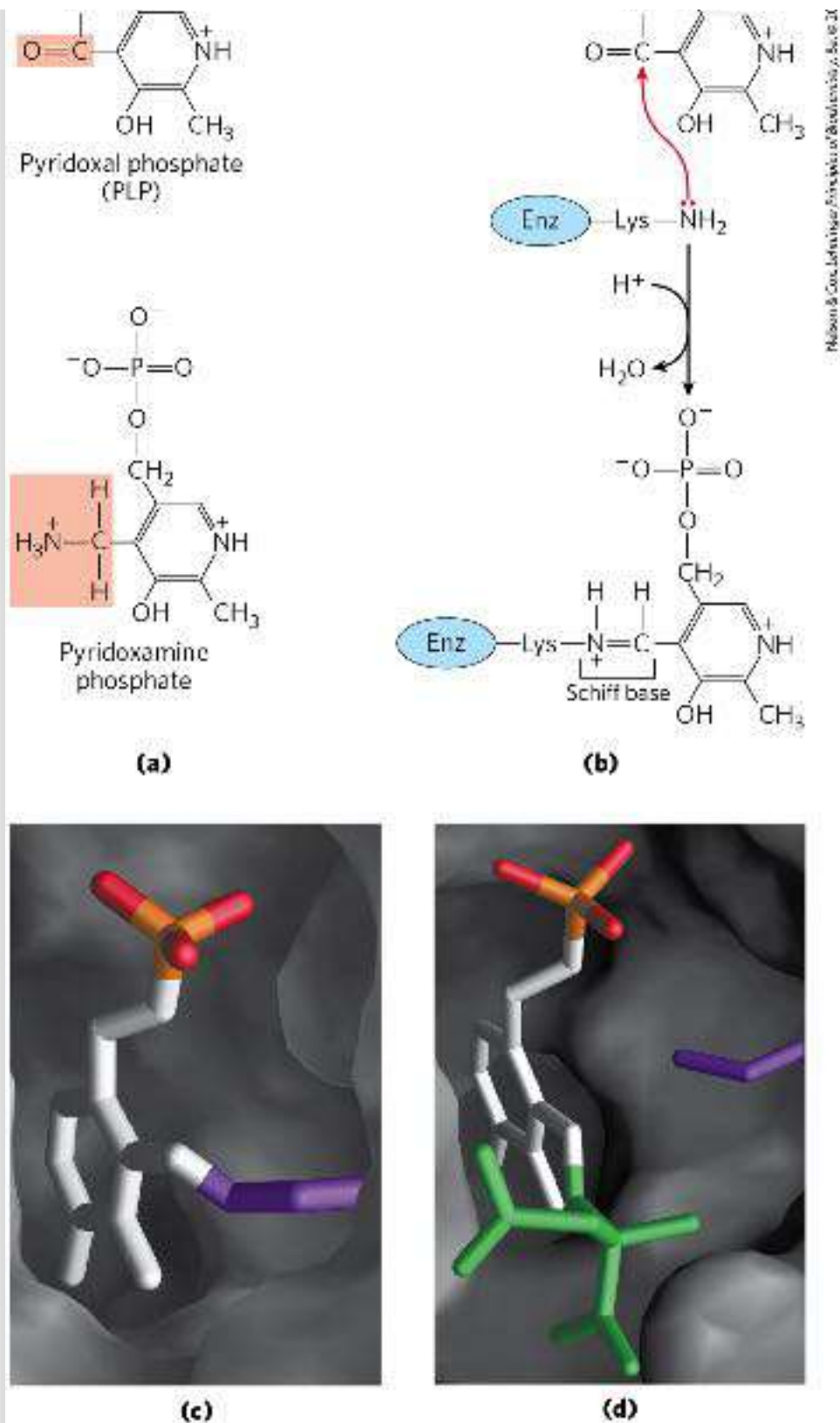
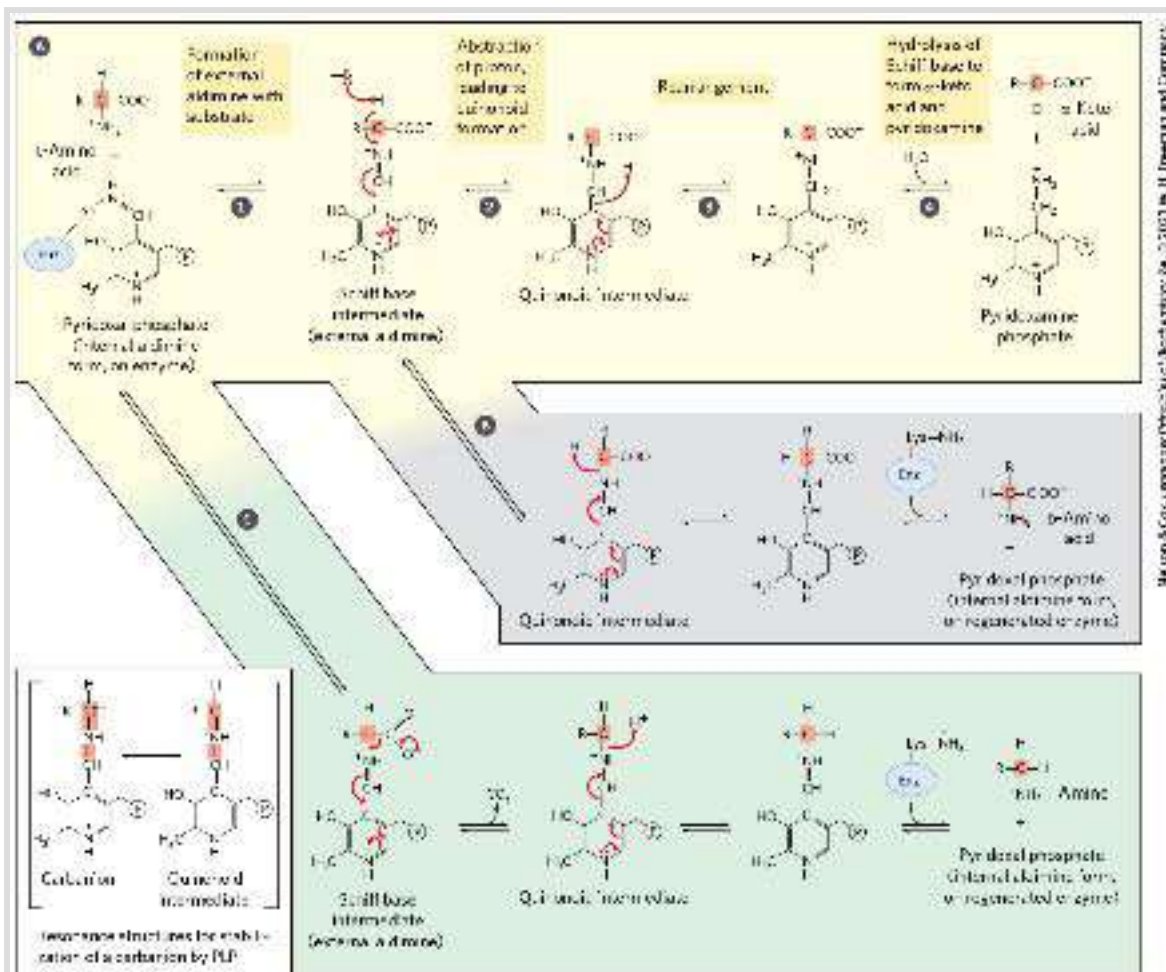


FIGURE 18-5 Pyridoxal phosphate, the prosthetic group of aminotransferases. (a) Pyridoxal phosphate (PLP) and its aminated form, pyridoxamine phosphate, are the tightly bound coenzymes of

aminotransferases. The functional groups are shaded. (b) Pyridoxal phosphate is bound to the enzyme through noncovalent interactions and a Schiff-base (aldimine) linkage to a Lys residue at the active site. The steps in the formation of a Schiff base from a primary amine and a carbonyl group are detailed in [Figure 14-5](#). (c, d) Close-up views of the active site of aspartate aminotransferase, with PLP (white, with the phosphoryl group in orange and red). In (c) PLP is in aldimine linkage with the side chain of Lys²⁵⁸ (purple). In (d) PLP is linked to the substrate analog 2-methylaspartate (green) via a Schiff base. [(c, d) Data from PDB ID 1AJS, S. Rhee et al., *J. Biol. Chem.* 272:17,293, 1997.]

Pyridoxal phosphate participates in a variety of reactions in amino acid metabolism ([Section 18.3](#) and [Chapter 22](#)), facilitating reactions at the α , β , and γ carbons (C-2 to C-4) of amino acids. Reactions at the α carbon ([Fig. 18-6](#)) include racemizations (interconverting L- and D-amino acids) and decarboxylations, as well as transaminations. Pyridoxal phosphate plays the same chemical role in each of these reactions. A bond to the α carbon of the substrate is broken, removing either a proton or a carboxyl group. Pyridoxal phosphate provides resonance stabilization of the otherwise unstable carbanion intermediate ([Fig. 18-6](#), inset). The highly conjugated structure of PLP (an electron sink) permits delocalization of the negative charge.



MECHANISM FIGURE 18-6 Some amino acid transformations at the α carbon that are facilitated by pyridoxal phosphate. Pyridoxal phosphate is generally bonded to the enzyme through a Schiff base, also called an internal aldimine. This activated form of PLP readily undergoes transamination to form a new Schiff base (external aldimine) with the α -amino group of the substrate amino acid (see [Fig. 18-5b, d](#)). Three alternative fates for the external aldimine are shown: **A** transamination, **B** racemization, and **C** decarboxylation. The PLP–amino acid Schiff base is in conjugation with the pyridine ring, an electron sink that permits delocalization of an electron pair to avoid formation of an unstable carbanion on the α carbon (inset). A quinonoid intermediate is involved in all three types of reactions. The transamination route **A** is especially important in the pathways described in this chapter. This pathway (shown left to right) represents only part of the overall reaction catalyzed by aminotransferases. To complete the process, a second α -keto acid replaces the one that is released, and this is converted to an amino acid in a reversal of the reaction steps (right to left). Pyridoxal phosphate is also involved in certain reactions at the β and γ carbons of some amino acids (not shown).

Aminotransferases ([Fig. 18-6a](#)) are classic examples of enzymes catalyzing bimolecular Ping-Pong reactions (see [Fig. 6-15b, d](#)), in which the first substrate reacts and the product must leave the active site before the second substrate can bind. Thus, the incoming amino acid binds to the active site, donates its amino group to pyridoxal phosphate, and departs in the form of an α -keto acid. The incoming α -keto acid then binds, accepts the amino group from pyridoxamine phosphate, and departs in the form of an amino acid.

Glutamate Releases Its Amino Group as Ammonia in the Liver

As we shall see, the urea cycle begins with free ammonia in the mitochondria of hepatocytes.  The delivery of NH_4^+ to these mitochondria is streamlined by collecting the amino groups of many different α -amino acids in the liver in one of two forms: the amino group of L-glutamate or the amide nitrogen of glutamine ([Fig. 18-2](#)).

As the product of many aminotransferase reactions, glutamate has a central role. In hepatocytes, glutamate is transported from the cytosol into mitochondria. Here, it undergoes **oxidative deamination** catalyzed by **L-glutamate dehydrogenase** (M_r 330,000) to produce NH_4^+ and α -ketoglutarate. In mammals, this enzyme is present in the mitochondrial matrix. It is unusual

in that it can use either NAD^+ or NADP^+ as the acceptor of reducing equivalents ([Fig. 18-7](#)).

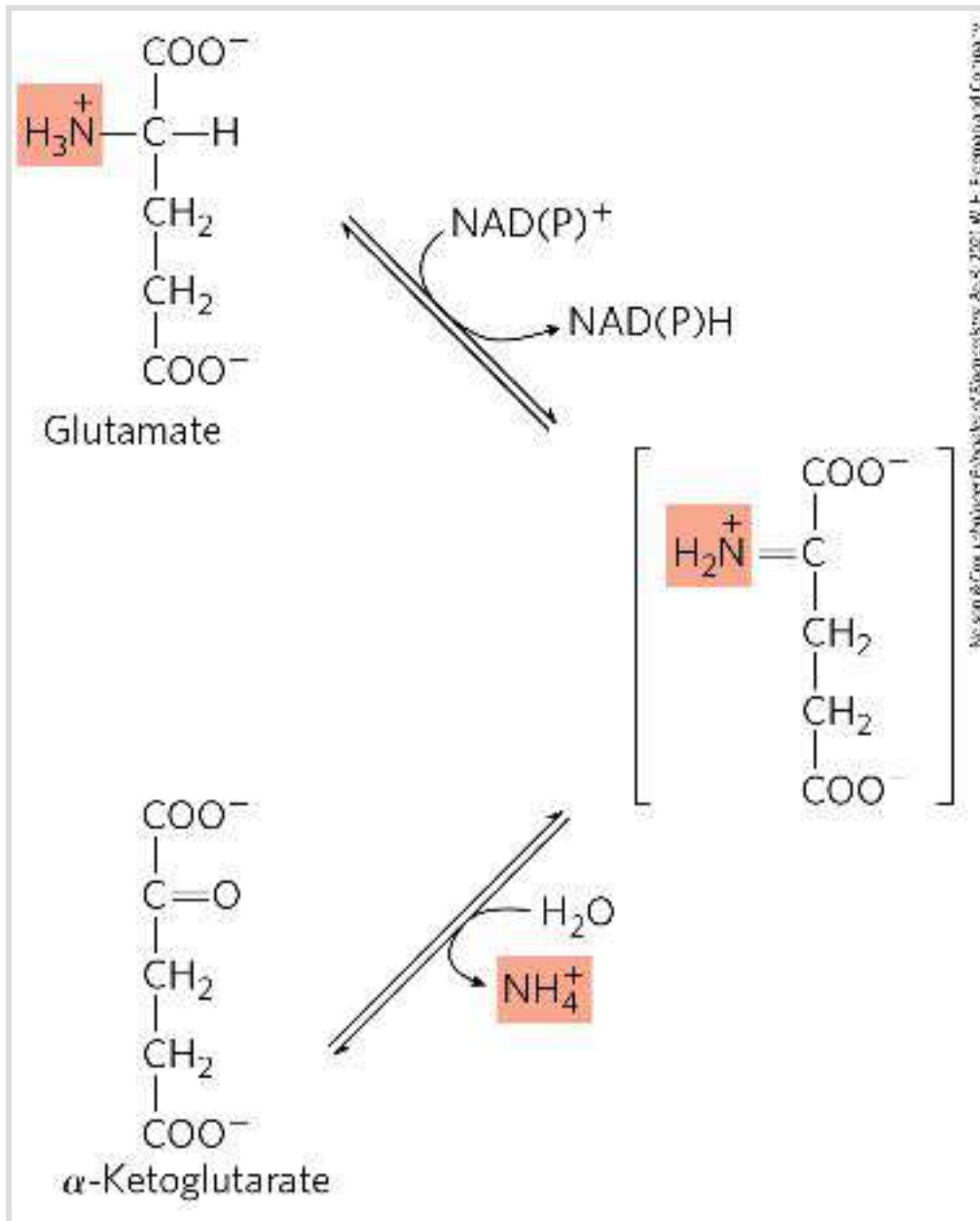


FIGURE 18-7 Reaction catalyzed by glutamate dehydrogenase. The glutamate dehydrogenase of mammalian liver has the unusual capacity to use either NAD^+ or NADP^+ as cofactor. The glutamate dehydrogenases of plants and microorganisms are generally specific for one or the other. The mammalian enzyme is allosterically regulated by GTP and ADP.

The combined action of an aminotransferase and glutamate dehydrogenase is referred to as **transdeamination**. A few amino acids bypass the transdeamination pathway and undergo direct oxidative deamination. The fate of the NH_4^+ produced by any of these deamination processes is discussed in detail in [Section 18.2](#).

 The α -ketoglutarate formed from glutamate deamination can be used either in the citric acid cycle or for glucose synthesis.

   Glutamate dehydrogenase operates at an important intersection of carbon and nitrogen metabolism. Its α -ketoglutarate product can be oxidized as fuel or serve as a glucose precursor in gluconeogenesis. An allosteric enzyme with six identical subunits, its activity is influenced by a complicated array of allosteric modulators. The best-studied of these are the positive modulator ADP and the negative modulator GTP. Although the metabolic rationale for this regulatory pattern has not been elucidated in detail, ADP can signal low glucose levels and GTP is a product of the citric acid cycle that can signal high levels of α -ketoglutarate (see [Fig. 16-7](#)).

 Mutations that alter the allosteric binding site for GTP or otherwise cause permanent activation of glutamate dehydrogenase lead to a human genetic disorder called hyperinsulinism-hyperammonemia syndrome, characterized by hypersecretion of insulin after a protein meal. This results in elevated levels of ammonia in the bloodstream and hypoglycemia. 

Glutamine Transports Ammonia in the Bloodstream

Glutamine is the second major source of ammonia in hepatocyte mitochondria and is particularly important for intercellular transport of ammonia.  Ammonia is quite toxic to animal tissues (we examine some possible reasons for this toxicity later). Significant amounts are present in blood, but the levels are tightly controlled. In many tissues, including the brain, some processes such as nucleotide degradation generate free ammonia. In most animals, much of the free ammonia is converted to a nontoxic compound before export from the extrahepatic tissues into the blood and transport to the liver or kidneys. For this transport function, glutamate, critical to *intracellular* amino group metabolism, is supplanted by L-glutamine. The free ammonia produced in tissues is combined with glutamate to yield glutamine by the action of **glutamine synthetase**. This reaction requires ATP and occurs in two steps ([Fig. 18-8](#)). First, glutamate and ATP react to form ADP and a γ -glutamyl phosphate intermediate, which then reacts with ammonia to produce glutamine and inorganic phosphate. In addition to its transport role, glutamine serves as a source of amino groups in a variety of biosynthetic reactions. Glutamine synthetase is found in all organisms, always playing a central metabolic role. In microorganisms, the enzyme serves as an essential portal for the entry of fixed nitrogen into biological systems. (The roles of glutamine and glutamine synthetase in metabolism are further discussed in [Chapter 22](#).)

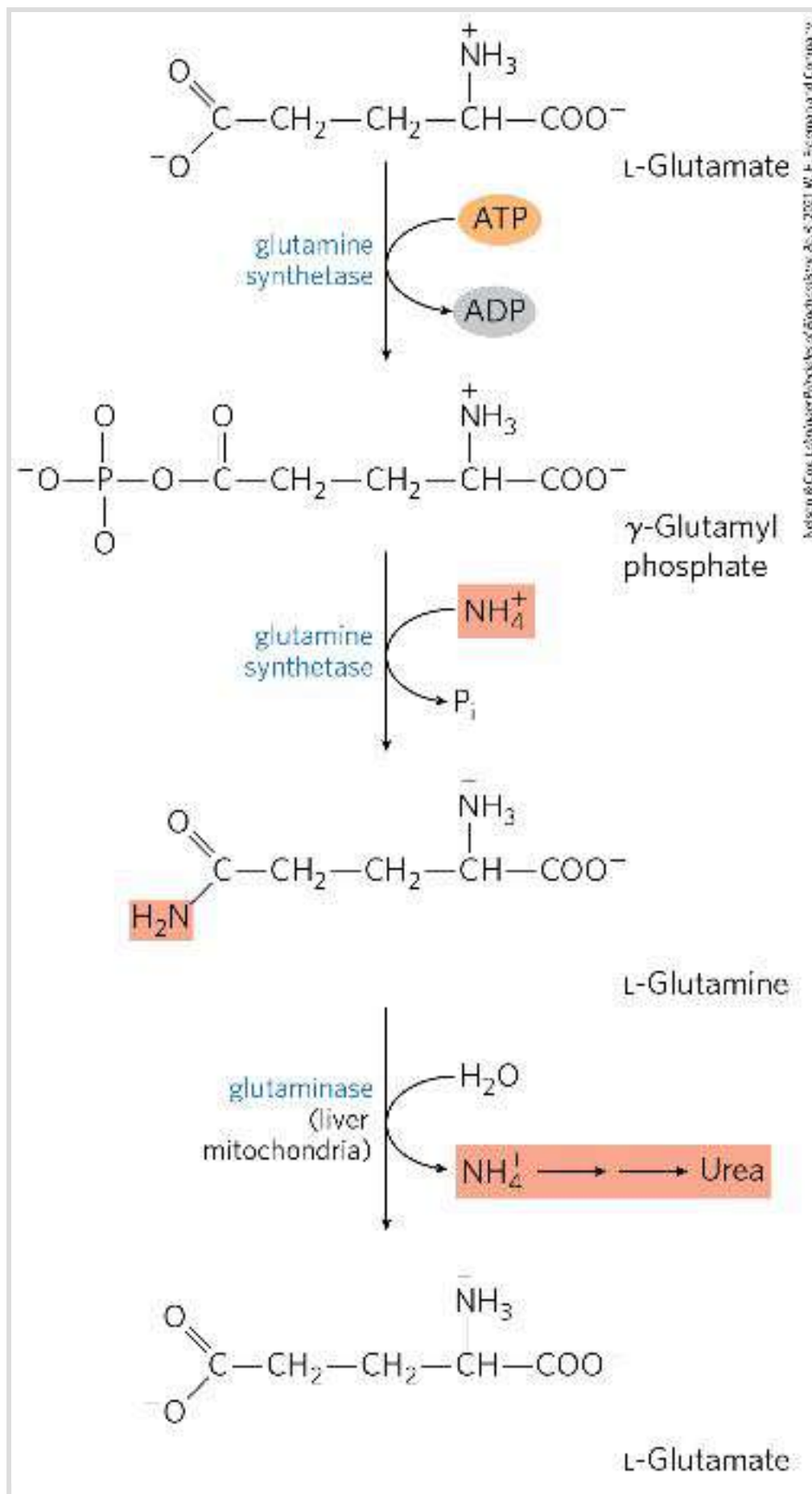


FIGURE 18-8 Ammonia transport in the form of glutamine. Excess ammonia in tissues is added to glutamate to form glutamine, a process catalyzed by glutamine synthetase. After transport in the bloodstream, the glutamine enters the liver, and NH_4^+ is liberated in mitochondria by the enzyme glutaminase.

In most terrestrial animals, glutamine in excess of that required for biosynthesis is transported in the blood to the intestine, liver, and kidneys for processing. In these tissues, the amide nitrogen is released as ammonium ion in the mitochondria, where the enzyme **glutaminase** converts glutamine to glutamate and NH_4^+ ([Fig. 18-8](#)). The NH_4^+ from intestine and kidney is transported in the blood to the liver. In the liver, the ammonia from all sources is disposed of by urea synthesis. Some of the glutamate produced in the glutaminase reaction may be further processed in the liver by glutamate dehydrogenase ([Fig. 18-7](#)), releasing more ammonia and producing carbon skeletons for metabolic fuel.



In metabolic acidosis ([p. 621](#)) there is a regulated increase in glutamine processing by the kidneys. Much of the excess NH_4^+ thus produced is not destined for the bloodstream or converted to urea; instead, it is excreted directly into the urine where it forms salts with metabolic acids. The glutamine breakdown thus facilitates removal of those acids in the urine. Bicarbonate produced by the decarboxylation of α -ketoglutarate in the citric acid cycle can also serve as a buffer in blood plasma. Taken together, these effects of glutamine metabolism in the kidney tend to counteract acidosis. ■

Alanine Transports Ammonia from Skeletal Muscles to the Liver

Vigorously contracting skeletal muscles operate anaerobically, producing large amounts of pyruvate and lactate from glycolysis as well as ammonia from protein breakdown. These products must find their way to the liver, where pyruvate and lactate are incorporated into glucose, which is returned to the muscles, and ammonia is converted to urea for excretion. Pyruvate and alanine are readily interconverted via transamination with glutamate by the action of **alanine aminotransferase**). Thus, alanine largely supplants glutamine in the transport of amino groups from muscle to the liver in a nontoxic form ([Fig. 18-2a](#)), ultimately delivering the free ammonia to hepatocyte mitochondria via glutamate in a pathway called the **glucose-alanine cycle** ([Fig. 18-2](#)). In the cytosol of hepatocytes, alanine aminotransferase acts in reverse to transfer the amino group from alanine to α -ketoglutarate, forming pyruvate and glutamate. Glutamate can then enter mitochondria, where the glutamate dehydrogenase reaction releases NH_4^+ ([Fig. 18-7](#)), or it can undergo transamination with oxaloacetate to form aspartate, another nitrogen donor in urea synthesis, as we shall see.

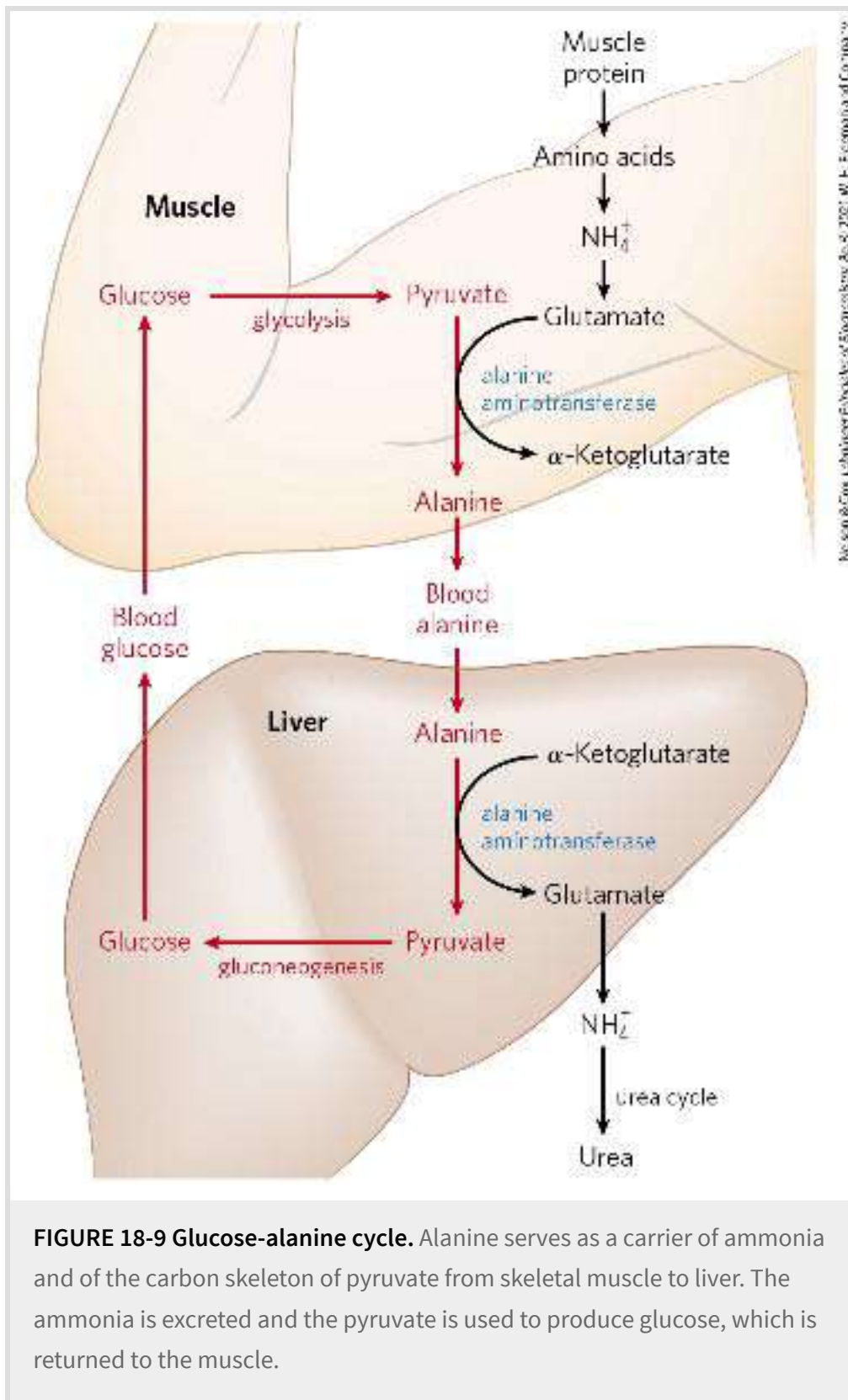


FIGURE 18-9 Glucose-alanine cycle. Alanine serves as a carrier of ammonia and of the carbon skeleton of pyruvate from skeletal muscle to liver. The ammonia is excreted and the pyruvate is used to produce glucose, which is returned to the muscle.

The use of alanine to transport ammonia from skeletal muscles to the liver is another example of the intrinsic economy of living

organisms. The energetic burden of gluconeogenesis is imposed on the liver rather than the muscle, and all available ATP in muscle is devoted to muscle contraction. The glucose-alanine cycle, in concert with the Cori cycle (see [Box 14-2](#) and [Fig. 23-17](#)), accomplishes this transaction.

Ammonia Is Toxic to Animals



The catabolic production of ammonia poses a serious biochemical problem, because ammonia is very toxic. The brain is particularly sensitive; damage from ammonia toxicity causes cognitive impairment, ataxia, and epileptic seizures. In extreme cases there is swelling of the brain leading to death. The molecular bases for this toxicity are gradually coming into focus. In the blood, about 98% of ammonia is in the protonated form (NH_4^+), which does not cross the plasma membrane. The small amount of NH_3 present readily crosses all membranes, including the blood-brain barrier, allowing it to enter cells, where much of it becomes protonated and can accumulate inside cells as NH_4^+ .

Ridding the cytosol of ammonia requires reductive amination of α -ketoglutarate to glutamate by glutamate dehydrogenase (the reverse of the reaction described earlier; [Fig. 18-7](#)) and conversion of glutamate to glutamine by glutamine synthetase. In the brain, only astrocytes — star-shaped cells of the nervous system that provide nutrients, support, and insulation for neurons — express glutamine synthetase. Glutamate and its

derivative γ -aminobutyrate (GABA; see [Fig. 22-31](#)) are important neurotransmitters; some of the sensitivity of the brain to ammonia may reflect depletion of glutamate in the glutamine synthetase reaction. However, glutamine synthetase activity is insufficient to deal with excess ammonia, or to fully explain its toxicity.

Increased $[\text{NH}_4^+]$ also alters the capacity of astrocytes to maintain potassium homeostasis across the membrane. NH_4^+ competes with K^+ for transport into the cell through the Na^+K^+ ATPase, resulting in elevated extracellular $[\text{K}^+]$. The excess extracellular K^+ enters neurons through a symporter, **$\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ cotransporter 1 (NKCC1)**, bringing Na^+ and 2Cl^- with it. Excess Cl^- in these neurons alters their response when the neurotransmitter GABA interacts with their GABA_A receptors, producing abnormal depolarization and increased neuronal activity that likely account for the neuromuscular incoordination and seizures that often result from ammonia poisoning. If extracellular $[\text{NH}_4^+]$ remains elevated, the perturbation of ion and aquaporin channels in astrocytes causes the cells to swell, resulting in fatal brain edema. 

As we close this discussion of amino group metabolism, note that we have described several processes that deposit excess ammonia in the mitochondria of hepatocytes ([Fig. 18-2](#)). We now look at the fate of that ammonia.

SUMMARY 18.1 *Metabolic Fates of Amino Groups*

- Humans derive a small fraction of their oxidative energy from the catabolism of amino acids. Proteases degrade ingested proteins in the stomach and small intestine. Most proteases are initially synthesized as inactive zymogens.
- An early step in the catabolism of amino acids is the separation of the amino group from the carbon skeleton. In most cases, the amino group is transferred to α -ketoglutarate to form glutamate. This transamination reaction requires pyridoxal phosphate, a coenzyme globally involved in amino acid metabolism.
- Glutamate, a central reservoir of metabolized amino groups, is transported to liver mitochondria, where glutamate dehydrogenase liberates the amino group as ammonium ion (NH_4^+).
- Ammonia formed in most other tissues is transported to the liver as the amide nitrogen of glutamine.
- To make use of the pyruvate and amino groups generated in hardworking skeletal muscle, the pyruvate is converted to alanine and transported to the liver within the glucose-alanine cycle.
- Free ammonia is toxic. Excess ammonia is often manifested in serious neurological damage.

18.2 Nitrogen Excretion and the Urea Cycle

In ureotelic organisms, the ammonia deposited in the mitochondria of hepatocytes is converted to urea in the [urea cycle](#) ([Fig. 18-10](#)). This pathway was discovered in 1932 by Hans Krebs (who later also discovered the citric acid cycle) and a medical student associate, Kurt Henseleit. Urea production occurs almost exclusively in the liver and is the fate of most of the ammonia channeled there. The urea passes into the bloodstream and thus to the kidneys and is excreted into the urine. The production of urea now becomes the focus of our discussion.

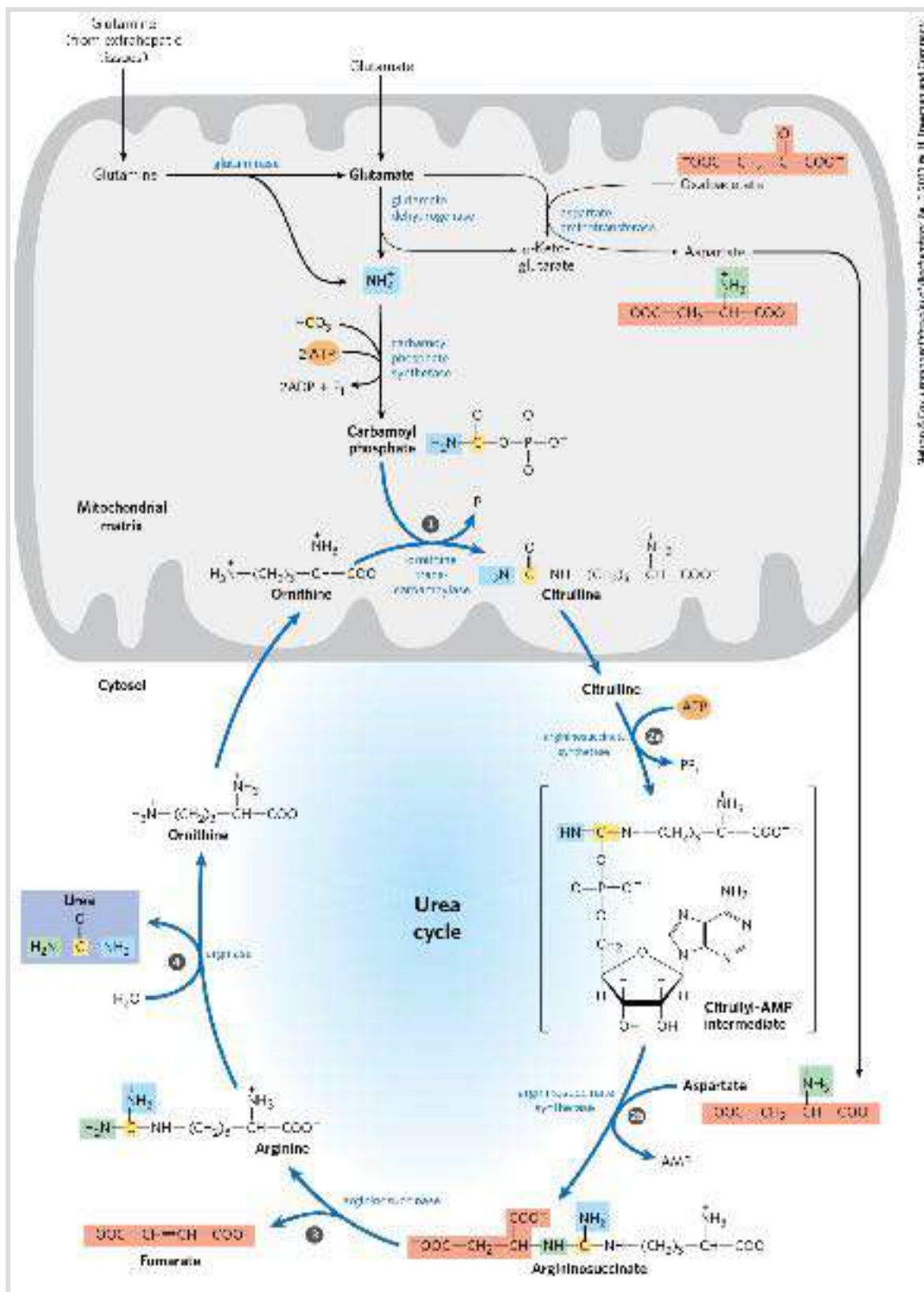
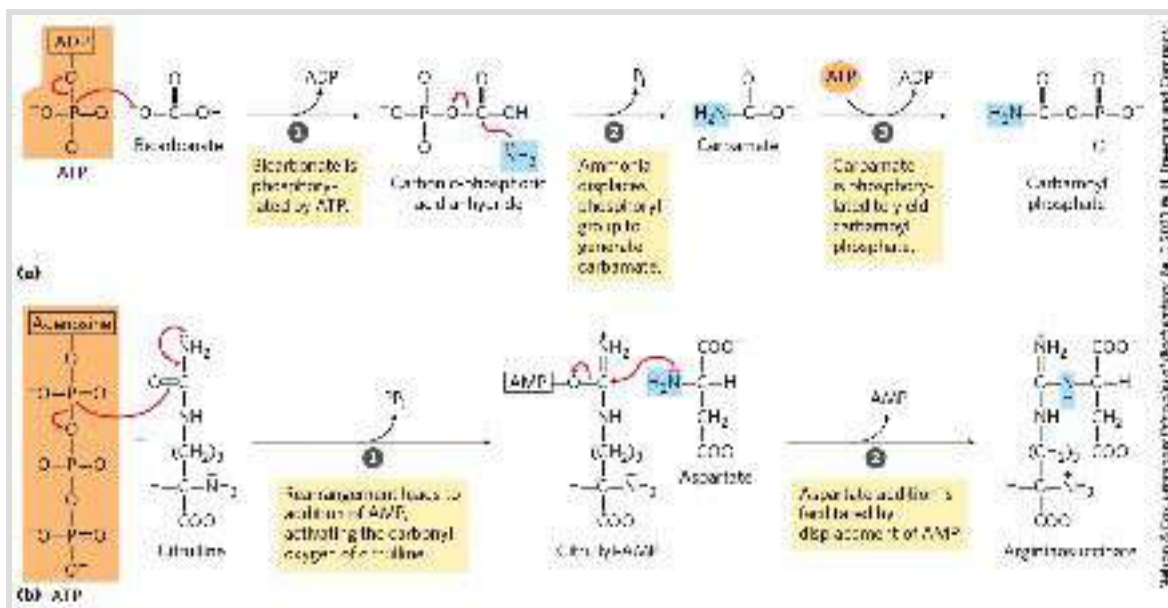


FIGURE 18-10 The urea cycle and reactions that feed amino groups into the cycle. The enzymes catalyzing these reactions (named in the text) are distributed between the mitochondrial matrix and the cytosol. One amino group enters the urea cycle as carbamoyl phosphate, formed in the matrix; the other enters as aspartate, formed in the

matrix by transamination of oxaloacetate and glutamate, catalyzed by aspartate aminotransferase. The urea cycle consists of four steps: **1** Formation of citrulline from ornithine and carbamoyl phosphate (entry of the first amino group); the citrulline passes into the cytosol. **2** Formation of argininosuccinate through a citrullyl-AMP intermediate (entry of the second amino group). **3** Formation of arginine from argininosuccinate; this reaction releases fumarate, which enters the citric acid cycle. **4** Formation of urea; this reaction also regenerates ornithine.

Urea Is Produced from Ammonia in Five Enzymatic Steps

The urea cycle begins inside liver mitochondria, but three of the subsequent steps take place in the cytosol; the cycle thus spans two cellular compartments ([Fig. 18-10](#)). The first amino group to enter the urea cycle is derived from ammonia in the mitochondrial matrix — most of this NH_4^+ arises by the pathways described in [Section 18.1](#). The liver also receives some ammonia via the portal vein from the intestine, from the bacterial oxidation of amino acids. Whatever its source, the NH_4^+ generated in liver mitochondria is immediately used, together with CO_2 (as HCO_3^-) produced by mitochondrial respiration, to form carbamoyl phosphate in the matrix (as shown in [Fig. 18-10](#) and explained in detail in [Fig. 18-11a](#)). This ATP-dependent reaction is catalyzed by **carbamoyl phosphate synthetase I**, a regulatory enzyme (see below). The mitochondrial form of the enzyme is distinct from the cytosolic (II) form, which has a separate function in pyrimidine biosynthesis ([Chapter 22](#)).



MECHANISM FIGURE 18-11 Nitrogen-acquiring reactions in the synthesis of urea. The urea nitrogens are acquired in two separate reactions, each requiring ATP. (a) In the first reaction, catalyzed by carbamoyl phosphate synthetase I, the first nitrogen enters from ammonia. The terminal phosphate groups of two molecules of ATP are used to form one molecule of carbamoyl phosphate. In other words, this reaction has two activation steps (1 and 3). (b) In the second reaction, catalyzed by argininosuccinate synthetase, the second nitrogen enters from aspartate. This reaction has two steps. Activation of the ureido oxygen of citrulline in step 1 sets up the addition of aspartate in step 2.

The carbamoyl phosphate, which functions as an activated carbamoyl group donor, now enters the urea cycle. The cycle has only four enzymatic steps. First, carbamoyl phosphate donates its carbamoyl group to ornithine to form citrulline, with the release of P_i ([Fig. 18-10](#), step 1). The reaction is catalyzed by **ornithine transcarbamoylase**. Ornithine is not one of the 20 common amino acids found in proteins, but it is a key intermediate in arginine biosynthesis and nitrogen metabolism in general. It is synthesized from glutamate in a five-step pathway described in [Chapter 22](#). Ornithine plays a role resembling that of oxaloacetate in the citric acid cycle, accepting material at each turn of the urea

cycle. The citrulline produced in the first step of the urea cycle passes from the mitochondrion to the cytosol.

The next two steps bring in the second amino group, featuring aspartate as the amino group donor. The aspartate is generated in the mitochondria by transamination between glutamate and oxaloacetate, and then transported into the cytosol. A condensation reaction between the amino group of aspartate and the ureido (carbonyl) group of citrulline forms argininosuccinate (step ② in [Fig. 18-10](#)). This cytosolic reaction, catalyzed by **argininosuccinate synthetase**, requires ATP and proceeds through a citrullyl-AMP intermediate ([Fig. 18-11b](#)). The argininosuccinate is then cleaved by **argininosuccinase** (step ③ in [Fig. 18-10](#)) to form free arginine and fumarate, the latter being converted to malate before entering mitochondria to join the pool of citric acid cycle intermediates. This is the only reversible step in the urea cycle. In the last reaction of the urea cycle (step ④), the cytosolic enzyme **arginase** cleaves arginine to yield **urea** and ornithine. Ornithine is transported into the mitochondrion to initiate another round of the urea cycle.

As we noted in [Chapter 16](#), the enzymes of many metabolic pathways are clustered in metabolons ([p. 595](#)), with the product of one enzyme reaction being channeled directly to the next enzyme in the pathway. In the urea cycle, the mitochondrial and cytosolic enzymes seem to be clustered in this way. The citrulline transported out of the mitochondrion is not diluted into the general pool of metabolites in the cytosol but is passed directly to the active site of argininosuccinate synthetase. This channeling

between enzymes continues for argininosuccinate, arginine, and ornithine. Only urea is released into the general cytosolic pool of metabolites.

The Citric Acid and Urea Cycles Can Be Linked

The fumarate produced in the argininosuccinase reaction is also an intermediate of the citric acid cycle. Thus, the cycles are, in principle, interconnected — in a process dubbed the “Krebs bicycle” ([Fig. 18-12](#)). However, each cycle can operate independently, and communication between them depends on the transport of key intermediates between the mitochondrion and cytosol. Major transporters in the inner mitochondrial membrane include the malate– α -ketoglutarate transporter, the glutamate-aspartate transporter, and the glutamate- OH^- transporter. Together, these transporters facilitate the movement of malate and glutamate into the mitochondrial matrix and the movement of aspartate and α -ketoglutarate out to the cytosol.

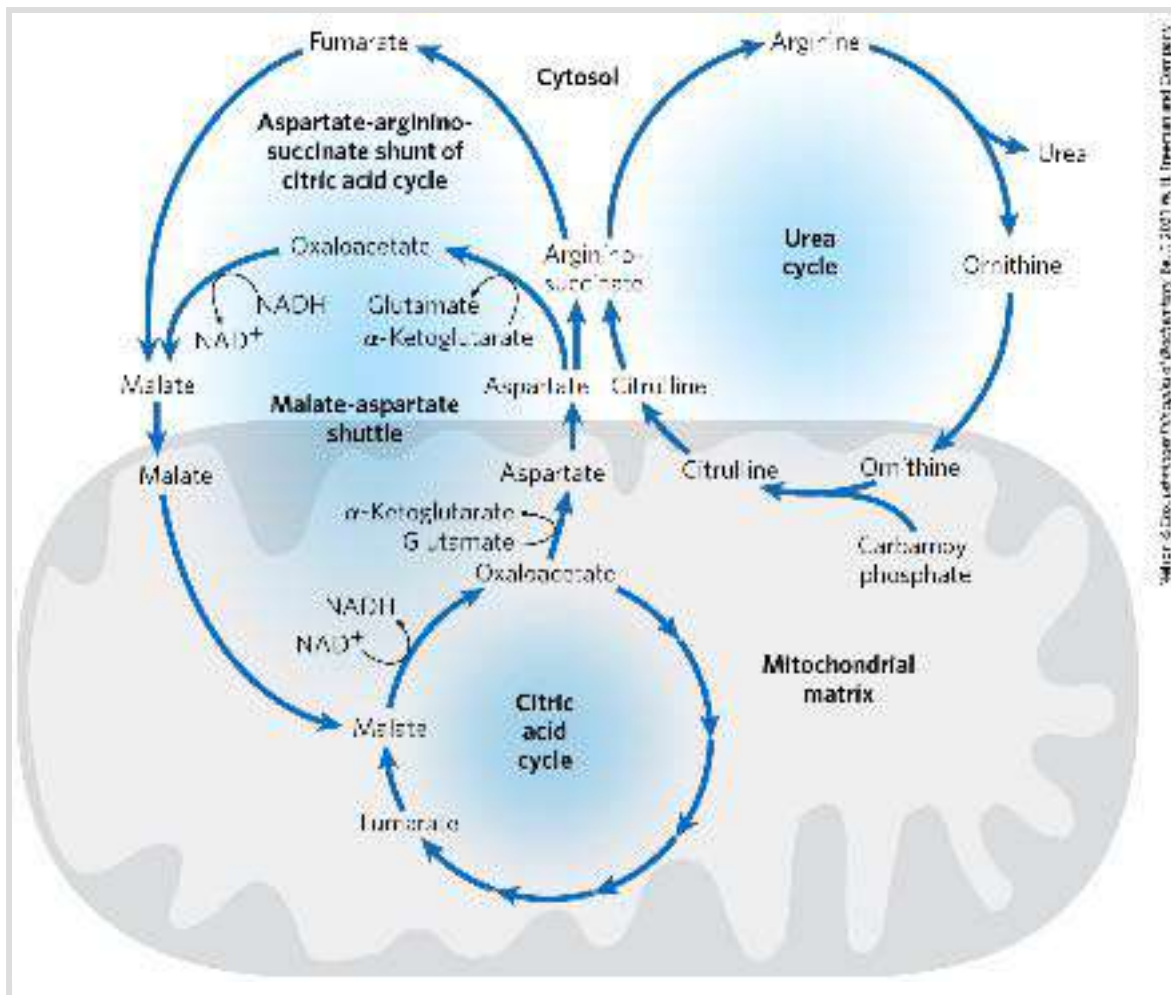


FIGURE 18-12 Links between the urea cycle and citric acid cycle. The interconnected cycles have been called the “Krebs bicycle.” The pathways linking the citric acid and urea cycles are known as the aspartate-argininosuccinate shunt; these effectively link the fates of the amino groups and the carbon skeletons of amino acids. The interconnections are quite elaborate. For example, some citric acid cycle enzymes, such as fumarase and malate dehydrogenase, have both cytosolic and mitochondrial isozymes. Fumarate produced in the cytosol — whether by the urea cycle, purine biosynthesis, or other processes — can be converted to cytosolic malate, which is used in the cytosol or transported into mitochondria to enter the citric acid cycle. These processes are further intertwined with the malate-aspartate shuttle, a set of reactions that brings reducing equivalents into the mitochondrion. These different cycles and processes rely on a limited number of transporters in the inner mitochondrial membrane.

Several enzymes of the citric acid cycle, including fumarase (fumarate hydratase) and malate dehydrogenase ([p. 586](#)), are also present as isozymes in the cytosol. There is no transporter to directly move the fumarate generated in cytosolic arginine synthesis back into the mitochondrial matrix. However, fumarate can be converted to malate in the cytosol. Fumarate and malate can be further metabolized in the cytosol, or malate can be transported into mitochondria for use in the citric acid cycle. Aspartate formed in mitochondria by transamination between oxaloacetate and glutamate can be transported to the cytosol, where it serves as nitrogen donor in the urea cycle reaction catalyzed by argininosuccinate synthetase. These reactions, making up the **aspartate-argininosuccinate shunt**, provide metabolic links between the separate pathways by which the amino groups and carbon skeletons of amino acids are processed.

The use of aspartate as a nitrogen donor in the urea cycle may appear to be a relatively complicated way to introduce the second amino group into urea. However, we shall see in [Chapter 22](#) that this pathway for nitrogen incorporation is one of the two common ways to introduce amino groups into biomolecules. In the urea cycle, additional pathway interconnections can help explain why aspartate is used as a nitrogen donor.  The urea and citric acid cycles are closely tied to an additional process that brings NADH, in the form of reducing equivalents, into the mitochondrion. As detailed in the next chapter, the NADH produced by glycolysis, fatty acid oxidation, and other processes cannot be transported across the mitochondrial inner membrane.

Reducing equivalents are instead brought into the mitochondrion by converting aspartate to oxaloacetate in the cytosol, reducing the oxaloacetate to malate with NADH, and transporting the malate into the mitochondrial matrix via the malate- α -ketoglutarate transporter. Once inside the mitochondrion, the malate can be reconverted to oxaloacetate while generating NADH. The oxaloacetate is converted to aspartate in the matrix and transported out of the mitochondrion by the aspartate-glutamate transporter. This malate-aspartate shuttle completes yet another cycle that functions to keep the mitochondrion supplied with NADH ([Fig. 18-12](#); see also [Fig. 19-31](#)).

These processes require that a balance be maintained in the cytosol between the concentrations of glutamate and aspartate. The enzyme that transfers amino groups between these key amino acids is aspartate aminotransferase, AAT (also called glutamate-oxaloacetate transaminase, GOT). This is one of the most active enzymes in hepatocytes and other tissues. When tissue damage occurs, this easily assayed enzyme and others leak into the blood. Thus, measuring blood levels of liver enzymes is important in diagnosing a variety of medical conditions ([Box 18-1](#)).

BOX 18-1 MEDICINE

Assays for Tissue Damage

Analyses of certain enzyme activities in blood serum give valuable diagnostic information for several disease conditions.

Blood levels of alanine aminotransferase (ALT; also called glutamate-pyruvate transaminase, GPT) and aspartate aminotransferase (AAT; also called glutamate-oxaloacetate transaminase, GOT) are important in the diagnosis of liver damage, toxicity associated with long-term drug use, or infection. When tissue is damaged, a variety of enzymes, including these aminotransferases, leak from injured cells into the bloodstream. Measurements of the blood serum concentrations of the two aminotransferases by the SGPT and SGOT tests (S for serum) — and of another enzyme, **creatinine kinase**, by the SCK test — can provide information about the severity of the damage.

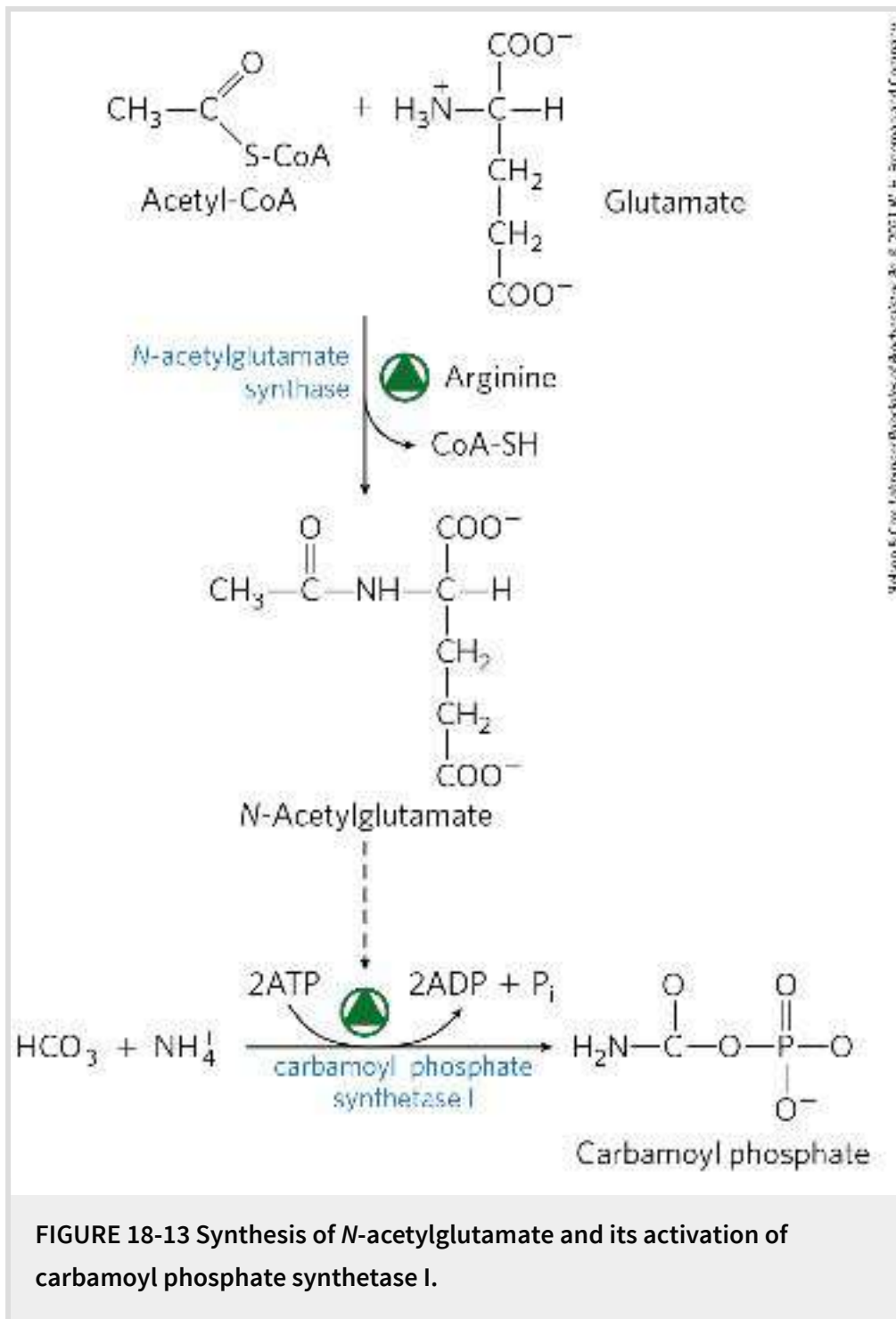
The SGOT and SGPT tests are used in occupational medicine to determine whether people exposed to carbon tetrachloride, chloroform, or other industrial solvents have suffered liver damage. Liver degeneration caused by these solvents is accompanied by leakage of various enzymes from injured hepatocytes into the blood. Aminotransferases are most useful in the monitoring of people exposed to these chemicals, because these enzyme activities are high in liver and thus are likely to be among the proteins leaked from damaged hepatocytes; also, they can be detected in the bloodstream in very small amounts.

The Activity of the Urea Cycle Is Regulated at Two Levels

The flux of nitrogen through the urea cycle in an individual animal varies with diet. When the dietary intake is primarily protein, the carbon skeletons of amino acids are used for fuel, producing much urea from the excess amino groups. During prolonged starvation, when breakdown of muscle protein begins to supply much of the organism's metabolic energy, urea production also increases substantially.

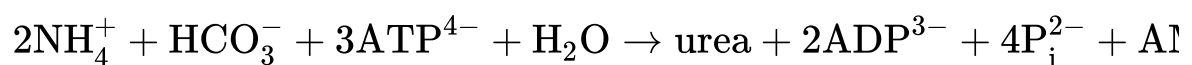
These changes in demand for urea cycle activity are met over the long term by regulation of the rates of synthesis of the four urea cycle enzymes and carbamoyl phosphate synthetase I in the liver. All five enzymes are synthesized at higher rates in starving animals and in animals on very-high-protein diets than in well-fed animals eating primarily carbohydrates and fats. Animals on protein-free diets produce lower levels of urea cycle enzymes.

On a shorter time scale, allosteric regulation of at least one key enzyme adjusts the flux through the urea cycle. The first enzyme in the pathway, carbamoyl phosphate synthetase I, is allosterically activated by ***N*-acetylglutamate**, which is synthesized from acetyl-CoA and glutamate by ***N*-acetylglutamate synthase** ([Fig. 18-13](#)). In plants and microorganisms, this enzyme catalyzes the first step in the de novo synthesis of arginine from glutamate (see [Fig. 22-12](#)), but in mammals, *N*-acetylglutamate synthase activity in the liver has a purely regulatory function (mammals lack the other enzymes needed to convert glutamate to arginine). The steady-state levels of *N*-acetylglutamate are determined by the concentrations of glutamate and acetyl-CoA (the substrates for *N*-acetylglutamate synthase) and arginine (an activator of *N*-acetylglutamate synthase, and thus an activator of the urea cycle).



Pathway Interconnections Reduce the Energetic Cost of Urea Synthesis

If we consider the urea cycle in isolation, we see that the synthesis of one molecule of urea requires four high-energy phosphate groups ([Fig. 18-10](#)). Two ATP molecules are required to make carbamoyl phosphate, and one ATP to make argininosuccinate — the latter ATP undergoing a pyrophosphate cleavage to AMP and PP_i , which is hydrolyzed to two P_i . The overall equation of the urea cycle is



However, this apparent cost is compensated for by the pathway interconnections detailed above. The fumarate generated by the urea cycle is converted to malate, and the malate is transported into the mitochondrion ([Fig. 18-12](#)). Inside the mitochondrial matrix, NADH is generated in the malate dehydrogenase reaction. Each NADH molecule can generate up to 2.5 ATP during mitochondrial respiration, greatly reducing the overall energetic cost of urea synthesis (mitochondrial respiration is discussed further in [Chapter 19](#)).

Genetic Defects in the Urea Cycle Can Be Life-Threatening



Infants with severe genetic defects in any enzyme involved in urea formation often appear normal at birth. However, they soon

develop symptoms of hyperammonemia, including cerebral edema, lethargy, and hyperventilation. Without treatment, early death usually results. Symptoms may be less severe in patients retaining partial enzyme activity. These patients cannot tolerate protein-rich diets. Amino acids ingested in excess of the minimum daily requirements for protein synthesis are deaminated in the liver, producing free ammonia that cannot be converted to urea and exported into the bloodstream, and, as we have seen, ammonia is highly toxic. Given that most urea cycle steps are irreversible, the absent enzyme activity can often be identified by determining which cycle intermediate is present in elevated concentration in the blood and/or urine. Although the breakdown of amino acids can have serious health consequences in individuals with urea cycle deficiencies, a protein-free diet is not a treatment option. Humans are incapable of synthesizing half of the 20 common amino acids; they must obtain these [essential amino acids](#) in the diet ([Table 18-1](#)).

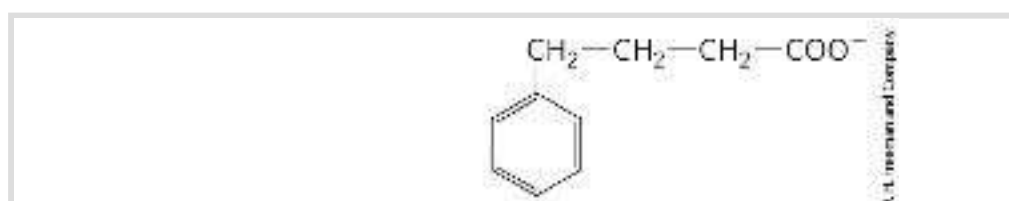
TABLE 18-1 Nonessential and Essential Amino Acids for Humans and the Albino Rat

Nonessential	Conditionally essential ^a	Essential
Alanine	Arginine	Histidine
Asparagine	Cysteine	Isoleucine
Aspartate	Glutamine	Leucine
Glutamate	Glycine	Lysine
Serine	Proline	Methionine

Tyrosine	Phenylalanine
	Threonine
	Tryptophan
	Valine

^aRequired to some degree in young, growing animals and/or sometimes during illness.

A variety of treatments are available for individuals with urea cycle defects. Careful administration of the aromatic acids benzoate or phenylbutyrate in the diet can help lower the level of ammonia in the blood. Benzoate is converted to benzoyl-CoA, which combines with glycine to form hippurate ([Fig. 18-14](#), left). The glycine used up in this reaction must be regenerated, and ammonia is thus taken up in the glycine synthase reaction. Phenylbutyrate is converted to phenylacetate by β oxidation. The phenylacetate is then converted to phenylacetyl-CoA, which combines with glutamine to form phenylacetylglutamine ([Fig. 18-14](#), right). The resulting removal of glutamine triggers its further synthesis by glutamine synthetase (see Eqn 22-1) in a reaction that takes up ammonia. Both hippurate and phenylacetylglutamine are nontoxic compounds that are excreted in the urine. The pathways shown in [Figure 18-14](#) make only minor contributions to normal metabolism, but they become prominent when aromatic acids are ingested.



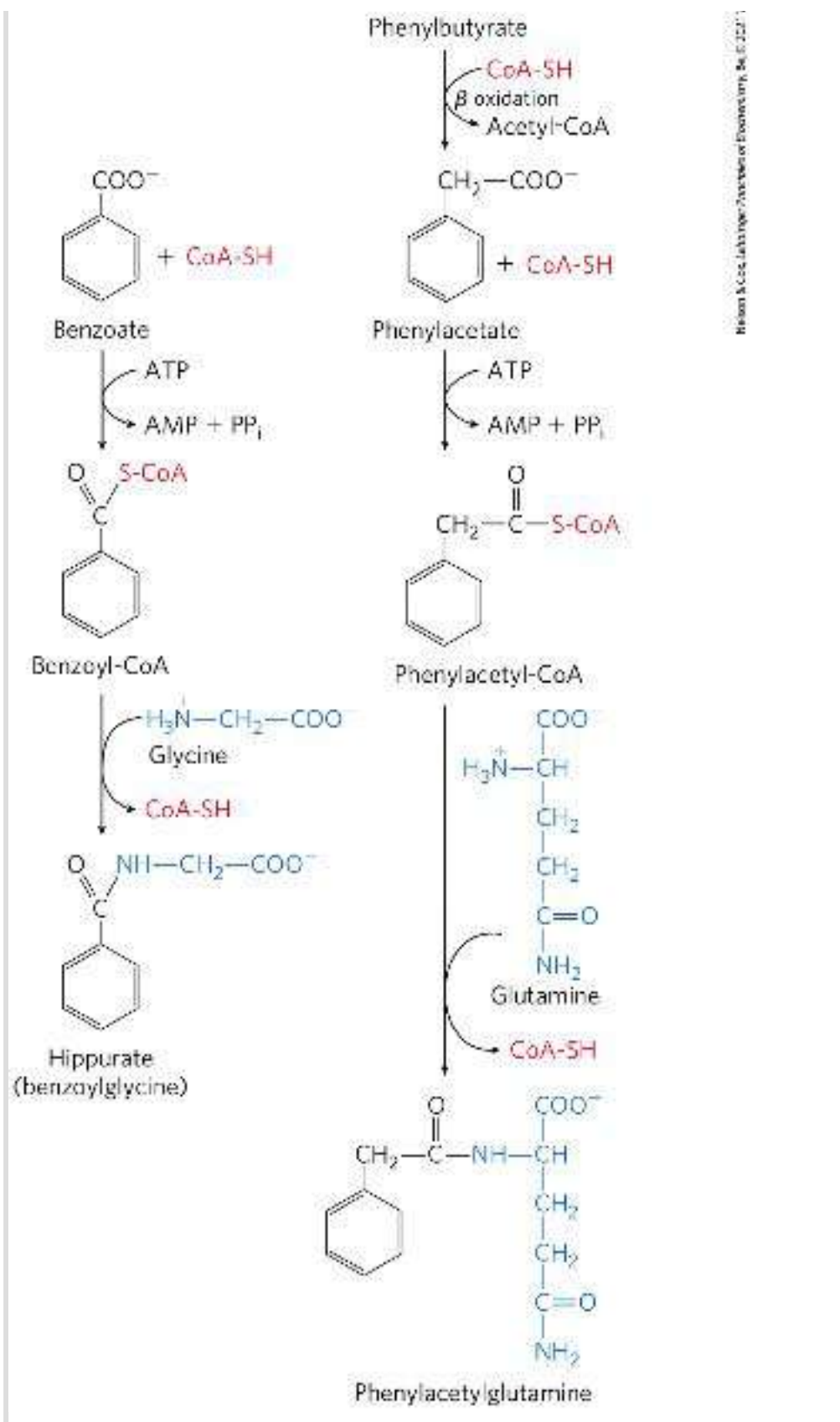
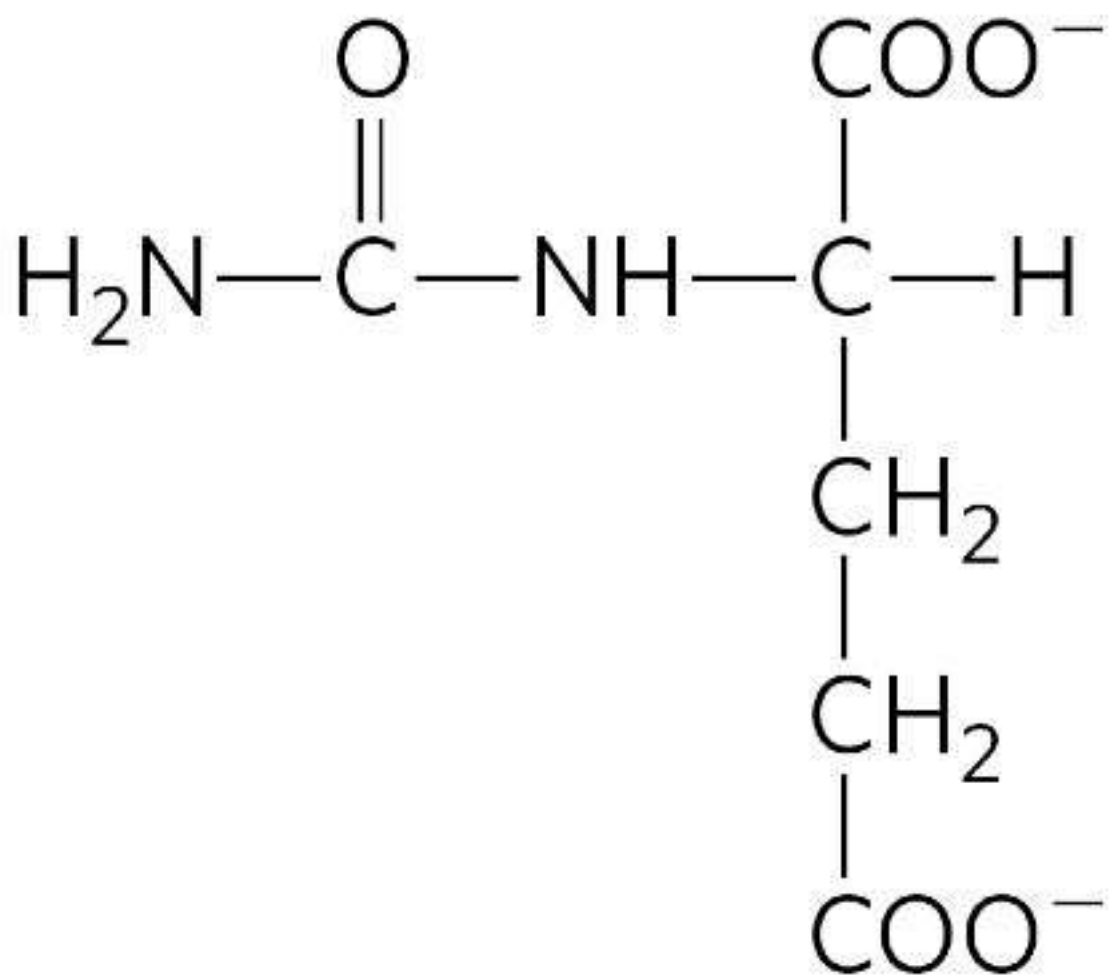


FIGURE 18-14 Treatment for deficiencies in urea cycle enzymes. The aromatic acids benzoate and phenylbutyrate, administered in the diet, are metabolized and combine with glycine and glutamine, respectively. The

products are excreted in the urine. Subsequent synthesis of glycine and glutamine to replenish the pool of these intermediates removes ammonia from the bloodstream.

Other therapies are more specific to a particular enzyme deficiency. Deficiency of *N*-acetylglutamate synthase results in the absence of the normal activator of carbamoyl phosphate synthetase I ([Fig. 18-13](#)). This condition can be treated by administering carbamoyl glutamate, an analog of *N*-acetylglutamate that is effective in activating carbamoyl phosphate synthetase I.



Carbamoyl glutamate

Supplementing the diet with arginine is useful in treating deficiencies of ornithine transcarbamoylase, argininosuccinate synthetase, and argininosuccinase. Many of these treatments must be accompanied by strict dietary control and supplements of essential amino acids. In the rare cases of arginase deficiency, arginine, the substrate of the defective enzyme, must be excluded from the diet. ■

SUMMARY 18.2 *Nitrogen Excretion and the Urea Cycle*

- In the urea cycle, ornithine combines with ammonia, in the form of carbamoyl phosphate, to form citrulline. A second amino group is transferred to citrulline from aspartate to form arginine — the immediate precursor of urea. Arginase catalyzes hydrolysis of arginine to urea and ornithine; ornithine is regenerated in each turn of the cycle.
- The urea cycle results in a net conversion of oxaloacetate to fumarate, both of which are intermediates in the citric acid cycle. The two cycles are thus interconnected.
- The activity of the urea cycle is regulated at the level of enzyme synthesis and by allosteric regulation of the enzyme that catalyzes the formation of carbamoyl phosphate.
- The energetic cost of the urea cycle is reduced by cycle interconnections.
- Genetic diseases involving urea cycle enzyme deficiencies have serious consequences but sometimes can be managed by dietary intervention.

18.3 Pathways of Amino Acid Degradation

Amino acid catabolism normally accounts for only 10% to 15% of the human body's energy production; these pathways are not nearly as active as glycolysis and fatty acid oxidation. Flux through these catabolic routes also varies greatly, depending on the balance between requirements for biosynthetic processes and the availability of a particular amino acid. The 20 catabolic pathways converge to form only six major products: pyruvate, acetyl-CoA, α -ketoglutarate, succinyl-CoA, fumarate, and oxaloacetate. All of these enter the citric acid cycle ([Fig. 18-15](#)). From here the carbon skeletons are diverted to gluconeogenesis or ketogenesis; alternatively, they are completely oxidized as fuel to CO_2 and H_2O .

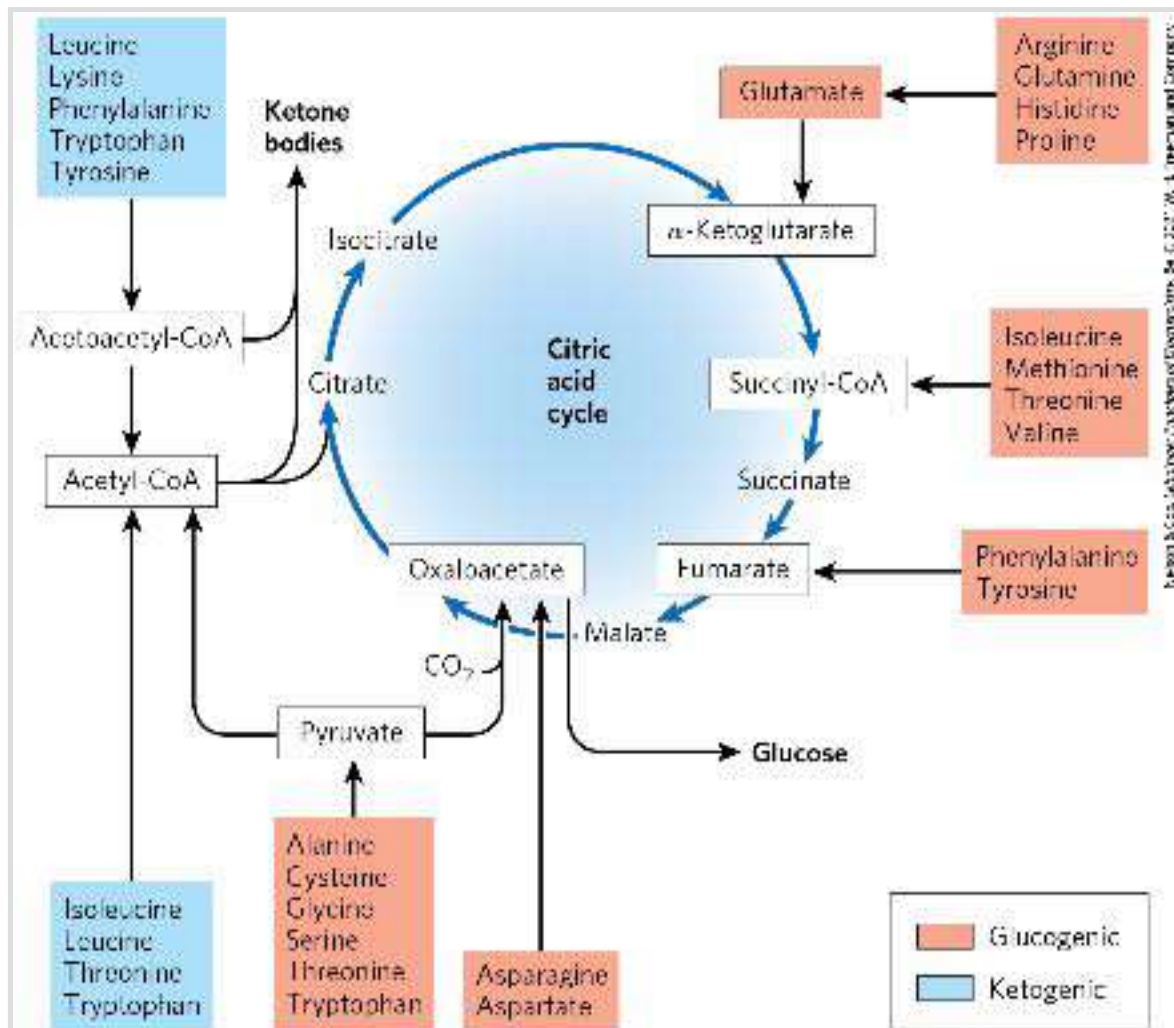


FIGURE 18-15 Summary of amino acid catabolism. Amino acids are grouped according to their major degradative end product. Some amino acids are listed more than once because different parts of their carbon skeletons are degraded to different end products. The figure shows the most important catabolic pathways in vertebrates, but there are minor variations among vertebrate species. Threonine, for instance, is degraded by at least two different pathways (see [Figs 18-19, 18-27](#)), and the importance of a given pathway can vary with the organism and its metabolic conditions. The glucogenic and ketogenic amino acids are also delineated in the figure, by color shading. Notice that five of the amino acids are both glucogenic and ketogenic. The amino acids degraded to pyruvate are also potentially ketogenic. Only two amino acids, leucine and lysine, are exclusively ketogenic.

We summarize the individual pathways for the 20 amino acids in flow diagrams, each leading to a specific point of entry into the citric acid cycle. In these diagrams, the carbon atoms that enter the citric acid cycle are shown in color. Note that some amino acids appear more than

once, reflecting different fates for different parts of their carbon skeletons. Rather than examining every step of every pathway in amino acid catabolism, we single out for special discussion some enzymatic reactions that are particularly noteworthy for their mechanisms or their medical significance.

Some Amino Acids Can Contribute to Gluconeogenesis, Others to Ketone Body Formation

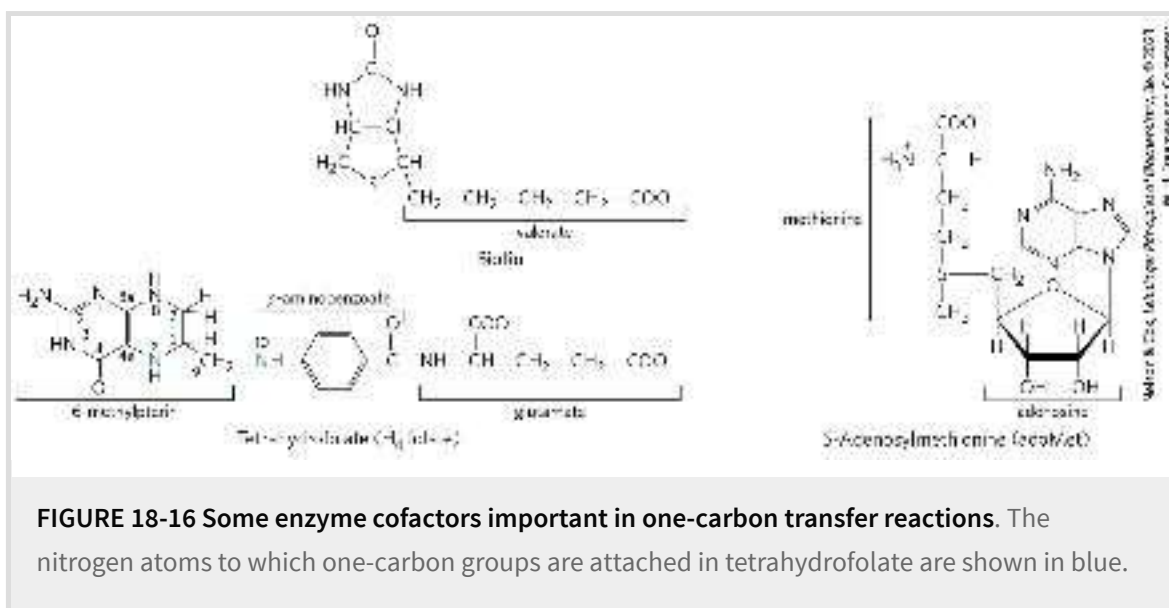
 The seven amino acids that are degraded entirely or in part to acetoacetyl-CoA and/or acetyl-CoA — phenylalanine, tyrosine, isoleucine, leucine, tryptophan, threonine, and lysine — can yield ketone bodies in the liver, where acetoacetyl-CoA is converted to acetoacetate and then to acetone and β -hydroxybutyrate (see [Fig. 17-16](#)). These are the **ketogenic** amino acids ([Fig. 18-15](#)). Their ability to form ketone bodies is particularly evident in uncontrolled diabetes mellitus, in which the liver produces large amounts of ketone bodies from both fatty acids and the ketogenic amino acids. Ketone bodies may also be metabolized in the brain as fuel in place of glucose in cases during starvation.

The amino acids that are degraded to pyruvate, α -ketoglutarate, succinyl-CoA, fumarate, and/or oxaloacetate can be converted to glucose and glycogen by pathways described in [Chapter 14](#). They are the **glucogenic** amino acids. The division between ketogenic and glucogenic amino acids is not sharp; five amino acids — tryptophan, phenylalanine, tyrosine, threonine, and isoleucine — are both ketogenic and glucogenic. All amino acids except for lysine and leucine can make some contribution to gluconeogenesis. Catabolism of amino acids is

particularly critical to the survival of animals with high-protein diets or during starvation. Leucine is an exclusively ketogenic amino acid that is very common in proteins. Its degradation makes a substantial contribution to ketosis under starvation conditions.

Several Enzyme Cofactors Play Important Roles in Amino Acid Catabolism

The structural diversity of amino acids is reflected in the varied reaction types encountered in their breakdown pathways. We begin our study of these pathways by noting important classes of reactions that recur and introducing their enzyme cofactors. We have already considered one important class: transamination reactions requiring pyridoxal phosphate. One-carbon transfers are another common type of reaction in amino acid catabolism. Such transfers usually involve one of three cofactors: biotin, tetrahydrofolate, or *S*-adenosylmethionine ([Fig. 18-16](#)). These cofactors transfer one-carbon groups in different oxidation states: biotin transfers carbon in its most oxidized state, CO_2 (see [Fig. 14-17](#)); tetrahydrofolate transfers one-carbon groups in intermediate oxidation states and sometimes as methyl groups; and *S*-adenosylmethionine transfers methyl groups, the most reduced state of carbon. The latter two cofactors are especially important in amino acid and nucleotide metabolism.



Tetrahydrofolate (H_4 folate), synthesized in bacteria, consists of substituted pterin (6-methylpterin), *p*-aminobenzoate, and glutamate moieties ([Fig. 18-16](#)).



The oxidized form, folate, is a vitamin for mammals; it is converted in two steps to tetrahydrofolate by the enzyme dihydrofolate reductase. The one-carbon group undergoing transfer, in any of three oxidation states, is bonded to N-5 or N-10 or both. The most reduced form of the cofactor carries a methyl group, a more oxidized form carries a methylene group, and the most oxidized forms carry a methenyl, formyl, or formimino group ([Fig. 18-17](#)). Most forms of tetrahydrofolate are interconvertible and serve as donors of one-carbon units in a variety

[illegible]

more stable and therefore a weaker donor of formyl groups. N^5 -Formyltetrahydrofolate is a minor byproduct of the cyclohydrolase reaction, and can also form spontaneously. Conversion of N^5 -formyltetrahydrofolate to N^5,N^{10} -methenyltetrahydrofolate requires ATP, because of an otherwise unfavorable equilibrium. Note that N^5 -formiminotetrahydrofolate is derived from histidine in a pathway shown in [Figure 18-26](#).

Although tetrahydrofolate can carry a methyl group at N-5, the transfer potential of this methyl group is insufficient for most biosynthetic reactions. **S-Adenosylmethionine (adoMet)** is the preferred cofactor for biological methyl group transfers. It is synthesized from ATP and methionine by the action of **methionine adenosyl transferase** ([Fig. 18-18](#), step ①). This reaction is unusual in that the nucleophilic sulfur atom of methionine attacks the 5' carbon of the ribose moiety of ATP rather than one of the phosphorus atoms. Triphosphate is released and is cleaved to P_i and PP_i on the enzyme, and the PP_i is cleaved by inorganic pyrophosphatase; thus three bonds, including two bonds of high-energy phosphate groups, are broken in this reaction. The only other known reaction in which triphosphate is displaced from ATP occurs in the synthesis of coenzyme B_{12} (see [Box 17-2](#), [Fig. 3](#)).

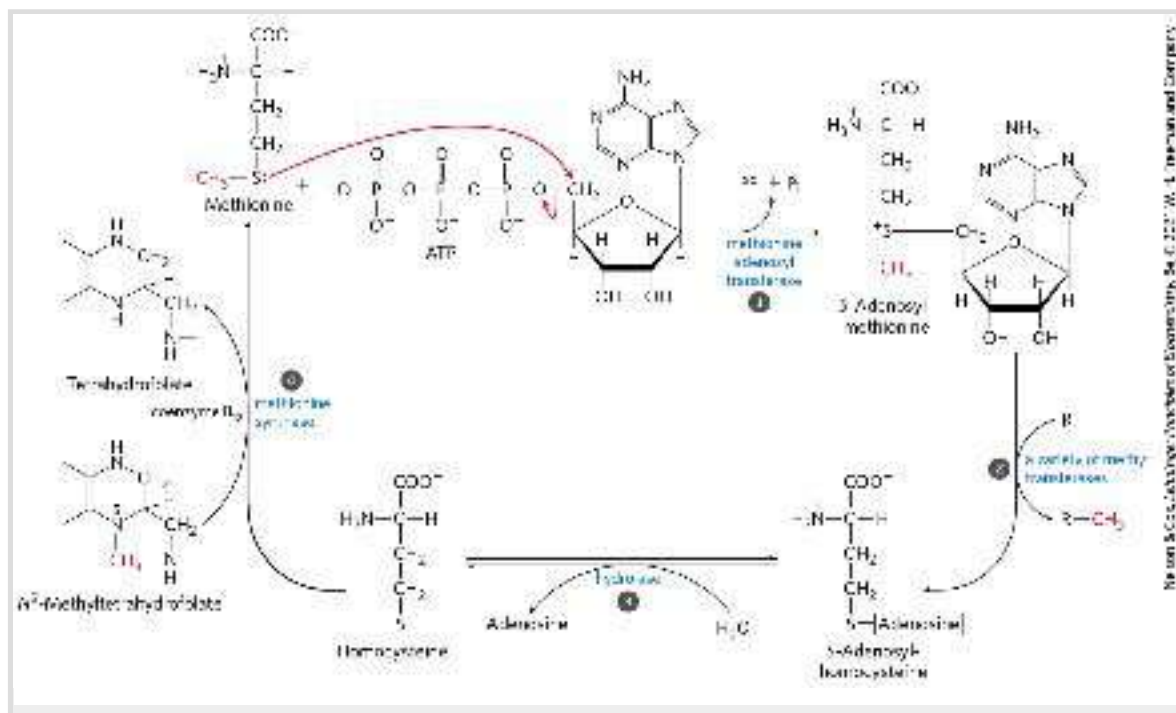


FIGURE 18-18 Synthesis of methionine and S-adenosylmethionine in an activated-methyl cycle. The steps are described in the text. In the methionine synthase reaction (step ④), the methyl group is transferred to cobalamin to form methylcobalamin, which is the methyl donor in the formation of methionine. S-Adenosylmethionine, which has a positively charged sulfur (and is thus a sulfonium ion), is a powerful methylating agent in several biosynthetic reactions. The methyl group acceptor (step ②) is designated R.

S-Adenosylmethionine is a potent alkylating agent by virtue of its destabilizing sulfonium ion. The methyl group is subject to attack by nucleophiles and is about 1,000 times more reactive than the methyl group of N^5 -methyltetrahydrofolate.

Transfer of the methyl group from S-adenosylmethionine to an acceptor yields **S-adenosylhomocysteine** (Fig. 18-18, step ②), which is subsequently broken down to homocysteine and adenosine (step ③). Methionine is regenerated by transfer of a methyl group to homocysteine in a reaction catalyzed by methionine synthase (step ④), and methionine is reconverted to S-adenosylmethionine to complete an activated-methyl cycle.

One form of methionine synthase common in bacteria uses N^5 -methyltetrahydrofolate as a methyl donor. Another form of the enzyme present in some bacteria and mammals uses N^5 -methyltetrahydrofolate, but the methyl group is first transferred to cobalamin, derived from coenzyme B₁₂, to form methylcobalamin as the methyl donor in methionine formation. This reaction and the rearrangement of L-methylmalonyl-CoA to succinyl-CoA (see Box 17-2, Fig. 1a) are the only known coenzyme B₁₂-dependent reactions in mammals.



The vitamins B₁₂ and folate are closely linked in these pathways.

The anemia observed in the rare B₁₂ deficiency disease **pernicious anemia** ([Box 17-2](#)) can be traced to the methionine synthase reaction. As noted above, the methyl group of methylcobalamin is derived from *N*⁵-methyltetrahydrofolate, and this is the only reaction in mammals that uses *N*⁵-methyltetrahydrofolate. The reaction converting the *N*⁵, *N*¹⁰-methylene form to the *N*⁵-methyl form of tetrahydrofolate is irreversible ([Fig. 18-17](#)). Thus, if coenzyme B₁₂ is not available for the synthesis of methylcobalamin, metabolic folates become trapped in the *N*⁵-methyl form.

The anemia associated with vitamin B₁₂ deficiency is called **megaloblastic anemia**. It manifests as a decline in the production of mature erythrocytes (red blood cells) and the appearance in the bone marrow of immature precursor cells, or **megaloblasts**. Erythrocytes are gradually replaced in the blood by smaller numbers of abnormally large erythrocytes called **macrocytes**. The defect in erythrocyte development is a direct consequence of the depletion of the *N*⁵, *N*¹⁰-methylenetetrahydrofolate, which is required for synthesis of the thymidine nucleotides needed for DNA synthesis (see [Chapter 22](#)). Folate deficiency, in which all forms of tetrahydrofolate are depleted, also produces anemia, for much the same reasons. The anemia symptoms of B₁₂ deficiency can be alleviated by administering either vitamin B₁₂ or folate.

However, it is dangerous to treat pernicious anemia by folate supplementation alone, because the neurological symptoms of B₁₂ deficiency will progress. These symptoms do not arise from the defect in the methionine synthase reaction. Instead, the impaired methylmalonyl-CoA mutase (see [Box 17-2](#) and [Fig. 17-12](#)) causes

accumulation of unusual, odd-number fatty acids in neuronal membranes. The anemia associated with folate deficiency is thus often treated by administering both folate and vitamin B₁₂, at least until the metabolic source of the anemia is unambiguously defined. Early diagnosis of B₁₂ deficiency is important because some of its associated neurological conditions may be irreversible.

Folate deficiency also reduces the availability of the *N*⁵-methyltetrahydrofolate required for methionine synthase function. This leads to a rise in homocysteine levels in blood, a condition linked to heart disease, hypertension, and stroke. High levels of homocysteine may be responsible for 10% of all cases of heart disease. The condition is treated with folate supplements. ■

[Tetrahydrobiopterin](#), another cofactor of amino acid catabolism, is similar to the pterin moiety of tetrahydrofolate, but it is not involved in one-carbon transfers; instead it participates in oxidation reactions. We consider its mode of action when we discuss phenylalanine degradation (see [Fig. 18-24](#)).

Six Amino Acids Are Degraded to Pyruvate

   The carbon skeletons of six amino acids — alanine, tryptophan, cysteine, serine, glycine, and threonine — are converted in whole or in part to pyruvate. The pyruvate can then be converted to acetyl-CoA and eventually oxidized via the citric acid cycle, or to oxaloacetate and shunted into gluconeogenesis ([Fig. 18-19](#)). **Alanine** yields pyruvate directly on transamination with α -ketoglutarate, and the side chain of **tryptophan** is cleaved to yield alanine and thus

pyruvate. **Cysteine** is converted to pyruvate in two steps; one removes the sulfur atom, the other is a transamination. **Serine** is converted to pyruvate by serine dehydratase. Both the β -hydroxyl and the α -amino groups of serine are removed in this single pyridoxal phosphate-dependent reaction ([Fig. 18-20a](#)).

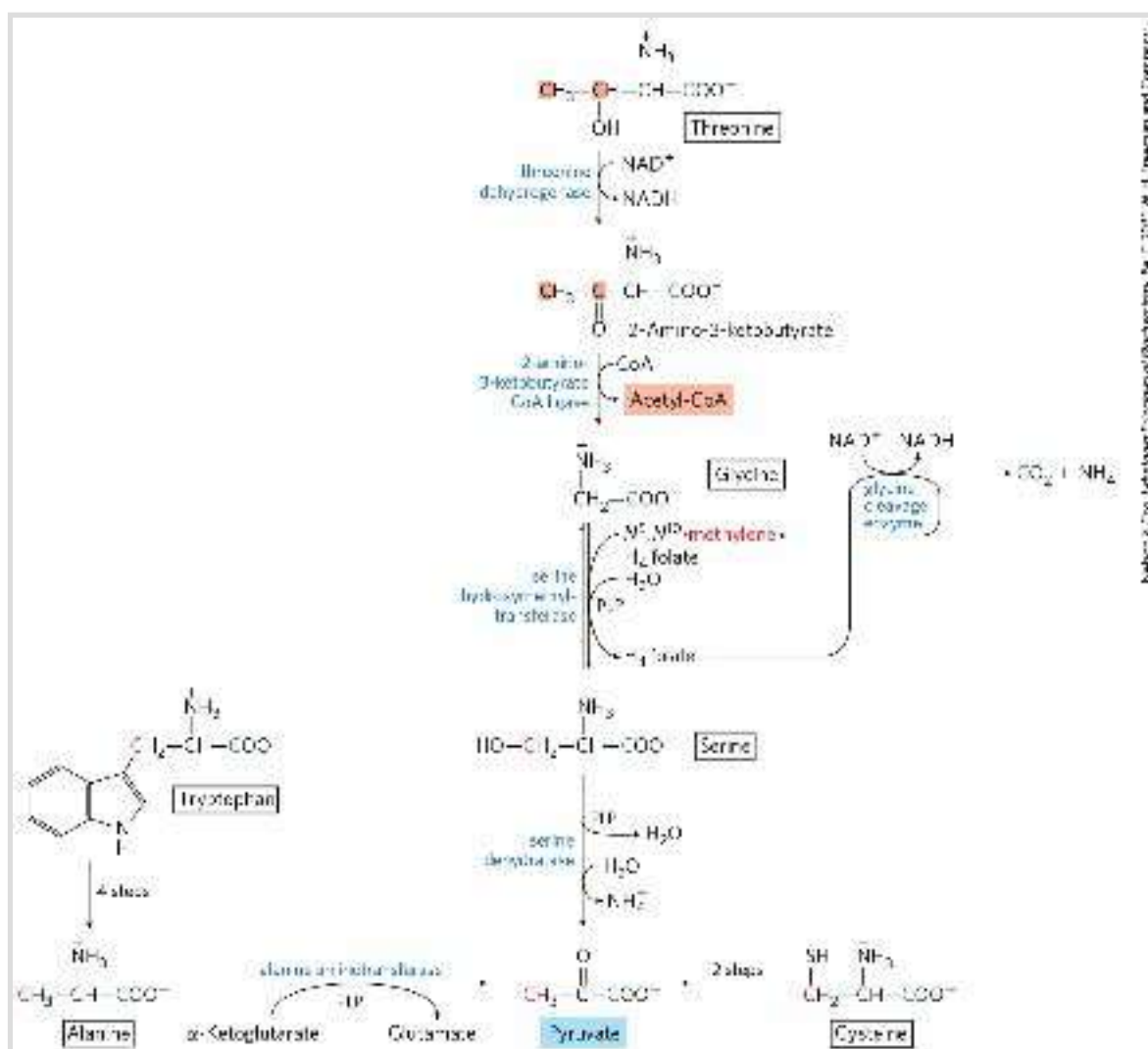
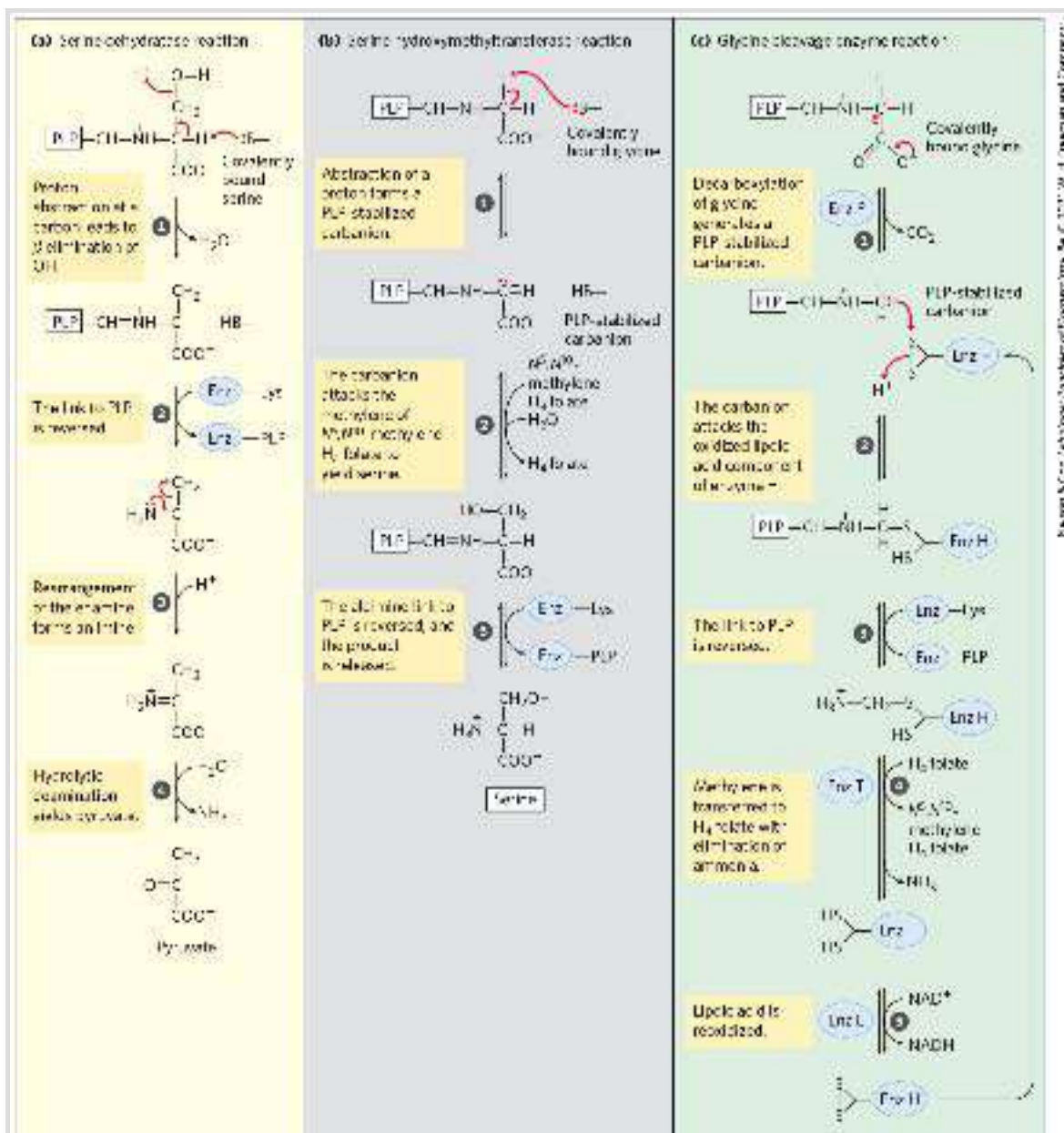


FIGURE 18-19 Catabolic pathways for alanine, tryptophan, cysteine, serine, glycine, and threonine. Carbon atoms here and in subsequent figures are color-coded as necessary to trace their fates. The fate of the indole group of tryptophan is shown in [Figure 18-21](#). Details of most of the reactions involving serine and glycine are shown in [Figure 18-20](#). Several pathways for cysteine degradation lead to pyruvate. The pathway for threonine degradation shown here accounts for only about a third of threonine catabolism (for the alternative pathway, see [Fig. 18-27](#)).



MECHANISM FIGURE 18-20 Interplay of the pyridoxal phosphate and tetrahydrofolate cofactors in serine and glycine metabolism. The first step in each of these reactions (not shown) involves the formation of a covalent imine linkage between enzyme-bound PLP and the substrate amino acid — serine in (a), glycine in (b) and (c). (a) A PLP-catalyzed elimination of water in the serine dehydratase reaction (step 1) begins the pathway to pyruvate. (b) In the serine hydroxymethyltransferase reaction, a PLP-stabilized carbanion (product of step 1) is a key intermediate in the reversible transfer of the methylene group (as $-\text{CH}_2-\text{OH}$) from N^5,N^{10} -methylenetetrahydrofolate to form serine. (c) The glycine cleavage enzyme is a multienzyme complex, with components P, H, T, and L. The overall reaction, which is reversible, converts glycine to CO_2 and NH_4^+ , with the second glycine carbon taken up by tetrahydrofolate to form N^5,N^{10} -methylenetetrahydrofolate. Pyridoxal phosphate activates the α carbon of amino acids at critical stages in all these reactions, and tetrahydrofolate carries a one-carbon unit in two of them (see [Figs 18-6, 18-17](#)).

Glycine is degraded via three pathways, only one of which leads to pyruvate. Glycine is converted to serine by enzymatic addition of a hydroxymethyl group ([Fig. 18-19](#), [18-20b](#)). This reaction, catalyzed by **serine hydroxymethyltransferase**, requires the coenzymes tetrahydrofolate and pyridoxal phosphate. The serine is converted to pyruvate as described above. In the second pathway, which predominates in animals, glycine undergoes oxidative cleavage to CO_2 , NH_4^+ , and a methylene group ($-\text{CH}_2-$) ([Figs. 18-19](#), [18-20c](#)). This readily reversible reaction, catalyzed by **glycine cleavage enzyme** (also called glycine synthase), also requires tetrahydrofolate, which accepts the methylene group. In this oxidative cleavage pathway, the two carbon atoms of glycine do not enter the citric acid cycle. One carbon is lost as CO_2 and the other becomes the methylene group of N^5, N^{10} -methylenetetrahydrofolate ([Fig. 18-17](#)), a one-carbon group donor in certain biosynthetic pathways.



This second pathway for glycine degradation seems to be critical in mammals. Humans with serious defects in glycine cleavage enzyme activity suffer from a condition known as nonketotic hyperglycinemia. The condition is characterized by elevated serum levels of glycine, leading to severe intellectual disability and death in very early childhood. At high levels, glycine is an inhibitory neurotransmitter, which may explain the neurological effects of the disease. Perhaps more important, high levels of glycine increase the levels of 2-amino-3-ketobutyrate, an unstable intermediate in the degradation of threonine in mitochondria ([Fig. 18-19](#)). 2-Amino-3-ketobutyrate decarboxylates spontaneously to form the toxic metabolite aminoacetone, which is readily metabolized to the highly reactive **methylglyoxal**, a molecule that modifies both protein and DNA.



Methylglyoxal is also a byproduct of glycolysis and is implicated in the progression of type 2 diabetes ([Box 7-2](#)).

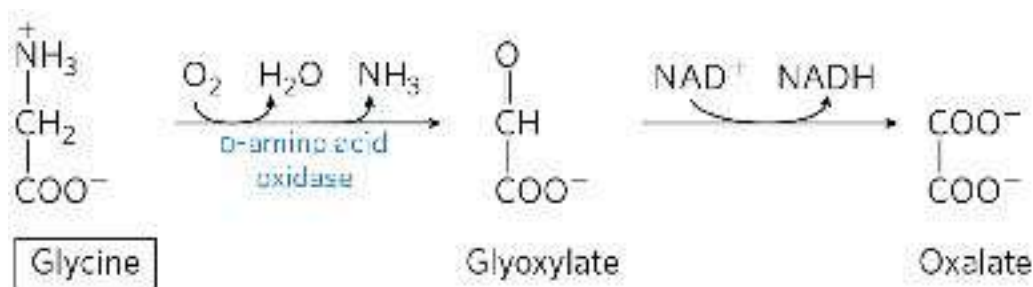
Many genetic defects of amino acid metabolism have been identified in humans ([Table 18-2](#)). We shall encounter several more in this chapter.

TABLE 18-2  **Some Human Genetic Disorders Affecting Amino Acid Catabolism**

Medical condition	Approximate incidence (per 100,000 births)	Defective process	Defective enzyme	Symptoms and effects
Albinism	<3	Melanin synthesis from tyrosine	Tyrosine 3-monooxygenase (tyrosinase)	Lack of pigmentation; white hair, pink skin
Alkaptonuria	<0.4	Tyrosine degradation	Homogentisate 1,2-dioxygenase	Dark pigment in urine; late-developing arthritis
Argininemia	<0.5	Urea synthesis	Arginase	Intellectual disability

Argininosuccinic acidemia	<1.5	Urea synthesis	Argininosuccinase	Vomiting; convulsions
Carbamoyl phosphate synthetase I deficiency	<0.5	Urea synthesis	Carbamoyl phosphate synthetase I	Lethargy; convulsions; early death
Homocystinuria	<0.5	Methionine degradation	Cystathionine β -synthase	Faulty bone development; intellectual disability
Maple syrup urine disease (branched-chain ketoaciduria)	<0.4	Isoleucine, leucine, and valine degradation	Branched-chain α -keto acid dehydrogenase complex	Vomiting; convulsions; intellectual disability; early death
Methylmalonic acidemia	<0.5	Conversion of propionyl-CoA to succinyl-CoA	Methylmalonyl-CoA mutase	Vomiting; convulsions; intellectual disability; early death
Phenylketonuria	<8	Conversion of phenylalanine to tyrosine	Phenylalanine hydroxylase	Neonatal vomiting; intellectual disability

In the third and final pathway of glycine degradation, the achiral glycine molecule is a substrate for the enzyme D-amino acid oxidase. The glycine is converted to glyoxylate, an alternative substrate for lactate dehydrogenase ([p. 526](#)). Glyoxylate is oxidized in an NAD^{+} -dependent reaction to oxalate:



 The function of D-amino acid oxidase, present at high levels in the kidney, is thought to be the detoxification of ingested D-amino acids derived from bacterial cell walls and from grilled foodstuffs (high heat causes some spontaneous racemization of the L-amino acids in proteins). Oxalate, whether obtained in foods or produced enzymatically in the kidneys, has medical significance. Crystals of calcium oxalate account for up to 75% of all kidney stones. ■

There are two significant pathways for **threonine** degradation. One pathway leads to pyruvate via glycine ([Fig. 18-19](#)). The conversion to glycine occurs in two steps, with threonine first converted to 2-amino-3-ketobutyrates by the action of threonine dehydrogenase. This pathway is important in a few classes of rapidly dividing human cells, such as embryonic stem cells. The glycine generated by this pathway is broken down primarily by the glycine cleavage enzyme ([Fig. 18-19](#)). The N^5, N^{10} -methylenetetrahydrofolate thus generated ([Fig. 18-20c](#)) is needed for the synthesis, via pathways described in [Chapter 22](#), of nucleotides used in DNA replication. However, in most human tissues, the degradation of threonine via glycine is a relatively minor pathway, accounting for 10% to 30% of threonine catabolism. It is more important in some other mammals. The major pathway in most human tissues leads to succinyl-CoA, as described later in this chapter.

Seven Amino Acids Are Degraded to Acetyl-CoA



Portions of the carbon skeletons of seven amino acids — **tryptophan, lysine, phenylalanine, tyrosine, leucine, isoleucine, and threonine** — yield acetyl-CoA and/or acetoacetyl-CoA, the latter being converted to acetyl-CoA ([Fig. 18-21](#)). Some of the final steps in the degradative pathways for leucine, lysine, and tryptophan resemble steps in the oxidation of fatty acids (see [Fig. 17-9](#)). Threonine (not shown in [Fig. 18-21](#)) yields some acetyl-CoA via the minor pathway illustrated in [Figure 18-19](#).

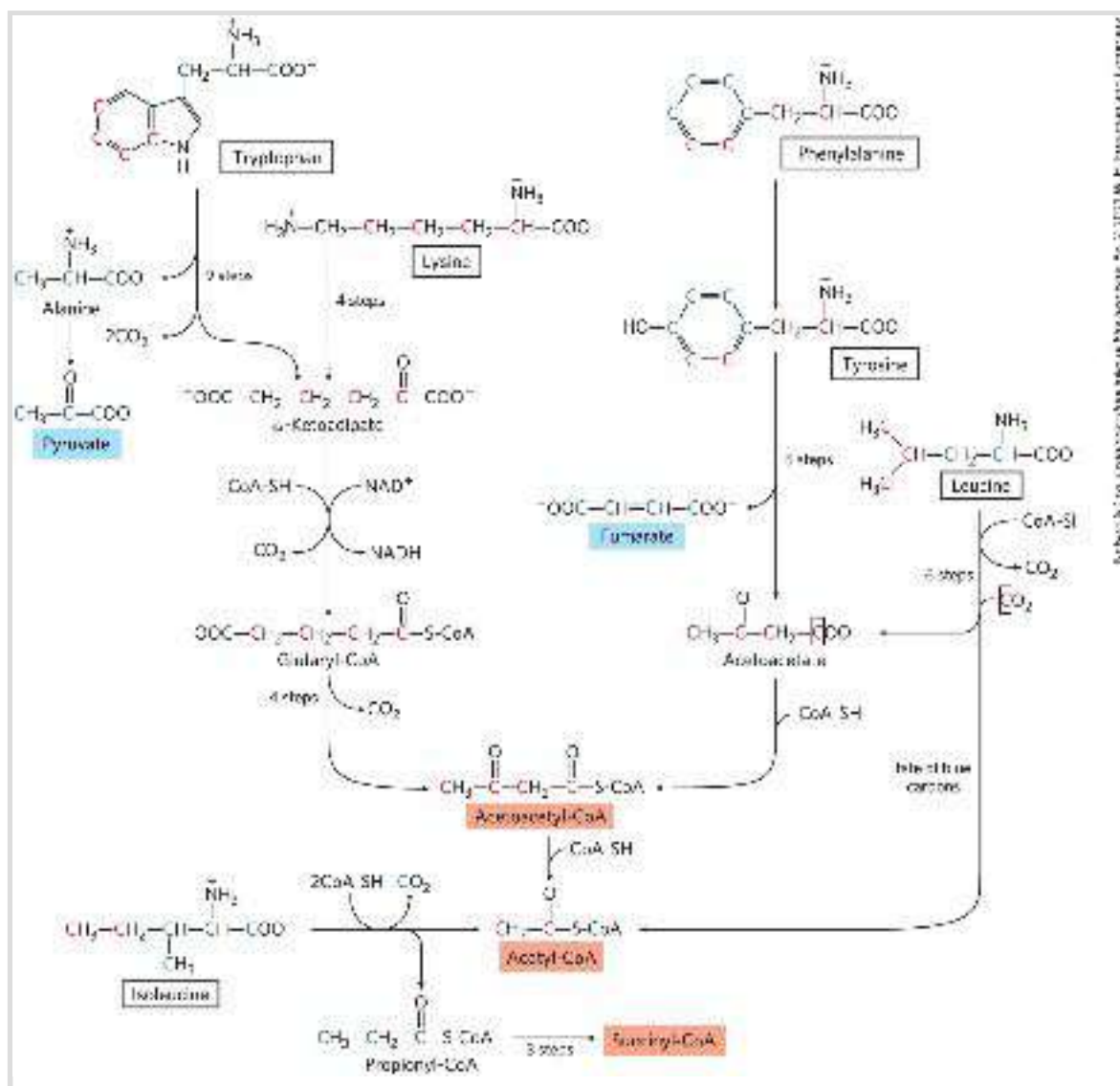
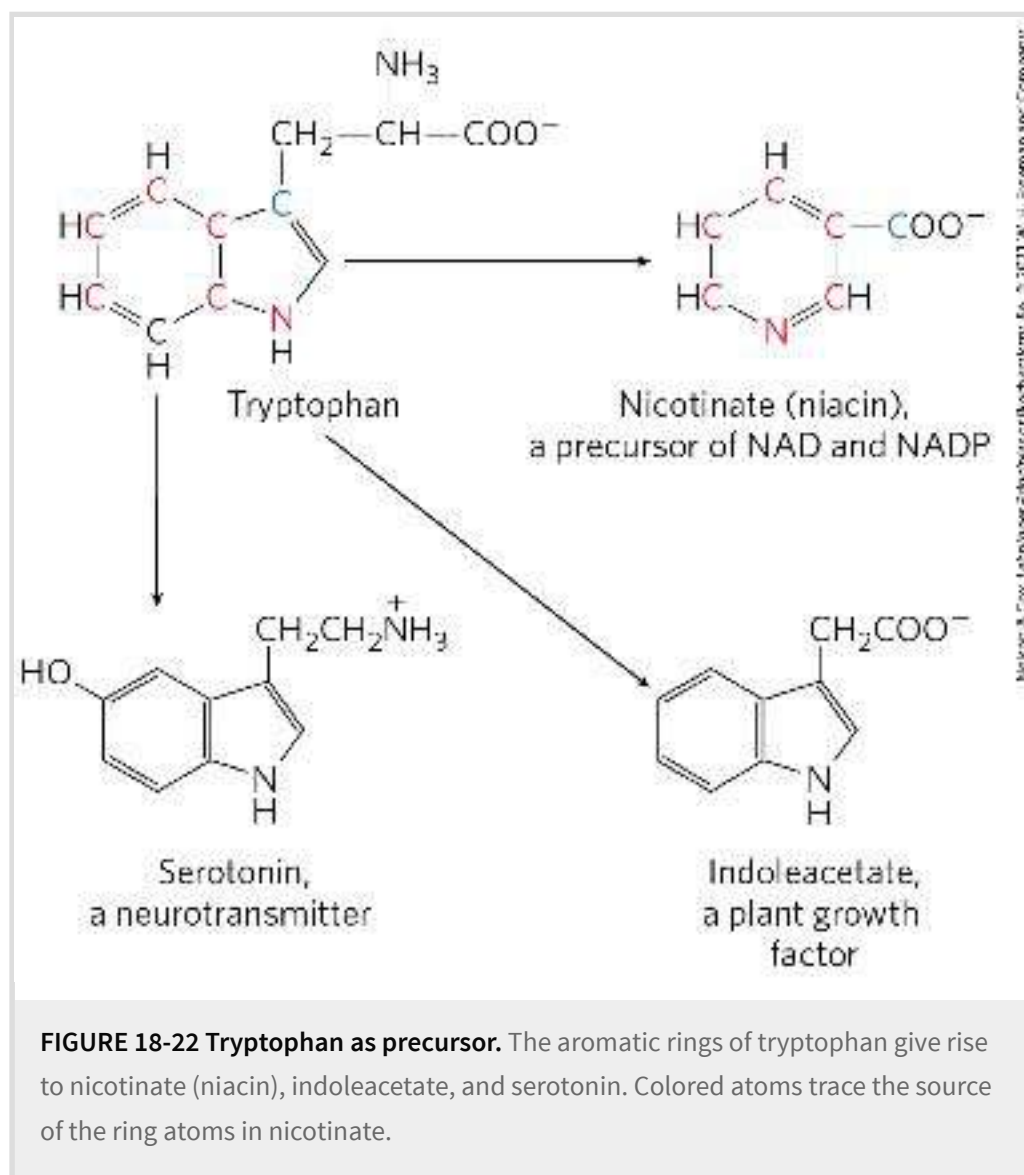


FIGURE 18-21 Catabolic pathways for tryptophan, lysine, phenylalanine, tyrosine, leucine, and isoleucine. These amino acids donate some of their carbons (red) to acetoacetate and acetyl-CoA. In the pathway for leucine catabolism, the carbon in acetoacetate that is donated by CO_2 is surrounded by a box to distinguish it from the three carbons that come from leucine itself. Tryptophan, phenylalanine, tyrosine, and isoleucine also contribute carbons (blue) to pyruvate or citric acid cycle intermediates. The phenylalanine pathway is described in more detail in [Figure 18-23](#). The fate of nitrogen atoms is not traced in this scheme; in most cases they are transferred to α -ketoglutarate to form glutamate.

The degradative pathways of two of these seven amino acids deserve special mention. Tryptophan breakdown is the most complex of all the pathways of amino acid catabolism in animal tissues; portions of tryptophan (four of its carbons) yield acetyl-CoA via acetoacetyl-CoA.

Some of the intermediates in tryptophan catabolism are precursors for the synthesis of other biomolecules ([Fig. 18-22](#)), including nicotinate, a precursor of NAD and NADP in animals; serotonin, a neurotransmitter in vertebrates; and indoleacetate, a growth factor in plants. Some of these biosynthetic pathways are described in more detail in [Chapter 22](#) (see [Figs. 22-30](#), [22-31](#)).



The breakdown of phenylalanine is noteworthy because genetic defects in the enzymes of this pathway lead to several inheritable human diseases ([Fig. 18-23](#)), as discussed below. Phenylalanine and its

oxidation product tyrosine (both with nine carbons) are degraded into two fragments, both of which can enter the citric acid cycle: four of the nine carbon atoms yield free acetoacetate, which is converted to acetoacetyl-CoA and thus acetyl-CoA, and a second four-carbon fragment is recovered as fumarate. Eight of the nine carbons of these two amino acids thus enter the citric acid cycle; the remaining carbon is lost as CO_2 . Phenylalanine, after its hydroxylation to tyrosine, is also the precursor of dopamine, a neurotransmitter, and of norepinephrine and epinephrine, hormones secreted by the adrenal medulla (see [Fig. 22-31](#)). Melanin, the black pigment of skin and hair, is also derived from tyrosine.

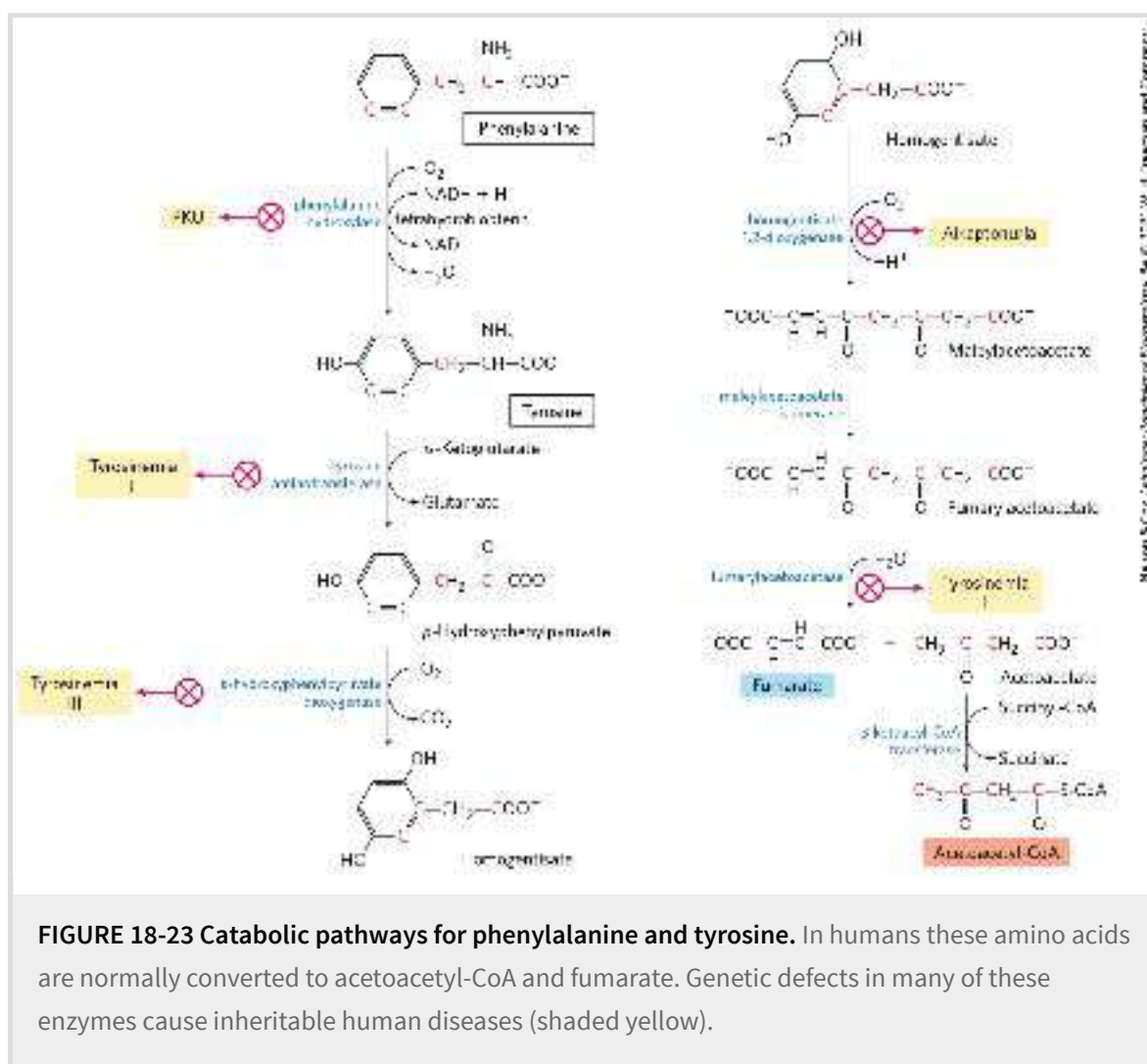


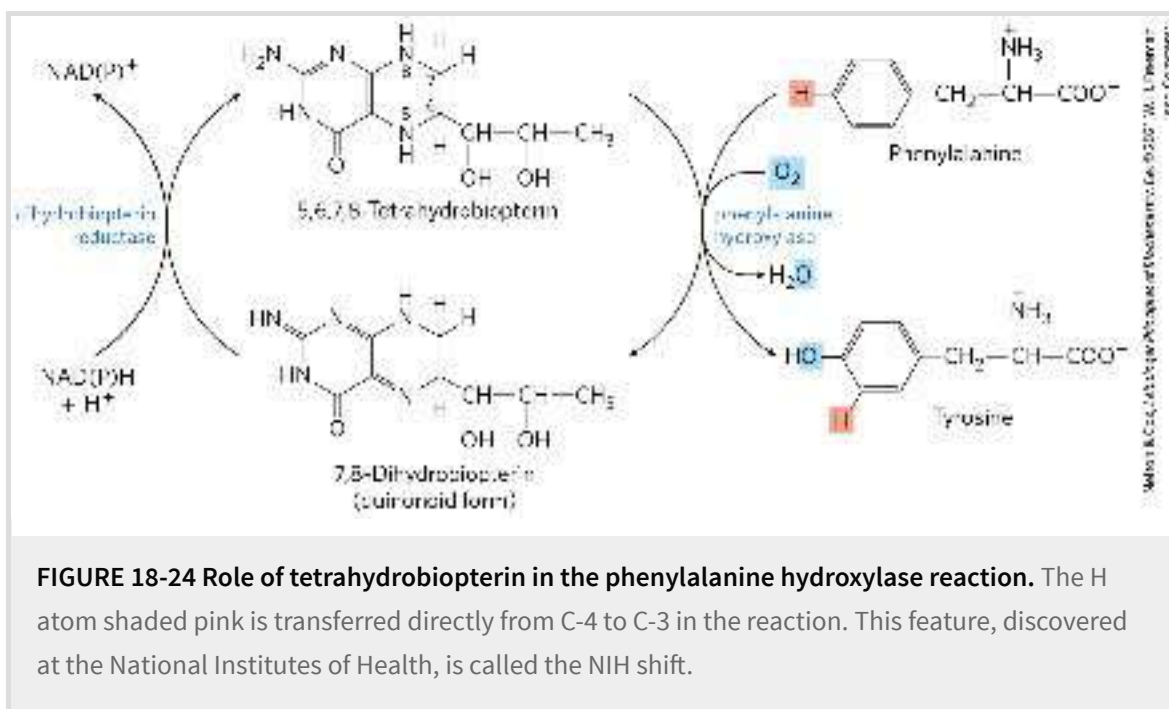
FIGURE 18-23 Catabolic pathways for phenylalanine and tyrosine. In humans these amino acids are normally converted to acetoacetyl-CoA and fumarate. Genetic defects in many of these enzymes cause inheritable human diseases (shaded yellow).

Phenylalanine Catabolism Is Genetically Defective in Some People



Given that many amino acids are either neurotransmitters or precursors or antagonists of neurotransmitters, it is not surprising that genetic defects of amino acid metabolism can cause defective neural development and intellectual deficits. In most such diseases, specific intermediates accumulate. For example, a genetic defect in **phenylalanine hydroxylase**, the first enzyme in the catabolic pathway for phenylalanine ([Fig. 18-23](#)), is responsible for the disease **phenylketonuria (PKU)**, the most common cause of elevated levels of phenylalanine in the blood (hyperphenylalaninemia).

Phenylalanine hydroxylase (also called phenylalanine-4-monooxygenase) is one of a general class of enzymes called [mixed-function oxygenases](#) (see [Box 21-1](#)), all of which catalyze simultaneous hydroxylation of a substrate by an oxygen atom of O_2 and reduction of the other oxygen atom to H_2O . Phenylalanine hydroxylase requires the cofactor tetrahydrobiopterin, which carries electrons from NADPH to O_2 and becomes oxidized to dihydrobiopterin in the process ([Fig. 18-24](#)). It is subsequently reduced by the enzyme **dihydrobiopterin reductase** in a reaction that requires NADPH.



In individuals with PKU, a secondary, normally little-used pathway of phenylalanine metabolism comes into play. In this pathway phenylalanine undergoes transamination with pyruvate to yield **phenylpyruvate** ([Fig. 18-25](#)). Phenylalanine and phenylpyruvate accumulate in the blood and tissues and are excreted in the urine — hence the name “phenylketonuria.” Much of the phenylpyruvate, rather than being excreted as such, is either decarboxylated to phenylacetate or reduced to phenyllactate. Phenylacetate imparts a characteristic honeylike odor to the urine, which nurses have traditionally used to detect PKU in infants. The accumulation of phenylalanine or its metabolites in early life impairs normal development of the brain, causing severe intellectual deficits. This may be caused by excess phenylalanine competing with other amino acids for transport across the blood-brain barrier, resulting in a deficit of required metabolites.

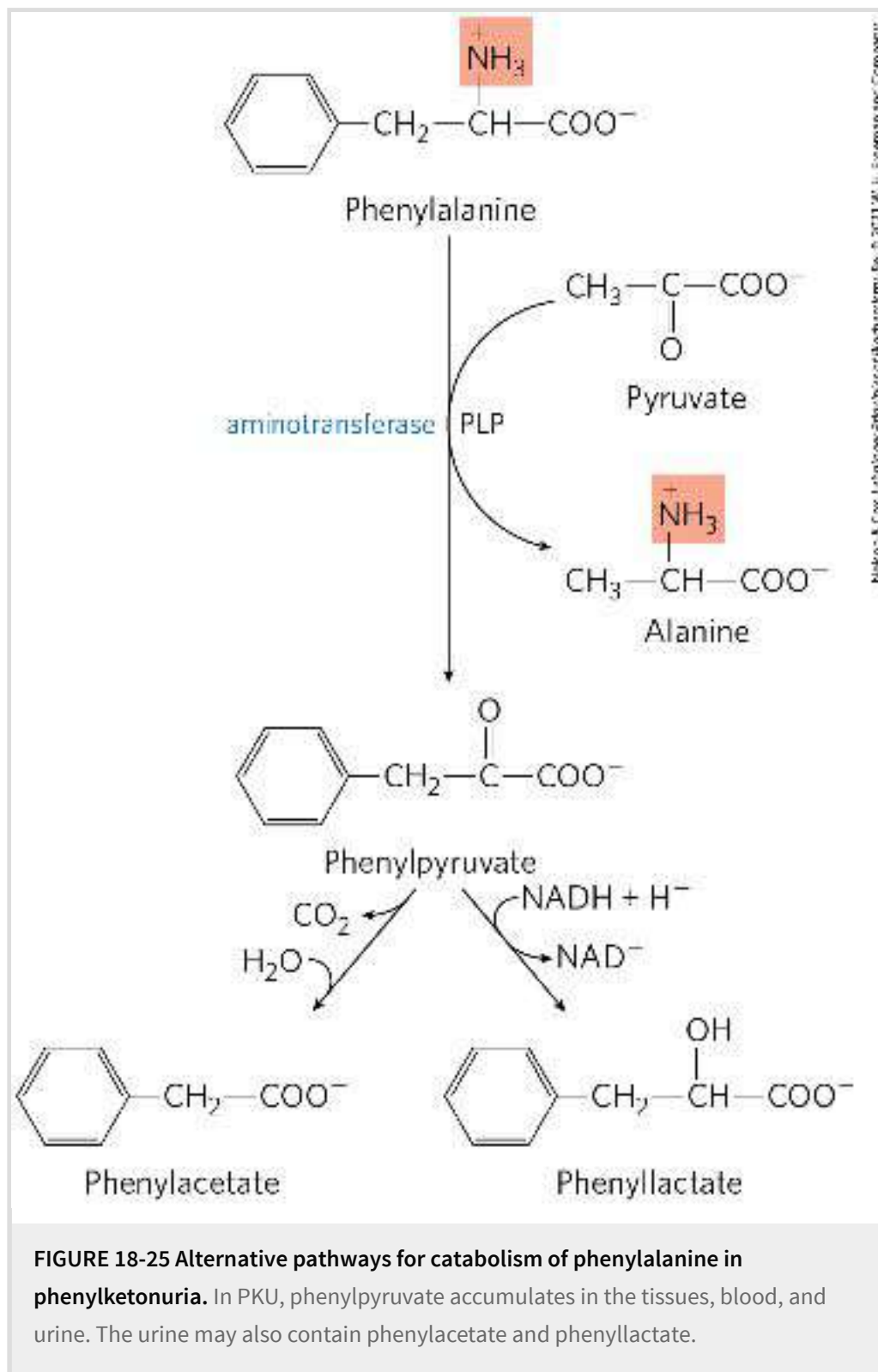


FIGURE 18-25 Alternative pathways for catabolism of phenylalanine in phenylketonuria. In PKU, phenylpyruvate accumulates in the tissues, blood, and urine. The urine may also contain phenylacetate and phenyllactate.

Phenylketonuria was among the first inheritable metabolic defects discovered in humans. When this condition is recognized early in infancy, intellectual disability can be prevented by rigid dietary control.

The diet must supply only enough phenylalanine and tyrosine to meet the needs for protein synthesis. Consumption of protein-rich foods must be curtailed. Natural proteins, such as casein of milk, must first be hydrolyzed and much of the phenylalanine removed to provide an appropriate diet, at least through childhood. Because the artificial sweetener aspartame is a dipeptide of aspartate and the methyl ester of phenylalanine (see [Fig. 1-23a](#)), foods sweetened with aspartame bear warnings addressed to individuals on phenylalanine-controlled diets.

The dietary restrictions are difficult to follow perfectly for a lifetime, and thus often do not completely eliminate neurological symptoms. An enzyme substitution treatment has been developed in which the enzyme phenylalanine ammonia lyase is modified with polyethylene glycol (PEGylated) and injected subcutaneously to degrade phenylalanine in proteins ingested as part of a somewhat less restricted diet. Derived from plants, bacteria, and many yeast and fungi, phenylalanine ammonia lyase normally contributes to the biosynthesis of polyphenol compounds such as flavonoids. It degrades phenylalanine to the harmless metabolite *trans*-cinnamic acid and ammonia; the small amounts of ammonia generated are not toxic. The treatment was approved for patients in 2018. The long-term effects of the treatment continue to be studied.

Phenylketonuria can also be caused by a defect in the enzyme that catalyzes the regeneration of tetrahydrobiopterin ([Fig. 18-24](#)). The treatment in this case is more complex than restricting the intake of phenylalanine and tyrosine. Tetrahydrobiopterin is also required for the formation of L-3,4-dihydroxyphenylalanine (L-dopa) and 5-hydroxytryptophan — precursors of the neurotransmitters norepinephrine and serotonin, respectively. In phenylketonuria of this

type, these precursors must be supplied in the diet, along with tetrahydrobiopterin.

Screening newborns for genetic diseases can be highly cost-effective, especially in the case of PKU. The tests (no longer relying on urine odor) are relatively inexpensive, and the detection and early treatment of PKU in infants (eight to ten cases per 100,000 newborns) saves millions of dollars in later health care costs each year. More importantly, the emotional trauma avoided by early detection with these simple tests is inestimable.

Another inheritable disease of phenylalanine catabolism is **alkaptonuria**, in which the defective enzyme is **homogentisate dioxygenase** ([Fig. 18-23](#)). Less serious than PKU, this condition produces few ill effects, although large amounts of homogentisate are excreted and its oxidation turns the urine black. Individuals with alkaptonuria are also prone to develop a form of arthritis. Alkaptonuria is of considerable historical interest. Archibald Garrod discovered in the early 1900s that this condition is inherited, and he traced the cause to the absence of a single enzyme. Garrod was the first to make a connection between an inheritable trait and an enzyme — a great advance on the path that ultimately led to our current understanding of genes and the information pathways described in Part III. 

Five Amino Acids Are Converted to α - Ketoglutarate

 The carbon skeletons of five amino acids (proline, glutamate, glutamine, arginine, and histidine) enter the citric acid cycle as α - ketoglutarate ([Fig. 18-26](#)). **Proline**, **glutamate**, and **glutamine** have five-

carbon skeletons. The cyclic structure of proline is opened by oxidation of the carbon most distant from the carboxyl group to create a Schiff base, then hydrolysis of the Schiff base to form a linear semialdehyde, glutamate γ -semialdehyde. This intermediate is further oxidized at the same carbon to produce glutamate. The action of glutaminase, or any of several enzyme reactions in which glutamine donates its amide nitrogen to an acceptor, converts glutamine to glutamate.

Transamination or deamination of glutamate produces α -ketoglutarate.

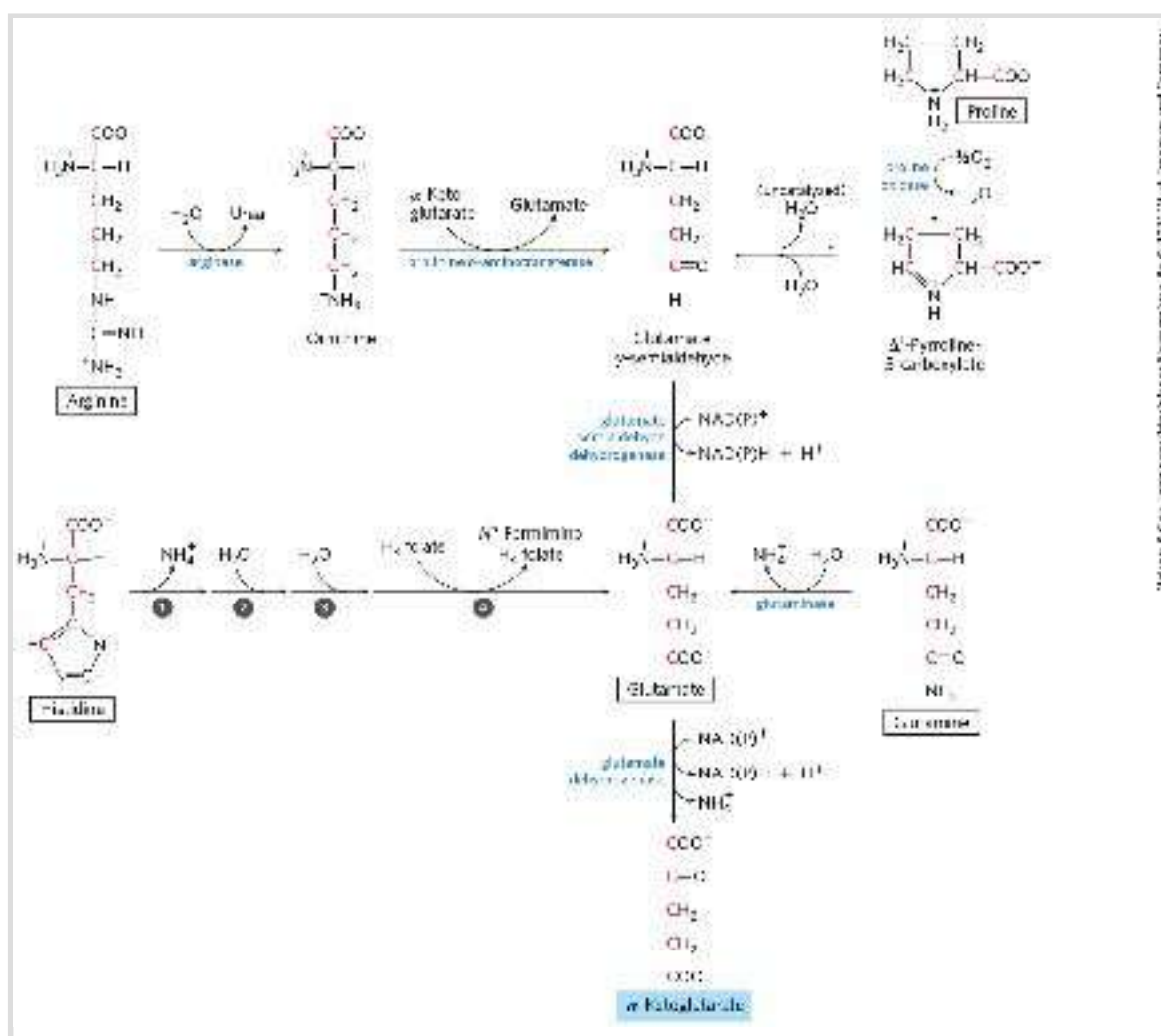


FIGURE 18-26 Catabolic pathways for arginine, histidine, glutamate, glutamine, and proline.

These amino acids are converted to α -ketoglutarate. The numbered steps in the histidine pathway are catalyzed by **1** histidine ammonia lyase, **2** urocanate hydratase, **3** imidazolonepropionase, and **4** glutamate formimino transferase.

Arginine and **histidine** contain five adjacent carbons and a sixth carbon attached through a nitrogen atom. The catabolic conversion of these amino acids to glutamate is therefore slightly more complex than the path from proline or glutamine ([Fig. 18-26](#)). Arginine is converted to the five-carbon skeleton of ornithine in the urea cycle ([Fig. 18-10](#)), and the ornithine is transaminated to glutamate γ -semialdehyde. Conversion of histidine to the five-carbon glutamate occurs in a multistep pathway; the extra carbon is removed in a step that uses tetrahydrofolate as cofactor.

Four Amino Acids Are Converted to Succinyl-CoA

 The carbon skeletons of methionine, isoleucine, threonine, and valine are degraded by pathways that yield succinyl-CoA ([Fig. 18-27](#)), an intermediate of the citric acid cycle. **Methionine** donates its methyl group to one of several possible acceptors through S-adenosyl methionine, and three of its four remaining carbon atoms are converted to the propionate of propionyl-CoA, a precursor of succinyl-CoA. **Isoleucine** undergoes transamination, followed by oxidative decarboxylation of the resulting α -keto acid. The remaining five-carbon skeleton is further oxidized to acetyl-CoA and propionyl-CoA. **Valine** undergoes transamination and decarboxylation, then a series of oxidation reactions that convert the remaining four carbons to propionyl-CoA. Some parts of the valine and isoleucine degradative pathways closely parallel steps in fatty acid degradation (see [Fig. 17-9](#)). In human tissues, **threonine** is also converted in two steps to propionyl-CoA. This is the primary pathway for threonine degradation in humans (see [Fig. 18-19](#) for the alternative pathway). The mechanism of the first

step is analogous to that catalyzed by serine dehydratase, and the serine and threonine dehydratases may actually be the same enzyme.

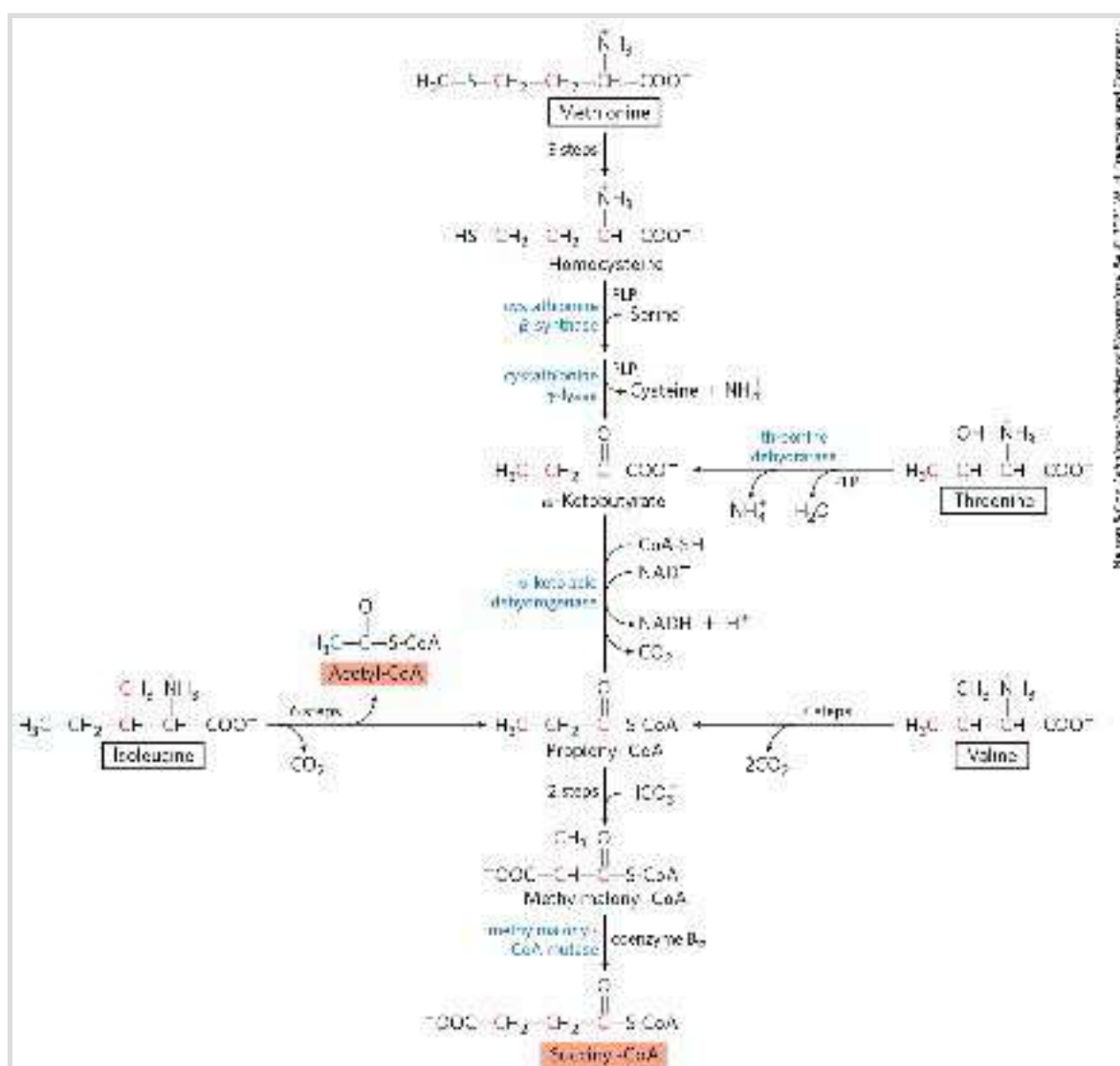


FIGURE 18-27 Catabolic pathways for methionine, isoleucine, threonine, and valine. These amino acids are converted to succinyl-CoA; isoleucine also contributes two of its carbon atoms to acetyl-CoA (see [Fig. 18-21](#)). The pathway of threonine degradation shown here occurs in humans; a pathway found in other organisms is shown in [Figure 18-19](#). The route from methionine to homocysteine is described in more detail in [Figure 18-18](#); the conversion of homocysteine to α -ketobutyrate, in [Figure 22-16](#); and the conversion of propionyl-CoA to succinyl-CoA, in [Figure 17-12](#).

The propionyl-CoA derived from these three amino acids is converted to succinyl-CoA by a pathway described in [Chapter 17](#): carboxylation to

methylmalonyl-CoA, epimerization of the methylmalonyl-CoA, and conversion to succinyl-CoA by the coenzyme B₁₂-dependent methylmalonyl-CoA mutase (see [Fig. 17-12](#)). In the rare genetic disease known as methylmalonic acidemia, methylmalonyl-CoA mutase is lacking — with serious metabolic consequences ([Table 18-2](#); [Box 18-2](#)).

BOX 18-2 MEDICINE

MMA: Sometimes More than a Genetic Disease

Without treatment, an infant born with a genetic deficiency involving the enzyme methylmalonyl-CoA mutase is subject to severe symptoms ranging from seizures and vomiting to lethargy, developmental disease, progressive encephalopathy (brain disease), and early death. A rare and recessive genetic disorder, the condition, methylmalonic acidemia or MMA, affects about 1 child in 48,000 ([Fig. 1](#)). For most parents of a child with MMA, the condition brings trials and some heartbreak. For Patricia Stallings, the condition of her undiagnosed child led to much worse. Biochemistry eventually came to the rescue.

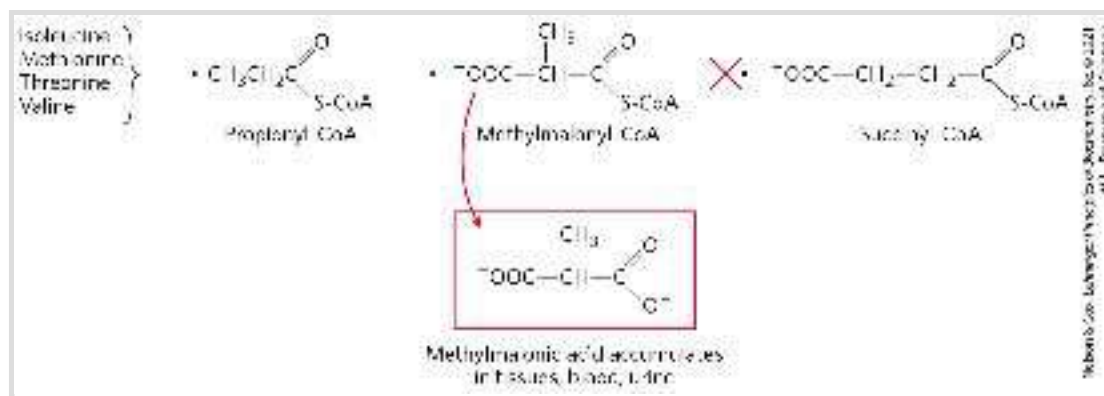


FIGURE 1 Children with a mutation (red X) that inactivates the enzyme methylmalonyl-CoA mutase cannot degrade isoleucine, methionine, threonine, and valine normally. Instead, a potentially fatal accumulation of methylmalonic acid occurs, with symptoms similar to those of ethylene glycol poisoning.

In the summer of 1989, Stallings took her baby son Ryan to the emergency room at the Cardinal Glennon Children's Hospital in St. Louis. To the attending physician, a toxicologist, Ryan's fit of vomiting and labored breathing suggested ingestion of the antifreeze ingredient ethylene glycol. The toxicologist suspected foul play and alerted

authorities. Analysis of Ryan's milk bottles by two laboratories seemed to confirm the physician's fears. Ryan was placed in foster care as soon as he recovered. Unfortunately for both Ryan and his mother, Ryan subsequently died immediately after a visit from Patricia. Patricia was arrested and charged with murder.

While awaiting trial, Patricia learned that she was again pregnant. She gave birth to a second son, David, who was placed in foster care. David developed symptoms similar to Ryan's and was diagnosed with MMA. Although the symptoms of MMA can mimic those observed in cases of ethylene glycol poisoning, the revelation did not help Patricia. The information about David's diagnosis was excluded at her trial for the murder of Ryan. In 1991, Patricia Stallings was sentenced to prison for life.

The situation was ultimately resolved when William Sly, chair of the Department of Biochemistry and Molecular Biology at St. Louis University, became interested in the case. Working with metabolic disease experts James Shoemaker and Piero Rinaldo, he had new analyses performed on Ryan's blood and the original milk bottles. The blood showed high levels of methylmalonic acid and ketones, diagnostic of MMA. Surprisingly, unlike the earlier lab reports, they found no ethylene glycol in any of the samples. Could both a hospital lab and a commercial lab have been mistaken? What the biochemists saw provided the final breakthrough. The biochemists presented all of their materials to the Missouri state prosecutor who had handled the case, George McElroy. Both of the original lab analyses had been so unusually shoddy that they were quickly dismissed. Patricia Stallings was cleared of charges and released on September 20, 1991.

Branched-Chain Amino Acids Are Not Degraded in the Liver

Although much of the catabolism of amino acids takes place in the liver, the three amino acids with branched side chains (leucine, isoleucine, and valine) are oxidized as fuels primarily in muscle, adipose, kidney, and brain tissue. These extrahepatic tissues contain an aminotransferase, absent in liver, that acts on all three branched-chain amino acids to produce the corresponding α -keto acids ([Fig. 18-28](#)).

The **branched-chain α -keto acid dehydrogenase complex** then catalyzes oxidative decarboxylation of all three α -keto acids, in each case releasing the carboxyl group as CO_2 and producing the acyl-CoA derivative. This reaction is formally analogous to two other oxidative decarboxylations encountered in [Chapter 16](#): oxidation of pyruvate to acetyl-CoA by the pyruvate dehydrogenase complex and oxidation of α -ketoglutarate to succinyl-CoA by the α -ketoglutarate dehydrogenase complex (see [Fig. 16-12](#)). In fact, all three enzyme complexes are similar in structure and share essentially the same reaction mechanism. Five cofactors (thiamine pyrophosphate, FAD, NAD, lipoate, and coenzyme A) participate, and the three proteins in each complex catalyze homologous reactions. This is clearly a case in which enzymatic machinery that evolved to catalyze one reaction was “borrowed” by gene duplication and further evolved to catalyze similar reactions in other pathways.

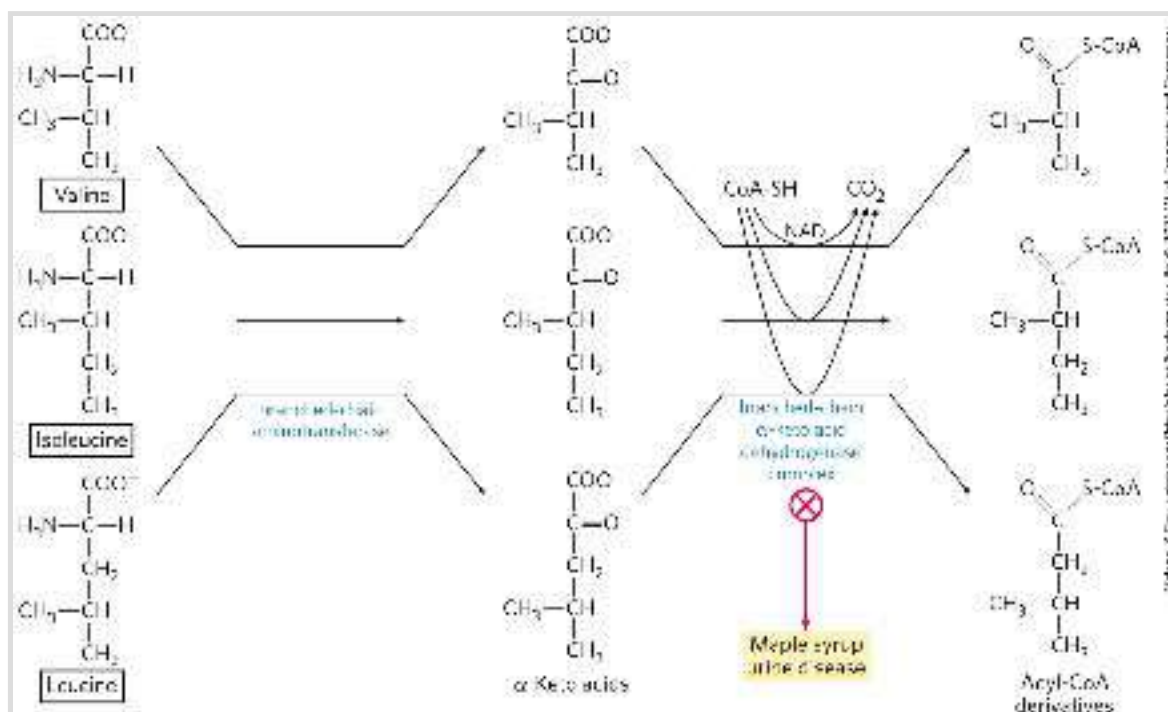


FIGURE 18-28 Catabolic pathways for the three branched-chain amino acids: valine, isoleucine, and leucine. All three pathways occur in extrahepatic tissues and share the first two enzymes, as shown here. The branched-chain α -keto acid dehydrogenase complex is analogous

to the pyruvate and α -ketoglutarate dehydrogenase complexes and requires the same five cofactors (some not shown here). This enzyme is defective in people with maple syrup urine disease.

Experiments with rats have shown that the branched-chain α -keto acid dehydrogenase complex is regulated by covalent modification in response to the content of branched-chain amino acids in the diet. With little or no excess dietary intake of branched-chain amino acids, the enzyme complex is phosphorylated by a protein kinase and thereby inactivated. Addition of excess branched-chain amino acids to the diet results in dephosphorylation and consequent activation of the enzyme. Recall that the pyruvate dehydrogenase complex is subject to similar regulation by phosphorylation and dephosphorylation (see [Fig. 16-19](#)).



There is a relatively rare genetic disease in which the three branched-chain α -keto acids (as well as their precursor amino acids, especially leucine) accumulate in the blood and “spill over” into the urine. This condition, called **maple syrup urine disease** because of the characteristic odor imparted to the urine by the α -keto acids, results from a defective branched-chain α -keto acid dehydrogenase complex. Untreated, the disease results in abnormal development of the brain and death in early infancy. Treatment entails rigid control of the diet, limiting the intake of valine, isoleucine, and leucine to the minimum required to permit normal growth. ■

Asparagine and Aspartate Are Degraded to Oxaloacetate

The carbon skeletons of **asparagine** and **aspartate** ultimately enter the citric acid cycle as malate in mammals or oxaloacetate in bacteria. The

enzyme **asparaginase** catalyzes the hydrolysis of asparagine to aspartate, which undergoes transamination with α -ketoglutarate to yield glutamate and oxaloacetate ([Fig. 18-29](#)). The oxaloacetate is converted to malate in the cytosol and then transported into the mitochondrial matrix through the malate- α -ketoglutarate transporter. In bacteria, the oxaloacetate produced in the transamination reaction can be used directly in the citric acid cycle.

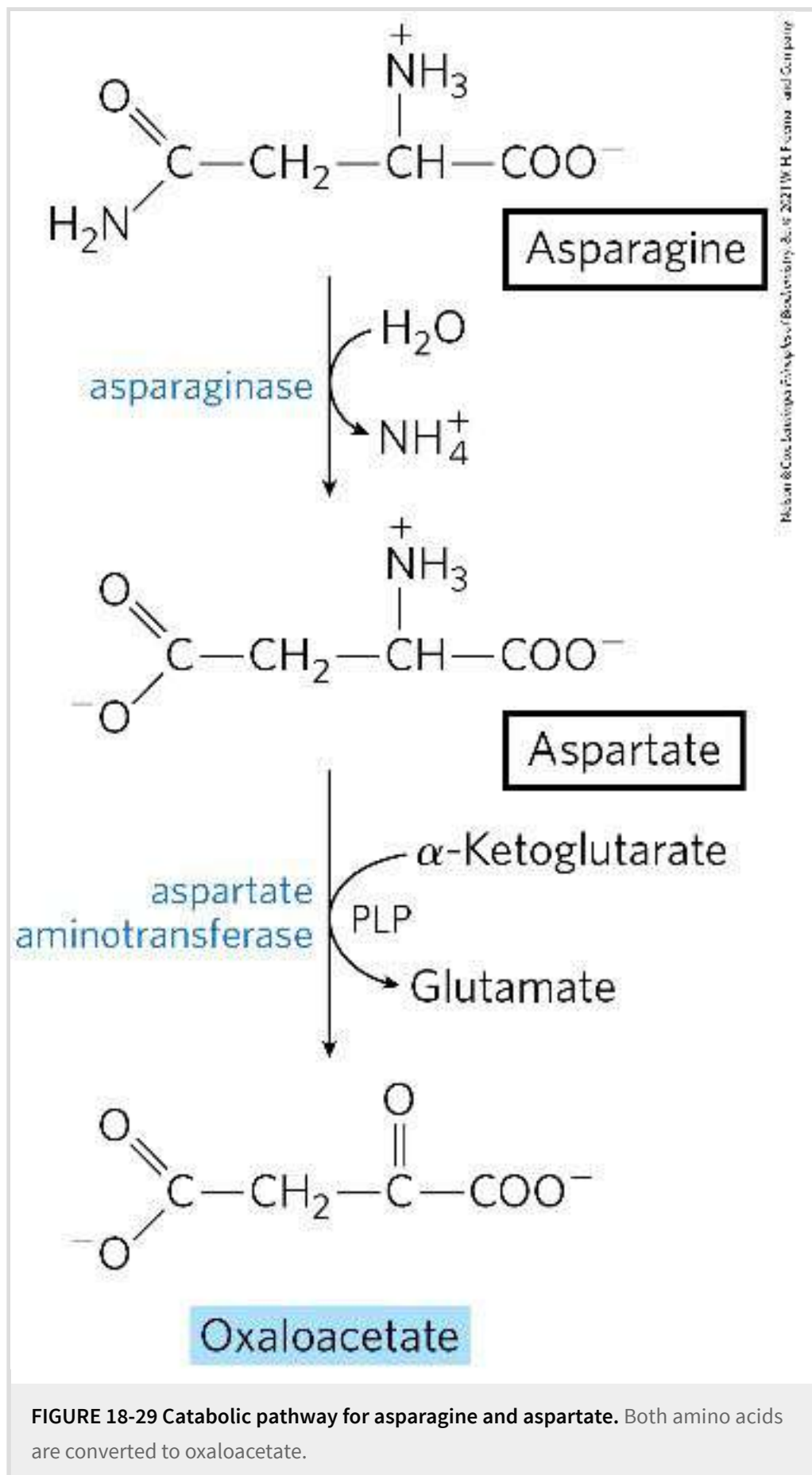


FIGURE 18-29 Catabolic pathway for asparagine and aspartate. Both amino acids are converted to oxaloacetate.

We have now seen how the 20 common amino acids, after losing their nitrogen atoms, are degraded by dehydrogenation, decarboxylation, and other reactions to yield portions of their carbon backbones in the form of six central metabolites that can enter the citric acid cycle. Those portions degraded to acetyl-CoA are completely oxidized to carbon dioxide and water, with generation of ATP by oxidative phosphorylation. Those portions degraded to other citric acid cycle intermediates can either be oxidized or contribute to gluconeogenesis, depending on the metabolic state.

As was the case for carbohydrates and lipids, the degradation of amino acids results ultimately in the generation of reducing equivalents (NADH and FADH₂) through the action of the citric acid cycle. Our survey of catabolic processes concludes in the next chapter with a discussion of respiration, in which these reducing equivalents fuel the ultimate oxidative and energy-generating process in aerobic organisms.

SUMMARY 18.3 *Pathways of Amino Acid Degradation*

- After the removal of amino groups, the carbon skeletons of amino acids undergo oxidation to compounds that can enter the citric acid cycle. The citric acid cycle products can be oxidized as fuel. Alternatively, depending on their degradative end product, some amino acids can be converted to ketone bodies, some to glucose, and some to both. Thus, amino acid degradation is integrated into intermediary metabolism and can be critical to survival under conditions in which amino acids are a significant source of metabolic energy.
- The reactions of these pathways require several cofactors, including tetrahydrofolate and S-adenosylmethionine in one-carbon transfer

reactions and tetrahydrobiopterin in the oxidation of phenylalanine by phenylalanine hydroxylase.

- Alanine, cysteine, glycine, serine, threonine, and tryptophan are converted in whole or in part to pyruvate.

- Leucine, lysine, phenylalanine, tyrosine, and tryptophan yield acetyl-CoA via acetoacetyl-CoA. Isoleucine, leucine, threonine, and tryptophan also form acetyl-CoA directly. Leucine and lysine are the only amino acids that cannot contribute to gluconeogenesis. Four carbon atoms of phenylalanine and tyrosine give rise to fumarate.

- Many human genetic diseases are traced to deficiencies in enzymes catalyzing steps in amino acid degradation. Two well-studied examples, phenylketonuria and alkaptonuria, feature defects in phenylalanine degradation. Most amino acid degradation deficiencies are treated with dietary intervention.

- Arginine, glutamate, glutamine, histidine, and proline are degraded to α -ketoglutarate.

- Isoleucine, methionine, threonine, and valine produce succinyl-CoA.

- The branched-chain amino acids (isoleucine, leucine, and valine), unlike the other amino acids, are degraded only in extrahepatic tissues.

- Asparagine and aspartate produce oxaloacetate.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

ammonotelic

ureotelic

uricotelic

aminotransferases

transaminases

transamination

pyridoxal phosphate (PLP)

oxidative deamination

L-glutamate dehydrogenase

glutamine synthetase

glutaminase

glucose-alanine cycle

urea cycle

urea

creatine kinase

essential amino acids

ketogenic

glucogenic

tetrahydrofolate

S-adenosylmethionine (adoMet)

tetrahydrobiopterin

phenylketonuria (PKU)

[mixed-function oxygenases](#)

[alkaptonuria](#)

[maple syrup urine disease](#)

PROBLEMS

1. Products of Amino Acid Transamination Name and draw the structure of the α -keto acid resulting when each of the four amino acids listed undergoes transamination with α -ketoglutarate: **(a)** aspartate, **(b)** glutamate, **(c)** alanine, **(d)** phenylalanine.

2. Measurement of Alanine Aminotransferase Activity The measurement of alanine aminotransferase activity (reaction rate) usually includes an excess of pure lactate dehydrogenase and NADH in the reaction system. The rate of alanine disappearance is equal to the rate of NADH disappearance measured spectrophotometrically. Explain how this assay works.

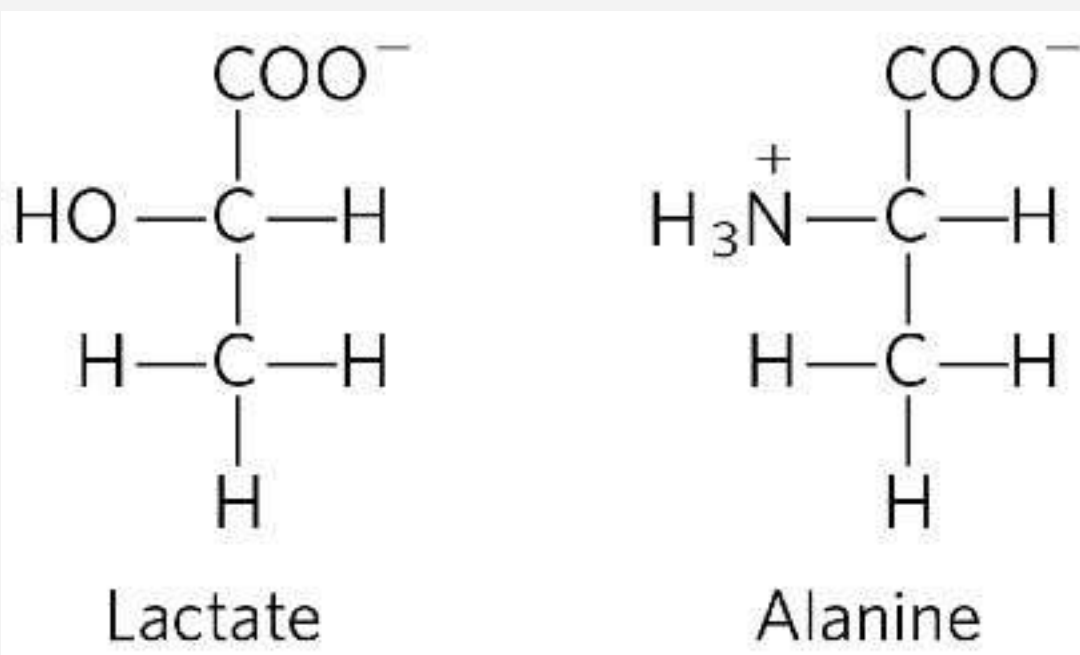
3. Alanine and Glutamine in the Blood Normal human blood plasma contains all the amino acids required for the synthesis of body proteins, but not in equal concentrations. Alanine and glutamine are present in much higher concentrations than any other amino acids. Suggest why.

4. Glutamate Dehydrogenase Function Increases in ATP levels in blood trigger insulin release by the pancreas, which

in turn stimulates the uptake of blood glucose (see [Chapter 23, p. 860](#)). Knowing this, suggest why a mutation that prevents inhibition of glutamate dehydrogenase by GTP results in insulin release and hypoglycemia.

5. Distribution of Amino Nitrogen If your diet is rich in alanine but deficient in aspartate, will you show signs of aspartate deficiency? Explain.

6. Lactate versus Alanine as Metabolic Fuel: The Cost of Nitrogen Removal The three carbons in lactate and alanine have identical oxidation states, and animals can use either carbon source as a metabolic fuel. Compare the net ATP yield (moles of ATP per mole of substrate) for the complete oxidation (to CO_2 and H_2O) of lactate versus alanine when the cost of nitrogen excretion as urea is included.



7. Ammonia Toxicity Resulting from an Arginine-Deficient Diet

In a study, cats were fasted overnight then given a single meal complete in all amino acids except arginine. Within 2 hours, blood ammonia levels increased from a normal level of $18 \mu\text{g/L}$ to $140 \mu\text{g/L}$, and the cats showed the clinical symptoms of ammonia toxicity. A control group fed a complete amino acid diet or an amino acid diet in which arginine was replaced by ornithine showed no unusual clinical symptoms.

- a. What was the role of fasting in the experiment?
- b. What caused the ammonia levels to rise in the experimental group? Why did the absence of arginine lead to ammonia toxicity? Is arginine an essential amino acid in cats? Why or why not?
- c. Why can ornithine be substituted for arginine?

8. Oxidation of Glutamate Write a series of balanced equations and an overall equation for the net reaction describing the oxidation of 2 mol of glutamate to 2 mol of α -ketoglutarate and 1 mol of urea.

9. Transamination and the Urea Cycle Aspartate aminotransferase has the highest activity of all the mammalian liver aminotransferases. Why?

10.  The Case against the Liquid Protein Diet A weight-reducing diet heavily promoted some years ago required the daily intake of a “liquid protein” soup made of hydrolyzed gelatin (derived from collagen), water, and an assortment of

vitamins. All other food and drink were to be avoided. People on this diet typically lost 10 to 14 lb in the first week.

- a. Opponents argued that the weight loss was almost entirely due to water loss and would be regained very soon after a normal diet was resumed. What is the biochemical basis for this argument?
- b. A few people on this diet died. What are some of the dangers inherent in the diet, and how can they lead to death?

11. Ketogenic Amino Acids Which amino acids are exclusively ketogenic?

12.  A Genetic Defect in Amino Acid Metabolism: A Case History A two-year-old child was taken to the hospital. His mother said that he vomited frequently, especially after feedings. The child's weight and physical development were below normal. His hair, although dark, contained patches of white. A urine sample treated with ferric chloride (FeCl_3) gave a green color characteristic of the presence of phenylpyruvate. Quantitative analysis of urine samples gave the results shown in the table.

Substance	Concentration (mM)	
	Patient's urine	Normal urine
Phenylalanine	7.0	0.01
Phenylpyruvate	4.8	0
Phenyllactate	10.3	0

- a. Suggest which enzyme might be deficient in this child. Propose a treatment.
- b. Why does phenylalanine appear in the urine in large amounts?
- c. What is the source of phenylpyruvate and phenyllactate? Why does this pathway (normally not functional) come into play when the concentration of phenylalanine rises?
- d. Why does the boy's hair contain patches of white?

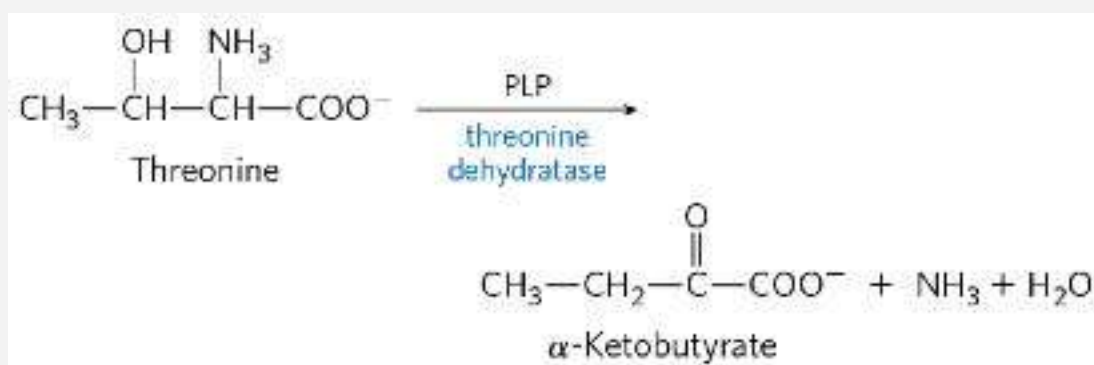
13. Role of Cobalamin in Amino Acid Catabolism

Pernicious anemia is caused by impaired absorption of vitamin B₁₂. What is the effect of this impairment on the catabolism of amino acids? Are all amino acids equally affected? (Hint: See [Box 17-2](#).)

14.  Vegetarian Diets Vegetarian diets can provide high levels of antioxidants and a lipid profile that can help prevent coronary disease. However, there can be some associated problems. Blood samples were taken from a large group of volunteer subjects who were vegans (strict vegetarians: no animal products), lactovegetarians (vegetarians who eat dairy products), or omnivores (individuals with a varied diet, including meat). In each case, the volunteers had followed the diet for several years. The blood levels of both homocysteine and methylmalonate were elevated in the vegan group, somewhat lower in the lactovegetarian group, and much lower in the omnivore group. Explain.

15.  Pernicious Anemia Vitamin B₁₂ deficiency can arise from a few rare genetic diseases that lead to low B₁₂ levels despite a normal diet that includes B₁₂-rich meat and dairy sources. These conditions cannot be treated with dietary B₁₂ supplements. Explain.

16. Pyridoxal Phosphate Reaction Mechanisms Threonine can be broken down by the enzyme threonine dehydratase, which catalyzes the conversion of threonine to α -ketobutyrate and ammonia. The enzyme uses PLP as a cofactor. Suggest a mechanism for this reaction, based on the mechanisms in [Figure 18-6](#). Note that this reaction includes an elimination at the β carbon of threonine.



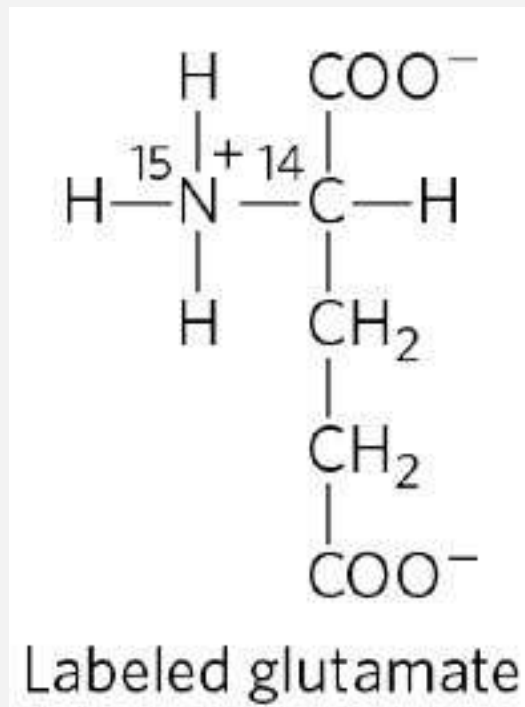
17. Pathway of Carbon and Nitrogen in Glutamate

Metabolism When $[2-^{14}\text{C}, ^{15}\text{N}]$ glutamate undergoes oxidative degradation in the liver of a rat, in which atoms of the following metabolites will each isotope be found?

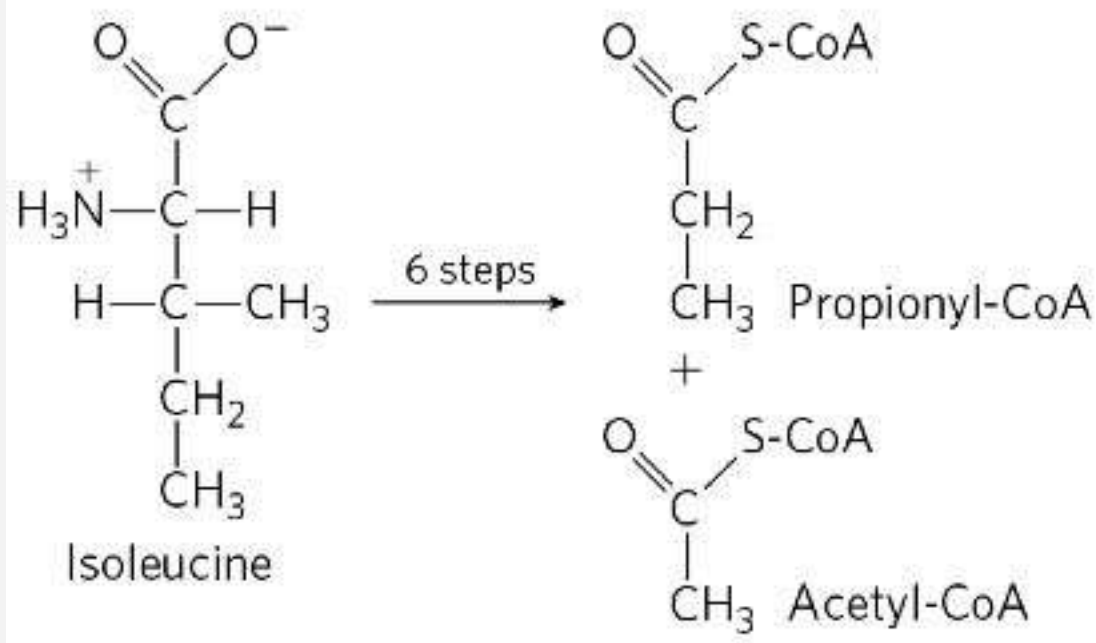
- a. urea
- b. succinate
- c. arginine
- d. citrulline

e. ornithine

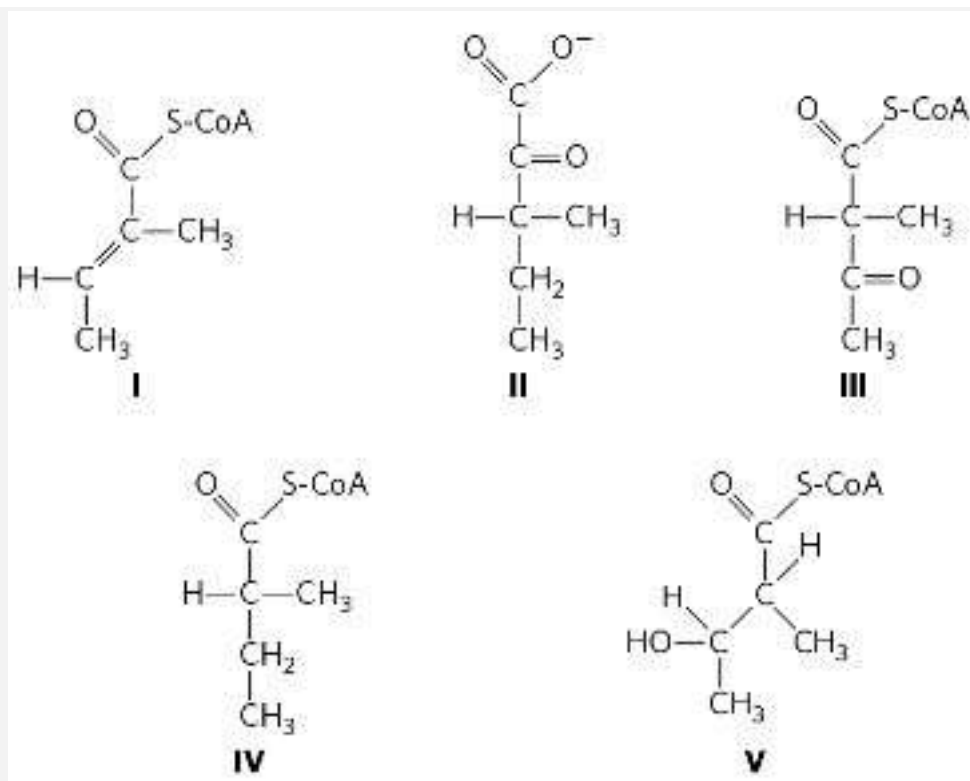
f. aspartate



18. Chemical Strategy of Isoleucine Catabolism Isoleucine is degraded in six steps to propionyl-CoA and acetyl-CoA.



- a. The chemical process of isoleucine degradation includes strategies analogous to those used in the citric acid cycle and the β oxidation of fatty acids. The intermediates of isoleucine degradation (I to V) shown below are not in the proper order. Use your knowledge and understanding of the citric acid cycle and β -oxidation pathway to arrange the intermediates in the proper metabolic sequence for isoleucine degradation.



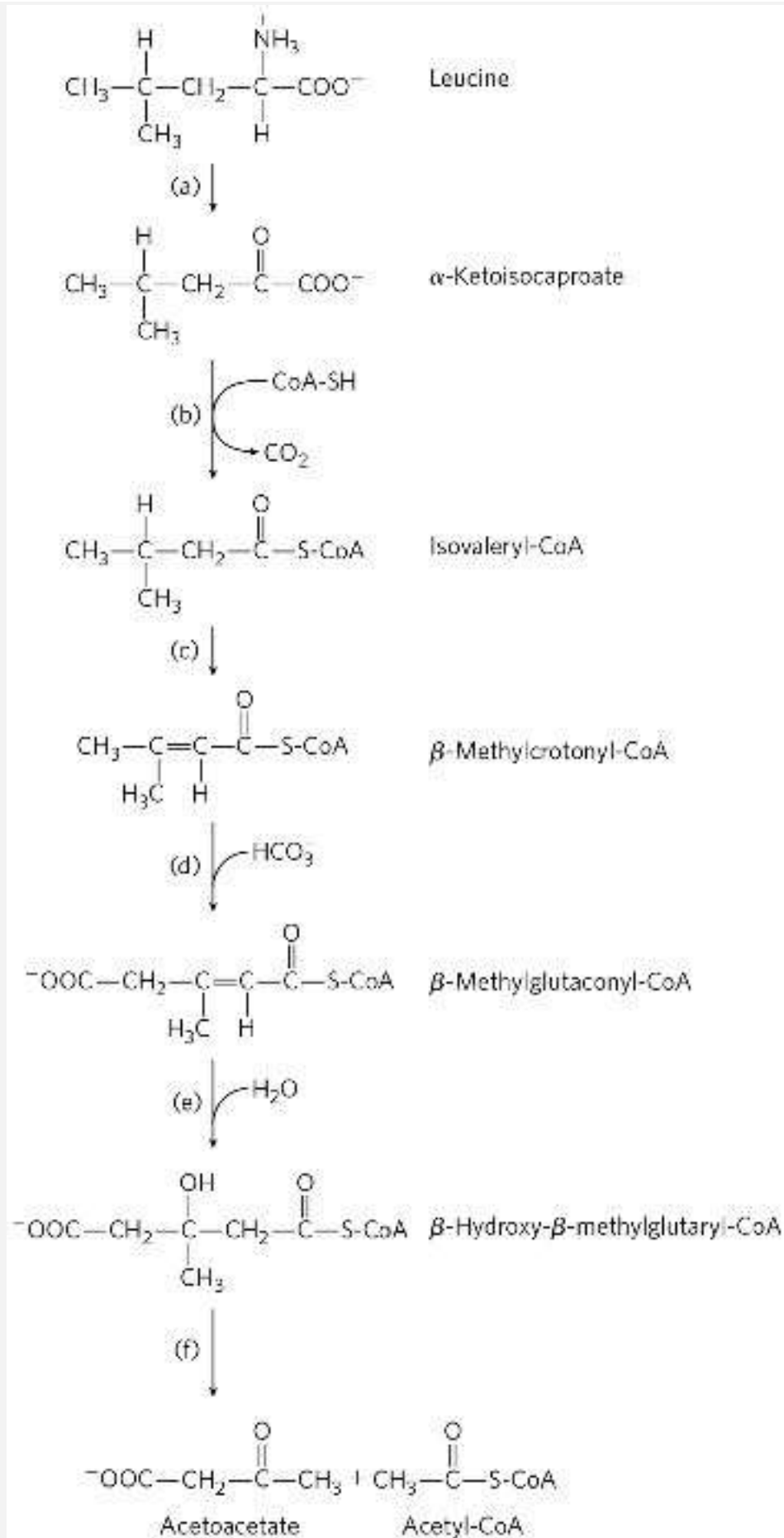
- b. For each step you propose, describe the chemical process, provide an analogous example from the citric acid cycle or β -oxidation pathway (where possible), and indicate any necessary cofactors.

19. Role of Pyridoxal Phosphate in Glycine Metabolism The enzyme serine hydroxymethyltransferase requires pyridoxal phosphate as a cofactor. Propose a mechanism for the reaction catalyzed by this enzyme, in the direction of serine degradation (glycine production). (Hint: See [Figs 18-19](#) and [18-20b](#).)

20. Parallel Pathways for Amino Acid and Fatty Acid

Degradation The carbon skeleton of leucine is degraded by a series of reactions closely analogous to those of the citric acid cycle and β oxidation. For each reaction, (a) through (f), shown below, indicate its type, provide an analogous

example from the citric acid cycle or β -oxidation pathway (where possible), and note any necessary cofactors.



21.  Treatments for a Genetic Disease The strict dietary controls required to stem the progress of maple syrup urine disease are difficult to follow for a lifetime, and patients may experience poor metabolic control that leads to neurological symptoms. In these cases, treatment can involve an organ transplant from a suitable donor. Organ transplantation involves considerable risk, but success can greatly alleviate this metabolic disorder and reduce the need for dietary restrictions. Which organ could be transplanted to gain this effect, and why?

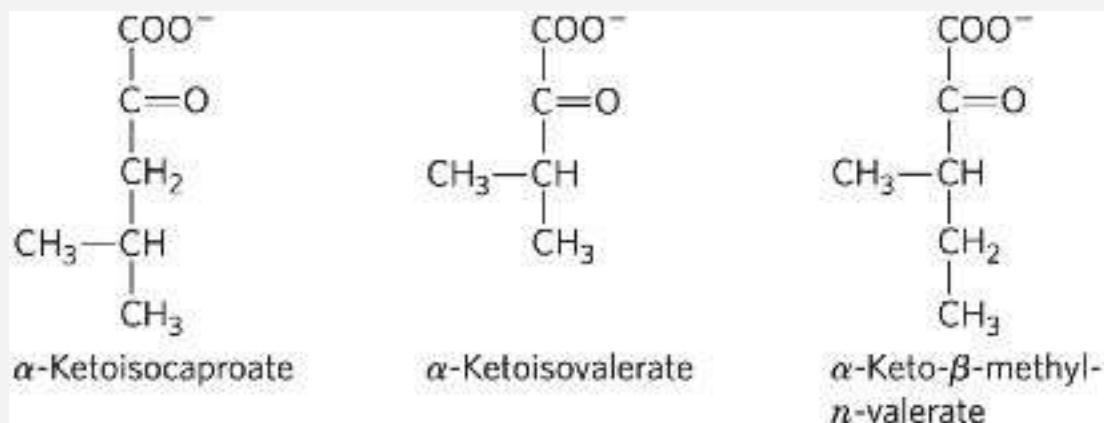
DATA ANALYSIS PROBLEM

22.  Maple Syrup Urine Disease [Figure 18-28](#) shows the pathway for the degradation of branched-chain amino acids and the site of the biochemical defect that causes maple syrup urine disease. The initial findings that eventually led to the discovery of the defect in this disease were presented in three papers published in the late 1950s and early 1960s. This problem traces the history of the findings from initial clinical observations to proposal of a biochemical mechanism.

Menkes, Hurst, and Craig (1954) presented the cases of four siblings, all of whom died following a similar course of symptoms. In all four cases, the mother's pregnancy and the birth had been normal. The first 3 to 5 days of each child's life were also normal. But soon thereafter each child began

having convulsions, and the children died between the ages of 11 days and 3 months. Autopsy showed considerable swelling of the brain in all cases. The children's urine had a strong, unusual "maple syrup" odor, starting from about the third day of life.

Menkes (1959) reported data collected from six more children. All showed symptoms similar to those described above and died within 15 days to 20 months of birth. In one case, Menkes was able to obtain urine samples during the last months of the infant's life. When he treated the urine with 2,4-dinitrophenylhydrazine, which forms colored precipitates with keto compounds, he found three α -keto acids in unusually large amounts:



- a. These α -keto acids are produced by the deamination of amino acids. For each of the α -keto acids above, draw and name the amino acid from which it was derived.

Dancis, Levitz, and Westall (1960) collected further data that led them to propose the biochemical defect

shown in [Figure 18-28](#). In one case, they examined a patient whose urine first showed the maple syrup odor when he was 4 months old. At the age of 10 months (March 1956), the child was admitted to the hospital because he had a fever, and he showed grossly delayed motor development. At the age of 20 months (January 1957), he was readmitted and was found to have the degenerative neurological symptoms seen in previous cases of maple syrup urine disease; he died soon after. Results of his blood and urine analyses are shown in the table, along with normal values for each component.

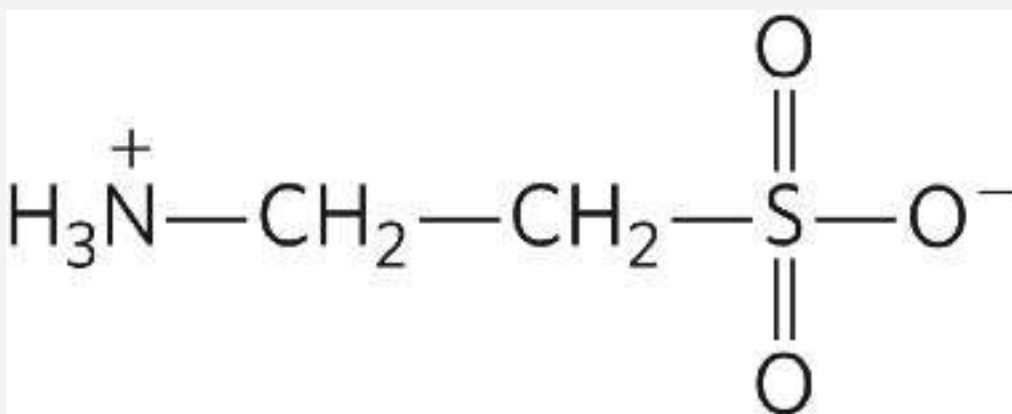
Amino acid(s)	Urine concentration (mg/24 h)			Plasma concentration (mg/mL)	
	Normal	Patient		Normal	Patient
		Mar. 1 1956	Jan. 1 1957		Jan. 1 1957
Alanine	5–15	0.2	0.4	3.0–4.8	0.6
Asparagine and glutamine	5–15	0.4	0	3.0–5.0	2.0
Aspartic acid	1–2	0.2	1.5	0.1–0.2	0.04
Arginine	1.5–3	0.3	0.7	0.8–1.4	0.8

e

Cystine	2–4	0.5	0.3	1.0–1.5	0
Glutamic acid	1.5–3	0.7	1.6	1.0–1.5	0.9
Glycine	20–40	4.6	20.7	1.0–2.0	1.5
Histidine	8–15	0.3	4.7	1.0–1.7	0.7
Isoleucine	2–5	2.0	13.5	0.8–1.5	2.2
Leucine	3–8	2.7	39.4	1.7–2.4	14.5
Lysine	2–12	1.6	4.3	1.5–2.7	1.1
Methionine	2–5	1.4	1.4	0.3–0.6	2.7
Ornithine	1–2	0	1.3	0.6–0.8	0.5
Phenylalanine	2–4	0.4	2.6	1.0–1.7	0.8
Proline	2–4	0.5	0.3	1.5–3.0	0.9
Serine	5–15	1.2	0	1.3–2.2	0.9
Taurine	1–10	0.2	18.7	0.9–1.8	0.4
Threonine	5–10	0.6	0	1.2–1.6	0.3

Tryptophan	3–8	0.9	2.3	Not measured	0
Tyrosine	4–8	0.3	3.7	1.5–2.3	0.7
Valine	2–4	1.6	15.4	2.0–3.0	13.1

- b. The table includes taurine, an amino acid not normally found in proteins. Taurine is often produced as a byproduct of cell damage. Its structure is



- Based on its structure and the information in this chapter, what is the most likely amino acid precursor of taurine? Explain your reasoning.
- c. Compared with the normal values given in the table, which amino acids showed significantly elevated levels in the patient's blood in January 1957? Which ones in the patient's urine?

Based on their results and their knowledge of the pathway shown in [Figure 18-28](#), Dancis and coauthors

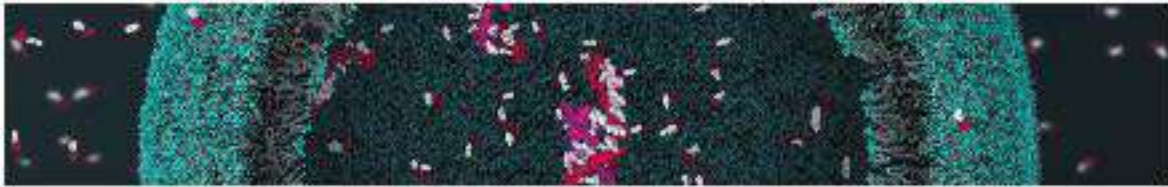
- concluded that “although it appears most likely to the authors that the primary block is in the metabolic degradative pathway of the branched-chain amino acids, this cannot be considered established beyond question.”
- d. How do the data presented here support this conclusion?
 - e. Which data presented here do *not* fit this model of maple syrup urine disease? How do you explain these seemingly contradictory data?
 - f. What data would you need to collect to be more secure in your conclusion?

References

- Dancis, J., M. Levitz, and R. Westall. 1960.** Maple syrup urine disease: branched-chain ketoaciduria. *Pediatrics* 25:72–79.
- Menkes, J.H. 1959.** Maple syrup disease: isolation and identification of organic acids in the urine. *Pediatrics* 23:348–353.
- Menkes, J.H., P.L. Hurst, and J.M. Craig. 1954.** A new syndrome: progressive familial infantile cerebral dysfunction associated with an unusual urinary substance. *Pediatrics* 14:462–466.

CHAPTER 19

OXIDATIVE PHOSPHORYLATION



[19.1 The Mitochondrial Respiratory Chain](#)

[19.2 ATP Synthesis](#)

[19.3 Regulation of Oxidative Phosphorylation](#)

[19.4 Mitochondria in Thermogenesis, Steroid Synthesis, and Apoptosis](#)

[19.5 Mitochondrial Genes: Their Origin and the Effects of Mutations](#)

Oxidative phosphorylation is the culmination of energy-yielding metabolism (catabolism) in aerobic organisms. All oxidative steps in the degradation of carbohydrates, fats, and amino acids converge at this final stage of cellular respiration, in which the energy of oxidation drives the synthesis of ATP. Oxidative phosphorylation accounts for most of the ATP synthesized by nonphotosynthetic organisms under most circumstances. In eukaryotes, oxidative phosphorylation occurs in mitochondria and involves huge protein complexes embedded in the mitochondrial membranes. The pathway to ATP synthesis in

mitochondria challenged and fascinated biochemists for much of the twentieth century. Our current exploration of the topic is guided by five principles:

P1 Mitochondria play a central role in eukaryotic aerobic metabolism. ATP production is not the only important mitochondrial function. Mitochondria play host to the citric acid cycle, the fatty acid β -oxidation pathway, and the pathways of amino acid oxidation. The mitochondria also act in thermogenesis, steroid synthesis, and apoptosis (programmed cell death). The discovery of these diverse and important roles of mitochondria has stimulated much current research on the biochemistry of this organelle.

P2 Mitochondria trace their evolutionary origin to bacteria. Over 1.45 billion years ago, an endosymbiotic relationship arose between bacteria and a primitive eukaryote or eukaryotic progenitor. Mitochondria are ubiquitous in modern eukaryotes, and their bacterial origin is evident in almost every aspect of their structure and function.

P3 Electrons flow from electron donors (oxidizable substrates) through a chain of membrane-bound carriers to a final electron acceptor with a large reduction potential. The final acceptor is molecular oxygen, O_2 . The appearance of oxygen in the atmosphere some 2.3 billion years ago, and its harnessing in living systems via the evolution of oxidative phosphorylation, made more complex life forms possible.

P4 The free energy made available by “downhill” (exergonic) electron flow is coupled to the “uphill” transport of protons across a proton-impermeable membrane. The free energy of fuel oxidation is thus conserved as a transmembrane electrochemical potential.

P5 The transmembrane flow of protons back down their electrochemical gradient through specific protein channels provides the free energy for synthesis of ATP. This process is catalyzed by a membrane protein complex (ATP synthase) that couples proton flow to phosphorylation of ADP.

Principles 3 through 5, illustrated in [Figure 19-1](#), encapsulate the theory, introduced by Peter Mitchell in 1961, that transmembrane differences in proton concentration are the reservoir for the energy extracted from biological oxidation reactions. This [chemiosmotic theory](#) has been accepted as one of the great unifying ideas of twentieth-century biology. It provides insight into the processes of oxidative phosphorylation and photophosphorylation in plants, and into such apparently disparate energy transductions as active transport across membranes and the motion of bacterial flagella.

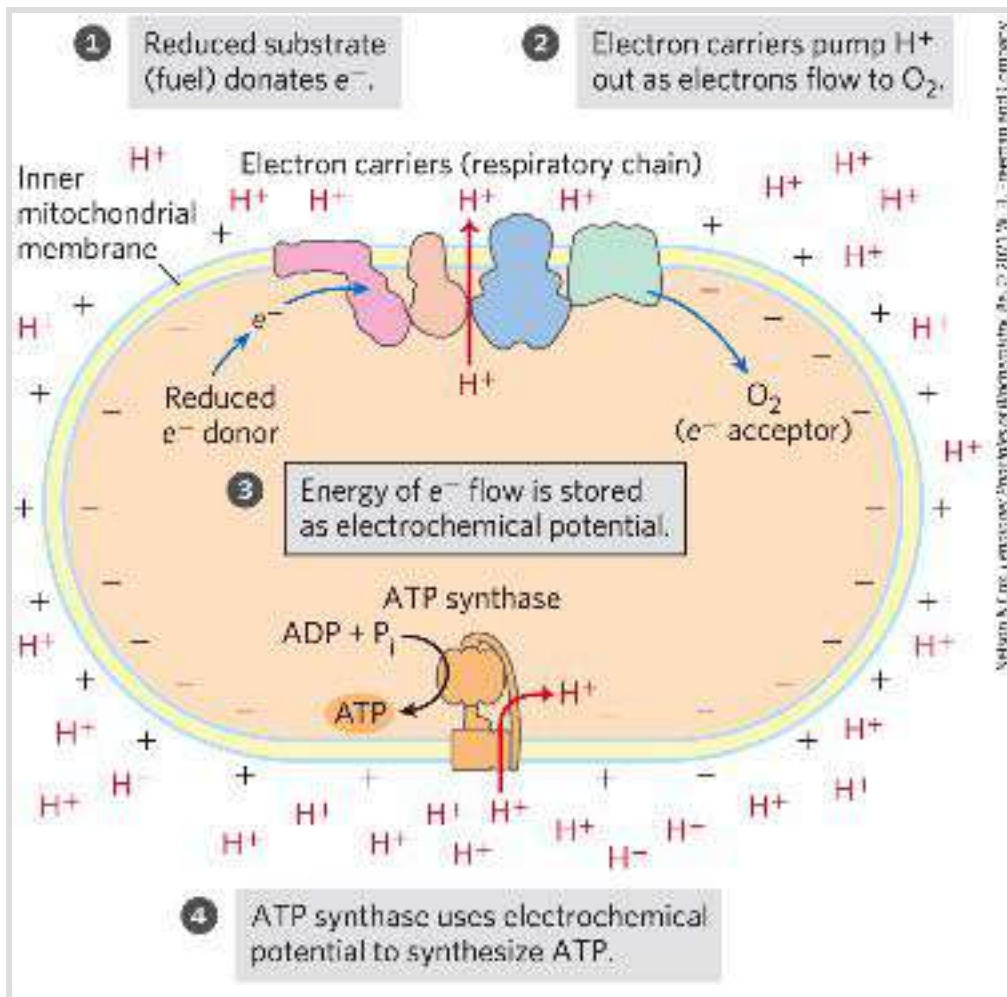


FIGURE 19-1 The chemiosmotic mechanism for ATP synthesis in mitochondria. Electrons move spontaneously through a chain of membrane-bound carriers, the respiratory chain, driven by the high reduction potential of oxygen and the relatively low reduction potentials of the various reduced substrates (fuels) that undergo oxidation in the mitochondrion. Electron flow creates an electrochemical potential by the transmembrane movement of protons and positive charge. This electrochemical potential drives ATP synthesis by a membrane-bound enzyme (ATP synthase) that is fundamentally similar in mitochondria and chloroplasts, and in bacteria and archaea as well.

This chapter begins with a description of the components of the mitochondrial electron-transfer chain — the respiratory chain — and their organization into large functional complexes in the inner mitochondrial membrane, the path of electron flow

through these complexes, and the proton movements that accompany this flow. We then consider the remarkable enzyme complex that captures, by “rotational catalysis,” the energy of proton flow in ATP, and the regulatory mechanisms that coordinate oxidative phosphorylation with the many catabolic pathways by which fuels are oxidized.



The metabolic role of mitochondria is so critical to cellular and organismal function that defects in mitochondrial function have serious medical consequences. Mitochondria are central to neuronal and muscular function, and to the regulation of whole-body energy metabolism and body weight. Human neurodegenerative diseases, as well as cancer, diabetes, and obesity, are recognized as possible results of compromised mitochondrial function, and one theory of aging is based on gradual loss of mitochondrial integrity. We consider the diverse functions of mitochondria, and the consequences of defective mitochondrial function in humans. ■

19.1 The Mitochondrial Respiratory Chain



Albert L. Lehninger, 1917–1986

The discovery in 1948, by Eugene Kennedy and Albert Lehninger, that mitochondria are the site of oxidative phosphorylation in eukaryotes marked the beginning of the enzymological studies of biological energy transductions. Mitochondria, like gram-negative bacteria, have two membranes ([Fig. 19-2a](#)). The outer mitochondrial membrane is readily permeable to small molecules ($M_r < 5,000$) and ions, which move freely through transmembrane channels formed by a family of integral membrane proteins called porins. The inner membrane is impermeable to most small molecules and ions, including protons (H^+); the only species that cross this membrane do so through specific transporters. The inner membrane bears the components of the respiratory chain and ATP synthase.

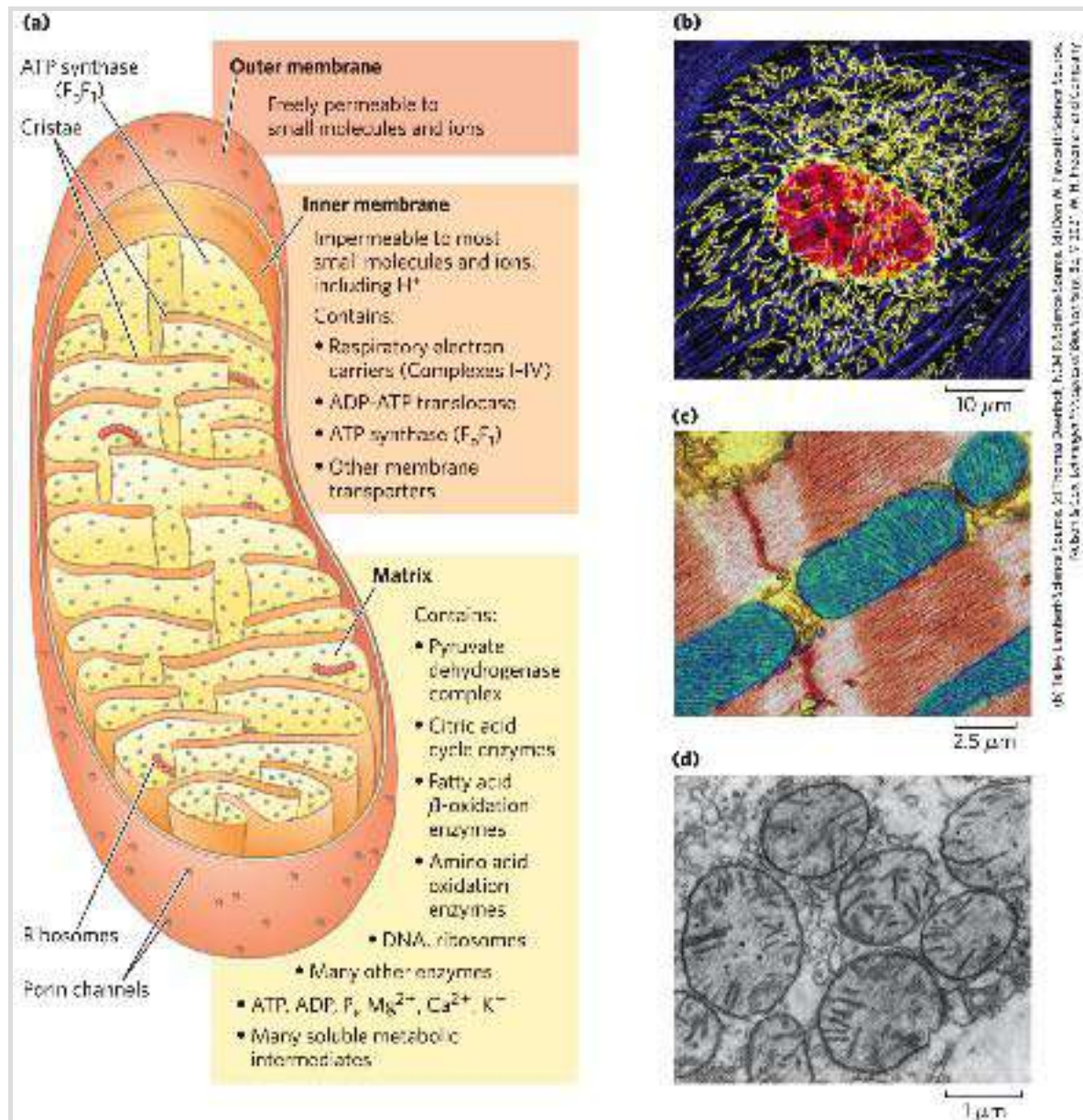


FIGURE 19-2 Biochemical anatomy of a mitochondrion. (a) The outer membrane has pores that make it permeable to small molecules and ions, but not to proteins. The cristae provide a very large surface area. The inner membrane of a single liver mitochondrion may have more than 10,000 sets of electron-transfer systems (respiratory chains) and ATP synthase molecules, distributed over the membrane surface. (b) A typical animal cell has hundreds or thousands of mitochondria. This endothelial cell from bovine pulmonary artery was stained with fluorescent probes for actin (blue), for DNA (red), and for mitochondria (yellow). Notice the variability in length of the mitochondria. (c) The mitochondria of heart muscle (blue in this colorized electron micrograph) have more profuse cristae and thus a much larger area of inner membrane, with more than three times as many sets of respiratory chains as (d) liver mitochondria. Muscle and liver mitochondria are about the size of a bacterium — 1 to $10\ \mu m$ long. The mitochondria of invertebrates, plants, and microbial eukaryotes are similar to those shown here, but with much variation in size, shape, and degree of convolution of the inner membrane.

 The mitochondrial matrix, enclosed by the inner membrane, contains the pyruvate dehydrogenase complex and the enzymes of the citric acid cycle, the fatty acid β -oxidation pathway, and the pathways of amino acid oxidation — all the pathways of fuel oxidation except glycolysis, which takes place in the cytosol. The selectively permeable inner mitochondrial membrane segregates the intermediates and enzymes of cytosolic metabolic pathways from those of metabolic processes occurring in the matrix. However, specific transporters carry pyruvate, fatty acids, and amino acids or their α -keto derivatives into the matrix for access to the machinery of the citric acid cycle. ADP and P_i are specifically transported into the matrix as newly synthesized ATP is transported out. Mammalian mitochondria have about 1,200 proteins, according to current best estimates. The functions of up to 25% of these remain partly or entirely enigmatic.

The bean-shaped representation of a mitochondrion in [Figure 19-2a](#) is an oversimplification, derived in part from early studies in which thin sections of cells were observed in the electron microscope. Three-dimensional images obtained either by reconstruction from serial sections or by confocal microscopy reveal great variation in mitochondrial size and shape. In living cells stained with mitochondrion-specific fluorescent dyes, large numbers of variously shaped mitochondria are seen, clustered about the nucleus ([Fig. 19-2b](#)).

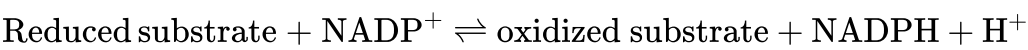
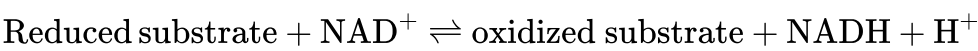
Tissues with a high demand for aerobic metabolism (brain, skeletal and heart muscle, liver, and eye, for example) contain many hundreds or thousands of mitochondria per cell, and in general, mitochondria of cells with high metabolic activity have more, and more densely packed, convolutions, or **cristae** ([Fig. 19-2c, d](#)).  During cell growth and division, mitochondria, like bacteria, divide by fission, and under some circumstances individual mitochondria fuse to form larger, more-extended structures. Stressful conditions, such as the presence of electron-transfer inhibitors or mutations in an electron carrier, trigger mitochondrial fission and sometimes **mitophagy** — the breakdown of mitochondria and recycling of the amino

acids, nucleotides, and lipids released. As stress is relieved, small mitochondria fuse to form long, thin, tubular organelles.

Electrons Are Funneled to Universal Electron Acceptors

Oxidative phosphorylation begins with the entry of electrons into the series of electron carriers called the [respiratory chain](#). Most of these electrons arise from the action of dehydrogenases that collect electrons from catabolic pathways and funnel them into universal electron acceptors — nicotinamide nucleotides (NAD⁺ or NADP⁺) or flavin nucleotides (FMN or FAD).

Nicotinamide nucleotide–linked dehydrogenases catalyze reversible reactions of the following general types:



Most dehydrogenases that act in catabolism are specific for NAD⁺ as electron acceptor ([Table 19-1](#)). Some are in the cytosol, many are in mitochondria, and still others have mitochondrial and cytosolic isozymes.

TABLE 19-1 Some Important Reactions Catalyzed by NAD(P)⁺-Linked Dehydrogenases

Reaction ^a	Location ^b
NAD⁺-linked	
$\alpha\text{-Ketoglutarate} + \text{CoA} + \text{NAD}^+ \rightleftharpoons \text{succinyl-CoA} + \text{CO}_2 + \text{NADH} + \text{H}^+$	M

$\text{L-Malate} + \text{NAD}^+ \rightleftharpoons \text{oxaloacetate} + \text{NADH} + \text{H}^+$	M and C
$\text{Pyruvate} + \text{CoA} + \text{NAD}^+ \rightleftharpoons \text{acetyl-CoA} + \text{CO}_2 + \text{NADH} + \text{H}^+$	M
$\text{Glyceraldehyde 3-phosphate} + \text{P}_i + \text{NAD}^+ \rightleftharpoons \text{1,3-bisphosphoglycerate} + \text{NADH} + \text{H}^+$	C
$\text{Lactate} + \text{NAD}^+ \rightleftharpoons \text{pyruvate} + \text{NADH} + \text{H}^+$	C
$\beta\text{-Hydroxyacyl-CoA} + \text{NAD}^+ \rightleftharpoons \beta\text{-ketoacyl-CoA} + \text{NADH} + \text{H}^+$	M
NADP⁺-linked	
$\text{Glucose 6-phosphate} + \text{NADP}^+ \rightleftharpoons \text{6-phosphogluconate} + \text{NADPH} + \text{H}^+$	C
$\text{L-Malate} + \text{NADP}^+ \rightleftharpoons \text{pyruvate} + \text{CO}_2 + \text{NADPH} + \text{H}^+$	C
NAD⁺ - or NADP⁺-linked	
$\text{L-Glutamate} + \text{H}_2\text{O} + \text{NAD(P)}^+ \rightleftharpoons \alpha\text{-ketoglutarate} + \text{NH}_4^+ + \text{NAD(P)H}$	M
$\text{Isocitrate} + \text{NAD(P)}^+ \rightleftharpoons \alpha\text{-ketoglutarate} + \text{CO}_2 + \text{NAD(P)H} + \text{H}^+$	M and C
^a These reactions and their enzymes are discussed in Chapters 14, 16, 17 , and 18 .	
^b M designates mitochondria; C designates cytosol.	

NAD⁺-linked dehydrogenases remove two hydrogen atoms from their substrates. One of these is transferred as a hydride ion (:H⁻) to NAD⁺, and the other is released as H⁺ in the medium (see [Fig. 13-24](#)). NADH and NADPH are water-soluble electron carriers that associate *reversibly* with dehydrogenases. About 70% of the cellular NAD pool is in the mitochondria. NADH carries electrons from catabolic reactions to their point of entry into the respiratory chain, the NADH dehydrogenase complex described below. NADPH is primarily involved in biosynthetic (anabolic) reactions, and much of it is concentrated in the cytosol. Cells maintain separate pools of NADPH and NADH, with different redox potentials. This is accomplished by holding the ratio [reduced form]/[oxidized form] relatively high for NADPH and relatively low for NADH. Neither NADH nor NADPH can cross the inner mitochondrial membrane, but the electrons they carry can be shuttled across indirectly, as we shall see.

Flavoproteins contain a very tightly, sometimes covalently, bound flavin nucleotide, either FMN or FAD (see [Fig. 13-27](#)).  The oxidized flavin nucleotide can accept either one electron (yielding the semiquinone form) or two (yielding FADH₂ or FMNH₂). Electron transfer occurs because the flavoprotein has a higher reduction potential than the compound oxidized. Recall that reduction potential is a quantitative measure of the relative tendency of a given chemical species to accept electrons in an oxidation-reduction reaction ([p. 490](#)). The standard reduction potential of a flavin nucleotide, unlike that of NAD or NADP, depends on the protein with which it is associated. Local interactions with functional groups in the protein distort the electron orbitals in the flavin ring, changing the relative stabilities of oxidized and reduced forms. The relevant standard reduction potential is therefore that of the particular flavoprotein, not that of isolated FAD or FMN. The flavin nucleotide should be considered part of the flavoprotein's active site rather than a reactant or product in the electron-transfer reaction. Because flavoproteins can participate in either one- or two-electron transfers, they can serve as intermediates between reactions in which two electrons are donated (as in dehydrogenations) and those in which only one electron is accepted (as in the reduction of a quinone to a hydroquinone, described below).

Electrons Pass through a Series of Membrane-Bound Carriers

The mitochondrial respiratory chain consists of a series of sequentially acting electron carriers, most of which are integral proteins with prosthetic groups capable of accepting and donating either one or two electrons. Three types of electron transfers occur in oxidative phosphorylation: (1) direct transfer of electrons, as in the reduction of Fe³⁺ to Fe²⁺; (2) transfer as a hydrogen atom (H⁺ + e⁻); and (3) transfer as a hydride ion (:H⁻), which bears two electrons. The term **reducing equivalent** is used to designate a single electron equivalent transferred in an oxidation-reduction reaction.

P3 In addition to NAD and flavoproteins, three other types of electron-carrying molecules function in the respiratory chain: a hydrophobic quinone called ubiquinone, and two different types of iron-containing proteins, cytochromes and iron-sulfur proteins. **Ubiquinone** (also called **coenzyme Q**, or simply **Q**) is a lipid-soluble benzoquinone with a long isoprenoid side chain ([Fig. 19-3](#)). Ubiquinone can accept one electron to become the semiquinone radical ($\bullet\text{QH}$ or ubisemiquinone) or two electrons to form ubiquinol (QH_2) and, like flavoprotein carriers, it can act at the junction between a two-electron donor and a one-electron acceptor. Small and hydrophobic, ubiquinone is not bound to proteins but instead is freely diffusible within the lipid bilayer of the inner mitochondrial membrane and can shuttle reducing equivalents between other, less mobile electron carriers in the membrane. And because it carries both electrons and protons, it plays a central role in coupling electron flow to proton movement.

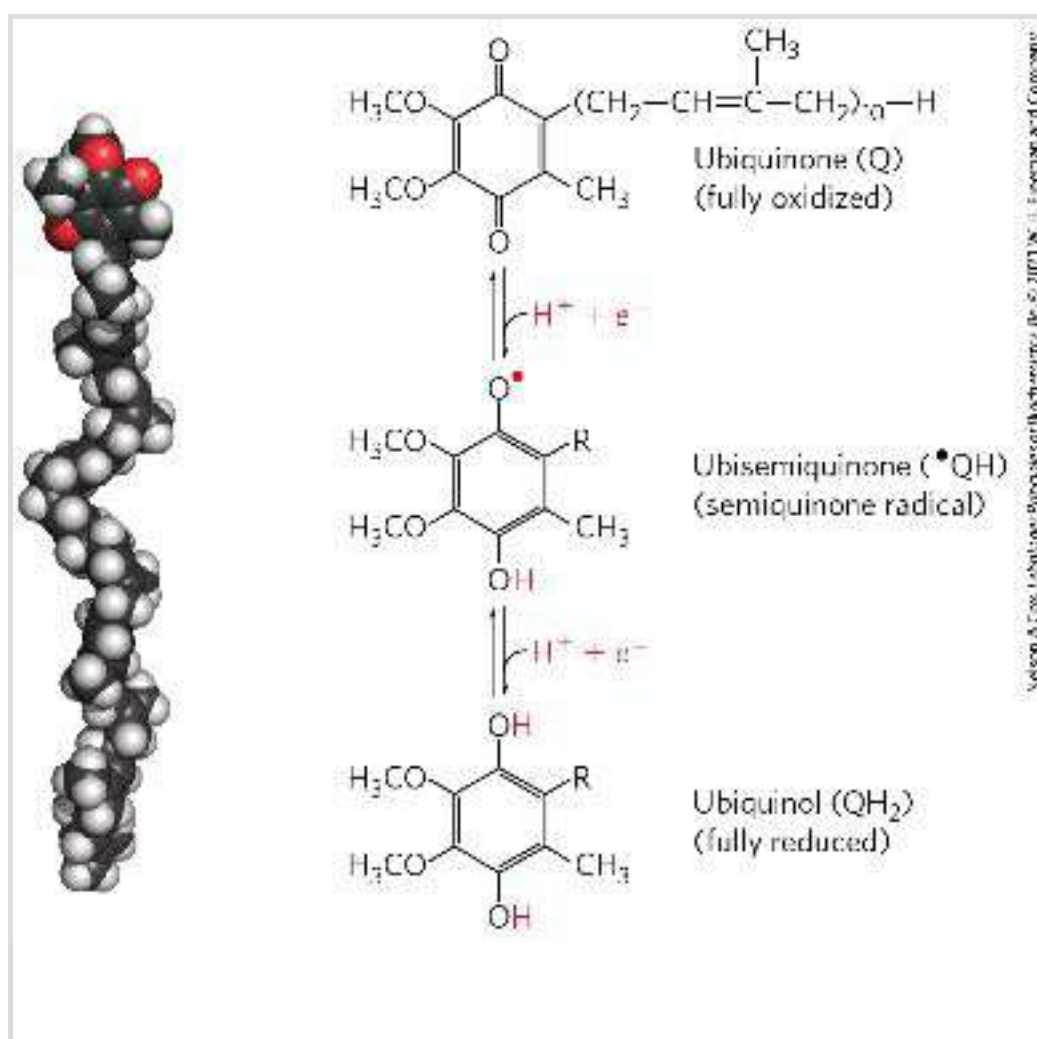


FIGURE 19-3 Ubiquinone (Q, or coenzyme Q). Complete reduction of ubiquinone requires two electrons and two protons, and occurs in two steps through the semiquinone radical intermediate.

The **cytochromes** are proteins with characteristic strong absorption of visible light, due to their iron-containing heme prosthetic groups (**Fig. 19-4a**). Mitochondria contain three classes of cytochromes, designated *a*, *b*, and *c*, which are distinguished by differences in their light-absorption spectra. Each type of cytochrome in its reduced (Fe^{2+}) state has three absorption bands in the visible range (**Fig. 19-4b**). The longest-wavelength band is near 600 nm in type *a* cytochromes, near 560 nm in type *b*, and near 550 nm in type *c*. To distinguish among closely related cytochromes of one type, the exact absorption maximum is sometimes used in the names, as in cytochrome *b*₅₆₂.

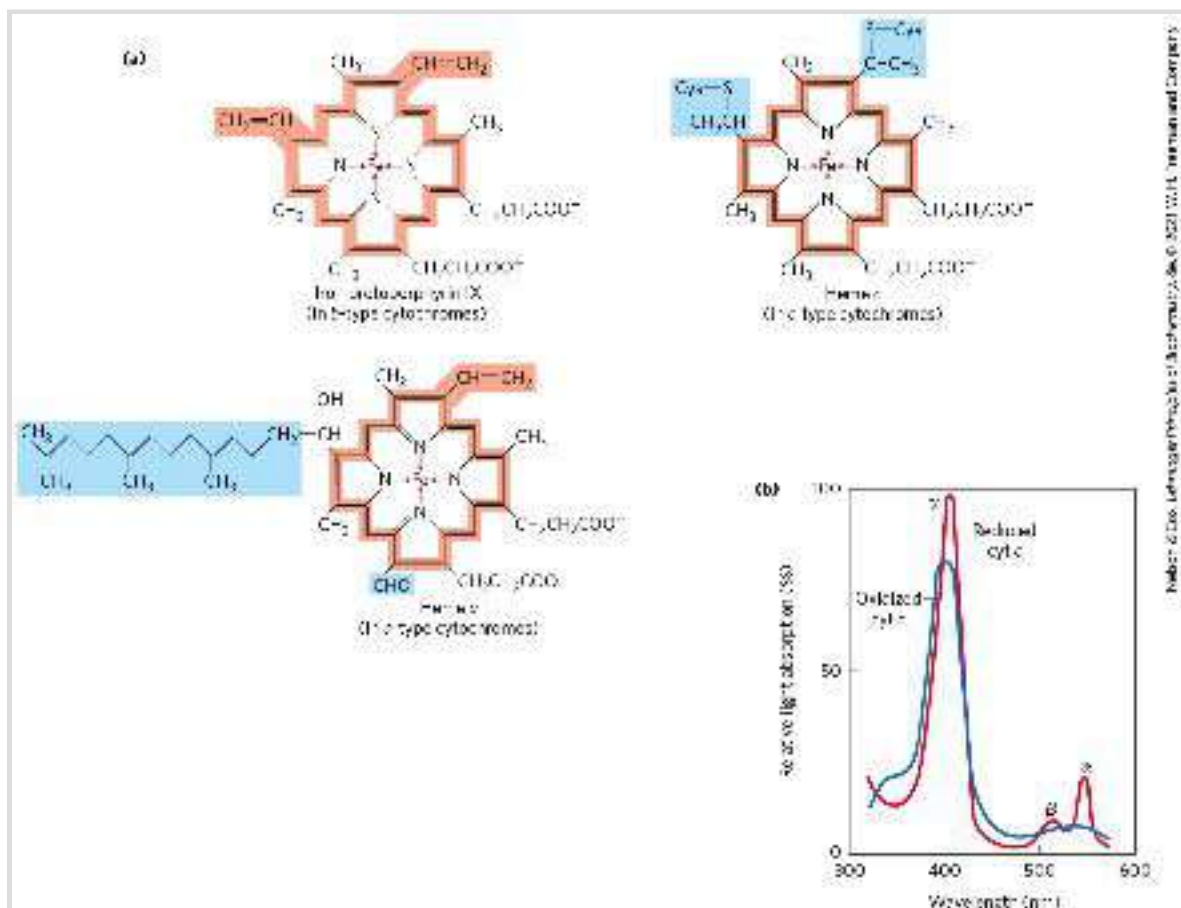


FIGURE 19-4 Prosthetic groups of cytochromes. (a) Each group consists of four five-membered, nitrogen-containing rings in a cyclic structure called a porphyrin. The four nitrogen atoms are coordinated with a central Fe ion, either Fe^{2+} or Fe^{3+} . Iron protoporphyrin IX is found in *b*-type cytochromes and in

hemoglobin and myoglobin (see [Fig. 5-1b](#)). Heme *c* is covalently bound to the protein of cytochrome *c* through thioether bonds to two Cys residues. Heme *a*, found in *a*-type cytochromes, has a long isoprenoid tail attached to one of the five-membered rings. The conjugated double-bond system (shaded light red) of the porphyrin ring has delocalized π electrons that are relatively easily excited by photons with the wavelengths of visible light, which accounts for the strong absorption by hemes (and related compounds) in the visible region of the spectrum. (b) Absorption spectra of cytochrome *c* (cyt *c*) in its oxidized (blue) and reduced (red) forms. The characteristic α , β , and γ bands of the reduced form are labeled.

The hemes of *a* and *b* cytochromes are tightly, but not covalently, bound to their associated proteins; the hemes of *c*-type cytochromes are covalently attached through Cys residues ([Fig. 19-4](#)). As with the flavoproteins, the standard reduction potential of the heme iron atom of a cytochrome depends on its interaction with protein side chains and is therefore different for each cytochrome. The cytochromes of type *a* and *b* and some of type *c* are integral proteins of the inner mitochondrial membrane. One striking exception is the soluble cytochrome *c* that associates through electrostatic interactions with the outer surface of the inner membrane.

In [iron-sulfur proteins](#), the iron is present not in heme but in association with inorganic sulfur atoms or with the sulfur atoms of Cys residues in the protein, or both. These iron-sulfur (Fe-S) centers range from simple structures with a single Fe atom coordinated to four Cys OSH groups to more complex Fe-S centers with two or four Fe atoms ([Fig. 19-5](#)). [Rieske iron-sulfur proteins](#) (named after their discoverer, John S. Rieske) are a variation on this theme, in which one Fe atom is coordinated to two His residues rather than two Cys residues. All iron-sulfur proteins participate in one-electron transfers in which one iron atom of the Fe-S cluster is oxidized or reduced. At least eight Fe-S proteins function in mitochondrial electron transfer. The reduction potential of Fe-S proteins varies from -0.65 V to $+0.45$ V, depending on the microenvironment of the iron within the protein.

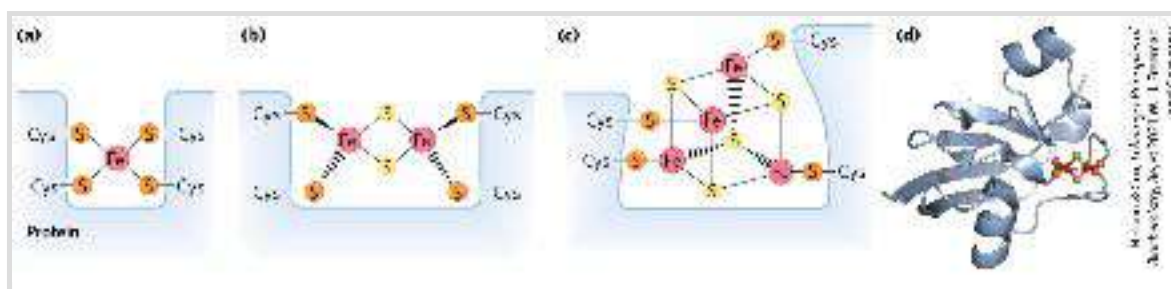


FIGURE 19-5 Iron-sulfur centers. The Fe-S centers of iron-sulfur proteins may be as simple as shown in (a), with a single Fe ion surrounded by the S atoms of four Cys residues; inorganic S is yellow and the S of Cys is orange. Other centers include both inorganic and Cys S atoms, as in (b) 2Fe-2S or (c) 4Fe-4S centers. (d) The ferredoxin of the cyanobacterium *Anabaena* 7120 has one 2Fe-2S center. (Note that in these designations, only the inorganic S atoms are counted. For example, in the 2Fe-2S center (b), each Fe ion is actually surrounded by four S atoms.) The exact standard reduction potential of the iron in these centers depends on the type of center and its interaction with the associated protein. [(d) Data from PDB ID 1FRD, B. L. Jacobson et al., *Biochemistry* 32:6788, 1993.]

In the overall reaction catalyzed by the mitochondrial respiratory chain, electrons move from NADH, succinate, or some other primary electron donor through flavoproteins, ubiquinone, iron-sulfur proteins, and cytochromes, and finally to O_2 . A look at the methods used to determine the sequence in which the carriers act is instructive, as the same general approaches have been used to study other electron-transfer chains, such as those of chloroplasts (see [Fig. 20-12](#)).

First, the standard reduction potentials of the individual electron carriers have been determined experimentally ([Table 19-2](#)). Electrons tend to flow spontaneously from carriers of lower E'° to carriers of higher E'° . The order of carriers deduced by this method is $NADH \rightarrow Q \rightarrow \text{cytochrome } b \rightarrow \text{cytochrome } c_1 \rightarrow \text{cytochrome } c \rightarrow \text{cytochrome } a \rightarrow \text{cytochrome } a_3 \rightarrow O_2$. Note, however, that the order of standard reduction potentials is not necessarily the same as the order of *actual* reduction potentials under cellular conditions, which depend on the concentrations of reduced and oxidized forms (see [Eqn 13-5, p. 491](#)). A second method for determining the sequence of electron carriers involves reducing the entire chain of carriers experimentally by providing an electron source but no electron acceptor (no O_2). When O_2 is suddenly introduced into the system, the rate at which each electron carrier

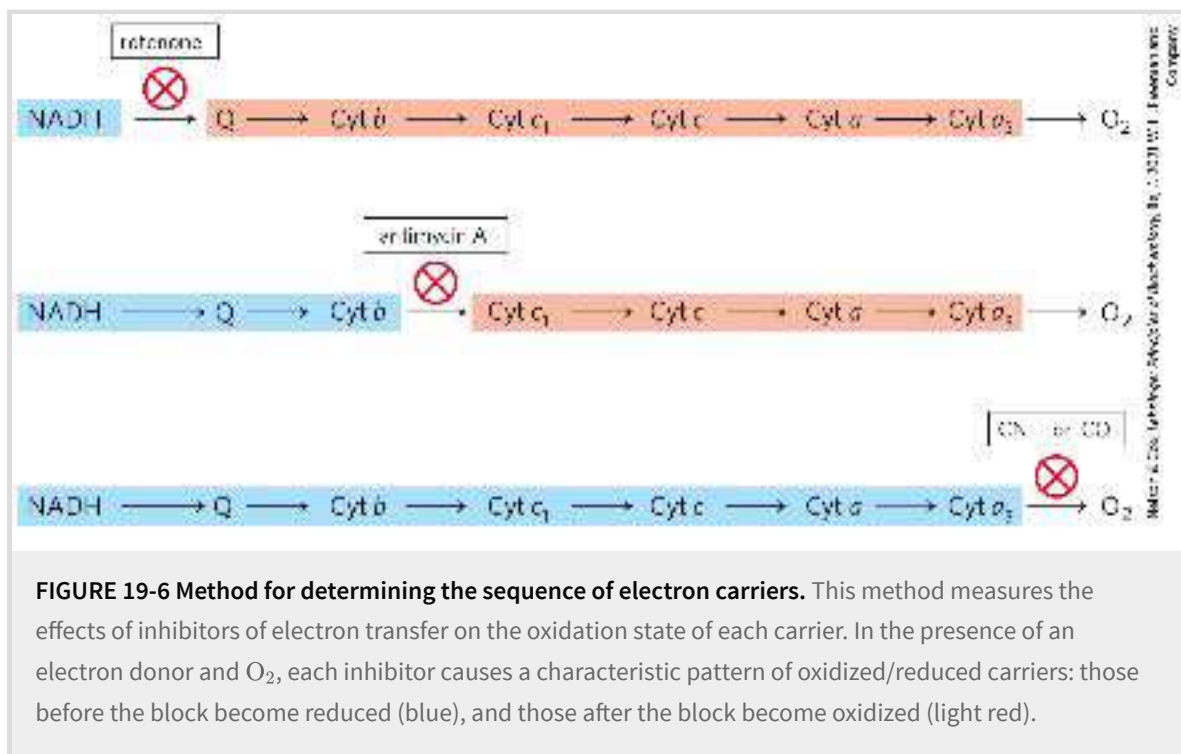
becomes oxidized, measured spectroscopically, reveals the order in which the carriers function. The carrier nearest O₂ (at the end of the chain) gives up its electrons first, the second carrier from the end is oxidized next, and so on. Such experiments have confirmed the sequence deduced from standard reduction potentials.

TABLE 19-2 Standard Reduction Potentials of Respiratory Chain and Related Electron Carriers

Redox reaction (half-reaction)	E'° (V)
$2\text{H}^{+} + 2e^{-} \rightarrow \text{H}_2$	-0.414
$\text{NAD}^{+} + \text{H}^{+} + 2e^{-} \rightarrow \text{NADH}$	-0.320
$\text{NADP}^{+} + \text{H}^{+} + 2e^{-} \rightarrow \text{NADPH}$	-0.324
$\text{NADH dehydrogenase (FMN)} + 2\text{H}^{+} + 2e^{-} \rightarrow \text{NADH dehydrogenase (FMNH}_2\text{)}$	-0.30
$\text{Ubiquinone} + 2\text{H}^{+} + 2e^{-} \rightarrow \text{ubiquinol}$	0.045
$\text{Cytochrome } b (\text{Fe}^{3+}) + e^{-} \rightarrow \text{cytochrome } b (\text{Fe}^{2+})$	0.077
$\text{Cytochrome } c_1 (\text{Fe}^{3+}) + e^{-} \rightarrow \text{cytochrome } c_1 (\text{Fe}^{2+})$	0.22
$\text{Cytochrome } c (\text{Fe}^{3+}) + e^{-} \rightarrow \text{cytochrome } c (\text{Fe}^{2+})$	0.254
$\text{Cytochrome } a (\text{Fe}^{3+}) + e^{-} \rightarrow \text{cytochrome } a (\text{Fe}^{2+})$	0.29
$\text{Cytochrome } a_3 (\text{Fe}^{3+}) + e^{-} \rightarrow \text{cytochrome } a_3 (\text{Fe}^{2+})$	0.35
$\frac{1}{2}\text{O}_2 + 2\text{H}^{+} + 2e^{-} \rightarrow \text{H}_2\text{O}$	0.817

In a final confirmation, agents that inhibit the flow of electrons through the chain have been used in combination with measurements of the degree of oxidation of each carrier. In the presence of O₂ and an electron donor, carriers that function before the inhibited step become fully reduced, and those that function after this step are completely oxidized ([Fig. 19-6](#)). By using several inhibitors that block different steps in the chain, investigators have

determined the entire sequence; it is the same as deduced in the first two approaches.



Electron Carriers Function in Multienzyme Complexes

The electron carriers of the respiratory chain are organized into membrane-embedded supramolecular complexes that can be physically separated. Gentle treatment of the inner mitochondrial membrane with detergents allows the resolution of four unique electron-carrier complexes, each capable of catalyzing electron transfer through a portion of the chain ([Fig. 19-7](#); [Table 19-3](#)). **P3** Complexes I and II catalyze electron transfer to ubiquinone from two different electron donors: NADH (Complex I) and succinate (Complex II). Complex III carries electrons from reduced ubiquinone to cytochrome *c*, and Complex IV completes the sequence by transferring electrons from cytochrome *c* to O_2 .

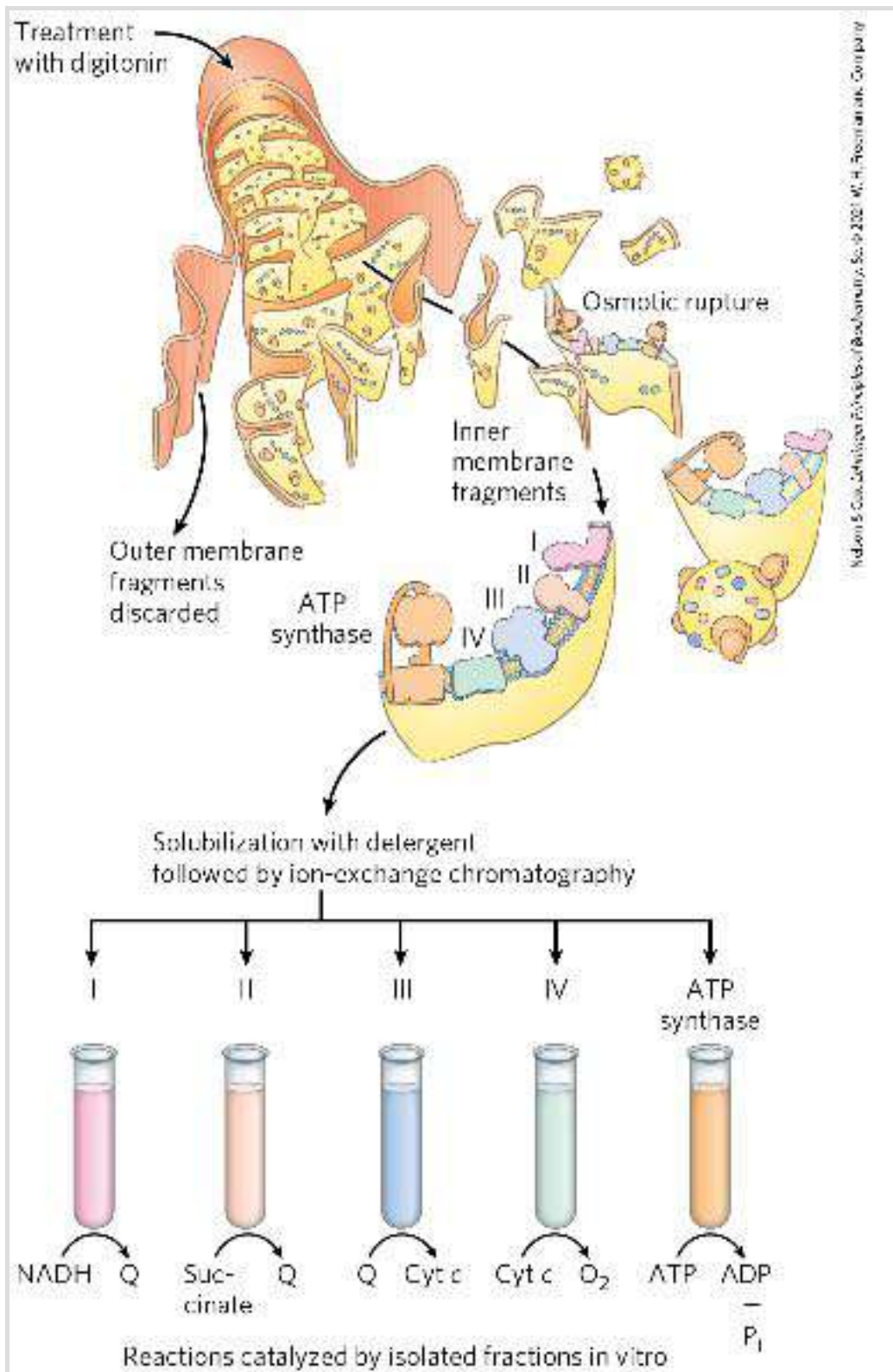


FIGURE 19-7 Separation of functional complexes of the respiratory chain. The outer mitochondrial membrane is first removed by treatment with the detergent digitonin. Fragments of inner membrane are then obtained by osmotic rupture of the membrane, and the fragments are gently dissolved in a second detergent. The resulting mixture of inner membrane proteins is resolved by ion-exchange chromatography into several complexes (I

through IV) of the respiratory chain, each with its unique protein composition (see [Table 19-3](#)), and the enzyme ATP synthase (sometimes called Complex V). The isolated Complexes I through IV catalyze electron transfers between donors (NADH and succinate), intermediate carriers (Q and cytochrome c), and O₂, as shown. In vitro, isolated ATP synthase has only ATP-hydrolyzing (ATPase), not ATP-synthesizing, activity.

TABLE 19-3 The Protein Components of the Mitochondrial Respiratory Chain

Enzyme complex/protein	Mass (kDa)	Number of subunits ^a	Prosthetic group(s)
I NADH dehydrogenase	850	45 (14)	FMN, Fe-S
II Succinate dehydrogenase	140	4	FAD, Fe-S
III Ubiquinone:cytochrome c oxidoreductase ^b	250	11	Hemes, Fe-S
Cytochrome c ^c	13	1	Heme
IV Cytochrome oxidase ^b	204	13 (3–4)	Hemes; Cu _A , Cu _B

^aNumber of subunits in bacterial complexes is shown in parentheses.

^bMass and subunit data are for the monomeric form.

^cCytochrome c is not part of an enzyme complex; it moves between Complexes III and IV as a freely soluble protein.

We now look in more detail at the structure and function of each complex of the mitochondrial respiratory chain.

Complex I: NADH to Ubiquinone

In mammals, [Complex I](#), also called **NADH:ubiquinone oxidoreductase** or **NADH dehydrogenase**, is a large enzyme composed of 45 different polypeptide chains, including an FMN-containing flavoprotein and at least 8 iron-sulfur centers. Complex I is L-shaped, with one arm embedded in the inner membrane and the other extending into the matrix. Comparative studies of Complex I in bacteria and other organisms show that 7

polypeptides in the membrane arm and 7 in the matrix arm are conserved and essential ([Fig. 19-8](#)).

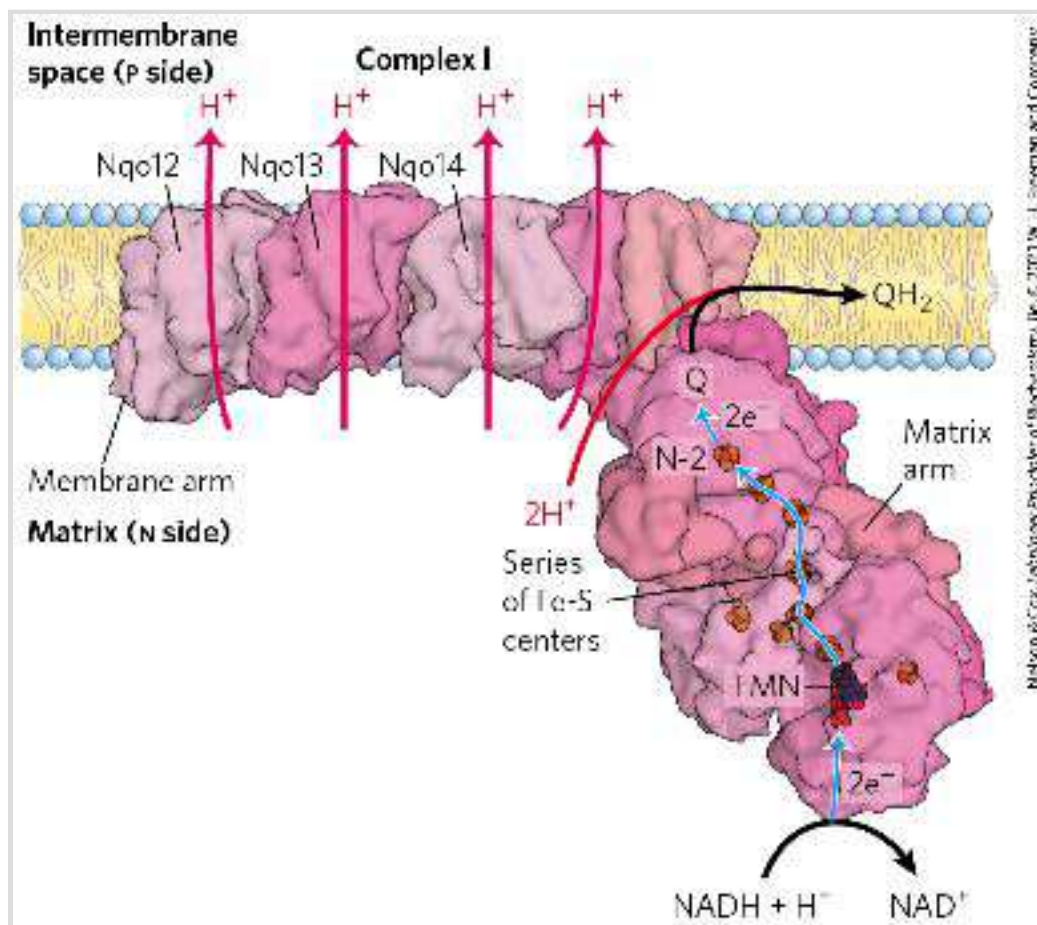
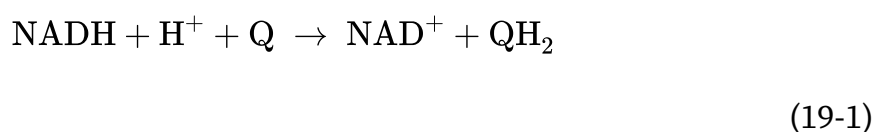


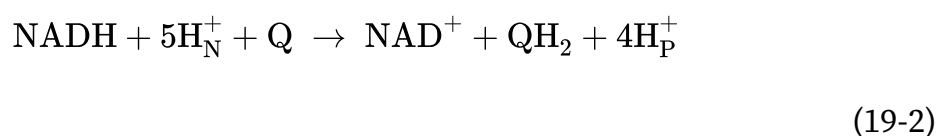
FIGURE 19-8 Structure of Complex I (NADH:ubiquinone oxidoreductase). Complex I catalyzes the transfer of a hydride ion from NADH to FMN. From the FMN, two electrons pass through a series of Fe-S centers to the Fe-S center N-2 in the matrix arm of the complex. Electron transfer from N-2 to ubiquinone on the membrane arm forms QH₂, which diffuses into the lipid bilayer. The protons travel a path dictated by subunit conformation changes triggered by the electron flow. Proton flux produces an electrochemical potential across the inner mitochondrial membrane (N side negative, P side positive). Three of the membrane subunits (subunits Nqo12, Nqo13, and Nqo14) are structurally related to a known Na⁺-H⁺ antiporter, and the path of proton movement may be similar in both cases. The fourth putative proton pathway is through an integral subunit closest to the Q-binding site. A long helix (not visible in this view) lying along the surface of the membrane arm may coordinate the action of all four proton pumps when Q is reduced. [Data from PDB ID 4HEA, R. Baradaran et al., *Nature* 494:443, 2013.]

Complex I catalyzes two simultaneous and obligately coupled processes: (1) the exergonic transfer to ubiquinone of a hydride ion from NADH and a

proton from the matrix, expressed by



and (2) the endergonic transfer of four protons from the matrix to the intermembrane space. Protons are moved against a transmembrane proton gradient in this process.  Complex I is therefore a proton pump driven by the energy of electron transfer, and the reaction it catalyzes is **vectorial**: it moves protons in a specific direction from one location (the matrix, which becomes negatively charged with the departure of protons) to another (the intermembrane space, which becomes positively charged). To emphasize the vectorial nature of the process, the overall reaction is often written with subscripts that indicate the location of the protons: _P for the positive side of the inner membrane (the intermembrane space), _N for the negative side (the matrix):



Amytal (a barbiturate drug), rotenone (a plant product commonly used as an insecticide), and piericidin A (an antibiotic) inhibit electron flow from the Fe-S centers of Complex I to ubiquinone ([Table 19-4](#)) and therefore block the overall process of oxidative phosphorylation.

TABLE 19-4 Agents That Interfere with Oxidative Phosphorylation

Type of interference	Compound ^a	Target/mode of action
Inhibition of electron transfer	Cyanide	Inhibit cytochrome oxidase
	Carbon monoxide	

	Antimycin A	Blocks electron transfer from cytochrome <i>b</i> to cytochrome <i>c</i> ₁
	Myxothiazol	Prevent electron transfer from Fe-S center to ubiquinone
	Rotenone	
	Amytal	
	Piericidin A	
Inhibition of ATP synthase	Aurovertin	Inhibit F ₁
	Oligomycin	Inhibit F _o
	Venturicidin	
	DCCD	Blocks proton flow through F _o
Uncoupling of phosphorylation from electron transfer	FCCP	Hydrophobic proton carriers
	DNP	
	Valinomycin	K ⁺ ionophore
	Uncoupling protein 1	In brown adipose tissue, forms proton-conducting pores in inner mitochondrial membrane
Inhibition of ATP-ADP exchange	Atractyloside	Inhibits adenine nucleotide translocase

^aDCCD, dicyclohexylcarbodiimide; FCCP, cyanide-*p*-trifluoromethoxyphenylhydrazine; DNP, 2,4-dinitrophenol.

Three of the seven integral protein subunits of the membrane arm are related to a Na⁺-H⁺ antiporter and are believed to be responsible for pumping three protons; a fourth subunit in the membrane arm, that nearest the Q-binding site, is probably responsible for pumping the fourth proton ([Fig. 19-8](#)).

How is the reduction of ubiquinone coupled to proton pumping? Reduction of Q occurs far away from the membrane arm of the protein, where proton

pumping occurs, so the coupling is clearly indirect. The high-resolution view of Complex I from crystallographic and cryo-EM studies suggests that reduction of Q is coupled to a long-range conformational change conducted to all subunits along the hydrophilic core of the transmembrane arm. It seems likely that all four protons are pumped simultaneously, so that the energy from a strongly exergonic reaction (Q reduction) is broken into smaller packets, a common strategy employed by living organisms.

Complex II: Succinate to Ubiquinone

We encountered [Complex II](#) in [Chapter 16](#) as **succinate dehydrogenase**, the only membrane-bound enzyme in the citric acid cycle ([p. 586](#)). 

Complex II couples the oxidation of succinate at one site with the reduction of ubiquinone at another site about 40 Å away. Although smaller and simpler than Complex I, Complex II contains five prosthetic groups of two types and four different protein subunits ([Fig. 19-9](#)). Subunits C and D are integral membrane proteins, each with three transmembrane helices. They contain a heme group, heme *b*, and a binding site for Q, the final electron acceptor in the reaction catalyzed by Complex II. Subunits A and B extend into the matrix; they contain three 2Fe-2S centers, bound FAD, and a binding site for the substrate, succinate. Although the overall path of electron transfer is long (from the succinate-binding site to FAD, then through the Fe-S centers to the Q-binding site), none of the individual electron-transfer distances exceeds about 11 Å — a reasonable distance for rapid electron transfer ([Fig. 19-9](#)). Electron transfer through Complex II is not accompanied by proton pumping across the inner membrane, although the QH₂ produced by succinate oxidation will be used by Complex III to drive proton transfer. Because Complex II functions in the citric acid cycle, factors that affect its activity (such as the availability of oxidized Q) probably serve to coordinate that cycle with mitochondrial electron transfer.

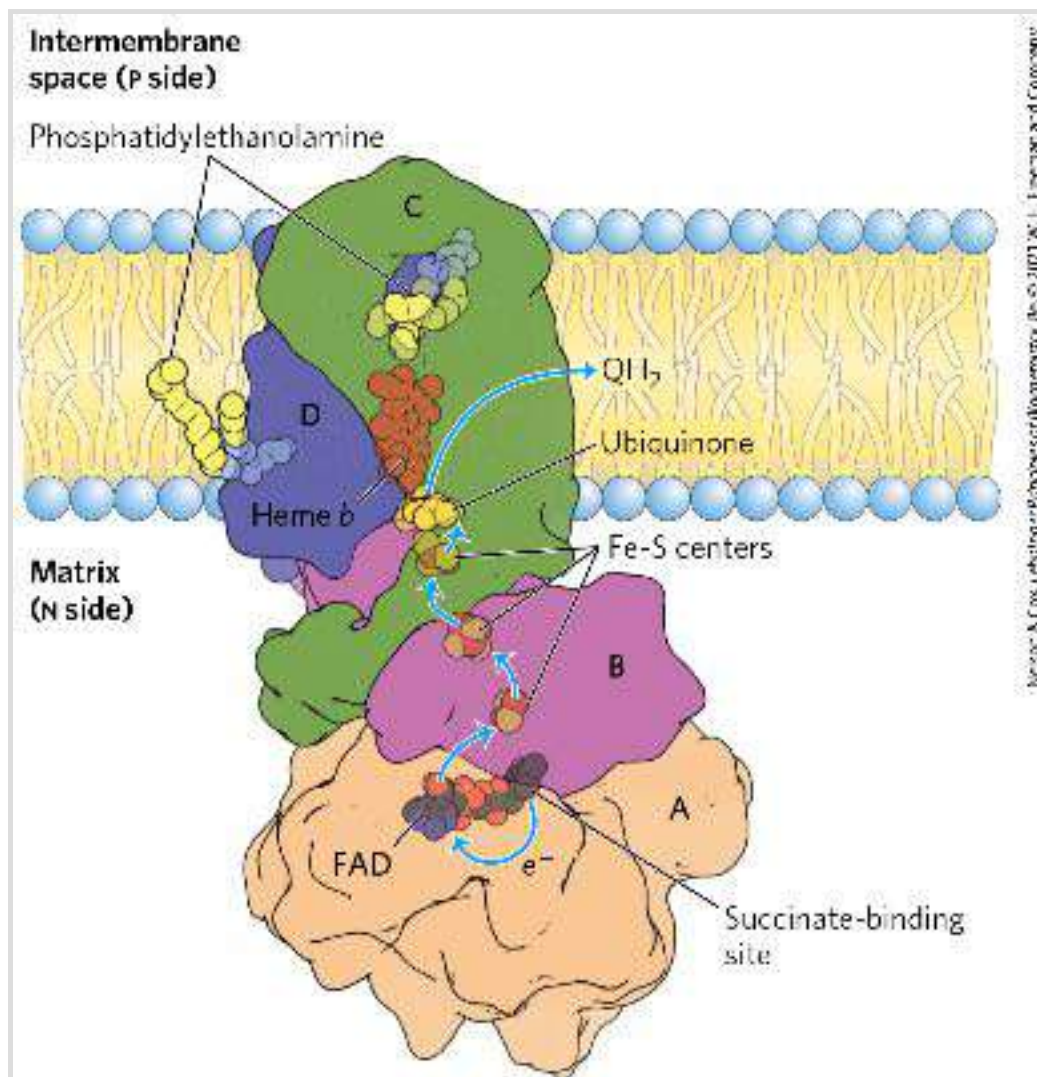


FIGURE 19-9 Structure of Complex II (succinate dehydrogenase). This complex (porcine) has two transmembrane subunits, C and D; subunits A and B extend into the matrix. Just behind the FAD in subunit A is the binding site for succinate. Subunit B has three Fe-S centers, ubiquinone is bound to subunit B, and heme *b* is sandwiched between subunits C and D. Two phosphatidylethanolamine molecules are so tightly bound to subunit D that they show up in the crystal structure. Electrons move (blue arrows) from succinate to FAD, then through the three Fe-S centers to ubiquinone. The heme *b* is not on the main path of electron transfer but protects against the formation of reactive oxygen species (ROS) by electrons that go astray. [Data from PDB ID 1ZOY, F. Sun et al., *Cell* 121:1043, 2005.]



The heme *b* of Complex II is apparently not in the direct path of electron transfer; it serves instead to reduce the frequency with which electrons “leak” out of the system, moving from succinate to molecular oxygen to produce the **reactive oxygen species (ROS)** hydrogen peroxide (H_2O_2) and the **superoxide radical** ($\bullet\text{O}_2^-$), as described below. Some individuals with point mutations in

Complex II subunits near heme *b* or the ubiquinone-binding site suffer from hereditary paraganglioma, characterized by benign tumors of the head and neck, commonly in the carotid body, an organ that senses O₂ levels in the blood. These mutations result in greater production of ROS, which cause DNA damage and genome instability that can lead to cancer. Mutations that affect the succinate-binding region in Complex II may lead to degenerative changes in the central nervous system, and some mutations are associated with tumors of the adrenal medulla. ■

Complex III: Ubiquinone to Cytochrome *c*

Electrons from reduced ubiquinone (ubiquinol, QH₂) pass through two more large protein complexes in the inner mitochondrial membrane before reaching the ultimate electron acceptor, O₂.  **Complex III** (also called **cytochrome *bc*₁ complex** or **ubiquinone:cytochrome *c* oxidoreductase**) couples the transfer of electrons from ubiquinol to cytochrome *c* with the vectorial transport of protons from the matrix to the intermembrane space. The functional unit of Complex III ([Fig. 19-10](#)) is a dimer. Each monomer consists of three proteins central to the action of the complex: cytochrome *b*, cytochrome *c*₁, and the Rieske iron-sulfur protein. (Several other proteins associated with Complex III in vertebrates are not conserved across the phyla and presumably play subsidiary roles.) The two cytochrome *b* monomers surround a cavern in the middle of the membrane, in which ubiquinone is free to move from the matrix side of the membrane (site Q_N on one monomer) to the intermembrane space (site Q_P on the other monomer) as it shuttles electrons and protons across the inner mitochondrial membrane.

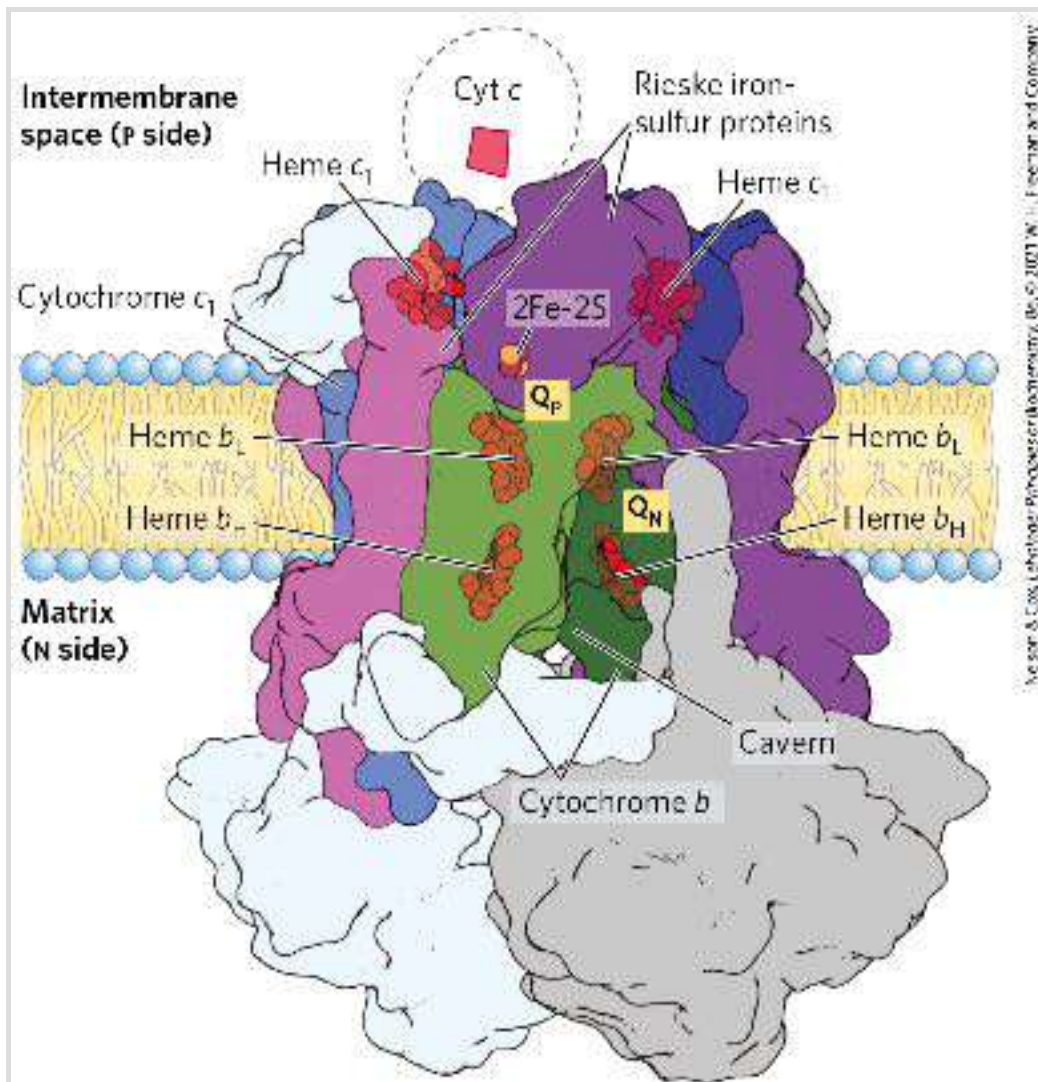


FIGURE 19-10 Structure of Complex III (cytochrome bc_1 complex). The complex (bovine) is a dimer of identical monomers, each with 11 different subunits. The functional core of each monomer consists of three subunits: cytochrome b (green), with its two hemes (b_H and b_L); the Rieske iron-sulfur protein (purple), with its 2Fe-2S centers; and cytochrome c_1 (blue), with its heme. Cytochrome c_1 and the Rieske iron-sulfur protein project from the P surface and can interact with cytochrome c (not part of the functional complex) in the intermembrane space. The complex has two distinct binding sites for ubiquinone, Q_N and Q_P , which correspond to the sites of inhibition by two drugs that block oxidative phosphorylation. Antimycin A, which blocks electron flow from cytochrome b to cytochrome c_1 , specifically from heme b_H to Q , binds at Q_N , close to heme b_H on the N (matrix) side of the membrane. Myxothiazol, which prevents electron flow from QH_2 to the Rieske iron-sulfur protein, binds at Q_P , near the 2Fe-2S center and heme b_L on the P side. The dimeric structure is essential to the function of Complex III. The interface between monomers forms two caverns, each containing a Q_P site from one monomer and a Q_N site from the other. The ubiquinone intermediates move within these sheltered caverns. [Data from PDB ID 1BGY, S. Iwata et al., *Science* 281:64, 1998.]

To account for the role of Q in energy conservation, Mitchell proposed the **Q cycle** ([Fig. 19-11](#)). As electrons move from QH₂ through Complex III, QH₂ is oxidized with the release of protons on one side of the membrane (at Q_P), while at the other site (Q_N), Q is reduced and protons are taken up.

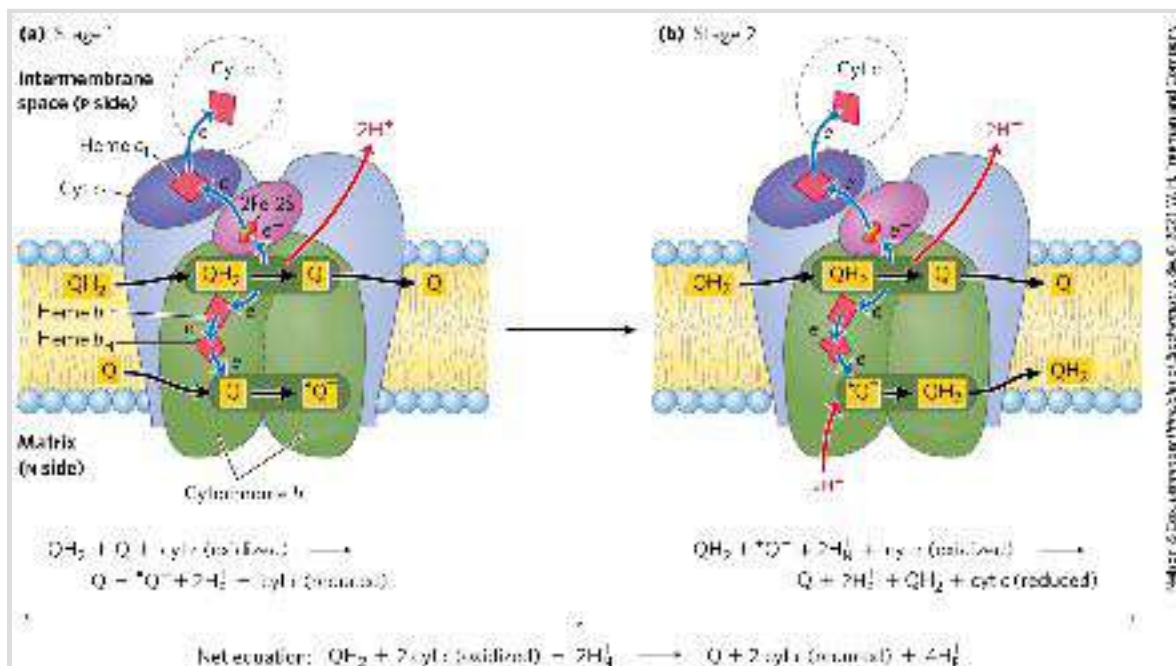
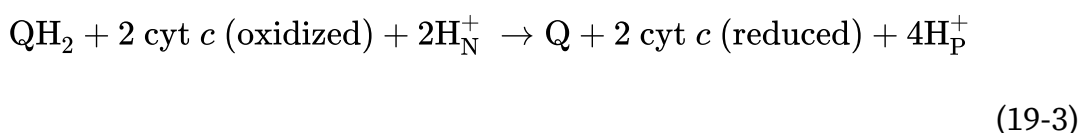


FIGURE 19-11 The Q cycle, shown in two stages. The path of electrons through Complex III is shown with blue arrows; the movement of various forms of ubiquinone, with black arrows. (a) In the first stage, Q on the N side is reduced to the semiquinone radical, which moves back into position (dotted line) to accept another electron. (b) In the second stage, the semiquinone radical is converted to QH₂. Meanwhile, on the P side of the membrane, two molecules of QH₂ are oxidized to Q, releasing two protons per Q molecule (four protons in all) into the intermembrane space. Each QH₂ donates one electron (via the Rieske Fe-S center) to cytochrome *c*₁, and one electron (via cytochrome *b*) to a molecule of Q near the N side, reducing it in two steps to QH₂. This reduction also consumes two protons per Q, which are taken up from the matrix (N side). Reduced cytochrome *c*₁ passes electrons one at a time to cytochrome *c*, which dissociates and carries electrons to Complex IV. In each cycle, one reduction of Q at the Q_N site is coupled with two oxidations of QH₂ at the Q_P site by consuming two protons from the matrix and releasing four protons into the intermembrane space.

The Q cycle is most easily understood as occurring in two stages, with two active sites where ubiquinone is either oxidized or reduced. In both stages, one QH₂ is oxidized at active site 1, shedding two H⁺ and two electrons. The protons are released into the intermembrane space. The two electrons take different paths, with one reducing cytochrome *c* and the other reducing a

molecule of Q at active site 2. Two electrons are required at active site 2 to fully reduce the Q to QH₂, one in each stage. Reducing one Q at one site while oxidizing two QH₂ at another may seem counterproductive at first glance. However, the two processes have complementary functions. The oxidation of two QH₂ is moving four protons to the intermembrane space and two electrons to cytochrome *c*. At the same time, the reduction of Q at the other site (using the other two electrons from the oxidation of QH₂ at site 1) is pulling protons from the matrix, creating a net movement of protons from the matrix to the intermembrane space. The QH₂ produced at active site 2 becomes a substrate for oxidation at active site 1 in subsequent turns of the cycle, and vice versa. The net equation for the redox reactions of the Q cycle is



The Q cycle accommodates the switch between the two-electron carrier ubiquinol (the reduced form of ubiquinone) and the one-electron carriers — hemes *b_L* and *b_H* of cytochrome *b*, and cytochromes *c₁* and *c* — and results in the uptake of two protons on the *N* side and the release of four protons on the *P* side, per pair of electrons passing through Complex III to cytochrome *c*. Two of the protons released on the *P* side are electrogenic; the other two are electroneutral, balanced by the two charges (electrons) passed to cytochrome *c* on the *P* side. Although the path of electrons through this segment of the respiratory chain is complicated, the net effect of the transfer is simple: QH₂ is oxidized to Q, two molecules of cytochrome *c* are reduced, and two protons are moved from the *N* side to the *P* side of the inner mitochondrial membrane.

Cytochrome *c* is a soluble protein of the intermembrane space, which associates reversibly with the *P* side of the inner membrane. After its single heme accepts an electron from Complex III, cytochrome *c* moves in the

intermembrane space to Complex IV to donate the electron to a binuclear copper center.

Complex IV: Cytochrome *c* to O₂

In the final step of the respiratory chain, [Complex IV](#), also called **cytochrome oxidase**, carries electrons from cytochrome *c* to molecular oxygen, reducing it to H₂O.  Complex IV is a large, dimeric enzyme of the inner mitochondrial membrane, each monomer having 13 subunits and M_r of 204,000. Bacteria contain a form that is much simpler, with only 3 or 4 subunits per monomer, but still capable of catalyzing both electron transfer and proton pumping. Comparison of the mitochondrial and bacterial complexes suggests that these 3 subunits have been conserved in evolution; in multicellular organisms, the other 10 subunits contribute to the assembly or stability of Complex IV ([Fig. 19-12](#)).

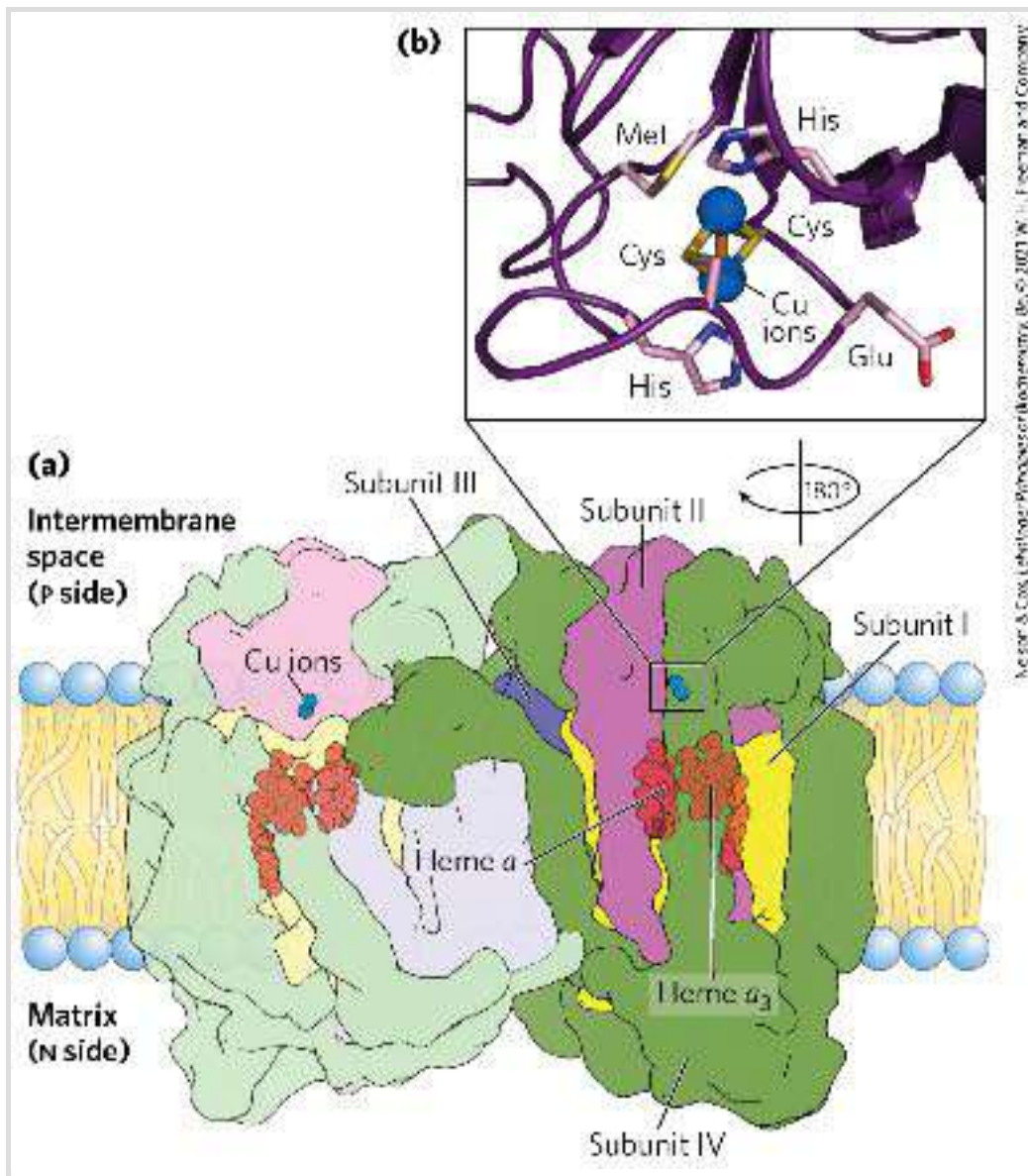
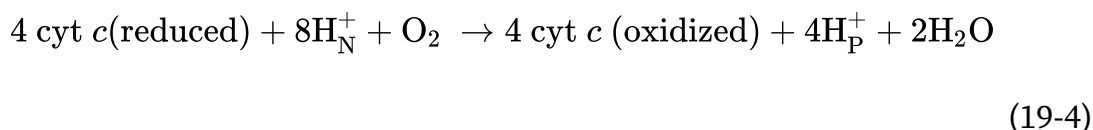


FIGURE 19-12 Structure of Complex IV (cytochrome oxidase). (a) This complex (bovine) has 13 subunits in each identical monomer of its dimeric structure. Subunit I has two heme groups, a and a_3 , near a single copper ion, Cu_B (not visible here). Heme a_3 and Cu_B form a binuclear Fe-Cu center. Subunit II contains two Cu ions complexed with the —SH groups of two Cys residues in a binuclear center, Cu_A , that resembles the 2Fe-2S centers of iron-sulfur proteins. This binuclear center and the cytochrome c -binding site are located in a domain of subunit II that protrudes from the P side of the inner membrane (into the intermembrane space). Subunit III is essential for rapid proton movement through subunit II. The roles of the other 10 subunits in mammalian Complex IV are not fully understood, although some function in assembly or stabilization of the complex. (b) The binuclear center of Cu_A . The Cu ions (blue spheres) share electrons equally. When the center is reduced, the ions have the formal charges $Cu^{1+}Cu^{1+}$; when oxidized, $Cu^{1.5+}Cu^{1.5+}$. Six amino acid residues are ligands around the Cu ions: Glu, Met, two His, and two Cys. [Data from PDB ID 1OCC, T. Tsukihara et al., *Science* 272:1136, 1996.]

Subunit II of Complex IV contains two Cu ions complexed with the —SH groups of two Cys residues in a binuclear center (Cu_A; [Fig. 19-12b](#)) that resembles the 2Fe-2S centers of iron-sulfur proteins. Subunit I contains two heme groups, designated *a* and *a*₃, and another copper ion (Cu_B). Heme *a*₃ and Cu_B form a second binuclear center that accepts electrons from heme *a* and transfers them to O₂ bound to heme *a*₃. The detailed role of subunit III is not clear, but its presence is essential to Complex IV function.

Electron transfer through Complex IV is from cytochrome *c* to the Cu_A center, to heme *a*, to the heme *a*₃—Cu_B center, and finally to O₂ ([Fig. 19-13a](#)). For every four electrons passing through this complex, the enzyme consumes four “substrate” H⁺ from the matrix (N side) in converting O₂ to two H₂O. It also uses the energy of this redox reaction to pump four protons outward into the intermembrane space (P side) for each four electrons that pass through, adding to the electrochemical potential produced by redox-driven proton transport through Complexes I and III. The overall reaction catalyzed by Complex IV is



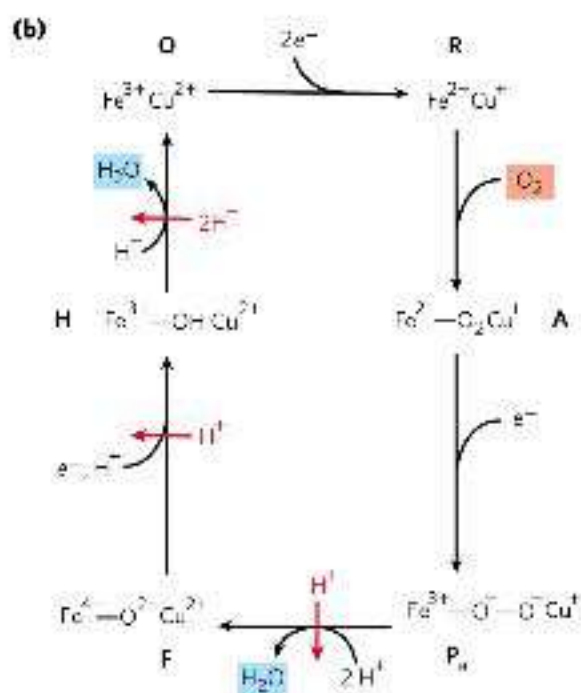
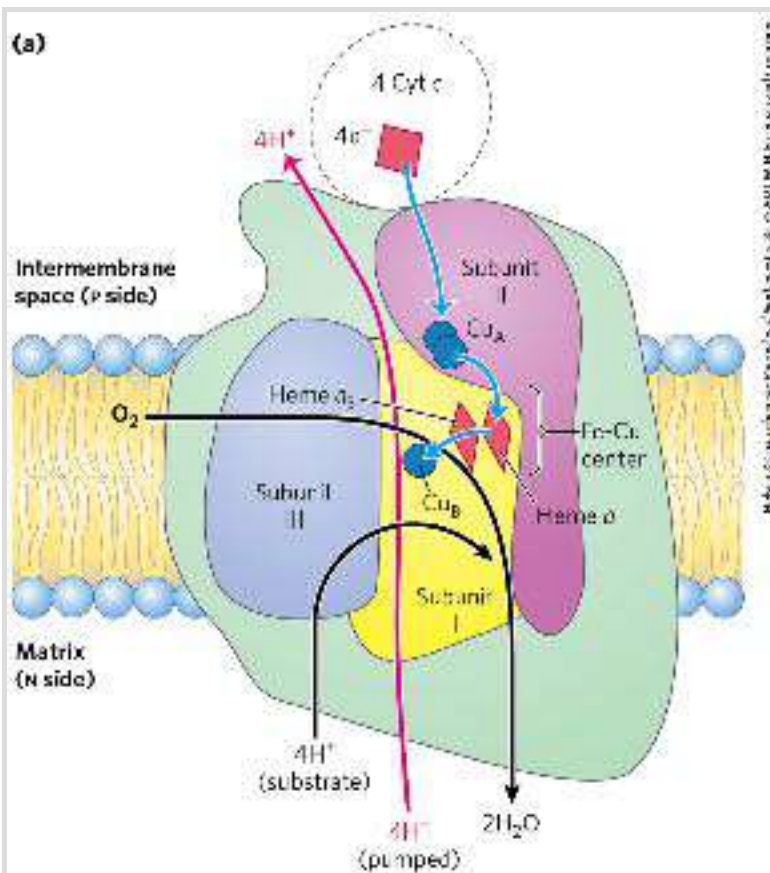


FIGURE 19-13 Path of electrons through Complex IV. (a) For simplicity, only one monomer of the dimeric bovine Complex IV is shown. The three proteins critical to electron flow are subunits I, II, and III. The larger green structure includes the other 10 proteins in each monomer of the dimeric complex. Electron transfer through Complex IV begins with cytochrome c (top). Two molecules of reduced cytochrome c each donate an electron to the

binuclear center Cu_A . From here, electrons pass through heme a to the Fe-Cu center (heme a_3 and Cu_B). Oxygen now binds to heme a_3 and is reduced to its peroxy derivative (O_2^{2-} ; not shown here) by two electrons from the Fe-Cu center. Delivery of two more electrons from cytochrome c (top), for a total of four electrons, converts the O_2^{2-} to two molecules of water, with consumption of four “substrate” protons from the matrix. At the same time, four protons are pumped from the matrix for every four electrons passing through Complex IV. A simplified reaction sequence is presented in (b). Intermediate complexes O, R, A, P_R , F, and H represent only a prominent subset of the species for which there is experimental evidence, with some steps and intermediate structures still being debated. The four electrons are introduced in separate steps, and H_2O is released in two separate steps.

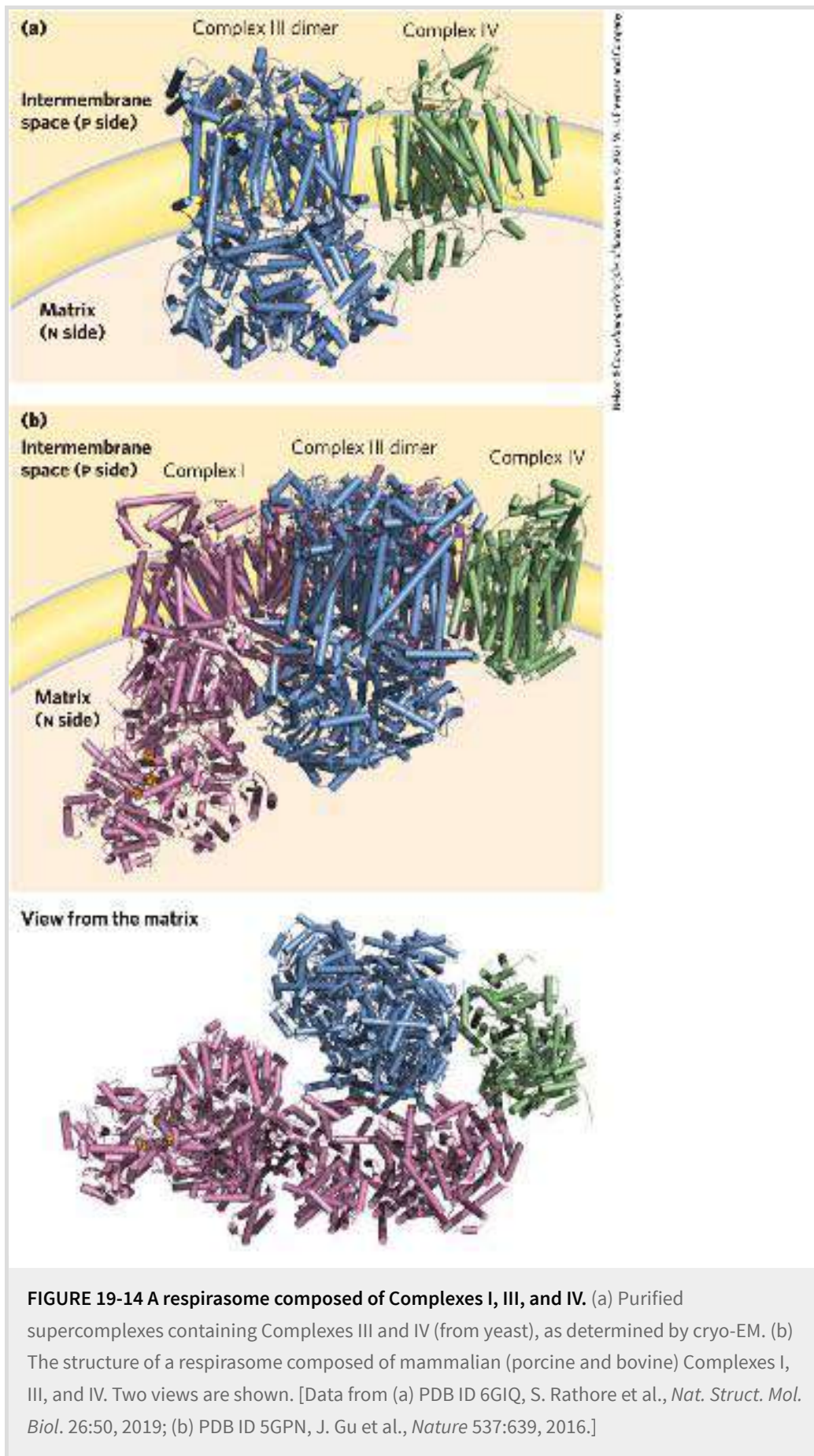
Note that the O_2 in this reaction is the final acceptor for electrons originating from the many sources already described, and the stoichiometries between electron sources and O_2 molecules consumed help to define the energetics of the systems. In this chapter, stoichiometries are sometimes presented, as here, in terms of one molecule of O_2 . For calculation simplicity, the stoichiometries will be presented in terms of $\frac{1}{2}\text{O}_2$ in some examples to come.

At Complex IV, O_2 is reduced at redox centers that carry only one electron at a time. A reaction scheme is presented in [Figure 19-13b](#). Normally the incompletely reduced oxygen intermediates remain tightly bound to the complex until completely converted to water. However, a small fraction of oxygen intermediates escape. These intermediates are reactive oxygen species that can damage cellular components unless eliminated by defense mechanisms described below.

Mitochondrial Complexes Associate in Respirasomes

Although the four electron-transferring complexes can be separated in the laboratory, in the intact mitochondrion, three of the four respiratory complexes associate with each other in the inner membrane. Combinations of Complexes I and III, III and IV, and I, III, and IV are formed in organisms ranging from yeast to plants to mammals. The supercomplex containing

Complexes I, III, and IV has been called the [respirasome](#). Unlike the other three complexes, Complex II is generally found free-floating within the membrane. Structural characterization of the various supercomplexes has been advanced by cryo-EM ([Fig. 19-14](#)). The functional significance of supercomplexes has not been determined. Researchers have suggested that they may facilitate electron transfers or limit the production of reactive oxygen species. Local pools of the electron carriers cytochrome c and ubiquinone are not constrained within supercomplexes, but instead readily diffuse between them.



Other Pathways Donate Electrons to the Respiratory Chain via Ubiquinone

Several other electron-transfer reactions can reduce ubiquinone in the inner mitochondrial membrane ([Fig. 19-15](#)). In the first step of the β oxidation of fatty acyl-CoA, catalyzed by the flavoprotein **acyl-CoA dehydrogenase** (see [Fig. 17-8](#)), electrons pass from the substrate to the FAD of the dehydrogenase, then to electron-transferring flavoprotein (ETF). ETF passes its electrons to **ETF: ubiquinone oxidoreductase**, which reduces Q in the inner mitochondrial membrane to QH₂. Glycerol 3-phosphate, formed either from glycerol released by triacylglycerol breakdown or from the reduction of dihydroxyacetone phosphate from glycolysis, is oxidized by **glycerol 3-phosphate dehydrogenase** (see [Fig. 17-4](#)), a flavoprotein located on the outer face of the inner mitochondrial membrane. The electron acceptor in this reaction is Q; the QH₂ produced enters the pool of QH₂ in the membrane. The important role of glycerol 3-phosphate dehydrogenase in shuttling reducing equivalents from cytosolic NADH into the mitochondrial matrix is described in [Section 19.2](#) (see [Fig. 19-32](#)). **Dihydroorotate dehydrogenase**, which acts in the synthesis of pyrimidines (see [Fig. 22-38](#)), is also on the outside of the inner mitochondrial membrane and donates electrons to Q in the respiratory chain. The reduced QH₂ passes its electrons through Complex III and ultimately to O₂.

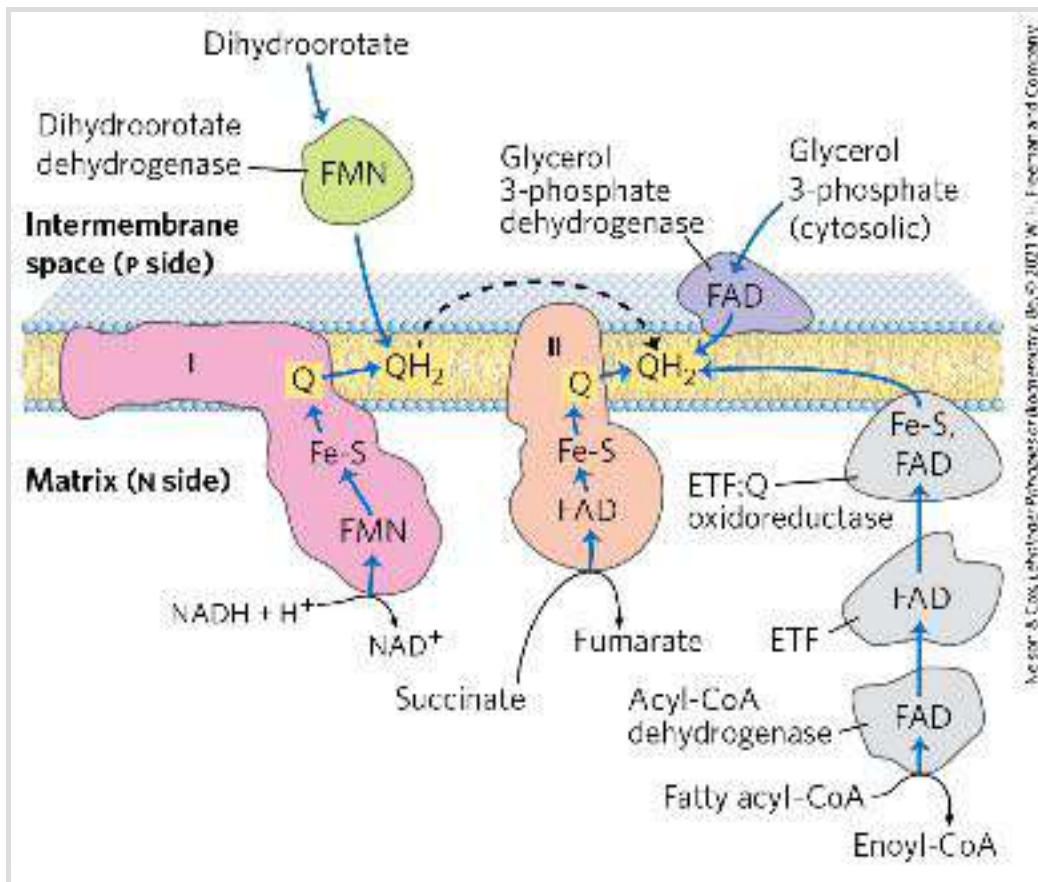
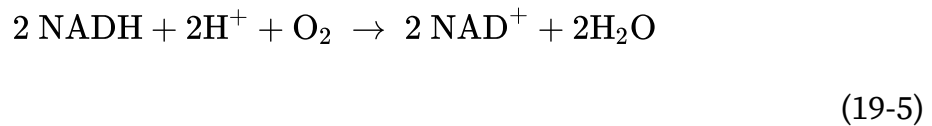


FIGURE 19-15 Paths of electron transfer to ubiquinone in the respiratory chain. Electrons from NADH in the matrix pass through the FMN of a flavoprotein (NADH dehydrogenase) to a series of Fe-S centers (in Complex I) and then to Q. Electrons from succinate oxidation in the citric acid cycle pass through a flavoprotein with several Fe-S centers (Complex II) on the way to Q. Acyl-CoA dehydrogenase, the first enzyme of fatty acid β oxidation, transfers electrons to electron-transferring flavoprotein (ETF), from which they pass to Q via ETF:ubiquinone oxidoreductase. Dihydroorotate, an intermediate in the biosynthetic pathway to pyrimidine nucleotides, donates two electrons to Q through a flavoprotein (dihydroorotate dehydrogenase). And glycerol 3-phosphate, an intermediate of glycolysis in the cytosol, donates electrons to a flavoprotein (glycerol 3-phosphate dehydrogenase) on the outer face of the inner mitochondrial membrane, from which they pass to Q. QH_2 freely diffuses through the membrane (black dashed arrow), and can interact with several additional complexes.

The Energy of Electron Transfer Is Efficiently Conserved in a Proton Gradient

The transfer of two electrons from NADH through the respiratory chain to molecular oxygen can be summarized as

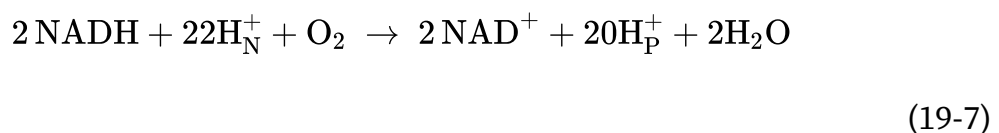


This net reaction is highly exergonic. For the redox pair NAD^+/NADH , E'° is -0.320 V , and for the pair $\text{O}_2/\text{H}_2\text{O}$, E'° is 0.816 V . The $\Delta E'^\circ$ for this reaction is therefore 1.14 V , and the standard free-energy change (see [Eqn 13-7, p. 492](#)) is

$$\begin{aligned} \Delta G'^\circ &= -nF \Delta E'^\circ \\ &= -2(96.5 \text{ kJ/V} \cdot \text{mol})(1.14 \text{ V}) \\ &= -220 \text{ kJ/mol (of NADH)} \end{aligned} \quad (19-6)$$

This *standard* free-energy change is based on the assumption of equal concentrations (1 M) of NADH and NAD^+ . In actively respiring mitochondria, the actions of many dehydrogenases keep the actual $[\text{NADH}]/[\text{NAD}^+]$ ratio well above unity, and the real free-energy change for the reaction shown in [Equation 19-5](#) is therefore substantially greater (more negative) than -220 kJ/mol . A similar calculation for the oxidation of succinate shows that electron transfer from succinate (E'° for fumarate/succinate = 0.031 V) to O_2 has a smaller, but still negative, standard free-energy change of about -150 kJ/mol .

Much of this energy is used to pump protons out of the matrix. For each pair of electrons transferred to O_2 , four protons are pumped out by Complex I, four by Complex III, and two by Complex IV ([Fig. 19-16](#)). The *vectorial* equation for the process is therefore



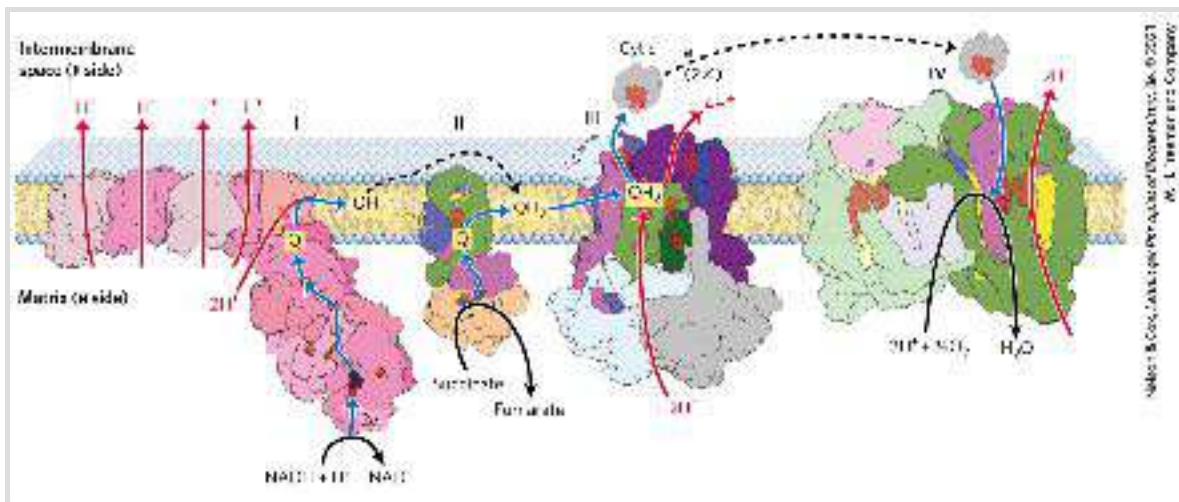
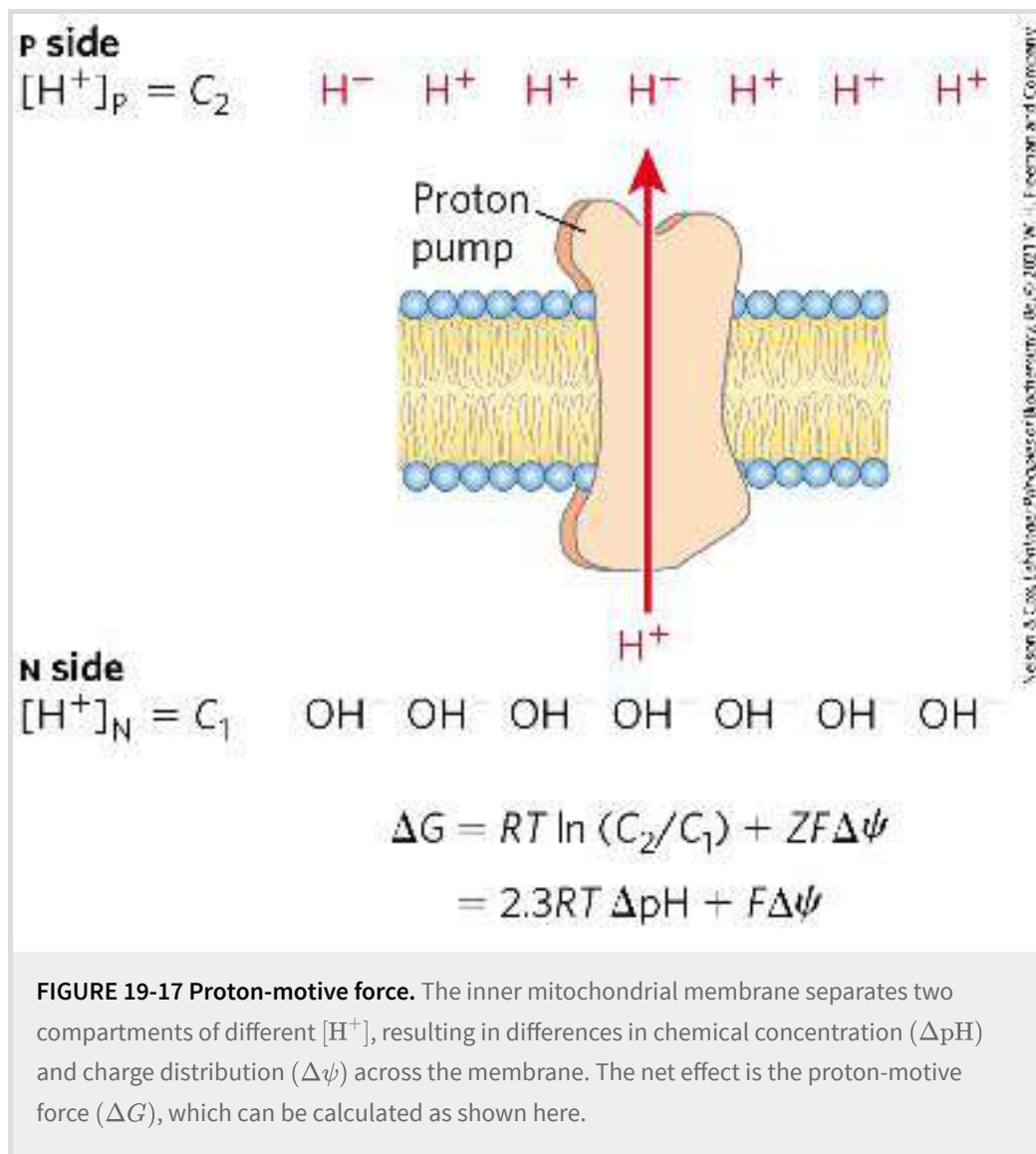


FIGURE 19-16 Summary of the flow of electrons and protons through the four complexes of the respiratory chain. Electrons reach Q through Complexes I and II (as well as through several other paths shown in [Fig. 19-15](#)). Reduced Q (QH₂) serves as a mobile carrier of electrons and protons. It passes electrons to Complex III, which passes them to another mobile connecting link, cytochrome c. Complex IV then transfers electrons from reduced cytochrome c to O₂. Electron flow through Complexes I, III, and IV is accompanied by proton efflux from the matrix into the intermembrane space. In bovine heart, the approximate ratios of Complexes I:II:III:IV are 1.1:1.3:3.0:6.7. Broken lines indicate the diffusion of Q in the plane of the inner membrane, and of cytochrome c through the intermembrane space. [Data from Complex I: PDB ID 4HEA, R. Baradaran et al., *Nature* 494:443, 2013; Complex II: PDB ID 1ZOY, F. Sun et al., *Cell* 121:1043, 2005; Complex III: PDB ID 1BGY, S. Iwata et al., *Science* 281:64, 1998; cytochrome c: PDB ID 1HRC, G. W. Bushnell et al., *J. Mol. Biol.* 214:585, 1990; Complex IV: PDB ID 1OCC, T. Tsukihara et al., *Science* 272:1136, 1996.]

P4 The electrochemical energy inherent in this difference in proton concentration and separation of charge represents a temporary conservation of much of the energy of electron transfer. The energy stored in such a gradient, termed the **proton-motive force**, has two components: (1) the *chemical potential energy* due to the difference in concentration of a chemical species (H⁺) in the two regions separated by the membrane, and (2) the *electrical potential energy* that results from the separation of charge when a proton moves across the membrane without a counterion ([Fig. 19-17](#)).



As we saw in [Chapter 11](#), the free-energy change for the creation of an electrochemical gradient by an ion pump is

$$\Delta G = RT \ln(C_2/C_1) + ZF \Delta\psi \quad (19-8)$$

where C_2 and C_1 are the concentrations of an ion in two regions, and $C_2 > C_1$; Z is the absolute value of its electrical charge (1 for a proton); and $\Delta\psi$ is the transmembrane difference in electrical potential, measured in volts.

For protons,

$$\begin{aligned}\ln(C_2/C_1) &= 2.3 (\log[\text{H}^+]_{\text{P}} - \log[\text{H}^+]_{\text{N}}) \\ &= 2.3 (\text{pH}_{\text{N}} - \text{pH}_{\text{P}}) = 2.3 \Delta\text{pH}\end{aligned}$$

and [Equation 19-8](#) reduces to

$$\Delta G = 2.3RT \Delta\text{pH} + F\Delta\psi \quad (19-9)$$

In actively respiring mitochondria, the measured $\Delta\psi$ is 0.15 to 0.20 V, and the pH of the matrix is about 0.75 units more alkaline than that of the intermembrane space.

WORKED EXAMPLE 19-1 *Energetics of Electron Transfer*

Calculate the amount of energy conserved in the proton gradient across the inner mitochondrial membrane per pair of electrons transferred through the respiratory chain from NADH to oxygen. Assume $\Delta\psi$ is 0.15 V and the pH difference is 0.75 unit at body temperature of 37 °C.

SOLUTION:

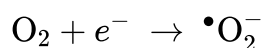
[Equation 19-9](#) gives the free-energy change when *one* mole of protons moves across the inner membrane. Substituting the values of the constants R and F , 310 K for T , and the measured values for ΔpH (0.75 unit) and $\Delta\psi$ (0.15 V) in this equation gives $\Delta G = 19 \text{ kJ/mol}$ (of protons). Because the transfer of two electrons from NADH to O_2 is accompanied by the outward pumping of 10

protons ([Eqn 19-7](#)), roughly 190 kJ (of the 220 kJ released by oxidation of 1 mol of NADH) is conserved in the proton gradient.

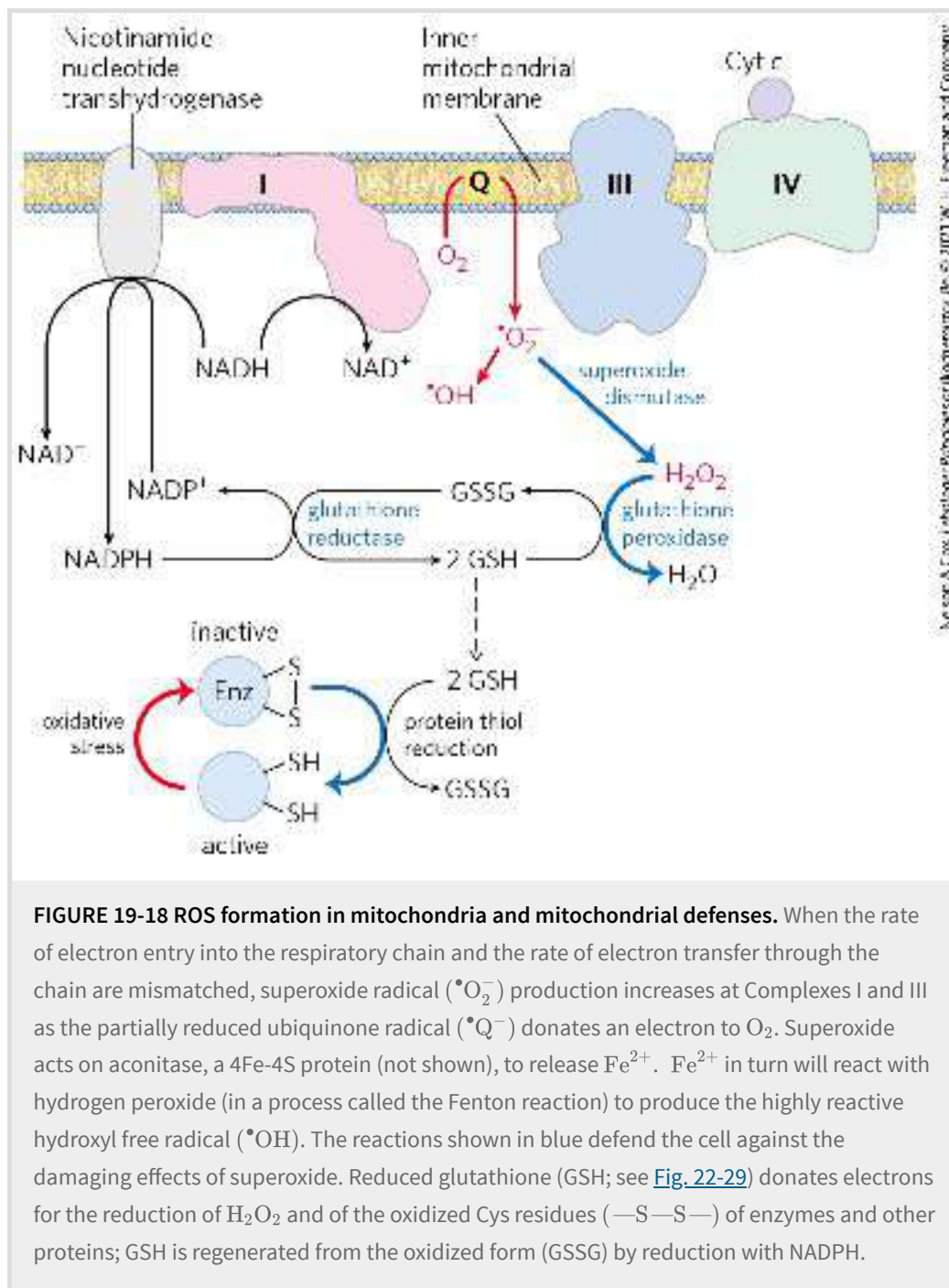
When protons flow spontaneously *down* their electrochemical gradient, energy is made available to do work. In mitochondria, chloroplasts, and aerobic bacteria, the electrochemical energy in the proton gradient drives the synthesis of ATP from ADP and P_i. We return to the energetics and stoichiometry of ATP synthesis driven by the electrochemical potential of the proton gradient in [Section 19.2](#).

Reactive Oxygen Species Are Generated during Oxidative Phosphorylation

Several steps in the path of oxygen reduction in mitochondria have the potential to produce reactive oxygen species (superoxide, hydrogen peroxide, and hydroxyl radicals) that can damage cells. Some intermediates in the electron-transfer system, such as the partially reduced ubisemiquinone, can react directly with oxygen to form the superoxide radical ($\bullet\text{Q}^-$) as an intermediate. The $\bullet\text{Q}^-$ radical is formed when a single electron is passed to O₂ in the reaction



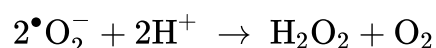
Successive reduction of the superoxide radical with additional electrons produces H₂O₂, hydroxyl radicals ($\bullet\text{OH}$), and finally H₂O. The very reactive hydroxyl radical can be especially damaging ([Fig. 19-18](#)).



Reactive oxygen species (ROS) can wreak havoc, reacting with and damaging enzymes, membrane lipids, and nucleic acids. In actively respiring mitochondria, 0.2% to as much as 2% of the O_2 used in respiration forms $\bullet\text{O}_2^-$ — more than enough to have lethal effects unless the free radical is quickly disposed of. Factors that slow the flow of electrons through the respiratory chain increase the formation of superoxide, perhaps by prolonging the lifetime of $\bullet\text{O}_2^-$ generated in the Q cycle. The formation of ROS is favored

when two conditions are met: (1) mitochondria are not making ATP (for lack of ADP or O₂) and therefore have a large proton-motive force and a high QH₂/Q ratio, and (2) there is a high NADH/NAD⁺ ratio in the matrix. In these situations, the mitochondrion is under oxidative stress — more electrons are available to enter the respiratory chain than can be immediately passed through to oxygen. When the supply of electron donors (NADH) is matched with that of electron acceptors, there is less oxidative stress, and ROS production is much reduced. Although overproduction of ROS is clearly detrimental, *low* levels of ROS are used by the cell as a signal reflecting the insufficient supply of oxygen (hypoxia), triggering metabolic adjustments (see [Fig. 19-34](#)).

To prevent oxidative damage by •O₂⁻, cells have the enzyme **superoxide dismutase**, which catalyzes the reaction



The hydrogen peroxide (H₂O₂) thus generated is rendered harmless by **glutathione peroxidase** ([Fig. 19-18](#)). Glutathione reductase recycles the oxidized glutathione to its reduced form, using electrons from the NADPH generated by nicotinamide nucleotide transhydrogenase (in the mitochondrion) or by the pentose phosphate pathway (in the cytosol; see [Fig. 14-30](#)). Reduced glutathione also serves to keep protein sulfhydryl groups in their reduced state, preventing some of the deleterious effects of oxidative stress.

SUMMARY 19.1 *The Mitochondrial Respiratory Chain*

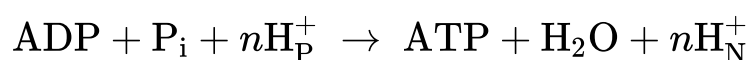
- Chemiosmotic theory provides the intellectual framework for understanding many biological energy transductions, including oxidative phosphorylation and photophosphorylation. The energy of electron flow is conserved by the concomitant pumping of protons across the membrane, producing an electrochemical gradient, the proton-motive force.
- In mitochondria, hydride ions removed from substrates (such as α -ketoglutarate and malate) by NAD-linked dehydrogenases donate electrons to the respiratory chain, which transfers the electrons to molecular O_2 , reducing it to H_2O .
- Reducing equivalents from NADH are passed through a series of carrier types that include ubiquinone, Fe-S centers, and cytochromes.
- Electron carriers are arranged in multiprotein complexes, which couple electron transfer to the generation of proton gradients. Complex I transfers electrons from NADH to ubiquinone. Complex II harvests electrons from the oxidation of succinate, also transferring them to ubiquinone. Ubiquinone diffuses through the inner mitochondrial membrane, and transfers the electrons to cytochrome *b*, the first carrier in Complex III. In this complex, electrons move first to an Fe-S center. The Fe-S center passes electrons, one at a time, through cytochrome *c* and into Complex IV, cytochrome oxidase. This copper-containing enzyme, which also contains cytochromes *a* and a_3 , accumulates electrons, then passes them to O_2 , reducing it to H_2O .
- Complexes I, III, and IV form supercomplexes that facilitate electron flow between them.
- Some electrons enter this chain of carriers through alternative paths. Electrons derived from the oxidation of fatty acids pass to ubiquinone via the electron-transferring flavoprotein. The oxidation of glycerol phosphate and of dihydroorotate also sends electrons into the respiratory chain at the level of QH_2 .
- Much of the free energy generated by electron transfer and the reduction of oxygen to form water is recovered and stored in the form of an electrochemical proton gradient across the mitochondrial inner membrane.
- Potentially harmful reactive oxygen species produced in mitochondria are inactivated by a set of protective enzymes, including superoxide dismutase and glutathione peroxidase. Low levels of ROS serve as signals coordinating mitochondrial oxidative phosphorylation with other metabolic pathways.

19.2 ATP Synthesis

How is a concentration gradient of protons transformed into ATP? We have seen that electron transfer releases, and the proton-motive force conserves, more than enough free energy (about 190 kJ) per “mole” of electron pairs to drive the formation of a mole of ATP, which requires about 50 kJ ([p. 480](#)). Mitochondrial oxidative phosphorylation therefore poses no thermodynamic problem. But what is the chemical mechanism that couples proton flux with phosphorylation?

In the Chemiosmotic Model, Oxidation and Phosphorylation Are Obligately Coupled

The **chemiosmotic model**, proposed by Peter Mitchell, is the paradigm for energy coupling. According to the model ([Fig. 19-19](#)), the electrochemical energy inherent in the difference in proton concentration and the separation of charge across the inner mitochondrial membrane — the proton-motive force — drives the synthesis of ATP as protons flow passively back into the matrix through a proton pore in [ATP synthase](#). To emphasize this crucial role of the proton-motive force, the equation for ATP synthesis is sometimes written



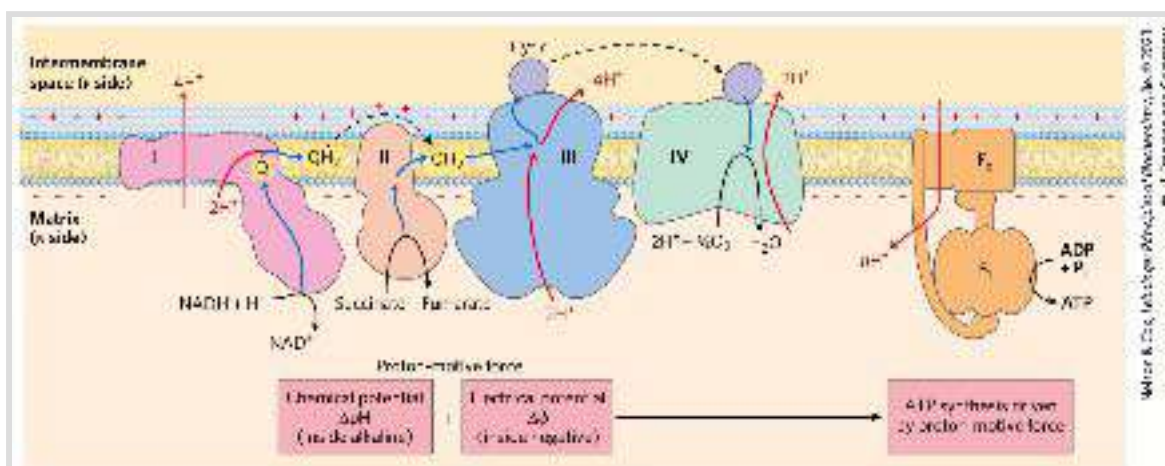
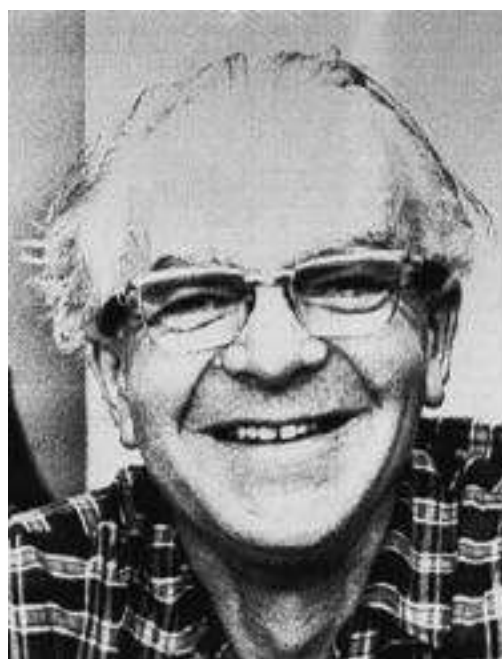


FIGURE 19-19 Chemiosmotic model. In this simple representation of the chemiosmotic theory applied to mitochondria, electrons from NADH and other oxidizable substrates pass through a chain of carriers arranged asymmetrically in the inner membrane. Electron flow is accompanied by proton transfer across the membrane, producing both a chemical gradient (ΔpH) and an electrical gradient ($\Delta\psi$), which, combined, create the proton-motive force. The inner mitochondrial membrane is impermeable to protons; protons can reenter the matrix only through proton-specific channels (F_0). The proton-motive force that drives protons back into the matrix provides the energy for ATP synthesis, catalyzed by the F_1 complex associated with F_0 .



AP Photo/AP Images.

Mitchell used the term “chemiosmotic” to describe enzymatic reactions that involve, simultaneously, a chemical reaction and a transport process, and the overall process is sometimes referred to as “chemiosmotic coupling.” Here, “coupling” refers to the *obligate* connection between mitochondrial ATP synthesis and electron flow through the respiratory chain; neither of the two processes can proceed without the other. The operational definition of *coupling* is shown in [Figure 19-20](#).  When isolated mitochondria are suspended in a buffer containing ADP, P_i , and an oxidizable substrate such as succinate, three easily measured processes occur: (1) the substrate is oxidized (succinate yields fumarate), (2) O_2 is consumed, and (3) ATP is synthesized. Oxygen consumption and ATP synthesis depend on the presence of an oxidizable substrate (succinate in this case) as well as ADP and P_i .

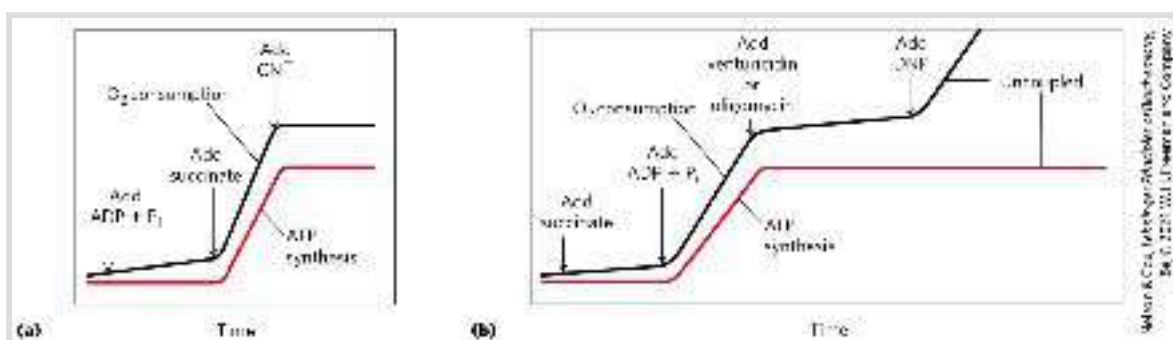


FIGURE 19-20 Coupling of electron transfer and ATP synthesis in mitochondria. In experiments to demonstrate coupling, mitochondria are suspended in a buffered medium, and an O_2 electrode monitors O_2 consumption. At intervals, samples are removed and assayed for the presence of ATP. (a) Addition of ADP and P_i alone results in little or no increase in either respiration (O_2 consumption; black) or ATP synthesis (red).

When succinate is added, respiration begins immediately, and ATP is synthesized. Addition of cyanide (CN^-), which blocks electron transfer between cytochrome oxidase (Complex IV) and O_2 , inhibits both respiration and ATP synthesis. (b) Mitochondria provided with succinate respire and synthesize ATP only when ADP and P_i are added. Subsequent addition of venturicidin or oligomycin, inhibitors of ATP synthase, blocks both ATP synthesis and respiration. Dinitrophenol (DNP) is an uncoupler, allowing respiration to continue without ATP synthesis.

Because substrate oxidation drives ATP synthesis, inhibitors of electron transfer block ATP synthesis ([Fig. 19-20a](#)). The converse is also true: inhibition of ATP synthesis blocks electron transfer in intact mitochondria. When isolated mitochondria are given O_2 and oxidizable substrates, but not ADP ([Fig. 19-20b](#)), no ATP synthesis can occur and electron transfer to O_2 does not proceed. Henry Lardy, who pioneered the use of antibiotics to explore mitochondrial function, showed coupling of oxidation and phosphorylation by using oligomycin and venturicidin. These toxic antibiotics bind to the ATP synthase in mitochondria, inhibiting both ATP synthesis *and* the transfer of electrons through the chain of carriers to O_2 ([Fig. 19-20b](#)). As oligomycin does not interact with the electron carriers, it follows that electron transfer and ATP synthesis are obligately coupled: neither reaction occurs without the other.

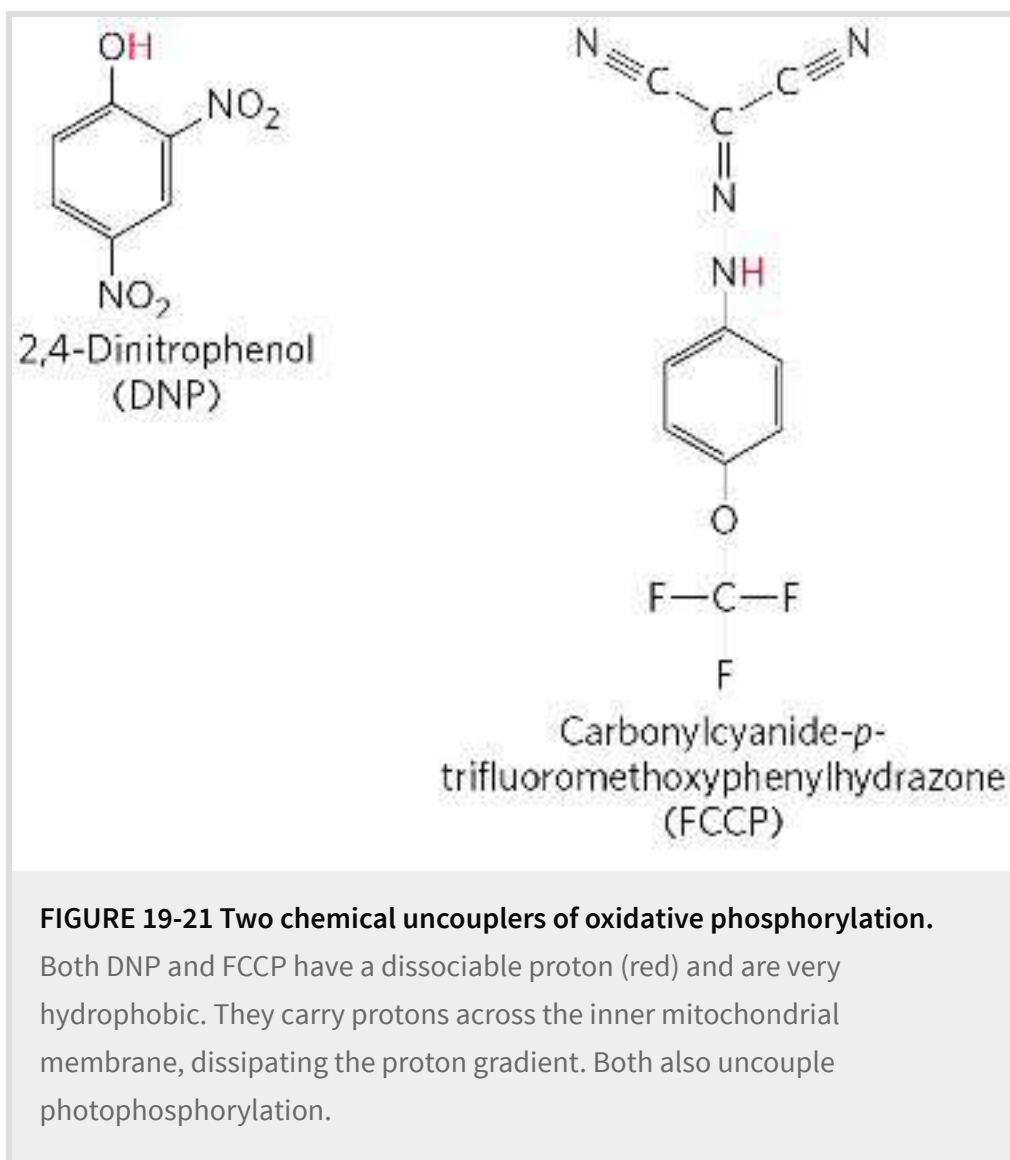


Henry Lardy, 1917–2010

Chemiosmotic theory readily explains the dependence of electron transfer on ATP synthesis in mitochondria. When the flow of protons into the matrix through the proton channel of ATP synthase is blocked (with oligomycin, for example), no path exists for the return of protons to the matrix, and the continued extrusion of protons driven by the activity of the respiratory chain generates a large proton gradient. The proton-motive force builds up until the cost (free energy) of pumping protons out of the matrix against this gradient equals or exceeds the energy released by the transfer of electrons from NADH to O_2 . At this point electron flow must stop; the free energy for the overall process of

electron flow coupled to proton pumping becomes zero, and the system is at equilibrium.

Certain conditions and reagents, however, can uncouple oxidation from phosphorylation. When intact mitochondria are disrupted by treatment with detergent or by physical shear, the resulting membrane fragments can still catalyze electron transfer from succinate or NADH to O_2 , but no ATP synthesis is coupled to this respiration. Certain chemical compounds cause uncoupling without physically disrupting mitochondrial structure. Chemical uncouplers include 2,4-dinitrophenol (DNP) and carbonylcyanide-*p*-trifluoromethoxyphenylhydrazone (FCCP) ([Table 19-4](#); [Fig. 19-21](#)), weak acids with hydrophobic properties that permit them to diffuse readily across mitochondrial membranes. After entering the matrix in the protonated form, they can release a proton, thus dissipating the proton gradient. Resonance stabilization delocalizes the charge on the anionic forms, making them sufficiently hydrophobic to diffuse back across the membrane, where they can pick up a proton and repeat the process. Ionophores such as valinomycin (see [Fig. 11-43](#)) allow inorganic ions to pass easily through membranes. Ionophores uncouple electron transfer from oxidative phosphorylation by dissipating the electrical contribution to the electrochemical gradient across the mitochondrial membrane.



A prediction of the chemiosmotic theory is that, because the role of electron transfer in mitochondrial ATP synthesis is simply to pump protons to create the electrochemical potential of the proton-motive force, an artificially created proton gradient should be able to replace electron transfer in driving ATP synthesis. This has been experimentally confirmed ([Fig. 19-22](#)). *In the absence of an oxidizable substrate*, the proton-motive force alone suffices to drive ATP synthesis.

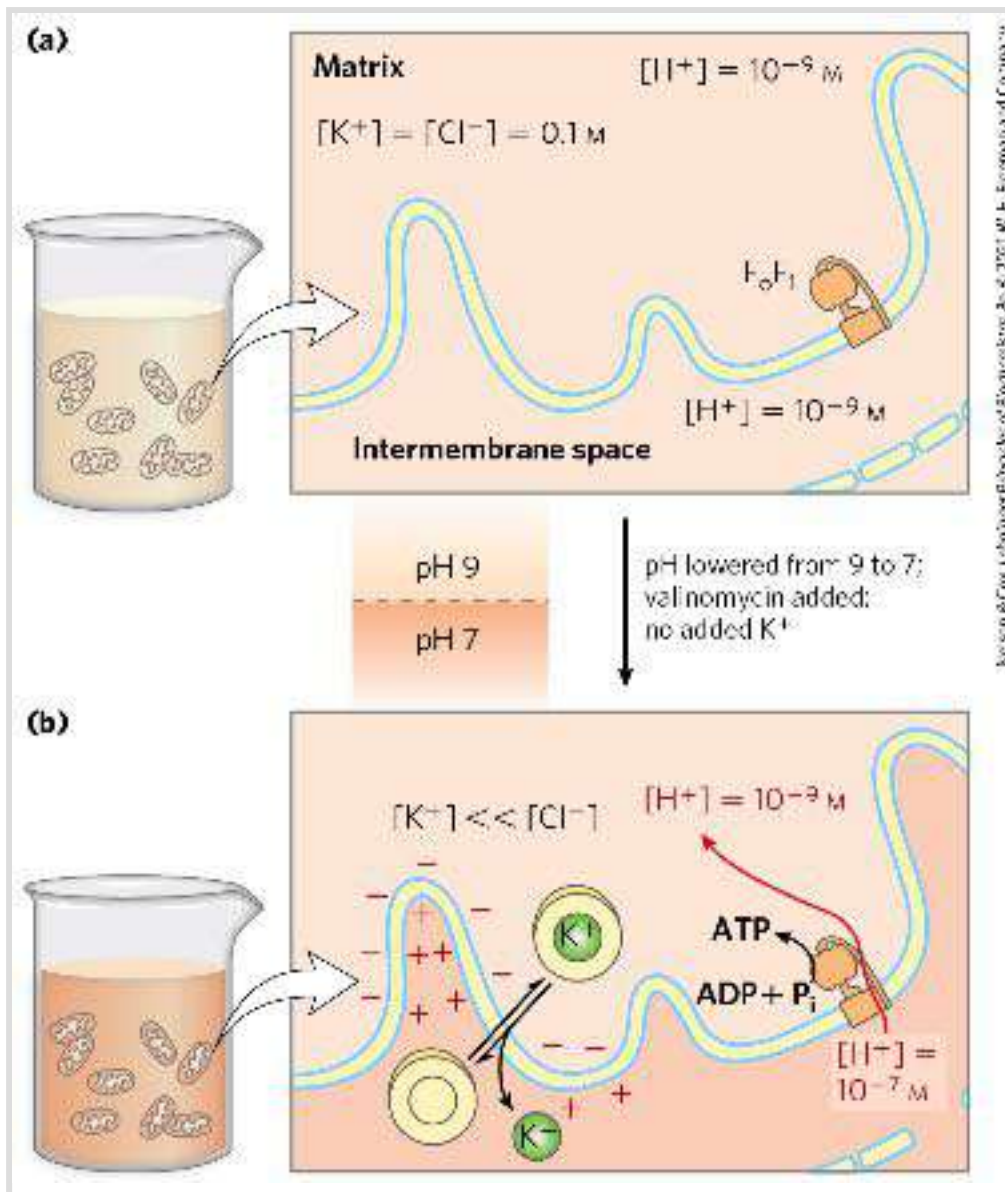


FIGURE 19-22 Evidence for the role of a proton gradient in ATP synthesis.

An artificially imposed electrochemical gradient can drive ATP synthesis in the absence of an oxidizable substrate as electron donor. In this two-step experiment, (a) isolated mitochondria are first incubated in a pH 9 buffer containing 0.1 M KCl. Slow leakage of buffer and KCl into the mitochondria eventually brings the matrix into equilibrium with the surrounding medium. No oxidizable substrates are present. (b) Mitochondria are now removed from the pH 9 buffer and resuspended in pH 7 buffer containing valinomycin but no KCl. The change in buffer creates a difference of two pH units across the inner mitochondrial membrane. The outward flow of K^+ , carried by valinomycin down the K^+ ion concentration gradient without a counterion, creates a charge imbalance across the membrane (matrix negative). The sum of the chemical potential provided by the pH difference

and the electrical potential provided by the separation of charges is a proton-motive force large enough to support ATP synthesis in the absence of an oxidizable substrate.

ATP Synthase Has Two Functional Domains, F_0 and F_1

 Mitochondrial ATP synthase is an F-type ATPase (see [Fig. 11-40b](#)) similar in structure and mechanism to the ATP synthases of bacteria and (as we will see in [Chapter 20](#)) chloroplasts. This large enzyme complex of the inner mitochondrial membrane catalyzes the formation of ATP from ADP and P_i , driven by the flow of protons from the P to the N side of the membrane ([Eqn 19-10](#)). ATP synthase, also called Complex V to relate it to the electron-transfer complexes described in the last section, has two distinct components. These are F_1 , a peripheral membrane protein, and F_o (*o* denoting oligomycin-sensitive), which is integral to the membrane. F_1 , the first factor recognized as essential for oxidative phosphorylation, was identified and purified by Efraim Racker and his colleagues in the early 1960s.

In the laboratory, small membrane vesicles formed from inner mitochondrial membranes carry out ATP synthesis coupled to electron transfer. When F_1 is gently extracted, the “stripped” vesicles still contain intact respiratory chains and the F_o portion of ATP synthase.

 The vesicles can catalyze electron transfer from NADH to O_2 but cannot produce a proton gradient:

F_o has a proton pore through which protons leak as fast as they are pumped by electron transfer, and without a proton gradient the F_1 -depleted vesicles cannot make ATP. Isolated F_1 catalyzes ATP hydrolysis (the reversal of synthesis) and was therefore originally called **F_1 ATPase**. When purified F_1 is added back to the depleted vesicles, it reassociates with F_o , plugging its proton pore and restoring the membrane's capacity to couple electron transfer and ATP synthesis.

ATP Is Stabilized Relative to ADP on the Surface of F_1

Isotope exchange experiments using purified F_1 reveal an extraordinary fact about the enzyme's catalytic mechanism: on the enzyme surface, the reaction $ADP + P_i \rightleftharpoons ATP + H_2O$ is readily reversible — the free-energy change for ATP synthesis is close to zero. When ATP is hydrolyzed by F_1 in the presence of ^{18}O -labeled water, the P_i released contains an ^{18}O atom. Careful measurement of the ^{18}O content of P_i formed in vitro by F_1 -catalyzed hydrolysis of ATP reveals that the P_i has not one but three or four ^{18}O atoms (**Fig. 19-23**). This indicates that the terminal pyrophosphate bond in ATP is cleaved and re-formed repeatedly before P_i leaves the enzyme surface. This exchange reaction occurs in unenergized F_oF_1 complexes (with no proton gradient) and with isolated F_1 — the exchange does not require the input of energy.

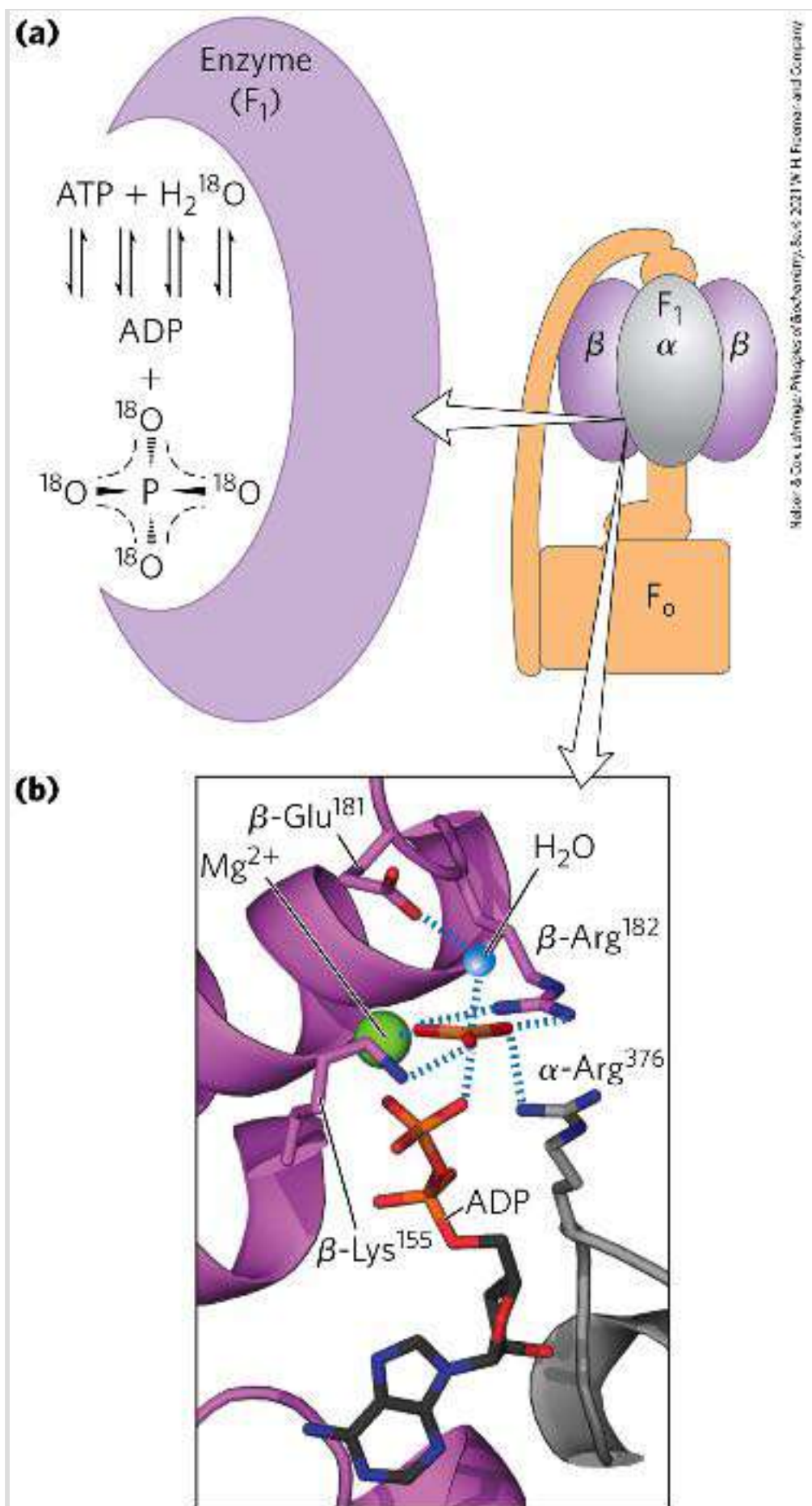


FIGURE 19-23 Catalytic mechanism of F_1 . (a) An ^{18}O -exchange experiment. F_1 solubilized from mitochondrial membranes is incubated with ATP in the presence of ^{18}O -labeled water. At intervals, a sample of the solution is withdrawn and analyzed for the incorporation of ^{18}O into the P_i produced from ATP hydrolysis. In minutes, the P_i contains three or four ^{18}O atoms, indicating that both ATP hydrolysis and ATP synthesis have occurred several times during the incubation. (b) The likely transition state complex for ATP hydrolysis and synthesis by ATP synthase. The α subunit is shown in gray, β in purple. The positively charged residues $\beta\text{-Arg}^{182}$ and $\alpha\text{-Arg}^{376}$ coordinate two oxygens of the pentavalent phosphate intermediate; $\beta\text{-Lys}^{155}$ interacts with a third oxygen, and the Mg^{2+} ion further stabilizes the intermediate. The blue sphere represents the leaving group (H_2O). These interactions result in ready equilibration of ATP and $\text{ADP} + \text{P}_i$ in the active site. [(b) Data from PDB ID 1BMF, J. P. Abrahams et al., *Nature* 370:621, 1994.]

Kinetic studies of the initial rates of ATP synthesis and hydrolysis confirm the conclusion that $\Delta G'^\circ$ for ATP synthesis on the enzyme is near zero. From the measured rates of hydrolysis ($k_1 = 10 \text{ s}^{-1}$) and synthesis ($k_{-1} = 24 \text{ s}^{-1}$), the calculated equilibrium constant for the reaction



is

$$K'_{\text{eq}} = \frac{k_{-1}}{k_1} = \frac{24 \text{ s}^{-1}}{10 \text{ s}^{-1}} = 2.4$$

From this K'_{eq} , the calculated apparent $\Delta G'^{\circ}$ is close to zero. This is much different from the K'_{eq} of about 10^5 ($\Delta G'^{\circ} = -30.5 \text{ kJ/mol}$) for the hydrolysis of ATP free in solution (i.e., not on the enzyme surface).

What accounts for the huge difference? ATP synthase stabilizes ATP relative to $\text{ADP} + \text{P}_i$ by binding ATP more tightly, releasing enough energy to counterbalance the cost of making ATP. Careful measurements of the binding constants show that F_0F_1 binds ATP with very high affinity ($K_d \leq 10^{-12} \text{ M}$) and ADP with much lower affinity ($K_d \approx 10^{-5} \text{ M}$). The difference in K_d corresponds to a difference of about 40 kJ/mol in binding energy, and this binding energy drives the equilibrium toward formation of the product ATP.

The Proton Gradient Drives the Release of ATP from the Enzyme Surface

Although ATP synthase equilibrates ATP with $\text{ADP} + \text{P}_i$, in the absence of a proton gradient the newly synthesized ATP does not leave the surface of the enzyme. Effectively, the enzyme cannot turn over and synthesize a second molecule of ATP. It is the proton gradient that causes the enzyme to release the ATP formed on its surface. The reaction coordinate diagram of the process ([Fig. 19-24](#)) illustrates the difference between the mechanism of

ATP synthase and that of many other enzymes that catalyze endergonic reactions.

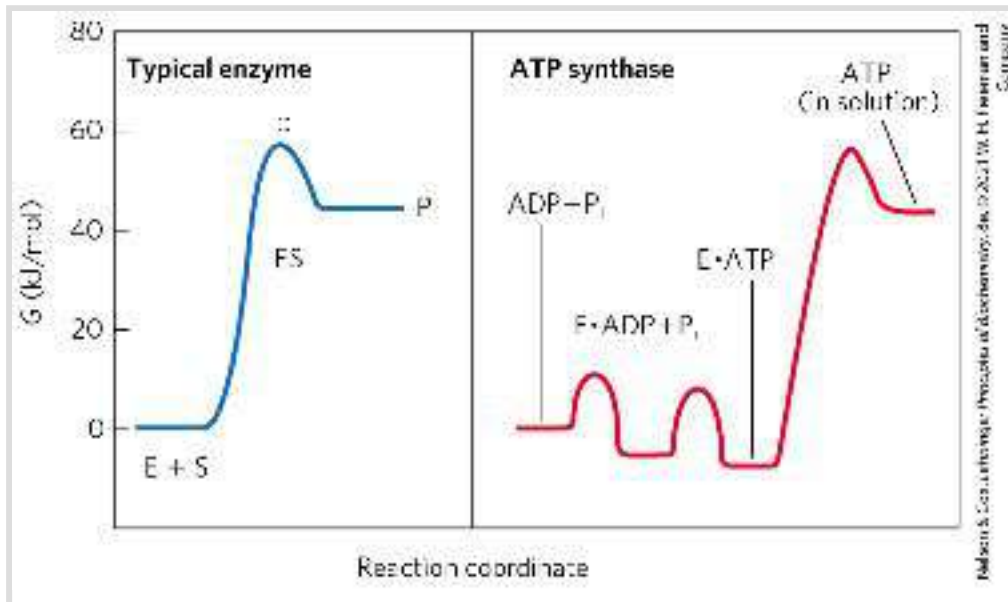


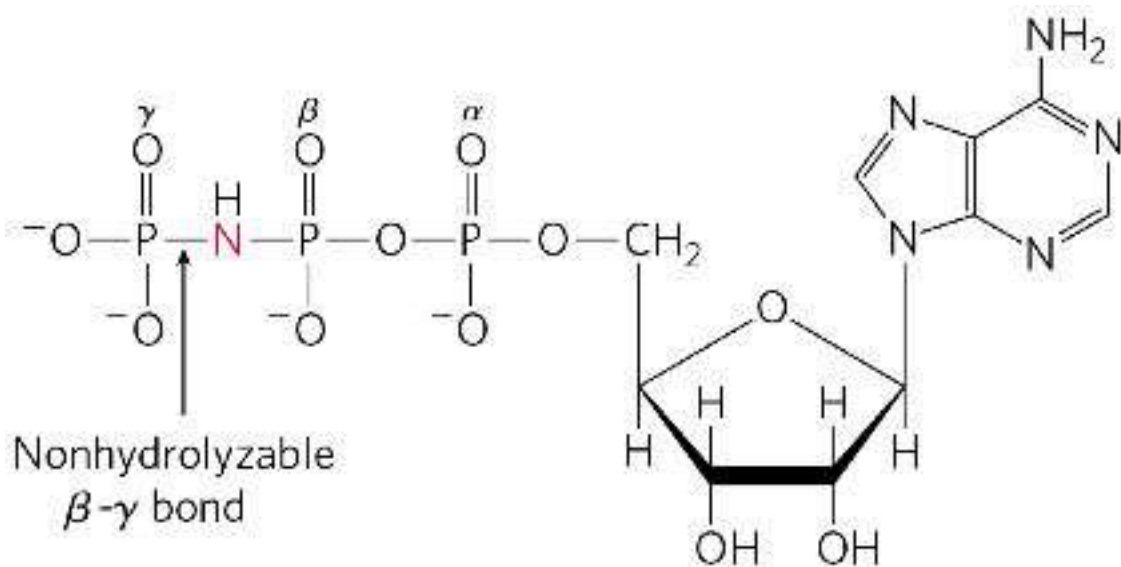
FIGURE 19-24 Reaction coordinate diagrams for ATP synthase and for a more typical enzyme. In a typical enzyme-catalyzed reaction (left), reaching the transition state ($\ddagger$) between substrate and product is the major energy barrier to overcome. In the reaction catalyzed by ATP synthase (right), release of ATP from the enzyme, not formation of ATP, is the major energy barrier. The free-energy change for the formation of ATP from ADP and P_i in aqueous solution is large and positive, but on the enzyme surface, the very tight binding of ATP provides sufficient binding energy to bring the free energy of the enzyme-bound ATP close to that of ADP + P_i, so the reaction is readily reversible. The equilibrium constant is near 1. The free energy required for the release of ATP is provided by the proton-motive force.

For the continued synthesis of ATP, the enzyme must cycle between a form that binds ATP very tightly and a form that releases ATP. Chemical and crystallographic studies of the ATP synthase have revealed the structural basis for this alternation in function.

Each β Subunit of ATP Synthase Can Assume Three Different Conformations

Mitochondrial F_1 has nine subunits of five different types, with the composition $\alpha_3\beta_3\gamma\delta\epsilon$. Each of the three β subunits has one catalytic site for ATP synthesis. The crystallographic determination of the F_1 structure by John E. Walker and colleagues revealed structural details that help explain the catalytic mechanism of the enzyme. The knoblike portion of F_1 is a flattened sphere, 8 nm by 10 nm, consisting of alternating α and β subunits arranged like the sections of an orange ([Fig. 19-25a-d](#)). Although the amino acid sequences of the three β subunits are identical, *their conformations differ*. The conformational differences extend to differences in their ATP/ADP-binding sites. When the protein is crystallized in the presence of ADP and App(NH)p, a close structural analog of ATP that cannot be hydrolyzed by the ATPase activity of F_1 , the binding site of one of the three β subunits is filled with App(NH)p, the second is filled with ADP, and the third is empty. The corresponding β subunit conformations are designated β -ATP, β -ADP, and β -empty ([Fig. 19-25b](#)). This difference in nucleotide binding among the three subunits is critical to the mechanism of the complex. The polypeptides that make up the stalk in the F_1 crystal structure are asymmetrically arranged. One domain of the single γ subunit makes up a central shaft that passes through F_1 . Another globular domain of γ helps to

stabilize the β -empty conformation in a β subunit it is transiently associated with ([Fig. 19-25c](#)).



App(NH)p (β , γ -imidoadenosine 5'-triphosphate)

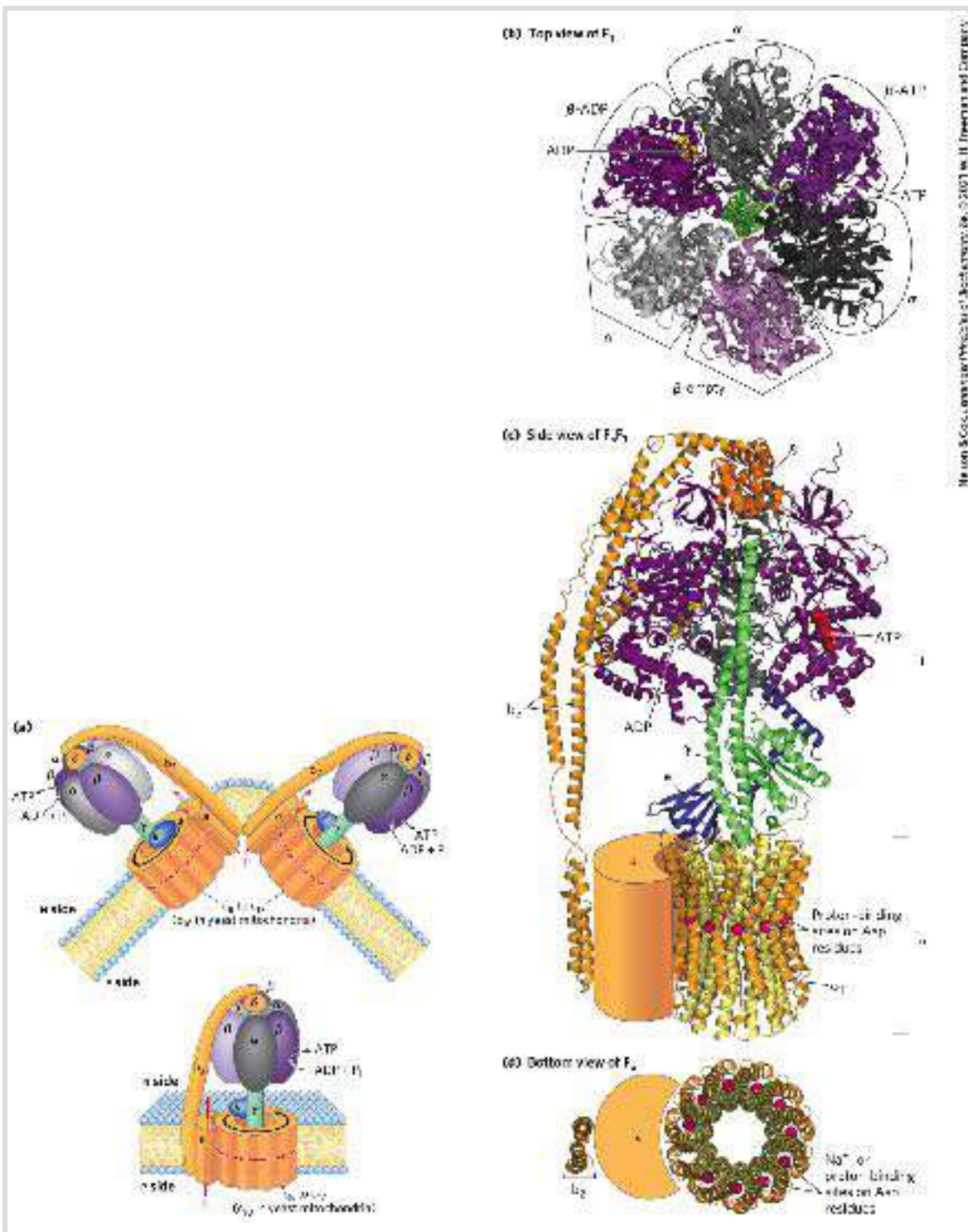


FIGURE 19-25 Mitochondrial ATP synthase complex. (a) A cartoon representation of the F_0F_1 complex. The dimeric form is found in eukaryotic mitochondria. The monomeric form is observed in bacteria. (b) F_1 viewed from above (that is, from the N side of the membrane), showing the three β (shades of purple) and three α (shades of gray) subunits and the central shaft (γ subunit, green). Each β subunit, near its interface with the neighboring α subunit, has a nucleotide-binding site critical to the catalytic activity. The single γ subunit associates primarily with one of the three $\alpha\beta$ pairs, forcing

each of the three β subunits into slightly different conformations, with different nucleotide-binding sites. In the crystalline enzyme, one subunit, β -ADP, has ADP (yellow) in its binding site; the next, β -ATP, has ATP (red); and the third, β -empty, has no bound nucleotide. (c) The entire enzyme viewed from the side (in the plane of the membrane). The F_1 portion has three α subunits and three β subunits arranged like the segments of an orange around a central shaft, the γ subunit (green). (Two α subunits and one β subunit have been omitted to reveal the γ subunit and the binding sites for ATP and ADP on the β subunits.) The δ subunit confers oligomycin sensitivity on the ATP synthase, and the ϵ subunit may serve to inhibit the enzyme's ATPase activity under some circumstances. The F_o subunit consists of one a subunit and two b subunits, which anchor the F_oF_1 complex in the membrane and act as a stator (the stationary part of a rotary system), holding the α and β subunits in place. F_o also includes the c ring, made up of a number (8 to 17, depending on the species) of identical c subunits, small, hydrophobic proteins. The c ring and the a subunit interact to provide a transmembrane path for protons. Each of the c subunits in F_o has a critical Asp residue near the middle of the membrane, which undergoes protonation/deprotonation during the catalytic cycle of the ATP synthase. Shown here is the homologous c_{11} ring of the Na^+ -ATPase of *Ilyobacter tartaricus*, for which the structure is well established. The Na^+ -binding sites, which correspond to the proton-binding sites of the F_oF_1 complex, are shown with their bound Na^+ ions (red spheres). (d) A view of F_o perpendicular to the membrane. As in (c), red spheres represent the Na^+ - or proton-binding sites in Asp residues. [(a) Information from W. Kühlbrandt and K. M. Davies, *Trends Biochem. Sci.* 41:106, 2016. (b, c, d) Data from F_1 : PDB ID 1BMF, J. P. Abrahams et al., *Nature* 370:621, 1994; PDB ID 1JNV, A. C. Hausrath et al., *J. Biol. Chem.* 276:47,227, 2001; PDB ID 2A7U, S. Wilkens et al., *Biochemistry* 44:11,786, 2005; PDB ID 2CLY, V. Kane Dickson et al., *EMBO J.* 25:2911, 2006; F_o : PDB ID 1B9U, O. Dmitriev et al., *J. Biol. Chem.* 274:15,598, 1999; c ring: PDB ID 1YCE, T. Meier et al., *Science* 308:659, 2005.]

The F_o complex, with its proton pore, is composed of three subunits, a, b, and c, in the proportion ab_2c_n , where n ranges from 8 to 17, depending on the species. Subunit c is a small (M_r 8,000), very hydrophobic polypeptide, consisting almost entirely of two transmembrane helices, with a small loop extending from the matrix side of the membrane. The crystal structure of the yeast F_oF_1 shows 10 c subunits, each with two

transmembrane helices roughly perpendicular to the plane of the membrane and arranged in two concentric circles to create the **c ring**. The inner circle is made up of the amino-terminal helices of each c subunit; the outer circle, about 55 Å in diameter, is made up of the carboxyl-terminal helices. *The c subunits in the c ring rotate together as a unit around an axis perpendicular to the membrane.* The ϵ and γ subunits of F_1 form a leg-and-foot that projects from the bottom (membrane) side of F_1 and stands firmly on the ring of c subunits. The a subunit consists of several hydrophobic helices that span the membrane in close association with one of the c subunits in the c ring.

Rotational Catalysis Is Key to the Binding-Change Mechanism for ATP Synthesis

On the basis of detailed kinetic and binding studies of the reactions catalyzed by F_0F_1 , Paul Boyer proposed a **rotational catalysis** mechanism in which the three active sites of F_1 take turns catalyzing ATP synthesis ([Fig. 19-26](#)). A given β subunit starts in the β -ADP conformation, which binds ADP and P_i from the surrounding medium. The subunit now changes conformation, assuming the β -ATP form that tightly binds and stabilizes ATP, bringing about the ready equilibration of $ADP + P_i$ with ATP on the enzyme surface. Finally, the subunit changes to the β -empty conformation, which has very low affinity for ATP, and the newly synthesized ATP leaves the enzyme surface.

Another round of catalysis begins when this subunit again assumes the β -ADP form and binds ADP and P_i .

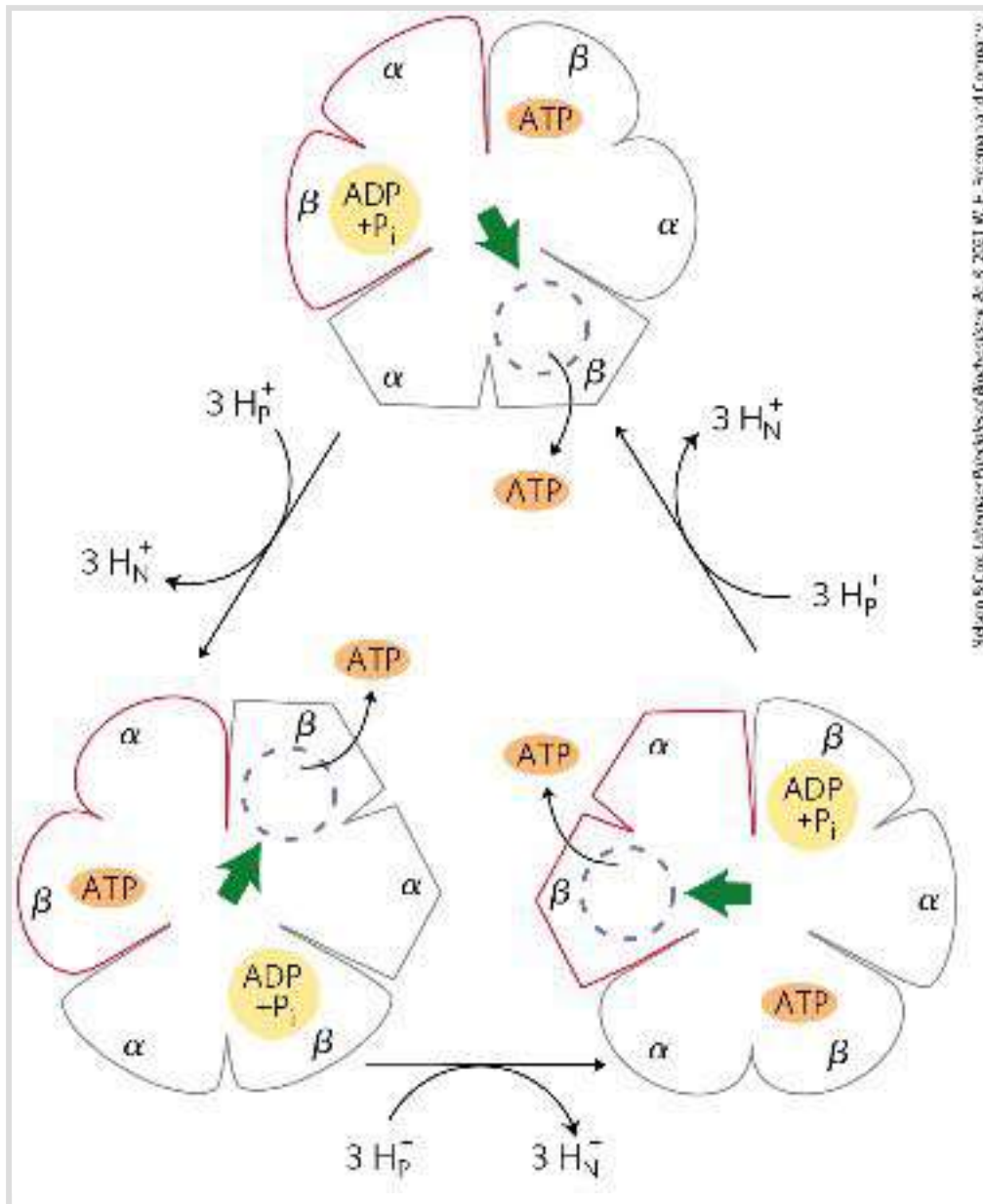


FIGURE 19-26 Binding-change model for ATP synthase. The F₁ complex has three nonequivalent adenine nucleotide-binding sites, one for each pair of α and β subunits. At any given moment, one of these sites is in the β -ATP conformation (which binds ATP tightly), a second is in the β -ADP (loose-binding) conformation, and a third is in the β -empty (very-loose-binding) conformation. In this view from the N side, the proton-motive force causes rotation of the central shaft — the γ subunit, shown as a green arrowhead — which comes into contact with each $\alpha\beta$ subunit pair in

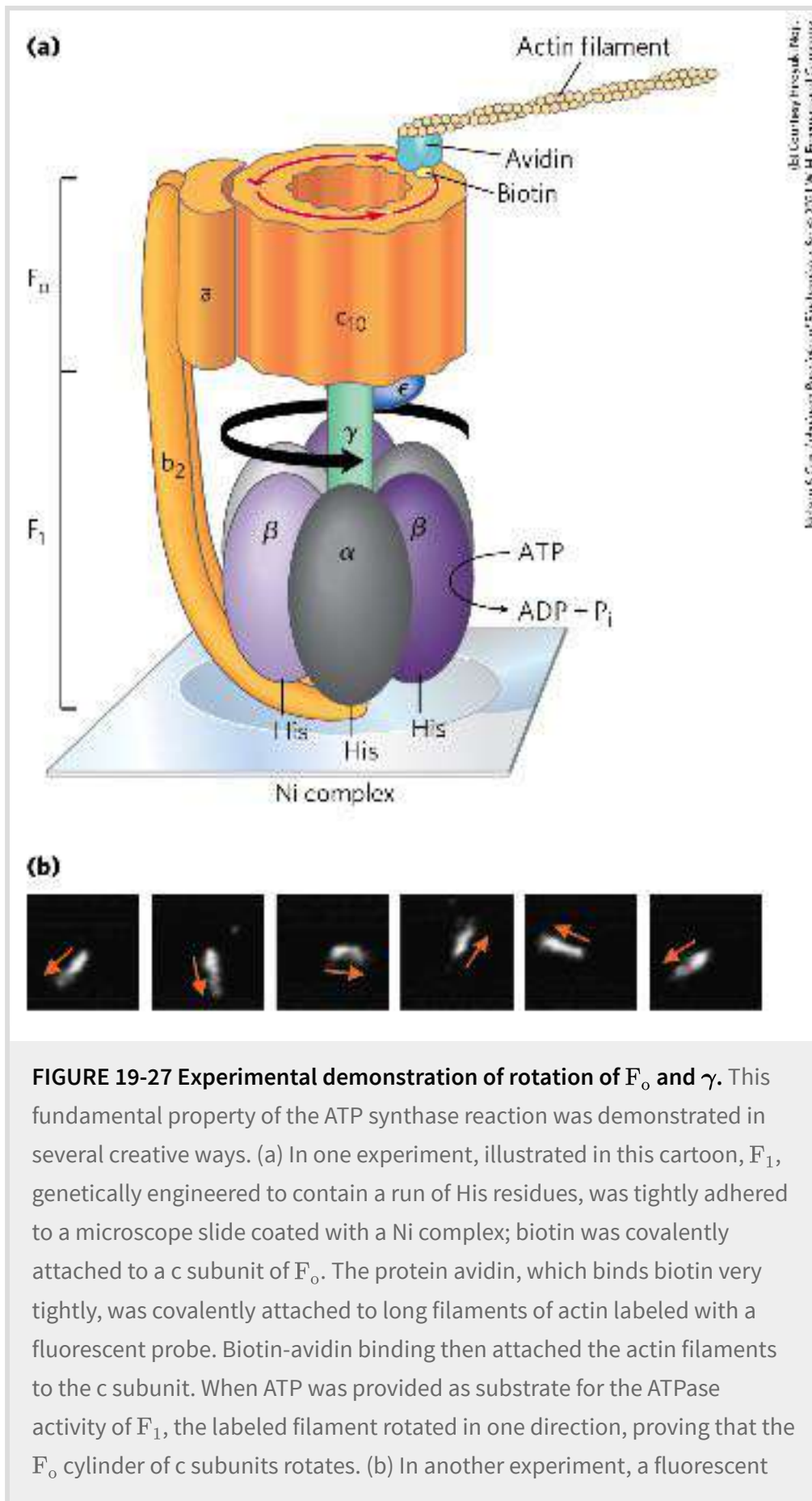
succession. This produces a cooperative conformational change in which the β -ATP site is converted to the β -empty conformation, and ATP dissociates; the β -ADP site is converted to the β -ATP conformation, which promotes condensation of bound $\text{ADP} + \text{P}_i$ to form ATP; and the β -empty site becomes a β -ADP site, which loosely binds $\text{ADP} + \text{P}_i$ entering from the solvent. Note that the direction of rotation reverses when the ATP synthase is acting as an ATPase, as in the experiment depicted in [Figure 19-27](#).



The conformational changes central to this mechanism are driven by the passage of protons through the F_o portion of ATP synthase. The streaming of protons through the F_o pore causes the c ring and the attached γ subunit to rotate about the long axis of γ , which is perpendicular to the plane of the membrane. The γ subunit passes through the center of the $\alpha_3\beta_3$ spheroid, which is held stationary relative to the membrane surface by the b_2 and δ subunits ([Fig. 19-25a](#)). With each rotation of 120° , γ comes into contact with a different β subunit, and the contact forces that β subunit into the β -empty conformation.

The three β subunits interact in such a way that when one assumes the β -empty conformation, its neighbor to one side *must* assume the β -ADP form, and the other neighbor the β -ATP form. Thus, one complete rotation of the γ subunit causes each β subunit to cycle through all three of its possible conformations, and for each rotation, three ATP are synthesized and released from the enzyme surface.

One strong prediction of this **binding-change model** is that the γ subunit should rotate in one direction when F_oF_1 is synthesizing ATP and in the opposite direction when the enzyme is hydrolyzing ATP. This prediction of rotation with ATP hydrolysis was confirmed in elegant experiments in the laboratories of Masasuke Yoshida and Kazuhiko Kinosita, Jr. The rotation of γ in a single F_1 molecule was observed microscopically by attaching a long, thin, fluorescent actin polymer to γ and watching it move relative to $\alpha_3\beta_3$ immobilized on a microscope slide as ATP was hydrolyzed. (The expected reversal of the rotation when ATP is being synthesized could not be tested in this experiment; there is no proton gradient to drive ATP synthesis.) When the entire F_oF_1 complex (not just F_1) was used in a similar experiment, the entire ring of c subunits rotated with γ ([Fig. 19-27](#)). The “shaft” rotated in the predicted direction through 360° . The rotation was not smooth but occurred in three discrete steps of 120° . As calculated from the known rate of ATP hydrolysis by one F_1 molecule and from the frictional drag on the long actin polymer, the efficiency of this mechanism in converting chemical energy into motion is close to 100%. It is, in Boyer’s words, “a splendid molecular machine!”



actin filament was attached directly to the γ subunit. The series of fluorescence micrographs (read left to right) shows the position of the actin filament at intervals of 133 ms. Note that as the filament rotated, it made discrete jumps rather than smooth rotation about the circle. The cylinder and shaft move as one unit. [(a) Information from Y. Sambongi et al., *Science* 286:1722, 1999.]



A model that illustrates how proton flow and rotary motion are coupled in the F_o complex is shown in [Figure 19-28](#). The a subunit is stationary, while the c ring rotates. Critical interactions occur between conserved amino acids in the a and c subunits. The individual subunits in the c ring are arranged in a circle with only a few in contact with the a subunit at any moment. Protons diffuse across the membrane through a path made up of both a and c subunits. Transient protonation of a key Glu residue in each c subunit elicits conformation changes that drive rotation and transmit protons between hydrophilic half channels positioned on each side of the membrane. The rotary movement of the c ring is made unidirectional by the large difference in proton concentration across the membrane. The number of protons that must be transferred to produce one complete rotation of the c ring is equal to the number of c subunits in the ring. Structural studies of the c ring have shown that the number of c subunits differs in different organisms ([Fig. 19-29](#)). In bovine mitochondria the number is 8, in yeast mitochondria and in *Escherichia coli* it is 10, and the number of c subunits can range as high as 17, as is seen in the soil bacterium *Burkholderia pseudomallei*. The rate of rotation in intact mitochondria has been estimated at about 6,000 rpm — 100 rotations per second.

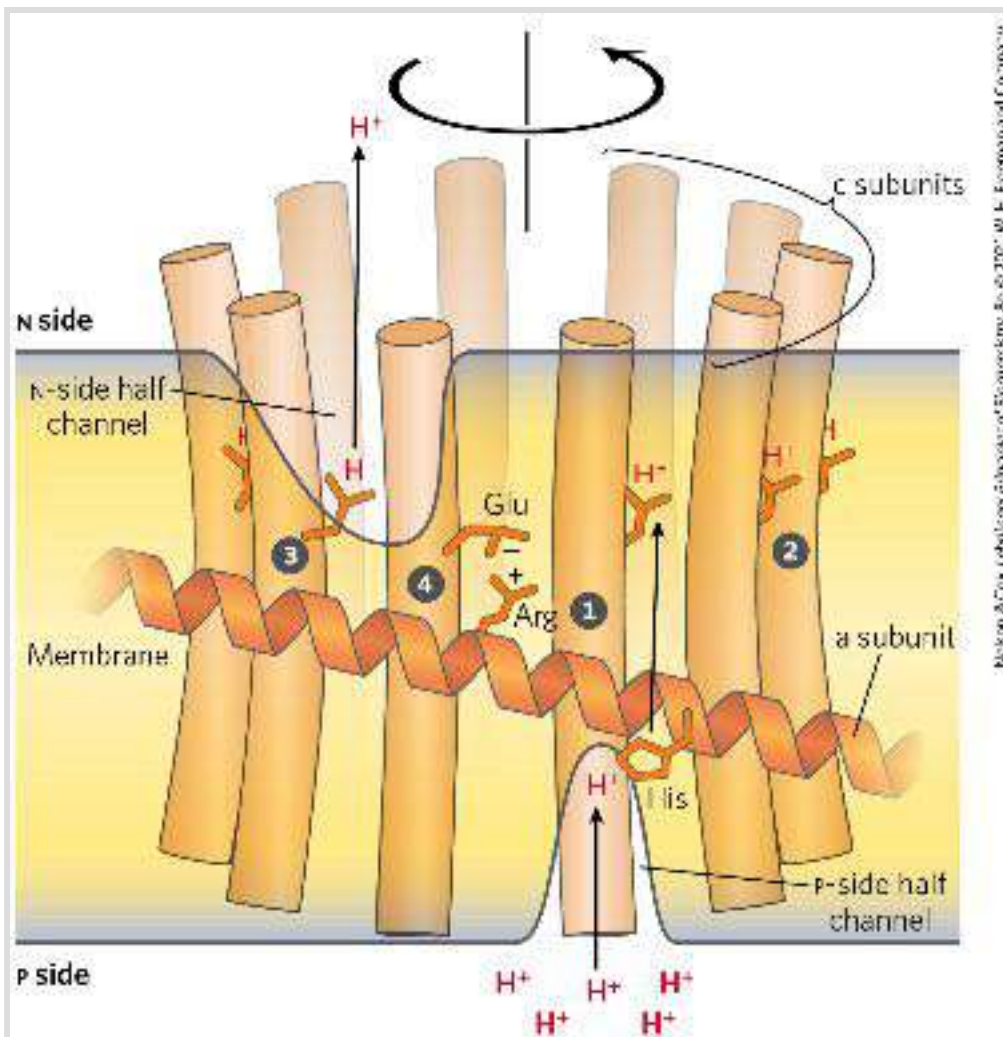


FIGURE 19-28 A model for proton-driven rotation of the c ring. The a subunit of the F_0 complex of the ATP synthase (see [Fig. 19-25a](#)) has two hydrophilic half-channels for protons, one leading from the P side to the middle of the membrane, the other leading from the middle of the membrane to the N side (matrix). The function of the stationary a subunit is to conduct protons to and from the c ring subunits to drive c ring rotational motion. Individual c subunits in F_0 (the total number varies from 8 to 17 in different species) are arranged in a circle about a central core. Each c subunit has a critical Glu residue (an Asp in some species) about midway across the membrane, with a perturbed pK_a allowing it to donate or accept a proton (red H^+) at pH near neutrality. The cycle that c subunits go through is illustrated. One c subunit **1** is initially positioned so that a proton that enters the half-channel on the P side (where the proton concentration is relatively high) encounters and protonates a conserved His residue in the a subunit, transferring it to the c subunit Glu residue. This triggers a rotation-facilitating conformational change in the protonated c

subunit as the Glu loses its negative charge. The now neutral Glu residue is sequestered in the hydrophobic membrane layer as it rotates as part of the c ring ❷. As the c ring rotates, the c subunit we are following eventually makes contact with the channel to the N side of the membrane, where the environment is relatively alkaline, and the proton is released ❸. As the Glu reacquires its negative charge, another rotation-facilitating conformational change occurs such that the Glu interacts transiently with a conserved Arg residue in the immobile a subunit ❹. The interaction with the Arg is disrupted as the Glu is again protonated by the His residue, in motions that again facilitate rotation. The c subunits positioned near the half channels are providing the rotational driving force at any given moment, as they are the ones undergoing conformation changes associated with protonation and deprotonation. The orientation of the proton gradient dictates the direction of proton flow and makes rotation of the c ring essentially unidirectional. [Information from W. Kühlbrandt and K. M. Davies, *Trends Biochem. Sci.* 41:106, 2016.]

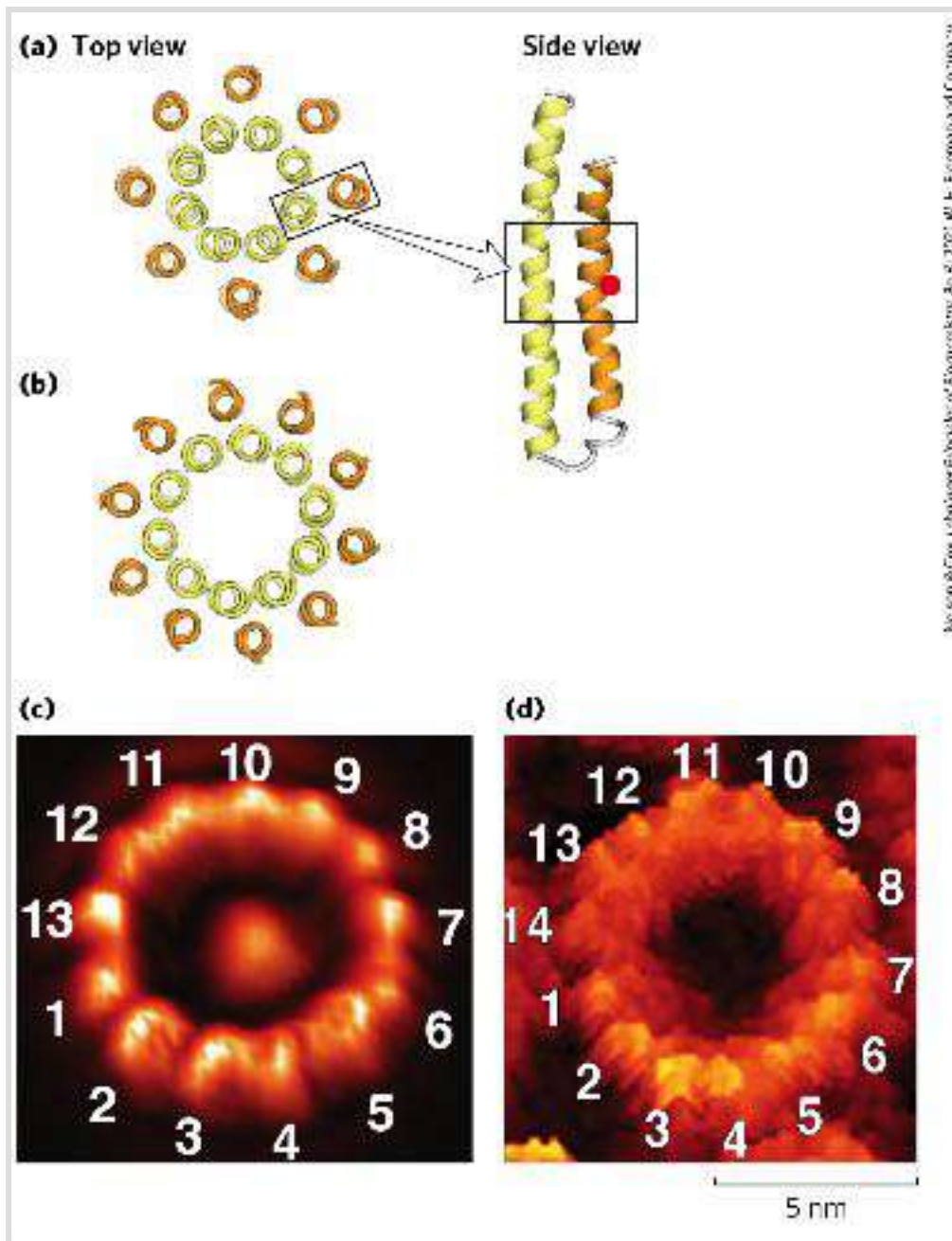
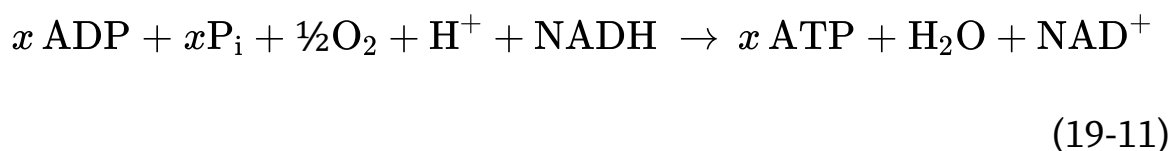


FIGURE 19-29 Species differences in number of c subunits in the c ring of the F_0 complex. The structures of the c rings from several species have been determined by x-ray crystallography. Each helix in the inner ring is half of a hairpin-shaped c subunit; the outer ring of helices forms the other half of the hairpin structure. The essential Glu residue (Asp in some species) is shown as a red dot. Views of the c ring perpendicular to the membrane show the number of c subunits for (a) bovine mitochondria (8 subunits) and (b) yeast mitochondria (10). Atomic force microscopy has been used to visualize the c rings of (c) a thermophilic bacterium, *Bacillus* species TA2.A1 (13 subunits), and (d) spinach (14). According to the model in [Figure 19-28](#),

different numbers of c subunits in the c ring should result in different ratios of ATP formed per pair of electrons passing through the respiratory chain (i.e., different P/O ratios). [(a) Data from PDB ID 1OHH, E. Cabezon et al., *Nat. Struct. Biol.* 10:744, 2003. (c) D. Matthies et al., *J. Mol. Biol.* 388:611, 2009. (d) H. Seelert et al., *Nature* 405:418, 2000. Republished with permission of Elsevier from *J. Mol. Biol.*, Matthies et al., Vol. 388(3), ©2009; permission conveyed through Copyright Clearance Center, Inc.]

Chemiosmotic Coupling Allows Nonintegral Stoichiometries of O₂ Consumption and ATP Synthesis

The overall reaction equation for ATP synthesis has the following form:



The value of x is sometimes called the **P/O ratio** or the **P/2e⁻ ratio**. When a proton gradient is coupled to ATP synthesis as described above, there is no theoretical requirement for P/O to be integral. The relevant questions about stoichiometry become these: How many protons are pumped outward by electron transfer from one NADH to O₂? and How many protons must flow inward through the F_oF₁ complex to drive the synthesis of one ATP? The measurement of proton fluxes is technically complicated; the investigator must take into account the

buffering capacity of mitochondria, nonproductive leakage of protons across the inner membrane, and use of the proton gradient for functions other than ATP synthesis, such as driving the transport of substrates across the inner mitochondrial membrane (described below). When NADH or succinate (which sends electrons into the respiratory chain at the level of ubiquinone) is the oxidizable substrate, the consensus experimental values for number of protons pumped out per pair of electrons are 10 and 6, respectively. The most widely accepted experimental value for number of protons required to drive the synthesis of an ATP molecule is 4, of which 1 is used in transporting P_i , ATP, and ADP across the mitochondrial membrane (see below). If 10 protons are pumped out per NADH and 4 must flow in to produce 1 ATP, the proton-based P/O ratio is 2.5 for NADH as the electron donor and 1.5 ($6/4$) for succinate. However, as we will see in [Worked Example 19-2](#), the proton stoichiometry of ATP synthesis by ATP synthase depends upon the number of c units in F_o , which ranges from 8 to 17, depending on the species.

WORKED EXAMPLE 19-2

Stoichiometry of ATP Production: Effect of c Ring Size

(a) If the ATP synthase of *bovine* mitochondria has 8 c subunits per c ring, what is the predicted ratio of ATP formed per NADH oxidized? (b) What is the predicted value for *yeast* mitochondria,

with 10 c subunits per ATP synthase? (c) What are the comparable values for electrons entering the respiratory chain from FADH_2 ?

SOLUTION:

(a) Here we are asked to determine how many ATP molecules are produced per NADH. This is another way of asking us to calculate the P/O ratio, or x , in [Equation 19-11](#). If the c ring has 8 c subunits, then one full rotation will transfer 8 protons to the matrix and produce 3 ATP molecules. But this synthesis also requires the transport of 3 P_i into the matrix, at a cost of 1 proton each, adding 3 more protons to the total number required. This brings the total cost to $(11 \text{ protons})/(3 \text{ ATP}) = 3.7 \text{ protons/ATP}$. The generally agreed value for the number of protons pumped out per pair of electrons transferred from NADH is 10 ([Eqn 19-7](#)). So, oxidizing 1 NADH produces

$$(10 \text{ protons})/(3.7 \text{ protons/ATP}) = 2.7 \text{ ATP}.$$

(b) If the c ring has 10 c subunits, then one full rotation will transfer 10 protons to the matrix and produce 3 ATP molecules. Adding in the 3 protons to transport the 3 P_i into the matrix brings the total cost to $(13 \text{ protons})/(3 \text{ ATP}) = 4.3 \text{ protons/ATP}$. Oxidizing 1 NADH produces

$$(10 \text{ protons})/(4.3 \text{ protons/ATP}) = 2.3 \text{ ATP}.$$

(c) When electrons enter the respiratory chain from FADH_2 (at ubiquinone), only 6 protons are available to drive ATP synthesis. This changes the calculation for bovine mitochondria to

$(6 \text{ protons}) / (3.7 \text{ protons/ATP}) = 1.6 \text{ ATP per pair of electrons}$ from FADH_2 . For yeast mitochondria, the calculation is $(6 \text{ protons}) / (4.3 \text{ protons/ATP}) = 1.4 \text{ ATP per pair of electrons}$ from FADH_2 .

These calculated values of x , or the P/O ratio, define a range that includes the experimental values of 2.5 ATP/NADH and 1.5 ATP/ FADH_2 , and we therefore use these values throughout this book.

The Proton-Motive Force Energizes Active Transport

Although the primary role of the proton gradient in mitochondria is to furnish energy for the synthesis of ATP, the proton-motive force also drives several transport processes essential to oxidative phosphorylation. The inner mitochondrial membrane is generally impermeable to charged species, but two specific systems transport ADP and P_i into the matrix and ATP out to the cytosol ([Fig. 19-30](#)).

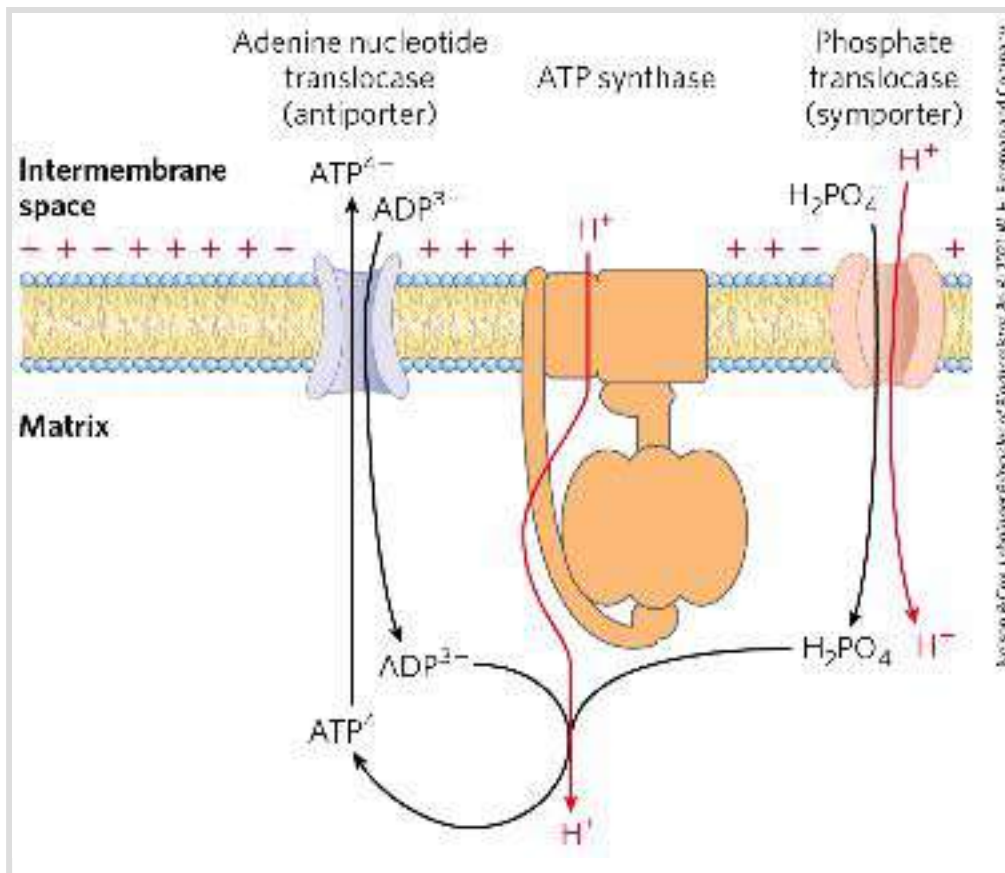


FIGURE 19-30 Adenine nucleotide and phosphate translocases. Transport systems of the inner mitochondrial membrane carry ADP and P_i into the matrix and newly synthesized ATP into the cytosol. The adenine nucleotide translocase is an antiporter; the same protein moves ADP into the matrix and ATP out. The effect of replacing ATP^{4-} with ADP^{3-} in the matrix is the net efflux of one negative charge, which is favored by the charge difference across the inner membrane (outside positive). At pH 7, P_i is present as both HPO_4^{2-} and $H_2PO_4^-$; the phosphate translocase is specific for $H_2PO_4^-$. There is no net flow of charge during symport of $H_2PO_4^-$ and H^+ , but the relatively low proton concentration in the matrix favors the inward movement of H^+ . Thus the proton-motive force is responsible both for providing the energy for ATP synthesis and for transporting substrates (ADP and P_i) into and product (ATP) out of the mitochondrial matrix. All three of these transport systems can be isolated as a single membrane-bound complex (ATP synthasome).

The **adenine nucleotide translocase**, integral to the inner membrane, binds ADP^{3-} in the intermembrane space and

transports it into the matrix in exchange for an ATP^{4-} molecule simultaneously transported outward (see [Fig. 13-11](#) for the ionic forms of ATP and ADP). Because this antiporter moves four negative charges out for every three moved in, its activity is favored by the transmembrane electrochemical gradient, which gives the matrix a net negative charge; the proton-motive force drives ATP-ADP exchange. Adenine nucleotide translocase is specifically inhibited by atractyloside, a toxic glycoside produced by a species of thistle. If the transport of ADP into and ATP out of mitochondria is inhibited, cytosolic ATP cannot be regenerated from ADP, explaining the toxicity of atractyloside.

A second membrane transport system essential to oxidative phosphorylation is the **phosphate translocase**, which promotes symport of one H_2PO_4^- and one H^+ into the matrix. This transport process, too, is favored by the transmembrane proton gradient ([Fig. 19-30](#)). Notice that the process requires movement of one proton from the P side to the N side of the inner membrane, consuming some of the energy of electron transfer. A complex of the ATP synthase and both translocases, the **ATP synthasome**, can be isolated from mitochondria by gentle dissection with detergents, suggesting that the functions of these three proteins are very tightly integrated.

ATP and ADP cross the outer mitochondrial membrane via the voltage-dependent anion channel (VDAC), a 19-stranded β barrel with an opening about 27 Å wide, connecting the cytosol and the intermembrane space. Each VDAC, when open, can move 10^5 ATP

molecules per second. The opening is gated by voltage, as its name indicates, and under some conditions VDAC is closed to ATP.

Shuttle Systems Indirectly Convey Cytosolic NADH into Mitochondria for Oxidation

The NADH dehydrogenase of the inner mitochondrial membrane of animal cells can accept electrons only from NADH in the matrix. Given that the inner membrane is not permeable to NADH, how can the NADH generated by glycolysis in the cytosol be reoxidized to NAD^+ by O_2 via the respiratory chain? Special shuttle systems carry reducing equivalents from cytosolic NADH into mitochondria by an indirect route. The most active NADH shuttle, which functions in liver, kidney, and heart mitochondria, is the **malate-aspartate shuttle** ([Fig. 19-31](#)). The reducing equivalents of cytosolic NADH are first transferred to cytosolic oxaloacetate to yield malate, catalyzed by cytosolic malate dehydrogenase. The malate thus formed passes through the inner membrane via the malate- α -ketoglutarate transporter. Within the matrix, the reducing equivalents are passed to NAD^+ by the action of matrix malate dehydrogenase, forming NADH; this NADH can pass electrons directly to the respiratory chain. About 2.5 molecules of ATP are generated as this pair of electrons passes to O_2 . Cytosolic oxaloacetate must be regenerated by

transamination reactions and the activity of membrane transporters to start another cycle of the shuttle.

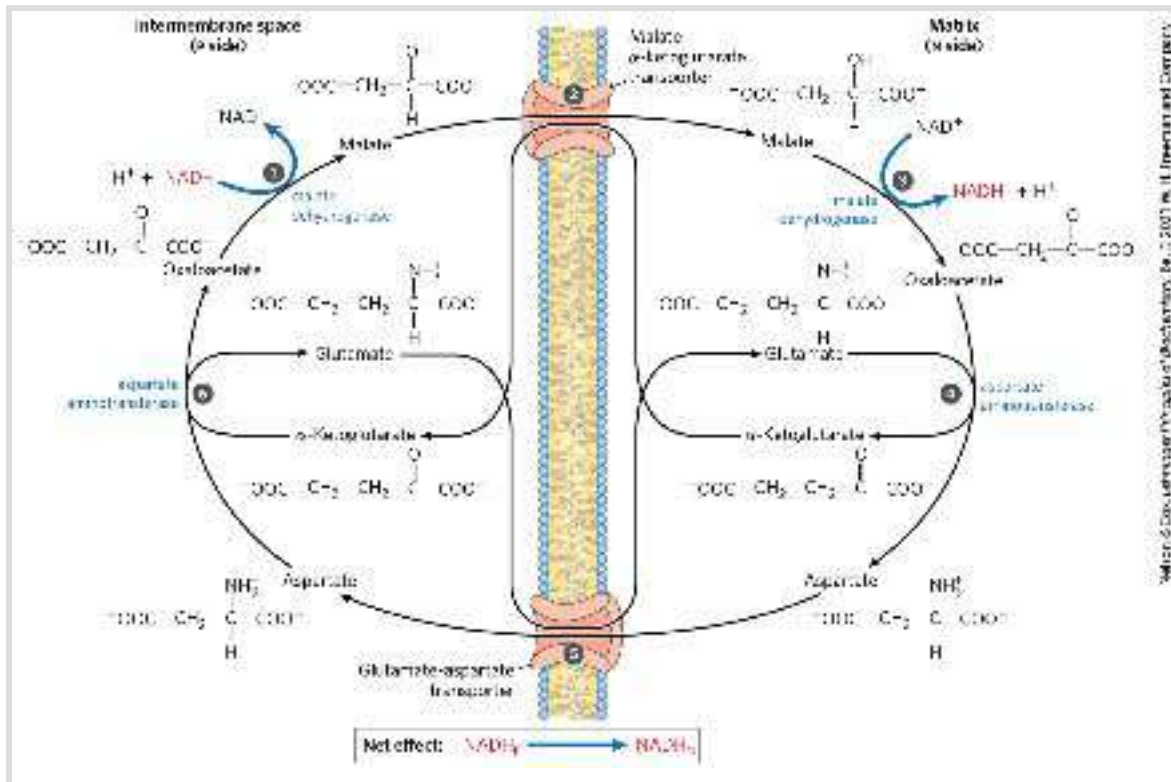


FIGURE 19-31 Malate-aspartate shuttle. This shuttle for transporting reducing equivalents from cytosolic NADH into the mitochondrial matrix is used in liver, kidney, and heart. **1** NADH in the cytosol enters the intermembrane space through openings in the outer membrane (porins), then passes two reducing equivalents to oxaloacetate, producing malate. **2** Malate crosses the inner membrane via the malate- α -ketoglutarate transporter. **3** In the matrix, malate passes two reducing equivalents to NAD⁺, and the resulting NADH is oxidized by the respiratory chain; the oxaloacetate formed from malate cannot pass directly into the cytosol. **4** Oxaloacetate is first transaminated to aspartate, and **5** aspartate can leave via the glutamate-aspartate transporter. **6** Oxaloacetate is regenerated in the cytosol, completing the cycle, and glutamate produced in the same reaction enters the matrix via the glutamate-aspartate transporter.

Skeletal muscle and brain use a different NADH shuttle, the **glycerol 3-phosphate shuttle** ([Fig. 19-32](#)). It differs from the malate-aspartate shuttle in that it delivers the reducing

equivalents from NADH through FAD in glycerol 3-phosphate dehydrogenase to ubiquinone and thus into Complex III, not Complex I ([Fig. 19-15](#)), providing enough energy to synthesize only 1.5 ATP molecules per pair of electrons.

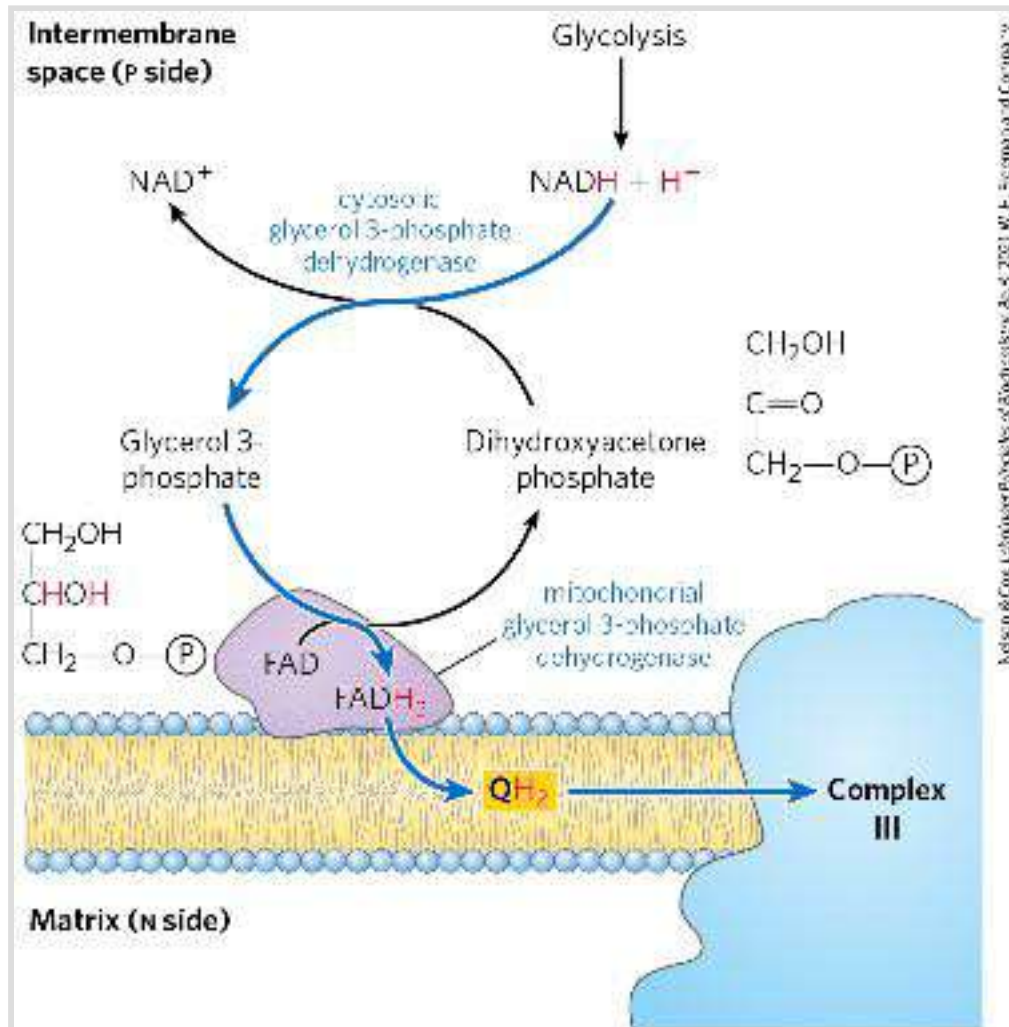


FIGURE 19-32 Glycerol 3-phosphate shuttle. This alternative means of moving reducing equivalents from the cytosol to the respiratory chain operates in skeletal muscle and the brain. In the cytosol, dihydroxyacetone phosphate accepts two reducing equivalents from NADH in a reaction catalyzed by cytosolic glycerol 3-phosphate dehydrogenase. An isozyme of glycerol 3-phosphate dehydrogenase bound to the outer face of the inner membrane then transfers two reducing equivalents from glycerol 3-phosphate in the intermembrane space to ubiquinone. Note that this shuttle does not involve membrane transport systems.

The mitochondria of plants have an *externally* oriented NADH dehydrogenase that can transfer electrons directly from cytosolic NADH into the respiratory chain at the level of ubiquinone. Because this pathway bypasses the NADH dehydrogenase of Complex I and the associated proton movement, the yield of ATP from cytosolic NADH is less than that from NADH generated in the matrix ([Box 19-1](#)).

BOX 19-1

Hot, Stinking Plants and Alternative Respiratory Pathways

Many flowering plants attract insect pollinators by releasing odorant molecules that mimic an insect's natural food sources or potential egg-laying sites. Plants pollinated by flies or beetles that normally feed on or lay their eggs in dung or carrion sometimes use foul-smelling compounds to attract these insects.

One family of stinking plants is the Araceae, which includes philodendrons, arum lilies, and skunk cabbages. These plants have tiny flowers densely packed on an erect structure, the spadix, surrounded by a modified leaf, the spathe. The spadix releases odors of rotting flesh or dung. Before pollination the spadix also heats up, in some species to as much as 20 to 40 °C above ambient temperatures. Heat production (thermogenesis) helps evaporate odorant molecules for better dispersal, and because rotting flesh and dung are usually warm from the hyperactive metabolism of scavenging microbes, the heat itself might also attract insects. In the case of the eastern skunk cabbage ([Fig. 1](#)), which flowers in late winter or early spring when snow still covers the ground, thermogenesis allows the spadix to grow up through the snow.



FIGURE 1 Eastern skunk cabbage.

How does a skunk cabbage heat its spadix? The mitochondria of plants, fungi, and unicellular eukaryotes have respiratory chains that are essentially the same as those in animals, but they also have an alternative respiratory pathway. A QH_2 oxidase transfers electrons from the ubiquinone pool directly to oxygen, bypassing the two proton-translocating steps of Complexes III and IV ([Fig. 2](#)). Energy that might have been conserved as ATP is instead released as heat. Plant mitochondria also have an alternative NADH dehydrogenase, insensitive to the Complex I inhibitor rotenone (see [Table 19-4](#)), that transfers electrons from NADH in the matrix directly to ubiquinone, bypassing Complex I and its associated proton pumping. And plant mitochondria have yet another NADH dehydrogenase, on the external face of the inner membrane, that transfers electrons from NADPH or NADH in the intermembrane space to ubiquinone, again bypassing Complex I. Thus when electrons enter the alternative respiratory pathway through the rotenone-insensitive NADH dehydrogenase, the external NADH dehydrogenase, or succinate dehydrogenase (Complex II), and pass to O_2 via the cyanide-resistant alternative oxidase, energy is not conserved as ATP but is released as heat. A

skunk cabbage can use the heat to melt snow, produce a foul stench, or attract beetles or flies.

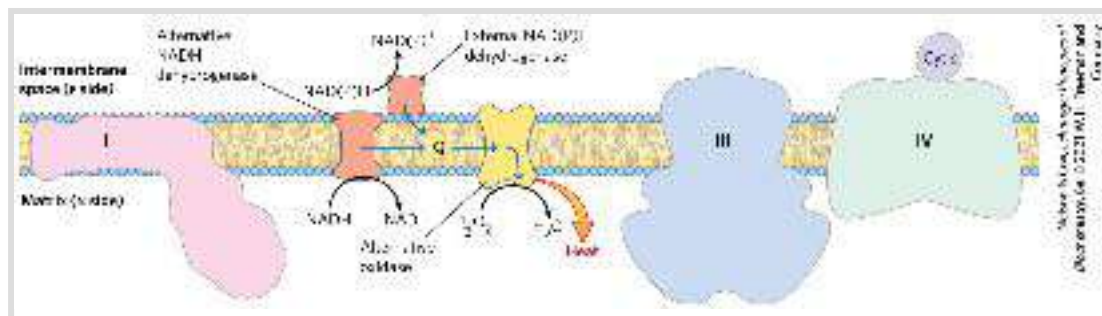


FIGURE 2 Electron carriers of the inner membrane of plant mitochondria. Electrons can flow through Complexes I, III, and IV, as in animal mitochondria, or through plant-specific alternative carriers by the paths shown with blue arrows.

SUMMARY 19.2 *ATP Synthesis*

- The chemiosmotic theory describes the coupling of ATP synthesis to an electrochemical proton gradient. The flow of electrons through Complexes I, III, and IV results in pumping of protons across the inner mitochondrial membrane, making the matrix alkaline relative to the intermembrane space. This proton gradient provides the energy, in the form of the proton-motive force, for ATP synthesis from ADP and P_i .
- ATP synthase has two major components, called F_o and F_1 . Both components have multiple subunits. The overall complex spans the inner mitochondrial membrane.
- ATP synthesis is reversible within the active site on the β subunits of the F_1 complex. Very tight binding to ATP offsets the negative ΔG for ATP hydrolysis in solution.
- Release of ATP from ATP synthase is promoted by the transmembrane proton gradient.

- The subunits of the F_1 complex cycle from (ADP + P_i)-bound to ATP-bound to empty conformations.
- ATP synthase carries out “rotational catalysis,” in which the flow of protons through F_o causes the c ring to rotate and in turn trigger the subunit conformational changes in F_1 .
- The ratio of ATP synthesized per $\frac{1}{2}O_2$ reduced to H_2O (the P/O ratio) is about 2.5 when electrons enter the respiratory chain at Complex I, and 1.5 when electrons enter at ubiquinone. This ratio varies among species, depending on the number of c subunits in the F_o complex.
- Energy conserved in a proton gradient can drive solute transport uphill across a membrane.
- The inner mitochondrial membrane is impermeable to NADH and NAD^+ , but NADH equivalents are moved from the cytosol to the matrix by either of two shuttles. NADH equivalents moved in by the malate-aspartate shuttle enter the respiratory chain at Complex I and yield a P/O ratio of 2.5; those moved in by the glycerol 3-phosphate shuttle enter at ubiquinone and give a P/O ratio of 1.5.

19.3 Regulation of Oxidative Phosphorylation

Oxidative phosphorylation produces most of the ATP made in aerobic cells. Complete oxidation of a molecule of glucose to CO_2 yields 30 or 32 ATP ([Table 19-5](#)). By comparison, glycolysis under anaerobic conditions (lactate fermentation) yields only 2 ATP per glucose. Clearly, the evolution of oxidative phosphorylation provided a tremendous increase in the energy efficiency of catabolism. Complete oxidation to CO_2 of the coenzyme A derivative of palmitate (16:0), which also occurs in the mitochondrial matrix, yields 108 ATP per palmitoyl-CoA (see [Table 17-1](#)). A similar calculation can be made for the ATP yield from oxidation of each of the amino acids ([Chapter 18](#)). Aerobic oxidative pathways that result in electron transfer to O_2 accompanied by oxidative phosphorylation therefore account for the vast majority of the ATP produced in catabolism, so the regulation of ATP production by oxidative phosphorylation to match the cell's fluctuating needs for ATP is absolutely essential.

TABLE 19-5 ATP Yield from Complete Oxidation of Glucose

Process	Direct product	Final ATP
Glycolysis	2 NADH (cytosolic)	3 or 5 ^a
	2 ATP	2
Pyruvate oxidation (two per glucose)	2 NADH (mitochondrial matrix)	5

Acetyl-CoA oxidation in citric acid cycle (two per glucose)	6 NADH (mitochondrial matrix)	15
	2 FADH ₂	3
	2 ATP or 2 GTP	2
Total yield per glucose		30 or 32

[a](#)If the malate/aspartate shuttle is used to transfer reducing equivalents into the mitochondrion, the yield is 5 ATP. If the glycerol 3-phosphate shuttle is used, the yield is 3 ATP.

Oxidative Phosphorylation Is Regulated by Cellular Energy Needs

The rate of respiration (O₂ consumption) in mitochondria is generally limited by the availability of ADP as a substrate for phosphorylation. Dependence of the rate of O₂ consumption on the availability of the P_i acceptor, ADP ([Fig. 19-20b](#)), the **acceptor control** of respiration, can be remarkable. In some animal tissues, the **acceptor control ratio**, the ratio of the maximal rate of ADP-induced O₂ consumption to the basal rate in the absence of ADP, is at least 10.

The intracellular concentration of ADP is one measure of the energy status of cells. Another, related measure is the **mass-action ratio** of the ATP-ADP system, $[ATP]/([ADP][P_i])$. Usually this ratio is very high, so the ATP-ADP system is almost fully

phosphorylated. When the rate of some energy-requiring process (protein synthesis, for example) increases, the rate of breakdown of ATP to ADP and P_i increases, lowering the mass-action ratio. With more ADP available for oxidative phosphorylation, the rate of respiration increases, causing regeneration of ATP. This continues until the mass-action ratio returns to its normal high level, at which point respiration slows again. The rate of oxidation of cellular fuels is regulated with such sensitivity and precision that the $[ATP]/([ADP][P_i])$ ratio fluctuates only slightly in most tissues, even during extreme variations in energy demand. In short, ATP is formed only as fast as it is used in energy-requiring cellular activities.

An Inhibitory Protein Prevents ATP Hydrolysis during Hypoxia

We have already encountered ATP synthase as an ATP-driven proton pump (see [Fig. 11-40](#)), catalyzing the reverse of ATP synthesis under some experimental conditions. When a cell is hypoxic (deprived of oxygen), as in a heart attack or a stroke, electron transfer to oxygen slows, and so does the pumping of protons. The proton-motive force soon collapses.  Under these conditions, the ATP synthase could operate in reverse, hydrolyzing ATP made by glycolysis to pump protons outward and causing a disastrous drop in ATP levels. This is prevented by a small (84 amino acids) protein inhibitor, IF_1 . IF_1 simultaneously binds to two ATP synthase molecules, inhibiting the enzyme's

Hypoxia Leads to ROS Production and Several Adaptive Responses

In hypoxic cells there is an imbalance between the input of electrons from fuel oxidation in the mitochondrial matrix and transfer of electrons to molecular oxygen, leading to increased formation of reactive oxygen species. In addition to the glutathione peroxidase system ([Fig. 19-18](#)), cells have two other lines of defense against ROS ([Fig. 19-34](#)). One is regulation of pyruvate dehydrogenase (PDH), the enzyme that delivers acetyl-CoA to the citric acid cycle ([Chapter 16](#)). Under hypoxic conditions, PDH kinase phosphorylates mitochondrial PDH, inactivating it and slowing the delivery of FADH_2 and NADH from the citric acid cycle to the respiratory chain. A second means of preventing ROS formation is the replacement of one subunit of Complex IV, known as COX4-1, with another subunit, COX4-2, that is better suited to hypoxic conditions. With COX4-1, the catalytic properties of Complex IV are optimal for respiration at normal oxygen concentrations; with COX4-2, Complex IV is optimized for operation under hypoxic conditions.

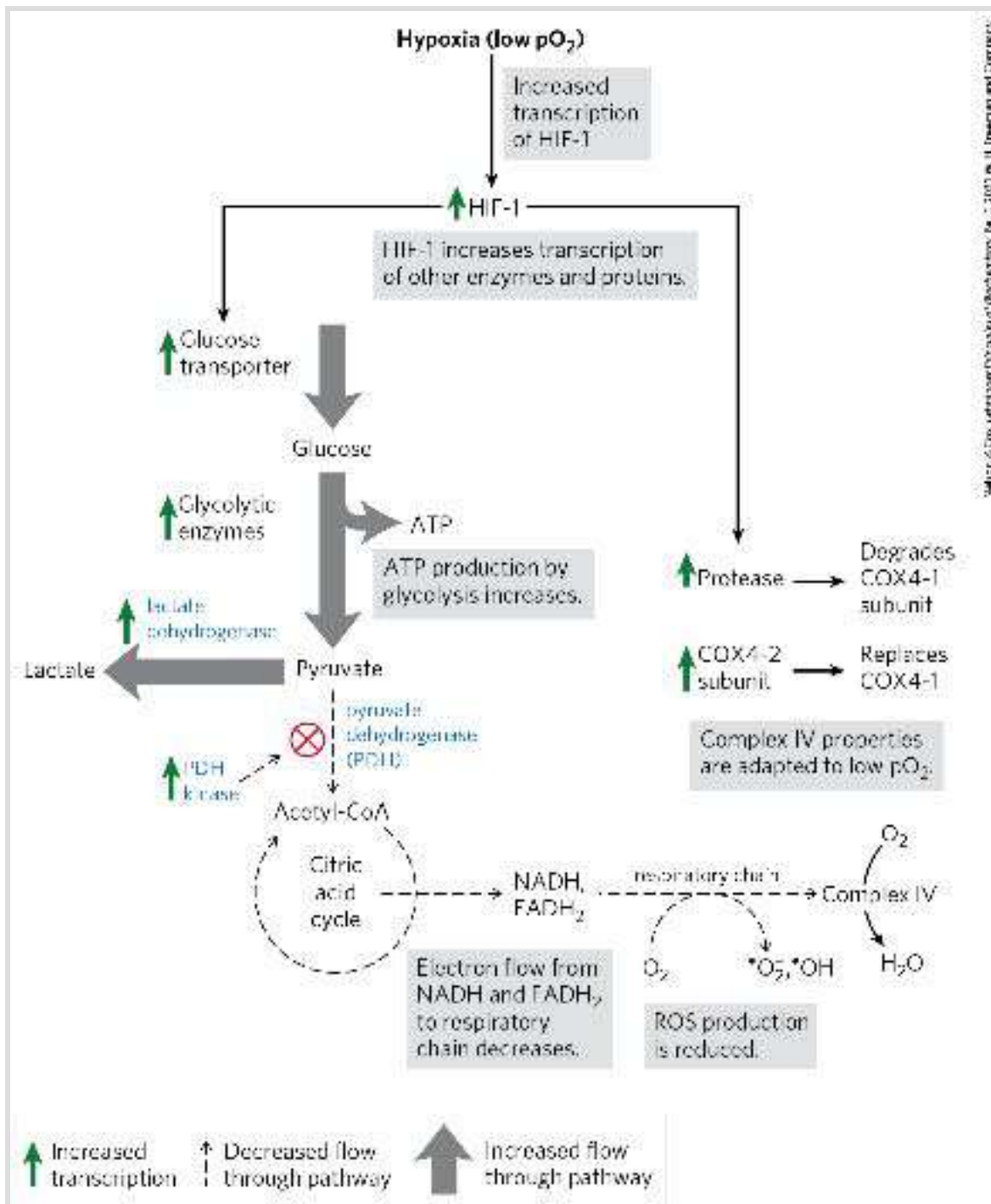


FIGURE 19-34 Regulation of gene expression by hypoxia-inducible factor (HIF-1) to reduce ROS formation. Under conditions of low oxygen (hypoxia), HIF-1 is synthesized in greater amounts and acts as a transcription factor, increasing synthesis of the glucose transporter, glycolytic enzymes, pyruvate dehydrogenase kinase (PDH kinase), lactate dehydrogenase, a protease that degrades the cytochrome oxidase subunit COX4-1, and cytochrome oxidase subunit COX4-2. These changes counter the formation of ROS by decreasing the supply of NADH and FADH₂ and making cytochrome oxidase of

Complex IV more effective. [Information from D. A. Harris, *Bioenergetics at a Glance*, p. 36, Blackwell Science, 1995.]

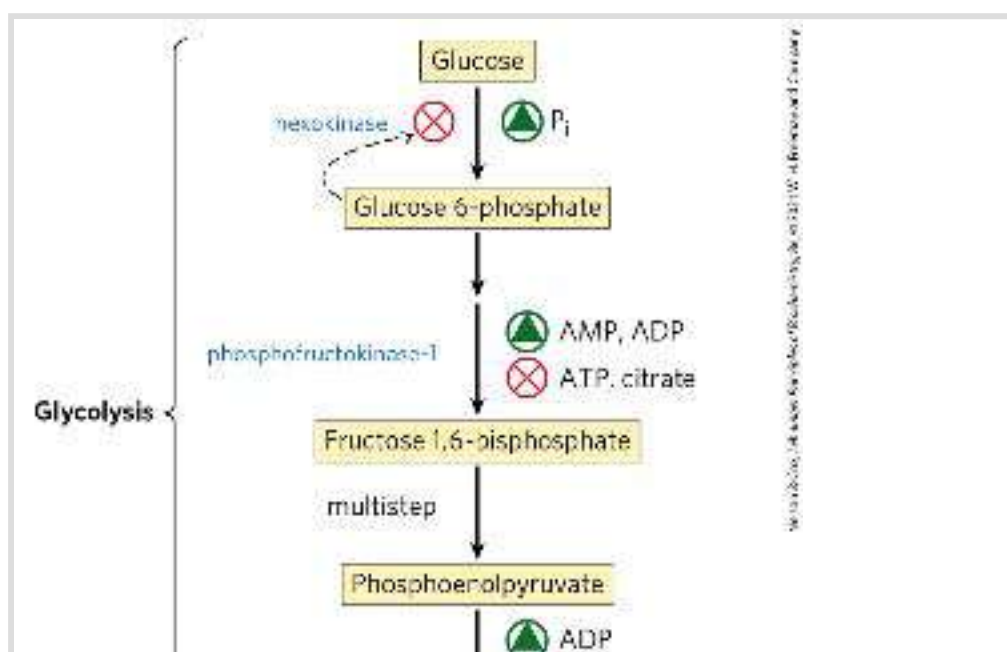
The changes in PDH activity and the COX4-2 content of Complex IV are both mediated by HIF-1, the hypoxia-inducible factor. HIF-1 (another intrinsically disordered protein) accumulates in hypoxic cells and, acting as a transcription factor, triggers increased synthesis of PDH kinase, COX4-2, and a protease that degrades COX4-1. HIF-1 is a master regulator of O₂ homeostasis. Recall that it also mediates the changes in glucose transport and glycolytic enzymes that produce the Warburg effect, the dependence on glycolysis (not mitochondrial respiration) for ATP production, even in the presence of sufficient oxygen (see [Box 14-1](#)).



When these mechanisms for dealing with ROS are insufficient, either due to genetic mutation affecting one of the protective proteins or under conditions of very high rates of ROS production, mitochondrial function is compromised. Mitochondrial damage is thought to be involved in aging, heart failure, certain rare cases of diabetes (described below), and several maternally inherited genetic diseases that affect the nervous system. ■

ATP-Producing Pathways Are Coordinately Regulated

The major catabolic pathways have overlapping and concerted regulatory mechanisms that allow them to function together in an economical and self-regulating manner to produce ATP and biosynthetic precursors. The relative concentrations of ATP and ADP control not only the rates of electron transfer and oxidative phosphorylation but also the rates of the citric acid cycle, pyruvate oxidation, and glycolysis ([Fig. 19-35](#)). Whenever ATP consumption increases, the rate of electron transfer and oxidative phosphorylation increases. Simultaneously, the rate of pyruvate oxidation via the citric acid cycle increases, increasing the flow of electrons into the respiratory chain. These events, in turn, can evoke an increased rate of glycolysis, increasing the rate of pyruvate formation. When conversion of ADP to ATP lowers the ADP concentration, acceptor control slows electron transfer and thus oxidative phosphorylation. Glycolysis and the citric acid cycle are also slowed, because ATP is an allosteric inhibitor of the glycolytic enzyme phosphofructokinase-1 (see [Fig. 14-23](#)) and of pyruvate dehydrogenase (see [Fig. 16-18](#)).



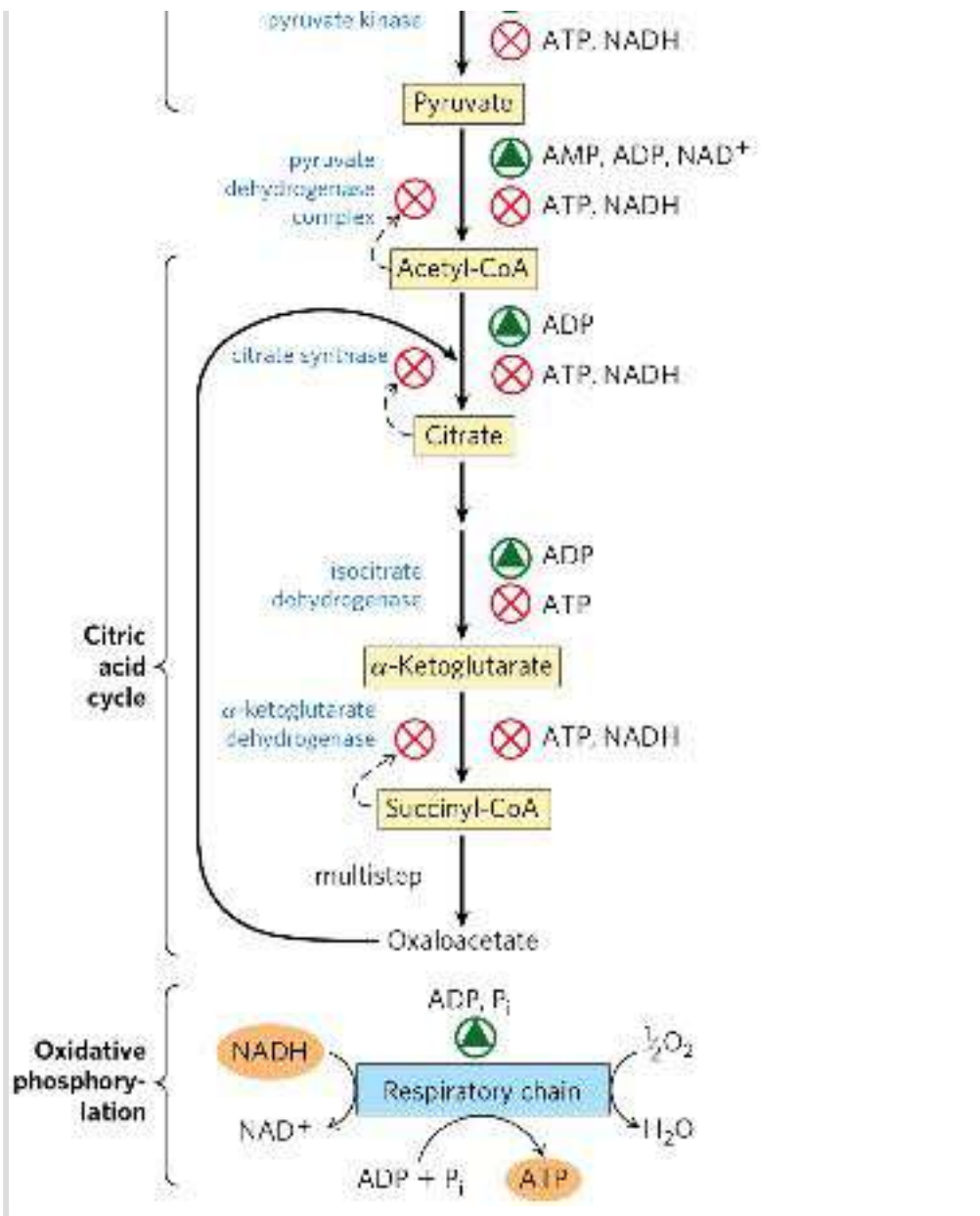


FIGURE 19-35 Regulation of ATP-producing pathways. This diagram shows the coordinated regulation of glycolysis, pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation by the relative concentrations of ATP, ADP, and AMP, and by NADH. High [ATP] (or low [ADP] and [AMP]) produces low rates of glycolysis, pyruvate oxidation, acetate oxidation via the citric acid cycle, and oxidative phosphorylation. All four pathways are accelerated when the use of ATP and the formation of ADP, AMP, and P_i increase. The ability of citrate to inhibit both glycolysis and the citric acid cycle reinforces the action of the adenine nucleotide system. In addition, increased [NADH] and [acetyl-CoA] also inhibit the oxidation of pyruvate to acetyl-CoA, and a high [NADH]/[NAD⁺] ratio inhibits the dehydrogenase reactions of the citric acid cycle (see [Fig. 16-18](#)).

Phosphofructokinase-1 is also inhibited by citrate, the first intermediate of the citric acid cycle. When the cycle is “idling,” citrate accumulates within mitochondria, then is transported into the cytosol. When the cytosolic concentrations of both ATP and citrate rise, they produce a concerted allosteric inhibition of phosphofructokinase-1 that is greater than the sum of their individual effects, slowing glycolysis.

SUMMARY 19.3 *Regulation of Oxidative Phosphorylation*

- Oxidative phosphorylation is regulated by cellular energy demands. Intracellular $[ADP]$ and the mass-action ratio $[ATP]/([ADP][P_i])$ are measures of a cell's energy status.
- In hypoxic (oxygen-deprived) cells, a protein inhibitor blocks ATP hydrolysis by the reverse activity of ATP synthase, preventing a drastic drop in $[ATP]$.
- The adaptive responses to hypoxia, mediated by HIF-1, slow electron transfer into the respiratory chain and modify Complex IV to act more efficiently under low-oxygen conditions.
- ATP and ADP concentrations set the rate of electron transfer through the respiratory chain via a series of coordinated controls on respiration, glycolysis, and the citric acid cycle.

19.4 Mitochondria in Thermogenesis, Steroid Synthesis, and Apoptosis

Although ATP production is a central role for the mitochondrion, this organelle has other functions that, in specific tissues or under specific circumstances, are also crucial. In adipose tissue, mitochondria generate heat to protect vital organs from low ambient temperature; in the adrenal glands and the gonads, mitochondria are the sites of steroid hormone synthesis; and in most or all tissues, they are key participants in apoptosis (programmed cell death).

Uncoupled Mitochondria in Brown Adipose Tissue Produce Heat

We noted above that respiration slows when the cell is adequately supplied with ATP. There is a remarkable and instructive exception to this general rule. Most newborn mammals, including humans, have a type of adipose tissue called [brown adipose tissue \(BAT\)](#) (BAT; p. 852), in which  fuel oxidation serves not to produce ATP but to generate heat to keep the newborn warm. This specialized adipose tissue is brown because of the presence of large numbers of mitochondria and thus high concentrations of cytochromes, with heme groups that are strong absorbers of visible light.

There are at least two mechanisms of thermogenesis. The mitochondria of brown adipocytes are much like those of other mammalian cells, except in having a unique protein in their inner membrane. **Uncoupling protein 1 (UCP1)**, a long-chain fatty acid/ H^+ symporter, provides a path for protons to return to the matrix without passing through the F_0F_1 complex (**Fig. 19-36**).

P5 As a result of this short-circuiting of protons, the energy of oxidation is not conserved by ATP formation but is dissipated as heat, which contributes to maintaining body temperature. However, UCP1 is only part of the story, and heat generation occurs in mammals even when it is absent. The thermogenic action of UCP1 is supplemented by a futile cycle involving creatine and phosphocreatine (**Fig. 19-36**). ATP synthesized in the mitochondrial matrix is exported to the intermembrane space via the adenine nucleotide translocator, an ADP/ATP antiporter. There, the ATP is used to phosphorylate creatine to create phosphocreatine and ADP. The ADP is transported back into the matrix. Hydrolysis of phosphocreatine completes a futile cycle that liberates heat.

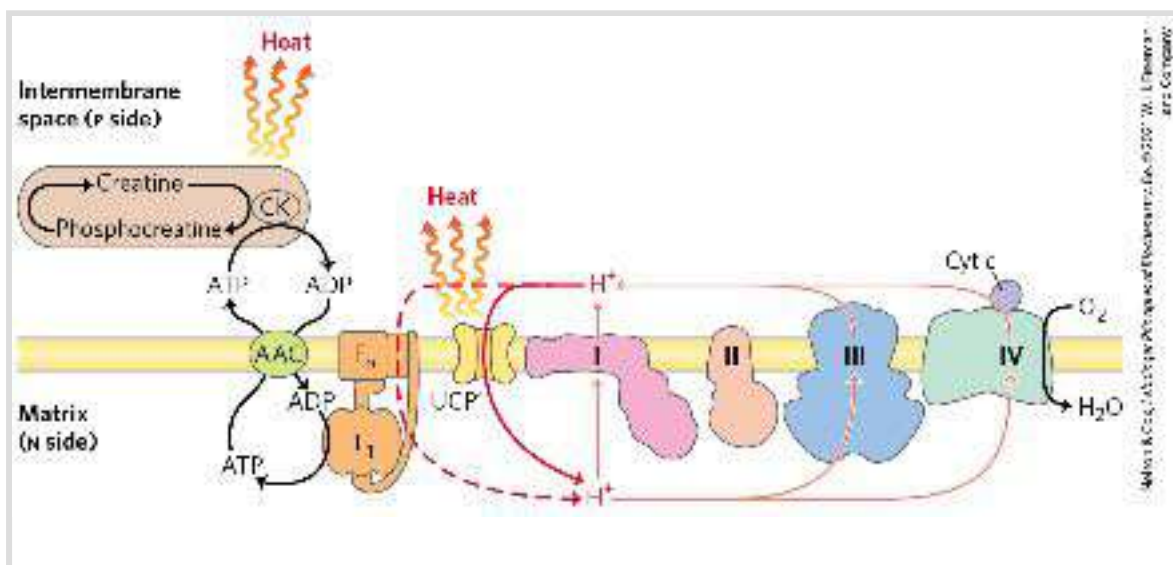
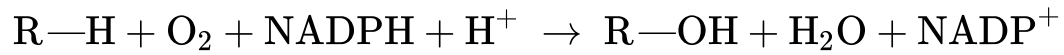


FIGURE 19-36 Two mechanisms of thermogenesis in mitochondria. UCP1, an uncoupling protein in the mitochondria of brown adipose tissue, causes the energy conserved by proton pumping to be dissipated as heat by providing an alternative route for protons to reenter the mitochondrial matrix. A futile cycle in which creatine is phosphorylated by creatine kinase (CK), using ATP and producing ADP transported by the ATP/ADP carrier (AAC), also generates heat.

Hibernating animals also depend on the activity of uncoupled BAT mitochondria to generate heat during their long dormancy (see [Box 17-1](#)). We will return to the role of UCP1 when we discuss the regulation of body mass in [Chapter 23 \(pp. 867–869\)](#).

Mitochondrial P-450 Monooxygenases Catalyze Steroid Hydroxylations

 Mitochondria are the site of biosynthetic reactions that produce steroid hormones, including the sex hormones, glucocorticoids, mineralocorticoids, and vitamin D hormone. These compounds are synthesized from cholesterol or a related sterol in a series of hydroxylations catalyzed by enzymes of the [cytochrome P-450](#) family (see [Box 21-1](#)), all of which have a critical heme group (its absorption at 450 nm gives this family its name). In the hydroxylation reactions, one atom of molecular oxygen is incorporated into the substrate and the second is reduced to H₂O, making cytochrome P-450 enzymes monooxygenases:



In this reaction, two species are oxidized: NADPH and R—H.

There are dozens of P-450 enzymes, all situated in the inner mitochondrial membrane with their catalytic site exposed to the matrix. Steroidogenic cells are packed with mitochondria specialized for steroid synthesis; the mitochondria are generally larger than those in other tissues and have more extensive and highly convoluted inner membranes ([Fig. 19-37](#)).

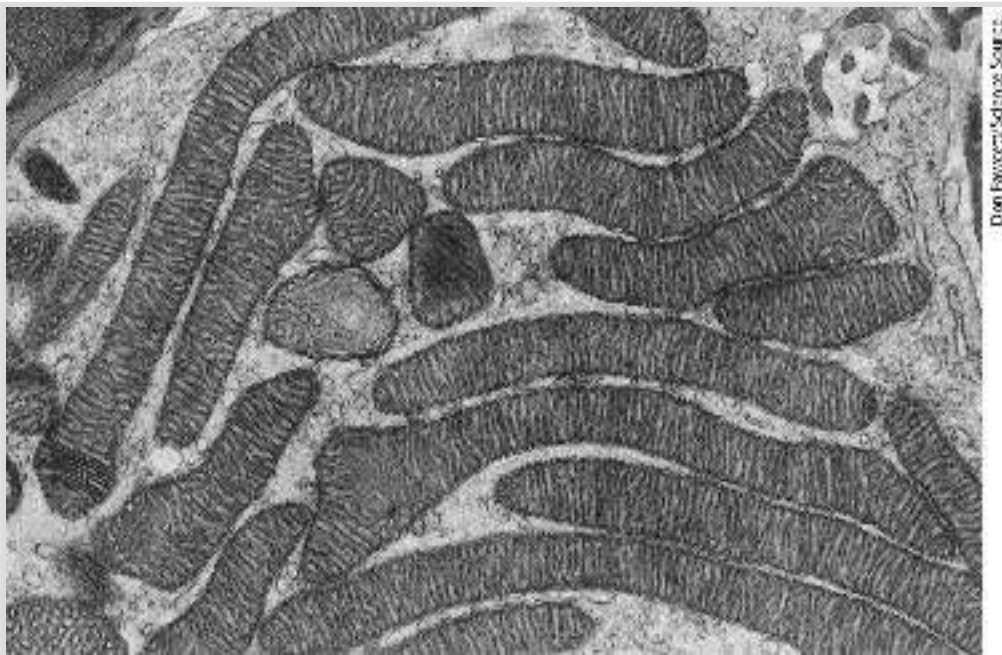


FIGURE 19-37 Mitochondria of adrenal gland, specialized for steroid synthesis. As seen in this electron micrograph of a thin section of adrenal gland, mitochondria are profuse and have extensive cristae, providing a large surface for the P-450 enzymes of the inner membrane.

The path of electron flow in the mitochondrial P-450 system is complex, involving a flavoprotein and an iron-sulfur protein that carry electrons from NADPH to the P-450 heme ([Fig. 19-38](#)). All P-450 enzymes have a heme that interacts with O_2 and a substrate-binding site that confers specificity.

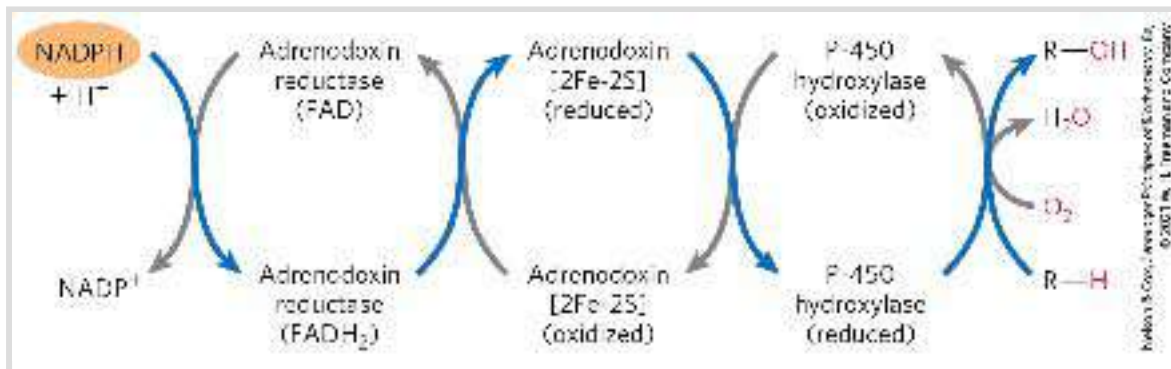


FIGURE 19-38 Path of electron flow in mitochondrial cytochrome P-450 reactions in adrenal gland. Two electrons are transferred from NADPH to the FAD-containing flavoprotein adrenodoxin reductase, which passes the electrons, one at a time, to adrenodoxin, a small, soluble 2Fe-2S protein. Adrenodoxin passes single electrons to the cytochrome P-450 hydroxylase, which interacts directly with O_2 and the substrate ($R-H$) to form the products, H_2O and $R-OH$.



Another large family of P-450 enzymes is found in the endoplasmic reticulum of hepatocytes. These enzymes catalyze reactions similar to the mitochondrial P-450 reactions, but their substrates include a wide variety of hydrophobic compounds, many of which are **xenobiotics** — compounds not found in nature but synthesized industrially. The P-450 enzymes of the ER have very broad and overlapping substrate specificities. Hydroxylation of the hydrophobic compounds makes them more water-soluble, and they can then be cleared by the kidneys and excreted in urine. Among the substrates for these P-450 oxygenases are many

commonly used prescription drugs. Metabolism by P-450 enzymes limits a drug's lifetime in the bloodstream and thus its therapeutic effects. Humans differ in their genetic complement of P-450 enzymes in the ER, as well as in the extent to which certain P-450 enzymes have been induced, such as by a history of ethanol ingestion. In principle, therefore, an individual's genetics and personal history should be considered in determining therapeutic drug doses. In practice, this precise tailoring of dosage is not yet economically feasible, but it may become so. 

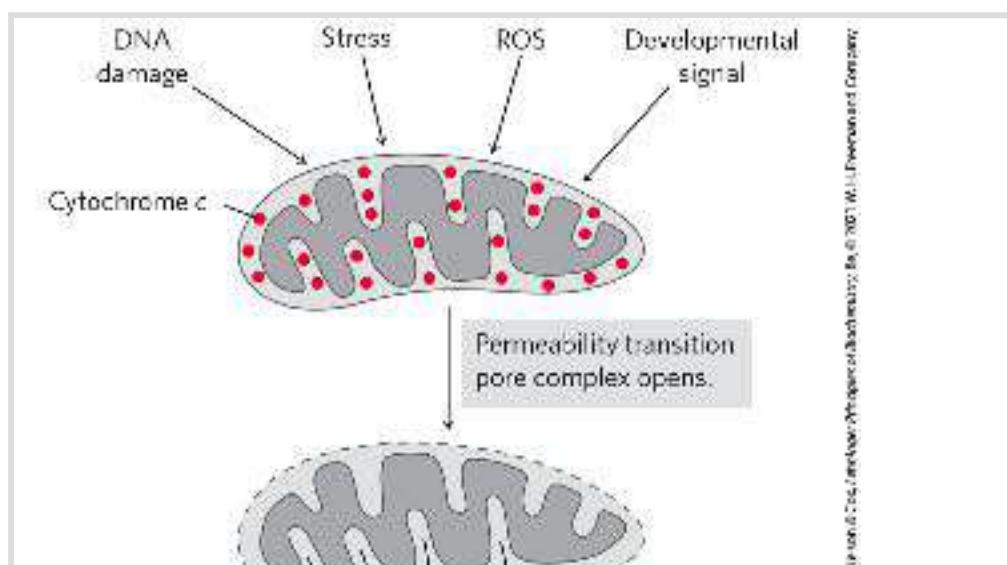
Mitochondria Are Central to the Initiation of Apoptosis

Apoptosis, also called **programmed cell death**, is a process in which individual cells die for the good of the organism, such as in the course of normal embryonic development, and the organism conserves the cells' molecular components (amino acids, nucleotides, and so forth). Apoptosis may be triggered by an external signal, acting at a plasma membrane receptor, or by internal events such as DNA damage, viral infection, oxidative stress from the accumulation of ROS, or other stress such as a heat shock.



Mitochondria play a critical role in triggering apoptosis. When a stressor gives the signal for cell death, one early consequence is an increase in the permeability of the outer mitochondrial membrane, allowing cytochrome c to escape from

the intermembrane space into the cytosol ([Fig. 19-39](#)). The increased permeability is due to the opening of the **permeability transition pore complex (PTPC)**, a multisubunit protein in the outer membrane; its opening and closing are affected by several proteins that stimulate or suppress apoptosis. When released into the cytosol, cytochrome *c* interacts with monomers of the protein **Apaf-1 (apoptosis protease activating factor-1)**, causing the formation of an **apoptosome** composed of seven Apaf-1 and seven cytochrome *c* molecules. The apoptosome provides the platform on which the proenzyme procaspase-9 is activated to caspase-9, a member of a family of highly specific proteases, called the **caspases**, involved in apoptosis. These cysteine proteases cleave proteins only on the carboxyl-terminal side of *Asp* residues, thus the name “caspases.” Caspase-9 initiates a cascade of proteolytic activations, with one caspase activating a second, and this in turn activating a third, and so forth (see [Fig. 12-42](#)). Note that this role of cytochrome *c* in apoptosis is a clear case of “moonlighting,” in that one protein plays two very different roles in the cell (see [Box 16-1](#)).



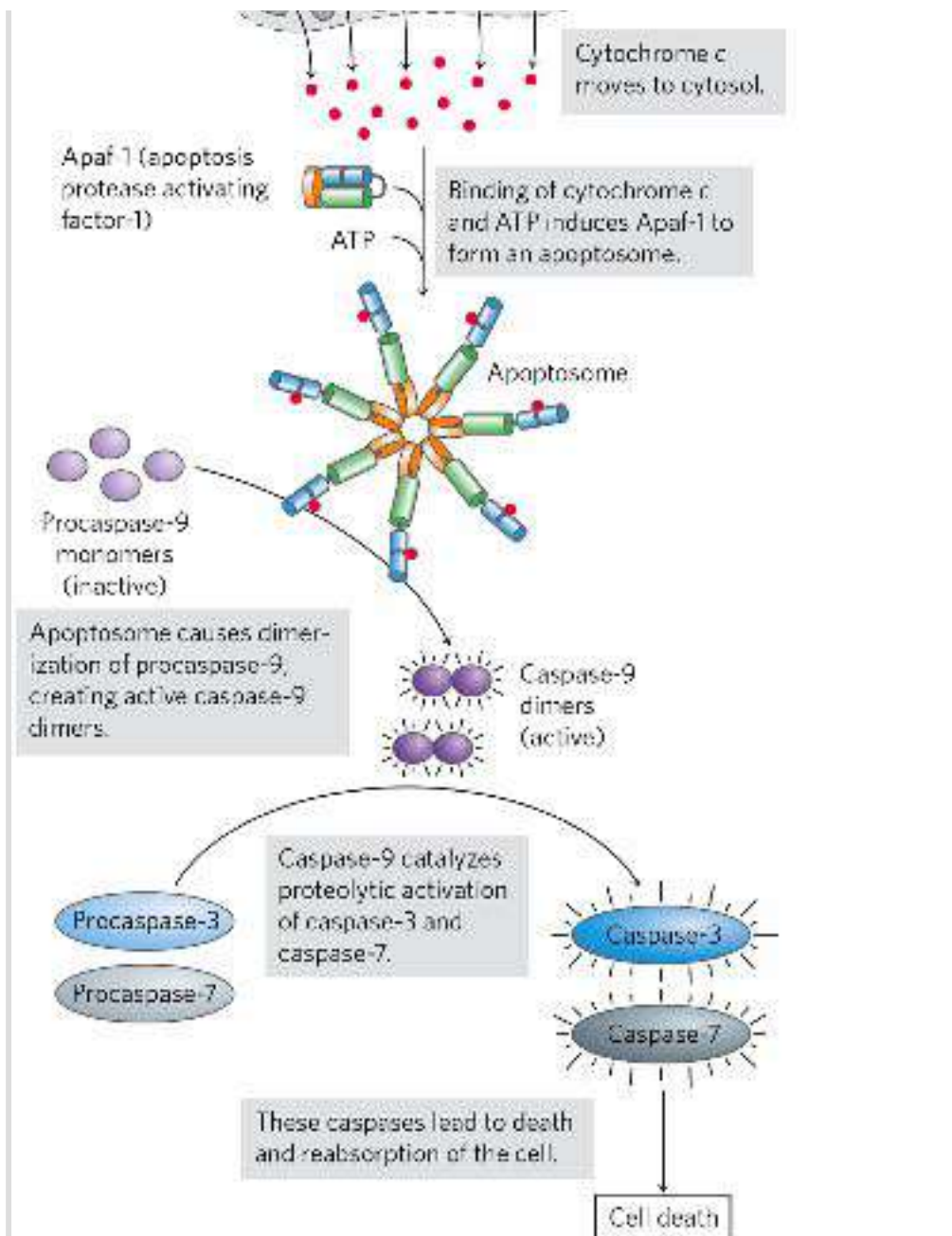


FIGURE 19-39 Role of cytochrome c in apoptosis. Cytochrome c is a small, soluble, mitochondrial protein, located in the intermembrane space, that carries electrons between Complex III and Complex IV during respiration. In a completely separate role, as outlined here, it acts as a trigger for apoptosis by stimulating the activation of a family of proteases called caspases. [Information from S. J. Riedl and G. S. Salvesen, *Nat. Rev. Mol. Cell Biol.* 8:409, 2007, Fig. 3.]

SUMMARY 19.4 *Mitochondria in Thermogenesis, Steroid Synthesis, and Apoptosis*

■ In the brown adipose tissue of newborns, electron transfer is uncoupled from ATP synthesis, and the energy of fuel oxidation is dissipated as heat. Hibernating animals use this strategy to avoid freezing.

■ Hydroxylation reaction steps in the synthesis of steroid hormones in steroidogenic tissues (adrenal gland, gonads, liver, and kidney) take place in specialized mitochondria. Key reactions are catalyzed by a family of P-450 monooxygenases.

■ Mitochondria play a central role in apoptosis. Mitochondrial cytochrome c, released into the cytosol, participates in activation of caspase-9, one of the proteases involved in apoptosis.

19.5 Mitochondrial Genes: Their Origin and the Effects of Mutations

Mitochondria contain their own genome, a circular, double-stranded DNA (mtDNA) molecule. Each of the hundreds or thousands of mitochondria in a typical cell has about five copies of this genome. The human mitochondrial chromosome ([Fig. 19-40](#)) contains 37 genes (16,569 bp), including 13 that encode subunits of proteins of the respiratory chain ([Table 19-6](#)); the remaining genes code for rRNA and tRNA molecules essential to the protein-synthesizing machinery of mitochondria. To synthesize these 13 protein subunits, mitochondria have their own ribosomes, distinctly different from those in the cytoplasm. The great majority of mitochondrial proteins — about 1,200 different types — are encoded by nuclear genes, synthesized on cytoplasmic ribosomes, then imported into and assembled in the mitochondria ([Chapter 27](#)).

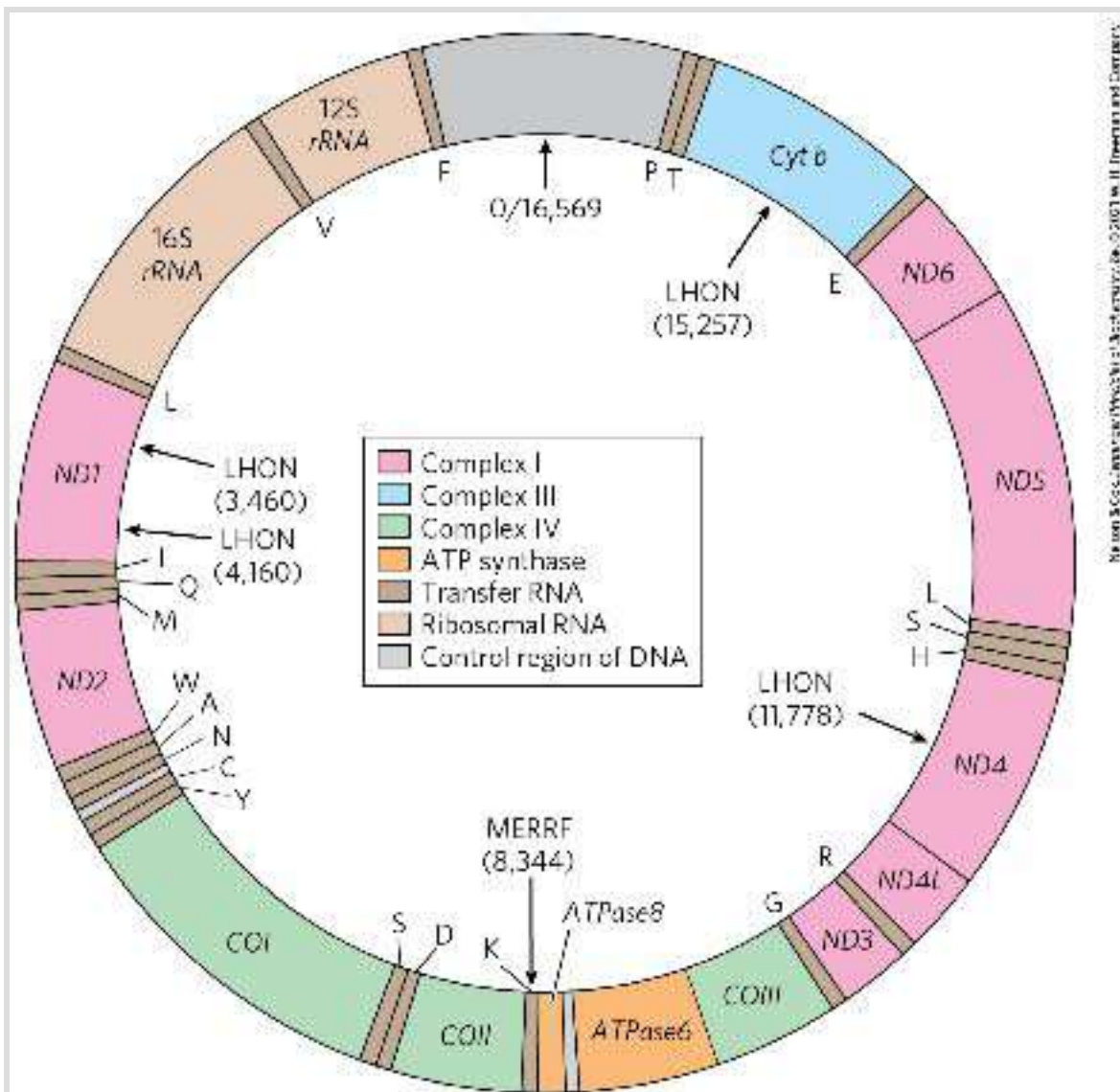


FIGURE 19-40 Mitochondrial genes and mutations. A map of human mitochondrial DNA, showing the genes that encode proteins of Complex I, the NADH dehydrogenase (ND1 to ND6); the cytochrome *b* of Complex III (Cyt *b*); the subunits of cytochrome oxidase, Complex IV (COI to COIII); and two subunits of ATP synthase (ATPase6 and ATPase8). The colors of the genes correspond to those of the complexes shown in [Figure 19-7](#). Also included here are the genes for ribosomal RNAs (*rRNA*) and for some mitochondrion-specific transfer RNAs; tRNA specificity is indicated by the one-letter codes for amino acids. Arrows indicate the positions of mutations that cause Leber hereditary optic neuropathy (LHON) and myoclonic epilepsy with ragged-red fibers (MERRF) syndrome. Numbers in parentheses indicate the position of the altered nucleotides (nucleotide 1 is at the top of the circle, and numbering proceeds counterclockwise). [Information from M. A. Morris, *J. Clin. Neuroophthalmol.* 10:159, 1990.]

TABLE 19-6 Respiratory Proteins Encoded by Mitochondrial Genes in Humans

Complex	Number of subunits	Number of subunits encoded by mtDNA
I NADH dehydrogenase	45	7
II Succinate dehydrogenase	4	0
III Ubiquinone:cytochrome c oxidoreductase	11	1
IV Cytochrome oxidase	13	3
V ATP synthase	8	2

Mitochondria Evolved from Endosymbiotic Bacteria

The existence of mitochondrial DNA, ribosomes, and tRNAs supports the theory of the endosymbiotic origin of mitochondria (see [Fig. 1-37](#)), which holds that the first organisms capable of aerobic metabolism, including respiration-linked ATP production, were bacteria.  Primitive eukaryotes that lived anaerobically (by fermentation) acquired the ability to carry out oxidative phosphorylation when they established a symbiotic relationship with bacteria living in their cytosol. After a long period of evolution and the movement of many bacterial genes into the nucleus of the “host” eukaryote, the endosymbiotic bacteria eventually became mitochondria.

This hypothesis presumes that early free-living bacteria had the enzymatic machinery for oxidative phosphorylation. And it predicts that their modern bacterial descendants must have respiratory chains closely similar to those of modern eukaryotes. They do. Aerobic bacteria carry out NAD-linked electron transfer from substrates to O_2 , coupled to the phosphorylation of cytosolic ADP. The dehydrogenases are located in the bacterial cytosol, and the respiratory chain in the plasma membrane. The electron carriers translocate protons outward across the plasma membrane as electrons are transferred to O_2 . Bacteria such as *E. coli* have F_oF_1 complexes in their plasma membranes; the F_1 portion protrudes into the cytosol and catalyzes ATP synthesis from ADP and P_i as protons flow back into the cell through the proton channel of F_o .

The respiration-linked extrusion of protons across the bacterial plasma membrane also provides the driving force for other processes. Certain bacterial transport systems bring about uptake of extracellular nutrients (lactose, for example) against a concentration gradient, in symport with protons. And the rotary motion of bacterial flagella is provided by “proton turbines,” molecular rotary motors driven not by ATP but directly by the transmembrane electrochemical potential generated by respiration-linked proton pumping ([Fig. 19-41](#)).  It seems likely that the chemiosmotic mechanism evolved early, before the emergence of eukaryotes.

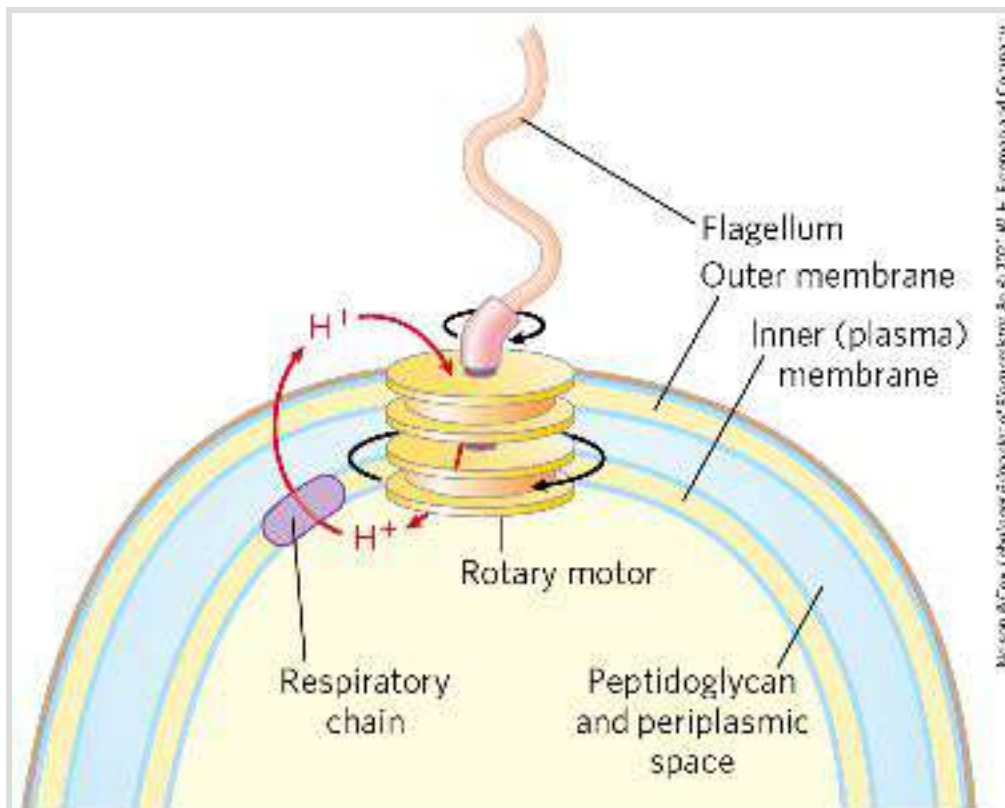


FIGURE 19-41 Rotation of bacterial flagella by proton-motive force. The shaft and rings at the base of the flagellum make up a rotary motor that has been called a “proton turbine.” Protons ejected by electron transfer flow back into the cell through the turbine, causing rotation of the shaft of the flagellum. This motion differs fundamentally from the motion of muscle and of eukaryotic flagella and cilia, for which ATP hydrolysis is the energy source.

Mutations in Mitochondrial DNA Accumulate throughout the Life of the Organism

The respiratory chain is the major producer of reactive oxygen species in cells, so mitochondrial contents, including the mitochondrial genome, suffer the greatest exposure to, and

damage by, ROS. Moreover, the mitochondrial DNA replication system is less effective than the nuclear system at correcting mistakes made during replication and at repairing DNA damage. As a consequence, defects in mtDNA accumulate over time. One theory of aging is that this gradual accumulation of defects is the primary cause of many of the “symptoms” of aging, which include, for example, progressive weakening of skeletal and heart muscle.

A unique feature of mitochondrial inheritance is the variation among individual cells, and between one individual organism and another, in the effects of a mtDNA mutation. A typical cell has hundreds or thousands of mitochondria, each with multiple copies of its own genome ([Fig. 19-2b](#)). Animals inherit essentially all of their mitochondria from the female parent. Eggs are large and contain 10^5 or 10^6 mitochondria; sperm are much smaller and contain far fewer mitochondria — perhaps 100 to 1,000. Furthermore, there is an active mechanism for targeting sperm-derived mitochondria for degradation in the fertilized egg. Just after fertilization, maternal phagosomes migrate to the site of sperm entry, engulf sperm mitochondria, and degrade them.

Suppose that, in a female organism, damage to one mitochondrial genome occurs in a germ cell from which oocytes develop, such that the germ cell contains mainly mitochondria with wild-type genes but one mitochondrion with a mutant gene. During the course of oocyte maturation, as this germ cell and its descendants repeatedly divide, the defective mitochondrion replicates and its progeny, all defective, are randomly distributed to daughter cells.

Eventually, the mature egg cells contain different proportions of the defective mitochondria. When an egg cell is fertilized and undergoes the many divisions of embryonic development, the resulting somatic cells differ in their proportion of mutant mitochondria ([Fig. 19-42a](#)). This **heteroplasmy** (in contrast to **homoplasmy**, in which every mitochondrial genome in every cell is the same) results in mutant phenotypes of varying degrees of severity. Cells (and tissues) containing mostly wild-type mitochondria have the wild-type phenotype; they are essentially normal. Other heteroplasmic cells have intermediate phenotypes, some almost normal, others (with a high proportion of mutant mitochondria) abnormal ([Fig. 19-42b](#)). If the abnormal phenotype is associated with a disease, individuals with the same mtDNA mutation may have disease symptoms of differing severity — depending on the number and distribution of affected mitochondria.

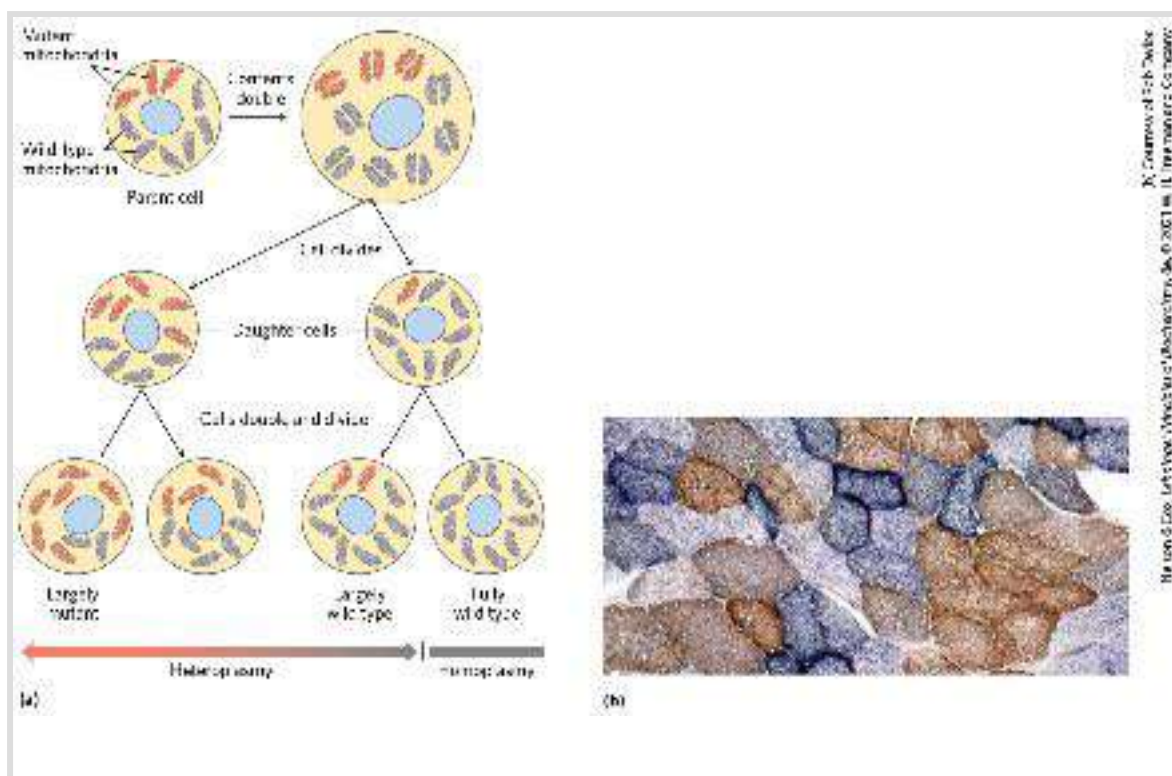


FIGURE 19-42 Heteroplasmy in mitochondrial genomes. (a) When a mature egg cell is fertilized, all of the mitochondria in the resulting diploid cell (zygote) are maternal; none come from the sperm. If some fraction of the maternal mitochondria have a mutant gene, the random distribution of mitochondria during subsequent cell divisions yields some daughter cells with mostly mutant mitochondria, some with mostly wild-type mitochondria, and some in between. Thus daughter cells show a varying degree of heteroplasmy. (b) Different degrees of heteroplasmy produce different cellular phenotypes. This section of human muscle tissue is from an individual with defective cytochrome oxidase. The cells were stained so that wild-type cells are blue and cells with mutant cytochrome oxidase are brown. As the micrograph shows, different cells in the same tissue are affected to different degrees by the mitochondrial mutation. [(b) Courtesy of Rob Taylor. Reprinted with permission from R. W. Taylor and D. M. Turnbull, *Nat. Rev. Genet.* 6:389, 2005, Fig. 2a.]

Some Mutations in Mitochondrial Genomes Cause Disease



About 1 in 5,000 people have a disease-causing mutation in a mitochondrial protein that reduces the cell's capacity to produce ATP. A growing number of these diseases have been attributed to mutations in mitochondrial genes. Some tissues and cell types — neurons, myocytes of both skeletal and cardiac muscle, and β cells of the pancreas — are less able than others to tolerate lowered ATP production and are therefore more affected by mutations in mitochondrial proteins.

A group of genetic diseases known as the **mitochondrial encephalomyopathies** affect primarily the brain and skeletal muscle. These diseases are invariably inherited from the mother, because, as noted above, a developing embryo derives all its

mitochondria from the egg. The rare disease **Leber hereditary optic neuropathy (LHON)** affects the central nervous system, including the optic nerves, causing bilateral loss of vision in early adulthood. A single base change in the mitochondrial gene *ND* ([Fig. 19-40](#)) changes an Arg residue to a His residue in a polypeptide of Complex I, and the result is mitochondria partially defective in electron transfer from NADH to ubiquinone.

Although these mitochondria can produce some ATP by electron transfer from succinate, they apparently cannot supply sufficient ATP to support the very active metabolism of neurons, including the optic nerve. A single base change in the mitochondrial gene for cytochrome *b*, a component of Complex III, also produces LHON, demonstrating that the pathology results from a general reduction of mitochondrial function, not specifically from a defect in electron transfer through Complex I.

A mutation in the mitochondrial gene *ATP* affects the proton pore in ATP synthase, leading to low rates of ATP synthesis while leaving the respiratory chain intact. Oxidative stress due to the continued supply of electrons from NADH increases the production of ROS, and the damage to mitochondria caused by ROS sets up a vicious cycle. Half of individuals with this mutant gene die within days or months of birth.

Myoclonic epilepsy with ragged-red fibers (MERRF) syndrome is caused by a mutation in the mitochondrial gene that encodes a tRNA specific for lysine (tRNA^{Lys}). This disease, characterized by uncontrollable muscular jerking, results from defective

production of several of the proteins that require mitochondrial tRNAs for their synthesis. Skeletal muscle fibers of individuals with MERRF syndrome have abnormally shaped mitochondria that sometimes contain paracrystalline structures ([Fig. 19-43](#)). Other mutations in mitochondrial genes are believed to be responsible for the progressive muscular weakness that characterizes mitochondrial myopathy and for enlargement and deterioration of the heart muscle in hypertrophic cardiomyopathy.

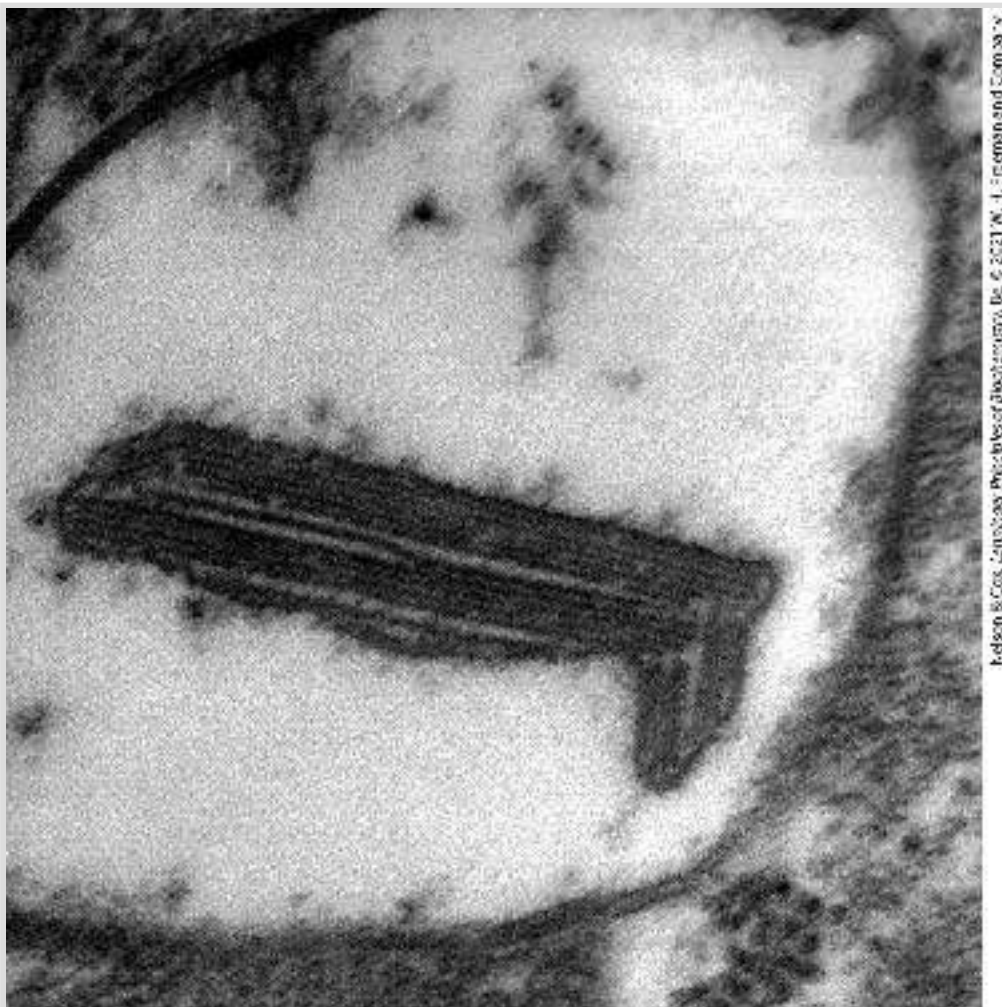


FIGURE 19-43 Paracrystalline inclusions in MERRF syndrome mitochondrion. Electron micrograph of an abnormal mitochondrion from the muscle of an individual with MERRF syndrome, showing the

paracrystalline protein inclusions sometimes present in the mutant mitochondria. [From Regionalized Pathology Correlates with Augmentation of mtDNA Copy Numbers in a Patient with Myoclonic Epilepsy with Ragged-Red Fibers (MERRF-Syndrome). *PLOS ONE*, Anja Brinckmann et al., October 20, 2010. <https://doi.org/10.1371/journal.pone.0013513>]

If a prospective mother is known to carry a pathogenic mitochondrial gene, the technique of mitochondrial donation can circumvent the passage of that mutant gene to her offspring. A prospective mother's nuclear genes are microscopically transplanted into an enucleated ovum from a donor with healthy mitochondria, then the ovum is fertilized in vitro and the resulting embryo is transplanted into the mother's uterus. This and similar "three-parent baby" procedures, which were approved in the United Kingdom in 2015, raise ethical issues that are being vigorously debated.

Mitochondrial disease can also result from mutations in any of the ~1,200 nuclear genes that encode mitochondrial proteins. For example, a mutation in one of the nuclear-encoded proteins of Complex IV, COX6B1, results in severe defects in brain development and thickened walls of the heart muscle. Other nuclear genes encode proteins essential for the *assembly* of mitochondrial complexes. Mutations in these genes can also lead to serious mitochondrial disease. 

A Rare Form of Diabetes Results from Defects in the Mitochondria of

Pancreatic β Cells



The insulin so important to glucose homeostasis in all humans is produced and exported from pancreatic β cells. Insulin export hinges on the ATP concentration in those cells. When blood glucose is high, β cells take up glucose and oxidize it by glycolysis and the citric acid cycle, raising [ATP] above a threshold level ([Fig. 19-44](#)). When [ATP] exceeds this threshold, an ATP-gated K^+ channel in the plasma membrane closes, depolarizing the membrane and triggering insulin release (see [Fig. 23-24](#)).

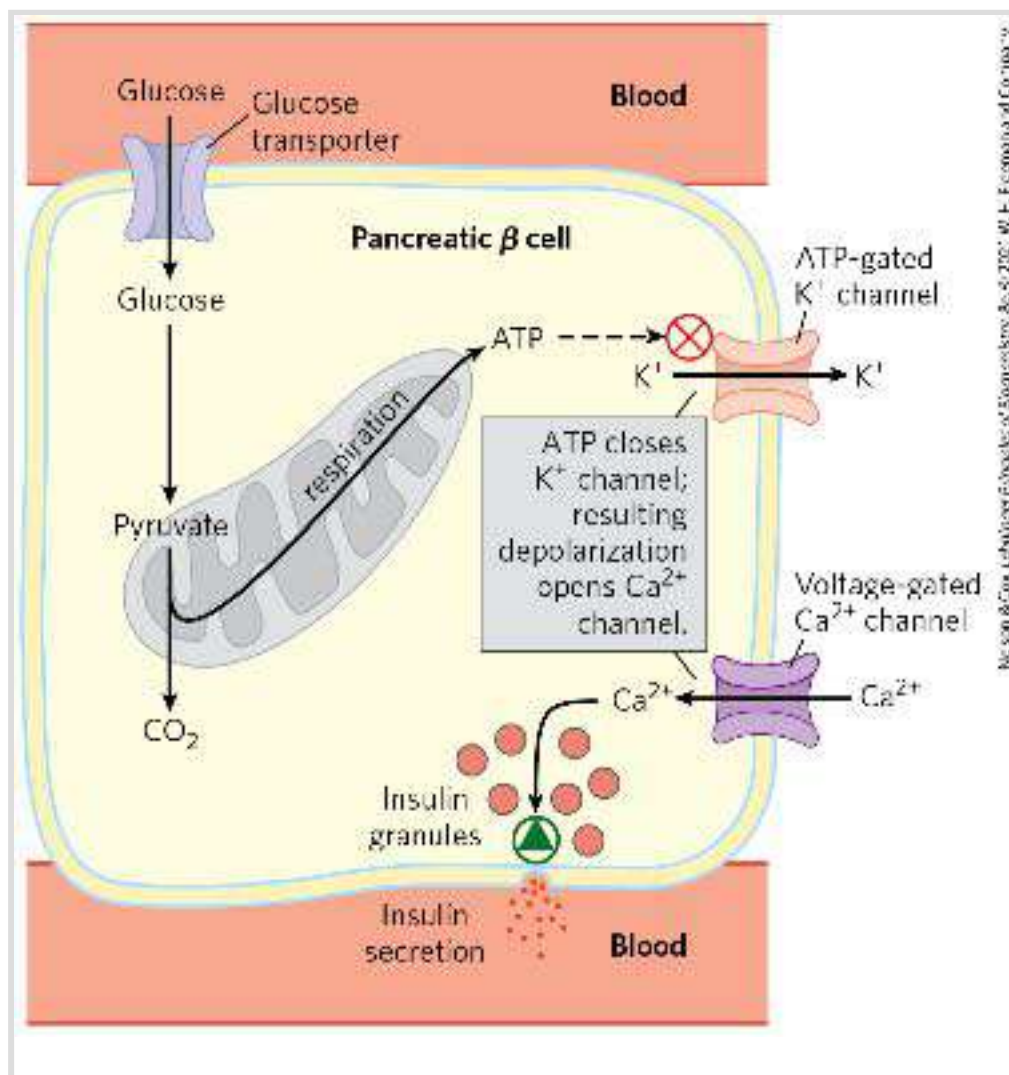


FIGURE 19-44 A mitochondrial defect prevents insulin secretion. In the normal situation, as depicted here, when the blood glucose level rises, production of ATP in β cells increases. ATP, by blocking K^+ channels, depolarizes the plasma membrane and thus opens voltage-gated Ca^{2+} channels. The resulting influx of Ca^{2+} triggers exocytosis of insulin-containing secretory vesicles, releasing insulin. When oxidative phosphorylation in β cells is defective, [ATP] is never sufficient to trigger this process, and insulin is not released.

Normal insulin release can be compromised in several ways. Pancreatic β cells with defects in any aspect of oxidative phosphorylation may not be able to increase [ATP] above this threshold, and the resulting failure of insulin release effectively produces diabetes. For example, defects in the gene for glucokinase, the hexokinase IV isozyme present in β cells, lead to a rare form of diabetes called MODY2 (*maturity onset diabetes of the young*); low glucokinase activity prevents the generation of ATP concentrations above the threshold needed to trigger insulin secretion. Mutations in the mitochondrial tRNA^{Lys} or tRNA^{Leu} genes also compromise mitochondrial ATP production by limiting the expression of electron transfer components encoded in the mitochondrial DNA. Type 2 diabetes mellitus is common among individuals with these defects (although such cases make up a very small fraction of all cases of diabetes).

When nicotinamide nucleotide transhydrogenase, which is part of the mitochondrial defense against ROS ([Fig. 19-18](#)), is genetically defective, the accumulation of ROS damages mitochondria, slowing ATP production and blocking insulin

release by β cells ([Fig. 19-44](#)). Damage caused by ROS, including damage to mtDNA, may also underlie other human diseases; there is some evidence for its involvement in Alzheimer, Parkinson, and Huntington diseases and in heart failure, as well as in aging. ■

SUMMARY 19.5 *Mitochondrial Genes: Their Origin and the Effects of Mutations*

■ A small proportion of human mitochondrial proteins, 13 in all, are encoded by the mitochondrial genome and synthesized in mitochondria. About 1,200 mitochondrial proteins are encoded by nuclear genes and imported into mitochondria after their synthesis.

■ Mitochondria arose from aerobic bacteria that entered into an endosymbiotic relationship with ancestral eukaryotes.

■ Mutations in the mitochondrial genome accumulate over the life of the organism. Mutations in nuclear or mitochondrial genes that encode components of the respiratory chain, ATP synthase, and the ROS-scavenging system, and even in tRNA genes, can cause a variety of human diseases, which often most severely affect muscle, heart, pancreatic β cells, and brain.

■ It is possible to combine the mitochondria from one woman with the nuclear genes of another to create an ovum free of a mutation that would have led to a mitochondrial disease.

■ Mitochondrial defects in pancreatic β cells that limit ATP production when glucose levels are high can compromise normal insulin release and give rise to a form of type 2 diabetes.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

chemiosmotic theory

cristae

respiratory chain

flavoprotein

reducing equivalent

ubiquinone (coenzyme Q, Q)

cytochromes

iron-sulfur protein

Rieske iron-sulfur protein

Complex I

NADH dehydrogenase

vectorial

Complex II

succinate dehydrogenase

reactive oxygen species (ROS)

superoxide radical ($\bullet\text{O}_2^-$)

Complex III

cytochrome b_{c1} complex

Q cycle

Complex IV

cytochrome oxidase

respirasome

proton-motive force

ATP synthase

F₁ ATPase

c ring

rotational catalysis

binding-change model

P/O ratio

P/2e⁻ ratio

malate-aspartate shuttle

glycerol 3-phosphate shuttle

acceptor control

mass-action ratio (Q)

brown adipose tissue (BAT)

uncoupling protein 1 (UCP1)

cytochrome P-450

xenobiotics

apoptosis

apoptosome

caspase

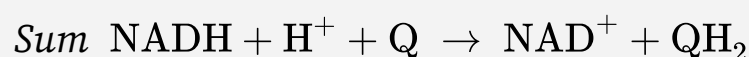
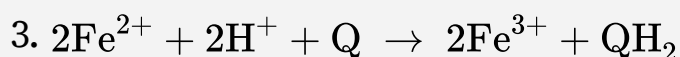
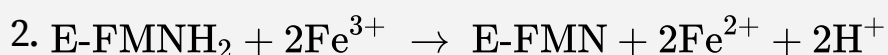
heteroplasmy

homoplasmy

PROBLEMS

1. Oxidation-Reduction Reactions Complex I, the NADH dehydrogenase complex of the mitochondrial respiratory chain, promotes the following series of oxidation-reduction reactions, in which Fe³⁺ and Fe²⁺ represent the iron in iron-

sulfur centers, Q is ubiquinone, QH₂ is ubiquinol, and E is the enzyme:



For each of the three reactions catalyzed by Complex I, identify (a) the electron donor, (b) the electron acceptor, (c) the conjugate redox pair, (d) the reducing agent, and (e) the oxidizing agent.

2. All Parts of Ubiquinone Have a Function In electron transfer, only the quinone portion of ubiquinone undergoes oxidation-reduction; the isoprenoid side chain remains unchanged. What is the function of this chain?

3. Use of FAD Rather Than NAD⁺ in Succinate Oxidation All the dehydrogenases of glycolysis and the citric acid cycle use NAD⁺ (E'° for NAD⁺/NADH is -0.32 V) as electron acceptor except succinate dehydrogenase, which uses covalently bound FAD (E'° for FAD⁺/FADH₂ in this enzyme is 0.050 V). Suggest why FAD is a more appropriate electron acceptor than NAD⁺ in the dehydrogenation of succinate, based on the E'° values of fumarate/succinate ($E'^\circ = 0.031\text{ V}$), NAD⁺/NADH, and the succinate dehydrogenase FAD/FADH₂.

4. Degree of Reduction of Electron Carriers in the Respiratory Chain

Mitochondrial conditions determine the degree of reduction of each carrier in the respiratory chain. For example, when NADH and O_2 are abundant, the steady-state degree of reduction of the carriers decreases as electrons pass from the substrate to O_2 . When electron transfer is blocked, the carriers before the block become more reduced and those beyond the block become more oxidized (see [Fig. 19-6](#)). For each of the mitochondrial conditions listed, predict the state of oxidation of ubiquinone and cytochromes b , c_1 , c , and $a + a_3$.

- a. Abundant NADH and O_2 , but cyanide added
- b. Abundant NADH, but O_2 exhausted
- c. Abundant O_2 , but NADH exhausted
- d. Abundant NADH and O_2

5. Effect of Rotenone and Antimycin A on Electron Transfer

Rotenone, a toxic natural product from plants, strongly inhibits NADH dehydrogenase of insect and fish mitochondria. Antimycin A, a toxic antibiotic, strongly inhibits the oxidation of ubiquinol.

- a. Explain why rotenone ingestion is lethal to some insect and fish species.
- b. Explain why antimycin A is a poison.
- c. Given that rotenone and antimycin A are equally effective in blocking their respective sites in the electron-transfer chain, which would be a more potent poison? Explain.

6.  Uncouplers of Oxidative Phosphorylation In normal mitochondria, the rate of electron transfer is tightly coupled to the demand for ATP. When the rate of ATP use is relatively low, the rate of electron transfer is low; when demand for ATP increases, the electron-transfer rate increases. Under these conditions of tight coupling, the number of ATP molecules produced per atom of oxygen consumed when NADH is the electron donor — the P/O ratio — is about 2.5.

- a. Predict the effect of a relatively low and a relatively high concentration of uncoupling agent on the rate of electron transfer and the P/O ratio.
- b. Ingestion of uncouplers causes profuse sweating and an increase in body temperature. Explain this phenomenon in molecular terms. What happens to the P/O ratio in the presence of uncouplers?
- c. Physicians used to prescribe the uncoupler 2,4-dinitrophenol (DNP) as a weight-reducing drug. How could this agent, in principle, serve as a weight-reducing aid? Physicians no longer prescribe uncoupling agents, because some deaths occurred following their use. Why might the ingestion of uncouplers cause death?

7. Effects of Valinomycin on Oxidative Phosphorylation

When investigators add the antibiotic valinomycin (see [Fig. 11-43](#)) to actively respiring mitochondria, several things happen: the yield of ATP decreases, the rate of O₂ consumption increases, heat is released, and the pH gradient across the inner mitochondrial membrane increases. Does valinomycin act as an uncoupler or as an inhibitor of

oxidative phosphorylation? Explain the experimental observations in terms of the antibiotic's ability to transfer K^+ ions across the inner mitochondrial membrane.

8. Cellular ADP Concentration Controls ATP Formation

Although ATP synthesis requires both ADP and P_i , the rate of synthesis depends mainly on the concentration of ADP, not P_i . Why?

9. Reactive Oxygen Species Describe the role played by superoxide dismutase in ameliorating the effects of reactive oxygen species.

10. How Many Protons in a Mitochondrion? Electron transfer translocates protons from the mitochondrial matrix to the external medium, establishing a pH gradient across the inner membrane (outside more acidic than inside). The tendency of protons to diffuse back into the matrix is the driving force for ATP synthesis by ATP synthase. During oxidative phosphorylation by a suspension of mitochondria in a medium of pH 7.4, the measured pH of the matrix is 7.7.

- Calculate $[H^+]$ in the external medium and in the matrix under these conditions.
- What is the outside-to-inside ratio of $[H^+]$? How much energy for ATP synthesis is available in this concentration difference. (Hint: See [Eqn 11-4, p. 392.](#))
- Calculate the number of protons in a respiring liver mitochondrion, assuming its inner matrix compartment is a sphere of diameter $1.5 \mu m$.

- d. From these data, is the pH gradient alone sufficient to generate ATP?
- e. If not, suggest how the necessary energy for synthesis of ATP arises.

11. Rate of ATP Turnover in Rat Heart Muscle Rat heart muscle operating aerobically fills more than 90% of its ATP needs by oxidative phosphorylation. Each gram of tissue consumes O_2 at the rate of $10.0 \mu\text{mol}/\text{min}$, with glucose as the fuel source.

- a. Calculate the rate at which the heart muscle consumes glucose and produces ATP.
- b. For a steady-state ATP concentration of $5.0 \mu\text{mol}/\text{g}$ of heart muscle tissue, calculate the time required (in seconds) to completely turn over the cellular pool of ATP. What does this result indicate about the need for tight regulation of ATP production? (Note: Concentrations are expressed as micromoles per gram of muscle tissue because the tissue is mostly water.)

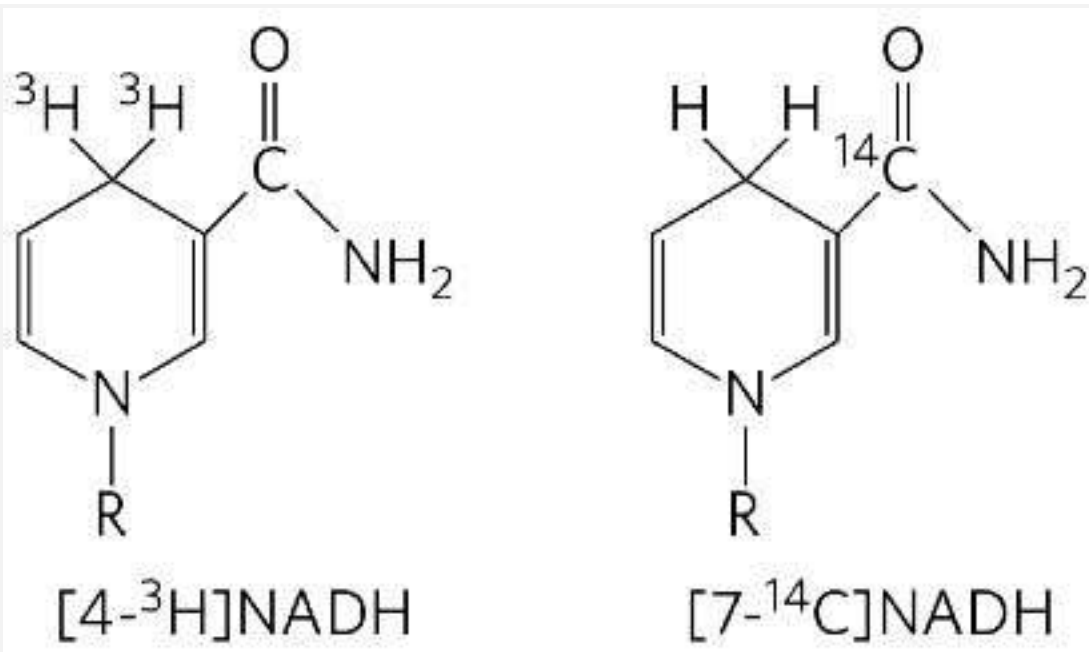
12. Rate of ATP Breakdown in Insect Flight Muscle ATP production in the flight muscle of the fly *Lucilia sericata* results almost exclusively from oxidative phosphorylation. During flight, maintaining an ATP concentration of $7.0 \mu\text{mol}/\text{g}$ of flight muscle requires $187 \text{ mL of O}_2/\text{h} \bullet \text{g}$ of body weight. Assuming that flight muscle makes up 20% of the fly's weight, calculate the rate at which the flight-muscle ATP pool turns over. How long would the reservoir of ATP last in the absence of oxidative phosphorylation? Assume

that the glycerol 3-phosphate shuttle transfers the reducing equivalents and that O_2 is at 25 °C and 101.3 kPa (1 atm).

13.  High Blood Alanine Level Associated with Defects in Oxidative Phosphorylation Most individuals with genetic defects in oxidative phosphorylation have relatively high concentrations of alanine in their blood. Explain this in biochemical terms.

14. Compartmentalization of Citric Acid Cycle Components Isocitrate dehydrogenase is found only in mitochondria, but malate dehydrogenase is found in both the cytosol and mitochondria. What is the role of cytosolic malate dehydrogenase?

15. Transmembrane Movement of Reducing Equivalents Under aerobic conditions, extramitochondrial NADH must undergo oxidation by the mitochondrial respiratory chain. Consider a preparation of rat hepatocytes containing mitochondria and all the cytosolic enzymes. After the introduction of $[4-^3H]NADH$, radioactivity soon appears in the mitochondrial matrix. Conversely, no radioactivity appears in the matrix after the introduction of $[7-^{14}C]NADH$. What do these observations reveal about the oxidation of extramitochondrial NADH by the respiratory chain?



16. NAD Pools and Dehydrogenase Activities Although both pyruvate dehydrogenase and glyceraldehyde 3-phosphate dehydrogenase use NAD^+ as their electron acceptor, the two enzymes do not compete for the same cellular NAD pool. Why?

17. The Malate- α -Ketoglutarate Transport System *n*-Butylmalonate inhibits the transport system that conveys malate and α -ketoglutarate across the inner mitochondrial membrane (see [Fig. 19-31](#)). Suppose you add *n*-butylmalonate to an aerobic suspension of kidney cells using exclusively glucose as fuel. Predict the effect of this inhibitor on

- glycolysis,
- oxygen consumption,
- lactate formation, and
- ATP synthesis.

18. Time Scales of Regulatory Events in Mitochondria

Compare the likely time scales for the adjustments in respiratory rate caused by

- a. increased [ADP] and
- b. reduced pO_2 . What accounts for the difference?

19. The Pasteur Effect When investigators add O_2 to an anaerobic suspension of cells consuming glucose at a high rate, the rate of glucose consumption declines greatly as the cells consume the O_2 , and accumulation of lactate ceases. This effect, first observed by Louis Pasteur in the 1860s, is characteristic of most cells capable of both aerobic and anaerobic glucose catabolism.

- a. Why does the accumulation of lactate cease after the addition of O_2 ?
- b. Why does the presence of O_2 decrease the rate of glucose consumption?
- c. How does the onset of O_2 consumption slow down the rate of glucose consumption? Explain in terms of specific enzymes.

20. Respiration-Deficient Yeast Mutants and Ethanol

Production Researchers can produce respiration-deficient yeast mutants (p^- ; “petites”) from wild-type parents by treatment with mutagenic agents. The mutants lack cytochrome oxidase, a deficit that markedly affects their metabolic behavior. One striking effect is that fermentation is not suppressed by O_2 — that is, the mutants do not experience the Pasteur effect (see [Problem 19](#)). Some companies are very interested in using these mutants to

ferment wood chips to ethanol for energy use. Why does the absence of cytochrome oxidase eliminate the Pasteur effect? Explain the advantages of using these mutants rather than wild-type yeast for large-scale ethanol production.

21.  **Mitochondrial Disease and Cancer** Mutations in the genes that encode certain mitochondrial proteins are associated with a high incidence of some types of cancer. How might defective mitochondria lead to cancer?

22.  **Variable Severity of a Mitochondrial Disease** Different individuals with a disease caused by the same specific defect in the mitochondrial genome may have symptoms ranging from mild to severe. Explain why.

23.  **Diabetes as a Consequence of Mitochondrial Defects** Glucokinase is essential in the metabolism of glucose in pancreatic β cells. Humans with two defective copies of the glucokinase gene exhibit a severe, neonatal diabetes, whereas those with only one defective copy of the gene have a much milder form of the disease (maturity onset diabetes of the young, MODY2). Explain this difference in terms of the biology of the β cell.

24.  **Effects of Mutations in Mitochondrial Complex II** Single nucleotide changes in the gene for succinate dehydrogenase (Complex II) are associated with midgut carcinoid tumors. Suggest a mechanism to explain this observation.

DATA ANALYSIS PROBLEM

25. Membrane Fluidity and Respiration Rate

The mitochondrial electron transfer complexes and the F_0F_1 ATP synthase are embedded in the inner mitochondrial membrane in eukaryotes and in the inner membrane of bacteria. Electrons are shuttled between complexes in part by coenzyme Q, or ubiquinone, a factor that migrates within the membrane. Jay Keasling and coworkers explored the effect of membrane fluidity on rates of respiration in *E. coli*.

E. coli naturally adjusts its membrane lipid content to maintain membrane fluidity at different temperatures.

Workers in the Keasling lab bioengineered an *E. coli* strain to allow them to control expression of the enzyme FabB, which catalyzes the limiting step in the synthesis of unsaturated fatty acids in *E. coli*.

- a. How does the content of unsaturated fatty acids affect membrane fluidity?
- b. The researchers were able to modulate the content of unsaturated fatty acids in the membrane lipid from 15% to 80%. They did not try to completely block synthesis of unsaturated fatty acids to extend the experimental range in the membrane to 0%. Why not?
- c. When the cells were grown under aerobic conditions, the researchers found that bacterial growth rate increased as the concentration of unsaturated fatty acids in the membrane increased. However, when oxygen was very limited, the unsaturated fatty acid

content of the membrane had no effect on growth rate. How might you explain this observation?

- d. The researchers measured rates of respiration, finding a strong correlation between those rates and the fraction of membrane fatty acids that was unsaturated. When the unsaturated fatty acid content of the membranes was kept low, the cells accumulated pyruvate and lactate. Explain these observations.
- e. Next, they measured rates of diffusion of membrane phospholipids and ubiquinone in vesicles derived from *E. coli* membranes. The diffusion rates increased as a function of the content of unsaturated fatty acids. These measured rates were consistent with simulations carried out to model the effects of ubiquinone diffusion on respiration. What overall conclusion can be drawn from this work?

Reference

Budin, I, T. de Rond, Y. Chen, L.J.G. Chan, C.J. Petzold, and J.D. Keasling. 2018. Viscous control of cellular respiration by membrane lipid composition. *Science* 362:1186–1189.

CHAPTER 20

PHOTOSYNTHESIS AND CARBOHYDRATE SYNTHESIS IN PLANTS



20.1 Light Absorption

20.2 Photochemical Reaction Centers

20.3 Evolution of a Universal Mechanism of ATP Synthesis

20.4 CO₂-Assimilation Reactions

20.5 Photorespiration and the C₄ and CAM Pathways

20.6 Biosynthesis of Starch, Sucrose, and Cellulose

We have now reached a turning point in our study of cellular metabolism. Thus far in Part II we have described how the major metabolic fuels — carbohydrates, fatty acids, and amino acids — are degraded through *converging catabolic* pathways that lead to the citric acid cycle and yield their electrons to the respiratory chain, driving ATP synthesis by *oxidative* phosphorylation. We now turn to *reductive, anabolic, divergent* processes fueled by energy from the sun that take place in photosynthetic organisms, and in all other organisms, driven ultimately by the photosynthetic reduction of CO₂.

As we examine this process, these principles will emerge:

P1 The capture of solar energy by photosynthetic organisms and its conversion to the chemical energy of reduced organic compounds is the ultimate source of nearly all biological energy and organic nutrients for all of the nonphotosynthetic organisms, including humans. It is arguably *the most important biochemical process in the biosphere*.

P2 Photosynthetic organisms use tightly organized light-harvesting complexes to absorb sunlight and capture its energy in chemical form: a separation of positive and negative charge leading to electron flow. The energy from an absorbed photon moves from one antenna chlorophyll to another and another until it arrives at the reaction center where it promotes the photochemical reaction that sends electrons through a series of electron carriers.

P3 The light-driven flow of electrons through specialized protein carriers is coupled to ATP synthesis. A strong reducing agent (NADPH) is also produced, and simultaneously, water is oxidized to O₂, which is released into the atmosphere.

P4 Evolution yielded a universal mechanism for coupling ATP synthesis to the flow of electrons. A proton gradient created by electron flow is used to energize the ATP-synthesizing enzyme in microorganisms, animals, and plants.

P5 The ATP and NADPH produced in the light-dependent reactions of photosynthesis provide the energy and the

reducing power to convert atmospheric CO_2 into simple organic compounds. High concentrations of ATP and NADPH allow the chloroplast to carry out redox reactions that are thermodynamically unfavorable.

Photosynthesis encompasses two processes: the **light-dependent reactions**, in which sunlight provides the energy for the synthesis of ATP and NADPH, and the **CO_2 -assimilation reactions**, in which ATP and NADPH are used to reduce CO_2 to form triose phosphates via a set of reactions known as the Calvin cycle (**Fig. 20-1**). We heterotrophs are alive because the enormous energy of sunlight has been captured and tamed by autotrophs by photosynthesis and made available to us as fuel, vitamins, and building blocks. How do they do it?

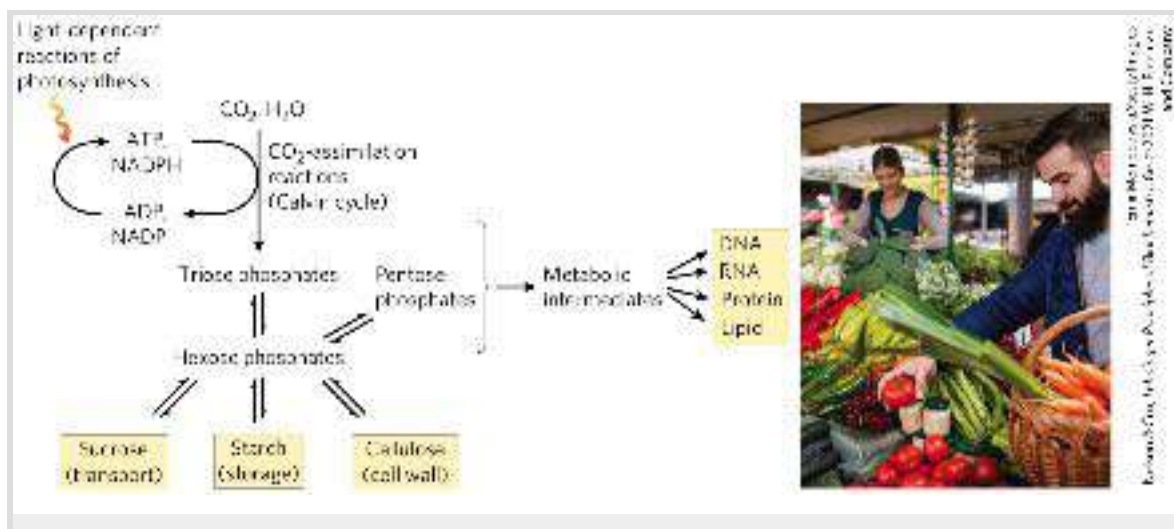


FIGURE 20-1 Assimilation of CO_2 provides all of the carbon a plant needs. The light-driven synthesis of ATP and NADPH provides energy and reducing power for the fixation of CO_2 into trioses in the Calvin cycle. All of the carbon-containing compounds of the plant cell are synthesized from this fixation of CO_2 .

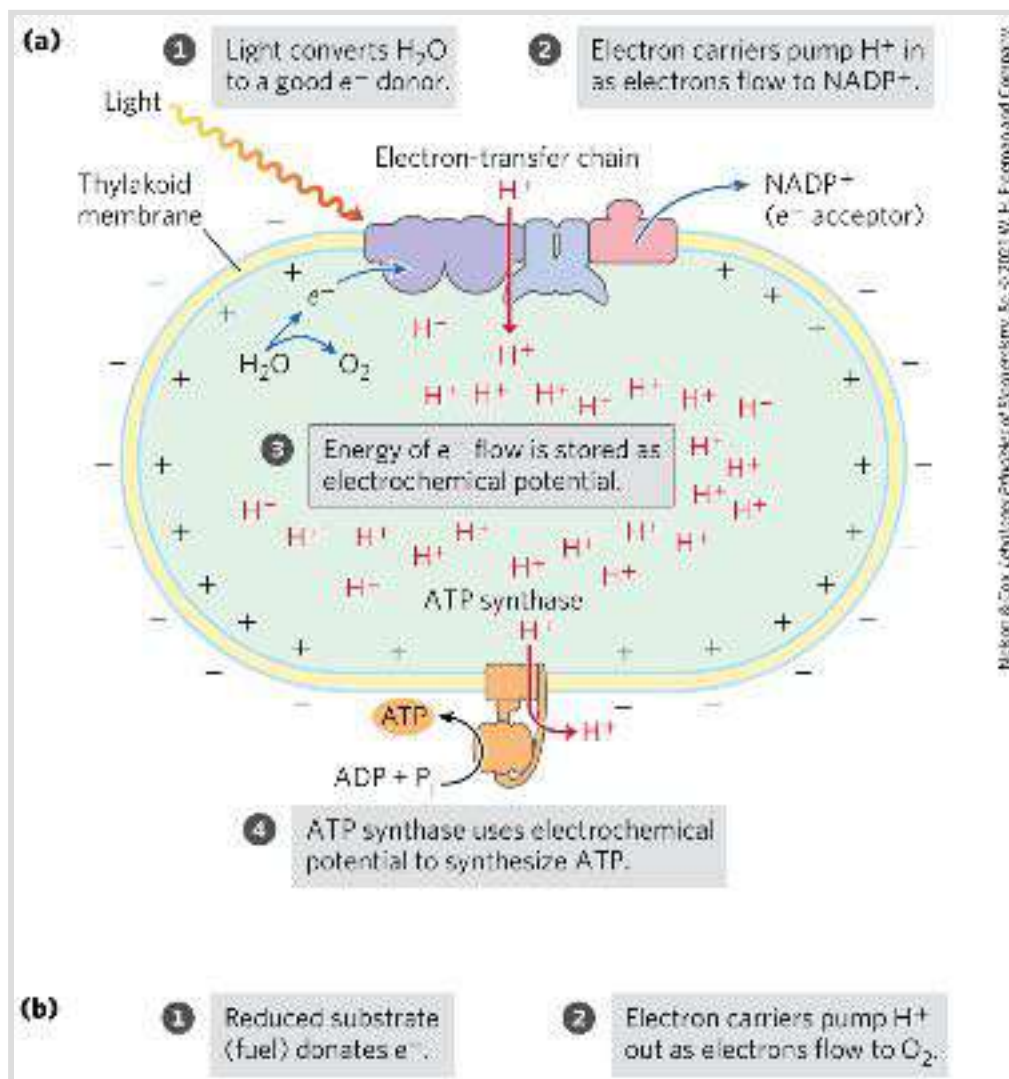
All vascular plants, as well as algae and cyanobacteria, carry out the same basic process of photosynthesis, but some are more amenable to study than others. Algae and cyanobacteria have been extensively studied because of the relative ease of culturing and manipulating them in the laboratory. Spinach is a vascular plant commonly used for studies of photosynthesis because of the ease of obtaining large amounts of material; and for genetic approaches, the small plant *Arabidopsis thaliana* is a favorite. What we say here about photosynthesis is essentially true of photosynthesis in all of these organisms.

After looking at photosynthesis, we will discuss the conversion of trioses produced in the Calvin cycle to sucrose (for sugar transport) and starch (for energy storage) (see [Fig. 20-1](#)). This conversion is accomplished by mechanisms analogous to those used by animal cells to make glycogen. We also describe the synthesis of the cellulose of plant cell walls. Finally, we consider how carbohydrate metabolism is integrated within a plant cell and throughout the plant.

Although strikingly different on the surface, the processes of photophosphorylation in the chloroplast and oxidative phosphorylation in the mitochondrion are closely similar at the molecular level, and the mechanism for ATP synthesis is virtually identical: a proton gradient drives rotary catalysis by a remarkable ATP synthase.

20.1 Light Absorption

Photophosphorylation (ATP synthesis driven by light) resembles oxidative phosphorylation in that electron flow through a series of membrane carriers is coupled to proton pumping, producing the proton motive force that powers ATP formation. The processes are compared in [Figure 20-2](#). In oxidative phosphorylation, the electron donor is NADH and the ultimate electron acceptor is O_2 , forming H_2O . In photophosphorylation, electrons flow in the opposite direction: H_2O is the electron *donor* and NADPH is *formed*. How is this endergonic process possible?



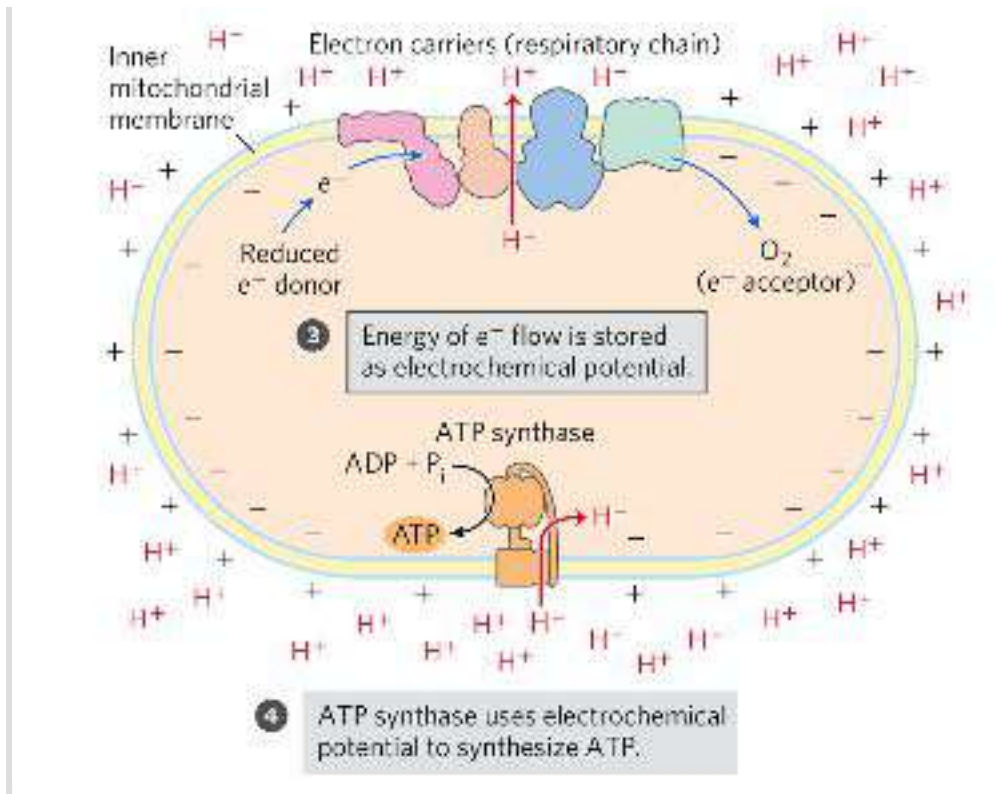


FIGURE 20-2 The chemiosmotic mechanism for ATP synthesis in

chloroplasts and mitochondria. (a) Movement of electrons through a chain

of membrane-bound carriers in the chloroplast membrane is driven by the energy of photons absorbed by the green pigment chlorophyll. Electron flow leads to the movement of protons and positive charge across the membrane, creating an electrochemical potential. This electrochemical potential drives ATP synthesis by the membrane-bound ATP synthase, which is fundamentally similar in structure and mechanism to (b) the mitochondrial machinery for oxidative phosphorylation of mitochondria. In mitochondria, the force that moves electrons through the complexes is a large difference in the reduction potentials of electron donor and acceptor. In both systems, the energy made available by electron transfer is captured as a transmembrane proton gradient, which drives ATP synthesis by an ATP synthase.

Water is a poor donor of electrons; its standard reduction potential is 0.816 V, compared with -0.320 V for NADH, a good electron donor. **P1** Photosynthesis requires the input of energy in the form of light to *create* a good electron donor and a

good electron acceptor.  Electrons flow from the electron donor through a series of membrane-bound carriers, including cytochromes, quinones, and iron-sulfur proteins, while protons are pumped across a membrane to create an electrochemical potential. Electron transfer and proton pumping are catalyzed by a membrane complex that is homologous in structure and function to Complex III of mitochondria. The electrochemical potential so produced is the driving force for ATP synthesis from ADP and P_i , catalyzed by a membrane-bound ATP synthase complex closely similar to that of mitochondria and bacteria.

Chloroplasts Are the Site of Light-Driven Electron Flow and Photosynthesis in Plants

In photosynthetic eukaryotic cells, both the light-dependent and the CO_2 -assimilation reactions take place in [chloroplasts](#) ([Fig. 20-3](#)), organelles that are variable in shape and generally a few micrometers in diameter. Like mitochondria, chloroplasts are surrounded by two membranes: an outer membrane that is permeable to small molecules and ions, and an impermeable inner membrane that bears specific transporters for a variety of ions and metabolites. The space enclosed by the inner membrane is called the [stroma](#) in chloroplasts and is analogous to the mitochondrial matrix; it is an aqueous phase containing most of the soluble enzymes required for the CO_2 -assimilation reactions. Throughout the stroma is a highly organized set of topologically

continuous internal membranes, forming a single compartment or lumen. This complex membrane system forms flattened sacks called **thylakoids**. **Granal thylakoids** are disk-like pouches arranged in stacks; they are connected by **stromal thylakoids**, which are flatter and spiral around a stack of grana. The thylakoid membranes provide a large area for the machinery of photophosphorylation — the photosynthetic pigments and enzyme complexes that carry out the light-dependent reactions and ATP synthesis. Traffic across these membranes is also mediated by specific transporters.

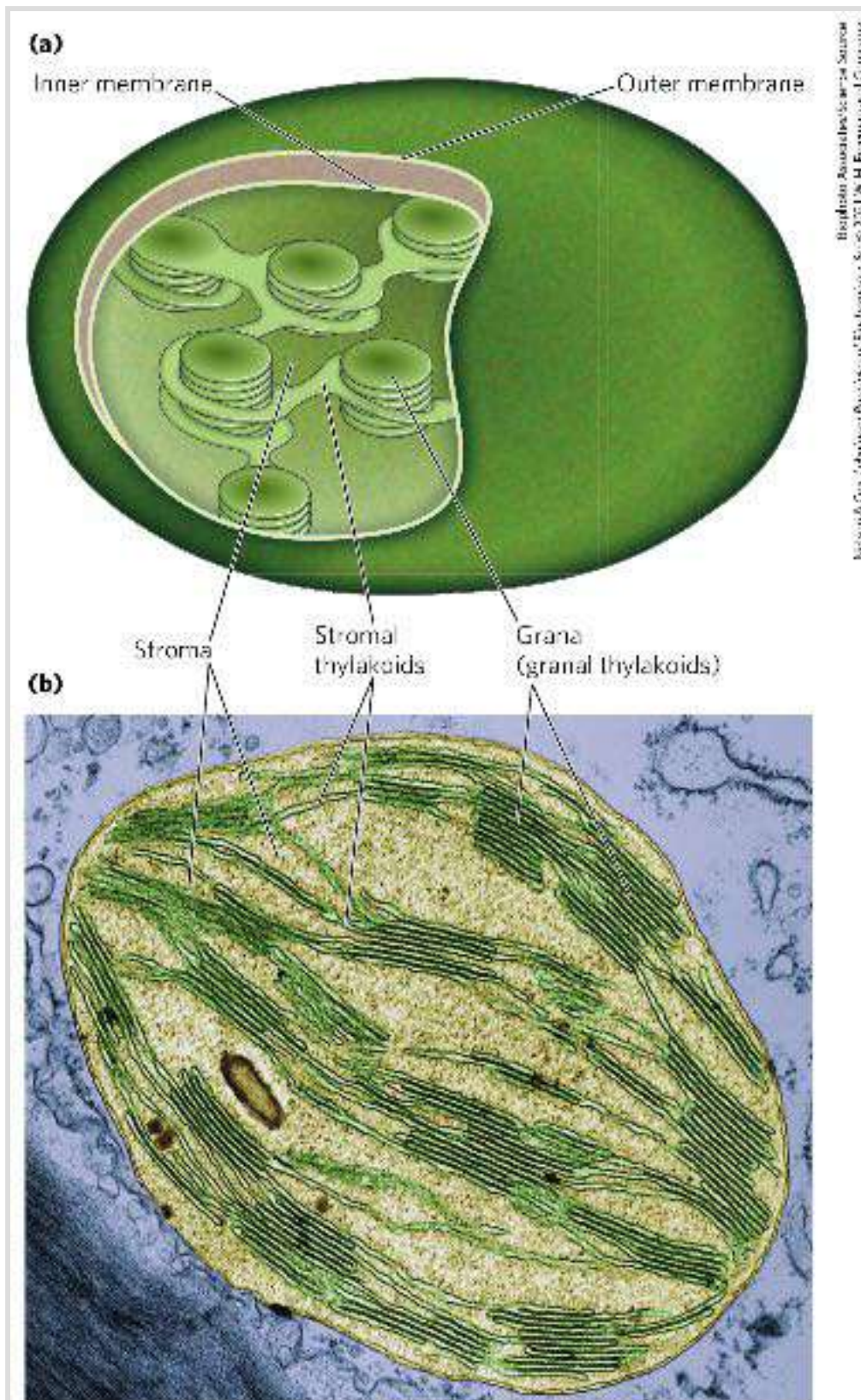
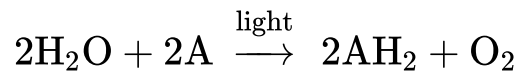
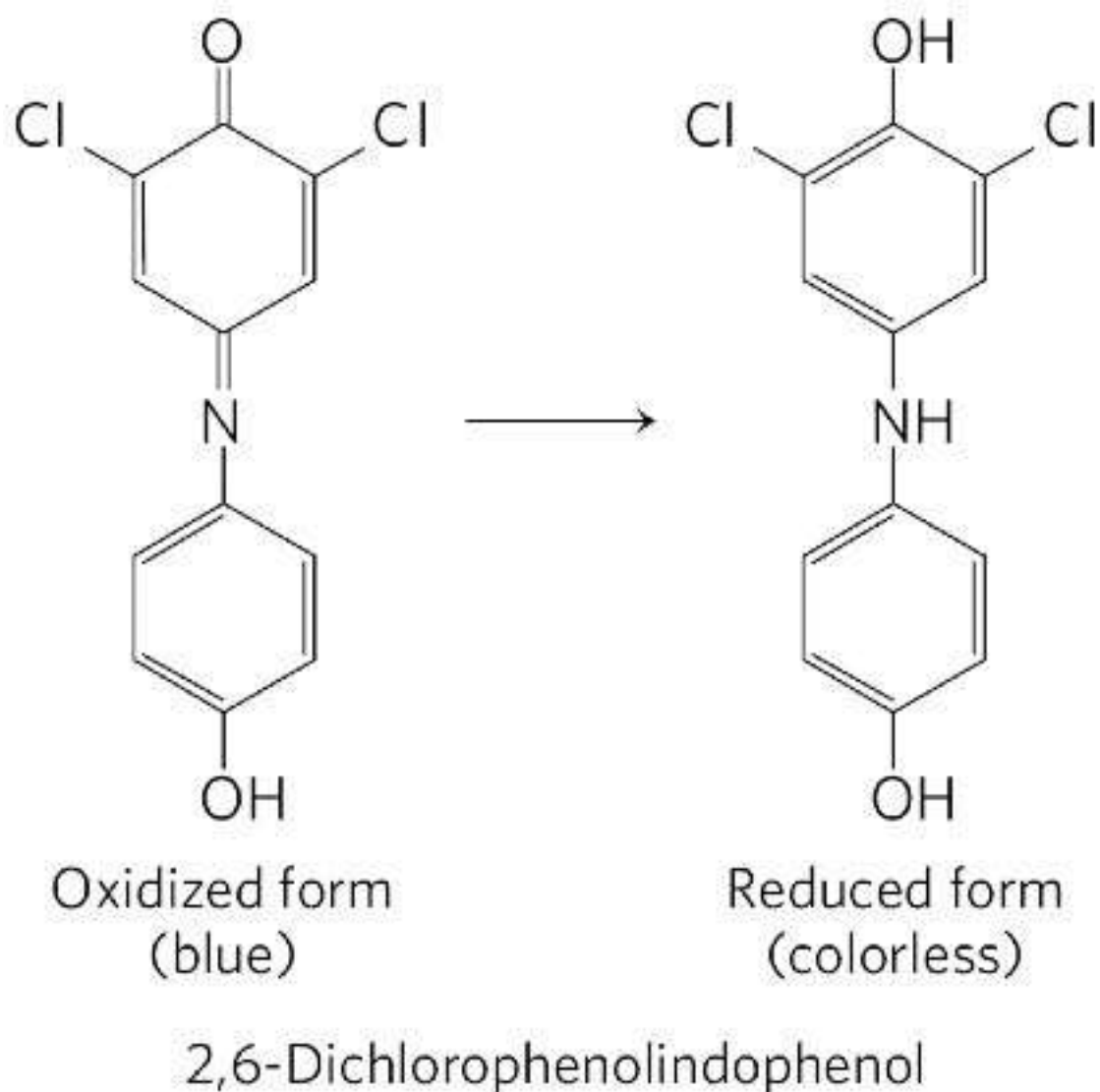


FIGURE 20-3 Chloroplast structure. (a) Schematic diagram. (b) Colorized electron micrograph at high magnification, showing the highly organized thylakoid membrane system.

In 1937, Robert Hill found that when leaf extracts containing chloroplasts were illuminated, they (1) evolved O₂ and (2) reduced a nonbiological electron acceptor added to the medium, according to the Hill reaction

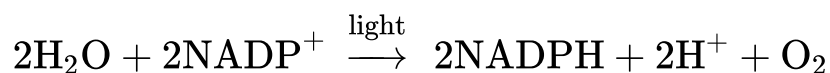


where A is an artificial electron acceptor, or Hill reagent. One Hill reagent, the dye 2,6-dichlorophenolindophenol, is blue when oxidized (A) and colorless when reduced (AH₂), making the reaction easy to follow.



When a leaf extract supplemented with the dye was illuminated, the blue dye became colorless and O_2 was evolved. In the dark, no O_2 evolution or dye reduction took place. **P2** This was the first evidence that absorbed light energy causes electrons to flow from some electron donor (now known to be H_2O) to an electron acceptor. Moreover, Hill found that CO_2 was neither required nor reduced to a stable form under these conditions; O_2 production could be dissociated from CO_2 reduction. Several years later,

Severo Ochoa showed that NADP^+ is the biological electron acceptor in chloroplasts, according to the equation

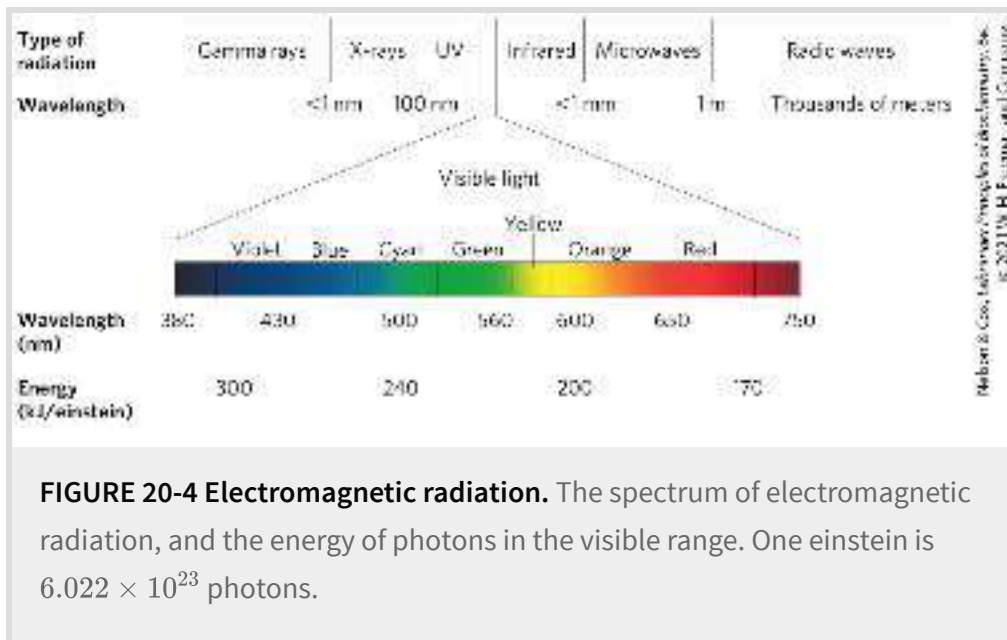


To understand this photochemical process, we must first consider the more general topic of the effects of light absorption on molecular structure.

Visible light is electromagnetic radiation of wavelengths 400 to 700 nm, a small part of the electromagnetic spectrum ([Fig. 20-4](#)), ranging from violet to red. The energy of a single [photon](#) (a quantum of light) is greater at the violet end of the spectrum than at the red end; shorter wavelength (and higher frequency) corresponds to higher energy. The energy, E , in a single photon of visible light is given by the Planck equation:

$$E = h\nu = hc/\lambda$$

where h is Planck's constant ($6.626 \times 10^{-34} \text{ J} \cdot \text{s}$), ν is the frequency of the light in cycles/s, c is the speed of light ($3.00 \times 10^8 \text{ m/s}$), and λ is the wavelength of the light in meters. The energy of a photon of visible light ranges from 150 kJ/einstein for red light to ~ 300 kJ/einstein for violet light.



WORKED EXAMPLE 20-1 *Energy of a Photon*

The light used by vascular plants for photosynthesis has a wavelength of about 700 nm. Calculate the energy in a “mole” of photons (an einstein) of light of this wavelength, and compare this with the energy needed to synthesize a mole of ATP.

SOLUTION:

The energy in a single photon is given by the Planck equation. At a wavelength of 700×10^{-9} m, the energy of a photon is

$$\begin{aligned}
 E &= hc/\lambda \\
 &= \frac{[(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})]}{(7.00 \times 10^{-7} \text{ m})} \\
 &= 2.84 \times 10^{-19} \text{ J}
 \end{aligned}$$

An einstein of light is Avogadro's number of photons (6.022×10^{23}); thus the energy of one einstein of photons at 700 nm is given by

$$\begin{aligned}
 (2.84 \times 10^{-19} \text{ J/photon})(6.022 \times 10^{23} \text{ photons/einstein}) \\
 &= 17.1 \times 10^4 \text{ J/einstein} \\
 &= 171 \text{ kJ/einstein}
 \end{aligned}$$

So, a “mole” of photons of red light has about five times the energy needed to produce a mole of ATP from ADP and P_i (30.5 kJ/mol).

When a photon is absorbed, an electron in the absorbing molecule (chromophore) is lifted to a higher energy level. This is an all-or-nothing event: to be absorbed, the photon must contain a quantity of energy, called a **quantum**, that exactly matches the energy of the electronic transition. A molecule that has absorbed a photon is in an **excited state**, which is generally unstable. An electron lifted into a higher-energy orbital usually returns rapidly to its lower-energy orbital; that is, the excited molecule decays to the stable **ground state**, giving up the absorbed quantum as light

or heat or using it to do chemical work. Light emission accompanying decay of excited molecules, [fluorescence](#), is always at a longer wavelength (lower energy) than that of the absorbed light (see [Box 12-1](#)). An alternative mode of decay, central to photosynthesis, involves direct transfer of excitation energy from an excited molecule to a neighboring molecule. Just as the photon is a quantum of light energy, so the **exciton** is a quantum of energy passed from an excited molecule to another molecule in a process called **exciton transfer**.

Chlorophylls Absorb Light Energy for Photosynthesis

The most important light-absorbing pigments in the thylakoid membranes are the [chlorophylls](#), green pigments with polycyclic, planar structures resembling the protoporphyrin of hemoglobin, except that Mg^{2+} , not Fe^{2+} , occupies the central position ([Fig. 20-5a](#); compare to [Fig. 5-1](#)). The four inward-oriented nitrogen atoms of chlorophyll are coordinated with the Mg^{2+} . All chlorophylls have a long phytol side chain, esterified to a carboxyl-group substituent in ring IV, and chlorophylls also have a fifth five-membered ring not present in heme.

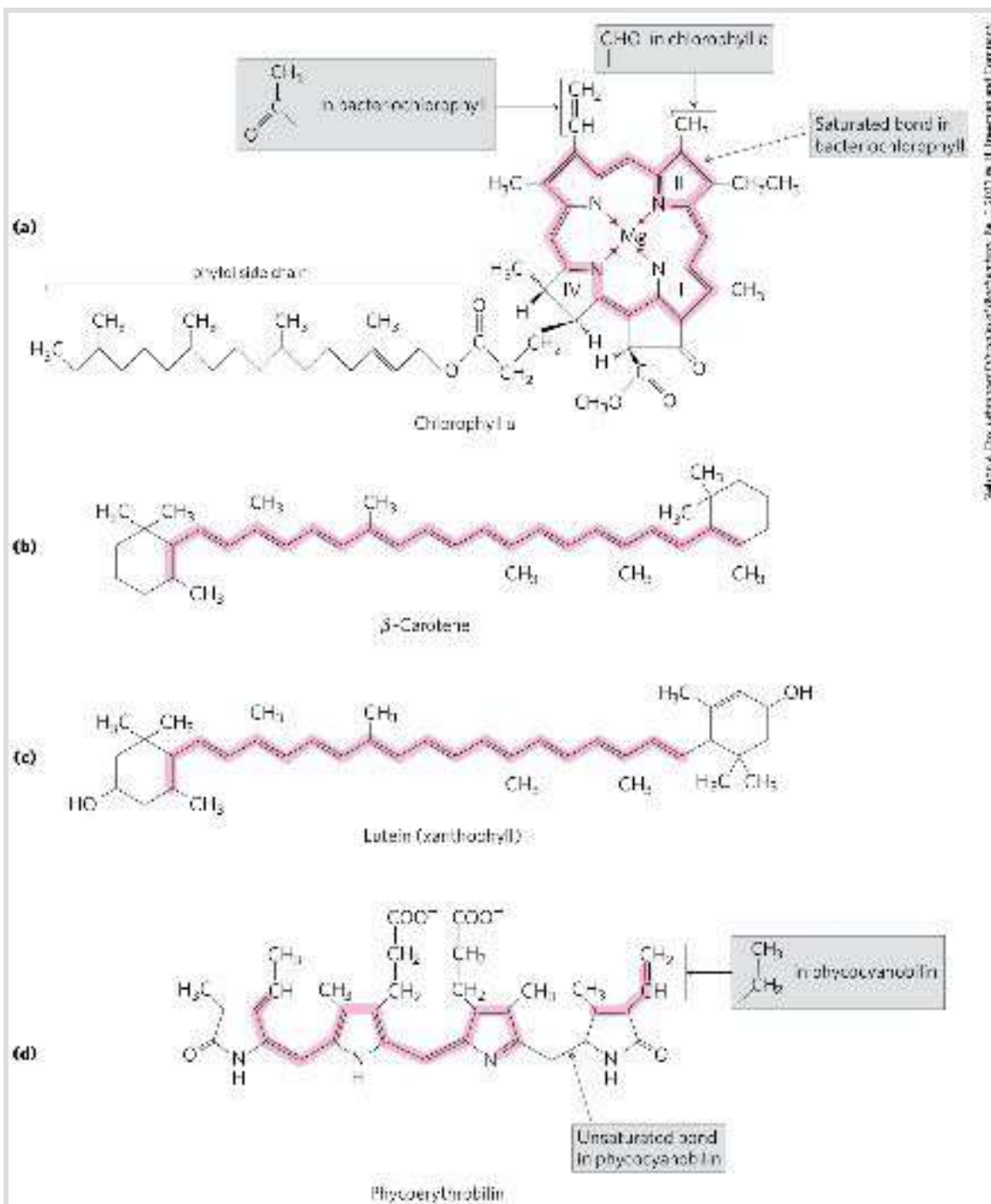


FIGURE 20-5 Primary and secondary photopigments. (a) Chlorophylls *a* and *b* and bacteriochlorophyll are the primary gatherers of light energy. (b) β -Carotene (a carotenoid) and (c) lutein (a xanthophyll) are accessory pigments in plants. (d) Phycoerythrobilin and phycocyanobilin (phycobilins) are accessory pigments in cyanobacteria and red algae. The conjugated bond systems in these molecules (alternating single and double bonds, shaded) have delocalized electrons that are easily excited by photons with the wavelengths of visible light.

The heterocyclic five-ring system that surrounds the Mg^{2+} has an extended polyene structure, with alternating single and double bonds. Such polyenes characteristically show strong absorption in the visible region of the spectrum ([Fig. 20-6](#)); the chlorophylls have unusually high molar extinction coefficients (see [Box 3-1](#)) and are therefore particularly well-suited for absorbing visible light during photosynthesis.

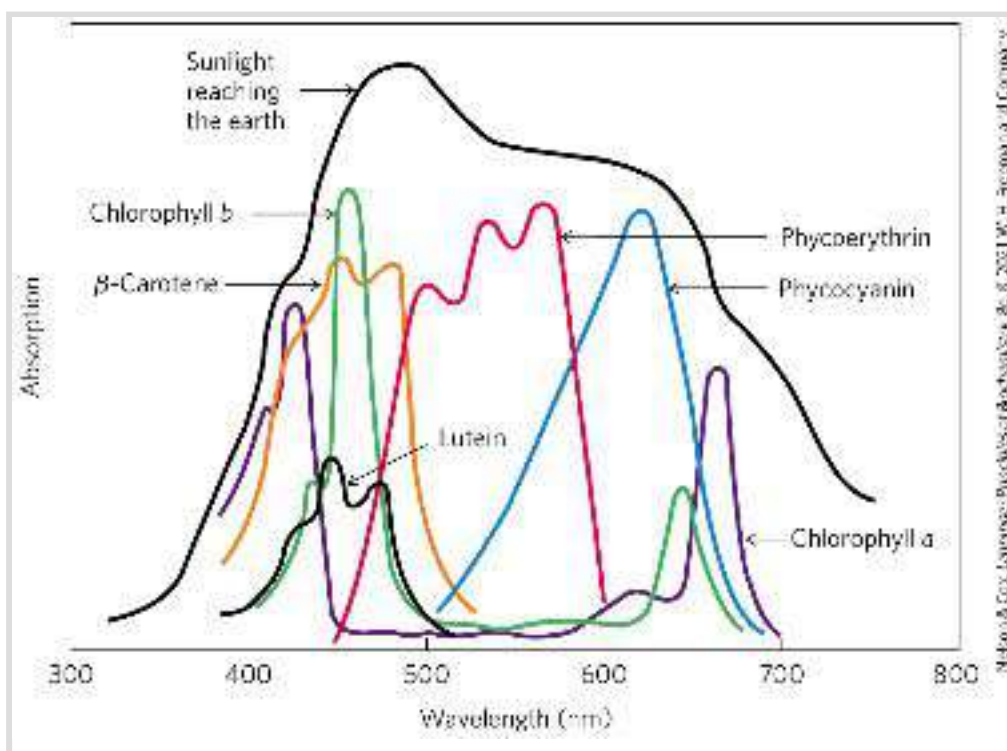


FIGURE 20-6 Absorption of visible light by photopigments. Plants are green because their pigments absorb light from the red and blue regions of the spectrum, leaving primarily green light to be reflected. Compare the absorption spectra of the pigments with the spectrum of sunlight reaching the earth's surface; the combination of chlorophylls (*a* and *b*) and accessory pigments enables plants to harvest most of the energy available in sunlight. The relative amounts of chlorophylls and accessory pigments are characteristic of a particular plant species. Variation in the proportions of these pigments is responsible for the range of colors of photosynthetic organisms, from the deep blue-green of spruce needles, to the greener green of maple leaves, to the red, brown, or purple color of some species of

multicellular algae and the leaves of some foliage plants favored by gardeners.

Chloroplasts always contain both chlorophyll *a* and chlorophyll *b* ([Fig. 20-5a](#)). Although both are green, their absorption spectra are sufficiently different ([Fig. 20-6](#)) that they complement each other's range of light absorption in the visible region. Most plants contain about twice as much chlorophyll *a* as chlorophyll *b*. The chlorophyll in cyanobacteria differs only slightly from those of plants.

In addition to chlorophylls, thylakoid membranes of plants contain secondary light-absorbing pigments, or [accessory pigments](#), called carotenoids. [Carotenoids](#) may be yellow, red, or purple. The two most prominent in plant leaves are β -**carotene**, a red-orange isoprenoid, and the yellow carotenoid lutein ([Fig. 20-5b, c](#)). Cyanobacteria and red algae use the accessory pigments phycocyanobilin and phycoerythrobilin ([Fig. 20-5d](#)). Accessory pigments absorb light at wavelengths not absorbed by the chlorophylls ([Fig. 20-6](#)) and thus are supplementary light receptors. They also protect downstream components from a highly reactive form of oxygen (singlet oxygen) that is formed when intense light exceeds the system's capacity to accept electrons.

Experimental determination of the effectiveness of light of different colors in promoting photosynthesis yields an [action spectrum](#) ([Fig. 20-7](#)), often useful in identifying the pigment

primarily responsible for a biological effect of light. By capturing light in a region of the spectrum not used by other organisms, a photosynthetic organism can claim a unique ecological niche.

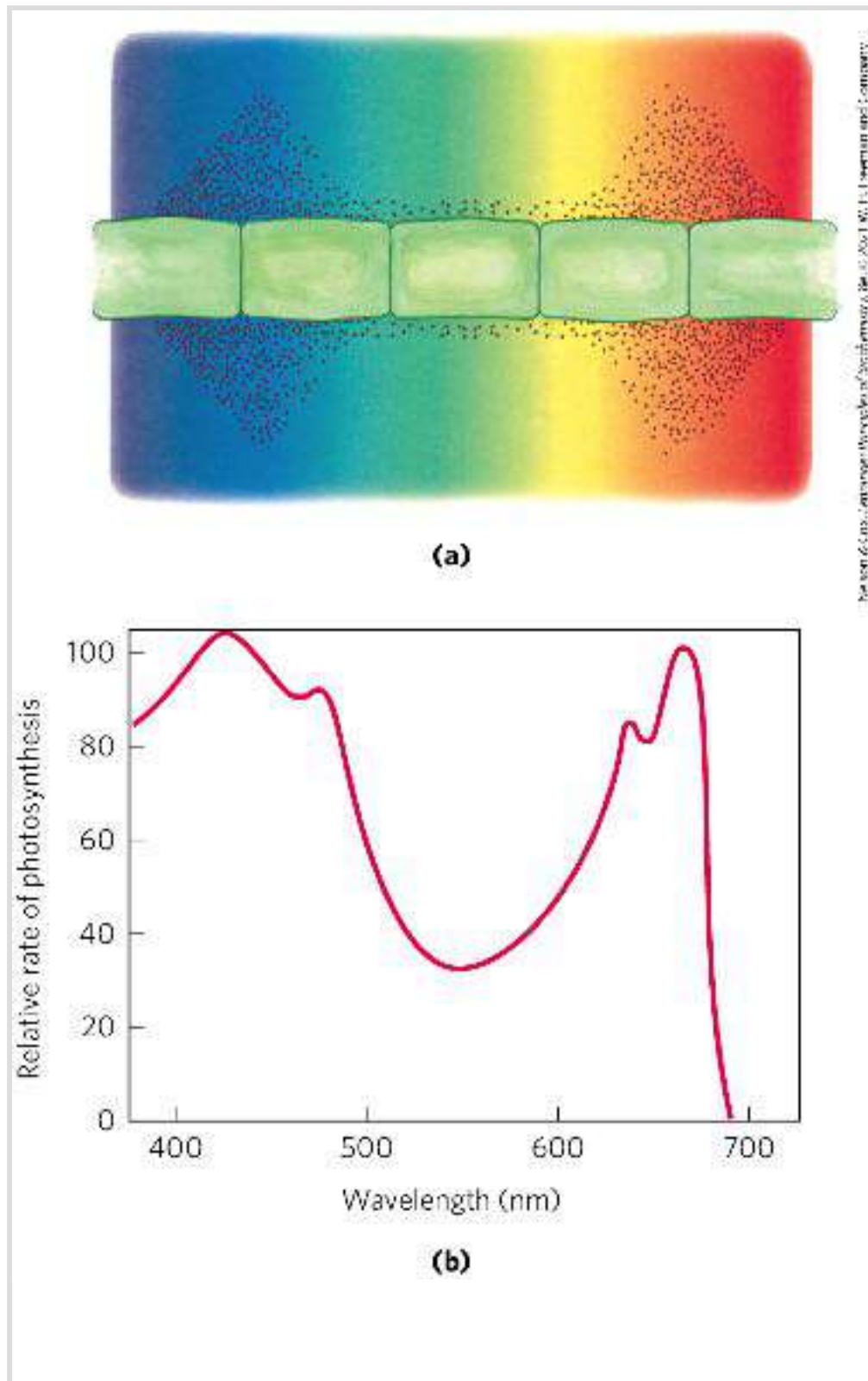
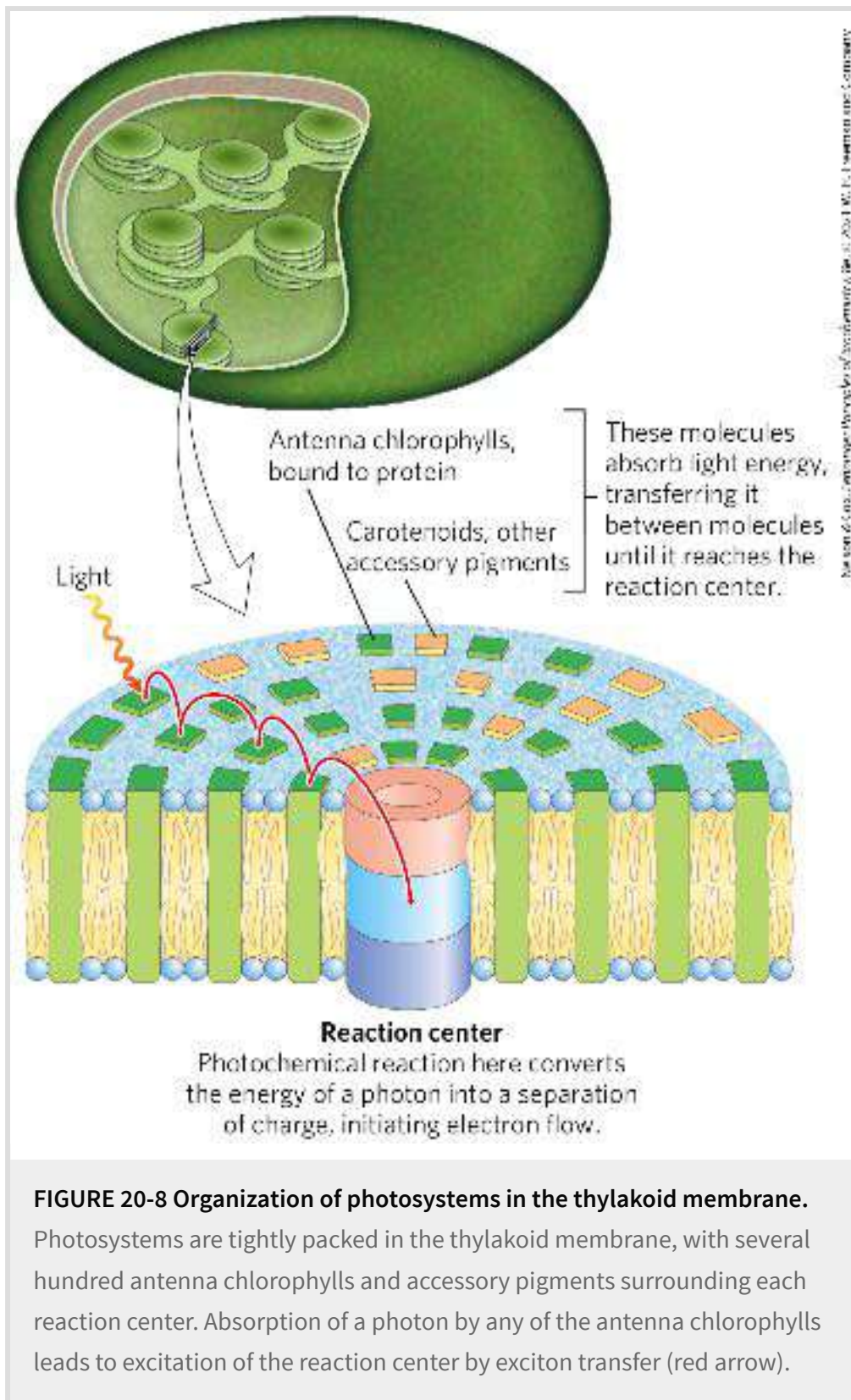


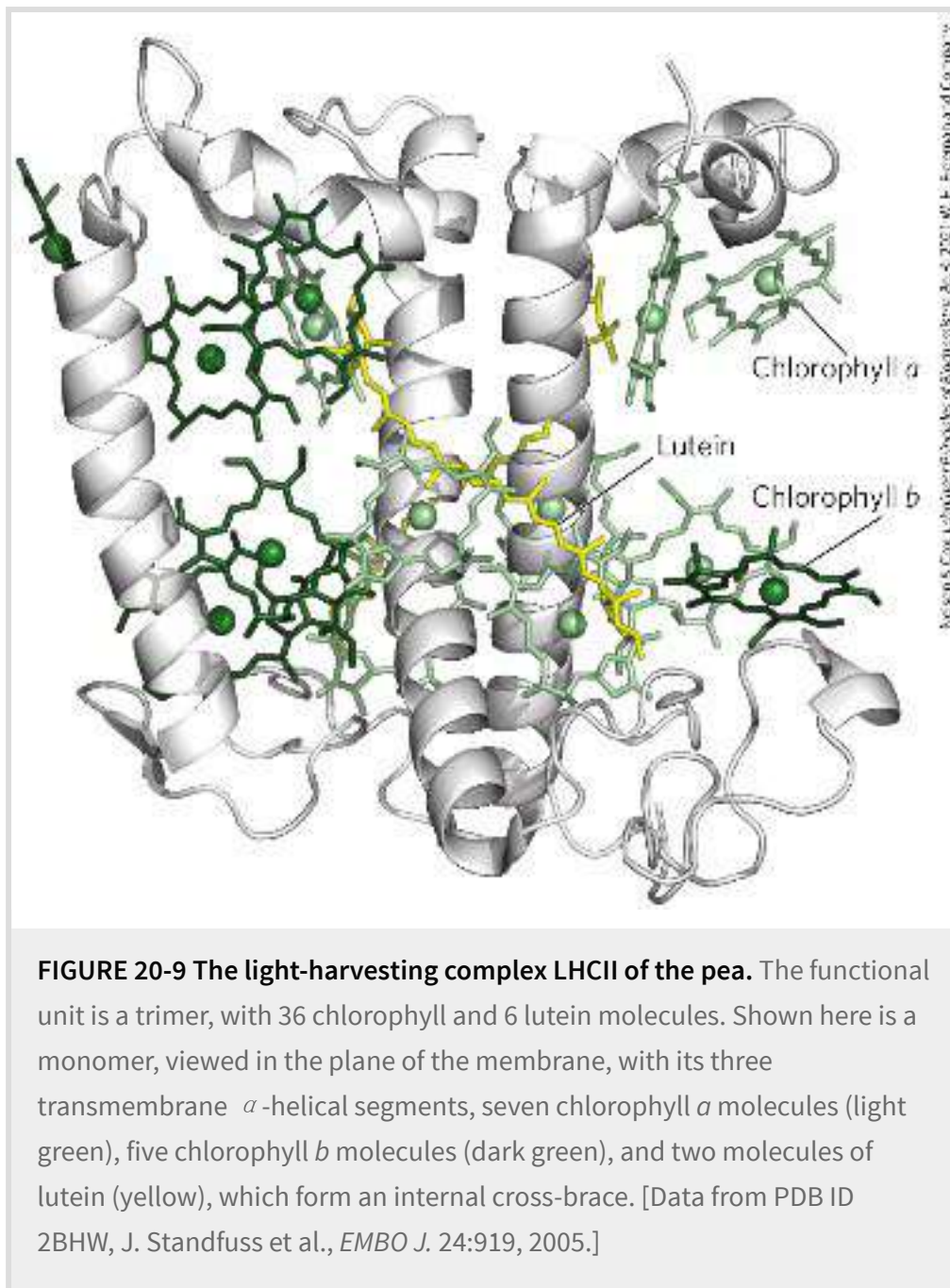
FIGURE 20-7 Two ways to determine the action spectrum for photosynthesis. (a) Results of a classic experiment performed by T. W. Engelmann in 1882 to determine the wavelength of light that is most effective in supporting photosynthesis. Engelmann placed cells of a filamentous photosynthetic alga on a microscope slide and illuminated them with light from a prism, so that one part of the algal filament received mainly blue light, another part yellow, another red. To determine which cells carried out photosynthesis most actively, Engelmann also placed on the microscope slide bacteria known to migrate toward regions of high O₂ concentration. After a period of illumination, the distribution of bacteria showed highest O₂ levels (produced by photosynthesis) in the regions illuminated with violet and red light. (b) Results of a similar experiment that used modern techniques (an oxygen electrode) for the measurement of O₂ production. An action spectrum, as shown here, describes the relative rate of photosynthesis for illumination with a constant number of photons of different wavelengths. An action spectrum is useful because, by comparison with absorption spectra (such as those in [Fig. 20-6](#)), it suggests which pigments can channel energy into photosynthesis.

Chlorophylls Funnel Absorbed Energy to Reaction Centers by Exciton Transfer

The light-absorbing pigments of thylakoid or bacterial membranes are arranged in functional arrays called [photosystems](#). In spinach chloroplasts, for example, each photosystem contains about 200 chlorophyll and 50 carotenoid molecules. All the pigment molecules in a photosystem can absorb photons, but  only one pair of chlorophyll molecules associated with the [photochemical reaction center](#) is specialized to transduce light into chemical energy. The other

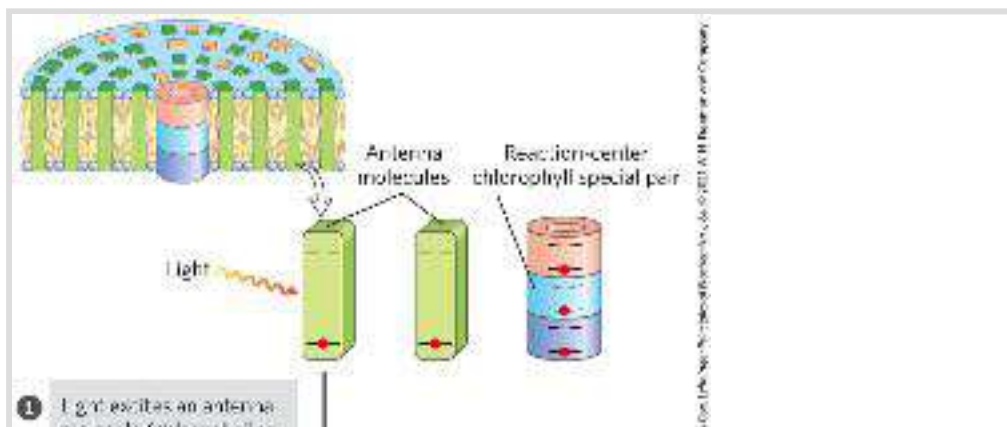
pigment molecules in a photosystem serve as **antenna molecules**. They absorb light energy and transmit it rapidly and efficiently to the reaction center ([Fig. 20-8](#)). Some chlorophylls are part of a core complex around the reaction center. Others form **light-harvesting complexes (LHCs)** around the periphery of the core complex. Chlorophyll and other pigments are always associated with specific binding proteins, which fix the chromophores in relation to each other, to other protein complexes, and to the membrane. For example, each monomer of the trimeric light-harvesting complex LHCII ([Fig. 20-9](#)) contains seven molecules of chlorophyll *a*, five of chlorophyll *b*, and two of lutein.





The chlorophyll molecules in light-harvesting complexes and other chlorophyll-binding proteins have light-absorption properties that are subtly different from those of free chlorophyll. When isolated chlorophyll molecules are excited by light, the absorbed energy is quickly released as fluorescence and heat; but when chlorophyll in intact leaves is excited by visible light ([Fig.](#)

[20-10](#), step ❶), very little fluorescence is observed. Instead, the excited antenna chlorophyll transfers energy directly to a neighboring chlorophyll molecule, which becomes excited as the first molecule returns to its ground state (step ❷). This transfer of energy, exciton transfer, extends to a third, fourth, or subsequent neighbor, until one of a “special pair” of chlorophyll *a* molecules at the photochemical reaction center is excited (step ❸). The special pair of chlorophyll molecules, often designated (Chl)₂, are held close enough to each other to share bonding orbitals, and to react as a single compound when excited. In this excited chlorophyll pair, an electron is promoted to a higher-energy orbital. This electron then passes to a nearby electron acceptor that is part of the photosynthetic electron-transfer chain, leaving the reaction-center chlorophyll pair with a missing electron (an “electron hole,” denoted by + in [Fig. 20-10](#)) (step ❹). The electron acceptor acquires a negative charge in this transaction. The electron lost by the reaction-center chlorophyll pair is replaced by an electron from a neighboring electron-donor molecule (step ❺), which thereby becomes positively charged. ➡ P1 ➡ P2 In this way, *excitation by light causes electric charge separation and initiates an oxidation-reduction chain.*



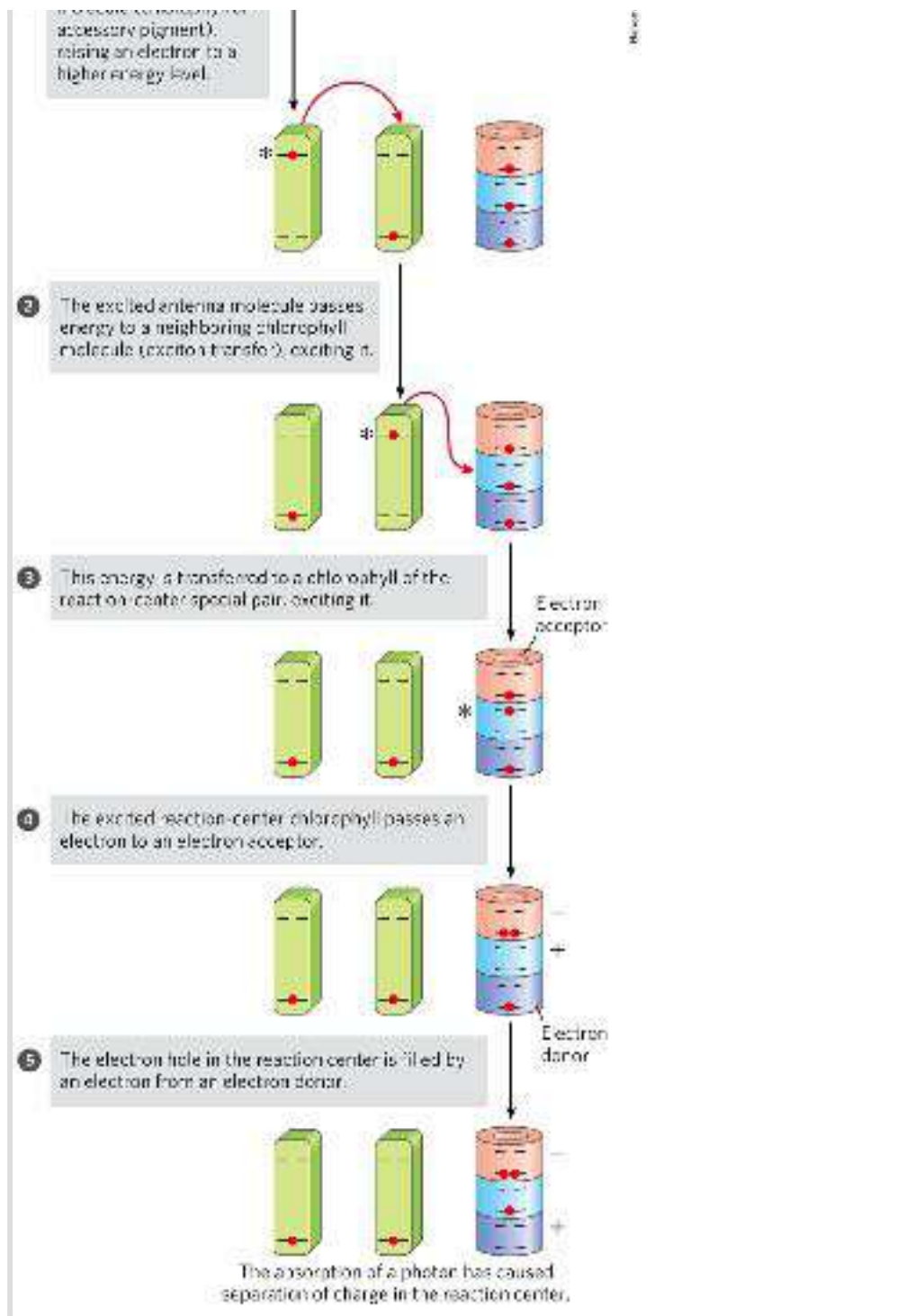


FIGURE 20-10 Exciton and electron transfer. This generalized scheme shows conversion of the energy of an absorbed photon into separation of charges at the reaction center. Note that step 1 may repeat between successive antenna molecules until the exciton reaches the special pair of chlorophylls in the reaction center. An asterisk (*) denotes the excited state of a molecule.

SUMMARY 20.1 *Light Absorption*

- Photosynthesis takes place in plant chloroplasts, structures enclosed in double membranes and filled with an elaborate system of thylakoid membranes containing the photosynthetic machinery.
- Chlorophyll molecules and other light-absorbing pigments are associated with proteins in light-harvesting complexes arrayed around photochemical reaction centers. The proteins are embedded in thylakoid membranes.
- The many chlorophyll molecules that surround the reaction center serve as antennas for light. When they absorb light, they pass its energy (exciton) to the reaction center. There the energy is used to create a charge separation that initiates electron flow through a series of oxidation-reduction reactions.

20.2 Photochemical Reaction Centers

Studies on a variety of bacteria that carry out photosynthesis have been helpful in understanding the mechanisms of photosynthesis in cyanobacteria, algae and vascular plants. Photosynthetic bacteria have relatively simple phototransduction machinery, with one of two general types of photosystems. Both systems send electrons through a cytochrome complex that pumps protons, producing the electrochemical gradient that drives ATP synthesis.

Photosynthetic Bacteria Have Two Types of Reaction Center

The **type II photosystem** in purple bacteria consists of three basic modules ([Fig. 20-11a](#)): a single P870 reaction center; a cytochrome bc_1 electron-transfer complex similar to Complex III of the mitochondrial electron-transfer chain; and an ATP synthase, also similar to that of mitochondria. Illumination lifts an electron in the reaction center to its excited state ($P870^*$), from which it passes through **pheophytin** (chlorophyll *a* lacking its central Mg^{2+}) and a quinone to the cytochrome bc_1 complex. After passing through the bc_1 complex, electrons flow through cytochrome c_2 back to the reaction center, restoring its preillumination state and completing one cycle. This light-driven [cyclic electron transfer](#) provides the energy for proton pumping

by the cytochrome bc_1 complex. Powered by the resulting proton gradient, ATP synthase produces ATP, exactly as in mitochondria.

The **type I photosystem** in green sulfur bacteria involves the same three modules as in purple bacteria, but the process differs in several respects and includes additional enzymatic reactions ([Fig. 20-11b](#)). Excitation by light causes an electron to move from the excited reaction center to the cytochrome bc_1 complex via a quinone carrier. Electron transfer through this complex powers proton transport and creates the proton-motive force used for ATP synthesis, just as in purple bacteria and in mitochondria. However, in contrast to the cyclic electron transfer path in purple bacteria, some electrons follow a [linear electron transfer](#) path from the reaction center to the soluble iron-sulfur protein [ferredoxin](#) (see [Fig. 19-5](#)), which then passes electrons via **ferredoxin: NAD^+ reductase** to NAD^+ , producing NADH. The electrons taken from the reaction center to reduce NAD^+ are replaced by the oxidation of H_2S to elemental S in the reaction that defines the green sulfur bacteria. This oxidation of H_2S by bacteria is chemically analogous to the oxidation of H_2O by oxygenic plants. Note that the path of electrons in the purple bacteria is cyclic; the path in the green sulfur bacteria can be either cyclic or linear, leading to NAD^+ and producing NADH.

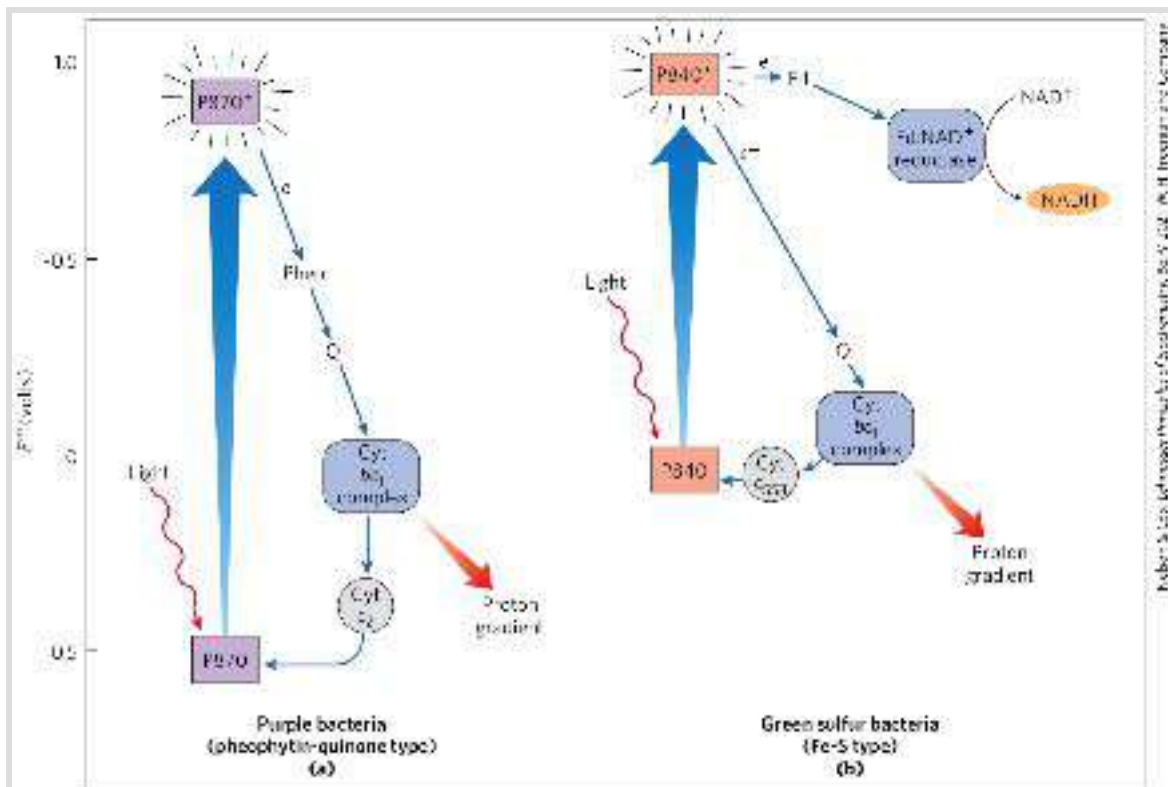


FIGURE 20-11 Functional Modules of Photosynthetic Machinery in Purple Bacteria and Green Sulfur Bacteria. The position on the vertical scale of each electron carrier reflects its standard reduction potential. (a) In purple bacteria, light energy excites an electron in the reaction center P870. The electron passes through pheophytin (Pheo), a quinone (Q), and the cytochrome bc_1 complex, then through cytochrome c_2 and thus back to the reaction center. Electron transfer through the cytochrome bc_1 complex causes proton pumping, creating an electrochemical potential that powers ATP synthesis. (b) Green sulfur bacteria have two routes for electrons driven by excitation of P840. A cyclic electron transfer route that goes through a quinone to the cytochrome bc_1 complex and back to the reaction center via cytochrome c_{553} causes proton pumping. A linear electron transfer route that goes from the reaction center through the iron-sulfur protein ferredoxin (Fd) reduces NAD^+ to $NADH$ in a reaction catalyzed by ferredoxin: NAD^+ reductase.

In Vascular Plants, Two Reaction Centers Act in Tandem

The photosynthetic apparatus of cyanobacteria, algae, and vascular plants is more complex than the one-center bacterial systems, and it most likely evolved through the combination of two simpler bacterial photosystems. The [Z scheme](#) diagram in [Figure 20-12](#) outlines the path of electron flow between the two photosystems and the energy relationships in the light-dependent reactions. (The Z scheme takes its name from the zigzag pattern of the pathways in the diagram.) The thylakoid membranes of chloroplasts have two different kinds of photosystems, each with its own type of photochemical reaction center and set of antenna molecules. The two systems have distinct and complementary functions. **Photosystem II (PSII)** is a pheophytin-quinone type of system (like the single photosystem of purple bacteria) containing roughly equal amounts of chlorophylls *a* and *b*. Excitation of the P680 special pair in its reaction center drives electrons through the cytochrome *b₆f* complex discussed below, with concomitant pumping of protons across the thylakoid membrane and ATP synthesis. **Photosystem I (PSI)** is structurally and functionally related to the photosynthetic machinery of green sulfur bacteria. It has a P700 reaction center and a high ratio of chlorophyll *a* to chlorophyll *b*. The excited P700 passes electrons through a linear chain of carriers to ferredoxin, then to NADP⁺, producing NADPH. An alternative pathway for electrons is cyclic: instead of following the linear path that leads to NADP⁺ reduction, electrons pass to plastoquinone (PQ) through a membrane-embedded protein complex, **cytochrome *b₆f*** (again, with the movement of protons into the chloroplast lumen). The thylakoid

membranes of a single spinach chloroplast have many hundreds of each kind of photosystem.

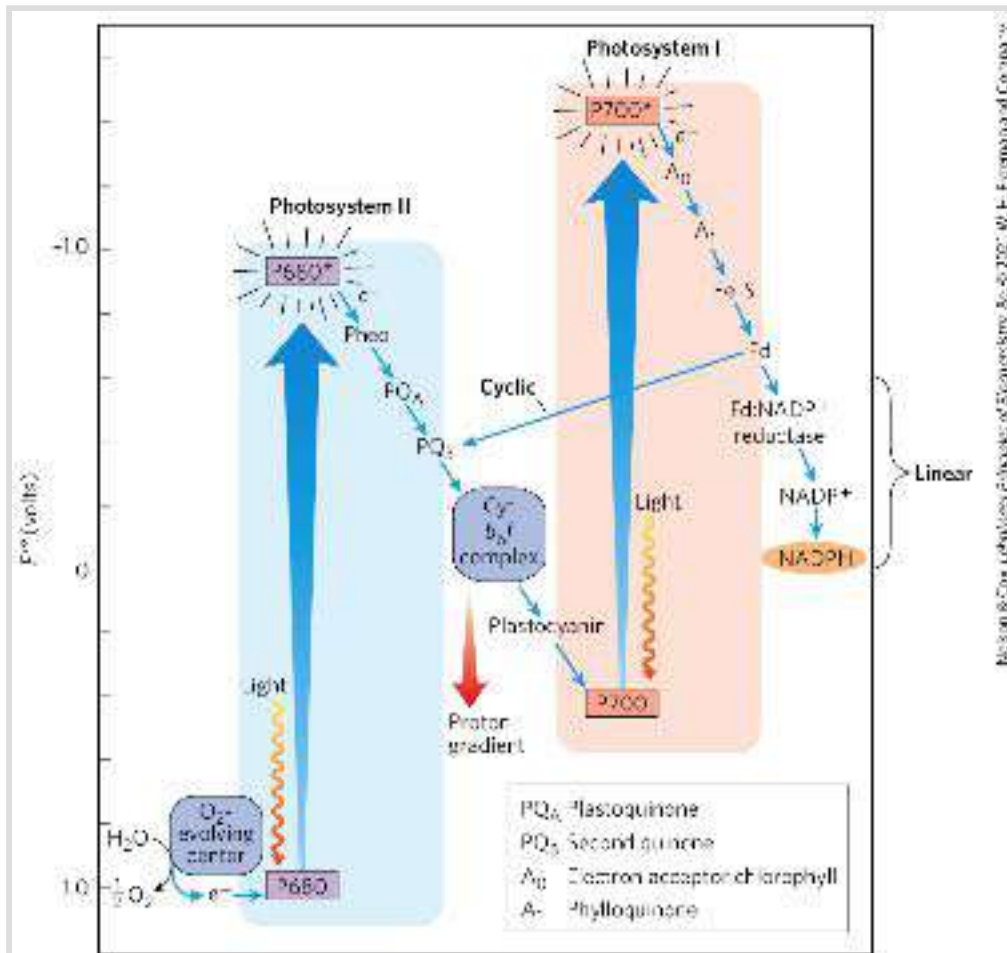


FIGURE 20-12 Integration of photosystems I and II in chloroplasts. This “Z scheme” shows the pathway of linear electron transfer from H_2O (lower left) to $NADP^+$ (far right). The position on the vertical scale of each electron carrier reflects its standard reduction potential. To raise the energy of electrons derived from H_2O to the energy level required to reduce $NADP^+$ to NADPH, each electron must be “lifted” twice (heavy arrows) by photons absorbed in PSII and PSI. One photon is required per electron in each photosystem. After excitation, the high-energy electrons flow “downhill” through the carrier chains as shown. Protons move across the thylakoid membrane during the water-splitting reaction and during electron transfer through the cytochrome b_6f complex, producing the proton gradient that is essential to ATP formation. An alternative path of electrons is cyclic electron transfer, in which electrons move from ferredoxin back to the plastoquinone and cytochrome b_6f complex, instead

of reducing NADP^+ to NADPH. The cyclic pathway produces more ATP and less NADPH than the linear pathway.

These two photosystems in plants act in tandem to catalyze the light-driven movement of electrons from H_2O to NADP^+ . The electron carriers include large, integral protein complexes (PSI, PSII, and the proton-pumping complex cytochrome b_6f); quinones that are lipid-soluble and move through the membrane between the protein complexes; and two soluble proteins, plastocyanin (analogous to cytochrome c in mitochondria) and ferredoxin.

To replace the electrons that move from PSII through PSI to NADP^+ , H_2O is oxidized, producing O_2 ([Fig. 20-12](#), bottom left). All O_2 -evolving photosynthetic cells — those of plants, algae, and cyanobacteria — contain both PSI and PSII.  The Z scheme thus describes the complete route by which electrons flow from H_2O to NADP^+ , according to the equation



For every two photons absorbed (one by each photosystem), one electron is transferred from H_2O to NADP^+ . To form one molecule of O_2 , which requires transfer of four electrons from two H_2O to two NADP^+ , a total of eight photons must be absorbed, four by each photosystem.

Having seen the overall process, we'll now look at how the structure of the photosystems informs our understanding of the electrochemistry.

Photosystem II

PSII is dimeric ([Fig. 20-13](#)). Each monomer is a huge complex of 19 proteins, including the accessory proteins CP47 and CP43, and the core complex of P680 reaction-center proteins D1 and D2; 2 chlorophyll-binding proteins; and associated chromophores, including carotenoids, a nonheme iron, and the critically important inorganic cofactor, Mn_4CaO_5 . Of the proteins in PSII, 16 have transmembrane segments, but 3 are peripheral proteins on the lumenal side that stabilize the Mn_4CaO_5 cofactor.

Surrounding PSII are additional chlorophyll-binding proteins and light-harvesting complexes.  When a photon is absorbed by any of these antenna molecules, the resulting exciton moves very rapidly from one to another of the antenna chlorophylls until it reaches the reaction center and excites P680, the special pair of chlorophyll *a* molecules ($\text{Chl } a$)₂, to initiate the photochemistry.

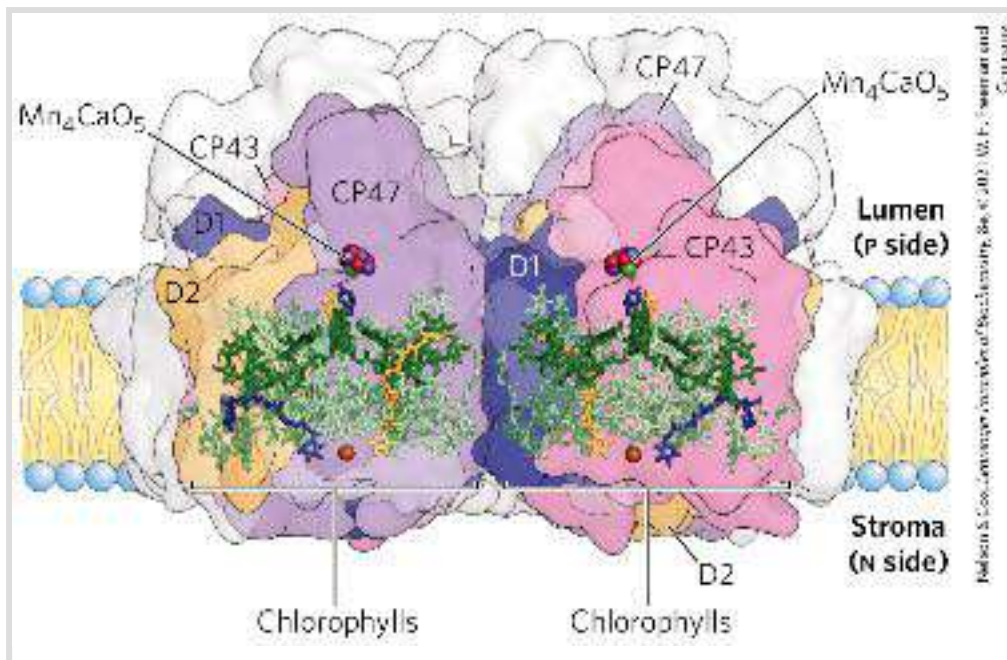
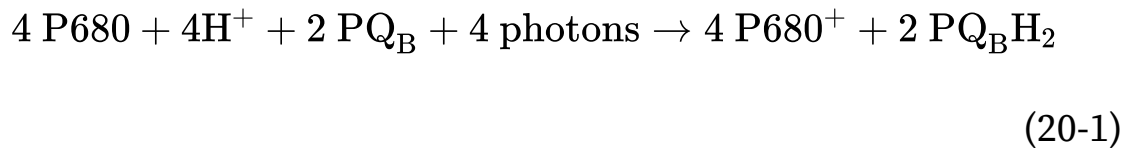


FIGURE 20-13 Structure of photosystem II of the cyanobacterium

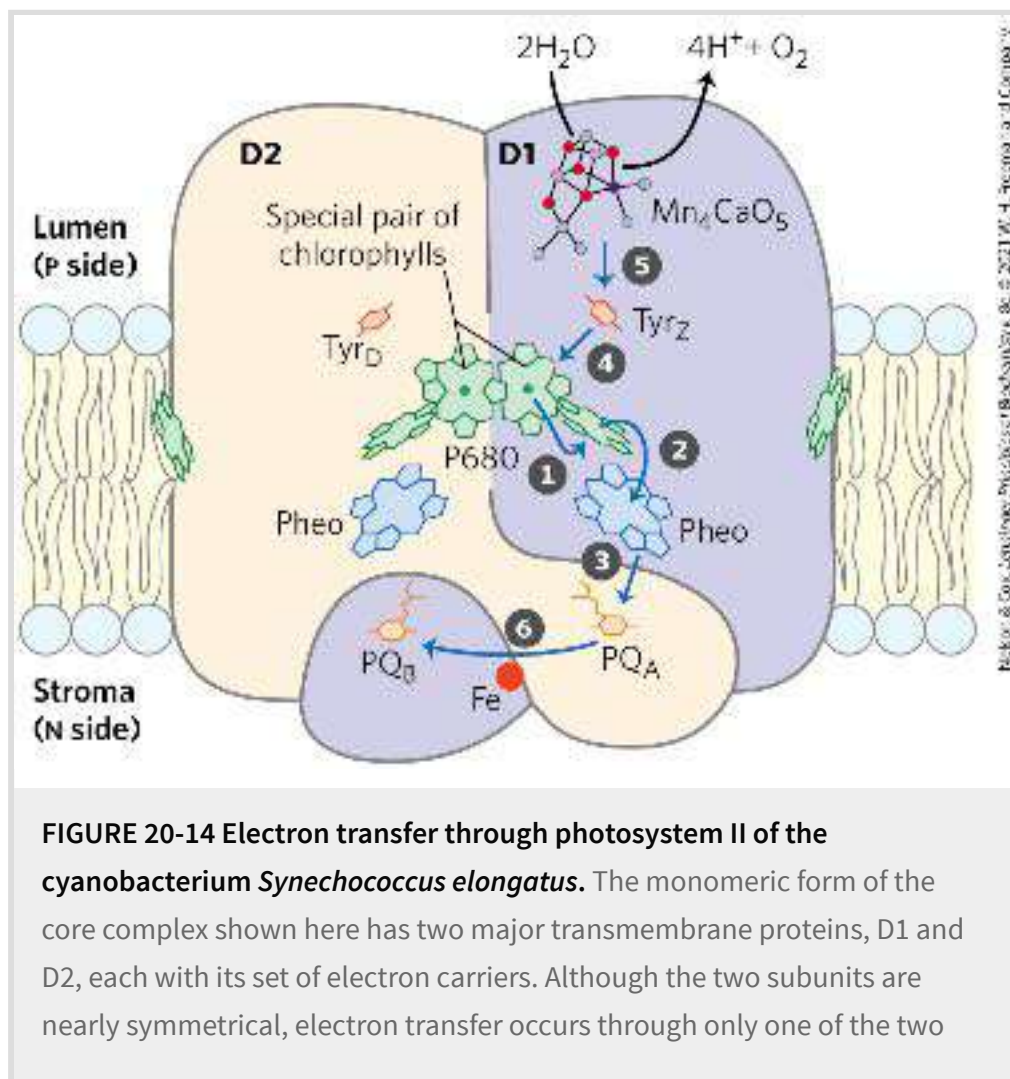
Thermosynechococcus vulcanus. The enormous complex, visualized by x-ray crystallography, is a dimer; each monomer has its own reaction center. Chlorophyll-binding proteins CP43 and CP47 form the core antenna, directly associated with the PSII reaction-center proteins D1 and D2. Each PSII monomer contains 35 chlorophylls, 2 pheophytins, 11 β -carotenes, 2 plastoquinones, and 1 each of *b*-type cytochrome, *c*-type cytochrome, and nonheme iron. Water is oxidized to form O_2 at the oxygen-evolving center (Mn_4CaO_5). [Data from PDB ID 3WU2, Y. Umena et al., *Nature* 473:55, 2011.]

Excitation of P680 in PSII ([Fig. 20-14](#)) produces $P680^*$, an excellent electron donor that, within picoseconds, transfers an electron to pheophytin, giving it a negative charge ($\bullet Pheo^-$). With the loss of its electron, $P680^*$ is transformed into a radical cation, $P680^+$. $\bullet Pheo^-$ very rapidly passes its extra electron to a protein-bound **plastoquinone, PQ_A** , which in turn passes its electron to another, more loosely bound plastoquinone, PQ_B . When PQ_B has acquired two electrons in two such transfers from PQ_A and two

protons from the solvent water, it is in its fully reduced quinol form, PQ_BH_2 . The overall reaction initiated by light in PSII is



Eventually, the electrons in PQ_BH_2 pass through the cytochrome b_6f complex (see [Fig. 20-12](#)). The electron initially removed from P680 is replaced with an electron obtained from the oxidation of water, as described below.



branches of electron carriers: that on the right (in D1). The arrows show the path of electron transfer from the Mn_4CaO_5 ion cofactor of the oxygen-evolving complex to plastoquinone PQ_B . The photochemical events occur in the sequence indicated by the step numbers. The role of the Tyr residues and the detailed structure of the Mn_4CaO_5 cofactor are discussed below (see [Fig. 20-20b](#)).

Photosystem I

PSI and its antenna molecules are part of a supramolecular complex composed of at least 16 proteins, including 4 chlorophyll-binding proteins arranged around the periphery of the reaction center ([Fig. 20-15](#)). The complex also includes 35 carotenoids of several types, three 4Fe-4S clusters, and two phylloquinones.

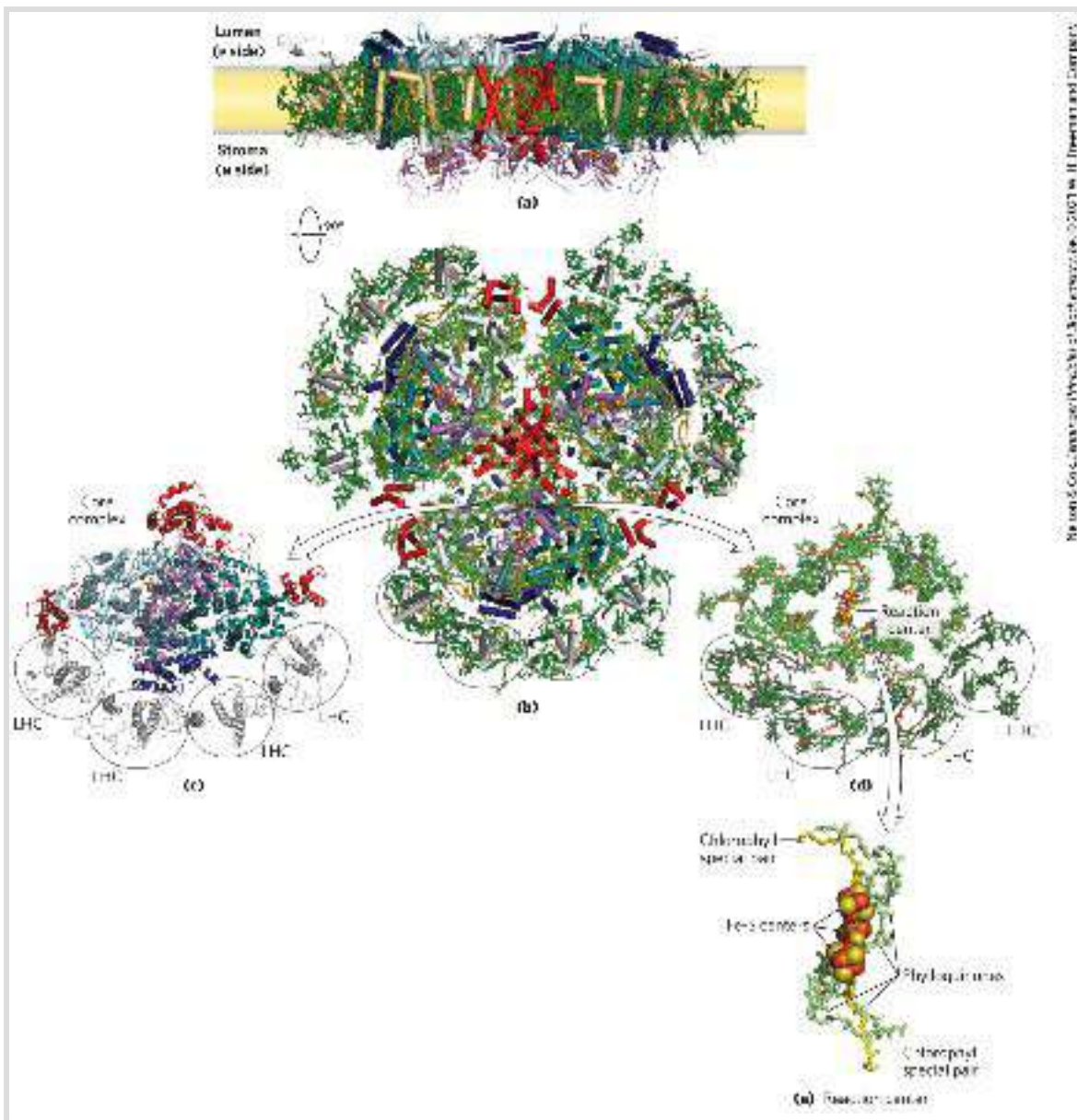
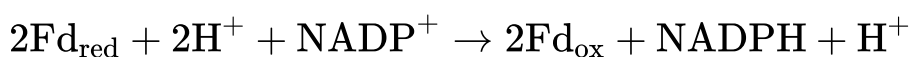
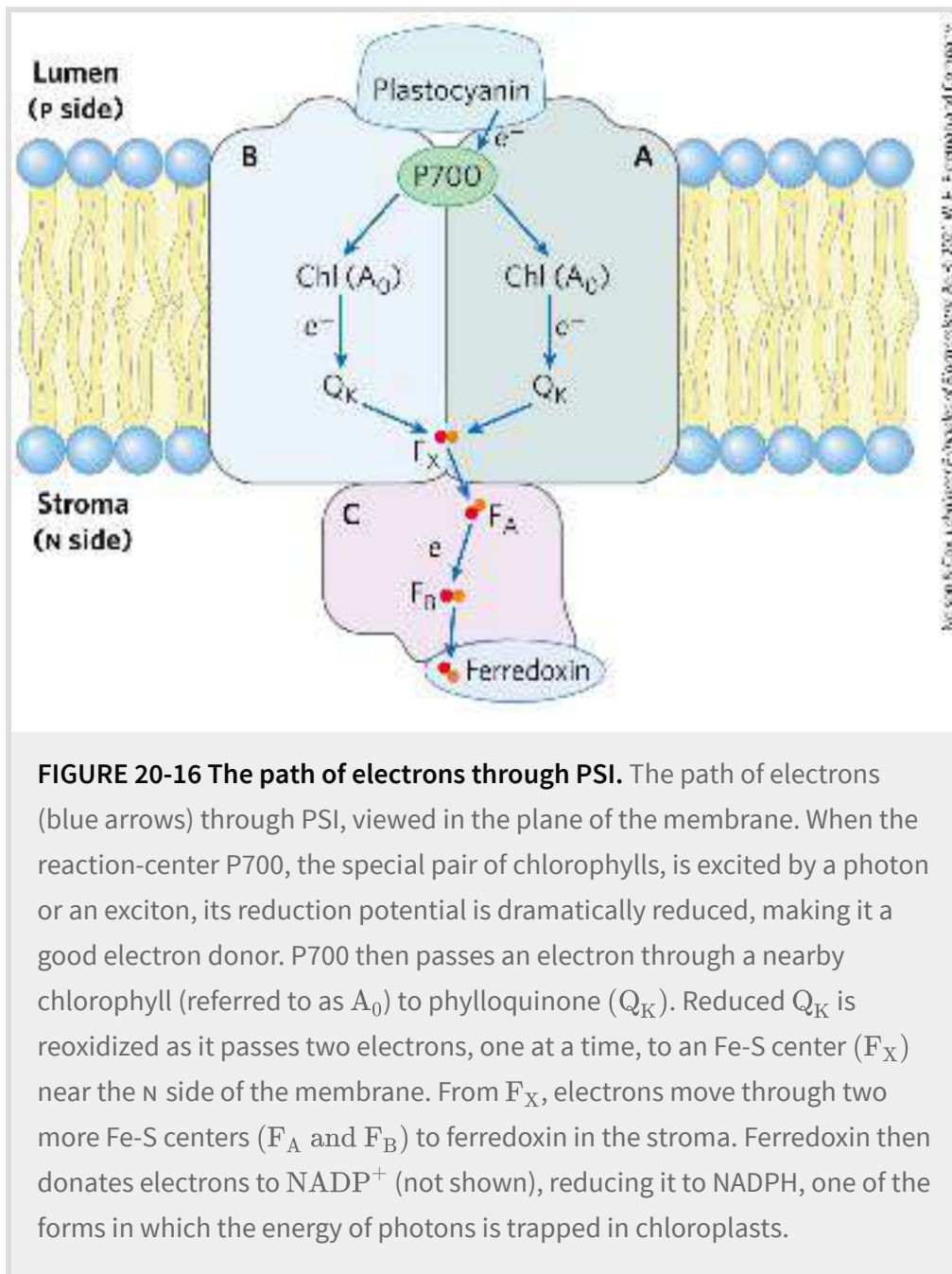


FIGURE 20-15 Structure of photosystem I in the cyanobacterium *Synechococcus elongatus*. PSI is a symmetric trimer, viewed here (a) in the plane of the thylakoid membrane and (b) from the stroma (N side of the membrane). (c) One of the three core complexes in PSI, displayed as the protein without its ligands and (d) the ligands alone. Note the four peripheral light-harvesting complexes (LHC) and the many chlorophyll molecules surrounding the reaction center. (e) Close-up view of the reaction center without the surrounding chlorophylls, showing the chlorophyll special pair, phylloquinones, and Fe-S centers. [Data from PDB ID 1JBO, P. Jordan et al., *Nature* 411:909, 2001; PDB ID 4RKU, Y. Mazor et al.]

The photochemical events that follow the excitation of PSI at the reaction-center P700 ([Fig. 20-16](#)) are formally similar to those occurring in PSII. The excited reaction-center P700* loses an electron to an acceptor, designated A₀ (a chlorophyll *a* molecule, functionally homologous to the pheophytin of PSII), creating A₀⁻ and P700⁺. Again, excitation results in charge separation at the photochemical reaction center. P700⁺ is a strong oxidizing agent, which quickly acquires an electron from [plastocyanin](#), a soluble Cu-containing electron-transfer protein. A₀⁻ is an exceptionally strong reducing agent that passes its electron through a chain of carriers that leads to NADP⁺ ([Fig. 20-12](#), right side).

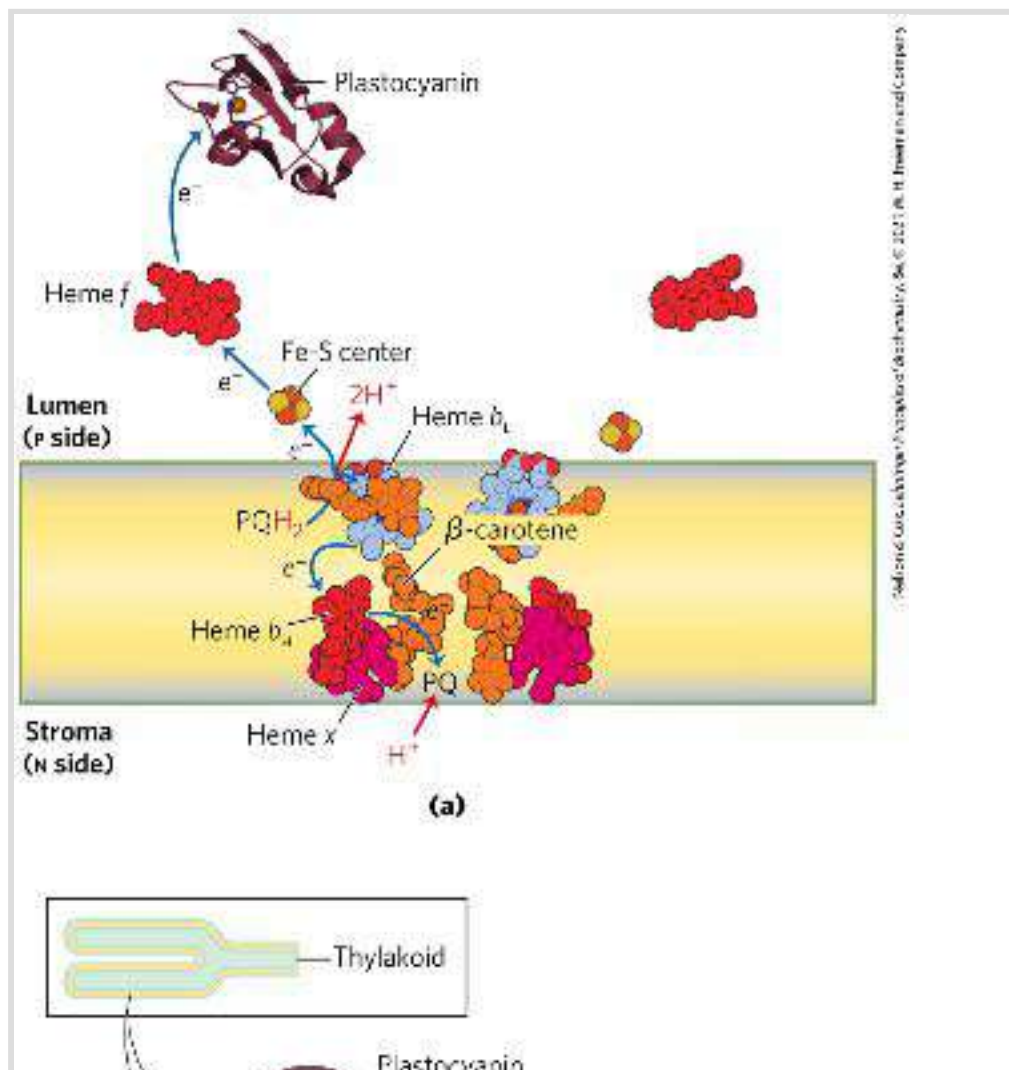
Phylloquinone (Q_K) accepts the electron and passes it to an iron-sulfur protein through three Fe-S centers in PSI. From here, the electron moves to ferredoxin (Fd). Recall that ferredoxin contains a 2Fe-2S center (see [Fig. 19-5](#)) that undergoes one-electron oxidation and reduction reactions. The fourth electron carrier in the chain is the flavoprotein **ferredoxin:NADP⁺ reductase**, which transfers electrons from reduced ferredoxin (Fd_{red}) to NADP⁺:





The Cytochrome b_6f Complex Links Photosystems II and I, Conserving the Energy of Electron Transfer

Electrons temporarily held in plastoquinol as a result of the excitation of P680 in PSII are carried to P700 of PSI via the cytochrome b_6f complex and the soluble protein plastocyanin (see [Fig. 20-12](#), center). With a structure and role analogous with that of Complex III in mitochondria, the cytochrome b_6f complex ([Fig. 20-17](#)) contains a b -type cytochrome with two heme groups (designated b_H and b_L), a Rieske iron-sulfur protein (M_r 20,000), and cytochrome f (named for the Latin *frons*, “leaf”). Electrons flow through the cytochrome b_6f complex from PQ_BH_2 to cytochrome f , then to plastocyanin, and finally to $P700^+$, thereby reducing it.



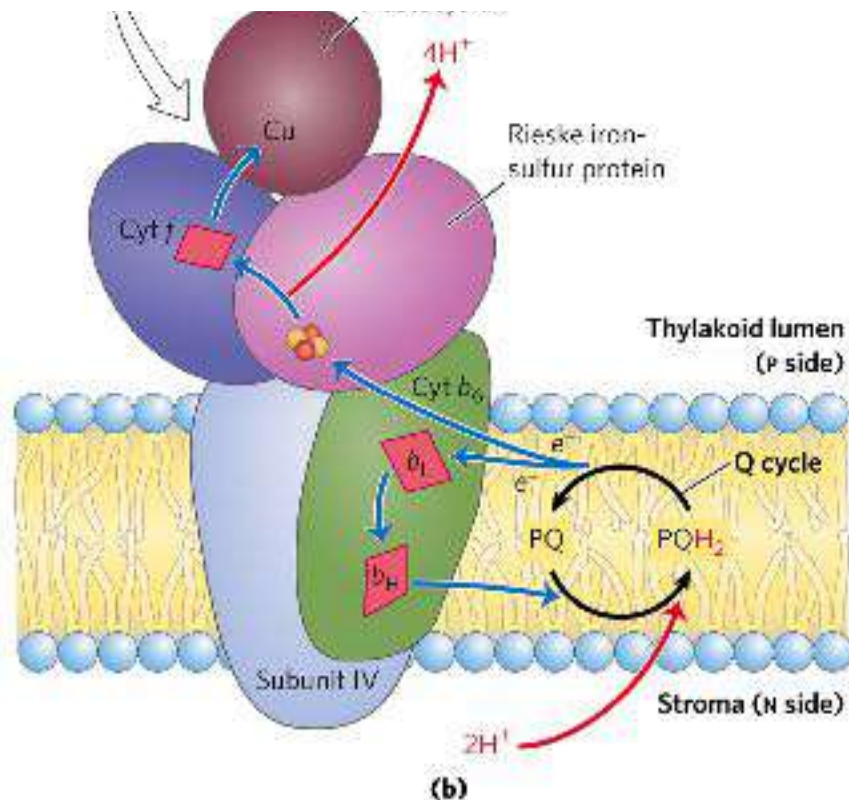


FIGURE 20-17 Electron and proton flow through the cytochrome b_6f complex.

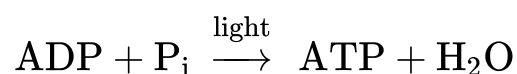
(a) In addition to the hemes of cytochrome b (heme b_H and b_L ; also called heme b_N and b_P , respectively, because of their proximity to the N and P sides of the bilayer) and cytochrome f (heme f), there is a fourth heme (heme x) near heme b_H ; also present is a β -carotene of unknown function. Two sites bind plastoquinone: the PQH_2 site near the P side of the bilayer, and the PQ site near the N side. The Fe-S center of the Rieske protein lies just outside the bilayer on the P side, and the heme f site is on a protein domain that extends well into the thylakoid lumen. The electron path is shown for just one of the monomers, but both sets of carriers in the dimer carry electrons to plastocyanin. (b) Plastoquinol (PQH_2), formed in PSII, is oxidized by the cytochrome b_6f complex in a series of steps like those of the Q cycle in Complex III of mitochondria (see [Fig. 19-11](#)). One electron from PQH_2 passes to the Fe-S center of the Rieske protein, the other to heme b_L of cytochrome b_6 . The net effect is passage of electrons from PQH_2 to the soluble protein plastocyanin, which carries them to PSI. [Data from PDB ID 1VF5, G. Kurisu et al., *Science* 302:1009, 2003; PDB ID 2Q5B, Y. S. Bukhman-DeRuyter et al.]

Like Complex III of mitochondria, cytochrome b_6f conveys electrons from a reduced quinone — a mobile, lipid-soluble carrier of two electrons (Q in mitochondria, PQ_B in chloroplasts; P for *plastoquinone*) — to a water-soluble protein that carries one electron (cytochrome *c* in mitochondria, plastocyanin in chloroplasts) ([Fig. 20-17a](#)). As in mitochondria, the function of this complex involves a Q cycle ([Fig. 20-17b](#); see [Fig. 19-11](#)) in which electrons pass, one at a time, from PQ_BH_2 to cytochrome b_6 . This cycle results in the pumping of protons across the membrane, from the stromal compartment to the thylakoid lumen. Up to four protons enter the lumen for each pair of electrons that passes through the cytochrome b_6f complex.  The result is production of a proton gradient across the thylakoid membrane as electrons pass from PSII to PSI. Because the volume of the flattened thylakoid lumen is small, the influx of a small number of protons has a relatively large effect on lumenal pH. The measured difference in pH between the stroma (pH 8) and the thylakoid lumen (pH 5) represents a 1,000-fold difference in proton concentration — a powerful driving force for ATP synthesis.

Cyclic Electron Transfer Allows Variation in the Ratio of ATP/NADPH Synthesized

Cyclic electron flow between PSI and cytochrome b_6f increases the production of ATP relative to NADPH. The linear path of

electrons from water, through PSII, cytochrome b_6f , and PSI to NADP^+ produces both a proton gradient, which is used to drive ATP synthesis, and NADPH, which is used in reductive biosynthetic processes (see [Fig. 20-12](#)). Some fraction of electrons passing from P700* to ferredoxin do not continue to NADP^+ , but cycle back through plastoquinone and the cytochrome b_6f complex to plastocyanin. Plastocyanin then donates electrons to P700. In this way, electrons are repeatedly recycled through the cytochrome b_6f complex and the reaction center of PSI, each electron propelled around the cycle by the energy of one photon. Cyclic electron flow is not accompanied by net formation of NADPH or evolution of O_2 . However, it is accompanied by proton pumping by the cytochrome b_6f complex and by phosphorylation of ADP to ATP, referred to as [cyclic photophosphorylation](#). The overall equation for cyclic electron flow and photophosphorylation is simply



 By regulating the partitioning of electrons between NADP^+ reduction and cyclic photophosphorylation, a plant adjusts the ratio of ATP to NADPH produced in the light-dependent reactions to match its needs for these products in the CO_2 -assimilation reactions and other biosynthetic processes. As we shall see in [Section 20.4](#), the CO_2 -assimilation reactions require ATP and NADPH in the ratio 3:2. This regulation of

electron-transfer pathways is part of a short-term adaptation to changes in light color (wavelength) and quantity (intensity).

State Transitions Change the Distribution of LHCII between the Two Photosystems

Photosynthetic organisms are exposed to light of highly variable intensity and wavelength in the course of a day or a season, and, although they can alter their growth patterns somewhat, they cannot uproot themselves and move to optimize their light exposure. Instead, cellular mechanisms have evolved that allow plants to accommodate changing light conditions. The energy needed to excite PSI (P700) is less (light of longer wavelength, lower energy) than the energy needed to excite PSII (P680). If PSI and PSII were physically contiguous, excitons originating in the antenna system of PSII would migrate to the reaction center of PSI, leaving PSII chronically underexcited and thus interfering with the operation of the two-center system. This imbalance in the supply of excitons is prevented by physically separating the two photosystems in the thylakoid membrane ([Fig. 20-18](#)). PSII is located almost exclusively in the tightly appressed membrane stacks of granal thylakoids; its associated light-harvesting complex (LHCII) mediates the tight association of adjacent membranes in the grana. PSI and the ATP synthase complex are located almost exclusively in the nonappressed membranes of the stromal thylakoids, where they have access to the contents of the

stroma, including ADP and NADP^+ . The cytochrome b_6f complex is present primarily in the granal thylakoids.

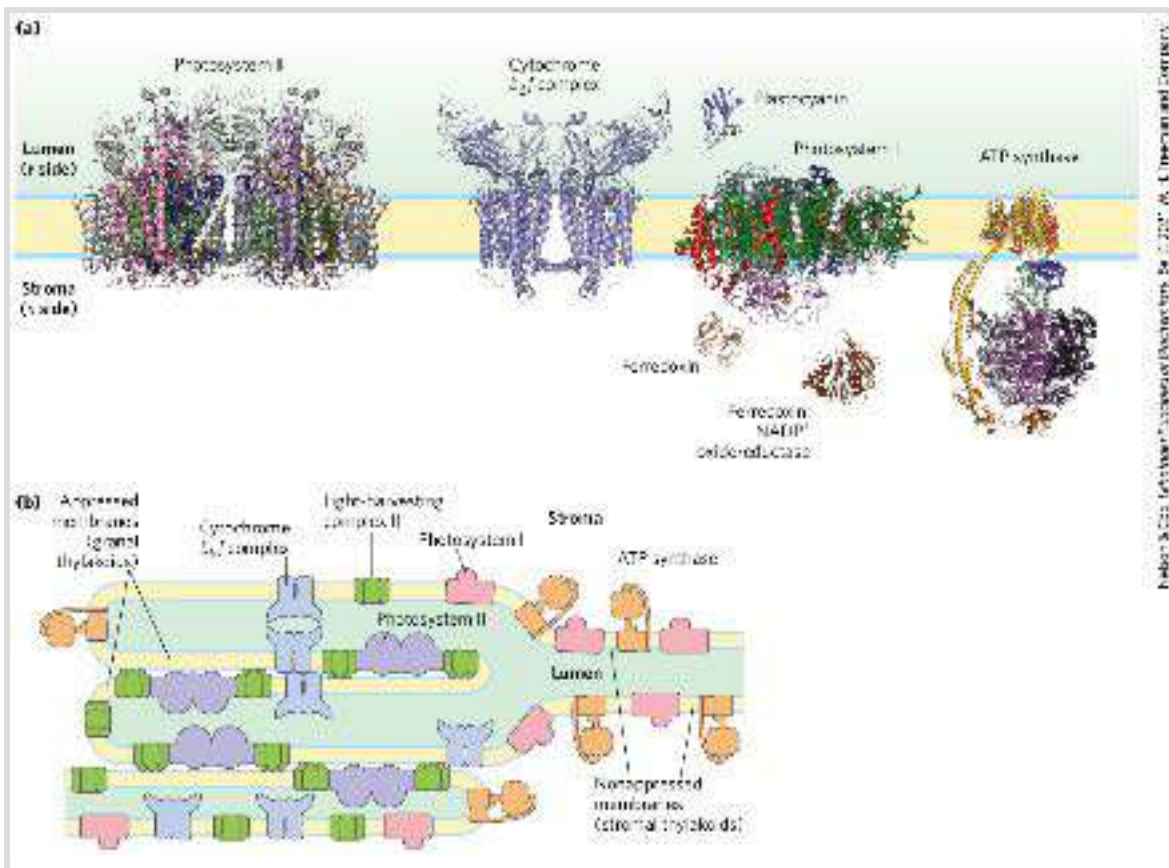
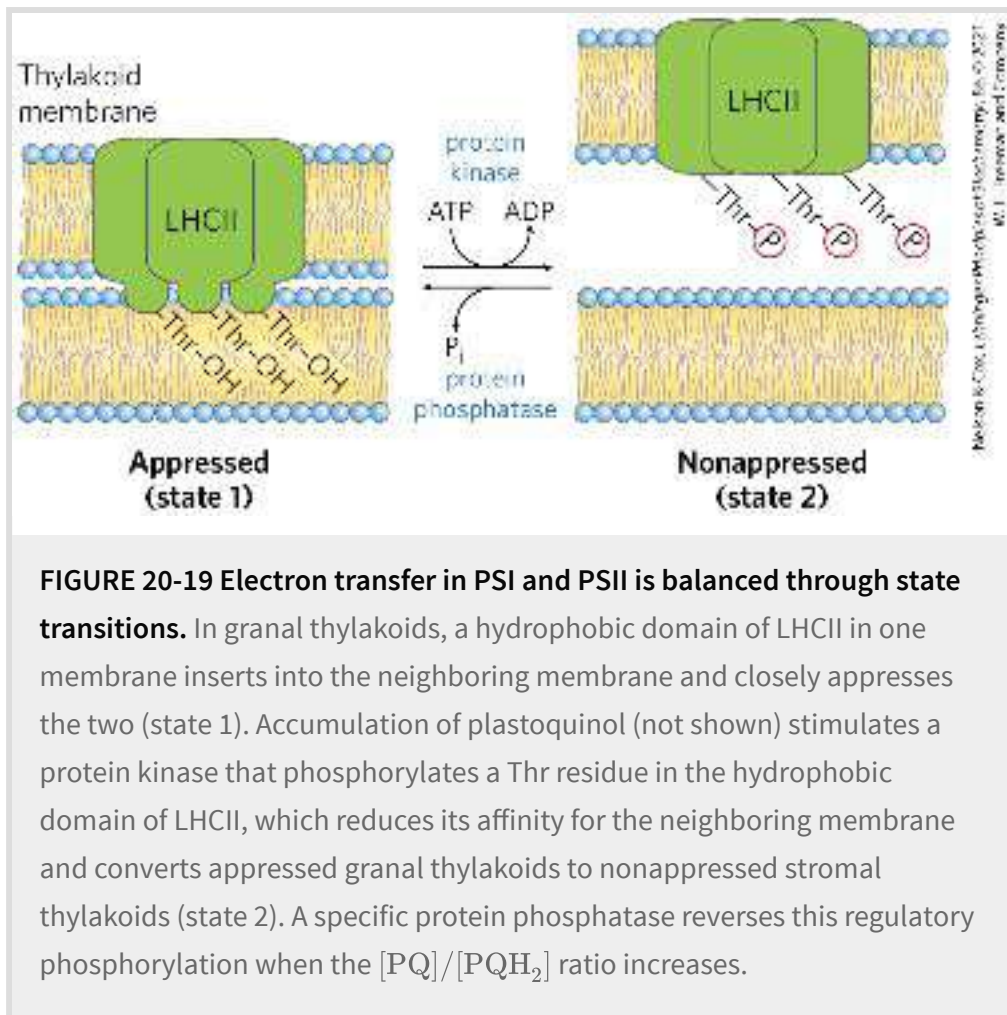


FIGURE 20-18 Localization of PSI and PSII in thylakoid membranes. (a) Structures of the complexes and soluble proteins of the photosynthetic apparatus of a vascular plant or alga, drawn to the same scale. The bovine ATP synthase is shown. (b) Light-harvesting complex LHCII and ATP synthase are located both in appressed regions of the thylakoid membrane (granal thylakoids, in which several membranes are in contact) and in nonappressed regions (stromal thylakoids), and have ready access to ADP and NADP^+ in the stroma. PSII is present almost exclusively in the appressed granal regions, and PSI almost exclusively in nonappressed stromal regions. LHCII is the “adhesive” that holds appressed thylakoid membranes together (see [Fig. 20-19](#)). [(a) Data from PSII: PDB ID 3WU2, Y. Umena et al., *Nature* 473:55, 2011; cyt b_6f complex: PDB ID 2E74, E. Yamashita et al., *J. Mol. Biol.* 370:39, 2007; plastocyanin: PDB ID 1AG6, Y. Xue et al., *Protein Sci.* 7:2099, 1998; PSI: PDB ID 4RKU, Y. Mazar et al.; ferredoxin: PDB ID 1A70, C. Binda et al., *Acta Crystallogr. D Biol. Crystallogr.* 54:1353, 1998; ferredoxin:NADP reductase: PDB ID 1QG0, Z. Deng et al., *Nat. Struct. Biol.* 6:847, 1999; ATP synthase: PDB ID 5ARA, A. Zhou et al., *eLife* 4:e10180, 2015.]



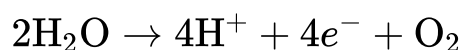
The association of LHCII with PSI and PSII depends on light intensity and wavelength, which can change in the short term and lead to **state transitions** in the chloroplast. In state 1, LHCII, PSII, and PSI are poised to maximize the capture of light energy. A critical Thr residue in LHCII is unphosphorylated, and LHCII associates with PSII. Under conditions of intense or blue light, which favor absorption by PSII, that photosystem reduces plastoquinone to plastoquinol (PQH_2) faster than PSI can oxidize it. The resulting accumulation of PQH_2 activates a protein kinase that triggers the transition to state 2 by phosphorylating a Thr residue on LHCII ([Fig. 20-19](#)). Phosphorylation weakens the interaction of LHCII with the appressed membrane and with PSII; some LHCII dissociates and moves to the stromal thylakoids. Here it captures photons (excitons) for PSI, speeding the oxidation of PQH_2 and reversing the imbalance between electron flow in PSI and PSII. In less intense light (in the shade, with more red light), PSI oxidizes PQH_2 faster than PSII can make it, and the resulting increase in [PQ] triggers dephosphorylation of LHCII, reversing the effect of phosphorylation. The state transition in LHCII localization and the transition from cyclic to linear electron transfer are coordinately regulated: the path of electrons is primarily linear in state 1 and primarily cyclic in state 2.



When light is so intense that the combined activity of PSII and PSI cannot synthesize ATP and NADPH fast enough to keep up with the supply of photons, carotenoids in LHCII absorb the excitons and very rapidly quench the excited chlorophyll before it can create damaging reactive oxygen species (ROS). The trigger for switching from an efficient light-harvesting state to an energy-dissipating state is the lowering of pH in the lumenal space, but the detailed mechanism for this transition is not yet known.

Water Is Split at the Oxygen-Evolving Center

The ultimate source of the electrons passed to NADPH in plant (oxygenic) photosynthesis is water. Having given up an electron to pheophytin, $P680^+$ (of PSII) must acquire an electron to return to its ground state in preparation for capture of another photon. In principle, the required electron might come from any number of organic or inorganic compounds. Photosynthetic bacteria use a variety of electron donors for this purpose — acetate, succinate, malate, or sulfide — depending on what is available in a particular ecological niche. About 2.5 billion years ago, evolution of primitive photosynthetic bacteria (progenitors of the modern cyanobacteria) produced a photosystem capable of taking electrons from a donor that is always available: water. Two water molecules are oxidized, yielding four electrons, four protons, and molecular oxygen:



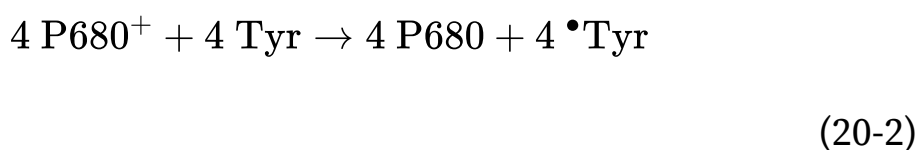
A single photon of visible light does not have enough energy to break the bonds in water; four photons are required in this photolytic cleavage reaction.



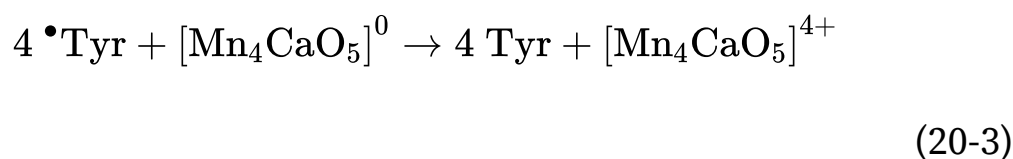
The four electrons abstracted from water do not pass directly to $P680^+$, which can accept only one electron at a time. Instead, a remarkable molecular device, the [oxygen-evolving](#)

center, passes four electrons *one at a time* to P680^+ ([Fig. 20-20a](#)).

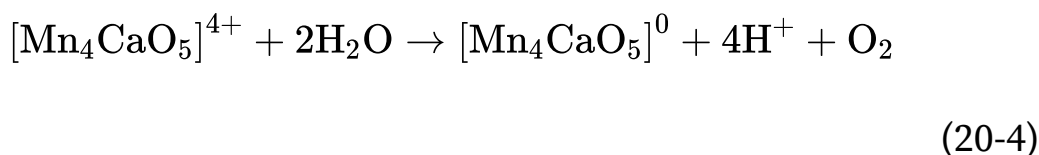
The immediate electron donor to P680^+ is a Tyr residue (sometimes designated Z or Tyr_Z) in subunit D1 of the PSII reaction center. The Tyr residue loses both a proton and an electron, generating the electrically neutral Tyr free radical, $\bullet\text{Tyr}$:



The Tyr radical regains its missing electron and proton by oxidizing a cofactor of four manganese ions and one calcium ion in the oxygen-evolving center. With each single-electron transfer, the Mn_4CaO_5 cofactor becomes more oxidized; four single-electron transfers, each corresponding to the absorption of one photon, produce a charge of 4+ on the Mn_4CaO_5 cofactor ([Fig. 20-20a](#)):



In this state, the Mn_4CaO_5 cofactor can take four electrons from a pair of water molecules, releasing four H^+ and O_2 :

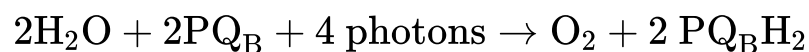


P4 Because the four protons produced in this reaction are released into the thylakoid lumen, the oxygen-evolving center acts as a proton pump, driven by electron transfer.

We saw in [Equation 20-1](#) that the overall reaction initiated by light in PSII is



The sum of [Equations 20-1](#) through [20-4](#) is



(20-5)

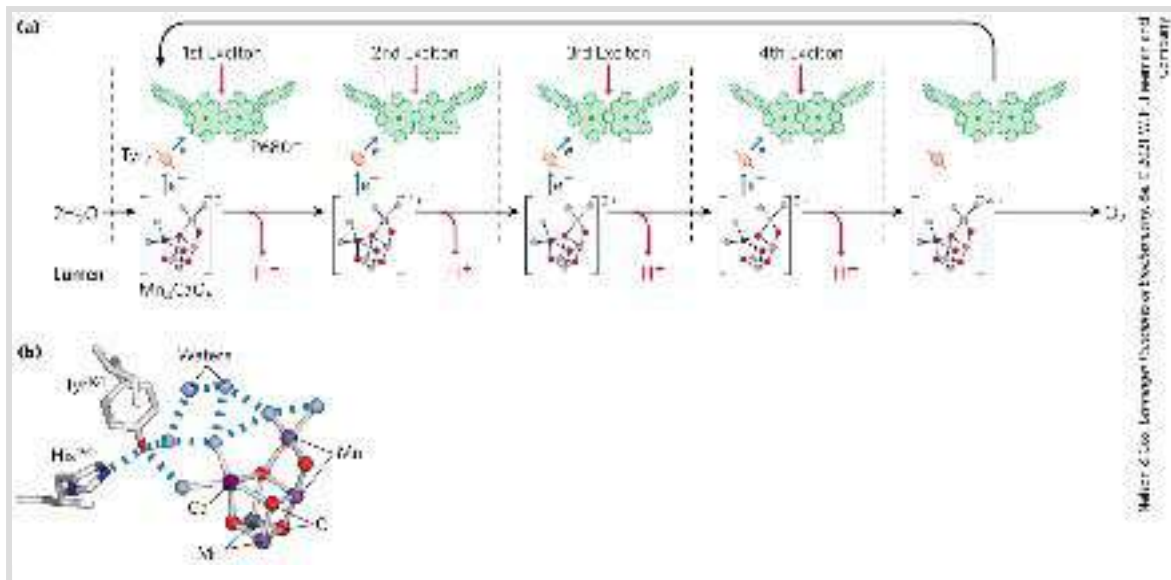


FIGURE 20-20 Water-splitting activity of the oxygen-evolving center. (a) The process that produces a four-electron oxidizing agent — a multinuclear center with four Mn ions, one Ca ion, and five oxygen atoms — in the oxygen-evolving center of PSII. The sequential absorption of four photons (excitons), each absorption causing the loss of

one electron from the Mn_4CaO_5 cofactor, produces an oxidizing agent that can remove four electrons from two molecules of water, producing O_2 . The electrons lost from the Mn_4CaO_5 cofactor pass one at a time to an oxidized Tyr residue in a PSII protein, then to P680^+ . (b) The chair-shaped metallic center of the oxygen-evolving center. Tyr^{161} , known to participate in the oxidation of water, is seen hydrogen-bonded to a network of water molecules, including several directly in contact with the Mn_4CaO_5 cofactor. This is the site of one of the most important reactions in the biosphere. [(b) Data from PDB ID 3WU2, Y. Umena et al., *Nature* 473:55, 2011.]



The oxygen-evolving cofactor takes the shape of a chair ([Fig. 20-20b](#)). The seat and legs of the chair are made up of three Mn ions, one Ca ion, and four O atoms; the fourth Mn and another O form the back of the chair. Four water molecules are also seen in the crystal structure, two associated with one of the Mn ions, the other two with the Ca ion. It may be one (or more) of these water molecules that undergoes oxidation to produce O_2 . This metal cofactor is associated with several peripheral membrane proteins on the luminal side of the thylakoid membrane that are believed to stabilize the cofactor. The Tyr residue designated Z, through which electrons move between water and the PSII reaction center, is connected with a network of hydrogen-bonded water molecules that includes the four associated with the Mn_4CaO_5 cofactor. The detailed mechanism of water oxidation by the Mn_4CaO_5 cofactor is not known but is under intense investigation. The reaction is central to life on Earth and may involve novel bioinorganic chemistry. Determination of the structure of the polymetallic center has inspired several reasonable and testable hypotheses. Stay tuned.

SUMMARY 20.2 *Photochemical Reaction Centers*

■ Bacteria have a single photochemical reaction center. Purple bacteria have a type II photosystem where electrons from an excited special pair of chlorophyll molecules (P870*) flow through pheophytin, quinones, and a proton-pumping cytochrome complex, back to the special pair of chlorophylls. Green sulfur bacteria have a type I photosystem that can send electrons through a similar cyclic path or through a linear path that reduces NAD^+ to NADH.

■ In cyanobacteria, algae, and plants, two different reaction centers are arranged in tandem. In the reaction center of PSII, when the special pair of chlorophylls (P680) is excited by light, it passes electrons to plastoquinone, and the electrons lost from P680 are replaced by electrons from H_2O . PSI passes electrons from the excited special pair (P700*) in its reaction center through a series of carriers to ferredoxin, which then reduces NADP^+ to NADPH.

■ Electron flow from either photosystem through the cytochrome b_6f complex drives protons across the thylakoid membrane, creating a proton-motive force that provides the energy for ATP synthesis by an ATP synthase.

■ Linear electron transfer through the photosystems produces NADPH and ATP. Cyclic electron transfer produces only ATP and allows variability in the proportions of NADPH and ATP formed.

■ The distribution of PSI and PSII between the granal and stromal thylakoids can change and is indirectly controlled by light intensity, optimizing the distribution of excitons between PSI and PSII for efficient energy capture.

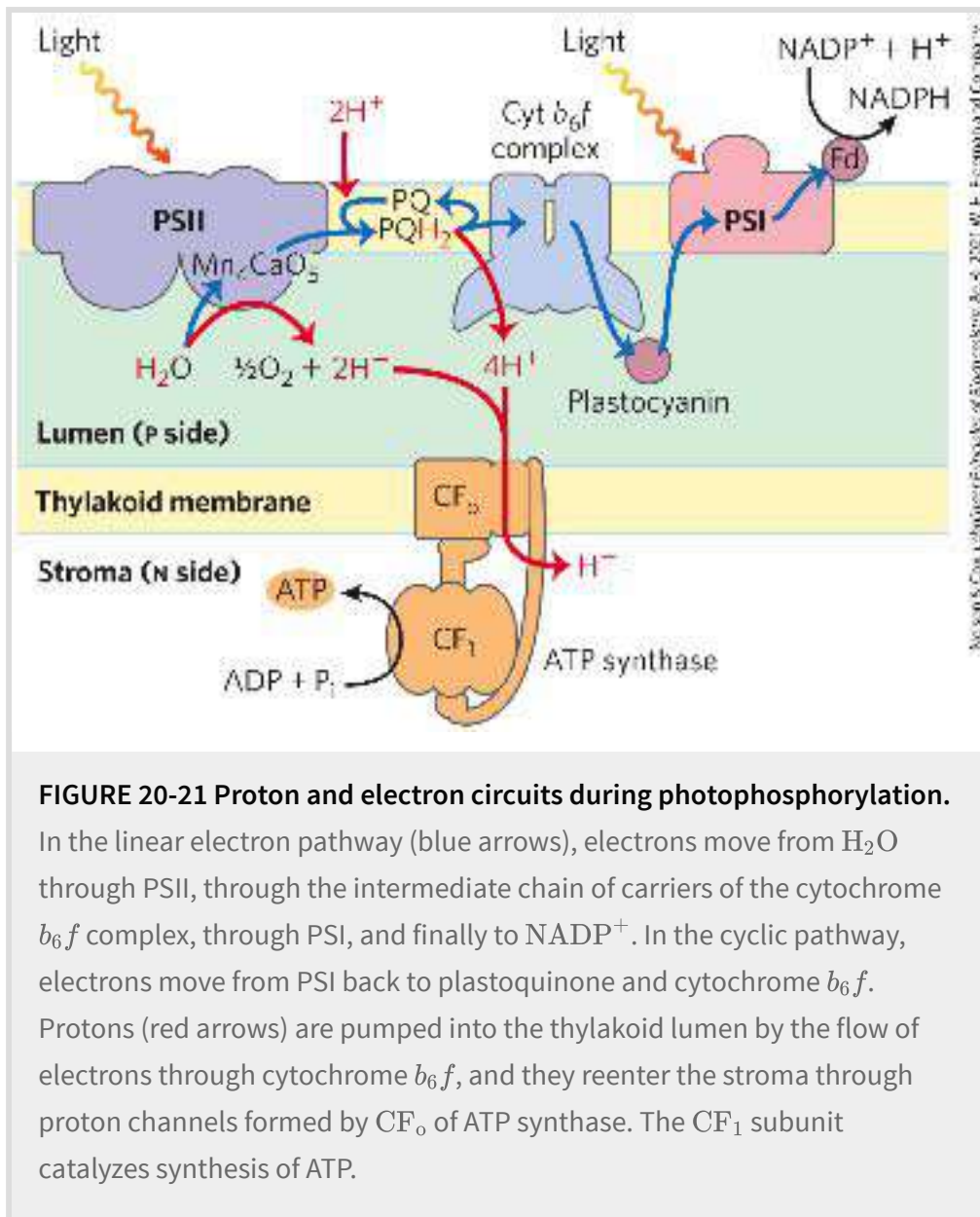
■ The oxygen-evolving center, which contains a Mn_4CaO_5 cofactor, uses energy from light to split water, producing O_2 . For each O_2 formed at the oxygen-evolving center, four protons are pumped into the thylakoid lumen, contributing to the proton motive force.

20.3 Evolution of a Universal Mechanism for ATP Synthesis

The combined activities of the two plant photosystems move electrons from water to NADP^+ , conserving some of the energy of absorbed light as NADPH ([Fig. 20-12](#)). Simultaneously, protons are pumped across the thylakoid membrane and energy is conserved as an electrochemical potential. We turn now to the process by which this proton gradient drives the synthesis of ATP, the other energy-conserving product of the light-dependent reactions.

A Proton Gradient Couples Electron Flow and Phosphorylation

 Although the energy source and electron carriers in photophosphorylation in chloroplasts differ from those of oxidative phosphorylation in mitochondria, they use essentially the same mechanism to capture the energy of the proton gradient. Electron-transferring molecules in the chain of carriers connecting PSII and PSI are oriented asymmetrically in the thylakoid membrane, so photoinduced electron flow results in the net movement of protons across the membrane, from the stromal side to the thylakoid lumen ([Fig. 20-21](#)).



The Approximate Stoichiometry of Photophosphorylation Has Been Established

As electrons move from water to $NADP^+$ in chloroplasts, about 12 protons move from the stroma into the thylakoid lumen per 4 electrons passed (that is, per O_2 formed). Of these protons, 4 are

moved by the oxygen-evolving center, and up to 8 are moved by the cytochrome b_6f complex. The measurable result is a 1,000-fold difference in H^+ concentration across the thylakoid membrane ($\Delta pH = 3$). Recall that the energy stored in a proton gradient (the electrochemical potential) has two components: a proton concentration difference (ΔpH) and an electrical potential ($\Delta\psi$) due to charge separation. In chloroplasts, ΔpH is the dominant component; counterion movement apparently dissipates most of the electrical potential. In illuminated chloroplasts, the energy stored in the proton gradient per mole of protons is

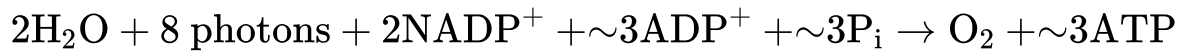
$$\Delta G = 2.3RT \Delta pH + ZF \Delta\psi = -17 \text{ kJ/mol}$$

so the movement of 12 mol of protons across the thylakoid membrane represents conservation of about 200 kJ of energy — enough energy to drive the synthesis of several moles of ATP ($\Delta G'^{\circ} = 30.5 \text{ kJ/mol}$). Experimental measurements yield values of about 3 ATP per O_2 produced.

At least 8 photons must be absorbed to drive 4 electrons from 2 H_2O to 2 NADPH (one photon per electron at each reaction center). The energy in 8 photons of visible light is more than enough for the synthesis of three molecules of ATP.

ATP synthesis is not the only energy-conserving reaction of photosynthesis in plants; the NADPH formed in the final electron

transfer is also energetically rich. The overall equation for this linear photophosphorylation is



(20-6)

The ATP Synthase Structure and Mechanism Are Nearly Universal

The enzyme responsible for ATP synthesis in chloroplasts is a large complex with two functional components, CF_o and CF_1 (C denoting its location in chloroplasts). CF_o is a transmembrane proton pore composed of several integral membrane proteins and is homologous to mitochondrial F_o . CF_1 is a peripheral membrane protein complex very similar in subunit composition, structure, and function to mitochondrial F_1 .

Electron microscopy of sectioned chloroplasts shows ATP synthase complexes as projections on the *outside* (stromal, or N) surface of thylakoid membranes; these complexes correspond to the ATP synthase complexes that project on the *inside* (matrix, or N) surface of the inner mitochondrial membrane. Thus, the relationship between the orientation of the ATP synthase and the direction of proton pumping is the same in chloroplasts and mitochondria. In both cases, the F_1 portion of ATP synthase is located on the more alkaline (N) side of the membrane through

which protons flow down their concentration gradient; the direction of proton flow relative to F_1 is the same in both cases: P to N ([Fig. 20-22](#)).

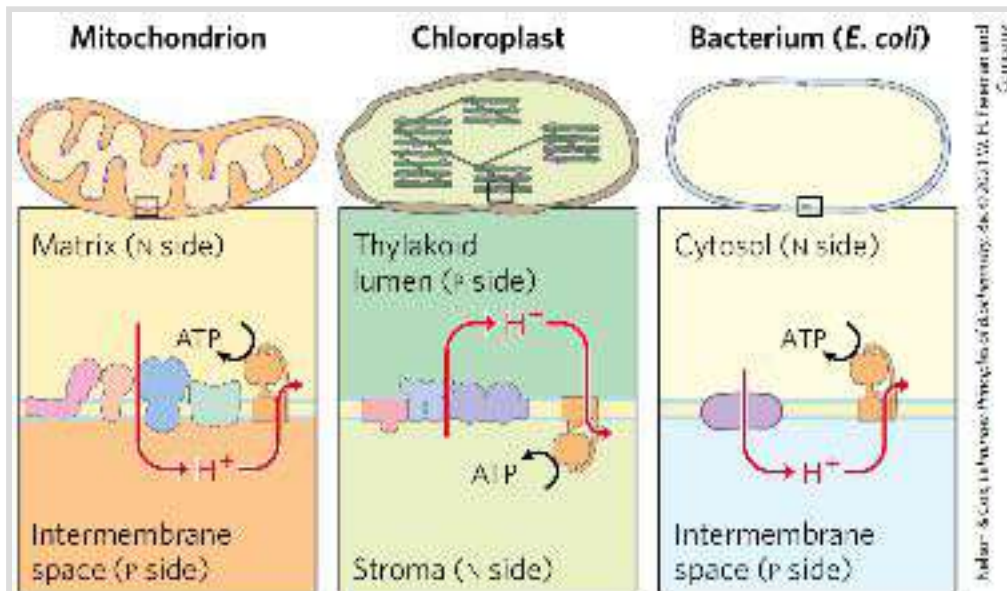


FIGURE 20-22 Orientation of ATP synthase is fixed relative to the proton gradient.

Superficially, the direction of proton pumping in chloroplasts may seem to be opposite to that in mitochondria and bacteria. In mitochondria and bacteria, protons are pumped *out of* the organelle or cell, and F_1 is on the *inside* of the membrane; in chloroplasts, protons are pumped *into* the thylakoid lumen, and CF_1 is on the *outside* of the thylakoid membrane. However, exactly the same mechanism of energy conversion (from proton gradient to ATP) occurs in all three cases. ATP is synthesized in the matrix of mitochondria, the stroma of chloroplasts, and the cytosol of bacteria.



The mechanism of chloroplast ATP synthase is essentially identical to that of its mitochondrial analog; ADP and P_i readily condense to form ATP on the enzyme surface, and the release of this enzyme-bound ATP requires a proton-motive force. Rotational catalysis sequentially engages each of the three β

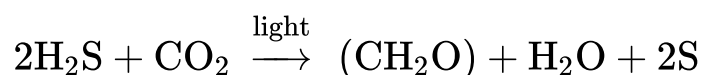
subunits of the ATP synthase in ATP synthesis, ATP release, and ADP + P_i binding (see [Figs. 19-26](#) and [19-27](#)).

The appearance of oxygenic photosynthesis on Earth about 2.5 billion years ago was a crucial event in the evolution of the biosphere. Before that, Earth's atmosphere was composed of methane, CO₂, and N₂. The planet was essentially devoid of molecular oxygen and lacked the ozone layer that protects organisms from solar UV radiation. Oxygenic photosynthesis made available a nearly limitless supply of reducing agent (H₂O) to drive the production of organic compounds by reductive biosynthetic reactions. And mechanisms evolved that allowed organisms to use O₂ as a terminal electron acceptor in highly energetic electron transfers from organic substrates, employing the energy of oxidation to support metabolism. The complex photosynthetic apparatus of a modern vascular plant is the culmination of a series of evolutionary events, the most recent of which was the acquisition by eukaryotic cells of a cyanobacterial endosymbiont.

The chloroplasts of modern organisms share several properties with mitochondria and originated by the same mechanism that gave rise to mitochondria: endosymbiosis. Like mitochondria, chloroplasts contain their own DNA and protein-synthesizing machinery. Some of the polypeptides of chloroplast proteins are encoded by chloroplast genes and synthesized in the chloroplast; others are encoded by nuclear genes, synthesized outside the chloroplast, and imported ([Chapter 27](#)). When plant cells grow

and divide, chloroplasts give rise to new chloroplasts by division, during which their DNA is replicated and divided between daughter chloroplasts. The machinery and mechanisms for light capture, electron flow, and ATP synthesis in modern cyanobacteria are similar in many respects to those in plant chloroplasts. These observations led to the now widely accepted hypothesis that the evolutionary progenitors of modern plant cells were primitive eukaryotes that engulfed photosynthetic cyanobacteria and established stable endosymbiotic relationships with them (see [Fig. 1-37](#)).

At least half of the photosynthetic activity on Earth now occurs in microorganisms — algae, other photosynthetic eukaryotes, and photosynthetic bacteria. Cyanobacteria have PSII and PSI in tandem, and the PSII has an associated oxygen-evolving activity resembling that of plants. However, the other groups of photosynthetic bacteria have single reaction centers and do not split H₂O or produce O₂. Many are obligate anaerobes and cannot tolerate O₂; they must use some compound other than H₂O as an electron donor. Some photosynthetic bacteria use inorganic compounds as electron (and hydrogen) donors. For example, green sulfur bacteria use hydrogen sulfide:

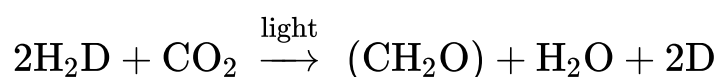


These bacteria, instead of producing molecular O₂, form elemental sulfur as the oxidation product of H₂S. (They further

oxidize the S to SO_4^{2-} .) Other photosynthetic bacteria use organic compounds such as lactate as electron donors:



The fundamental similarity of photosynthesis in plants and bacteria, despite the differences in the electron donors they employ, becomes more obvious when the equation of photosynthesis is written in the more general form



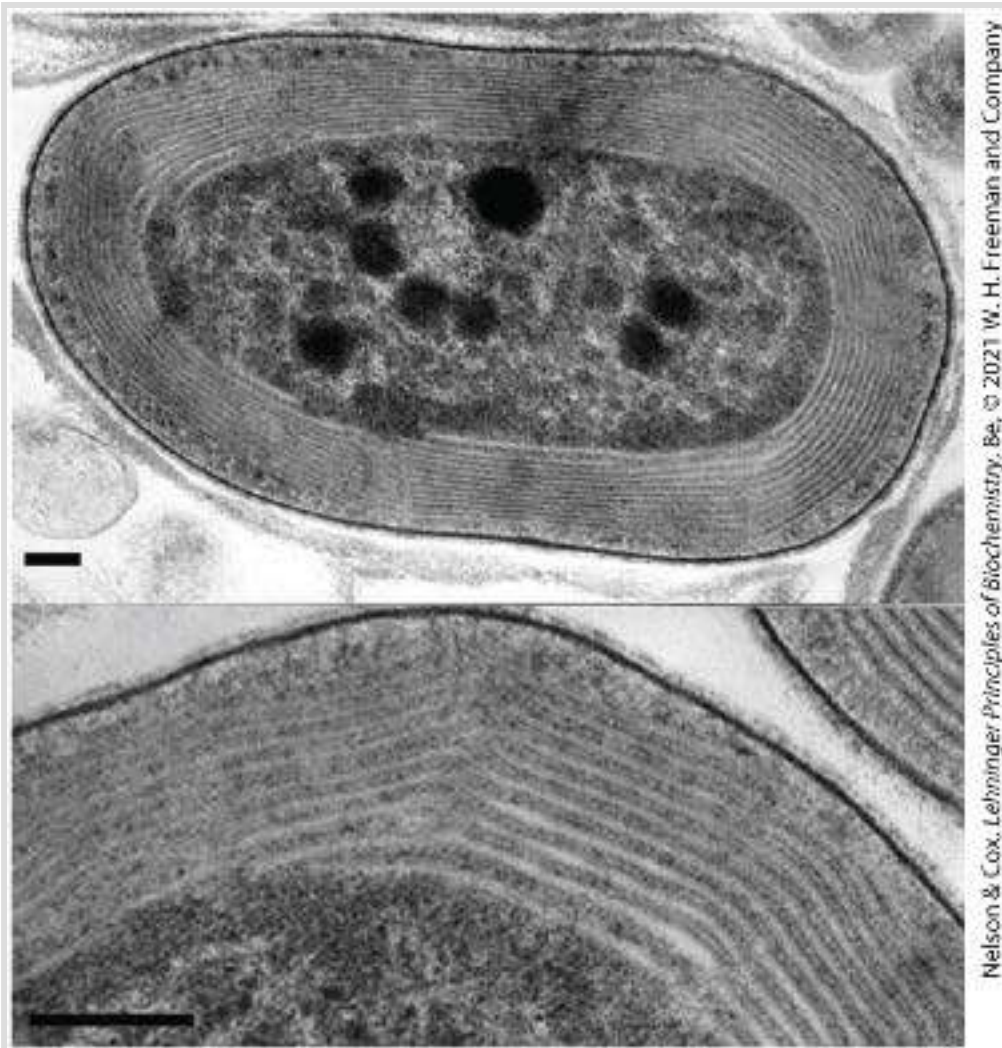
in which H_2D is an electron (and hydrogen) donor and D is its oxidized form. H_2D may be water, hydrogen sulfide, lactate, or some other organic compound, depending on the species. Most likely, the bacteria that first developed photosynthetic ability used H_2S as their electron source.

Modern cyanobacteria can synthesize ATP by oxidative phosphorylation or by photophosphorylation, although they have neither mitochondria nor chloroplasts. The enzymatic machinery for both processes is in a highly convoluted plasma membrane ([Fig. 20-23](#)). Three protein components function in both processes, giving evidence that the processes have a common evolutionary origin ([Fig. 20-24](#)). First, the proton-pumping

cytochrome b_6f complex carries electrons from plastoquinone to cytochrome c_6 in photosynthesis, and also carries electrons from ubiquinone to cytochrome c_6 in oxidative phosphorylation — the role played by cytochrome bc_1 in mitochondria. Second, cytochrome c_6 , homologous to mitochondrial cytochrome c , carries electrons from Complex III to Complex IV in cyanobacteria; it can also carry electrons from the cytochrome b_6f complex to PSI — a role performed in plants by plastocyanin.



We therefore see the functional homology between the cyanobacterial cytochrome b_6f complex and the mitochondrial cytochrome bc_1 complex, and between cyanobacterial cytochrome c_6 and plant plastocyanin. The third conserved component is the ATP synthase, which functions in oxidative phosphorylation and photophosphorylation in cyanobacteria, and in the mitochondria and chloroplasts of photosynthetic eukaryotes. The structure and remarkable mechanism of this enzyme have been strongly conserved throughout evolution.



Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021 W. H. Freeman and Company

FIGURE 20-23 The photosynthetic membranes of a cyanobacterium. In these thin sections of a cyanobacterium, viewed with a transmission electron microscope, the multiple layers of the internal membranes are seen to fill half the total volume of the cell. The extensive membrane system serves the same role as the thylakoid membranes of vascular plants, providing a large surface area containing all of the photosynthetic machinery. (Bar = 100 nm.) [S. R. Miller et al. Discovery of a free-living chlorophyll *d*-producing cyanobacterium with a hybrid proteobacterial/cyanobacterial small-subunit rRNA gene. *Proc. Natl. Acad. Sci. USA* 102:850, 2005, Fig. 2. © 2005 National Academy of Sciences.]

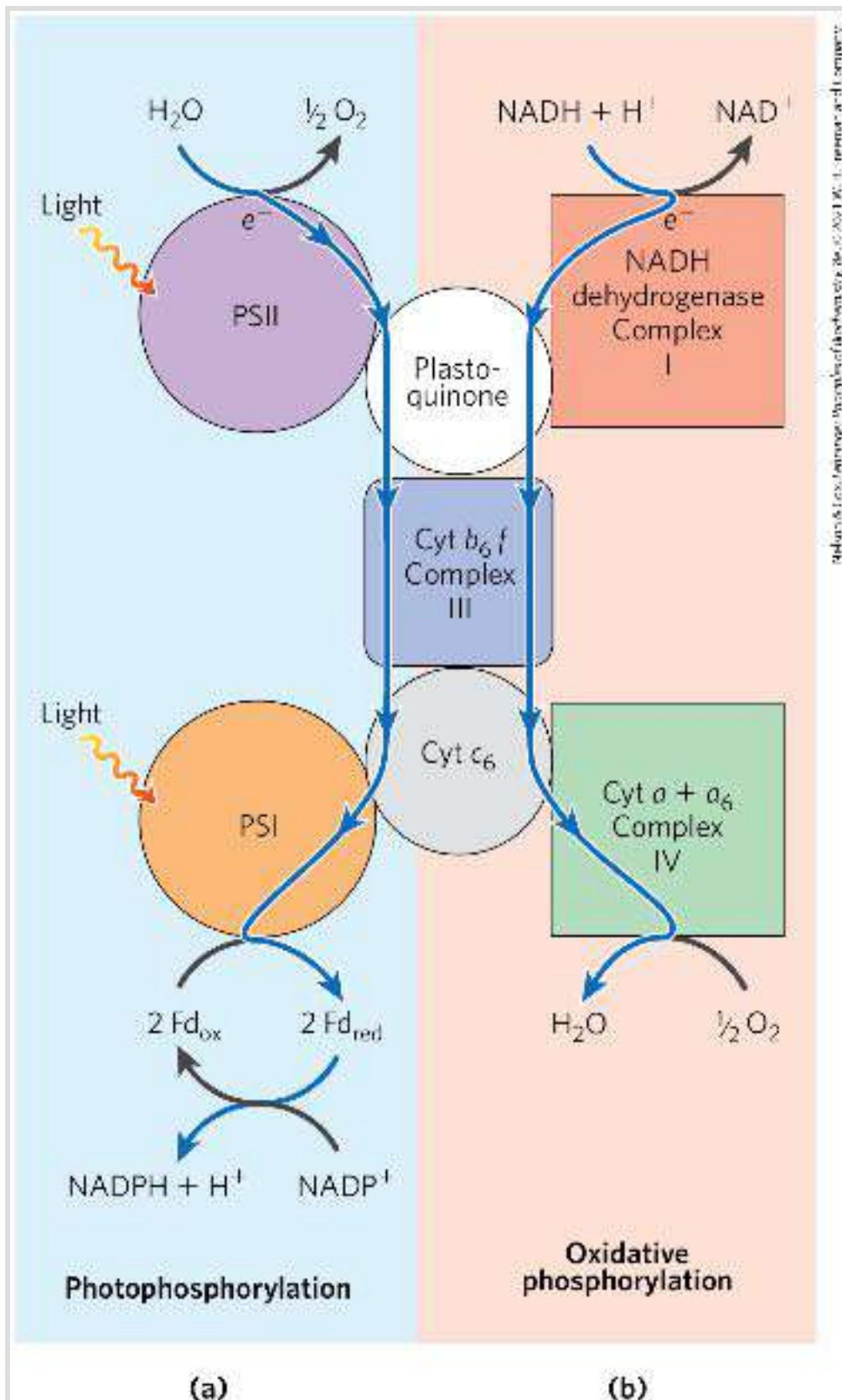


FIGURE 20-24 Dual roles of cytochrome b_6f and cytochrome c_6 in cyanobacteria reflect evolutionary origins. Cyanobacteria use cytochrome b_6f , cytochrome c_6 , and plastoquinone for both oxidative phosphorylation

and photophosphorylation. (a) In photophosphorylation, electrons flow (blue arrows) from water to NADP^+ . (b) In oxidative phosphorylation, electrons flow from NADH to O_2 . Both processes are accompanied by proton movement across the membrane, accomplished by a Q cycle.

SUMMARY 20.3 *Evolution of a Universal Mechanism for ATP Synthesis*

■ In plants, both the water-splitting reaction and electron flow through the cytochrome b_6f complex are accompanied by proton pumping across the thylakoid membrane. The proton-motive force thus created drives ATP synthesis by a CF_0CF_1 complex similar to the mitochondrial F_0F_1 complex in both structure and catalytic mechanism.

■ Direct measurements show that eight photons drive the production of one O_2 from oxidation of two H_2O , making three ATP molecules.

■ About 2.5 billion years ago, cyanobacteria appeared on Earth. They had acquired two photosystems — one of the type now found in purple bacteria, the other of the type found in green sulfur bacteria — that operated in tandem, and a water-splitting activity that released oxygen into the atmosphere.

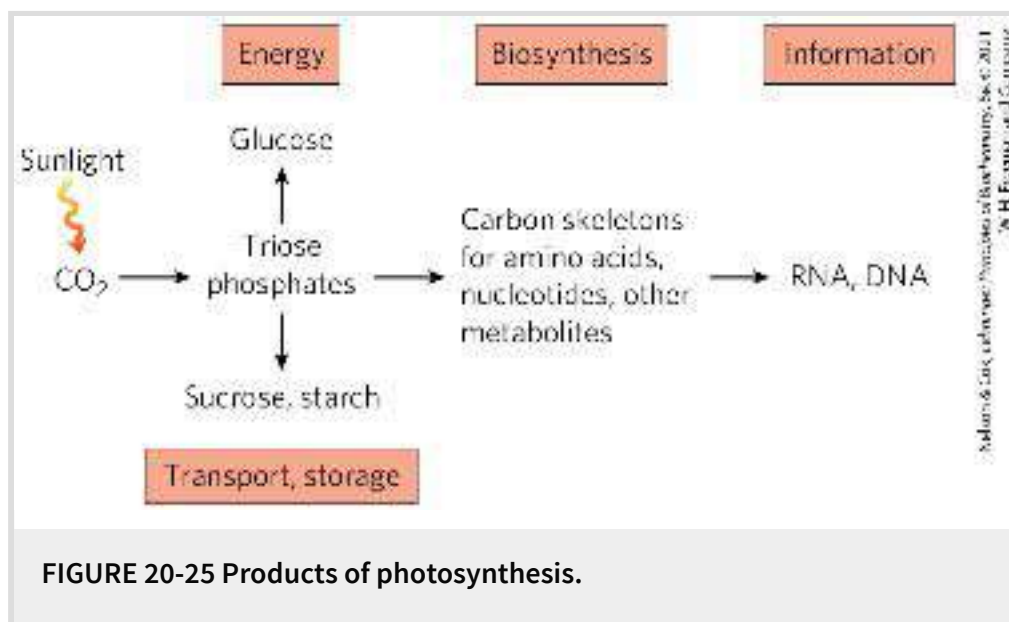
■ Many photosynthetic microorganisms obtain electrons for photosynthesis not from water but from donors such as H_2S , forming an oxidized product such as elemental sulfur (not oxygen).

■ Chloroplasts, like mitochondria, evolved from bacteria living as endosymbionts in early eukaryotic cells. The ATP synthases of bacteria, cyanobacteria, mitochondria, and chloroplasts share a common evolutionary precursor and a common enzymatic mechanism.

20.4 CO₂-Assimilation Reactions

Photosynthetic organisms use the ATP and NADPH produced in the light-dependent reactions of photosynthesis to synthesize all of the thousands of components that make up the organism.

P5 Plants (and other autotrophs) can reduce atmospheric CO₂ to trioses, then use the trioses as precursors for the synthesis of sucrose and starch, lipids and proteins, and the many other organic components of plant cells ([Fig. 20-25](#)). Lacking these synthetic capacities, humans and other animals are ultimately dependent on photosynthetic organisms to provide the reduced fuels and organic precursors essential to life.



Green plants contain in their chloroplasts the enzymatic machinery that catalyzes the conversion of CO₂ to simple (reduced) organic compounds, a process called **CO₂ assimilation**. This process has also been called **CO₂ fixation**, but

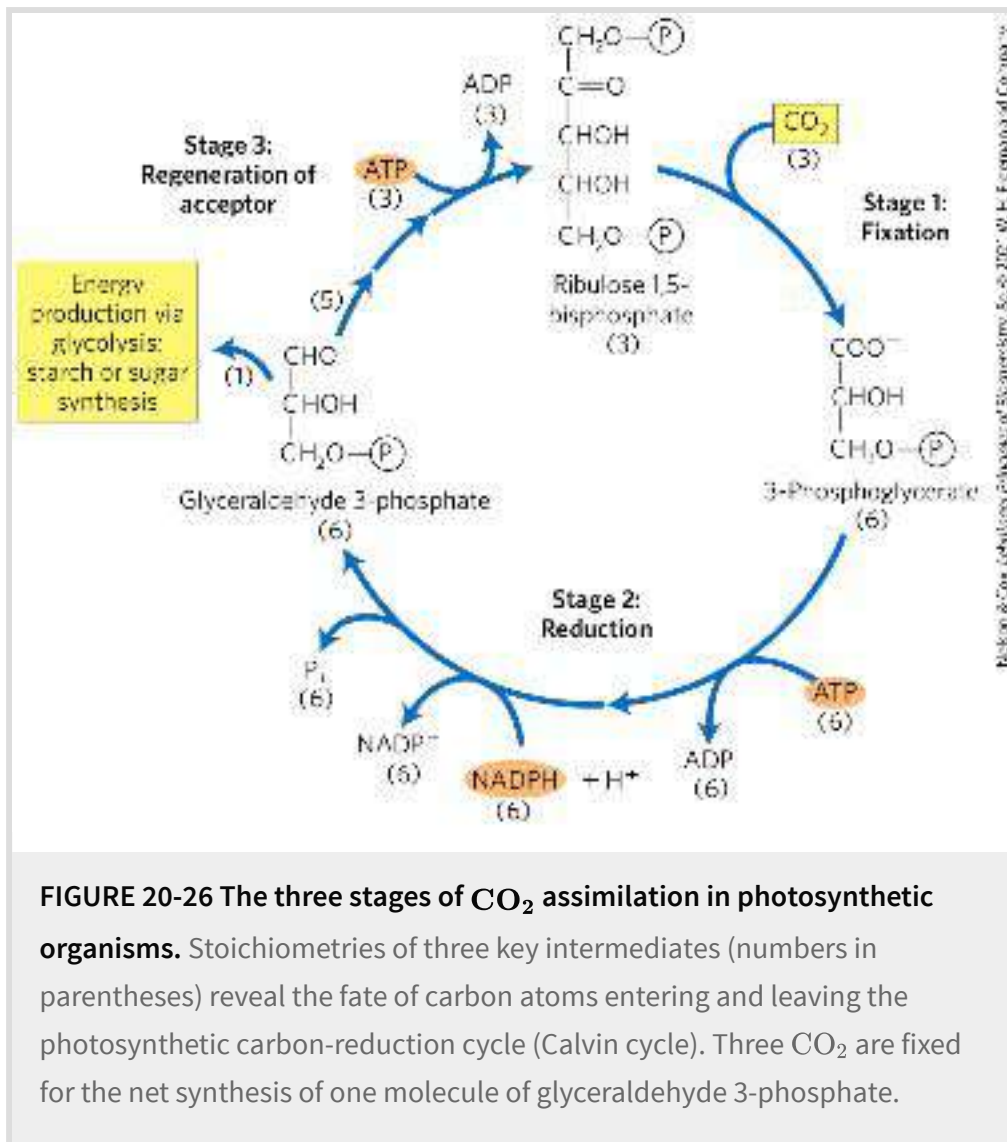
we reserve this term for the specific reaction in which CO_2 is incorporated (fixed) into a three-carbon organic compound, the triose phosphate 3-phosphoglycerate. This simple product of photosynthesis is the precursor of more-complex biomolecules, including sugars, polysaccharides, and the metabolites derived from them, all of which are synthesized by metabolic pathways similar to those of animal tissues. Carbon dioxide is assimilated via a cyclic pathway, its key intermediates constantly regenerated. The pathway was elucidated in the early 1950s by Melvin Calvin, Andrew Benson, and James A. Bassham and is often called the [Calvin cycle](#) or, more descriptively, the **reductive pentose phosphate pathway**. It is essentially the reversal of a central pathway of glucose oxidation, the pentose phosphate pathway, which we described in [Section 14.6](#).

Carbohydrate metabolism is more complex in plant cells than in animal cells or in nonphotosynthetic microorganisms.  *In addition to the universal pathways of glycolysis, gluconeogenesis, and the pentose phosphate pathway, plants have the unique reaction sequences for reduction of CO_2 to triose phosphates and the associated reductive pentose phosphate pathway — all of which must be coordinately regulated to ensure proper allocation of carbon to energy production and synthesis of starch and sucrose. Key enzymes are regulated, as we shall see, by (1) reduction of disulfide bonds by electrons flowing from photosystem I and (2) changes in pH and Mg^{2+} concentration that result from illumination. When we look at other aspects of plant carbohydrate metabolism, we also find enzymes that are*

modulated by (3) conventional allosteric regulation by one or more metabolic intermediates and (4) covalent modification (phosphorylation).

Carbon Dioxide Assimilation Occurs in Three Stages

The first stage in the assimilation of CO_2 into biomolecules ([Fig. 20-26](#)) is the CO_2 -fixation reaction: condensation of CO_2 with a five-carbon acceptor, **ribulose 1,5-bisphosphate**, to form two molecules of **3-phosphoglycerate**. In the second stage, the 3-phosphoglycerate is reduced to triose phosphates.  Overall, three molecules of CO_2 are fixed to three molecules of ribulose 1,5-bisphosphate to form six molecules of glyceraldehyde 3-phosphate (18 carbons). In the third stage, five of the six molecules of triose phosphate (15 carbons) are used to regenerate three molecules of ribulose 1,5-bisphosphate (15 carbons), the starting material. The sixth molecule of triose phosphate, the net product of photosynthesis, can be used to make hexoses for fuel and building materials, sucrose for transport to nonphotosynthetic tissues, or starch for storage. Thus, the overall process is cyclical, with the continuous conversion of CO_2 to triose and hexose phosphates.



Fructose 6-phosphate is a key intermediate in stage 3 of CO₂ assimilation; it stands at a branch point, leading either to regeneration of ribulose 1,5-bisphosphate or to synthesis of starch. The pathway from hexose phosphate to pentose biphosphate involves many of the same reactions used in animal cells for the conversion of pentose phosphates to hexose phosphates during the nonoxidative phase of the pentose phosphate pathway (see [Fig. 14-31](#)). In the photosynthetic assimilation of CO₂, essentially the same set of reactions operates in the reverse direction, converting hexose phosphates to pentose

phosphates. This reductive pentose phosphate cycle uses the same enzymes as the oxidative pathway, and several additional enzymes that make the reductive cycle irreversible. All 13 enzymes of the pathway are in the chloroplast stroma.

Stage 1: Fixation of CO₂ into 3-Phosphoglycerate

An important clue to the nature of the CO₂-assimilation mechanisms in photosynthetic organisms came in the late 1940s. Calvin and his associates illuminated a suspension of green algae in the presence of radioactive carbon dioxide (¹⁴CO₂) for just a few seconds, then quickly killed the cells, extracted their contents, and used chromatographic methods to search for the metabolites in which the labeled carbon first appeared. The first compound that became labeled was 3-phosphoglycerate, with the ¹⁴C predominantly located in the carboxyl carbon atom. These experiments strongly suggested that 3-phosphoglycerate is an early intermediate in photosynthesis.

The many plants in which this three-carbon compound is the first intermediate are called **C₃ plants**, in contrast to the C₄ plants described below. Most plant species — 80% to 90% — are C₃, including most trees, wheat, oats, rice, beans, peas, and spinach. The enzyme that catalyzes incorporation of CO₂ into an organic form is [ribulose 1,5-bisphosphate carboxylase/oxygenase](#), a name mercifully shortened to **rubisco**.  As a carboxylase,

rubisco catalyzes the covalent attachment of CO₂ to the five-carbon sugar ribulose 1,5-bisphosphate and cleavage of the unstable six-carbon intermediate to form two molecules of 3-phosphoglycerate, one of which bears the carbon introduced as CO₂ in its carboxyl group ([Fig. 20-26](#)). The enzyme's oxygenase activity is discussed in [Section 20.5](#).

There are two distinct forms of rubisco. The rubisco of vascular plants, algae, and cyanobacteria is a crucial enzyme in the production of biomass from CO₂. It has a complex form I structure ([Fig. 20-27a](#)), with eight identical large catalytic subunits (M_r 53,000; encoded in the chloroplast genome), and eight identical small subunits (M_r 14,000; encoded in the nuclear genome) of uncertain function. The form II rubisco of photosynthetic bacteria is simpler, having two subunits that in many respects resemble the large subunits of the plant enzyme ([Fig. 20-27b](#)). The plant enzyme has an exceptionally low turnover number; only three molecules of CO₂ are fixed per second per molecule of rubisco at 25 °C. To achieve high rates of CO₂ fixation, plants therefore need large amounts of this enzyme. Rubisco is present at about 250 mg/mL in the chloroplast stroma, corresponding to an extraordinarily high concentration of active sites (~4 mM). In fact, rubisco makes up almost 50% of soluble protein in chloroplasts and is probably one of the most abundant enzymes in the biosphere.

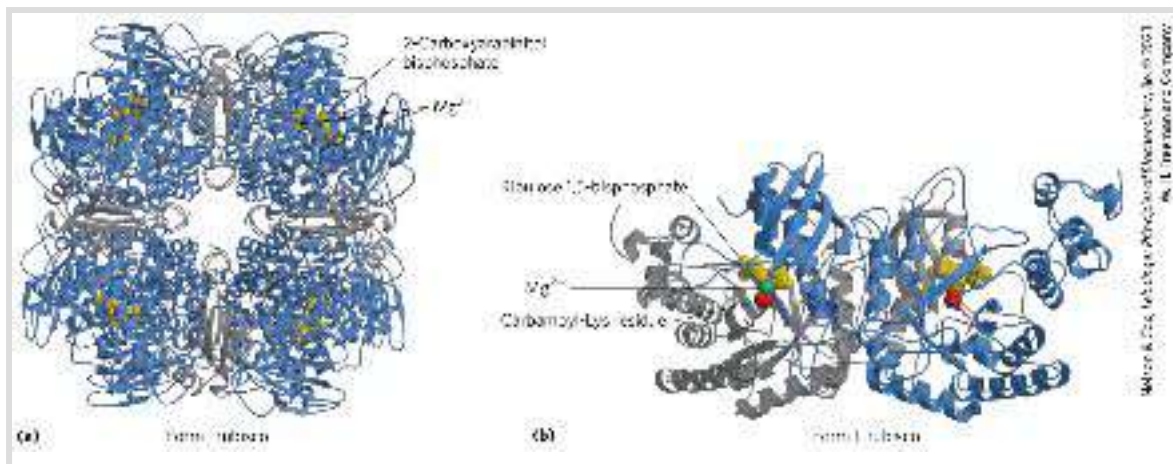


FIGURE 20-27 Structure of ribulose 1,5-bisphosphate carboxylase/oxygenase (rubisco). (a) A ribbon model of form I rubisco from spinach. The enzyme has eight large (blue) and eight small (gray) subunits, tightly packed into a structure of $M_r > 500,000$. A transition-state analog, 2-carboxyarabinitol biphosphate (yellow), is shown bound to each of the eight substrate-binding sites. Mg^{2+} in the active site is shown in green. (b) Ribbon model of form II rubisco from the bacterium *Rhodospirillum rubrum*. The identical subunits are in gray and blue. [Data from (a) PDB ID 8RUC, I. Andersson, *J. Mol. Biol.* 259:160, 1996; (b) PDB ID 9RUB, T. Lundqvist and G. Schneider, *J. Biol. Chem.* 266:12,604, 1991.]

Central to the proposed mechanism for plant rubisco is a carbamoylated Lys side chain with a bound Mg^{2+} ion. The Mg^{2+} ion brings together and orients the reactants at the active site ([Fig. 20-28](#)), setting up for a nucleophilic attack by the five-carbon enediolate reaction intermediate formed on the enzyme ([Fig. 20-29](#)). The resulting six-carbon intermediate breaks down to yield two molecules of 3-phosphoglycerate.

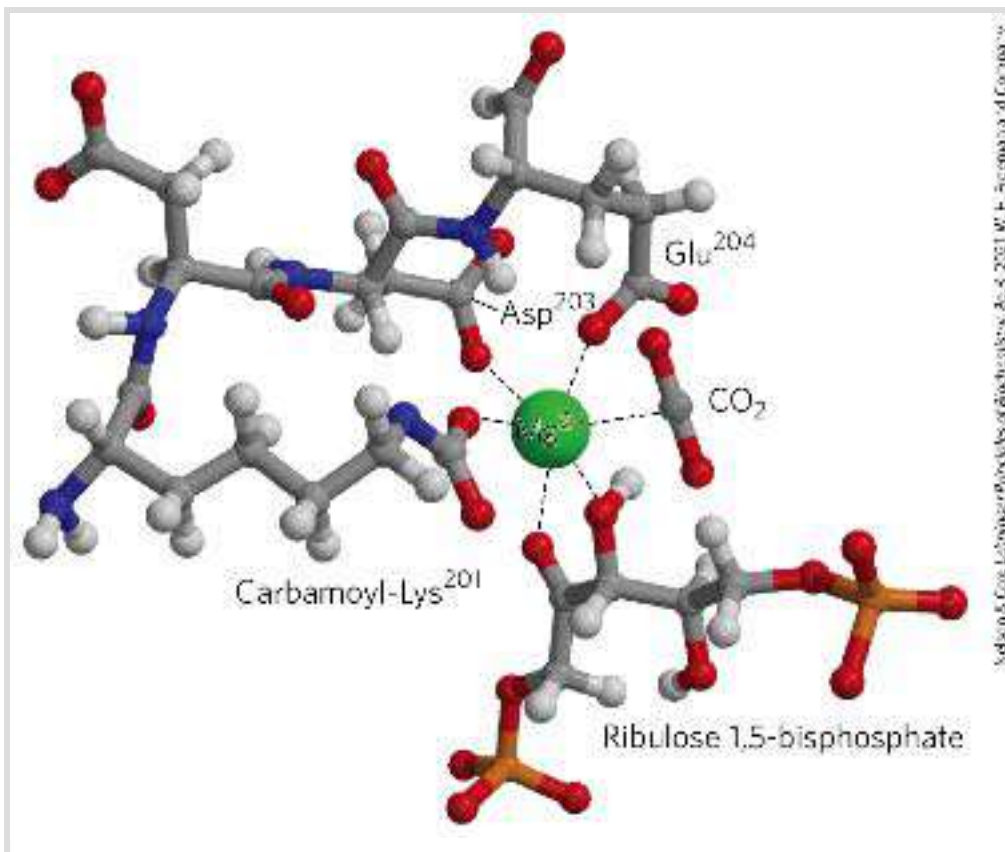
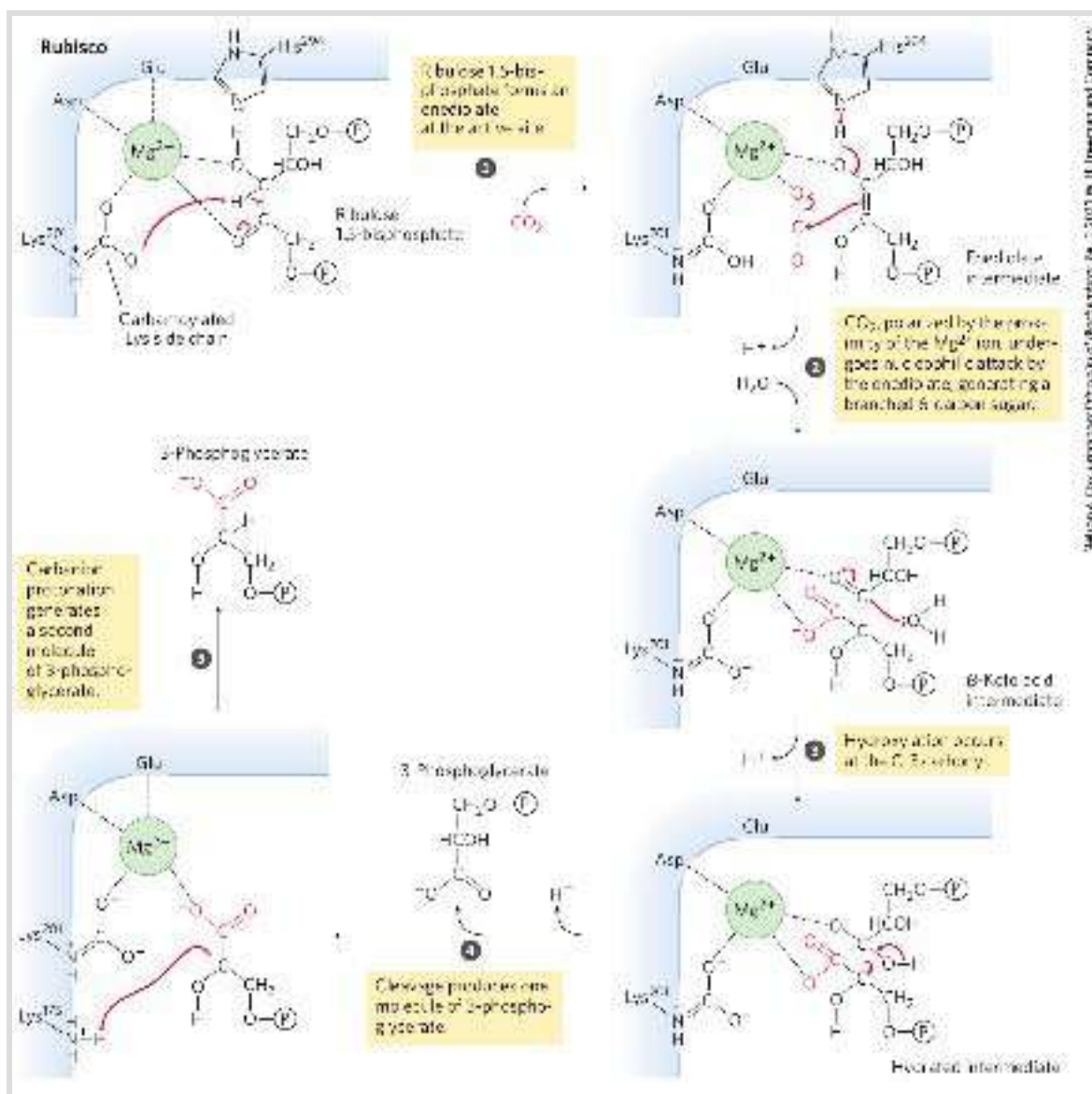


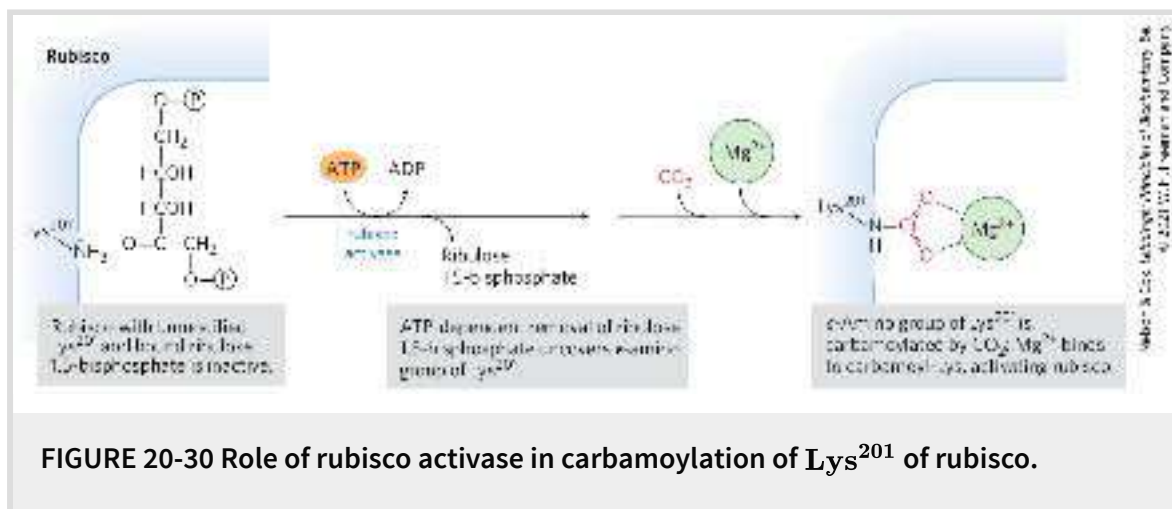
FIGURE 20-28 Central role of Mg^{2+} in the active site of rubisco. Mg^{2+} is coordinated in a roughly octahedral complex with six oxygen atoms: one oxygen in the carbamate on Lys^{201} ; two in the carboxyl groups of Glu^{204} and Asp^{203} ; two at C-2 and C-3 of the substrate, ribulose 1,5-bisphosphate; and one in the other substrate, CO_2 . A water molecule occupies the CO_2 -binding site in the crystal structure. In this figure, a CO_2 molecule is modeled in its place. (Residue numbers refer to the spinach enzyme.) [Data from PDB ID 1RXO, T. C. Taylor and I. Andersson, *J. Mol. Biol.* 265:432, 1997.]



MECHANISM FIGURE 20-29 First stage of CO₂ assimilation: rubisco's carboxylase activity. The CO₂-fixation reaction is catalyzed by ribulose 1,5-bisphosphate carboxylase/oxygenase. The overall reaction accomplishes the combination of one CO₂ and one ribulose 1,5-bisphosphate to form two molecules of 3-phosphoglycerate, one of which contains the carbon atom from CO₂ (red). Additional proton transfers (not shown), involving Lys²⁰¹, Lys¹⁷⁵, and His²⁹⁴, occur in several of these steps.

As the catalyst for the first step of photosynthetic CO₂ assimilation, rubisco is a prime target for regulation. The enzyme is inactive until carbamoylated on the ε-amino group of Lys²⁰¹ ([Fig. 20-30](#)). Ribulose 1,5-bisphosphate inhibits carbamoylation

by binding tightly to the active site and locking the enzyme in the “closed” conformation, in which Lys²⁰¹ is inaccessible. **Rubisco activase** overcomes the inhibition by promoting ATP-dependent release of the ribulose 1,5-bisphosphate, exposing the Lys amino group to nonenzymatic carbamoylation by CO₂; this is followed by Mg²⁺ binding, which activates the rubisco.



Stage 2: Conversion of 3-Phosphoglycerate to Glyceraldehyde 3-Phosphate

Stage 2 begins as stromal 3-phosphoglycerate kinase catalyzes the transfer of a phosphoryl group from ATP to 3-phosphoglycerate, yielding 1,3-bisphosphoglycerate. Next, NADPH donates electrons in a reduction catalyzed by the chloroplast-specific isozyme of glyceraldehyde 3-phosphate dehydrogenase, producing glyceraldehyde 3-phosphate and P_i. The high concentrations of NADPH and ATP in the chloroplast stroma allow this thermodynamically unfavorable pair of reactions to proceed in

the direction of glyceraldehyde 3-phosphate formation. Triose phosphate isomerase then interconverts glyceraldehyde 3-phosphate and dihydroxyacetone phosphate, producing the two substrates for aldolase, which condenses them into fructose 1,6-bisphosphate. Thus far, the process has employed the same enzymes we saw in glycolysis, but operating in the reverse direction.

Most of the triose phosphate and fructose 1,6-bisphosphate produced by photosynthesis is used to regenerate ribulose 1,5-bisphosphate, the essential starting material for photosynthesis. Any excess triose phosphate is either converted to starch in the chloroplast and stored for later use or immediately exported to the cytosol and converted to sucrose for transport to growing regions of the plant.

Stage 3: Regeneration of Ribulose 1,5-Bisphosphate from Triose Phosphates

For the continuous flow of CO₂ into carbohydrate, ribulose 1,5-bisphosphate must be constantly regenerated. This is accomplished in a series of mostly reversible reactions ([Fig. 20-31](#)) that, together with stages 1 and 2, constitute the reductive pentose phosphate pathway summarized in [Figure 20-26](#). Three exergonic reactions, shown with blue arrows in [Figure 20-31](#), make the whole process irreversible. These are the reactions catalyzed by ② fructose 1,6-bisphosphatase, ⑤ sedoheptulose 1,7-bisphosphatase, and ⑨ ribulose 5-phosphate kinase.

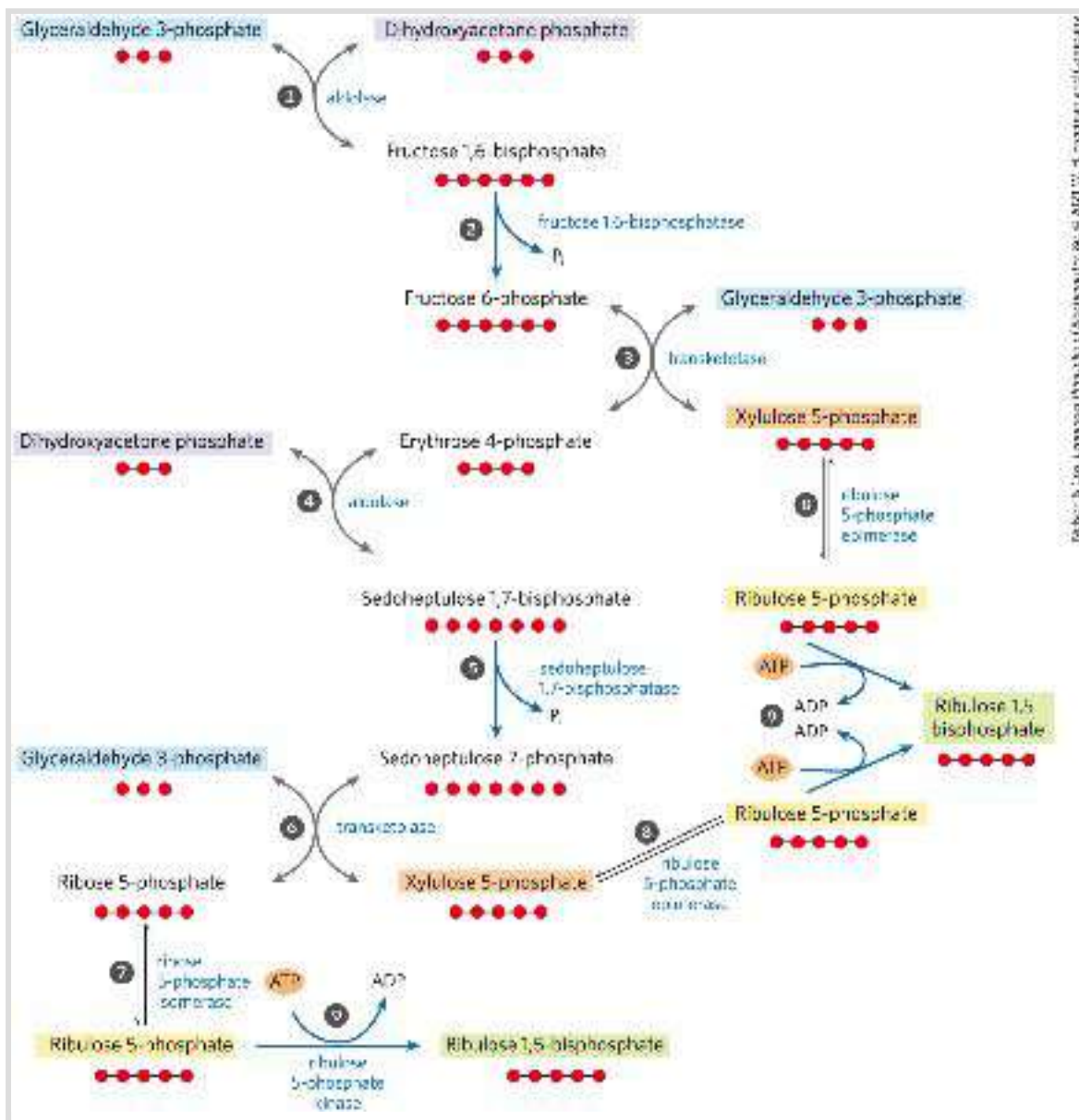
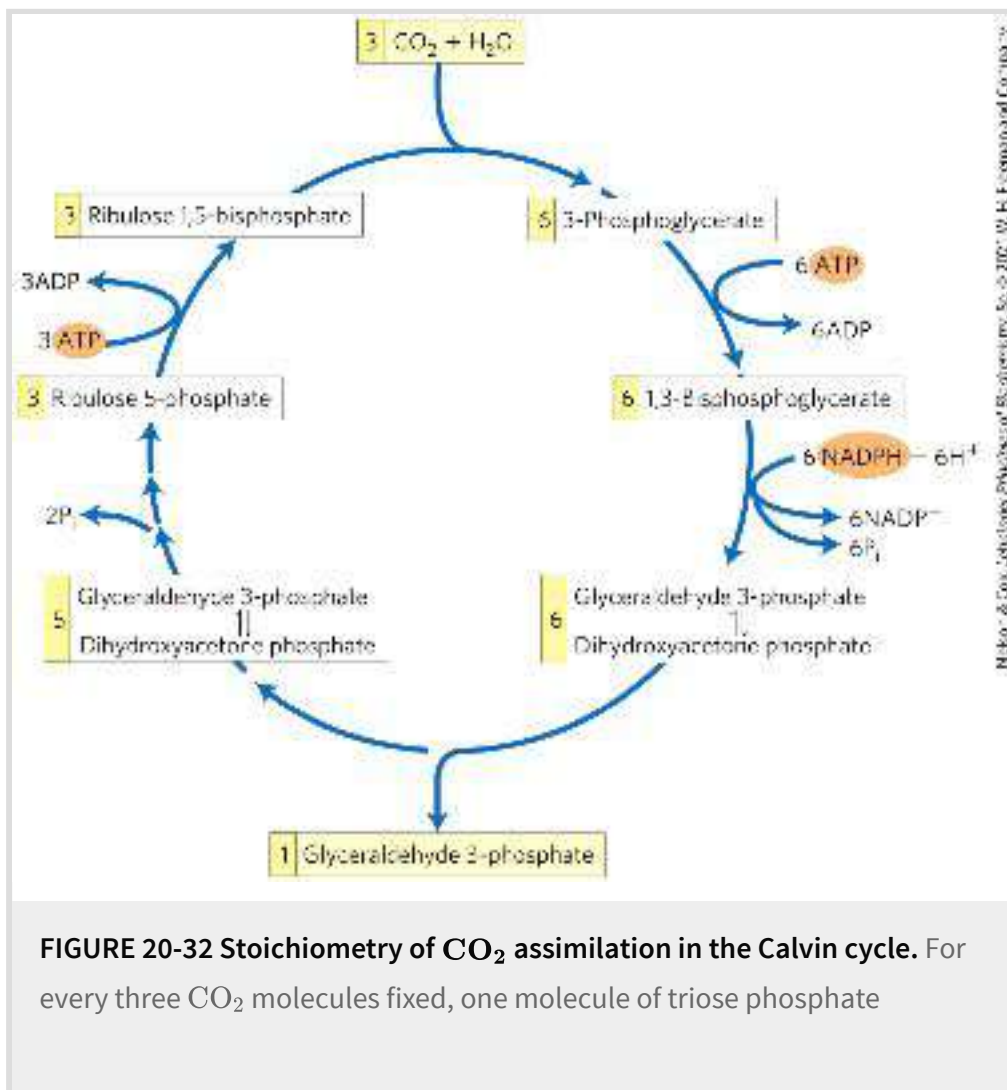


FIGURE 20-31 Third stage of CO₂ assimilation. This schematic diagram shows the interconversions of triose phosphates and pentose phosphates. Red dots represent the number of carbons in each compound. Compounds that appear more than once are highlighted. The starting materials are glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. Reactions catalyzed by aldolase (1 and 4) and transketolase (3 and 6) produce pentose phosphates that are converted to ribulose 1,5-bisphosphate — ribose 5-phosphate by 7 ribose 5-phosphate isomerase and xylulose 5-phosphate by 8 ribulose 5-phosphate epimerase. Ribulose 5-phosphate is phosphorylated 9, regenerating ribulose 1,5-bisphosphate. The reactions with blue arrows are exergonic and make the whole process irreversible: 2 fructose 1,6-bisphosphatase, 5 sedoheptulose 1,7-bisphosphatase, and 9 ribulose 5-phosphate kinase.

Synthesis of Each Triose Phosphate from CO₂ Requires Six NADPH and Nine ATP

P5 The net result of three turns of the Calvin cycle is the conversion of three molecules of CO₂ and one molecule of phosphate to a molecule of triose phosphate. The stoichiometry of the overall path from CO₂ to triose phosphate, with regeneration of ribulose 1,5-bisphosphate, is shown in [Figure 20-32](#).



(glyceraldehyde 3-phosphate) is produced and nine ATP and six NADPH are consumed.

One molecule of glyceraldehyde 3-phosphate is the net product of the CO₂-assimilation pathway. The other five triose phosphate molecules (15 carbons) are rearranged in steps ❶ to ❹ of [Figure 20-31](#) to form three molecules of ribulose 1,5-bisphosphate (15 carbons). The last step in this conversion requires one ATP per ribulose 1,5-bisphosphate, or a total of three ATP. Thus, in summary, for every molecule of triose phosphate produced by photosynthetic CO₂ assimilation, six NADPH and nine ATP are required.



NADPH and ATP are produced in the light-dependent reactions of photosynthesis in about the same ratio (2:3) as they are consumed in the Calvin cycle. Nine ATP molecules are converted to ADP and phosphate in the generation of a molecule of triose phosphate; eight of the phosphates are released as P_i and combined with eight ADP to regenerate ATP. The ninth phosphate is incorporated into the triose phosphate itself. To convert the ninth ADP to ATP, a molecule of P_i must be imported from the cytosol, as we shall see.

In the dark, the production of ATP and NADPH by photophosphorylation and the incorporation of CO₂ into triose phosphate (once referred to as the dark reactions) cease. The “dark reactions” of photosynthesis were so named to distinguish them from the *primary* light-driven reactions of electron transfer

to NADP^+ and synthesis of ATP. They do not, in fact, occur at significant rates in the dark and are thus more appropriately called the CO_2 -assimilation reactions. Later in this section we describe the regulatory mechanisms that turn CO_2 assimilation on in the light and turn it off in the dark.

The chloroplast stroma contains all the enzymes necessary to convert the triose phosphates produced by CO_2 assimilation (glyceraldehyde 3-phosphate and dihydroxyacetone phosphate) to starch, which is temporarily stored in the chloroplast as insoluble granules. Aldolase condenses the triose phosphates to fructose 1,6-bisphosphate; fructose 1,6-bisphosphatase produces fructose 6-phosphate; phosphohexose isomerase yields glucose 6-phosphate; and phosphoglucomutase produces glucose 1-phosphate, the starting material for starch synthesis (see [Section 20.6](#)).

All the reactions of the Calvin cycle except those catalyzed by rubisco, sedoheptulose 1,7-bisphosphatase, and ribulose 5-phosphate kinase also take place in animal tissues. Lacking these three enzymes, animals cannot carry out significant conversion of CO_2 to glucose.

A Transport System Exports Triose Phosphates from the Chloroplast and Imports Phosphate

The inner chloroplast membrane is impermeable to most phosphorylated compounds, including fructose 6-phosphate, glucose 6-phosphate, and fructose 1,6-bisphosphate. It does, however, have a specific antiporter that catalyzes the one-for-one exchange of P_i with a triose phosphate, either dihydroxyacetone phosphate or 3-phosphoglycerate ([Fig. 20-33](#)). This antiporter simultaneously moves P_i into the chloroplast, where it is used in photophosphorylation, and moves triose phosphate into the cytosol, where it can be used to synthesize sucrose, the form in which the fixed carbon is transported to distant plant tissues.

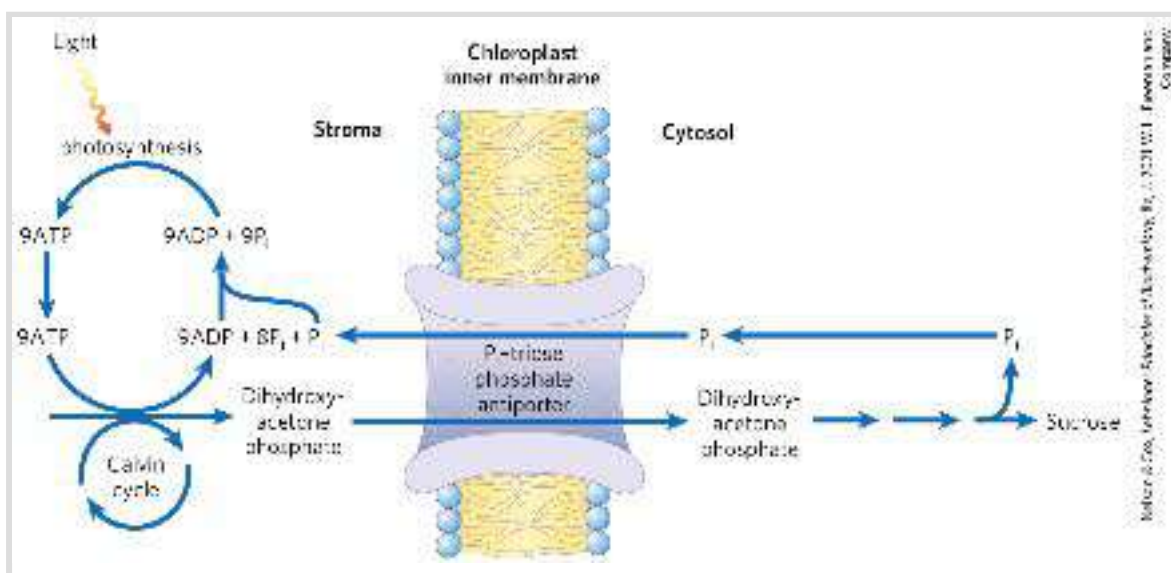


FIGURE 20-33 The P_i -triose phosphate antiport system of the chloroplast inner membrane. This transporter facilitates the exchange of cytosolic P_i for stromal dihydroxyacetone phosphate. The products of photosynthetic CO_2 assimilation are thus moved into the cytosol, where they serve as a starting point for sucrose biosynthesis, and P_i required for photophosphorylation is moved into the stroma. This same antiporter can transport 3-phosphoglycerate, and it acts indirectly in the export of ATP and reducing equivalents (see [Fig. 20-34](#)).

Sucrose synthesis in the cytosol and starch synthesis in the chloroplast are the major pathways by which the excess triose phosphate from photosynthesis is harvested. Sucrose synthesis (described later) releases four P_i molecules from the four triose phosphates required to make sucrose. For every molecule of triose phosphate removed from the chloroplast, one P_i is transported into the chloroplast, providing the ninth P_i mentioned above, to be used in regenerating ATP. If this exchange were blocked, triose phosphate synthesis would quickly deplete the available P_i in the chloroplast, slowing ATP synthesis and suppressing assimilation of CO_2 into starch.

The P_i -triose phosphate antiport system serves one additional function. ATP and reducing power are needed in the cytosol for a variety of synthetic and energy-requiring reactions. These requirements are met to an as-yet-undetermined degree by mitochondria, but a second potential source of energy is the ATP and NADPH generated in the chloroplast stroma during the light-dependent reactions. However, neither ATP nor NADPH can cross the chloroplast membrane. The P_i -triose phosphate antiport system has the indirect effect of moving ATP equivalents and reducing equivalents from the chloroplast to the cytosol ([Fig. 20-34](#)). Dihydroxyacetone phosphate formed in the stroma is transported to the cytosol, where it is converted by glycolytic enzymes to 3-phosphoglycerate, generating ATP and NADH. 3-Phosphoglycerate reenters the chloroplast, completing the cycle.

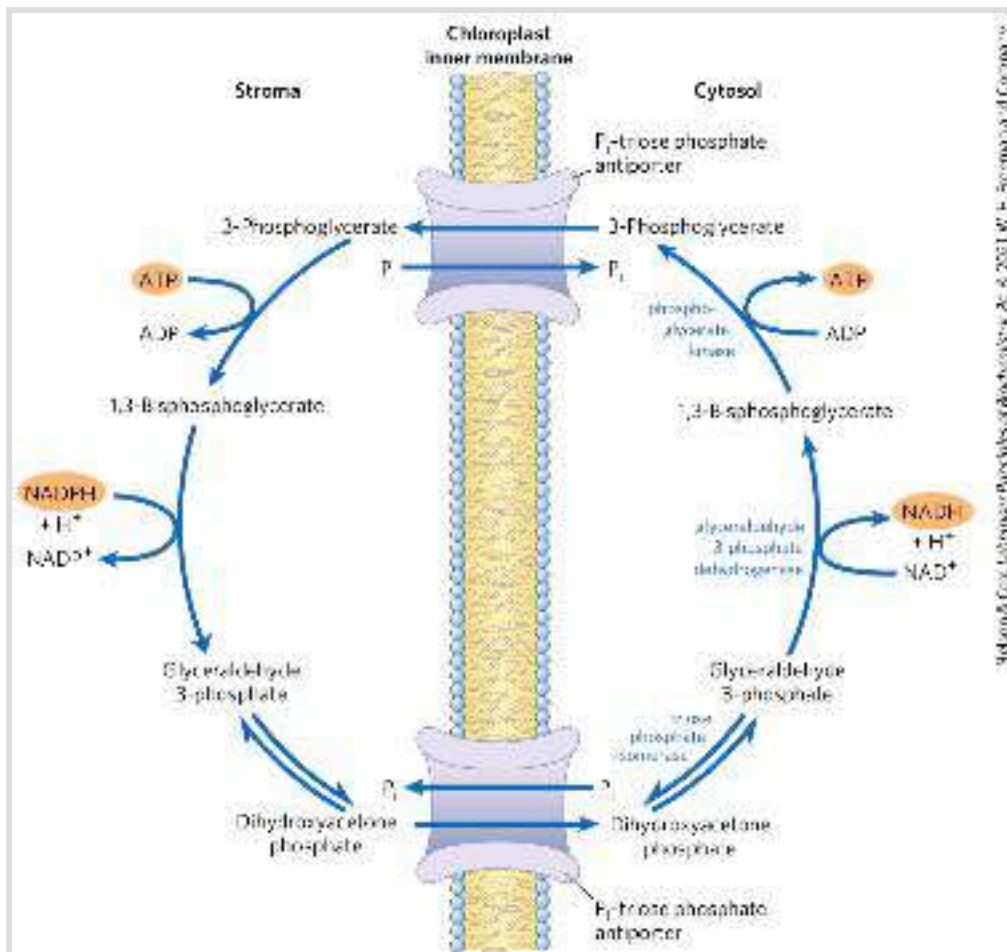


FIGURE 20-34 Role of the P_i-triose phosphate antiporter in the transport of ATP and reducing equivalents. Dihydroxyacetone phosphate leaves the chloroplast and is converted to glyceraldehyde 3-phosphate in the cytosol. The cytosolic glyceraldehyde 3-phosphate dehydrogenase and phosphoglycerate kinase reactions then produce NADH, ATP, and 3-phosphoglycerate. The latter reenters the chloroplast and is reduced to dihydroxyacetone phosphate, completing a cycle that effectively moves ATP and reducing equivalents (NAD(P)H) from chloroplast to cytosol.

Four Enzymes of the Calvin Cycle Are Indirectly Activated by Light

The reductive assimilation of CO_2 requires a lot of ATP and NADPH, and their stromal concentrations increase when chloroplasts are illuminated ([Fig. 20-35](#)). The light-induced transport of protons across the thylakoid membrane also increases the stromal pH from about 7 to about 8, and it is accompanied by a flow of Mg^{2+} from the thylakoid compartment into the stroma, raising the $[\text{Mg}^{2+}]$ from 1 to 3 mM to 3 to 6 mM.

 Several stromal enzymes have evolved to take advantage of these light-induced conditions, which signal the availability of ATP and NADPH: the enzymes are more active in an alkaline environment and at high $[\text{Mg}^{2+}]$. For example, activation of rubisco by formation of carbamoyllysine is faster at alkaline pH, and high stromal $[\text{Mg}^{2+}]$ favors formation of the enzyme's active Mg^{2+} complex. Fructose 1,6-bisphosphatase requires Mg^{2+} and is very dependent on pH ([Fig. 20-36](#)); its activity increases more than 100-fold when pH and $[\text{Mg}^{2+}]$ rise during chloroplast illumination.

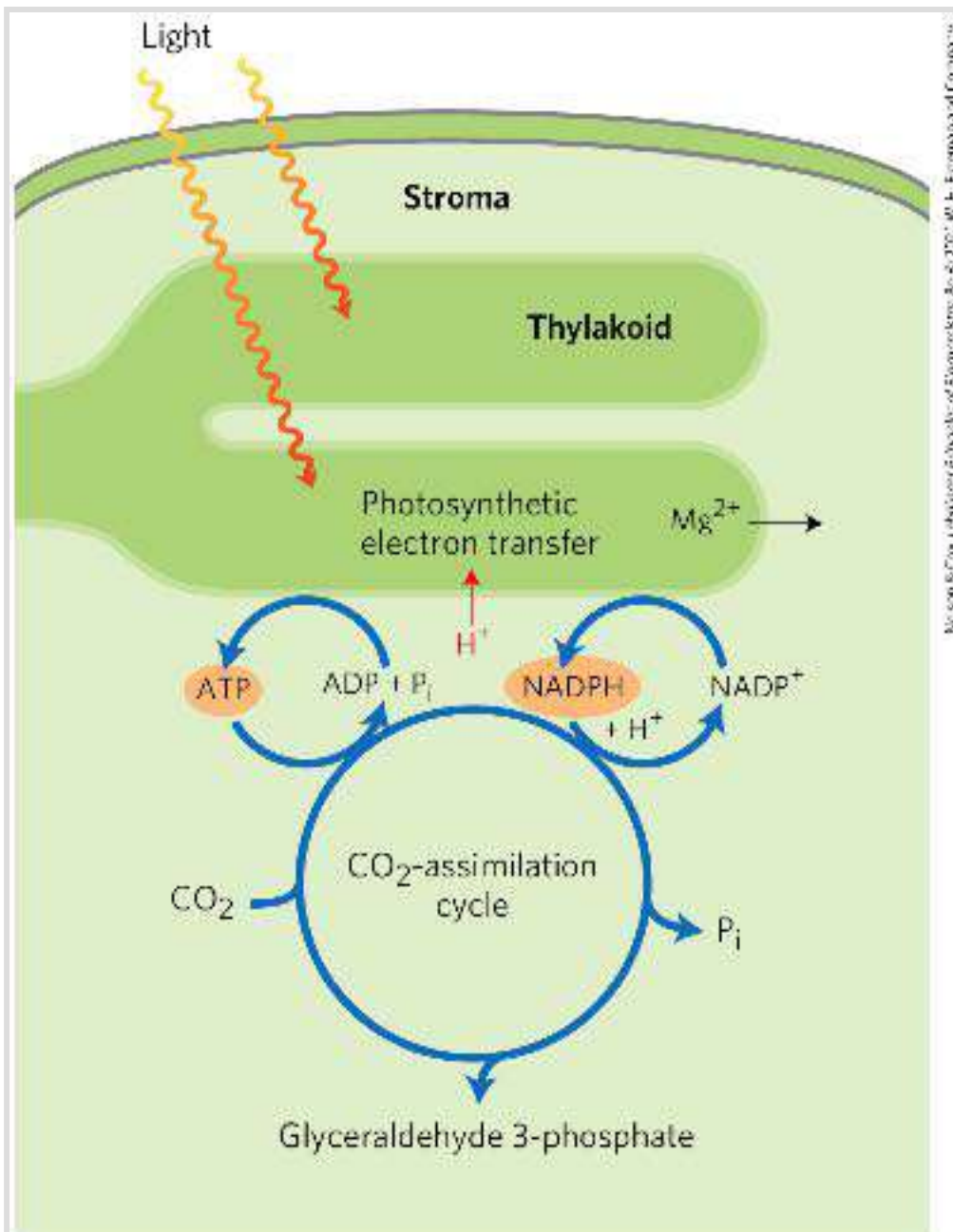


FIGURE 20-35 Source of ATP and NADPH. ATP and NADPH produced by the light-dependent reactions are essential substrates for the reduction of CO_2 . The photosynthetic reactions that produce ATP and NADPH are accompanied by movement of protons (red) from the stroma into the thylakoid, creating alkaline conditions in the stroma. Magnesium ions pass from the thylakoid into the stroma, increasing the stromal $[Mg^{2+}]$.

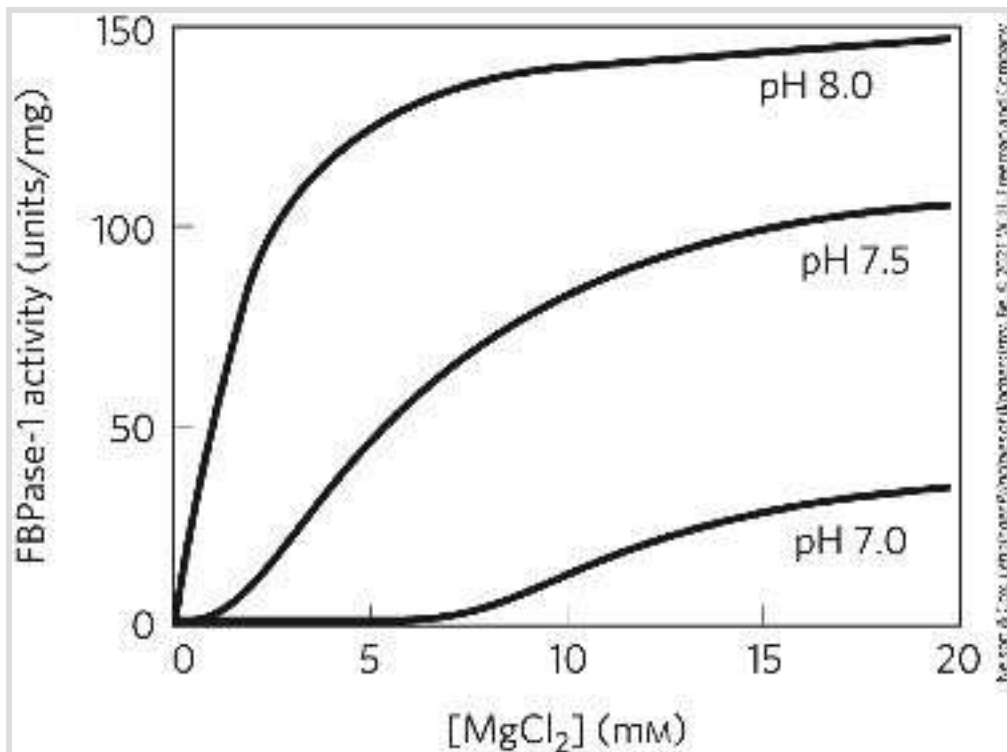


FIGURE 20-36 Activation of chloroplast fructose 1,6-bisphosphatase.

Reduced fructose 1,6-bisphosphatase (FBPase-1) is activated by light and by the combination of high pH and high $[\text{Mg}^{2+}]$ in the stroma, both of which are results of illumination. [Information from B. Halliwell, *Chloroplast Metabolism: The Structure and Function of Chloroplasts in Green Leaf Cells*, p. 97, Clarendon Press, 1984.]

P5 Four Calvin cycle enzymes are subject to a special type of regulation by light. Ribulose 5-phosphate kinase, fructose 1,6-bisphosphatase, sedoheptulose 1,7-bisphosphatase, and glyceraldehyde 3-phosphate dehydrogenase are activated by light-driven reduction of disulfide bonds between two Cys residues critical to their catalytic activities. When these Cys residues are disulfide-bonded (oxidized), the enzymes are inactive; this is the normal situation in the dark. With illumination, electrons flow from photosystem I to ferredoxin ([Fig. 20-12](#)), which passes electrons to a small, soluble, disulfide-containing protein called

thioredoxin (Fig. 20-37), in a reaction catalyzed by **ferredoxin: thioredoxin reductase**. Reduced thioredoxin donates electrons for the reduction of the disulfide bonds of the light-activated enzymes, and these reductive cleavage reactions are accompanied by conformational changes that increase enzyme activities. At nightfall, the Cys residues in the four enzymes are reoxidized to their disulfide forms, the enzymes are inactivated, and ATP is not expended in CO₂ assimilation. Instead, starch synthesized and stored during the daytime is degraded to fuel glycolysis and oxidative phosphorylation at night.

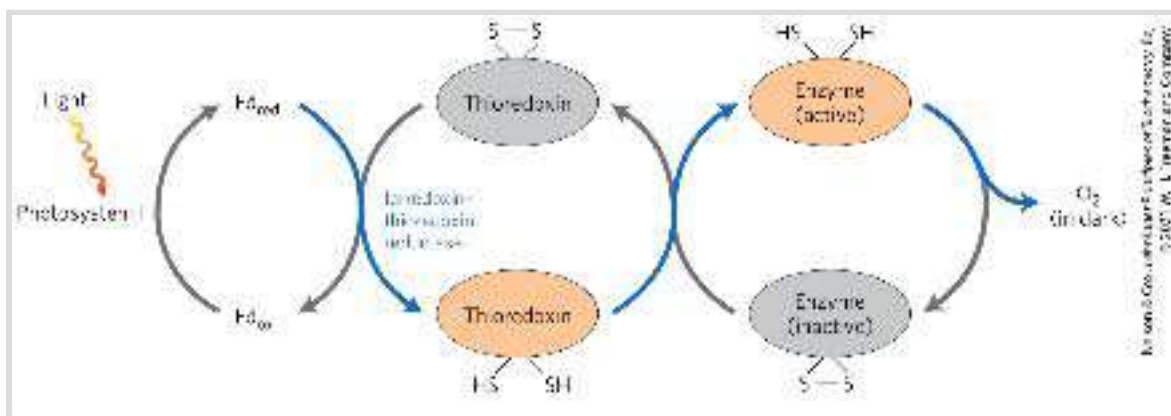


FIGURE 20-37 Light activation of several enzymes of the Calvin cycle. The light activation is mediated by thioredoxin, a small, disulfide-containing protein. In the light, thioredoxin is reduced by electrons moving from photosystem I through ferredoxin (Fd) (blue arrows), then thioredoxin reduces critical disulfide bonds in each of the enzymes sedoheptulose 1,7-bisphosphatase, fructose 1,6-bisphosphatase, ribulose 5-phosphate kinase, and glyceraldehyde 3-phosphate dehydrogenase, activating these enzymes. In the dark, the OSH groups undergo reoxidation to disulfides, inactivating the four enzymes.

Glucose 6-phosphate dehydrogenase, the first enzyme in the *oxidative* pentose phosphate pathway, is also regulated by this light-driven reduction mechanism, but in the opposite sense.

During the day, when photosynthesis produces plenty of NADPH, this enzyme is not needed for NADPH production. Reduction of a critical disulfide bond by electrons from ferredoxin *inactivates* the enzyme.

SUMMARY 20.4 *Carbon-Assimilation Reactions*

■ Photosynthesis in eukaryotes takes place in chloroplasts. In the CO₂-assimilating reactions (the Calvin cycle), ATP and NADPH are used to reduce CO₂ to triose phosphates. These reactions occur in three stages: the fixation reaction itself, catalyzed by rubisco; reduction of the resulting 3-phosphoglycerate to glyceraldehyde 3-phosphate; and regeneration of ribulose 1,5-bisphosphate from triose phosphates.

■ Stromal enzymes rearrange the carbon skeletons of triose phosphates to generate intermediates of three, four, five, six, and seven carbons, eventually yielding pentose phosphates. The pentose phosphates are converted to ribulose 5-phosphate, which is phosphorylated to ribulose 1,5-bisphosphate to complete the Calvin cycle.

■ The cost of fixing three CO₂ into one triose phosphate is nine ATP and six NADPH, which are provided by the light-dependent reactions of photosynthesis.

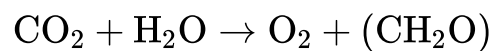
■ An antiporter in the inner chloroplast membrane exchanges P_i in the cytosol for 3-phosphoglycerate or dihydroxyacetone phosphate molecules produced by CO₂ assimilation in the stroma. Oxidation of dihydroxyacetone phosphate in the cytosol

generates ATP and NADH, thus moving ATP and reducing equivalents from the chloroplast to the cytosol.

■ Four enzymes of the Calvin cycle are activated indirectly by light and are inactive in the dark, so that hexose synthesis does not compete with glycolysis — which is required to provide energy in the dark.

20.5 Photorespiration and the C₄ and CAM Pathways

As we have seen, photosynthetic cells produce O₂ (by the splitting of H₂O) during the light-driven reactions and use CO₂ during the light-independent processes, so the net gaseous change during photosynthesis is the uptake of CO₂ and release of O₂:



In the dark, plants also carry out **mitochondrial respiration**, the oxidation of substrates to CO₂ and the conversion of O₂ to H₂O. And there is another process in plants that, like mitochondrial respiration, consumes O₂ and produces CO₂ and, like photosynthesis, is driven by light. This process, photorespiration, is a costly side reaction of photosynthesis, a result of the lack of specificity of the enzyme rubisco. In this section we describe this side reaction and the strategies plants use to minimize its metabolic consequences.

Photorespiration Results from Rubisco's Oxygenase Activity



Rubisco is not absolutely specific for CO₂ as a substrate. Molecular oxygen (O₂) competes with CO₂ at the active site, and about once in every three or four turnovers, rubisco catalyzes the condensation of O₂ with ribulose 1,5-bisphosphate to form 3-phosphoglycerate and **2-phosphoglycolate** (Fig. 20-38), a

metabolically unneeded product. This is the oxygenase activity referred to in the full name of rubisco: *ribulose 1,5-bisphosphate carboxylase/oxygenase*. The reaction with O_2 results in no fixation of CO_2 and is presumably a net liability to the cell; salvaging the carbons from 2-phosphoglycolate (by the pathway outlined below) consumes significant amounts of cellular energy and releases some previously fixed CO_2 .

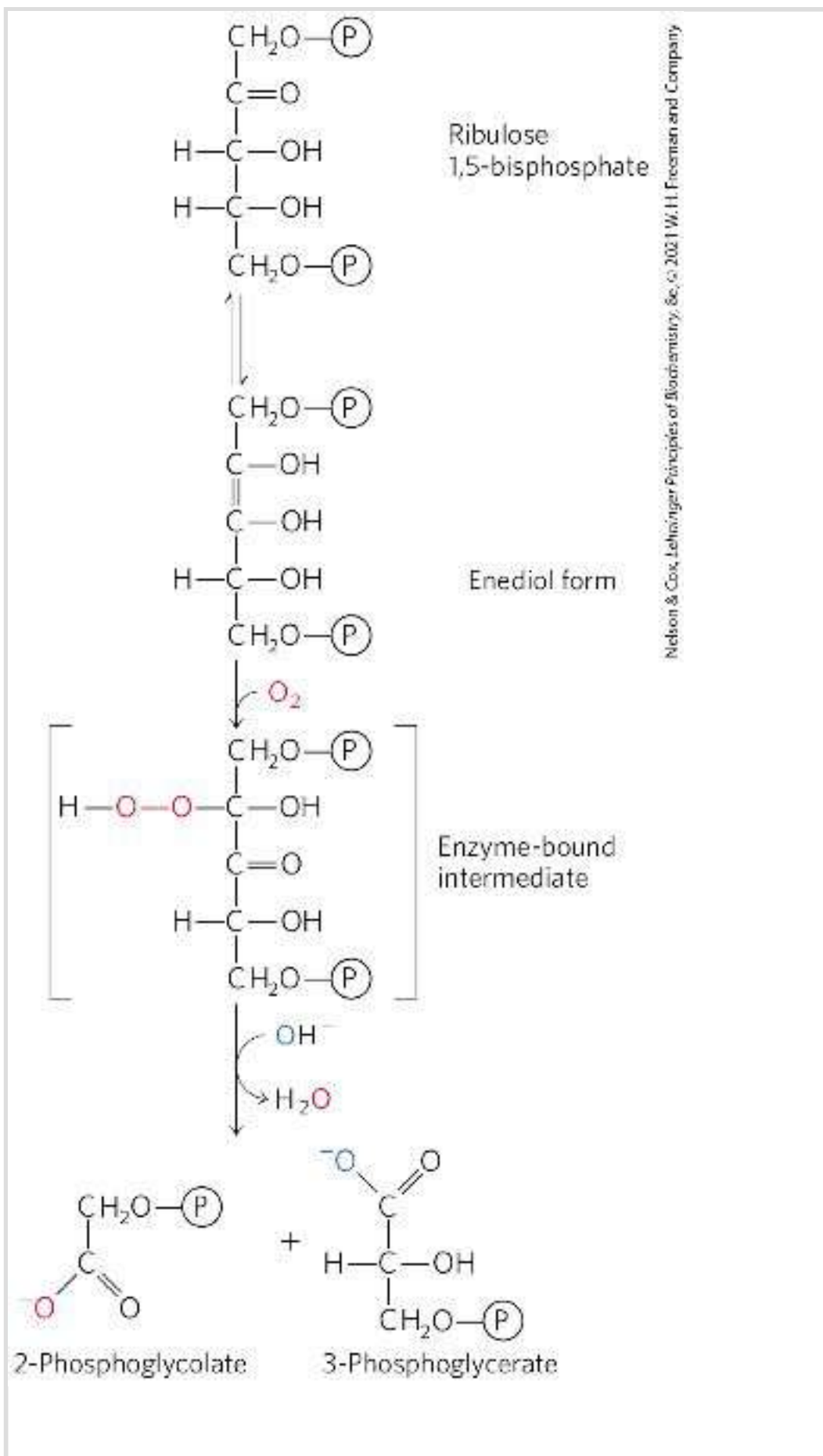


FIGURE 20-38 Oxygenase activity of rubisco. Rubisco can incorporate O_2 rather than CO_2 into ribulose 1,5-bisphosphate. The unstable intermediate thus formed splits into 2-phosphoglycolate (recycled as described in [Fig. 20-39](#)) and 3-phosphoglycerate, which can reenter the Calvin cycle.

Phosphoglycolate Is Salvaged in a Costly Set of Reactions in C_3 Plants

The [glycolate pathway](#) converts two molecules of 2-phosphoglycolate to a molecule of serine (three carbons) and a molecule of CO_2 ([Fig. 20-39](#)). In the chloroplast, a phosphatase converts 2-phosphoglycolate to glycolate, which is exported to the peroxisome. There, glycolate is oxidized by molecular oxygen, and the resulting aldehyde (glyoxylate) undergoes transamination to glycine. The hydrogen peroxide formed as a side product of glycolate oxidation is rendered harmless by peroxidases in the peroxisome. Glycine passes from the peroxisome to the mitochondrial matrix, where it undergoes oxidative decarboxylation by the glycine decarboxylase complex, an enzyme similar in structure and mechanism to two mitochondrial complexes we have already encountered: the pyruvate dehydrogenase complex and the α -ketoglutarate dehydrogenase complex ([Chapter 16](#)). The **glycine decarboxylase complex** oxidizes glycine to CO_2 and NH_3 , with the concomitant reduction of NAD^+ to NADH and transfer of the remaining carbon from glycine to the cofactor tetrahydrofolate. The one-carbon unit carried on tetrahydrofolate is then transferred to a second glycine by serine hydroxymethyltransferase, producing serine. The net reaction catalyzed by the glycine decarboxylase complex and serine hydroxymethyltransferase is



The serine is converted to hydroxypyruvate, then to glycerate, and finally to 3-phosphoglycerate, which is used to regenerate ribulose 1,5-bisphosphate, completing the long, expensive cycle ([Fig. 20-39](#)).

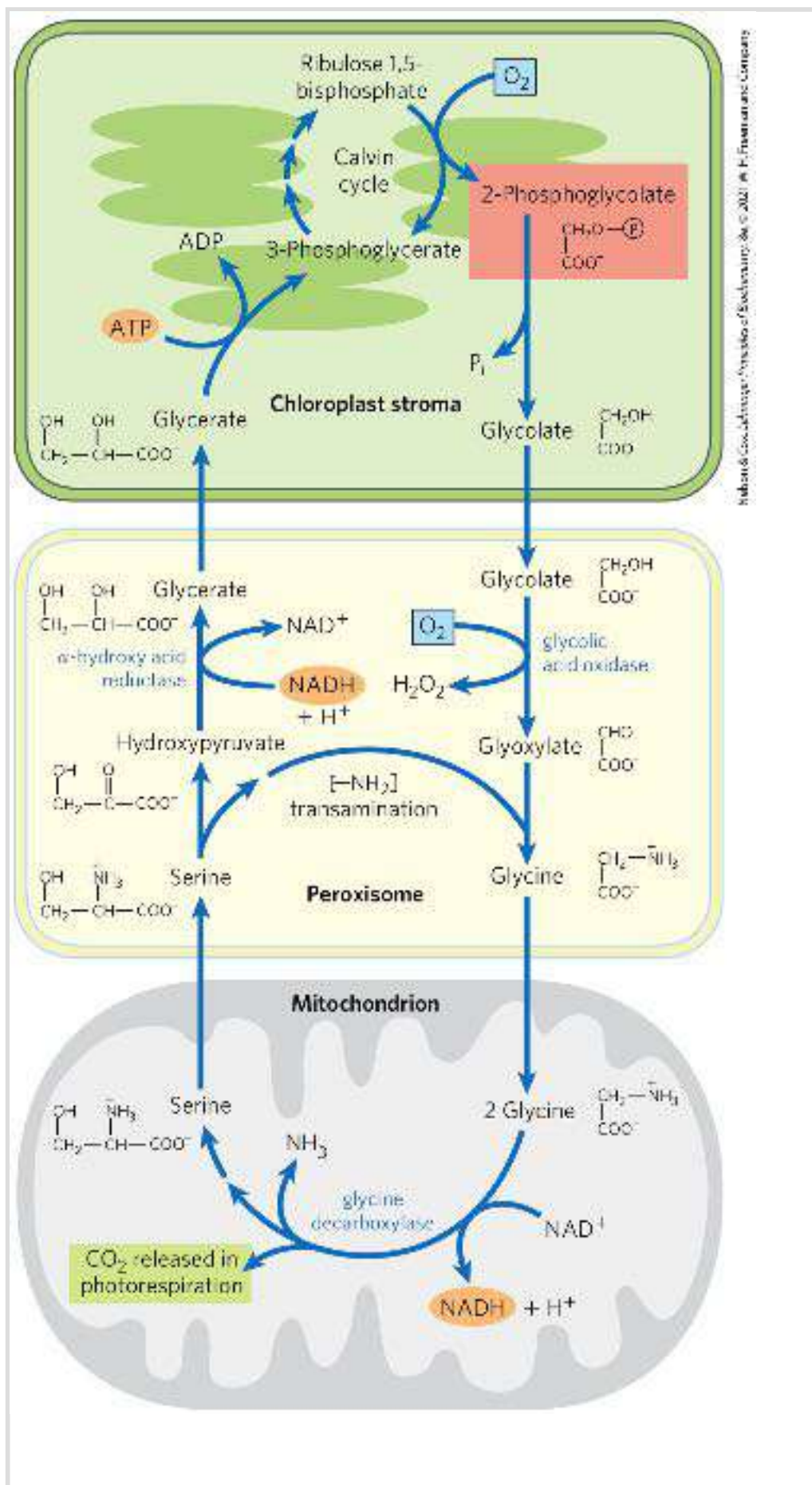


FIGURE 20-39 Glycolate pathway. This pathway, which salvages 2-phosphoglycolate (shaded light red) by converting it to serine and, eventually, to 3-phosphoglycerate, involves three cellular compartments. Glycolate formed by dephosphorylation of 2-phosphoglycolate in chloroplasts is oxidized to glyoxylate and transaminated to glycine in peroxisomes. In mitochondria, two glycine molecules condense to form serine and CO_2 , released in photorespiration. This reaction is catalyzed by glycine decarboxylase, an enzyme present at very high levels in the mitochondria of C_3 plants. The serine is converted to hydroxypyruvate and then to glycerate in peroxisomes; glycerate reenters the chloroplasts to be phosphorylated, rejoining the Calvin cycle. Oxygen is consumed at two steps during photorespiration.

In bright sunlight, the carbon flux through the glycolate salvage pathway can be very high, producing about five times more CO_2 than is typically produced by all the oxidations of the citric acid cycle. To generate this large flux, mitochondria contain prodigious amounts of the glycine decarboxylase complex: the four proteins of the complex make up *half* of all the protein in the mitochondrial matrix in the leaves of pea and spinach plants. In nonphotosynthetic parts of a plant, such as potato tubers, mitochondria have very low concentrations of the glycine decarboxylase complex.

The practical effects of this inefficiency are large and costly. The average yield of soybeans and wheat in the United States is reduced by an estimated 36% and 20%, respectively, by the necessity of recycling glycolate from photorespiration.



The combined activity of the rubisco oxygenase and the glycolate salvage pathway consumes O_2 and produces CO_2 — hence the name photorespiration. Unlike mitochondrial respiration, photorespiration does not conserve energy and actually inhibits net biomass formation. This inefficiency has led to evolutionary

adaptations in the CO₂-assimilation processes, particularly in plants that have evolved in warm climates. The apparent inefficiency of rubisco, and its effect in limiting biomass production, has inspired efforts to genetically engineer a “better” rubisco, but this goal is not, as yet, within reach ([Box 20-1](#)).

BOX 20-1

Will Genetic Engineering of Photosynthetic Organisms Increase Their Efficiency?

Three pressing world problems have prompted serious attention to the possibility of engineering plants to be more efficient in converting sunlight into biomass: the greenhouse effect of increasing levels of atmospheric CO₂ on climate change, the finite supply of oil for generating energy, and the need for more and better food for the world's growing population.

The concentration of CO₂ in the earth's atmosphere has risen steadily over the past 50 years ([Fig. 1](#)), a combined effect of the use of fossil fuels for energy and the clearing and burning of tropical forests to allow use of the land for agriculture. As atmospheric CO₂ increases, the atmosphere absorbs more heat radiated from the earth's surface and reradiates more heat toward the surface of the planet (and in all other directions). Retention of heat raises the temperature at the surface of the earth; this is the greenhouse effect. One way to limit the increase in atmospheric CO₂ would be to engineer plants or microorganisms with a greater capacity for sequestering CO₂.

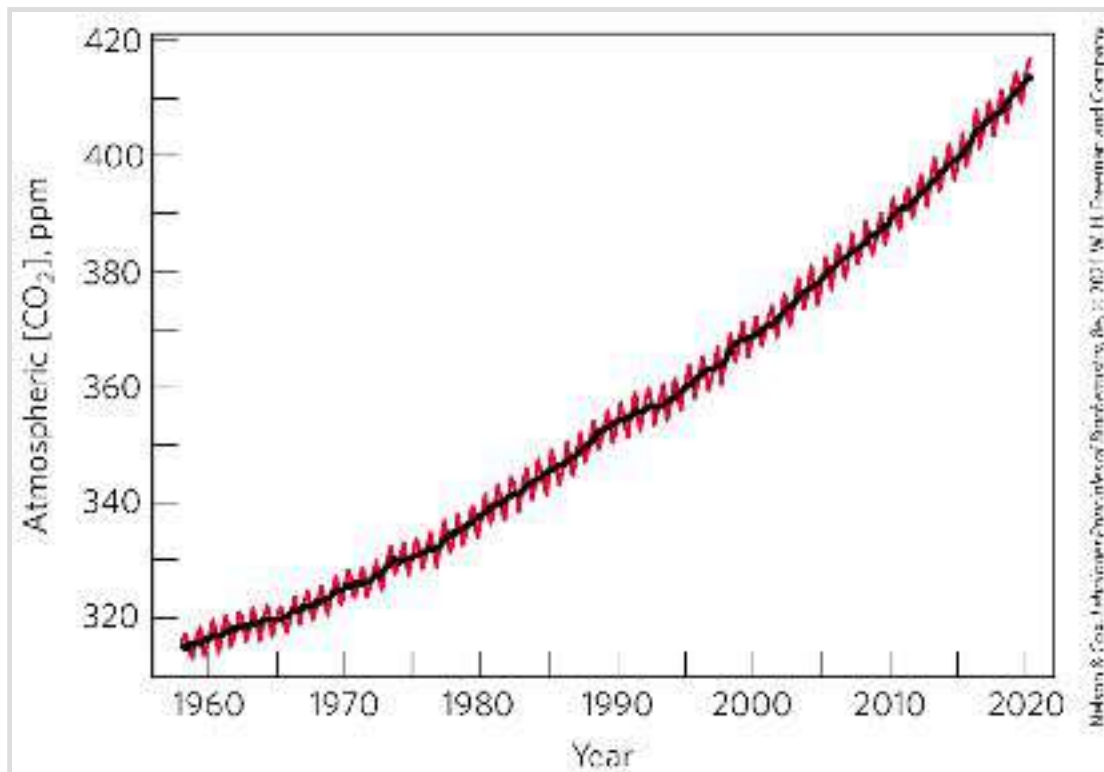


FIGURE 1 The concentration of CO₂ in the atmosphere measured at the Mauna Loa Observatory in Hawaii. [Data from the National Oceanic and Atmospheric Administration and the Scripps Institution of Oceanography CO₂ Program.]

The estimated amount of total carbon in all terrestrial systems (atmosphere, soil, biomass) is about 3,200 gigatons (GT), or 3,200 billion metric tons. The atmosphere contains another 760 GT of CO₂.

The flux of carbon through these terrestrial reservoirs ([Fig. 2](#)) is largely due to the photosynthetic activities of plants and the degradative activities of microorganisms. Plants fix some 123 GT of carbon annually, then immediately release about half of that to the atmosphere as they respire. Much of the remainder is gradually released to the atmosphere by microbial action on dead plant materials, but biomass is sequestered in woody plants and trees for decades or centuries. Anthropogenic carbon flux, the amount of CO₂ released into the atmosphere by human activities, is 9 GT per year — small compared with total biomass, but enough to tip the balance toward increased CO₂ in the atmosphere. Estimates indicate that the forests of North America sequester 0.7 GT of carbon annually, which represents about a tenth of the annual *global* production of CO₂ from fossil fuels. Clearly, preservation of forests and reforestation are effective ways to limit the flow of CO₂ back into the atmosphere.

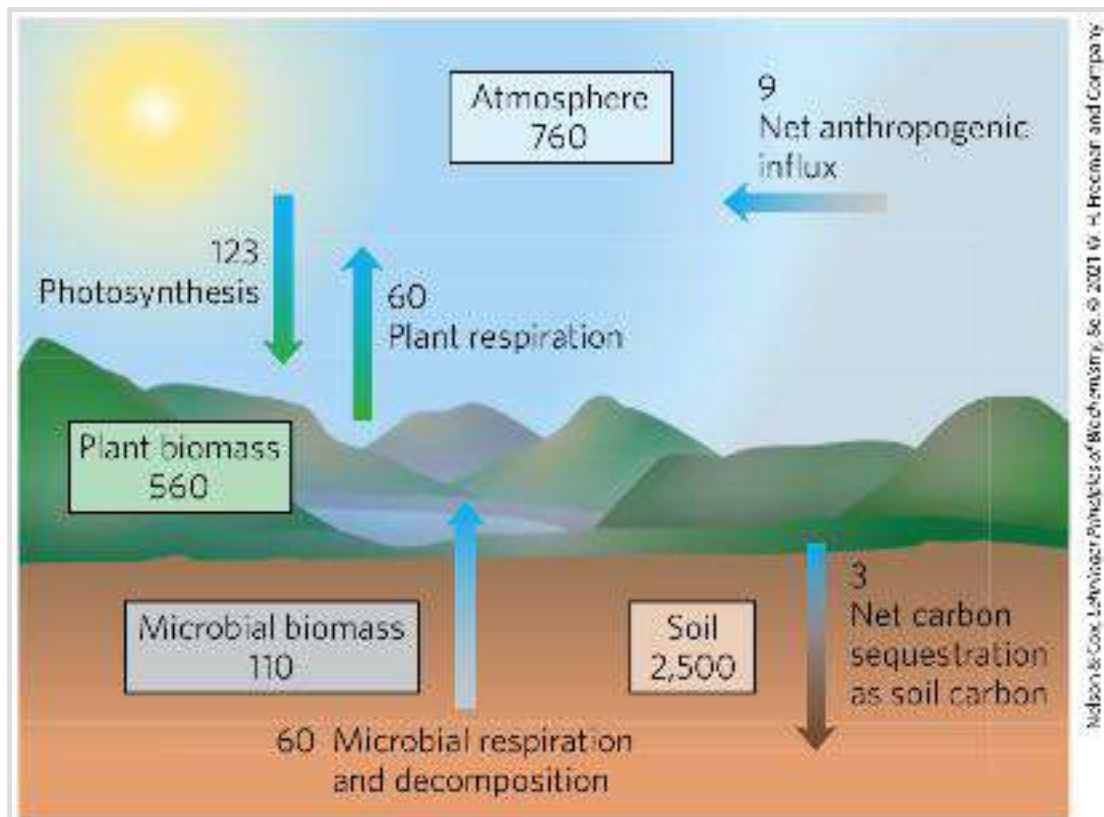


FIGURE 2 The terrestrial carbon cycle. Carbon stocks (boxed text) are shown as gigatons (GT), and fluxes (arrows) are shown in GT per year. Animal biomass is negligible here — less than 0.5 GT. [Information from C. Jansson et al., *BioScience* 60:683, 2010, Fig. 1.]

A second approach to limiting the increase of atmospheric CO₂, while also addressing the need to replace dwindling fossil fuels, is to use renewable biomass as a source of ethanol to replace fossil fuels in internal combustion engines. This reduces the *unidirectional* movement of carbon from fossil fuels into the atmospheric pool of CO₂, replacing it with the *cyclic* flow of CO₂ from ethanol to CO₂ and back to biomass. When maize, wheat, or switchgrass is fermented to ethanol for fuel, every increase in biomass production brought about by more efficient photosynthesis should result in a corresponding decrease in the use of fossil fuels.

Finally, engineering of food crops to yield more food per acre of land, or per hour of work, could improve human nutrition worldwide.

In principle, these goals might be accomplished by developing a rubisco that didn't also catalyze the wasteful reaction with O₂, or by increasing the turnover number for rubisco, or by increasing the level of rubisco or other enzymes in the pathway for CO₂ fixation. Rubisco, as we have noted, is an unusually inefficient enzyme, with a

turnover number of 3 s^{-1} at 25°C ; most enzymes have turnover numbers orders of magnitude larger. It also catalyzes the wasteful reaction with oxygen, which further reduces its efficiency in fixing CO_2 and producing biomass. If rubisco could be genetically engineered to turn over faster or to be more selective for CO_2 relative to O_2 , would the effect be greater photosynthetic production of biomass and thus greater sequestration of CO_2 , greater production of nonfossil fuel, and improved nutrition?

The traditional view of metabolic pathways held that one step in any pathway was the slowest and therefore the limiting factor in material flow through the pathway. However, efforts to engineer cells or organisms to produce more of the “limiting” enzyme in a pathway have often given discouraging results; the organisms often show little or no change in the flux through that pathway. The Calvin cycle is an instructive case in point. Increasing the amount of rubisco in plant cells through genetic engineering has little or no effect on the rate of CO_2 conversion into carbohydrate. Similarly, changes in the levels of enzymes known to be regulated by light and therefore suspected of playing key roles in the regulation of the CO_2 -assimilation pathway (fructose 1,6-bisphosphatase, 3-phosphoglycerate kinase, and glyceraldehyde 3-phosphate dehydrogenase) also produce little or no significant improvement in photosynthetic rate. This should probably not be surprising; in the living organism, pathways can be limited by more than one enzymatic step, because every change in one step results in compensating changes in other steps. **Metabolic control analysis** is the science of measuring, understanding, and eventually altering the factors that govern the overall flux through a pathway. Its application will be essential to the success of engineering plants for higher efficiency or greater yield.

In C_4 Plants, CO_2 Fixation and Rubisco Activity Are Spatially Separated

In many plants that grow in the tropics (and in temperate-zone crop plants native to the tropics, such as maize, sugarcane, and sorghum) a

mechanism has evolved to circumvent the problem of wasteful photorespiration. The step in which CO_2 is fixed into a three-carbon product, 3-phosphoglycerate, is preceded by several steps, one of which is temporary fixation of CO_2 into oxaloacetate, a four-carbon compound. Plants that use this process are referred to as **C₄ plants**, and the assimilation process is known as the **C₄ pathway**, by comparison to the C₃ pathway in which CO_2 is first fixed in the three-carbon compound 3-phosphoglycerate.



The C₄ plants, which typically grow at high light intensity and high temperatures, have several important characteristics: high photosynthetic rates, high growth rates, low photorespiration rates, low rates of water loss, and a specialized leaf structure.

Photosynthesis in the leaves of C₄ plants involves two cell types: mesophyll and bundle-sheath cells (**Fig. 20-40a**).

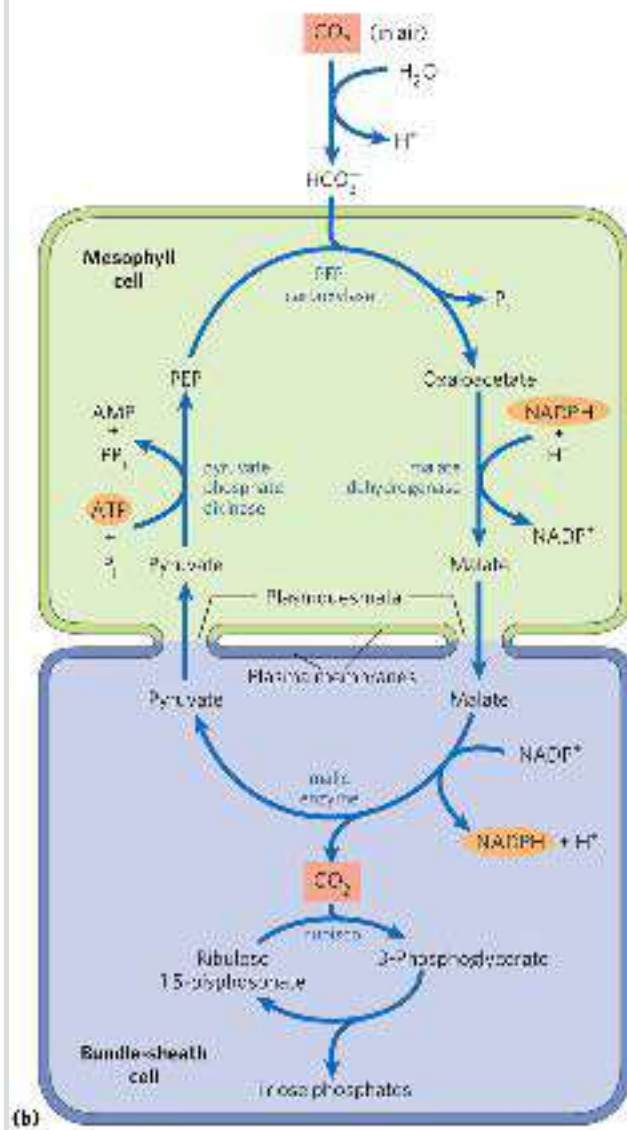
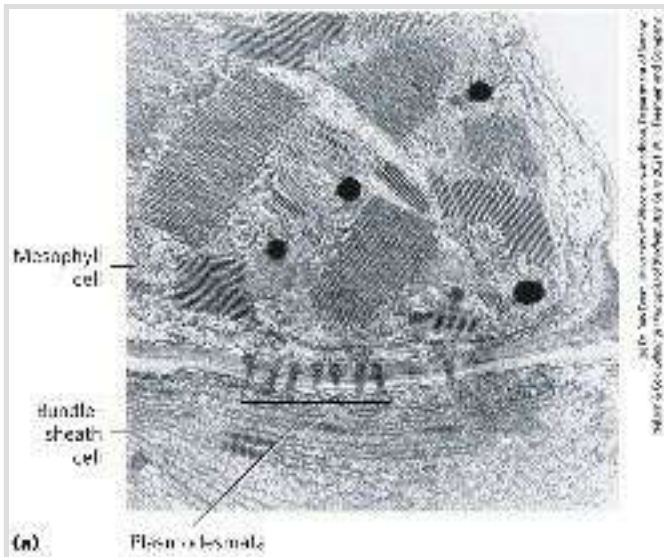
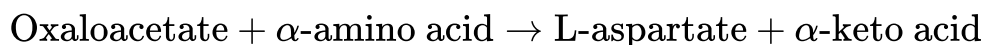


FIGURE 20-40 CO₂ assimilation in C₄ plants. The C₄ pathway, involving mesophyll cells and bundle-sheath cells, predominates in plants of tropical origin. (a) Electron micrograph showing chloroplasts of adjacent mesophyll and bundle-sheath cells. The bundle-sheath cell contains starch granules. Plasmodesmata connecting the two cells are visible. (b) The C₄ pathway of CO₂ assimilation, which occurs through a four-carbon intermediate.

The fixation of CO₂ into the four-carbon oxaloacetate occurs in the cytosol of leaf mesophyll cells. The reaction is catalyzed by **phosphoenolpyruvate (PEP) carboxylase**, for which the substrate is HCO₃⁻, not CO₂. The oxaloacetate thus formed is either reduced to malate at the expense of NADPH (as shown in [Fig. 20-40b](#)) or converted to aspartate by transamination:



The malate or aspartate formed in the mesophyll cells then passes into neighboring bundle-sheath cells through plasmodesmata, protein-lined channels that connect two plant cells and provide a path for movement of metabolites and even small proteins between cells. In the bundle-sheath cells, malate is oxidized and decarboxylated to yield pyruvate and CO₂ by the action of **malic enzyme**, reducing NADP⁺. In plants that use aspartate as the CO₂ carrier, aspartate arriving in bundle-sheath cells is transaminated to form oxaloacetate and reduced to malate, then the CO₂ is released by malic enzyme or PEP carboxykinase. Labeling experiments show that the free CO₂ released in the bundle-sheath cells is the same CO₂ molecule originally fixed into oxaloacetate in the mesophyll cells. This CO₂ is now fixed again, this time by rubisco, in exactly the same reaction

that occurs in C_3 plants: incorporation of CO_2 into C-1 of 3-phosphoglycerate.

The pyruvate formed by decarboxylation of malate in bundle-sheath cells is transferred back to the mesophyll cells, where it is converted to PEP by an unusual enzymatic reaction catalyzed by **pyruvate phosphate dikinase** ([Fig. 20-40b](#)). This enzyme is called a dikinase because two different molecules are simultaneously phosphorylated by one molecule of ATP: pyruvate to PEP, and phosphate to pyrophosphate. The pyrophosphate is subsequently hydrolyzed to phosphate, so two high-energy phosphate groups of ATP are used in regenerating PEP. The PEP is now ready to receive another molecule of CO_2 in the mesophyll cell.

The PEP carboxylase of mesophyll cells has a high affinity for HCO_3^- (which is favored relative to CO_2 in aqueous solution) and can fix CO_2 more efficiently than can rubisco. Unlike rubisco, it does not use O_2 as an alternative substrate, so there is no competition between CO_2 and O_2 . The PEP carboxylase reaction, then, serves to fix and concentrate CO_2 in the form of malate.  Release of CO_2 from malate in the bundle-sheath cells yields a sufficiently high local concentration of CO_2 for rubisco to function near its maximal rate, and for suppression of the enzyme's oxygenase activity.

Once CO_2 is fixed into 3-phosphoglycerate in the bundle-sheath cells, the other reactions of the Calvin cycle take place exactly as described earlier. Thus in C_4 plants, mesophyll cells carry out CO_2 assimilation by the C_4 pathway and bundle-sheath cells synthesize starch and sucrose by the C_3 pathway.

Three enzymes of the C_4 pathway are regulated by light, becoming more active in daylight. Malate dehydrogenase is activated by the thioredoxin-dependent reduction mechanism shown in [Figure 20-37](#); PEP carboxylase is activated by phosphorylation of a Ser residue; and pyruvate phosphate dikinase is activated by dephosphorylation.

The pathway of CO_2 assimilation has a greater energy cost in C_4 plants than in C_3 plants. For each molecule of CO_2 assimilated in the C_4 pathway, a molecule of PEP must be regenerated at the expense of two phosphoanhydride bonds in ATP.  Thus C_4 plants need five ATP molecules to assimilate one molecule of CO_2 , whereas C_3 plants need only three (nine per triose phosphate). As the temperature increases (and the affinity of rubisco for CO_2 decreases, as noted above), a point is reached, at about 28 to 30 °C, at which the gain in efficiency from the elimination of photorespiration more than compensates for this energetic cost. C_4 plants (crabgrass, for example) outgrow most C_3 plants during the summer, as any experienced gardener can attest.

In CAM Plants, CO_2 Capture and Rubisco Action Are Temporally Separated

Succulent plants such as cactus and pineapple, which are native to very hot, very dry environments, have another variation on photosynthetic CO_2 fixation, which reduces loss of water vapor through the pores (stomata) by which CO_2 and O_2 must enter leaf tissue. Instead of separating the initial trapping of CO_2 and its fixation

by rubisco across space (as do the C_4 plants), they separate these two events over time. At night, when the air is cooler and moister, the stomata open to allow entry of CO_2 , which is then fixed into oxaloacetate by PEP carboxylase. The oxaloacetate is reduced to malate and stored in the vacuoles, to protect cytosolic and plastid enzymes from the low pH produced by malic acid dissociation. During the day the stomata close, preventing the water loss that would result from high daytime temperatures, and the CO_2 trapped overnight in malate is released as CO_2 by the NADP-linked malic enzyme. This CO_2 is now assimilated by the action of rubisco and the Calvin cycle enzymes. Because this method of CO_2 fixation was first discovered in stonecrops, perennial flowering plants of the family Crassulaceae, it is called crassulacean acid metabolism, and the plants are called **CAM plants**. [Table 20-1](#) compares characteristics of C_3 , C_4 , and CAM plants.

TABLE 20-1 Comparison of C_3 , C_4 , and CAM Plants

	C_3 Plants	C_4 Plants	CAM Plants
Examples	Spinach, pea, rice, wheat, beans, most trees	Maize (corn), sugarcane, crabgrass	Cactus, prickly pear, orchid, pineapple
Most efficient environment	15 to 25 °C	Hot and dry; 30 to 47 °C	Extremely dry; 35 °C
Path of CO_2 fixation	C_3 photosynthesis only	Sequential C_4 and C_3 cycles spatially separated: C_4 in mesophyll cells followed by C_3 in bundle-sheath cells	C_3 and C_4 cycles, separated spatially and temporally
Cell type	Mesophyll cells	C_4 in mesophyll cells,	C_3 and C_4 in the

involved		C ₃ in bundle-sheath cells	same mesophyll cells
Light conditions	Light	Light	C ₃ in light; C ₄ in dark
Initial CO ₂ acceptor	Ribulose 1,5-bisphosphate	Phosphoenolpyruvate	Ribulose 1,5-bisphosphate in light; phosphoenolpyruvate in dark
CO ₂ -fixing enzyme	Rubisco	PEP carboxylase, then rubisco	Rubisco in light; PEP carboxylase at night
First stable product of CO ₂ fixation	3-Phosphoglycerate	Oxaloacetate in C ₄ cycle	3-Phosphoglycerate in light; oxaloacetate in dark
Energy needed for complete reduction of one molecule of CO ₂	3 ATP, 2 NADPH	5 ATP, 2 NADPH	6.5 ATP, 2 NADPH
Photorespiration	Present	Absent or suppressed	Absent or suppressed

SUMMARY 20.5 *Photorespiration and the C₄ and CAM Pathways*

■ Rubisco is not completely specific for CO₂ as its substrate; it can also use O₂, producing 2-phosphoglycolate, which must be disposed of in an oxygen-dependent pathway. The result is increased consumption of O₂ — photorespiration.

■ The 2-phosphoglycolate is converted to glyoxylate, to glycine, and then to serine in a pathway that involves enzymes in the chloroplast stroma, peroxisomes, and mitochondria.

■ In C₄ plants, the CO₂-assimilation pathway minimizes photorespiration: CO₂ is first fixed in mesophyll cells into a four-

carbon compound, which passes into bundle-sheath cells and releases CO_2 in high concentrations. The released CO_2 is fixed by rubisco, and the remaining reactions of the Calvin cycle occur as in C_3 plants.

■ In CAM plants, CO_2 is fixed into malate in the dark and stored in vacuoles until daylight, when the stomata are closed (minimizing water loss), and the stored malate serves as a source of CO_2 for rubisco.

20.6 Biosynthesis of Starch, Sucrose, and Cellulose

During active photosynthesis in bright light, a plant leaf produces more carbohydrate (as triose phosphates) than it needs for generating energy or synthesizing precursors.  The excess is converted to sucrose and transported to other parts of the plant, to be used as fuel or stored. In most plants, starch is the main storage form of carbohydrate, but in a few plants, such as sugar beet and sugarcane, sucrose is the primary storage form. The synthesis of sucrose and starch occurs in different cellular compartments (cytosol and plastids, respectively), and these processes are coordinated by a variety of regulatory mechanisms that respond to changes in light level and photosynthetic rate. The synthesis of sucrose and starch is important to the plant but also to humans: starch provides more than 80% of human dietary calories worldwide.

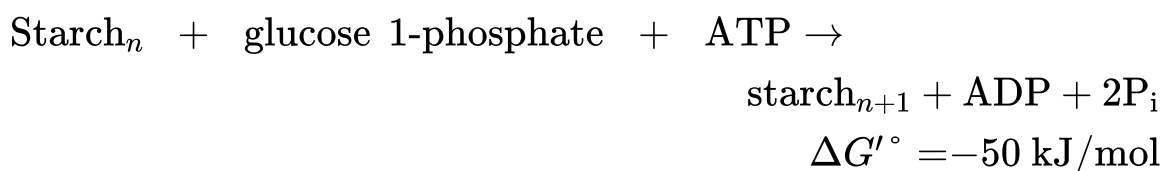
ADP-Glucose Is the Substrate for Starch Synthesis in Plant Plastids and for Glycogen Synthesis in Bacteria

Starch, like glycogen, is a high molecular weight polymer of D-glucose in ($\alpha 1 \rightarrow 4$) linkage. It is synthesized in chloroplasts for temporary storage as one of the stable end products of

photosynthesis, and for long-term storage it is synthesized in amyloplasts of the nonphotosynthetic parts of plants: seeds, roots, and tubers (underground stems).

The mechanism of glucose activation in starch synthesis is similar to that in glycogen synthesis, described in [Chapter 15](#). An activated **sugar nucleotide**, in this case ADP-glucose, is formed by condensation of glucose 1-phosphate with ATP in a reaction made essentially irreversible by the presence in plastids of inorganic pyrophosphatase. **Starch synthase** then transfers glucose residues from ADP-glucose to preexisting starch molecules. The monomeric units are almost certainly added to the nonreducing end of the growing polymer, as they are in glycogen synthesis.

The amylose of starch is unbranched, but amylopectin has numerous ($\alpha 1 \rightarrow 6$)-linked branches (see [Fig. 7-13](#)). Chloroplasts contain a branching enzyme, similar to the glycogen-branching enzyme that introduces the ($\alpha 1 \rightarrow 6$) branches of amylopectin. Taking into account the hydrolysis by inorganic pyrophosphatase of the PP_i produced during ADP-glucose synthesis, the overall reaction for starch formation from glucose 1-phosphate is



Starch synthesis is regulated at the level of ADP-glucose formation, as discussed below.

UDP-Glucose Is the Substrate for Sucrose Synthesis in the Cytosol of Leaf Cells

Most of the triose phosphate generated by CO₂ fixation in plants is converted to sucrose ([Fig. 20-41](#)) or starch. In the course of evolution, sucrose may have been selected as the transport form of carbon because of its unusual linkage between the anomeric C-1 of glucose and the anomeric C-2 of fructose. This bond is not hydrolyzed by amylases or other common carbohydrate-cleaving enzymes, and the unavailability of the sucrose molecule's anomeric carbons prevents it from reacting nonenzymatically (as does glucose) with amino acids and proteins.

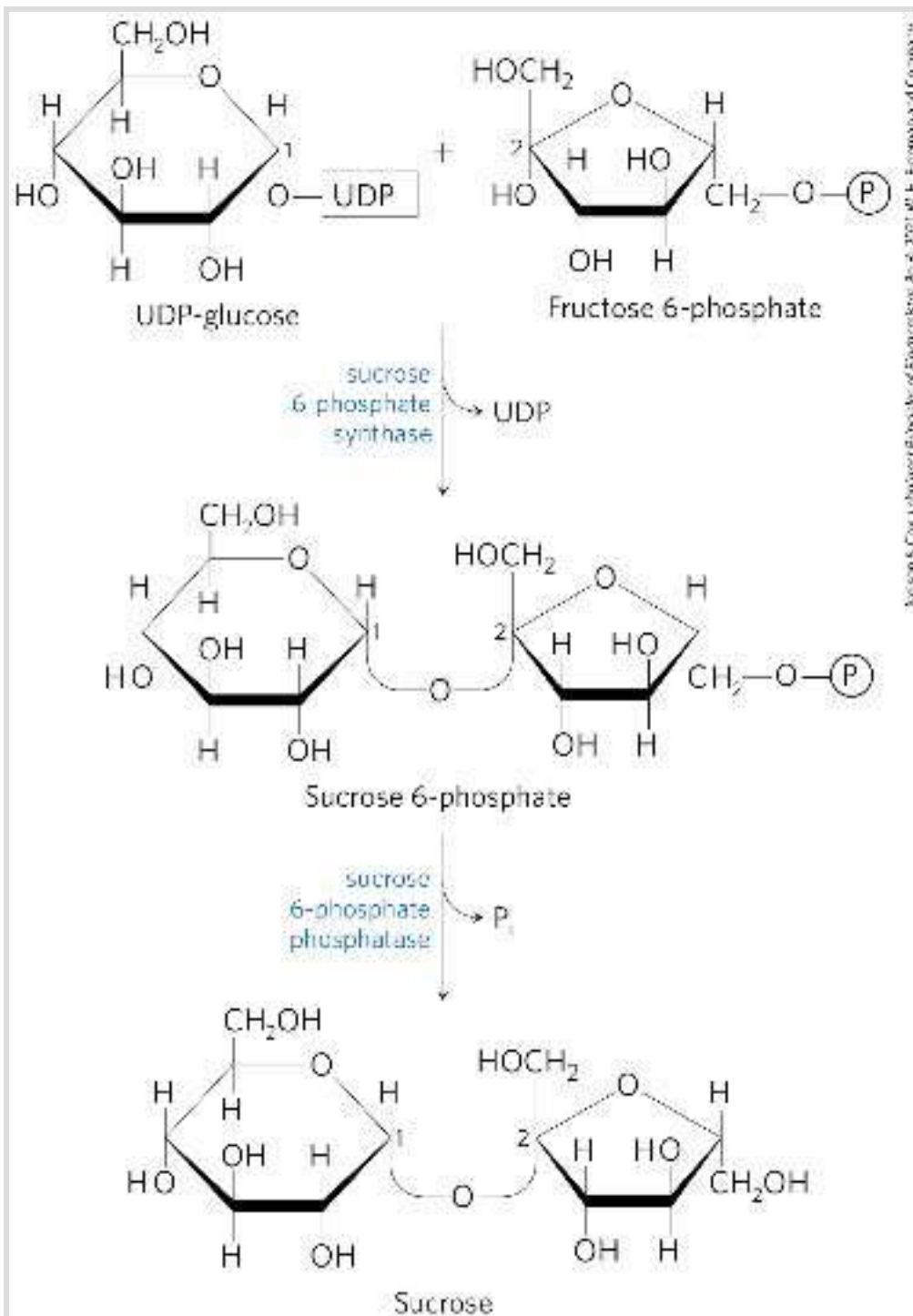


FIGURE 20-41 Sucrose synthesis. Sucrose is synthesized from UDP-glucose and fructose 6-phosphate, which are synthesized from triose phosphates in the plant cell cytosol. The sucrose 6-phosphate synthase of most plant species is allosterically regulated by glucose 6-phosphate and P_i .

Sucrose is synthesized in the cytosol, beginning with dihydroxyacetone phosphate and glyceraldehyde 3-phosphate exported from the chloroplast. After condensation of two triose phosphates to form fructose 1,6-bisphosphate (catalyzed by aldolase), hydrolysis by fructose 1,6-bisphosphatase yields fructose 6-phosphate. **Sucrose 6-phosphate synthase** then catalyzes the reaction of fructose 6-phosphate with UDP-glucose to form sucrose 6-phosphate ([Fig. 20-41](#)). Finally, sucrose 6-phosphate phosphatase removes the phosphate group, making sucrose available for export to other tissues. The reaction catalyzed by sucrose 6-phosphate synthase is a low-energy process ($\Delta G'^{\circ} = -5.7 \text{ kJ/mol}$), but the hydrolysis of sucrose 6-phosphate to sucrose is sufficiently exergonic ($\Delta G'^{\circ} = -16.5 \text{ kJ/mol}$) to make the overall synthesis of sucrose thermodynamically favorable. Sucrose synthesis is regulated and closely coordinated with starch synthesis, as we shall see.

One remarkable difference between the cells of plants and animals is the absence in the plant cell cytosol of the enzyme inorganic pyrophosphatase, which catalyzes the reaction



For many biosynthetic reactions that liberate PP_i , pyrophosphatase activity makes the process more favorable energetically, tending to make these reactions irreversible. In plants, this enzyme is present in plastids but absent from the

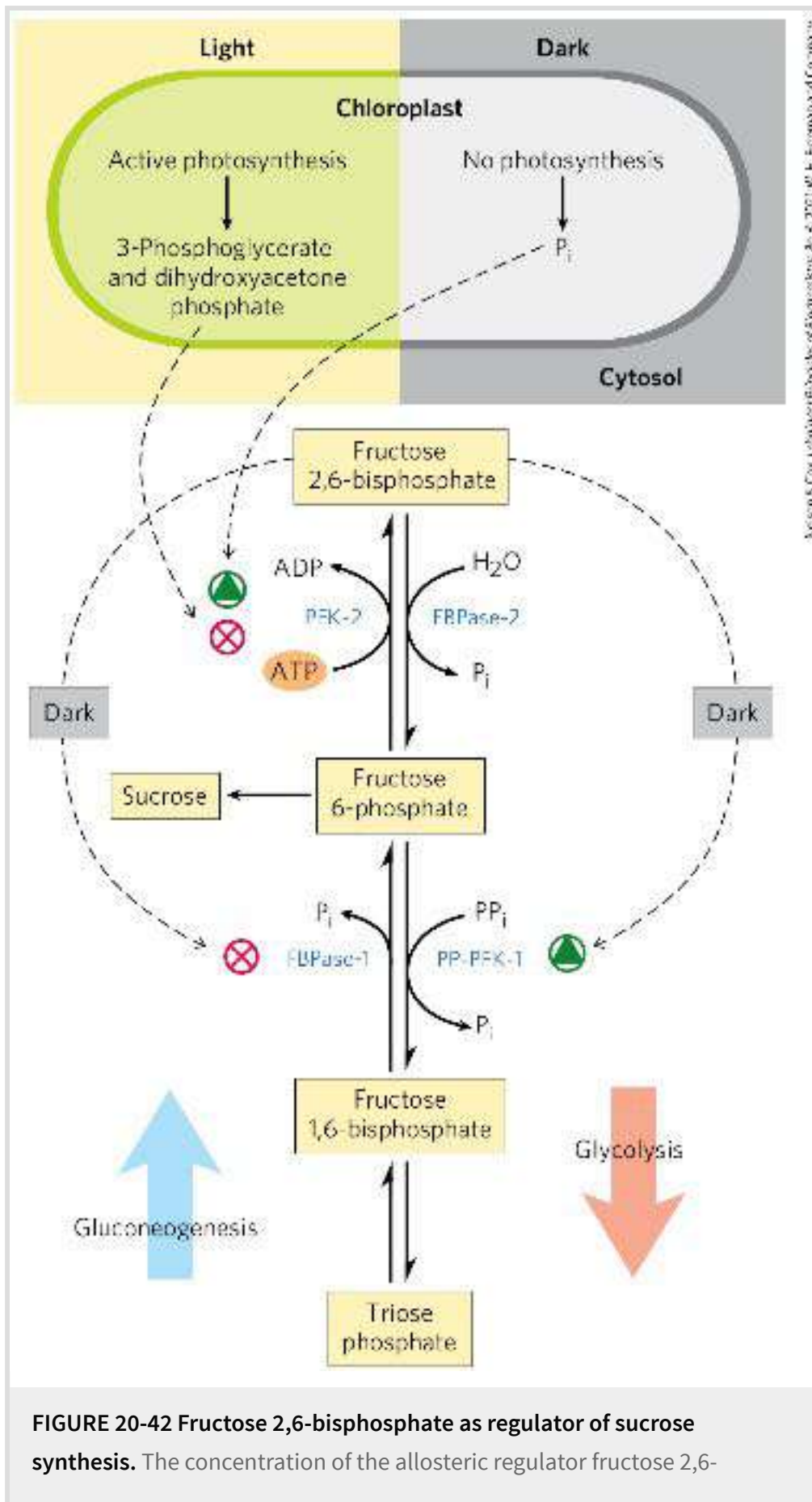
cytosol. As a result, the cytosol of leaf cells contains a substantial concentration of PP_i — enough ($\sim 0.3 \text{ mM}$) to make reactions such as that catalyzed by UDP-glucose pyrophosphorylase (see [Fig. 15-7](#)) readily reversible.

Conversion of Triose Phosphates to Sucrose and Starch Is Tightly Regulated

Triose phosphates produced by the Calvin cycle in bright sunlight, as we have noted, may be stored temporarily in the chloroplast as starch, or converted to sucrose and exported to nonphotosynthetic parts of the plant, or both.  The balance between the two processes is tightly regulated, and both must be coordinated with the rate of CO_2 fixation. Five-sixths of the triose phosphate formed in the Calvin cycle must be recycled to ribulose 1,5-bisphosphate ([Fig. 20-32](#)); if more than one-sixth of the triose phosphate is drawn out of the cycle to make sucrose and starch, the cycle will slow or stop. However, *insufficient* conversion of triose phosphate to starch or sucrose would tie up phosphate, leaving a chloroplast deficient in P_i , which is also essential for operation of the Calvin cycle.

The flow of triose phosphates into sucrose is regulated by the activity of fructose 1,6-bisphosphatase (FBPase-1) and the enzyme that effectively reverses its action, PP_i -dependent phosphofructokinase (PP-PFK-1). These enzymes are therefore

critical points for determining the fate of triose phosphates produced by photosynthesis. Both enzymes are regulated by **fructose 2,6-bisphosphate (F26BP)**, which inhibits FBPase-1 and stimulates PP-PFK-1. In vascular plants, the concentration of F26BP varies inversely with the rate of photosynthesis ([Fig. 20-42](#)). Phosphofructokinase-2, responsible for F26BP synthesis, is inhibited by dihydroxyacetone phosphate or 3-phosphoglycerate and is stimulated by fructose 6-phosphate and P_i . During active photosynthesis, dihydroxyacetone phosphate is produced and P_i is consumed, resulting in inhibition of PFK-2 and lowered concentrations of F26BP. This favors greater flux of triose phosphate into fructose 6-phosphate formation and sucrose synthesis. With this regulatory system, sucrose synthesis occurs when the level of triose phosphate produced by the Calvin cycle exceeds that needed to maintain operation of the cycle.



bisphosphate in plant cells is regulated by the products of photosynthetic CO_2 assimilation and by P_i . Dihydroxyacetone phosphate and 3-phosphoglycerate produced by CO_2 assimilation inhibit phosphofructokinase-2 (PFK-2), the enzyme that synthesizes the regulator; P_i stimulates PFK-2. The concentration of the regulator is therefore inversely proportional to the rate of photosynthesis. In the dark, the concentration of fructose 2,6-bisphosphate increases and stimulates the glycolytic enzyme PP_i -dependent phosphofructokinase-1 (PP-PFK-1), while inhibiting the gluconeogenic enzyme fructose 1,6-bisphosphatase (FBPase-1). When photosynthesis is active (in the light), the concentration of the regulator drops and the synthesis of fructose 6-phosphate and sucrose is favored.

Sucrose synthesis is also regulated at the level of sucrose 6-phosphate synthase, which is allosterically activated by glucose 6-phosphate and inhibited by P_i . This enzyme is further regulated by phosphorylation and dephosphorylation; a protein kinase phosphorylates the enzyme on a specific Ser residue, making it less active, and a phosphatase reverses this inactivation by removing the phosphate ([Fig. 20-43](#)). Inhibition of the kinase by glucose 6-phosphate, and of the phosphatase by P_i , enhances the effects of these two compounds on sucrose synthesis. When hexose phosphates are abundant, sucrose 6-phosphate synthase is activated by glucose 6-phosphate; when P_i is elevated (as when photosynthesis is slow), sucrose synthesis is slowed. During active photosynthesis, triose phosphates are converted to fructose 6-phosphate, which is rapidly equilibrated with glucose 6-phosphate by phosphohexose isomerase. Because the equilibrium lies far toward glucose 6-phosphate, as soon as fructose 6-phosphate accumulates, the level of glucose 6-phosphate rises and sucrose synthesis is stimulated.

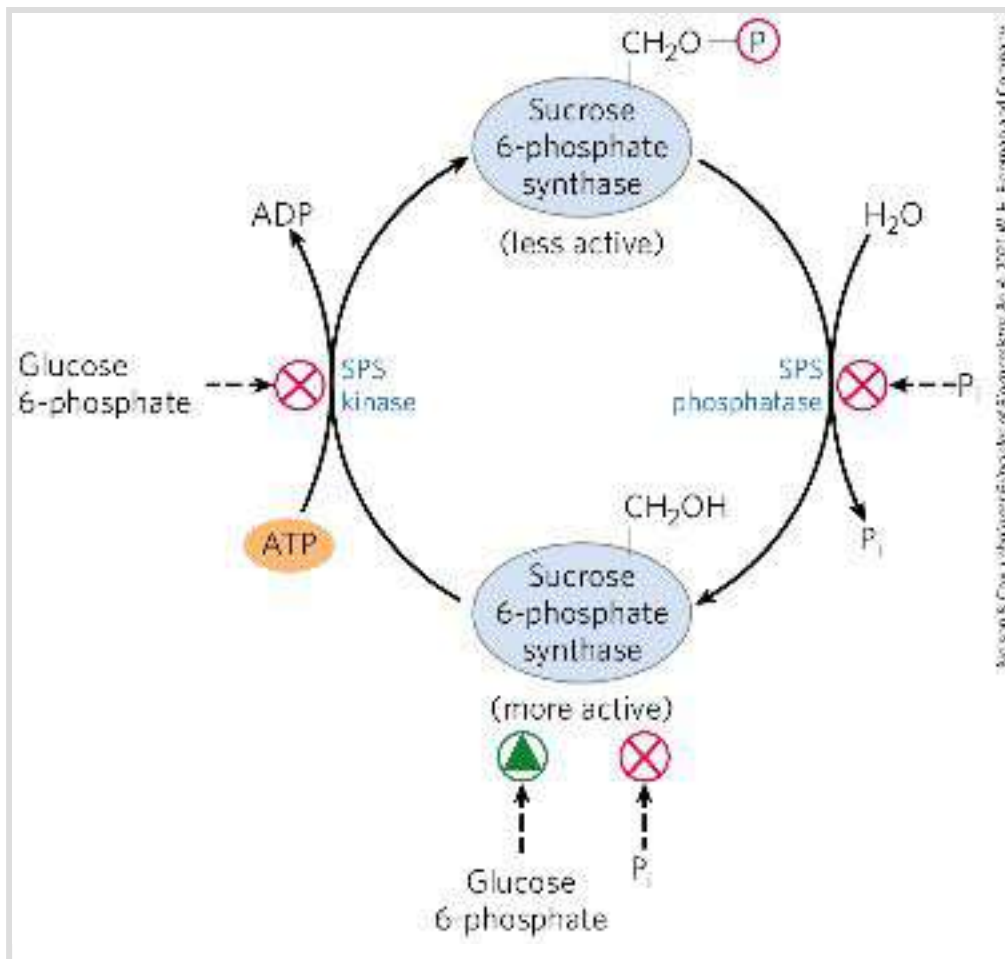


FIGURE 20-43 Regulation of sucrose phosphate synthase by phosphorylation. A protein kinase (SPS kinase) specific for sucrose phosphate synthase (SPS) phosphorylates a Ser residue in SPS, inactivating it; a specific phosphatase (SPS phosphatase) reverses this inhibition. The kinase is inhibited allosterically by glucose 6-phosphate, which also activates SPS allosterically. The phosphatase is inhibited by P_i, which also inhibits SPS directly. Thus, when the concentration of glucose 6-phosphate is high as a result of active photosynthesis, SPS is activated and produces sucrose phosphate. A high P_i concentration, which occurs when photosynthetic conversion of ADP to ATP is slow, inhibits sucrose phosphate synthesis.

The key regulatory enzyme in starch synthesis is **ADP-glucose pyrophosphorylase** ([Fig. 20-44](#)); it is activated by 3-phosphoglycerate, which accumulates during active

photosynthesis, and inhibited by P_i , which accumulates when light-driven condensation of ADP and P_i slows. When sucrose synthesis slows, 3-phosphoglycerate formed by CO_2 fixation accumulates, activating this enzyme and stimulating the synthesis of starch.

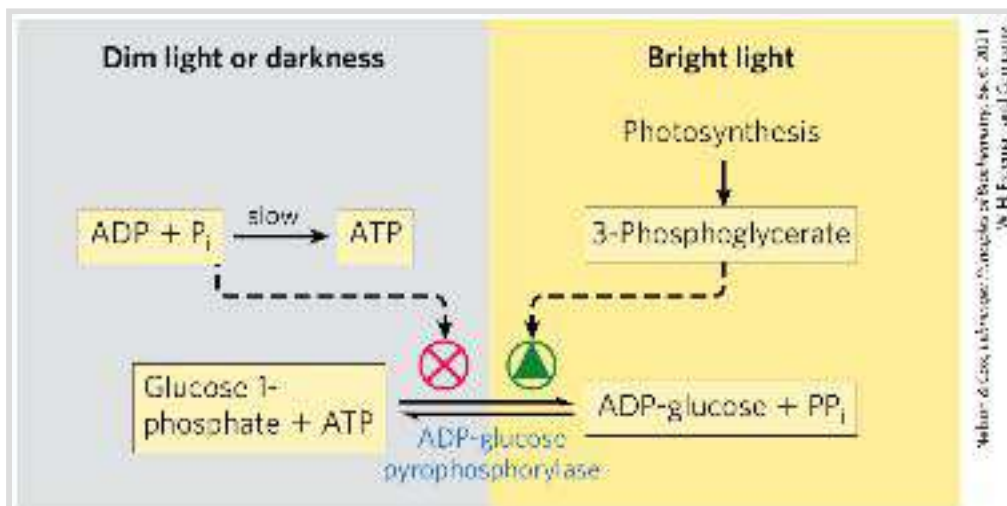


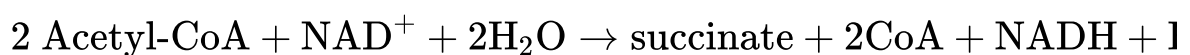
FIGURE 20-44 Regulation of ADP-glucose pyrophosphorylase by 3-phosphoglycerate and P_i . This enzyme, which produces the precursor for starch synthesis, is rate-limiting in starch production. The enzyme is stimulated allosterically by 3-phosphoglycerate (3-PGA) and inhibited by P_i ; in effect, the ratio $[3\text{-PGA}]/[P_i]$, which rises with increasing rates of photosynthesis, controls starch synthesis at this step.

The Glyoxylate Cycle and Gluconeogenesis Produce Glucose in Germinating Seeds

Many plants store lipids (oils) and proteins in their seeds, to be used as sources of energy and as biosynthetic precursors during

germination, before photosynthetic capacity has developed. These stored components are converted to carbohydrates by the combined action of several pathways. Glucogenic amino acids (see [Table 14-4](#)) derived from the breakdown of stored seed proteins are transaminated and oxidized to succinyl-CoA, pyruvate, oxaloacetate, fumarate, and α -ketoglutarate ([Chapter 18](#)) – all good starting materials for gluconeogenesis. Active gluconeogenesis in germinating seeds provides glucose for the synthesis of sucrose, polysaccharides, and many metabolites derived from hexoses. In plant seedlings, sucrose provides much of the chemical energy needed for initial growth.

Triacylglycerols stored in seeds also provide fuel for the germinating plants. They are hydrolyzed to free fatty acids, which undergo β oxidation to acetyl-CoA in specialized peroxisomes called [glyoxysomes](#) that develop during seed germination (see [Fig. 17-14](#)). The acetyl-CoA formed from seed oils enters the [glyoxylate cycle](#) ([Fig. 20-45](#)), which brings about the net conversion of acetate to succinate or other four-carbon intermediate of the citric acid cycle:



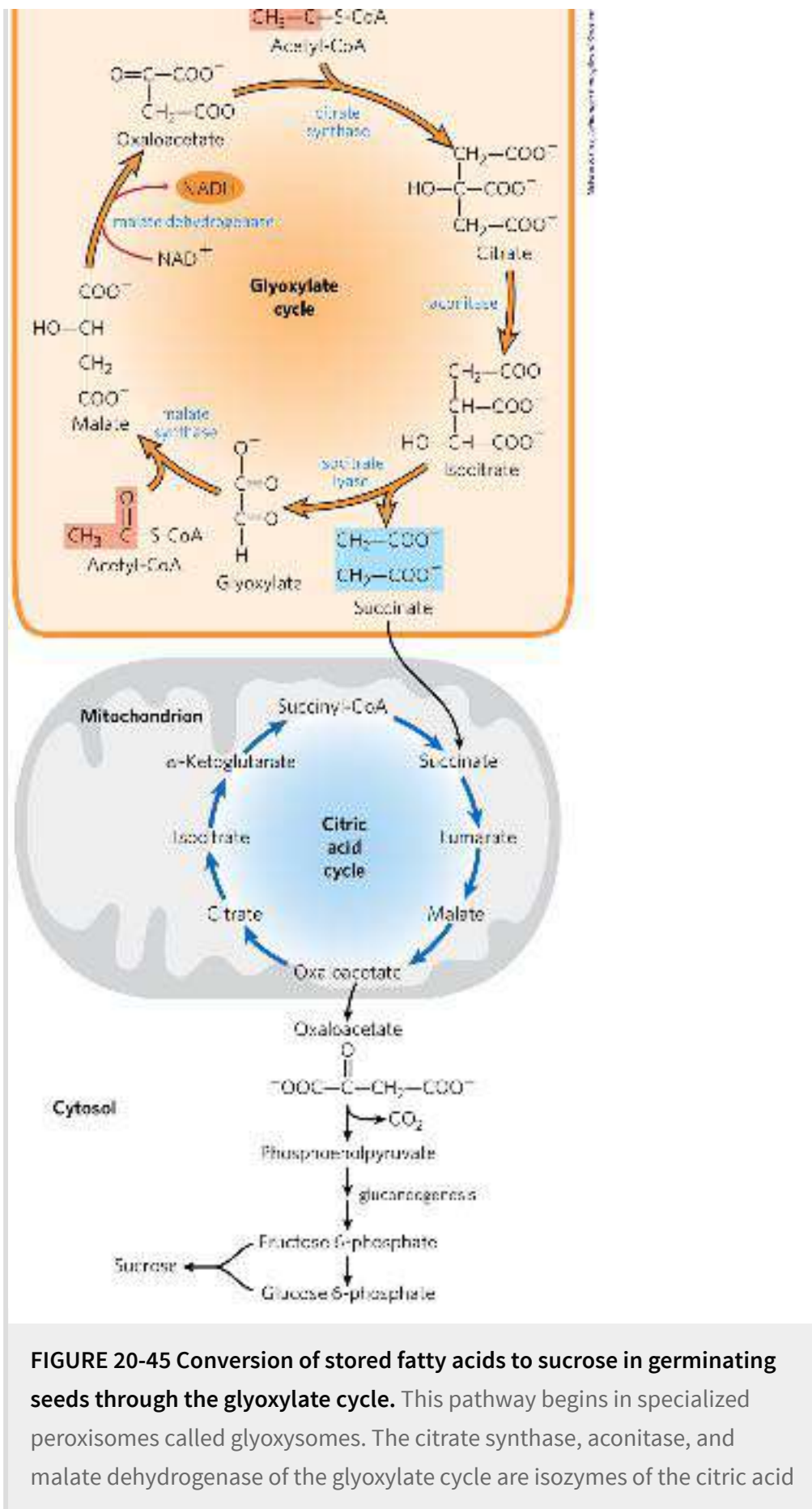


FIGURE 20-45 Conversion of stored fatty acids to sucrose in germinating seeds through the glyoxylate cycle. This pathway begins in specialized peroxisomes called glyoxysomes. The citrate synthase, aconitase, and malate dehydrogenase of the glyoxylate cycle are isozymes of the citric acid

cycle enzymes; isocitrate lyase and malate synthase are unique to the glyoxylate cycle. Notice that two acetyl groups enter the cycle and four carbons leave as succinate. Succinate is exported to mitochondria, where it is converted to oxaloacetate by enzymes of the citric acid cycle. Oxaloacetate enters the cytosol and serves as the starting material for gluconeogenesis and for synthesis of sucrose, the transport form of carbon in plants.

In the glyoxylate cycle, acetyl-CoA condenses with oxaloacetate to form citrate, and citrate is converted to isocitrate, exactly as in the citric acid cycle. The next step, however, is not the breakdown of isocitrate by isocitrate dehydrogenase but the cleavage of isocitrate by **isocitrate lyase**, forming succinate and **glyoxylate**. The glyoxylate then condenses with a second molecule of acetyl-CoA to yield malate, in a reaction catalyzed by **malate synthase**. The malate is subsequently oxidized to oxaloacetate, which can condense with another molecule of acetyl-CoA to start another turn of the cycle. The succinate passes into the mitochondrial matrix, where it is converted by citric acid cycle enzymes to oxaloacetate. The oxaloacetate moves into the cytosol and can be converted to phosphoenolpyruvate by PEP carboxykinase, then to fructose 6-phosphate, the precursor of sucrose, by gluconeogenesis. Thus, reaction sequences carried out in three subcellular compartments (glyoxysomes, mitochondria, and cytosol) are integrated for the production of fructose 6-phosphate or sucrose from stored lipids.

Enzymes common to the citric acid and glyoxylate cycles have two isozymes, one specific to mitochondria, the other to glyoxysomes. Physical separation of the glyoxylate cycle and β -oxidation

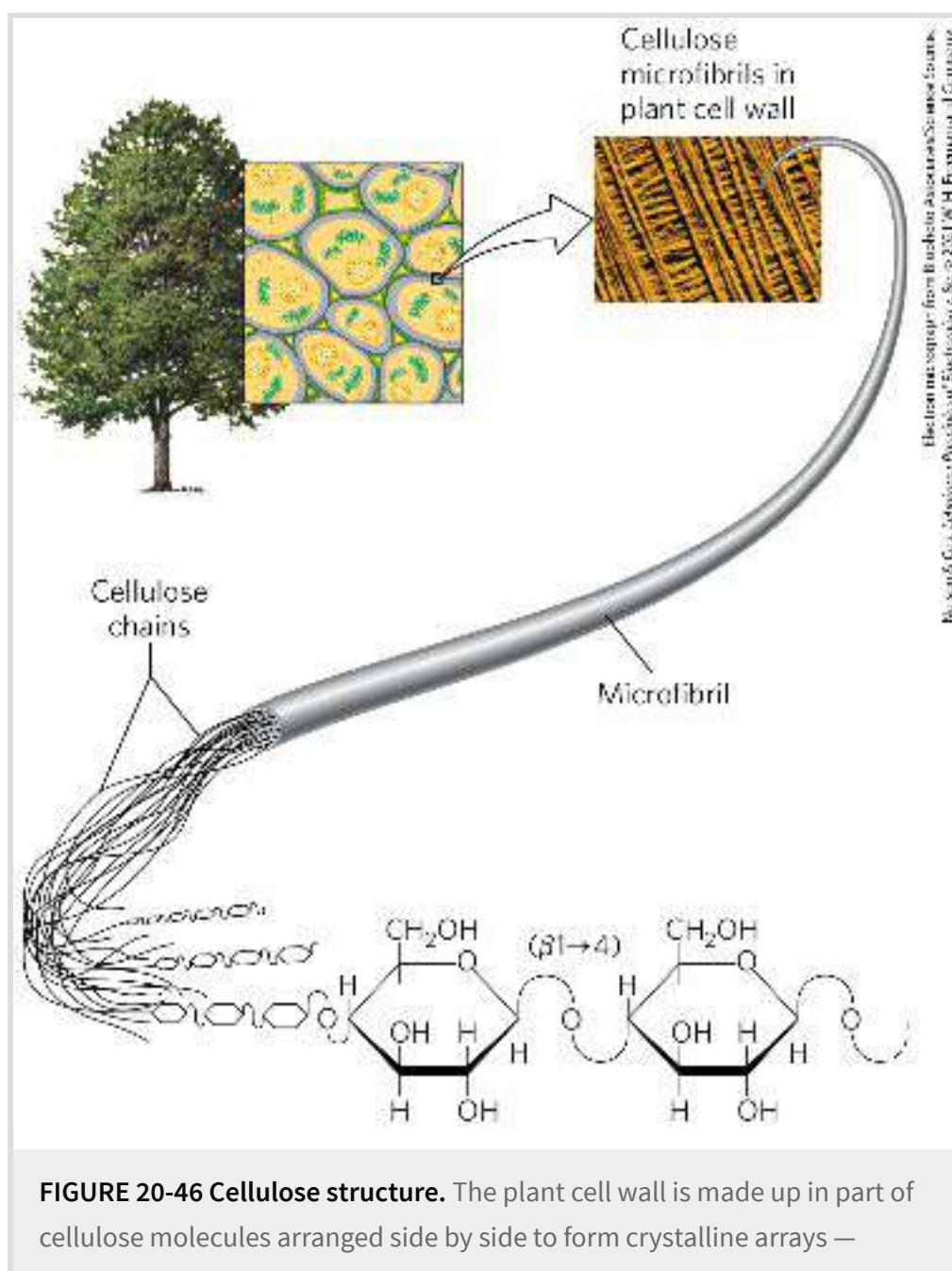
enzymes from the mitochondrial citric acid cycle enzymes prevents further oxidation of acetyl-CoA to CO₂. Each turn of the glyoxylate cycle consumes two molecules of acetyl-CoA and produces one molecule of succinate, which is then available for biosynthetic purposes. Hydrolysis of stored triacylglycerols also produces glycerol 3-phosphate, which can enter the gluconeogenic pathway, after its oxidation to dihydroxyacetone phosphate (see [Fig. 14-16](#)).

We noted in [Chapter 14](#) that animal cells can carry out gluconeogenesis from three- and four-carbon precursors, but not from the two acetyl carbons of acetyl-CoA. Because the pyruvate dehydrogenase reaction is effectively irreversible (see [Section 16.1](#)) and animals do not have the enzymes specific to the glyoxylate cycle (isocitrate lyase and malate synthase), they have no way to convert acetyl-CoA to pyruvate or oxaloacetate. So, unlike vascular plants, animals cannot bring about the net synthesis of glucose from fatty acids.

Cellulose Is Synthesized by Supramolecular Structures in the Plasma Membrane

Cellulose is a major constituent of plant cell walls, providing strength and rigidity and preventing the swelling of the cell and rupture of the plasma membrane that might result when osmotic conditions favor water entry into the cell. Each year, worldwide,

plants synthesize more than 10^{11} metric tons of cellulose, making this simple polymer one of the most abundant compounds in the biosphere. The structure of cellulose in the plant cell wall is simple: linear polymers of thousands of ($\beta 1 \rightarrow 4$)-linked D-glucose units, assembled into bundles of at least 18 chains, which co-crystallize to form microfibrils, which may in turn be assembled into larger macrofibrils. ([Fig. 20-46](#)).



cellulose microfibrils. Several microfibrils may combine to form larger cellulose macrofibrils. The scanning electron microscope shows macrofibrils, 5 to 12 nm in diameter, laid down on the cell surface in several layers distinguishable by the different orientations of the fibrils.



As a major component of the plant cell wall, cellulose must be synthesized from intracellular precursors but deposited and assembled *outside* the plasma membrane. The enzymatic machinery for initiation, elongation, and export of cellulose chains is therefore more complicated than that used to synthesize starch or glycogen (which are not exported).

The complex enzymatic machinery that assembles cellulose chains spans the plasma membrane, with one part on the cytoplasmic side positioned to bind the substrate, UDP-glucose, and elongate the chains, and another part extending to the outside, responsible for exporting the cellulose molecules to the extracellular space. Freeze-fracture electron microscopy shows a cellulose synthesis complex, or rosette, composed of six large particles arranged in a regular hexagon with a diameter of about 30 nm ([Fig. 20-47a](#)). Several proteins, including the catalytic subunit of **cellulose synthase**, make up this structure. The structure of the plant cellulose synthase is similar to that of the bacterium *Rhodobacter sphaeroides*, which has been determined by x-ray crystallography ([Fig. 20-47b](#)).

surface of the plasma membrane, joins growing chains from neighboring cellulose synthase molecules to form a cellulose microfibril. Polymers of more than 6 to 8 glucose units are insoluble in water, promoting microfibril crystallization. There is no definite length for a cellulose polymer; synthesis is highly processive, and some polymers are as long as 15,000 glucose units.

The UDP-glucose used for cellulose synthesis (step ❶ in [Fig. 20-47](#)) is generated from sucrose produced during photosynthesis, in a reaction catalyzed by sucrose synthase (named for the reverse reaction):



A membrane-bound form of sucrose synthase may produce a high local concentration of UDP-glucose for cellulose synthesis.

Each of the six particles of the rosette most likely contains three cellulose synthase molecules, each synthesizing a single cellulose chain (step ❷). The large enzyme complex that catalyzes this process moves along the plasma membrane with directionality often related to the course of microtubules in the cell cortex, the cytoplasmic layer just below the membrane (step ❸). When these microtubules lie perpendicular to the axis of the plant's growth, the cellulose microfibrils are laid down similarly to promote elongation. The motion of the cellulose synthase complexes is

believed to be driven by energy released in the polymerization reaction, not by a molecular motor such as kinesin.

The fundamental cellulose microfibril made by one rosette-type cellulose synthesis complex is thought to be composed of 18 chains lying side by side with the same (parallel) orientation of nonreducing and reducing ends. The 18 separate polymers coalesce on the outer surface of the cell and crystallize soon after they are polymerized (step 4), just prior to integrating into the cell wall.

In UDP-glucose, the glucose is α -linked to the nucleotide, but in cellulose, the glucose residues are ($\beta 1 \rightarrow 4$)-linked, so there is an inversion of configuration at the anomeric carbon (C-1) as the glycosidic bond forms. Glycosyltransferases that invert configuration are generally assumed to use a single-displacement mechanism, with nucleophilic attack by the acceptor species at the anomeric carbon of the donor sugar (in this case, UDP-glucose).

Pools of Common Intermediates Link Pathways in Different Organelles

Although we have described metabolic transformations in plant cells in terms of individual pathways, these pathways interconnect so completely that we should instead consider pools of metabolic intermediates shared among these pathways and

connected by readily reversible reactions ([Fig. 20-48](#)). One such **metabolite pool** includes the hexose phosphates glucose 1-phosphate, glucose 6-phosphate, and fructose 6-phosphate; a second includes the 5-phosphates of the pentoses ribose, ribulose, and xylulose; a third includes the triose phosphates dihydroxyacetone phosphate and glyceraldehyde 3-phosphate. Metabolite fluxes through these pools change in magnitude and direction in response to changes in the circumstances of the plant, and they vary with tissue type. Transporters in the membranes of each organelle move specific compounds in and out, and the regulation of these transporters presumably influences the degree to which the pools mix.

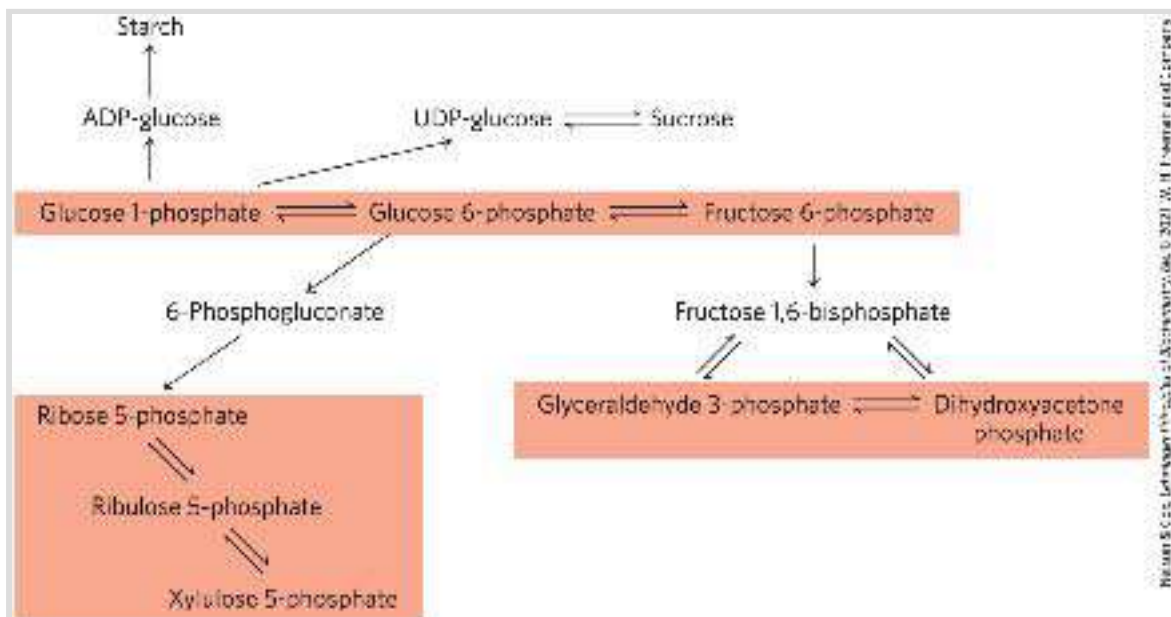


FIGURE 20-48 Pools of hexose phosphates, pentose phosphates, and triose phosphates. The compounds in each pool are readily interconvertible by reactions that have small standard free-energy changes. When one component of the pool is temporarily depleted, a new equilibrium is quickly established to replenish it. Movement of the sugar phosphates between intracellular compartments is limited; specific transporters must be present in an organelle membrane.

During daylight hours, triose phosphates produced in photosynthetic leaf tissue (“source” tissues, in which there is a net fixation of CO_2) move out of the chloroplast and into the cytosolic hexose phosphate pool, where they are converted to sucrose for transport via the plant phloem (sap) to nonphotosynthetic “sink” tissues ([Fig. 20-49](#)). In sink tissues such as roots, tubers, and bulbs, sucrose is converted to starch for storage or is used as an energy source via glycolysis. In growing plants, hexose phosphates are also withdrawn from the pool for the synthesis of cell walls. At night, starch is metabolized by glycolysis and oxidative phosphorylation to provide energy for both source and sink tissues.

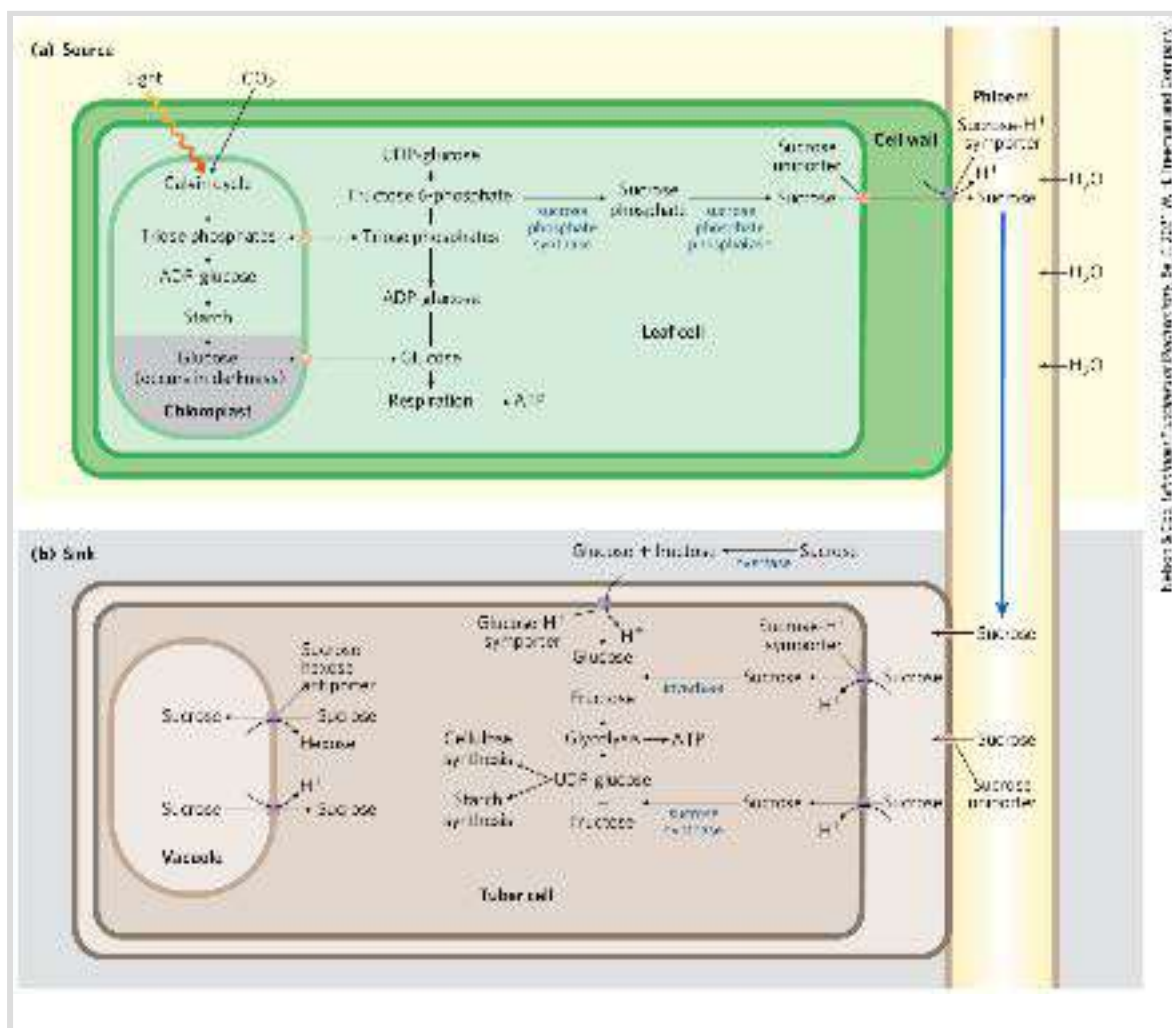


FIGURE 20-49 Movement of sucrose between source and sink tissues. (a) In daylight, photosynthetic leaves (source tissue) fix CO_2 into triose phosphates via the Calvin cycle in chloroplasts. Some of the triose phosphate is used in the chloroplasts to synthesize starch; the rest is exported to the cytosol, where it can be converted via gluconeogenesis to fructose 6-phosphate and glucose 1-phosphate. Sucrose, synthesized from UDP-glucose and fructose, is exported from leaf mesophyll cells to the plant phloem; the resulting high sucrose content draws water into the phloem by osmosis. The resulting increased turgor pressure ([p. 52](#)) pushes the solution in the phloem toward sink tissues. (b) Sucrose moves from the phloem into the sink tissues, where it is converted to starch or cell wall cellulose, or is used as fuel for glycolysis, the citric acid cycle, and oxidative phosphorylation to provide ATP for these nonphotosynthetic tissues. Sugar transport across the plasma membrane and between intracellular compartments is catalyzed by several symporters and antiporters coupled to a proton gradient. [Information from Dr. Gerald Edwards, School of Biological Sciences, Washington State University.]

SUMMARY 20.6 *Biosynthesis of Starch, Sucrose, and Cellulose*

- Starch synthase in chloroplasts and amyloplasts catalyzes the addition of single glucose residues, donated by ADP-glucose, to the growing polymer chain.
- Sucrose is synthesized in the cytosol from UDP-glucose and fructose 1-phosphate, in two steps.
- The partitioning of triose phosphates between sucrose synthesis and starch synthesis is regulated by fructose 2,6-bisphosphate (F26BP). [F26BP] varies inversely with the rate of photosynthesis, and F26BP inhibits the synthesis of fructose 6-phosphate, the precursor of sucrose.
- The glyoxylate cycle, taking place in the glyoxysomes of germinating seeds of some plants, uses several citric acid cycle enzymes and two additional enzymes: isocitrate lyase and malate synthase. The two decarboxylation steps of the citric acid cycle

are bypassed, making possible the *net* formation of succinate, oxaloacetate, and other cycle intermediates from acetyl-CoA.

■ Cellulose synthase has a glycosyl transferase activity in its cytoplasmic domain and forms a transmembrane channel through which the growing cellulose chain is extruded. Glucose units are transferred from UDP-glucose to the nonreducing end of the growing chain.

■ The plant cell shares pools of common intermediates, including hexose-, pentose-, and triose-phosphates. Transporters in the membranes of chloroplasts, mitochondria, and amyloplasts mediate the movement of sugar phosphates between organelles. The direction of metabolite flow through the pools within a leaf changes from day to night.

■ Sucrose produced in a photosynthetic (source) tissue is exported to nonphotosynthetic (sink) tissue such as roots and tubers via the plant phloem.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

photosynthesis

light-dependent reactions

photophosphorylation

chloroplast

stroma

thylakoid

photon

excited state

ground state

exciton

exciton transfer

chlorophylls

accessory pigments

carotenoids

β -carotene

action spectrum

photosystem

photochemical reaction center

light-harvesting complexes (LHCs)

cyclic electron transfer

linear electron transfer

ferredoxin

Z scheme

photosystem II (PSII)

photosystem I (PSI)

cytochrome b_6f

plastoquinone (PQ_A)

plastocyanin

phylloquinone (PQ_K)

cyclic photophosphorylation

state transition

oxygen-evolving center

CO₂ assimilation

CO₂ fixation

Calvin cycle

reductive pentose phosphate pathway

ribulose 1,5-bisphosphate

3-phosphoglycerate

C₃ plants ribulose 1,5-bisphosphate carboxylase/oxygenase (rubisco)

rubisco activase

thioredoxin

ferredoxin:thioredoxin reductase

photorespiration

2-phosphoglycolate

glycolate pathway

C₄ plants

C₄ pathway

phosphoenolpyruvate carboxylase

malic enzyme

pyruvate phosphate dikinase

CAM plants

[sugar nucleotide](#)

[starch synthase](#)

[glyoxysome](#)

[glyoxylate cycle](#)

[isocitrate lyase](#)

[glyoxylate](#)

[malate synthase](#)

[cellulose synthase](#)

PROBLEMS

1. Photochemical Efficiency of Light at Different

Wavelengths The rate of photosynthesis in a green plant, measured by O₂ production, is higher when illuminated with light of wavelength 680 nm than with light of wavelength 700 nm. However, illumination by a combination of light of 680 nm and 700 nm gives a higher rate of photosynthesis than light of either wavelength alone. Explain.

2. Balance Sheet for Photosynthesis In 1804, Nicolas-Théodore de Saussure observed that the total weight of oxygen and dry organic matter produced by plants is greater than the weight of carbon dioxide consumed during photosynthesis. Where does the extra weight come from?

3. Role of H₂S in Some Photosynthetic Bacteria Illuminated purple sulfur bacteria carry out photosynthesis in the presence of H₂O and ¹⁴CO₂, but only if H₂S is added and O₂

is absent. During photosynthesis, measured by formation of [^{14}C] carbohydrate, the bacteria convert H_2S to elemental sulfur but do not produce O_2 . What is the role of the conversion of H_2S to sulfur? Why doesn't photosynthesis produce O_2 in these bacteria?

4. Electron Transfer through Photosystems I and II Predict how an inhibitor of electron passage through pheophytin would affect electron transfer through (a) photosystem II and (b) photosystem I. Explain your reasoning.

5. Limited ATP Synthesis in the Dark In a laboratory experiment, a researcher illuminates spinach chloroplasts in the absence of ADP and P_i . Then, the researcher turns the light off and adds ADP and P_i . ATP synthesis occurs for a short time in the dark. Explain this finding.

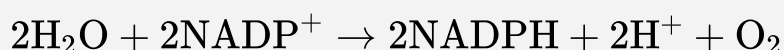
6. Mode of Action of the Herbicide DCMU Treating chloroplasts with 3-(3,4-dichlorophenyl)-1,1-dimethylurea (DCMU, or diuron), a potent herbicide, causes oxygen evolution and photophosphorylation to cease. Adding a Hill reagent (an external electron acceptor) restores oxygen evolution, but not photophosphorylation. How does DCMU act as a weed killer? Suggest a location for the inhibitory action of this herbicide in the scheme shown in [Figure 20-12](#). Explain.

7. Effect of Venturicidin on Oxygen Evolution Venturicidin is a powerful inhibitor of the chloroplast ATP synthase, interacting with CF_o and blocking proton passage through

the CF_0CF_1 complex. How would venturicidin affect oxygen evolution in a suspension of well-illuminated chloroplasts? Would your answer change if the experiment were done in the presence of an uncoupling reagent such as 2,4-dinitrophenol (DNP)? Explain.

8. Light Energy for a Redox Reaction Suppose you have isolated a new photosynthetic microorganism that oxidizes H_2S and passes the electrons to NAD^+ . What wavelength of light would provide enough energy for H_2S to reduce NAD^+ under standard conditions? Assume 100% efficiency in the photochemical event, and use an E'° of -243 mV for H_2S and -320 mV for NAD^+ . See [Figure 20-4](#) for the energy equivalents of wavelengths of light.

9. Equilibrium Constant for Water-Splitting Reactions The coenzyme NADP^+ is the terminal electron acceptor in chloroplasts, according to the reaction



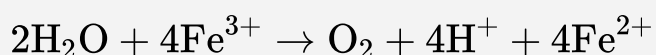
Use information in [Chapter 19 \(Table 19-2\)](#) to calculate the equilibrium constant for this reaction at 25°C . (The relationship between K'_{eq} and $\Delta G'^\circ$ is discussed on [p. 468](#).) How can the chloroplast overcome this unfavorable equilibrium?

10. Energetics of Phototransduction During photosynthesis, pigment molecules in chloroplasts must absorb eight photons (four by each photosystem) for every O₂ molecule they produce, according to the equation



The $\Delta G'^\circ$ for the light-independent production of O₂ is 400 kJ/mol. Assuming that these photons have a wavelength of 700 nm (red) and that the light absorption and use of light energy are 100% efficient, calculate the free-energy change for the process.

11. Electron Transfer to a Hill Reagent Isolated spinach chloroplasts evolve O₂ when illuminated in the presence of potassium ferricyanide (a Hill reagent), according to the equation



where Fe³⁺ represents ferricyanide and Fe²⁺ represents ferrocyanide. Does this process produce NADPH? Explain.

12. How Often Does a Chlorophyll Molecule Absorb a Photon? The amount of chlorophyll *a* (*M_r* 892) in a spinach leaf is about 20 μg/cm² of leaf surface. In noonday sunlight (average energy reaching the leaf is 5.4 J/cm² • min), the leaf

absorbs about 50% of the radiation. How often does a single chlorophyll molecule absorb a photon? Given that the average lifetime of an excited chlorophyll molecule in vivo is 1 ns, what fraction of the chlorophyll molecules are excited at any one time?

13. Effect of Monochromatic Light on Electron Flow Using a spectrophotometer, researchers can sometimes directly observe the extent of oxidation or reduction of an electron carrier during photosynthetic electron transfer. Illuminating chloroplasts with 700 nm light oxidizes cytochrome *f*, plastocyanin, and plastoquinone. Illuminating chloroplasts with 680 nm light, however, reduces these electron carriers. Explain.

14. Function of Cyclic Photophosphorylation When the $[\text{NADPH}]/[\text{NADP}^+]$ ratio in chloroplasts is high, photophosphorylation is predominantly cyclic (see [Fig. 20-12](#)). Does cyclic electron transfer evolve O_2 ? Does cyclic electron transfer produce NADPH? Explain. What is the main function of cyclic electron transfer?

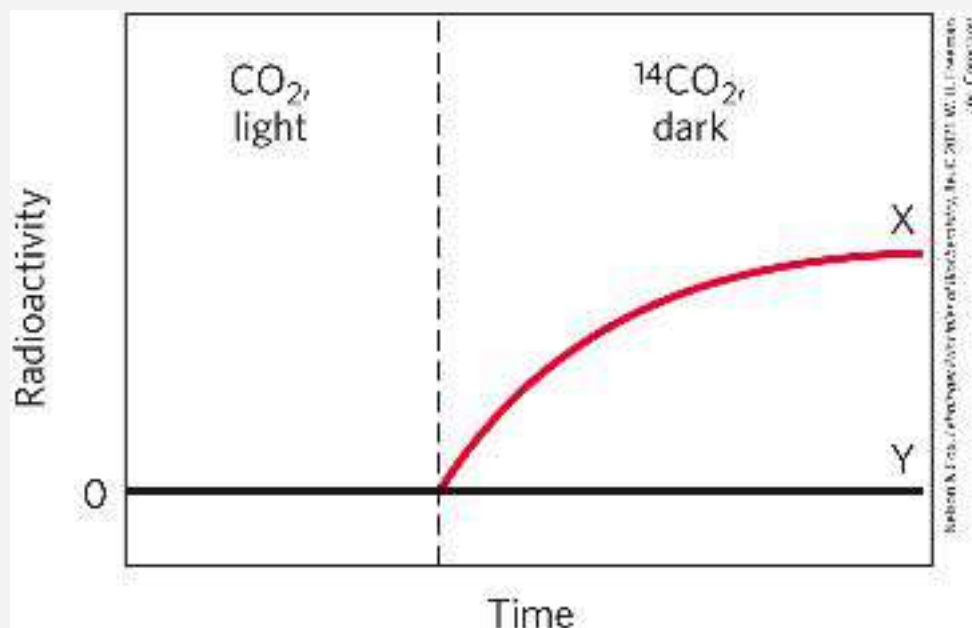
15. Phases of Photosynthesis A researcher illuminates a suspension of green algae in the absence of CO_2 . He then incubates the algae with $^{14}\text{CO}_2$ in the dark and observes the conversion of $^{14}\text{CO}_2$ to $[^{14}\text{C}]$ glucose for a brief time. What is the significance of this observation with regard to the CO_2 -assimilation process, and how is it related to the light-

dependent reactions of photosynthesis? Why does the conversion of $^{14}\text{CO}_2$ to $[^{14}\text{C}]$ glucose stop after a brief time?

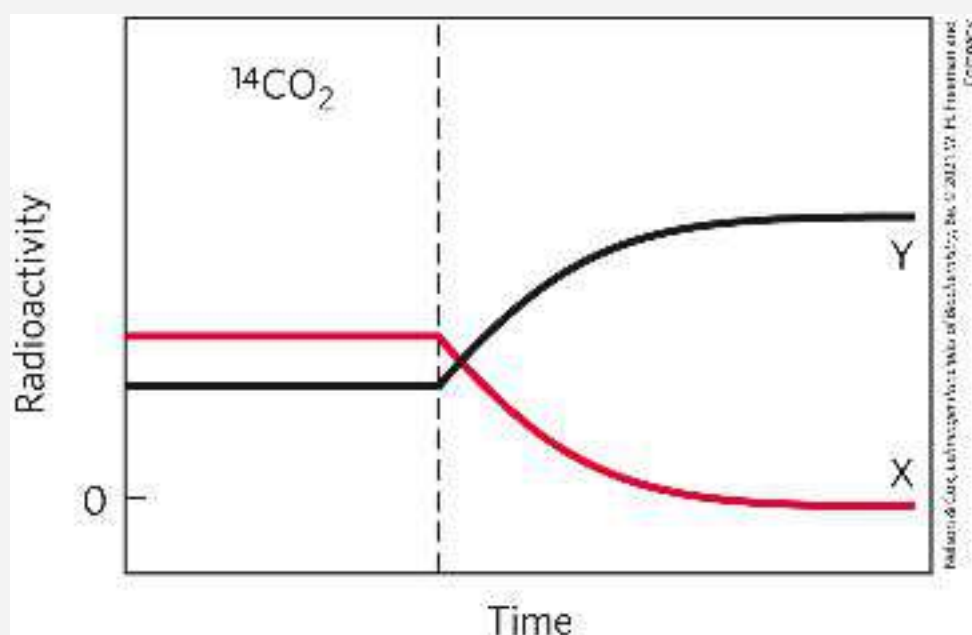
16. Identification of Key Intermediates in CO_2 Assimilation

Calvin and his colleagues used the unicellular green alga *Chlorella* to study the CO_2 -assimilation reactions of photosynthesis. They incubated $^{14}\text{CO}_2$ with illuminated suspensions of algae and followed the time course of appearance of ^{14}C in two compounds, X and Y, under two sets of conditions. Suggest the identities of X and Y, based on your understanding of the Calvin cycle.

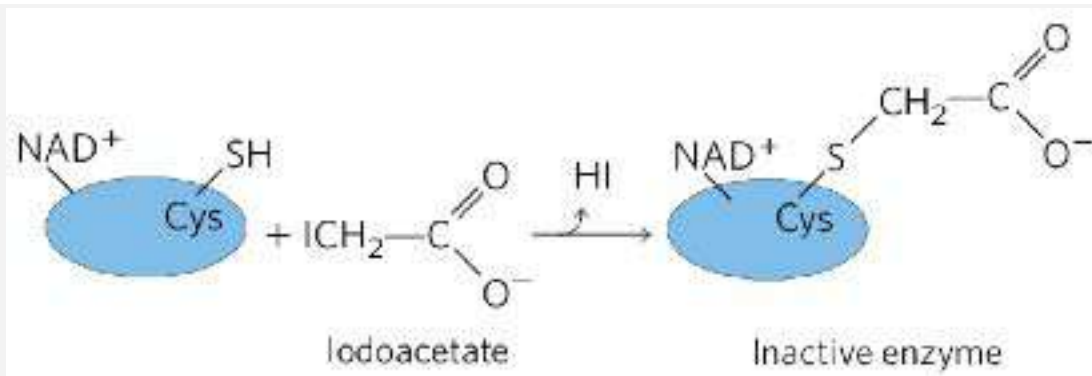
- They grew illuminated *Chlorella* with unlabeled CO_2 , then turned off the light and added $^{14}\text{CO}_2$ (vertical dashed line in the graph below). Under these conditions, X was the first compound to become labeled with ^{14}C ; Y was unlabeled.



- b. They grew illuminated *Chlorella* cells with $^{14}\text{CO}_2$. Illumination was continued until all the $^{14}\text{CO}_2$ had been taken up (vertical dashed line in the graph below). Under these conditions, X became labeled quickly but lost its radioactivity with time, whereas Y became more radioactive with time.



17. Regulation of the Calvin Cycle Iodoacetate reacts irreversibly with the free — SH groups of Cys residues in proteins. Predict which Calvin cycle enzyme(s) would be inhibited by iodoacetate, and explain why.



18. Comparison of the Reductive and Oxidative Pentose

Phosphate Pathways The *reductive* pentose phosphate pathway generates several intermediates identical to those of the *oxidative* pentose phosphate pathway ([Chapter 14](#)). What role does each pathway play in cells where it is active?

19. Photorespiration and Mitochondrial Respiration

Compare the oxidative photosynthetic carbon cycle, also called *photorespiration*, with the *mitochondrial respiration* that drives ATP synthesis. Why are both processes referred to as respiration? Where in the cell do they occur, and under what circumstances? What is the path of electron flow in each?

20. Pathway of CO₂ Assimilation in Maize

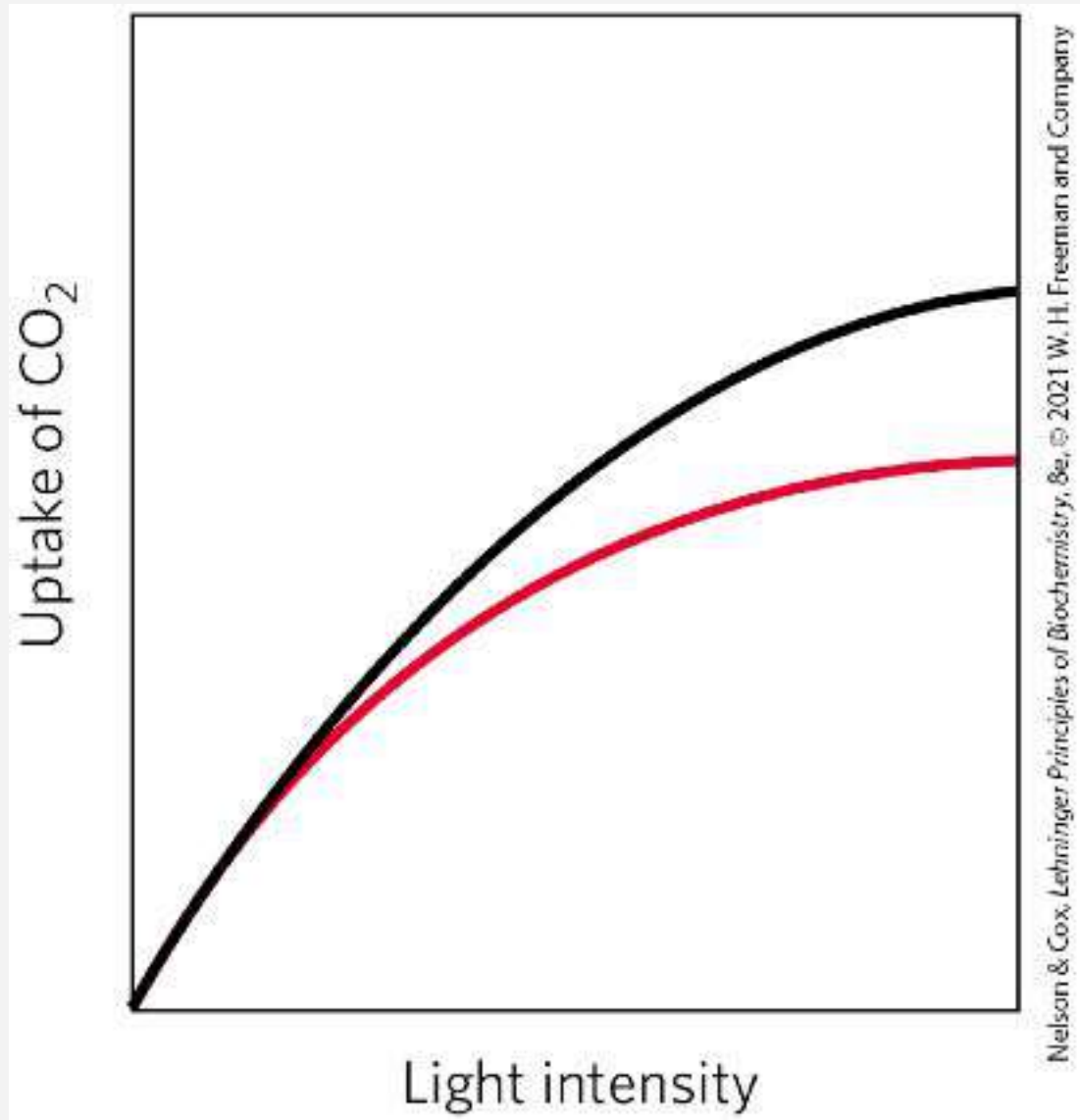
Researchers illuminate a maize (corn) plant in the presence of ¹⁴CO₂. After about 1 second of illumination, they find more than 90% of all the radioactivity incorporated in the leaves at C-4 of malate, aspartate, and oxaloacetate. Only after 60 seconds does ¹⁴C appear at C-1 of 3-phosphoglycerate. Explain.

21. **Identifying CAM Plants** Given some ¹⁴CO₂ and all the tools typically present in a biochemistry research lab, how

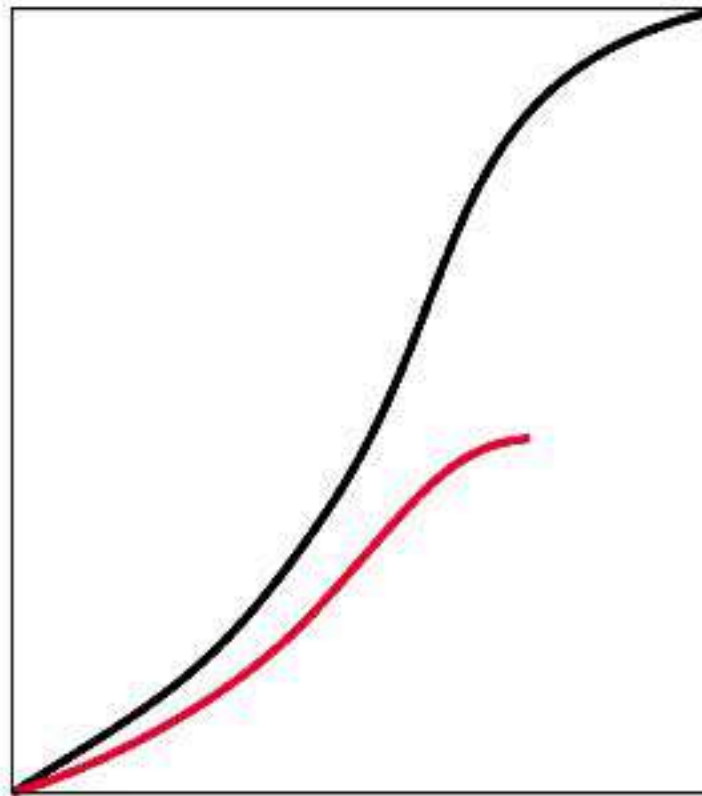
would you design a simple experiment to determine whether a plant is a typical C_4 plant or a CAM plant?

22. Chemistry of Malic Enzyme: Variation on a Theme Malic enzyme, found in the bundle-sheath cells of C_4 plants, carries out a reaction that has a counterpart in the citric acid cycle. What is the analogous reaction? Explain your choice.

23. Differences between C_3 and C_4 Plants The plant genus *Atriplex* includes some C_3 and some C_4 species. In the plots, the black curve represents species 1; the red curve represents species 2. From the data in the plots, identify which is a C_3 plant and which is a C_4 plant. Justify your answer in molecular terms that account for the data in all three plots.



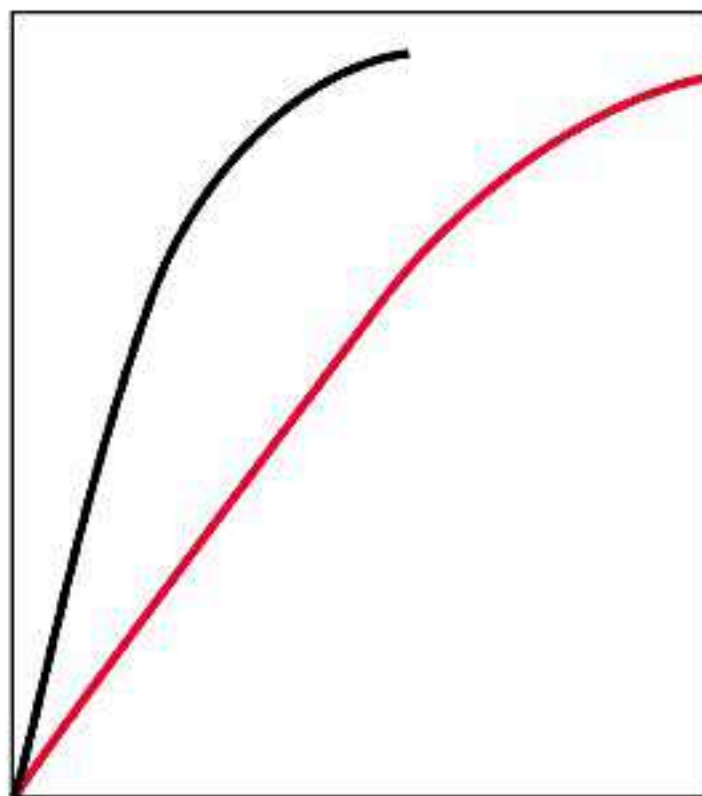
Uptake of CO_2



Leaf temperature

Melzer & Cox, Lehninger Principles of Biochemistry, 8e, © 2021 W. H. Freeman and Company

Uptake of CO_2



[CO_2] in intracellular space

24. Inorganic Pyrophosphatase The enzyme inorganic pyrophosphatase contributes to making many biosynthetic reactions that generate inorganic pyrophosphate essentially irreversible in cells. By keeping the concentration of PP_i very low, the enzyme “pulls” these reactions in the direction of PP_i formation. The synthesis of ADP-glucose in chloroplasts is one such reaction. However, the synthesis of UDP-glucose in the plant cytosol, which also produces PP_i , is readily reversible in vivo. How do you reconcile these two facts?

25. Regulation of Starch and Sucrose Synthesis Sucrose synthesis occurs in the cytosol and starch synthesis occurs in the chloroplast stroma, yet the two processes are intricately balanced. What factors shift the reactions in favor of (a) starch synthesis and (b) sucrose synthesis?

26. Regulation of Sucrose Synthesis In the regulation of sucrose synthesis from the triose phosphates produced during photosynthesis, 3-phosphoglycerate and P_i play critical roles (see [Fig. 20-42](#)). Explain why the concentrations of these two regulators reflect the rate of photosynthesis.

27. Sucrose and Dental Caries The most prevalent infection in humans worldwide is dental caries, which stems from the colonization and destruction of tooth enamel by a variety of acidifying microorganisms. These organisms synthesize and live within a water-insoluble network of dextrans, called dental plaque, composed of ($\alpha 1 \rightarrow 6$)-linked polymers of

glucose with many ($\alpha 1 \rightarrow 3$) branch points. Polymerization of dextran requires dietary sucrose, and the bacterial enzyme dextran-sucrose glucosyltransferase catalyzes the reaction.

- a. Write the overall reaction for dextran polymerization.
- b. In addition to providing a substrate for the formation of dental plaque, how does dietary sucrose also provide oral bacteria with an abundant source of metabolic energy?

28. Partitioning between the Citric Acid and Glyoxylate

Cycles In an organism (such as *Escherichia coli*) that has both the citric acid cycle and the glyoxylate cycle, what determines which of these pathways isocitrate will enter?

DATA ANALYSIS PROBLEM

29. Photophosphorylation: Discovery, Rejection, and

Rediscovery In the 1930s and 1940s, researchers were beginning to make progress toward understanding the mechanism of photosynthesis. At the time, the role of “energy-rich phosphate bonds” (today, “ATP”) in glycolysis and cellular respiration was just becoming known. There were many theories about the mechanism of photosynthesis, especially about the role of light. This problem focuses on what was then called the “primary photochemical process” — that is, on what, exactly, the energy from captured light produces in the photosynthetic cell. Interestingly, one important part of the modern model of photosynthesis was

proposed early on, only to be rejected, ignored for several years, then finally revived and accepted.

In 1944, Emerson, Stauffer, and Umbreit proposed that “the function of light energy in photosynthesis is the formation of ‘energy-rich’ phosphate bonds” ([p. 107](#)). In their model (hereafter, the “Emerson model”), the free energy necessary to drive both CO₂ fixation *and* reduction came from these “energy-rich phosphate bonds” (i.e., ATP), produced as a result of light absorption by a chlorophyll-containing protein.

This model was explicitly rejected by Rabinowitch (1945). After summarizing Emerson and coauthors’ findings, Rabinowitch stated: “Until more positive evidence is provided, we are inclined to consider as more convincing a general argument against this hypothesis, which can be derived from energy considerations. Photosynthesis is eminently a problem of energy *accumulation*. What good can be served, then, by converting light quanta (even those of red light, which amount to about 43 kcal per Einstein) into ‘phosphate quanta’ of only 10 kcal per mole? This appears to be a start in the wrong direction — toward *dissipation* rather than toward accumulation of energy” ([p. 228](#)). This argument, along with other evidence, led to abandonment of the Emerson model until the 1950s, when it was found to be correct — albeit in a modified form.

For each piece of information from Emerson and coauthors’ article presented in (a) through (d), answer the following three questions:

1. How does this information support the Emerson model, in which light energy is used directly by chlorophyll *to make ATP*, and the ATP then provides the energy to drive CO₂ fixation and reduction?
2. How would Rabinowitch explain this information, based on his model (and most other models of the day), in which light energy is used directly by chlorophyll *to make reducing compounds*? Rabinowitch wrote: “Theoretically, there is no reason why *all* electronic energy contained in molecules excited by the absorption of light should not be available for oxidation-reduction” ([p. 152](#)). In this model, the reducing compounds are then used to fix and reduce CO₂, and the energy for these reactions comes from the large amounts of free energy released by the reduction reactions.
3. How is this information explained by our modern understanding of photosynthesis?
 - a. Chlorophyll contains a Mg²⁺ ion, which is known to be an essential cofactor for many enzymes that catalyze phosphorylation and dephosphorylation reactions.
 - b. A crude “chlorophyll protein” isolated from photosynthetic cells showed phosphorylating activity.

- c. The phosphorylating activity of the “chlorophyll protein” was inhibited by light.
- d. The levels of several different phosphorylated compounds in photosynthetic cells changed dramatically in response to light exposure. (Emerson and coworkers were not able to identify the specific compounds involved.)

As it turned out, the Emerson and Rabinowitch models were both partly correct and partly incorrect.

- e. Explain how the two models relate to our current model of photosynthesis.

In his rejection of the Emerson model, Rabinowitch went on to say: “The difficulty of the phosphate storage theory appears most clearly when one considers the fact that, in weak light, eight or ten quanta of light are sufficient to reduce one molecule of carbon dioxide. If each quantum should produce one molecule of high-energy phosphate, the accumulated energy would be only 80–100 kcal per Einstein — while photosynthesis requires *at least* 112 kcal per mole, and probably more, because of losses in irreversible partial reactions” ([p. 228](#)).

- f. How does Rabinowitch’s value of 8 to 10 photons per molecule of CO₂ reduced compare with the value accepted today?

g. How would you rebut Rabinowitch's argument, based on our current knowledge about photosynthesis?

References

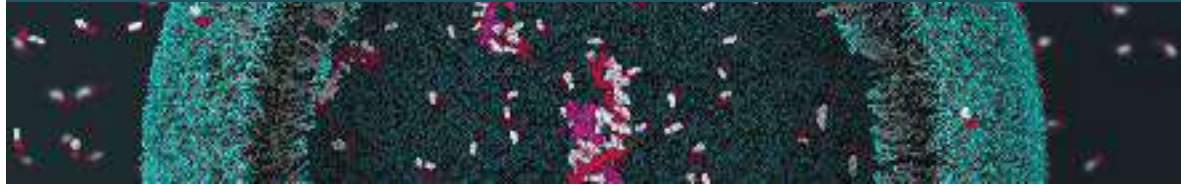
Emerson, R.L., J.F. Stauffer, and W.W. Umbreit. 1944.

Relationships between phosphorylation and photosynthesis in *Chlorella*. *Am. J. Botany* 31:107–120.

Rabinowitch, E.I. 1945. *Photosynthesis and Related Processes*, Vol. I. New York: Interscience Publishers.

CHAPTER 21

LIPID BIOSYNTHESIS



21.1 Biosynthesis of Fatty Acids and Eicosanoids

21.2 Biosynthesis of Triacylglycerols

21.3 Biosynthesis of Membrane Phospholipids

21.4 Cholesterol, Steroids, and Isoprenoids: Biosynthesis, Regulation, and Transport

Lipids play a variety of cellular roles, some recognized only recently. This chapter describes the biosynthetic pathways for some of the most common cellular lipids, illustrating the strategies employed in assembling these water-insoluble products from water-soluble precursors such as acetate. We first describe the biosynthesis of fatty acids, the primary components of both triacylglycerols and phospholipids, then examine the assembly of fatty acids into triacylglycerols and the simpler membrane phospholipids. Finally, we consider the synthesis of cholesterol, a component of some membranes and the precursor of steroids such as bile acids, sex hormones, and adrenocortical hormones.

Our coverage of lipid biosynthesis is organized around the following principles:

P1 Lipids are the principal form of stored energy in most higher organisms, as well as the major constituents of membranes.

P2 Anabolism is not simply the reverse of catabolism. Biosynthetic pathways typically diverge from breakdown pathways to overcome irreversible steps in catabolism.

P3 Like other anabolic pathways, the reaction sequences in lipid biosynthesis are endergonic and reductive. They use ATP as a source of metabolic energy and a reduced electron carrier (usually NADPH) as a reductant.

P4 Lipid biosynthesis, like other anabolic pathways, is subject to regulation to respond to cellular and organismal requirements. The places where catabolic and anabolic pathways diverge (Principle 2) provide opportunities to impose metabolic regulation to conserve resources and avoid futile cycles.

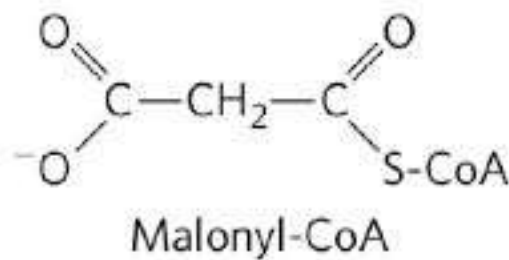
P5 Like other major classes of biological molecules, lipids have a plethora of cellular functions. Specialized lipids serve as pigments (retinal, carotene), cofactors (vitamin K), detergents (bile salts), transporters (dolichols), hormones (vitamin D derivatives, sex hormones), extracellular and intracellular messengers (eicosanoids, phosphatidylinositol derivatives), and anchors for membrane proteins (covalently attached fatty acids, prenyl groups, phosphatidylinositol). The

ability to synthesize a variety of lipids is essential to all organisms.

We will focus on eukaryotes, with occasional digressions to highlight important distinctions in bacteria and plants.

21.1 Biosynthesis of Fatty Acids and Eicosanoids

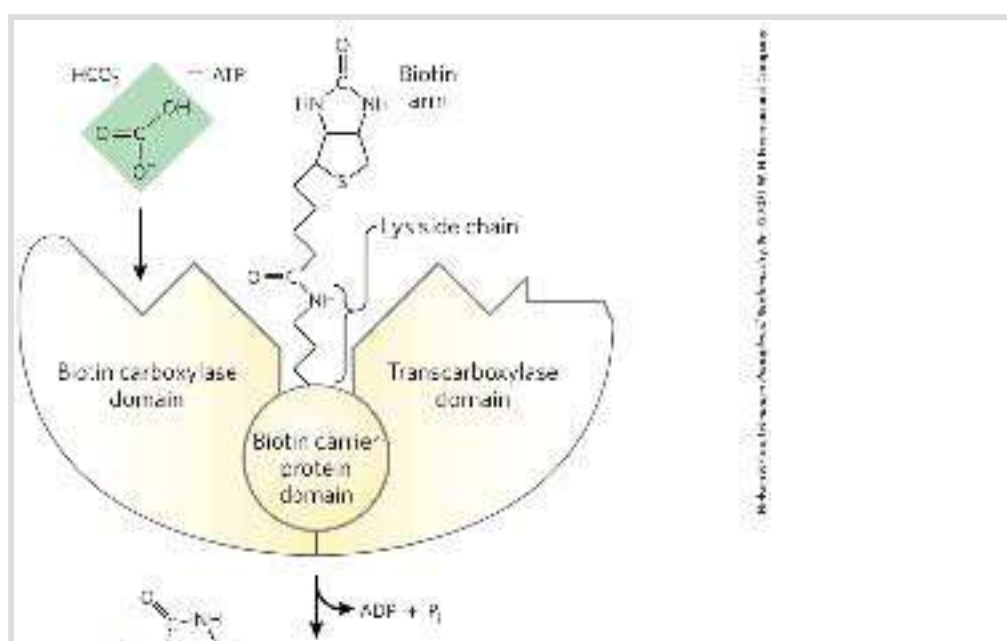
P2 Even when compared to other major classes of metabolites, the division between fatty acid biosynthesis and breakdown is particularly striking. The two processes occur by different pathways, catalyzed by different sets of enzymes, and, in eukaryotes, occur in different cellular compartments. Fatty acid breakdown occurs in the mitochondria, whereas biosynthesis occurs in the cytosol. Moreover, biosynthesis requires the participation of a three-carbon intermediate, **malonyl-CoA**, that does not appear in the path of fatty acid breakdown.



The general pathway of fatty acid synthesis and its regulation now take center stage. We consider the biosynthesis of longer-chain fatty acids, unsaturated fatty acids, and their eicosanoid derivatives at the end of this section.

Malonyl-CoA Is Formed from Acetyl-CoA and Bicarbonate

The formation of malonyl-CoA from acetyl-CoA is an irreversible three-step process, catalyzed by **acetyl-CoA carboxylase**. In animal cells, all three steps are catalyzed in the cytoplasm by a single multifunctional polypeptide ([Fig. 21-1](#)). The enzyme contains a biotin prosthetic group covalently bound in amide linkage to the ϵ -amino group of a Lys residue in one of the domains of the enzyme molecule. The reaction catalyzed by this enzyme is very similar to other biotin-dependent carboxylation reactions, such as those catalyzed by pyruvate carboxylase (see [Fig. 16-16](#)) and propionyl-CoA carboxylase (see [Fig. 17-12](#)). A carboxyl group, derived from bicarbonate (HCO_3^-), is first transferred to biotin in an ATP-dependent reaction. In a second step, the carboxyl group is carried by the biotin to a different active site, where the CO_2 is transferred to acetyl-CoA in the third and final step to yield malonyl-CoA. As we will see, this carboxylation has the same function as the carboxylation of pyruvate by pyruvate carboxylase — it renders the next step in the reaction sequence much more favorable thermodynamically.



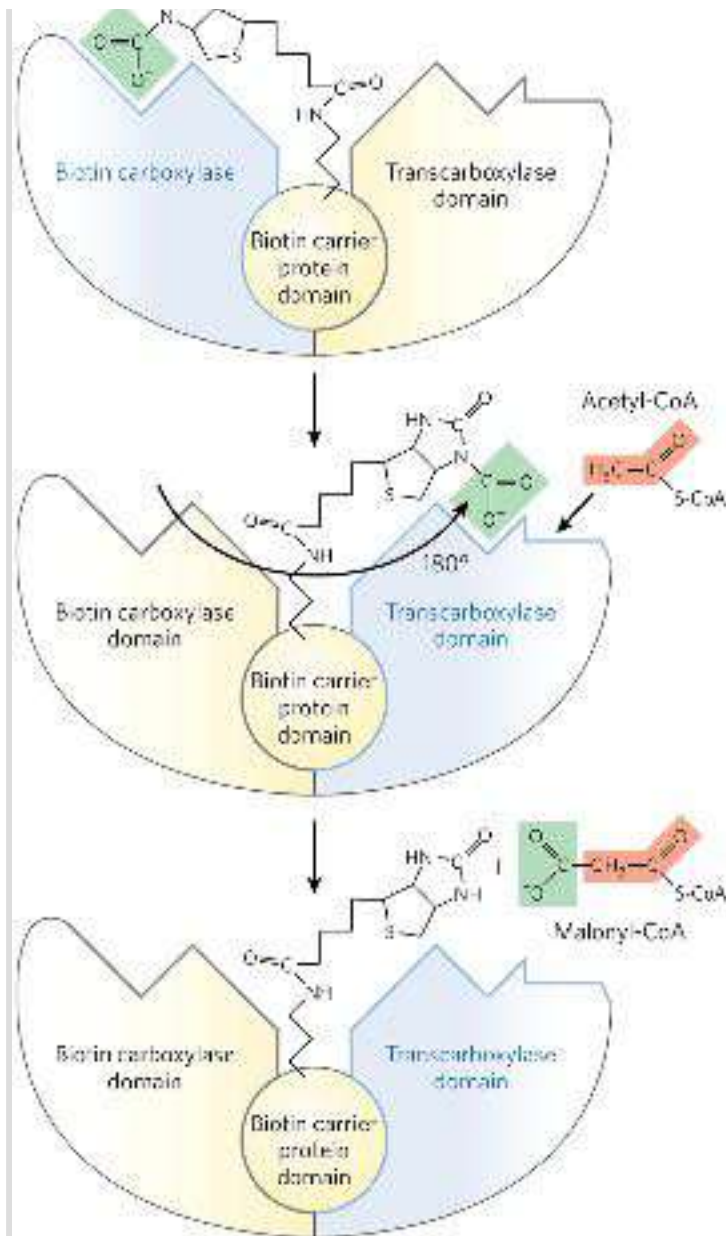
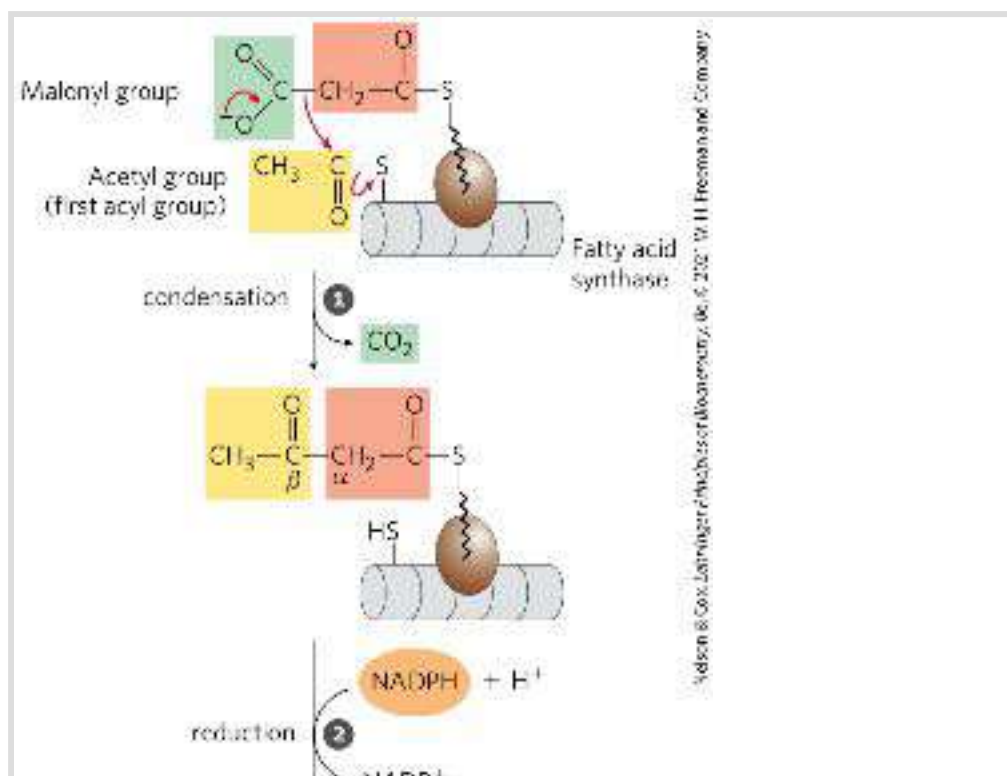


FIGURE 21-1 The acetyl-CoA carboxylase reaction. The mammalian acetyl-CoA carboxylase of the cytoplasm has three functional domains with distinct functions: biotin carrier protein; biotin carboxylase, which activates CO_2 by attaching it to a nitrogen in the biotin ring in an ATP-dependent reaction; and transcarboxylase, which transfers activated CO_2 (shaded green) from biotin to acetyl-CoA, producing malonyl-CoA. Part of the biotin carrier protein domain and the long, flexible biotin arm rotate to carry the activated CO_2 from the biotin carboxylase active site to the transcarboxylase active site. The active domain in each step is shaded in blue.

The bacterial version of acetyl-CoA carboxylase is similar but has three separate polypeptide subunits (including a separate biotin carrier protein) that catalyze the three steps. Plant cells contain both types of acetyl-CoA carboxylase.

Fatty Acid Synthesis Proceeds in a Repeating Reaction Sequence

The long carbon chains of fatty acids are assembled in the cytosol in a repeating four-step sequence ([Fig. 21-2](#)), catalyzed by a system collectively referred to as **fatty acid synthase**. A saturated acyl group produced by each four-step series of reactions becomes the substrate for subsequent condensation with an activated malonyl group. With each passage through the cycle, the fatty acyl chain is extended by two carbons.



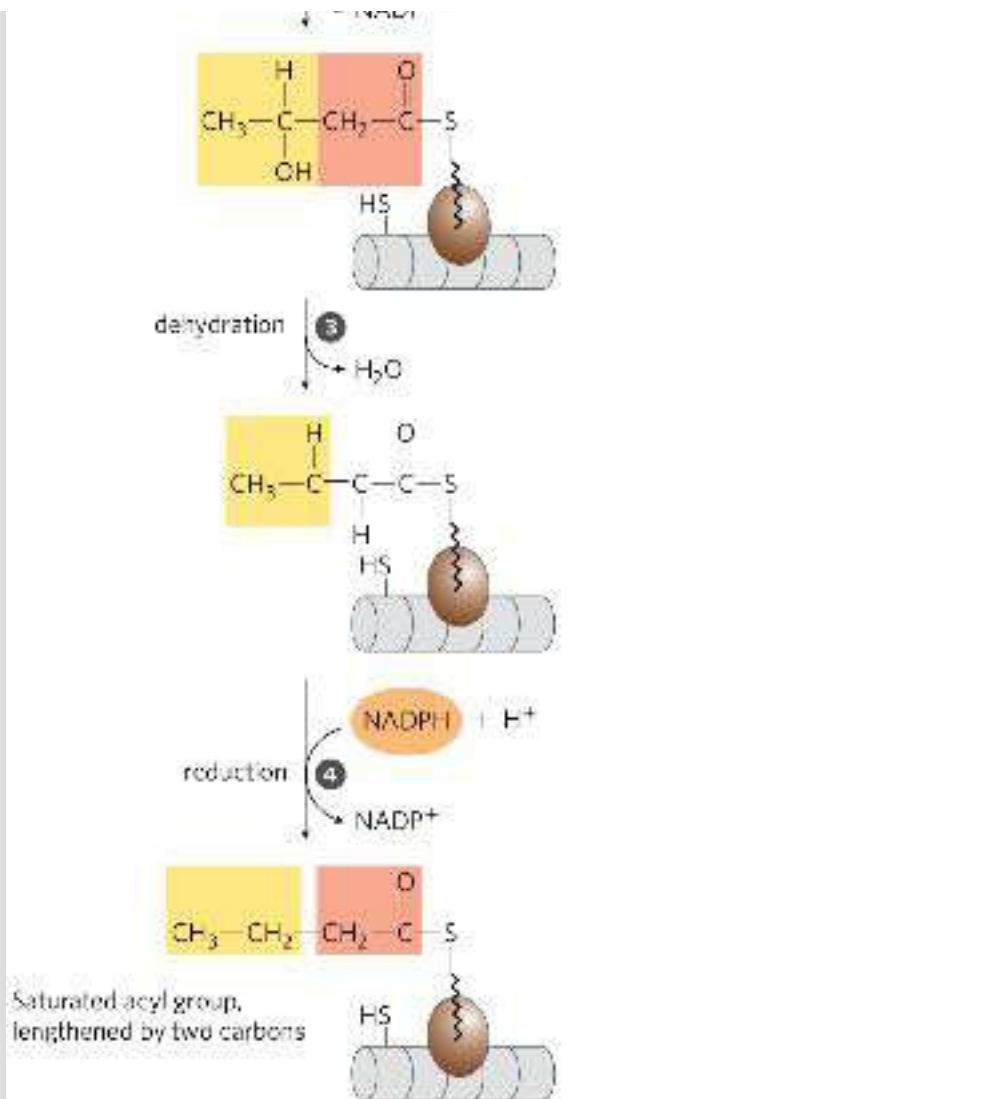


FIGURE 21-2 Addition of two carbons to a growing fatty acyl chain: a four-step sequence. Each malonyl group and acetyl (or longer acyl) group is activated by a thioester that links it to fatty acid synthase, a multienzyme system. **1** Condensation of an activated acyl group (an acetyl group from acetyl-CoA is the first acyl group) and two carbons derived from malonyl-CoA, with elimination of CO_2 from the malonyl group, extends the acyl chain by two carbons. The mechanism of the first step of this reaction is given to illustrate the role of decarboxylation in facilitating condensation. The β -keto product of the condensation is then reduced in three more steps nearly identical to the reactions of β oxidation, but in the reverse sequence: **2** the β -keto group is reduced to an alcohol, **3** elimination of H_2O creates a double bond, and **4** the double bond is reduced to form the corresponding saturated fatty acyl group.

In the four-step pathway, a condensation reaction is followed by a reduction-dehydration-reduction sequence to convert the C-3 carbonyl to a methylene. The latter three steps are the chemical reverse of the oxidation-hydration-oxidation sequence in the β oxidation of fatty acids ([Fig. 17-8a](#)).  However, both the electron-carrying cofactor and the activating groups in the reductive anabolic sequence differ from those in the oxidative catabolic process. Recall that in β oxidation, NAD^+ and FAD serve as electron acceptors and the activating group is the thiol (—SH) group of coenzyme A. By contrast, the reducing agent in the synthetic sequence is NADPH and the activating groups are two different enzyme-bound —SH groups, as described below.

In mammals, this synthetic system is called fatty acid synthase I (FAS I). There are seven active sites to catalyze the four-step cycle plus the charging steps described below. The active sites for different reactions lie in separate domains ([Fig. 21-3a](#)), all within a single multifunctional polypeptide chain (M_r 240,000). Two of these multifunctional polypeptides function as a homodimer (M_r 480,000; [Fig. 21-3b](#)). The two subunits seem to function independently. When all the active sites in one subunit are inactivated by mutation, fatty acid synthesis is only moderately reduced.

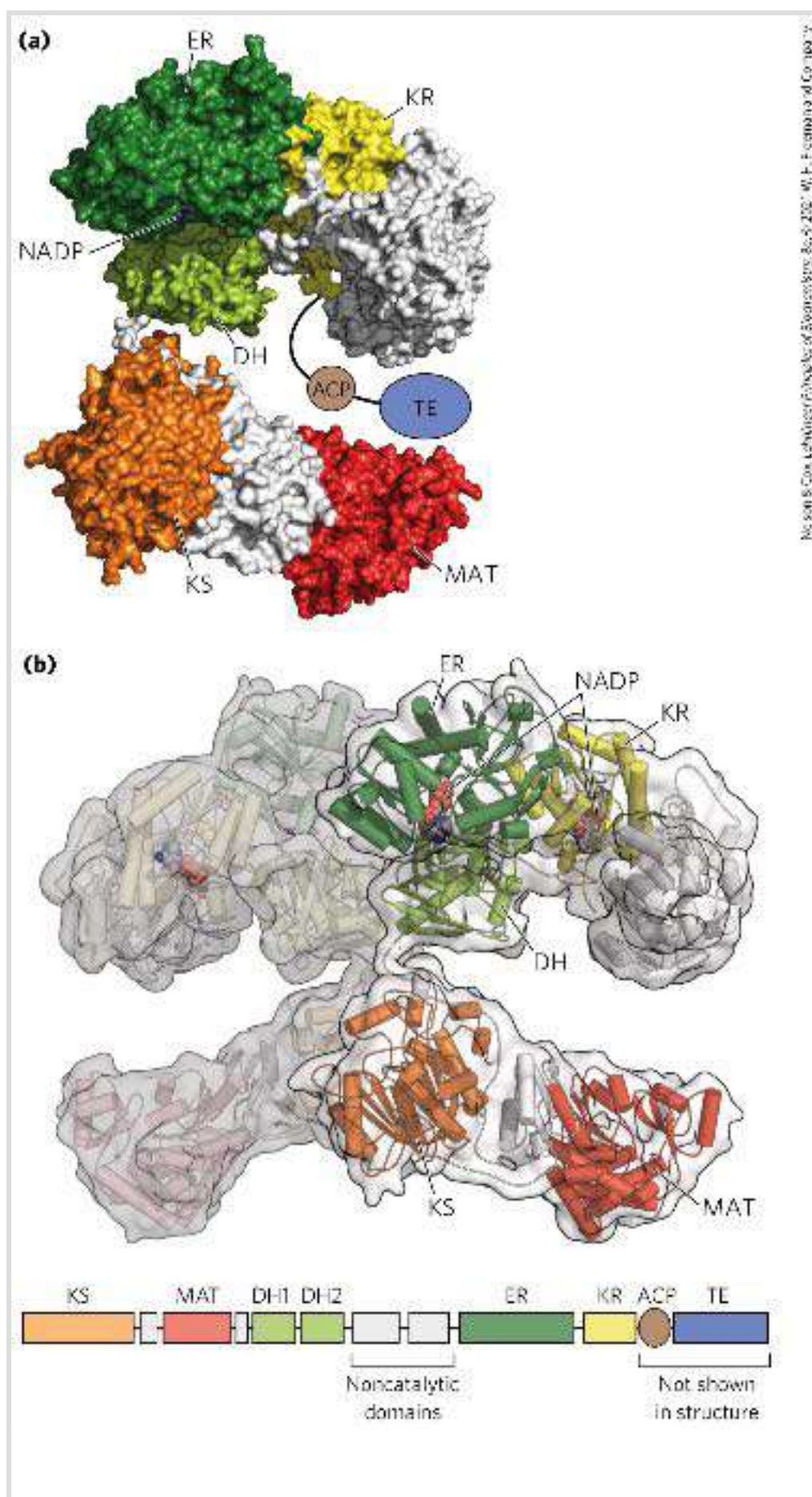


FIGURE 21-3 The structure of a fatty acid synthase type I system. Shown here is the structure of a single (monomeric) polypeptide chain of the mammalian (porcine) enzyme system. (a) All of the active sites in the mammalian system are located in different domains within a single large polypeptide chain. The different enzymatic activities are β -ketoacyl-ACP synthase (KS), malonyl/acetyl-CoA-ACP transferase (MAT), β -hydroxyacyl-ACP dehydratase (DH), enoyl-ACP reductase (ER), and β -ketoacyl-ACP reductase (KR). ACP is the acyl carrier protein. The seventh domain is a thioesterase (TE) that releases the palmitate product from ACP when synthesis is completed. The ACP and TE domains are disordered in the crystal and are therefore not shown in the structure. (b) The native dimeric structure is shown, with one polypeptide transparent to show how the two independently operating subunits come together. The linear arrangement of the domains in the polypeptide is shown below the structure. [Data from PDB ID 2CF2, T. Maier et al., *Science* 311:1258, 2006.]

With FAS I systems, fatty acid synthesis leads to a single product. As it goes through the cycle, the acyl group is covalently linked to **acyl carrier protein (ACP)**, which shuttles it from one active site to another in sequence. Acyl carrier protein is yet another contiguous part of the single FAS I polypeptide. No intermediates are released. When the chain length reaches 16 carbons, that product (palmitate, 16:0; see [Table 10-1](#)) leaves the cycle. Carbons C-16 and C-15 of the palmitate are derived from the methyl and carboxyl carbon atoms, respectively, of an acetyl-CoA used directly to prime the system at the outset ([Fig. 21-4](#)); the rest of the carbon atoms in the chain are derived from acetyl-CoA via malonyl-CoA.

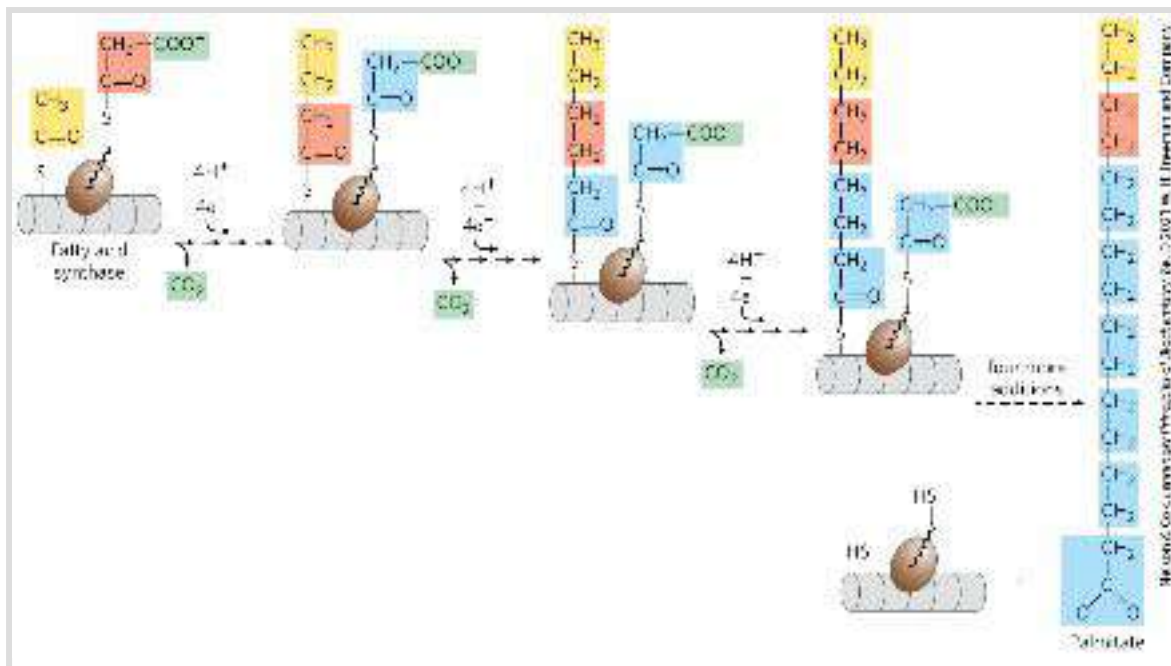


FIGURE 21-4 The overall process of palmitate synthesis. The fatty acyl chain grows by two-carbon units donated by activated malonate, with loss of CO₂ at each step. The initial acetyl group is shaded yellow; C-1 and C-2 of malonate, light red; and the carbon released as CO₂, green. After each two-carbon addition, reductions convert the growing chain to a saturated fatty acid of four, then six, then eight carbons, and so on. The final product is palmitate (16:0).

A somewhat different FAS I is found in yeast and other fungi, and is made up of two multifunctional polypeptides that form a complex with an architecture distinct from that of the vertebrate systems. Three of the seven required active sites are found on the α subunit and four on the β subunit. A different system, called FAS II, is found in plants and most bacteria. FAS II is a dissociated system; each step in the synthesis is catalyzed by a separate enzyme. Unlike FAS I, FAS II generates a variety of products, including saturated fatty acids of several lengths, as well as unsaturated, branched, and hydroxy fatty acids. An FAS II system is also found in vertebrate mitochondria, yet another indication of the bacterial origins of mitochondria in evolution.

The Mammalian Fatty Acid Synthase Has Multiple Active Sites

The multiple domains of mammalian FAS I function as distinct but linked enzymes. The active site for each enzyme is found in a separate domain within the larger polypeptide. Throughout the process of fatty acid synthesis, the intermediates remain covalently attached as thioesters to one of two thiol groups. One point of attachment is the —SH group of a Cys residue in one of the synthase domains (β -ketoacyl-ACP synthase; KS); the other is the —SH group of acyl carrier protein, a separate domain of the same polypeptide. Hydrolysis of thioesters is highly exergonic, and the energy released helps to make two steps (❶ and ❺ in [Fig. 21-6](#)) in fatty acid synthesis thermodynamically favorable.

Acyl carrier protein is the shuttle that holds the system together, containing the prosthetic group **4'-phosphopantetheine**, also found in coenzyme A ([Fig. 21-5](#)). The 4'-phosphopantetheine serves as a flexible arm, tethering the growing fatty acyl chain to the surface of the fatty acid synthase complex while carrying the reaction intermediates from one enzyme active site to the next. The same prosthetic group is used in FAS II systems.

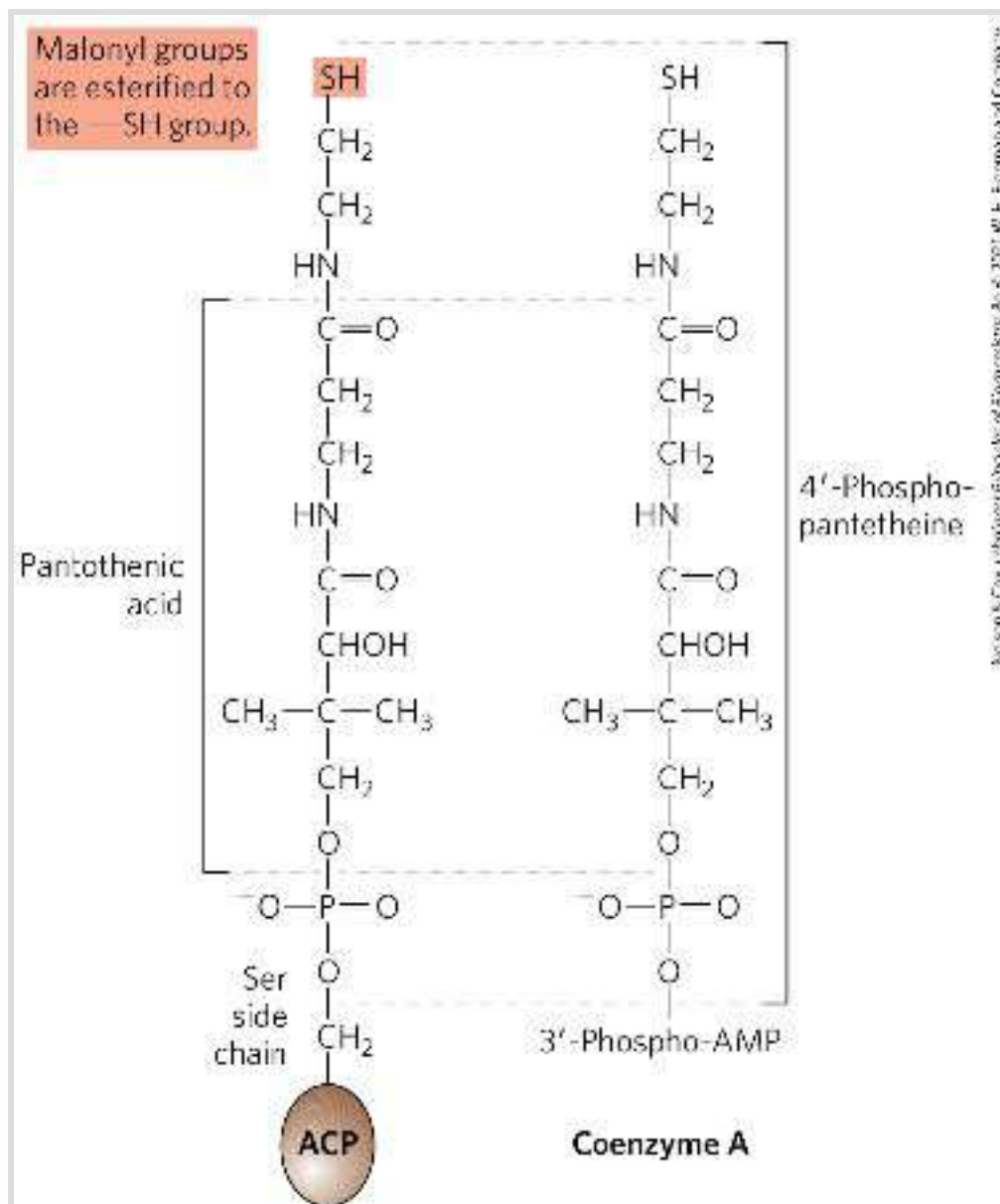


FIGURE 21-5 Acyl carrier protein (ACP). The prosthetic group is 4'-phosphopantetheine, which is covalently attached to the hydroxyl group of a Ser residue in ACP. Phosphopantetheine, also found in the coenzyme A molecule, contains the B vitamin pantothenic acid. Its —SH group is the site of entry of malonyl groups during fatty acid synthesis. Coenzyme A, shown here for comparison, serves a similar chemical purpose in general metabolism.

Fatty Acid Synthase Receives the Acetyl and Malonyl Groups

Before the condensation reactions that build up the fatty acid chain can begin, the two thiol groups on the enzyme complex must be charged with the correct acyl groups ([Fig. 21-6a](#)). First, the acetyl group of acetyl-CoA is transferred to ACP in a reaction catalyzed by the **malonyl/acetyl-CoA-ACP transferase** (MAT) domain of the multifunctional polypeptide. The acetyl group is then transferred to the Cys —SH group of the β -**ketoacyl-ACP synthase** (KS). The second reaction, transfer of the malonyl group from malonyl-CoA to the —SH group of ACP, is also catalyzed by malonyl/acetyl-CoA-ACP transferase. In the charged synthase complex, the acetyl and malonyl groups are activated for the chain-lengthening process. We now consider the first four steps of this process in some detail; all step numbers refer to [Figure 21-6](#).

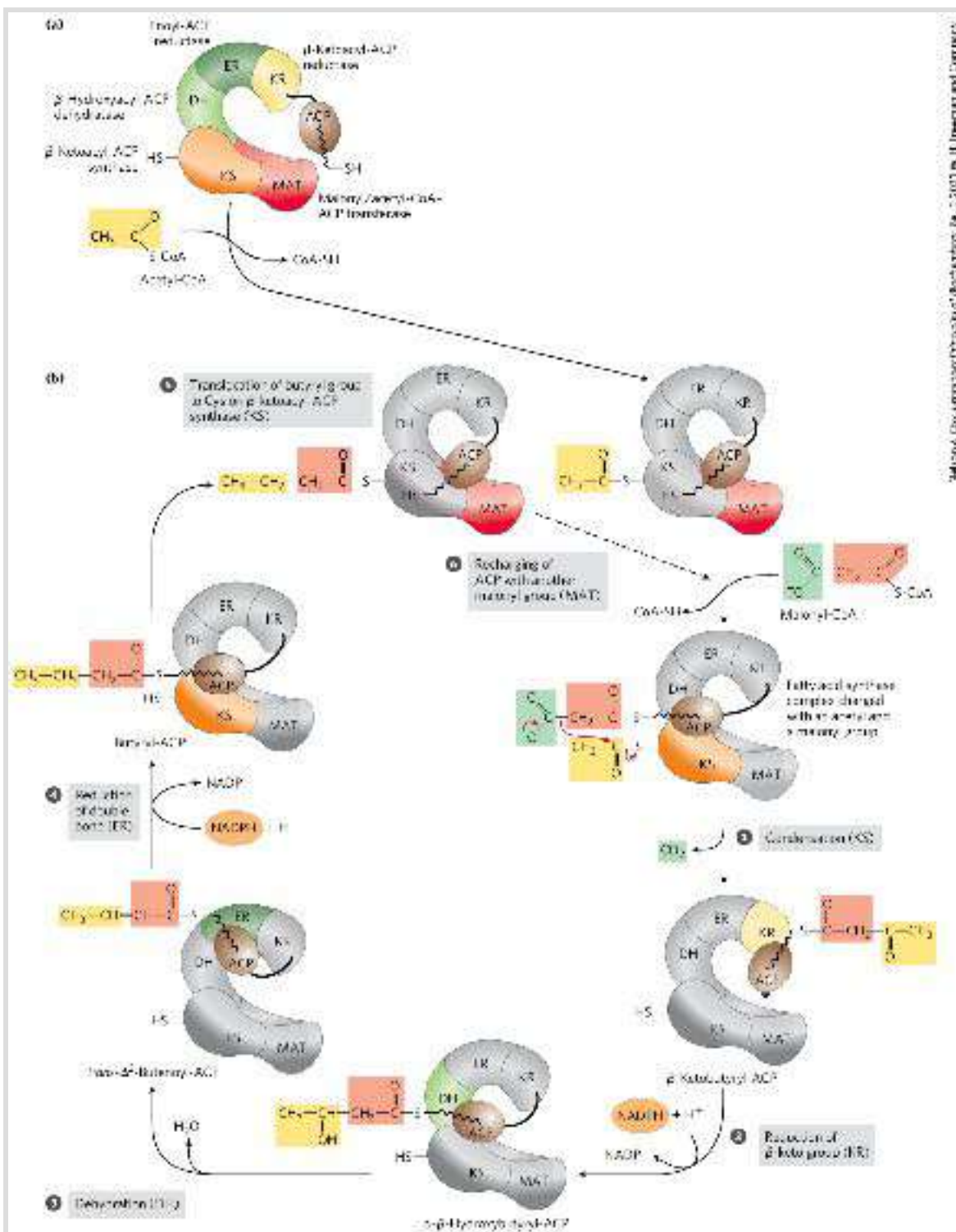


FIGURE 21-6 Sequence of events during synthesis of a fatty acid. (a) The mammalian FAS I complex is shown schematically, with catalytic domains colored as in [Figure 21-3](#). Each domain of the larger polypeptide represents one of the six enzymatic activities of the complex, arranged in a large, tight S shape. The acyl carrier protein (ACP) is not resolved in the crystal structure shown in [Figure 21-3](#), but it is attached to the KR domain. The phosphopantetheine arm of ACP ends in an —SH. (b) The enzyme shown in

color is the one that will act in the next step. As in [Figure 21-4](#), the initial acetyl group is shaded yellow; C-1 and C-2 of malonate, light red; and the carbon released as CO_2 , green. Steps **1** to **6** are described in the text.

Step **1** Condensation

The first reaction in the formation of a fatty acid chain is a formal Claisen condensation (see reaction class in [Fig. 13-4](#)) of the activated acetyl and malonyl groups to form **acetoacetyl-ACP**, an acetoacetyl group bound to ACP through the phosphopantetheine —SH group; simultaneously, a molecule of CO_2 is produced. In this reaction, catalyzed by β -ketoacyl-ACP synthase, the acetyl group is transferred from the Cys —SH group of the enzyme to the malonyl group on the —SH of ACP, becoming the methyl-terminal two-carbon unit of the new acetoacetyl group.

The carbon atom of the CO_2 formed in this reaction is the same carbon originally introduced into malonyl-CoA from HCO_3^- in the acetyl-CoA carboxylase reaction ([Fig. 21-1](#)). Thus, CO_2 is only transiently in covalent linkage during fatty acid biosynthesis; it is removed as each two-carbon unit is added.

Why do cells go to the trouble of adding CO_2 to make a malonyl group from an acetyl group, only to lose the CO_2 during the formation of acetoacetate?  The use of activated malonyl groups rather than acetyl groups makes the condensation reactions thermodynamically favorable. The methylene carbon (C-2) of the malonyl group, sandwiched between carbonyl and

carboxyl carbons, forms a good nucleophile. In the condensation step, decarboxylation of the malonyl group facilitates nucleophilic attack of the methylene carbon on the thioester linking the acetyl group to β -ketoacyl-ACP synthase, displacing the enzyme's —SH group. Coupling the condensation to the decarboxylation of the malonyl group renders the overall process highly exergonic. A similar carboxylation-decarboxylation sequence facilitates the formation of phosphoenolpyruvate from pyruvate in gluconeogenesis (see [Figs. 14-17](#) and [14-18](#)).

  By using activated malonyl groups in the synthesis of fatty acids and activated acetate in their degradation, the cell makes both processes energetically favorable, although one is effectively the reversal of the other. The extra energy required to make fatty acid synthesis favorable is provided by the ATP used to synthesize malonyl-CoA from acetyl-CoA and HCO_3^- ([Fig. 21-1](#)).

Step 2 Reduction of the Carbonyl Group

The acetoacetyl-ACP formed in the condensation step now undergoes reduction of the carbonyl group at C-3 to form D- β -hydroxybutyryl-ACP. This reaction is catalyzed by **β -ketoacyl-ACP reductase** (KR), and the electron donor is NADPH. Notice that the D- β -hydroxybutyryl group does not have the same stereoisomeric form as the L- β -hydroxyacyl intermediate in fatty acid oxidation (see [Fig. 17-8](#)).

Step 3 Dehydration

The elements of water are now removed from C-2 and C-3 of D- β -hydroxybutyryl-ACP to yield a double bond in the product, ***trans*- Δ^2 -butenoyl-ACP**. The enzyme that catalyzes this dehydration is **β -hydroxyacyl-ACP dehydratase (DH)**.

Step 4 Reduction of the Double Bond

Finally, the double bond of *trans*- Δ^2 -butenoyl-ACP is reduced (saturated) to form **butyryl-ACP** by the action of **enoyl-ACP reductase (ER)**; again, NADPH is the electron donor.

The Fatty Acid Synthase Reactions Are Repeated to Form Palmitate

Production of the four-carbon, saturated fatty acyl-ACP marks completion of one pass through the fatty acid synthase complex. In step 5, the butyryl group is transferred from the phosphopantetheine —SH group of ACP to the Cys —SH group of β -ketoacyl-ACP synthase, which initially bore the acetyl group ([Fig. 21-6](#)). To start the next cycle of four reactions that lengthens the chain by two more carbons (step 6), another malonyl group is linked to the now unoccupied phosphopantetheine —SH group of ACP ([Fig. 21-7](#)). Condensation occurs as the butyryl group, acting like the acetyl group in the first cycle, is linked to two carbons of the malonyl-ACP group with concurrent loss of CO₂.

The product of this condensation is a six-carbon acyl group, covalently bound to the phosphopantetheine —SH group. Its β -keto group is reduced in the next three steps of the synthase cycle to yield the saturated acyl group, exactly as in the first round of reactions — in this case forming the six-carbon product.

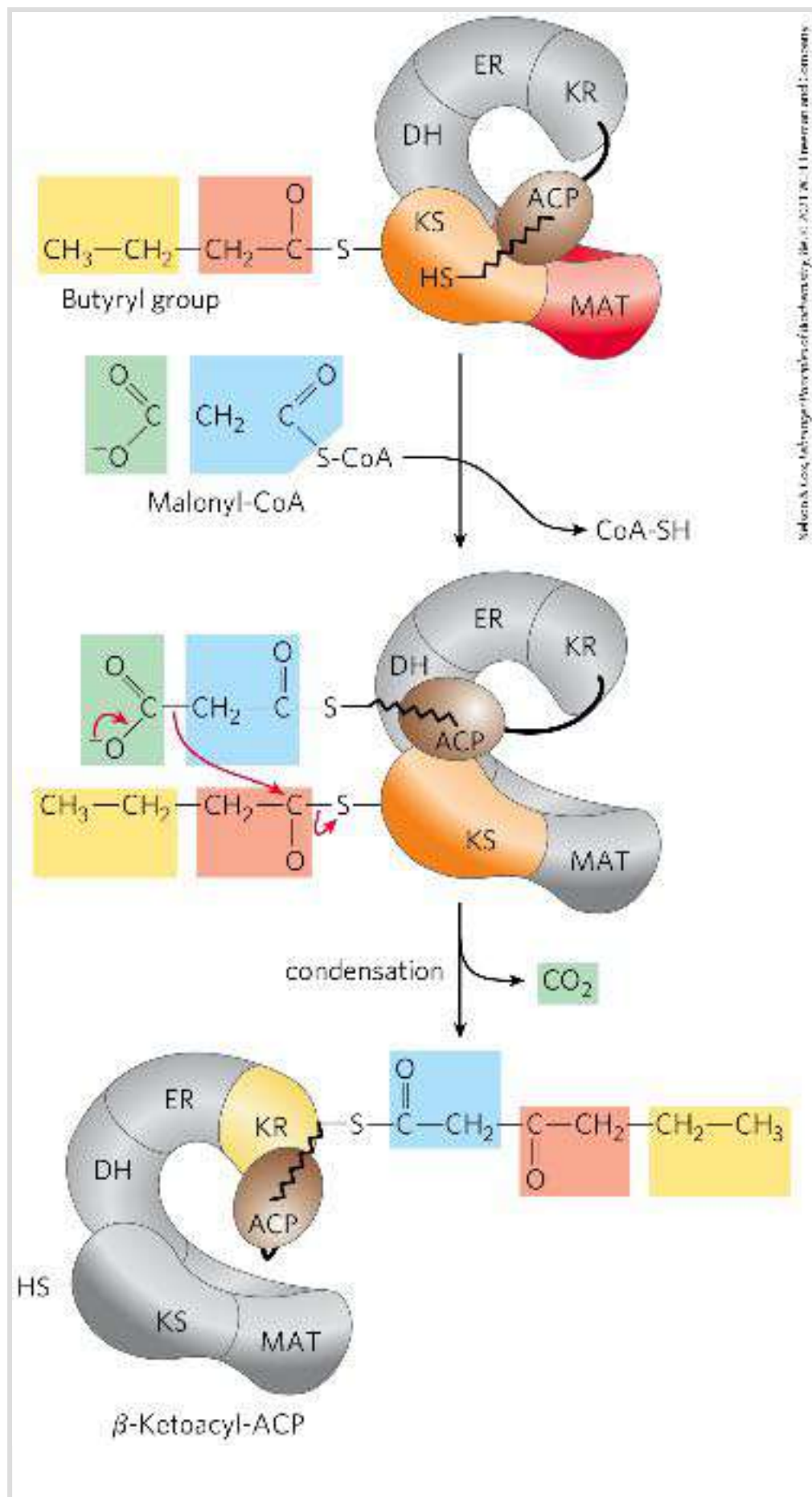
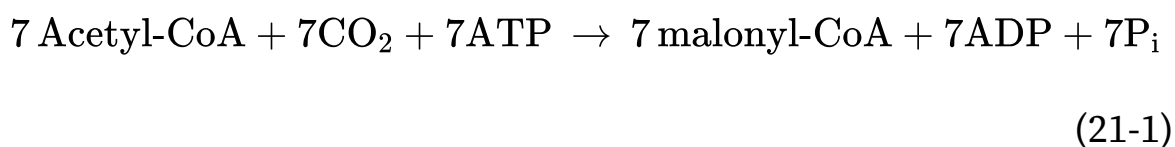


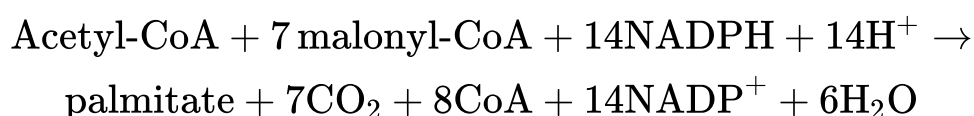
FIGURE 21-7 Beginning of the second round of the fatty acid synthesis cycle. The butyryl group is on the Cys —SH group. The incoming malonyl group is first attached to the phosphopantetheine —SH group. Then, in the condensation step, the entire butyryl group on the Cys —SH is exchanged for the carboxyl group of the malonyl residue, which is lost as CO₂ (green). This step is analogous to step ① in [Figure 21-6](#). The product, a six-carbon β -ketoacyl group, now contains four carbons derived from malonyl-CoA and two derived from the acetyl-CoA that started the reaction. The β -ketoacyl group then undergoes steps ② through ④ in [Figure 21-6](#).

Seven cycles of condensation and reduction produce the 16-carbon saturated palmitoyl group, still bound to ACP. For reasons not well understood, chain elongation by the synthase complex generally stops at this point, and free palmitate is released from the ACP by a hydrolytic activity (**thioesterase**; TE) in the multifunctional protein.

We can consider the overall reaction for the synthesis of palmitate from acetyl-CoA in two parts. First, the formation of seven malonyl-CoA molecules:

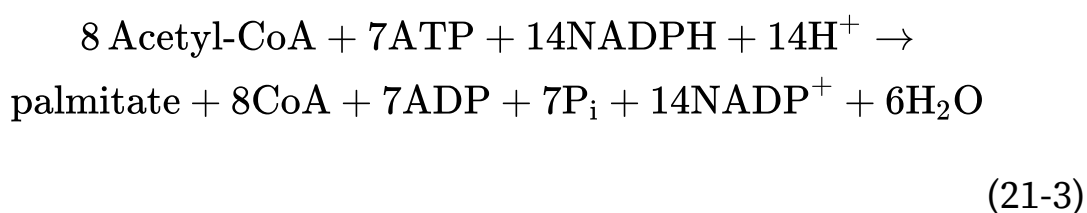


then seven cycles of condensation and reduction:



(21-2)

Notice that only six net water molecules are produced, because one is used to hydrolyze the thioester linking the palmitate product to the enzyme. The overall process (the sum of [Eqns 21-1](#) and [21-2](#)) is



 The biosynthesis of fatty acids such as palmitate thus requires acetyl-CoA and the input of chemical energy in two forms: the group transfer potential of ATP and the reducing power of NADPH. The ATP is required to attach CO₂ to acetyl-CoA to make malonyl-CoA; the NADPH molecules are required to reduce the β -keto group and the double bond.

In nonphotosynthetic eukaryotes there is an additional cost to fatty acid synthesis, because acetyl-CoA is generated in the mitochondria and must be transported to the cytosol. As we will see, this extra step consumes two ATP per molecule of acetyl-CoA transported, increasing the energetic cost of fatty acid synthesis to three ATP per two-carbon unit.

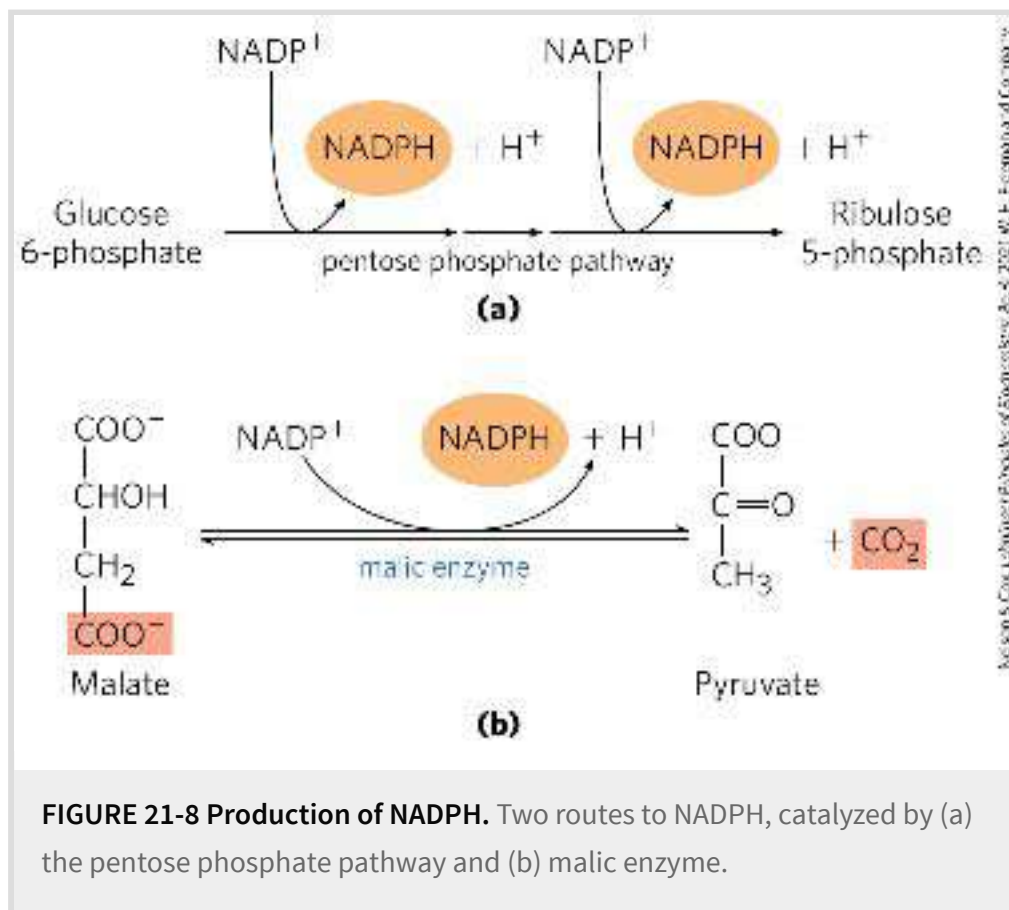
Fatty Acid Synthesis Is a Cytosolic Process in Most Eukaryotes but Takes Place in the Chloroplasts in Plants

In most eukaryotes, the fatty acid synthase complex (FAS I) is found in the cytosol, as are the biosynthetic enzymes for nucleotides, amino acids, and glucose. This location segregates synthetic processes from degradative reactions, many of which take place in the mitochondrial matrix.  There is a corresponding segregation of the electron-carrying cofactors used in anabolism (generally a reductive process) and those used in catabolism (generally oxidative).

Usually, NADPH is the electron carrier for anabolic reactions, and NAD^+ serves in catabolic reactions. The liver, the largest mammalian internal organ and a key metabolic center responding to large changes during feasting and fasting, provides a useful focus for this discussion. In hepatocytes, the $[\text{NADPH}]/[\text{NADP}^+]$ ratio is very high (~ 75) in the cytosol, furnishing a strongly reducing environment for the reductive synthesis of fatty acids and other biomolecules. The cytosolic $[\text{NADH}]/[\text{NAD}^+]$ ratio is much lower ($\sim 8 \times 10^{-4}$), so the NAD^+ -dependent oxidative catabolism of glucose can take place in the same compartment, and at the same time, as fatty acid synthesis. The $[\text{NADH}]/[\text{NAD}^+]$ ratio in the mitochondrion is much higher than that in the cytosol, because of the flow of electrons to NAD^+ from the oxidation of fatty acids, amino acids, pyruvate, and

acetyl-CoA. This high mitochondrial $[\text{NADH}]/[\text{NAD}^+]$ ratio favors the reduction of oxygen via the respiratory chain.

In hepatocytes and adipocytes, cytosolic NADPH is largely generated by the pentose phosphate pathway ([Fig. 21-8a](#); also see [Fig. 14-30](#)) and by **malic enzyme** ([Fig. 21-8b](#)). The pyruvate produced by the action of malic enzyme reenters the mitochondrion.



In the photosynthetic cells of plants, fatty acid synthesis occurs not in the cytosol but in the chloroplast stroma ([Fig. 21-9](#)). This makes sense, given that NADPH is produced in chloroplasts by the light-dependent reactions of photosynthesis:

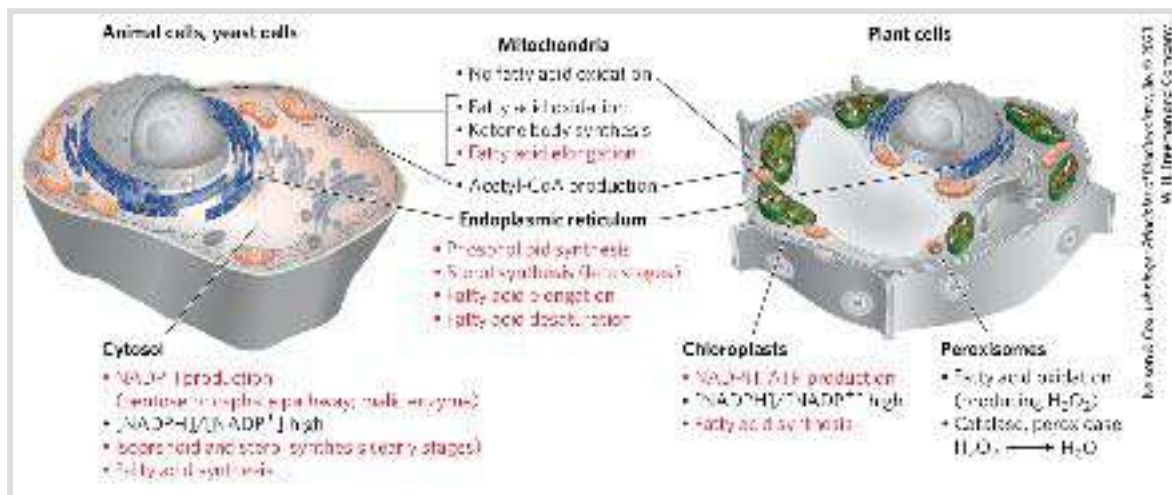
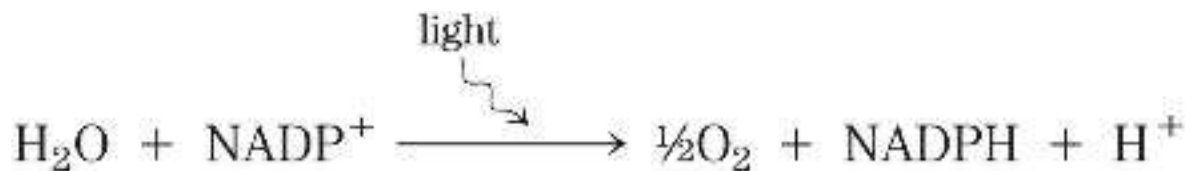


FIGURE 21-9 Subcellular localization of lipid metabolism. Yeast and animal cells differ from higher plant cells in the compartmentation of lipid metabolism. Fatty acid synthesis takes place in the compartment in which NADPH is available for reductive synthesis (i.e., where the [NADPH]/[NADP⁺] ratio is high); this is the cytosol in animals and yeast, and the chloroplast in plants. Processes in red type are addressed in this chapter.

Acetate Is Shuttled out of Mitochondria as Citrate

In nonphotosynthetic eukaryotes, nearly all the acetyl-CoA used in fatty acid synthesis is formed in mitochondria from pyruvate oxidation and from catabolism of the carbon skeletons of amino acids. Acetyl-CoA arising from the oxidation of fatty acids is not a significant source of acetyl-CoA for fatty acid biosynthesis in animals, because the two pathways are reciprocally regulated, as described below.

The inner mitochondrial membrane is impermeable to acetyl-CoA, so an indirect shuttle transfers acetyl group equivalents across the membrane ([Fig. 21-10](#)). Intramitochondrial acetyl-CoA first reacts with oxaloacetate to form citrate, in the citric acid cycle reaction catalyzed by **citrate synthase** (see [Fig. 16-7](#)). Citrate then passes through the inner membrane on the **citrate transporter**. In the cytosol, citrate cleavage by **citrate lyase** regenerates acetyl-CoA and oxaloacetate in an ATP-dependent reaction. Oxaloacetate cannot return to the mitochondrial matrix directly, as there is no oxaloacetate transporter. Instead, cytosolic malate dehydrogenase reduces the oxaloacetate to malate, which can return to the mitochondrial matrix on the **malate- α -ketoglutarate transporter**, in exchange for citrate. In the matrix, malate is reoxidized to oxaloacetate to complete the shuttle. However, most of the malate produced in the cytosol is used to generate cytosolic NADPH through the activity of malic enzyme ([Fig. 21-8b](#)). The pyruvate produced is transported into the mitochondria by the pyruvate transporter ([Fig. 21-10](#)), then converted back into oxaloacetate by pyruvate carboxylase in the matrix. In the resulting cycle, two ATP molecules are consumed (by citrate lyase and pyruvate carboxylase) for every molecule of acetyl-CoA delivered to fatty acid synthesis.

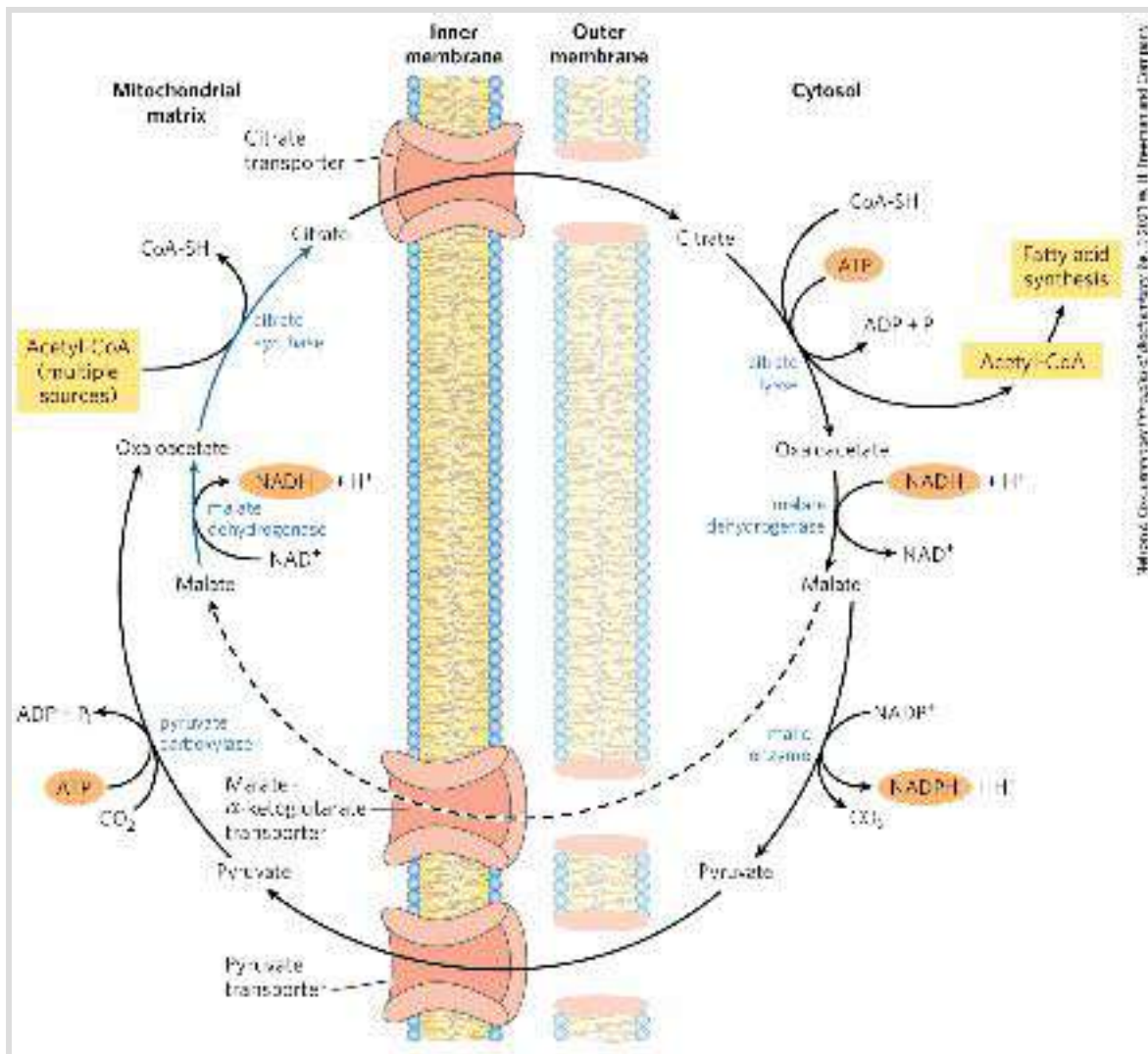


FIGURE 21-10 Shuttle for transfer of acetyl groups from mitochondria to the cytosol.

The outer mitochondrial membrane is freely permeable to all these compounds. Pyruvate derived from amino acid catabolism in the mitochondrial matrix, or from glucose by glycolysis in the cytosol, is converted to acetyl-CoA in the matrix. Acetyl groups pass out of the mitochondrion as citrate; in the cytosol they are delivered as acetyl-CoA for fatty acid synthesis. Oxaloacetate is reduced to malate, which can be returned to the mitochondrial matrix. The steps converting malate to oxaloacetate and oxaloacetate plus acetyl-CoA to citrate (indicated by blue arrows) are part of the citric acid cycle. The major fate of cytosolic malate, however, is oxidation by malic enzyme to generate cytosolic NADPH; the pyruvate produced returns to the mitochondrial matrix.

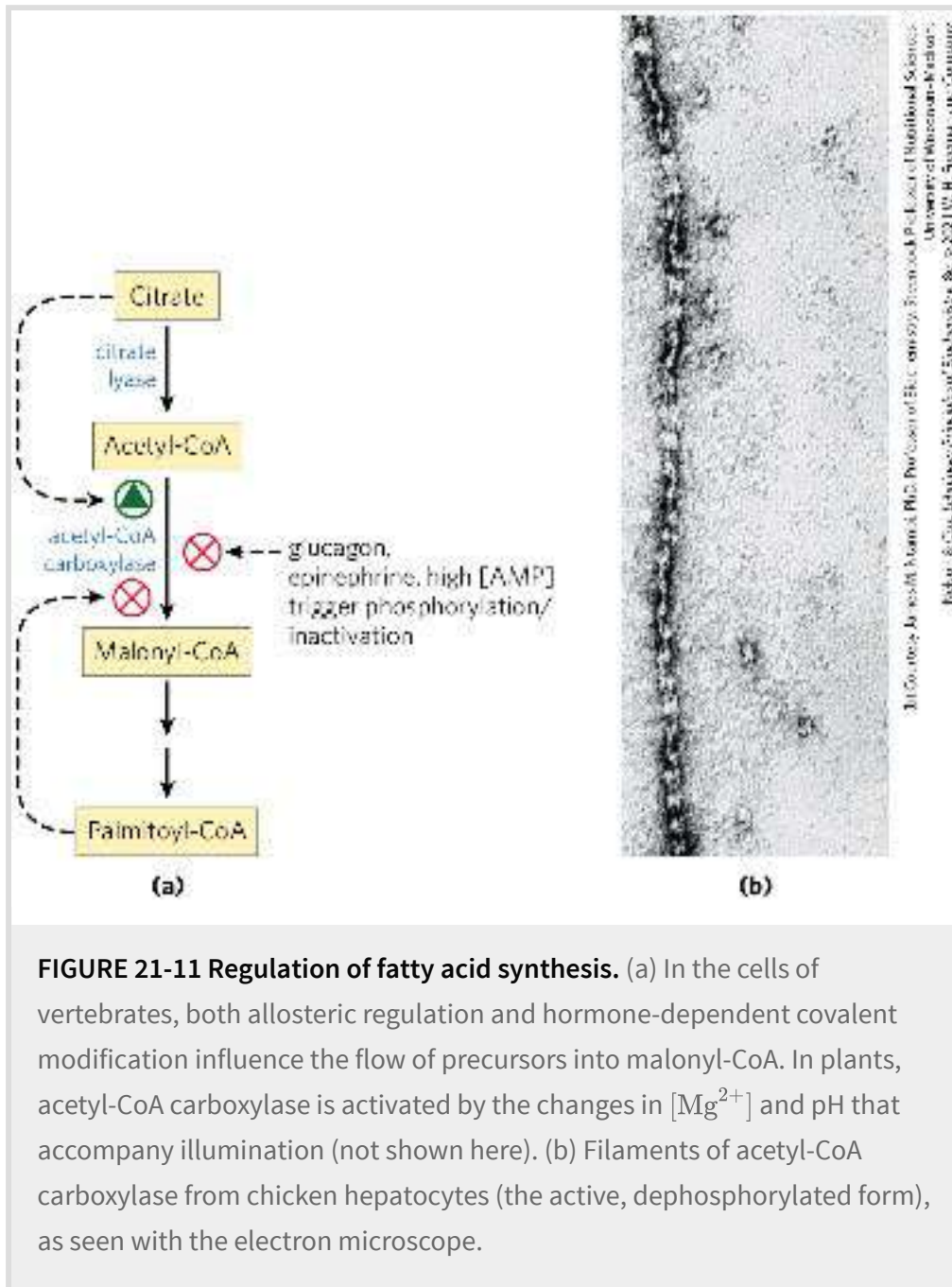
Malate thus has two fates in metabolism. In the mitochondrial matrix, malate is part of the citric acid cycle. In the cytosol,

malate degradation becomes a significant source of NADPH. After citrate cleavage to generate acetyl-CoA, conversion of the four remaining carbons to pyruvate and CO₂ by malic enzyme generates about half the NADPH required for fatty acid synthesis. The pentose phosphate pathway contributes the rest of the needed NADPH.

Fatty Acid Biosynthesis Is Tightly Regulated

P1 When a cell or an organism has more than enough metabolic fuel to meet its energy needs, the excess is generally converted to fatty acids and stored as lipids such as triacylglycerols. **P4** The reaction catalyzed by acetyl-CoA carboxylase is the rate-limiting step in the biosynthesis of fatty acids, and this enzyme is an important site of regulation. In vertebrates, palmitoyl-CoA, the principal product of fatty acid synthesis, is a feedback inhibitor of the enzyme; citrate is an allosteric activator ([Fig. 21-11a](#)), increasing $V_{\max}$. Citrate plays a central role in diverting cellular metabolism from the consumption (oxidation) of metabolic fuel to the storage of fuel as fatty acids. When the concentrations of mitochondrial acetyl-CoA and ATP increase, citrate is transported out of mitochondria; it then becomes both the precursor of cytosolic acetyl-CoA and an allosteric signal for the activation of acetyl-CoA carboxylase. At the same time, citrate inhibits the activity of

phosphofructokinase-1 (see [Fig. 14-23](#)), reducing the flow of carbon through glycolysis.

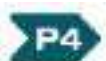


P4 Acetyl-CoA carboxylase is also regulated by covalent modification. Phosphorylation, triggered by the hormones glucagon and epinephrine or by high [AMP], inactivates the

enzyme and reduces its sensitivity to activation by citrate, thereby slowing fatty acid synthesis. Phosphorylation occurs on at least three Ser residues and is catalyzed primarily by the AMP-activated protein kinase (AMPK). In its active (dephosphorylated) form, acetyl-CoA carboxylase polymerizes into long filaments ([Fig. 21-11b](#)); phosphorylation is accompanied by dissociation into monomeric subunits and loss of activity.

The acetyl-CoA carboxylase of plants and bacteria is not regulated by citrate or by a phosphorylation-dephosphorylation cycle.

Instead,  the plant enzyme is activated by an increase in stromal pH and $[\text{Mg}^{2+}]$, which occurs on illumination of the plant (see [Fig. 20-35](#)). Bacteria do not use triacylglycerols as energy stores. In *Escherichia coli*, the primary role of fatty acid synthesis is to provide precursors for membrane lipids; the regulation of this process is complex, employing guanine nucleotides (such as ppGpp; see [Fig. 8-42](#)) that coordinate cell growth with membrane formation.

 In addition to the moment-by-moment regulation of enzymatic activity, these pathways are regulated at the level of gene expression. For example, when animals ingest an excess of certain polyunsaturated fatty acids, the expression of genes encoding many lipogenic enzymes in the liver is suppressed. This gene regulation is mediated by a family of nuclear receptor proteins called PPARs, which we encountered in [Section 17.2](#).



If fatty acid synthesis and β oxidation were to proceed simultaneously, the two processes would constitute a futile cycle, wasting energy. We noted earlier (see [Fig. 17-13](#)) that β oxidation is blocked by malonyl-CoA, which inhibits carnitine acyltransferase I. Thus, during fatty acid synthesis, production of the first intermediate, malonyl-CoA, shuts down β oxidation at the level of a transport system in the inner mitochondrial membrane. This control mechanism illustrates another advantage of segregating synthetic and degradative pathways in different cellular compartments.

Long-Chain Saturated Fatty Acids Are Synthesized from Palmitate

Palmitate, the principal product of the fatty acid synthase system in animal cells, is the precursor of other long-chain fatty acids ([Fig. 21-12](#)). It may be lengthened to form stearate (18:0) or even longer saturated fatty acids by further additions of acetyl groups, through the action of **fatty acid elongation systems** present in the smooth endoplasmic reticulum (smooth ER) and in mitochondria. The more active elongation system of the ER extends the 16-carbon chain of palmitoyl-CoA by two carbons, forming stearyl-CoA. Although different enzyme systems are used, and coenzyme A rather than ACP is the acyl carrier in the reaction, the mechanism of elongation in the ER is otherwise identical to that in palmitate synthesis: donation of two carbons

by malonyl-CoA, followed by reduction, dehydration, and reduction to the saturated 18-carbon product, stearoyl-CoA.

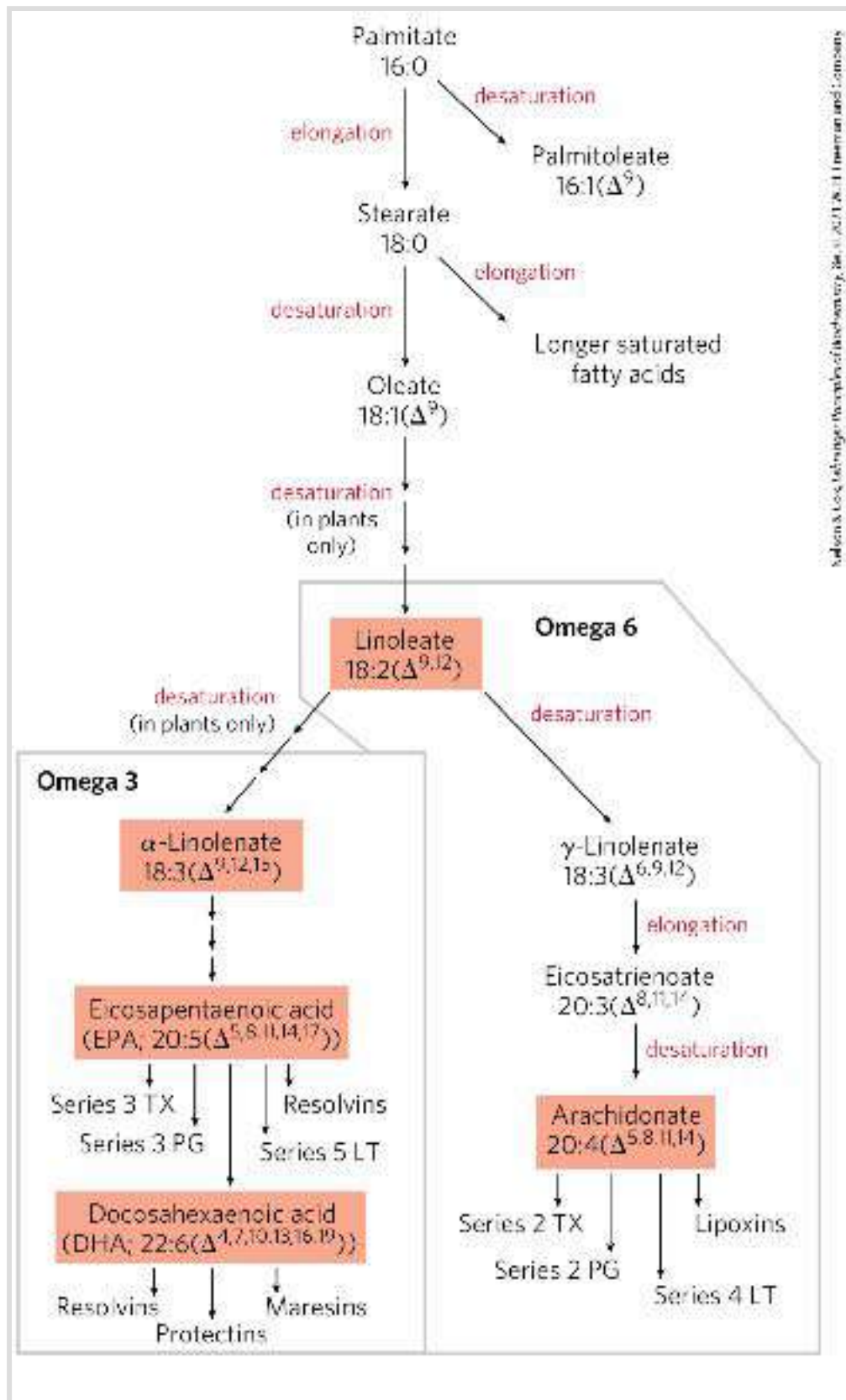


FIGURE 21-12 Routes of synthesis of unsaturated fatty acids and their derivatives. Palmitate is the precursor of stearate and longer-chain saturated fatty acids, as well as the monounsaturated acids palmitoleate and oleate. Mammals cannot convert oleate to linoleate or α -linolenate (shaded), which are therefore required in the diet as essential fatty acids. Conversion of linoleate to other polyunsaturated fatty acids and eicosanoids is outlined. Unsaturated fatty acids are symbolized by indicating the number of carbons and the number and position of double bonds, as in [Table 10-1](#). Linoleate and α -linolenate are important omega-6 and omega-3 fatty acids, respectively; they are also precursors for a wide range of unsaturated fatty acids that act as signaling molecules. Two-letter abbreviations specify the eicosanoid prostaglandins (PG), thromboxanes (TX), and leukotrienes (LT). Particular classes of unsaturated fatty acids are further delineated by the number of double bonds, which defines subclasses referred to as series. For example, series 2 TX are thromboxanes with two double bonds in the hydrocarbon chain.

Two key products of elongation pathways are linoleate, an omega-6 fatty acid (see [Chapter 10](#) for the alternative nomenclature), and α -linolenate, an omega-3 fatty acid. These are precursors for two extensive families of derivative unsaturated fatty acids, the omega-6 and omega-3 families. Humans cannot synthesize linoleate and α -linolenate and must obtain them in the diet. The ratio of omega-6 to omega-3 fatty acids in the diet, if too high, can lead to cardiovascular disease. The importance of this ratio may reflect the multitude of signaling molecules in the omega-6 and omega-3 families ([Fig. 21-12](#)), with their equally complex physiological effects. Several of these derivative unsaturated fatty acids are considered below.

Desaturation of Fatty Acids Requires a Mixed-Function Oxidase

Palmitate and stearate serve as precursors of the two most common monounsaturated fatty acids of animal tissues: palmitoleate, 16:1(Δ^9), and oleate, 18:1(Δ^9); both of these fatty acids have a single *cis* double bond between C-9 and C-10 (see [Table 10-1](#)). The double bond is introduced into the fatty acid chain by an oxidative reaction catalyzed by **fatty acyl-CoA desaturase** ([Fig. 21-13](#)), a **mixed-function oxidase** ([Box 21-1](#)). Two different substrates, the fatty acid and NADPH, simultaneously undergo two-electron oxidations. The path of electron flow includes a cytochrome (cytochrome b_5) and a flavoprotein (cytochrome b_5 reductase), both of which, like fatty acyl-CoA desaturase, are in the smooth ER. In plants, oleate is produced by a **stearoyl-ACP desaturase (SCD)** that uses reduced ferredoxin as the electron donor in the chloroplast stroma.

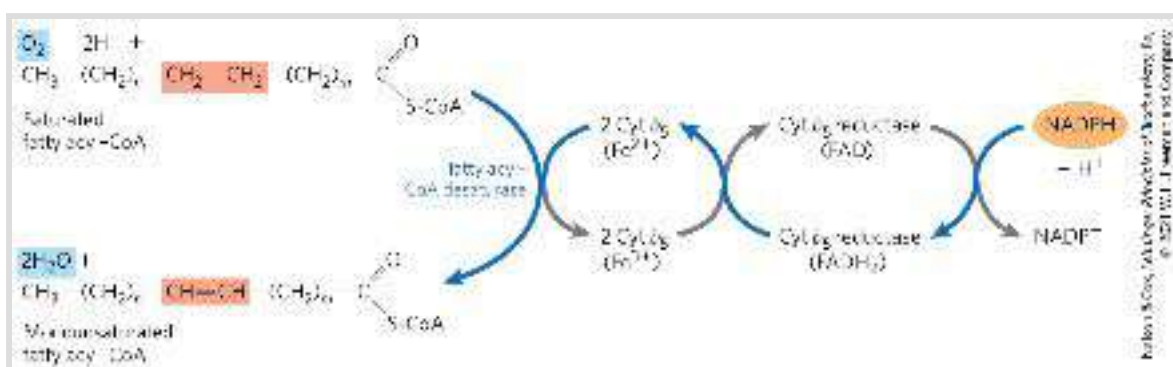


FIGURE 21-13 Electron transfer in the desaturation of fatty acids in vertebrates. Blue arrows show the path of electrons as two substrates — a fatty acyl-CoA and NADPH — undergo oxidation by molecular oxygen. These reactions take place on the luminal face of the smooth ER. A similar pathway, but with different electron carriers, occurs in plants.

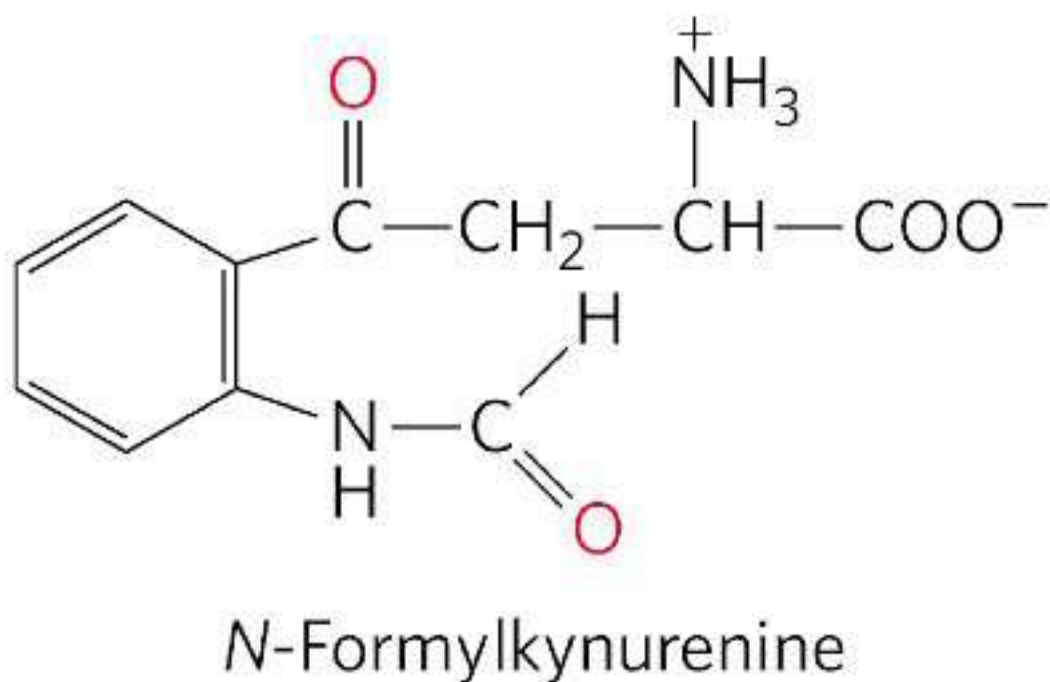
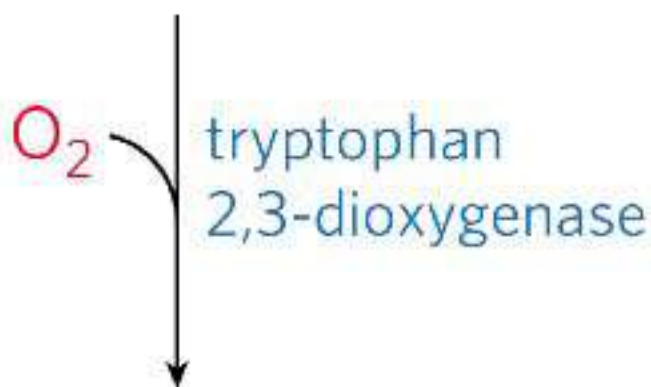
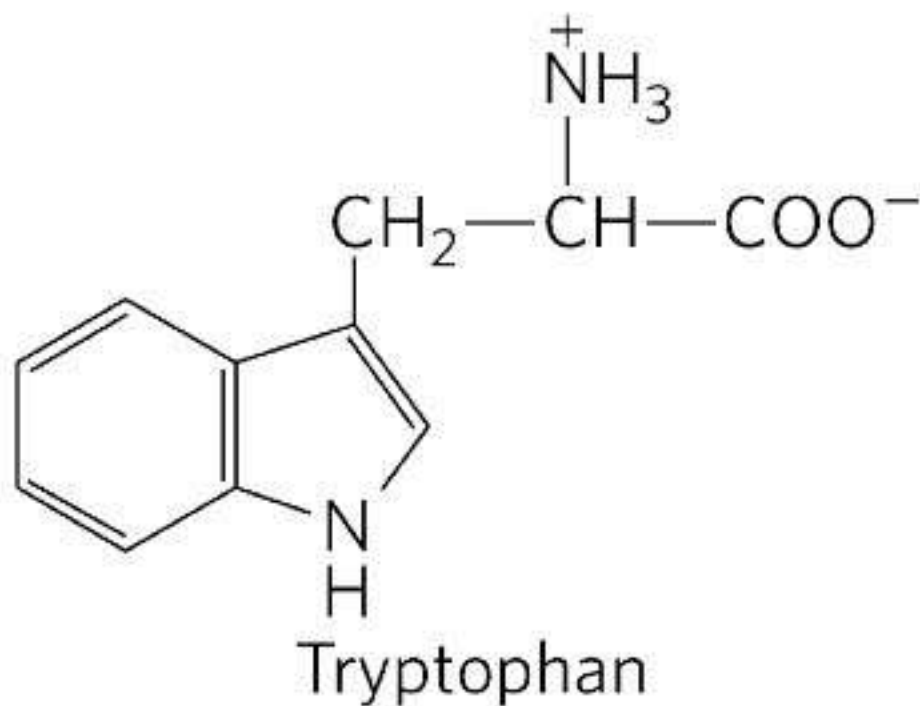
BOX 21-1 MEDICINE

Oxidases, Oxygenases, Cytochrome P-450 Enzymes, and Drug Overdoses

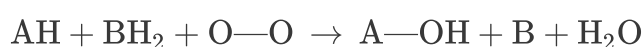
In this chapter we encounter several enzymes that carry out oxidation-reduction reactions in which molecular oxygen is a participant. The stearoyl-CoA desaturase (SCD) that introduces a double bond into a fatty acyl chain (see [Fig. 21-13](#)) is one such enzyme.

The nomenclature for enzymes that catalyze reactions of this general type can be confusing. **Oxidase** is the general name for enzymes that catalyze oxidations in which molecular oxygen is the electron acceptor but oxygen atoms do not appear in the oxidized product. The enzyme that creates a double bond in fatty acyl-CoA during the oxidation of fatty acids in peroxisomes (see [Fig. 17-14](#)) is an oxidase of this type; a second example is the cytochrome oxidase of the mitochondrial respiratory chain (see [Fig. 19-13](#)). In the first case, the transfer of two electrons to H_2O produces hydrogen peroxide, H_2O_2 ; in the second, two electrons reduce $\frac{1}{2}\text{O}_2$ to H_2O . Many, but not all, oxidases are flavoproteins. **Mixed-function oxidases** oxidize two different substrates simultaneously; again, the molecular oxygen atoms do not appear in the oxidized products. Mixed-function oxidases act in fatty acid desaturation (fatty acyl-CoA desaturase; see [Fig. 21-13](#)) and in the last step of plasmalogen synthesis (see [Fig. 21-30](#)).

Oxygenases catalyze oxidative reactions in which oxygen atoms *are* directly incorporated into the product molecule, forming a new hydroxyl or carboxyl group, for example. **Dioxygenases** catalyze reactions in which both oxygen atoms of O_2 are incorporated into the organic product. An example of a dioxygenase is tryptophan 2,3-dioxygenase, which catalyzes the opening of the five-membered ring of tryptophan in the catabolism of this amino acid. When the reaction takes place in the presence of $^{18}\text{O}_2$, the isotopic oxygen atoms are found in the two carbonyl groups of the product (shown in red):



Monooxygenases, more common and more complex in their action, catalyze reactions in which only one of the two oxygen atoms of O₂ is incorporated into the organic product, the other being reduced to H₂O; an example is squalene monooxygenase (see [Fig. 21-37](#)). Monooxygenases require two substrates to serve as reductants of the two oxygen atoms of O₂. The main substrate accepts one of the two oxygen atoms, and a cosubstrate furnishes hydrogen atoms to reduce the other oxygen atom to H₂O. The general reaction equation for monooxygenases is



where AH is the main substrate and BH₂ is the cosubstrate. Because most monooxygenases catalyze reactions in which the main substrate becomes hydroxylated, they are also called **hydroxylases**. They are also sometimes called **[mixed-function oxygenases](#)** to indicate that they oxidize two different substrates simultaneously.

Monooxygenases are divided into several classes, depending on the nature of the cosubstrate. Some use reduced flavin nucleotides (FMNH₂ or FADH₂), others use NADH or NADPH, and still others use α -ketoglutarate as cosubstrate. The enzyme that hydroxylates the phenyl ring of phenylalanine to form tyrosine is a monooxygenase that uses tetrahydrobiopterin as cosubstrate (see [Fig. 18-23](#)). (This is the enzyme that is defective in the human genetic disease phenylketonuria.)

The most numerous and most complex monooxygenation reactions are those employing a type of heme protein called **[cytochrome P-450](#)**. Like mitochondrial cytochrome oxidase, enzymes containing a cytochrome P-450 domain can react with O₂ and bind carbon monoxide, but they can be differentiated from cytochrome oxidase because the carbon monoxide complex of their reduced form absorbs light strongly at 450 nm — thus the name P-450.

Cytochrome P-450 enzymes catalyze hydroxylation reactions in which an organic substrate, RH, is hydroxylated to R—OH, incorporating one oxygen atom of O₂; the other oxygen atom is reduced to H₂O by reducing equivalents that are furnished by NADH or NADPH but are usually passed to cytochrome P-450 by an iron-sulfur protein. [Figure 1](#) shows a simplified outline of the action of cytochrome P-450.

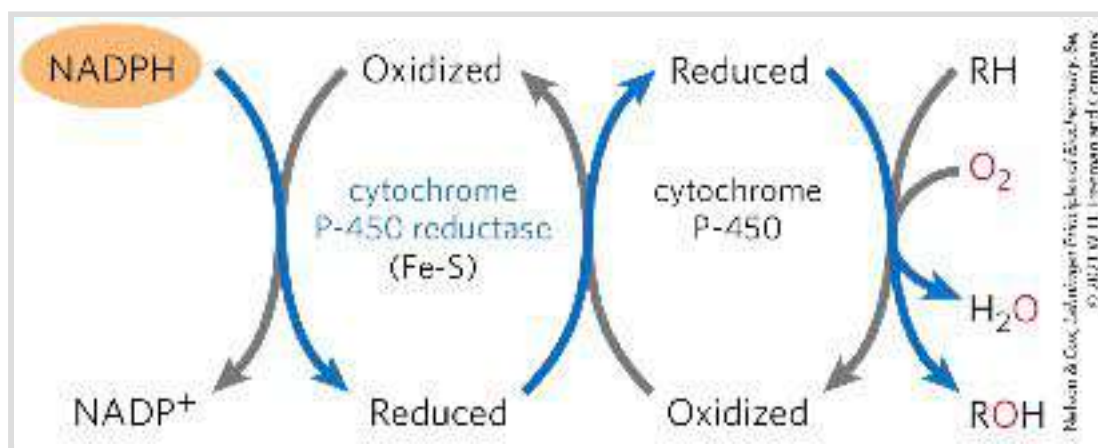
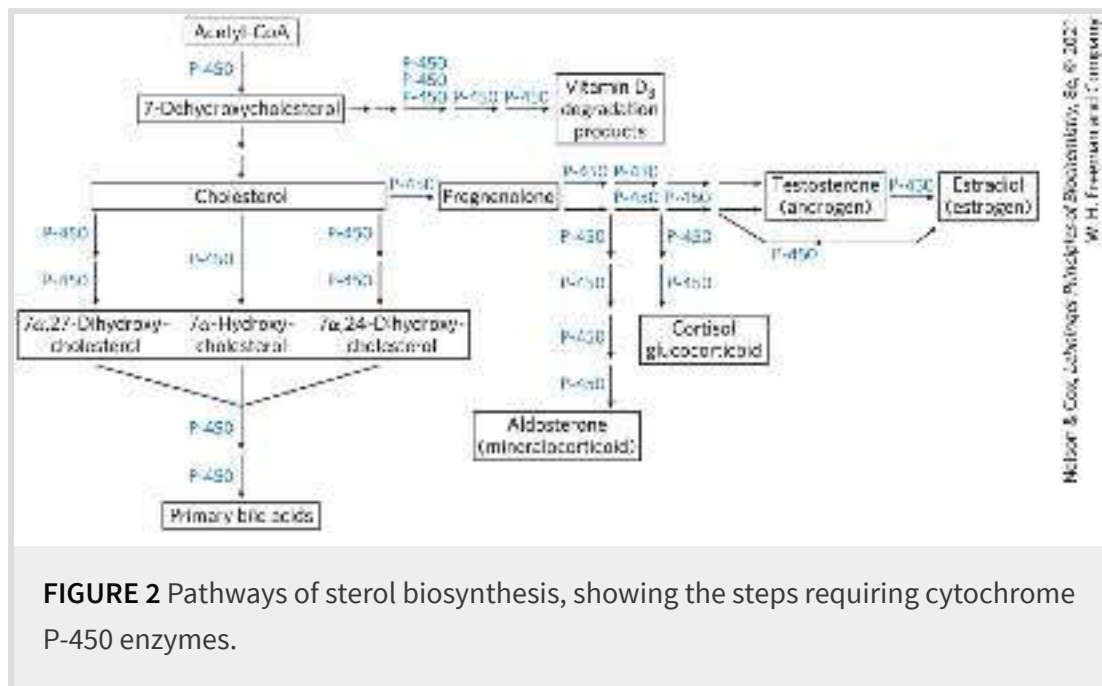


FIGURE 1 Simplified cytochrome P-450 reaction cycle.

One large family of P-450-containing proteins consists of two general types: those highly specific for a single substrate (like typical enzymes) and those with more promiscuous binding sites that accept a variety of substrates, generally similar in being hydrophobic. In the adrenal cortex, for example, a specific cytochrome P-450 participates in the hydroxylation of steroids to yield the adrenocortical hormones (see [Fig. 21-49](#)). There are dozens of P-450 enzymes that act on specific substrates in the biosynthetic pathways to steroid hormones and eicosanoids ([Fig. 2](#)). Cytochrome P-450 enzymes with broader specificity are important in the hydroxylation of many different drugs, such as barbiturates and other xenobiotics (substances foreign to the organism), particularly if they are hydrophobic and relatively insoluble. The environmental carcinogen benzo[*a*]pyrene, found in cigarette smoke, undergoes cytochrome P-450-dependent hydroxylation during detoxification. Hydroxylation of xenobiotics, sometimes combined with the attachment of a polar compound such as glucuronic acid to the hydroxyl group, makes them more soluble in water and allows their excretion in urine. Hydroxylation (and glucuronidation)

inactivates most drugs, and the rate at which it occurs can determine how long a given dose of a medication remains in the blood at therapeutic levels.



Humans differ in their levels of drug-metabolizing enzymes, both because of their genetics and because past exposure to substrates can induce the synthesis of higher levels of P-450 enzymes. Ethanol and barbiturate drugs share a P-450 enzyme. Long-term heavy drinking induces synthesis of this enzyme. Then, because the barbiturate is inactivated and cleared faster, larger doses are required to get the same therapeutic effect. If an individual takes this larger-than-usual dose of barbiturate and then also drinks alcohol, competition between the alcohol and the barbiturate for the limited amount of enzyme means that both alcohol and barbiturate are cleared more slowly. The resulting high levels of these two central nervous system depressants can be lethal. Similar complications arise when an individual takes two drugs that happen to be inactivated by the same P-450 enzyme; each drug increases the effective dose of the other by slowing its inactivation. It is therefore essential for physicians and pharmacists to know about all of a patient's prescribed and over-the-counter drugs and supplements, as well as a history of heavy drinking, or smoking, or exposure to environmental toxins.



The SCD of animals (as studied in mice) has an important role in the development of obesity and the insulin resistance that often accompanies obesity and precedes development of type 2 diabetes mellitus. Mice have four isozymes, SCD1 through SCD4, of which SCD1 is the best understood. Its synthesis is induced by dietary saturated fatty acids, and also by the action of SREBP and LXR, two protein regulators of lipid metabolism that activate transcription of lipid-synthesizing enzymes (described in [Section 21.4](#)). Mice with mutant forms of SCD1 are resistant to diet-induced obesity and do not develop diabetes under conditions that cause both obesity and diabetes in mice with normal SCD1. ■

Mammalian hepatocytes can readily introduce double bonds at the Δ^9 position of fatty acids but cannot introduce additional double bonds between C-10 and the methyl-terminal end. Thus, as noted above, mammals cannot synthesize the omega-6 family precursor linoleate, $18:2(\Delta^{9,12})$, or the omega-3 family precursor α -linolenate, $18:3(\Delta^{9,12,15})$. Plants, however, can synthesize both; the desaturases that introduce double bonds at the Δ^{12} and Δ^{15} positions are located in the ER and in chloroplasts. The ER enzymes act not on free fatty acids but on a phospholipid, phosphatidylcholine, that contains at least one oleate linked to the glycerol ([Fig. 21-14](#)). Both plants and bacteria must synthesize polyunsaturated fatty acids to ensure membrane fluidity at reduced temperatures.

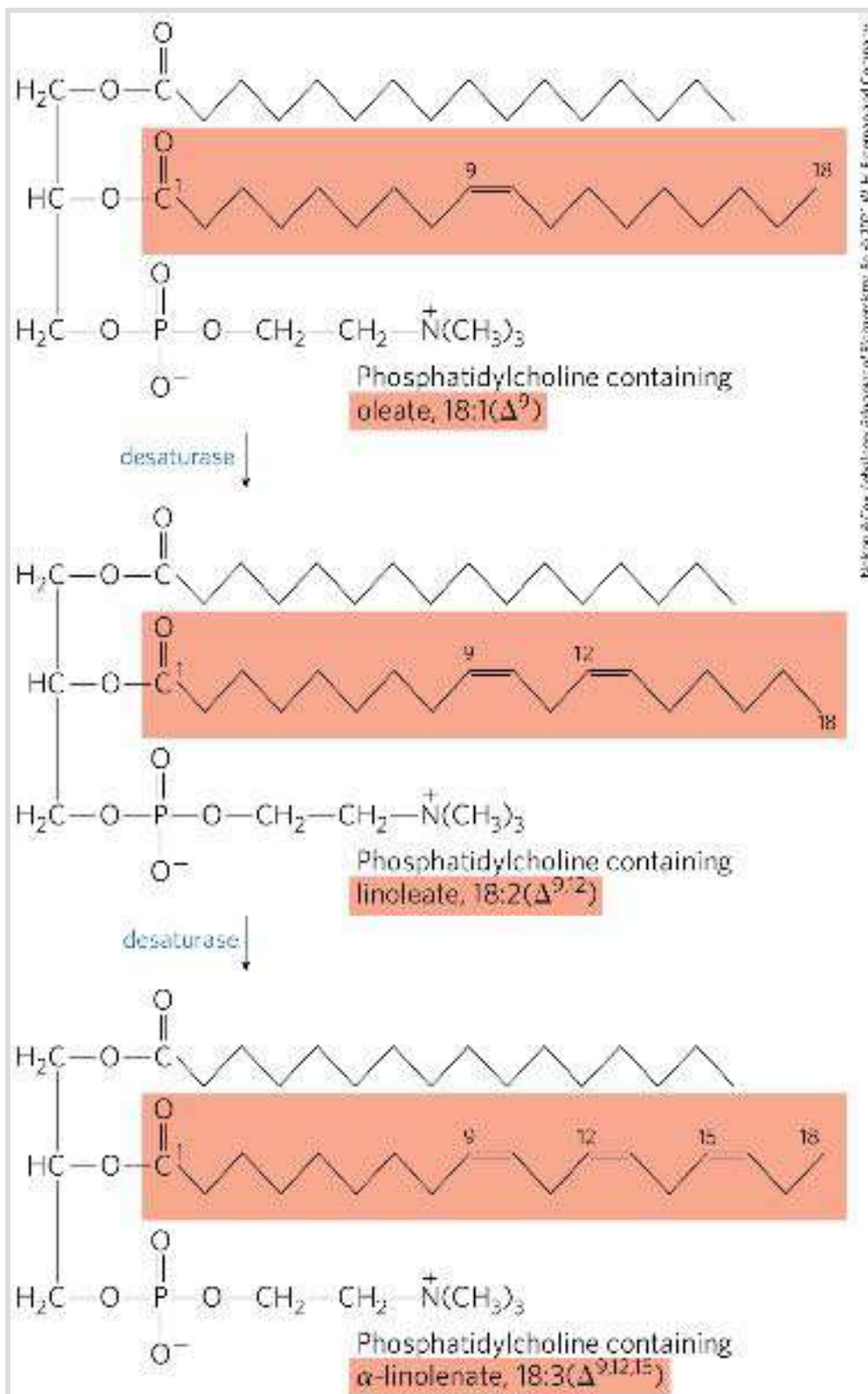


FIGURE 21-14 Action of plant desaturases. Desaturases in plants oxidize phosphatidylcholine-bound oleate to polyunsaturated fatty acids. Some of the products are released from the phosphatidylcholine by hydrolysis.

Because they are necessary precursors for the synthesis of other products, linoleate and α -linolenate are **essential fatty acids** for mammals; they must be obtained from dietary plant material. Once ingested, linoleate may be converted to certain other polyunsaturated acids, particularly γ -linolenate, eicosatrienoate, and **arachidonate (eicosatetraenoate)**, all of which can be made only from linoleate ([Fig. 21-12](#)). Similarly, α -linolenate is converted to two important derivatives, **eicosapentaenoic acid (EPA)** and **docosa-hexaenoic acid (DHA)**. Arachidonate, 20:4($\Delta^{5,8,11,14}$), EPA, 20:5($\Delta^{5,8,11,14,17}$), and DHA, 22:6($\Delta^{4,7,10,13,16,19}$), are essential precursors of distinct classes of eicosanoids, lipids with important regulatory functions. The 20- and 22-carbon fatty acids are synthesized from linoleate and α -linolenate by fatty acid elongation reactions analogous to those described on [page 753](#).

Eicosanoids Are Formed from 20- and 22-Carbon Polyunsaturated Fatty Acids



Eicosanoids are a family of very potent biological signaling molecules that act as short-range messengers, affecting tissues near the cells that produce them.

In response to hormonal or other stimuli, phospholipase A₂, present in most types of mammalian cells, attacks membrane phospholipids, releasing arachidonate from the middle carbon of glycerol. Enzymes of the smooth ER then convert arachidonate to

prostaglandins, beginning with the formation of prostaglandin H_2 (PGH_2), the immediate precursor of many other prostaglandins and thromboxanes (**Fig. 21-15a**). The two reactions that lead to PGH_2 are catalyzed by a bifunctional enzyme, **cyclooxygenase (COX)**, also called H_2 **prostaglandin synthase**. In the first step, the cyclooxygenase activity introduces molecular oxygen to convert arachidonate to PGG_2 . The second step, catalyzed by the peroxidase activity of COX, converts PGG_2 to PGH_2 .

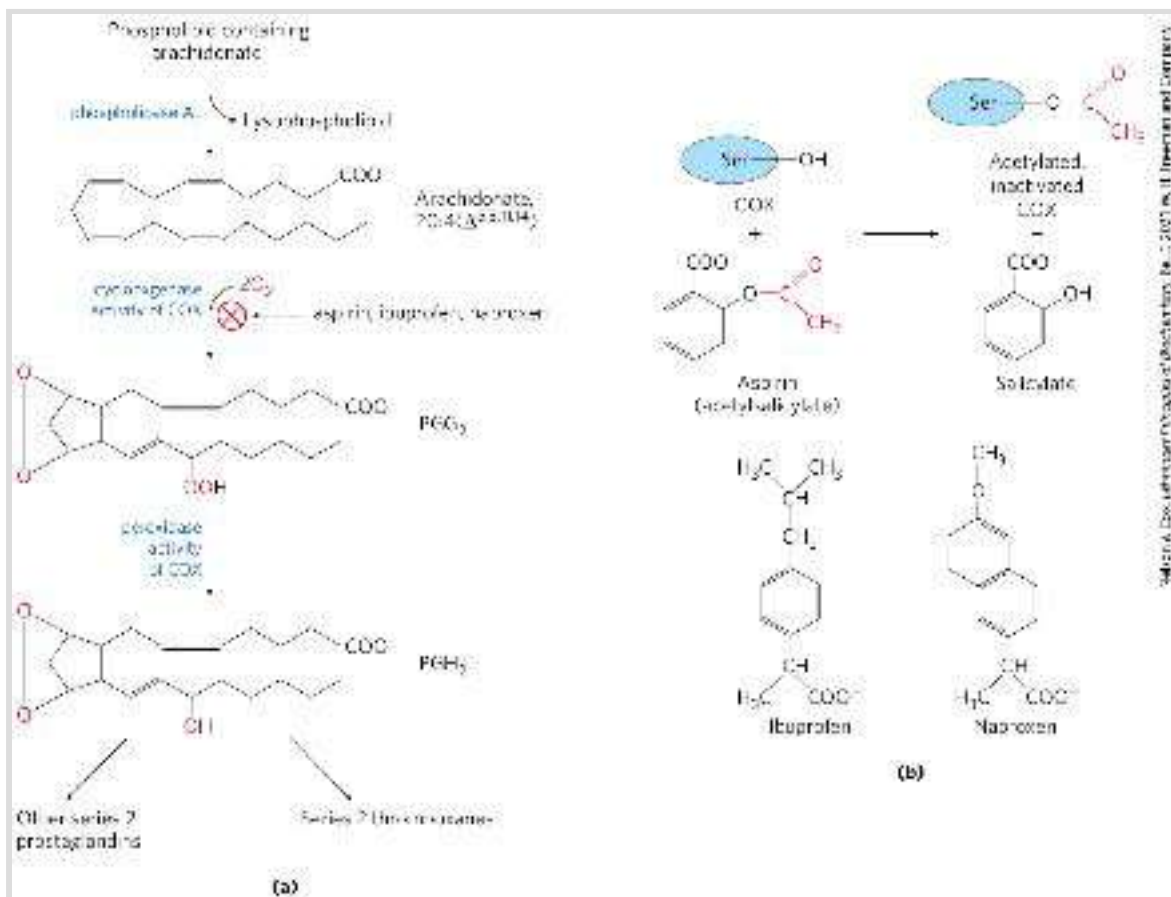


FIGURE 21-15 The “cyclic” pathway from arachidonate to prostaglandins and thromboxanes. (a) After arachidonate is released from phospholipids by the action of phospholipase A₂, the cyclooxygenase and peroxidase activities of COX (also called prostaglandin H_2 synthase) catalyze the production of PGH_2 , the precursor of other prostaglandins and of thromboxanes. (b) Aspirin inhibits the first reaction by acetylating

an essential Ser residue on the enzyme. Ibuprofen and naproxen inhibit the same step, probably by mimicking the structure of the substrate or an intermediate in the reaction.

KEY CONVENTION

Prostaglandins with different functional groups on the ring are given different letter designations: A, B, C, D, E, F, G, H, and R. The subscript number following the letter, as in PGH_2 and PGG_2 , indicates the number of double bonds. Prostaglandins with two double bonds, all of which are derived from arachidonate, are referred to as series 2 prostaglandins; those with three double bonds, derived from EPA, as series 3 ([Fig. 21-12](#)). Similar naming patterns are used for other classes of eicosanoids described below. ■

Series 2 prostaglandins have important roles in the immediate response to stress or injury, including inflammation, pain, swelling, and dilation of blood vessels. Series 3 prostaglandins, in general, act more slowly and usually moderate the responses associated with series 2 prostaglandins.

Mammals have two isozymes of prostaglandin H_2 synthase, COX-1 and COX-2. These have different functions but closely similar amino acid sequences (60% to 65% sequence identity) and similar reaction mechanisms at both of their catalytic centers. COX-1 is responsible for synthesis of the prostaglandins that regulate the

secretion of gastric mucin, and COX-2 is responsible for synthesis of the prostaglandins that mediate inflammation, pain, and fever.



Pain can be relieved by inhibiting COX-2. The first drug widely marketed for this purpose was aspirin (acetylsalicylate; [Fig. 21-15b](#)). The name “aspirin” (from *a* for acetyl and *spir* for *Spirsa re*, the German word for the salicylates prepared from the plant *Spiraea ulmaria*) appeared in 1899 when the drug was introduced by the Bayer company. Aspirin irreversibly inactivates the cyclooxygenase activity of both COX isozymes, by acetylating a Ser residue and blocking each enzyme’s active site. The synthesis of prostaglandins and thromboxanes is thereby inhibited. Additional widely used *nonsteroidal anti-inflammatory drugs* (NSAIDs; [Fig. 21-15b](#)), ibuprofen and naproxen, inhibit the same pair of enzymes. However, the inhibition of COX-1 can result in undesired side effects, including stomach irritation and more serious conditions. In the 1990s, NSAID compounds that had a greater specificity for COX-2 were developed as advanced therapies for severe pain. Three of these drugs were approved for use worldwide: rofecoxib (Vioxx), valdecoxib (Bextra), and celecoxib (Celebrex). Though initially considered a success, Vioxx and Bextra were withdrawn as field reports and clinical studies connected the drugs with an increased risk of heart attack and stroke. Celebrex is still on the market but is being used with increased caution. The detailed reasons for the problems with these drugs are still not clear but serve as a cautionary note. We are increasingly aware of the complexity of the web of these signaling interactions, and predicting the consequences of

targeting specific components with pharmaceutical agents remains an imperfect process.

Thromboxane synthase, present in blood platelets (thrombocytes), converts PGH_2 to thromboxane A_2 , from which other series 2 **thromboxanes** are derived ([Fig. 21-15a](#)). The series 2 thromboxanes induce constriction of blood vessels and platelet aggregation, early steps in blood clotting. Low doses of aspirin, taken regularly, reduce the probability of heart attacks and strokes by reducing thromboxane production. ■

Thromboxanes, like prostaglandins, contain a ring of five or six atoms; the pathway from arachidonate to the series 2 prostaglandins and thromboxanes is sometimes called the “cyclic” pathway, to distinguish it from the “linear” pathway that leads from arachidonate to the **leukotrienes**, which are linear compounds ([Fig. 21-16](#)). Leukotriene synthesis begins with the action of several lipoxygenases that catalyze the incorporation of molecular oxygen into arachidonate. These enzymes, found in leukocytes and in heart, brain, lung, and spleen, are mixed-function oxidases of the cytochrome P-450 family (see [Box 21-1](#)). The various leukotrienes differ in the position of the peroxide group introduced by the lipoxygenases. The linear pathway from arachidonate, unlike the cyclic pathway, is not inhibited by aspirin or other NSAIDs.

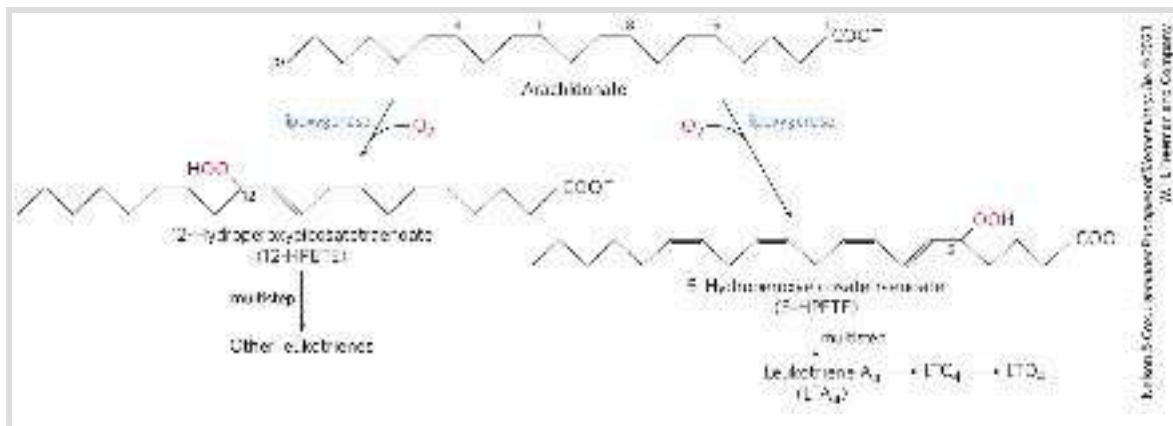


FIGURE 21-16 The “linear” pathway from arachidonate to leukotrienes.



Pathogenic organisms, as well as irritants such as air pollution and tobacco smoke, trigger an inflammatory response in the affected tissue, which consists of two phases: initiation and resolution. Eicosanoids of the omega-6 family are critical to initiation — playing key roles in recruiting leukocytes, making blood vessels more permeable, and stimulating chemotaxis and migration of immune system cells. As the source of tissue damage is brought under control, the inflammation must be resolved and the tissue brought back to its normal state. Resolution of inflammation is called **catabasis**, and it is promoted by several classes of signaling molecules; prominent among these are several leukotrienes and prostaglandins. Many eicosanoids of the omega-3 family (including series 3 prostaglandins and thromboxanes) are anti-inflammatory, although the classification is not absolute; individual eicosanoids can be inflammatory in one tissue and anti-inflammatory in another.

Catabasis is also promoted by a set of eicosanoids termed **specialized pro-resolving mediators (SPMs)**. The first family of

SPMs to be discovered was the [lipoxins](#), followed more recently by resolvins, protectins, and maresins. All SPMs are derived from essential fatty acids ([Fig. 21-12](#)). They affect different target cells and tissues in different ways. The sum of their action is to promote removal of debris, microbes, and dead cells, to restore blood vessel integrity, and to regenerate tissue. Particular SPMs also reduce pain and fever, and play roles in resolving the tissue inflammation leading to diabetes, obesity, and asthma. Further research on SPMs thus has potential for the development of new pharmaceutical targets. ■

Plants also derive important signaling molecules from fatty acids. As in animals, a key step in the initiation of signaling is activation of a specific phospholipase. In plants, the fatty acid substrate released by phospholipase action is α -linolenate. A lipoxygenase then catalyzes the first step in a pathway that converts α -linolenate to jasmonate, a substance known to have signaling roles in defense against insects, resistance to fungal pathogens, and maturation of pollen. Jasmonate also affects seed germination, root growth, and fruit and seed development.

SUMMARY 21.1 *Biosynthesis of Fatty Acids and Eicosanoids*

- Malonyl-CoA, a key precursor of fatty acids, is synthesized by the action of acetyl-CoA carboxylase.
- Beginning with malonyl-CoA and acetyl-CoA, fatty acids are synthesized in a repeating cycle of four steps.

■ Long-chain saturated fatty acids are synthesized from acetyl-CoA by a cytosolic system of six enzymatic activities plus acyl carrier protein (ACP). There are two types of fatty acid synthase. FAS I, found in vertebrates and fungi, consists of multifunctional polypeptides. FAS II is a dissociated system found in bacteria and plants. Both contain two types of OSH groups (one furnished by the phosphopantetheine of ACP, the other by a Cys residue of β -ketoacyl-ACP synthase) that function as carriers of the fatty acyl intermediates.

■ Malonyl-ACP, formed from acetyl-CoA (shuttled out of mitochondria) and CO_2 , condenses with an acetyl bound to the Cys OSH to yield acetoacetyl-ACP, with release of CO_2 . This is followed by reduction to the D- β -hydroxy derivative, dehydration to the *trans*- Δ^2 -unsaturated acyl-ACP, and reduction to butyryl-ACP. NADPH is the electron donor for both reductions. Fatty acid synthesis is regulated at the level of malonyl-CoA formation.

■ Six more molecules of malonyl-ACP react successively at the carboxyl end of the growing fatty acid chain to form palmitoyl-ACP — the end product of the fatty acid synthase reaction. Free palmitate is released by hydrolysis.

■ Fatty acid synthesis occurs in the cytosol of animal cells, and in chloroplasts in plants.

■ Acetate is exported from the mitochondria as citrate.

■ Fatty acid synthesis is tightly regulated, principally by regulation of acetyl-CoA carboxylase.

■ Palmitate may be elongated to the 18-carbon stearate.

■ Palmitate and stearate can be desaturated to yield palmitoleate and oleate, respectively, by the action of mixed-function oxidases.

■ Mammals cannot make linoleate and must obtain it from plant sources; they convert exogenous linoleate to arachidonate, the

parent compound of eicosanoids (prostaglandins, thromboxanes, leukotrienes, and specialized pro-resolving mediators), a family of very potent signaling molecules. The synthesis of prostaglandins and thromboxanes is inhibited by NSAIDs that act on the cyclooxygenase activity of prostaglandin H₂ synthase.

21.2 Biosynthesis of Triacylglycerols



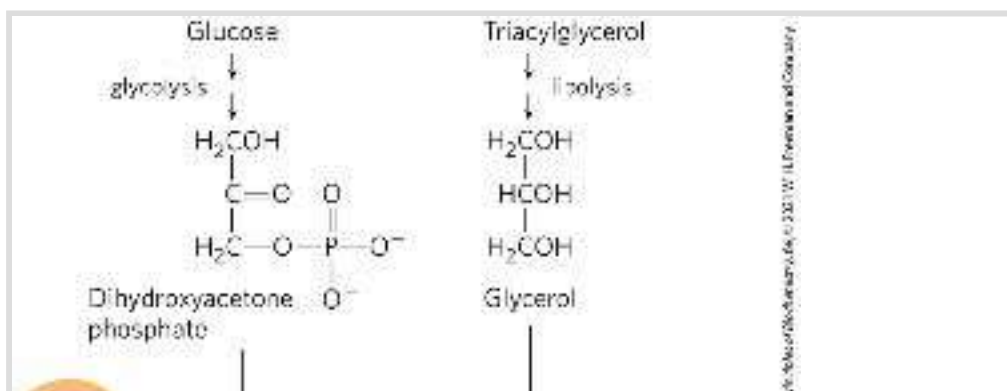
Most of the fatty acids synthesized or ingested by an organism have one of two fates, depending on the organism's needs: incorporation into triacylglycerols for the storage of metabolic energy or incorporation into the phospholipid components of membranes. During rapid growth, synthesis of new membranes requires the production of membrane phospholipids; when an organism has a plentiful food supply but is not actively growing, it shunts most of its fatty acids into storage fats. Both pathways begin at the same point: the formation of fatty acyl esters of glycerol. In this section we examine the route to triacylglycerols and its regulation, and the production of glycerol 3-phosphate in the process of glyceroneogenesis.

Triacylglycerols and Glycerophospholipids Are Synthesized from the Same Precursors

Animals can synthesize and store large quantities of triacylglycerols, to be used later as fuel (see [Box 17-1](#)). Humans can store only a few hundred grams of glycogen in liver and muscle, barely enough to supply the body's energy needs for 12

hours. However, a 70 kg human stores about 15 kg of triacylglycerol in its tissues, enough to support basal energy needs for as long as 12 weeks (see [Table 23-5](#)). Triacylglycerols have the highest energy content of all stored nutrients — more than 38 kJ/g. Whenever carbohydrate is ingested in excess of the organism's capacity to store glycogen, the excess is converted to triacylglycerols and stored in adipose tissue. Plants also manufacture triacylglycerols as an energy-rich fuel, mainly stored in fruits, nuts, and seeds.

In animal tissues, triacylglycerols and glycerophospholipids such as phosphatidylethanolamine share two precursors, fatty acyl-CoA and L-glycerol 3-phosphate, and several biosynthetic steps. The vast majority of the glycerol 3-phosphate is derived from the glycolytic intermediate dihydroxyacetone phosphate (DHAP) by the action of the cytosolic NAD-linked **glycerol 3-phosphate dehydrogenase**; in liver and kidney, a small amount of glycerol 3-phosphate is also formed from glycerol by the action of **glycerol kinase** ([Fig. 21-17](#)). The other precursors of triacylglycerols are fatty acyl-CoAs, formed from fatty acids by **acyl-CoA synthetases**, the same enzymes responsible for the activation of fatty acids for β oxidation (see [Fig. 17-5](#)).



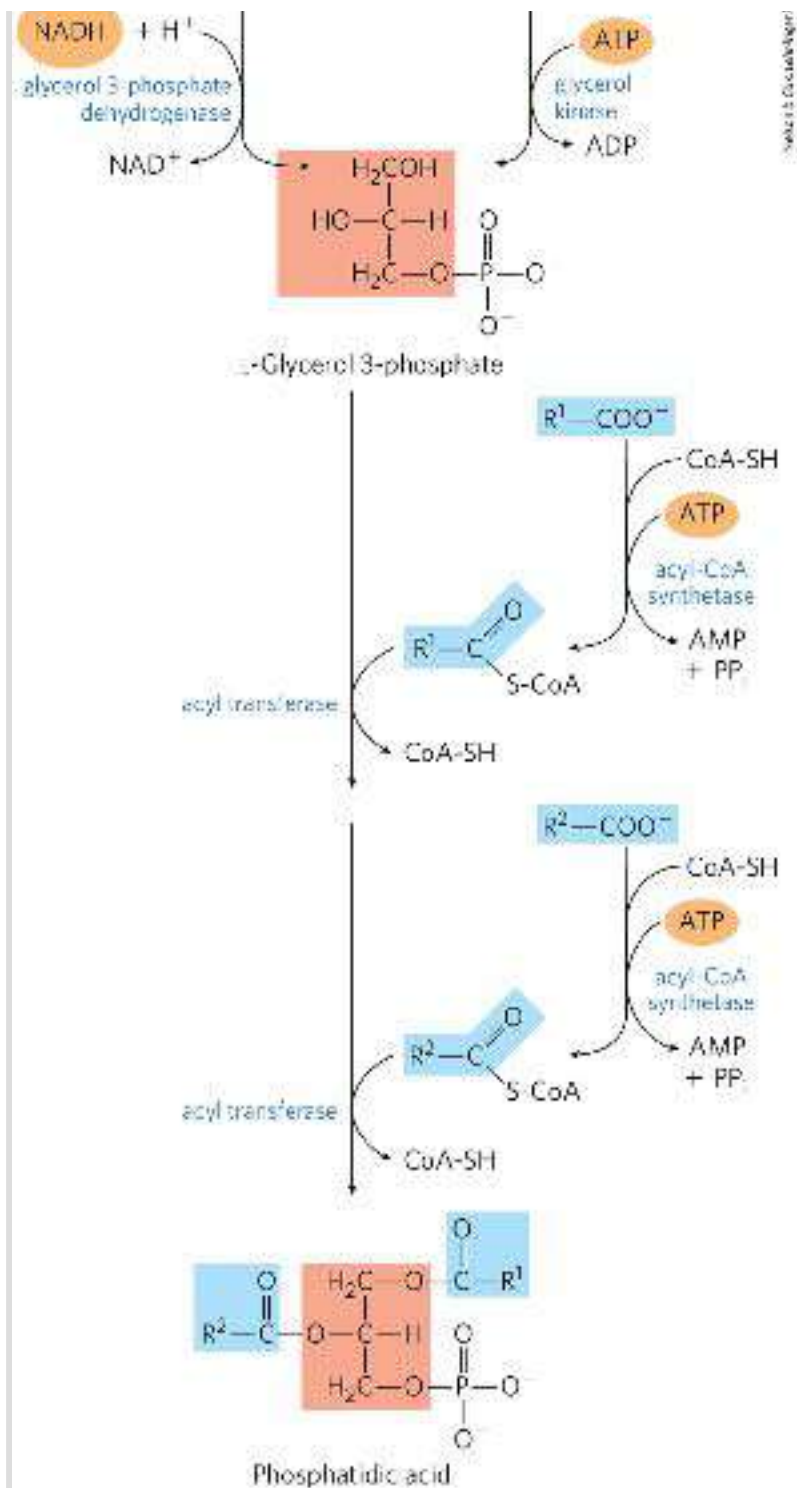
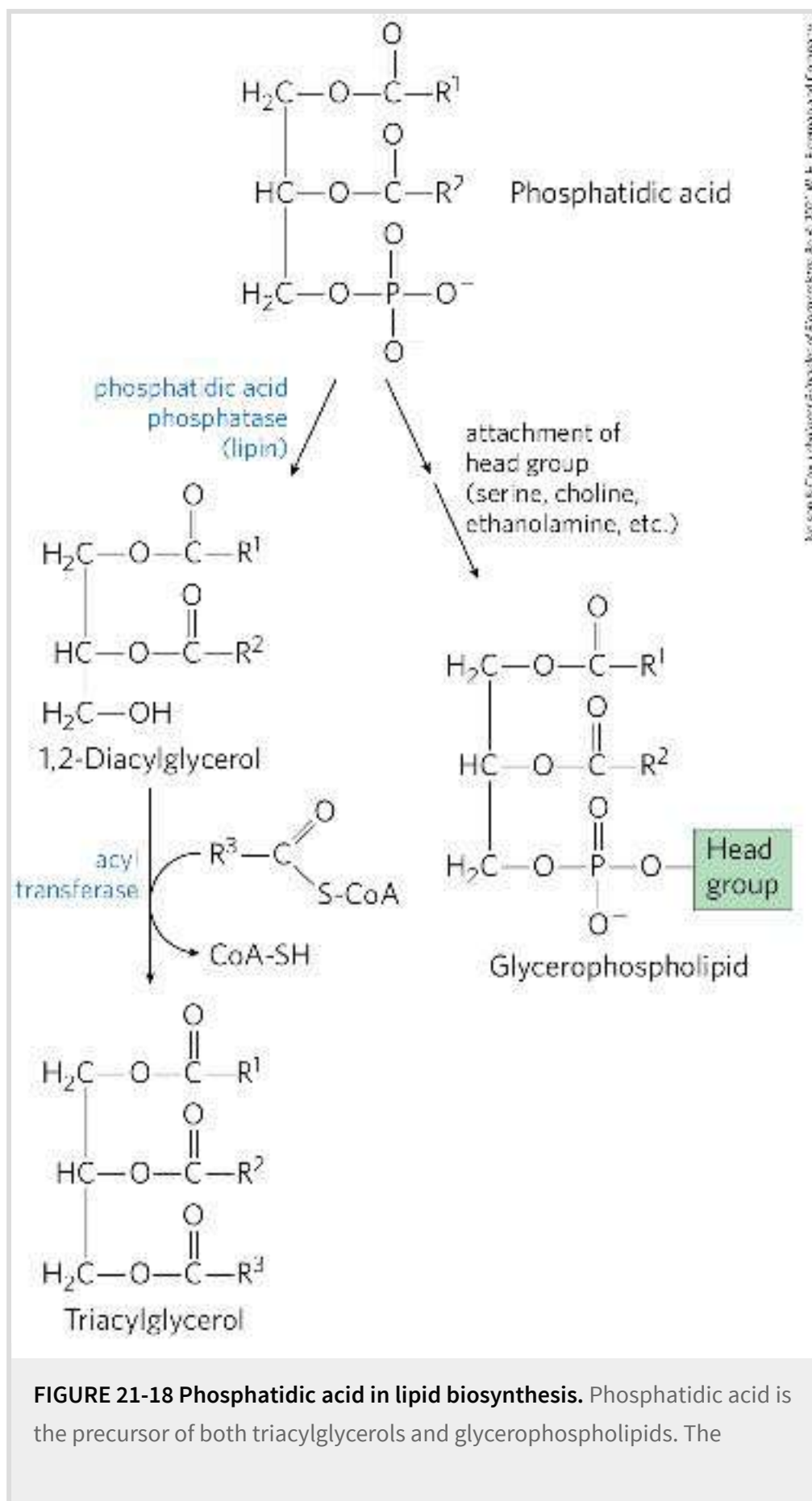


FIGURE 21-17 Biosynthesis of phosphatidic acid. A fatty acyl group is activated by formation of the fatty acyl-CoA, then transferred to ester linkage with L-glycerol 3-phosphate, formed in either of the two ways shown. Phosphatidic acid is shown here with the correct stereochemistry (L) at C-2 of the glycerol molecule. (The intermediate product with only one esterified fatty acyl group is lysophosphatidic acid.) To conserve space in subsequent figures (and in [Fig. 21-14](#)), both fatty acyl groups of

glycerophospholipids, and all three acyl groups of triacylglycerols, are shown projecting to the right.

The first stage in the biosynthesis of triacylglycerols is acylation of the two free hydroxyl groups of L-glycerol 3-phosphate by two molecules of fatty acyl-CoA to yield **diacylglycerol 3-phosphate**, more commonly called **phosphatidic acid**, or phosphatidate ([Fig. 21-17](#)). Phosphatidic acid is present in only trace amounts in cells but is a central intermediate in lipid biosynthesis; it can be converted either to a triacylglycerol or to a glycerophospholipid. In the pathway to triacylglycerols, phosphatidic acid is hydrolyzed by **phosphatidic acid phosphatase** (also called lipin) to form a 1,2-diacylglycerol ([Fig. 21-18](#)). Diacylglycerols are then converted to triacylglycerols by transesterification with a third fatty acyl-CoA.

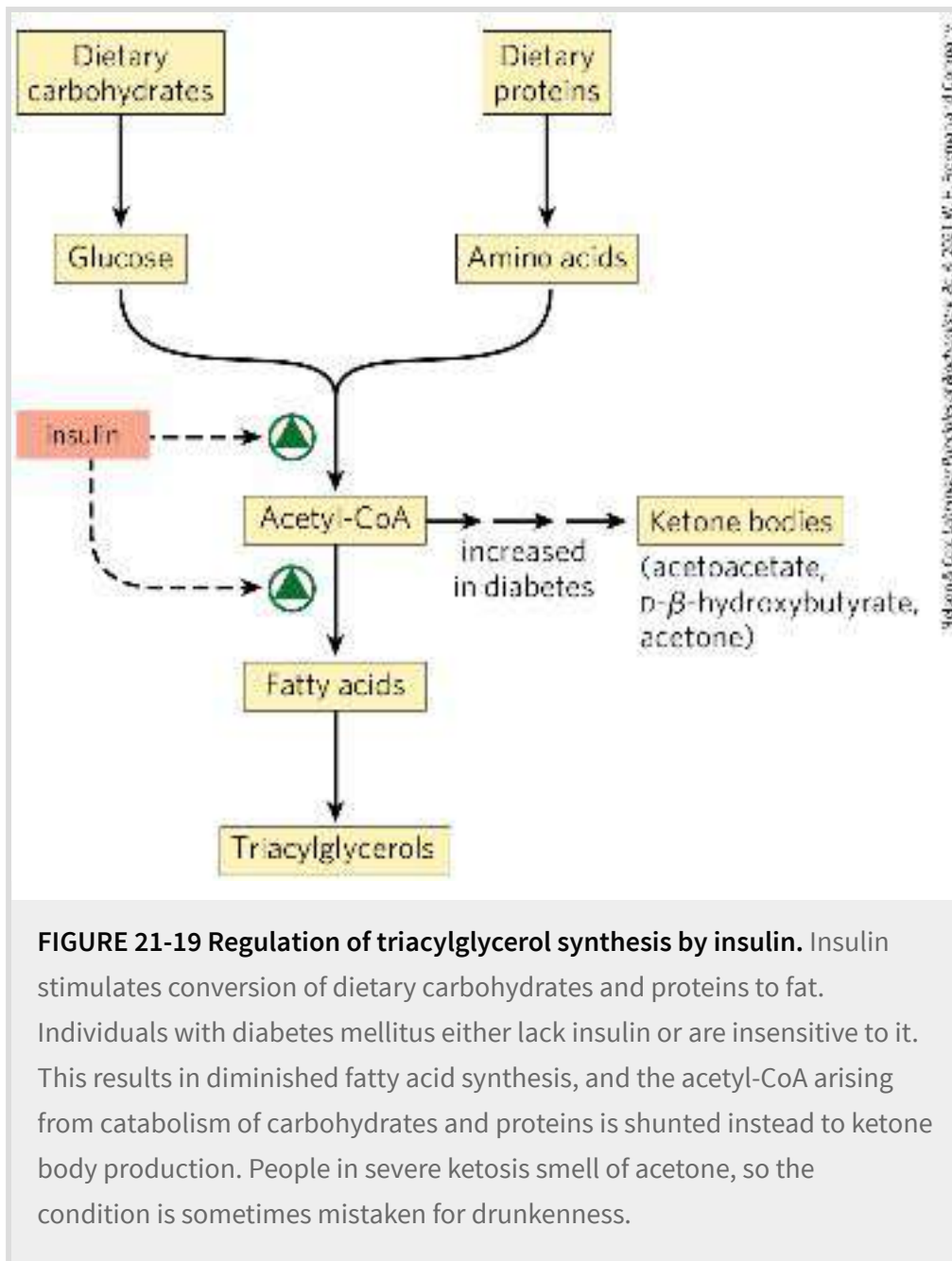


mechanisms for head-group attachment in phospholipid synthesis are described later in this section.

Triacylglycerol Biosynthesis in Animals Is Regulated by Hormones



In humans, the amount of body fat stays relatively constant over long periods, although there may be minor short-term changes as caloric intake fluctuates. Biosynthesis and degradation of triacylglycerols are regulated to meet the metabolic requirements of the moment. The rate of triacylglycerol biosynthesis is profoundly altered by the action of several hormones. Insulin, for example, promotes the conversion of carbohydrate to triacylglycerols ([Fig. 21-19](#)). People with severe diabetes mellitus, due to failure of insulin secretion or action, not only are unable to use glucose properly but also fail to synthesize fatty acids from carbohydrates or amino acids. If the diabetes is untreated, these individuals have increased rates of fat oxidation and ketone body formation ([Chapter 17](#)) and therefore lose weight. ■



Approximately 75% of all fatty acids released by triacylglycerol breakdown (lipolysis) are reesterified to form triacylglycerols rather than used for fuel. This ratio persists even under starvation conditions, when energy metabolism is shunted from the use of carbohydrate to the oxidation of fatty acids. Some of this fatty acid recycling takes place in adipose tissue, with the reesterification occurring before release into the bloodstream;

some takes place via a systemic cycle in which free fatty acids are transported to the liver, recycled to triacylglycerol, exported again into the blood (transport of lipids in the blood is discussed in [Section 21.4](#)), and taken up again by adipose tissue, after release from triacylglycerol by extracellular lipoprotein lipase ([Fig. 21-20](#); see also [Fig. 17-1](#)). Flux through this **triacylglycerol cycle** between adipose tissue and liver may be low when other fuels are available and the release of fatty acids from adipose tissue is limited, but, as noted above, the proportion of released fatty acids that are reesterified remains roughly constant at 75% under all metabolic conditions. The level of free fatty acids in the blood thus reflects both the rate of release of fatty acids and the balance between the synthesis and breakdown of triacylglycerols in adipose tissue and liver.

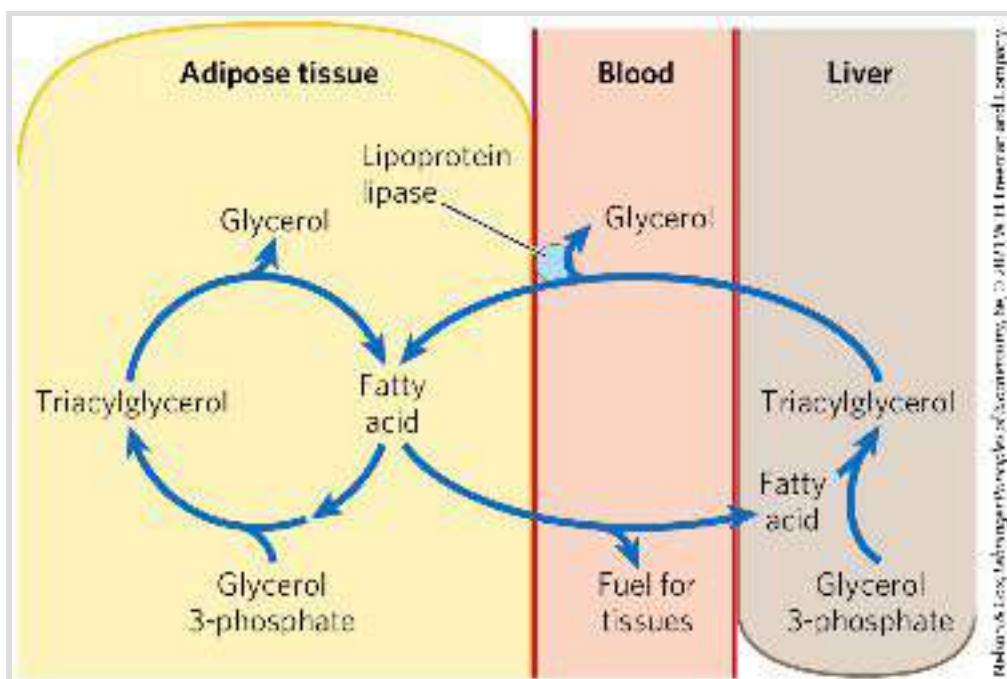


FIGURE 21-20 The triacylglycerol cycle. In mammals, triacylglycerol molecules are broken down and resynthesized in a triacylglycerol cycle during starvation. Some of the fatty acids released by lipolysis of

triacylglycerol in adipose tissue pass into the bloodstream, and the remainder are used for resynthesis of triacylglycerol. Some of the fatty acids released into the blood are used for energy (in muscle, for example), and some are taken up by the liver and used in triacylglycerol synthesis. The triacylglycerol formed in the liver is transported in the blood back to adipose tissue, where the fatty acid is released by extracellular lipoprotein lipase, taken up by adipocytes, and reesterified into triacylglycerol.

When the mobilization of fatty acids is required to meet energy needs, release from adipose tissue is stimulated by the hormones glucagon and epinephrine (see [Figs. 17-2, 17-13](#)). Simultaneously, these hormonal signals decrease the rate of glycolysis and increase the rate of gluconeogenesis in the liver (providing glucose for the brain, as further elaborated in [Chapter 23](#)). The released fatty acid is taken up by several tissues, including muscle, where it is oxidized to provide energy. Much of the fatty acid taken up by liver is not oxidized but is recycled to triacylglycerol and returned to adipose tissue.

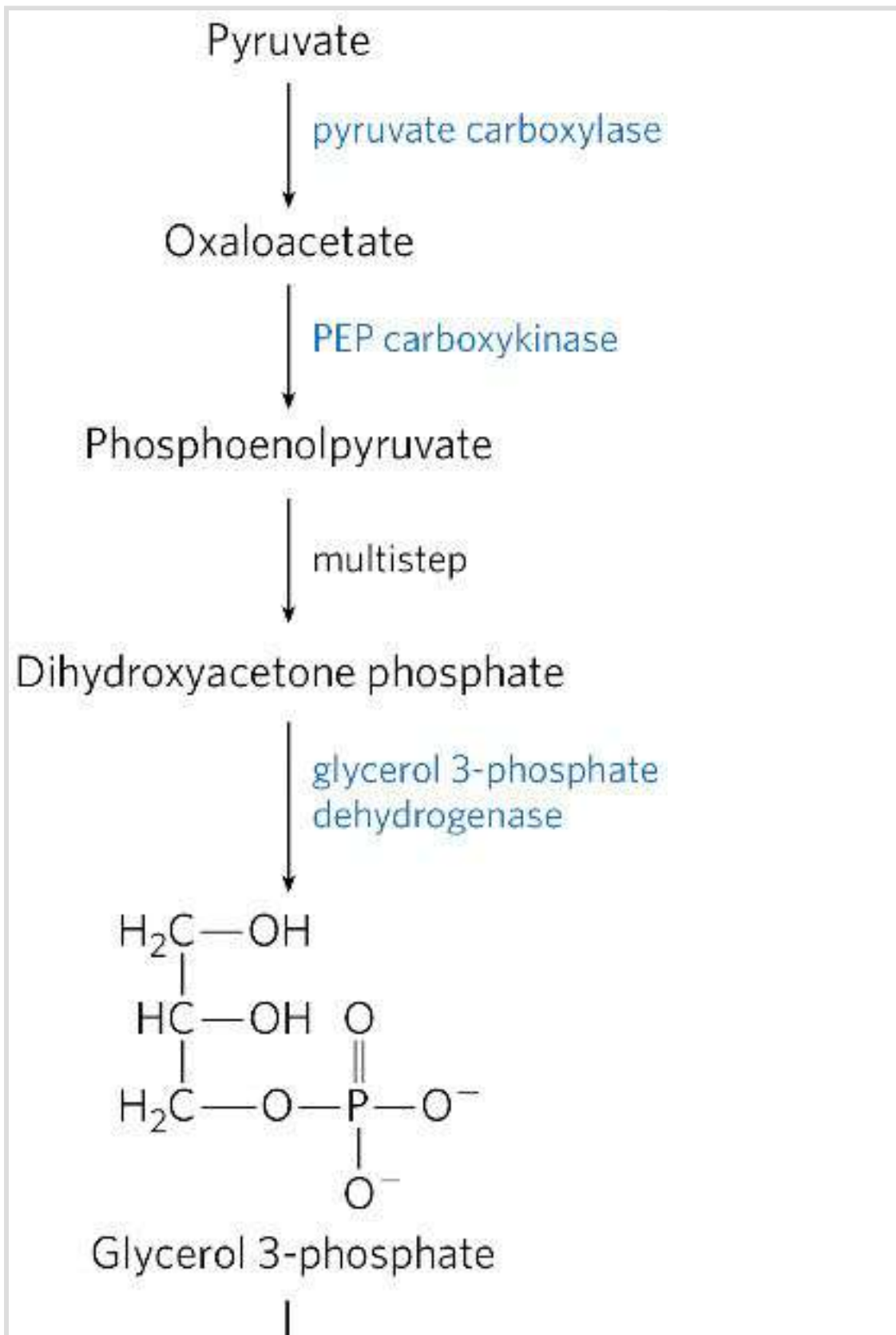
The function of the apparently futile triacylglycerol cycle is not well understood, but as we learn more about how the cycle is sustained via metabolism in two separate organs and is coordinately regulated, some possibilities emerge. For example, the excess capacity in the triacylglycerol cycle — the fatty acid that is eventually reconverted to triacylglycerol rather than oxidized as fuel — could represent an energy reserve in the bloodstream during fasting, one that could be more rapidly mobilized in a “fight or flight” emergency than stored triacylglycerol could be.

The constant recycling of triacylglycerols in adipose tissue even during starvation raises a second question: what is the source of the glycerol 3-phosphate required for this process? As noted above, glycolysis is suppressed under these conditions by the action of glucagon and epinephrine, so little DHAP is available. And glycerol released during lipolysis cannot be converted directly to glycerol 3-phosphate in adipose tissue, which lacks glycerol kinase ([Fig. 21-17](#)). So, how is sufficient glycerol 3-phosphate produced? The answer lies in the pathway of glyceroneogenesis, discovered in the 1960s by Lea Reshef, Richard Hanson, and John Ballard, and simultaneously by Eleazar Shafrir and his coworkers. The investigators were intrigued by the presence of two gluconeogenic enzymes, pyruvate carboxylase and phosphoenolpyruvate (PEP) carboxykinase, in adipose tissue, where glucose is not synthesized. Yet, the importance of this pathway was not appreciated until decades later. Glyceroneogenesis is intimately linked to the triacylglycerol cycle and, in a larger sense, to the balance between fatty acid and carbohydrate metabolism.

Adipose Tissue Generates Glycerol 3-Phosphate by Glyceroneogenesis

[Glyceroneogenesis](#) is a shortened version of gluconeogenesis, from pyruvate to DHAP (see [Fig. 14-16](#)), followed by conversion of the DHAP to glycerol 3-phosphate by cytosolic NAD-linked glycerol 3-phosphate dehydrogenase ([Fig. 21-21](#)). Glycerol 3-phosphate is subsequently used in triacylglycerol synthesis. There

is a link between glyceroneogenesis and type 2 diabetes, as we shall see.





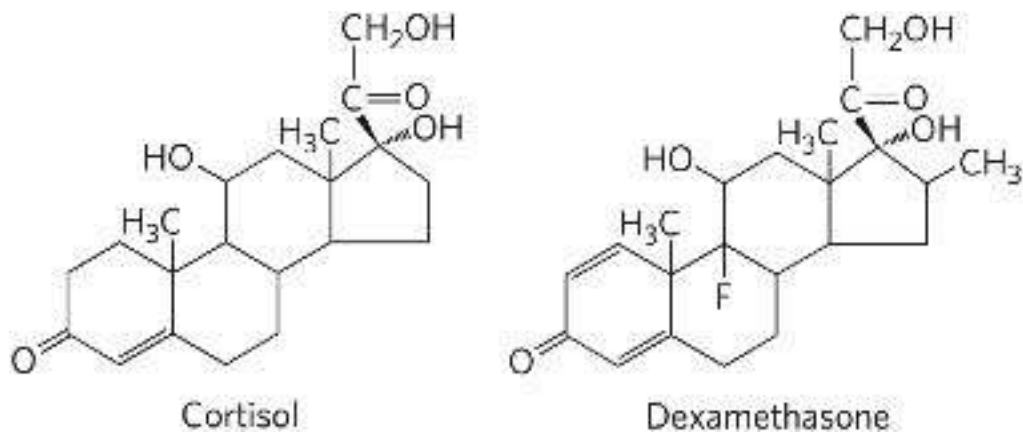
Triacylglycerol synthesis

FIGURE 21-21 Glyceroneogenesis. The pathway is essentially an abbreviated version of gluconeogenesis, from pyruvate to dihydroxyacetone phosphate (DHAP), followed by conversion of DHAP to glycerol 3-phosphate, which is used for the synthesis of triacylglycerol.

Glyceroneogenesis has multiple roles. In adipose tissue, glyceroneogenesis coupled with reesterification of free fatty acids controls the rate of fatty acid release to the blood. In brown adipose tissue, the same pathway may control the rate at which free fatty acids are delivered to mitochondria for use in thermogenesis. And in fasting humans, glyceroneogenesis in the liver alone supports the synthesis of enough glycerol 3-phosphate to account for up to 65% of fatty acids reesterified to triacylglycerol.

Flux through the triacylglycerol cycle between liver and adipose tissue is controlled to a large degree by the activity of PEP carboxykinase, which limits the rate of both gluconeogenesis and glyceroneogenesis. Glucocorticoid hormones such as cortisol (a biological steroid derived from cholesterol; see [Fig. 21-48](#)) and dexamethasone (a synthetic glucocorticoid) regulate the levels of PEP carboxykinase reciprocally in the liver and adipose tissue. Acting through the glucocorticoid receptor, these steroid hormones increase the expression of the gene encoding PEP

carboxykinase in the liver, thus increasing gluconeogenesis and glyceroneogenesis ([Fig. 21-22](#)).



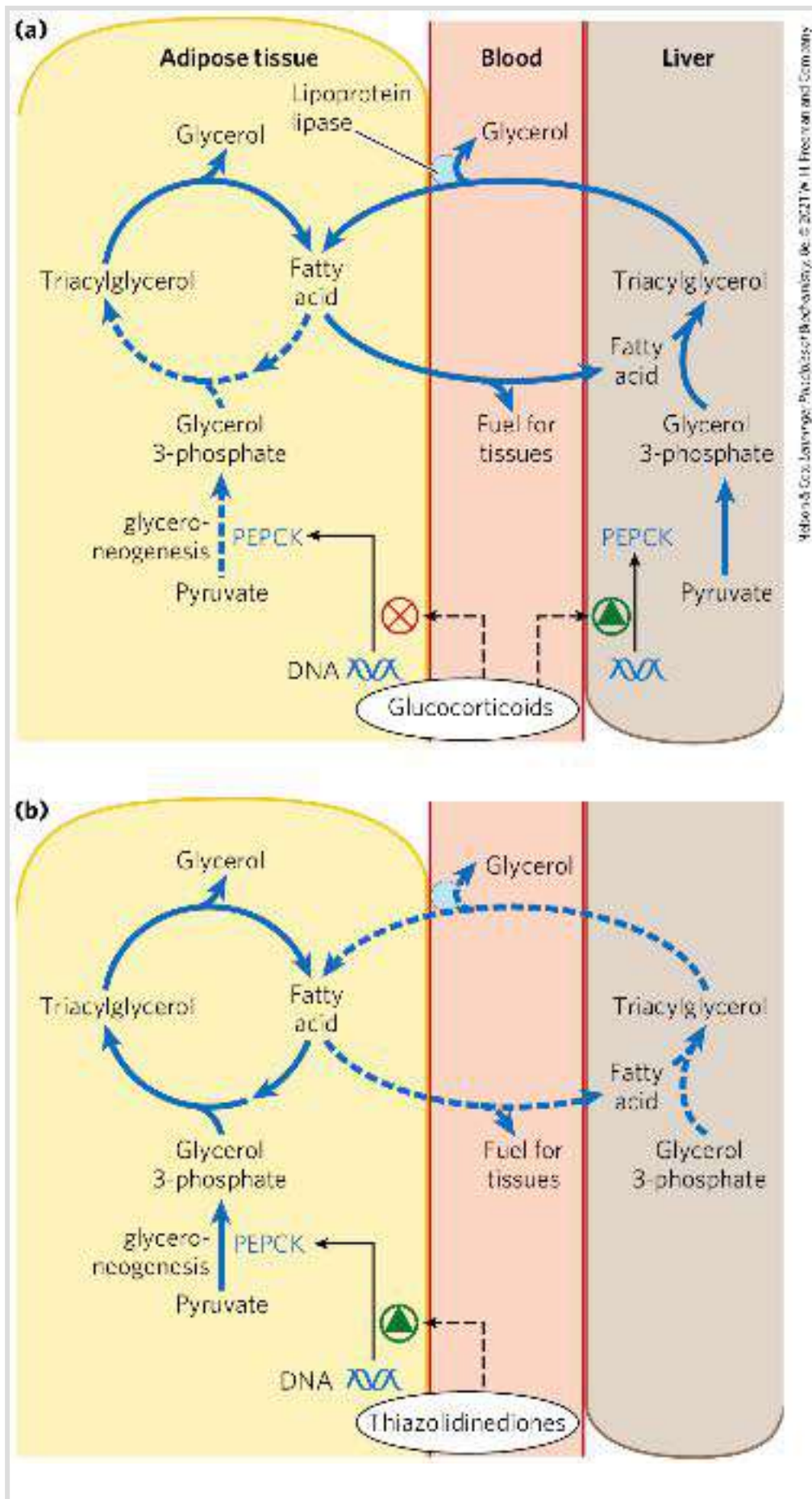


FIGURE 21-22 Regulation of glyceroneogenesis. (a) Glucocorticoid hormones stimulate glyceroneogenesis and gluconeogenesis in the liver, while suppressing glyceroneogenesis in adipose tissue (by reciprocal regulation of the gene expressing PEP carboxykinase (PEPCK) in the two tissues); this increases the flux through the triacylglycerol cycle. The glycerol freed by the breakdown of triacylglycerol in adipose tissue is released to the blood and transported to the liver, where it is primarily converted to glucose, although some is converted to glycerol 3-phosphate by glycerol kinase. (b) A class of drugs called thiazolidinediones is used to treat type 2 diabetes. In this disease, high levels of free fatty acids in the blood interfere with glucose utilization in muscle and promote insulin resistance. Thiazolidinediones activate a nuclear receptor called peroxisome proliferator-activated receptor γ (PPAR γ), which induces the activity of PEP carboxykinase. Therapeutically, thiazolidinediones increase the rate of glyceroneogenesis, thus increasing the resynthesis of triacylglycerol in adipose tissue and reducing the amount of free fatty acid in the blood. In both panels, dashed lines identify pathways in which flux declines under the conditions indicated.

Stimulation of glyceroneogenesis leads to an increase in the synthesis of triacylglycerol molecules in the liver and their release into the blood. At the same time, glucocorticoids suppress expression of the gene encoding PEP carboxykinase in adipose tissue. This results in a decrease in glyceroneogenesis in adipose tissue; recycling of fatty acids declines as a result, and more free fatty acids are released into the blood. Thus, regulation of glyceroneogenesis in the liver and adipose tissue affects lipid metabolism in opposite ways: a lower rate of glyceroneogenesis in adipose tissue leads to more fatty acid release (rather than recycling), whereas a higher rate in the liver leads to more synthesis and export of triacylglycerols. The net result is an increase in flux through the triacylglycerol cycle. When the

glucocorticoids are no longer present, flux through the cycle declines as the expression of PEP carboxykinase increases in adipose tissue and decreases in the liver.

Thiazolidinediones Treat Type 2 Diabetes by Increasing Glyceroneogenesis



The connection between glyceroneogenesis and diabetes has stimulated new interest. High levels of free fatty acids in the blood interfere with glucose utilization in muscle and promote the insulin resistance that leads to type 2 diabetes. A class of drugs called [thiazolidinediones](#) reduces the levels of fatty acids circulating in the blood and increases sensitivity to insulin. Thiazolidinediones promote the increased expression of PEP carboxykinase in adipose tissue ([Fig. 21-22](#)), leading to increased synthesis of the precursors of glyceroneogenesis. The therapeutic effect of thiazolidinediones is thus due, at least in part, to the increase in glyceroneogenesis, which in turn increases the resynthesis of triacylglycerol in adipose tissue and reduces the release of free fatty acid from adipose tissue into the blood. Two thiazolidinediones have been available for treatment of type 2 diabetes: rosiglitazone (Avandia) and pioglitazone (Actos). Large-scale trials of rosiglitazone have indicated an increased risk of heart attack, so rosiglitazone has been withdrawn in the United Kingdom, India, South Africa, and many European countries. It remains available in the United States with limitations. ■

SUMMARY 21.2 *Biosynthesis of Triacylglycerols*

■ Triacylglycerols are formed by reaction of two molecules of fatty acyl-CoA with glycerol 3-phosphate to form phosphatidic acid; this product is dephosphorylated to a diacylglycerol, then acylated by a third molecule of fatty acyl-CoA to yield a triacylglycerol.

■ The synthesis and degradation of triacylglycerols are hormonally regulated.

■ Mobilization and recycling of triacylglycerol molecules result in a triacylglycerol cycle. Triacylglycerols are resynthesized from free fatty acids and glycerol 3-phosphate even during starvation. The dihydroxyacetone phosphate precursor of glycerol 3-phosphate is derived from pyruvate via glyceroneogenesis.

■ Thiazolidinediones stimulate glyceroneogenesis and can be used to treat type 2 diabetes.

21.3 Biosynthesis of Membrane Phospholipids

In [Chapter 10](#) we introduced two major classes of membrane phospholipids: glycerophospholipids and sphingolipids. Many different phospholipid species can be constructed by combining various fatty acids and polar head groups with the glycerol or sphingosine backbone (see [Figs. 10-8, 10-11](#)). All the biosynthetic pathways follow a few basic patterns. In general, the assembly of phospholipids from simple precursors requires (1) synthesis of the backbone molecule (glycerol or sphingosine); (2) attachment of fatty acid(s) to the backbone through an ester or amide linkage; (3) addition of a hydrophilic head group to the backbone through a phosphodiester linkage; and, in some cases, (4) alteration or exchange of the head group or the fatty acids to yield the final phospholipid product.

In eukaryotic cells, phospholipid synthesis occurs primarily on the surfaces of the smooth ER and the inner mitochondrial membrane. Some newly formed phospholipids remain at the site of synthesis, but most are destined for other cellular membranes. Once they arrive, phospholipids can be remodeled within membranes to alter the fatty acid constituents. The process by which water-insoluble phospholipids move from the site of synthesis to the point of their eventual function is not fully understood, but we discuss some mechanisms that have emerged in recent years.

Cells Have Two Strategies for Attaching Phospholipid Head Groups

Stage 1 of glycerophospholipid synthesis is shared with the pathway to triacylglycerols, the formation of glycerol 3-phosphate by one of the two paths shown in [Fig. 21-17](#). In stage 2, fatty acyl groups are esterified to C-1 and C-2 of L-glycerol 3-phosphate to form phosphatidic acid. Usually, the fatty acid at C-1 is saturated and the one at C-2 is unsaturated. A second route to phosphatidic acid is the phosphorylation of a diacylglycerol by a specific kinase.



Eugene P. Kennedy, 1919–2011

In stages 3 and 4, the polar head group of glycerophospholipids is attached through a phosphodiester bond, in which each of two alcohol hydroxyls (one on the polar head group and one on C-3 of

glycerol) forms an ester with phosphoric acid ([Fig. 21-23](#)). In the biosynthetic process, one of the hydroxyls is first activated by attachment of a nucleotide, cytidine diphosphate (CDP). Cytidine monophosphate is then displaced in a nucleophilic attack by the other hydroxyl ([Fig. 21-24](#)). Two strategies are employed by mammals. The CDP is attached either to the diacylglycerol, forming the activated phosphatidic acid **CDP-diacylglycerol** (strategy 1), or to the hydroxyl of the head group (strategy 2). The central importance of cytidine nucleotides in lipid biosynthesis was discovered by Eugene P. Kennedy in the late 1950s, and this pathway is commonly referred to as the Kennedy pathway. In bacteria, only strategy 1 is used to generate glycerophospholipids.

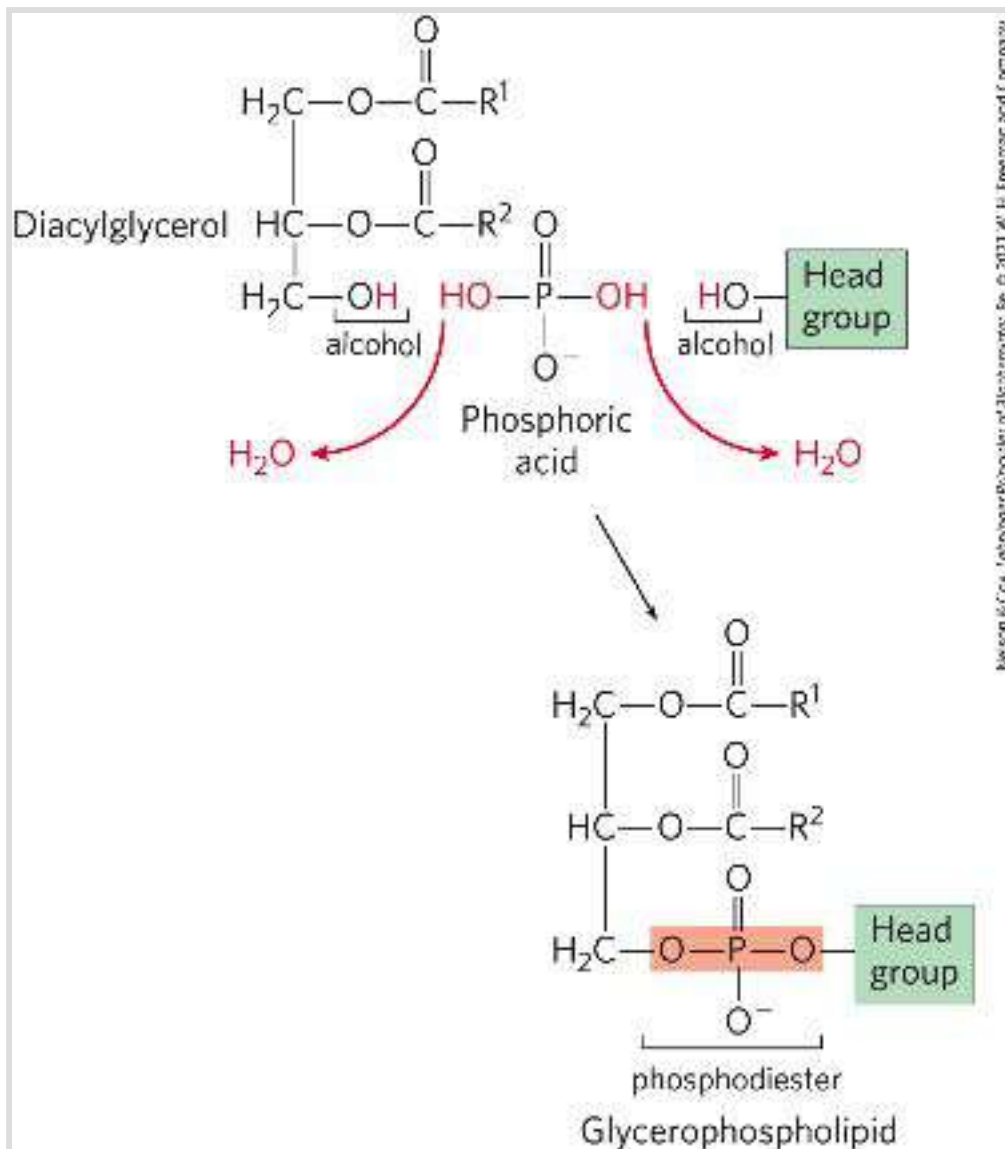


FIGURE 21-23 Final stages of glycerophospholipid biosynthesis: head-group attachment. The phospholipid head group is attached to a diacylglycerol by a phosphodiester bond (shaded light red), formed when phosphoric acid condenses with two alcohols, eliminating two molecules of H_2O .

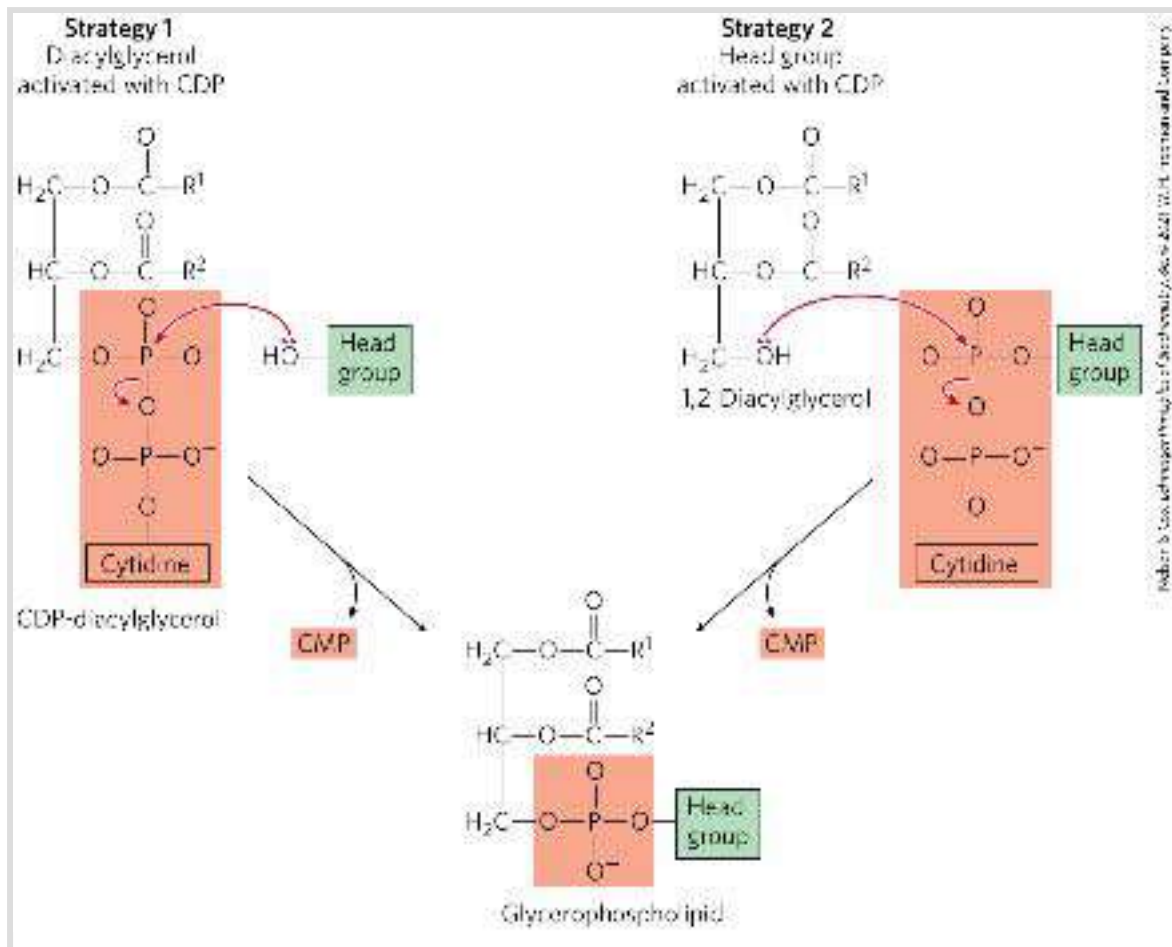


FIGURE 21-24 Two general strategies for forming the phosphodiester bond of phospholipids. In both cases, CDP supplies the phosphate group of the phosphodiester bond.

Pathways for Phospholipid Biosynthesis Are Interrelated

In eukaryotes, many phospholipids are synthesized using strategy 1 in [Figure 21-24](#), and many of the pathways begin with CDP-diacylglycerol. The synthesis of **phosphatidylglycerol** provides our first example. Beginning with CDP-diacylglycerol ([Fig. 21-25](#)), displacement of CMP through nucleophilic attack by the C-1 hydroxyl of glycerol 3-phosphate yields phosphatidylglycerol 3-

phosphate. Phosphatidylglycerol 3-phosphate is processed further by cleavage of the phosphate monoester (with release of P_i) to yield phosphatidylglycerol.

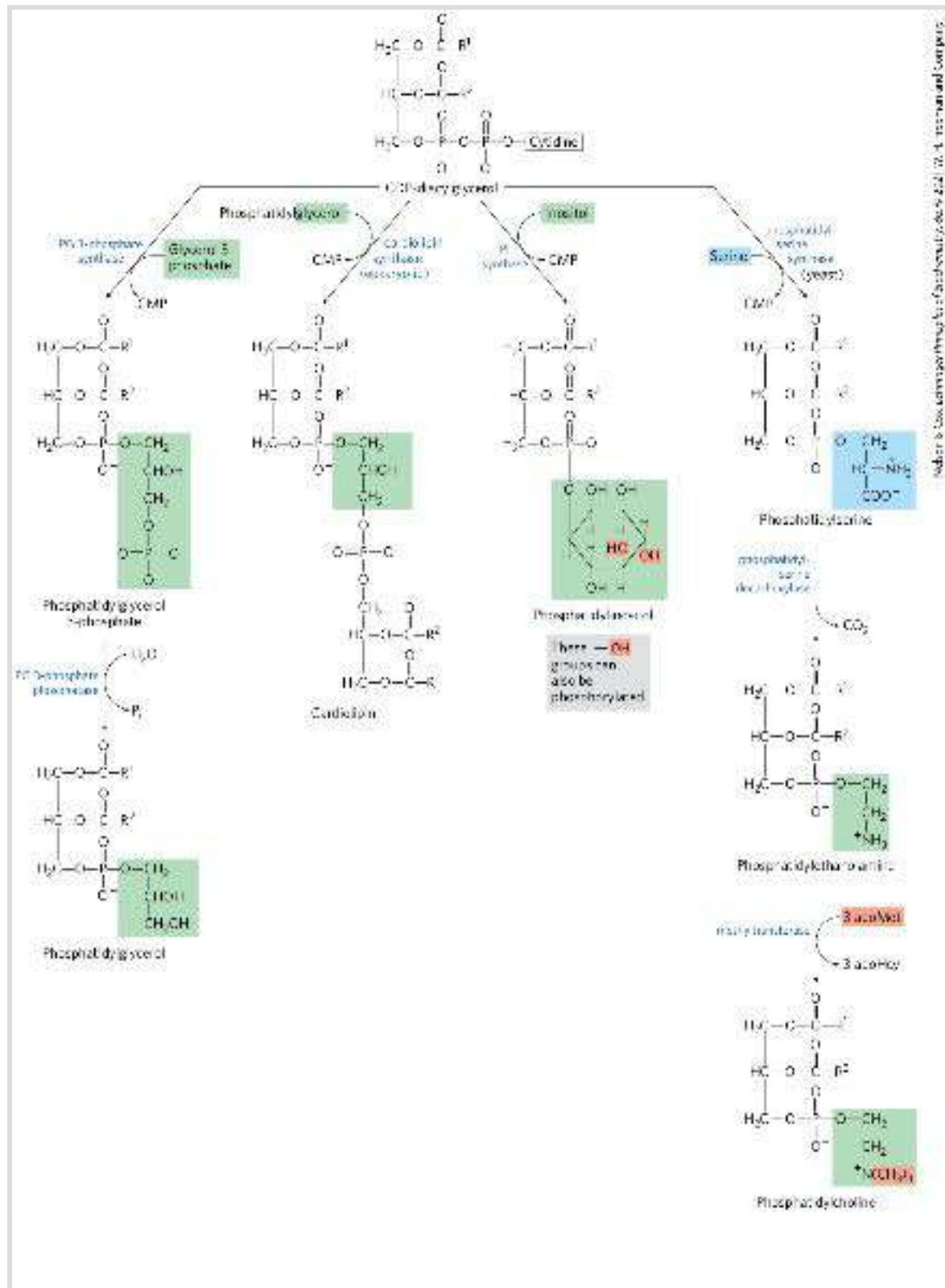


FIGURE 21-25 Synthesis of glycerophospholipids in eukaryotes using CDP-diacylglycerol. These glycerophospholipids are synthesized using strategy 1 in [Figure 21-24](#). Phosphatidylglycerol is synthesized by reaction of CDP-diacylglycerol with glycerol-3-phosphate, followed by dephosphorylation. Phosphatidylglycerol can react with CDP-diacylglycerol to generate cardiolipin. Phosphatidylinositol is generated from CDP-diacylglycerol in a single step. The pathway from CDP-diacylglycerol to phosphatidylserine is used in yeast but not in mammals. The pathway from phosphatidylserine to phosphatidylethanolamine and phosphatidylcholine is common to all eukaryotes.

In eukaryotes, [cardiolipin](#) is a relatively uncommon phospholipid, found almost exclusively in the inner membranes of mitochondria. As described below, cardiolipin is important in bacteria, and its presence in mitochondria is likely yet another relic of the bacterial origin of these organelles. Cardiolipin is essential for the function of some mitochondrial enzymes. It is also synthesized using strategy 1, by condensation of CDP-diacylglycerol with phosphatidylglycerol ([Fig. 21-25](#)).

Phosphatidylinositol is similarly synthesized by condensation of CDP-diacylglycerol with inositol ([Fig. 21-25](#)). Specific **phosphatidylinositol kinases** then convert phosphatidylinositol to its phosphorylated derivatives. Phosphatidylinositol and its phosphorylated products in the plasma membrane play a central role in signal transduction in eukaryotes (see [Figs. 12-11, 12-15, 12-23](#)).

Yeast (but not mammals) use a similar path to produce **phosphatidylserine** by condensation of CDP-diacylglycerol and serine, and they can synthesize **phosphatidylethanolamine** from

phosphatidylserine in the reaction catalyzed by phosphatidylserine decarboxylase ([Fig. 21-25](#)). This pathway for synthesis of phosphatidylethanolamine occurs primarily in the mitochondria, although this lipid is transported from there to other cellular membranes. Phosphatidylethanolamine may be converted to **phosphatidylcholine** (lecithin) by the addition of three methyl groups to its amino group; S-adenosylmethionine is the methyl group donor (see [Fig. 18-18](#)) for all three methylation reactions.

In mammals, strategy 2 ([Fig. 21-24](#)) is utilized for the synthesis of phosphatidylethanolamine and phosphatidylcholine in the membranes of the ER and nucleus. The activation of the head group to the CDP derivative is followed by condensation with diacylglycerol as shown for phosphatidylcholine ([Fig. 21-26a](#)). These pathways serve to salvage free ethanolamine and choline. In contrast, mammalian phosphatidylserine biosynthesis does not utilize either strategy shown in [Figure 21-24](#); instead, it is derived from phosphatidylethanolamine or phosphatidylcholine via one of two head-group exchange reactions carried out in the ER ([Fig. 21-26b](#)). These reactions generate free ethanolamine and choline, respectively. The major sources of phosphatidylethanolamine and phosphatidylcholine in all eukaryotic cells are summarized in [Figure 21-27](#).

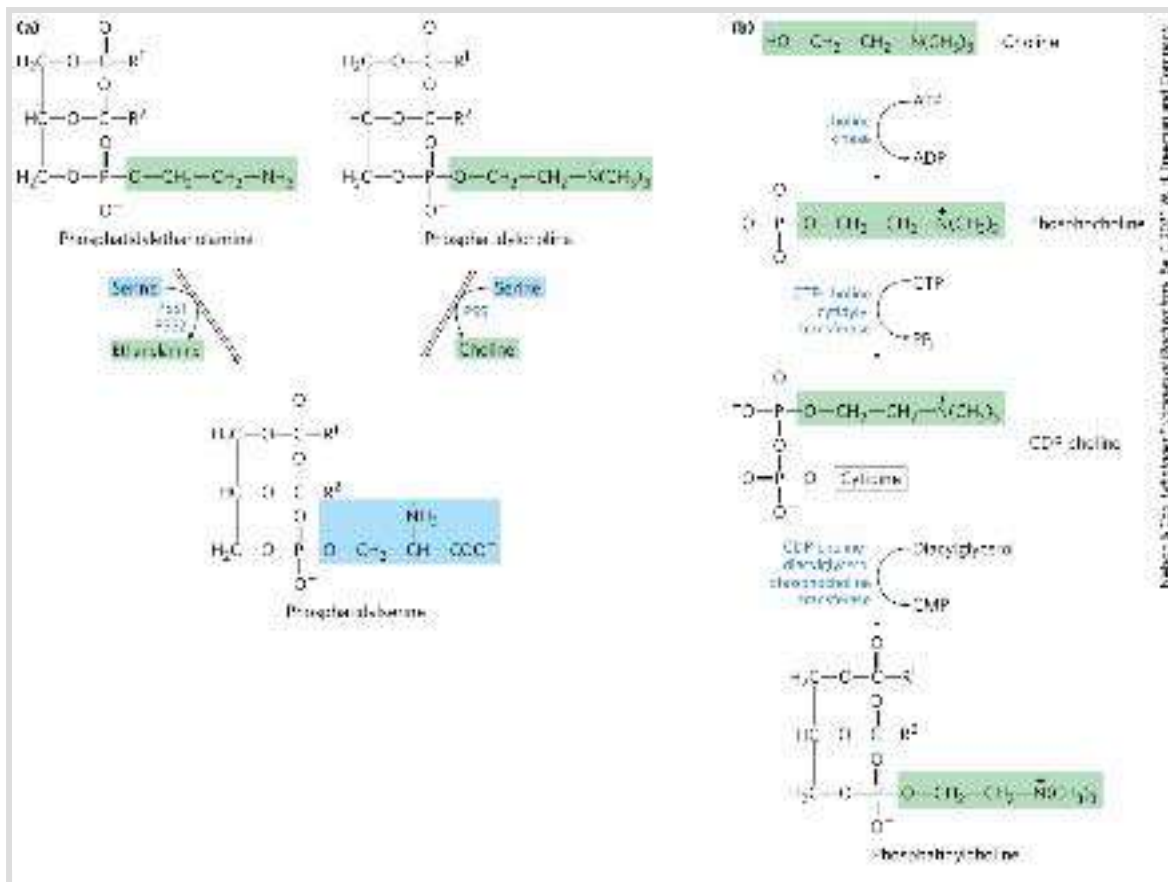


FIGURE 21-26 Pathways for phosphatidylserine and phosphatidylcholine synthesis in mammals. (a) Phosphatidylserine is synthesized by Ca^{2+} -dependent head-group exchange reactions promoted by phosphatidylserine synthase 1 (PSS1) or phosphatidylserine synthase 2 (PSS2). PSS1 can use either phosphatidylethanolamine or phosphatidylcholine as a substrate. (b) The same strategy shown here for phosphatidylcholine synthesis (strategy 2 in [Fig. 21-24](#)) is also used for salvaging ethanolamine in phosphatidylethanolamine synthesis.

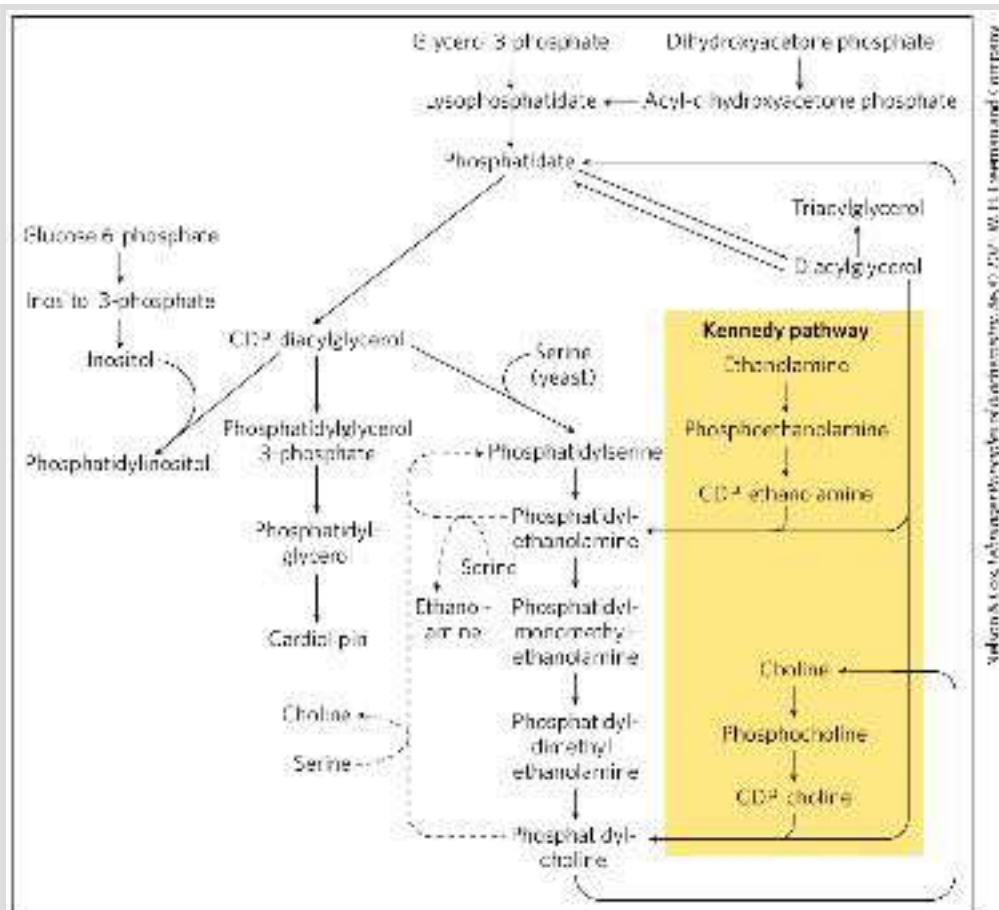


FIGURE 21-27 Summary of the pathways for synthesis of major phospholipids and triacylglycerols in eukaryotes.

Phosphatidic acid is formed by transacylation of L-glycerol 3-phosphate with two fatty acyl groups donated from fatty acyl-CoA. The enzyme phosphatidic acid phosphatase (lipin) converts phosphatidic acid to diacylglycerol, which in the Kennedy pathway condenses with a CDP-activated head group (ethanolamine or choline) to form phosphatidylethanolamine or phosphatidylcholine. Alternatively, phosphatidic acid can be activated with a CDP moiety, which is displaced by condensation with a head-group alcohol — inositol, glycerol 3-phosphate, or serine, forming phosphatidylinositol, phosphatidylglycerol, or (only in yeast and fungi) phosphatidylserine. Decarboxylation of phosphatidylserine yields phosphatidylethanolamine, and methylation of phosphatidylethanolamine produces phosphatidylcholine. In mammals, phosphatidylserine and phosphatidylcholine are generated via the head-group exchange pathways, detailed in [Fig. 21-26](#). Lysophosphatidic acid is phosphatidic acid missing one of the two fatty acyl groups. [Information from G. M. Carman and G.-S. Han, *Annu. Rev. Biochem.* 80:859, 2011, Fig. 2.]

The most prominent phospholipids in bacteria are phosphatidylglycerol, phosphatidylethanolamine, and cardiolipin. The pathway for phosphatidylglycerol synthesis ([Fig. 21-28](#)) is identical to the path employed in mammals (compare to [Fig. 21-25](#)), beginning with CDP-diacylglycerol and using strategy 1. Phosphatidylethanolamine is produced in a similar pathway, with phosphatidylserine an intermediate. In bacteria, there are multiple biosynthetic paths to the third prominent phospholipid, cardiolipin, in which two diacylglycerols are joined through a common head group ([Fig. 21-28](#)).

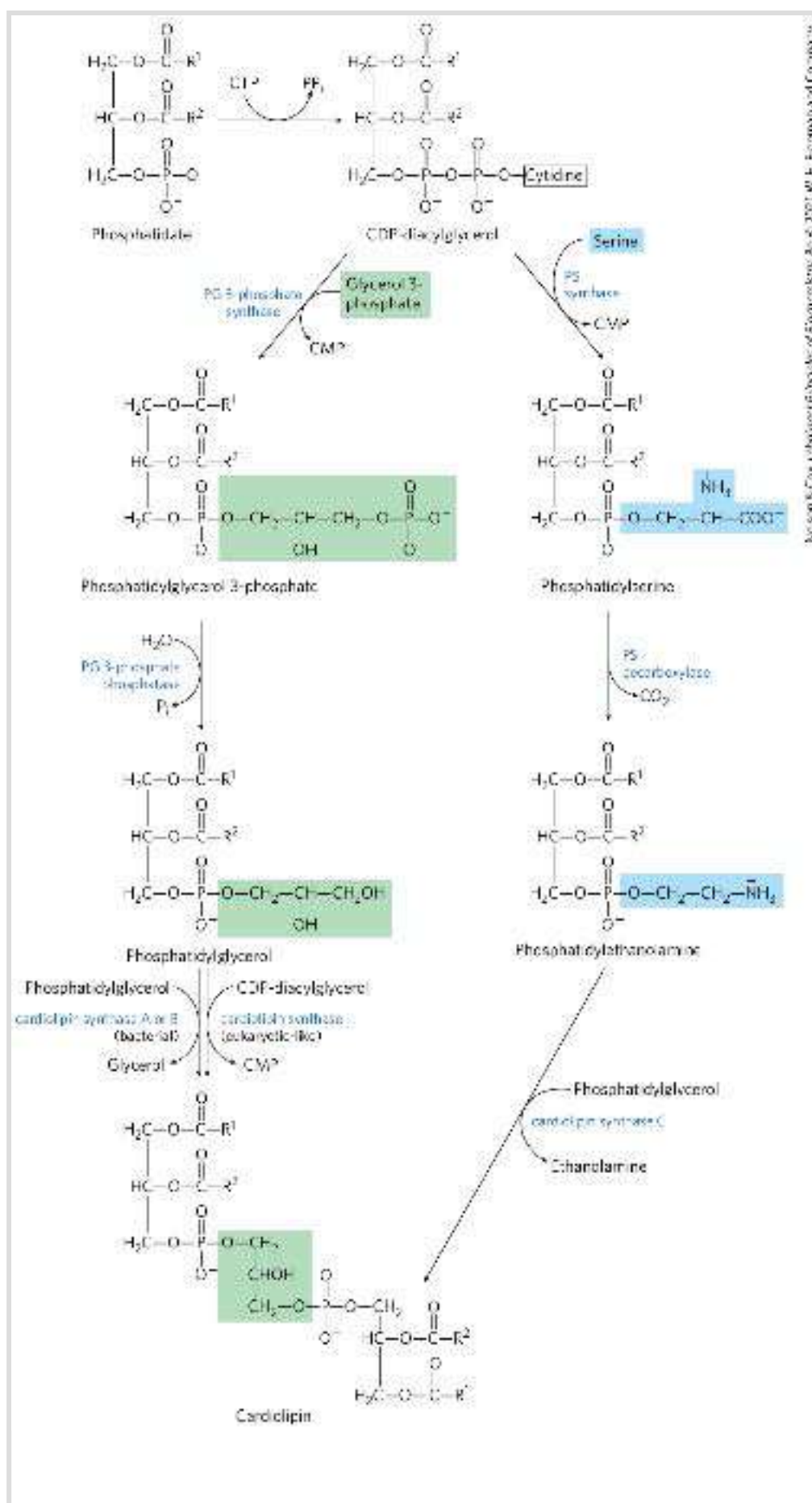


FIGURE 21-28 Origin of the polar head groups of phospholipids in *E. coli*.

Initially, a head group (either serine or glycerol 3-phosphate) is attached via a CDP-diacylglycerol intermediate (strategy 1 in [Fig. 21-24](#)). For phospholipids other than phosphatidylserine, the head group is further modified, as shown here. In the enzyme names, PG represents phosphatidylglycerol; PS, phosphatidylserine. Cardiolipin can be generated from either phosphatidylglycerol or phosphatidylethanolamine via multiple pathways, as shown. One of these pathways, in which phosphatidylglycerol is condensed with CDP-diacylglycerol, is identical to the pathway used in eukaryotes.

Eukaryotic Membrane Phospholipids Are Subject to Remodeling

In principle, the two fatty acyl groups esterified to C-1 and C-2 in a phospholipid may vary in length and degree of desaturation, thus altering the properties of the membrane of which it is a part.

Phosphatidylcholine is the major structural phospholipid of mammalian membranes, often representing 40% to 50% of the total. Thus, much of the remodeling centers on phosphatidylcholine. The remodeling occurs largely by a process called the [Lands cycle](#) ([Fig. 21-29](#)), which replaces the polyunsaturated fatty acyl group at C-2. The fatty acyl moieties are first hydrolyzed by phospholipase A₂ ([Fig. 10-14](#)) to generate 1-acyl lysophospholipids. The fatty acid is then replaced by a class of enzymes called [lysophosphatidylcholine acyltransferases](#), or [LPCATs](#). There are at least four LPCATs in humans, each with distinct tissue distributions and substrate specificities.

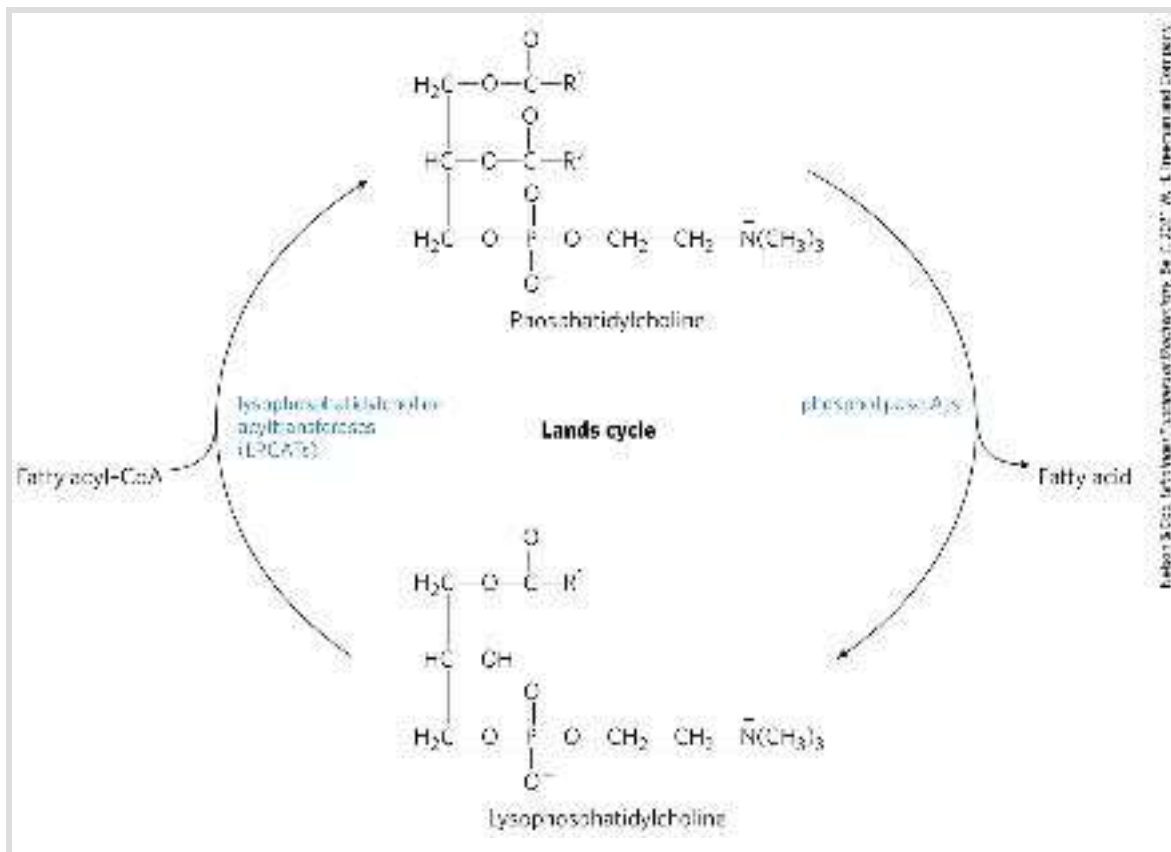


FIGURE 21-29 The Lands cycle for phospholipid remodeling. Phosphatidylcholine, which is the most common phospholipid of eukaryotic membranes, is a major target of this process. Fatty acids at C-2 are removed by phospholipase A₂ enzymes. A new fatty acid is then introduced by the action of a lysophosphatidylphocholine acyltransferase (LPCAT). The four mammalian LPCATs differ in tissue distribution and substrate specificity.



The physiological effects of LPCAT enzymes go far beyond the alteration of the lipid composition of membranes. LPCAT3, the most widely distributed version of the enzyme, helps to regulate lipogenesis and secretion of very-low-density lipoproteins (VLDLs), described later in this chapter. Mice lacking LPCAT3 have a greatly reduced intake of fatty acids in the intestine, resulting in the release of gut hormones that control appetite. They are unable to survive on a high-fat diet, they resist eating, and they die of starvation unless the diet is changed.

LPCATs play a demonstrable but still often mysterious role in processes from atherosclerosis to obesity to cancer, making these enzymes the subjects of increasing interest as research and drug targets. ■

Plasmalogen Synthesis Requires Formation of an Ether-Linked Fatty Alcohol

 The biosynthetic pathway to ether lipids, including **plasmalogens** and the **platelet-activating factor** (see [Fig. 10-9](#)), requires displacement of an esterified fatty acyl group by a long-chain alcohol to form the ether linkage ([Fig. 21-30](#)). Head-group attachment follows, by mechanisms essentially like those used in formation of the common ester-linked phospholipids. Finally, the characteristic double bond of plasmalogens is introduced by the action of a mixed-function oxidase similar to that responsible for desaturation of fatty acids ([Fig. 21-13](#)). The peroxisome is the primary site of plasmalogen synthesis.

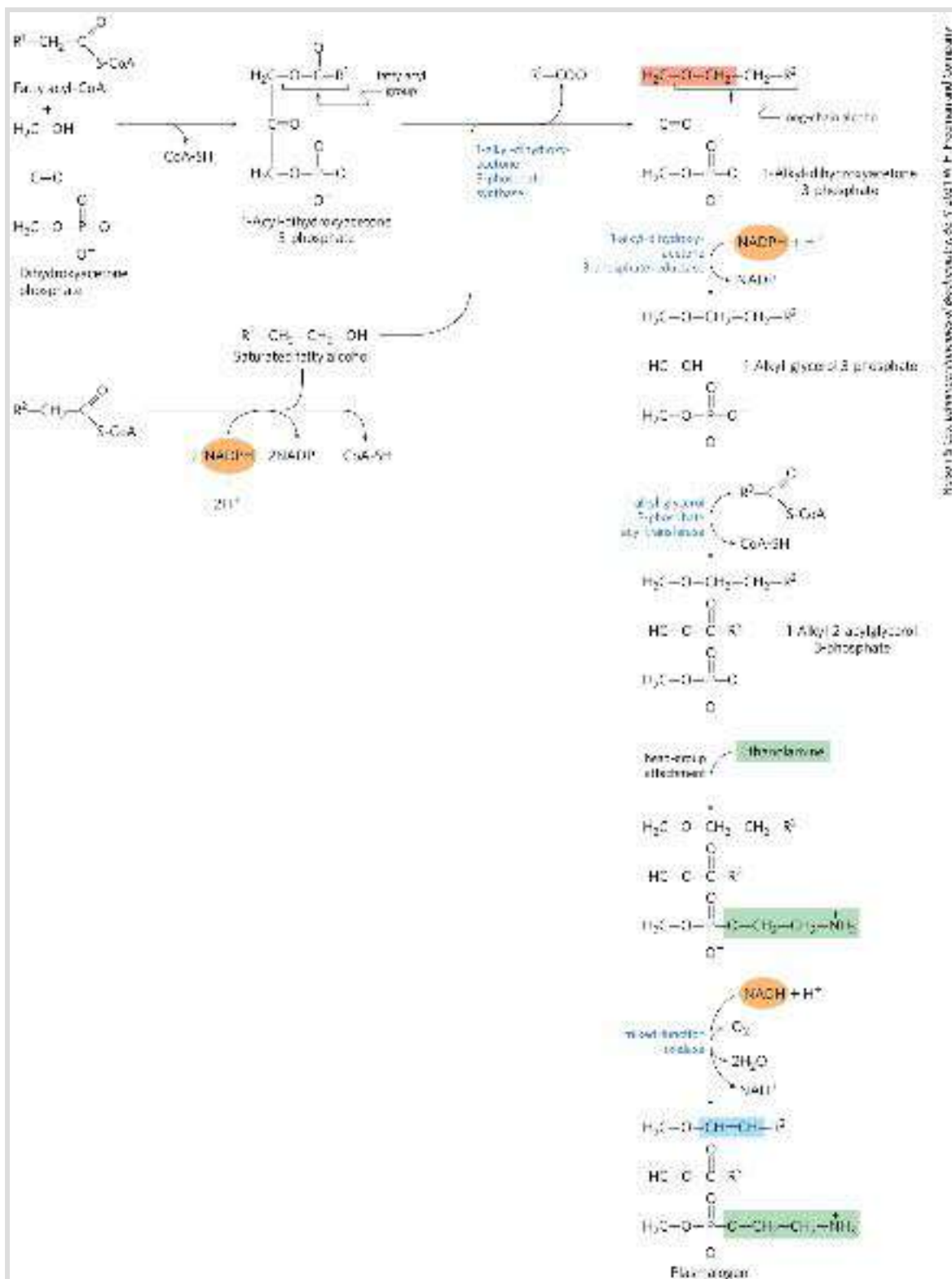


FIGURE 21-30 Synthesis of ether lipids and plasmalogens. The newly formed ether linkage is shaded light red. The intermediate 1-alkyl-2-acylglycerol 3-phosphate is the ether analog of phosphatidic acid. Mechanisms for attaching head groups to ether lipids are essentially the same as for their ester-linked analogs. The characteristic double bond

of plasmalogens (shaded blue) is introduced in a final step by a mixed-function oxidase system similar to fatty acyl-CoA desaturase.

Sphingolipid and Glycerophospholipid Synthesis Share Precursors and Some Mechanisms

 The biosynthesis of sphingolipids takes place in four stages: (1) synthesis of the 18-carbon amine **sphinganine** from palmitoyl-CoA and serine; (2) attachment of a fatty acid in amide linkage to yield **N-acylsphinganine**; (3) desaturation of the sphinganine moiety to form **N-acylsphingosine** (ceramide); and (4) attachment of a head group to produce a sphingolipid such as a **cerebroside** or **sphingomyelin** ([Fig. 21-31](#)). The first few steps of this pathway occur in the ER; the attachment of head groups in stage 4 occurs in the Golgi complex. The pathway shares several features with the pathways leading to glycerophospholipids: NADPH provides reducing power, and fatty acids enter as their activated CoA derivatives. In cerebroside formation, sugars enter as their activated nucleotide derivatives. Head-group attachment in sphingolipid synthesis has several novel aspects. For example, phosphatidylcholine, rather than CDP-choline, serves as the donor of phosphocholine in the synthesis of sphingomyelin.

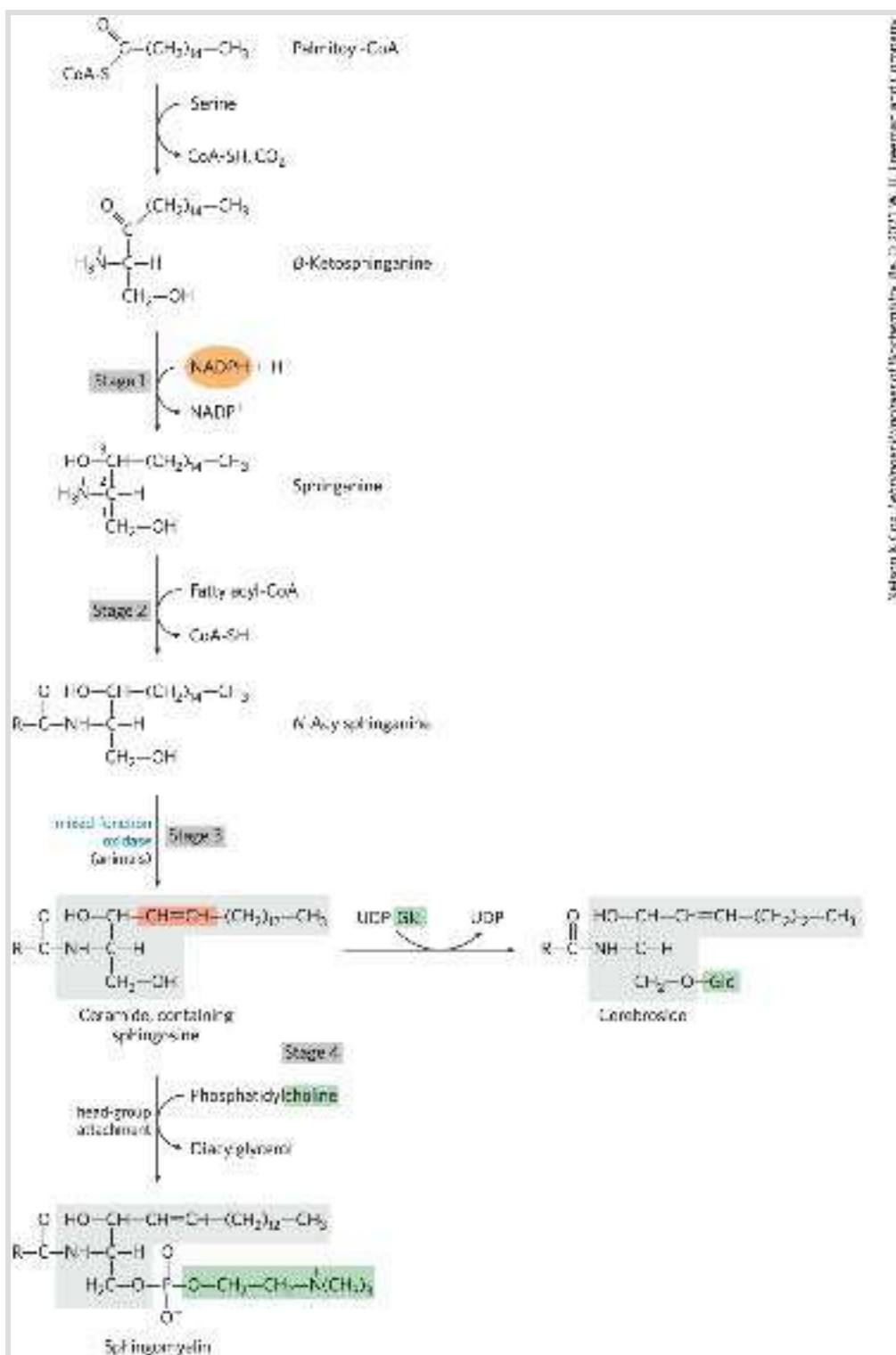


FIGURE 21-31 Biosynthesis of sphingolipids. Condensation of palmitoyl-CoA and serine, forming β -ketosphinganine, followed by reduction with NADPH, yields sphinganine, which is then acylated to *N*-acylsphinganine (a ceramide). The sphingosine is shaded gray. In animals, a double bond (shaded light red) is created by a mixed-function oxidase before the final

addition of a head group: phosphatidylcholine, to form sphingomyelin, or glucose, to form a cerebroside.

P5 In glycolipids — the cerebroside and **gangliosides** (see [Fig. 10-11](#)) — the head-group sugar is attached directly to the C-1 hydroxyl of sphingosine in glycosidic linkage rather than through a phosphodiester bond. The sugar donor is a UDP-sugar (UDP-glucose or UDP-galactose).

Polar Lipids Are Targeted to Specific Cellular Membranes

Membrane lipids are insoluble in water, so they cannot simply diffuse from their point of synthesis (the ER) to their point of insertion. Instead, they are transported from the ER to the Golgi complex, where additional synthesis can take place. They are then delivered in membrane vesicles that bud from the Golgi complex and then move to and fuse with the target membrane (see [Fig. 11-4](#)). Sphingolipid transfer proteins carry ceramide from the ER to the Golgi complex, where sphingomyelin synthesis occurs. Cytosolic proteins also bind phospholipids and sterols and transport them between cellular membranes (see [Fig. 11-7](#)). These mechanisms contribute to establishment of the characteristic lipid compositions of organelle membranes (see [Fig. 11-5](#)).

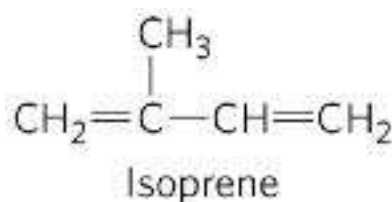
SUMMARY 21.3 *Biosynthesis of Membrane Phospholipids*

- Beginning with diacylglycerol precursors, there are two pathways for adding head groups to phospholipids. Either the diacylglycerol (strategy 1) or the head group (strategy 2) is activated by CDP.
- In eukaryotes, phospholipid biosynthetic strategies vary with subcellular location. The major phospholipids phosphatidylethanolamine and phosphatidylcholine are synthesized using strategy 1 in the mitochondria and strategy 2 in the ER and nucleus. Phosphatidylserine is derived from head-group exchange with phosphatidylethanolamine or phosphatidylcholine.
- In mammals, phospholipids are remodeled in membranes via the Lands cycle. Remodeling is facilitated by lysophosphatidylcholine acyltransferases.
- The characteristic double bond in plasmalogens is introduced by a mixed-function oxidase.
- The head groups of sphingolipids are attached by unique mechanisms.
- Phospholipids travel to their intracellular destinations via transport vesicles or specific proteins.

21.4 Cholesterol, Steroids, and Isoprenoids: Biosynthesis, Regulation, and Transport

Cholesterol is doubtless the most publicized lipid, notorious because of the strong correlation between high levels of cholesterol in the blood and the incidence of human cardiovascular diseases. Less well advertised is cholesterol's crucial role as a component of cellular membranes and as a precursor of steroid hormones and bile acids. Cholesterol is an essential molecule in many animals, including humans, but is not required in the mammalian diet — all cells can synthesize it from simple precursors.

The structure of this 27-carbon compound suggests a complex biosynthetic pathway, but all of its carbon atoms are provided by a single precursor — acetate. The **isoprene** units that are the essential intermediates in the pathway from acetate to cholesterol are also precursors to many other natural lipids, and the mechanisms by which isoprene units are polymerized are similar in all these pathways.



We begin with an account of the main steps in the biosynthesis of cholesterol from acetate, and then discuss the transport of cholesterol in the blood, its uptake by cells, the normal regulation of cholesterol synthesis, and its regulation in those with defects in cholesterol uptake or transport. We next consider other cellular components derived from cholesterol, such as bile acids and steroid hormones. Finally, an outline of the biosynthetic pathways to some of the many compounds derived from isoprene units, which share early steps with the pathway to cholesterol, illustrates the extraordinary versatility of isoprenoid condensations in biosynthesis.

Cholesterol Is Made from Acetyl-CoA in Four Stages

 Cholesterol, like long-chain fatty acids, is made from acetyl-CoA. But the assembly plan of cholesterol is quite different from that of long-chain fatty acids. In

early experiments, animals were fed acetate labeled with ^{14}C in either the methyl carbon or the carboxyl carbon. The pattern of labeling in the cholesterol isolated from the two groups of animals in these tracer experiments ([Fig. 21-32](#)) provided the blueprint for working out the enzymatic steps in cholesterol biosynthesis.

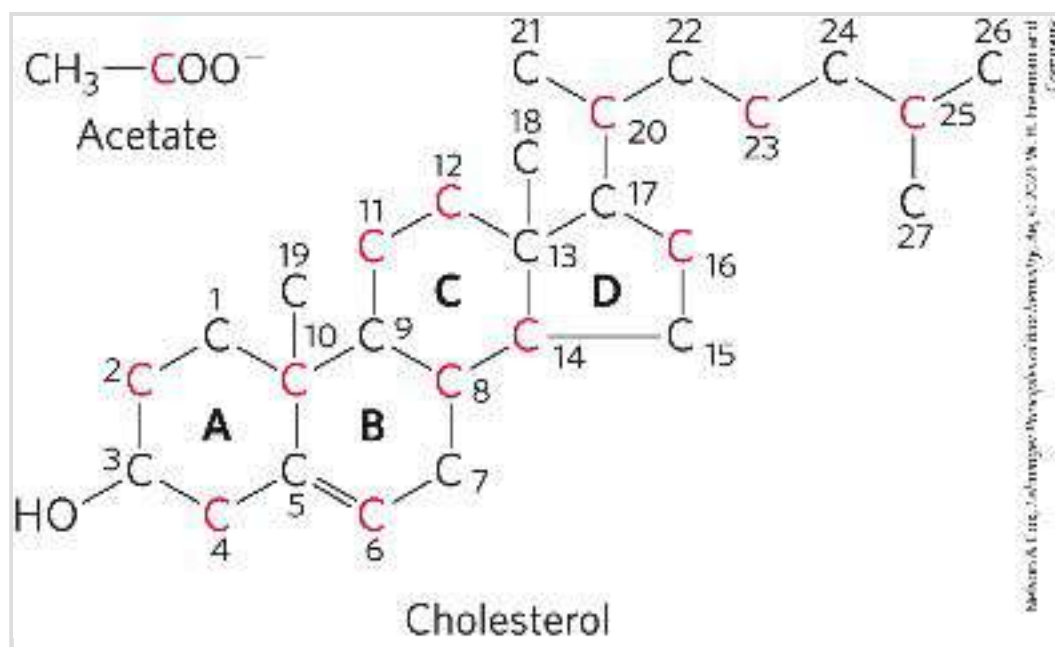
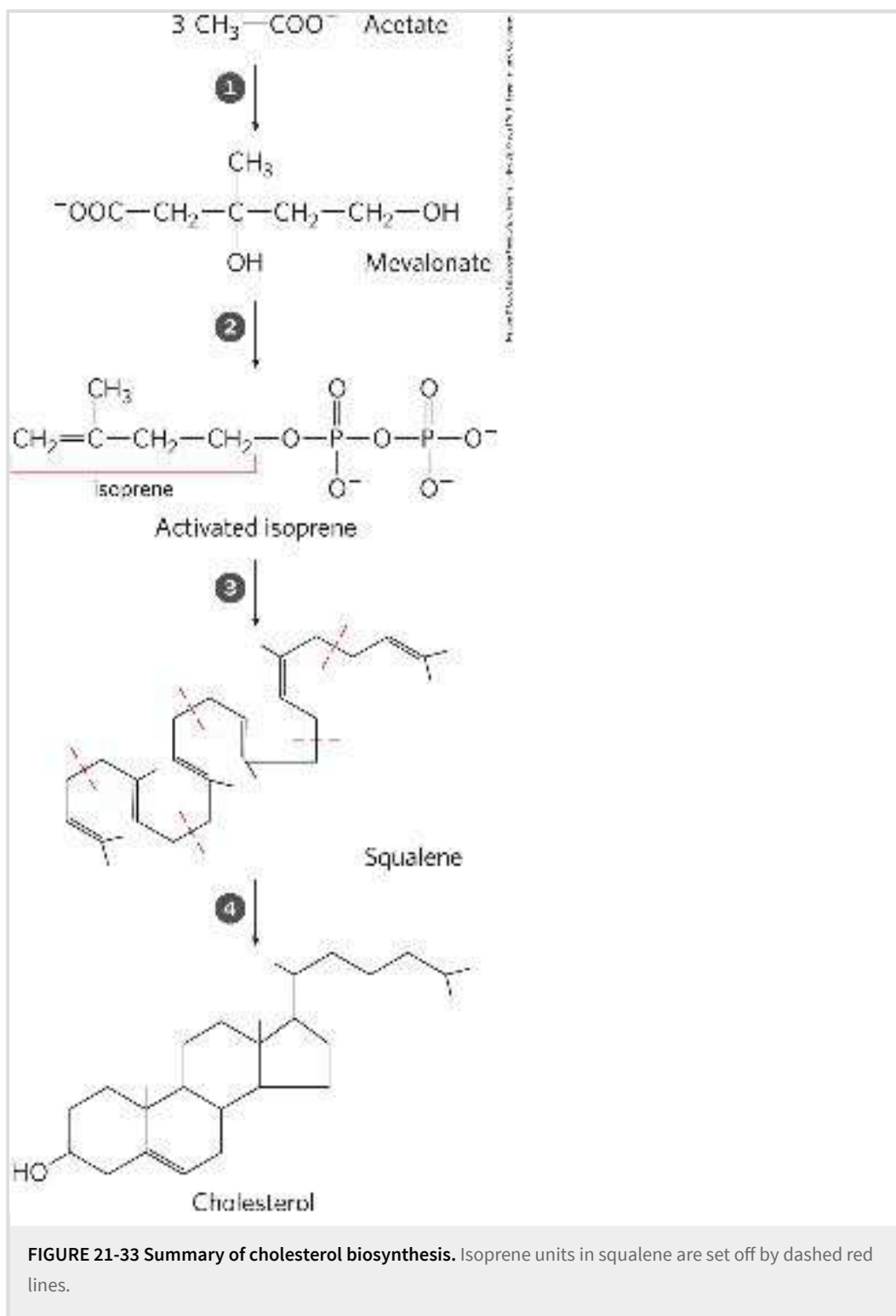


FIGURE 21-32 Origin of the carbon atoms of cholesterol. This was deduced from tracer experiments with acetate labeled in the methyl carbon (black) or the carboxyl carbon (red). The individual rings in the fused-ring system are designated A through D.

Synthesis takes place in four stages, as shown in [Figure 21-33](#): ❶ condensation of three acetate units to form a six-carbon intermediate, mevalonate; ❷ conversion of mevalonate to activated isoprene units; ❸ polymerization of six 5-carbon isoprene units to form the 30-carbon linear squalene; and ❹ cyclization of squalene to form the four rings of the steroid nucleus, with a further series of changes (oxidations, removal or migration of methyl groups) to produce cholesterol.



Stage 1 Synthesis of Mevalonate from Acetate

The first stage in cholesterol biosynthesis leads to the intermediate **mevalonate** ([Fig. 21-34](#)). Two molecules of acetyl-CoA condense to form acetoacetyl-CoA, which condenses with a third molecule of acetyl-CoA to yield the six-carbon compound β -

hydroxy- β -methylglutaryl-CoA (HMG-CoA). These first two reactions are catalyzed by **acetyl-CoA acetyl transferase** and **HMG-CoA synthase**, respectively. Both reactions are Claisen condensations, and the standard equilibrium in each case favors degradation to acetyl-CoA. However, in cells, the synthetic reactions are facilitated by the rapid utilization of the product HMG-CoA in subsequent reactions. The cytosolic HMG-CoA synthase in this pathway is distinct from the mitochondrial isozyme that catalyzes HMG-CoA synthesis in ketone body formation (see [Fig. 17-16](#)).

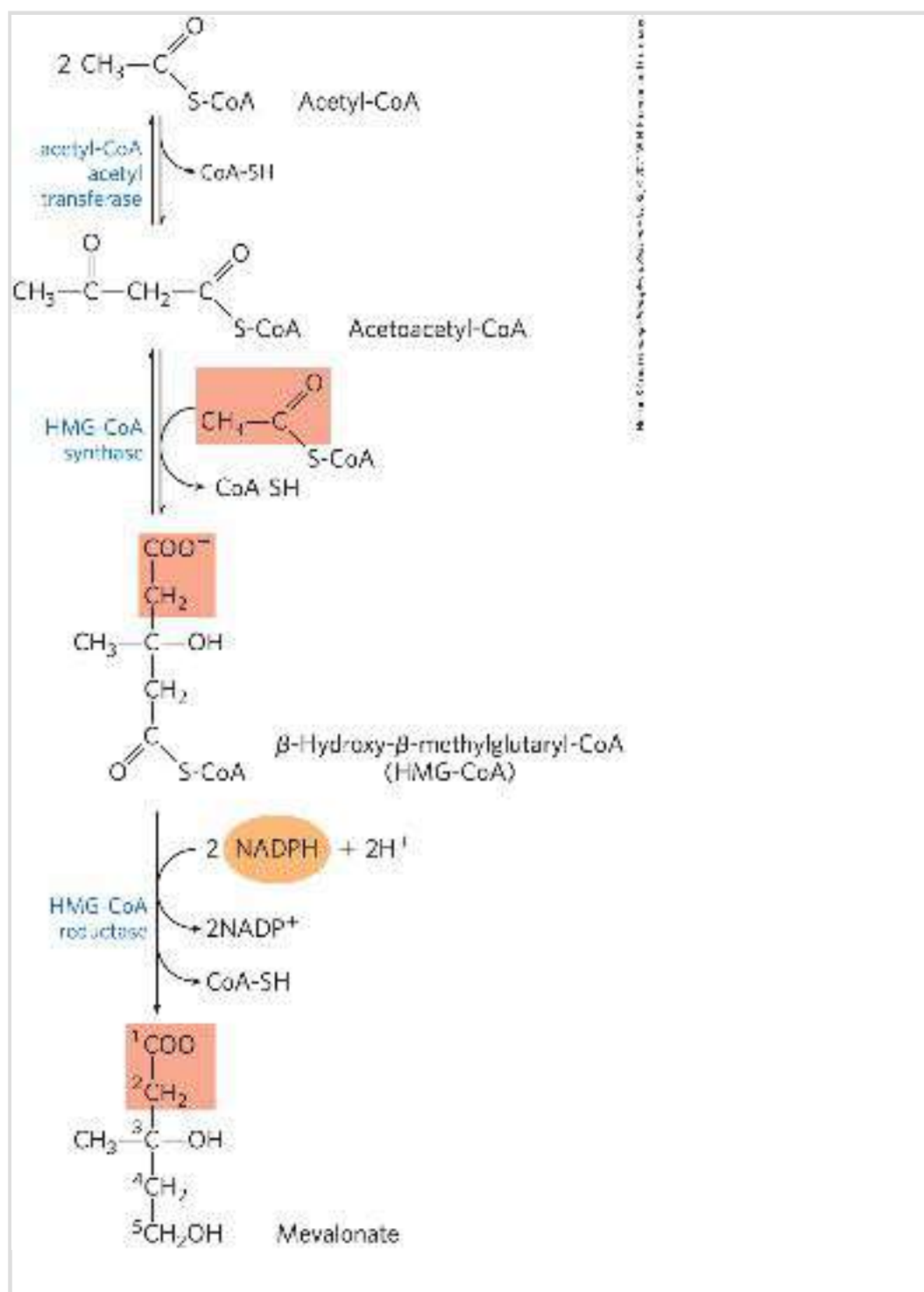


FIGURE 21-34 Formation of mevalonate from acetyl-CoA. The origin of C-1 and C-2 of mevalonate from acetyl-CoA is shaded light red.

 The third reaction is the committed step: reduction of HMG-CoA to mevalonate, for which two molecules of NADPH each donate two electrons. **HMG-CoA reductase**, an integral membrane protein of the smooth ER, is the major point of regulation on the pathway to cholesterol, as we shall see.

Stage 2 Conversion of Mevalonate to Two Activated Isoprenes

In the next stage, three phosphate groups are transferred from three ATP molecules to mevalonate ([Fig. 21-35](#)). The phosphate attached to the C-3 hydroxyl group of mevalonate in the intermediate 3-phospho-5-pyrophosphomevalonate is a good leaving group; in the next step, both this phosphate and the nearby carboxyl group leave, producing a double bond in the five-carbon product, **Δ^3 -isopentenyl pyrophosphate**. This is the first of the two activated isoprenes central to cholesterol formation.

Isomerization of Δ^3 -isopentenyl pyrophosphate yields the second activated isoprene, **dimethylallyl pyrophosphate**. Synthesis of isopentenyl pyrophosphate in the cytoplasm of plant cells follows the pathway described here. However, plant chloroplasts and many bacteria use a mevalonate-independent pathway. This alternative pathway does not occur in animals, so it is an attractive target for the development of new antibiotics.

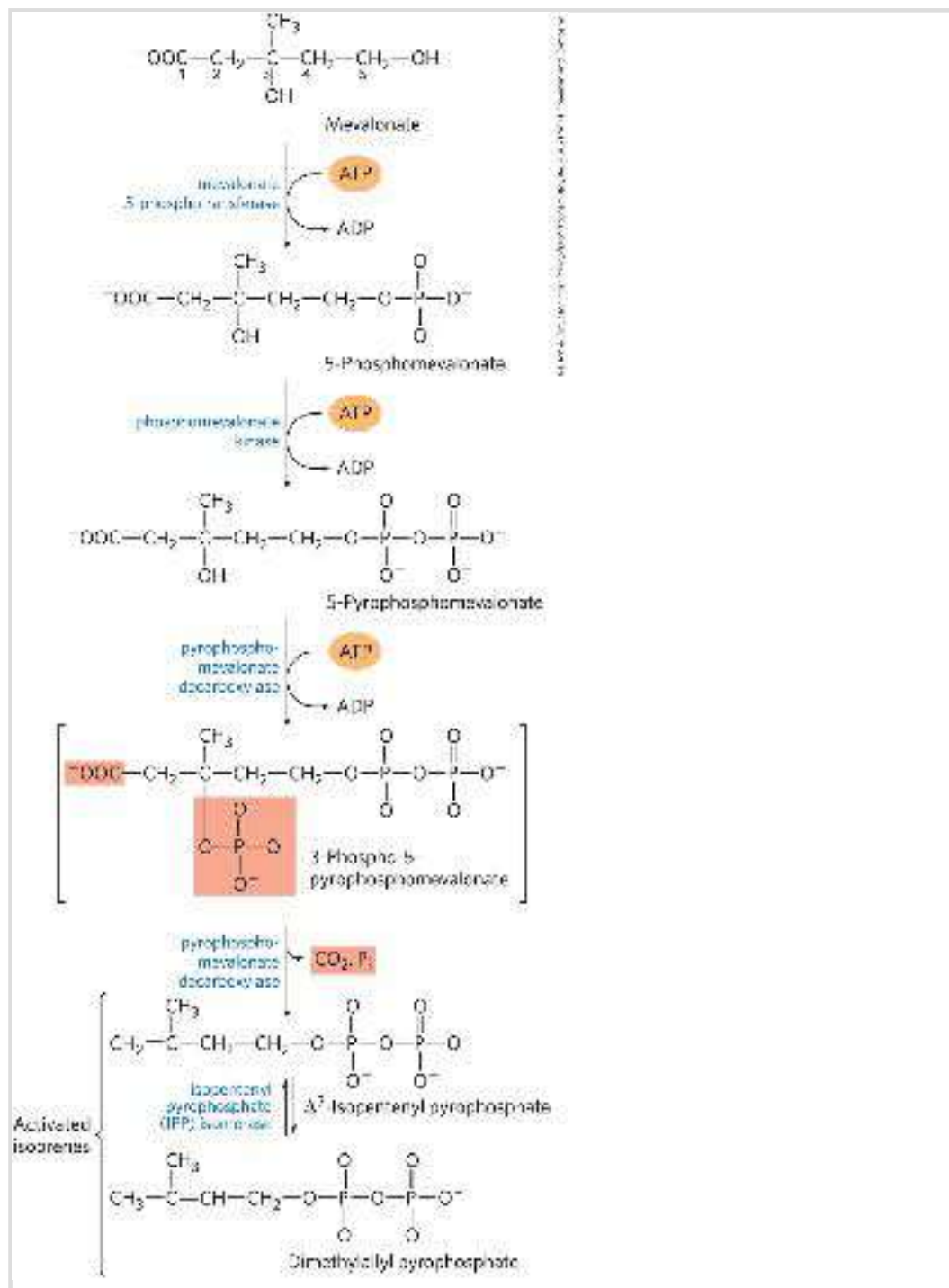
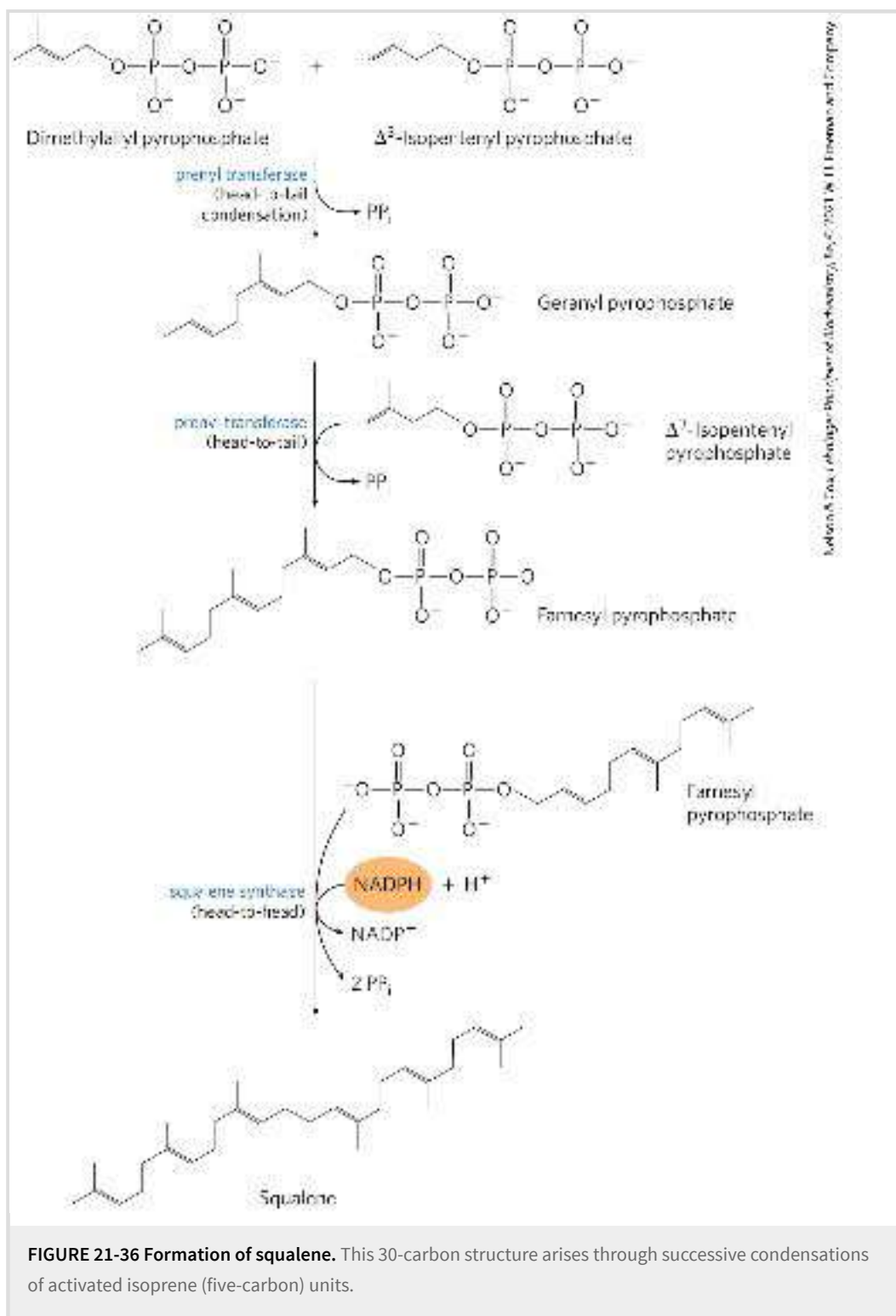


FIGURE 21-35 Conversion of mevalonate to activated isoprene units. Six of these activated units combine to form squalene (see Fig. 21-36). The leaving groups of 3-phospho-5-pyrophosphomevalonate are shaded light red. The bracketed intermediate is hypothetical. Both of these isoprene products are required for the next stage in cholesterol biosynthesis.

Stage 3 Condensation of Six Activated Isoprene Units to Form Squalene

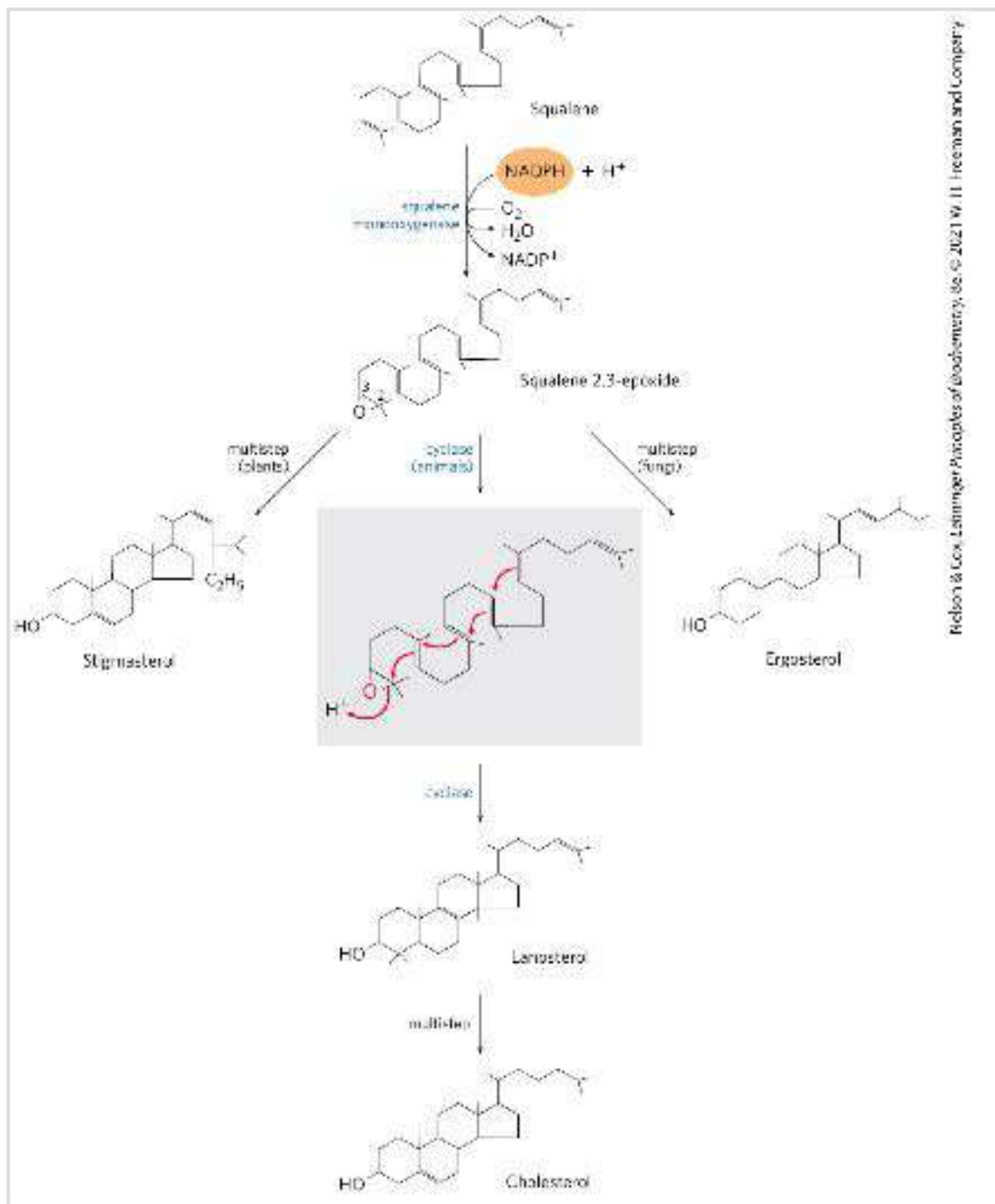
Isopentenyl pyrophosphate and dimethylallyl pyrophosphate now undergo a head-to-tail condensation, in which one pyrophosphate group is displaced and a 10-carbon chain, **geranyl pyrophosphate**, is formed ([Fig. 21-36](#)). (The “head” is the end to which pyrophosphate is joined.) Geranyl pyrophosphate undergoes another head-to-tail condensation with isopentenyl pyrophosphate, yielding the 15-carbon intermediate **farnesyl pyrophosphate**. Finally, two molecules of farnesyl pyrophosphate join head to head, with the elimination of both pyrophosphate groups, to form **squalene**. Squalene has 30 carbons: 24 in the main chain and 6 in the form of methyl group branches.



The common names of these intermediates derive from the sources from which they were first isolated. Geraniol, a component of rose oil, has the aroma of geraniums, and farnesol is an aromatic compound found in flowers of the Farnese acacia tree. Many natural scents of plant origin are synthesized from isoprene units. Squalene was first isolated from the liver of sharks (genus *Squalus*).

Stage 4 Conversion of Squalene to the Four-Ring Steroid Nucleus

When the squalene molecule is represented as in [Figure 21-37](#), the relationship of its linear structure to the cyclic structure of the sterols becomes apparent. All sterols have the four fused rings that form the steroid nucleus, and all are alcohols, with a hydroxyl group at C-3 — thus the name “sterol.” The action of **squalene monooxygenase** adds one oxygen atom from O_2 to the end of the squalene chain, forming an epoxide. This enzyme is another mixed-function oxidase; NADPH reduces the other oxygen atom of O_2 to H_2O . The double bonds of the product, **squalene 2,3-epoxide**, are positioned so that a remarkable concerted reaction can convert the linear squalene epoxide to a cyclic structure. In animal cells, this cyclization results in the formation of **lanosterol**, which contains the four rings characteristic of the steroid nucleus. Lanosterol is finally converted to cholesterol in a series of about 20 reactions that include the migration of some methyl groups and the removal of others. Elucidation of this extraordinary biosynthetic pathway, one of the most complex known, was accomplished by Konrad Bloch, Feodor Lynen, John Cornforth, and George Popják in the late 1950s.



Reelson & Cox, Lehninger Principles of Biochemistry, 8e, © 2021 W. H. Freeman and Company

FIGURE 21-37 Ring closure converts linear squalene to the condensed steroid nucleus. The first step in this sequence is catalyzed by a mixed-function oxygenase, for which the cosubstrate is NADPH. The product is an epoxide, which in the next step is cyclized to the steroid nucleus. The final product of these reactions in animal cells is cholesterol; in other organisms, slightly different sterols are produced, as shown.

Cholesterol is the sterol characteristic of animal cells; plants, fungi, and protists make other, closely related sterols instead. They use the same synthetic pathway as far as squalene 2,3-epoxide, at which point the pathways diverge slightly, yielding other sterols, such as stigmasterol in many plants and ergosterol in fungi ([Fig. 21-37](#)).

WORKED EXAMPLE 21-1 *Energetic Cost of Squalene Synthesis*

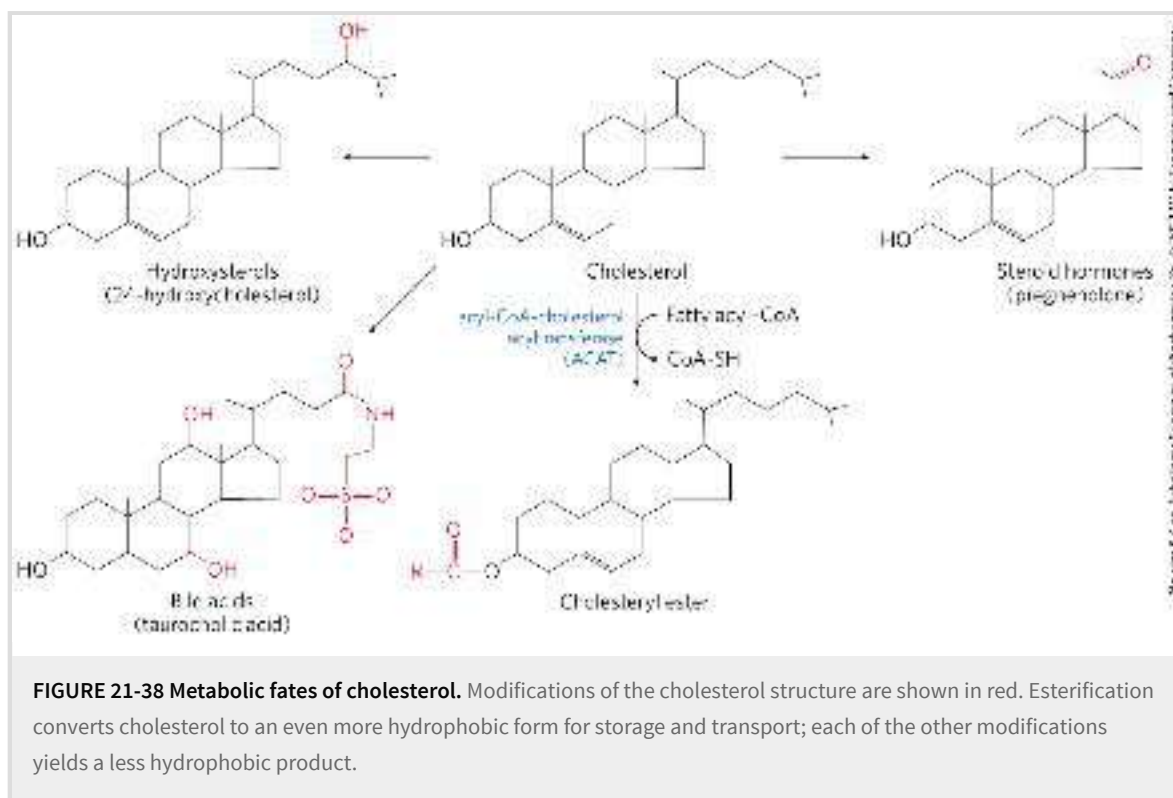
What is the energetic cost of the synthesis of squalene from acetyl-CoA, in number of ATPs per molecule of squalene synthesized?

SOLUTION:

In the pathway from acetyl-CoA to squalene, ATP is consumed only in the steps that convert mevalonate to the activated isoprene precursors of squalene. Three ATP molecules are used to create each of the six activated isoprenes required to construct squalene, for a total cost of 18 ATP molecules.

Cholesterol Has Several Fates

Most of the cholesterol synthesis in vertebrates takes place in the liver. A small fraction of the cholesterol made there is incorporated into the membranes of hepatocytes, but most of it is exported in one of three forms: as bile acids, as biliary cholesterol, or as cholesteryl esters ([Fig. 21-38](#)). Small quantities of oxysterols such as 25-hydroxycholesterol are formed in the liver and act as regulators of cholesterol synthesis (see below). In other tissues, cholesterol is converted into steroid hormones (in the adrenal cortex and gonads, for example; see [Fig. 10-18](#)) or into vitamin D hormone (in the liver and kidney; see [Fig. 10-19](#)). Such hormones are extremely potent biological signals acting through nuclear receptor proteins.



Bile acids, one of the three forms of cholesterol exported from the liver, are the principal components of bile, a fluid stored in the gallbladder and excreted into the small intestine to aid in the digestion of fat-containing meals. Bile acids and their salts are relatively hydrophilic cholesterol derivatives that serve as emulsifiers in the intestine, converting large particles of fat into tiny micelles and thereby greatly increasing the surface at which digestive lipases can act (see [Fig. 17-1](#)). Bile also contains much smaller amounts of cholesterol (biliary cholesterol). Bile helps remove excess cholesterol from the intestine and facilitates excretion. Dietary fiber can enhance this effect by binding to bile and interfering with bile reabsorption in the intestines, leading to increased bile excretion in the feces. More cholesterol is then used to make bile. The soluble fiber available in oats (oatmeal) and barley is especially effective.

Cholesteryl esters are formed in the liver through the action of **acyl-CoA-cholesterol acyltransferase (ACAT)**. This enzyme catalyzes the transfer of a fatty acid from coenzyme A to the hydroxyl group of cholesterol ([Fig. 21-38](#)), converting the cholesterol to a more hydrophobic form that is no longer sufficiently amphipathic to function appropriately in membranes. Cholesteryl esters are transported in secreted lipoprotein particles to other tissues that use cholesterol, or they are stored in the liver in lipid droplets.

Cholesterol and Other Lipids Are Carried on Plasma Lipoproteins

Cholesterol and cholesteryl esters, like triacylglycerols and phospholipids, are essentially insoluble in water, yet they must be moved from the tissue of origin to the tissues in which they will be stored or consumed. They are carried in the blood plasma as **plasma lipoproteins**, macromolecular complexes of specific carrier proteins, called **apolipoproteins**, and various combinations of phospholipids, cholesterol, cholesteryl esters, and triacylglycerols.

Apolipoproteins (“apo” designates the protein in its lipid-free form) combine with lipids to form several classes of lipoprotein particles, spherical complexes with hydrophobic lipids in the core and hydrophilic amino acid side chains at the surface (**Fig. 21-39a, b**). Different combinations of lipids and proteins produce particles of different densities, ranging from chylomicrons to high-density lipoproteins. These particles can be separated by ultracentrifugation (**Table 21-1**) and visualized by electron microscopy (**Fig. 21-39c**).

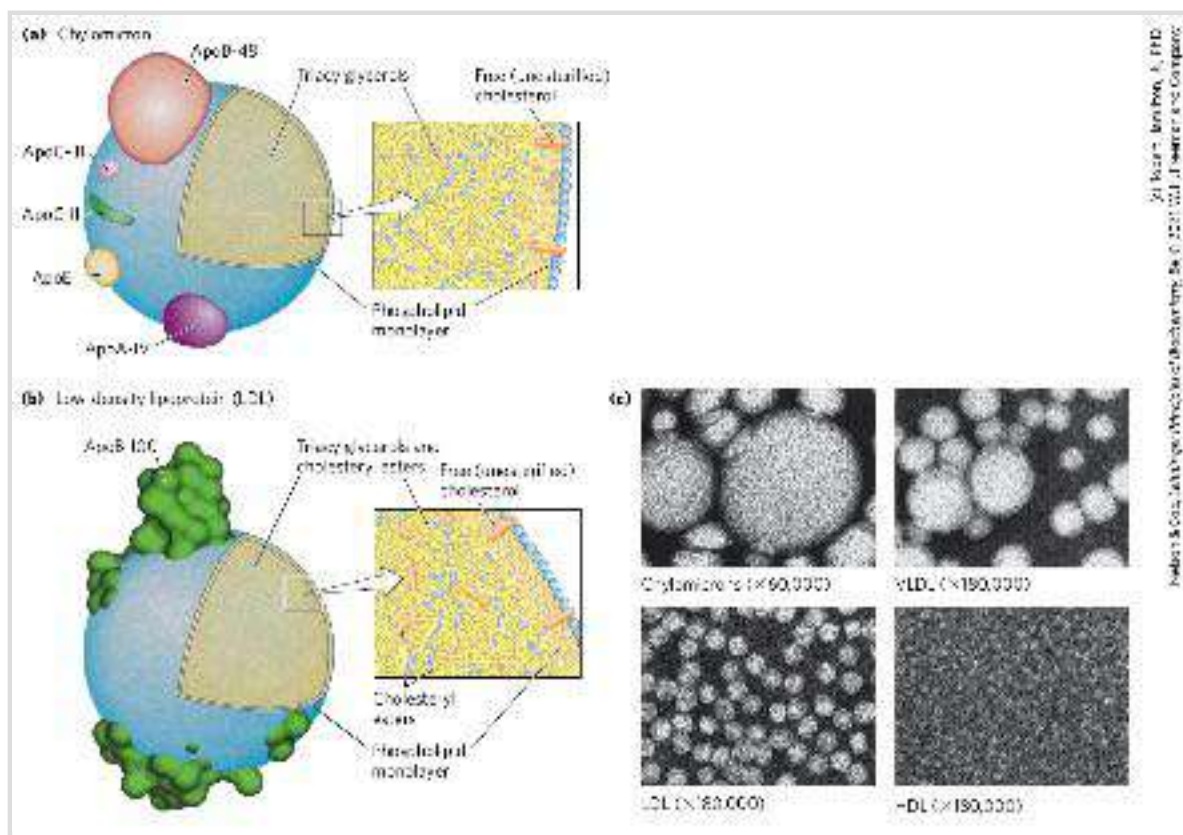


FIGURE 21-39 Lipoproteins. (a) Structure of a chylomicron. Apolipoprotein B-48 (apoB-48) defines the chylomicron. Through the life cycle of a chylomicron other apolipoproteins, including apoC-II, apoC-III, and apoE, become part of the particle, acting as signals in the uptake and metabolism of chylomicron contents. Chylomicrons range from about 100

to 500 nm in diameter. (b) Structure of a low-density lipoprotein (LDL). Apolipoprotein B-100 (apoB-100) is one of the largest single polypeptide chains known, with 4,636 amino acid residues (M_r 512,000). One particle of LDL contains a core with about 1,500 molecules of cholesteryl esters, surrounded by a shell composed of about 500 more molecules of cholesterol, 800 molecules of phospholipids, and one molecule of apoB-100. (c) Four classes of lipoproteins, visualized in the electron microscope after negative staining. The chylomicrons shown here are 50 to 200 nm in diameter; VLDL, 28 to 70 nm; LDL, 20 to 25 nm; and HDL, 8 to 11 nm. Particle sizes given are those measured for these samples; particle sizes vary considerably in different preparations. For properties of lipoproteins, see [Table 21-1](#). [(b) Data for apoB-100 from A. Johs et al., *J. Biol. Chem.* 281:19,732, 2006. (c) Robert Hamilton, Jr., PhD.]

TABLE 21-1 Major Classes of Human Plasma Lipoproteins: Some Properties

Lipoprotein	Density (g/mL)	Composition (wt %)				
		Protein	Phospholipids	Free cholesterol	Cholesteryl esters	Triacylglycerols
Chylomicrons	<1.006	2	9	1	3	85
VLDL	0.95–1.006	10	18	7	12	50
LDL	1.006–1.063	23	20	8	37	10
HDL	1.063–1.210	55	24	2	15	4

Data from D. Kritchevsky, *Nutr. Int.* 2:290, 1986.

Each class of lipoprotein has a specific function, determined by its point of synthesis, lipid composition, and apolipoprotein content. At least 10 distinct apolipoproteins are found in the lipoproteins of human plasma ([Table 21-2](#)), distinguishable by their size, their reactions with specific antibodies, and their characteristic distribution in the lipoprotein classes. These protein components act as signals, targeting lipoproteins to specific tissues or activating enzymes that act on the lipoproteins. [Figure 21-40](#) provides an overview of the formation and transport of the lipoproteins in mammals. The numbered steps in the following discussion refer to this figure.

TABLE 21-2 Apolipoproteins of the Human Plasma Lipoproteins

Apolipoprotein	Polypeptide molecular weight	Lipoprotein association	Function (if known)
ApoA-I	28,100	HDL	Activates LCAT; interacts with ABC transporter
ApoA-II	17,400	HDL	Inhibits LCAT

ApoA-IV	44,500	Chylomicrons, HDL	Activates LCAT; cholesterol transport/clearance
ApoB-48	242,000	Chylomicrons	Cholesterol transport/clearance
ApoB-100	512,000	VLDL, LDL	Binds to LDL receptor
ApoC-I	7,000	VLDL, HDL	
ApoC-II	9,000	Chylomicrons, VLDL, HDL	Activates lipoprotein lipase
ApoC-III	9,000	Chylomicrons, VLDL, HDL	Inhibits lipoprotein lipase
ApoD	32,500	HDL	
ApoE	34,200	Chylomicrons, VLDL, HDL	Triggers clearance of VLDL and chylomicron remnants
ApoH	50,000	Possibly VLDL, binds phospholipids such as cardiolipin	Roles in coagulation, lipid metabolism, apoptosis, inflammation

Information from D. E. Vance and J. E. Vance (eds), *Biochemistry of Lipids and Membranes*, 5th edn, Elsevier Science Publishing, 2008.

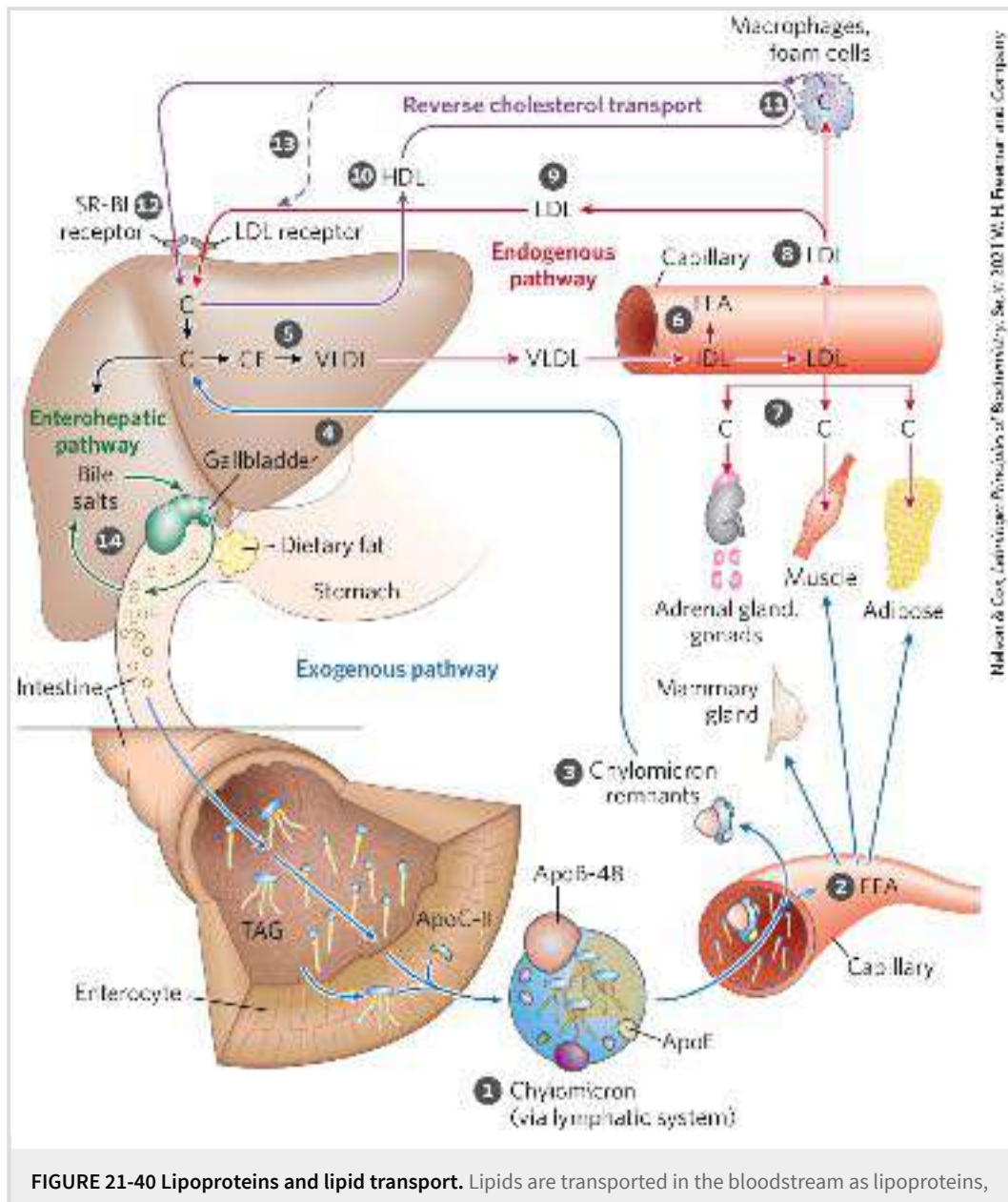


FIGURE 21-40 Lipoproteins and lipid transport. Lipids are transported in the bloodstream as lipoproteins, which exist as several variants that have different functions, different protein and lipid compositions (see [Tables 21-1](#) and [21-2](#)), and thus different densities. Numbered steps are described in the text. In the exogenous pathway (blue arrows), dietary lipids are packaged into chylomicrons; fatty acids from triacylglycerol (TAG) are released by lipoprotein lipase to adipose and muscle tissues, during transport through capillaries. Chylomicron remnants (containing largely protein and cholesterol) are taken up by the liver. Bile salts produced in the liver aid in dispersing dietary fats and are then reabsorbed in the enterohepatic pathway (green arrows). In the endogenous pathway (red arrows), lipids synthesized or packaged in the liver are delivered to peripheral tissues by VLDL. Extraction of lipid from VLDL (along with loss of some apolipoproteins) gradually converts some of it to LDL, which delivers cholesterol to extrahepatic tissues or returns to the liver. Excess cholesterol in extrahepatic tissues is transported back to the liver as HDL in reverse cholesterol transport (purple arrows). C represents cholesterol; CE, cholesteryl ester.

Chylomicrons, discussed in [Chapter 17](#) in connection with the movement of dietary triacylglycerols from the intestine to other tissues, are the first of four classes of

lipoproteins we will discuss. These are the largest of the lipoproteins and the least dense, containing a high proportion of triacylglycerols. ❶ Chylomicrons are synthesized from dietary fats in the ER of enterocytes, epithelial cells that line the small intestine. The chylomicrons then move through the lymphatic system and enter the bloodstream via the left subclavian vein. The apolipoproteins of chylomicrons include apoA-IV, apoB-48 (unique to this class of lipoproteins), apoE, apoC-II, and apoC-III ([Table 21-2](#)). ❷ ApoC-II activates lipoprotein lipase in the capillaries of adipose, heart, skeletal muscle, and lactating mammary tissues, allowing the release of free fatty acids (FFA) to these tissues. Chylomicrons thus carry dietary fatty acids to tissues where they will be consumed or stored as fuel. ❸ The remnants of chylomicrons, depleted of most of their triacylglycerols but still containing cholesterol, apoE, and apoB-48, move through the bloodstream to the liver. Receptors in the liver bind to the apoE in the chylomicron remnants and mediate uptake of these remnants by endocytosis. ❹ In the liver, the remnants release their cholesterol and are degraded in lysosomes. This pathway from dietary cholesterol to the liver is the **exogenous pathway** (blue arrows in [Fig. 21-40](#)).

Very-low-density lipoprotein (VLDL) is the second of the four classes. When the diet contains more fatty acids and cholesterol than are needed immediately as fuel or as precursors to other molecules, ❺ they are converted to triacylglycerols or cholesteryl esters in the liver and packaged with specific apolipoproteins into VLDL. Excess carbohydrate in the diet can also be converted to triacylglycerols in the liver and exported as VLDL. In addition to triacylglycerols and cholesteryl esters, VLDL contains apoB-100, apoC-I, apoC-II, apoC-III, and apoE ([Table 21-2](#)). VLDL is transported in the blood from the liver to muscle and adipose tissue. ❻ In the capillaries of these tissues, apoC-II activates lipoprotein lipase, which catalyzes the release of free fatty acids from triacylglycerols in the VLDL. Adipocytes take up these fatty acids, reconvert them to triacylglycerols, and store the products in intracellular lipid droplets; myocytes, in contrast, primarily oxidize the fatty acids to supply energy. When the insulin level is high (after a meal), VLDL serves primarily to convey lipids from the diet to adipose tissue for storage. In the fasting state between meals, the fatty acids used to produce VLDL in the liver originate primarily from adipose tissue, and the principal VLDL target is myocytes of the heart and skeletal muscle.

Low-density lipoprotein (LDL), the third class of lipoprotein, is formed when triacylglycerol loss converts some VLDL to VLDL remnants, also called intermediate-density lipoprotein (IDL). Further removal of triacylglycerol from IDL (remnants) produces LDL. Rich in cholesterol and cholesteryl esters, and containing apoB-100 as its

major apolipoprotein, ⑦ LDL carries cholesterol to extrahepatic tissues such as muscle, adrenal glands, and adipose tissue. These tissues have plasma membrane LDL receptors that recognize apoB-100 and mediate uptake of cholesterol and cholesteryl esters. ⑧ LDL also delivers cholesterol to macrophages, sometimes converting them into foam cells (see [Fig. 21-46](#)). ⑨ LDL not taken up by peripheral tissues and cells returns to the liver and is taken up via **LDL receptors** in the hepatocyte plasma membrane. Cholesterol that enters hepatocytes by this path may be incorporated into membranes, converted to bile acids, or reesterified by ACAT ([Fig. 21-38](#)) for storage within cytosolic lipid droplets. This pathway, from VLDL formation in the liver to LDL return to the liver, is the **endogenous pathway** of cholesterol metabolism and transport (red arrows in [Fig. 21-40](#)). Accumulation of excess intracellular cholesterol is prevented by reducing the rate of cholesterol synthesis when sufficient cholesterol is available from LDL in the blood. Regulatory mechanisms to accomplish this are described below.

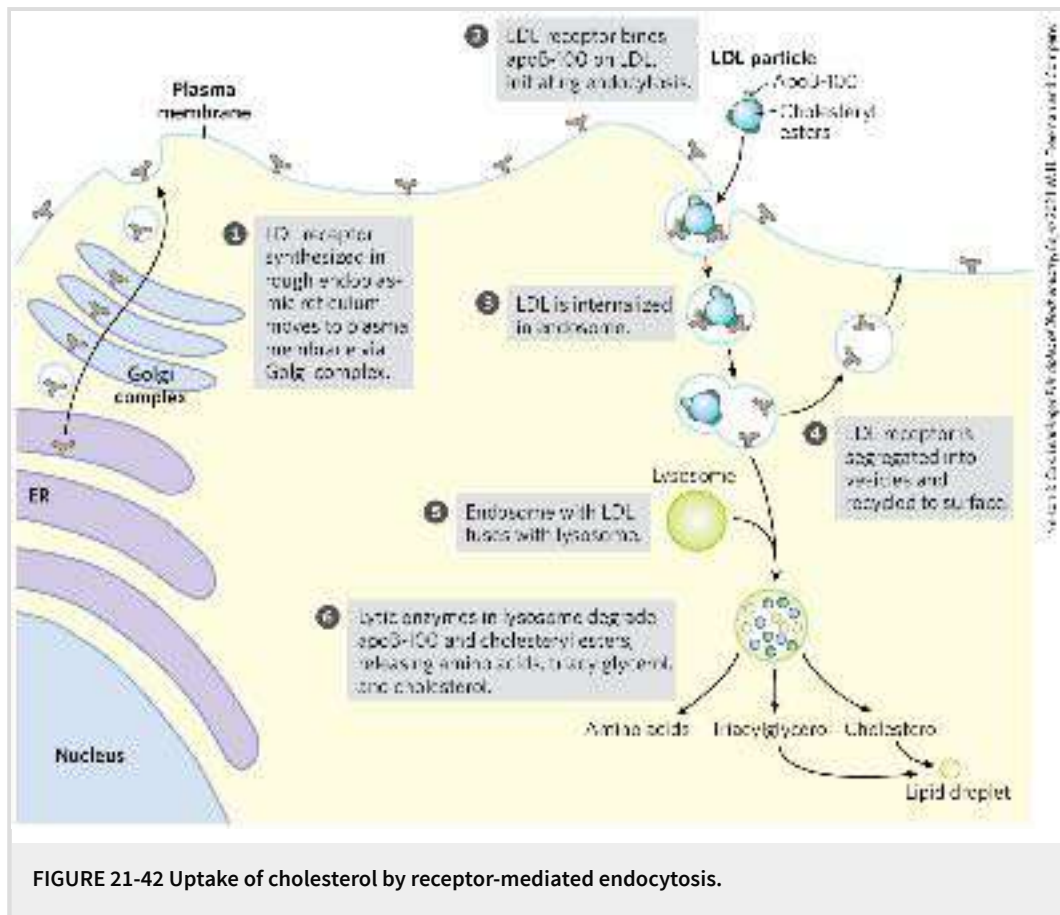
HDL Carries Out Reverse Cholesterol Transport

High-density lipoprotein (HDL), the fourth major lipoprotein in mammals, ⑩ originates in the liver and small intestine as small, protein-rich particles that contain relatively little cholesterol and no cholesteryl esters ([Fig. 21-40](#)). HDLs contain primarily apoA-I and other apolipoproteins ([Table 21-2](#)). They also contain the enzyme **lecithin-cholesterol acyltransferase (LCAT)**, which catalyzes the formation of cholesteryl esters from lecithin (phosphatidylcholine) and cholesterol ([Fig. 21-41](#)). LCAT on the surface of nascent (newly forming) HDL particles converts the cholesterol and phosphatidylcholine of chylomicron and VLDL remnants encountered in the bloodstream to cholesteryl esters, which begin to form a core, transforming the disk-shaped nascent HDL to a mature, spherical HDL particle. ⑪ Nascent HDL can also pick up cholesterol from cholesterol-rich extrahepatic cells (including macrophages and foam cells, formed from macrophages; see below). ⑫ Mature HDL then returns to the liver, where the cholesterol is unloaded via the scavenger receptor SR-BI. ⑬ Some of the cholesteryl esters in HDL can also be transferred to LDL by the cholesteryl ester transfer protein. The HDL circuit is **reverse cholesterol transport** (purple arrows in [Fig. 21-40](#)). Much of this cholesterol is converted to bile salts by enzymes sequestered in hepatic peroxisomes; the bile salts are stored in the gallbladder and excreted into the intestine when a meal is ingested. ⑭ Bile salts are reabsorbed by the liver and recirculate through the gallbladder in this **enterohepatic circulation** (green arrows in [Fig. 21-40](#)).

other lipids in HDL into the cell. Depleted HDL then dissociates to recirculate in the bloodstream and extract more lipids from remnants of chylomicrons and VLDL, and from cells overloaded with cholesterol, as described below.

Cholesteryl Esters Enter Cells by Receptor-Mediated Endocytosis

Each LDL particle in the bloodstream contains apoB-100, which is recognized by LDL receptors present in the plasma membranes of cells that need to take up cholesterol. [Figure 21-42](#) shows such a cell. ❶ LDL receptors are synthesized in the ER and transported to the plasma membrane, after modification in the Golgi complex. At the plasma membrane, they are available to bind apoB-100. ❷ Binding of LDL to an LDL receptor initiates endocytosis, which ❸ conveys the LDL and its receptor into the cell within an endosome. ❹ The receptor-containing portions of the endosome membrane bud off and are returned to the cell surface, to function again in LDL uptake. ❺ The endosome fuses with a lysosome, which ❻ contains enzymes that hydrolyze the cholesteryl esters, releasing cholesterol and fatty acids into the cytosol. The apoB-100 protein is also degraded to amino acids that are released to the cytosol. ApoB-100 is also present in VLDL, but its receptor-binding domain is not available for binding to the LDL receptor; conversion of VLDL to LDL exposes the receptor-binding domain of apoB-100.



This pathway for the transport of cholesterol in blood and its **receptor-mediated endocytosis** by target tissues was elucidated by Michael Brown and Joseph Goldstein. They discovered that individuals with the genetic disease **familial hypercholesterolemia (FH)** have mutations in the LDL receptor that prevent the normal uptake of LDL by liver and peripheral tissues. The result of defective LDL uptake is very high blood levels of LDL (and of the cholesterol it carries). Individuals with FH have a greatly increased probability of developing atherosclerosis, a disease of the cardiovascular system in which blood vessels are occluded by cholesterol-rich plaques (see [Fig. 21-46](#)).



Michael Brown and Joseph Goldstein

Niemann-Pick type-C (NPC) disease is an inherited defect in lipid storage. In this disorder, cholesterol is not transported out of the lysosomes and instead accumulates in lysosomes of liver, brain, and lung, bringing about early death. NPC is the result of a mutation in either of two genes, *NPC1* and *NPC* , essential to moving cholesterol out of the lysosome and into the cytosol, where it can be further metabolized. *NPC1* encodes a transmembrane lysosomal protein, and *NPC* encodes a soluble protein. These proteins act in tandem to transfer cholesterol out of the lysosome and into the cytosol for further processing or metabolism. ■

Cholesterol Synthesis and Transport Are Regulated at Several Levels

Cholesterol synthesis is a complex and energy-expensive process. Excess cholesterol cannot be catabolized for use as fuel and must be excreted. Therefore, it is clearly advantageous to an organism to regulate the biosynthesis of cholesterol to complement dietary intake. In mammals, cholesterol production is regulated by intracellular cholesterol concentration, by the supply of ATP, and by the hormones glucagon and insulin. The committed step in the pathway to cholesterol (and a major site of regulation) is the conversion of HMG-CoA to mevalonate ([Fig. 21-34](#)), the reaction catalyzed by HMG-CoA reductase.

Short-term (minute-to-minute) regulation of the *activity* of existing HMG-CoA reductase is accomplished by reversible covalent alteration: phosphorylation by the AMP-dependent protein kinase (AMPK), which senses high AMP concentration (indicating low ATP concentration). Thus, when ATP levels drop, the synthesis of cholesterol slows, and catabolic pathways for the generation of ATP are stimulated ([Fig. 21-43](#)). Hormones that mediate global regulation of lipid and carbohydrate metabolism also act on HMG-CoA reductase; glucagon stimulates its phosphorylation (inactivation), and insulin promotes dephosphorylation, activating the enzyme and favoring cholesterol synthesis. These covalent regulatory mechanisms are probably not as important, quantitatively, as the mechanisms that affect the synthesis and degradation of the enzyme.

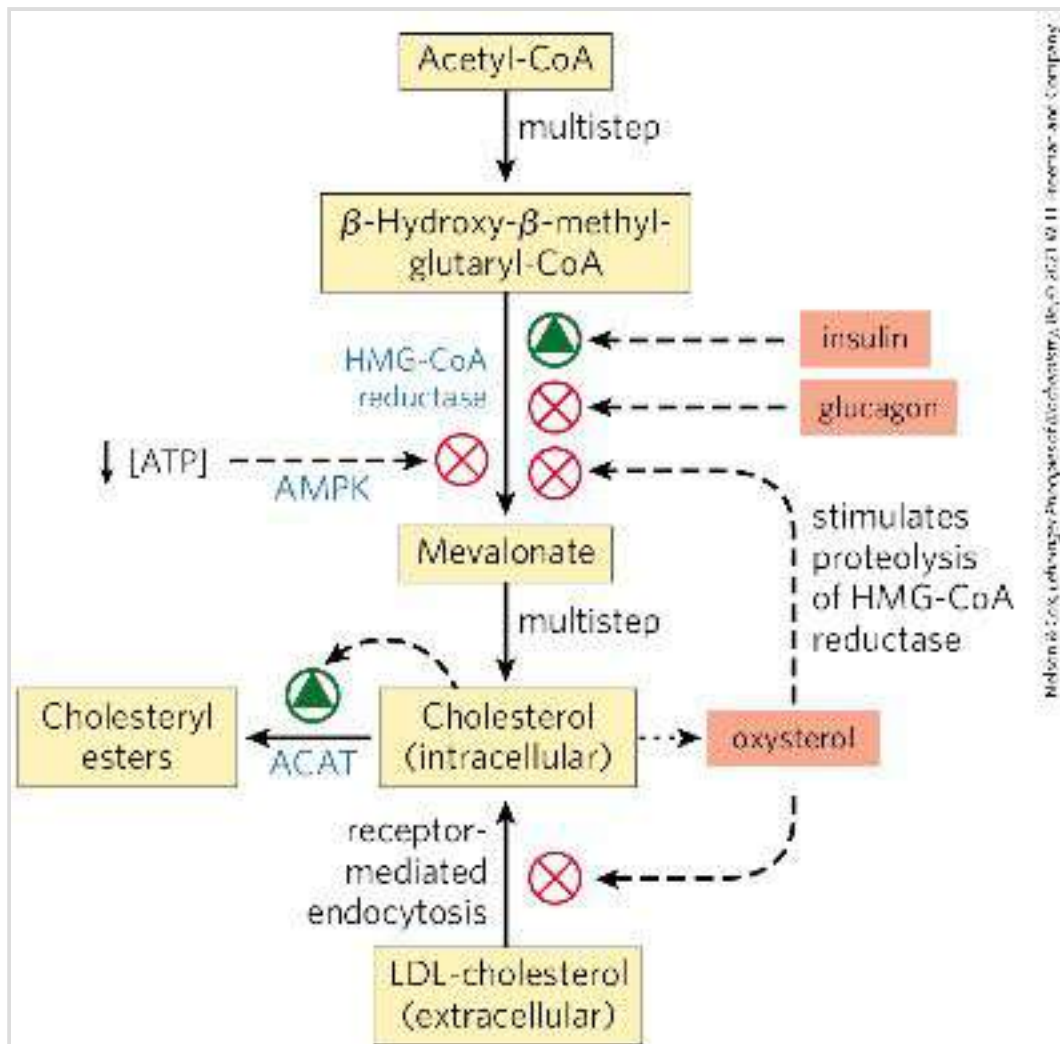
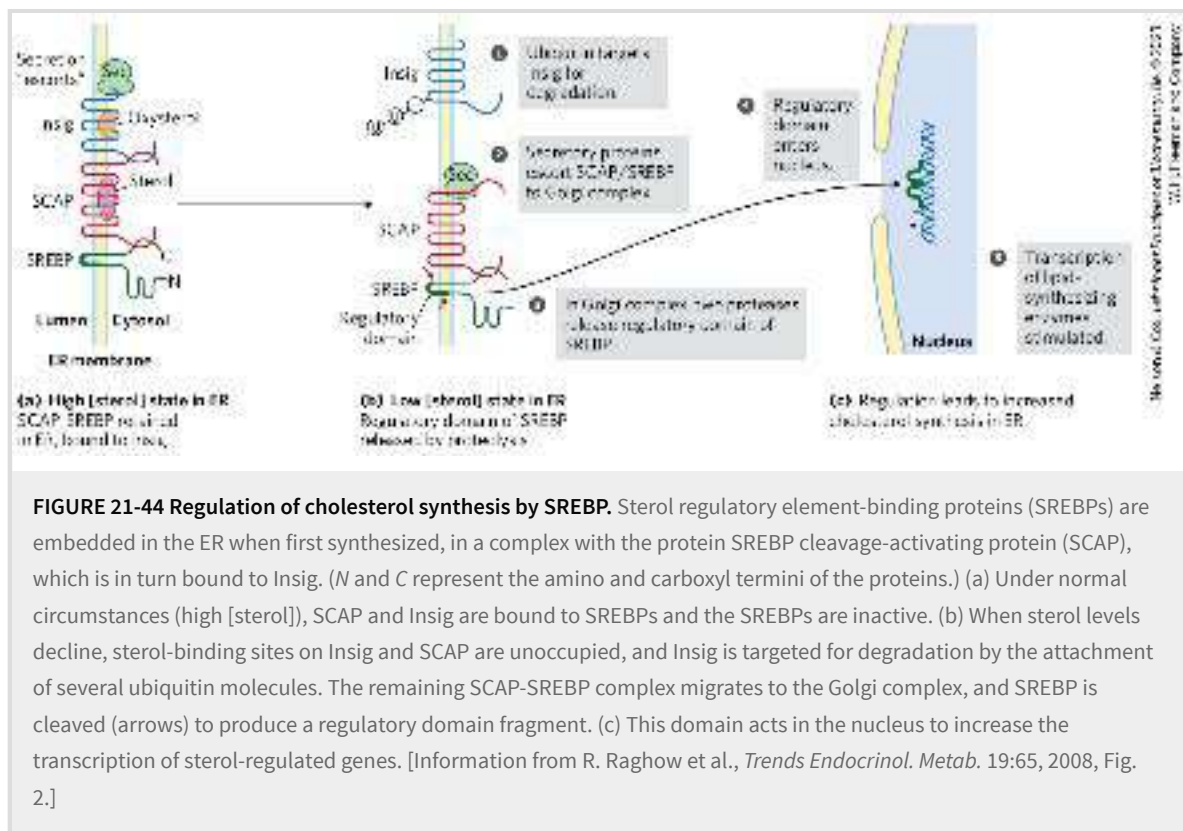


FIGURE 21-43 Regulation of cholesterol formation balances synthesis with dietary uptake and energy state. Insulin promotes dephosphorylation (activation) of HMG-CoA reductase; glucagon promotes its phosphorylation (inactivation); and the AMP-dependent protein kinase AMPK, when activated by low [ATP] relative to [AMP], phosphorylates and inactivates HMG-CoA reductase. Oxysterol metabolites of cholesterol (for example, 24(S)-hydroxycholesterol) stimulate proteolysis of HMG-CoA reductase.

In the longer term, the *number of molecules* of HMG-CoA reductase is increased or decreased in response to cellular concentrations of cholesterol. Regulation of HMG-CoA reductase synthesis by cholesterol is mediated by an elegant system of transcriptional regulation of the HMG-CoA gene ([Fig. 21-44](#)). This gene, along with more than 20 other genes encoding enzymes that mediate the uptake and synthesis of cholesterol and unsaturated fatty acids, is controlled by a small family of proteins called **sterol regulatory element-binding proteins (SREBPs)**. When newly synthesized, these proteins are embedded in the ER. Only the soluble regulatory domain fragment of an SREBP functions as a transcription activator, through mechanisms discussed in [Chapter 28](#). When cholesterol and oxysterol levels are high, SREBPs are held in the ER

in a complex with another protein called **SREBP cleavage-activating protein (SCAP)**, which in turn is anchored in the ER membrane by its interaction with a third membrane protein, **Insig (insulin-induced gene protein)** ([Fig. 21-44a](#)). SCAP and Insig act as sterol sensors. When sterol levels are high, the Insig-SCAP-SREBP complex is retained in the ER membrane. When the level of sterols in the cell declines ([Fig. 21-44b](#)), the SCAP-SREBP complex is escorted by secretory proteins to the Golgi complex. There, two proteolytic cleavages of SREBP release a regulatory fragment, which enters the nucleus and activates transcription of its target genes, including those for HMG-CoA reductase, the LDL receptor protein, and other proteins needed for lipid synthesis. When sterol levels increase sufficiently, the proteolytic release of SREBP amino-terminal domains is again blocked, and proteasome degradation of the existing active domains results in rapid shutdown of the gene targets.



The level of HMG-CoA reductase is also regulated by proteolytic degradation of the enzyme itself. High levels of cellular cholesterol are sensed by Insig, which triggers attachment of ubiquitin molecules to HMG-CoA reductase, leading to its degradation by proteasomes.

Liver X receptor (LXR) is a nuclear transcription factor activated by oxysterol ligands (reflecting high cholesterol levels), which integrates the metabolism of fatty acids,

sterols, and glucose. LXR α is expressed primarily in liver, adipose tissue, and macrophages; LXR β is present in all tissues. When bound to an oxysterol ligand, LXRs form heterodimers with a second type of nuclear receptor, the **retinoid X receptors (RXR)**, and the LXR-RXR dimer activates transcription from a set of genes ([Fig. 21-45](#)), including those for acetyl-CoA carboxylase, the first enzyme in fatty acid synthesis; fatty acid synthase; the cytochrome P-450 enzyme CYP7A1, required for sterol conversion to bile acids; apoproteins that participate in cholesterol transport (apoC-I, apoC-II, apoD, and apoE); the ATP-binding cassette (ABC) transporters ABCA1 and ABCG1, required for reverse cholesterol transport (see below); GLUT4, the insulin-stimulated glucose transporter of muscle and adipose tissue; and an SREBP called SREBP1C. The transcriptional regulators LXR and SREBP therefore work together to achieve and maintain cholesterol homeostasis; SREBPs are activated by low levels of cellular cholesterol, and LXRs are activated by high cholesterol levels.

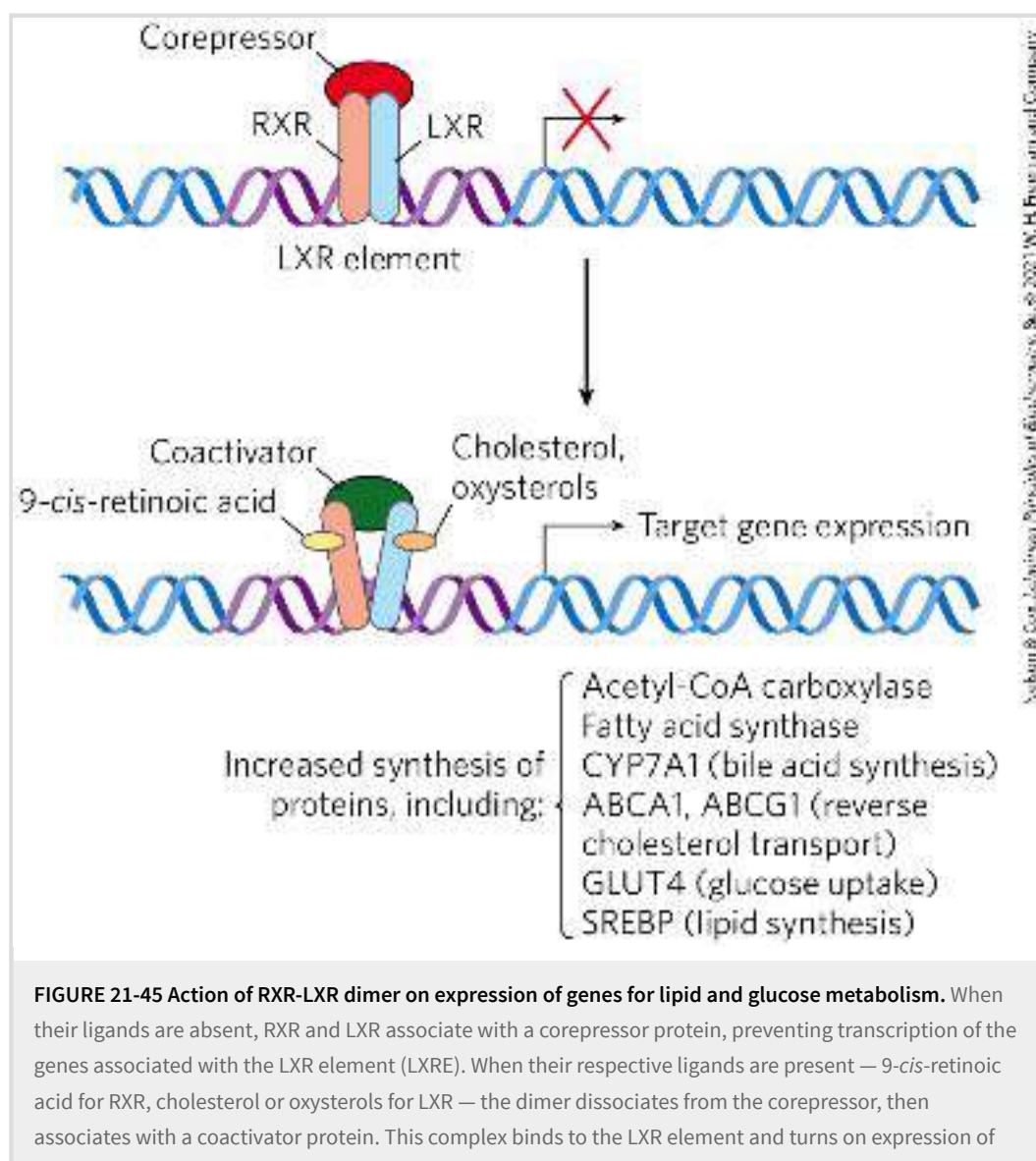


FIGURE 21-45 Action of RXR-LXR dimer on expression of genes for lipid and glucose metabolism. When their ligands are absent, RXR and LXR associate with a corepressor protein, preventing transcription of the genes associated with the LXR element (LXRE). When their respective ligands are present — 9-*cis*-retinoic acid for RXR, cholesterol or oxysterols for LXR — the dimer dissociates from the corepressor, then associates with a coactivator protein. This complex binds to the LXR element and turns on expression of

the associated genes. Regulation of gene expression is a topic discussed in more detail in [Chapter 28](#).
[Information from A. C. Calkin and P. Tontonoz, *Nat. Rev. Mol. Cell Biol.* 13:213, 2012, Fig. 1.]

Regulation by LXRs is complemented by the activity of **farnesoid X receptor (FXR)**, which also forms a heterodimer with RXR, with an effect that is often reciprocal to that of LXR-RXR. Although farnesol is a ligand for this receptor, FXR responds primarily to bile acids. High levels of bile acids can be toxic. FXR, expressed mainly in the intestine, liver, kidney, and adrenal glands, provides essential control of bile acid levels by increasing or decreasing expression of multiple genes. FXR represses many of the genes that are activated by LXR.

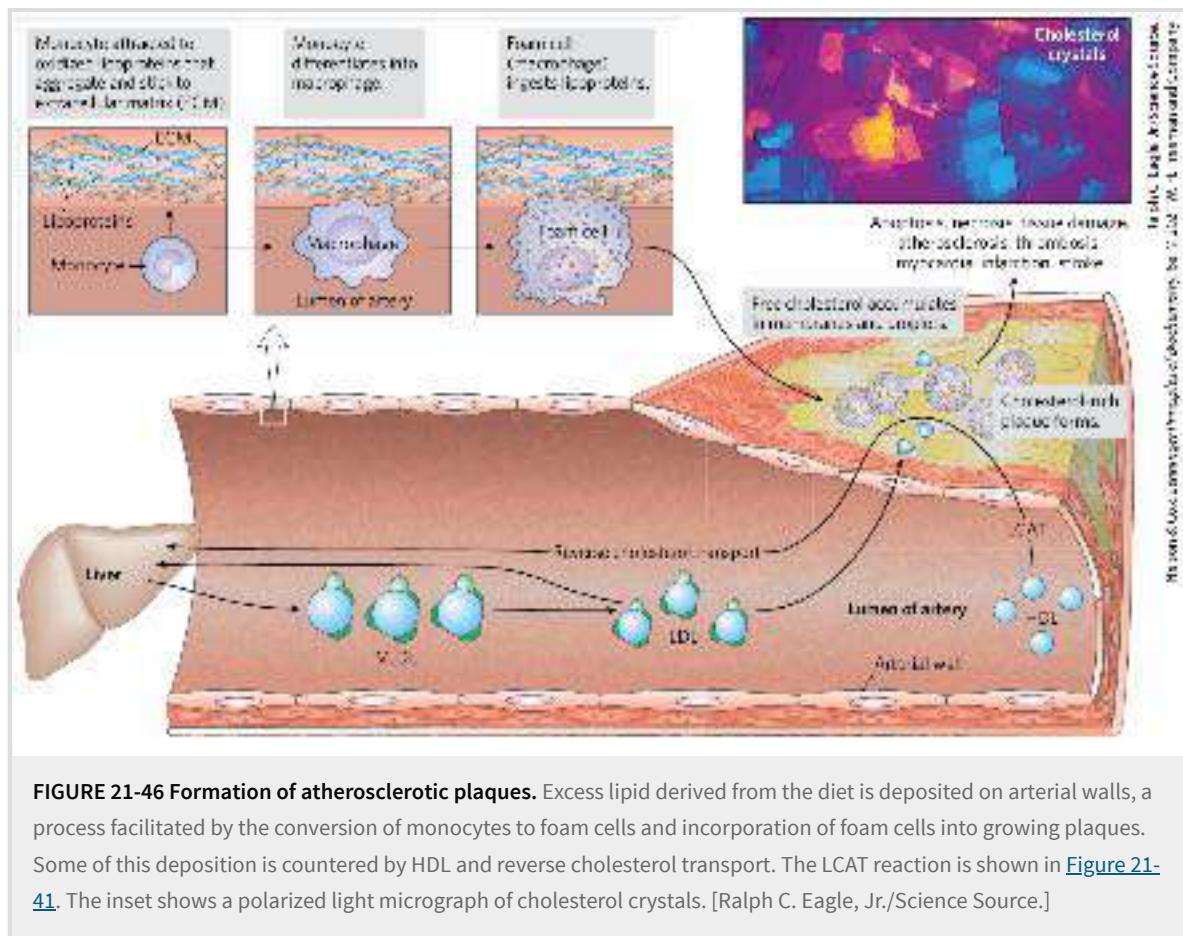
Finally, two other regulatory mechanisms influence cellular cholesterol level: (1) high cellular concentrations of cholesterol activate ACAT, which increases esterification of cholesterol for storage, and (2) high cellular cholesterol levels diminish (via SREBP) transcription of the gene that encodes the LDL receptor, reducing production of the receptor and thus the uptake of cholesterol from the blood.

Dysregulation of Cholesterol Metabolism Can Lead to Cardiovascular Disease



As noted earlier, cholesterol cannot be catabolized by animal cells. Excess cholesterol can be removed only by excretion or by conversion to bile salts. When the sum of cholesterol synthesized and cholesterol obtained in the diet exceeds the amount required for the synthesis of membranes, bile salts, and steroids, pathological accumulations of cholesterol (plaques) can obstruct blood vessels, a condition called **atherosclerosis**. Heart failure due to occluded coronary arteries is a leading cause of death in industrialized societies. Atherosclerosis is linked to high levels of cholesterol in the blood, and particularly to high levels of LDL-cholesterol (“bad cholesterol”); there is a *negative* correlation between HDL (“good cholesterol”) levels and arterial disease. Plaque formation in blood vessels is initiated when LDL containing partially oxidized fatty acyl groups adheres to and accumulates in the extracellular matrix of epithelial cells lining arteries ([Fig. 21-46](#)). Immune cells (monocytes) are attracted to regions with such LDL accumulations, and they differentiate into macrophages, which take up the oxidized LDL and the cholesterol they contain. Macrophages cannot limit their uptake of sterols, and with increasing accumulation of cholesteryl esters and free cholesterol, the macrophages become **foam cells** (they appear foamy when viewed under the microscope). As excess free cholesterol accumulates in foam cells and their

membranes, the cells undergo apoptosis. Over long periods of time, arteries become progressively occluded as plaques consisting of extracellular matrix material, scar tissue formed from smooth muscle tissue, and foam cell remnants gradually become larger. Within the cholesterol-rich plaques, cholesterol can crystallize (Fig. 21-46, inset). Occasionally, a plaque breaks loose from the site of its formation and is carried through the blood to a narrowed region of an artery in the brain or the heart, causing a stroke or a heart attack. Cholesterol crystals that break loose from plaques can cause vascular injury.



In familial hypercholesterolemia, blood levels of cholesterol are extremely high and severe atherosclerosis develops in childhood. Affected individuals have a defective LDL receptor and lack receptor-mediated uptake of cholesterol carried by LDL. Consequently, cholesterol is not cleared from the blood; it accumulates in foam cells and contributes to the formation of atherosclerotic plaques. Endogenous cholesterol synthesis continues despite the excessive cholesterol in the blood, because extracellular cholesterol cannot enter cells to regulate intracellular synthesis (Fig. 21-44). Drugs in a class called [statins](#), some isolated from natural sources and some synthesized industrially, are used to treat patients with elevated serum cholesterol caused by

familial hypercholesterolemia and other conditions. The statins resemble mevalonate ([Box 21-2](#)) and are competitive inhibitors of HMG-CoA reductase.

BOX 21-2 MEDICINE

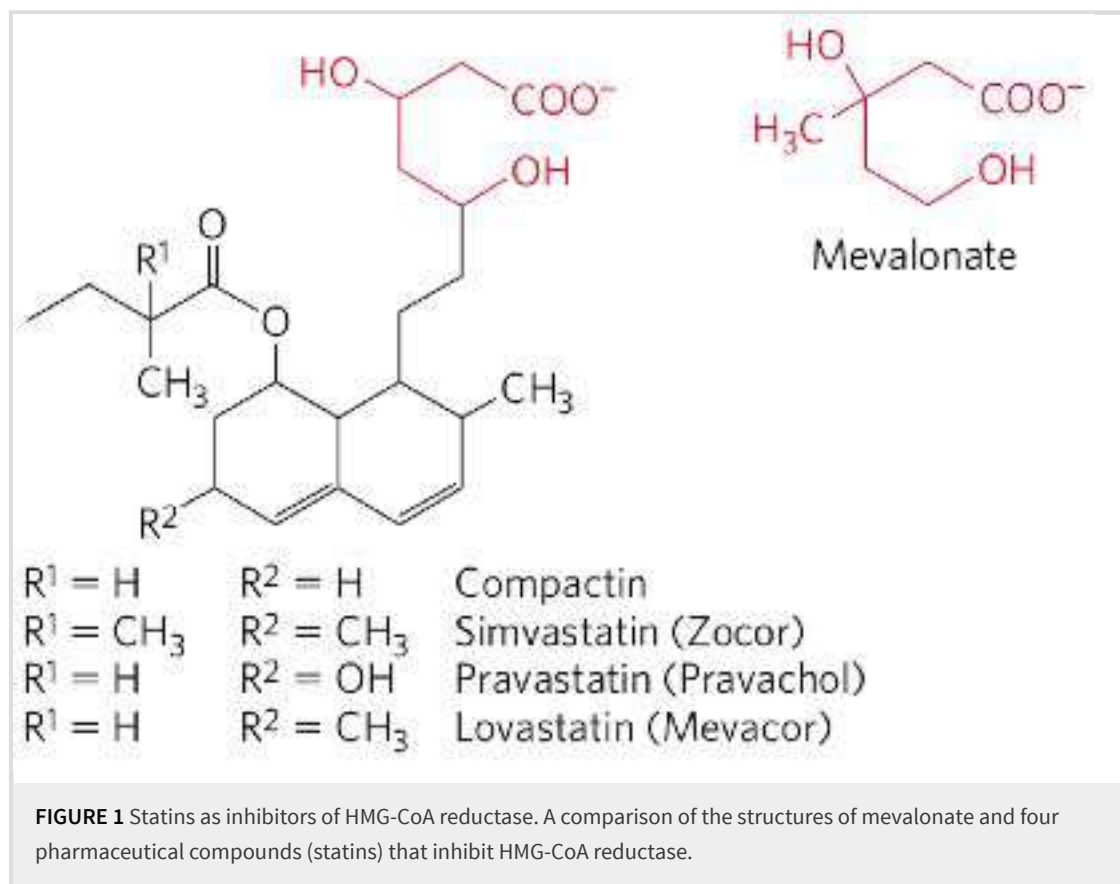
The Lipid Hypothesis and the Development of Statins

Coronary heart disease is the leading cause of death in developed countries. The coronary arteries that bring blood to the heart become narrowed due to the formation of fatty deposits called atherosclerotic plaques, containing cholesterol, fibrous proteins, calcium deposits, blood platelets, and cell debris. Developing the link between artery occlusion (atherosclerosis) and blood cholesterol levels was a project of the twentieth century, triggering a dispute that was resolved only with the development of effective cholesterol-lowering drugs. The Framingham Heart Study, a longitudinal study begun in 1948 and continuing today, was aimed at identifying factors correlated with cardiovascular disease. About 5,000 participants from the city of Framingham, Massachusetts, underwent periodic physical examinations and lifestyle interviews. By 2002, participants of the third generation were included in the study. This monumental study led to the identification of risk factors for cardiovascular disease, including smoking, obesity, physical inactivity, diabetes, high blood pressure, and high blood cholesterol.

In 1913, N. N. Anitschkov, an experimental pathologist in Saint Petersburg, Russia, published a study showing that rabbits fed a diet rich in cholesterol developed lesions very similar to the atherosclerotic plaques seen in aging humans. Anitschkov continued his work over the next few decades, publishing it in prominent western journals. Nevertheless, the work was not accepted as a model for human atherosclerosis, due to a prevailing view that the disease was simply a consequence of aging and could not be prevented. The link between serum cholesterol and atherosclerosis (the lipid hypothesis) was gradually strengthened, however, until researchers in the 1960s openly suggested that therapeutic intervention might be helpful. Controversy persisted until the results of a large study of cholesterol lowering, sponsored by the U.S. National Institutes of Health, was published in 1984: the Coronary Primary Prevention Trial. This study conclusively showed a statistically significant decrease in heart attacks and strokes as a result of decreasing blood cholesterol level. The study made use of a bile acid-binding resin, cholestyramine, to control cholesterol. The results triggered a search for more effective therapeutic interventions. Some controversy persisted until development of the statins in the late 1980s and 1990s.

Dr. Akira Endo, working at the Sankyo company in Tokyo, discovered the first statin and reported the work in 1976. Endo had been interested in cholesterol metabolism for some time, and he speculated in 1971 that the fungi being screened at that time for new antibiotics might also contain an inhibitor of cholesterol synthesis. Over a period of several years, he screened more than 6,000 fungal cultures until a positive result emerged. The compound that resulted was named compactin ([Fig. 1](#)). This compound eventually proved effective in reducing cholesterol levels in dogs and monkeys, and the work came to the attention of Michael Brown and Joseph Goldstein at the University of Texas–Southwestern Medical School. Brown and Goldstein began to work with Endo, and they confirmed his results. Some dramatic results in the first limited clinical trials convinced several pharmaceutical firms to join the hunt for statins. A team at Merck, led by Alfred Alberts and P. Roy Vagelos, began screening fungal cultures and found a positive result after screening just 18 cultures. The new statin was eventually called lovastatin ([Fig. 1](#)). In 1980, a rumor that compactin, at very high doses, was carcinogenic in dogs almost sidelined the race to develop statins, but the benefits to people with familial hypercholesterolemia were already evident. After much consultation with experts around the world and with the U.S. Food and Drug Administration, Merck proceeded carefully

to develop lovastatin. Extensive testing over the next two decades revealed no carcinogenic effects from lovastatin, or from the newer generations of statins that have appeared since.





Courtesy of Akira Endo, Ph.D.

Akira Endo



Courtesy of Eli Alberts.

Alfred Alberts (1931–2018)



Courtesy of P. Roy Vagelos.

P. Roy Vagelos

Statins inhibit HMG-CoA reductase, in part, by mimicking the structure of mevalonate ([Fig. 1](#)), and thus inhibit cholesterol synthesis. Lovastatin treatment lowers serum cholesterol by as much as 30% in individuals with hypercholesterolemia resulting from one defective copy of the gene for the LDL receptor. When combined with an edible resin that binds bile acids and prevents their reabsorption from the intestine, the statin is even more effective.

Statins are now the most widely used drugs for lowering serum cholesterol levels. Side effects are always a concern with drugs, but in the case of statins, many of the side effects are positive. These drugs can improve blood flow, enhance the stability of atherosclerotic plaques (so they don't rupture and obstruct blood flow), reduce platelet aggregation, and reduce vascular inflammation. In patients taking statins for the first time, some of these effects occur before cholesterol levels drop and may be related to a secondary inhibition of isoprenoid synthesis. Not all effects of statins are positive. Some individuals, usually among those taking statins in combination with other cholesterol-lowering drugs, experience muscle pain or weakness that can become severe and even debilitating. For these patients, the effects may sometimes be ameliorated by diet supplementation with coenzyme Q₁₀, a coenzyme that statin therapy depletes in the serum. A fairly long list of other side effects has been documented; most are rare. However, for the vast majority of people, the statin-mediated decrease in risks associated with coronary heart disease can be dramatic. As with all medications, statins should be used only in consultation with a physician.

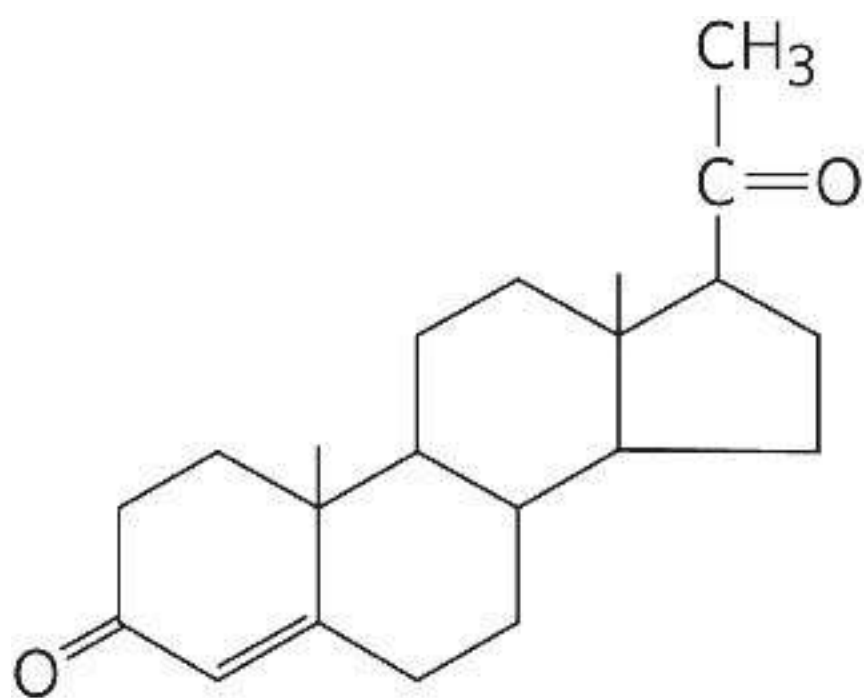
Cholesterol movement out of cells requires transporters. The human genome encodes 48 transporters of the ATP-binding cassette (ABC) class, and about half of these promote lipid transport. Two of them transport cholesterol out of cells. In this process, apoA-I interacts with the transporter ABCA1 in a cholesterol-rich cell. ABCA1 transports a load of cholesterol from inside the cell to the outer surface of the plasma membrane, where lipid-free or lipid-poor apoA-I picks it up, then transports it to the liver. Another transporter, ABCG1, interacts with mature HDL, facilitating the movement of cholesterol out of the cell and into the HDL. This efflux process is particularly critical to reverse cholesterol transport away from foam cells at the sites of plaques in the blood vessels of individuals with cardiovascular disease.



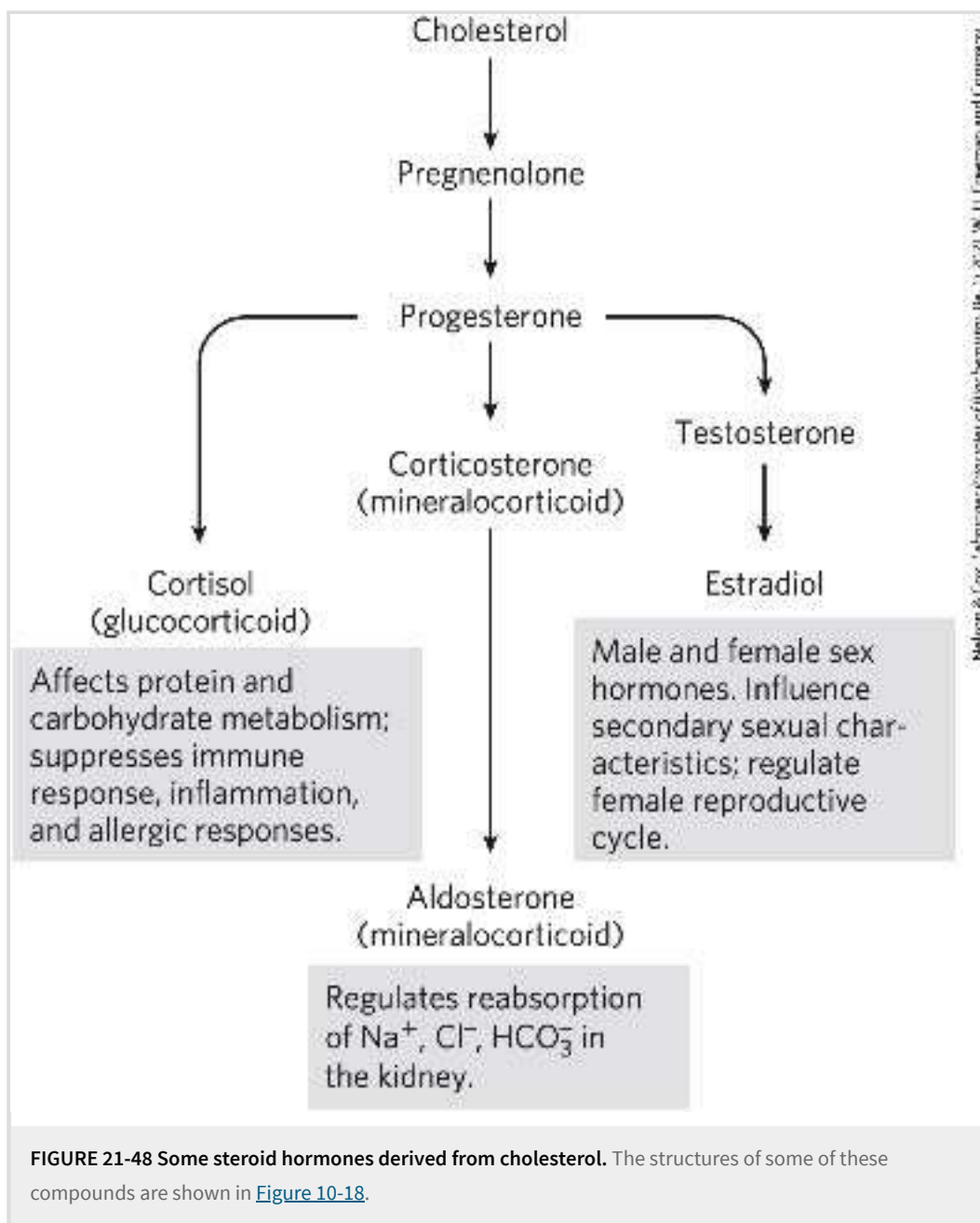
In **familial HDL deficiency**, HDL levels are very low, and in **Tangier disease** they are almost undetectable ([Fig. 21-47](#)). Both genetic disorders are the result of mutations in the ABCA1 protein. ApoA-I in cholesterol-depleted HDL cannot take up cholesterol from cells that lack ABCA1 protein, and apoA-I and cholesterol-poor HDL are rapidly removed from the blood and destroyed. Both familial HDL deficiency and Tangier disease are very rare (worldwide, fewer than 100 families with Tangier disease are known), but the existence of these diseases establishes a role for ABCA1 and ABCG1 proteins in the regulation of plasma HDL levels. ■

Steroid Hormones Are Formed by Side-Chain Cleavage and Oxidation of Cholesterol

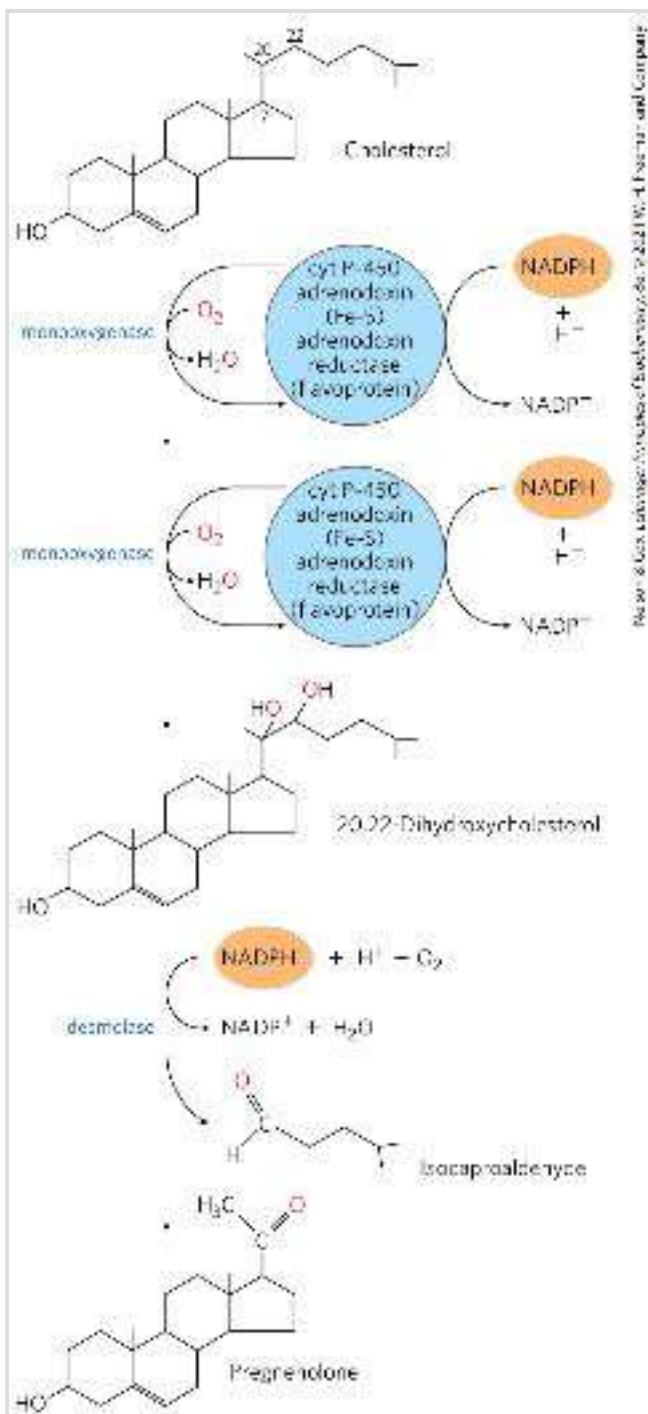
Humans derive all their steroid hormones from cholesterol ([Fig. 21-48](#)).  Two classes of steroid hormones are synthesized in the cortex of the adrenal gland: **mineralocorticoids**, which control the reabsorption of inorganic ions (Na^+ , Cl^- , and HCO_3^-) by the kidney, and **glucocorticoids**, which help regulate gluconeogenesis and reduce the inflammatory response. Sex hormones are produced in male and female gonads and the placenta. They include **progesterone**, which regulates the female reproductive cycle, and **androgens** (such as testosterone) and **estrogens** (such as estradiol), which influence the development of secondary sexual characteristics in males and females, respectively. Steroid hormones are effective at very low concentrations and are therefore synthesized in relatively small quantities. In comparison with the bile salts, their production consumes relatively little cholesterol.



Progesterone



Synthesis of steroid hormones requires removal of some or all of the carbons in the “side chain” on C-17 of the D ring of cholesterol. Side-chain removal takes place in the mitochondria of steroidogenic tissues. Removal requires the hydroxylation of two adjacent carbons in the side chain (C-20 and C-22) followed by cleavage of the bond between them ([Fig. 21-49](#)). Formation of the various hormones also requires the introduction of oxygen atoms. All the hydroxylation and oxygenation reactions in steroid biosynthesis are catalyzed by mixed-function oxygenases (see [Box 21-1](#)) that use NADPH, O_2 , and mitochondrial cytochrome P-450.

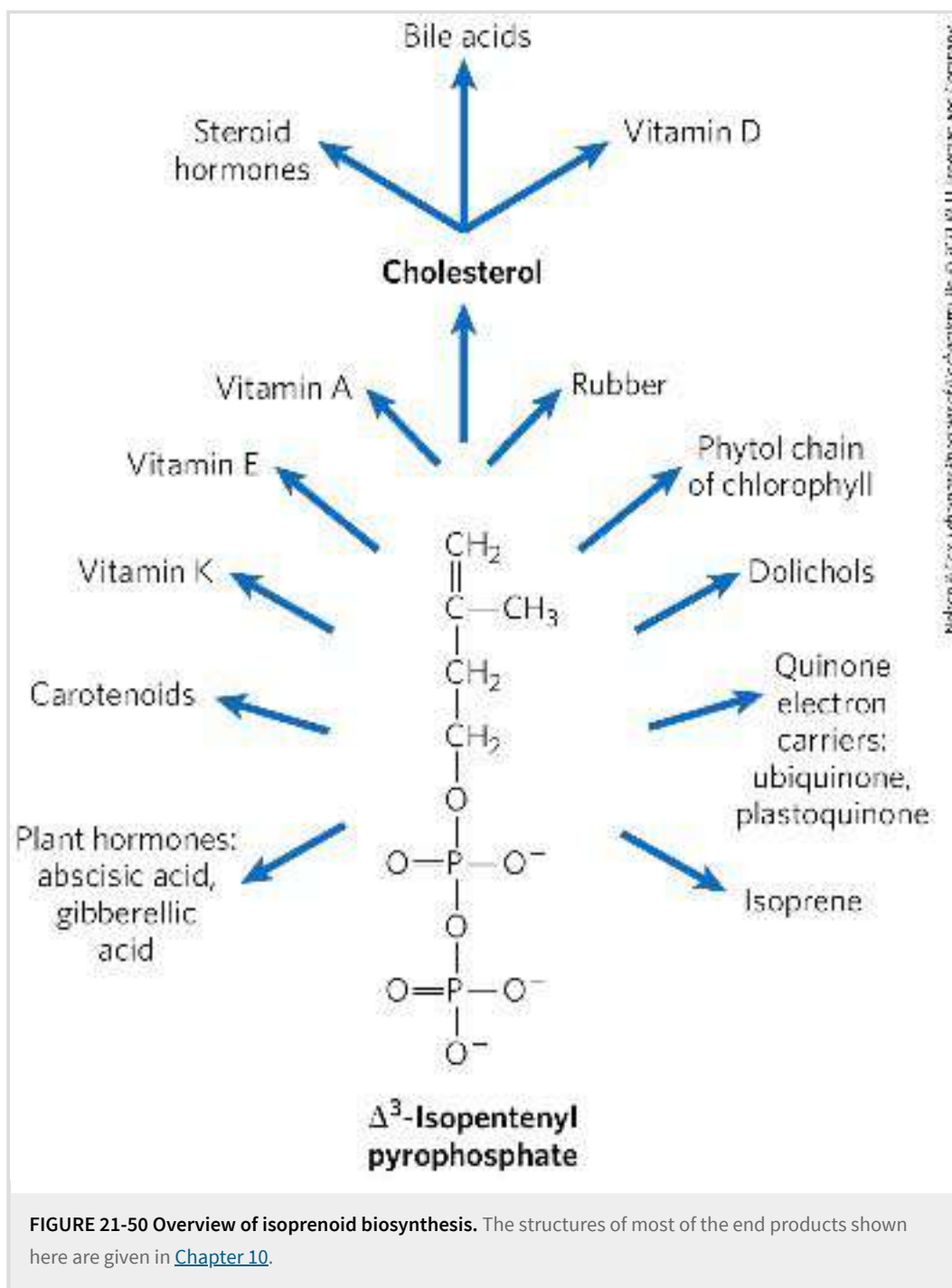


Reprinted with permission from: Biochemistry, 7e, 2011 W. H. Freeman and Company.

FIGURE 21-49 Side-chain cleavage in the synthesis of steroid hormones. Cytochrome P-450 acts as electron carrier in this monooxygenase system that oxidizes adjacent carbons. The process also requires the electron-transferring proteins adrenodoxin and adrenodoxin reductase. This system for cleaving side chains is found in mitochondria of the adrenal cortex, where active steroid production occurs. Pregnenolone is the precursor of all other steroid hormones (see [Fig. 21-48](#)).

Intermediates in Cholesterol Biosynthesis Have Many Alternative Fates

In addition to its role as an intermediate in cholesterol biosynthesis, isopentenyl pyrophosphate is the activated precursor of a huge array of biomolecules with diverse biological roles ([Fig. 21-50](#)).  They include vitamins A, E, and K; plant pigments such as carotene and the phytol chain of chlorophyll; natural rubber; many essential oils, such as the fragrant principles of lemon oil, eucalyptus, and musk; insect juvenile hormone, which controls metamorphosis; dolichols, which serve as lipid-soluble carriers in complex polysaccharide synthesis; and ubiquinone and plastoquinone, electron carriers in mitochondria and chloroplasts. Collectively, these molecules are called isoprenoids. More than 20,000 different isoprenoid molecules have been discovered in nature, and hundreds of new ones are reported each year.



Prenylation, the covalent attachment of an isoprenoid (see [Fig. 27-35](#)), is a common mechanism by which proteins are anchored to the inner surface of cellular membranes in mammals (see [Fig. 11-16](#)). In some of these proteins, the attached lipid is the 15-carbon farnesyl group; others have the 20-carbon geranylgeranyl group. Different enzymes attach the two types of lipids. It is possible that prenylation targets proteins to different membranes, depending on which lipid is attached. Protein prenylation is another important role for the isoprene derivatives of the pathway to cholesterol.

SUMMARY 21.4 *Cholesterol, Steroids, and Isoprenoids: Biosynthesis, Regulation, and Transport*

■ Cholesterol is formed from acetyl-CoA in a complex series of reactions, through the intermediates β -hydroxy- β -methylglutaryl-CoA (HMG-CoA), mevalonate, and two activated isoprenes, dimethylallyl pyrophosphate and isopentenyl pyrophosphate. Condensation of isoprene units produces the noncyclic squalene, which is cyclized to yield the steroid ring system and side chain.

■ Synthesized primarily in the liver, cholesterol is exported as bile acids, biliary cholesterol, or cholesteryl esters.

■ Cholesterol and cholesteryl esters are carried in the blood as plasma lipoproteins. VLDL carries cholesterol, cholesteryl esters, and triacylglycerols from the liver to other tissues, where the triacylglycerols are degraded by lipoprotein lipase, converting VLDL to LDL. Cholesterol scavenging and transport back to the liver are mediated by HDL. Much of the lipid carried to the liver is utilized in bile salts.

■ The LDL, rich in cholesterol and its esters, is taken up by receptor-mediated endocytosis, in which the apolipoprotein B-100 of LDL is recognized by receptors in the plasma membrane.

■ Cholesterol synthesis and transport are under complex regulation by hormones, cellular cholesterol content, and energy level (AMP concentration). HMG-CoA reductase is regulated allosterically and by covalent modification. Furthermore, a complex of three proteins—Insig, SCAP, and SREBP — sense cholesterol levels and trigger increased synthesis or degradation of HMG-CoA reductase in response. The number of LDL receptors per cell is also regulated by cholesterol content.

■ Dietary conditions or genetic defects in cholesterol metabolism may lead to atherosclerosis and heart disease.

■ In reverse cholesterol transport, HDL removes cholesterol from peripheral tissues, carrying it to the liver. By reducing the cholesterol content of foam cells, HDL protects against atherosclerosis.

■ The steroid hormones (glucocorticoids, mineralocorticoids, and sex hormones) are produced from cholesterol by alteration of the side chain and introduction of oxygen atoms into the steroid ring system. In addition to cholesterol, a wide variety of isoprenoid compounds are derived from mevalonate through condensations of isopentenyl pyrophosphate and dimethylallyl pyrophosphate.

■ Prenylation of certain proteins targets them for association with cellular membranes and is essential for their biological activity.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

malonyl-CoA

acetyl-CoA carboxylase

fatty acid synthase

acyl carrier protein (ACP)

4'-phosphopantetheine

fatty acid elongation systems

fatty acyl-CoA desaturase

mixed-function oxidases

stearoyl-ACP desaturase (SCD)

essential fatty acids

arachidonate

eicosatetraenoate

prostaglandin (PG)

cyclooxygenase (COX)

prostaglandin H₂ synthase

mixed-function oxygenases

cytochrome P-450

thromboxane synthase

thromboxane (TX)

leukotriene (LT)

catabasis

specialized pro-resolving mediator (SPM)

lipoxin

glycerol 3-phosphate dehydrogenase

phosphatidic acid

triacylglycerol cycle

glyceroneogenesis

thiazolidinediones

phosphatidylglycerol

cardiolipin

phosphatidylinositol

phosphatidylserine

phosphatidylethanolamine

phosphatidylcholine

Lands cycle

lysophosphatidylcholine acyltransferases (LPCATs)

plasmalogen

platelet-activating factor

cerebroside

sphingomyelin

ganglioside

isoprene

mevalonate

β -hydroxy- β -methylglutaryl-CoA (HMG-CoA)

HMG-CoA synthase

HMG-CoA reductase

squalene

bile acids

cholesteryl esters

apolipoprotein

chylomicron

exogenous pathway

[very-low-density lipoprotein \(VLDL\)](#),
[low-density lipoprotein \(LDL\)](#),
[LDL receptors](#)
[endogenous pathway](#),
[high-density lipoprotein \(HDL\)](#),
[reverse cholesterol transport](#)
[enterohepatic circulation](#)
[receptor-mediated endocytosis](#)
[sterol regulatory element-binding proteins \(SREBPs\)](#),
[SREBP cleavage-activating protein \(SCAP\)](#),
[insulin-induced gene protein \(Insig\)](#),
[liver X receptor \(LXR\)](#),
[retinoid X receptor \(RXR\)](#),
[farnesoid X receptor \(FXR\)](#),
[atherosclerosis](#)
[foam cell](#)
[statin](#)

PROBLEMS

- 1. Pathway of Carbon in Fatty Acid Synthesis** Using your knowledge of fatty acid biosynthesis, provide an explanation for the two experimental observations.
- A biochemist adds uniformly labeled $[^{14}\text{C}]$ acetyl-CoA to a soluble liver fraction, which yields palmitate uniformly labeled with ^{14}C .
 - In a second experiment, the biochemist adds a *trace* of uniformly labeled $[^{14}\text{C}]$ acetyl-CoA in the presence of

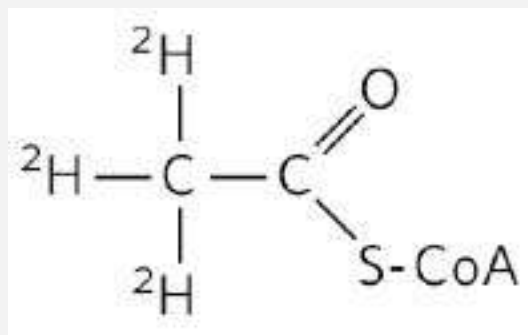
an excess of unlabeled malonyl-CoA to a soluble liver fraction, which yields palmitate labeled with ^{14}C only in C-15 and C-16.

2. Synthesis of Fatty Acids from Glucose After a person has ingested large amounts of sucrose, the body transforms the glucose and fructose that exceed caloric requirements to fatty acids for triacylglycerol synthesis. This fatty acid synthesis consumes acetyl-CoA, ATP, and NADPH. How do cells produce acetyl-CoA, ATP, and NADPH from glucose?

3. Net Equation of Fatty Acid Synthesis Write the net equation for the biosynthesis of palmitate in rat liver, starting from mitochondrial acetyl-CoA and cytosolic NADPH, ATP, and CO_2 .

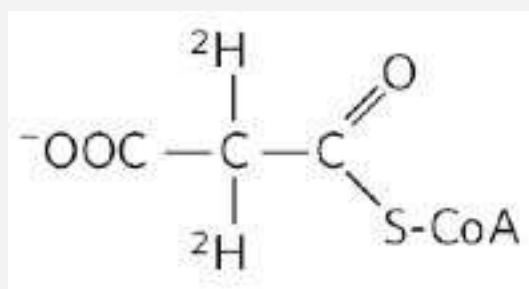
4. Pathway of Hydrogen in Fatty Acid Synthesis A researcher has prepared a solution that contains all the enzymes and cofactors necessary for fatty acid biosynthesis from added acetyl-CoA and malonyl-CoA.

- a. She then adds $[2\text{-}^2\text{H}]\text{acetyl-CoA}$ (labeled with deuterium, the heavy isotope of hydrogen) and an excess of unlabeled malonyl-CoA as substrates.



How many deuterium atoms incorporate into every molecule of palmitate? What are their locations? Explain.

- b. In a separate experiment, the researcher adds unlabeled acetyl-CoA and [2- ^2H]malonyl-CoA as substrates.



How many deuterium atoms incorporate into every molecule of palmitate? What are their locations? Explain.

5. Energetics of β -Ketoacyl-ACP Synthase The condensation reaction catalyzed by β -ketoacyl-ACP synthase (see [Fig. 21-6](#)) synthesizes a four-carbon unit by combining a two-carbon unit and a three-carbon unit, with the release of CO_2 . What is the thermodynamic rationale for this process relative to one that simply combines two two-carbon units?

6. Modulation of Acetyl-CoA Carboxylase Acetyl-CoA carboxylase is the principal regulation point in the biosynthesis of fatty acids. Following are some of the properties of the enzyme:

- a. Addition of citrate or isocitrate raises the V_{max} of the enzyme as much as 10-fold.

- b. The enzyme exists in two interconvertible forms that differ markedly in their activities:



Citrate and isocitrate bind preferentially to the filamentous form, and palmitoyl-CoA binds preferentially to the protomer.

Explain how these properties are consistent with the regulatory role of acetyl-CoA carboxylase in the biosynthesis of fatty acids.

7. Shuttling of Acetyl Groups across the Inner

Mitochondrial Membrane The acetyl group shuttle transfers acetyl-CoA, produced by oxidative decarboxylation of pyruvate in the mitochondrion, to the cytosol as outlined in [Figure 21-10](#).

- Write the overall equation for the transfer of one acetyl group from the mitochondrion to the cytosol.
- What is the cost of this process in ATPs per acetyl group?
- In [Chapter 17](#) we encountered an acyl group shuttle in the transfer of fatty acyl-CoA from the cytosol to the mitochondrion in preparation for β oxidation (see [Fig. 17-6](#)). One result of that shuttle is separation of the mitochondrial and cytosolic pools of CoA. Does the acetyl group shuttle also accomplish this? Explain.

8. Oxygen Requirement for Desaturases The biosynthesis of palmitoleate (see [Fig. 21-12](#)), a common unsaturated fatty acid with a cis double bond in the Δ^9 position, uses palmitate as a precursor. Can palmitoleate synthesis be carried out under strictly anaerobic conditions? Explain.

9. Energy Cost of Triacylglycerol Synthesis Use a net equation for the biosynthesis of tripalmitoylglycerol (tripalmitin) from glycerol and palmitate to show how many ATPs are required per molecule of tripalmitin formed.

10. Turnover of Triacylglycerols in Adipose Tissue A researcher adds [^{14}C]glucose to the balanced diet of adult rats. She finds no increase in the total amount of stored triacylglycerols, but the triacylglycerols become labeled with ^{14}C . Explain.

11. Energy Cost of Phosphatidylcholine Synthesis Write the sequence of steps and the net reaction for the biosynthesis of phosphatidylcholine by the salvage pathway from oleate, palmitate, dihydroxyacetone phosphate, and choline. Starting from these precursors, what is the cost (in number of ATPs) of the synthesis of phosphatidylcholine by the salvage pathway?

12. Salvage Pathway for Synthesis of Phosphatidylcholine A young rat maintained on a diet deficient in methionine fails to thrive unless choline is included in the diet. Explain.

13. Energetics of Acetyl-CoA Condensation to Form

Acetoacetyl-CoA The formation of a thioester of acetoacetate is catalyzed by fatty acid synthase during fatty acid synthesis, and by acetyl-CoA acetyltransferase in the first step of cholesterol biosynthesis. Both are Claisen condensations. However, in fatty acid synthesis, malonyl-CoA forms in an earlier step so that decarboxylation facilitates the condensation. In the cholesterol biosynthesis pathway, the condensation occurs between two acetyl-CoA molecules, and no decarboxylation occurs to facilitate the reaction. Suggest a reason why the thermodynamic augmentation of decarboxylation is needed in fatty acid synthesis, but not in the first steps of cholesterol biosynthesis.

14. Synthesis of Isopentenyl Pyrophosphate A researcher adds [2-¹⁴C]acetyl-CoA to a rat liver homogenate that is synthesizing cholesterol. Where will the ¹⁴C label appear in Δ^3 -isopentenyl pyrophosphate, the activated form of an isoprene unit?

15. Activated Donors in Lipid Synthesis In the biosynthesis of complex lipids, components are assembled by transfer of the appropriate group from an activated donor. For example, the activated donor of acetyl groups is acetyl-CoA. For each of the following groups, give the form of the activated donor:

- a. phosphate;
- b. D-glucosyl;
- c. phosphoethanolamine;
- d. D-galactosyl;

- e. fatty acyl;
- f. methyl;
- g. the two-carbon group in fatty acid biosynthesis;
- h. Δ^3 -isopentenyl.

16. Importance of Fats in the Diet When young rats are placed on a completely fat-free diet, they grow poorly, develop a scaly dermatitis, lose hair, and soon die. These symptoms can be prevented if linoleate or plant material is included in the diet. What makes linoleate an essential fatty acid? Why can plant material be substituted?

17. Regulation of Cholesterol Biosynthesis Cholesterol in humans can be obtained from the diet or synthesized de novo. An adult human on a low-cholesterol diet typically synthesizes 600 mg of cholesterol per day in the liver. If the amount of cholesterol in the diet is large, de novo synthesis of cholesterol is drastically reduced. How is this regulation brought about?

18. Lowering Serum Cholesterol Levels with Statins Patients treated with a statin drug generally exhibit a dramatic lowering of serum cholesterol. However, the amount of the enzyme HMG-CoA reductase present in cells can increase substantially. Suggest a simple explanation for this effect.

19. Roles of Thiol Esters in Cholesterol Biosynthesis Draw a mechanism for each of the three reactions shown in [Figure 21-34](#), detailing the pathway for the synthesis of mevalonate from acetyl-CoA.

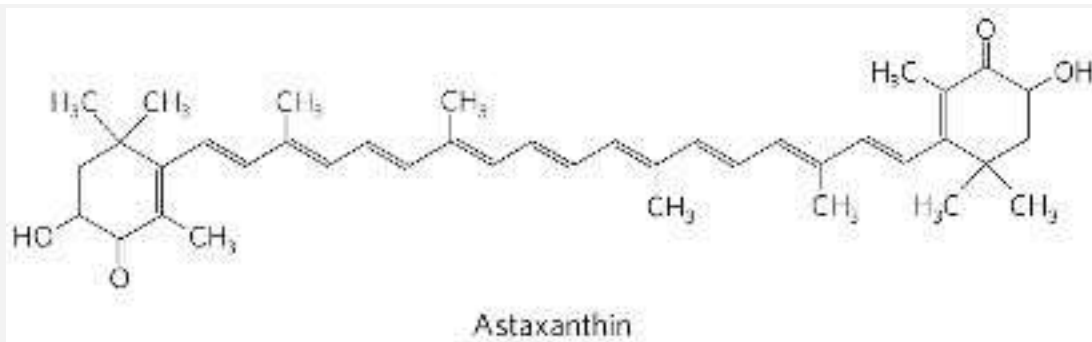
20. Potential Side Effects of Treatment with Statins

Although the benefits of taking statins have become clear, side effects have yet to be documented in detail. Some physicians have suggested that patients being treated with statins should also take a supplement of coenzyme Q. Suggest a rationale for this recommendation.

DATA ANALYSIS PROBLEM

21. Engineering *E. coli* to Produce Large Quantities of an Isoprenoid

There are more than 20,000 naturally occurring isoprenoids, some of which are medically or commercially important and produced industrially. The production methods include in vitro enzymatic synthesis, an expensive and low-yield process. In 1999, Wang, Oh, and Liao reported on their experiments to engineer the easily grown bacterium *E. coli* to produce large amounts of astaxanthin, a commercially important isoprenoid. Astaxanthin is a red-orange carotenoid pigment (an antioxidant) produced by marine algae. Marine animals such as shrimp, lobster, and some fish that feed on the algae get their orange color from the ingested astaxanthin. Astaxanthin is composed of eight isoprene units; its molecular formula is $C_{40}H_{52}O_4$.



- a. Circle the eight isoprene units in the astaxanthin molecule. Hint: Use the projecting methyl groups as a guide.

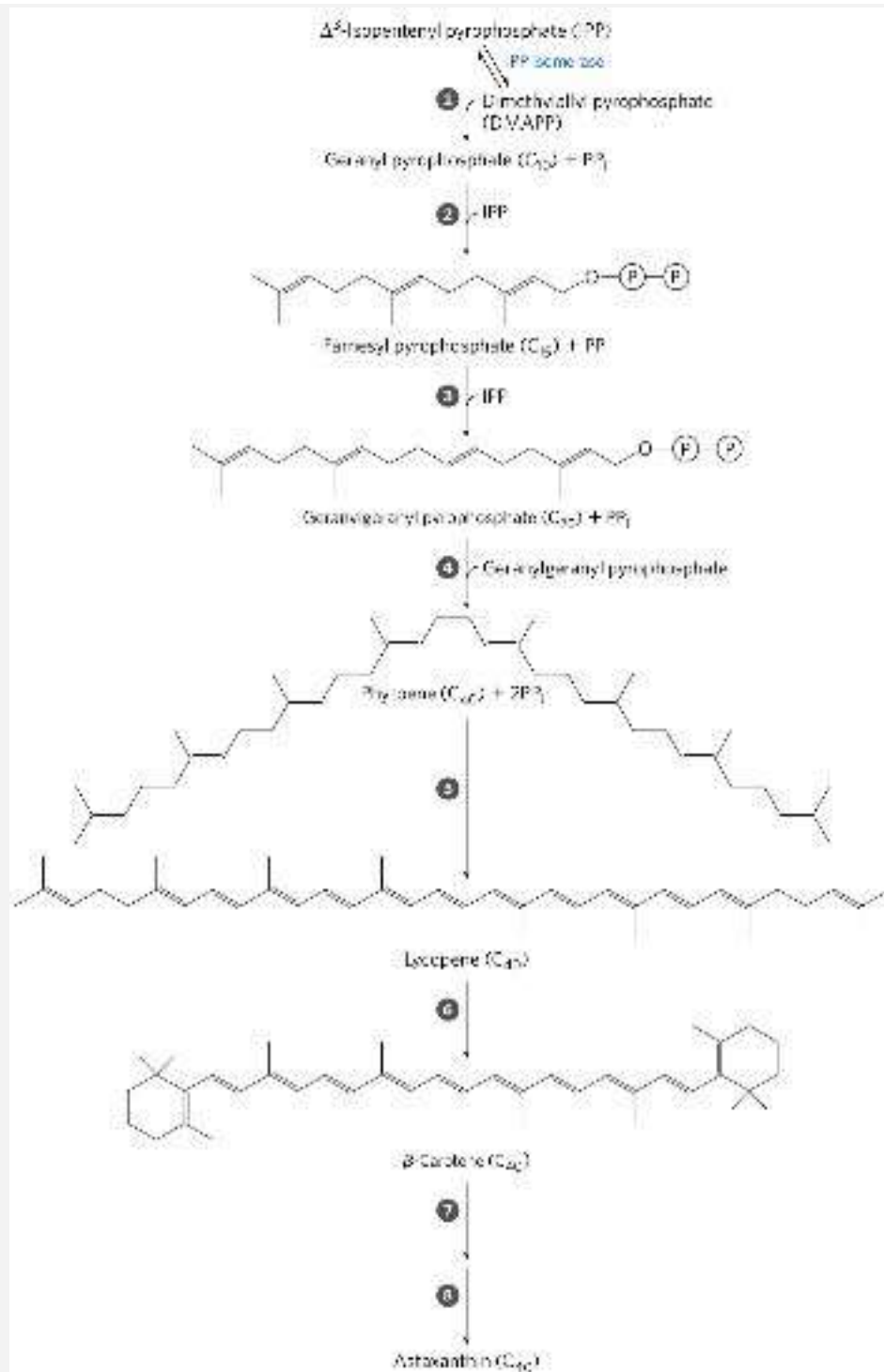
Astaxanthin is synthesized by the pathway shown on the next page, starting with Δ^3 -isopentenyl pyrophosphate (IPP). Steps ① and ② are shown in [Figure 21-36](#), and the reaction catalyzed by IPP isomerase is shown in [Figure 21-35](#).

- b. In step ④ of the pathway, two molecules of geranylgeranyl pyrophosphate are linked to form phytoene. Is this a head-to-head joining or is it a head-to-tail joining? (See [Figure 21-36](#) for details.)
- c. Briefly describe the chemical transformation in step ⑤.
- d. The synthesis of cholesterol ([Fig. 21-37](#)) includes a cyclization (ring closure) that requires a net oxidation by O_2 . Does the cyclization in step ⑥ of the astaxanthin synthetic pathway require a net oxidation of the substrate (lycopene)? Explain your reasoning.

E. coli does not make large quantities of many isoprenoids, and does not synthesize astaxanthin. It is known to synthesize small amounts of IPP, DMAPP,

geranyl pyrophosphate, farnesyl pyrophosphate, and geranylgeranyl pyrophosphate. Wang and colleagues cloned several of the *E.coli* genes that encode enzymes needed for astaxanthin synthesis, in plasmids that allowed their overexpression. These genes included *idi*, which encodes IPP isomerase, and *ispA*, which encodes a prenyl transferase that catalyzes steps ❶ and ❷.

To engineer an *E. coli* capable of the complete astaxanthin pathway, Wang and colleagues cloned several genes from other bacteria into plasmids that would allow their overexpression in *E. coli*. These genes included *crtE* from *Erwinia uredovora*, which encodes an enzyme that catalyzes step ❸; and *crtB*, *crtI*, *crtY*, *crtZ*, and *crtW* from *Agrobacterium aurantiacum*, which encode enzymes for steps ❹, ❺, ❻, ❼, and ❽, respectively.



The investigators also cloned the gene *gps* from *Archaeoglobus fulgidus*, overexpressed this gene in *E. coli*, and extracted the gene product. When this extract was reacted with [¹⁴C]IPP and DMAPP or geranyl

pyrophosphate or farnesyl pyrophosphate, only ^{14}C -labeled geranylgeranyl pyrophosphate was produced in all cases.

- e. Based on these data, which step(s) in the pathway are catalyzed by the enzyme encoded by *gps*? Explain your reasoning.

Wang and coworkers then constructed several *E. coli* strains overexpressing different genes; they measured the orange color of the colonies (wild-type *E. coli* colonies are off-white) and the amount of astaxanthin produced (as measured by its orange color). Their results are shown below (ND indicates not determined).

Strain	Gene(s) overexpressed	Orange color	Astaxanthin yield ($\mu\text{g/g}$ dry weight)
1	<i>crtBIZYW</i>	—	ND
2	<i>crtBIZYW, ispA</i>	—	ND
3	<i>crtBIZYW, idi</i>	—	ND
4	<i>crtBIZYW, idi, ispA</i>	—	ND
5	<i>crtBIZYW, crtE</i>	+	32.8
6	<i>crtBIZYW, crtE, ispA</i>	+	35.3
7	<i>crtBIZYW, crtE, idi</i>	++	234.1

8	<i>crtBIZYW, crtE, idi, ispA</i>	+ + +	390.3
9	<i>crtBIZYW, gps</i>	+	35.6
10	<i>crtBIZYW, gps, idi</i>	+ + +	1,418.8

- f. Comparing the results for strains 1 through 4 with those for strains 5 through 8, what can you conclude about the expression level of an enzyme capable of catalyzing step 3 of the astaxanthin synthetic pathway in wild-type *E. coli*? Explain your reasoning.
- g. Based on the data above, which enzyme is rate-limiting in this pathway, IPP isomerase or the enzyme encoded by *idi*? Explain your reasoning.

Reference

Wang, C.-W., M.-K. Oh, and J.C. Liao. 1999. Engineered isoprenoid pathway enhances astaxanthin production in *Escherichia coli*. *Biotechnol. Bioeng.* 62:235–241.

CHAPTER 22

BIOSYNTHESIS OF AMINO ACIDS, NUCLEOTIDES, AND RELATED MOLECULES



[22.1 Overview of Nitrogen Metabolism](#)

[22.2 Biosynthesis of Amino Acids](#)

[22.3 Molecules Derived from Amino Acids](#)

[22.4 Biosynthesis and Degradation of Nucleotides](#)

Nitrogen ranks behind only carbon, hydrogen, and oxygen in its contribution to the mass of living systems. Most of this nitrogen is bound up in amino acids and nucleotides. In this chapter we address all aspects of the metabolism of these nitrogen-containing compounds except amino acid catabolism, which is the subject of [Chapter 18](#).

In this final chapter covering anabolic processes, some of the underlying principles are unique, while others will be familiar from [Chapter 21](#) and the consideration of gluconeogenesis in [Chapter 14](#):

P1 Amino acids and nucleotides are the precursors to proteins and nucleic acids, respectively. They also give rise to numerous neurotransmitters, metabolic cofactors, and other molecules of biological importance.

P2 The supply of biologically available nitrogen can be limiting in many environments. Atmospheric N_2 is relatively inert and must be converted to other forms, such as ammonia and nitrate, to accommodate the requirements of life. A complex web of enzymatic processes, based primarily in microorganisms, interconverts the various molecular forms that comprise the global inventory of reactive nitrogen.

P3 Oxygen and nitrogen metabolism are interlinked. The oxidation and reduction of the various forms of nitrogen in the biosphere often involve oxygen.

P4 Regulation again plays an important role. Many of the processes covered in this chapter are regulated to carefully conserve a critical and limited resource. Tight regulation is needed to maintain balanced supplies of amino acids and nucleotides. The metabolic flux through most of these pathways is far less than for carbohydrate or lipid biosynthetic pathways; most amino acids and nucleotides are not stored, but are synthesized as they are needed.

P5 The amino acids glutamate and glutamine represent the entry point where reactive forms of nitrogen are incorporated into biological systems. Reflecting their

importance, the concentrations of these amino acids are sufficiently elevated in many tissues that they are major contributors to the electrochemical environment of cells. The prominent role of these two amino acids is a universal feature of animate nitrogen metabolism, yet another molecular manifestation of the shared evolutionary history of all organisms on the planet.



Like other anabolic pathways, the reaction sequences in amino acid and nucleotide biosynthesis are endergonic and reductive. They use ATP as a source of metabolic energy and they use a reduced electron carrier (usually NADPH) as a reductant.

Discussing the biosynthetic pathways for amino acids and nucleotides together is a sound approach, not only because both classes of molecules contain nitrogen, but also because the two sets of pathways are extensively intertwined, with several key intermediates in common. Certain amino acids or parts of amino acids are incorporated into the structure of purines and pyrimidines, and in one case, part of a purine ring is incorporated into an amino acid (histidine). The two sets of pathways also share much common chemistry, in particular a preponderance of reactions involving the transfer of nitrogen or one-carbon groups.

The sheer number of steps and variety of intermediates in the pathways described here can be intimidating to the beginning biochemistry student. These pathways are best approached by

maintaining a focus on metabolic principles we have already discussed, on key intermediates and precursors, and on common classes of reactions. Even a cursory look at the chemistry can be rewarding, for some of the most unusual chemical transformations in biological systems occur in these pathways; for instance, we find prominent examples of the rare biological use of the metals molybdenum, selenium, and vanadium. The effort also offers a practical dividend, especially for students of human or veterinary medicine. Many genetic diseases of humans and animals have been traced to an absence of one or more enzymes of amino acid and nucleotide metabolism, and many pharmaceuticals in common use to combat infectious diseases are inhibitors of enzymes in these pathways — as are a number of the most important agents in cancer chemotherapy.

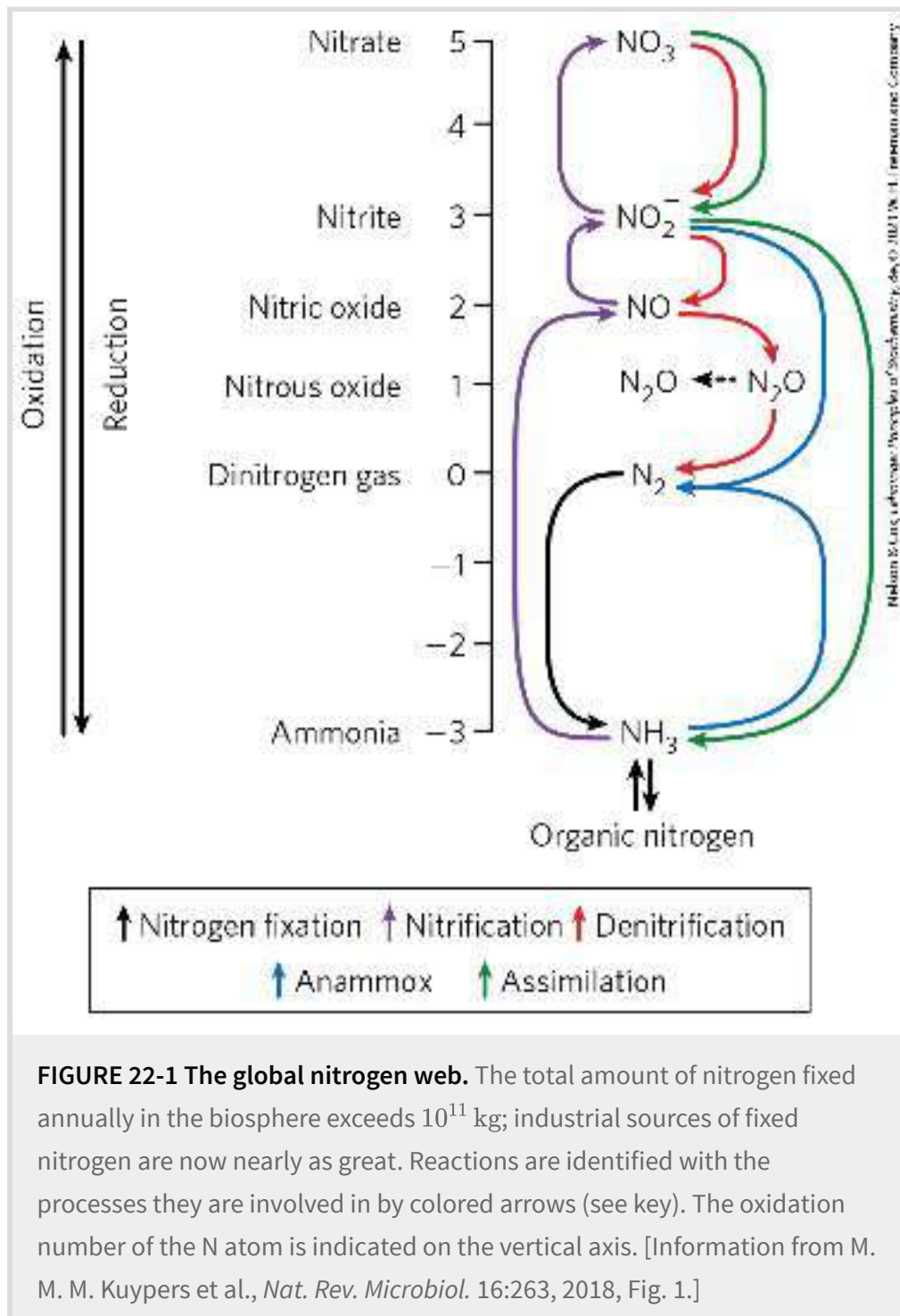
22.1 Overview of Nitrogen Metabolism

The biosynthetic pathways leading to amino acids and nucleotides share a requirement for nitrogen. Soluble, biologically useful nitrogen compounds are generally scarce in natural environments; thus, most organisms use ammonia, amino acids, and nucleotides economically.  Available amino acids, purines, and pyrimidines formed during metabolic turnover of proteins and nucleic acids are often salvaged and reused. We begin by examining the pathways by which nitrogen from the environment is introduced into biological systems.

A Global Nitrogen Cycling Network Maintains a Pool of Biologically Available Nitrogen

The movement of nitrogen through the biosphere has been viewed historically as a cycle. However, our evolving understanding of the complexity of nitrogenous interconversions makes it clear that nitrogen moves through a complex web, rather than in a neat cycle ([Fig. 22-1](#)). Earth's atmosphere is four-fifths molecular nitrogen (N_2). However, N_2 is too unreactive to be of use to living organisms. Conversion of N_2 to forms that can support life (NH_3 , NO_2^- , NO_3^-) is called [nitrogen fixation](#). The reduction of N_2 to NH_3 plays such a central role in making N_2

available that this one reaction is often considered synonymous with nitrogen fixation. In the biosphere, the metabolic processes of countless species function interdependently to salvage and reuse biologically available nitrogen. Most of the key reactions are carried out by bacteria and archaea.



The reduction of atmospheric nitrogen (N_2) by nitrogen-fixing bacteria and archaea to yield ammonia (NH_3 or NH_4^+), provides a useful anchor for our discussion. This critical process, described in detail in [Section 22.2](#), provides most of the reduced nitrogen for incorporation into biomolecules.

Free ammonia does not build up, and reduction is balanced by oxidation.  Bacteria that derive their energy by oxidizing ammonia to nitrite (NO_2^-) and ultimately nitrate (NO_3^-) are abundant and active in both terrestrial and marine environments. The processes of converting ammonia to nitric oxide, nitrite, and finally nitrate are known as **nitrification** ([Fig. 22-1](#), purple arrows). Individual bacterial or archaeal species may promote one or more of these steps.

Atmospheric N_2 must be replaced in order to maintain levels at a steady-state concentration. Some of the replacement comes from reduction of nitrate and nitrite. The reduction of nitrate and nitrite to N_2 under anaerobic conditions, a process called **denitrification** ([Fig. 22-1](#), red arrows), is carried out by specialized microorganisms in all three domains of life. These organisms use NO_3^- or NO_2^- rather than O_2 as the ultimate electron acceptor in a series of reactions that (like oxidative phosphorylation) generates a transmembrane proton gradient, which is used to synthesize ATP. These microorganisms exist in all anoxic environments where nitrate is present, including soils, marine sediments, and eutrophic marine zones. An alternative

path back to atmospheric N_2 is provided by a group of bacteria that promote anaerobic ammonia oxidation, or [anammox](#) ([Fig. 22-1](#), blue arrows). Anammox converts ammonia and nitrite to N_2 . As much as 50% to 70% of the NH_3 -to- N_2 conversion in the biosphere may occur through this pathway, which went undetected until the 1980s. The obligate anaerobes that promote anammox are fascinating in their own right and are providing some useful solutions to waste-treatment problems ([Box 22-1](#)).

BOX 22-1 UNUSUAL LIFESTYLES OF THE OBSCURE BUT ABUNDANT

Air-breathers that we are, we can easily overlook the bacteria and archaea that thrive in anaerobic environments. Although rarely featured in introductory biochemistry textbooks, these organisms constitute much of the biomass of this planet, and their contributions to the balance of carbon and nitrogen in the biosphere are essential to all forms of life.

As detailed in earlier chapters, the energy used to maintain living systems relies on the generation of proton gradients across membranes. Electrons derived from a reduced substrate are made available to electron carriers in membranes and pass through a series of electron transfers to a final electron acceptor. As a byproduct of this process, protons are released on one side of the membrane, generating the transmembrane proton gradient. The proton gradient is used to synthesize ATP or to drive other energy-requiring processes. For all eukaryotes, the reduced substrate is generally a carbohydrate (glucose or pyruvate) or a fatty acid and the electron acceptor is oxygen.

Many bacteria and archaea are much more versatile. In anaerobic environments such as marine and freshwater sediments, the variety of life strategies is extraordinary. Almost any available redox pair can be an energy

source for some specialized organism or group of organisms. For example, a large number of lithotrophic bacteria (a lithotroph is a chemotroph that uses inorganic energy sources) have a hydrogenase that uses molecular hydrogen to reduce NAD^+ :



The NADH is a source of electrons for a variety of membrane-bound electron acceptors, generating the proton gradient needed for ATP synthesis. Other lithotrophs oxidize sulfur compounds (H_2S , elemental sulfur, or thiosulfate) or ferrous iron. A widespread group of archaea called methanogens, all strict anaerobes, extract energy from the reduction of CO_2 to methane. And this is just a small sampling of what anaerobic organisms do for a living. Their metabolic pathways are replete with interesting reactions and highly specialized cofactors unknown in our own world of obligate aerobic metabolism. Study of these organisms can yield practical dividends. It can also provide clues about the origins of life on an early Earth, in an atmosphere that lacked molecular oxygen.

The web of reactions that defines nitrogen utilization in the biosphere depends on a wide range of specialized bacteria. Some nitrifying bacteria oxidize ammonia to nitrites, and some oxidize the resulting nitrites to nitrates (see [Fig. 22-1](#), purple arrows). Nitrate is second only to O_2 as a biological electron acceptor, and a great many bacteria and archaea can catalyze the denitrification of nitrates and nitrites to nitrogen, which the nitrogen-fixing bacteria then convert back into ammonia. Ammonia is a major pollutant in sewage and in farm animal waste, and it is a byproduct of fertilizer manufacture and oil refining. Waste-treatment plants have generally made use of communities of nitrifying and denitrifying bacteria to convert ammonia waste to atmospheric nitrogen. The process is expensive, requiring inputs of organic carbon and oxygen.

In the 1960s and 1970s, a few articles appeared in the research literature suggesting that ammonia could be oxidized to nitrogen anaerobically, using

nitrite as an electron acceptor; this process was called anammox. The reports received little notice until bacteria promoting anammox were discovered in a waste-treatment system in Delft, the Netherlands, in the mid-1980s. A team of Dutch microbiologists led by Gijs Kuenen and Mike Jetten began to study these bacteria, which were soon identified as belonging to an unusual bacterial phylum, Planctomycetes. Some surprises were to follow.

The biochemistry underlying the anammox process was slowly unraveled (Fig. 1). Hydrazine (N_2H_4), a highly reactive molecule used as a rocket fuel, was an unexpected intermediate. As a small molecule, hydrazine is both highly toxic and difficult to contain. It readily diffuses across typical phospholipid membranes. The anammox bacteria solve this problem by sequestering hydrazine in a specialized organelle, dubbed the **anammoxosome**. The membrane of this organelle is composed of lipids known as **ladderanes** (Fig. 2), never before encountered in biology. The fused cyclobutane rings of ladderanes stack tightly to form a very dense barrier, greatly slowing the release of hydrazine. Cyclobutane rings, with their unusual bond angles, are strained and difficult to synthesize; the bacterial mechanisms for synthesizing these lipids are not yet known.

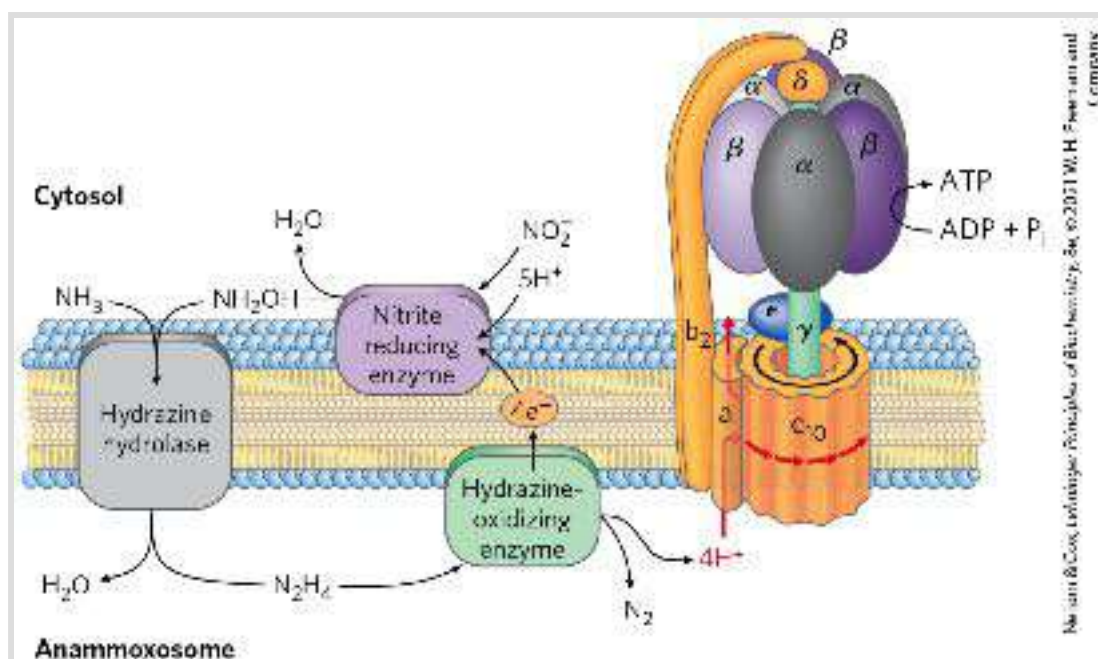


FIGURE 1 The anammox reactions. Ammonia and hydroxylamine are converted to hydrazine and H_2O by hydrazine hydrolase, and the hydrazine is oxidized by hydrazine-oxidizing enzyme, generating N_2 and protons. The protons create a

proton gradient for ATP synthesis. On the anammoxosome exterior, protons are used by the nitrite-reducing enzyme, producing hydroxylamine and completing the cycle. All of the anammox enzymes are embedded in the anammoxosome membrane. [Information from L. A. van Niftrik et al., *FEMS Microbiol. Lett.* 233:10, 2004, Fig. 4.]

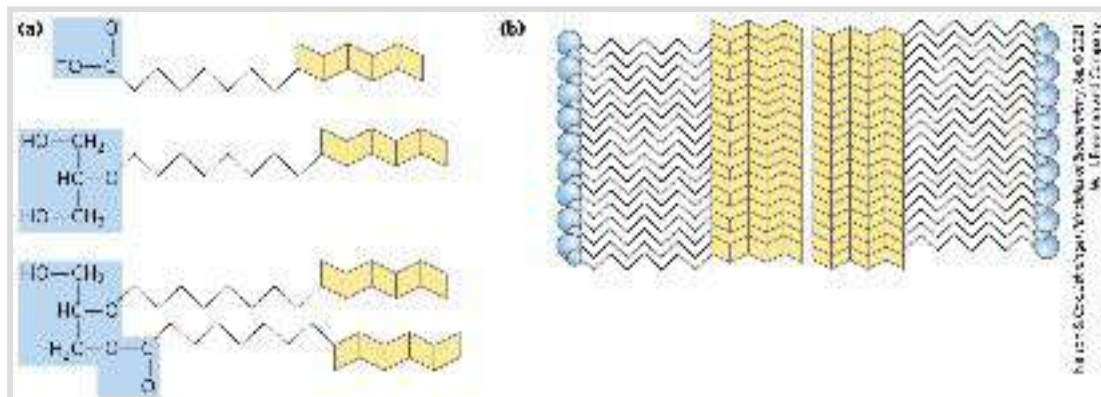


FIGURE 2 (a) Ladderane lipids of the anammoxosome membrane. The mechanism for synthesis of the unstable fused cyclobutane ring structures is unknown. (b) Ladderanes can stack to form a very dense, impermeable, hydrophobic membrane structure, allowing sequestration of the hydrazine produced in the anammox reactions. [Information from L. A. van Niftrik et al., *FEMS Microbiol. Lett.* 233:7, 2004, [Fig. 3](#).]

The anammoxosome was a surprising finding. Bacterial cells generally do not have compartments, and the lack of a membrane-enclosed nucleus is often cited as the primary distinction between eukaryotes and bacteria. One type of organelle in a bacterium was interesting enough, but microbiologists also found that planctomycetes have a nucleus: their chromosomal DNA is contained within a membrane ([Fig. 3](#)). Planctomycetes are an ancient bacterial line with multiple genera, three of which are known to carry out the anammox reactions. Discovery of this subcellular organization has prompted further research to trace the origin of the planctomycetes and the evolution of eukaryotic nuclei. Further study of this group may ultimately bring us closer to a key goal of evolutionary biology: a description of the organism affectionately referred to as LUCA — the last universal common ancestor of all life on our planet.

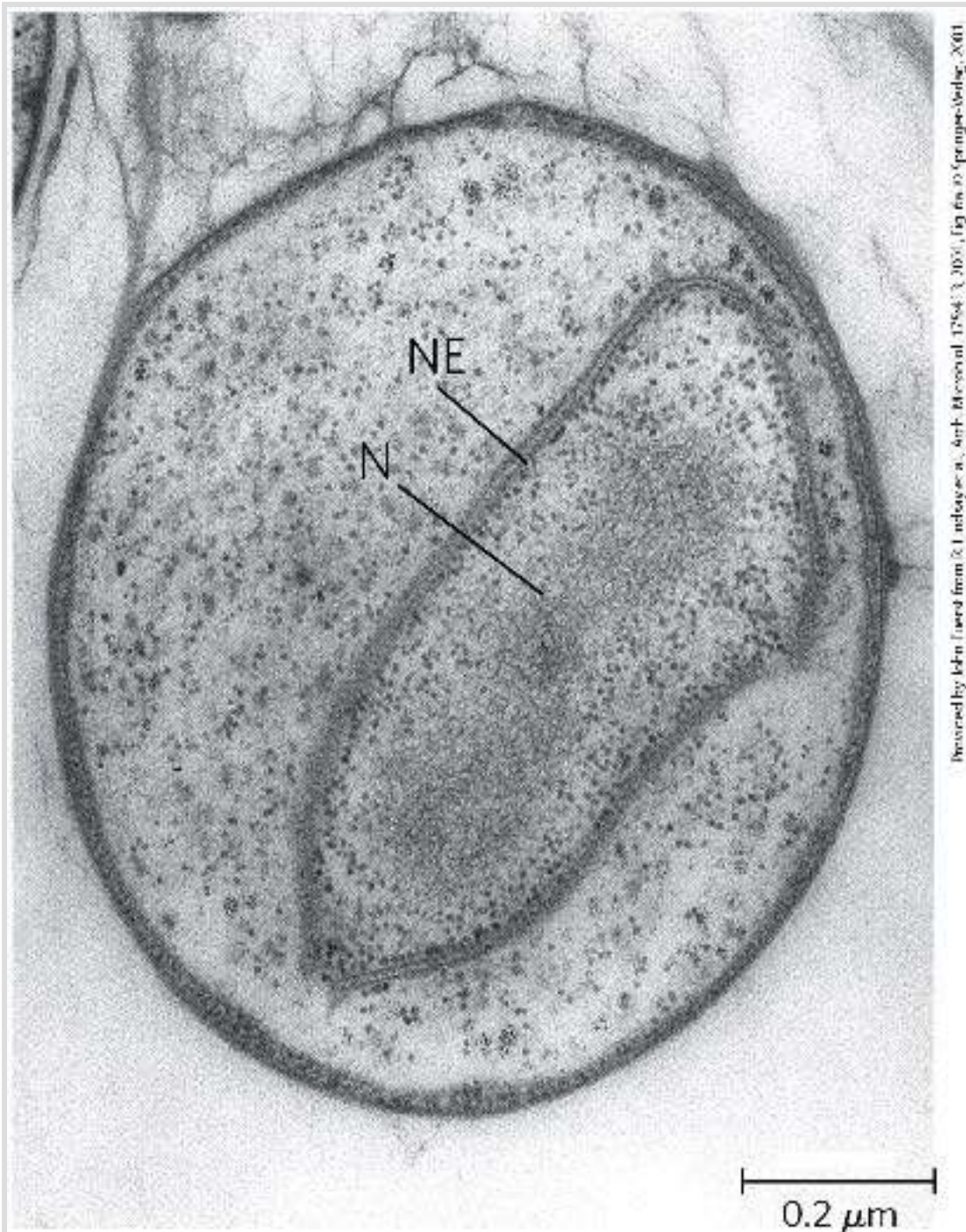


FIGURE 3 Transmission electron micrograph of a cross section through *Gemmata obscuriglobus*, showing the DNA in a nucleus (N) with enclosing nuclear envelope (NE). Bacteria of the *Gemmata* genus (phylum Planctomycetes) do not promote the anammox reactions.

For now, the anammox bacteria offer a major advance in waste treatment, reducing the cost of ammonia removal by as much as 90% (the conventional denitrification steps are eliminated completely, and the aeration costs

associated with nitrification are lower) and reducing the release of polluting byproducts. Clearly, a greater familiarity with the bacterial underpinnings of the biosphere can pay big dividends as we deal with the environmental challenges of the twenty-first century.

Fixation of atmospheric N_2 is not the only source of reduced ammonia for biological systems. Much of it comes from an alternative fate of nitrate that circumvents denitrification. More than 90% of the NH_4^+ generated by vascular plants, algae, and microorganisms comes from **nitrate assimilation**, a two-step reductive process that bypasses atmospheric N_2 . First NO_3^- is reduced to NO_2^- by **nitrate reductase**, then the NO_2^- is reduced to NH_4^+ in a six-electron transfer catalyzed by **nitrite reductase** (Fig. 22-2). Both reactions involve chains of electron carriers and cofactors we have not yet encountered. Nitrate reductase is a large, soluble protein (M_r 220,000). Within the enzyme, a pair of electrons, donated by NADH, flows through —SH groups of cysteine, FAD, and a cytochrome (cyt b_{557}), then to a novel cofactor containing molybdenum, before reducing the substrate NO_3^- to NO_2^- .

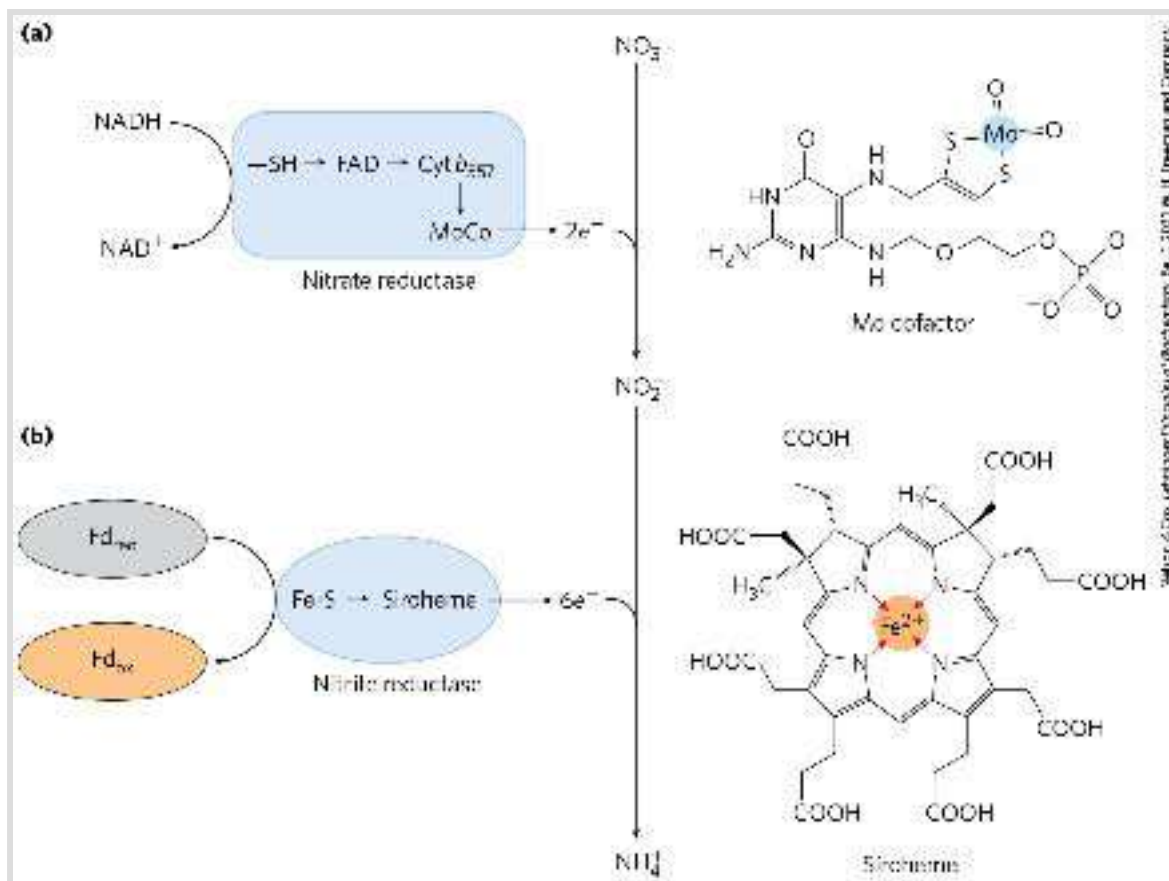


FIGURE 22-2 Nitrate assimilation by nitrate reductase and nitrite reductase. (a) Nitrate reductases of plants and bacteria catalyze the two-electron reduction of NO_3^- to NO_2^- , in which a novel Mo-containing cofactor plays a central role. NADH is the electron donor. (b) Nitrite reductase converts the product of nitrate reductase into NH_4^+ in a six-electron, eight-proton transfer process in which the metallic center in siroheme carries electrons and the carboxyl groups of siroheme may donate protons. The initial source of electrons is reduced ferredoxin.

The nitrite reductase of plants is located in the chloroplasts and receives its electrons from ferredoxin (which is reduced in the light-dependent reactions of photosynthesis; see [Section 20.2](#)). Six electrons, donated one at a time by ferredoxin, pass through a 4Fe-4S center in the enzyme, then through a novel hemelike molecule (siroheme) before reducing NO_2^- to NH_4^+ ([Fig. 22-2](#)).

 Nonphotosynthetic microbes possess a distinct nitrite reductase for which NADPH is the electron donor.

Human activity presents an increasing challenge to the global nitrogen balance, and to all life in the biosphere supported by that balance.  Fixed nitrogen is increasingly necessary to boost production in agriculture. Industrial nitrogen-based fertilizers now contribute as much ammonia and other reactive nitrogen species to the biosphere as do natural processes. Nonfarming manufacturing activity releases additional reactive nitrogen into the atmosphere, including nitric oxide, a prominent greenhouse gas. Controlling the damaging effects of agricultural runoff and industrial pollutants will remain an important component of the continuing effort to expand the food supply for a growing human population.

Nitrogen Is Fixed by Enzymes of the Nitrogenase Complex

The availability of fixed nitrogen, an essential nutrient, may have limited the size of the primordial biosphere. As early cells acquired a capacity to fix atmospheric nitrogen, the biosphere expanded. Evidence for biological nitrogen fixation has been found in sedimentary rocks more than 3 billion years old.

In the biosphere of today, only certain bacteria and archaea can fix atmospheric N_2 . These organisms, called diazotrophs, include

the cyanobacteria of soils and fresh and salt waters, methanogenic archaea (strict anaerobes that obtain energy and carbon by converting H₂ and CO₂ to methane), other kinds of free-living soil bacteria such as *Azotobacter* species, and the nitrogen-fixing bacteria that live as [symbionts](#) in the root nodules of leguminous plants. The most important product of nitrogen fixation is ammonia, which can be used by all organisms either directly or after its conversion to other soluble compounds such as nitrites, nitrates, or amino acids.

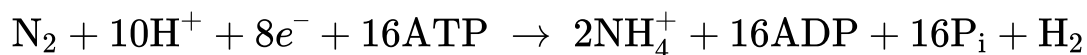
The reduction of nitrogen to ammonia is an exergonic reaction:



The N≡N triple bond, however, is very stable, with a bond energy of 930 kJ/mol. Nitrogen fixation therefore has an extremely high activation energy, and atmospheric nitrogen is almost chemically inert under normal conditions. Ammonia is produced industrially by the Haber process (named for its inventor, Fritz Haber), which requires temperatures of 400 to 500 °C and nitrogen and hydrogen at pressures of tens of thousands of kilopascals (several hundred atmospheres) to provide the necessary activation energy.

Biological nitrogen fixation must occur at biological temperatures and at 0.8 atm of nitrogen, and the high activation barrier is overcome by other means. This is accomplished, at least in part,

by the binding and hydrolysis of ATP. The overall reaction can be written



Biological nitrogen fixation to produce ammonia is carried out by a highly conserved complex of proteins called the **nitrogenase complex**; its central components are **dinitrogenase reductase** and **dinitrogenase** (**Fig. 22-3a**). Dinitrogenase reductase (M_r 60,000) is a dimer of two identical subunits. It contains a single 4Fe-4S redox center (see **Fig. 19-5**), bound between the subunits, and can be oxidized and reduced by one electron. It also has two binding sites for ATP/ADP (one site on each subunit). Dinitrogenase (M_r 240,000), an $\alpha_2\beta_2$ tetramer, has two Fe-containing cofactors that transfer electrons (**Fig. 22-3b**). One, the **P cluster**, has a pair of 4Fe-4S centers; these share a sulfur atom, making an 8Fe-7S center. The second cofactor in dinitrogenase, the **FeMo cofactor**, is a novel structure composed of 7 Fe atoms, 9 inorganic S atoms, a Cys side chain, and a single carbon atom in the center of the FeS cluster. Also part of the cofactor is a molybdenum atom, with ligands that include three inorganic S atoms, a His side chain, and two oxygen atoms from a molecule of homocitrate that is an intrinsic part of the FeMo cofactor.

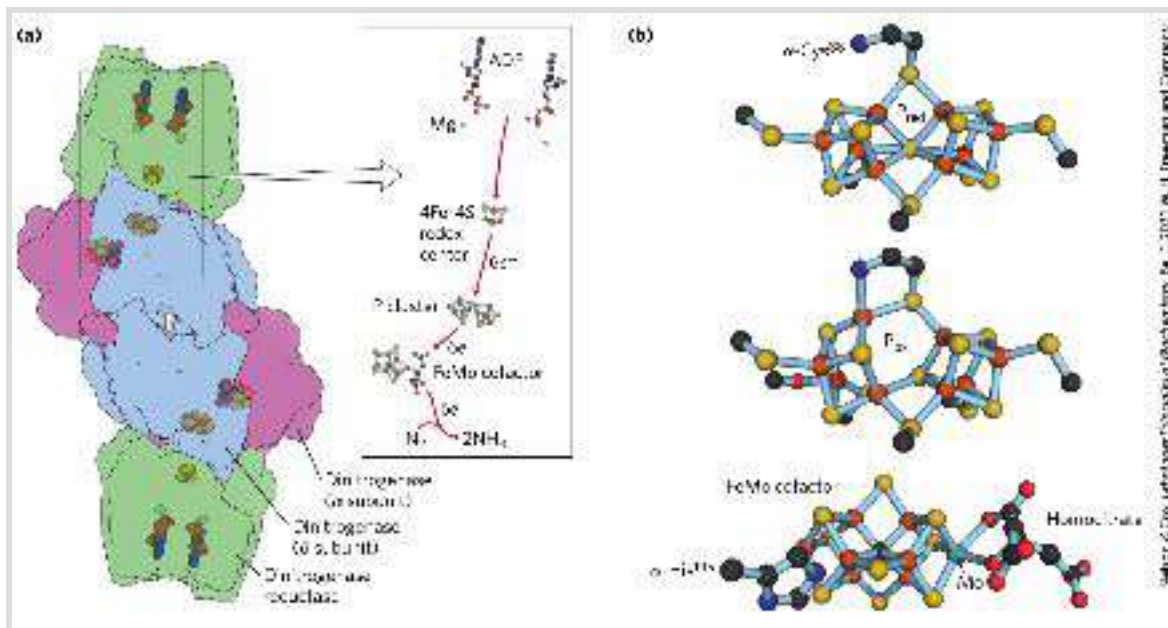


FIGURE 22-3 Enzymes and cofactors of the nitrogenase complex. (a) The holoenzyme consists of two identical dinitrogenase reductase molecules (green), each with a 4Fe-4S redox center and binding sites for two ATP, and two identical dinitrogenase heterodimers (purple and blue), each with a P cluster (Fe-S center) and an FeMo cofactor. In this structure, ADP is bound in the ATP site, to make the crystal more stable. (b) The electron-transfer cofactors. A P cluster is shown here in its reduced (top) and oxidized (middle) forms. The FeMo cofactor (bottom) has a Mo atom with three S ligands, a His ligand, and two oxygen ligands from a molecule of homocitrate. In some organisms, the Mo atom is replaced with a vanadium atom. (Fe is shown in orange, S in yellow.) [Data from (a) PDB ID 1N2C, H. Schindelin et al., *Nature* 387:370, 1997; (b) P_{red} : PDB ID 3MIN, and P_{ox} : PDB ID 2MIN, J. W. Peters et al., *Biochemistry* 36:1181, 1997; FeMo cofactor: PDB ID 1M1N, O. Einsle et al., *Science* 297:1696, 2002.]

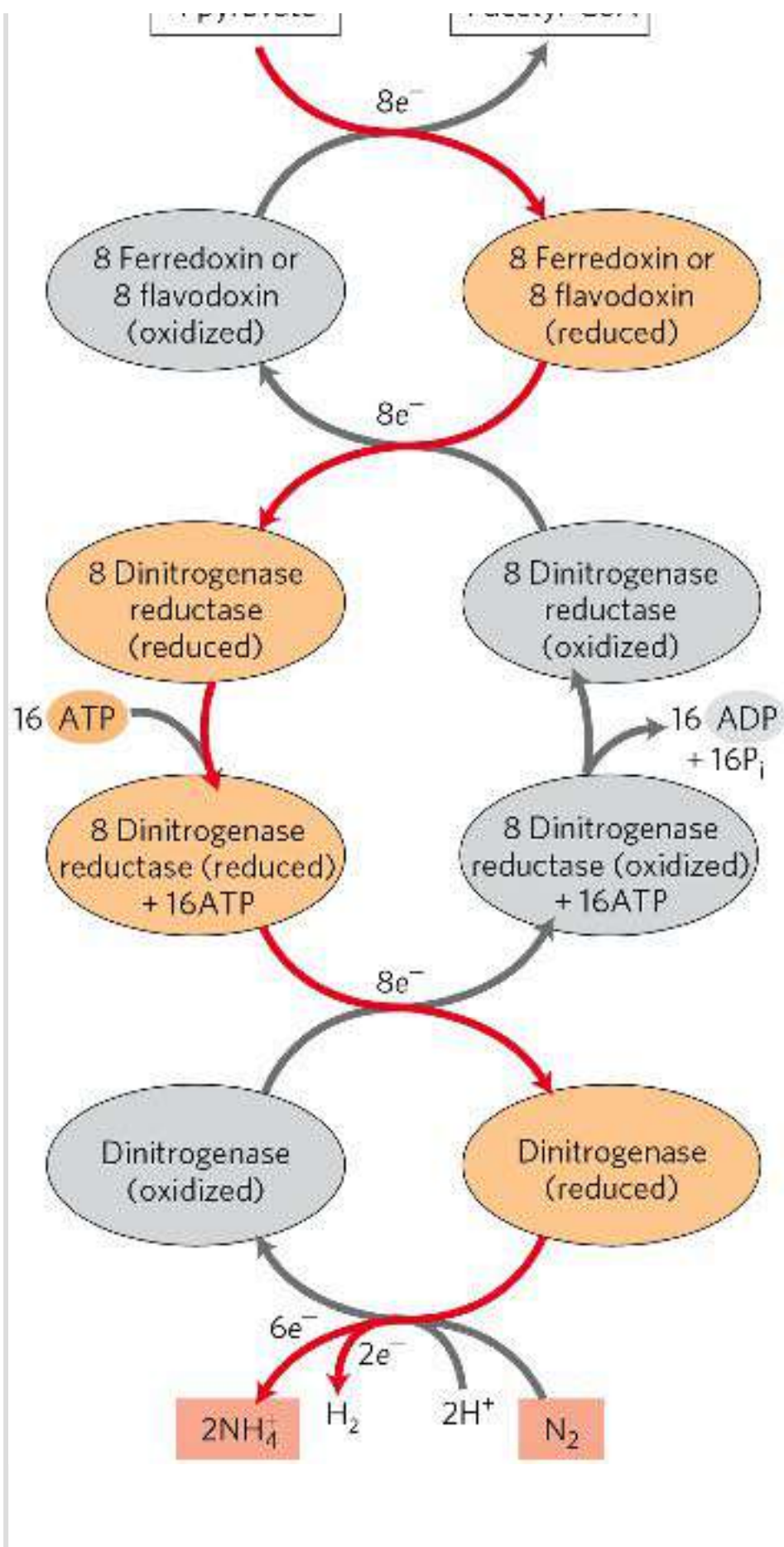
There are two additional forms of nitrogenase. One includes a dinitrogenase with a vanadium-containing cofactor rather than molybdenum (VFe); the other contains a second Fe atom (FeFe). Each of the nitrogenase complexes is encoded by a separate set of genes. The FeMo nitrogenase complex is the ancestral type, and all nitrogen-fixing bacteria and archaea contain it. Some species can produce one or both of the alternative VFe or FeFe types. Although the alternative enzymes are somewhat less efficient,

they may play important roles in environments in which molybdenum is limiting or absent. They may also permit some additional reactions to occur. The vanadium nitrogenase system of *Azotobacter vinelandii* has the remarkable capacity to catalyze the reduction of carbon monoxide (CO) to ethylene (C₂H₄), ethane, and propane.

Nitrogen fixation to produce ammonia is carried out by a highly reduced form of dinitrogenase and requires eight electrons: six for the reduction of N₂ and two to produce one molecule of H₂. Production of H₂ is an obligate part of the reaction mechanism, but the biological role of H₂ in the process is not understood.

Dinitrogenase is reduced by the transfer of electrons from dinitrogenase reductase ([Fig. 22-4](#)). The dinitrogenase tetramer has two binding sites for the reductase. The required eight electrons are transferred from reductase to dinitrogenase one at a time: a reduced reductase molecule binds to the dinitrogenase and transfers a single electron, then the oxidized reductase dissociates from dinitrogenase, in a repeating cycle. Each turn of the cycle requires the hydrolysis of two ATP molecules by the dimeric reductase. The immediate source of electrons to reduce dinitrogenase reductase varies, with reduced **ferredoxin** (see [Section 20.2](#)), reduced flavodoxin, and perhaps other sources playing a role. In at least one species, the ultimate source of electrons to reduce ferredoxin is pyruvate.





Miller & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2007 W. H. Freeman & Co.

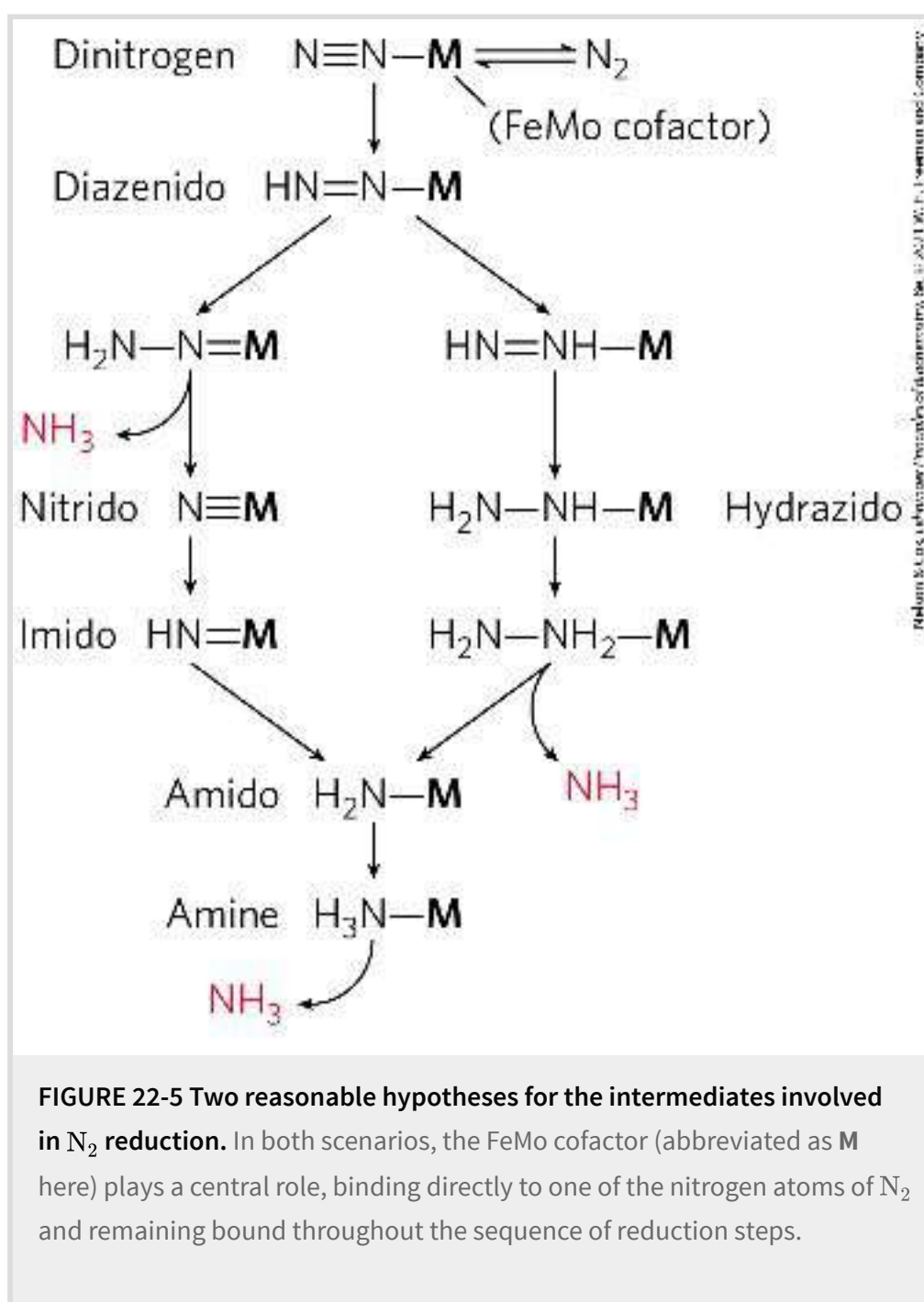
FIGURE 22-4 Electron path in nitrogen fixation by the nitrogenase

complex. Electrons are transferred from pyruvate to dinitrogenase via ferredoxin (or flavodoxin) and dinitrogenase reductase. Dinitrogenase reductase reduces dinitrogenase one electron at a time, with at least six electrons required to fix one molecule of N_2 . Two additional electrons are used to reduce 2H^+ to H_2 in a process that obligatorily accompanies nitrogen fixation in anaerobes, making a total of eight electrons required per N_2 molecule. The subunit structures and metal cofactors of the dinitrogenase reductase and dinitrogenase proteins are described in the text and in [Figure 22-3](#).

In the reaction carried out by dinitrogenase reductase, both ATP binding and ATP hydrolysis bring about protein conformational changes that help overcome the high activation energy of nitrogen fixation. The binding of two ATP molecules to the reductase shifts the reduction potential (E'°) of this protein from -300 to -420 mV, an enhancement of its reducing power that is required to transfer electrons through dinitrogenase to N_2 ; the standard reduction potential for the half-reaction $\text{N}_2 + 6\text{H}^+ + 6e^- \rightarrow 2\text{NH}_3$ is -0.34 V. The ATP molecules are then hydrolyzed just before the actual transfer of one electron to dinitrogenase.

ATP binding and hydrolysis change the conformation of nitrogenase reductase in two regions, which are structurally homologous with the switch 1 and switch 2 regions of the GTP-binding proteins involved in biological signaling (see [Fig. 12-12](#)). ATP binding produces a conformational change that brings the 4Fe-4S center of the reductase closer to the P cluster of dinitrogenase (from $18 \text{ \AA}$ to $14 \text{ \AA}$ away), which facilitates electron

transfer between the reductase and dinitrogenase. The details of electron transfer from the P cluster to the FeMo cofactor, and the means by which eight electrons are accumulated by nitrogenase, are not yet known in detail. Two pathways that conform to available data, both involving the Mo atom as a central player, are illustrated in [Figure 22-5](#).

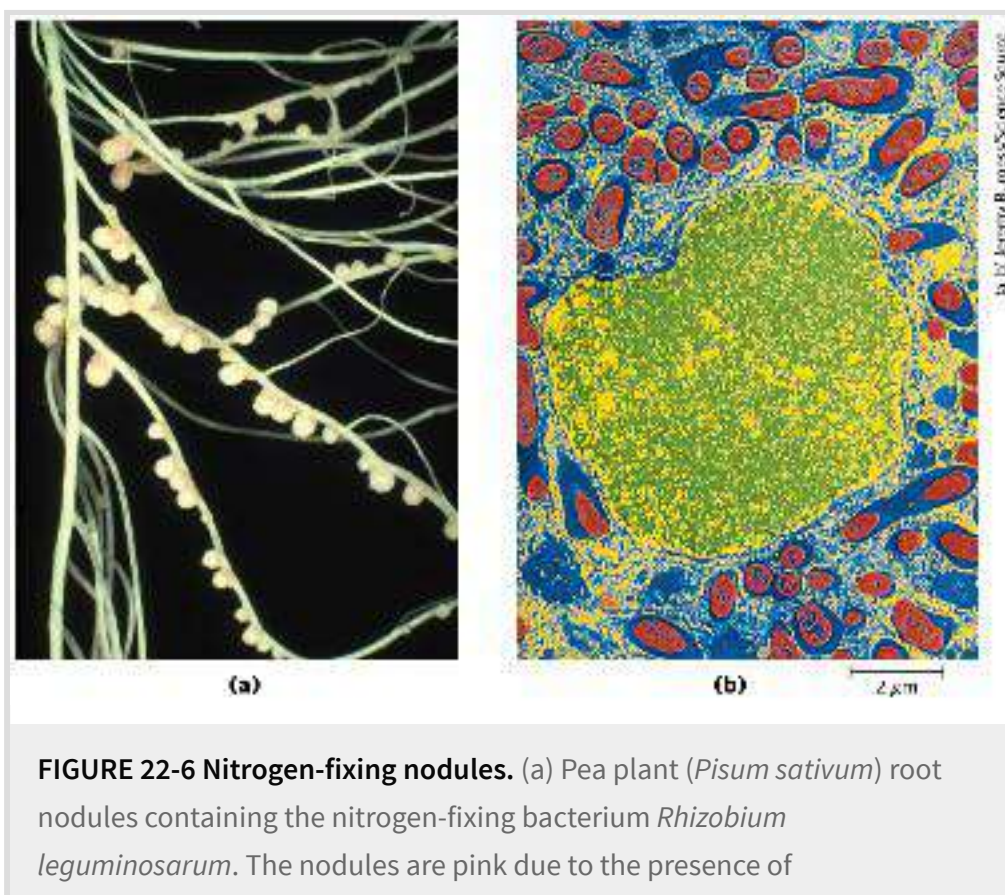


[Information from L. C. Seefeldt et al., *Annu. Rev. Biochem.* 78:701, 2009, Fig. 9.]

 The nitrogenase complex is remarkably unstable in the presence of oxygen. The reductase is inactivated in air, with a half-life of 30 seconds; dinitrogenase has a half-life of only 10 minutes in air. Free-living bacteria that fix nitrogen cope with this problem in a variety of ways. Some live only anaerobically or repress nitrogenase synthesis when oxygen is present. Some aerobic species, such as *A. vinelandii*, partially uncouple electron transfer from ATP synthesis so that oxygen is burned off as rapidly as it enters the cell (see [Box 19-1](#)). When fixing nitrogen, cultures of these bacteria increase in temperature as a result of their efforts to rid themselves of oxygen.

The symbiotic relationship between leguminous plants and the nitrogen-fixing bacteria in their root nodules ([Fig. 22-6](#)) takes care of both the energy requirements and the oxygen lability of the nitrogenase complex. The energy required for nitrogen fixation was probably the evolutionary driving force for this plant-bacteria association. The bacteria in root nodules have access to a large reservoir of energy in the form of abundant carbohydrate and citric acid cycle intermediates made available by the plant. This may allow the bacteria to fix hundreds of times more nitrogen than do their free-living cousins under conditions generally encountered in soils. To solve the oxygen-toxicity problem, the bacteria in root nodules are bathed in a solution of the oxygen-binding heme protein **leghemoglobin**, produced by the plant

(although the heme may be contributed by the bacteria). Leghemoglobin binds all available oxygen so that it cannot interfere with nitrogen fixation, and it efficiently delivers the oxygen to the bacterial electron-transfer system. The benefit to the plant, of course, is a ready supply of reduced nitrogen. In fact, the bacterial symbionts typically produce far more NH_3 than is needed by their symbiotic partner; the excess is released into the soil. The efficiency of the symbiosis between plants and bacteria is evident in the enrichment of soil nitrogen brought about by leguminous plants. This enrichment of NH_3 in the soil is the basis of crop rotation methods, in which plantings of nonleguminous plants (such as maize) that extract fixed nitrogen from the soil are alternated with plantings of legumes such as alfalfa, peas, or clover.



leghemoglobin; this heme protein has a very high binding affinity for oxygen, which strongly inhibits nitrogenase. (b) Artificially colorized electron micrograph of a thin section through a pea root nodule. Symbiotic nitrogen-fixing bacteria, or bacteroids (red), live inside the nodule cell, surrounded by the peribacteroid membrane (blue). Bacteroids produce the nitrogenase complex that converts atmospheric nitrogen (N_2) to ammonium (NH_4^+); without the bacteroids, the plant is unable to utilize N_2 . (The cell nucleus is shown in yellow/green. Not visible in this micrograph are other organelles of the infected root cell that are normally found in plant cells.)

Nitrogen fixation is energetically costly: 16 ATP and 8 electrons yield only 2 NH_3 . It is therefore not surprising that the process is tightly regulated so that NH_3 is produced only when needed. High [ADP], an indicator of low [ATP], is a strong inhibitor of nitrogenase. NH_4^+ represses the expression of the ~20 nitrogen fixation (*nif*) genes, effectively shutting down the pathway. Covalent alteration of nitrogenase is also used in some diazotrophs to control nitrogen fixation in response to the availability of NH_4^+ in the surroundings. Transfer of an ADP-ribosyl group from NADH to a specific Arg residue in the nitrogenase reductase shuts down N_2 fixation in *Rhodospirillum*, for example. This is the same covalent modification that we saw in the case of G protein inhibition by the toxins of cholera and pertussis (see [Fig. 12-14](#)).

Nitrogen fixation is the subject of intense study because of its immense practical importance. Industrial production of ammonia for use in fertilizers requires a large and expensive input of energy, and this has spurred a drive to develop

recombinant or transgenic organisms that can fix nitrogen. In principle, recombinant DNA techniques ([Chapter 9](#)) might be used to transfer the DNA that encodes the enzymes of nitrogen fixation into non-nitrogen-fixing bacteria and plants. However, those genes alone will not suffice. About 20 genes are essential to nitrogenase activity in bacteria, many of them needed for the synthesis, assembly, and insertion of the cofactors. There is also the problem of protecting the enzyme in its new setting from destruction by oxygen. In all, there are formidable challenges in engineering new nitrogen-fixing plants. Success in these efforts will depend on overcoming the problem of oxygen toxicity in any cell that produces nitrogenase.

Ammonia Is Incorporated into Biomolecules through Glutamate and Glutamine

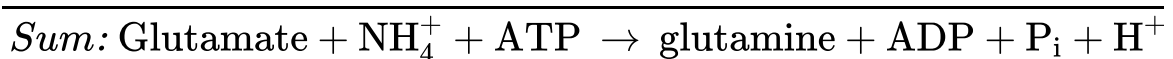
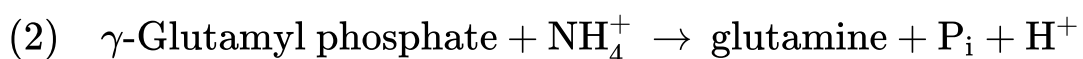
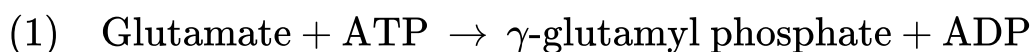
Reduced nitrogen in the form of NH_4^+ is assimilated into amino acids and then into other nitrogen-containing biomolecules.

 Two amino acids, **glutamate** and **glutamine**, provide the critical entry point. Recall that these same two amino acids play central roles in the catabolism of ammonia and amino groups in amino acid oxidation ([Chapter 18](#)). Glutamate is the source of amino groups for most other amino acids, through transamination reactions (the reverse of the reaction shown in [Fig. 18-4](#)). The amide nitrogen of glutamine is a source of amino groups in a wide range of biosynthetic processes. In most types of

cells, and in extracellular fluids in higher organisms, one or both of these amino acids are present at higher concentrations — sometimes an order of magnitude or more higher — than other amino acids. An *Escherichia coli* cell requires so much glutamate that this amino acid is one of the primary solutes in the cytosol. Its concentration is regulated not only in response to the cell's nitrogen requirements but also to maintain an osmotic balance between the cytosol and the external medium.

The biosynthetic pathways to glutamate and glutamine are simple, and all or some of the steps occur in most organisms. The most important pathway for the assimilation of NH_4^+ into glutamate requires two reactions. The net effect is to convert glutamate, α -ketoglutarate, and ammonia into two molecules of glutamate.

First, NH_4^+ is reacted with glutamate to produce glutamine, using the enzyme **glutamine synthetase**. This reaction takes place in two steps, with enzyme-bound γ -glutamyl phosphate as an intermediate (see [Fig. 18-8](#)):



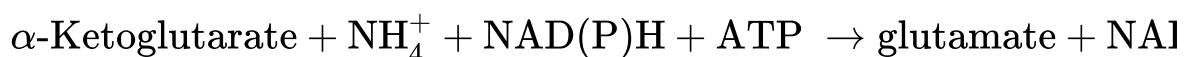
(22-1)

Glutamine synthetase is found in all organisms. In addition to its importance for NH_4^+ assimilation in bacteria, it has a central role in amino acid metabolism in mammals, converting free NH_4^+ , which is toxic, to glutamine for transport in the blood ([Chapter 18](#)).

In the second reaction needed for NH_4^+ assimilation, the glutamine reacts with α -ketoglutarate to generate two molecules of glutamate. In bacteria and plants, this reaction is catalyzed by **glutamate synthase**. (An alternative name for this enzyme, glutamate:oxoglutarate aminotransferase, yields the acronym GOGAT, by which the enzyme also is known.) α -Ketoglutarate, an intermediate of the citric acid cycle, undergoes reductive amination with glutamine as nitrogen donor:



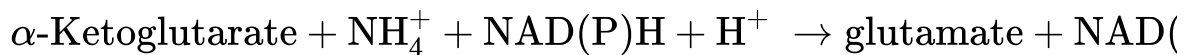
The net reaction of glutamine synthetase and glutamate synthase ([Eqns 22-1](#) and [22-2](#)) is



Glutamate synthase is not present in animals, which instead maintain high levels of glutamate by processes such as the

transamination of α -ketoglutarate during amino acid catabolism. Plants possess a second alternative form of glutamate synthase that uses reduced ferredoxin rather than NADPH as a source of reducing electrons.

Glutamate can also be formed in yet another, albeit minor, pathway: the reaction of α -ketoglutarate and NH_4^+ to form glutamate in one step.  This is catalyzed by glutamate dehydrogenase, an enzyme present in all organisms. Reducing power is furnished by NADPH:



We encountered this reaction in the catabolism of amino acids (see [Fig. 18-7](#)). In eukaryotic cells, glutamate dehydrogenase is located in the mitochondrial matrix. The reaction equilibrium favors the reactants, and the K_m for NH_4^+ (~ 1 mM) is so high that the reaction is not important for NH_4^+ assimilation in mammals. (Recall that the glutamate dehydrogenase reaction, in reverse (see [Fig. 18-10](#)), is one source of NH_4^+ destined for the urea cycle.) In microorganisms and plants, concentrations of NH_4^+ high enough for the glutamate dehydrogenase reaction to make a significant contribution to glutamate levels generally occur only when NH_3 is added artificially to the growth environment. In general, soil bacteria and plants rely on the two-enzyme pathway outlined above ([Eqns 22-1, 22-2](#)).

Glutamine Synthetase Is a Primary Regulatory Point in Nitrogen Metabolism

There are three known classes of glutamine synthetases. The class I enzyme (GSI, found in bacteria) has 12 identical subunits of M_r 50,000 ([Fig. 22-7](#)) and is regulated both allosterically and by covalent modification. The class II enzyme (GSII, found in eukaryotes and some bacteria) has 10 identical subunits. The third class of glutamine synthetases (GSIII), so far found only in two bacterial species, are much larger enzymes, consisting of a double-ringed dodecamer of identical chains.

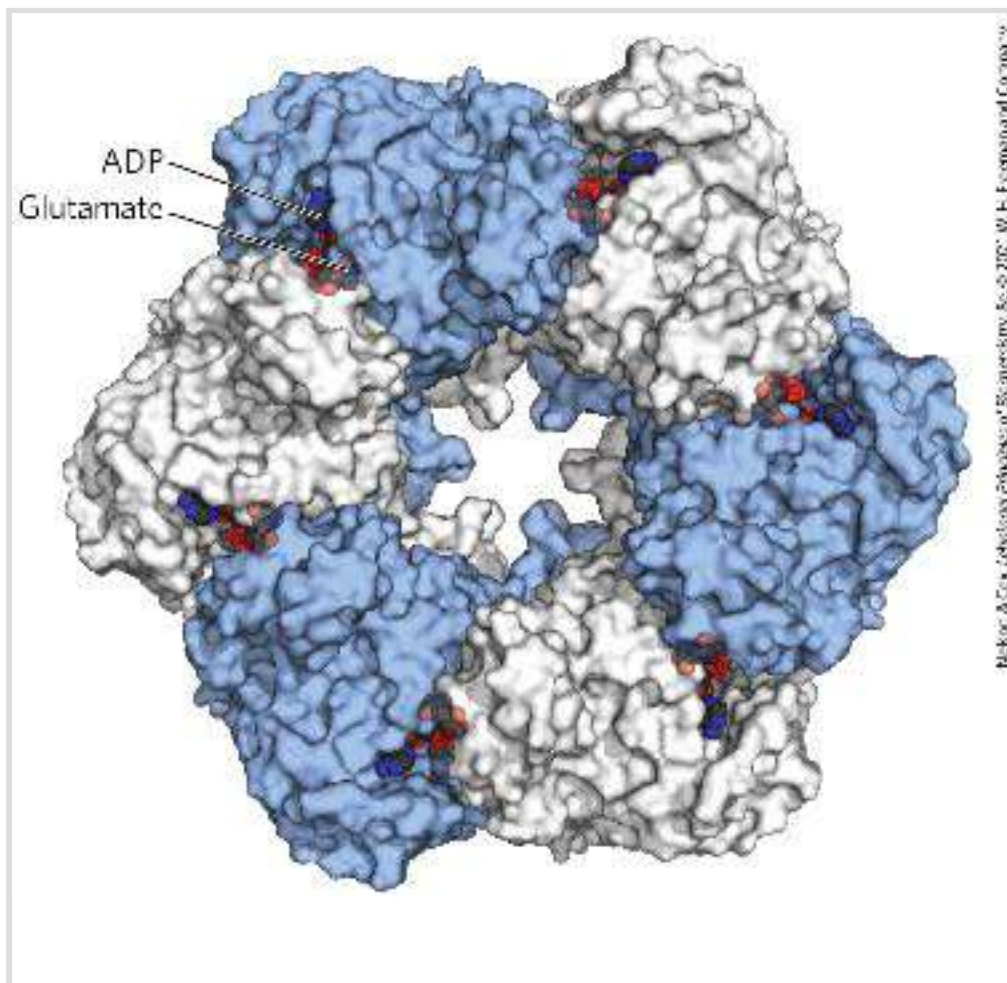
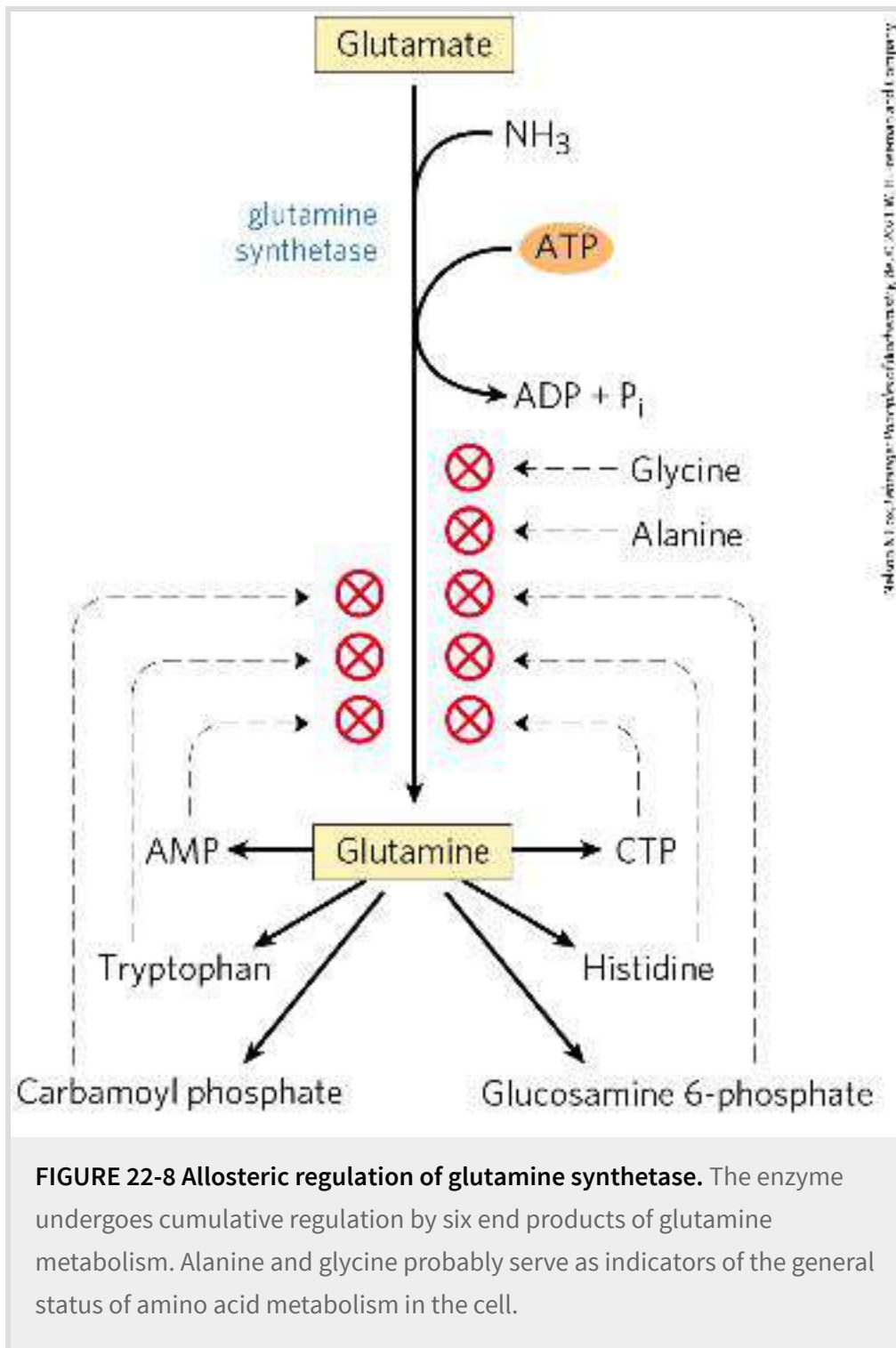


FIGURE 22-7 Subunit structure of bacterial type I glutamine synthetase.

This view shows 6 of the 12 identical subunits; a second layer of 6 subunits lies directly beneath those shown. Each of the 12 subunits has an active site, where ATP and glutamate are bound in orientations that favor transfer of a phosphoryl group from ATP to the side-chain carboxyl of glutamate. In this crystal structure, ADP occupies the ATP site. [Data from PDB ID 2GLS, M. M. Yamashita et al., *J. Biol. Chem.* 264:17,681, 1989.]

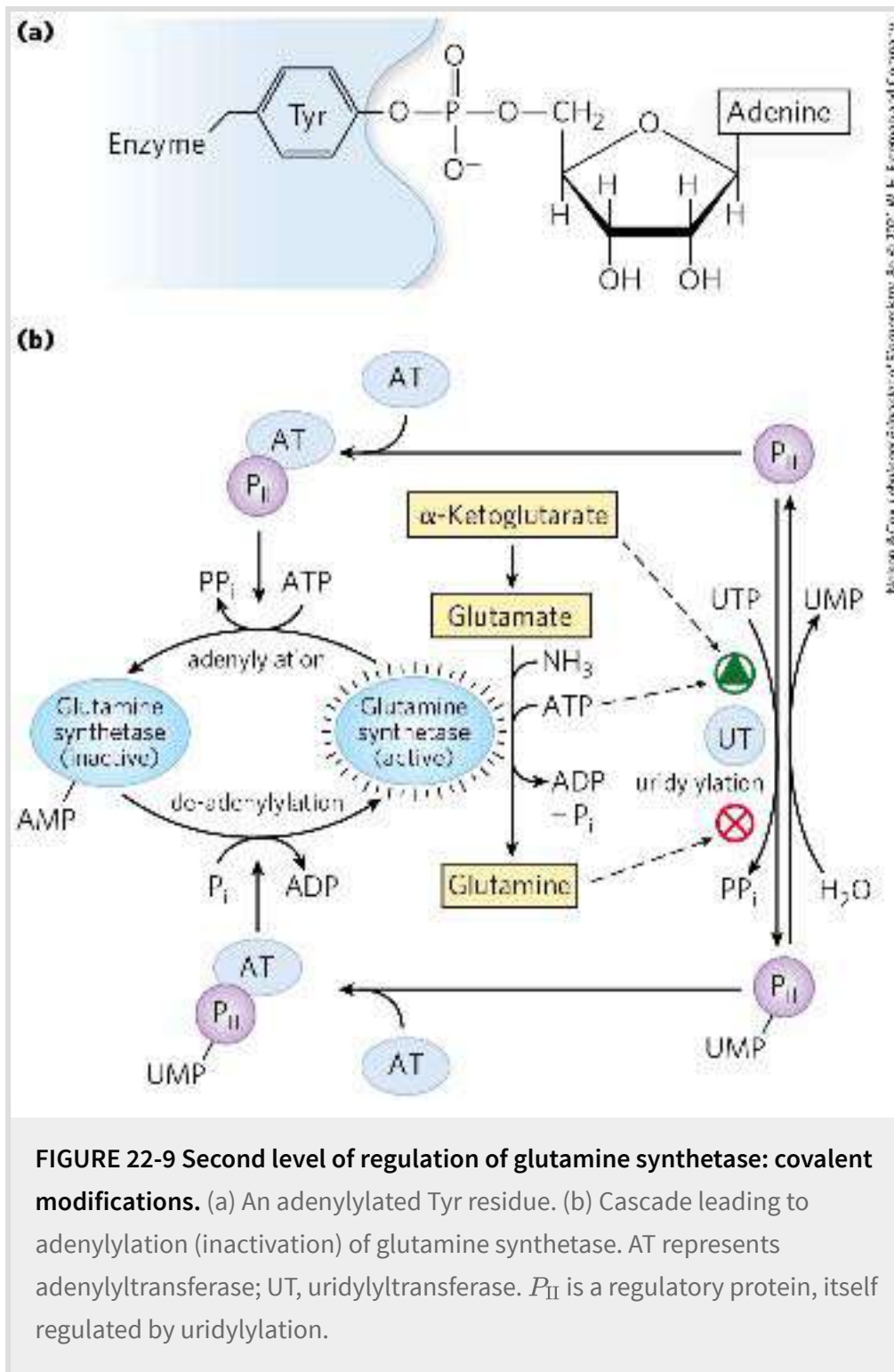


Befitting their central metabolic role as an entry point for reduced nitrogen, GSI glutamine synthetases are highly regulated. In enteric bacteria such as *E. coli*, the regulation is unusually complex. Alanine, glycine, and at least six end products of glutamine metabolism are allosteric inhibitors ([Fig. 22-8](#)). Each inhibitor alone produces only partial inhibition, but the effects of multiple inhibitors are more than additive, and all eight together virtually shut down the enzyme. This is an example of cumulative feedback inhibition. This control mechanism provides a constant adjustment of glutamine levels to match immediate metabolic requirements.



Superimposed on the allosteric regulation is inhibition by adenylation of (addition of AMP to) Tyr^{397} , located near the enzyme's active site ([Fig. 22-9](#)). This covalent modification increases sensitivity to the allosteric inhibitors, and activity

decreases as more subunits are adenylylated. Both adenylylation and de-adenylylation are promoted by **adenylyltransferase** (AT in [Fig. 22-9](#)), part of a complex enzymatic cascade that responds to levels of glutamine, α -ketoglutarate, ATP, and P_i . The activity of adenylyltransferase is modulated by binding to a regulatory protein called P_{II} , and the activity of P_{II} , in turn, is regulated by covalent modification (uridylylation), again at a Tyr residue. The adenylyltransferase complex with uridylylated P_{II} (P_{II} -UMP) stimulates de-adenylylation, whereas the same complex with deuridylylated P_{II} stimulates adenylylation of glutamine synthetase. Both uridylylation and deuridylylation of P_{II} are brought about by a single enzyme, **uridylyltransferase**. Uridylylation is inhibited by binding of glutamine and P_i to uridylyltransferase and is stimulated by binding of α -ketoglutarate and ATP to P_{II} .



The regulation does not stop there. The uridylylated P_{II} also mediates the activation of transcription of the gene encoding glutamine synthetase, thus increasing the cellular concentration

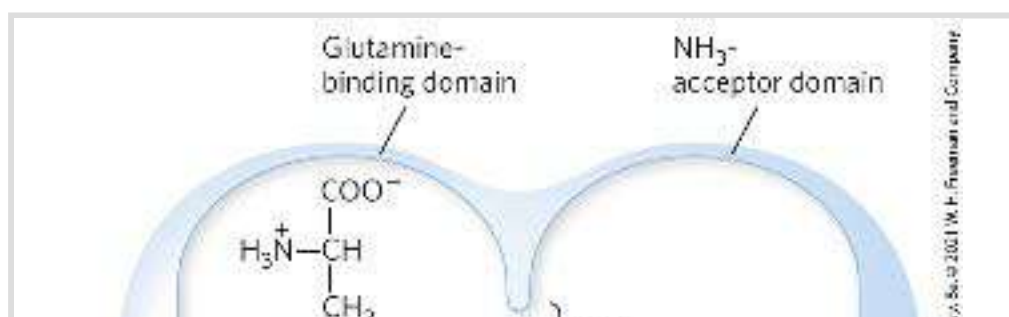
of the enzyme; the deacylated P_{II} brings about a decrease in transcription of the same gene. This mechanism involves an interaction of P_{II} with additional proteins involved in gene regulation, of a type described in [Chapter 28](#). The net result of this elaborate system of controls is a decrease in glutamine synthetase activity when glutamine levels are high, and an increase in activity when glutamine levels are low and α -ketoglutarate and ATP (substrates for the synthetase reaction) are available. The multiple layers of regulation permit a sensitive response in which glutamine synthesis is tailored to cellular needs.

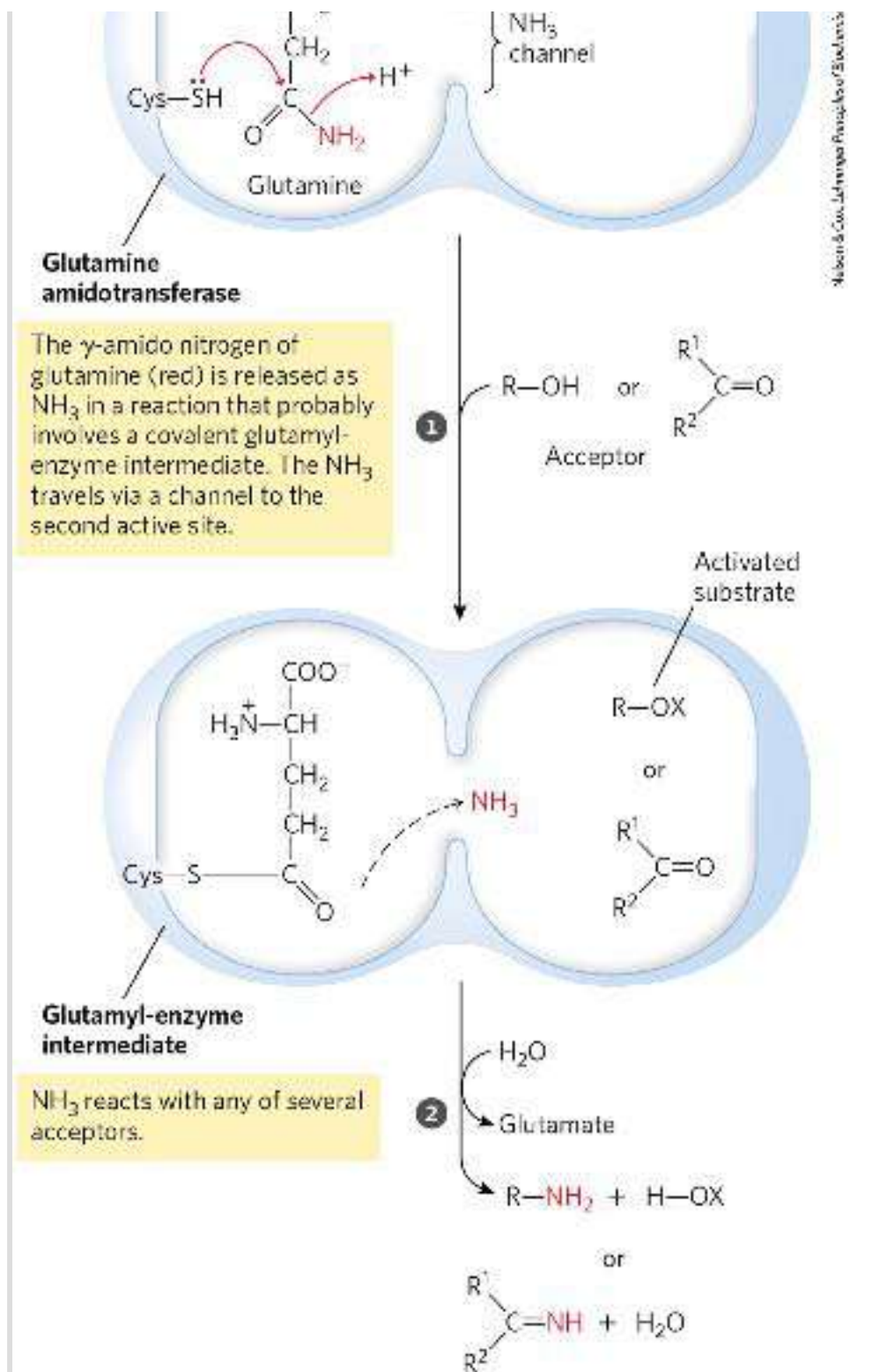
Several Classes of Reactions Play Special Roles in the Biosynthesis of Amino Acids and Nucleotides

The pathways described in this chapter include a variety of interesting chemical rearrangements. Several of these recur and deserve special note before we progress to the pathways themselves. These are (1) transamination reactions and other rearrangements promoted by enzymes containing pyridoxal phosphate; (2) transfer of one-carbon groups, with either tetrahydrofolate (usually at the $-\text{CHO}$ and $-\text{CH}_2\text{OH}$ oxidation levels) or *S*-adenosylmethionine (at the $-\text{CH}_3$ oxidation level) as cofactor; and (3) transfer of amino groups derived from the amide nitrogen of glutamine. Pyridoxal phosphate (PLP), tetrahydrofolate (H_4 folate), and *S*-adenosylmethionine (adoMet)

are described in some detail in [Chapter 18](#) (see [Figs. 18-6](#), [18-17](#), and [18-18](#)). Here we focus on amino group transfer involving the amide nitrogen of glutamine.

More than a dozen known biosynthetic reactions use glutamine as the major physiological source of amino groups, and most of these occur in the pathways outlined in this chapter. As a class, the enzymes catalyzing these reactions are called **glutamine amidotransferases**. All have two structural domains: one binding glutamine, the other binding the second substrate, which serves as amino group acceptor ([Fig. 22-10](#)). A conserved Cys residue in the glutamine-binding domain is believed to act as a nucleophile, cleaving the amide bond of glutamine and forming a covalent glutamyl-enzyme intermediate. The NH_3 produced in this reaction is not released, but instead is transferred through an “ammonia channel” to a second active site, where it reacts with the second substrate to form the aminated product. The covalent intermediate is hydrolyzed to the free enzyme and glutamate. If the second substrate must be activated, the usual method is the use of ATP to generate an acyl phosphate intermediate (R—OX in [Fig. 22-10](#), where X is a phosphoryl group). The enzyme glutaminase acts similarly but uses H_2O as the second substrate, yielding NH_4^+ and glutamate (see [Fig. 18-8](#)).





MECHANISM FIGURE 22-10 Proposed mechanism for glutamine

amidotransferases. Each enzyme has two domains. The glutamine-binding domain contains structural elements conserved among many of these enzymes, including a Cys residue required for activity. The NH_3 -acceptor (second-substrate) domain varies. Two types of amino acceptors are shown.

X represents an activating group, typically a phosphoryl group derived from ATP, that facilitates displacement of a hydroxyl group from R—OH by NH₃.

SUMMARY 22.1 *Overview of Nitrogen Metabolism*

■ The molecular nitrogen that makes up 80% of Earth's atmosphere is unavailable to most living organisms until it is reduced. A complex web of reactions converts atmospheric N₂ to biologically useful forms and maintains a global balance between them. Prominent species that are interconverted within this web include ammonia (NH₃ or NH₄⁺; most reduced), nitrite (NO₂⁻), and nitrate (NO₃⁻; most oxidized). Conversion of N₂ to ammonia is fixation. Nitrification constitutes the steps converting ammonia to nitrate. Conversion of nitrate to N₂ constitutes denitrification. The alternative conversion of nitrate to ammonia is nitrate assimilation.

■ Fixation of N₂ as NH₃ is carried out by the nitrogenase complex, in a reaction that requires large investments of ATP and of reducing power. The nitrogenase complex is highly labile in the presence of O₂, and it is subject to regulation by the supply of NH₃.

■ In living systems, reduced nitrogen is incorporated first into amino acids and then into a variety of other biomolecules, including nucleotides. The key entry point is the amino acid glutamate. Glutamate and glutamine are the nitrogen donors in a wide range of biosynthetic reactions.

■ Glutamine synthetase, which catalyzes the formation of glutamine from glutamate, is a main regulatory enzyme of nitrogen metabolism.

■ The amino acid and nucleotide biosynthetic pathways make repeated use of the biological cofactors pyridoxal phosphate, tetrahydrofolate, and *S*-adenosylmethionine. Pyridoxal phosphate is required for transamination reactions involving glutamate and for other amino acid transformations. One-carbon transfers require *S*-adenosylmethionine and tetrahydrofolate. Glutamine amidotransferases catalyze reactions that incorporate nitrogen derived from the amide group of glutamine.

22.2 Biosynthesis of Amino Acids

All amino acids are derived from intermediates in glycolysis, the citric acid cycle, or the pentose phosphate pathway ([Fig. 22-11](#)). Nitrogen enters these biosynthetic pathways by way of glutamate and glutamine. Some pathways are simple, others are not. Ten of the amino acids are just one or several steps removed from the common metabolite from which they are derived. The biosynthetic pathways for others, such as the aromatic amino acids, are more complex.

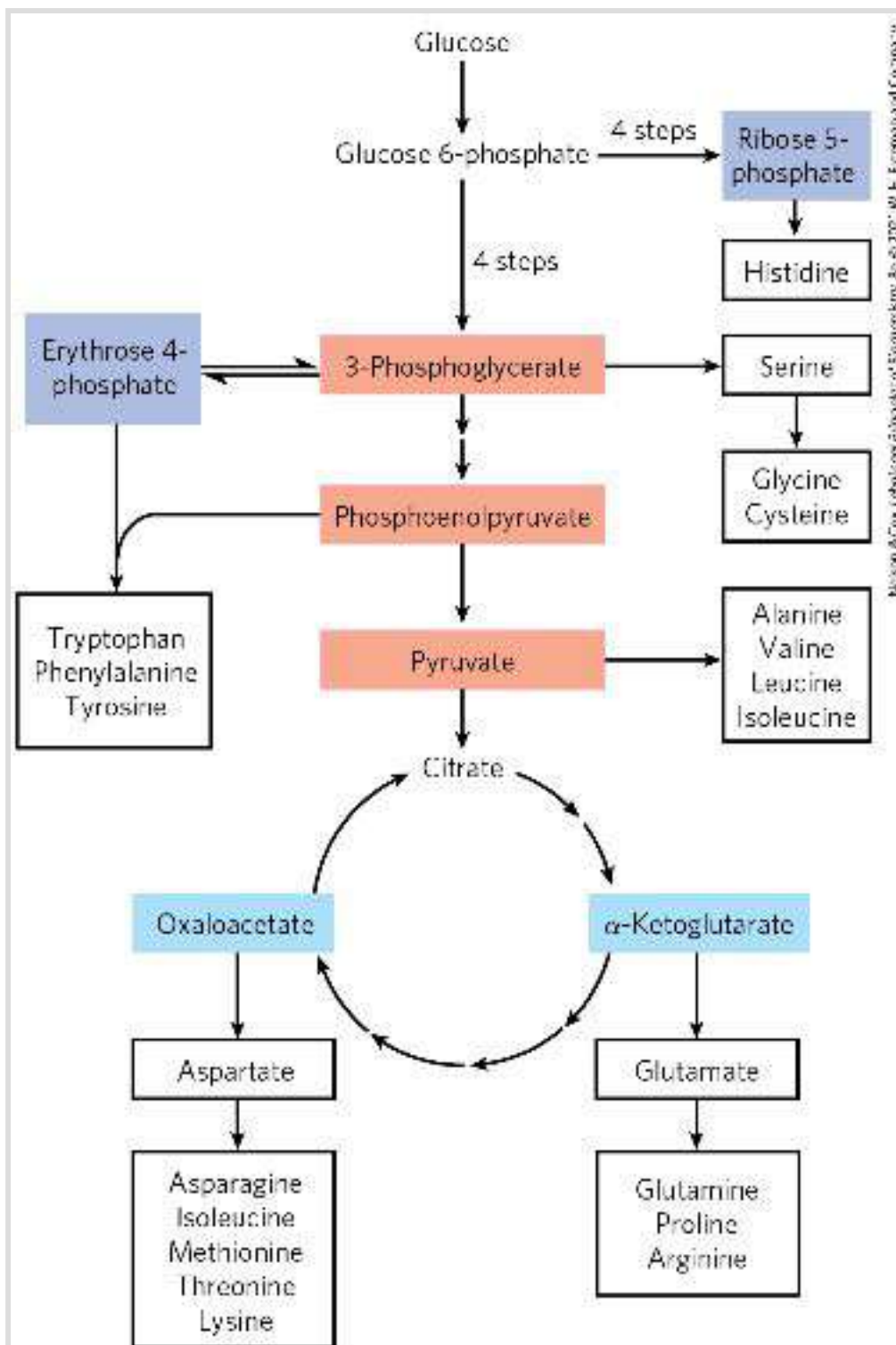


FIGURE 22-11 Overview of amino acid biosynthesis. The carbon skeleton precursors derive from three sources: glycolysis (light red), the citric acid cycle (blue), and the pentose phosphate pathway (purple).

Organisms Vary Greatly in Their Ability to Synthesize the 20 Common Amino Acids

Whereas most bacteria and plants can synthesize all 20 amino acids, mammals can synthesize only about half of them — generally those with simple pathways. These are often called **nonessential amino acids** (see [Table 18-1](#)). The label is somewhat misleading, however, because innate biosynthetic pathways often do not provide enough of these amino acids to support optimal growth and health. The remaining amino acids, the **essential amino acids**, cannot be synthesized by most mammals and must be obtained from food. A few amino acids are conditionally essential in mammals, required at particular stages of development. Unless otherwise indicated, the pathways for the 20 common amino acids presented below are those operative in bacteria.

A useful way to organize these biosynthetic pathways is to group them into six families corresponding to their metabolic precursors ([Table 22-1](#)), and we use this approach to structure the detailed descriptions that follow. In addition to these six precursors, there is a notable intermediate in several pathways of amino acid and nucleotide synthesis: **5-phosphoribosyl-1-pyrophosphate (PRPP)**:

TABLE 22-1 Amino Acid Biosynthetic Families, Grouped by Metabolic Precursor

α -Ketoglutarate

Glutamate
Glutamine
Proline
Arginine

Pyruvate

Alanine
Valine^a
Leucine^a
Isoleucine^a

3-Phosphoglycerate

Serine
Glycine
Cysteine

Phosphoenolpyruvate and erythrose 4-phosphate

Tryptophan^a
Phenylalanine^a
Tyrosine^b

Oxaloacetate

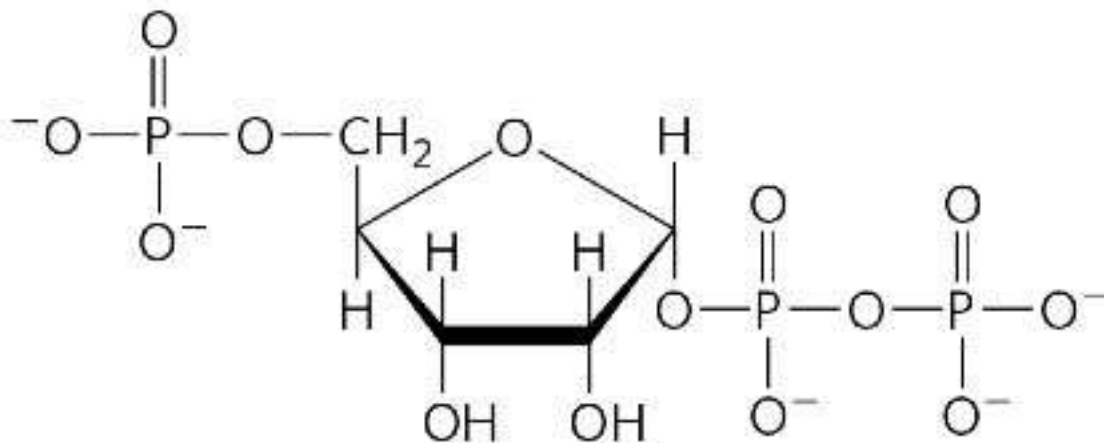
Aspartate
Asparagine
Isoleucine
Methionine^a
Threonine^a
Lysine^a

Ribose 5-phosphate

Histidine^a

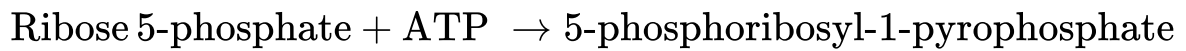
^aEssential amino acids in mammals.

^bDerived from phenylalanine in mammals.



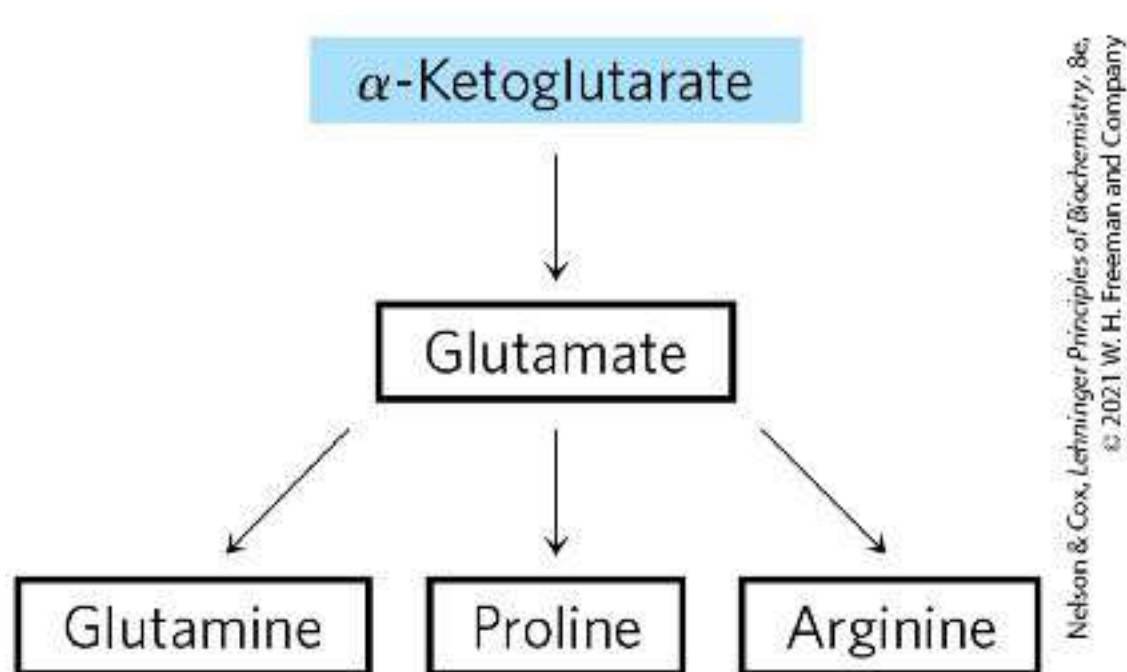
PRPP is synthesized from ribose 5-phosphate derived from the pentose phosphate pathway (see [Fig. 14-31](#)), in a reaction

catalyzed by **ribose phosphate pyrophosphokinase**:



This enzyme is allosterically regulated by many of the biomolecules for which PRPP is a precursor.

α -Ketoglutarate Gives Rise to Glutamate, Glutamine, Proline, and Arginine



We have already described the biosynthesis of **glutamate** and **glutamine**. **Proline** is a cyclized derivative of glutamate ([Fig. 22-](#)

12).  In the first step of proline synthesis, ATP reacts with the γ -carboxyl group of glutamate to form an acyl phosphate, which is reduced by NADPH or NADH to glutamate γ -semialdehyde. This intermediate undergoes rapid spontaneous cyclization and is then reduced further to yield proline.

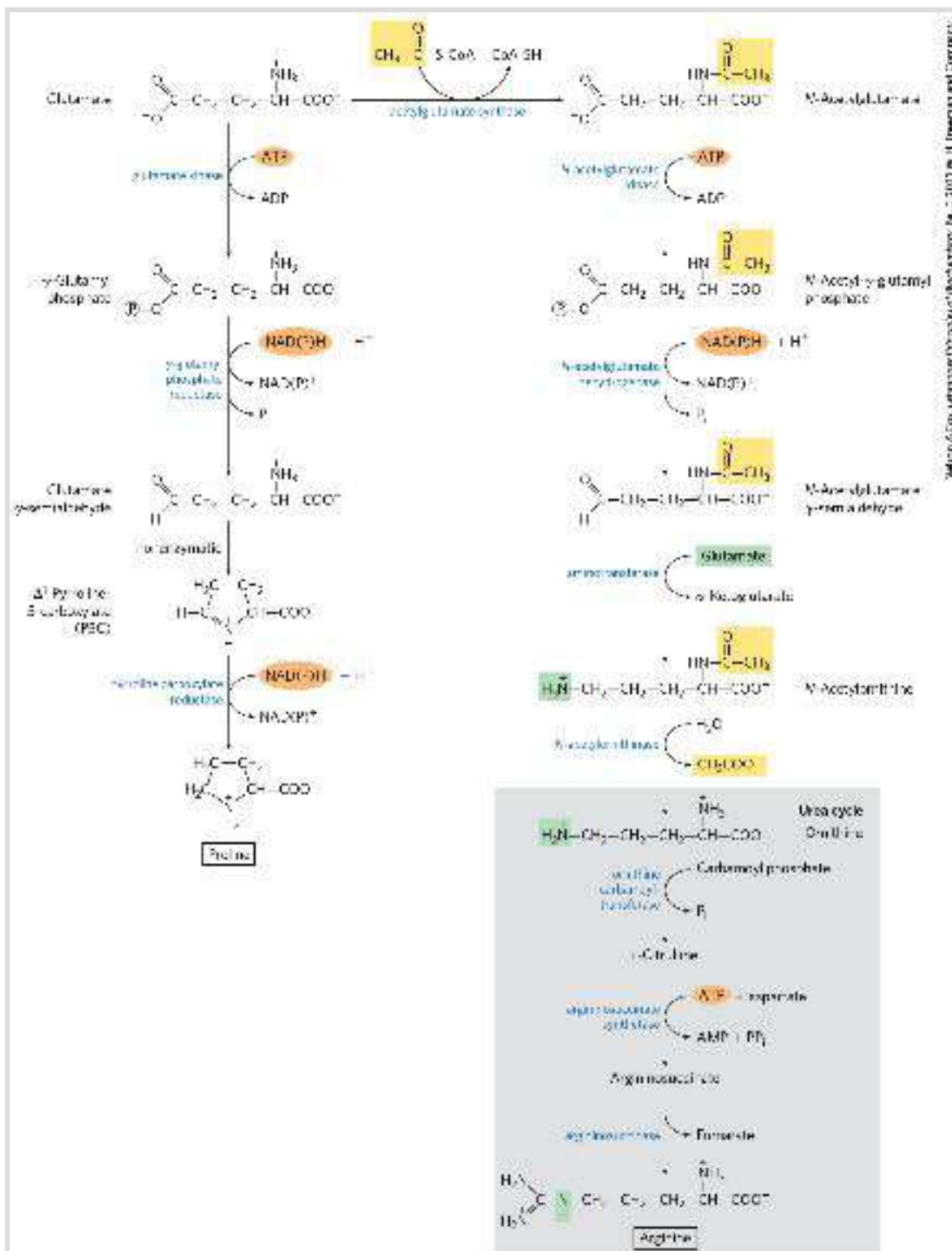


FIGURE 22-12 Biosynthesis of proline and arginine from glutamate in bacteria. All five carbon atoms of proline arise from glutamate. In many organisms, glutamate dehydrogenase is unusual in that it uses *either* NADH or NADPH as a cofactor. The same may be true of other enzymes in these pathways. The γ -semialdehyde in the proline pathway undergoes a rapid, reversible cyclization to Δ^1 -pyrroline-5-carboxylate (P5C),

with the equilibrium favoring P5C formation. Cyclization is averted in the ornithine/arginine pathway by acetylation of the α -amino group of glutamate in the first step and removal of the acetyl group after the transamination. This acetyl group is highlighted in yellow. Although some bacteria lack arginase and thus the complete urea cycle, they can synthesize arginine from ornithine in steps that parallel the mammalian urea cycle, with citrulline and argininosuccinate as intermediates (see [Fig. 18-10](#)).

Here, and in subsequent figures in this chapter, the reaction arrows indicate the linear path to the final products, without considering the reversibility of individual steps. For example, the step of the pathway leading to arginine that is catalyzed by *N*-acetylglutamate dehydrogenase is chemically similar to the glyceraldehyde 3-phosphate dehydrogenase reaction in glycolysis (see [Fig. 14-7](#)) and is readily reversible.

Arginine is synthesized from glutamate via ornithine and the urea cycle in animals ([Chapter 18](#)). In principle, ornithine could also be synthesized from glutamate γ -semialdehyde by transamination, but the spontaneous cyclization of the semialdehyde in the proline pathway precludes a sufficient supply of this intermediate for ornithine synthesis. Bacteria have a *de novo* biosynthetic pathway for ornithine (and thus arginine) that parallels some steps of the proline pathway but includes two additional steps that avoid the problem of the spontaneous cyclization of glutamate γ -semialdehyde ([Fig. 22-12](#)). In the first step, the α -amino group of glutamate is blocked by an acetylation requiring acetyl-CoA; then, after the transamination step, the acetyl group is removed to yield ornithine.

The pathways to proline and arginine are somewhat different in mammals. Proline can be synthesized by the pathway shown in [Figure 22-12](#), but it is also formed from arginine obtained from dietary or tissue protein. Arginase, a urea cycle enzyme, converts

arginine to ornithine and urea (see [Figs. 18-10, 18-26](#)). The ornithine is converted to glutamate γ -semialdehyde by the enzyme **ornithine γ -aminotransferase** ([Fig. 22-13](#)). The semialdehyde cyclizes to Δ^1 -pyrroline-5-carboxylate, which is then converted to proline ([Fig. 22-12](#)). The pathway for arginine synthesis shown in [Figure 22-12](#) is absent in mammals. 

When arginine from dietary intake or protein turnover is insufficient for protein synthesis, the ornithine δ -aminotransferase reaction operates in the direction of ornithine formation. Ornithine is then converted to citrulline and arginine in the urea cycle.

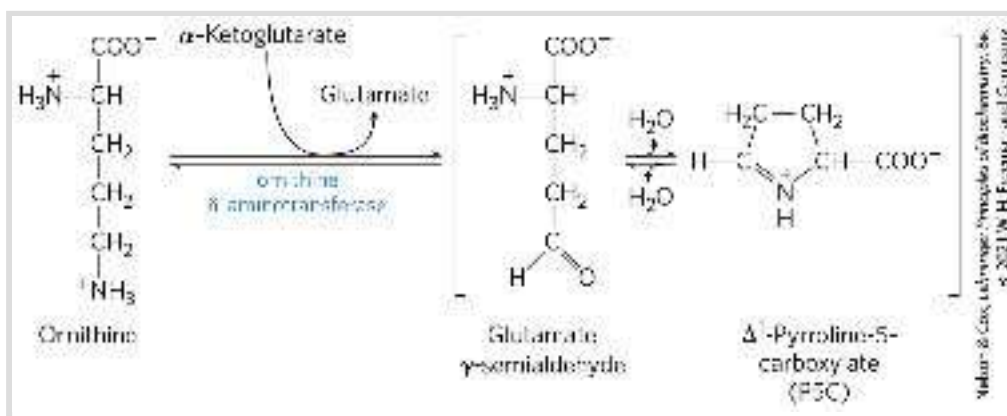
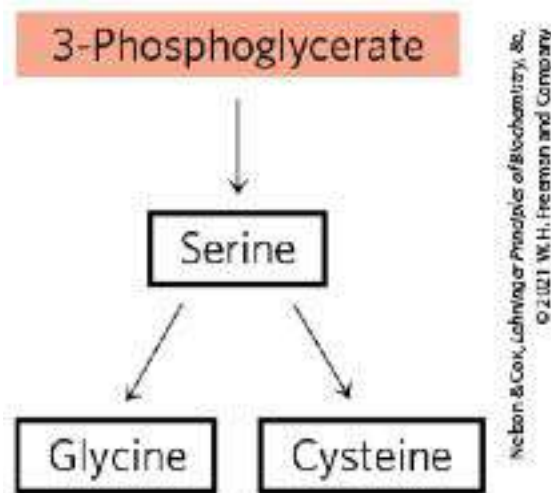
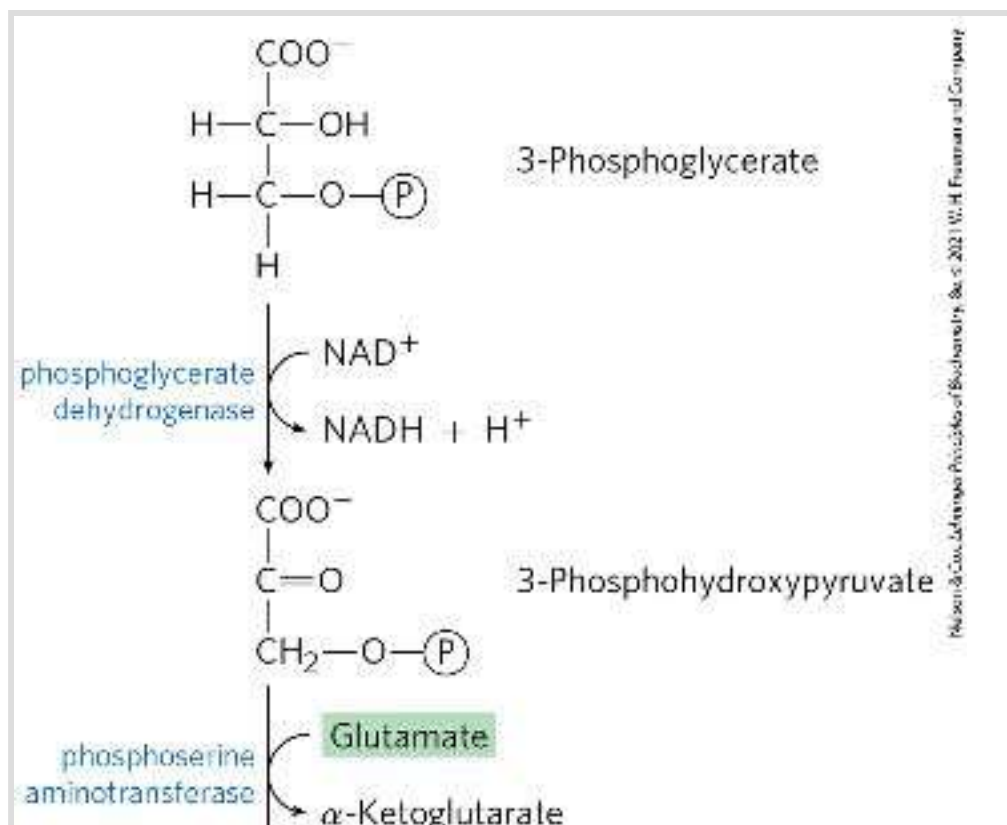


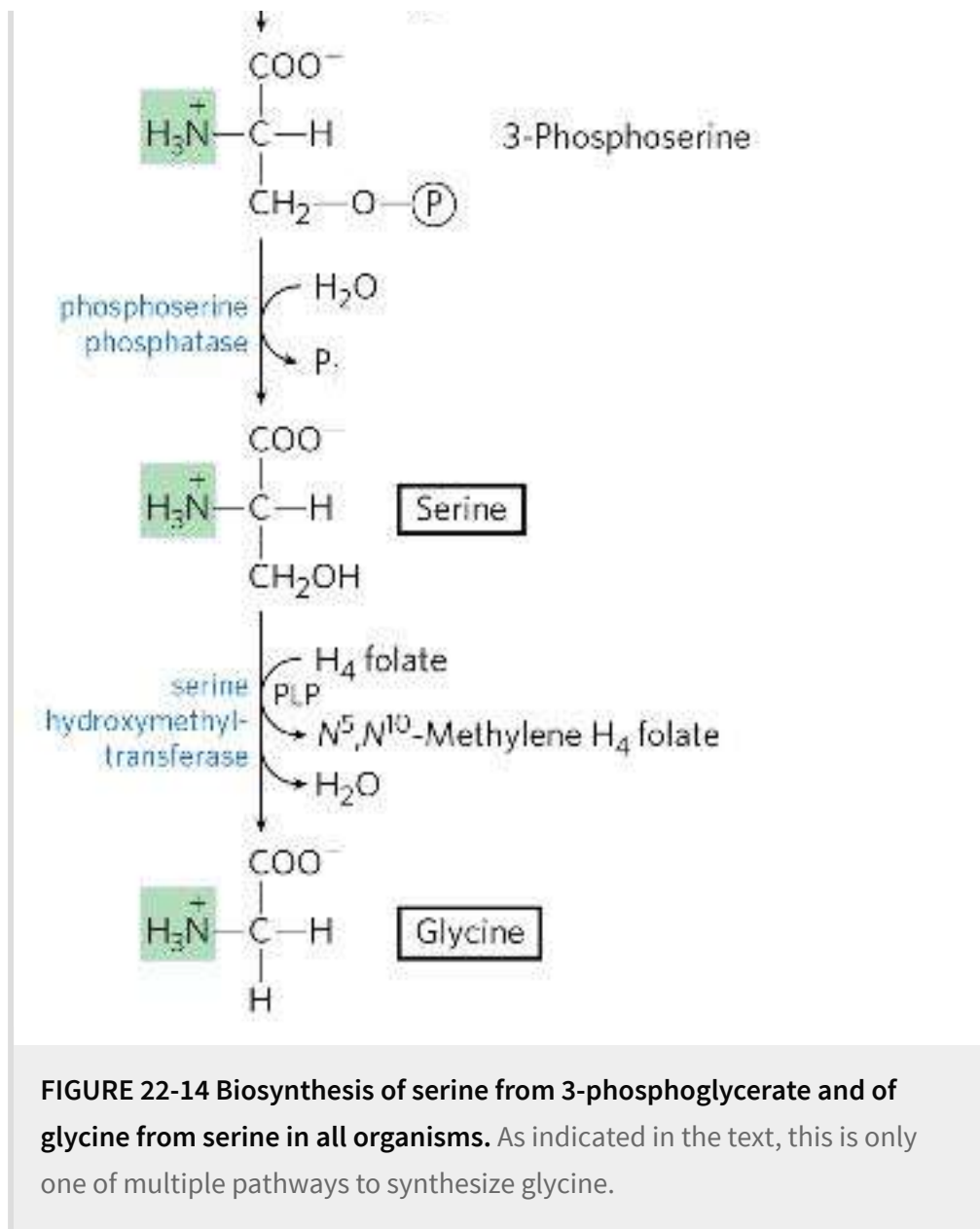
FIGURE 22-13 Ornithine δ -aminotransferase reaction: a step in the mammalian pathway to proline. This enzyme is found in the mitochondrial matrix of most tissues. Although the equilibrium favors P5C formation, the reverse reaction is the only mammalian pathway for synthesis of ornithine (and thus arginine) when arginine levels are insufficient for protein synthesis.

Serine, Glycine, and Cysteine Are Derived from 3-Phosphoglycerate



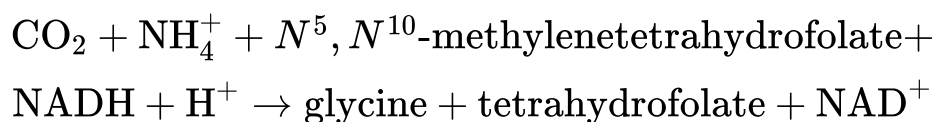
The major pathway for the formation of **serine** is the same in all organisms ([Fig. 22-14](#)). In the first step, the hydroxyl group of 3-phosphoglycerate is oxidized by a dehydrogenase (using NAD^+) to yield 3-phosphohydroxypyruvate. Transamination from glutamate yields 3-phosphoserine, which is hydrolyzed to free serine by phosphoserine phosphatase.





Serine (three carbons) is the precursor of **glycine** (two carbons) through removal of a carbon atom by **serine hydroxymethyltransferase** ([Fig. 22-14](#)). Tetrahydrofolate accepts the β carbon (C-3) of serine, which forms a methylene bridge between N-5 and N-10 to yield N^5, N^{10} -methylenetetrahydrofolate (see [Fig. 18-17](#)). The overall reaction, which is reversible, also requires pyridoxal phosphate. In the liver of vertebrates, glycine can be made by another route: the reverse of the reaction shown

in [Figure 18-20c](#), catalyzed by **glycine synthase** (also called **glycine cleavage enzyme**):



Plants and bacteria produce the reduced sulfur required for the synthesis of **cysteine** (and methionine, described later) from environmental sulfates; the pathway is shown on the right side of [Figure 22-15](#). Sulfate is activated in two steps to produce 3'-phosphoadenosine 5'-phosphosulfate (PAPS), which undergoes an eight-electron reduction to sulfide. The sulfide is then used in the formation of cysteine from serine in a two-step pathway.

Mammals synthesize cysteine from two amino acids: methionine furnishes the sulfur atom, and serine furnishes the carbon skeleton. Methionine is first converted to *S*-adenosylmethionine (see [Fig. 18-18](#)), which can lose its methyl group to any of a number of acceptors to form *S*-adenosylhomocysteine (adoHcy). This demethylated product is hydrolyzed to free homocysteine, which undergoes a reaction with serine, catalyzed by **cystathionine β -synthase**, to yield cystathionine ([Fig. 22-16](#)). Finally, **cystathionine δ -lyase**, a PLP-requiring enzyme, catalyzes removal of ammonia and cleavage of cystathionine to yield free cysteine.

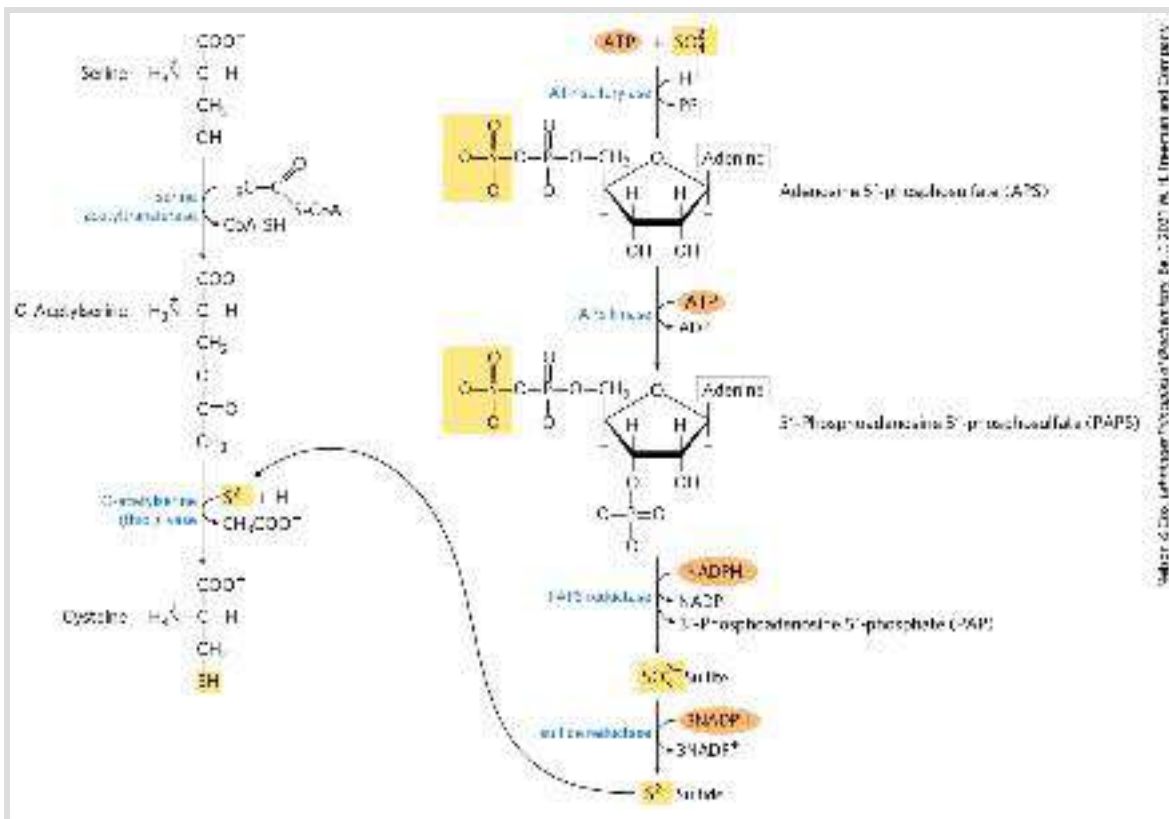
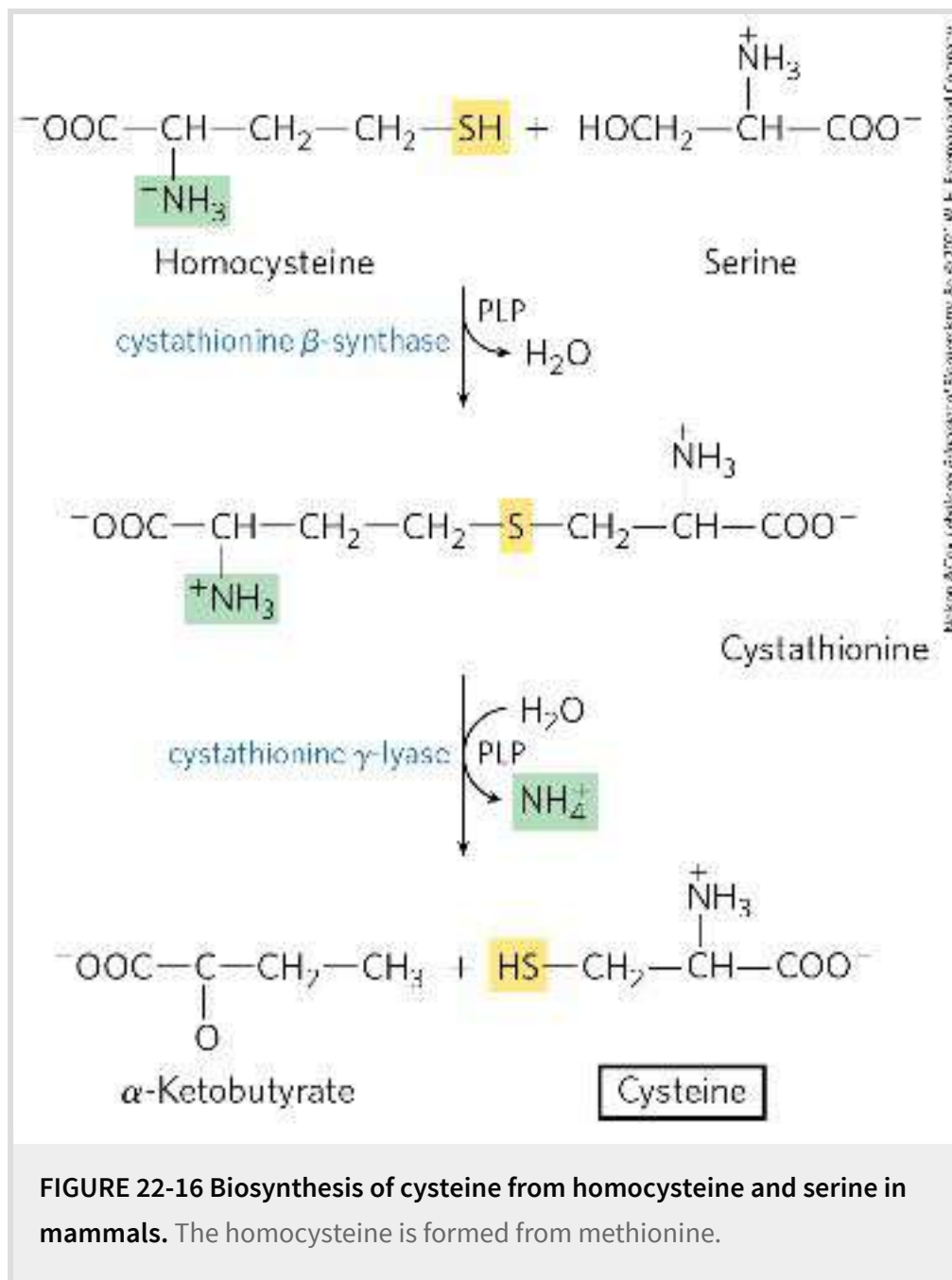
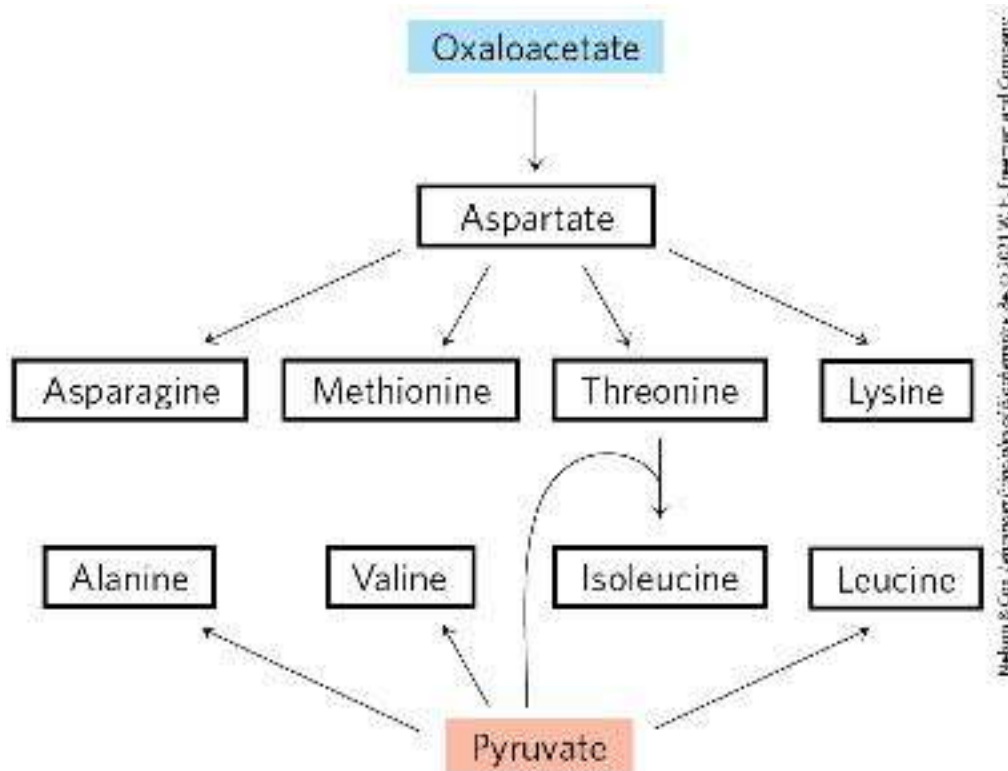


FIGURE 22-15 Biosynthesis of cysteine from serine in bacteria and plants. The origin of reduced sulfur is shown in the pathway on the right.



Three Nonessential and Six Essential Amino Acids Are Synthesized from Oxaloacetate and Pyruvate



Alanine and **aspartate** are synthesized from pyruvate and oxaloacetate, respectively, by transamination from glutamate. **Asparagine** is synthesized by amidation of aspartate, catalyzed by the enzyme asparagine synthetase. The NH_4^+ is donated by glutamine. These are nonessential amino acids, and their simple biosynthetic pathways occur in all organisms.



The malignant lymphocytes present in childhood acute lymphoblastic leukemia (ALL) produce little or no asparagine synthetase, and they are thus sensitive to asparagine depletion. The chemotherapy for ALL is administered together with an L-asparaginase derived from bacteria, with the enzyme functioning to reduce serum asparagine. The combined treatment results in a greater than 95% remission rate in cases of childhood ALL (L-asparaginase treatment alone produces remission in 40% to 60%

of cases). However, the asparaginase treatment has some deleterious side effects, and about 10% of patients who achieve remission eventually suffer relapse, with tumors resistant to drug therapy. Researchers are now developing inhibitors of human asparagine synthetase to augment these therapies for childhood ALL. ■

Methionine, threonine, lysine, isoleucine, valine, and leucine are essential amino acids; humans cannot synthesize them. Their biosynthetic pathways in bacteria are complex and interconnected ([Fig. 22-17](#)). In some cases, the pathways in bacteria, fungi, and plants differ significantly.

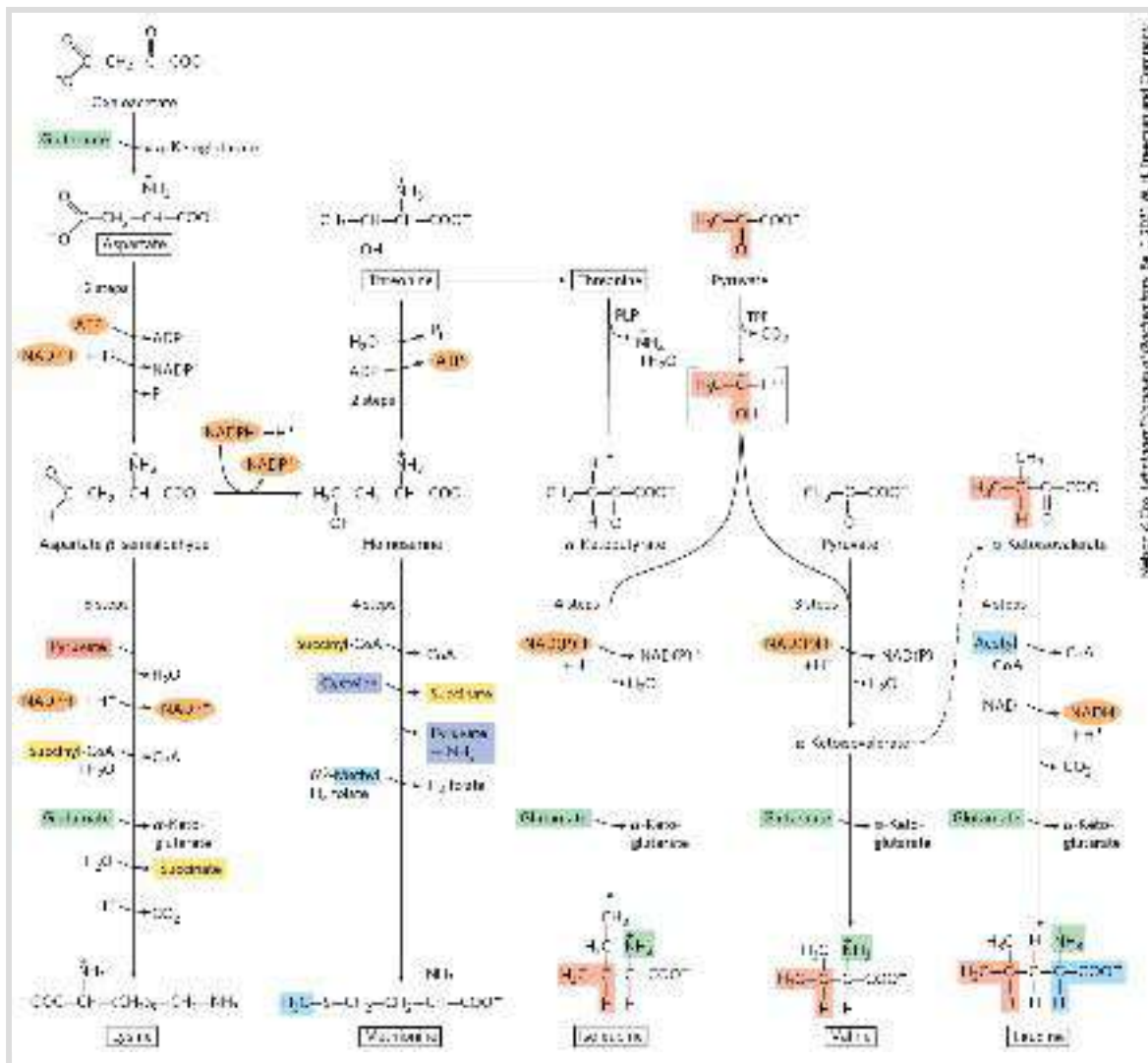
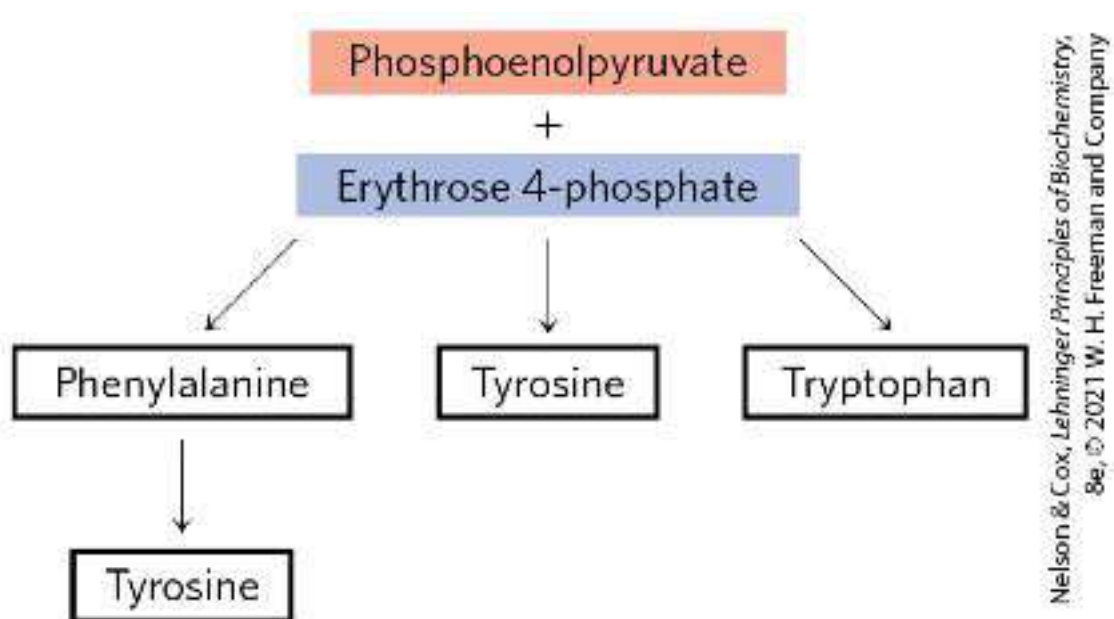


FIGURE 22-17 Biosynthesis of six essential amino acids from oxaloacetate and pyruvate in bacteria: methionine, threonine, lysine, isoleucine, valine, and leucine. Some of the most complex pathways for amino acid biosynthesis are found here. Pathways are abbreviated to emphasize precursors and pathway products.

Aspartate gives rise to **methionine**, **threonine**, and **lysine**. Branch points occur at aspartate β -semialdehyde, an intermediate in all three pathways, and at homoserine, a precursor of threonine and methionine. Threonine, in turn, is one of the precursors of isoleucine. The **valine** and **isoleucine** pathways share four enzymes ([Fig. 22-17](#)). Pyruvate gives rise to valine and isoleucine in pathways that begin with condensation of

two carbons of pyruvate (in the form of hydroxyethyl thiamine pyrophosphate; see [Fig. 14-13b](#)) with another molecule of pyruvate (the valine path) or with α -ketobutyrate (the isoleucine path). The α -ketobutyrate is derived from threonine in a reaction that requires pyridoxal phosphate. An intermediate in the valine pathway, α -ketoisovalerate, is the starting point for a four-step branch pathway leading to **leucine**.

Chorismate Is a Key Intermediate in the Synthesis of Tryptophan, Phenylalanine, and Tyrosine



The three amino acid side chains that contain aromatic rings, tryptophan, phenylalanine, and tyrosine, present a special chemical problem for biosynthesis. Aromatic rings are not readily

available in the environment, even though the benzene ring is very stable. The branched pathway to tryptophan, phenylalanine, and tyrosine, occurring in bacteria, fungi, and plants, is the main biological route of aromatic ring formation. It proceeds through ring closure of an aliphatic precursor followed by stepwise addition of double bonds. The first four steps produce shikimate, a seven-carbon molecule derived from erythrose 4-phosphate and phosphoenolpyruvate ([Fig. 22-18](#)). Shikimate is converted to chorismate in three steps that include the addition of three more carbons from another molecule of phosphoenolpyruvate. Chorismate is the first branch point of the pathway, with one branch leading to tryptophan, the other to phenylalanine and tyrosine.

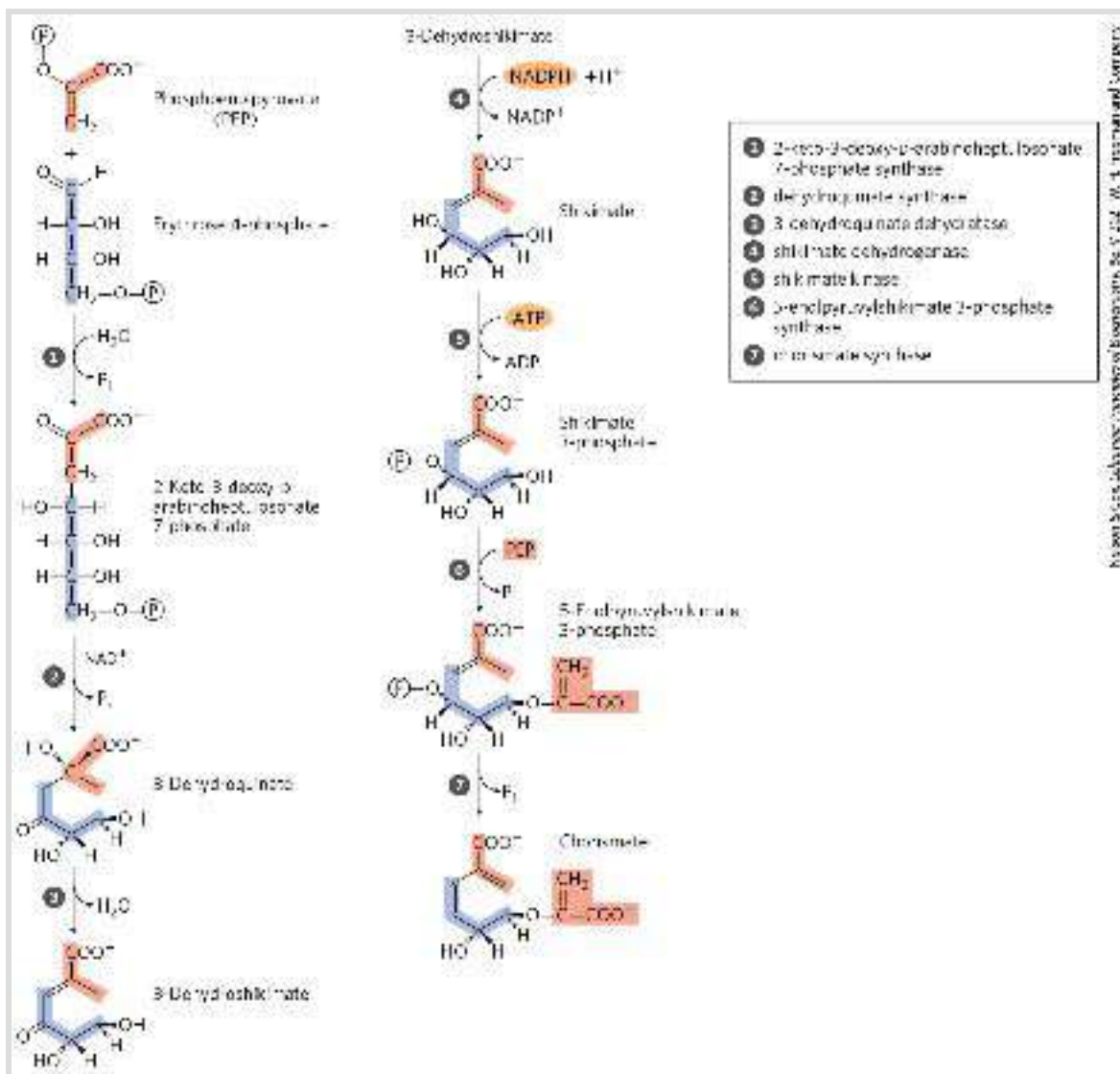
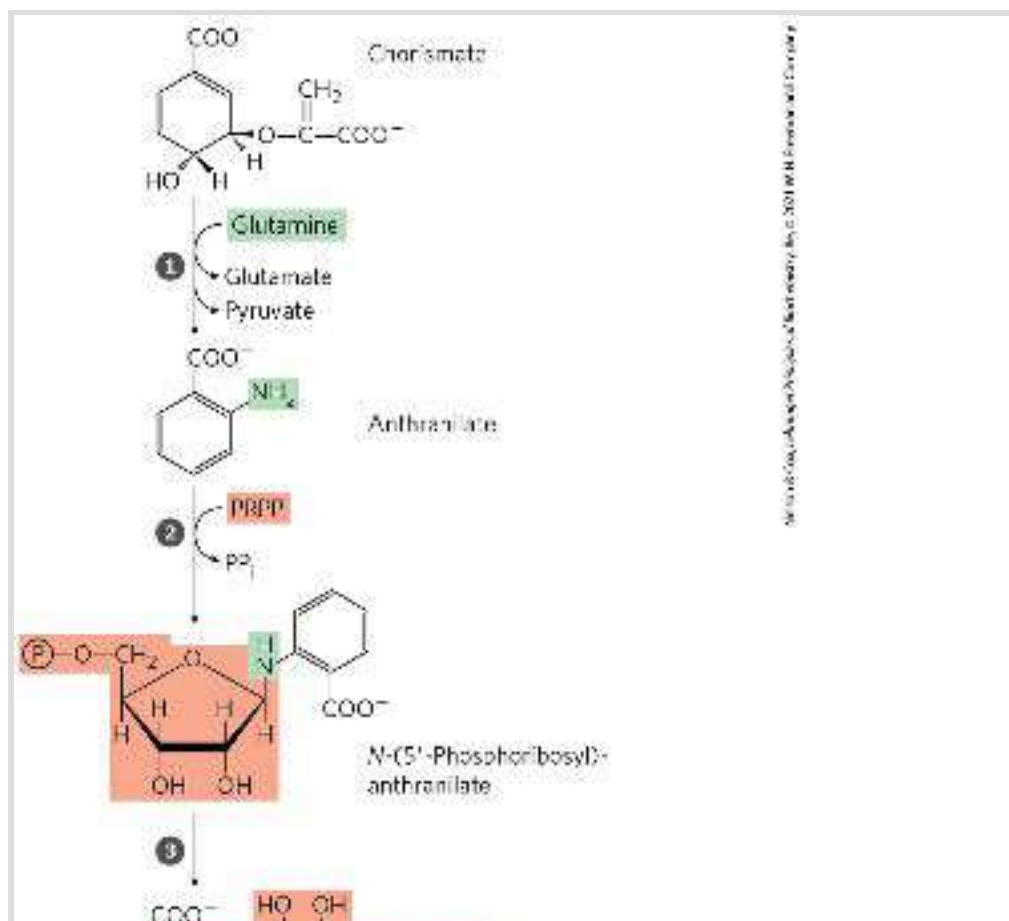
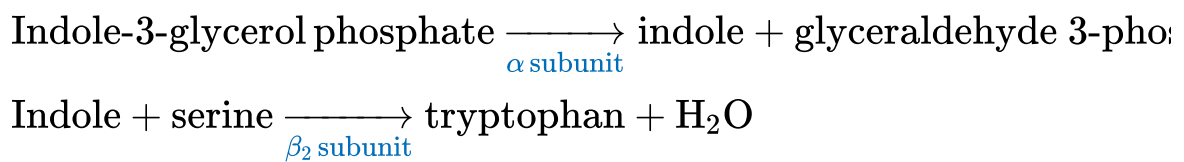
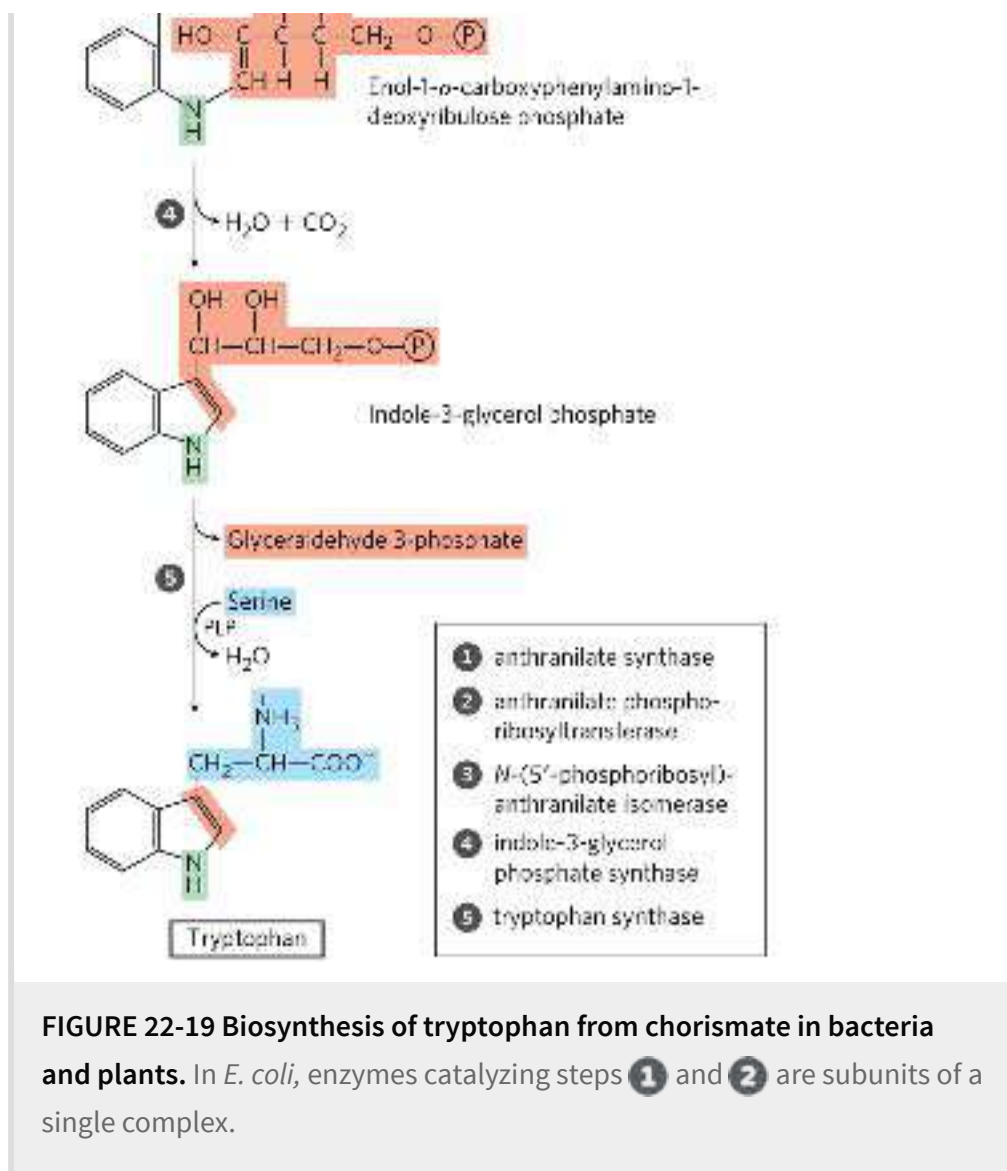


FIGURE 22-18 Biosynthesis of chorismate, an intermediate in the synthesis of aromatic amino acids in bacteria and plants. All carbons are derived from either erythrose 4-phosphate (light purple) or phosphoenolpyruvate (light red). Note that the NAD^+ required as a cofactor in step **2** is released unchanged; it may be transiently reduced to NADH during the reaction, with formation of an oxidized reaction intermediate. Step **6** is competitively inhibited by glyphosate ($^-\text{COO}-\text{CH}_2-\text{NH}-\text{CH}_2-\text{PO}_3^{2-}$), the active ingredient in the widely used herbicide Roundup. The herbicide is relatively nontoxic to mammals, which lack this biosynthetic pathway. The intermediates quinate and shikimate are named after the plants in which they have been found to accumulate.

In the **tryptophan** branch ([Fig. 22-19](#)), chorismate is converted to anthranilate in a reaction in which glutamine donates the

nitrogen that will become part of the indole ring. Anthranilate then condenses with PRPP. The indole ring of tryptophan is derived from the ring carbons and amino group of anthranilate plus two carbons derived from PRPP. The final reaction in the sequence is catalyzed by **tryptophan synthase**. This enzyme has an $\alpha_2\beta_2$ subunit structure and can be dissociated into two α subunits and a β_2 unit that catalyze different parts of the overall reaction:

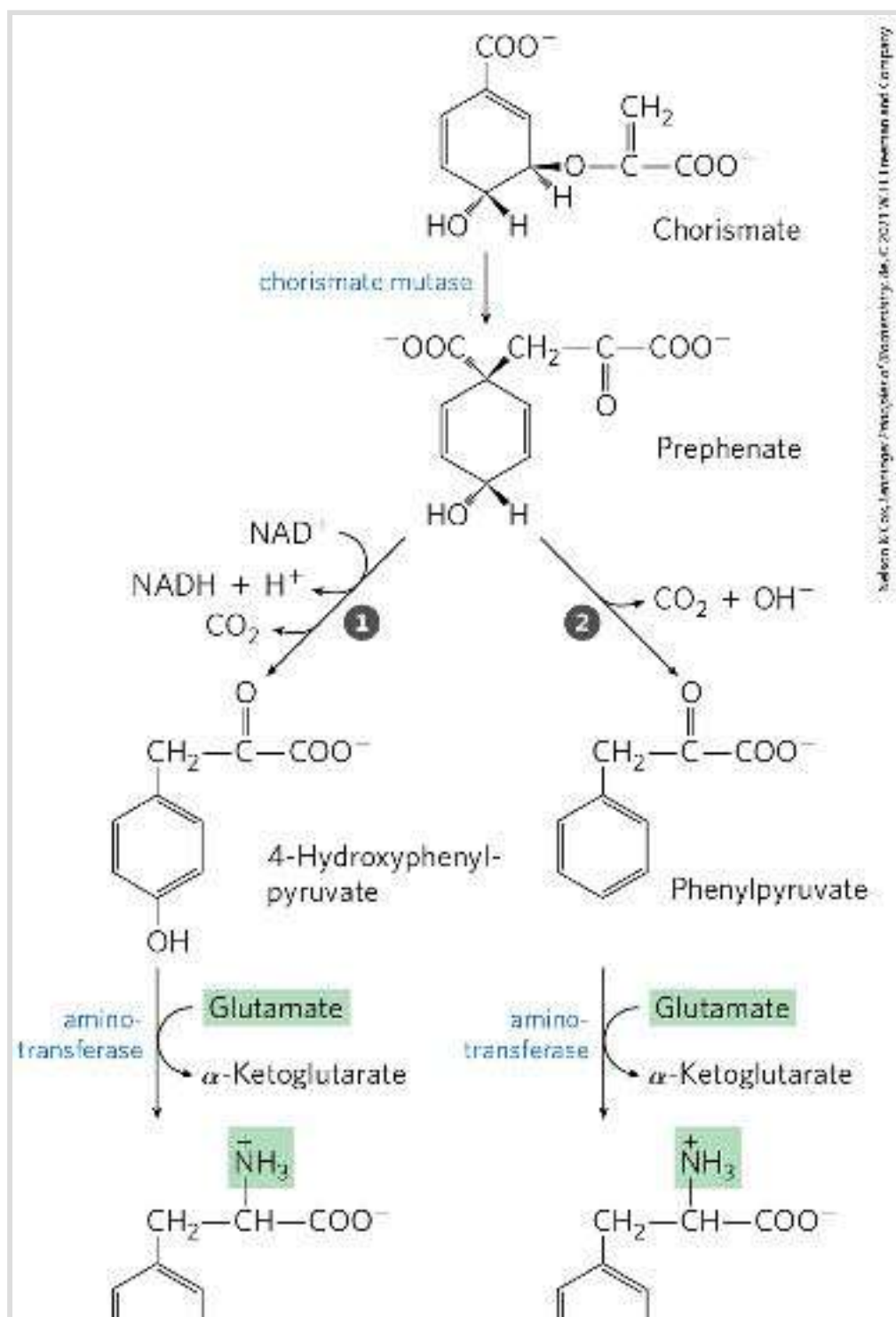


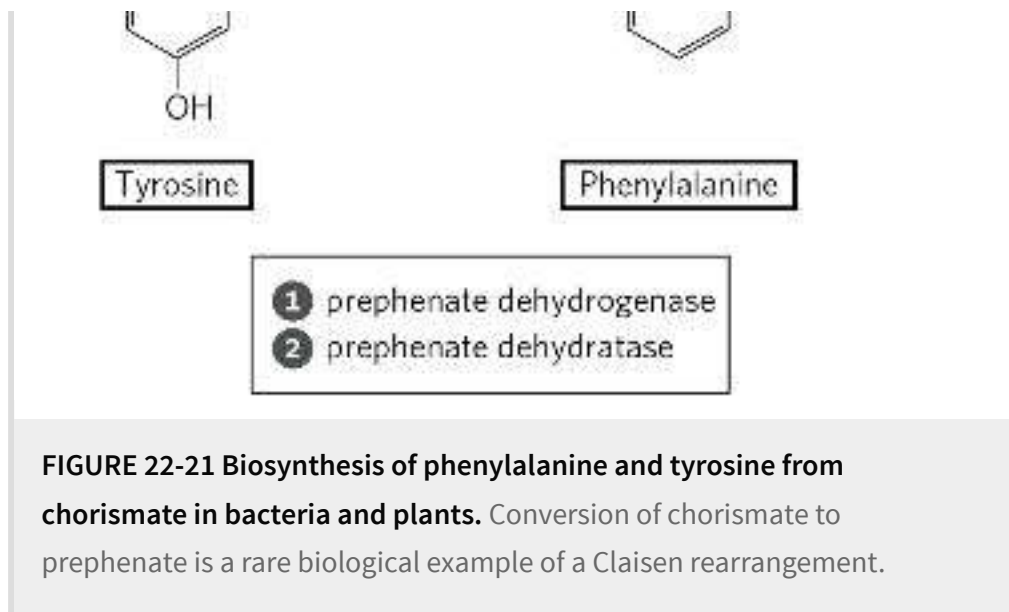


The second part of the reaction requires pyridoxal phosphate ([Fig. 22-20](#)). Indole formed in the first part is not released by the enzyme, but instead moves through a channel from the α -subunit active site to one of the β -subunit active sites, where it condenses with a Schiff base intermediate derived from serine and PLP. Intermediate channeling of this type may be a feature of the entire pathway from chorismate to tryptophan. Enzyme active sites catalyzing different steps (sometimes not sequential steps) of the pathway to tryptophan are found on single polypeptides in

some species of fungi and bacteria, but they are separate proteins in other species. In addition, the activity of some of these enzymes requires a noncovalent association with other enzymes of the pathway. These observations suggest that all the pathway enzymes are components of a large, multienzyme complex, a metabolon, in both bacteria and eukaryotes. Such complexes are generally not preserved intact when the enzymes are isolated using traditional biochemical methods (see [Section 16.4](#)).

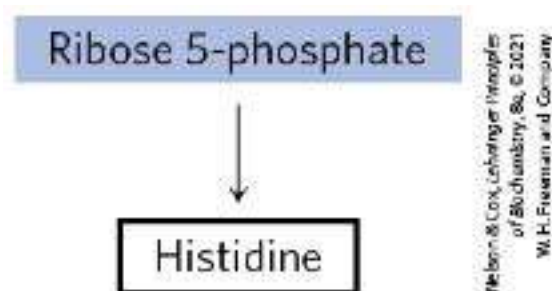
In plants and bacteria, **phenylalanine** and **tyrosine** are synthesized from chorismate in pathways much less complex than the tryptophan pathway. The common intermediate is prephenate ([Fig. 22-21](#)). The final step in both cases is transamination with glutamate.



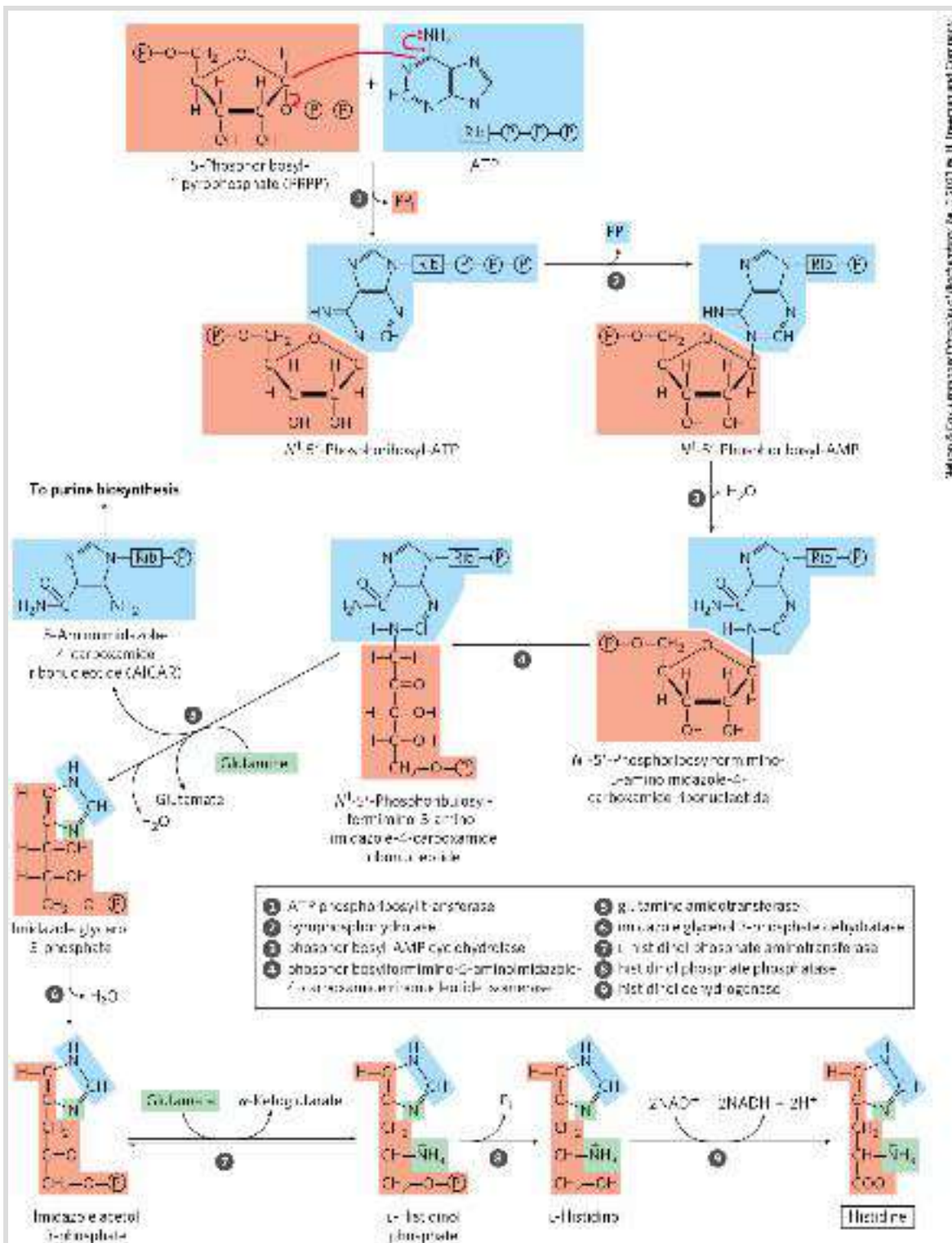


Animals can produce tyrosine directly from phenylalanine through hydroxylation at C-4 of the phenyl group by **phenylalanine hydroxylase**; this enzyme also participates in the degradation of phenylalanine (see [Figs. 18-23](#), [18-24](#)). Tyrosine is considered a conditionally essential amino acid, or as nonessential insofar as it can be synthesized from the essential amino acid phenylalanine.

Histidine Biosynthesis Uses Precursors of Purine Biosynthesis

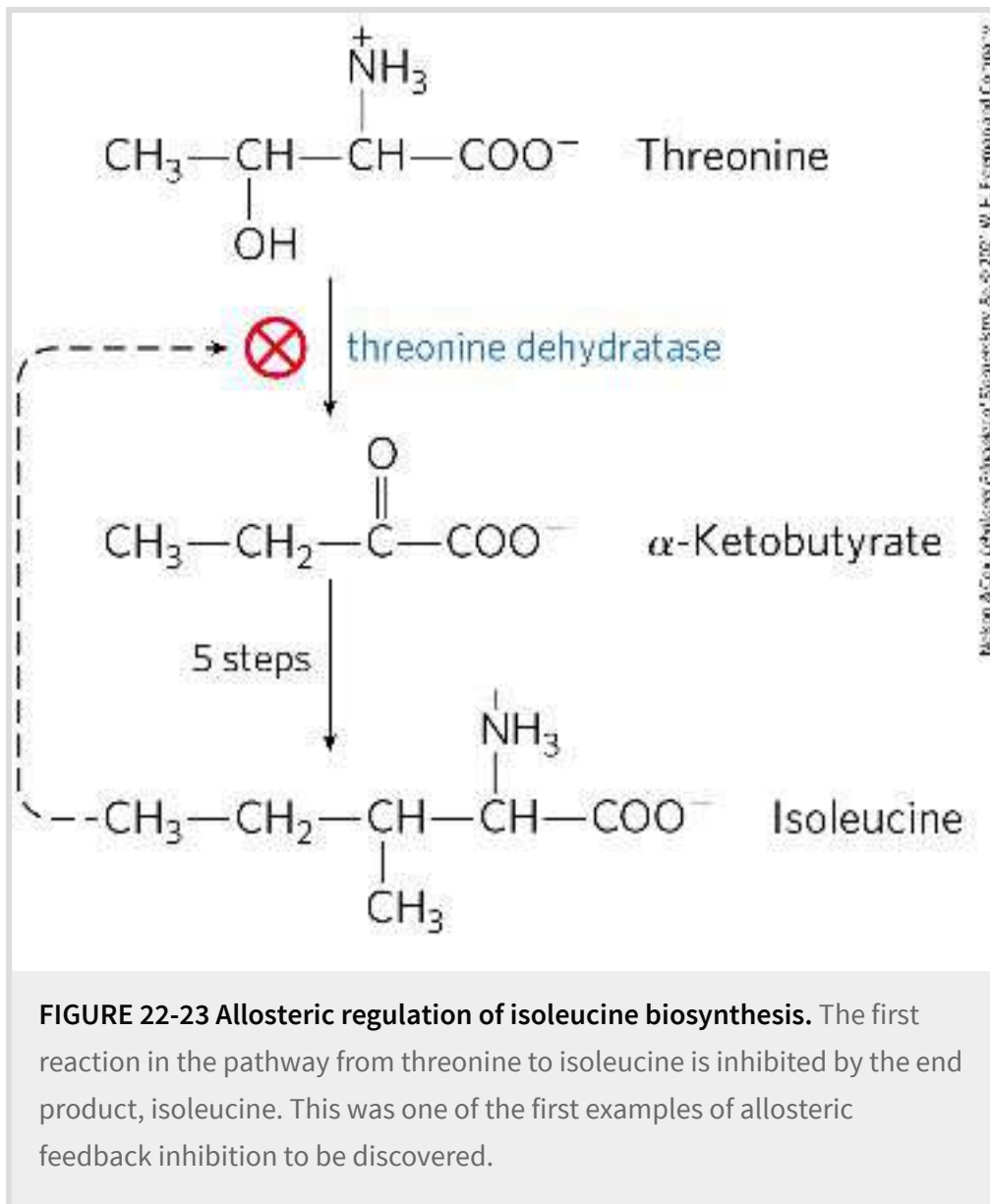


The pathway to **histidine** in all plants and bacteria differs in several respects from other amino acid biosynthetic pathways. Histidine is derived from three precursors ([Fig. 22-22](#)): PRPP contributes five carbons, the purine ring of ATP contributes a nitrogen and a carbon, and glutamine supplies the second ring nitrogen. The key steps are condensation of ATP and PRPP, in which N-1 of the purine ring is linked to the activated C-1 of the ribose of PRPP (step ① in [Fig. 22-22](#)); purine ring opening that ultimately leaves N-1 and C-2 of adenine linked to the ribose (step ③); and formation of the imidazole ring, a reaction in which glutamine donates a nitrogen (step ⑤). The use of ATP as a metabolite rather than a high-energy cofactor is unusual — but not wasteful, because it dovetails with the purine biosynthetic pathway. The remnant of ATP that is released after the transfer of N-1 and C-2 is 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR), an intermediate of purine biosynthesis (see [Fig. 22-35](#)) that is rapidly recycled to ATP.



Amino Acid Biosynthesis Is under Allosteric Regulation

As detailed in [Chapter 13](#), the control of flux through a metabolic pathway often reflects the activity of multiple enzymes in that pathway.  In the case of amino acid synthesis, regulation often takes place in part through feedback inhibition of the first reaction by the end product of the pathway. This first reaction is often catalyzed by an allosteric enzyme that plays an important role in the overall control of flux through that pathway. As an example, [Figure 22-23](#) shows the allosteric regulation of isoleucine synthesis from threonine. The end product, isoleucine, is an allosteric inhibitor of the first reaction in the sequence. In bacteria, such allosteric modulation of amino acid synthesis contributes to the minute-to-minute adjustment of pathway activity to cellular needs.



Allosteric regulation of an individual enzyme can be considerably more complex. An example is the remarkable set of allosteric controls exerted on glutamine synthetase of *E. coli* ([Fig. 22-8](#)). Six products derived from glutamine serve as negative feedback modulators of the enzyme, and the overall effects of these and other modulators are more than additive. Such regulation is called **concerted inhibition**.

Additional mechanisms contribute to the regulation of the amino acid biosynthetic pathways.  Because the 20 common amino acids must be made in the correct proportions for protein synthesis, cells have developed ways not only of controlling the rate of synthesis of individual amino acids but also of coordinating their formation. Such coordination is especially well developed in fast-growing bacterial cells. [Figure 22-24](#) shows how *E. coli* cells coordinate the synthesis of lysine, methionine, threonine, and isoleucine, all made from aspartate. Several important types of inhibition patterns are evident. The step from aspartate to aspartyl- β -phosphate is catalyzed by three isozymes, each independently controlled by different modulators. This **enzyme multiplicity** prevents one biosynthetic end product from shutting down key steps in a pathway when other products of the same pathway are required. The steps from aspartate β -semialdehyde to homoserine and from threonine to α -ketobutyrate (detailed in [Fig. 22-17](#)) are also catalyzed by dual, independently controlled isozymes. One isozyme for the conversion of aspartate to aspartyl- β -phosphate is allosterically inhibited by two different modulators, lysine and isoleucine, whose action is more than additive — another example of concerted inhibition. The sequence from aspartate to isoleucine undergoes multiple, overlapping negative feedback inhibitions; for example, isoleucine inhibits the conversion of threonine to α -ketobutyrate (as described above), and threonine inhibits its own formation at three points: from homoserine, from aspartate β -semialdehyde, and from aspartate (see [Fig. 22-17](#)). This overall regulatory mechanism is called **sequential feedback inhibition**.

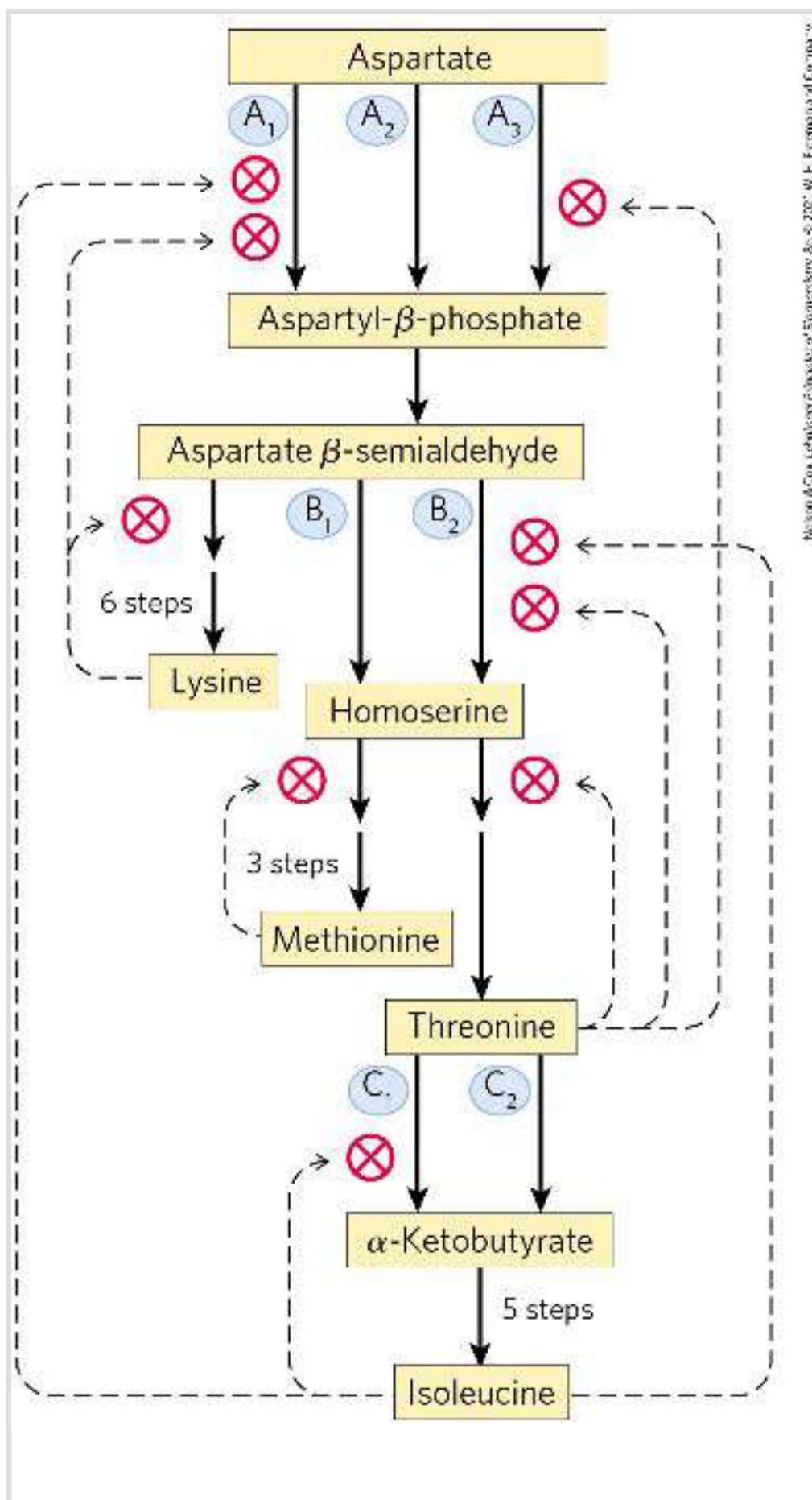


FIGURE 22-24 Interlocking regulatory mechanisms in the biosynthesis of several amino acids derived from aspartate in *E. coli*. Three enzymes (A, B, C) have either two or three isozyme forms, indicated by numerical subscripts. In each case, one isozyme (A_2 , B_1 , and C_2) has no allosteric regulation; these isozymes are regulated by changes in the amount of enzyme synthesized. Synthesis of isozymes A_2 and B_1 is repressed when methionine levels are high, and synthesis of isozyme C_2 is repressed when isoleucine levels are high. Enzyme A is aspartokinase; B, homoserine dehydrogenase; C, threonine dehydratase.

Similar patterns are evident in the pathways leading to the aromatic amino acids. The first step of the early pathway to the common intermediate chorismate is catalyzed by the enzyme 2-keto-3-deoxy-D1-arabinoheptulosonate 7-phosphate (DAHP) synthase (❶ in [Fig. 22-18](#)). Most microorganisms and plants have three DAHP synthase isozymes. One is allosterically inhibited (feedback inhibition) by phenylalanine, another by tyrosine, and the third by tryptophan. This scheme helps the overall pathway to respond to cellular requirements for one or more of the aromatic amino acids. Additional regulation takes place after the pathway branches at chorismate. For example, the enzymes catalyzing the first two steps of the tryptophan branch are subject to allosteric inhibition by tryptophan.

SUMMARY 22.2 *Biosynthesis of Amino Acids*

■ Plants and bacteria synthesize all 20 common amino acids. Mammals can synthesize about half; the others are required in the diet (essential or conditionally essential amino acids).

- Glutamate is formed by reductive amination of α -ketoglutarate and serves as the precursor of glutamine, proline, and arginine.
- The carbon chain of serine is derived from 3-phosphoglycerate. Serine is a precursor of glycine; the β -carbon atom of serine is transferred to tetrahydrofolate. In microorganisms, cysteine is produced from serine and from sulfide produced by the reduction of environmental sulfate. Mammals produce cysteine from methionine and serine by a series of reactions requiring S-adenosylmethionine and cystathionine.
- Alanine and aspartate (and thus asparagine) are formed from pyruvate and oxaloacetate, respectively, by transamination. Pyruvate and oxaloacetate also give rise to methionine, threonine, lysine, valine, isoleucine, and leucine via longer pathways.
- The aromatic amino acids (phenylalanine, tyrosine, and tryptophan) form by a pathway in which chorismate occupies a key branch point. Tyrosine can also be formed by hydroxylation of phenylalanine (and thus is considered conditionally essential). Phosphoribosyl pyrophosphate is a precursor of tryptophan and histidine.
- The pathway to histidine is interconnected with the purine synthetic pathway.
- The amino acid biosynthetic pathways are subject to allosteric end-product inhibition; the regulatory enzyme is usually the first in the sequence. Regulation of the various synthetic pathways is coordinated.

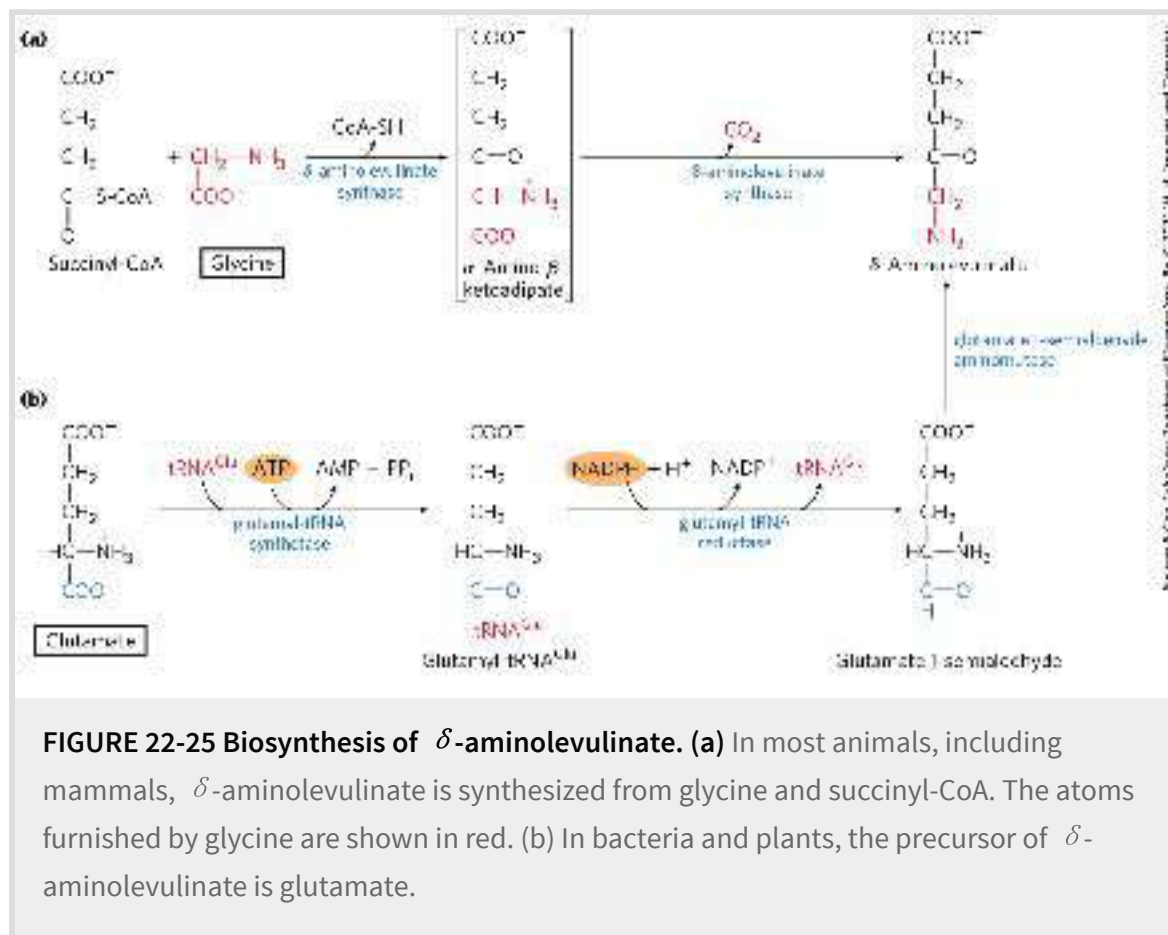
22.3 Molecules Derived from Amino Acids

In addition to their role as the building blocks of proteins, amino acids are precursors of many specialized biomolecules, including hormones, coenzymes, nucleotides, alkaloids, cell wall polymers, porphyrins, antibiotics, pigments, and neurotransmitters. We describe here the pathways to a number of these amino acid derivatives.

Glycine Is a Precursor of Porphyrins

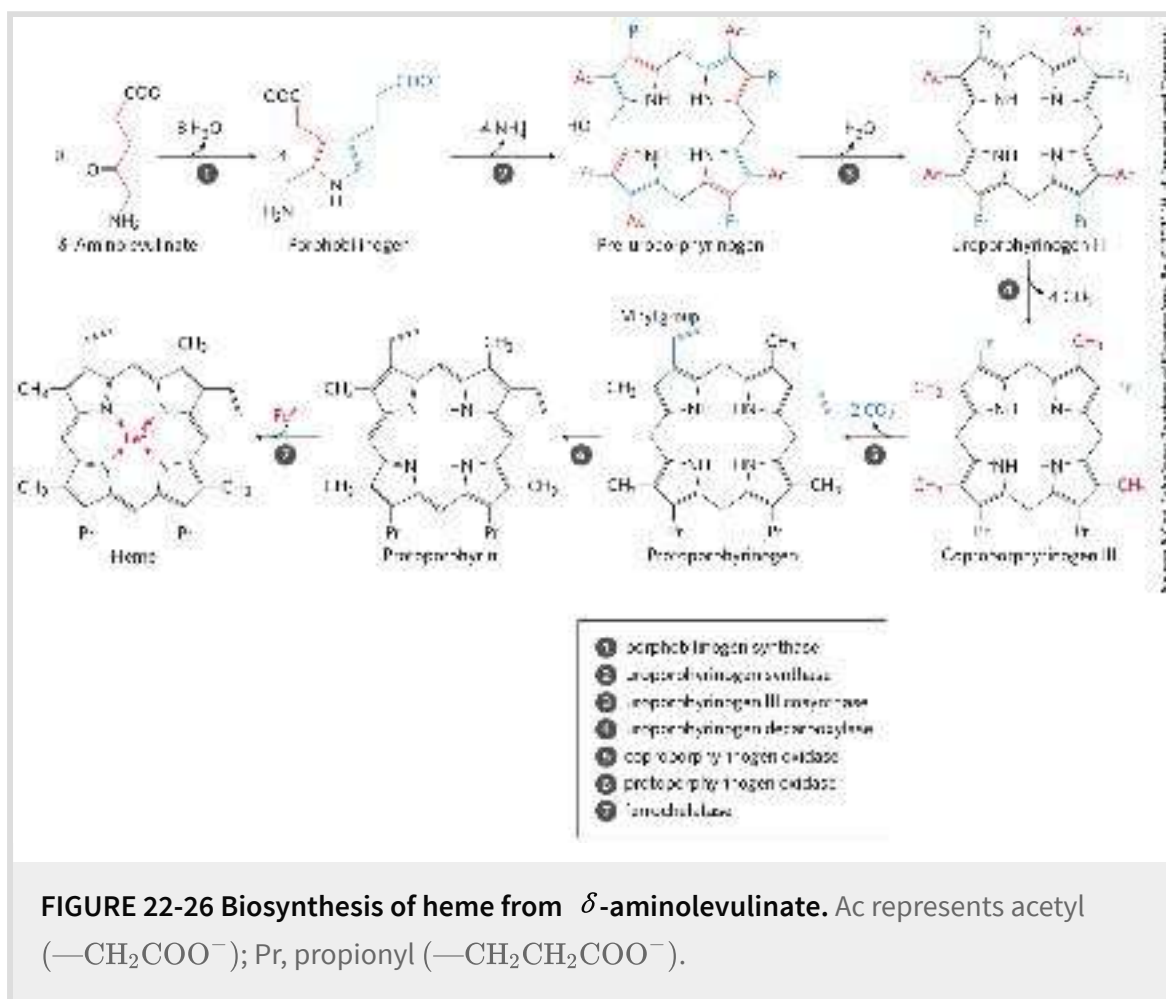
The biosynthesis of [porphyrins](#), for which glycine is a major precursor, is our first example because of the central importance of the porphyrin nucleus in heme proteins such as hemoglobin and the cytochromes. The porphyrins are constructed from four molecules of the monopyrrole derivative **porphobilinogen**, which itself is derived from two molecules of δ -aminolevulinate. There are two major pathways to δ -aminolevulinate. In higher eukaryotes ([Fig. 22-25a](#)), glycine reacts with succinyl-CoA in the first step to yield α -amino- β -ketoadipate, which is then decarboxylated to δ -aminolevulinate. In plants, algae, and most bacteria, δ -aminolevulinate is formed from glutamate ([Fig. 22-25b](#)). The glutamate is first esterified to glutamyl-tRNA^{Glu};  reduction by NADPH converts the glutamate to glutamate 1-semialdehyde, which is cleaved from the tRNA. An

aminotransferase converts the glutamate 1-semialdehyde to δ -aminolevulinate.



In all organisms, two molecules of δ -aminolevulinate condense to form porphobilinogen and, through a series of complex enzymatic reactions, four molecules of porphobilinogen come together to form **protoporphyrin** ([Fig. 22-26](#)). The iron atom is incorporated after the protoporphyrin has been assembled, in a step catalyzed by ferrochelatase. Porphyrin biosynthesis is regulated in higher eukaryotes by heme, which serves as a feedback inhibitor of early steps in the synthetic pathway. Genetic defects in the biosynthesis of porphyrins can lead to the

accumulation of pathway intermediates, causing a variety of human diseases known collectively as **porphyrias** ([Box 22-2](#)).



BOX 22-2 MEDICINE

On Kings and Vampires

Porphyrias are a group of genetic diseases that result from defects in enzymes of the biosynthetic pathway from glycine to porphyrins ([Fig. 1](#)); specific porphyrin precursors accumulate in erythrocytes, body fluids, and the liver. The most common form is acute intermittent porphyria. Most individuals inheriting this condition are heterozygotes and are usually asymptomatic, because the single copy of the normal gene provides a sufficient level of enzyme function. However, certain nutritional or environmental factors (as yet

poorly understood) can cause a buildup of δ -aminolevulinate and porphobilinogen, leading to attacks of acute abdominal pain and neurological dysfunction. King George III, British monarch during the American Revolution, suffered several episodes of apparent madness that tarnished the record of this otherwise accomplished man. The symptoms of his condition suggest that George III suffered from acute intermittent porphyria.

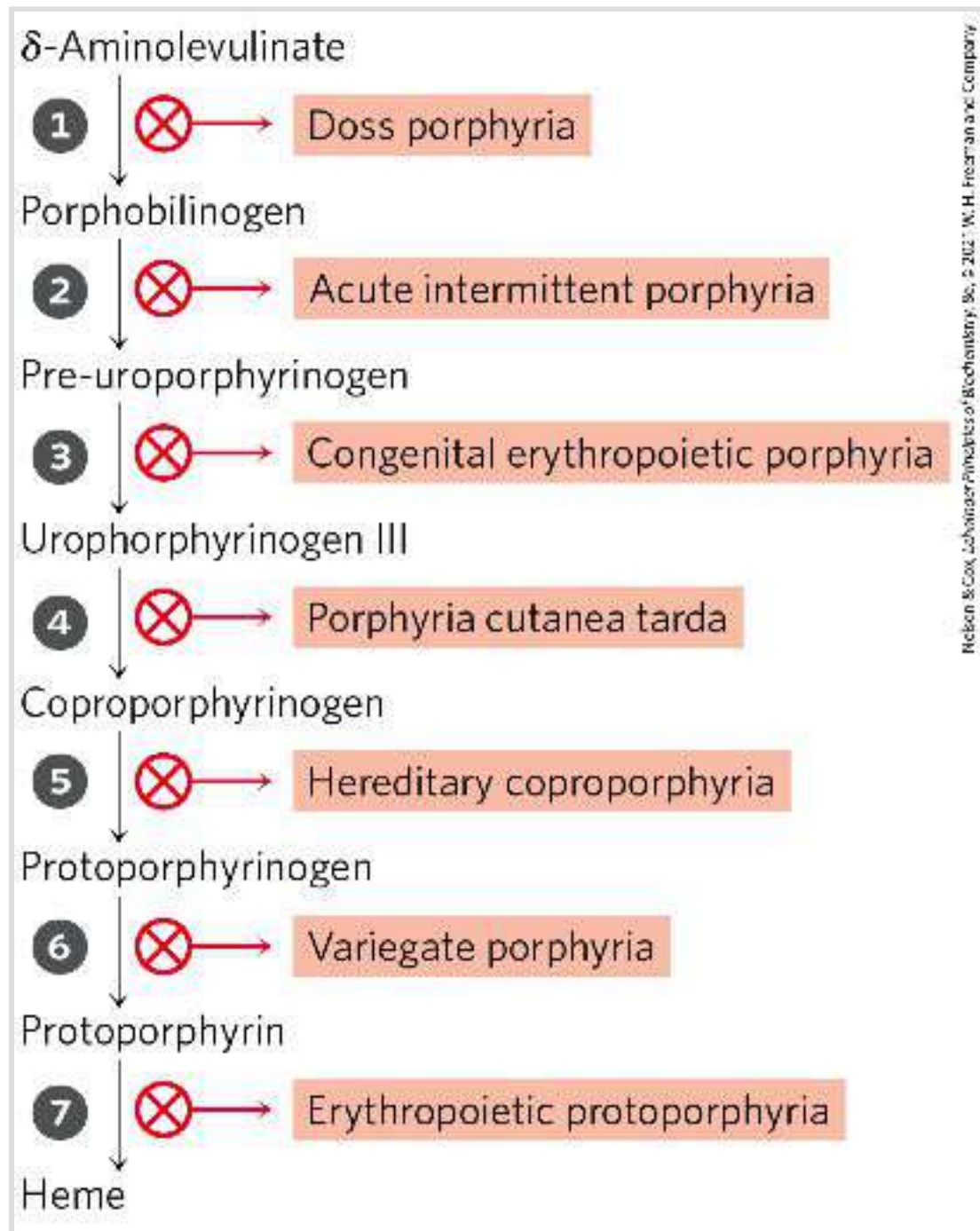


FIGURE 1 The key to [Figure 22-26](#) identifies the defective enzyme at each step.

About 5,000 to 10,000 humans worldwide suffer from a genetic condition in which the activity of ferrochelatase (step 7) is reduced, leading to abnormally high concentrations of protoporphyrin in tissues. When exposed to light, the excess protoporphyrin releases free radicals that damage cellular macromolecules, including proteins and lipids in cell membranes. The resulting cell damage and inflammation in the endothelial (skin) cells can be very painful. Patients are forced to adopt a lifestyle in which they avoid sunlight and even bright indoor light. A new drug called afamelanotide, slowly released over several months from a small implant under the skin, has shown success in alleviating the symptoms of sunlight exposure. The drug interacts with the melanocortin-1 receptor, stimulating melanin production and modulating the expression of antioxidants that eliminate the free radical species. Afamelanotide may also prove useful in treating vitiligo, a more common condition in which melanin production is not uniform and skin color is lost in blotches.

One of the rarer porphyrias results in an accumulation of uroporphyrinogen I, an abnormal isomer of a protoporphyrin precursor. This compound stains the urine red, causes the teeth to fluoresce strongly in ultraviolet light, and makes the skin abnormally sensitive to sunlight. Many individuals with this porphyria are anemic because insufficient heme is synthesized. This genetic condition may have given rise to the vampire stories of folk legend.

The symptoms of many porphyrias are now readily controlled with dietary changes or the administration of heme or heme derivatives.

Heme Degradation Has Multiple Functions



The iron-porphyrin (heme) group of hemoglobin, released from dying erythrocytes in the spleen, is degraded to yield free

Fe^{2+} and, ultimately, **bilirubin**. The pathway also contributes the pigment present in mixtures of the bile salts derived from cholesterol.

The first step in the two-step pathway to bilirubin, catalyzed by heme oxygenase, converts heme to biliverdin, a linear (open) tetrapyrrole derivative ([Fig. 22-27](#) on p. 820). The other products of the reaction are free Fe^{2+} and CO. The Fe^{2+} is quickly bound by ferritin. Carbon monoxide is a poison that binds to hemoglobin (see [Box 5-1](#)), and the production of CO by heme oxygenase ensures that, even in the absence of environmental exposure, about 1% of an individual's heme is complexed with CO.

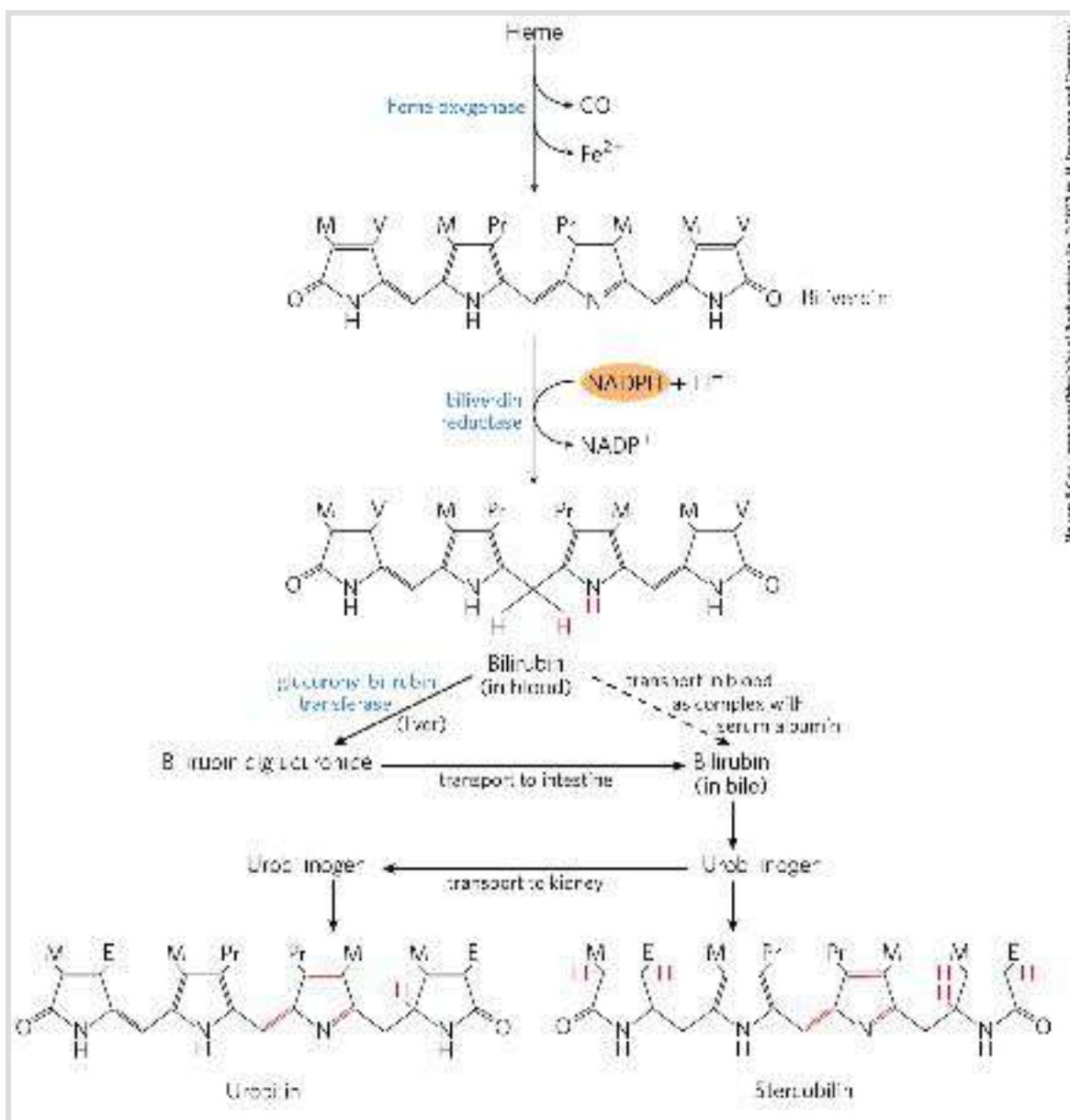


FIGURE 22-27 Bilirubin and its breakdown products. M represents methyl; V, vinyl; Pr, propionyl; E, ethyl. For ease of comparison, these structures are shown in linear form, rather than in their correct stereochemical conformations.

Biliverdin is converted to bilirubin in the second step, catalyzed by biliverdin reductase. You can monitor the reactions in the breakdown of heme ([Fig. 22-27](#)) colorimetrically in a familiar in situ experiment. When you are bruised, the black and/or purple color results from hemoglobin released from damaged

erythrocytes. Over time, the color changes to the green of biliverdin, and then to the yellow of bilirubin. Bilirubin is largely insoluble, and it travels in the bloodstream as one of many metabolites, fatty acids and others, complexed with serum albumin (see [Fig. 17-3](#)). In the liver, bilirubin is transformed to the bile pigment bilirubin diglucuronide. This product is sufficiently water-soluble to be secreted with other components of bile into the small intestine, where microbial enzymes convert it to several products, predominantly urobilinogen. Some urobilinogen is reabsorbed into the blood and transported to the kidney, where it is converted to urobilin, the compound that gives urine its yellow color. Urobilinogen remaining in the intestine is converted (in another microbe-dependent reaction) to stercobilin, which imparts the red-brown color to feces.

Impaired liver function or blocked bile secretion causes bilirubin to leak from the liver into the blood, resulting in a yellowing of the skin and eyeballs, a condition called jaundice. In cases of jaundice, determination of the concentration of bilirubin in the blood may be useful in the diagnosis of underlying liver disease. Newborn infants sometimes develop jaundice because they have not yet produced enough glucuronyl bilirubin transferase to process their bilirubin. A traditional treatment to reduce excess bilirubin, exposure to a fluorescent lamp, causes a photochemical conversion of bilirubin to compounds that are more soluble and easily excreted.

These pathways of heme breakdown play significant roles in protecting cells from oxidative damage and in regulating certain

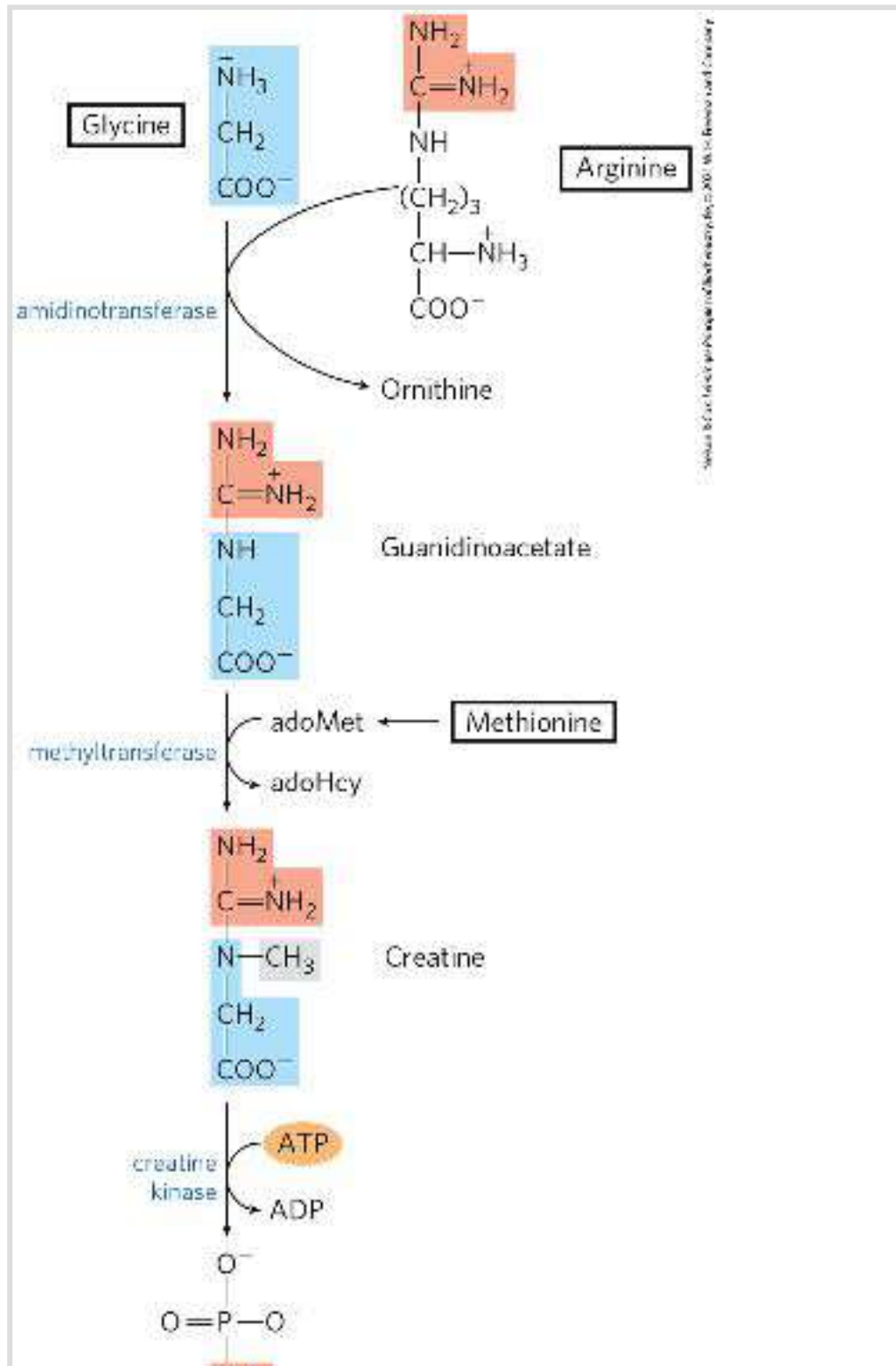
cellular functions. The CO produced by heme oxygenase is toxic at high concentrations, but at the very low concentrations generated during heme degradation it seems to have some regulatory and/or signaling functions. It acts as a vasodilator, much the same as (but less potent than) nitric oxide (discussed below). Low levels of CO also have some regulatory effects on neurotransmission. Bilirubin is the most abundant antioxidant in mammalian tissues and is responsible for most of the antioxidant activity in serum. Its protective effects seem to be especially important in the developing brain of newborn infants. The cell toxicity associated with jaundice may be due to bilirubin levels in excess of the serum albumin needed to solubilize it.



Given these varied roles of heme degradation products, the degradative pathway is subject to regulation, mainly at the first step. Humans have at least three isozymes of heme oxygenase (HO). HO-1 is highly regulated; the expression of its gene is induced by a wide range of stress conditions, including shear stress, uncontrolled angiogenesis (development of blood vessels), hypoxia, hyperoxia, heat shock, exposure to ultraviolet light, hydrogen peroxide, and many other metabolic insults. HO-2 is found mainly in the brain and the testes, where it is continuously expressed. The third isozyme, HO-3, is not catalytically active, but may play a role in oxygen sensing. ■

Amino Acids Are Precursors of Creatine and Glutathione

Phosphocreatine, derived from **creatine**, is an important energy buffer in skeletal muscle (see [Box 23-1](#)). Creatine is synthesized from glycine and arginine ([Fig. 22-28](#)); methionine, in the form of S-adenosylmethionine, acts as methyl group donor.



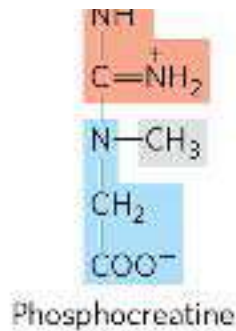
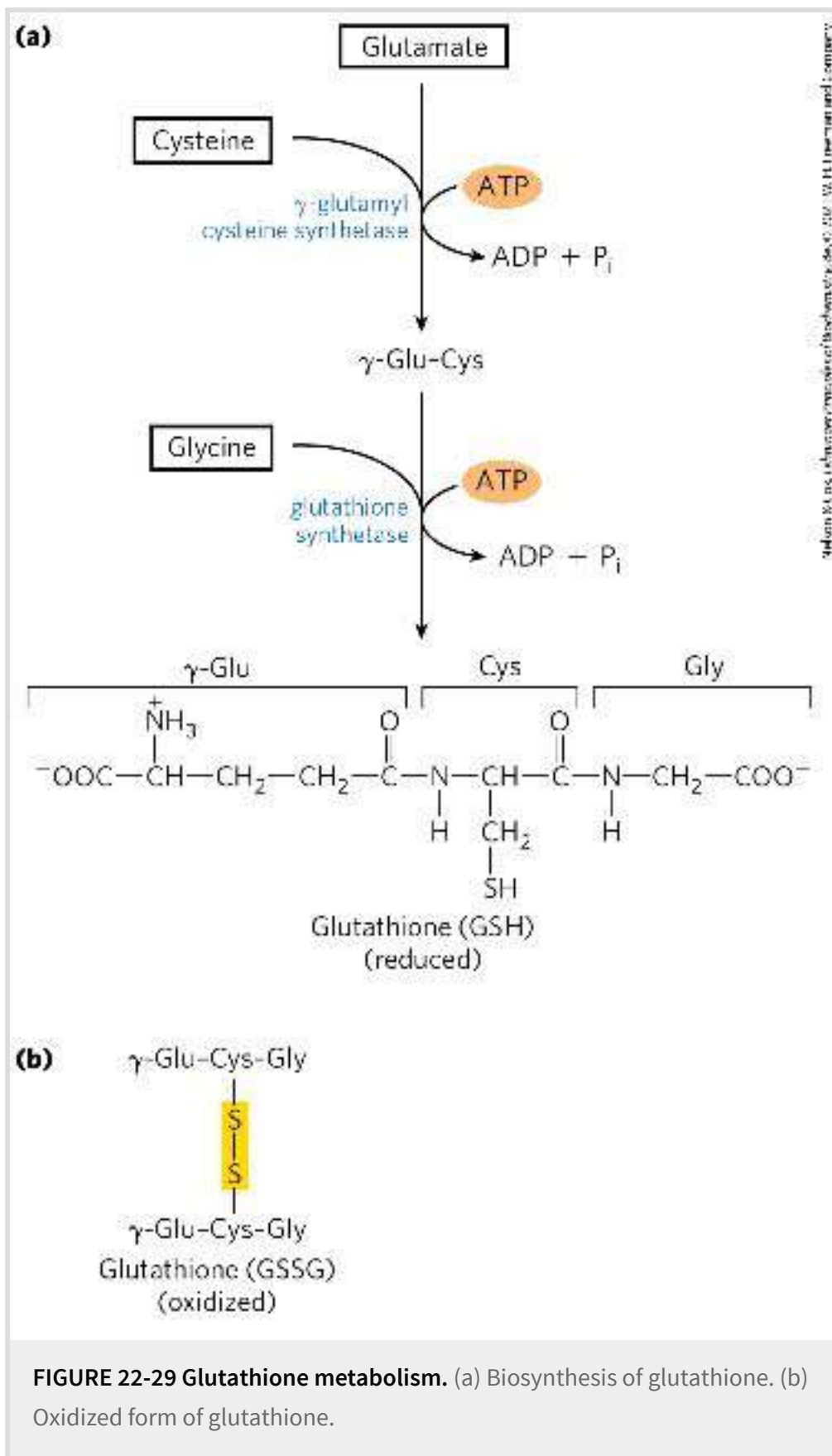


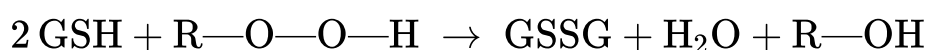
FIGURE 22-28 Biosynthesis of creatine and phosphocreatine. Creatine is made from three amino acids: glycine, arginine, and methionine. This pathway shows the versatility of amino acids as precursors of other nitrogenous biomolecules.

Glutathione (GSH), present in plants, animals, and some bacteria, often at high levels, can be thought of as a redox buffer. It is derived from glutamate, cysteine, and glycine ([Fig. 22-29](#)).

P6 The γ -carboxyl group of glutamate is activated by ATP to form an acyl phosphate intermediate, which is then attacked by the α -amino group of cysteine. A second condensation reaction follows, with the α -carboxyl group of cysteine activated to an acyl phosphate to permit reaction with glycine. The oxidized form of glutathione (glutathione disulfide, or GSSG), produced in the course of its redox activities, contains two glutathione molecules linked by a disulfide bond.



Glutathione helps maintain the sulfhydryl groups of proteins in the reduced state and the iron of heme in the ferrous (Fe^{2+}) state, and it serves as a reducing agent for glutaredoxin in deoxyribonucleotide synthesis (see [Fig. 22-41](#)). Its redox function is also used to remove toxic peroxides formed in the normal course of growth and metabolism under aerobic conditions:

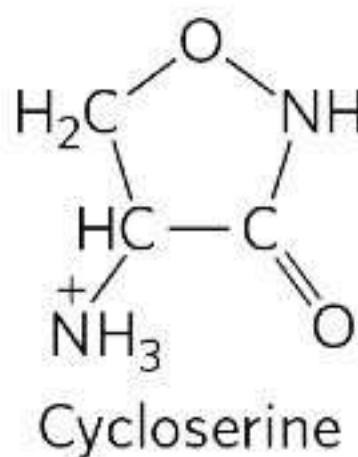
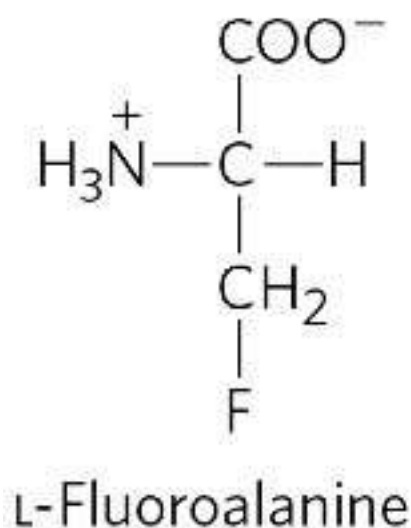


This reaction is catalyzed by **glutathione peroxidase**, a remarkable enzyme in that it contains a covalently bound selenium (Se) atom in the form of selenocysteine (see [Fig. 3-8a](#)), which is essential for its activity.

D-Amino Acids Are Found Primarily in Bacteria

Although D-amino acids do not generally occur in proteins, they do serve some special functions in the structure of bacterial cell walls and peptide antibiotics. Bacterial peptidoglycans (see [Fig. 6-32](#)) contain both D-alanine and D-glutamate. D-Amino acids arise directly from the L isomers by the action of amino acid racemases, which have pyridoxal phosphate as cofactor (see [Fig. 18-6](#)). Amino acid racemization is uniquely important to bacterial metabolism, and enzymes such as alanine racemase are prime targets for pharmaceutical agents. One such agent, L-

fluoroalanine, is being tested as an antibacterial drug. Another, **cycloserine**, is used to treat tuberculosis. Because these inhibitors also affect some PLP-requiring human enzymes, however, they have potentially undesirable side effects.



Aromatic Amino Acids Are Precursors of Many Plant Substances

Phenylalanine, tyrosine, and tryptophan are converted to a variety of important compounds in plants. The rigid polymer **lignin**, derived from phenylalanine and tyrosine, is second only to cellulose in abundance in plant tissues. The structure of the lignin polymer is complex and not well understood. Tryptophan is also the precursor of the plant growth hormone indole-3-acetate, or **auxin** ([Fig. 22-30a](#)), which is important in the regulation of a wide range of biological processes in plants.

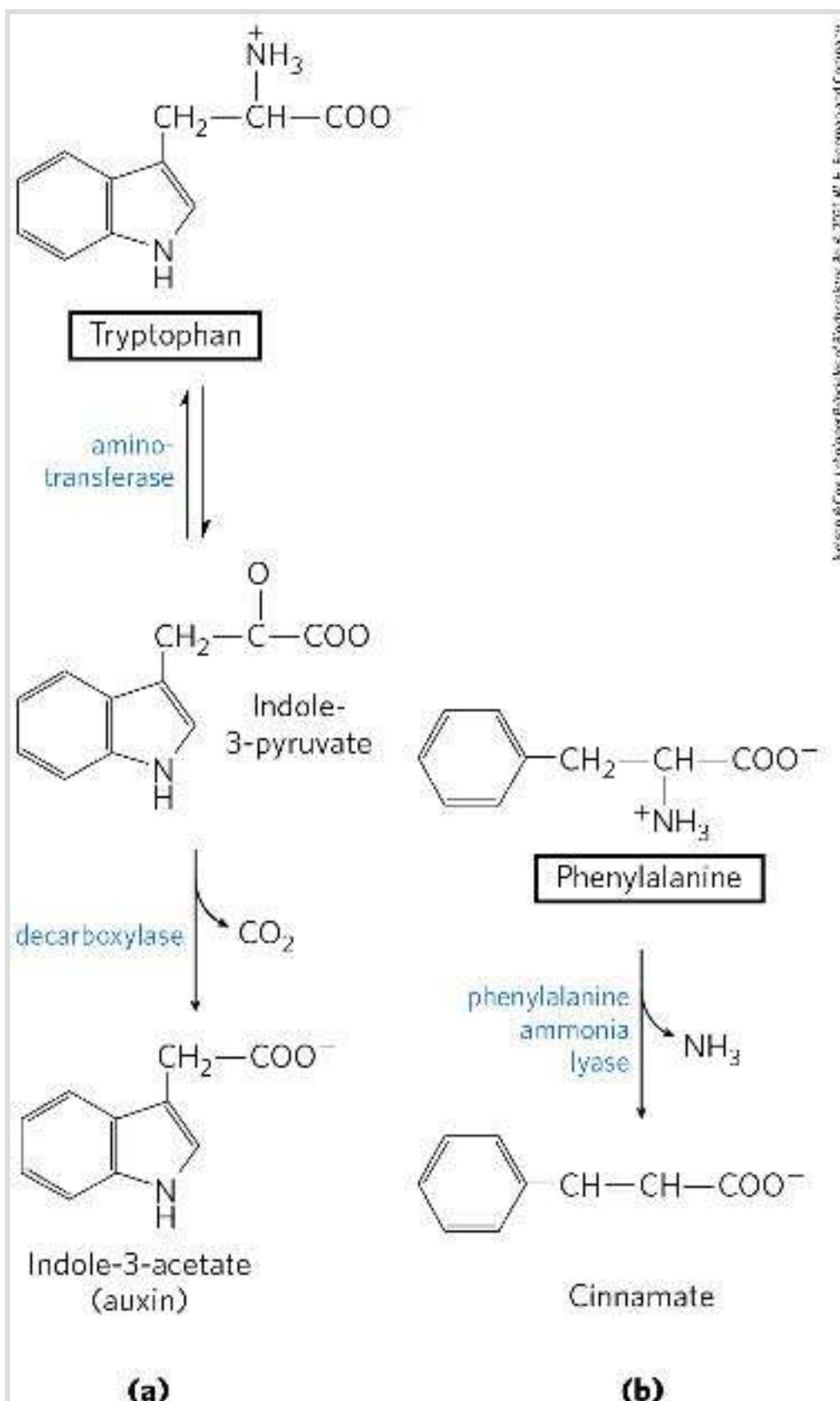


FIGURE 22-30 Biosynthesis of two plant substances from amino acids. (a) Indole-3-acetate (auxin) and (b) cinnamate (cinnamon flavor).

Phenylalanine and tyrosine also give rise to many commercially significant natural products, including the tannins that inhibit oxidation in wines; alkaloids such as morphine, which have potent physiological effects; and the flavoring of cinnamon oil ([Fig. 22-30b](#)), nutmeg, cloves, vanilla, cayenne pepper, and other products.

Biological Amines Are Products of Amino Acid Decarboxylation

Many important neurotransmitters are primary or secondary amines, derived from amino acids in simple pathways. In addition, some polyamines that form complexes with DNA are derived from the amino acid ornithine, a component of the urea cycle. A common denominator of many of these pathways is amino acid decarboxylation, another PLP-requiring reaction (see [Fig. 18-6](#)).

The synthesis of some neurotransmitters is illustrated in [Figure 22-31](#). Tyrosine gives rise to a family of catecholamines that includes **dopamine**, **norepinephrine**, and **epinephrine**. Levels of catecholamines are correlated with, among other things, changes in blood pressure. The neurological disorder Parkinson disease is associated with an underproduction of dopamine, and it has traditionally been treated by administering L-dopa. Overproduction of dopamine in the brain may be linked to psychological disorders such as schizophrenia.

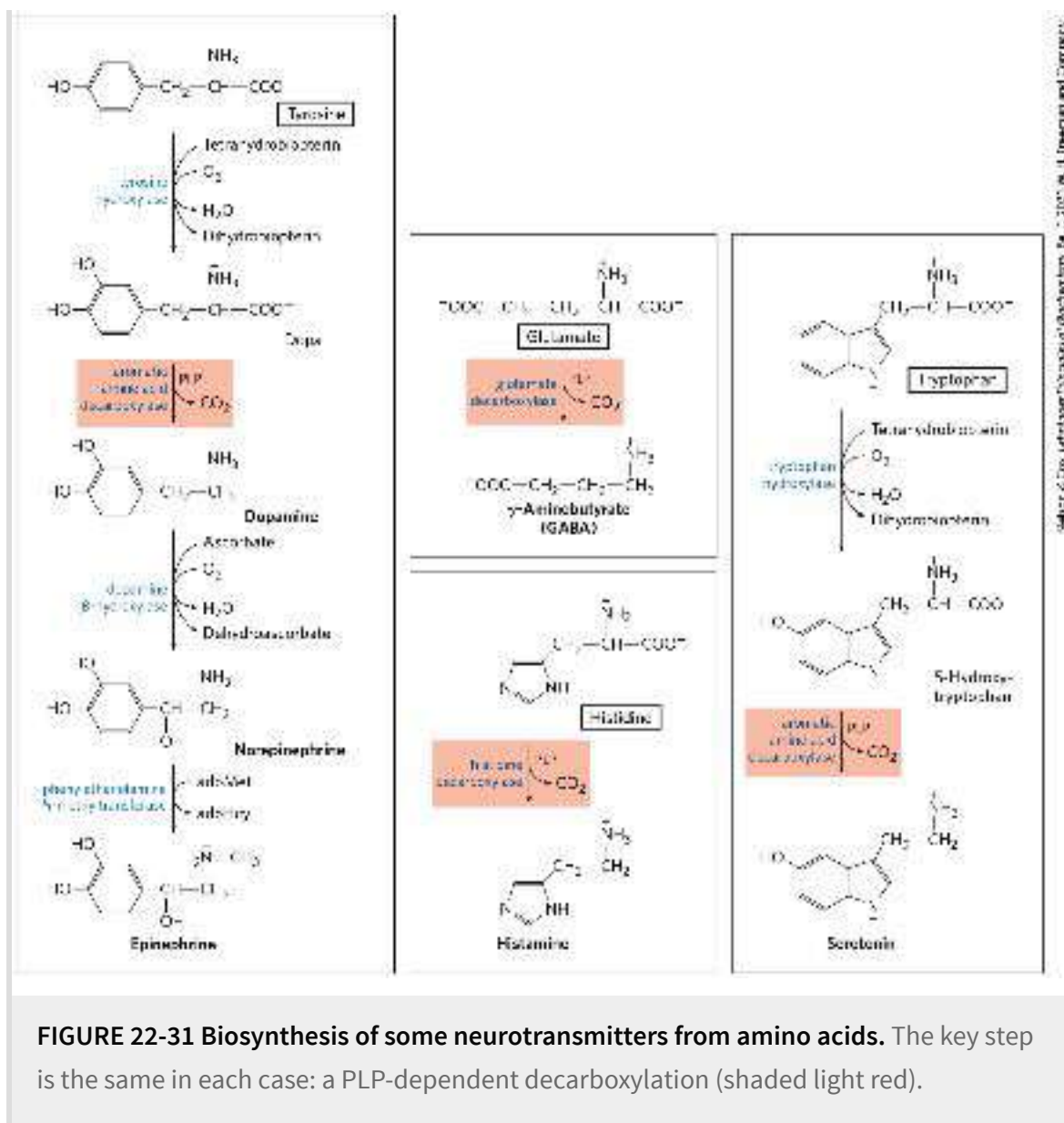
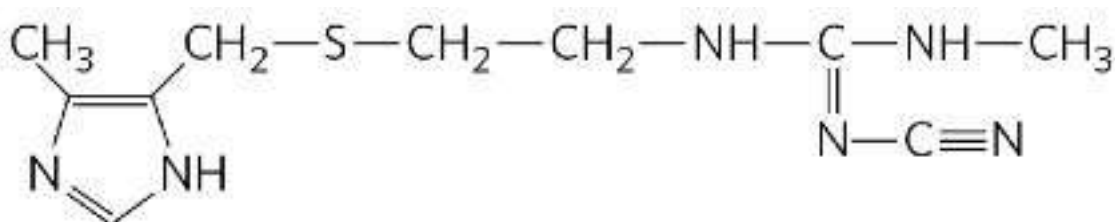


FIGURE 22-31 Biosynthesis of some neurotransmitters from amino acids. The key step is the same in each case: a PLP-dependent decarboxylation (shaded light red).

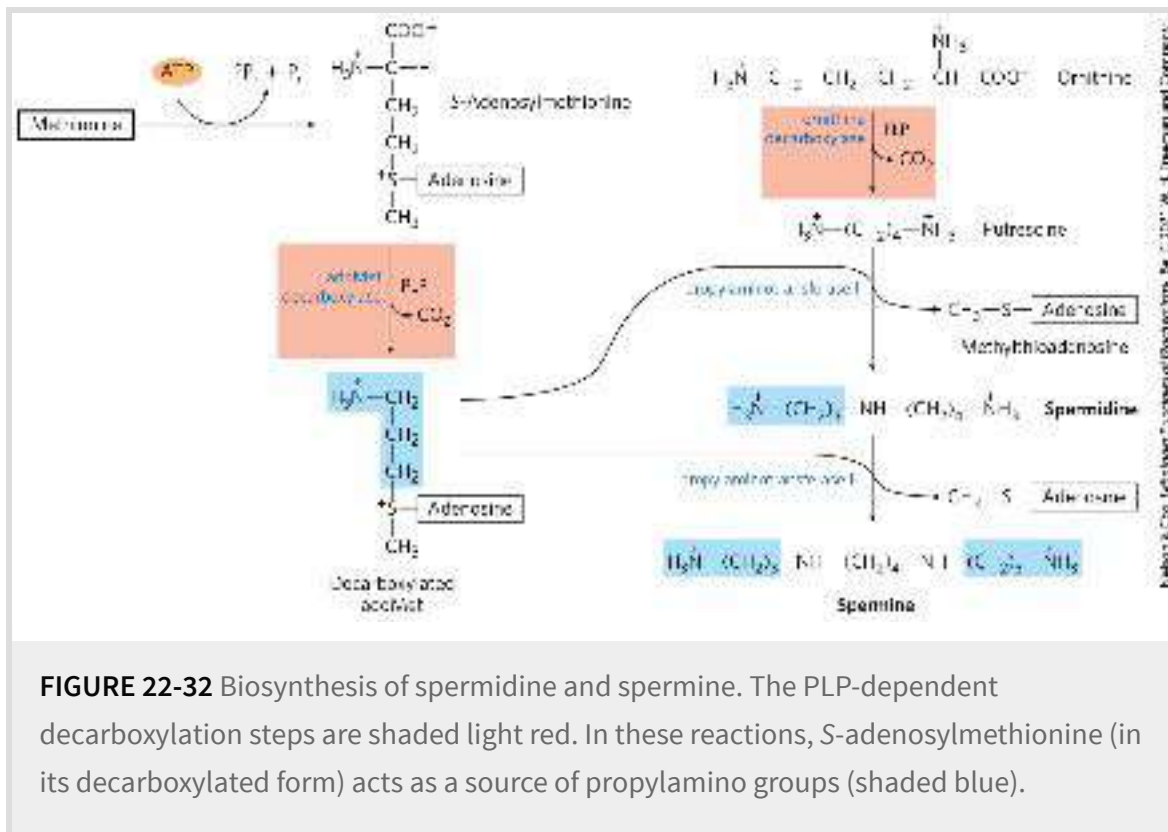
Glutamate decarboxylation gives rise to **γ -aminobutyrate (GABA)**, an inhibitory neurotransmitter. Its underproduction is associated with epileptic seizures. GABA analogs are used in the treatment of epilepsy and hypertension. Levels of GABA can also be increased by administering inhibitors of the GABA-degrading enzyme GABA aminotransferase. Another important neurotransmitter, **serotonin**, is derived from tryptophan in a two-step pathway.

Histidine undergoes decarboxylation to **histamine**, a powerful vasodilator in animal tissues. Histamine is released in large amounts as part of the allergic response, and it also stimulates acid secretion in the stomach. A growing array of pharmaceutical agents are being designed to interfere with either the synthesis or the action of histamine. A prominent example is the histamine receptor antagonist **cimetidine** (Tagamet), a structural analog of histamine:



It promotes the healing of duodenal ulcers by inhibiting secretion of gastric acid.

Polyamines such as **spermine** and **spermidine**, involved in DNA packaging, are derived from methionine and ornithine by the pathway shown in [Figure 22-32](#). The first step is decarboxylation of ornithine, a precursor of arginine ([Fig. 22-12](#)). **Ornithine decarboxylase**, a PLP-requiring enzyme, is the target of several powerful inhibitors used as pharmaceutical agents (see [Box 6-1](#)).



Arginine Is the Precursor for Biological Synthesis of Nitric Oxide

A surprise finding in the mid-1980s was the role of nitric oxide (NO) — previously known mainly as a component of smog — as an important biological messenger. This simple gaseous substance diffuses readily through membranes, although its high reactivity limits its range of diffusion to about a 1 mm radius from the site of synthesis. In humans NO plays a role in a range of physiological processes, including neurotransmission, blood clotting, and the control of blood pressure. Its mode of action is described in [Box 12-2](#).

P6 Nitric oxide is synthesized from arginine in an NADPH-dependent reaction catalyzed by nitric oxide synthase ([Fig. 22-33](#)), a dimeric enzyme structurally related to NADPH cytochrome P-450 reductase (see [Box 21-1](#)). The reaction is a five-electron oxidation. Each subunit of the enzyme contains one bound molecule of each of four different cofactors: FMN, FAD, tetrahydrobiopterin, and Fe^{3+} heme. NO is an unstable molecule and cannot be stored. Its synthesis is stimulated by interaction of nitric oxide synthase with Ca^{2+} -calmodulin (see [Fig. 12-17](#)).

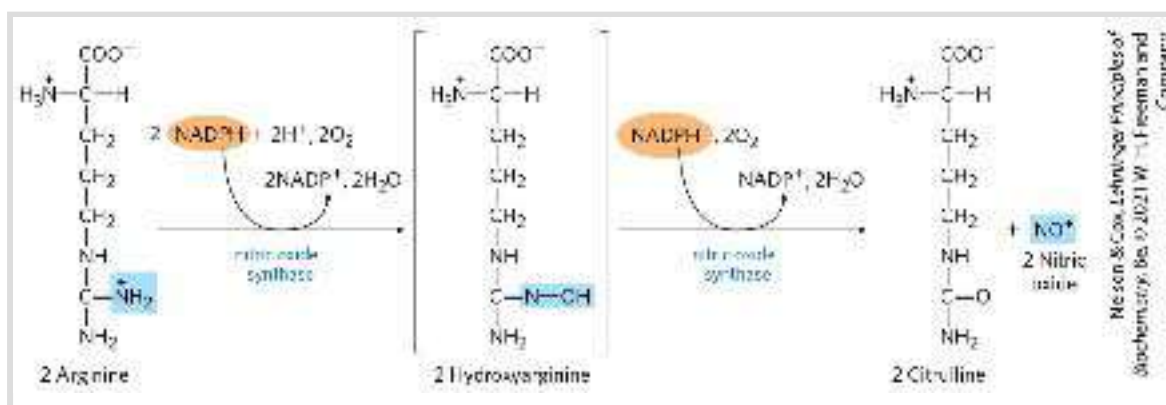


FIGURE 22-33 Biosynthesis of nitric oxide. The nitrogen of the NO is derived from the guanidinium group of arginine.

SUMMARY 22.3 Molecules Derived from Amino Acids

- Many important biomolecules are derived from amino acids. Glycine is a precursor of porphyrins.
- Degradation of iron-porphyrin (heme) generates bilirubin, which is converted to bile pigments with several physiological functions.

- Glycine and arginine give rise to creatine and phosphocreatine, an energy buffer. Glutathione, formed from three amino acids, is an important cellular reducing agent.
- Bacteria synthesize D-amino acids from L-amino acids in racemization reactions requiring pyridoxal phosphate. D-Amino acids are commonly found in certain bacterial walls and certain antibiotics.
- Plants make many substances from aromatic amino acids.
- The PLP-dependent decarboxylation of some amino acids yields important biological amines, including neurotransmitters and polyamines.
- Arginine is the precursor of nitric oxide, a biological messenger.

22.4 Biosynthesis and Degradation of Nucleotides

As discussed in [Chapter 8](#), nucleotides have a variety of important functions in all cells. They are the precursors of DNA and RNA. They are essential carriers of chemical energy — a role primarily of ATP and to some extent GTP. They are components of the cofactors NAD, FAD, S-adenosylmethionine, and coenzyme A, as well as of activated biosynthetic intermediates such as UDP-glucose and CDP-diacylglycerol. Some, such as cAMP and cGMP, are also cellular second messengers.

Two types of pathways lead to nucleotides: the [de novo pathways](#) and the [salvage pathways](#). De novo synthesis of nucleotides begins with their metabolic precursors: amino acids, ribose 5-phosphate, CO₂, and NH₃. Salvage pathways recycle the free bases and nucleosides released from nucleic acid breakdown. Both types of pathways are important in cellular metabolism, and both are discussed in this section.

The de novo pathways for purine and pyrimidine biosynthesis seem to be nearly identical in all living organisms. Notably, the free bases guanine, adenine, thymine, cytidine, and uracil are *not* intermediates in these pathways; that is, the bases are not synthesized and then attached to ribose, as might be expected. The purine ring structure is built up one or a few atoms at a time, attached to ribose throughout the process. The pyrimidine ring is synthesized as **orotate**, attached to ribose phosphate, and then

converted to the common pyrimidine nucleotides required in nucleic acid synthesis. Although the free bases are not intermediates in the de novo pathways, they are intermediates in some of the salvage pathways.

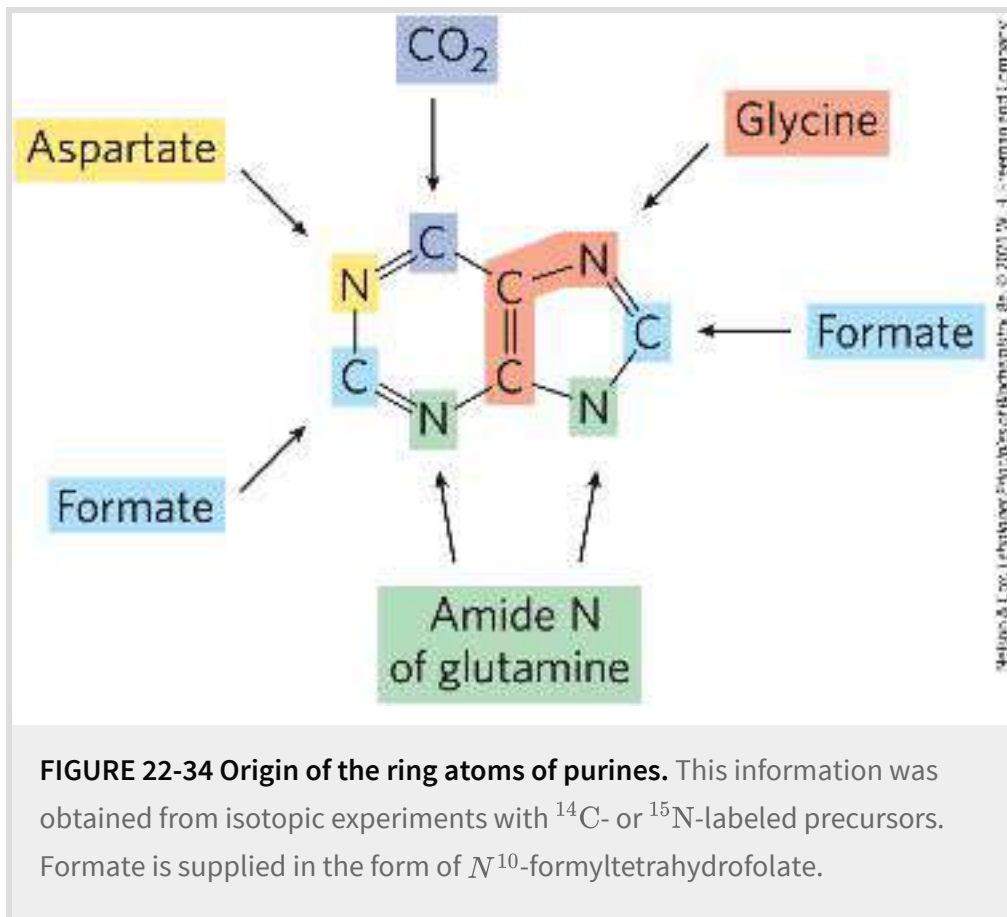
Several important precursors are shared by the de novo pathways for synthesis of pyrimidines and purines. Phosphoribosyl pyrophosphate (PRPP) is important in both, and in these pathways the structure of ribose is retained in the product nucleotide, in contrast to its fate in the tryptophan and histidine biosynthetic pathways discussed earlier. An amino acid is an important precursor in each type of pathway: glycine for purines and aspartate for pyrimidines. Glutamine again is the most important source of amino groups — in five different steps in the de novo pathways. Aspartate is also used as the source of an amino group in the purine pathways, in two steps.

Two other features deserve mention. First, there is evidence, especially in the de novo purine pathway, that the enzymes are present as large, multienzyme complexes or metabolons in the cell, a recurring theme in our discussion of metabolism. Second, the cellular pools of nucleotides (other than ATP) are quite small, perhaps 1% or less of the amounts required to synthesize the cell's DNA. Therefore, cells must continue to synthesize nucleotides during nucleic acid synthesis, and in some cases, nucleotide synthesis may limit the rates of DNA replication and transcription. Because of the importance of these processes in dividing cells, agents that inhibit nucleotide synthesis have become particularly important in medicine.

We examine here the biosynthetic pathways of purine and pyrimidine nucleotides and their regulation, the formation of the deoxynucleotides, and the degradation of purines and pyrimidines to uric acid and urea. We end with a discussion of chemotherapeutic agents that affect nucleotide synthesis.

De Novo Purine Nucleotide Synthesis Begins with PRPP

The two parent purine nucleotides of nucleic acids are adenosine 5'-monophosphate (AMP; adenylate) and guanosine 5'-monophosphate (GMP; guanylate), containing the purine bases adenine and guanine. [Figure 22-34](#) shows the origin of the carbon and nitrogen atoms of the purine ring system, as determined by John M. Buchanan using isotopic tracer experiments in birds (who conveniently excrete excess nitrogen as insoluble uric acid, a purine analog). The detailed pathway of purine biosynthesis was worked out primarily by Buchanan and G. Robert Greenberg in the 1950s.



In the first committed step of the pathway, an amino group donated by glutamine is attached at C-1 of PRPP ([Fig. 22-35](#)). The resulting **5-phosphoribosylamine** is highly unstable, with a half-life of 30 seconds at pH 7.5. This intermediate is rapidly funneled into the next biosynthetic step, and the purine ring is subsequently built up on this structure. The pathway described here is identical in all organisms, with the exception of one step that differs in higher eukaryotes, as noted below.

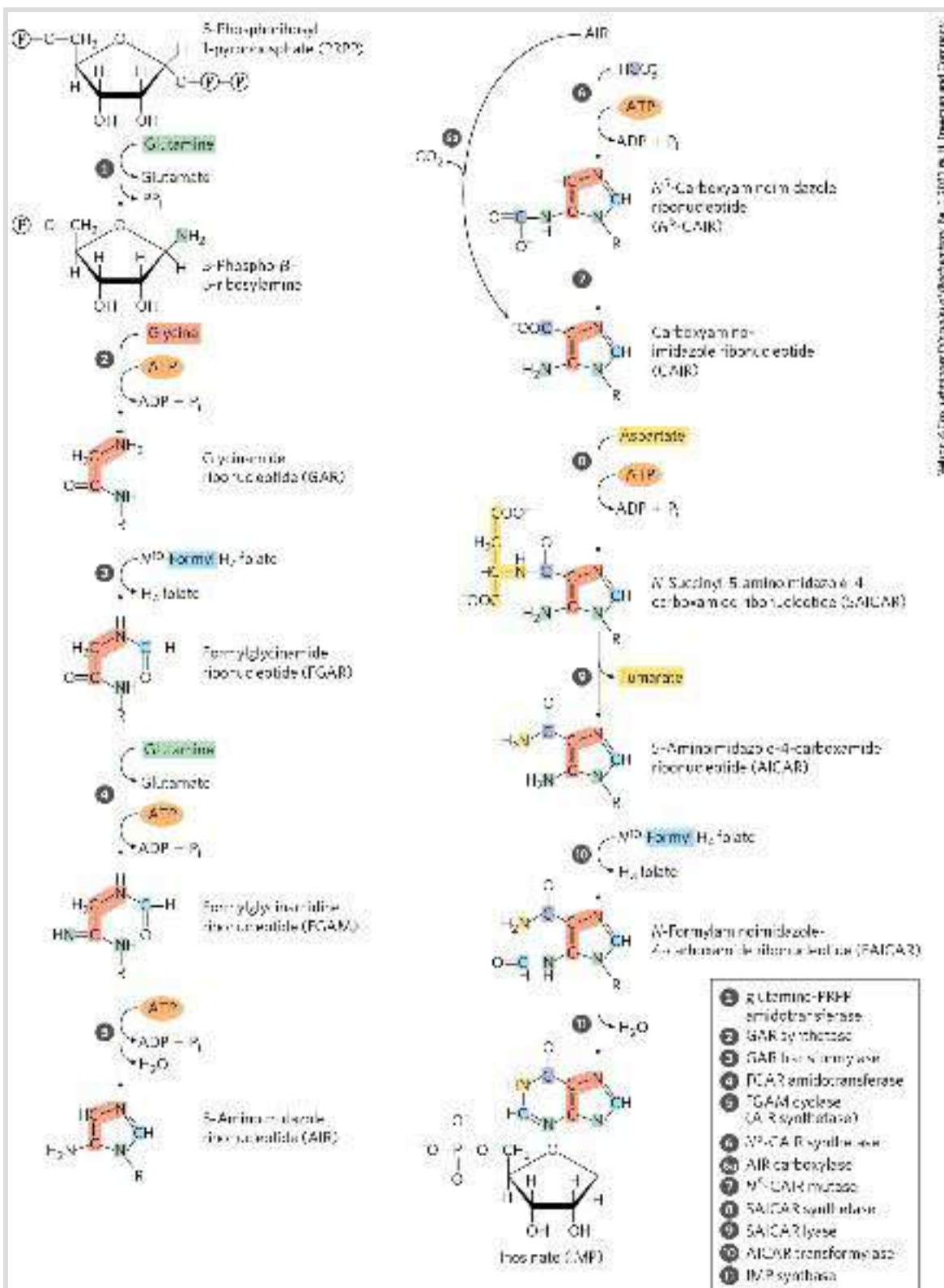


FIGURE 22-35 De novo synthesis of purine nucleotides: construction of the purine ring of inosinate (IMP). Each addition to the purine ring is shaded to match Figure 22-34.

After step 2, R symbolizes the 5-phospho-D-ribose group on which the purine ring is built. Formation of 5-phosphoribosylamine (step 1) is the first committed step in

purine synthesis. Note that the product of step 9, AICAR, is the remnant of ATP released during histidine biosynthesis (see [Fig. 22-22](#), step 5). Abbreviations are given for most intermediates to simplify the naming of the enzymes. Step 6a is the alternative path from AIR to CAIR occurring in higher eukaryotes.

The second step is the addition of three atoms from glycine ([Fig. 22-35](#), step 2). An ATP is consumed to activate the glycine carboxyl group (in the form of an acyl phosphate) for this condensation reaction. The added glycine amino group is then formylated by N^{10} -formyltetrahydrofolate (step 3), and a nitrogen is contributed by glutamine (step 4), before dehydration and ring closure yield the five-membered imidazole ring of the purine nucleus, as 5-aminoimidazole ribonucleotide (AIR; step 5).

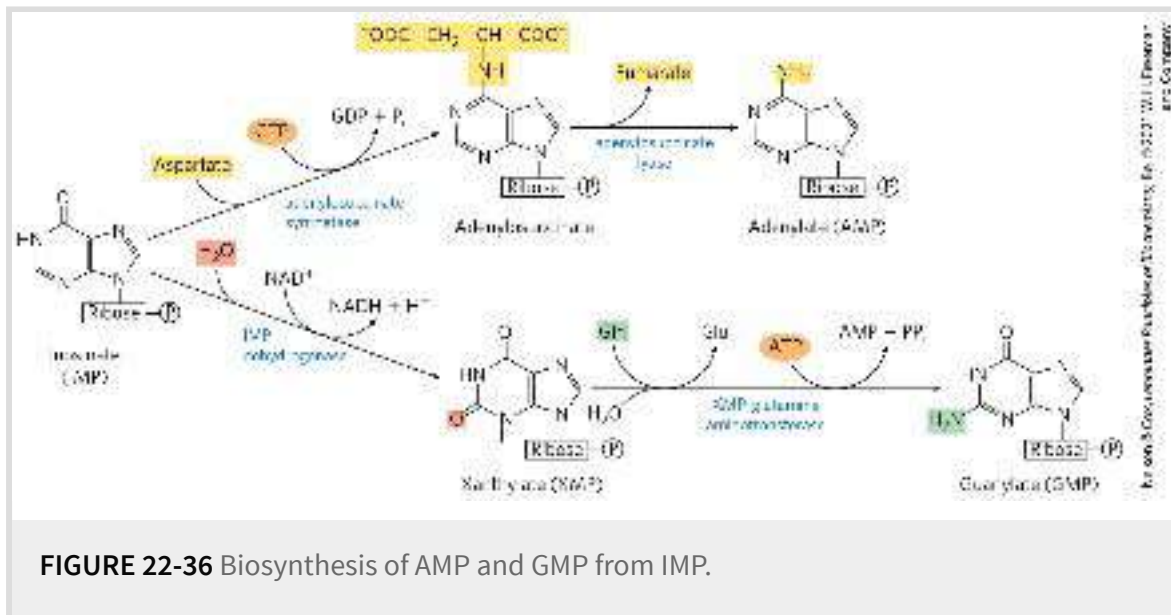
At this point, three of the six atoms needed for the second ring in the purine structure are in place. To complete the process, a carboxyl group is first added (step 6). This carboxylation is unusual in that it does not require biotin, but instead uses the bicarbonate generally present in aqueous solutions. A rearrangement transfers the carboxylate from the exocyclic amino group to position 4 of the imidazole ring (step 7). Steps 6 and 7 are found only in bacteria and fungi. In higher eukaryotes, including humans, the 5-aminoimidazole ribonucleotide product of step 5 is carboxylated directly to carboxyaminoimidazole ribonucleotide in one step instead of two (step 6a). The enzyme catalyzing this reaction is AIR carboxylase.

Aspartate now donates its amino group in two steps (8 and 9): formation of an amide bond, followed by elimination of the carbon skeleton of aspartate (as fumarate). (Recall that aspartate plays an analogous role in two steps of the urea cycle; see [Fig. 18-10](#).) The final carbon is contributed by N^{10} -formyltetrahydrofolate (step 10), and a second ring closure takes place to yield the second fused ring of the purine nucleus (step 11). The first intermediate with a complete purine ring is **inosinate (IMP)**.

As in the tryptophan and histidine biosynthetic pathways, the enzymes of IMP synthesis seem to be organized as large metabolons in the cell. Once again, evidence comes from the existence of single polypeptides with several functions, some catalyzing nonsequential steps in the pathway. In eukaryotic cells ranging from yeast to fruit flies to chickens, steps 1, 3, and 5 in [Figure 22-35](#) are catalyzed by a multifunctional protein. An additional multifunctional protein catalyzes steps 10 and 11. In humans, a multifunctional enzyme combines the activities of AIR carboxylase and SAICAR synthetase (steps 6a and 8). In bacteria, these activities are found on separate proteins, but the proteins may form a metabolon. The channeling of reaction intermediates from one enzyme to the next permitted by these complexes is probably especially important for unstable intermediates such as 5-phosphoribosylamine.

Conversion of inosinate to adenylate requires the insertion of an amino group derived from aspartate ([Fig. 22-36](#)); this takes place in two reactions similar to those used to introduce N-1 of the purine ring ([Fig. 22-35](#), steps 8 and 9). A crucial difference is

that GTP rather than ATP is the source of the high-energy phosphate in synthesizing adenylosuccinate. Guanylate is formed by the NAD^+ -requiring oxidation of inosinate at C-2, followed by addition of an amino group derived from glutamine. ATP is cleaved to AMP and PP_i in the final step (Fig. 22-36).



Purine Nucleotide Biosynthesis Is Regulated by Feedback Inhibition

P4 Four major feedback mechanisms cooperate in regulating the overall rate of de novo purine nucleotide synthesis and the relative rates of formation of the two end products, adenylosuccinate and guanylate (Fig. 22-37). The first mechanism is exerted on the first reaction that is unique to purine synthesis: transfer of an amino group to PRPP to form 5-phosphoribosylamine. This reaction is catalyzed by the allosteric enzyme glutamine-PRPP amidotransferase, which is inhibited by the end products IMP,

AMP, and GMP. AMP and GMP act synergistically in this concerted inhibition. Thus, whenever either AMP or GMP accumulates to excess, the first step in its biosynthesis from PRPP is partially inhibited.

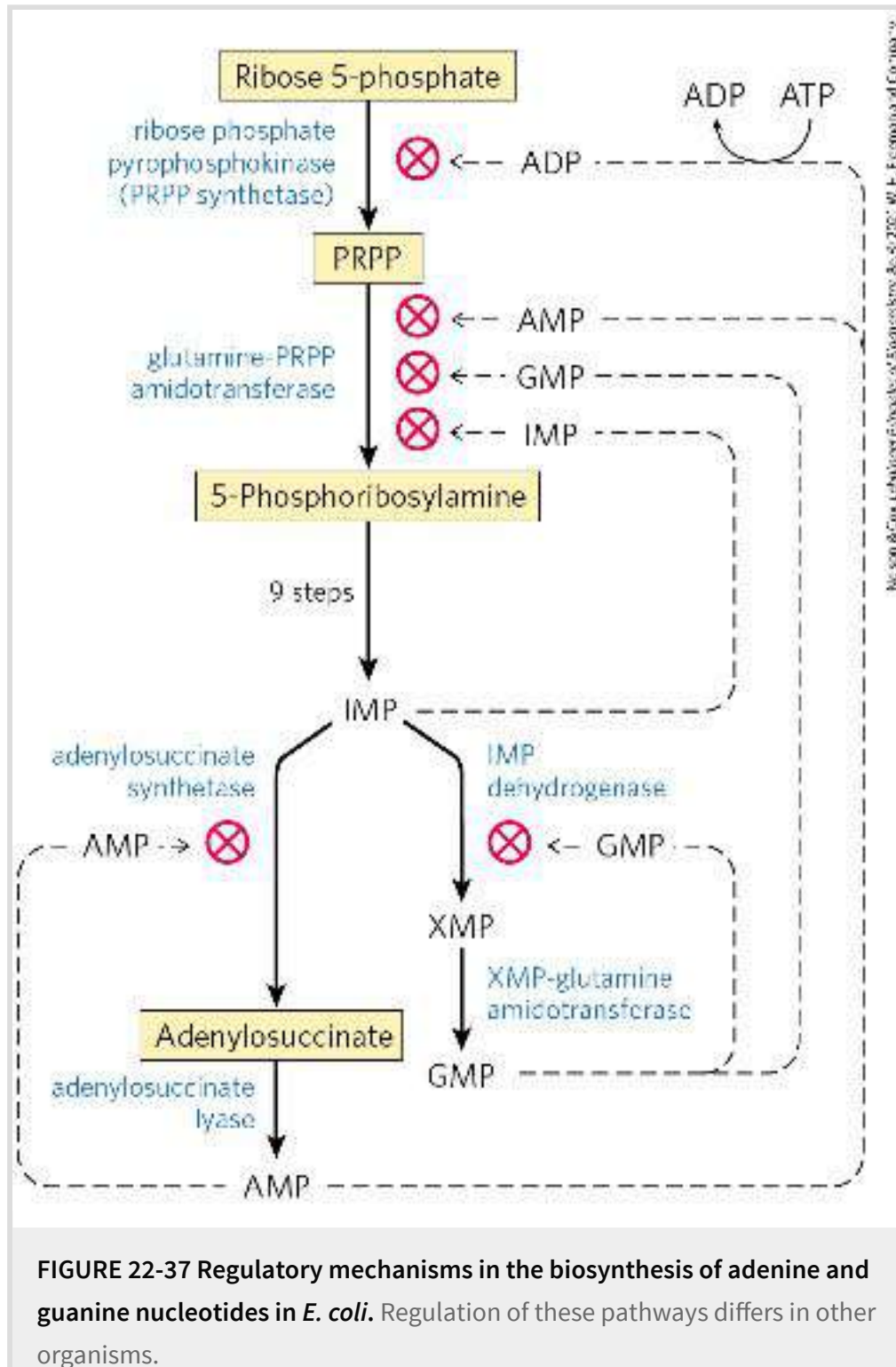
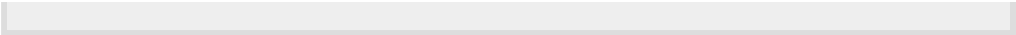


FIGURE 22-37 Regulatory mechanisms in the biosynthesis of adenine and guanine nucleotides in *E. coli*. Regulation of these pathways differs in other organisms.



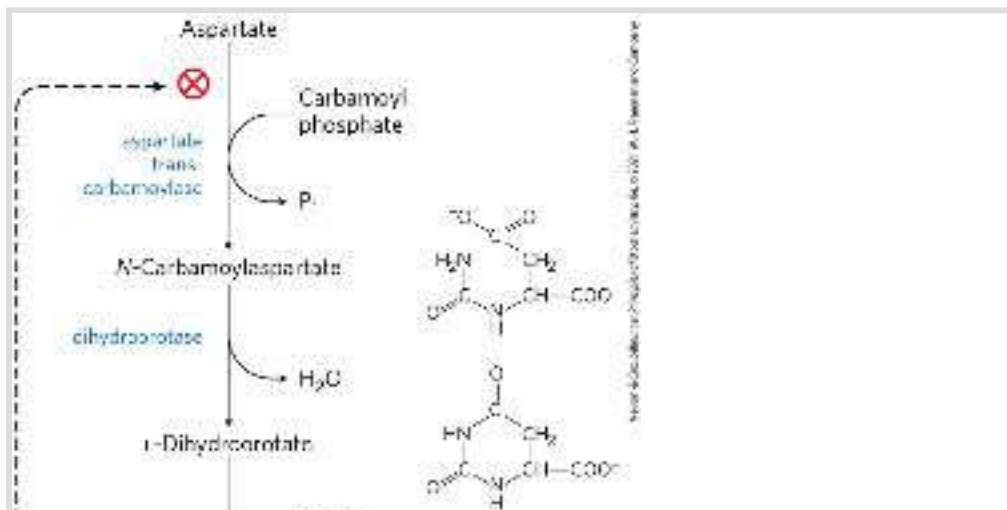
In the second control mechanism, exerted at a later stage, an excess of GMP in the cell inhibits formation of xanthylate from inosinate by IMP dehydrogenase, without affecting the formation of AMP. Conversely, an accumulation of adenyate inhibits formation of adenylosuccinate by adenylosuccinate synthetase, without affecting the biosynthesis of GMP. When both products are present in sufficient quantities, IMP builds up, and it inhibits an earlier step in the pathway; this is another example of the regulatory strategy called **sequential feedback inhibition**.

In the third mechanism, GTP is required in the conversion of IMP to AMP, whereas ATP is required for conversion of IMP to GMP ([Fig. 22-36](#)), a reciprocal arrangement that tends to balance the synthesis of the two ribonucleotides.

The fourth and final control mechanism is the inhibition of PRPP synthesis by the allosteric regulation of ribose phosphate pyrophosphokinase. This enzyme is inhibited by ADP and GDP, in addition to metabolites from other pathways for which PRPP is a starting point.

Pyrimidine Nucleotides Are Made from Aspartate, PRPP, and Carbamoyl Phosphate

The common pyrimidine ribonucleotides are cytidine 5'-monophosphate (CMP; cytidylate) and uridine 5'-monophosphate (UMP; uridylate), which contain the pyrimidines cytosine and uracil. De novo pyrimidine nucleotide biosynthesis ([Fig. 22-38](#)) proceeds in a somewhat different manner from purine nucleotide synthesis; the six-membered pyrimidine ring is made first and then attached to ribose 5-phosphate. Required in this process is carbamoyl phosphate, also an intermediate in the urea cycle. However, in animals the carbamoyl phosphate required in urea synthesis is made in mitochondria by carbamoyl phosphate synthetase I, whereas the carbamoyl phosphate required in pyrimidine biosynthesis is made in the cytosol by a different form of the enzyme, **carbamoyl phosphate synthetase II**. In bacteria, a single enzyme supplies carbamoyl phosphate for the synthesis of arginine and pyrimidines. The bacterial enzyme has three separate active sites, spaced along a channel nearly 100 Å long ([Fig. 22-39](#)). Bacterial carbamoyl phosphate synthetase provides a vivid illustration of the channeling of unstable reaction intermediates between active sites so that products are formed efficiently.



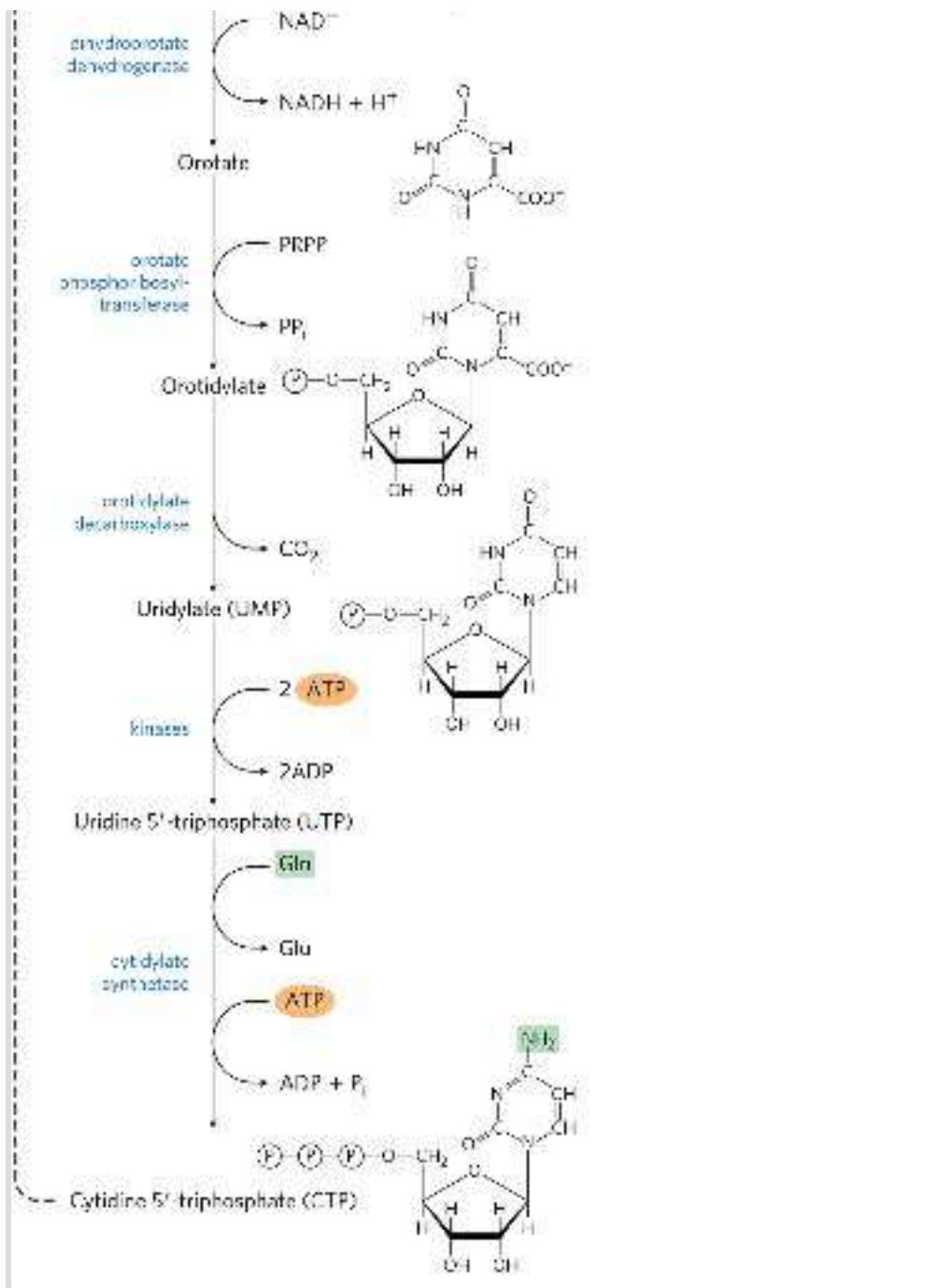


FIGURE 22-38 De novo synthesis of pyrimidine nucleotides: biosynthesis of UTP and CTP via orotidylate. The pyrimidine is constructed from carbamoyl phosphate and aspartate. The ribose 5-phosphate is then added to the completed pyrimidine ring by orotate phosphoribosyltransferase. The first step in this pathway (not shown here; see [Fig. 18-11a](#)) is the synthesis of carbamoyl phosphate from CO_2 , NH_4^+ , and ATP. In eukaryotes, the first step is catalyzed by carbamoyl phosphate synthetase II.

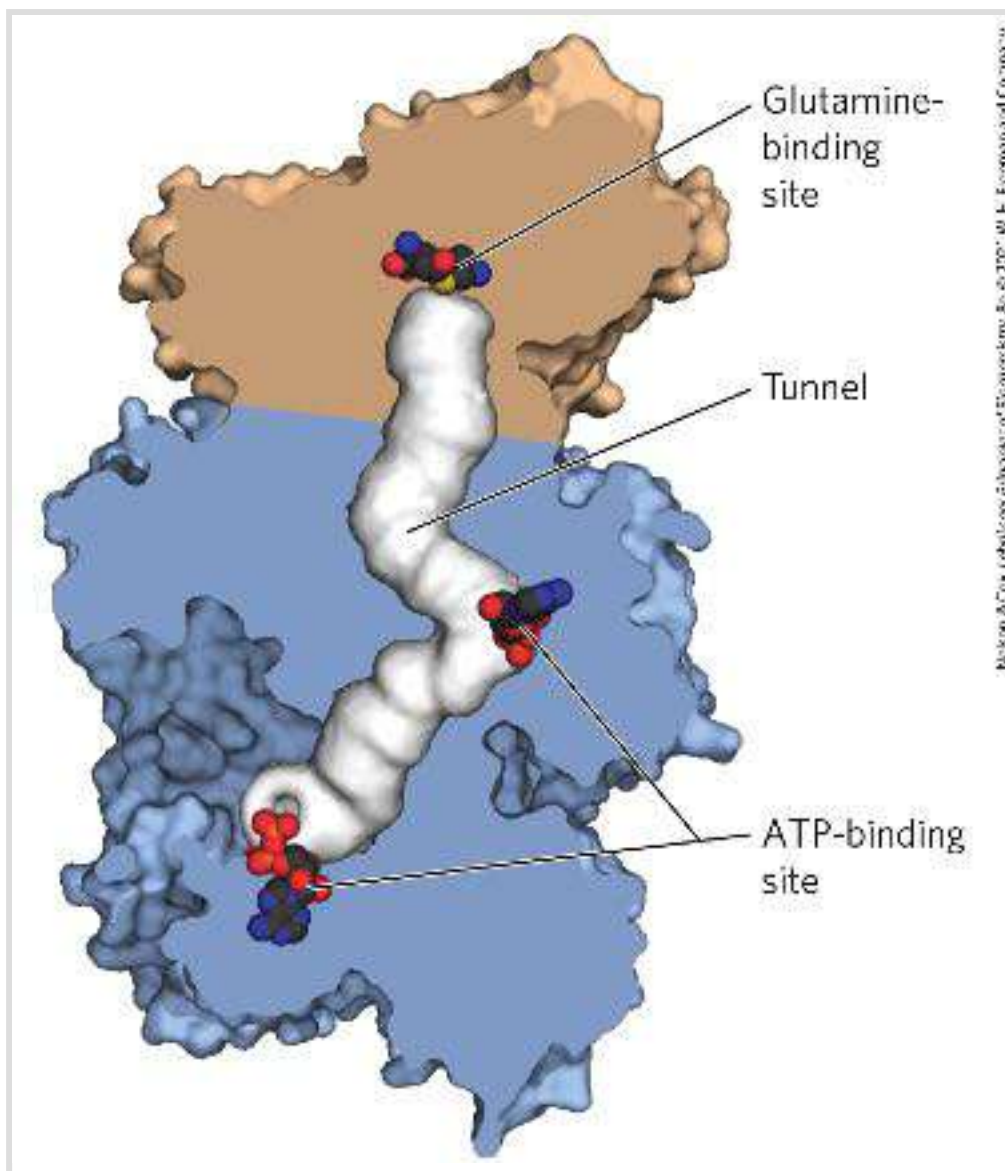


FIGURE 22-39 Channeling of intermediates in bacterial carbamoyl phosphate synthetase. The reaction catalyzed by this enzyme (and its mitochondrial counterpart) is illustrated in [Figure 18-11a](#). In this cutaway, the small and large subunits are shown in tan and blue, respectively; the tunnel between active sites (almost 100 Å long) is shown as white. In this reaction, a glutamine molecule binds to the small subunit, donating its amido nitrogen as NH_4^+ in a glutamine amidotransferase-type reaction. The NH_4^+ enters the tunnel, which takes it to a second active site, where it combines with bicarbonate in a reaction requiring ATP. The carbamate then reenters the tunnel to reach the third active site, where it is phosphorylated by ATP to carbamoyl phosphate. To solve this structure, the enzyme was crystallized with ornithine bound to the glutamine-binding site and ADP

bound to the ATP-binding sites. [Data from PDB ID 1M6V, J. B. Thoden et al., *J. Biol. Chem.* 277:39,722, 2002.]

Carbamoyl phosphate reacts with aspartate to yield *N*-carbamoylaspartate in the first committed step of pyrimidine biosynthesis ([Fig. 22-38](#)). This reaction is catalyzed by **aspartate transcarbamoylase**. In bacteria, this step is highly regulated, and bacterial aspartate transcarbamoylase is one of the most thoroughly studied allosteric enzymes (see below). By removal of water from *N*-carbamoylaspartate, a reaction catalyzed by **dihydroorotase**, the pyrimidine ring is closed to form *L*-dihydroorotate. This compound is oxidized to the pyrimidine derivative orotate, a reaction in which NAD^+ is the ultimate electron acceptor. In eukaryotes, the first three enzymes in this pathway — carbamoyl phosphate synthetase II, aspartate transcarbamoylase, and dihydroorotase — are part of a single trifunctional protein. The protein, known by the acronym CAD, contains three identical polypeptide chains (each of M_r 230,000), each with active sites for all three reactions. This suggests that metabolons may be the rule in this pathway.

Once orotate is formed, the ribose 5-phosphate side chain, provided once again by PRPP, is attached to yield orotidylate ([Fig. 22-38](#)). Orotidylate is then decarboxylated to uridylate, which is phosphorylated to UTP.  CTP is formed from UTP by the action of **cytidylate synthetase**, by way of an acyl phosphate intermediate (consuming one ATP). The nitrogen donor is

normally glutamine, although the cytidylate synthetases in many species can use NH_4^+ directly.

Pyrimidine Nucleotide Biosynthesis Is Regulated by Feedback Inhibition

 Regulation of the rate of pyrimidine nucleotide synthesis in bacteria occurs in large part through aspartate transcarbamoylase (ATCase), which catalyzes the first reaction in the sequence and is inhibited by CTP, the end product of the sequence ([Fig. 22-38](#)). The bacterial ATCase molecule consists of six catalytic subunits and six regulatory subunits (see [Fig. 6-36](#)). The catalytic subunits bind the substrate molecules, and the allosteric subunits bind the allosteric inhibitor, CTP. The entire ATCase molecule, as well as its subunits, exists in two conformations, active and inactive. When CTP is not bound to the regulatory subunits, the enzyme is maximally active. As CTP accumulates and binds to the regulatory subunits, they undergo a change in conformation. This change is transmitted to the catalytic subunits, which then also shift to an inactive conformation. ATP prevents the changes induced by CTP. [Figure 22-40](#) shows the effects of the allosteric regulators on the activity of ATCase.

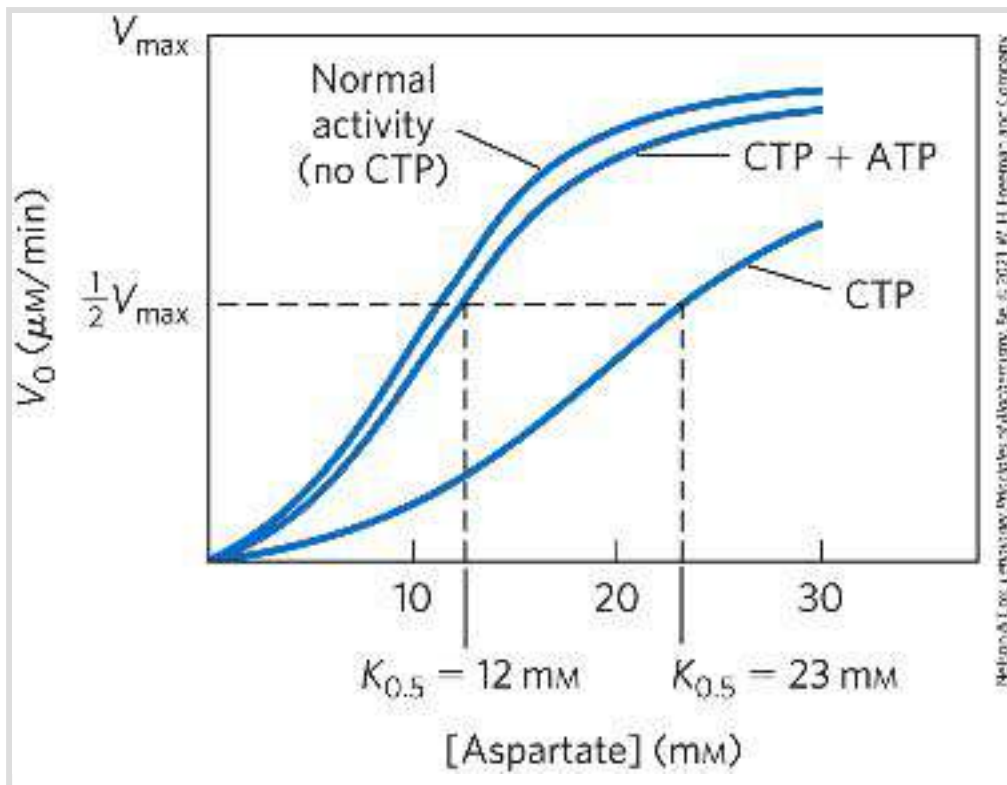


FIGURE 22-40 Allosteric regulation of aspartate transcarbamoylase by CTP and ATP. Addition of 0.8 mM CTP, the allosteric inhibitor of ATCase, increases the $K_{0.5}$ for aspartate (lower curve), thereby reducing the rate of conversion of aspartate to *N*-carbamoylaspartate. ATP at 0.6 mM fully reverses this inhibition by CTP (middle curve).

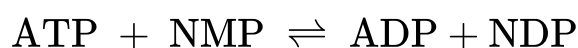
Nucleoside Monophosphates Are Converted to Nucleoside Triphosphates

Nucleotides to be used in biosynthesis are generally converted to nucleoside triphosphates. The conversion pathways are common to all cells. Phosphorylation of AMP to ADP is promoted by **adenylate kinase**, in the reaction



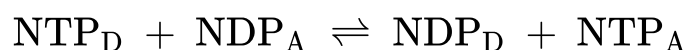
The ADP so formed is phosphorylated to ATP by the glycolytic enzymes or through oxidative phosphorylation.

ATP also brings about the formation of other nucleoside diphosphates by the action of a class of enzymes called nucleoside monophosphate kinases. These enzymes, which are generally specific for a particular base but nonspecific for the sugar (ribose or deoxyribose), catalyze the reaction



The efficient cellular systems for rephosphorylating ADP to ATP (ATP synthase; [Chapter 19](#)) tend to remove ADP and pull this reaction in the direction of products.

Nucleoside diphosphates are converted to triphosphates by the action of a ubiquitous enzyme, nucleoside diphosphate kinase, which catalyzes the reaction



This enzyme is notable in that it is not specific for the base (purines or pyrimidines) or the sugar (ribose or deoxyribose). This nonspecificity applies to both phosphate acceptor (A) and

donor (D), although the donor (NTP_D) is almost invariably ATP because it is present in higher concentration than other nucleoside triphosphates under aerobic conditions.

Ribonucleotides Are the Precursors of Deoxyribonucleotides

Deoxyribonucleotides, the building blocks of DNA, are derived from the corresponding ribonucleotides by direct reduction at the 2'-carbon atom of the D-ribose to form the 2'-deoxy derivative. For example, adenosine diphosphate (ADP) is reduced to 2'-deoxyadenosine diphosphate (dADP), and GDP is reduced to dGDP. This reaction is somewhat unusual in that the reduction occurs at a nonactivated carbon; no closely analogous chemical reactions are known. The reaction is catalyzed by **ribonucleotide reductase**, best characterized in *E. coli*, in which its substrates are ribonucleoside diphosphates.

The reduction of the D-ribose portion of a ribonucleoside diphosphate to 2'-deoxy-D-ribose requires a pair of hydrogen atoms, which are ultimately donated by NADPH via an intermediate hydrogen-carrying protein, [thioredoxin](#). This ubiquitous protein serves a similar redox function in photosynthesis (see [Fig. 20-37](#)) and other processes. Thioredoxin has pairs of —SH groups that carry hydrogen atoms from NADPH to the ribonucleoside diphosphate.  Its oxidized (disulfide) form is reduced by NADPH in a reaction catalyzed by **thioredoxin**

reductase ([Fig. 22-41](#)), and reduced thioredoxin is then used by ribonucleotide reductase to reduce the nucleoside diphosphates (NDPs) to deoxyribonucleoside diphosphates (dNDPs). A second source of reducing equivalents for ribonucleotide reductase is glutathione (GSH). Glutathione serves as the reductant for a protein closely related to thioredoxin, **glutaredoxin**, which then transfers the reducing power to ribonucleotide reductase.

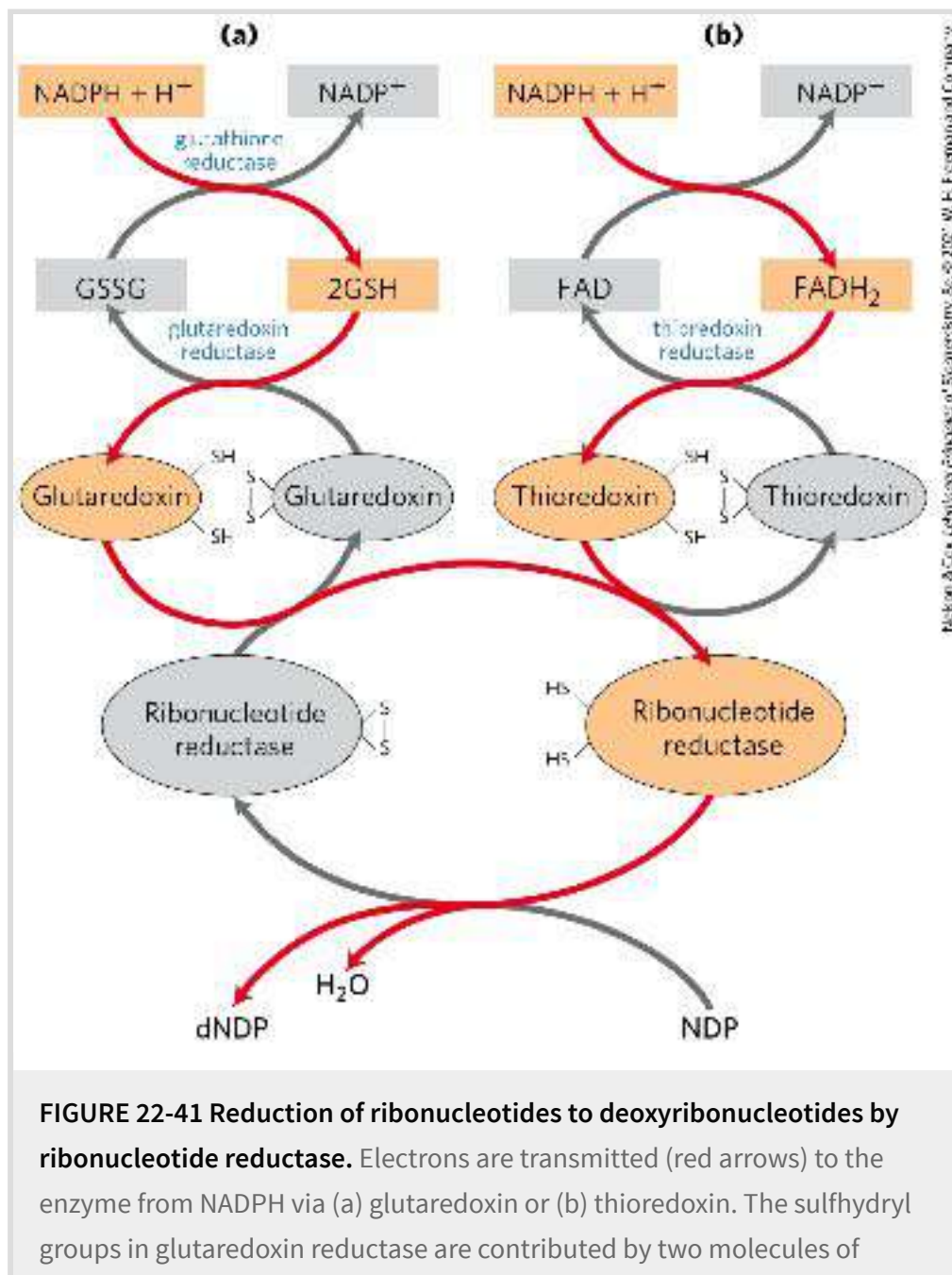


FIGURE 22-41 Reduction of ribonucleotides to deoxyribonucleotides by ribonucleotide reductase. Electrons are transmitted (red arrows) to the enzyme from NADPH via (a) glutaredoxin or (b) thioredoxin. The sulfhydryl groups in glutaredoxin reductase are contributed by two molecules of

bound glutathione (GSH; GSSG indicates oxidized glutathione). Note that thioredoxin reductase is a flavoenzyme, with FAD as a prosthetic group.

Ribonucleotide reductase is notable in that its reaction mechanism provides the best-characterized example of the involvement of free radicals in biochemical transformations, once thought to be rare in biological systems. The enzyme in *E. coli* and most eukaryotes is an $\alpha_2\beta_2$ dimer, with two catalytic subunits, α_2 , and two radical-generation subunits, β_2 ([Fig. 22-42](#)). Each catalytic subunit contains two kinds of regulatory sites, as described below. The two active sites of the enzyme are formed at the interface between the catalytic (α_2) and radical-generation (β_2) subunits. At each active site, an α subunit contributes two sulfhydryl groups required for activity, and the β_2 subunits contribute a stable tyrosyl radical. The β_2 subunits also have a binuclear iron (Fe^{3+}) cofactor that helps generate and stabilize the Tyr^{122} radical. The tyrosyl radical is too far from the active site to interact directly with the site, but several aromatic residues form a long-range radical-transfer pathway to the active site ([Fig. 22-42c](#)). A likely mechanism for the ribonucleotide reductase reaction is illustrated in [Figure 22-43](#). In *E. coli*, the sources of the required reducing equivalents for this reaction are thioredoxin and glutaredoxin, as noted above.

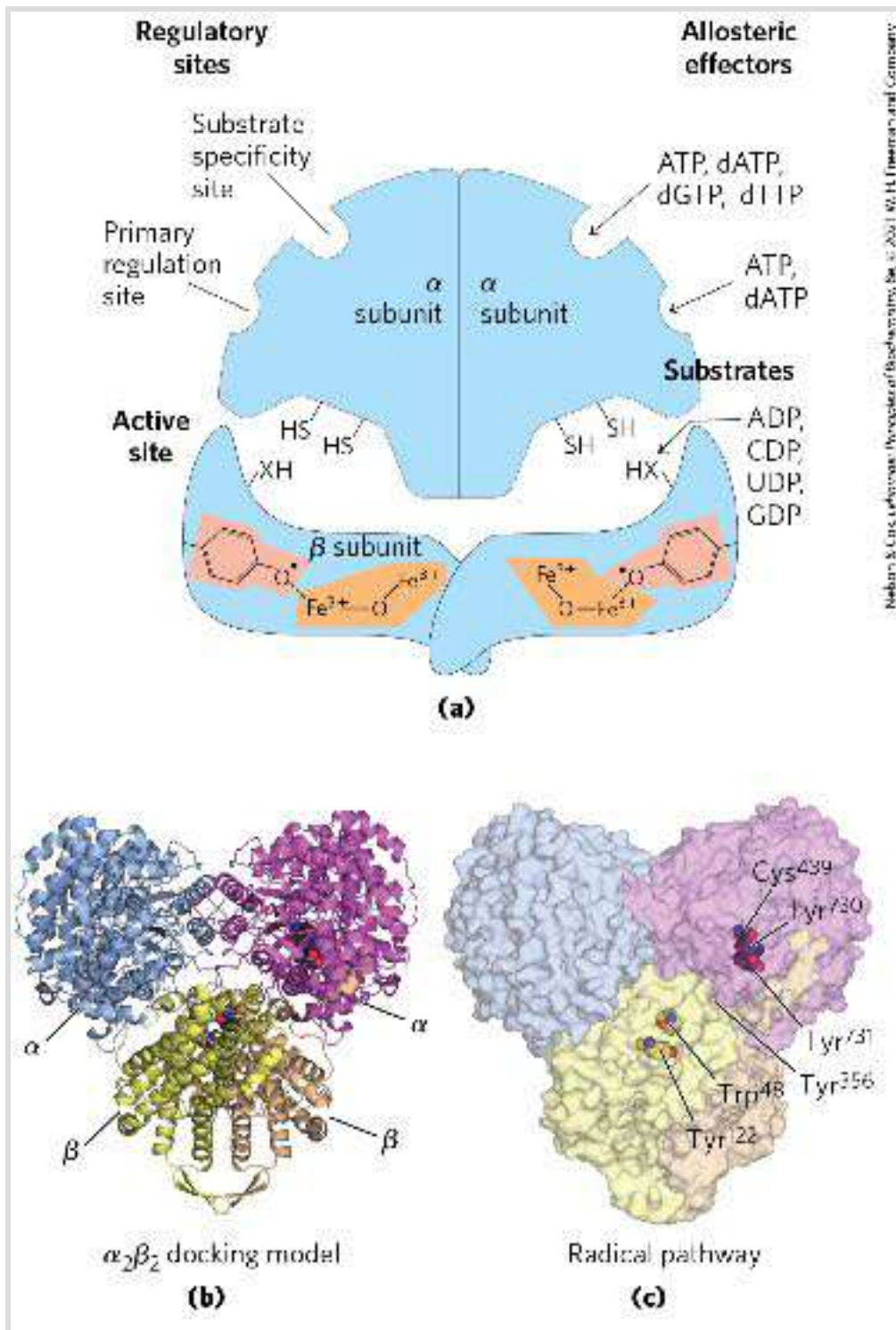
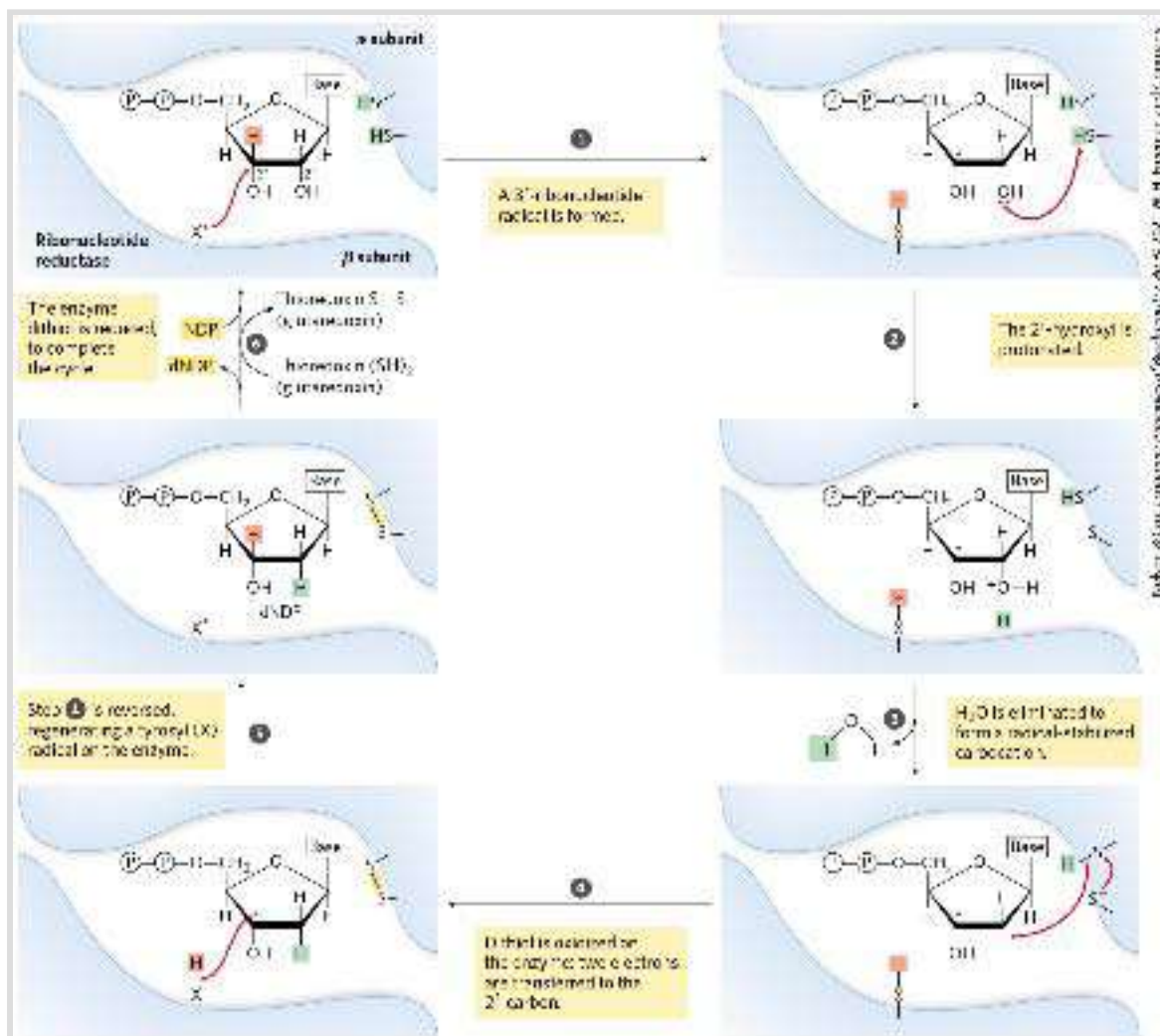


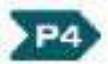
FIGURE 22-42 Ribonucleotide reductase. (a) A schematic diagram of the subunit structures. Each catalytic subunit (α ; also called R1) contains the two regulatory sites described in [Figure 22-44](#) and two Cys residues central to the reaction mechanism. The radical-generation subunits (β ; also called R2) each contain a critical Tyr¹²² residue and binuclear iron center. (b) The likely structure of $\alpha_2\beta_2$. (c) The likely path of radical formation from the initial Tyr¹²² in a β subunit to the active-site Cys⁴³⁹, which is used in the

mechanism shown in [Figure 22-43](#). Several aromatic amino acid residues participate in long-range transfer of the radical from the point of its formation at Tyr¹²² to the active site, where the nucleotide substrate is bound. [(a) Information from L. Thelander and P. Reichard, *Annu. Rev. Biochem.* 48:133, 1979. (b, c) Data from PDB ID 3UUS, N. Ando et al., *Proc. Natl. Acad. Sci. USA* 108:21,046, 2011.]



MECHANISM FIGURE 22-43 Proposed mechanism for ribonucleotide reductase. In the enzyme of *E. coli* and most eukaryotes, the active thiol groups are on the α subunit. The active-site radical ($\text{—X}^\bullet$) is on the β subunit and in *E. coli* is probably a thiyl radical of Cys⁴³⁹ (see [Fig. 22-42](#)).

Three classes of ribonucleotide reductase have been reported. Their mechanisms (where known) generally conform to the scheme in [Figure 22-43](#), but they differ in the identity of the group supplying the active-site radical and in the cofactors used to generate it. The *E. coli* enzyme (class I) requires oxygen to regenerate the tyrosyl radical if it is quenched, so this enzyme functions only in an aerobic environment. Class II enzymes, found in other microorganisms, have 5'-deoxyadenosylcobalamin (see [Box 17-2](#)) rather than a binuclear iron center. Class III enzymes have evolved to function in an anaerobic environment. *E. coli* contains a separate class III ribonucleotide reductase when grown anaerobically; this enzyme contains an iron-sulfur cluster (structurally distinct from the binuclear iron center of the class I enzyme) and requires NADPH and S-adenosylmethionine for activity. It uses nucleoside triphosphates rather than nucleoside diphosphates as substrates. The evolution of different classes of ribonucleotide reductase for production of DNA precursors in different environments reflects the importance of this reaction in nucleotide metabolism.

 Regulation of *E. coli* ribonucleotide reductase is unusual in that not only its *activity* but also its *substrate specificity* is regulated by the binding of effector molecules. Each α subunit has two types of regulatory sites ([Fig. 22-42](#)). One type affects overall enzyme activity and binds either ATP, which activates the enzyme, or dATP, which inactivates it. The second type alters substrate specificity in response to the effector molecule — ATP, dATP, dTTP, or dGTP — that is bound there ([Fig. 22-44](#)). When

ATP or dATP is bound, reduction of UDP and CDP is favored. When dTTP or dGTP is bound, reduction of GDP or ADP, respectively, is stimulated.  The scheme is designed to provide a balanced pool of precursors for DNA synthesis. ATP is also a general activator for biosynthesis and ribonucleotide reduction. The presence of dATP in small amounts increases the reduction of pyrimidine nucleotides. An oversupply of the pyrimidine dNTPs is signaled by high levels of dTTP. Abundant dTTP shifts the specificity to favor reduction of GDP. High levels of dGTP, in turn, shift the specificity to ADP reduction, and high levels of dATP shut the enzyme down. These effectors are thought to induce several distinct enzyme conformations with altered specificities.

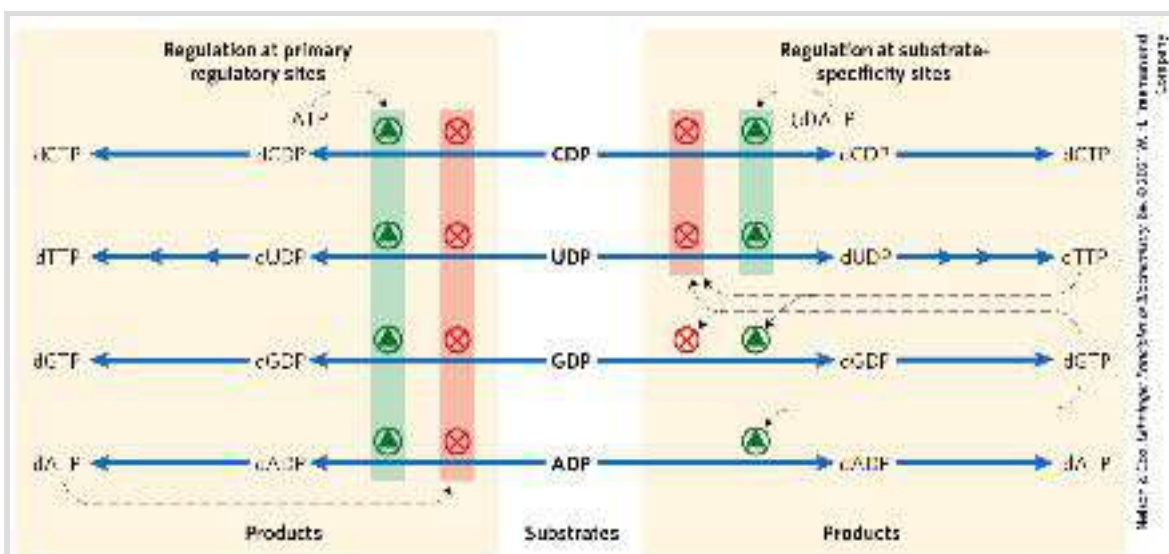


FIGURE 22-44 Regulation of ribonucleotide reductase by deoxynucleoside triphosphates.

The overall activity of the enzyme is affected by binding at the primary regulatory site (left). The substrate specificity of the enzyme is affected by the nature of the effector molecule bound at the second type of regulatory site, the substrate-specificity site (right). The diagram indicates inhibition or stimulation of enzyme activity with the four different substrates. The pathway from dUDP to dTMP is described below (see [Figs 22-46, 22-47](#)).

These regulatory effects are accompanied by, and presumably mediated by, large structural rearrangements in the enzyme. When the active form of the *E. coli* enzyme ($\alpha_2\beta_2$) is inhibited by the addition of the allosteric inhibitor dATP, a ringlike $\alpha_4\beta_4$ structure forms, with alternating α_2 and β_2 subunits ([Fig. 22-45](#)). In this altered structure, the radical-forming path from β to α is disrupted and the residues in the path are exposed to solvent, effectively preventing radical transfer and thus inhibiting the reaction. The formation of ringlike $\alpha_4\beta_4$ structures is reversed when dATP levels are reduced. The yeast ribonucleotide reductase also undergoes oligomerization in the presence of dATP, forming a hexameric ring structure, $\alpha_6\beta_6$.

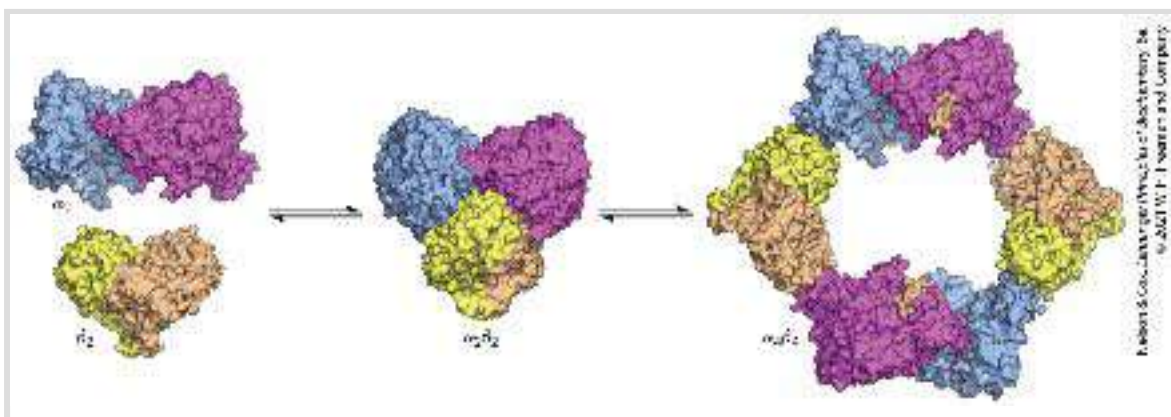
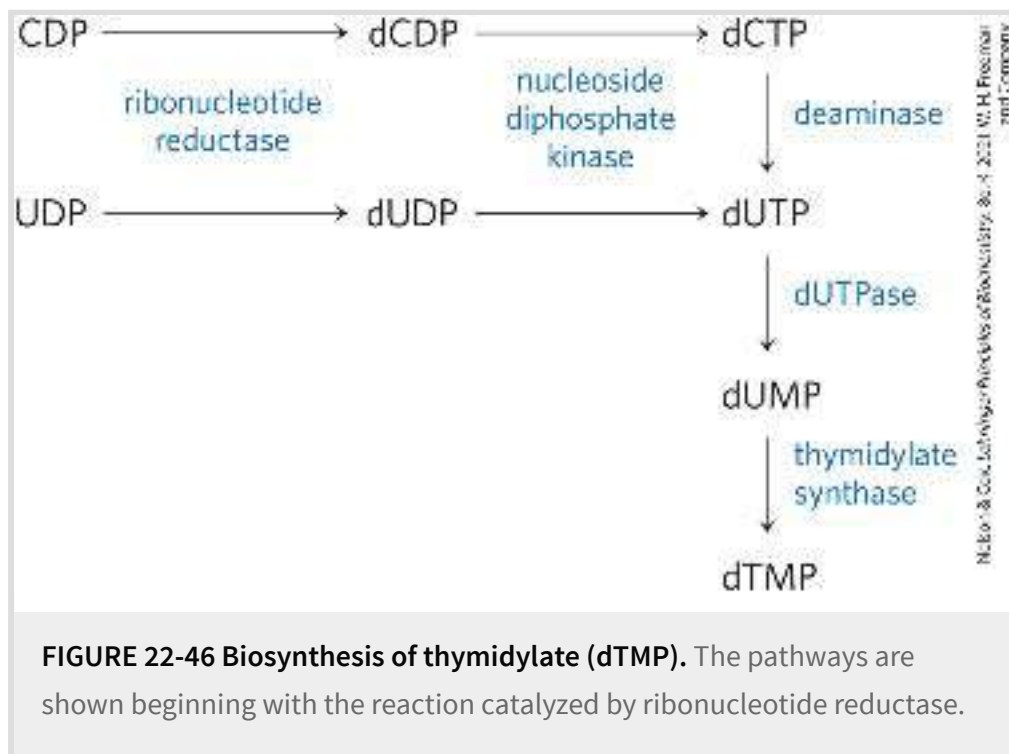


FIGURE 22-45 Oligomerization of ribonucleotide reductase induced by the allosteric inhibitor dATP. At high concentrations of dATP ($50 \mu\text{M}$), ring-shaped $\alpha_4\beta_4$ structures form. In this conformation, the residues in the radical-forming path are exposed to the solvent, blocking the radical reaction and inhibiting the enzyme. The oligomerization is reversed at lower dATP concentrations. [Data from PDB ID 3UUS, N. Ando et al., *Proc. Natl. Acad. Sci. USA* 108:21,046, 2011.]

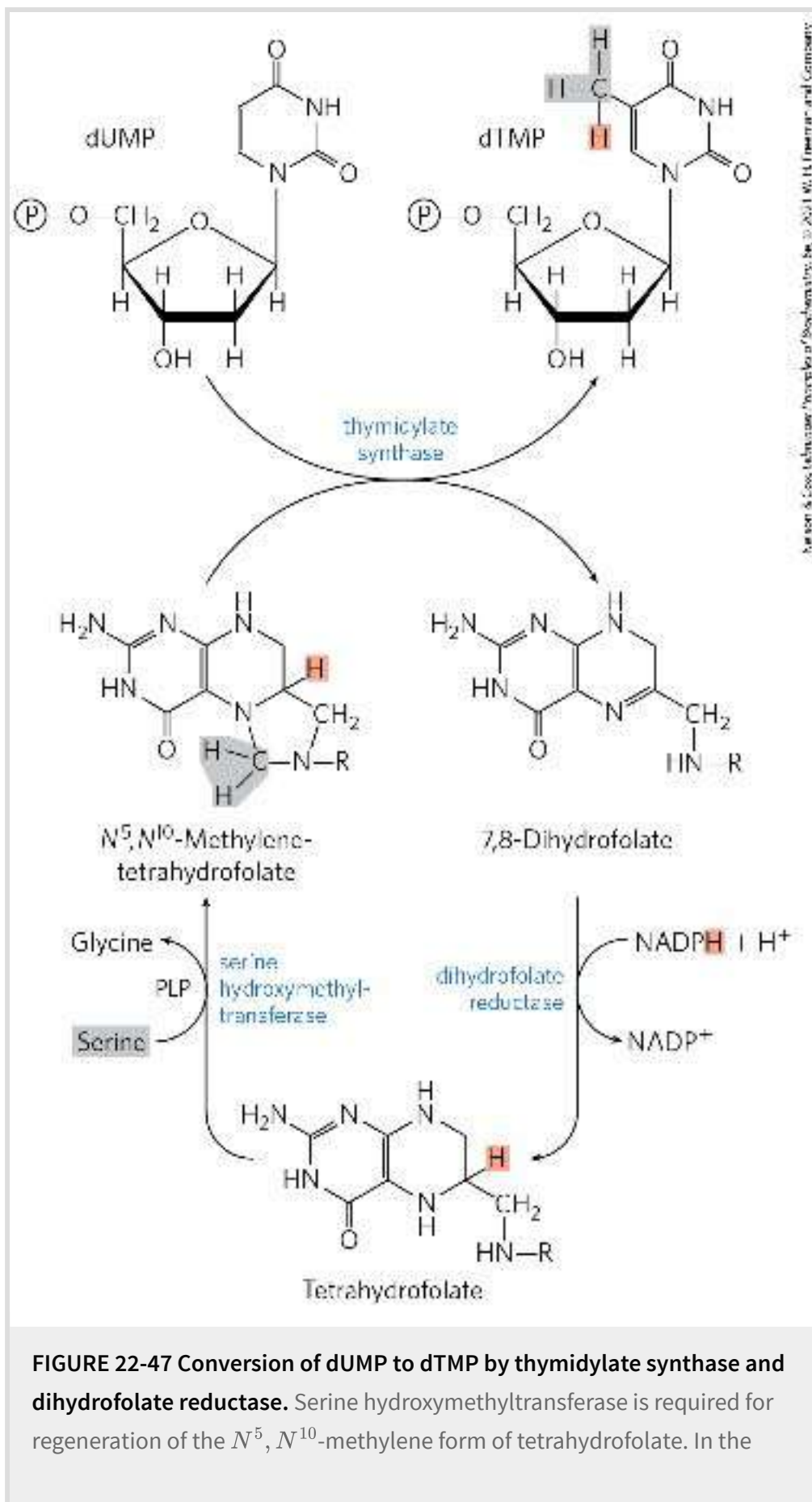
Thymidylate Is Derived from dCDP and dUMP

DNA contains thymine rather than uracil, and the de novo pathway to thymine involves only deoxyribonucleotides. The immediate precursor of thymidylate (dTMP) is dUMP. In bacteria, the pathway to dUMP begins with formation of dUTP, either by deamination of dCTP or by phosphorylation of dUDP (Fig. 22-46). The dUTP is converted to dUMP by a dUTPase. The latter reaction must be efficient to keep dUTP pools low and prevent incorporation of uridylate into DNA.



Conversion of dUMP to dTMP is catalyzed by **thymidylate synthase**. A one-carbon unit at the hydroxymethyl ($\text{—CH}_2\text{OH}$) oxidation level (see Fig. 18-17) is transferred from N^5, N^{10} -

methylenetetrahydrofolate to dUMP, then reduced to a methyl group ([Fig. 22-47](#)). The reduction occurs at the expense of oxidation of tetrahydrofolate to dihydrofolate, which is unusual in tetrahydrofolate-requiring reactions. (The mechanism of this reaction is shown in [Fig. 22-52](#).) The dihydrofolate is reduced to tetrahydrofolate by **dihydrofolate reductase** — a regeneration that is essential for the many processes that require tetrahydrofolate. In plants and at least one protist, thymidylate synthase and dihydrofolate reductase reside on a single, bifunctional protein.



synthesis of dTMP, all three hydrogens of the added methyl group are derived from N^5 , N^{10} -methylenetetrahydrofolate (light red and gray).



About 10% of the human population (and up to 50% of people in impoverished communities) suffers from folic acid deficiency. When the deficiency is severe, the symptoms can include heart disease, cancer, and some types of brain dysfunction. Folic acid deficiency during pregnancy can also produce neural tube defects in infants. At least some of these symptoms arise from a reduction in thymidylate synthesis, leading to an abnormal incorporation of uracil into DNA. Uracil is recognized by DNA repair pathways (described in [Chapter 25](#)) and is cleaved from the DNA. The presence of high levels of uracil in DNA leads to strand breaks that can greatly affect the function and regulation of nuclear DNA, ultimately causing the observed effects on the heart and brain, as well as increased mutagenesis that leads to cancer. ■

Degradation of Purines and Pyrimidines Produces Uric Acid and Urea, Respectively

Purine nucleotides are degraded by a pathway in which they lose their phosphate through the action of 5'-**nucleotidase** ([Fig. 22-48](#)). Adenylate yields adenosine, which is deaminated to inosine by **adenosine deaminase**, and inosine is hydrolyzed to hypoxanthine (its purine base) and D-ribose. Hypoxanthine is

oxidized successively to xanthine and then uric acid by **xanthine oxidase**, a flavoenzyme with an atom of molybdenum and four iron-sulfur centers in its prosthetic group. Molecular oxygen is the electron acceptor in this complex reaction.

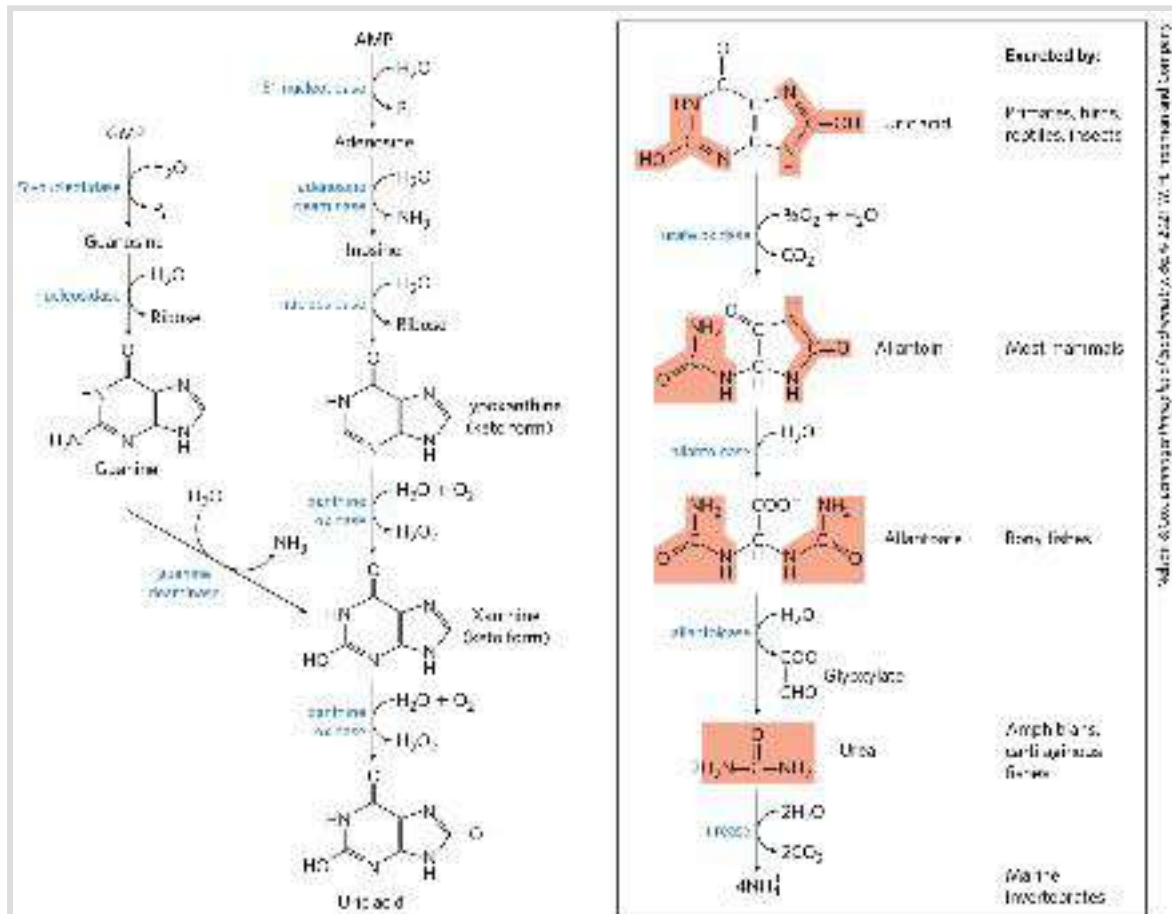


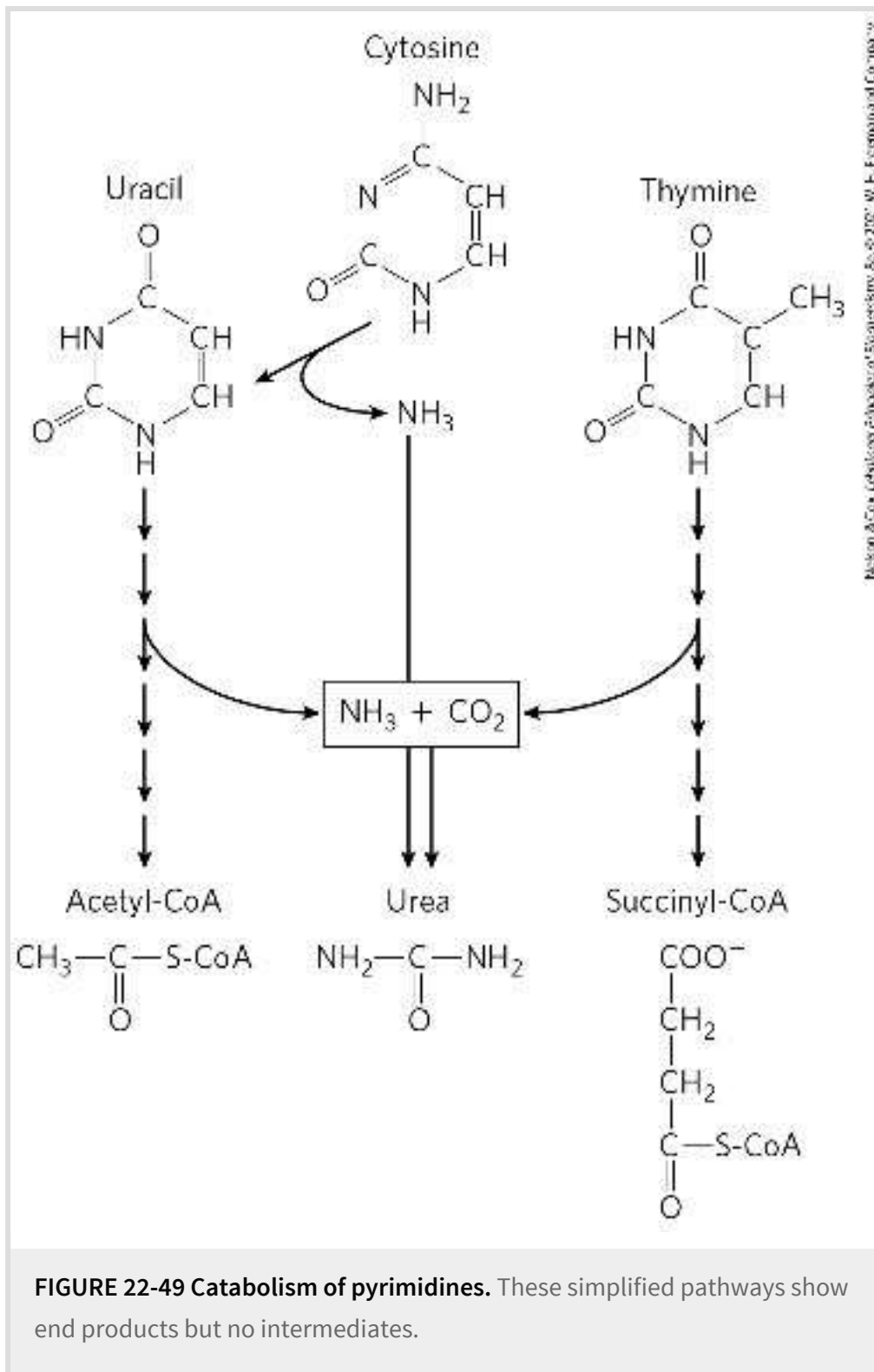
FIGURE 22-48 Catabolism of purine nucleotides. Note that primates excrete much more nitrogen as urea via the urea cycle ([Chapter 18](#)) than as uric acid from purine degradation. Similarly, fish excrete much more nitrogen as NH_4^+ than as urea produced by the pathway shown here.

GMP catabolism also yields uric acid as an end product. GMP is first hydrolyzed to guanosine, which is then cleaved to free guanine. Guanine undergoes hydrolytic removal of its amino

group to yield xanthine, which is converted to uric acid by xanthine oxidase.

Uric acid is the excreted end product of purine catabolism in primates, birds, and some other animals. A healthy adult human excretes uric acid at a rate of about 0.6 g/24 h; the excreted product arises in part from ingested purines and in part from turnover of the purine nucleotides of nucleic acids. In most mammals and many other vertebrates, uric acid is degraded to **allantoin** by the action of **urate oxidase**. In other organisms the pathway is further extended, as shown in [Figure 22-48](#).

The pathways for degradation of pyrimidines generally lead to NH_4^+ production and thus to urea synthesis. The carbons of thymine are degraded to succinyl-CoA; those of cytosine and uracil are degraded to acetyl-CoA ([Fig. 22-49](#)).



Genetic aberrations in human purine metabolism have been found, some with serious consequences. For example, **adenosine deaminase (ADA) deficiency** leads to severe immunodeficiency

disease in which T lymphocytes and B lymphocytes do not develop properly. Lack of ADA leads to a 100-fold increase in the cellular concentration of dATP, a strong inhibitor of ribonucleotide reductase ([Fig. 22-44](#)). High levels of dATP produce a general deficiency of other dNTPs in T lymphocytes. The basis for B-lymphocyte toxicity is less clear. Individuals with ADA deficiency lack an effective immune system and do not survive unless treated. Current therapies include bone marrow transplants from a matched donor to replace the hematopoietic stem cells that mature into B and T lymphocytes. However, transplant recipients often suffer a variety of cognitive and physiological problems. Enzyme replacement therapy, requiring once- or twice-weekly intramuscular injection of active ADA, is effective, but the therapeutic benefit often declines after 8 to 10 years; complications may then arise, including malignancies. For many people, a permanent cure requires replacing the defective gene with a functional one in bone marrow cells. ADA deficiency was one of the first targets of human gene therapy trials (in 1990). Mixed results in early trials have given way to significant successes, and gene therapy is rapidly becoming a viable path for long-term restoration of immune function for these patients. Newer approaches based on CRISPR-mediated gene editing (see [Fig. 9-21](#)) may eventually be even more effective. ■

Purine and Pyrimidine Bases Are Recycled by Salvage Pathways

Free purine and pyrimidine bases are constantly released in cells during the metabolic degradation of nucleotides. Free purines are in large part salvaged and reused to make nucleotides, in a pathway much simpler than the de novo synthesis of purine nucleotides described earlier. One of the primary salvage pathways consists of a single reaction catalyzed by **adenosine phosphoribosyltransferase**, in which free adenine reacts with PRPP to yield the corresponding adenine nucleotide:



Free guanine and hypoxanthine (the deamination product of adenine; [Fig. 22-48](#)) are salvaged in the same way by **hypoxanthine-guanine phosphoribosyltransferase**. A similar salvage pathway exists for pyrimidine bases in microorganisms, and possibly in mammals.



A genetic lack of hypoxanthine-guanine phosphoribosyltransferase activity, seen almost exclusively in young boys, results in a set of symptoms called **Lesch-Nyhan syndrome**. Children with this genetic disorder, which becomes manifest by the age of 2 years, are sometimes poorly coordinated and have intellectual deficits. In addition, they are extremely hostile and show compulsive self-destructive tendencies: they mutilate themselves by biting off their fingers, toes, and lips.

The devastating effects of Lesch-Nyhan syndrome illustrate the importance of the salvage pathways. Hypoxanthine and guanine arise constantly from the breakdown of nucleic acids. In the absence of hypoxanthine-guanine phosphoribosyltransferase, PRPP levels rise and purines are overproduced by the de novo pathway, resulting in high levels of uric acid production and goutlike damage to tissue (see below). The brain is especially dependent on the salvage pathways, and this may account for the central nervous system damage in children with Lesch-Nyhan syndrome. This syndrome is another potential target for gene therapy. ■

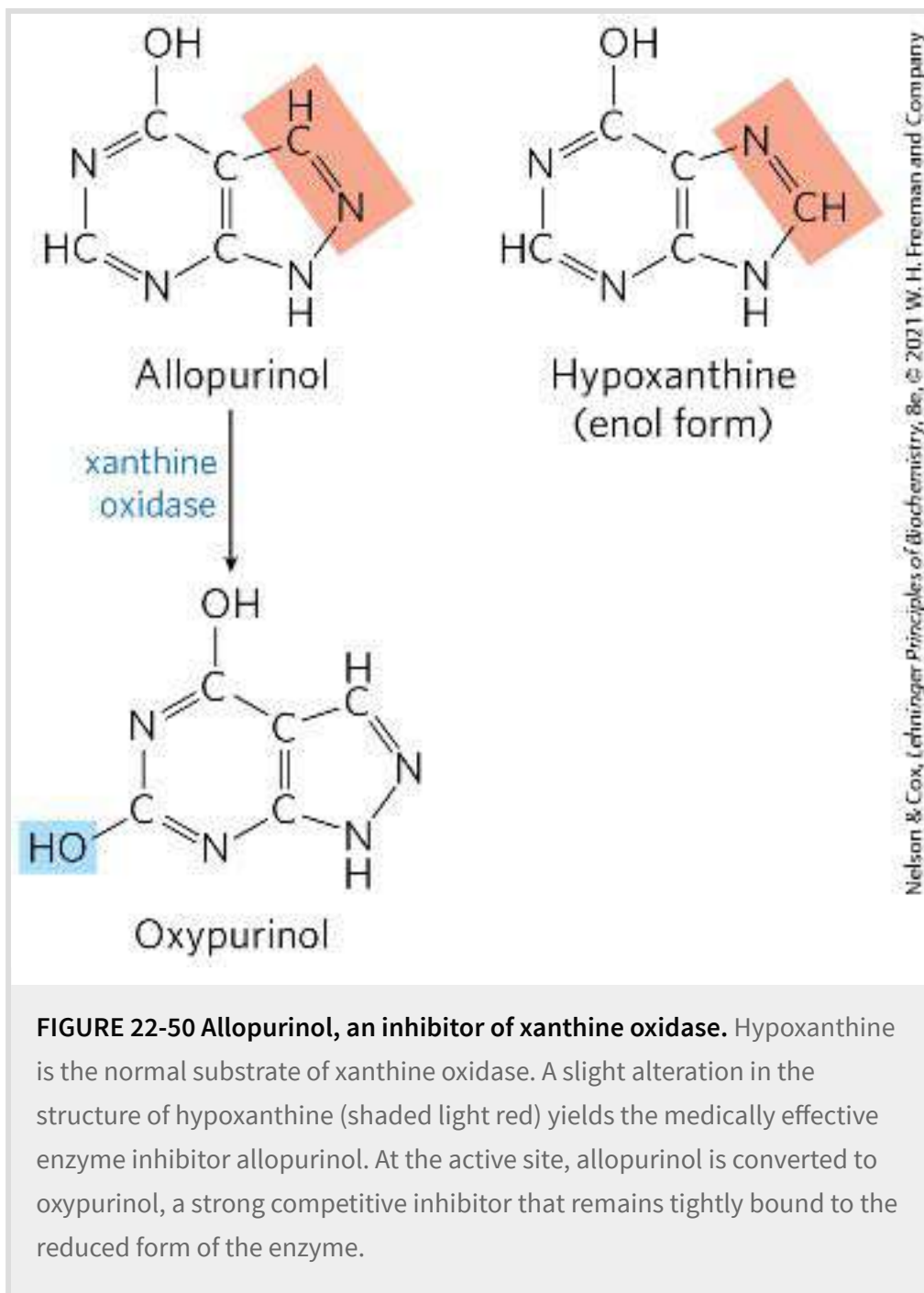
Excess Uric Acid Causes Gout

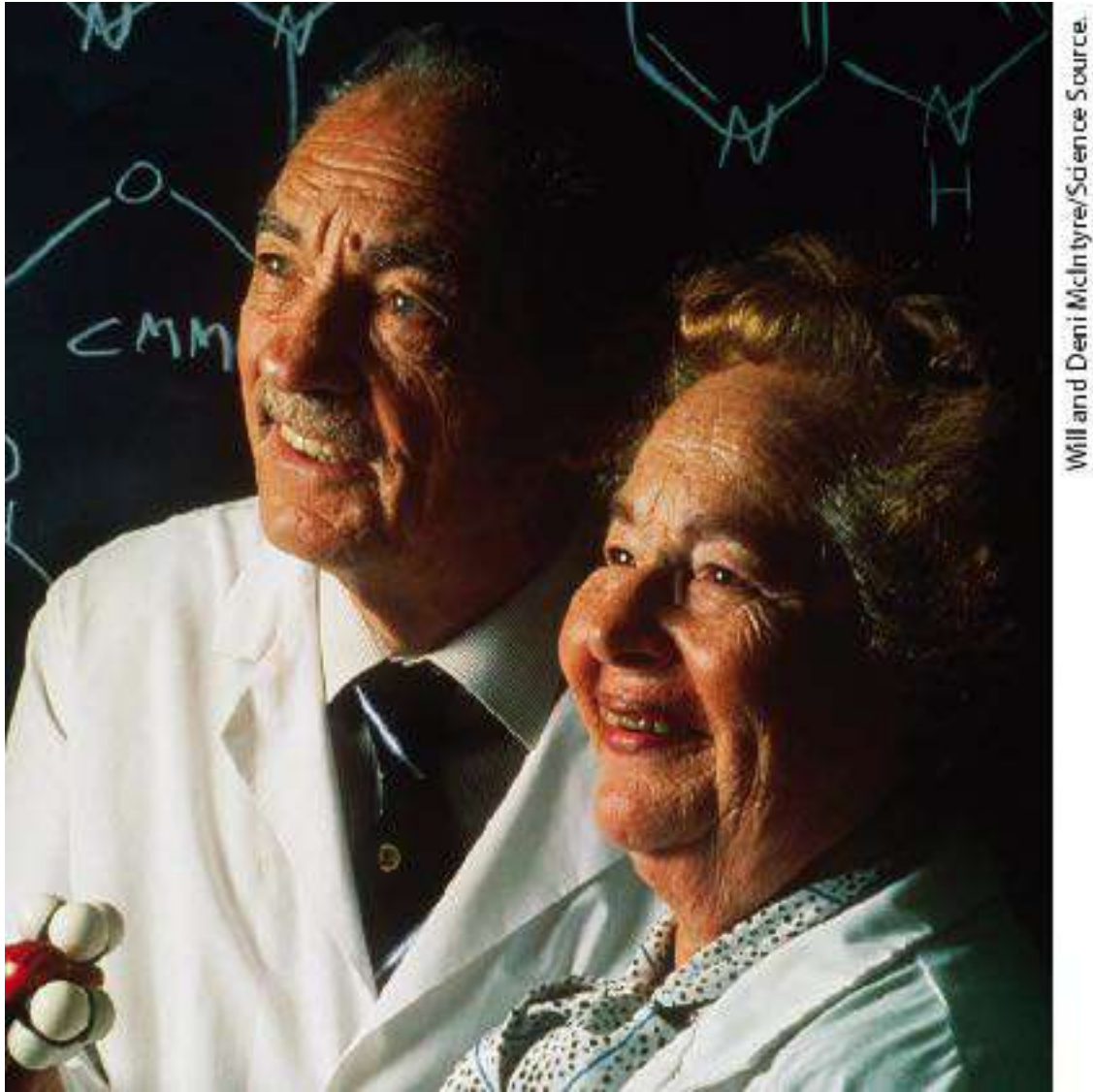


Long thought (erroneously) to be due to “high living,” gout is a disease of the joints caused by an elevated concentration of uric acid in the blood and tissues. The joints become inflamed, painful, and arthritic, owing to the abnormal deposition of sodium urate crystals. The kidneys are also affected, as excess uric acid is deposited in the kidney tubules. Gout occurs predominantly in males. Its precise cause is not known, but it often involves an underexcretion of urate. A genetic deficiency of one or another enzyme of purine metabolism may also be a factor in some cases.

Gout is effectively treated by a combination of nutritional and drug therapies. Patients exclude foods especially rich in nucleotides and nucleic acids, such as liver or glandular products,

from the diet. Major alleviation of the symptoms is provided by the drug **allopurinol** ([Fig. 22-50](#)), which inhibits xanthine oxidase, the enzyme that catalyzes the conversion of purines to uric acid. Allopurinol is a substrate of xanthine oxidase, which converts allopurinol to oxypurinol (alloxanthine). Oxypurinol inactivates the reduced form of the enzyme by remaining tightly bound in its active site. When xanthine oxidase is inhibited, the excreted products of purine metabolism are xanthine and hypoxanthine, which are more water-soluble than uric acid and less likely to form crystalline deposits. Allopurinol was developed by Gertrude Elion and George Hitchings, who also developed acyclovir, used in treating people with genital and oral herpes infections, and other purine analogs used in cancer chemotherapy. ■





Will and Deni McIntyre/Science Source

George Hitchings, 1905–1998, and Gertrude Elion, 1918–1999.

**Many Chemotherapeutic Agents
Target Enzymes in Nucleotide
Biosynthetic Pathways**



The growth of cancer cells is not controlled in the same way as cell growth in most normal tissues. Cancer cells have greater requirements for nucleotides as precursors of DNA and RNA, and consequently are generally more sensitive than normal cells to inhibitors of nucleotide biosynthesis. A number of important chemotherapeutic agents — for cancer and other diseases — act by inhibiting one or more enzymes in these pathways. Several well-studied examples can illustrate productive approaches to treatment and help us understand how these enzymes work.

Important targets for pharmaceutical agents include thymidylate synthase and dihydrofolate reductase, enzymes that provide the only cellular pathway for thymine synthesis ([Fig. 22-51](#)). One inhibitor that acts on thymidylate synthase, **fluorouracil**, is an important chemotherapeutic agent. Fluorouracil itself is not the enzyme inhibitor. In the cell, salvage pathways convert it to the deoxynucleoside monophosphate FdUMP, which binds to and inactivates the enzyme. Inhibition by FdUMP ([Fig. 22-52](#)) is a classic example of mechanism-based enzyme inactivation.

Another prominent chemotherapeutic agent, **methotrexate**, is an inhibitor of dihydrofolate reductase. This folate analog acts as a competitive inhibitor; the enzyme binds methotrexate with about 100 times higher affinity than dihydrofolate. **Aminopterin** is a related compound that acts similarly.

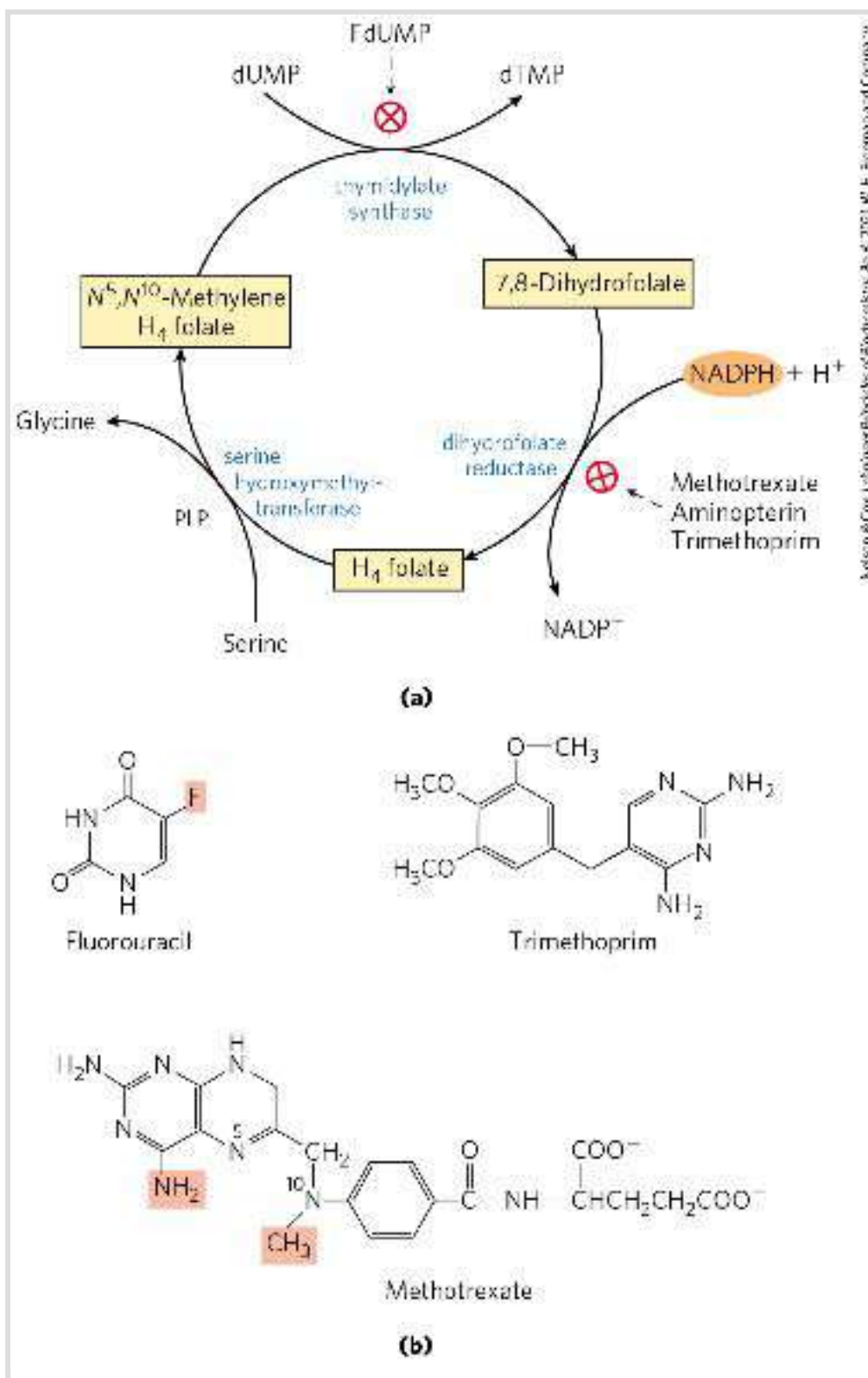
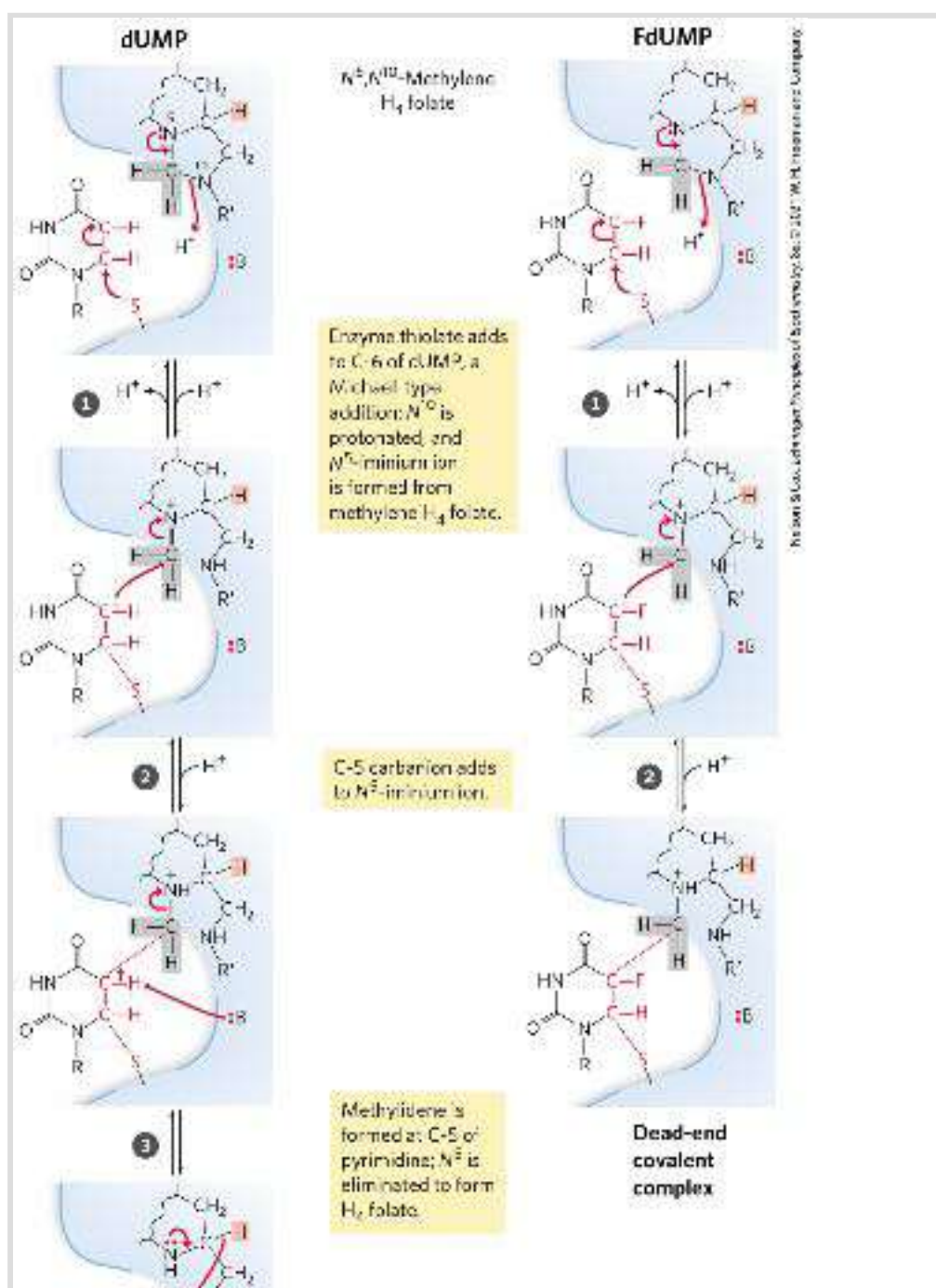
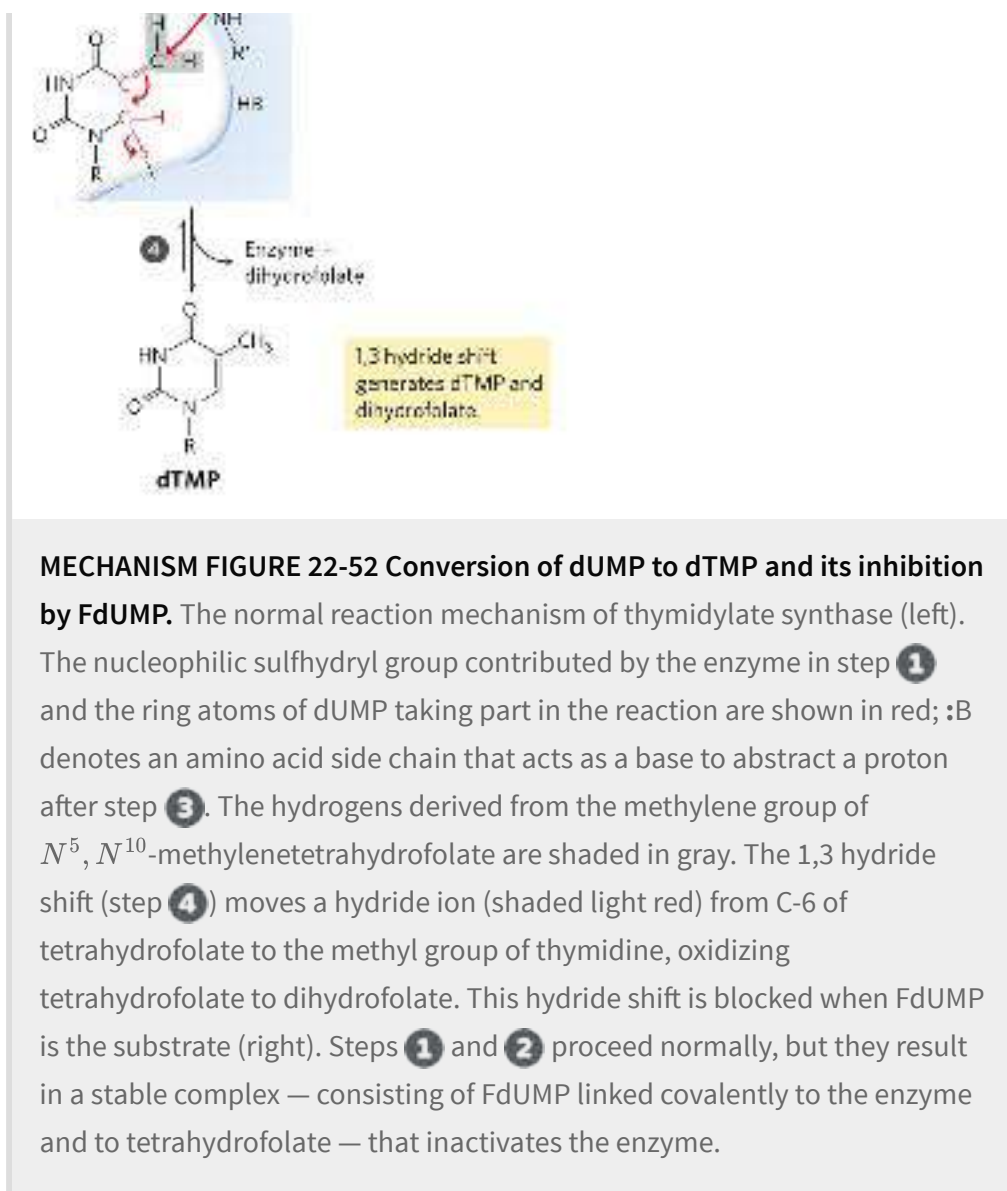


FIGURE 22-51 Thymidylate synthesis and folate metabolism as targets of chemotherapy. (a) During thymidylate synthesis, N^5, N^{10} -methylenetetrahydrofolate is converted to 7,8-dihydrofolate; the N^5, N^{10} -methylenetetrahydrofolate is regenerated in two steps (see [Fig. 22-47](#)). This cycle is a major target of several chemotherapeutic agents. (b) Fluorouracil

and methotrexate are important chemotherapeutic agents. In cells, fluorouracil is converted to FdUMP, which inhibits thymidylate synthase. Methotrexate, a structural analog of tetrahydrofolate, inhibits dihydrofolate reductase; the shaded amino and methyl groups replace a carbonyl oxygen and a proton, respectively, in folate. Another important folate analog, aminopterin, is identical to methotrexate except that it lacks the shaded methyl group. Trimethoprim, a tight-binding inhibitor of bacterial dihydrofolate reductase, was developed as an antibiotic.





The medical potential of inhibitors of nucleotide biosynthesis is not limited to cancer treatment. All fast-growing cells (including bacteria and protists) are potential targets. **Trimethoprim**, an antibiotic developed by Hitchings and Elion, binds to bacterial dihydrofolate reductase nearly 100,000 times better than to the mammalian enzyme. It is used to treat certain urinary and middle-ear bacterial infections. Parasitic protists, such as the trypanosomes that cause African sleeping sickness (African trypanosomiasis), lack pathways for de novo nucleotide

biosynthesis and are particularly sensitive to agents that interfere with their ability to use salvage pathways to scavenge nucleotides from the surrounding environment. Allopurinol ([Fig. 22-50](#)) and several similar purine analogs have shown promise for the treatment of African trypanosomiasis and related afflictions. See [Box 6-1](#) for another approach to combating African trypanosomiasis, made possible by advances in our understanding of metabolism and enzyme mechanisms. ■

SUMMARY 22.4 *Biosynthesis and Degradation of Nucleotides*

- The purine ring system is built up step by step, beginning with 5-phosphoribosylamine. The amino acids glutamine, glycine, and aspartate furnish all the nitrogen atoms of purines. Two ring-closure steps form the purine nucleus.
- Purine biosynthesis is regulated by an elaborate system of feedback inhibition.
- Pyrimidines are synthesized from carbamoyl phosphate and aspartate, and ribose 5-phosphate is then attached to yield the pyrimidine ribonucleotides.
- Pyrimidine biosynthesis is regulated by feedback inhibition of aspartate transcarbamoylase.
- Nucleoside monophosphates are converted to their triphosphates by enzymatic phosphorylation reactions.
- Ribonucleotides are converted to deoxyribonucleotides by ribonucleotide reductase, an enzyme with novel mechanistic and regulatory characteristics.
- The thymine nucleotides are derived from dCDP and dUMP.

- Uric acid and urea are the end products of purine and pyrimidine degradation.
- Free purines can be salvaged and rebuilt into nucleotides. Genetic deficiencies in certain salvage enzymes cause serious disorders such as Lesch-Nyhan syndrome.
- Accumulation of uric acid crystals in the joints, possibly caused by another genetic deficiency, results in gout.
- Enzymes of the nucleotide biosynthetic pathways are targets for an array of chemotherapeutic agents used to treat cancer and other diseases.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

nitrogen fixation

nitrification

denitrification

anammox

symbionts

nitrogenase complex

FeMo cofactor

ferredoxin

leghemoglobin

glutamate

glutamine

glutamine synthetase

glutamate synthase

glutamine amidotransferases

5-phosphoribosyl-1-pyrophosphate (PRPP)

porphyrin

porphyria

bilirubin

phosphocreatine

creatine

glutathione (GSH)

auxin

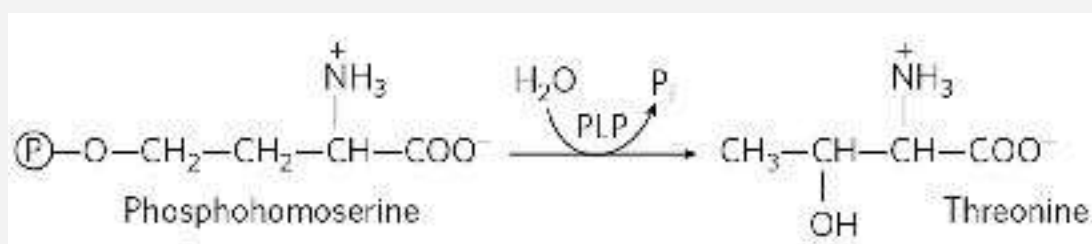
dopamine
norepinephrine
epinephrine
 γ -aminobutyrate (GABA)
serotonin
histamine
spermine
spermidine
ornithine decarboxylase
de novo pathway
salvage pathway
inosinate (IMP)
carbamoyl phosphate synthetase II
aspartate transcarbamoylase
nucleoside monophosphate kinase
nucleoside diphosphate kinase
ribonucleotide reductase
thioredoxin
thymidylate synthase
dihydrofolate reductase
adenosine deaminase (ADA) deficiency
Lesch-Nyhan syndrome
allopurinol
fluorouracil
methotrexate
aminopterin
trimethoprim

PROBLEMS

1. ATP Consumption by Root Nodules in Legumes Bacteria residing in the root nodules of the pea plant consume more than 20% of the ATP produced by the plant. Suggest why these bacteria consume so much ATP.

2. Nitrate Fertilizers and Oceanic Dead Zones Farmers apply industrially fixed nitrogen, in the form of ammonia or nitrate, to agricultural fields worldwide to increase crop yields. Agricultural runoff feeds into rivers and creates large hypoxic dead zones at the point where rivers meet oceans. How does an increase in soluble fixed nitrogen create dead zones?

3. PLP Reaction Mechanisms Pyridoxal phosphate (PLP) can help catalyze transformations one or two carbons removed from the α carbon of an amino acid. The enzyme threonine synthase promotes the PLP-dependent conversion of phosphohomoserine to threonine. Suggest a mechanism for this reaction.



4. Transformation of Aspartate to Asparagine There are two routes for transforming aspartate to asparagine at the expense of ATP. Many bacteria have an asparagine synthetase that uses ammonium ion as the nitrogen donor. Mammals

have an asparagine synthetase that uses glutamine as the nitrogen donor. Given that the latter requires an extra ATP (for the synthesis of glutamine), why do mammals use this route?

5. Equation for the Synthesis of Aspartate from Glucose

Write the net equation for the synthesis of aspartate (a nonessential amino acid) from glucose, carbon dioxide, and ammonia.

6. Asparagine Synthetase Inhibitors in Leukemia

Therapy Mammalian asparagine synthetase is a glutamine-dependent amidotransferase. Efforts to identify an effective inhibitor of human asparagine synthetase for use in chemotherapy for patients with leukemia have focused not on the amino-terminal glutaminase domain but on the carboxyl-terminal synthetase active site. Explain why the glutaminase domain is not a promising target for a useful drug.

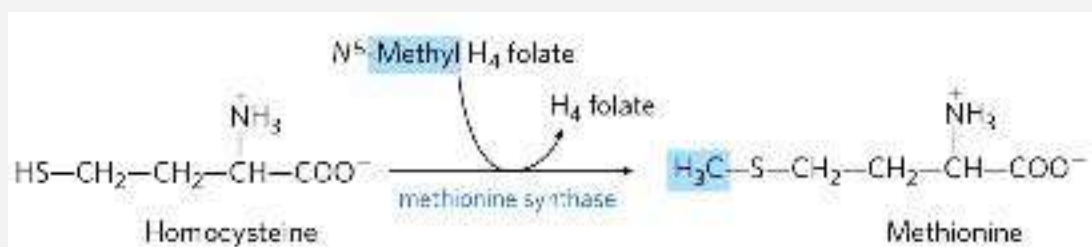
7. Phenylalanine Hydroxylase Deficiency and Diet

Tyrosine is normally a nonessential amino acid, but individuals with a genetic defect in phenylalanine hydroxylase require tyrosine in their diet for normal growth. Explain.

8. Arginine Biosynthesis The first step of arginine biosynthesis from glutamate acetylates glutamate on the α -amino group. A subsequent step late in the same pathway removes the added acetyl group. What chemical problem is

solved by adding and then removing an acetyl group, with none of the acetyl atoms appearing in the arginine product of the pathway?

9. Cofactors for One-Carbon Transfer Reactions Most one-carbon transfers are promoted by one of three cofactors: biotin, tetrahydrofolate, or *S*-adenosylmethionine. *S*-Adenosylmethionine generally serves as a methyl group donor; the transfer potential of the methyl group in N^5 -methyltetrahydrofolate is insufficient for most biosynthetic reactions. However, one example of the use of N^5 -methyltetrahydrofolate in methyl group transfer is in methionine formation by the methionine synthase reaction; methionine is the immediate precursor of *S*-adenosylmethionine (see [Fig. 18-18](#)).



Explain how the methyl group of *S*-adenosylmethionine can be derived from N^5 -methyltetrahydrofolate, even though the transfer potential of the methyl group in N^5 -methyltetrahydrofolate is one-thousandth of that of *S*-adenosylmethionine.

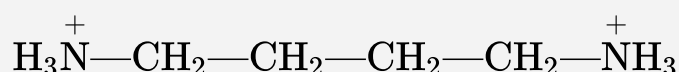
10. Concerted Regulation in Amino Acid Biosynthesis

Various products of glutamine metabolism independently

modulate the glutamine synthetase of *E. coli* (see [Fig. 22-8](#)). In this concerted inhibition, the extent of enzyme inhibition is greater than the sum of the separate inhibitions caused by each product. For *E. coli* grown in a medium rich in histidine, what is the advantage of concerted inhibition?

11.  Relationship between Folic Acid Deficiency and Anemia Folic acid deficiency, believed to be the most common vitamin deficiency, causes a type of anemia in which hemoglobin synthesis is impaired and erythrocytes do not mature properly. What is the metabolic relationship between hemoglobin synthesis and folic acid deficiency?

12. Synthesis of Polyamines The metabolic amino acid ornithine is a direct precursor of the polyamine putrescine, shown here.



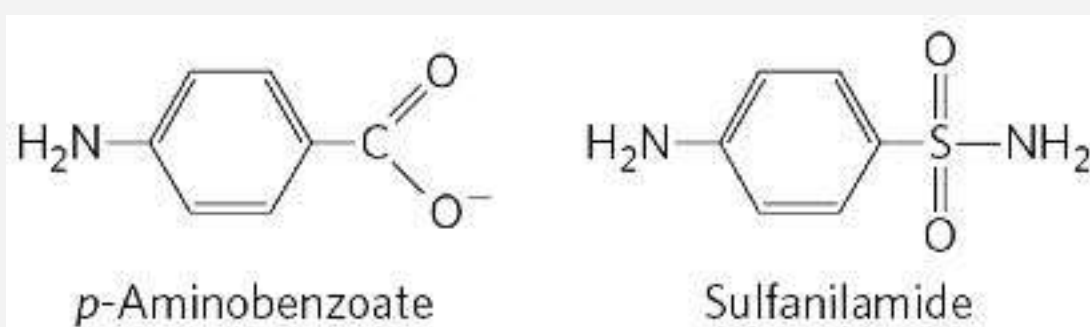
Subsequent reactions convert putrescine to spermine and spermidine. What type of reaction is required to convert ornithine to putrescine, and what enzymatic cofactor is needed?

13. Nucleotide Biosynthesis in Amino Acid Auxotrophic Bacteria Wild-type *E. coli* cells can synthesize all 20 common amino acids, but some mutants, called amino acid auxotrophs, are unable to synthesize a specific amino acid

and require its addition to the culture medium for optimal growth. Besides their role in protein synthesis, some amino acids are also precursors for other nitrogenous cell products. Consider the three amino acid auxotrophs that are unable to synthesize glycine, glutamine, and aspartate, respectively. For each mutant, what nitrogenous products other than proteins would the cell fail to synthesize?

14. Inhibitors of Nucleotide Biosynthesis Suggest a mechanism for the inhibition of alanine racemase by L-fluoroalanine.

15.  Mode of Action of Sulfa Drugs Some bacteria require *p*-aminobenzoate in the culture medium for normal growth, and their growth is severely inhibited by the addition of sulfanilamide, one of the earliest sulfa drugs. Moreover, in the presence of this drug, 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR; see [Fig. 22-35](#)) accumulates in the culture medium. Addition of excess *p*-aminobenzoate reverses these effects.



- a. What is the role of *p*-aminobenzoate in these bacteria?
(Hint: See [Fig. 18-16](#).)

- b. Why does AICAR accumulate in the presence of sulfanilamide?
- c. Why does addition of excess *p*-aminobenzoate reverse the inhibition and accumulation?

16. Purine Biosynthesis Which atoms of the purine ring derive from the amide nitrogen of glutamine?

- a. N-1
- b. N-3
- c. N-7
- d. N-9

17. Pathway of Carbon in Pyrimidine Biosynthesis Predict the locations of ^{14}C in orotate isolated from cells grown on a small amount of uniformly labeled $[^{14}\text{C}]$ succinate. Justify your prediction.

18. Nucleotides as Poor Sources of Energy Under starvation conditions, organisms can use proteins and amino acids as sources of energy. Deamination of amino acids produces carbon skeletons that can enter the glycolytic pathway and the citric acid cycle to produce energy in the form of ATP. Nucleotides are not similarly degraded for use as energy-yielding fuels. What observations about cellular physiology support this statement? What aspect of the structure of nucleotides makes them a relatively poor source of energy?

19.  Treatment of Gout Physicians use allopurinol (see [Fig. 22-50](#)), an inhibitor of xanthine oxidase, to treat chronic gout. Explain the biochemical basis for this treatment.

Patients treated with allopurinol sometimes develop xanthine stones in the kidneys, although the incidence of kidney damage is much lower than in untreated gout. Explain this observation in light of these solubilities in urine: uric acid, 0.15 g/L; xanthine, 0.05 g/L; and hypoxanthine, 1.4 g/L.

20. Antibiotics That Inhibit Dihydrofolate Reductase

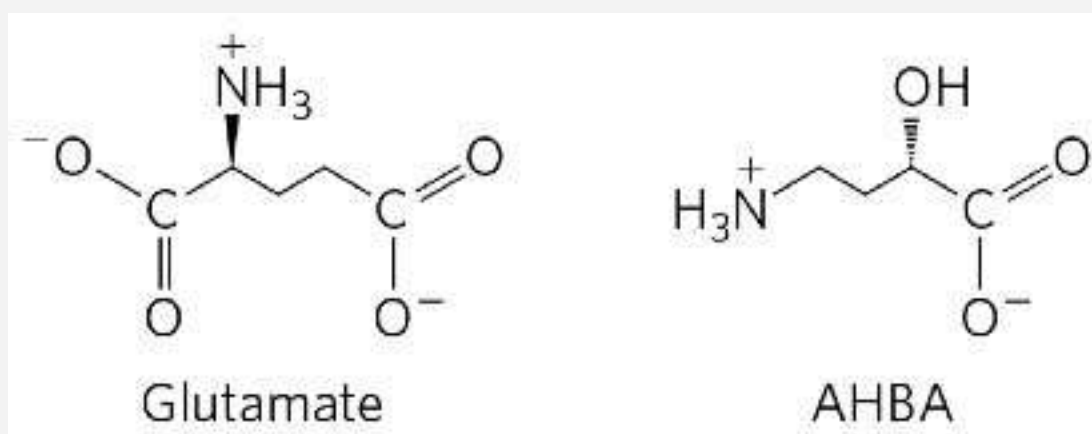
Trimethoprim, a commonly used antibiotic, inhibits the bacterial form of dihydrofolate reductase much more than it inhibits the mammalian enzyme. What metabolic processes described in this chapter are affected by depleting tetrahydrofolate?

DATA ANALYSIS PROBLEM

21. Use of Modern Molecular Techniques to Determine the Synthetic Pathway of a Novel Amino Acid Most of the biosynthetic pathways described in this chapter were determined before the development of recombinant DNA technology and genomics, so the techniques were quite different from those that researchers would use today. Here we explore an example of the use of modern molecular techniques to investigate the pathway of synthesis of a novel amino acid, (2S)-4-amino-2-hydroxybutyrate (AHBA). The techniques mentioned here are described in various places in the text; this problem is designed to show how they can be integrated in a comprehensive study.

AHBA is a γ -amino acid that is a component of some aminoglycoside antibiotics, including the antibiotic butirosin. Antibiotics modified by the addition of an AHBA residue are often more resistant to inactivation by bacterial antibiotic-resistance enzymes. As a result, understanding how AHBA is synthesized and added to antibiotics is useful in the design of pharmaceuticals.

In an article published in 2005, Li and coworkers describe how they determined the synthetic pathway of AHBA from glutamate.



- Briefly describe the chemical transformations needed to convert glutamate to AHBA. At this point, don't be concerned about the *order* of the reactions.

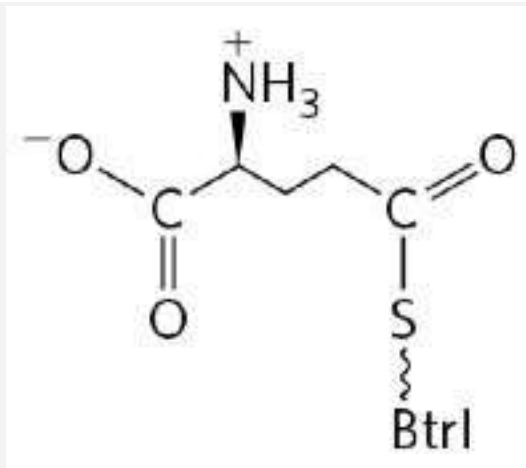
Li and colleagues began by cloning the butirosin biosynthetic gene cluster from the bacterium *Bacillus circulans*, which makes large quantities of butirosin. They identified five genes that are essential for the pathway: *btrI*, *btrJ*, *btrK*, *btrO*, and *btrV*. They cloned these genes into *E. coli* plasmids that allow

overexpression of the genes, producing proteins with “histidine tags” fused to their amino termini to facilitate purification (see [p. 313](#)).

The predicted amino acid sequence of the BtrI protein showed strong homology to known acyl carrier proteins (see [Fig. 21-5](#)). Using mass spectrometry, Li and colleagues found a molecular mass of 11,812 for the purified BtrI protein (including the His tag). When the purified BtrI was incubated with coenzyme A and an enzyme known to attach CoA to other acyl carrier proteins, the majority molecular species had an M_r of 12,153.

- b. How would you use these data to argue that BtrI can function as an acyl carrier protein with a CoA prosthetic group?

Using standard terminology, Li and coauthors called the form of the protein lacking CoA apo-BtrI and the form with CoA (linked as in [Fig. 21-5](#)) holo-BtrI. When holo-BtrI was incubated with glutamine, ATP, and purified BtrJ protein, the holo-BtrI species of M_r 12,153 was replaced with a species of M_r 12,281, corresponding to the thioester of glutamate and holo-BtrI. Based on these data, the authors proposed the following structure for the M_r 12,281 species, γ -glutamyl-S-BtrI:



- c. What other structure(s) is (are) consistent with the data above?
- d. Li and coauthors argued that the structure shown here (γ -glutamyl-S-BtrI) is likely to be correct because the α -carboxyl group must be removed at some point in the synthetic process. Explain the chemical basis of this argument. (Hint: See [Fig. 18-6](#), reaction C.)

The BtrK protein showed significant homology to PLP-dependent amino acid decarboxylases, and BtrK isolated from *E. coli* was found to contain tightly bound PLP. When γ -glutamyl-S-BtrI was incubated with purified BtrK, a molecular species of M_r 12,240 was produced.

- e. What is the most likely structure of this species?
- f. When the investigators incubated glutamate and ATP with purified BtrI, BtrJ, and BtrK, they found a molecular species of M_r 12,370. What is the most likely structure of this species? Hint: Remember that BtrJ can use ATP to γ -glutamylate nucleophilic groups.

Li and colleagues found that BtrO is homologous to monooxygenase enzymes (see [Box 21-1](#)) that hydroxylate alkanes, using FMN as a cofactor, and BtrV is homologous to an NAD(P)H oxidoreductase. Two other genes in the cluster, *btrG* and *btrH*, probably encode enzymes that remove the γ -glutamyl group and attach AHBA to the target antibiotic molecule.

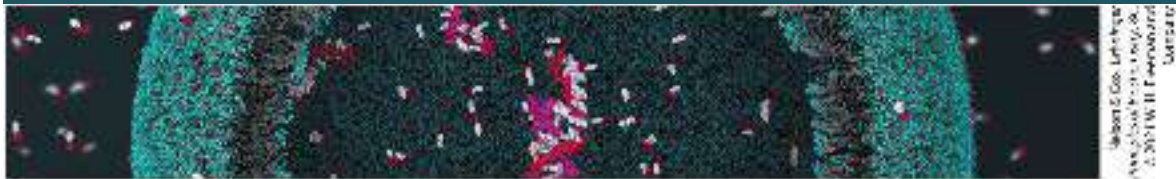
- g. Based on these data, propose a plausible pathway for the synthesis of AHBA and its addition to the target antibiotic. Include the enzymes that catalyze each step and any other substrates or cofactors needed (ATP, NAD, etc.).

References

- Li, Y., N.M. Llewellyn, R. Giri, F. Huang, and J.B. Spencer. 2005.** Biosynthesis of the unique amino acid side chain of butirosin: possible protective-group chemistry in an acyl carrier protein-mediated pathway. *Chem. Biol.* 12:665–675.

CHAPTER 23

HORMONAL REGULATION AND INTEGRATION OF MAMMALIAN METABOLISM



23.1 Hormone Structure and Action

23.2 Tissue-Specific Metabolism

23.3 Hormonal Regulation of Fuel Metabolism

23.4 Obesity and the Regulation of Body Mass

23.5 Diabetes Mellitus

In [Chapters 13](#) through [22](#) our focus has been on metabolism at the level of the individual cell, emphasizing central pathways common to almost all cells — bacterial, archaeal, and eukaryotic. We have seen how metabolic processes within cells are regulated at the level of individual enzyme reactions by substrate availability, by allosteric mechanisms, and by reversible covalent modification of enzymes.

To fully appreciate the significance of individual metabolic pathways and their regulation, we must view these pathways in the context of the whole organism. An essential characteristic of

multicellular organisms is cell differentiation and division of labor. Specialized functions of the tissues and organs require specialized fuels and patterns of metabolism. Hormonal and neuronal signals integrate and coordinate the metabolic activities of different tissues and optimize the allocation of fuels and precursors to each organ. Although our focus is on mammalian systems, mammals are hardly unique in possessing hormonal signaling systems. Insects and nematode worms have highly developed systems for hormonal regulation, with fundamental mechanisms similar to those in mammals. Plants, too, use hormonal signals to coordinate the activities of their differentiated, specialized tissues.

In this chapter we will look at the specialized metabolism of several major organs and tissues and the integration of metabolism in mammals. We begin with an overview of the broad range of hormones and hormonal mechanisms, then turn to the tissue-specific functions regulated by these mechanisms. We discuss the distribution of nutrients to various organs, emphasizing the central role of the liver, and the metabolic cooperation among these organs. To illustrate the integrative role of hormones, we describe the interplay of insulin, glucagon, and epinephrine in coordinating fuel metabolism in muscle, liver, and adipose tissue. We also introduce other hormones, produced in adipose tissue, muscle, gut, and brain, that play key roles in coordinating metabolism and behavior. We discuss the long-term hormonal regulation of body mass and the role of obesity in the development of metabolic syndrome and type 2 diabetes. Finally, we discuss interventions used to manage diabetes.

In this chapter, we illustrate the following principles:

P1 The tissues in a mammal are connected by a **neurosecretory system that coordinates their activities.**

Mammals use chemically diverse hormones in a highly specific and multidirectional signaling system, connecting the tissues with each other and with the central nervous system.

P2 Among the tissues and organs of an animal, there is a **striking division of labor.** The specialized role of each organ is reflected in its metabolic activities and capabilities. The circulatory system connects all of the tissues by carrying hormonal signals and metabolites between and among them.

P3 Because the brain requires a continuous supply of **glucose, maintaining an adequate concentration of glucose in the blood is a high priority in the activities of the other tissues.** The liver integrates the use of fuels (glucose, fatty acids, and amino acids) by each tissue to keep the blood glucose level within the optimal range. Hormones carried in the blood (insulin, glucagon, epinephrine, cortisol) mediate this regulation.

P4 **Maintaining an optimal body mass is an important priority in the adult mammal.** Body mass is a function of dietary intake, physical activity, and the choice of metabolic fuel, all of which are subject to hormonal regulation. Hormonal signals between the brain, the adipose tissue, and

the gastrointestinal tract help to set activity and feeding behavior.



The metabolic activities of cells and organisms are complex and intertwined; perturbation at one point in the system has far-reaching consequences for health. When the normal energy-yielding metabolism of glucose and fatty acids is impeded by defective insulin signaling, the result is the disease diabetes.

23.1 Hormone Structure and Action

Hormones are small molecules or proteins that are produced in one tissue, released into the bloodstream, and carried to other tissues, where they act through specific receptors to bring about changes in cellular activities. Hormones serve to coordinate the metabolic activities of several tissues or organs. Virtually every process in a complex organism is regulated by one or more hormones: maintenance of blood pressure, blood volume, and electrolyte balance; embryogenesis; sexual differentiation, development, and reproduction; hunger, eating behavior, digestion, and fuel allocation — to name but a few.

The coordination of metabolism in mammals is achieved by the **neuroendocrine system**. Individual cells in one tissue sense a change in the organism's circumstances and respond by secreting a chemical messenger that passes to another cell in the same or different tissue, where the messenger binds to a specific receptor molecule and triggers a change in this target cell. These chemical messengers may relay information over very short or very long distances. In neuronal signaling ([Fig. 23-1a](#)), the chemical messenger is a neurotransmitter (acetylcholine, for example) that may travel only a fraction of a micrometer across a synaptic cleft to the next neuron in a network. In endocrine signaling ([Fig. 23-1b](#)), the messenger is a hormone that is carried in the bloodstream to neighboring cells or to distant organs and tissues; it may travel a meter or more before encountering its target cell.

Except for this anatomic difference, the neuronal and endocrine signaling mechanisms are remarkably similar, and the same molecule can sometimes act as both neurotransmitter and hormone. Epinephrine and norepinephrine, for example, serve as neurotransmitters at certain synapses of the brain and at neuromuscular junctions of smooth muscle and as hormones that regulate fuel metabolism in liver and muscle. The following discussion of cellular signaling emphasizes hormone action, drawing on discussions of fuel metabolism in earlier chapters, but most of the fundamental mechanisms described here also occur in neurotransmitter action.

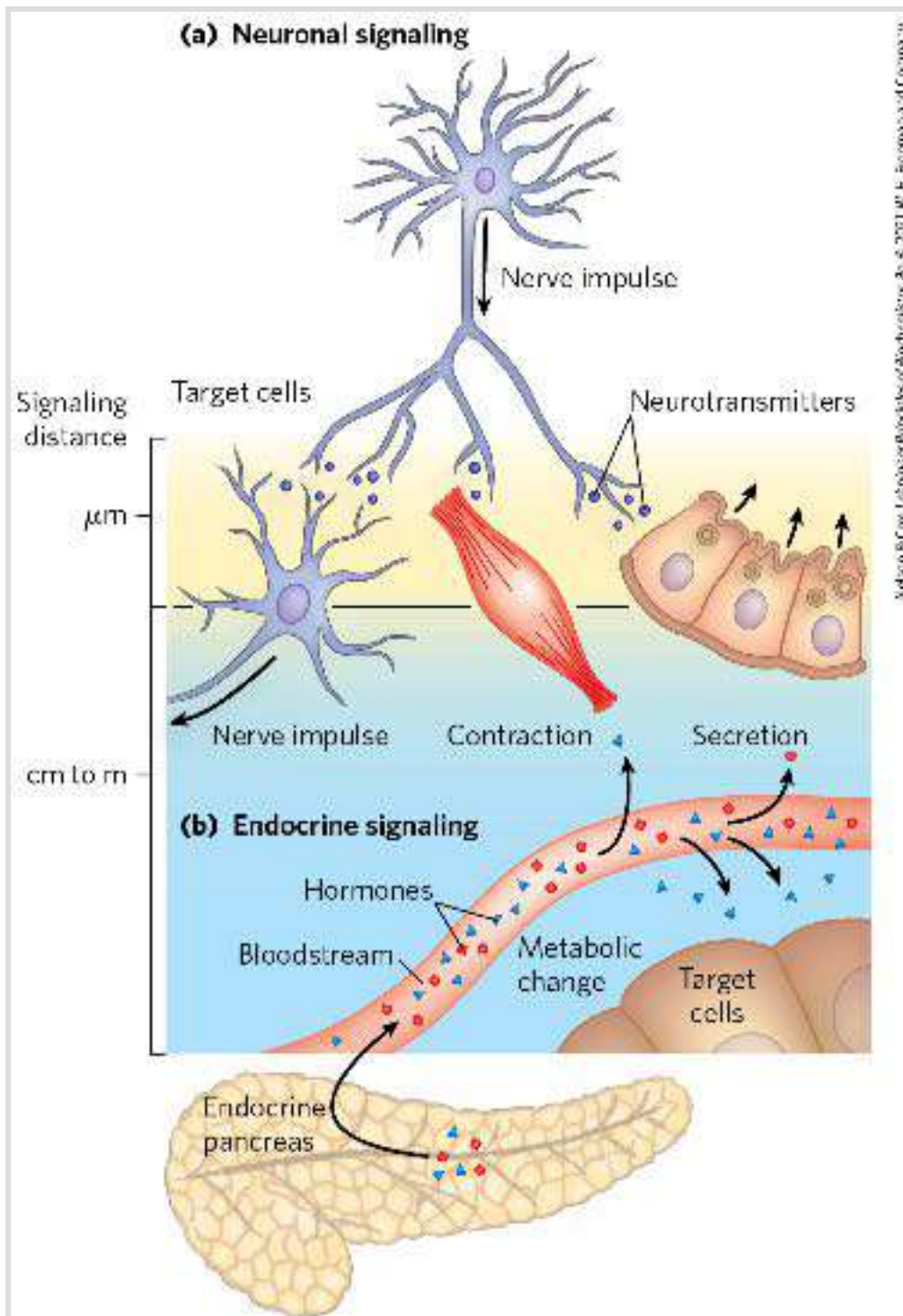


FIGURE 23-1 Signaling by the neuroendocrine system. (a) In neuronal signaling, electrical signals (nerve impulses) originate in the cell body of a neuron and travel very rapidly over long distances to the axon tip, where neurotransmitters are released and diffuse to the target cell. The target cell (another neuron, a myocyte, or a secretory cell) is only a fraction of a micrometer or a few micrometers away from the site of neurotransmitter release. (b) In endocrine signaling, hormones (such as insulin produced in

pancreatic β cells) are secreted into the bloodstream, which carries them throughout the body to target tissues that may be a meter or more away from the secreting cell. Both neurotransmitters and hormones interact with specific receptors on or in their target cells, triggering responses.

Hormones Act through Specific High-Affinity Cellular Receptors

Hormones exert their effects through specific receptors in target cells. The high affinity of the hormone-receptor interaction allows cells to respond to very low concentrations of hormone. Recall from [Chapter 12 \(Fig. 12-2\)](#) that there are four general types of intracellular consequences of ligand-receptor interaction: (1) a second messenger, such as cAMP, cGMP, or inositol trisphosphate, generated inside the cell, acts as an allosteric regulator of one or more enzymes; (2) a receptor tyrosine kinase is activated by the extracellular hormone; (3) a change in membrane potential results from the opening or closing of a hormone-gated ion channel; and (4) a steroid or steroidlike molecule causes a change in the level of expression (transcription of DNA into mRNA) of one or more genes, mediated by a nuclear hormone receptor protein. It can be helpful to characterize cell surface hormone receptors as **metabotropic**, those that activate or inhibit an enzyme downstream from the receptor, or **ionotropic**, those that open or close an ion channel in the plasma membrane, resulting in a change in membrane potential (ΔV_m) or in the concentration of an ion such as Ca^{2+} ([Fig. 23-2](#)).

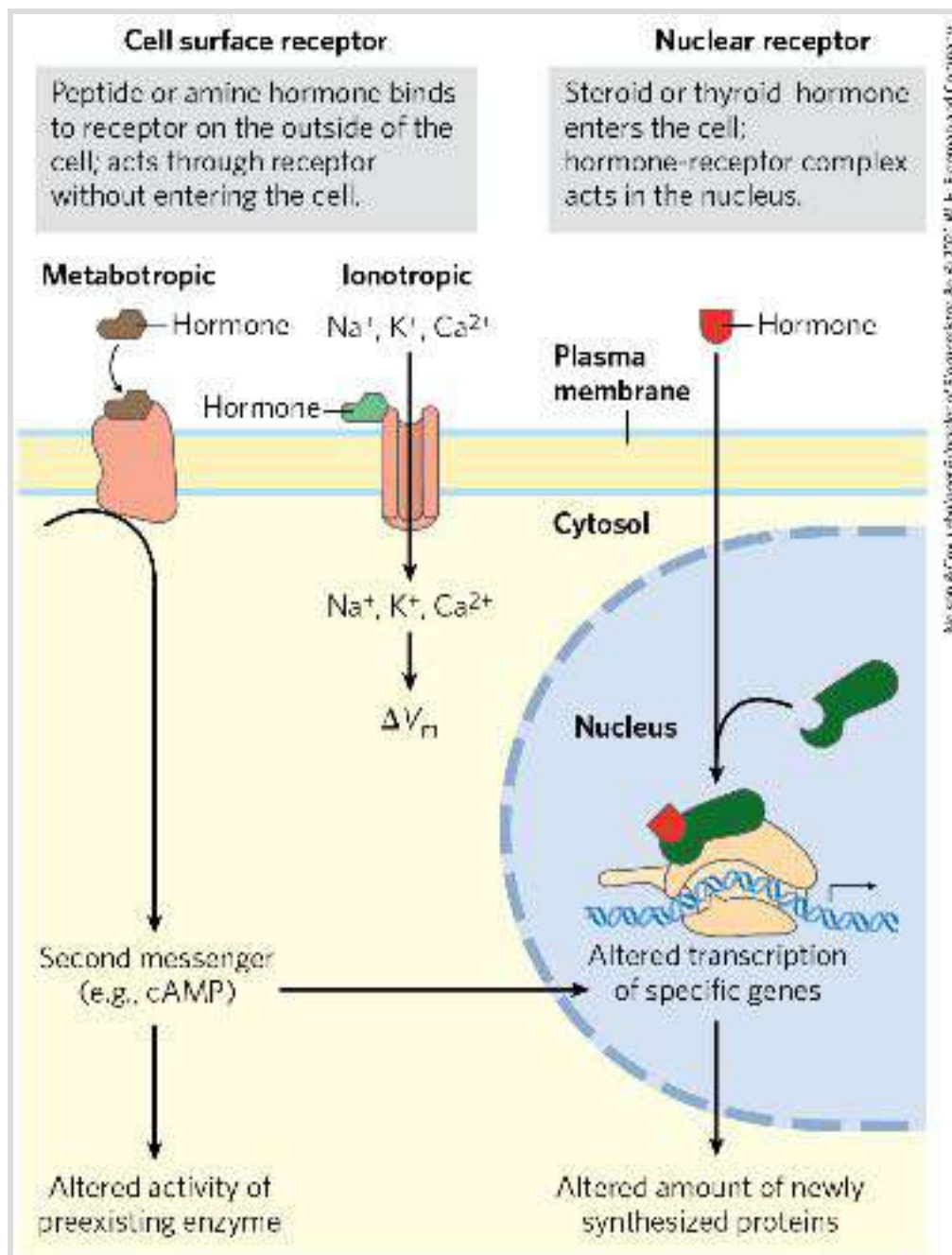


FIGURE 23-2 Two general mechanisms of hormone action. The peptide and amine hormones are faster-acting than steroid and thyroid hormones. Peptide hormones act from outside the cell, binding a plasma membrane hormone receptor. Steroid and thyroid hormones pass through the plasma membrane and enter the nucleus, where they regulate the expression of specific genes.

A single hormone molecule, in forming a hormone-receptor complex, activates a catalyst that produces many molecules of second messenger, so the receptor serves as both signal transducer and signal amplifier. The signal may be further amplified by a signaling cascade, such as we saw in the regulation of glycogen synthesis and breakdown by epinephrine (see [Fig. 12-7](#)). Signal amplification allows one epinephrine molecule to result in the production of many thousands or millions of molecules of glucose 1-phosphate from glycogen.

Water-insoluble hormones, including the steroid, retinoid, and thyroid hormones, readily pass through the plasma membrane of their target cells to reach their receptor proteins in the nucleus ([Fig. 23-2](#)). The hormone-receptor complex itself carries the message: it interacts with DNA to alter the expression of specific genes, changing the enzyme content of the cell and thereby changing cellular metabolism (see [Fig. 12-34](#)).

Hormones that act through plasma membrane receptors generally trigger very rapid physiological or biochemical responses. Just seconds after the adrenal medulla secretes epinephrine into the bloodstream, skeletal muscle responds by accelerating the breakdown of glycogen. By contrast, the thyroid hormones and the sex (steroid) hormones promote maximal responses in their target tissues only after hours or even days. These differences in response time correspond to different modes of action. In general, the fast-acting hormones lead to a change in the activity of one or more preexisting enzymes in the cell, by allosteric mechanisms or covalent modification. The

slower-acting hormones generally alter gene expression, resulting in the synthesis of more (upregulation) or less (downregulation) of the regulated protein(s).

Hormones Are Chemically Diverse

P1 Mammals have several classes of hormones, distinguishable by their diverse chemical structures and their modes of action ([Table 23-1](#)). Peptide, catecholamine, eicosanoid, and endocannabinoid hormones act from outside the target cell via cell surface receptors. Steroid, vitamin D, retinoid, and thyroid hormones enter the cell and act through nuclear receptors. Nitric oxide (a gas) also enters the cell, but activates a cytosolic enzyme, guanylyl cyclase.

TABLE 23-1 Classes of Hormones

Type	Example	Synthetic path	Mechanism of action
Protein	Insulin (Fig. 23-4)	Proteolytic processing of polypeptide	Plasma membrane RTK
Protein	Ghrelin		
Peptide	Oxytocin (p. 580)		
Catecholamine	Epinephrine (Fig. 23-11)	From tyrosine	Plasma membrane GPCR second messengers
Eicosanoid	Prostaglandin E ₂ (Fig. 10-17)	From arachidonic acid	
Endocannabinoid	Anandamide (Fig. 23-40)		
Steroid	Testosterone (Fig. 10-18)	From cholesterol	Nuclear receptors; transcriptional regulation
Glucocorticoid	Cortisol (Fig. 10-19)		
Vitamin D	Calcitriol (Fig. 10-19)		
Retinoid	Retinoic acid (Fig. 10-20)	From vitamin A	Nuclear receptors; transcriptional regulation
Thyroid	Triiodothyronine (T ₃)	From Tyr in thyroglobulin	
Humoral	NO (Fig. 23-32)	From arginine	Cytosolic receptor; second messenger

Hormones can also be classified by the way they get from their point of release to their target tissue. **Endocrine** (from the Greek *endon*, “within,” and *krinein*, “to release”) hormones are released

into the blood and carried to target cells throughout the body (insulin and glucagon are examples). **Paracrine** hormones are released into the extracellular space and diffuse to neighboring target cells (the eicosanoid hormones are of this type). **Autocrine** hormones affect the same cell that releases them, binding to receptors on the cell surface.

The **peptide hormones** vary in size from 3 (thyrotropin-releasing hormone; [Fig. 23-3](#)) to more than 200 (human chorionic gonadotropin) amino acid residues. They include the pancreatic hormones insulin, glucagon, and somatostatin; the parathyroid hormone calcitonin; and all the hormones of the hypothalamus and pituitary. Peptide hormones (some of which are actually small proteins) are synthesized as proproteins (prohormones) that are activated upon release by proteolytic cleavage, much as we saw with zymogen activation of pancreatic enzymes ([Fig. 6-42](#)) and the blood-clotting cascade ([Fig. 6-44](#)). In some cases, a preproprotein is synthesized and processed to create more than one active peptide product.

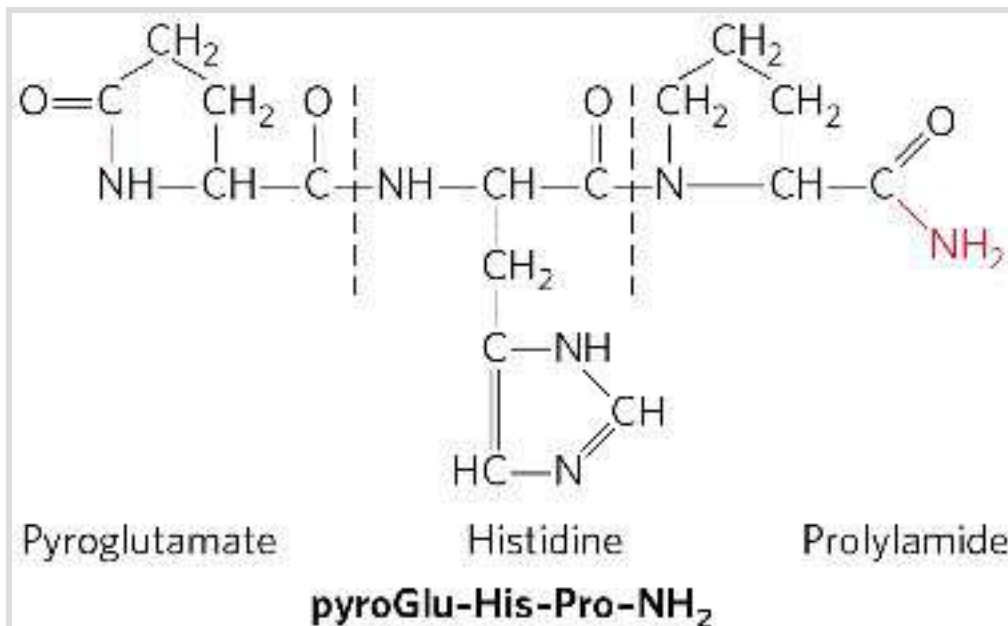
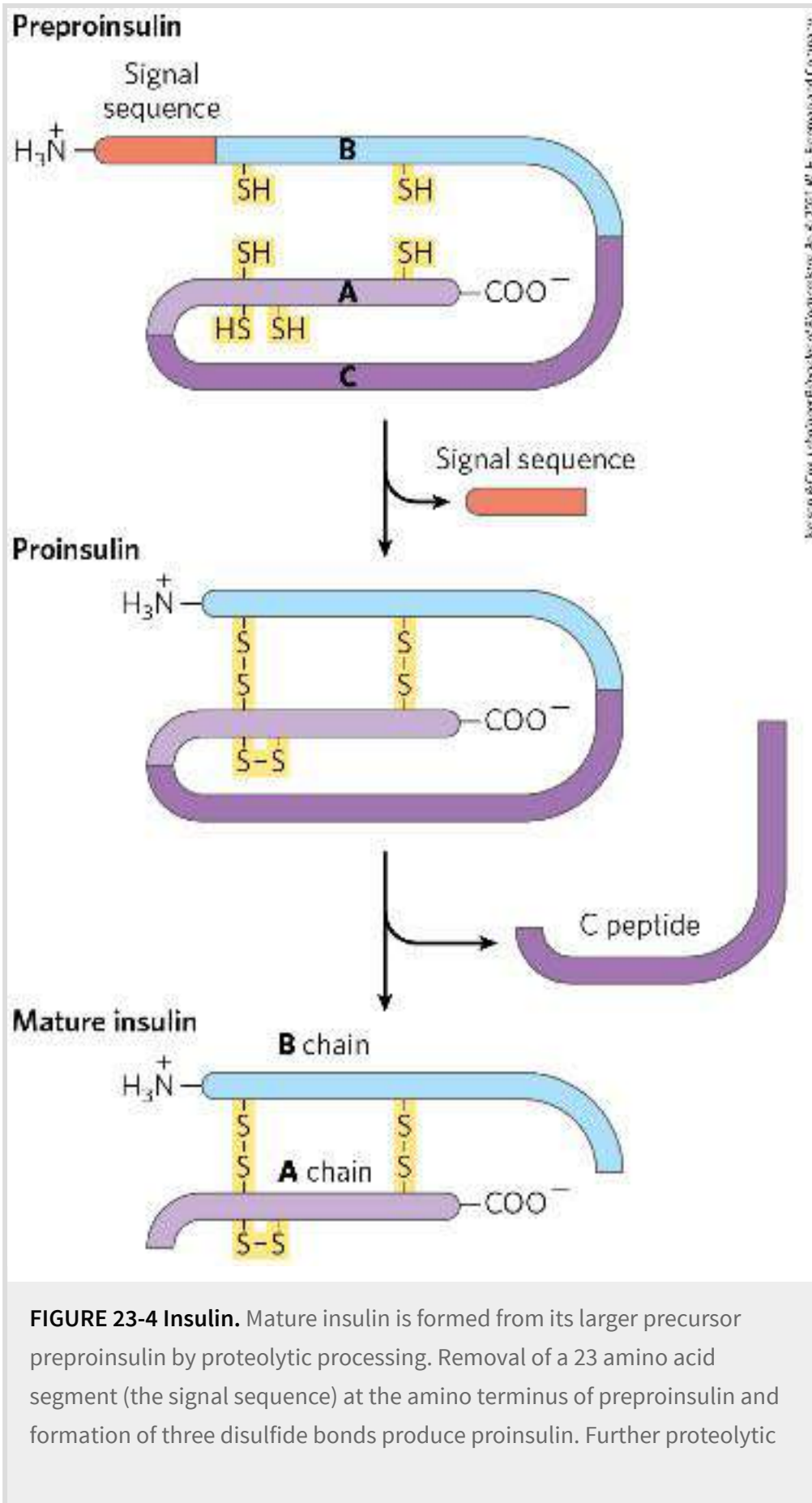


FIGURE 23-3 The structure of thyrotropin-releasing hormone (TRH).

Purified (through heroic efforts) from extracts of hypothalamus, TRH proved to be a derivative of the tripeptide Glu-His-Pro. The side-chain carboxyl group of the amino-terminal Glu forms an amide (red bond) with the residue's α -amino group, creating pyroglutamate, and the carboxyl group of the carboxyl-terminal Pro is converted to an amide (red —NH₂). Such modifications are common among the small peptide hormones. In a typical protein of $M_r \sim 50,000$, the charges on the amino- and carboxyl-terminal groups contribute relatively little to the overall charge on the molecule, but in a tripeptide these two charges dominate the properties of the molecule. Formation of the amide derivatives removes these charges.

Insulin is a small protein (M_r 5,800) with two polypeptide chains, A and B, joined by two disulfide bonds. (The amino acid sequence of bovine insulin is shown in [Figure 3-24](#).) It is synthesized in the pancreas as an inactive single-chain precursor, preproinsulin ([Fig. 23-4](#)), with an amino-terminal “signal sequence” that directs its passage into secretory vesicles. (Signal sequences are discussed in [Chapter 27](#); see [Fig. 27-38](#).) Proteolytic removal of the signal sequence and formation of three disulfide bonds produces

proinsulin, which is stored in secretory granules in pancreatic β cells. When blood glucose is elevated sufficiently to trigger insulin secretion, proinsulin is converted to active insulin by specific proteases, which cleave two peptide bonds to form the mature insulin molecule and C peptide, which are released into the blood by exocytosis. The capillaries that serve peptide-producing endocrine glands are fenestrated (punctuated with tiny holes or “windows”), so the hormone molecules readily enter the bloodstream for transport to target cells elsewhere. Insulin, acting through its receptor tyrosine kinase ([Fig. 12-21](#)) has profound effects on both anabolic and catabolic processes in many tissues, which we explore in detail below.



cuts remove the C peptide from proinsulin to produce mature insulin, composed of A and B chains.

Pro-opiomelanocortin (POMC) is a spectacular example of a proprotein that undergoes specific cleavage to produce several active hormones. The POMC gene encodes a large polypeptide that contains within it at least nine biologically active peptides ([Fig. 23-5](#)). The proprotein is processed differently in different tissues, depending on which proteases the cells express. The active products influence an astonishing number of physiological systems.

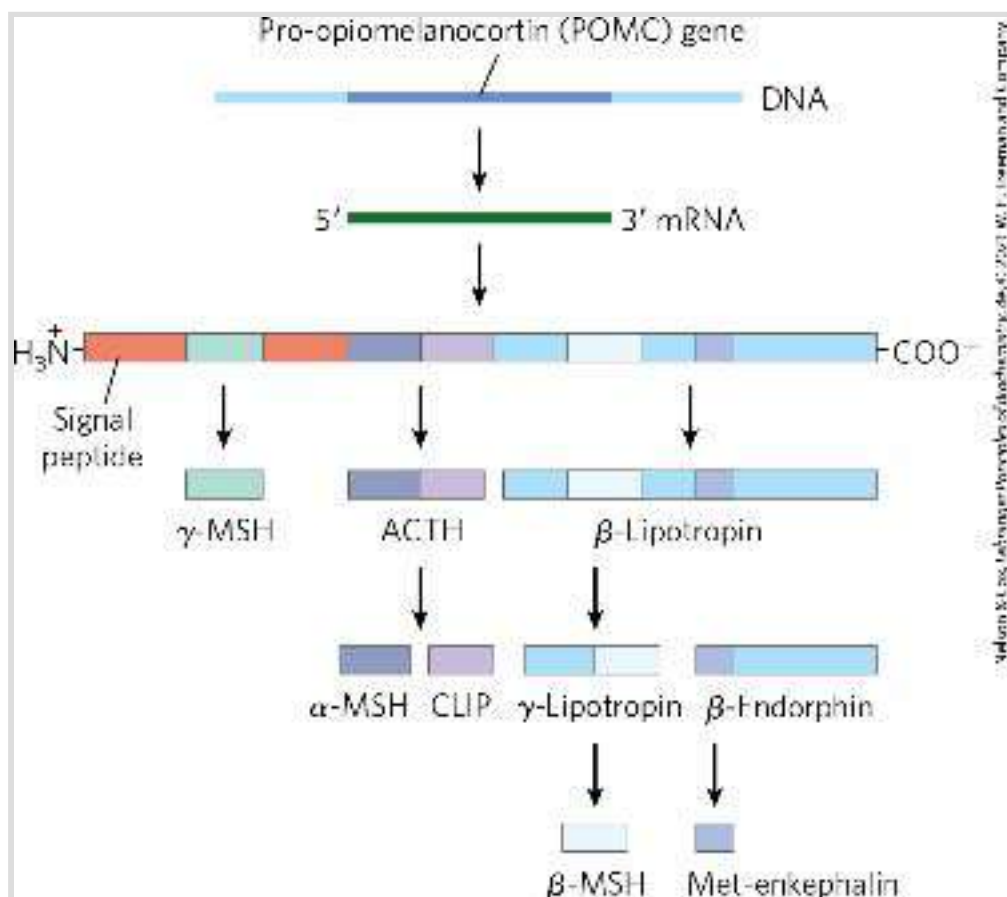


FIGURE 23-5 Proteolytic processing of the pro-opiomelanocortin (POMC) precursor. The initial gene product of the POMC gene is a long polypeptide that undergoes cleavage by a series of specific proteases to produce ACTH

(corticotropin), β - and γ -lipotropin, α -, β -, and γ -MSH (melanocyte-stimulating hormone, or melanocortin), CLIP (corticotropin-like intermediary peptide), β -endorphin, and Met-enkephalin. The points of cleavage are pairs of basic residues, Arg-Lys, Lys-Arg, or Lys-Lys.

Some Hormones Are Released by a “Top-Down” Hierarchy of Neuronal and Hormonal Signals

The changing levels of specific hormones regulate specific cellular processes, but what regulates the regulators — what sets the level of each hormone? The brief answer is that  the central nervous system receives input from many internal and external sensors — signals about danger, hunger, dietary intake, blood composition, for example — and orchestrates the production of appropriate hormonal signals by the endocrine tissues. For a more complete answer, consider the path of cortisol release by the adrenal gland, triggered by a stress detected by the central nervous system. [Figure 23-6](#) illustrates the “chain of command” in this top-down hormonal signaling hierarchy. The **hypothalamus**, a pea-size region of the brain ([Fig. 23-7](#)), is the coordination center of the endocrine system; it receives and integrates messages from the central nervous system. In response, the hypothalamus produces releasing factors, including corticotropin-releasing hormone (CRH), that pass directly to the nearby pituitary gland through blood vessels and neurons that connect the two glands. The pituitary gland secretes

adrenocorticotrophic hormone (also called corticotropin or ACTH), which travels through the blood to the adrenal cortex and triggers the release of cortisol. Cortisol, the ultimate hormone in this cascade, acts through its receptor in many types of target cells to alter their metabolism. In hepatocytes, one effect of cortisol is to increase the rate of gluconeogenesis.

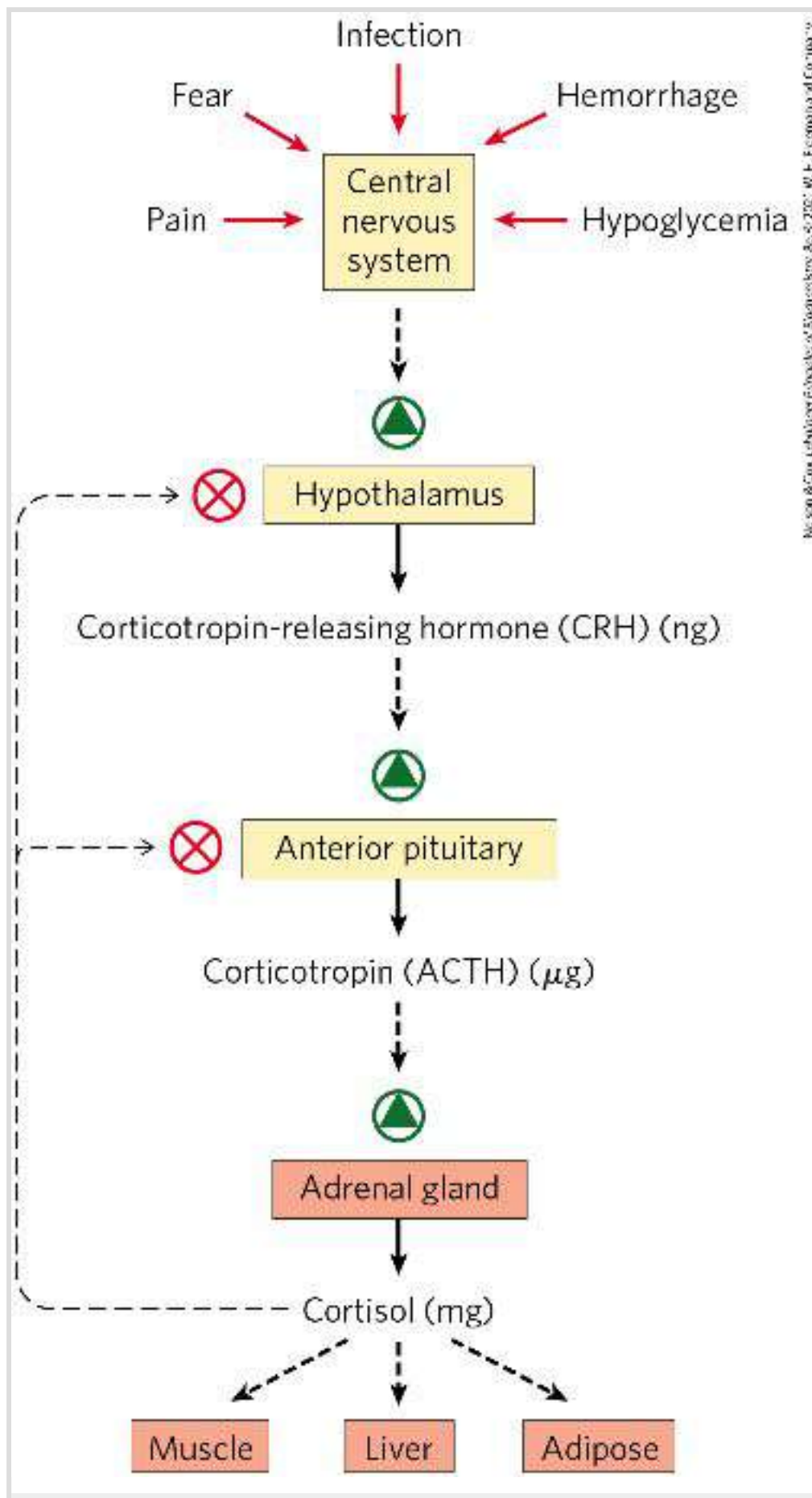


FIGURE 23-6 Cascade of top-down hormone release following central nervous system input to the hypothalamus. Solid black arrows indicate hormone production and release; broken black arrows indicate the action of hormones on target tissues. In each endocrine tissue along the pathway, a stimulus from the level above is received, amplified, and transduced into release of the next hormone in the cascade. The cascade is sensitive to regulation at several levels through feedback inhibition (thin, dashed arrows) by the ultimate hormone (in this case, cortisol). The product therefore regulates its own production, as in feedback inhibition of biosynthetic pathways within a single cell.

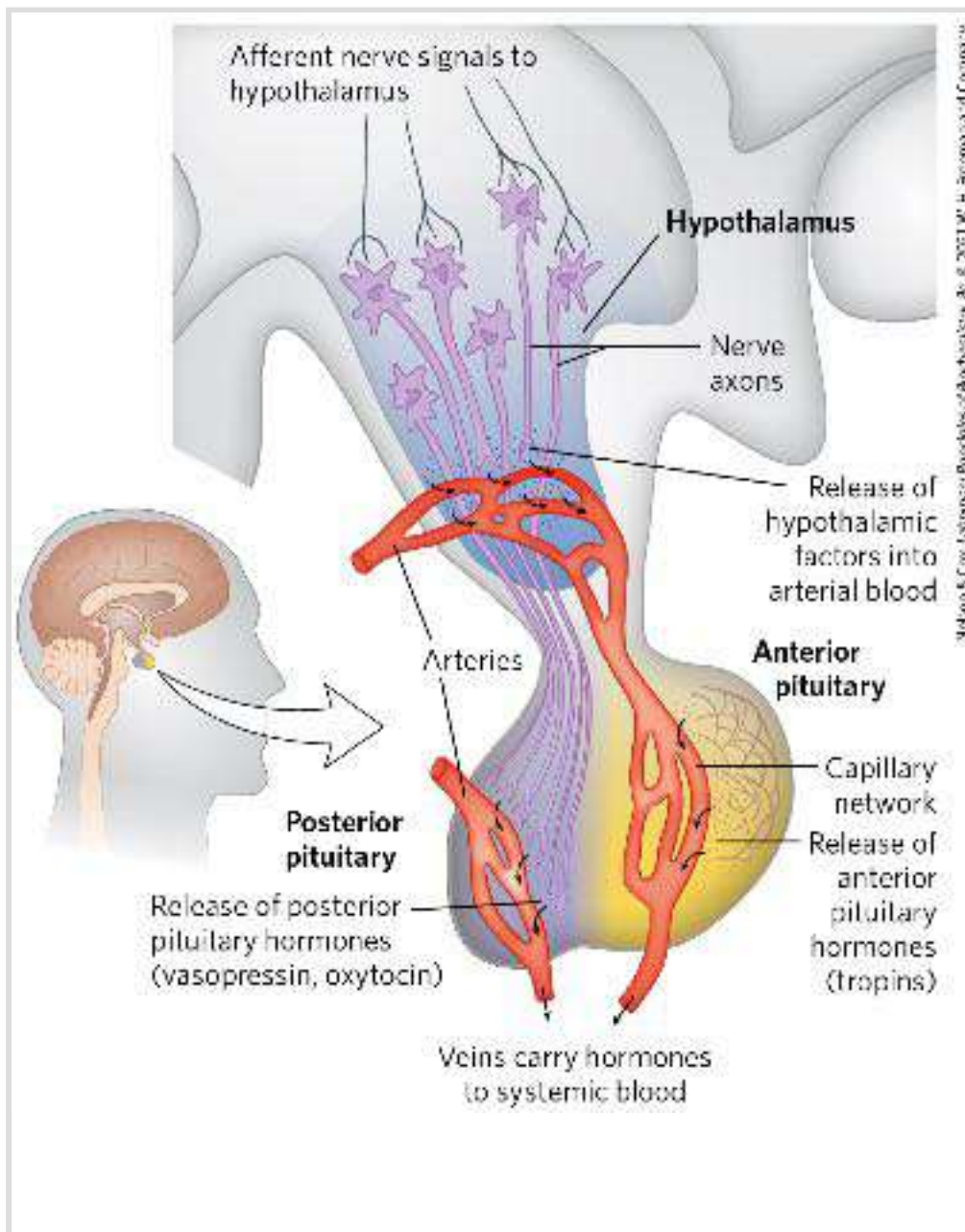


FIGURE 23-7 Neuroendocrine origins of hormone signals. Location of the hypothalamus and pituitary gland and details of the hypothalamus-pituitary system. Signals from connecting neurons stimulate the hypothalamus to secrete releasing factors into a blood vessel that carries the hormones directly to a capillary network in the anterior pituitary. In response to each releasing factor, the anterior pituitary releases the appropriate hormone into the general circulation. Posterior pituitary hormones are synthesized in neurons arising in the hypothalamus; the hormones are transported along axons to nerve endings in the posterior pituitary and stored there until released into the blood in response to a neuronal signal.

Hormonal cascades such as those responsible for the release of cortisol result in large amplifications of the initial signal and allow exquisite fine-tuning of the output of the ultimate hormone ([Fig. 23-6](#)). At each level in the cascade, a small signal elicits a larger response. For example, the initial electrical signal to the hypothalamus results in the release of a few *nanograms* of corticotropin-releasing hormone, which elicits the release of a few *micrograms* of corticotropin. Corticotropin acts on the adrenal cortex to cause the release of *milligrams* of cortisol, for an overall amplification of at least a millionfold.

At each level of a hormonal cascade, feedback inhibition of earlier steps in the cascade is possible; an unnecessarily elevated level of the ultimate hormone or of an intermediate hormone inhibits the release of earlier hormones in the cascade. These feedback mechanisms accomplish the same end as those that limit the output of a biosynthetic pathway (compare [Fig. 23-6](#) with [Fig. 22-37](#)): a product is synthesized (or released) only until the

necessary concentration is reached. In the blood-clotting cascade ([Fig. 6-44](#)) we saw a similar pattern: a signal stimulates a cascade of protein activations; feedback mechanisms limit their action and the duration of the response.

“Bottom-Up” Hormonal Systems Send Signals Back to the Brain and to Other Tissues

In addition to the top-down hierarchy of hormonal signaling shown in [Figure 23-6](#),  some hormones are produced in the digestive tract, muscle, and adipose tissue and communicate the current metabolic state to the hypothalamus ([Fig. 23-8](#)). These signals are integrated in the hypothalamus, and an appropriate neuronal or hormonal response is elicited.

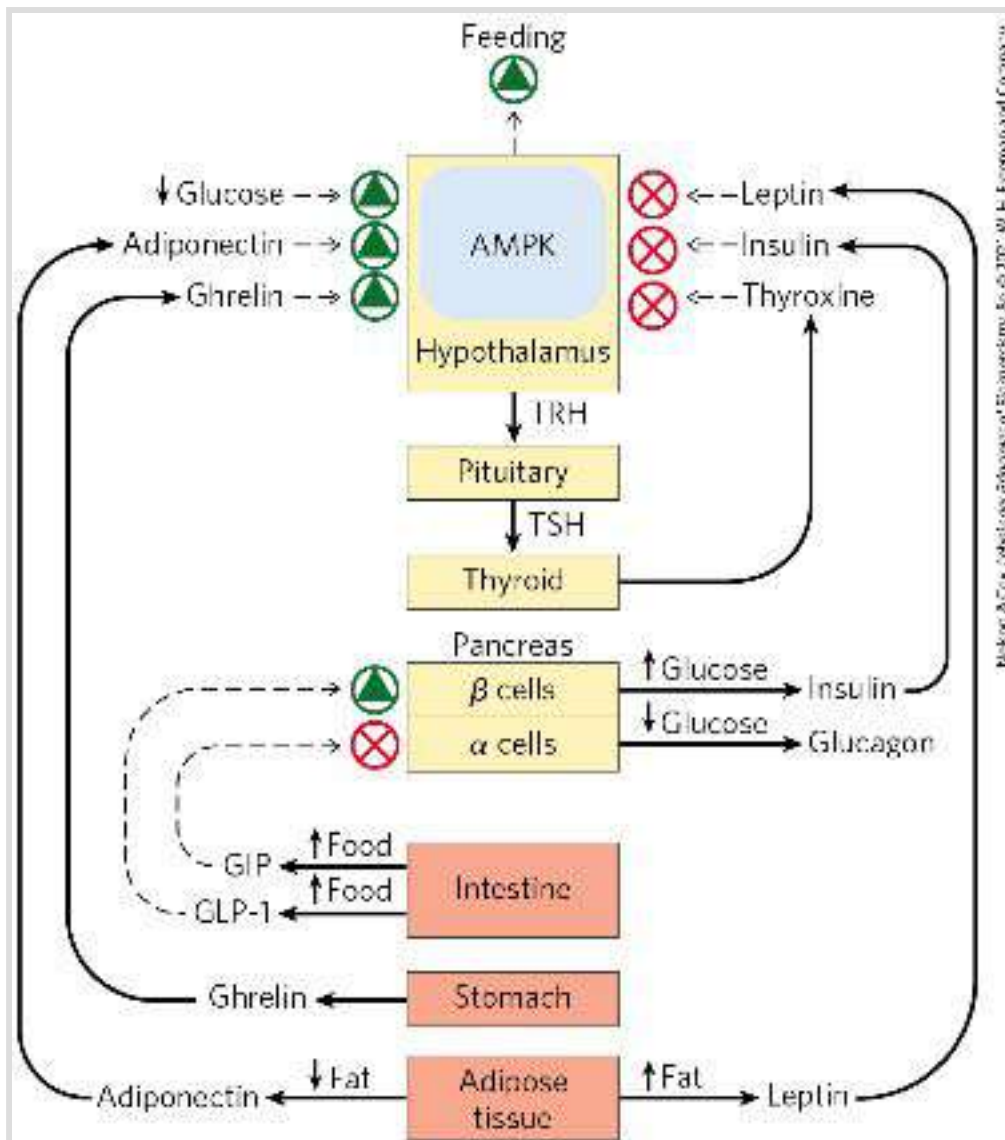


FIGURE 23-8 Regulation of feeding behavior by two-way information flow between tissues and the hypothalamus. When food intake and energy production are adequate, peptide hormones released by the stomach, intestine, and adipose tissue feed back on the hypothalamus to signal satiety and reduce feeding behavior. Other tissue-specific peptide hormones signal inadequate supplies of stored triacylglycerols or low blood glucose levels. All of these signals impinge, directly or indirectly, on AMP-activated protein kinase (AMPK) in the hypothalamus, which integrates these signals and influences feeding behavior and energy-yielding metabolism in the tissues. Nerves carry electrical signals from the brain to the other tissues to complete the information circuit and achieve homeostasis (not shown). TRH, thyrotropin-releasing hormone; GLP-1, glucagon-like peptide-1; GIP, gastric inhibitory polypeptide.

Adipokines, for example, are peptide hormones, produced *in adipose tissue*, that signal the adequacy of fat reserves. The adipokine **leptin**, released when adipose tissue is well-filled with triacylglycerols, acts in the brain to inhibit feeding behavior, whereas **adiponectin** signals depletion of fat reserves and stimulates feeding. Other tissues produce and release other hormones. **Ghrelin** is produced in the gastrointestinal tract when the stomach is empty and acts in the hypothalamus to stimulate feeding behavior; when the stomach fills, ghrelin release ceases. **Incretins** are peptide hormones produced in the gut after ingestion of a meal; they increase secretion of insulin and decrease secretion of glucagon from the pancreas. **Neuropeptide Y (NPY)** is a hormone produced in the hypothalamus and in the adrenal glands. Its release promotes feeding and reduces energy expenditure for nonessential activities. **Peptide YY (PYY_{3–36})**, produced in the intestine, signals satiety in the brain. **Irisin** is a peptide hormone produced in muscle as a result of exercise; it acts to convert white adipose tissue into beige adipose tissue, which dissipates energy as heat (see [Section 23.2](#)).

We return to these hormones (summarized in [Table 23-2](#)) when we discuss the regulation of body mass in humans ([Section 23.4](#)).

TABLE 23-2 Some Peptide Hormones That Act on Feeding Behavior and Fuel Selection in Mammals

Hormone	Production site(s)	Target tissue(s)	Action(s)
Insulin	Pancreatic β	Muscle,	Stimulates glucose uptake

	cells	adipose, liver	and synthesis of glycogen and fat
Glucagon	Pancreatic α cells	Liver, adipose	Stimulates gluconeogenesis and glucose release to blood
Leptin	Adipose tissue	Hypothalamus	Reduces hunger
Adiponectin	Adipose tissue	Muscle, liver, others	Stimulates catabolism and feeding behavior
Ghrelin	Stomach, intestine	Brain	Signals hunger
Incretins: GLP-1, GIP	Intestine	Pancreas	Stimulate insulin release
NPY	Hypothalamus, adrenals	Brain, autonomic nervous system	Stimulates feeding behavior
PYY ₃₋₃₆	Intestine	Brain	Signals satiety
Irisin	Muscle (after exercise)	Adipose	Turns white adipose tissue to beige

SUMMARY 23.1 *Hormone Structure and Action*

■ Hormones connect all of the organs in the body, carrying information and signals between the central nervous system and all of the tissues.

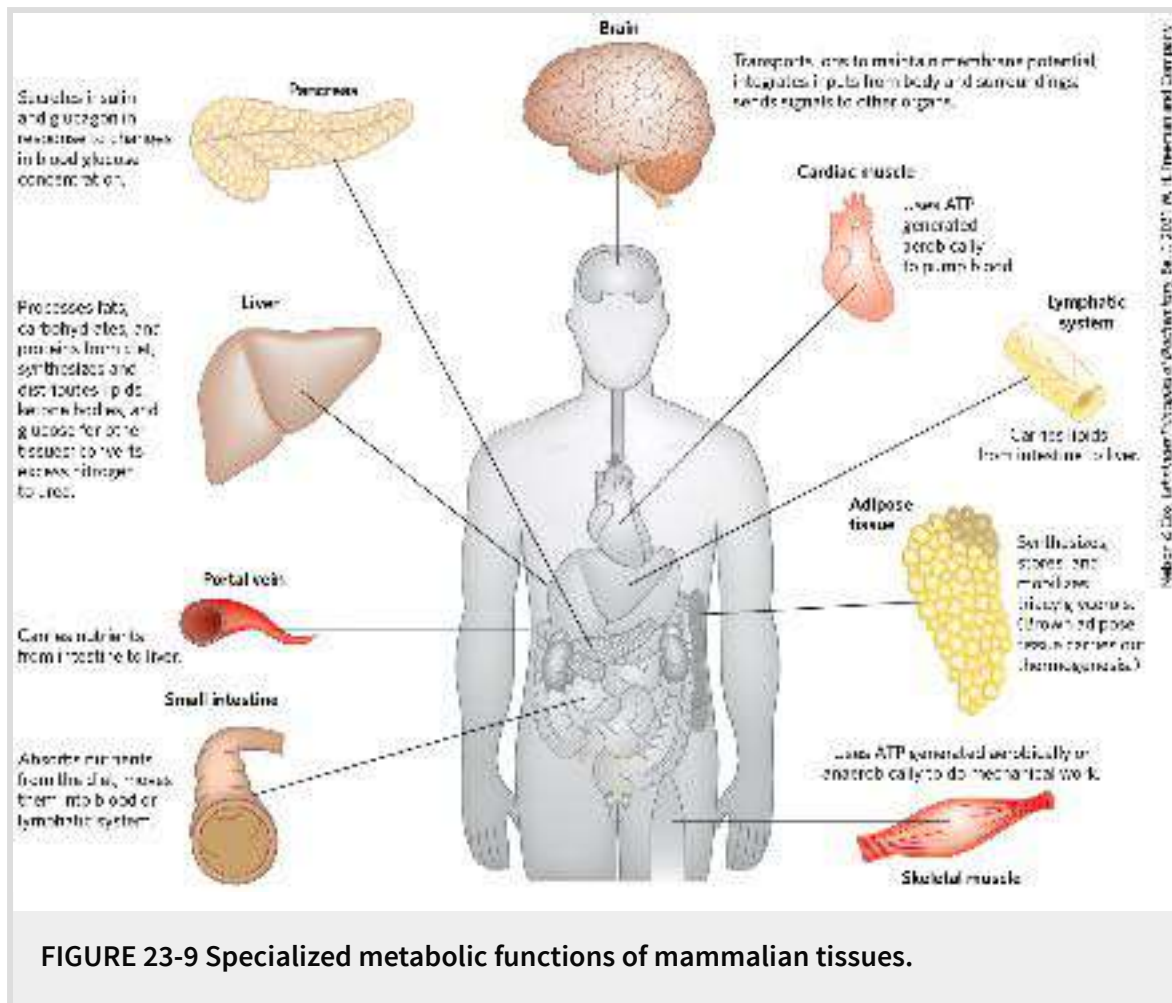
■ Hormones are chemically diverse with a wide range of biological roles. Peptide hormones are synthesized as proteins on ribosomes, then cleaved proteolytically to form the active peptide(s).

■ Hormones are regulated by a top-down hierarchy of interactions between the brain and endocrine glands: nerve impulses stimulate the hypothalamus to send specific hormones to the pituitary gland, thus stimulating (or inhibiting) the release of a second rank of hormones. The anterior pituitary hormones in turn stimulate other endocrine glands (adrenals, for example) to secrete their characteristic hormones, which in turn stimulate specific target tissues.

■ Some hormones act in bottom-up signaling: adipose tissue, muscle, and the gastrointestinal tract release peptide hormones that act on other tissues or in the central nervous system.

23.2 Tissue-Specific Metabolism

Each tissue of the human body has a specialized function, reflected in its anatomy and metabolic activity ([Fig. 23-9](#)). Skeletal muscle allows directed motion; adipose tissue stores and distributes energy in the form of fats, which serve as fuel throughout the body and as thermal insulation; in the brain, cells pump ions across their plasma membranes to produce electrical signals. The liver plays a central processing and distribution role in metabolism and furnishes all other organs and tissues with an appropriate mix of nutrients via the bloodstream. The functional centrality of the liver is indicated by the common reference to all other tissues and organs as “extrahepatic.” We therefore begin our discussion of the division of metabolic labor by considering the transformations of carbohydrates, amino acids, and fats in the mammalian liver. This is followed by brief descriptions of the primary metabolic functions of adipose tissue, muscle, brain, and the tissue that interconnects all others: the blood.



The Liver Processes and Distributes Nutrients

During digestion in mammals, the three main classes of nutrients (carbohydrates, fats, and proteins) undergo enzymatic hydrolysis into their simple constituents (monosaccharides, fatty acids, and amino acids). This breakdown is necessary because the epithelial cells lining the intestinal lumen absorb only relatively small molecules. Many of the fatty acids and monoacylglycerols released by digestion of fats in the intestine are reassembled within these epithelial cells into triacylglycerols (TAGs).

After being absorbed, most sugars and amino acids and some reconstituted TAGs pass from intestinal epithelial cells into blood capillaries and travel in the bloodstream to the liver; the remaining TAGs enter adipose tissue via the lymphatic system. The portal vein ([Fig. 23-9](#)) is a direct route from the digestive organs to the liver, and the liver therefore has first access to ingested nutrients. The liver has two main cell types. Kupffer cells are phagocytes, important in immune function. [Hepatocytes](#), of primary interest here, transform dietary nutrients into the fuels and precursors required by other tissues and export them via the blood. The kinds and amounts of nutrients supplied to the liver are determined by the composition of the diet, the interval between meals, and several other factors. The demand of extrahepatic tissues for fuels and precursors varies from one organ to another, and it also varies with the level of activity and overall nutritional state of the individual.

To meet these changing circumstances, the liver has remarkable metabolic flexibility. For example, when the diet is rich in protein, hepatocytes synthesize more of the enzymes needed for amino acid catabolism and gluconeogenesis. Within hours after a shift to a high-carbohydrate diet, the levels of these enzymes begin to drop and the hepatocytes increase their synthesis of enzymes essential to carbohydrate metabolism and fat synthesis. Liver enzymes turn over (that is, are synthesized and degraded) at 5 to 10 times the rate of enzyme turnover in other tissues, such as muscle. Extrahepatic tissues also can adjust their metabolism to prevailing conditions, but none of these tissues is as adaptable as

the liver, and none is so central to the organism's overall metabolism.

We turn now to a survey of the possible fates of sugars, amino acids, and lipids that enter the liver from the bloodstream.

Carbohydrates

The glucose transporter of hepatocytes (GLUT2) allows rapid, passive diffusion of glucose, so that the concentration of glucose in a hepatocyte is essentially the same as that in the blood.

Glucose entering hepatocytes is phosphorylated by glucokinase (hexokinase IV) to yield glucose 6-phosphate. Glucokinase has a much higher K_m for glucose (10 mM) than do the hexokinase isozymes in other cells ([p. 539](#)) and, unlike these other isozymes, it is not inhibited by its product, glucose 6-phosphate. The presence of glucokinase allows hepatocytes to continue phosphorylating glucose when the glucose concentration rises well above levels that would overwhelm other hexokinases. The high K_m of glucokinase also ensures that the phosphorylation of glucose in hepatocytes is minimal when the glucose concentration is low, preventing the liver from consuming glucose as fuel via glycolysis. This spares glucose for other tissues. Fructose, galactose, and mannose, all absorbed from the small intestine, are also converted to glucose 6-phosphate by enzymatic pathways examined in [Chapter 14 \(see Fig. 14-9\)](#). Glucose 6-phosphate is at the crossroads of carbohydrate metabolism in the liver. It may take any of several major metabolic routes ([Fig. 23-](#)

10), depending on the current metabolic needs of the organism. By the action of various allosterically regulated enzymes, and through hormonal regulation of enzyme synthesis and activity, the liver directs the flow of glucose into one or more of these pathways.

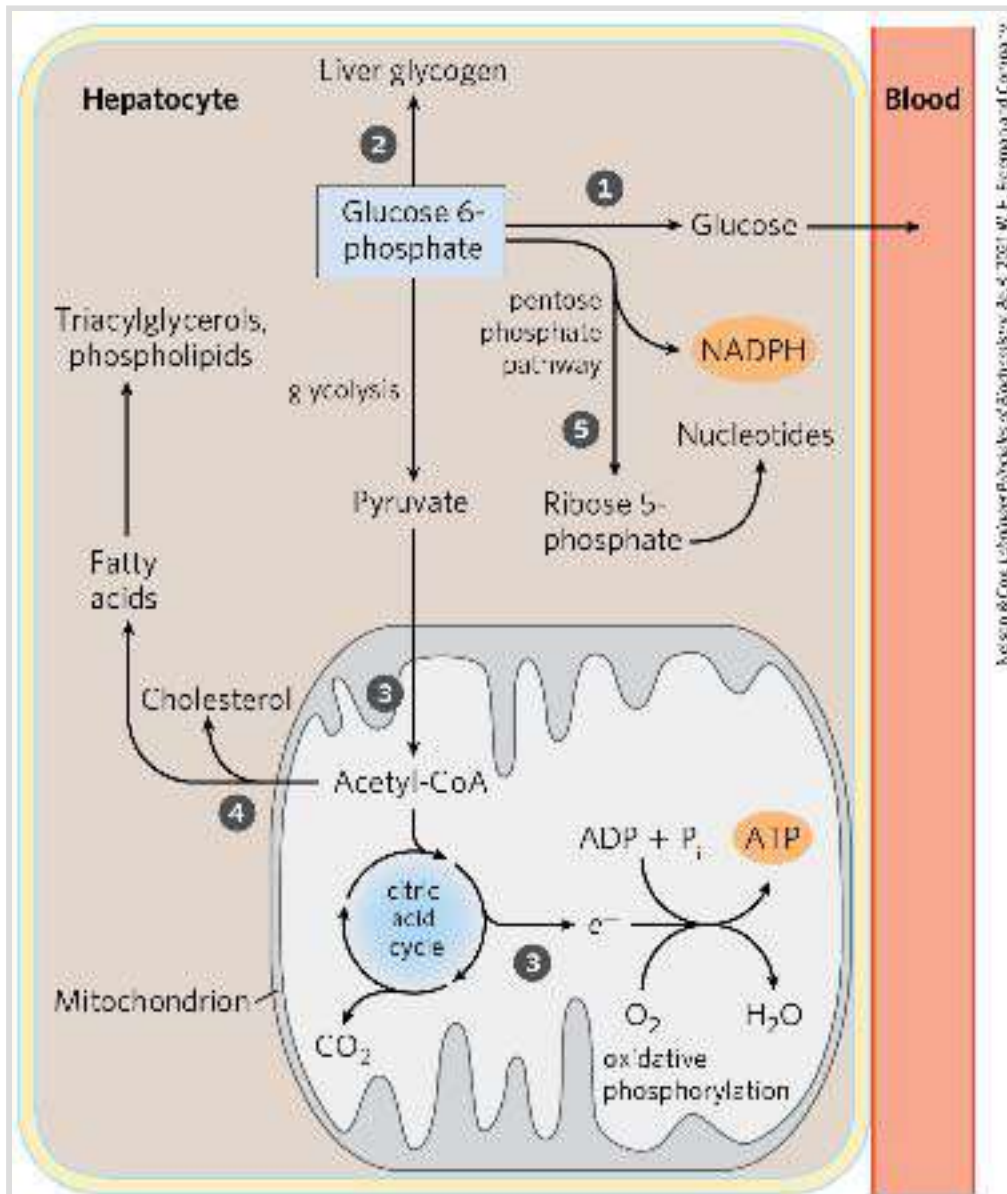


FIGURE 23-10 Metabolic pathways for glucose 6-phosphate in the liver.

Here and in [Figures 23-11](#) and [23-12](#), anabolic pathways are generally shown leading upward, catabolic pathways leading downward, and distribution to other organs leading horizontally. The numbered processes in each figure are described in the text.

① Glucose 6-phosphate is dephosphorylated by glucose 6-phosphatase to yield free glucose (see [Fig. 15-6](#)), which is exported to replenish blood glucose. Export is the predominant pathway when glucose 6-phosphate is in limited supply, because the blood glucose concentration must be kept sufficiently high (4 to 5 mM) to provide adequate energy for the brain and other tissues. ② Glucose 6-phosphate not immediately needed to form blood glucose is converted to liver glycogen, or has one of several other fates. Following glycolysis and the pyruvate dehydrogenase reaction, ③ the acetyl-CoA so formed can be oxidized for ATP production by the citric acid cycle, with ensuing electron transfer and oxidative phosphorylation yielding ATP. (Normally, however, fatty acids are the preferred fuel for ATP production in hepatocytes.) ④ Acetyl-CoA can also serve as the precursor of fatty acids, which are incorporated into TAGs and phospholipids, and of cholesterol. Much of the lipid synthesized in the liver is transported to other tissues by blood lipoproteins. ⑤ Alternatively, glucose 6-phosphate can enter the pentose phosphate pathway, yielding both reducing power (NADPH), needed for the biosynthesis of fatty acids and cholesterol, and ribose 5-phosphate, a precursor for nucleotide biosynthesis. NADPH is also an essential cofactor in the detoxification and elimination of many drugs and other **xenobiotics** (compounds that don't occur naturally but are the products of human activity, such as drugs, food additives, and preservatives) that are metabolized in the liver.

Amino Acids

Amino acids that enter the liver follow several important metabolic routes ([Fig. 23-11](#)). ❶ They are precursors for protein synthesis, a process discussed in [Chapter 27](#). The liver constantly renews its own proteins, which have a relatively high turnover rate (average half-life of hours to days), and it is also the site of biosynthesis of most plasma proteins. ❷ Alternatively, amino acids pass in the bloodstream to other organs to be used in the synthesis of tissue proteins. ❸ Other amino acids are precursors in the biosynthesis of nucleotides, hormones, and other nitrogenous compounds in the liver and other tissues.

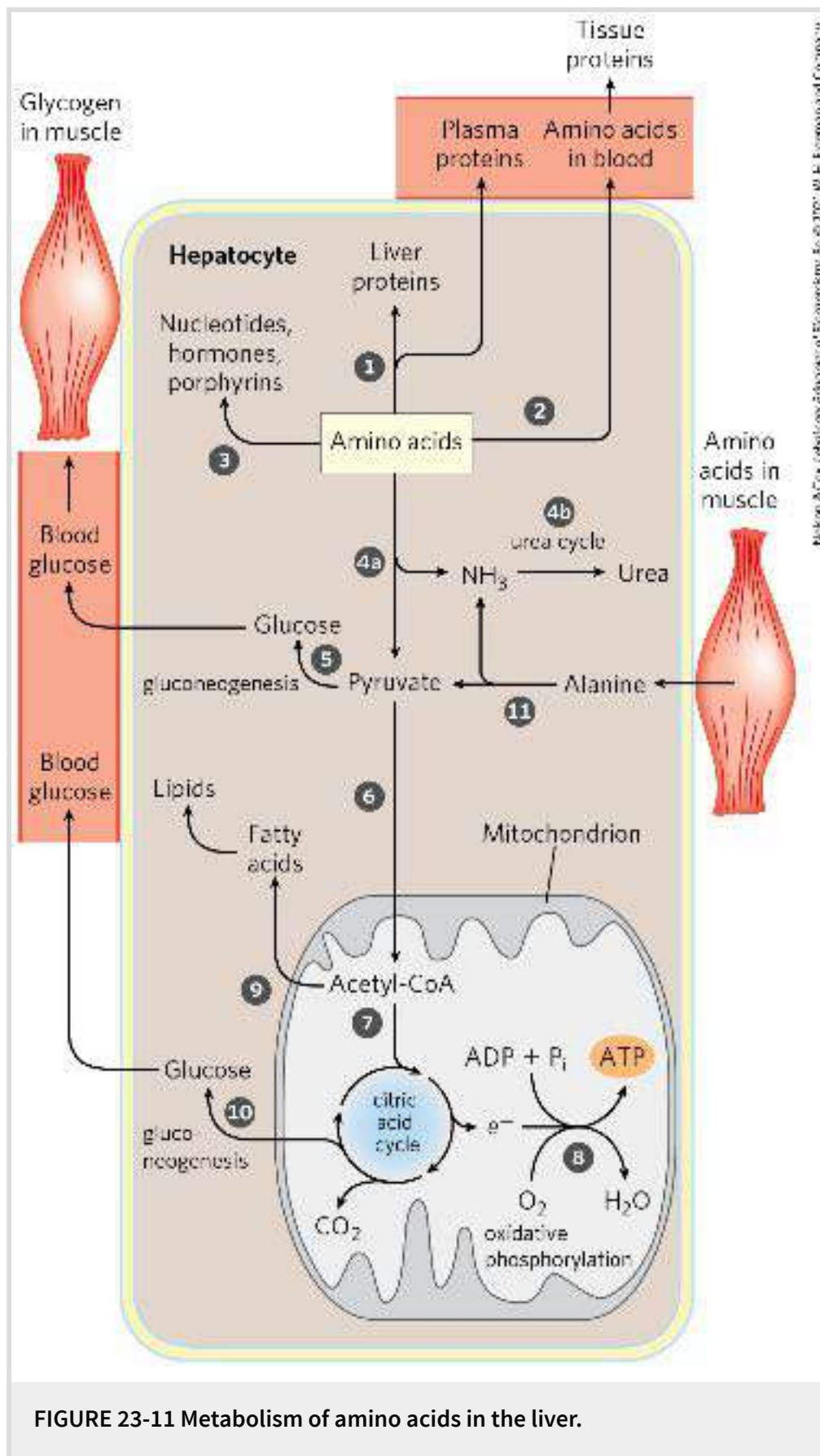


FIGURE 23-11 Metabolism of amino acids in the liver.

4a Amino acids not needed as biosynthetic precursors are transaminated or deaminated and degraded to yield pyruvate and citric acid cycle intermediates, with various fates; 4b the ammonia released is converted to the excretory product urea. 5 Pyruvate can be converted to glucose and glycogen via gluconeogenesis, or 6 can be converted to acetyl-CoA, which has several possible fates: 7 oxidation via the citric acid cycle and 8 oxidative phosphorylation to produce ATP, or 9 conversion to lipids for storage. 10 Citric acid cycle intermediates can be siphoned off into glucose synthesis by gluconeogenesis.

 The liver also metabolizes amino acids that arrive intermittently from other tissues. The blood is adequately supplied with glucose just after the digestion and absorption of dietary carbohydrate or, between meals, by the conversion of liver glycogen to blood glucose. During the interval between meals, especially if prolonged, some muscle protein is degraded to amino acids. These amino acids donate their amino groups (by transamination) to pyruvate, the product of glycolysis, to yield alanine, which 11 is transported to the liver and deaminated. Hepatocytes convert the resulting pyruvate to blood glucose via gluconeogenesis 5, and the ammonia to urea for excretion 4b. One benefit of this glucose-alanine cycle is the smoothing out of fluctuations in blood glucose between meals. The amino acid deficit incurred in muscles is made up after the next meal by incoming dietary amino acids.

Lipids

The fatty acid components of lipids entering hepatocytes also have several different fates ([Fig. 23-12](#)). ❶ Some are converted to liver lipids. ❷ Under most circumstances, fatty acids are the primary oxidative fuel in the liver. Free fatty acids may be activated and oxidized to yield acetyl-CoA and NADH. ❸ The acetyl-CoA is further oxidized via the citric acid cycle, and ❹ oxidations in the cycle drive the synthesis of ATP by oxidative phosphorylation. ❺ Excess acetyl-CoA, not required by the liver, is converted to acetoacetate and β -hydroxybutyrate; these ketone bodies circulate in the blood to other tissues to be used as fuel for the citric acid cycle. Ketone bodies, unlike fatty acids, can cross the blood-brain barrier, providing the brain with a source of acetyl-CoA for energy-yielding oxidation. Ketone bodies can supply a significant fraction of the energy in some extrahepatic tissues — up to one-third in the heart and as much as 60% to 70% in the brain during prolonged fasting. ❻ Some of the acetyl-CoA derived from fatty acids (and from glucose) is used for the biosynthesis of cholesterol, which is required for membrane synthesis. Cholesterol is also the precursor of all steroid hormones and of the bile salts, which are essential for the digestion and absorption of lipids.

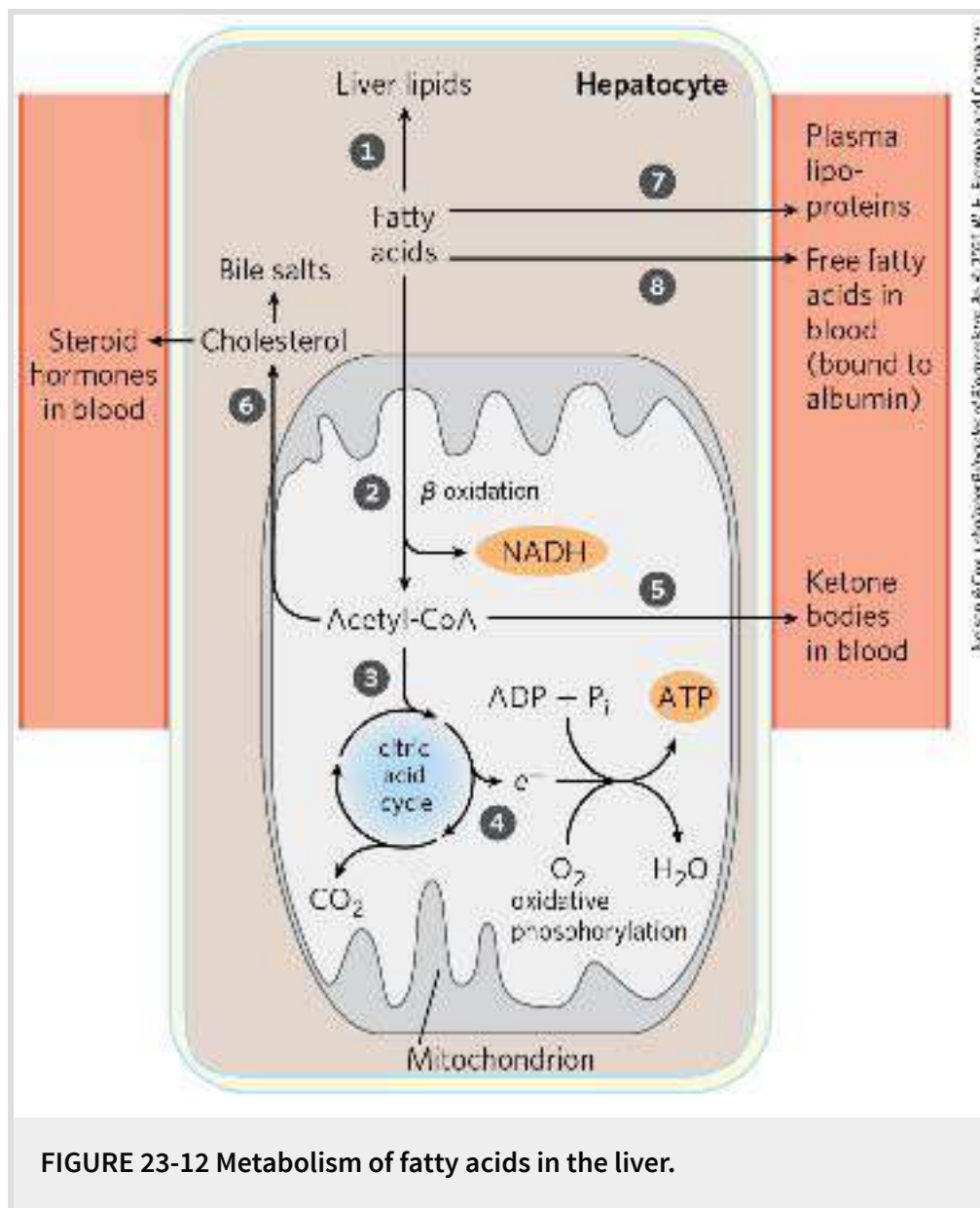


FIGURE 23-12 Metabolism of fatty acids in the liver.

The other two metabolic fates of lipids require specialized mechanisms for the transport of insoluble lipids in blood. **7** Fatty acids are converted to the phospholipids and TAGs of plasma lipoproteins, which carry lipids to adipose tissue for storage. **8** Some free fatty acids (FFAs) are bound to serum albumin and carried to the heart and skeletal muscles, which take up and oxidize FFAs as a major fuel. Serum albumin is the most abundant

plasma protein; one molecule can carry up to seven FFA molecules (see [Fig. 17-3](#)).

The liver thus serves as the body's distribution center, exporting nutrients in the correct proportions to other organs, smoothing out fluctuations in metabolism caused by intermittent food intake, and processing excess amino groups into urea and other products to be disposed of by the kidneys. Certain nutrients are stored in the liver, including iron ions and vitamin A. The liver also detoxifies xenobiotics. Detoxification often includes the cytochrome P-450–dependent hydroxylation of relatively insoluble organic compounds, making them sufficiently soluble for further breakdown and excretion (see [Box 21-1](#)).

Adipose Tissues Store and Supply Fatty Acids

There are two primary types of adipose tissue, white and brown ([Fig. 23-13](#)), with different roles, and we focus first on the more abundant of the two. [White adipose tissue \(WAT\)](#) is amorphous and widely distributed in the body: under the skin, around deep blood vessels, and in the abdominal cavity. The [adipocytes](#) of white adipose tissue are large (diameter 30 to 70 μm), spherical cells, completely filled with a single large lipid droplet that constitutes about 65% of the cell mass and squeezes the mitochondria and nucleus into a thin layer against the plasma membrane ([Fig. 23-13a, c](#)). The lipid droplet contains TAGs and

sterol esters and is coated with a monolayer of phospholipids, oriented with their polar head groups facing the cytosol. Specific proteins are associated with the surface of the droplets, including perilipin and the enzymes for synthesis and breakdown of TAGs (see [Fig. 17-2](#)). White adipose tissue typically makes up about 15% of the mass of a healthy young adult human. Adipocytes are metabolically active, responding quickly to hormonal stimuli in a metabolic interplay with the liver, skeletal muscles, and heart.

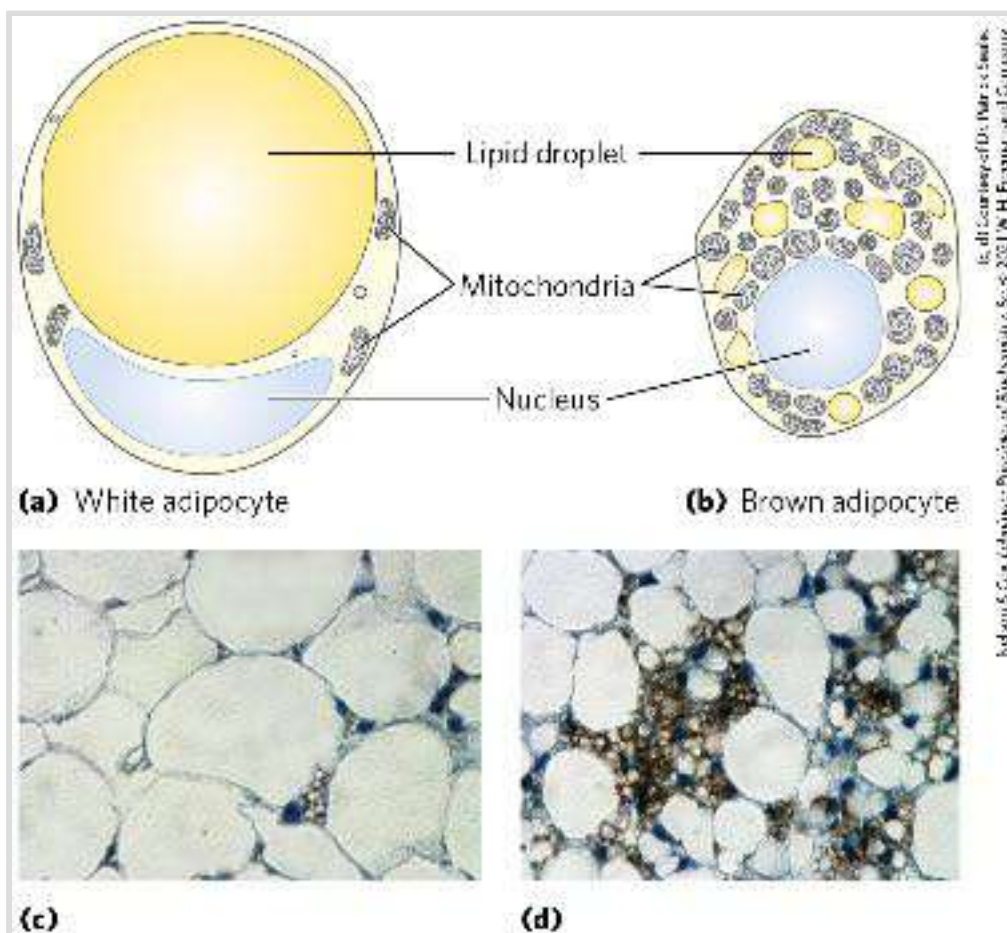


FIGURE 23-13 White and brown adipose tissue. Schematic views of typical mouse adipocytes from (a) white adipose tissue (WAT) and (b) brown adipose tissue (BAT). White adipocytes are larger and contain a single huge lipid droplet, which squeezes the mitochondria and nucleus against the plasma membrane. In brown adipocytes, mitochondria are much more prominent, the nucleus is near the center of the cell, and multiple small fat

droplets are present. Below are light micrographs of (c) adipocytes in white adipose tissue, stained to show nuclei, and (d) a region of mixed white and brown adipocytes, stained with an antibody specific to UCP1, the uncoupling protein responsible for thermogenesis.

Adipocytes have an active glycolytic metabolism, oxidize pyruvate and fatty acids via the citric acid cycle, and carry out oxidative phosphorylation. During periods of high carbohydrate intake, adipose tissue can convert glucose (via pyruvate and acetyl-CoA) to fatty acids, convert the fatty acids to TAGs, and store the TAGs as large lipid droplets — although in humans, much of the fatty acid synthesis occurs in hepatocytes. Adipocytes store TAGs arriving from the liver (carried in the blood as very-low-density lipoproteins) and from the intestinal tract (carried in chylomicrons), particularly after meals rich in fat.

When the demand for fuel rises (between meals, for example), lipases in adipocytes hydrolyze stored TAGs to release FFAs, which can travel in the bloodstream to skeletal muscle, the heart, and, during starvation, the liver. The release of fatty acids from adipocytes is greatly accelerated by epinephrine, which stimulates the cAMP-dependent phosphorylation of perilipin and thus gives lipases that are specific for tri-, di-, and monoacylglycerols access to TAGs in lipid droplets (see [Fig. 17-2](#)). Hormone-sensitive lipase is also stimulated by phosphorylation. Insulin counterbalances this effect of epinephrine, decreasing the activity of the lipase.

The breakdown and synthesis of TAGs in adipose tissue constitute a substrate cycle; up to 70% of the fatty acids released by the three lipases are reesterified in adipocytes, re-forming TAGs. Each cycle consumes ATP (used to activate the fatty acids as acyl-CoA esters), so the net effect of the substrate cycling is the breakdown of ATP and the accompanying release of heat. In adipose tissue, glycerol liberated by adipocyte lipases cannot be reused in the synthesis of TAGs, because adipocytes lack glycerol kinase. Instead, the glycerol phosphate required for TAG synthesis is made from pyruvate by glyceroneogenesis, requiring the action of the cytosolic PEP carboxykinase (see [Fig. 21-22](#)).

In addition to its central function as a fuel depot, adipose tissue plays an important role as an endocrine organ, producing and releasing hormones that signal the state of energy reserves and coordinate metabolism of fats and carbohydrates throughout the body. We return to this function in [Section 23.4](#) when we discuss the hormonal regulation of body mass.

Brown and Beige Adipose Tissues Are Thermogenic

In small vertebrates and hibernating animals, a significant proportion of the adipose tissue is **brown adipose tissue (BAT)**, distinguished from white adipose tissue by its smaller (diameter 20 to 40 μm), differently shaped (polygonal, not round) adipocytes ([Fig. 23-13b, d](#)). Like white adipocytes, brown

adipocytes store TAGs, but in several smaller lipid droplets per cell rather than as a single central droplet. Brown adipocytes have more mitochondria and a richer supply of capillaries and innervation than white adipocytes; it is the cytochromes of mitochondria and the hemoglobin in capillaries that give brown adipose tissue its characteristic color. A unique feature of brown adipocytes is their production of **uncoupling protein 1 (UCP1)**, also called thermogenin (see [Fig. 19-36](#)). This protein is responsible for one of the principal functions of brown adipose tissue: **thermogenesis**.

In brown adipocytes, fatty acids stored in lipid droplets are released, enter mitochondria, and undergo complete conversion to CO₂ by β oxidation and the citric acid cycle. The reduced FADH₂ and NADH so generated pass their electrons through the respiratory chain to molecular oxygen. In white adipocytes, protons pumped out of the mitochondria during electron transfer reenter the matrix through ATP synthase, with the energy of electron transfer conserved in ATP synthesis. In brown adipocytes, UCP1 provides an alternative route for the reentry of protons that bypasses ATP synthase. The energy of the proton gradient is thus dissipated as heat, which can maintain the body (especially the nervous system and viscera) at its optimal temperature when the ambient temperature is relatively low.

In the human fetus, differentiation of fibroblast preadipocytes into brown adipose tissue begins at the twentieth week of gestation, and at the time of birth, brown adipose tissue

represents 1% to 5% of total body mass. The brown fat deposits are located where the heat generated by thermogenesis can ensure that vital tissues — blood vessels to the head, major abdominal blood vessels, and the viscera, including the pancreas, adrenal glands, and kidneys — are not chilled as the newborn enters a world of lower ambient temperature ([Fig. 23-14a](#)).

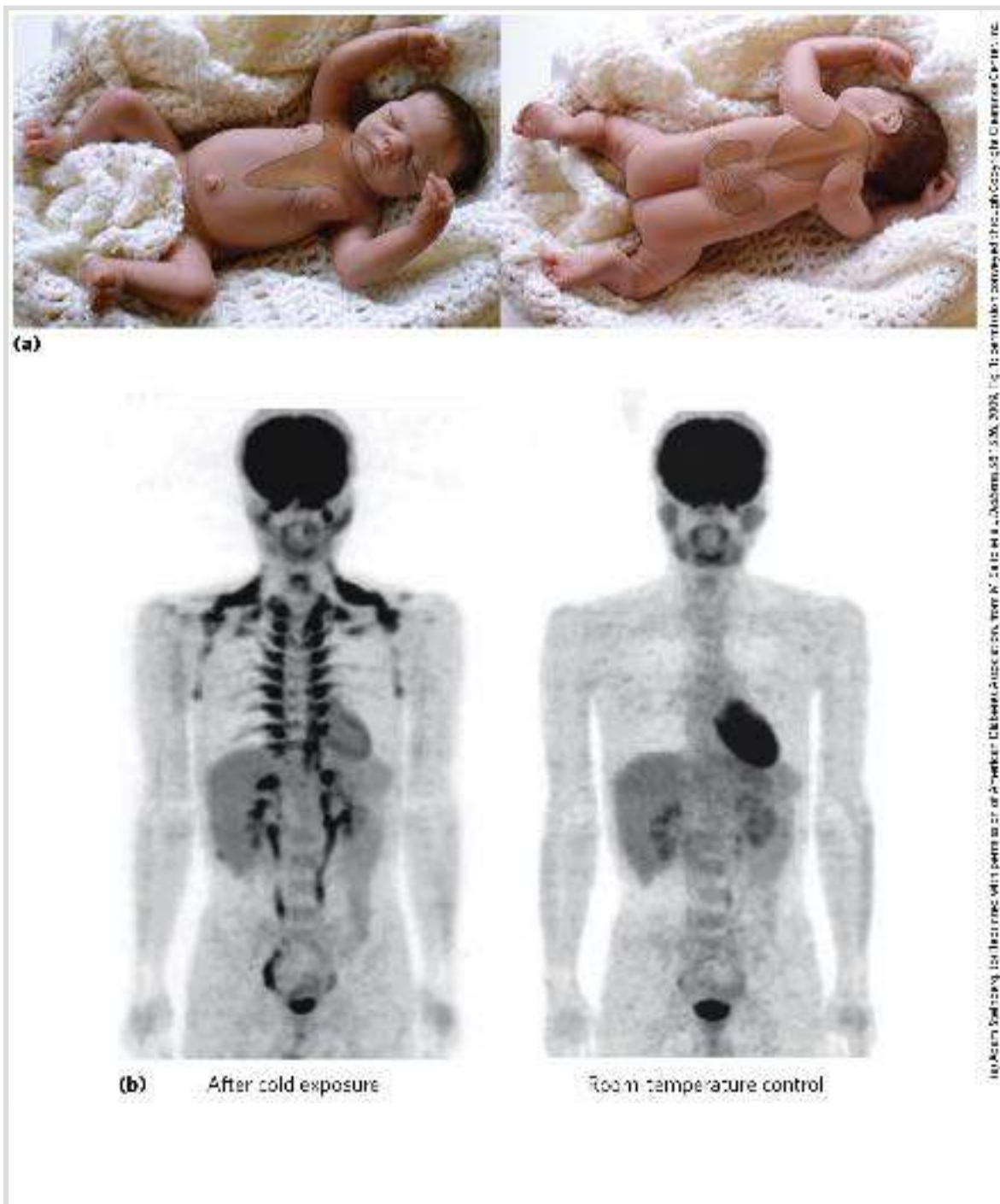


FIGURE 23-14 Brown adipose tissue in infants and adults. (a) At birth, human infants have brown fat distributed as shown here, to protect the spine, major blood vessels, and internal organs. (b) Positron emission tomography (PET) scanning can show metabolic activity in a living person, in real time. PET scans allow visualization of isotopically labeled glucose in precisely localized regions of the body. A positron-emitting glucose analog, 2- $[^{18}\text{F}]$ -fluoro-2-deoxy-D-glucose (FDG), is injected into the bloodstream; a short time later, a PET scan shows how much of the glucose has been taken up by each part of the body — a measure of metabolic activity. On the left is a PET scan of a healthy 25-year-old man who fasted for 12 hours, then stayed for 1 hour in a cold (19°C) room, with his legs on ice to thoroughly chill him. At the end of the hour, he was injected with $[^{18}\text{F}]$ -FDG, then remained under cold conditions for another hour. Whole-body PET scans were then done at 24°C . For the control scan, the same man underwent the same PET scan protocol two weeks later, but this time following 2 hours at 27°C instead of chilling (right). Intense labeling of the brain and heart shows high rates of glucose uptake; labeling of the kidneys and bladder indicates clearance of FDG. In the scan after chilling (left), $[^{18}\text{F}]$ -FDG labels brown adipose tissue in the region above the collarbone and along the vertebrae.

At birth, white adipose tissue development begins and brown adipose tissue begins to disappear. Young adult humans have much-diminished deposits of brown adipose tissue, ranging from 3% of all adipose tissue in males to 7% in females, making up less than 0.1% of body mass. However, adults have, distributed among their white adipose cells, significant numbers of adipocytes that can be converted by cold exposure or by β -adrenergic stimulation into cells very similar to brown adipocytes. These **beige adipocytes** have multiple lipid droplets, are richer in mitochondria than white adipocytes, and produce UCP1, so they function effectively as heat generators. Brown and beige adipocytes produce heat by oxidation of their own fatty acids, but they also take up and oxidize both fatty acids and glucose from the blood at rates out of proportion to their mass. In fact, the

detection of brown adipose tissue by PET scanning depends on the adipocytes' relatively high rate of *glucose* uptake and metabolism ([Fig. 23-14b](#)). In adaptation to warm or cold surroundings, and in the normal differentiation of white, brown, and beige adipose tissue, the nuclear transcription factor PPAR γ (described later in the chapter) plays a central role. And as we noted above, the peptide hormone irisin, produced in muscle by exercise, triggers the development of beige adipose tissue that continues to burn fuel long after the exercise ends.

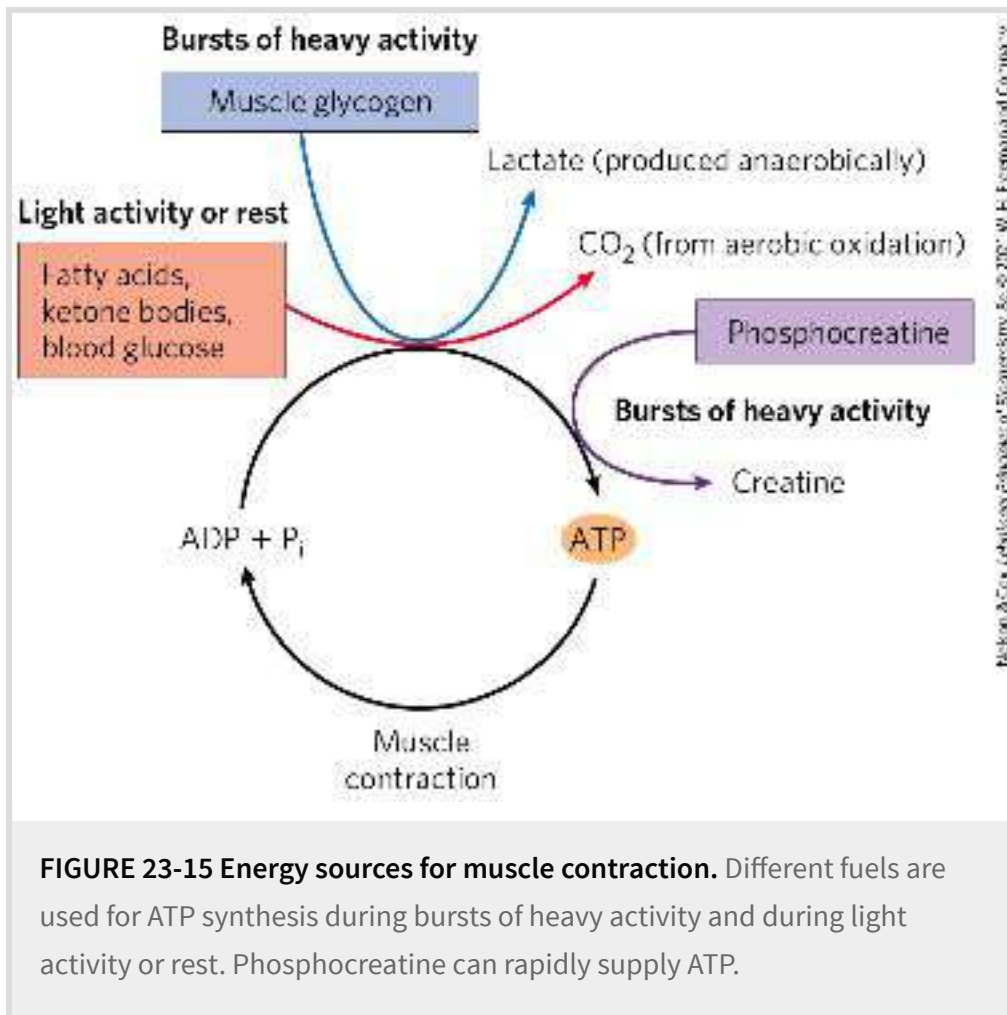
Muscles Use ATP for Mechanical Work

Metabolism in skeletal muscle cells — [myocytes](#) — is specialized to generate ATP as the immediate source of energy for contraction. Moreover, skeletal muscle is adapted to do its mechanical work intermittently, on demand. Sometimes skeletal muscles must work at their maximum capacity for a short time, as in a 100 m sprint; at other times more prolonged work is required, as in running a marathon or in prolonged physical labor.

There are two general classes of muscle tissue, which differ in physiological role and fuel utilization. **Slow-twitch muscle**, also called red muscle, provides relatively low tension but is highly resistant to fatigue. It produces ATP by the relatively slow but steady process of oxidative phosphorylation. Red muscle is very rich in mitochondria and is served by dense networks of blood vessels, which bring the oxygen essential to ATP production. **Fast-**

twitch muscle, or white muscle, has fewer mitochondria than red muscle and is less well supplied with blood vessels, but it can develop greater tension and do so faster. White muscle is quicker to fatigue because, when active, it uses ATP faster than it can replace it. There is a genetic component to the proportion of red and white muscle in any individual, but with training, an athlete can improve the endurance of fast-twitch muscle.

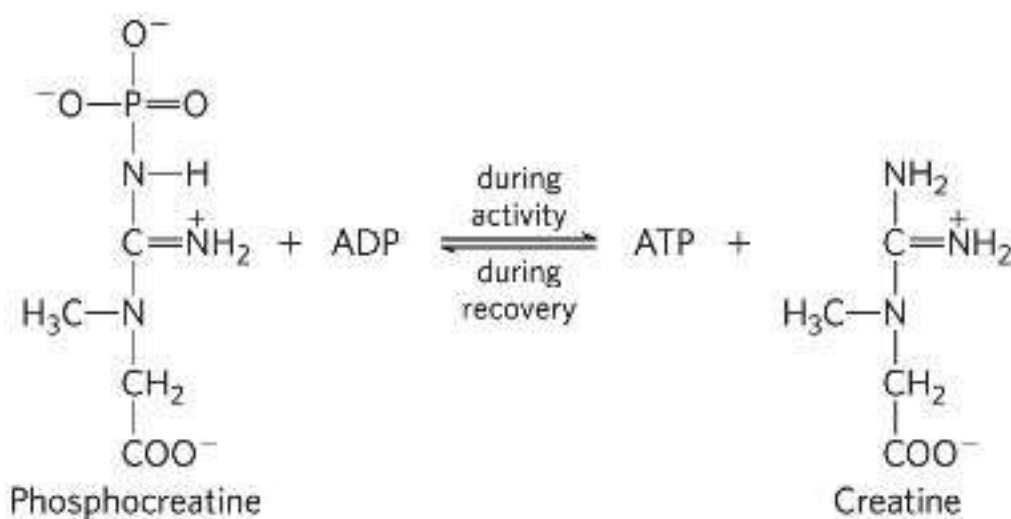
Skeletal muscle can use free fatty acids or glucose as fuel, depending on the degree of muscular activity ([Fig. 23-15](#)). In resting muscle, the primary fuels are FFAs from adipose tissue. These are oxidized and degraded to yield acetyl-CoA, which enters the citric acid cycle, ultimately yielding the energy for ATP synthesis by oxidative phosphorylation. Muscle in light activity uses blood glucose in addition to fatty acids. The glucose is phosphorylated, then degraded by glycolysis to pyruvate, which is converted to acetyl-CoA and oxidized via the citric acid cycle and oxidative phosphorylation.



In maximally active fast-twitch muscles, the demand for ATP is so great that the blood flow cannot provide O_2 and fuels fast enough to supply sufficient ATP by aerobic respiration alone. Under these conditions, stored muscle glycogen is broken down to lactate by fermentation ([p. 525](#)). Each glucose unit degraded yields three ATP, because phosphorolysis of glycogen produces glucose 6-phosphate (via glucose 1-phosphate), sparing the ATP normally consumed in the hexokinase reaction. Lactic acid fermentation thus responds more quickly than oxidative phosphorylation to an increased need for ATP, supplementing basal ATP production by aerobic oxidation of other fuels via the citric acid cycle and respiratory chain. The secretion of epinephrine, which stimulates

both the release of glucose from liver glycogen and the breakdown of glycogen in muscle tissue, greatly enhances the use of blood glucose and muscle glycogen as fuels for muscular activity. (Epinephrine mediates the so-called fight-or-flight response, discussed more fully below.)

The relatively small amount of glycogen (about 1% of the total weight of skeletal muscle) limits the glycolytic energy available during all-out exertion. Moreover, the accumulation of lactate and consequent decrease in pH in maximally active muscles reduces their efficiency. Skeletal muscle, however, contains another source of ATP, phosphocreatine (10 to 30 mM), which can rapidly regenerate ATP from ADP by the creatine kinase reaction:



During periods of active contraction and glycolysis, this reaction proceeds predominantly in the direction of ATP synthesis; during recovery from exertion, the same enzyme resynthesizes phosphocreatine from creatine and ATP. Because of the relatively high levels of ATP and phosphocreatine in muscle, these

compounds can be detected in intact muscle, in real time, by NMR spectroscopy ([Fig. 23-16](#)). Creatine serves to shuttle ATP equivalents from the mitochondrion to sites of ATP consumption and can be the limiting factor in the development of new muscle tissue ([Box 23-1](#)).

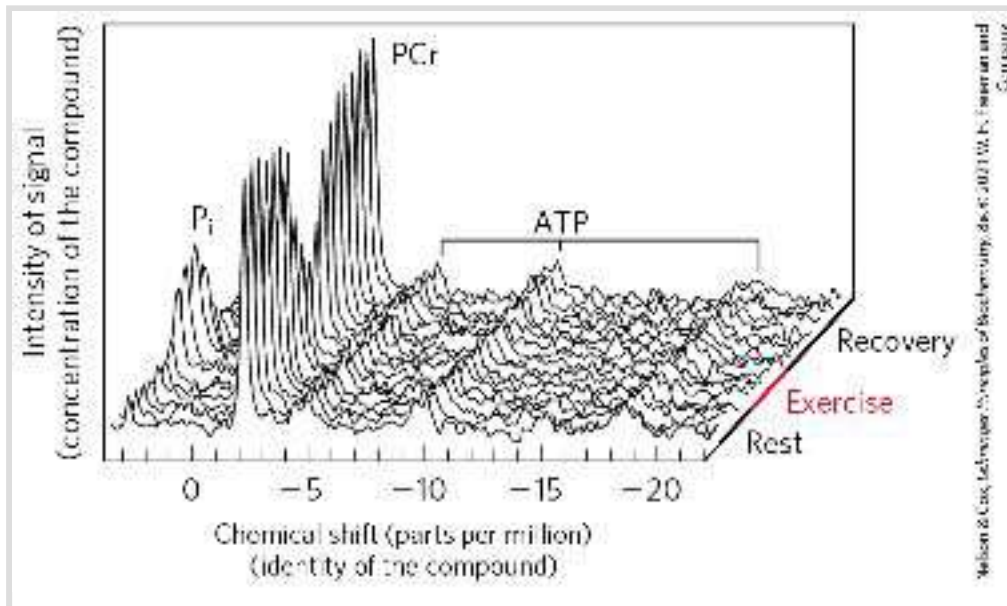
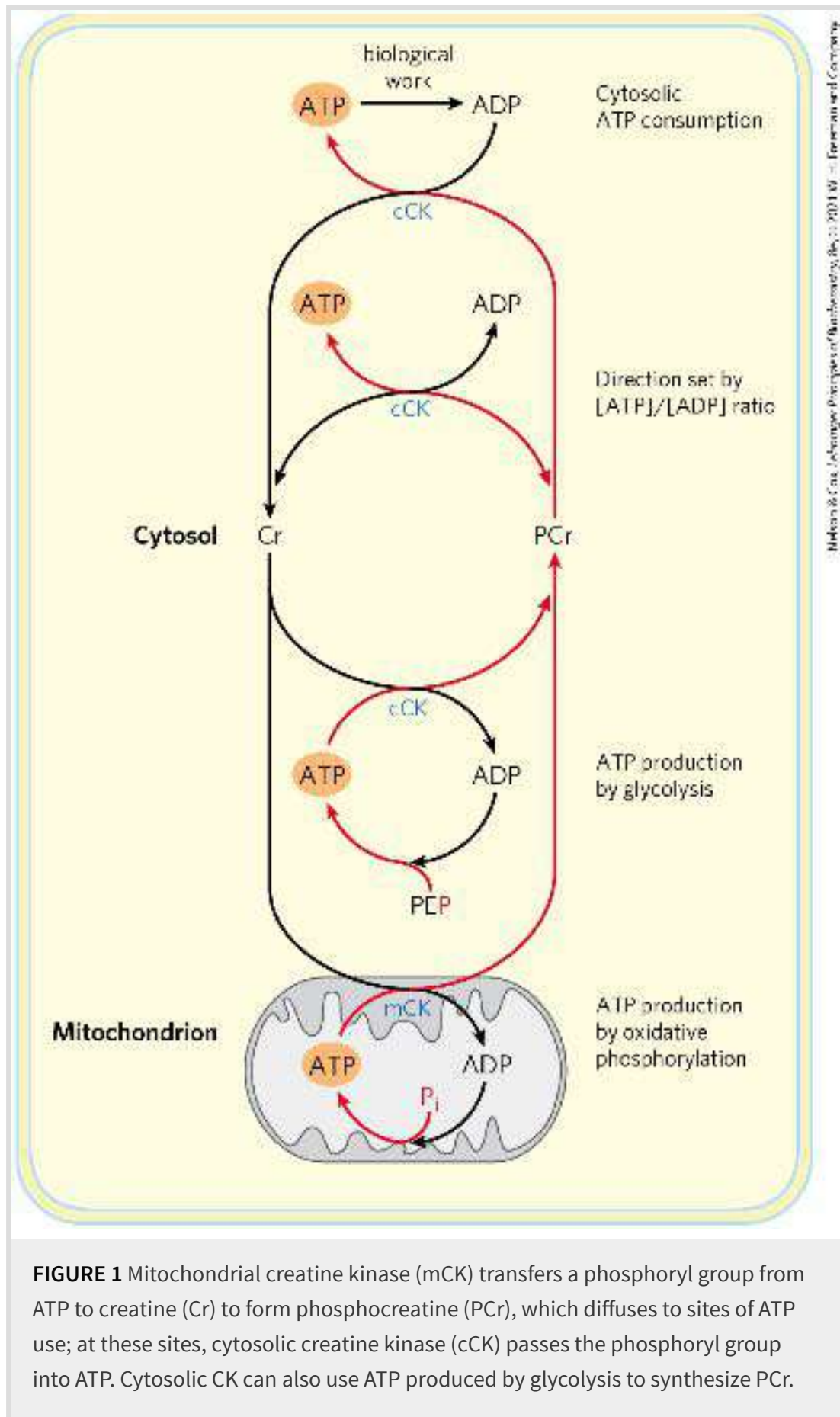


FIGURE 23-16 Phosphocreatine buffers ATP concentration during exercise. A “stack plot” of magnetic resonance spectra (of ^{31}P) shows inorganic phosphate (P_i), phosphocreatine (PCr), and ATP (each of its three phosphates giving a signal). The series of plots represents the passage of time, from a period of rest to one of exercise, and then of recovery. Notice that the ATP signal barely changes during exercise, kept high by continued respiration and by the reservoir of phosphocreatine, which diminishes during exercise. During recovery, when ATP production by catabolism is greater than ATP use by the (now resting) muscle, the phosphocreatine reservoir is refilled. [Data from M. L. Blei, K. E. Conley, and M. J. Kushmerick, *J. Physiol.* 465:203, 1993, Fig. 4.]

BOX 23-1

Creatine and Creatine Kinase: Invaluable Diagnostic Aids and the Muscle Builder's Friends

Animal tissues that have a high and fluctuating need for ATP, primarily skeletal muscle, cardiac muscle, and brain, contain several isozymes of creatine kinase. A cytosolic isozyme (cCK) is present in regions of high ATP use (myofibrils and sarcoplasmic reticulum, for example). By converting ADP produced during periods of high ATP use back to ATP, cCK prevents the accumulation of ADP to concentrations that could inhibit ATP-using enzymes by mass action. Another isozyme of creatine kinase is located in regions where the inner and outer membranes of mitochondria come into contact. This mitochondrial isozyme (mCK) probably serves to shuttle ATP equivalents produced in mitochondria to cytosolic sites of ATP use ([Fig. 1](#)). The species that diffuses from the mitochondrion to ATP-consuming activities in the cytosol is therefore creatine phosphate, not ATP. The mCK isozyme co-localizes with the adenine nucleotide transporter (in the inner mitochondrial membrane) and porin (in the outer mitochondrial membrane), which suggests that these three components may function together to transport ATP equivalents formed in mitochondria into the cytosol.



During periods of little ATP demand, the pools of ATP and PCr are replenished in preparation for the next period of intense demand for ATP. In resting muscle, the concentration of PCr is three to five times that of ATP, buffering the cell against rapid depletion of ATP during short bursts of ATP demand. [Information from U. Schlattner et al., *Biochim. Biophys. Acta* 1762:164, 2006, Fig. 1.]

In knockout mice lacking the mitochondrial isozyme, myocytes compensate by producing more mitochondria, closely associated with myofibrils and sarcoplasmic reticulum, allowing quick diffusion of mitochondrial ATP to the sites of ATP use. Nevertheless, these mice have a reduced capacity for running, indicating a defect in some aspect of energy-supplying metabolism.

Creatine and phosphocreatine spontaneously break down to form creatinine ([Fig. 2](#)). To maintain high creatine levels, these losses have to be compensated for, either by dietary creatine, obtained primarily from meat (muscle) and dairy products, or by de novo synthesis from glycine, arginine, and methionine (see [Fig. 22-28](#)), which occurs primarily in liver and kidney. De novo synthesis of creatine is a major consumer of these amino acids, particularly in vegans, for whom this is the only source of creatine; plants do not contain creatine. Muscle tissue has a specific system to take up creatine (exported by liver or kidney) from the blood, against a substantial concentration gradient. Efficient uptake of dietary creatine requires continuous exercise; without exercise, creatine supplementation is of little value.

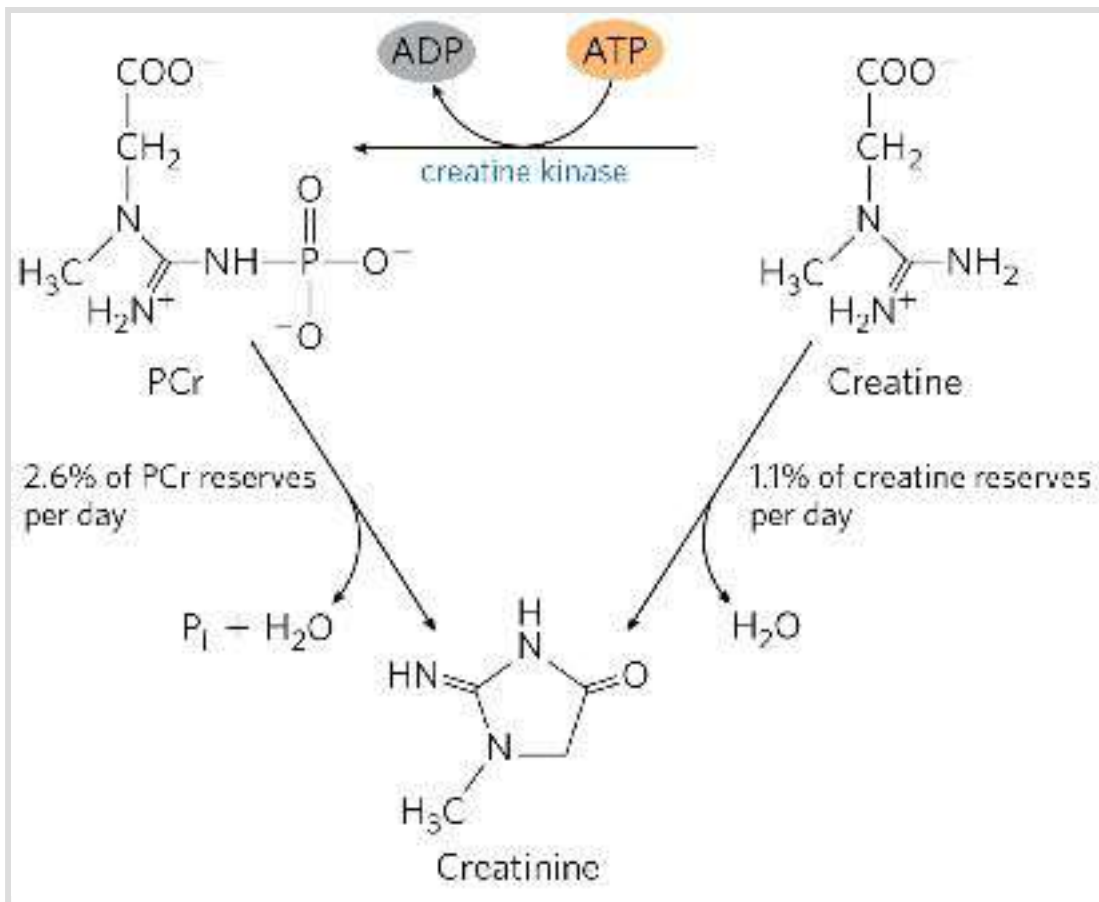


FIGURE 2 Spontaneous (nonenzymatic) formation of creatinine from phosphocreatine or creatine consumes a few percent of the body's total creatine per day, which must be replaced by biosynthesis or from the diet.



Heart muscle contains a unique isozyme of creatine kinase (MB, for myocardial band), which is not normally found in the blood but appears there when released from heart muscle damaged by a heart attack. The blood level of MB begins to rise within 2 hours of the heart attack, typically peaks 12 to 36 hours after the heart attack, and returns to normal levels in 3 to 5 days. Measurement of the MB isozyme in blood therefore confirms a diagnosis of heart attack and indicates approximately when it occurred.

Children with inborn errors in the enzymes of creatine synthesis or uptake suffer severe intellectual disability and seizures. They have much-reduced levels of brain creatine as measured by NMR (see [Fig. 23-16](#)). Creatine supplementation raises their brain creatine and creatine phosphate concentrations and brings about partial improvement of the symptoms.

In the healthy kidney, creatinine from creatine breakdown is efficiently cleared from the blood into the urine. When renal function is defective, creatinine levels in the blood rise above the normal range of 0.8 to 1.4 mg/dL. Elevated blood creatinine is associated with renal failure in diabetes and other conditions in which renal function is temporarily or permanently compromised. Renal clearance of creatinine varies slightly with age, ethnicity, and gender, so correcting the calculation for those factors yields a more sensitive measure of the extent of renal function, the **glomerular filtration rate (GFR)**. ■

Body builders who are adding muscle mass have a greater need for creatine and commonly take creatine supplements of up to 20 g per day for a few days, followed by lower maintenance doses. The combination of exercise and creatine supplementation increases muscle mass ([Fig. 3](#)) and improves performance in high-intensity, short-duration work.



FIGURE 3 Many body builders take supplemental creatine to supply phosphocreatine in new muscle tissue.

After a period of intense muscular activity, the individual continues breathing heavily for some time, using much of the extra O_2 for oxidative phosphorylation in the liver. The ATP produced is used for gluconeogenesis (in the liver) from lactate that has been carried in the blood from the muscles. The glucose thus formed returns to the muscles to replenish their glycogen, completing the Cori cycle ([Fig. 23-17](#); see also [Box 15-1](#)).

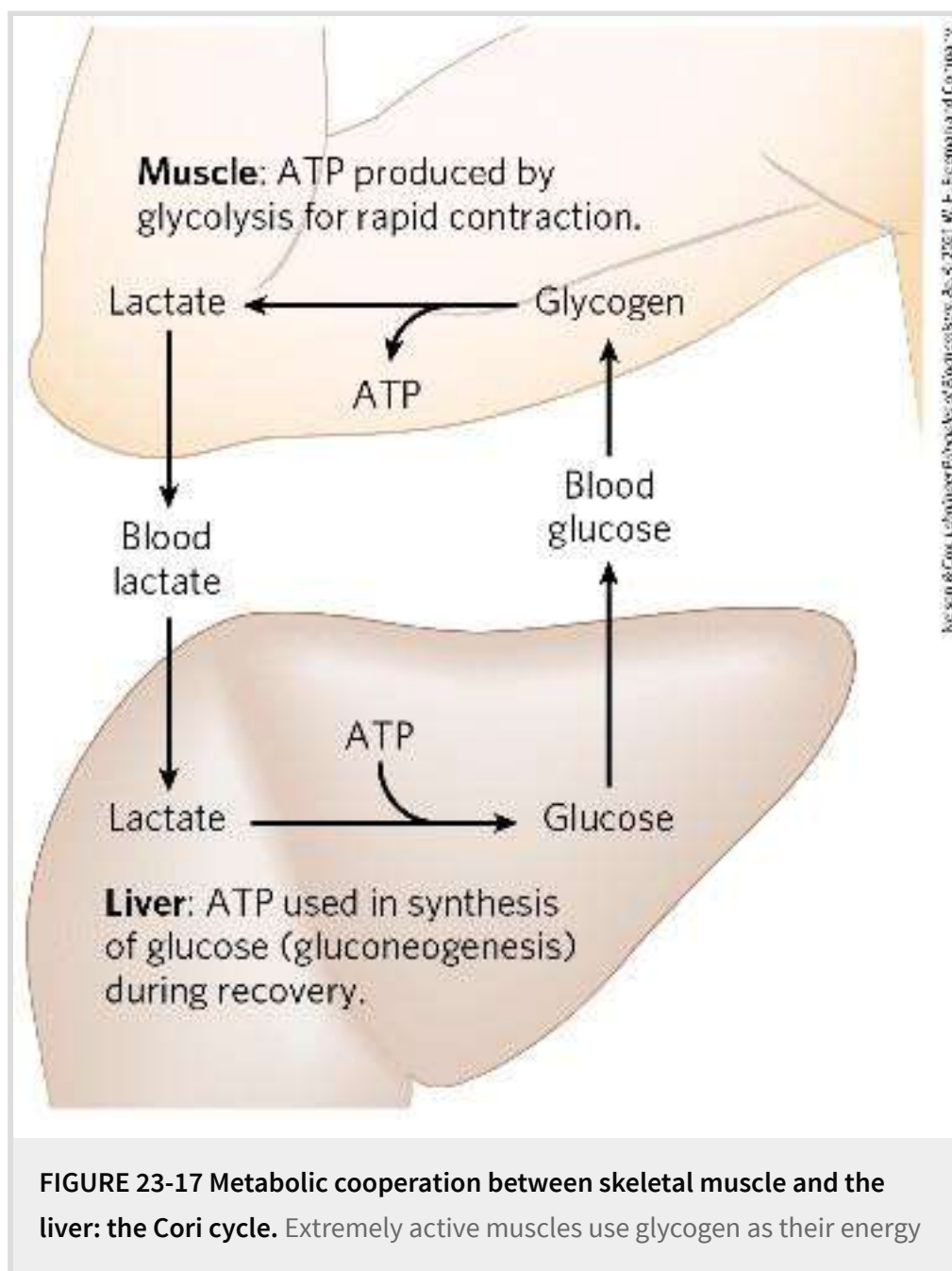


FIGURE 23-17 Metabolic cooperation between skeletal muscle and the liver: the Cori cycle. Extremely active muscles use glycogen as their energy

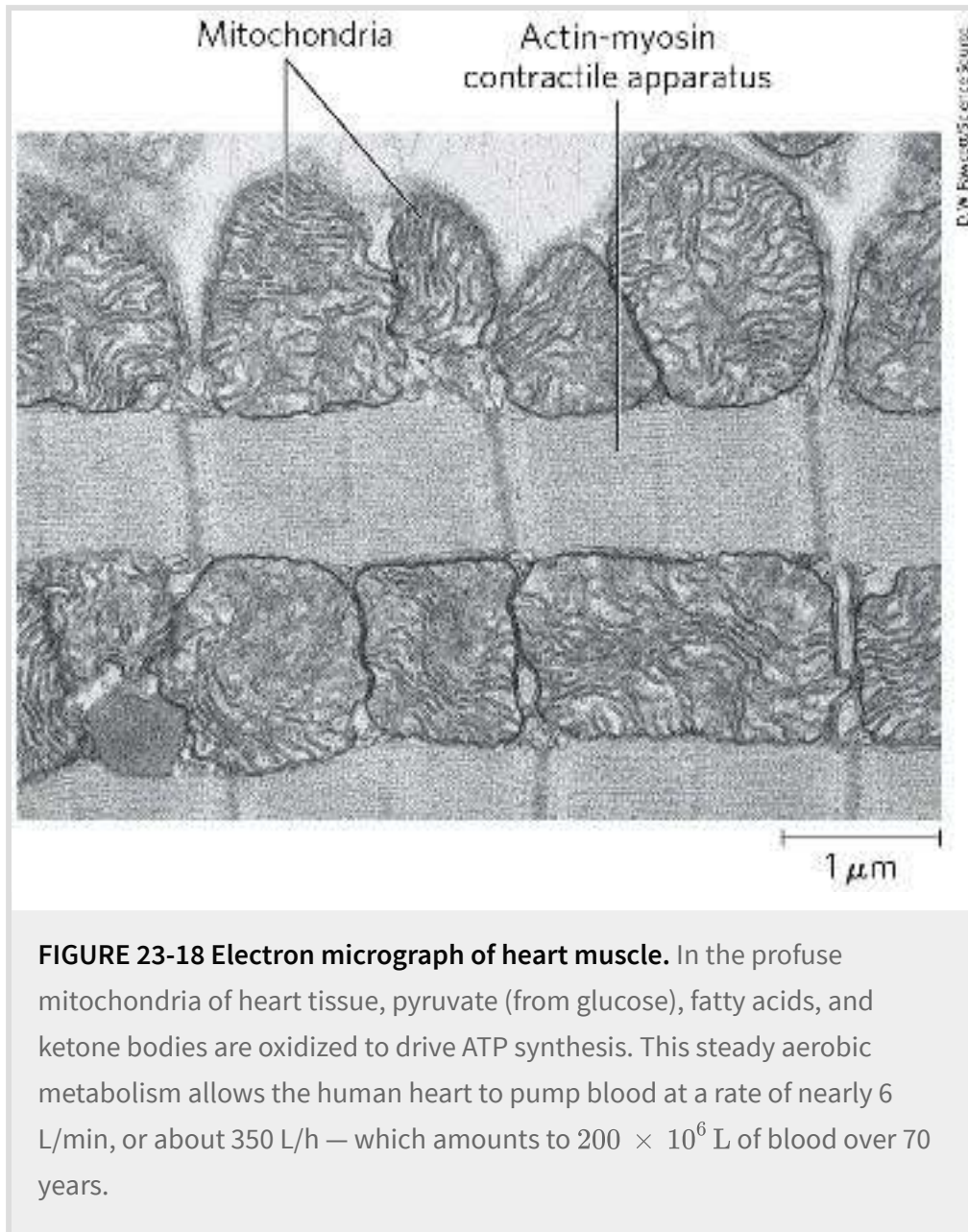
source, generating lactate by glycolysis. During recovery, some of the lactate is transported to the liver and converted to glucose by gluconeogenesis. The glucose is released to the blood and returned to the muscles to replenish their glycogen stores. The overall pathway, glucose → lactate → glucose, constitutes the Cori cycle.

Actively contracting skeletal muscle generates heat as a byproduct of imperfect coupling of the chemical energy of ATP with the mechanical work of contraction. This heat production can be put to good use when ambient temperature is low: skeletal muscle carries out **shivering thermogenesis**, rapidly repeated muscle contraction that produces heat but little motion, helping to maintain the body at its preferred temperature of 37 °C.



Heart muscle differs from skeletal muscle in that it is continuously active in a regular rhythm of contraction and relaxation, and has a completely aerobic metabolism at all times. Mitochondria are much more abundant in heart muscle than in skeletal muscle, making up almost half the volume of the cells ([Fig. 23-18](#)). The heart uses mainly FFAs as a source of energy, but also some glucose and ketone bodies taken up from the blood; these fuels are oxidized aerobically to generate ATP. Like skeletal muscle, heart muscle does not store lipids or glycogen in large amounts. It does have small amounts of reserve energy in the form of phosphocreatine, enough for a few seconds of contraction. Because the heart is normally aerobic and obtains its energy from oxidative phosphorylation, the failure of O₂ to reach part of the heart muscle when the blood vessels are blocked by lipid deposits (atherosclerosis) or blood clots (coronary

thrombosis) can cause that region of the heart muscle to die. This is what happens in myocardial infarction (heart attack). ■



The Brain Uses Energy for Transmission of Electrical Impulses

The metabolism of the brain is remarkable in several respects. The neurons of the adult mammalian brain normally use only glucose as fuel ([Fig. 23-19](#)). (Astrocytes, the other major cell type in the brain, can oxidize fatty acids.) The brain, which constitutes about 2% of total body mass, has a very active respiratory metabolism ([Fig. 23-15](#)); more than 90% of the ATP produced in the neurons comes from oxidative phosphorylation. The brain uses O_2 at a fairly constant rate, accounting for almost 20% of the total O_2 consumed by the body at rest. Because the brain contains very little glycogen, it is constantly dependent on incoming glucose in the blood. Should blood glucose fall significantly below a critical level for even a short time, severe and sometimes irreversible changes in brain function may result.

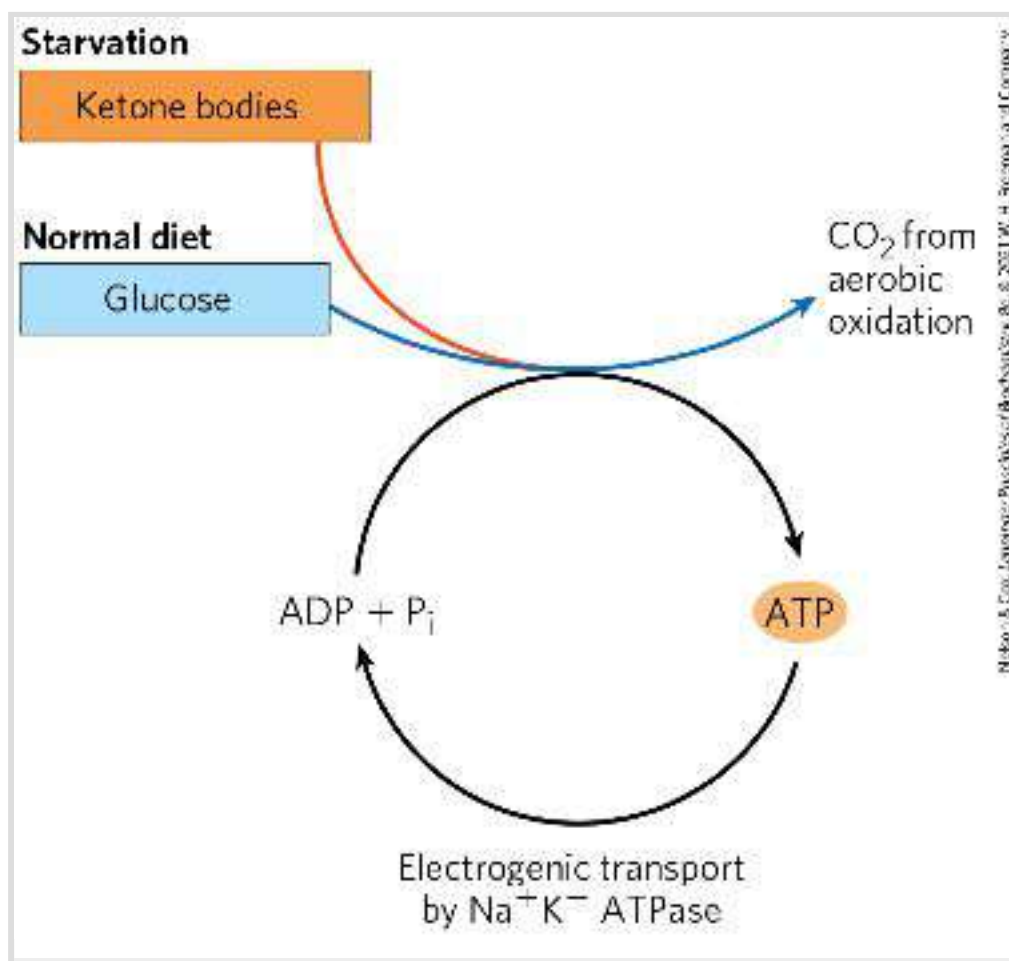


FIGURE 23-19 The fuels that supply ATP in the brain. The energy source used by the brain varies with nutritional state. The ketone body used during starvation is β -hydroxybutyrate. Electrogenic transport by the Na^+K^+ ATPase maintains the transmembrane potential essential to information transfer among neurons.



Although the neurons of the brain cannot directly use free fatty acids or lipids from the blood as fuels, they can, when necessary, get up to 60% of their energy requirement from the oxidation of β -hydroxybutyrate (a ketone body), formed in the liver from fatty acids. The capacity of the brain to oxidize β -hydroxybutyrate via acetyl-CoA becomes important during prolonged fasting or starvation, after liver glycogen has been depleted, because it allows the brain to use body fat as an energy source. This spares muscle proteins — until they become the brain's ultimate source of glucose, via gluconeogenesis in the liver, during severe starvation.

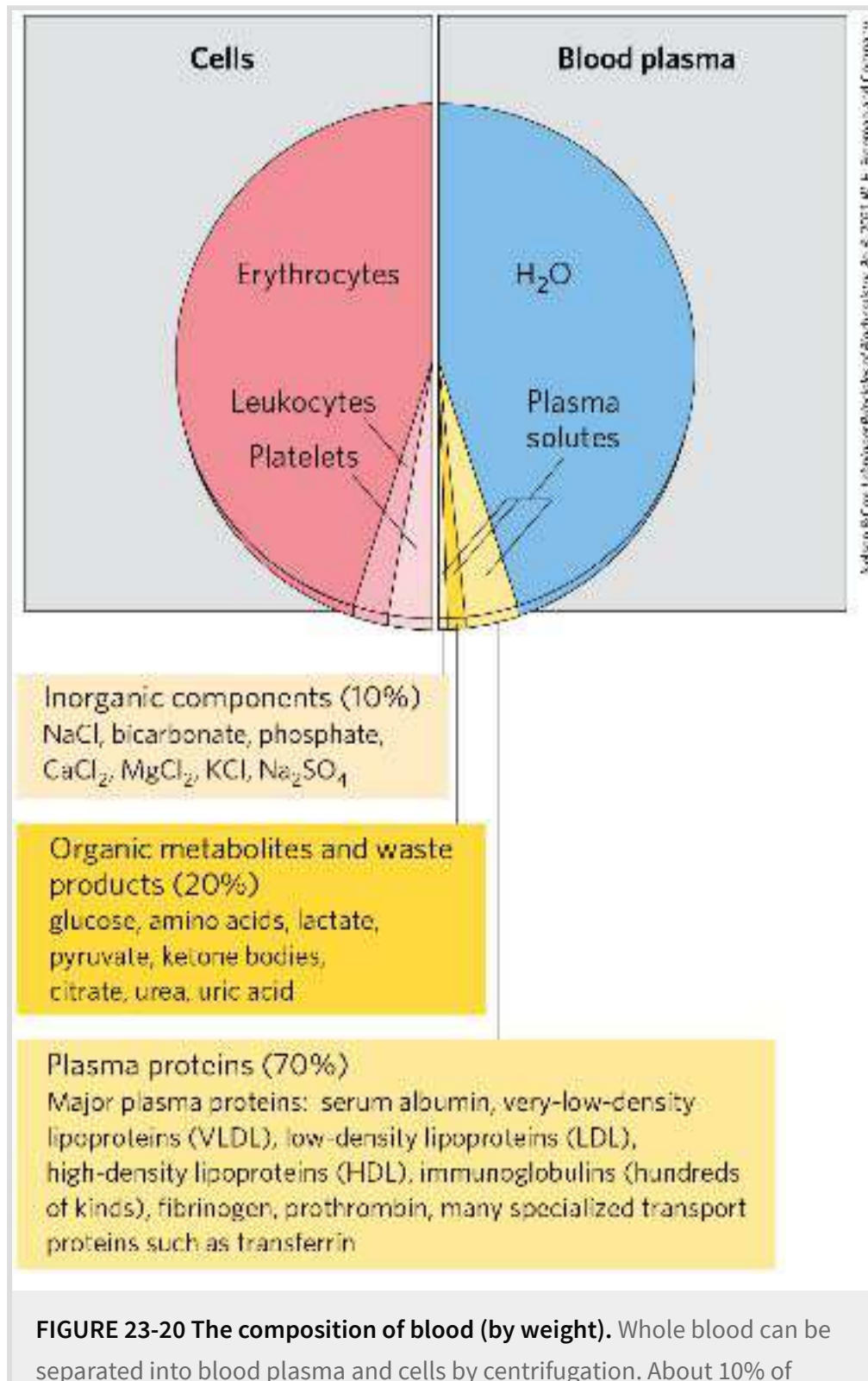
In neurons, energy is required to create and maintain an electrical potential across the plasma membrane. The membrane contains an electrogenic ATP-driven antiporter, the Na^+K^+ ATPase, which simultaneously pumps two K^+ ions into and three Na^+ ions out of the neuron (see [Fig. 11-39](#)). The resulting transmembrane potential changes transiently as an electrical signal, an **action potential**, sweeps from one end of a neuron to the other (see [Fig. 12-33](#)). Action potentials are the chief mechanism of information transfer in the nervous system, so depletion of ATP in neurons would have disastrous effects on all activities coordinated by neuronal signaling.

Blood Carries Oxygen, Metabolites, and Hormones

 Blood mediates the metabolic interactions among all tissues. It transports nutrients from the small intestine to the liver and from the liver and adipose tissue to other organs; it also transports waste products from extrahepatic tissues to the liver for processing and to the kidneys for excretion. Oxygen moves in the bloodstream from the lungs to the tissues, and CO₂ generated by tissue respiration returns via the bloodstream to the lungs for exhalation. Blood also carries hormonal signals from one tissue to another. In its role as signal carrier, the circulatory system resembles the nervous system: both regulate and integrate the activities of different organs.

The average adult human has 5 to 6 L of blood. Almost half of this volume is occupied by three types of blood cells ([Fig. 23-20](#)): **erythrocytes** (red cells), filled with hemoglobin and specialized for carrying O₂ and CO₂; much smaller numbers of **leukocytes** (white cells) of several types (including **lymphocytes**, also found in lymphatic tissue), which are central to the immune system to defend against infections; and **platelets** (cell fragments), which help to mediate blood clotting. The liquid portion is the **blood plasma**, which is 90% water and 10% solutes. Dissolved or suspended in the plasma are many proteins, lipoproteins, nutrients, metabolites, waste products, inorganic ions, and hormones. More than 70% of the plasma solids are [plasma proteins](#), primarily immunoglobulins (circulating antibodies),

serum albumin, apolipoproteins (for lipid transport), transferrin (for iron transport), and blood-clotting proteins such as fibrinogen and prothrombin.



blood plasma is solutes, about 10% of these consisting of inorganic salts, 20% small organic molecules, and 70% plasma proteins. The major dissolved components are listed here. Blood contains many other substances, often in trace amounts. These include other metabolites, enzymes, hormones, vitamins, trace elements, and bile pigments. Measurements of the concentrations of components in blood plasma are important in the diagnosis and treatment of many diseases.

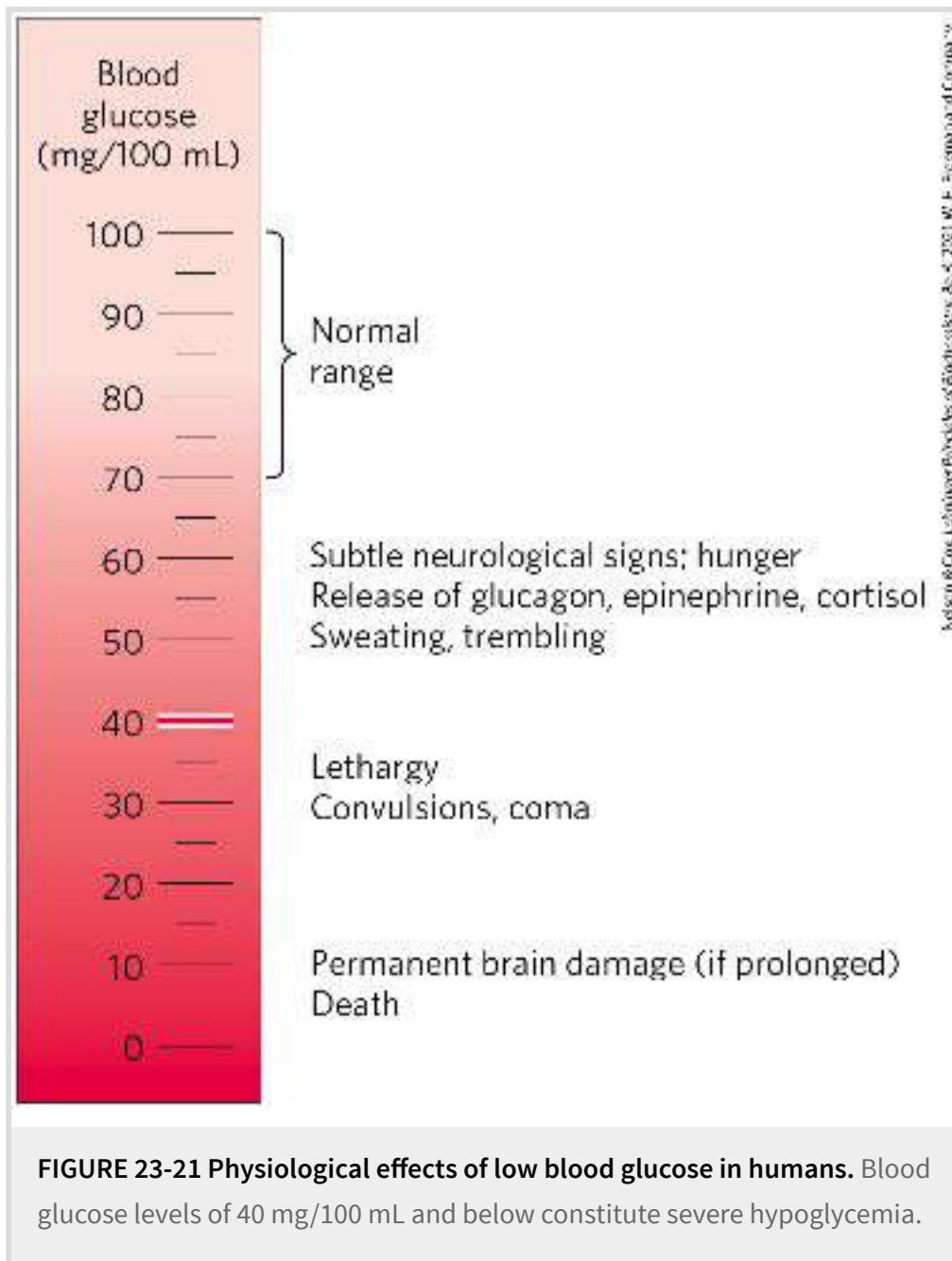
The ions and low molecular weight solutes in blood plasma are not fixed components; they are in constant flux between blood and various tissues. Dietary uptake of the inorganic ions that are the predominant electrolytes of blood and cytosol (Na^+ , K^+ , and Ca^{2+}) is, in general, counterbalanced by their excretion in the urine. For many blood components, something near a dynamic steady state is achieved: the concentration of a component changes little, although a continuous flux occurs between the digestive tract, blood, and urine. The plasma levels of Na^+ , K^+ , and Ca^{2+} remain close to 140, 5, and 2.5 mM, respectively, with little change in response to dietary intake. Any significant departure from these values can result in serious illness or death. The kidneys play an especially important role in maintaining ion balance by selectively filtering waste products and excess ions out of the blood while preventing the loss of essential nutrients and ions.

The human erythrocyte loses its nucleus and mitochondria during differentiation. It therefore relies on glycolysis alone for its supply of ATP. The lactate produced by glycolysis returns to the liver, where gluconeogenesis converts it to glucose, to be stored as

glycogen or recirculated to peripheral tissues. The erythrocyte has constant access to glucose in the bloodstream.



The concentration of glucose in plasma is subject to tight regulation. We have noted the constant requirement of the brain for glucose and the role of the liver in maintaining blood glucose in the normal range, 60 to 90 mg/100 mL of whole blood (~4.5 mM). (Because erythrocytes make up a significant fraction of blood volume, their removal by centrifugation leaves a supernatant fluid, the plasma, containing the “blood glucose” in a smaller volume. To convert blood glucose to plasma glucose concentration, multiply the blood glucose level by 1.14.) When blood glucose in a human drops to 70 mg/100 mL (the hypoglycemic condition), the person experiences discomfort and mental confusion ([Fig. 23-21](#)); further reductions lead to coma, convulsions, and, in extreme hypoglycemia, death. Maintaining the normal concentration of glucose in blood is therefore a high priority, and a variety of regulatory mechanisms have evolved to achieve that end. Among the most important regulators of blood glucose are the hormones insulin, glucagon, and epinephrine, as discussed in the next section. ■



SUMMARY 23.2 *Tissue-Specific Metabolism*

■ In mammals there is a division of metabolic labor among specialized tissues and organs. The liver is the central processing and distribution organ for nutrients. Glucose 6-phosphate may be used to synthesize glycogen or fatty acids, or it may be sent into

the citric acid cycle to produce ATP, or into the pentose phosphate pathway to yield NADPH and pentoses. Amino acids are used to synthesize liver and plasma proteins, or their carbon skeletons are converted to glucose and glycogen by gluconeogenesis; the ammonia formed by deamination is converted to urea. Fatty acids in the liver may undergo β oxidation, or be converted to triglycerides, phospholipids or cholesterol, or be converted into ketone bodies.

■ White adipose tissue stores large reserves of triacylglycerol molecules and releases them into the blood in response to epinephrine or glucagon. Brown and beige adipose tissue are specialized for thermogenesis, the result of fatty acid oxidation in uncoupled mitochondria.

■ Skeletal muscle is specialized to produce and use ATP for mechanical work, oxidizing fatty acids and glucose during low to moderate muscular activity. During strenuous muscular activity, glycogen is the ultimate fuel, supplying ATP through glycolysis and fermentation to lactate. Phosphocreatine directly replenishes ATP during active contraction. Heart muscle obtains nearly all its ATP from oxidative phosphorylation, with fatty acids as the primary fuel.

■ The neurons of the brain use only glucose and β -hydroxybutyrate as fuels, the latter being important during fasting or starvation. The brain uses most of its ATP for the active transport of Na^+ and K^+ to maintain the electrical potential across the neuronal membrane.

■ The blood transfers nutrients, oxygen, waste products, and hormonal signals among tissues and organs. It is made up of cells (erythrocytes, leukocytes, and platelets) and electrolyte-rich water (plasma) containing many dissolved proteins.

23.3 Hormonal Regulation of Fuel Metabolism



The minute-by-minute adjustments that keep the blood glucose level near 4.5 mM involve the combined actions of insulin, glucagon, epinephrine, and cortisol on metabolic processes in many body tissues, but especially in liver, muscle, and adipose tissue. Insulin signals to these tissues that blood glucose is higher than necessary; as a result, cells take up excess glucose from the blood and convert it to glycogen and triacylglycerols for storage. Glucagon signals that blood glucose is too low, and tissues respond by producing glucose through glycogen breakdown and (in the liver) gluconeogenesis, and by oxidizing fats to reduce the need for glucose. Epinephrine is released into the blood to prepare the muscles, lungs, and heart for a burst of activity. Cortisol mediates the body's response to longer-term stresses.

In this section, we discuss these hormonal regulations in the context of three normal metabolic states — well fed, fasted, and starving.

Insulin Counters High Blood Glucose in the Well-Fed State

After a high-carbohydrate meal, blood glucose rises several-fold. In response, insulin is released and acts through its plasma membrane receptors in muscle and liver (see [Figs. 12-22, 12-23](#)) to

stimulate glucose uptake, phosphorylation, and oxidation via glycolysis and the citric acid cycle ([Table 23-3](#)). In response to insulin, GLUT4 glucose transporters sequestered in intracellular vesicles move to the plasma membrane, dramatically increasing the uptake of glucose from the blood (see [Box 11-1](#)). In the liver, insulin activates glycogen synthase and inactivates glycogen phosphorylase, so that much of the glucose 6-phosphate derived from blood glucose is channeled into glycogen.

TABLE 23-3 Effects of Insulin on Blood Glucose: Uptake of Glucose by Cells and Storage as Triacylglycerols and Glycogen

Metabolic effect	Target enzyme
↑ Glucose uptake (muscle, adipose tissue)	↑ Glucose transporter (GLUT4)
↑ Glucose uptake (liver)	↑ Glucokinase (increased expression)
↑ Glycogen synthesis (liver, muscle)	↑ Glycogen synthase
↓ Glycogen breakdown (liver, muscle)	↓ Glycogen phosphorylase
↑ Glycolysis, acetyl-CoA production (liver, muscle)	↑ PFK-1 (by PFK-2) ↑ Pyruvate dehydrogenase complex
↑ Fatty acid synthesis (liver)	↑ Acetyl-CoA carboxylase
↑ Triacylglycerol synthesis (adipose tissue)	↑ Lipoprotein lipase

Insulin also stimulates the storage of excess fuel as fat in adipose tissue ([Fig. 23-22](#)). Excess acetyl-CoA not needed for energy production via the citric acid cycle is used for fatty acid synthesis.

These fatty acids are converted to TAGs in the liver and exported to adipose tissue as components of plasma lipoproteins (VLDL; see [Fig. 21-40](#)). In adipose tissue, TAGs are released from VLDL as fatty acids, which are taken up by adipocytes and reconverted to TAGs for storage in response to insulin stimulation.

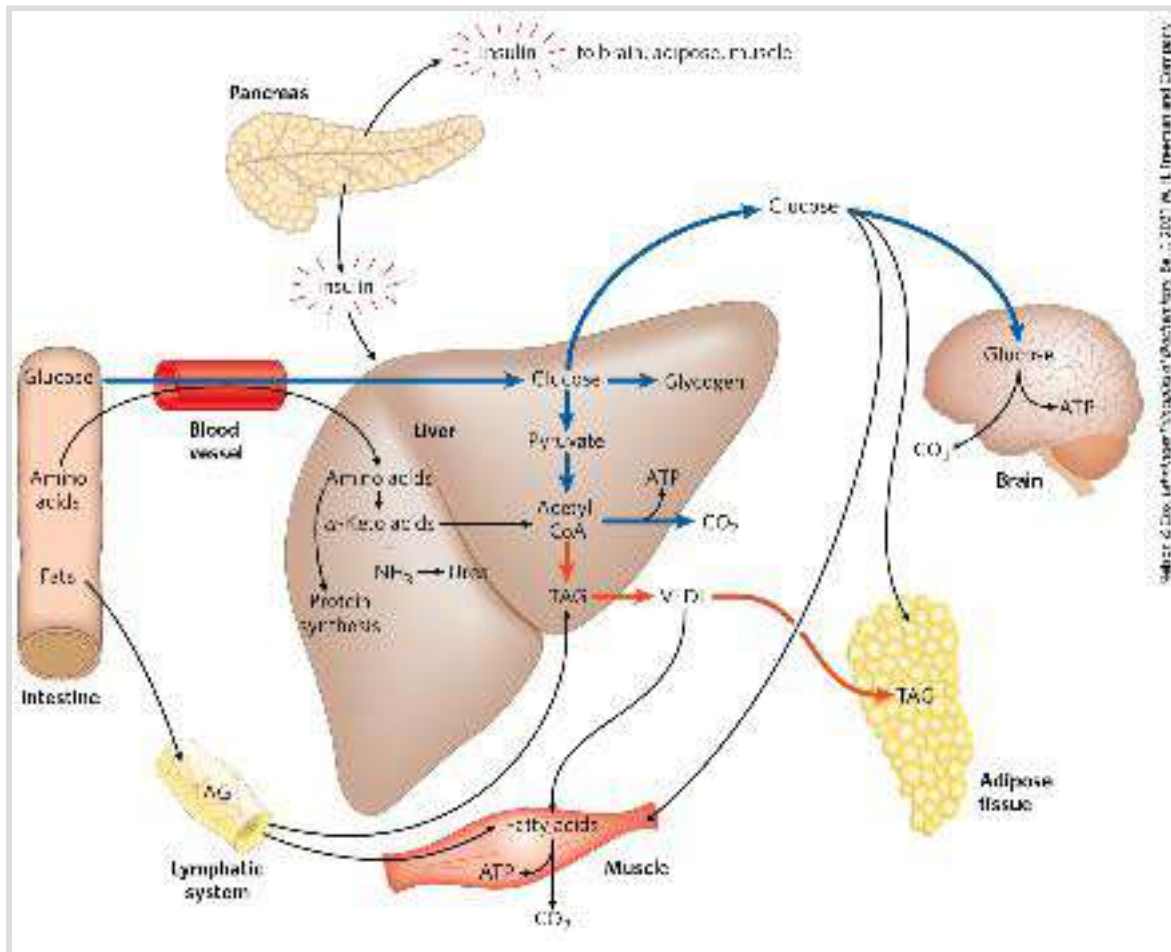


FIGURE 23-22 The well-fed state: the lipogenic liver. Immediately after a calorie-rich meal, glucose, fatty acids, and amino acids enter the liver. Blue arrows follow the path of glucose; orange arrows follow the path of lipids. Insulin released in response to the high blood glucose concentration stimulates glucose uptake by the tissues. Some glucose is exported to the brain for its energy needs, and some goes to adipose and muscle tissue. In the liver, excess glucose is oxidized to acetyl-CoA, which is used to make triacylglycerols for export to adipose and muscle tissue. The NADPH necessary for lipid synthesis is obtained by oxidation of glucose in the pentose phosphate pathway. Excess amino acids are converted to pyruvate and acetyl-CoA, which are also used for lipid

synthesis. Dietary fats move from the intestine as chylomicrons, via the lymphatic system, to the liver, muscle, and adipose tissues.

In summary, an important effect of insulin is to bring about the conversion of excess blood glucose after a meal to two storage forms: glycogen (in the liver and muscle) and TAGs (in adipose tissue).

Pancreatic β Cells Secrete Insulin in Response to Changes in Blood Glucose

The peptide hormones insulin, glucagon, and somatostatin are produced by clusters of specialized pancreatic cells, the islets of Langerhans ([Fig. 23-23](#)). Each cell type of the islets produces a single hormone: α cells produce glucagon; β cells, insulin; and δ cells, somatostatin. When glucose enters the bloodstream from the intestine after a carbohydrate-rich meal, the resulting increase in blood glucose causes the pancreas to secrete insulin (and to decrease the secretion of glucagon).

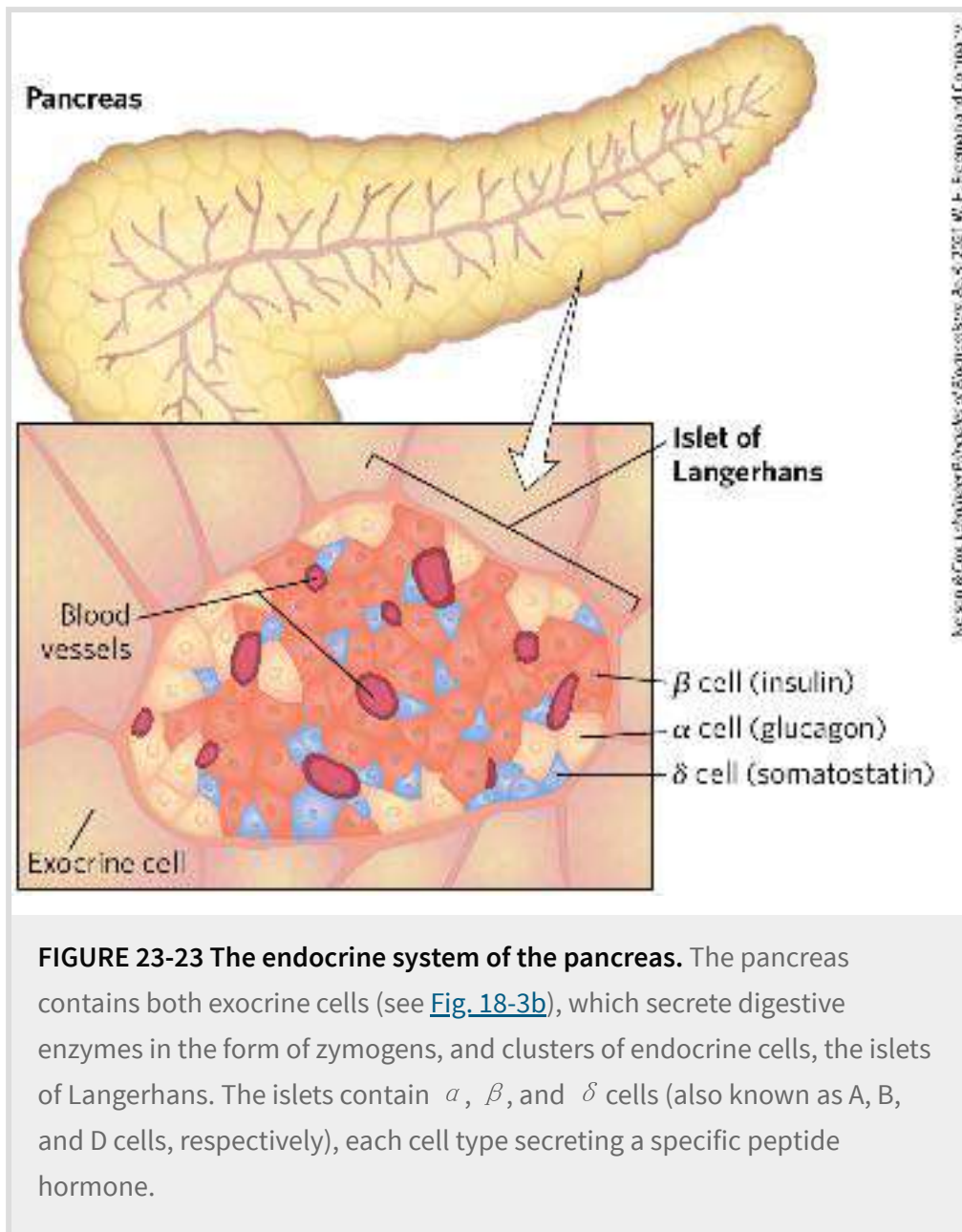


FIGURE 23-23 The endocrine system of the pancreas. The pancreas contains both exocrine cells (see [Fig. 18-3b](#)), which secrete digestive enzymes in the form of zymogens, and clusters of endocrine cells, the islets of Langerhans. The islets contain α , β , and δ cells (also known as A, B, and D cells, respectively), each cell type secreting a specific peptide hormone.

As shown in [Figure 23-24](#), when blood glucose rises, ❶ GLUT2 transporters carry glucose into the β cells, where it is immediately converted to glucose 6-phosphate by glucokinase and enters glycolysis. With the higher rate of glucose catabolism, ❷ [ATP] increases, causing **ATP-gated K^+ channels** in the plasma membrane to close. ❸ Reduced efflux of K^+ depolarizes the membrane. (Recall from [Section 12.6](#) that exit of K^+ through

an open K^+ channel hyperpolarizes the membrane; thus, closing the K^+ channel effectively depolarizes the membrane.)

Membrane depolarization opens voltage-gated Ca^{2+} channels, and ④ the resulting increase in cytosolic $[Ca^{2+}]$ triggers ⑤ the release of insulin by exocytosis. The brain integrates inputs on energy supply and demand, and signals from the parasympathetic and sympathetic nervous systems also affect (stimulate and inhibit, respectively) insulin release. A simple feedback loop limits hormone release: insulin lowers blood glucose by stimulating glucose uptake by the tissues; the reduced blood glucose is detected by the β cell as a diminished flux through the glucokinase reaction; this slows or stops the release of insulin. This feedback regulation holds blood glucose concentration nearly constant despite large fluctuations in dietary intake.

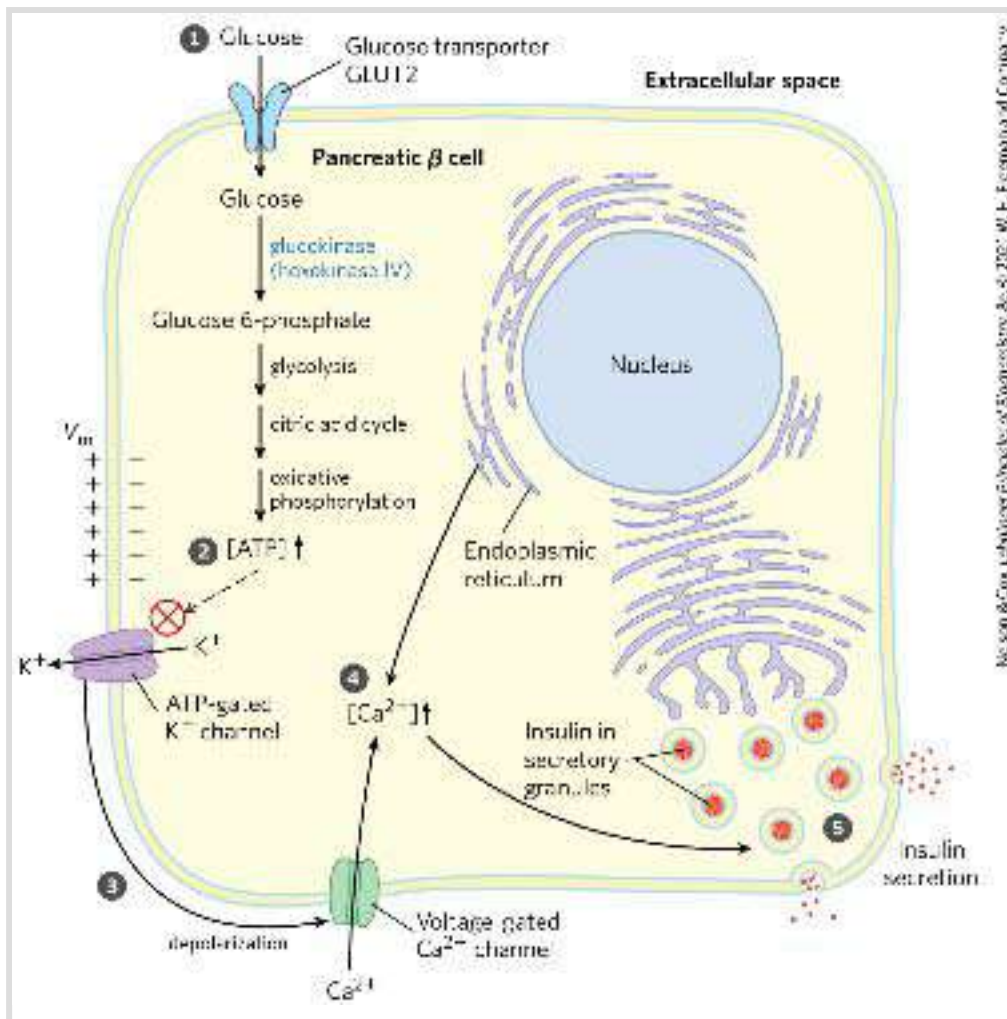


FIGURE 23-24 Glucose regulation of insulin secretion by pancreatic β cells. When the blood glucose level is high, active metabolism of glucose in the β cell raises intracellular [ATP], closing K^+ channels in the plasma membrane and thus depolarizing the membrane. In response to this membrane depolarization, voltage-gated Ca^{2+} channels open, allowing Ca^{2+} to flow into the cell. (Ca^{2+} is also released from the ER, in response to the initial elevation of [Ca^{2+}] in the cytosol.) Cytosolic [Ca^{2+}] is now high enough to trigger insulin release by exocytosis. The numbered processes are discussed in the text.



The activity of ATP-gated K^+ channels is central to the regulation of insulin secretion by β cells. The channels are octamers of four identical Kir6.2 subunits and four identical SUR1 subunits ([Fig. 23-25a](#)) and are constructed along the same lines as

the K^+ channels of bacteria and those of other eukaryotic cells (see [Fig. 11-45](#)). The four Kir6.2 subunits form a cone around the K^+ channel and function as the selectivity filter and ATP-gating mechanism ([Fig. 23-25b](#)). When [ATP] rises, indicating increased blood glucose, the K^+ channels close, thus depolarizing the plasma membrane and triggering insulin release as shown in [Figure 23-24](#). The **[sulfonylurea drugs](#)**, oral medications used in the treatment of type 2 diabetes mellitus, bind to the SUR1 (sulfonylurea receptor) subunits of the K^+ channels, closing the channels and stimulating insulin release.

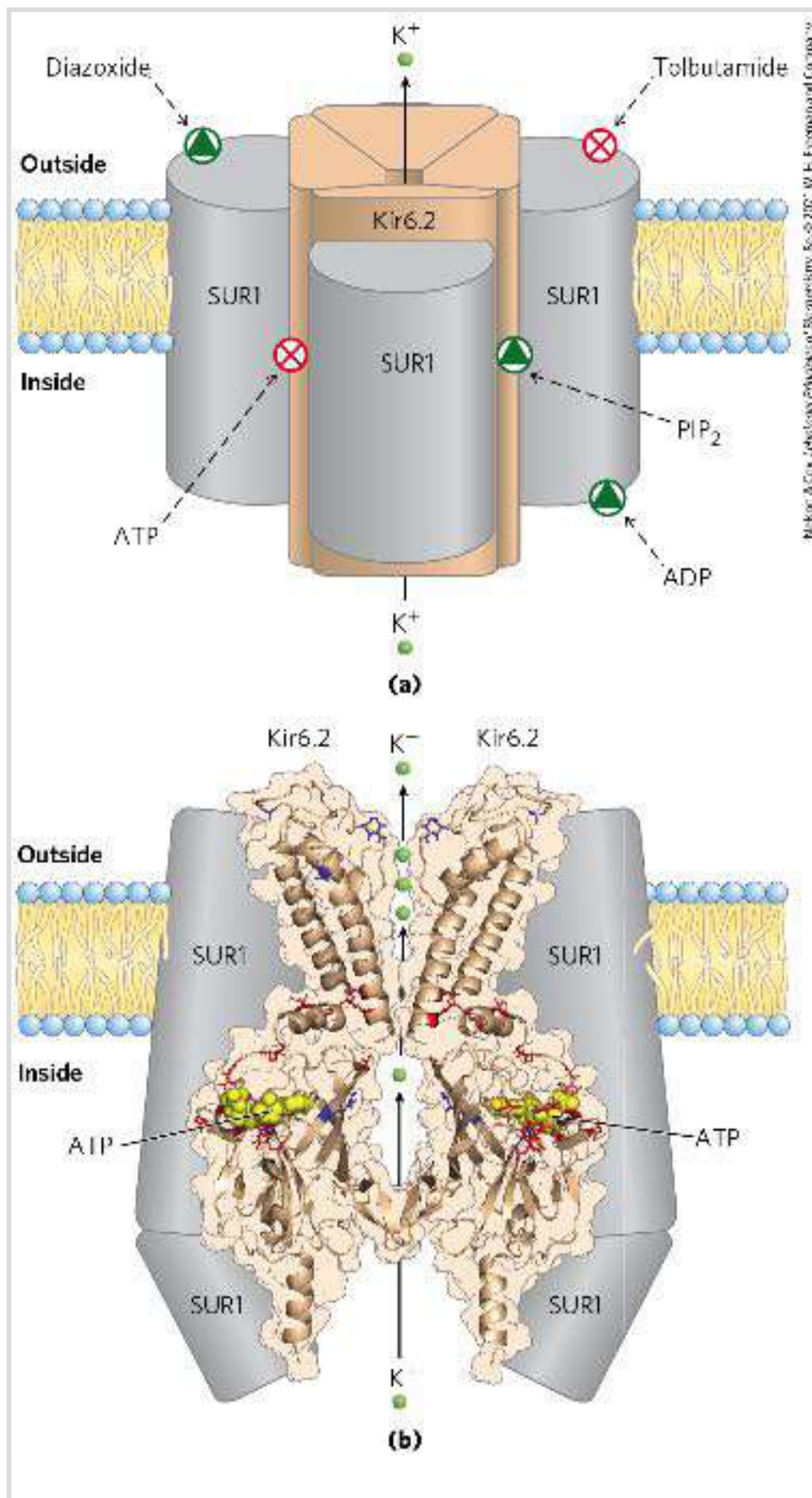


FIGURE 23-25 ATP-gated K^+ channels in β cells. (a) The ATP-gated channel, viewed in the plane of the membrane. The channel is formed by four identical Kir6.2 subunits, which are surrounded by four SUR1 (sulfonylurea receptor) subunits. The SUR1 subunits have binding sites for ADP and the drug diazoxide, both of which favor the open channel, and tolbutamide, a sulfonylurea drug that favors the closed channel. The Kir6.2 subunits constitute the channel, and they contain, on the cytosolic side, binding sites for ATP and phosphatidylinositol 4,5-bisphosphate (PIP_2), which favor the closed and the open channel, respectively. (b) The structure of the Kir6.2 portion of the channel, viewed in the plane of the membrane. For clarity, only two transmembrane domains and two cytosolic domains are shown. Three K^+ ions (green) are shown in the region of the selectivity filter. Mutation in certain amino acid residues (shown in red) leads to neonatal diabetes; mutation in others (shown in blue) leads to hyperinsulinism of infancy. This structure was obtained by mapping the known Kir6.2 sequence onto the crystal structures of a bacterial Kir channel (KirBac1.1) and the amino and carboxyl domains of another Kir protein, Kir3.1. [Data from (b) KirBac1.1: PDB ID 1P7B, A. Kuo et al., *Science* 300:1922, 2003; Kir3.1: PDB ID 1U4E, S. Pegan et al., *Nature Neurosci.* 8:279, 2005. Coordinates courtesy of Frances M. Ashcroft, Oxford University, used with permission of S. Haider and M. S. P. Sansom to re-create a model published in J. F. Antcliff et al., *EMBO J.* 24:229, 2005.]

Mutations in the ATP-gated K^+ channels of β cells are, fortunately, rare. Mutations in Kir6.2 that result in constantly *open* K^+ channels (red residues in [Fig. 23-25b](#)) lead to neonatal diabetes mellitus, with severe hyperglycemia that requires insulin therapy. Other mutations in Kir6.2 or SUR1 (blue residues in [Fig. 23-25b](#)) produce permanently *closed* K^+ channels and continuous release of insulin. If untreated, individuals with these mutations develop a condition in which excessive insulin causes severe hypoglycemia (low blood glucose) leading to irreversible brain

damage. One effective treatment is surgical removal of part of the pancreas to reduce insulin production. ■

Glucagon Counters Low Blood Glucose

 Several hours after the intake of dietary carbohydrate, blood glucose levels fall slightly because of the ongoing oxidation of glucose by the brain and other tissues. Lowered blood glucose triggers the pancreas to secrete **glucagon** and simultaneously decrease the release of insulin ([Fig. 23-26](#)).

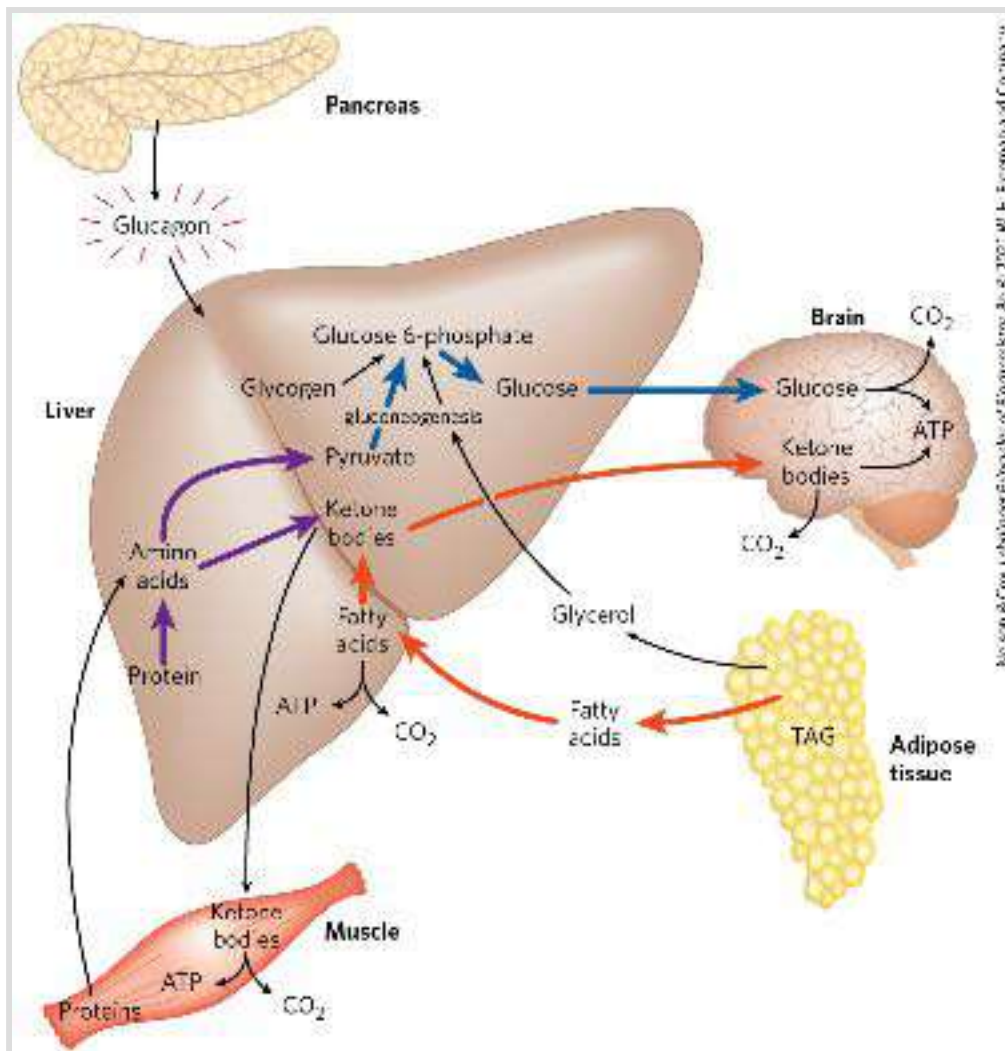


FIGURE 23-26 The fasting state: the glucogenic liver. After some hours without a meal, the liver becomes the principal source of glucose for the brain. Liver glycogen is broken down to glucose 1-phosphate, and this is converted to glucose 6-phosphate, then to free glucose, which is released into the bloodstream. Amino acids from the degradation of proteins in liver and muscle, and glycerol from the breakdown of TAGs in adipose tissue, are used for gluconeogenesis. The liver uses fatty acids as its principal fuel, and excess acetyl-CoA is converted to ketone bodies for export to other tissues; the brain is especially dependent on this fuel when glucose is in short supply (see [Fig. 23-19](#)). Blue arrows follow the path of glucose; orange arrows, the path of lipids; and purple arrows, the path of amino acids.

Glucagon causes an *increase* in blood glucose concentration in several ways ([Table 23-4](#)). Like epinephrine, it stimulates the net

breakdown of liver glycogen by activating glycogen phosphorylase and inactivating glycogen synthase; both effects are the result of phosphorylation of the regulated enzymes, triggered by cAMP. Glucagon inhibits glucose breakdown by glycolysis in the liver and stimulates glucose synthesis by gluconeogenesis. Both effects result from lowering the concentration of fructose 2,6-bisphosphate, an allosteric inhibitor of the gluconeogenic enzyme fructose 1,6-bisphosphatase (FBPase-1) and an activator of the glycolytic enzyme phosphofructokinase-1. Recall that [fructose 2,6-bisphosphate] is ultimately controlled by a cAMP-dependent protein phosphorylation reaction (see [Fig. 14-25](#)). Glucagon also inhibits the glycolytic enzyme pyruvate kinase, by promoting its cAMP-dependent phosphorylation, thus blocking the conversion of phosphoenolpyruvate to pyruvate and preventing oxidation of pyruvate via the citric acid cycle (see [Fig. 14-26](#)). The resulting accumulation of phosphoenolpyruvate favors gluconeogenesis. This effect is augmented by glucagon's stimulation of the synthesis of the gluconeogenic enzyme PEP carboxykinase.



By stimulating glycogen breakdown, preventing glycolysis and promoting gluconeogenesis in hepatocytes, glucagon enables the liver to export glucose, restoring blood glucose to its normal level.

TABLE 23-4 Effects of Glucagon on Blood Glucose: Production and Release of Glucose by the Liver

Metabolic effect	Effect on glucose metabolism	Target enzyme
------------------	------------------------------	---------------

↑ Glycogen breakdown (liver)	Glycogen → glucose	↑ Glycogen phosphorylase
↓ Glycogen synthesis (liver)	Less glucose stored as glycogen	↓ Glycogen synthase
↓ Glycolysis (liver)	Less glucose used as fuel in liver	↓ PFK-1
↑ Gluconeogenesis (liver)	Amino acids Glycerol Oxaloacetate } → glucose	↑ FBPase-2 ↓ Pyruvate kinase ↑ PEP carboxykinase
↑ Fatty acid mobilization (adipose tissue)	Less glucose used as fuel by liver, muscle	↑ Hormone-sensitive lipase ↑ PKA (perilipin- )
↑ Ketogenesis	Provides alternative to glucose as energy source for brain	↓ Acetyl-CoA carboxylase

Although its primary target is the liver, glucagon (like epinephrine) also affects adipose tissue, activating TAG breakdown by causing cAMP-dependent phosphorylation of perilipin and hormone-sensitive lipase. The activated lipase liberates free fatty acids, which are exported to the liver and other tissues as fuel, sparing glucose for the brain. The net effect of glucagon is therefore to stimulate glucose synthesis and release by the liver and to mobilize fatty acids from adipose tissue, to be used instead of glucose by tissues other than the brain. All these

effects of glucagon are mediated by cAMP-dependent protein phosphorylation.

During Fasting and Starvation, Metabolism Shifts to Provide Fuel for the Brain

A healthy adult human has three types of fuel reserves: glycogen stored in the liver and, in smaller quantities, in muscles; large quantities of TAG in adipose tissues; and tissue proteins, which can be degraded when necessary to provide fuel ([Table 23-5](#)).

TABLE 23-5 Available Metabolic Fuels in a Normal-Weight, 70 kg Man and in an Obese, 140 kg Man at the Beginning of a Fast

Type of fuel	Weight (kg)	Caloric equivalent (thousands of kcal (kJ))	Estimated survival (months) ^a
Normal-weight, 70 kg man			
Triacylglycerols (adipose tissue)	15	140 (590)	
Proteins (mainly muscle)	6	24 (100)	
Glycogen (muscle, liver)	0.23	0.90 (3.8)	
Circulating fuels (glucose, fatty acids, triacylglycerols, etc.)	0.023	0.10 (0.42)	
Total		165 (690)	3

Obese, 140 kg man			
Triacylglycerols (adipose tissue)	80	750 (3,100)	
Proteins (mainly muscle)	8	32 (130)	
Glycogen (muscle, liver)	0.23	0.92 (3.8)	
Circulating fuels	0.025	0.11 (0.46)	
Total		783 (3,200)	14

^aSurvival time is calculated on the assumption of a basal energy expenditure of 1,800 kcal/day.

P3 Two hours after a meal, the blood glucose level is diminished slightly, and tissues receive glucose released from liver glycogen. There is little or no synthesis of TAGs. By four hours after a meal, blood glucose has fallen further, insulin secretion has slowed, and glucagon secretion has increased. These hormonal signals mobilize TAGs from adipose tissue, which now become the primary fuel for muscle and liver. [Figure 23-27](#) shows the responses to prolonged fasting. ❶ To provide glucose for the brain, the liver degrades certain proteins — those most expendable in an organism not ingesting food. Their nonessential amino acids are transaminated or deaminated ([Chapter 18](#)), and ❷ the extra amino groups are converted to urea, which is exported via the bloodstream to the kidneys and excreted in the urine.

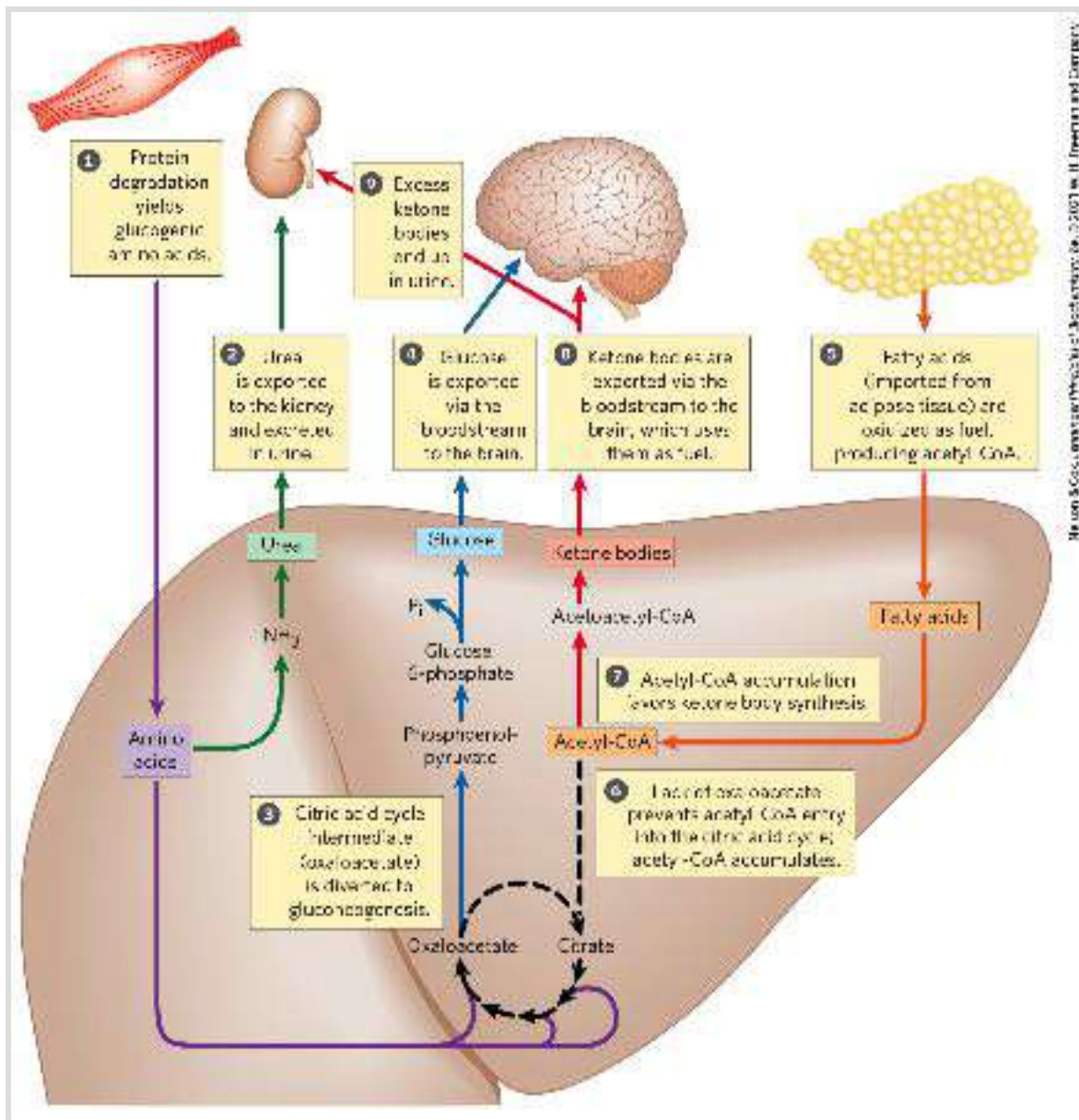


FIGURE 23-27 Fuel metabolism in the liver during prolonged fasting or in uncontrolled diabetes mellitus. The numbered steps are described in the text. After depletion of stored carbohydrates (glycogen), gluconeogenesis in the liver becomes the main source of glucose for the brain (blue arrows). NH₃ from amino acid deamination is converted into urea and excreted (green arrows). Glucogenic amino acids from protein breakdown (purple arrows) provide substrates for gluconeogenesis, and glucose is exported to the brain. Fatty acids from adipose tissue are imported into the liver and oxidized to acetyl-CoA (orange arrows), and acetyl-CoA is the starting material for ketone body formation in the liver and for export to the brain to serve as an energy source (red arrows). Excess ketone bodies are excreted in the urine.

Also in the liver, and to some extent in the kidneys, the carbon skeletons of glucogenic amino acids are converted to pyruvate or intermediates of the citric acid cycle. ③ These intermediates (as well as the glycerol derived from TAGs in adipose tissue) provide the starting materials for gluconeogenesis in the liver, ④ yielding glucose for export to the brain. ⑤ Fatty acids released from adipose tissue are oxidized to acetyl-CoA in the liver, but as oxaloacetate is depleted by the use of citric acid cycle intermediates for gluconeogenesis, ⑥ entry of acetyl-CoA into the cycle is inhibited and acetyl-CoA accumulates. ⑦ This favors the formation of acetoacetyl-CoA and ketone bodies. After a few days of fasting, the levels of ketone bodies in the blood rise ([Fig. 23-28](#)) as they are exported from the liver to the heart, skeletal muscle, and brain, which use these fuels instead of glucose ([Fig. 23-27](#), ⑧). When the concentration of ketone bodies in the blood exceeds the ability of the kidneys to reabsorb ketones ⑨, these compounds begin to appear in the urine.

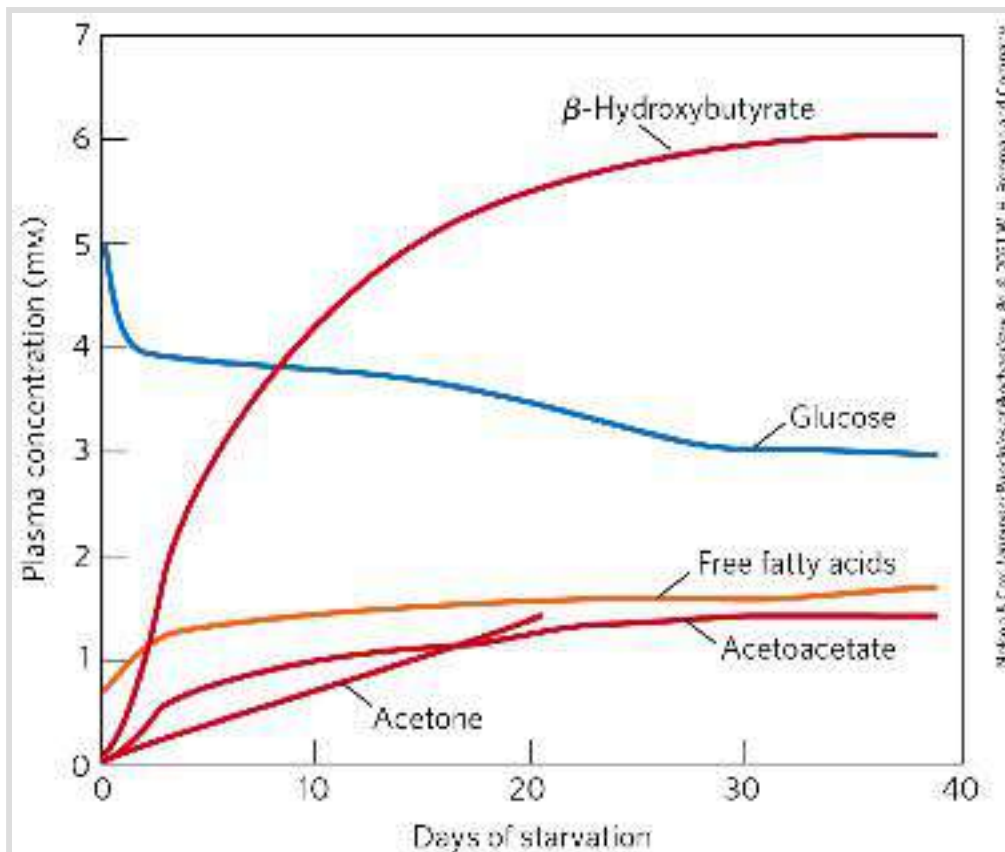


FIGURE 23-28 Plasma concentrations of fatty acids, glucose, and ketone bodies during six weeks of starvation. Despite the hormonal mechanisms for maintaining the glucose level in blood, glucose begins to diminish within 2 days of beginning a fast. The levels of ketone bodies, almost unmeasurable before the fast, rise dramatically after 2 to 4 days of fasting, with β -hydroxybutyrate as the major contributor. These water-soluble ketones, acetoacetate and β -hydroxybutyrate, supplement glucose as an energy source for the brain during a long fast. Acetone, a minor ketone body, is not metabolized but is eliminated in the breath. A much smaller rise in blood fatty acids also occurs, but this does not contribute to energy metabolism in the brain, as fatty acids do not cross the blood-brain barrier. [Data from G. F. Cahill, Jr., *Annu. Rev. Nutr.* 26:1, 2006, Fig. 2.]

Triacylglycerols stored in the adipose tissue of a normal-weight adult could provide enough fuel to maintain a basal rate of metabolism for about three months; a very obese adult has enough stored fuel to endure a fast of more than a year ([Table 23-](#)

5). When fat reserves are gone, the degradation of *essential* proteins begins; this leads to loss of heart and liver function and, in prolonged starvation, to death. Stored fat can provide adequate energy (calories) during a fast or a rigid diet, but vitamins and minerals must be provided, and sufficient dietary glucogenic amino acids are needed to replace those being used for gluconeogenesis. Rations for those on a weight-reduction diet are commonly fortified with vitamins, minerals, and amino acids or proteins.

Epinephrine Signals Impending Activity

When an animal is confronted with a stressful situation that requires increased activity — fighting or fleeing, in the extreme case — neuronal signals from the brain trigger the release of epinephrine and norepinephrine from the adrenal medulla. Both hormones dilate the respiratory passages to facilitate the uptake of O₂, increase the rate and strength of the heartbeat, and raise the blood pressure, thereby promoting the flow of O₂ and fuels to the tissues ([Table 23-6](#)). This is the “fight-or-flight” response.

TABLE 23-6 Physiological and Metabolic Effects of Epinephrine: Preparation for Action

Immediate effect	Overall effect
Physiological ↑ Heart rate ↑ Blood pressure ↑ Dilation of respiratory passages	Increase delivery of O_2 to tissues (muscle)
Metabolic ↑ Glycogen breakdown (muscle, liver) ↓ Glycogen synthesis (muscle, liver) ↑ Gluconeogenesis (liver)	
↑ Glycolysis (muscle)	
↑ Fatty acid mobilization (adipose tissue)	Increases ATP production in muscle
↑ Glucagon secretion	Increases availability of fatty acids as fuel
↓ Insulin secretion	Reinforce metabolic effects of epinephrine

Epinephrine acts primarily on muscle, adipose, and liver tissues. It activates glycogen phosphorylase and inactivates glycogen synthase by cAMP-dependent phosphorylation of the enzymes, thus stimulating the conversion of *liver* glycogen to blood glucose, the fuel for anaerobic muscular work. Epinephrine also promotes the anaerobic breakdown of *muscle* glycogen by lactic acid fermentation, stimulating glycolytic ATP formation. The stimulation of glycolysis is accomplished by raising the concentration of fructose 2,6-bisphosphate, a potent allosteric activator of the key glycolytic enzyme phosphofructokinase-1. Epinephrine also stimulates fat mobilization in adipose tissue, by activating hormone-sensitive lipase and moving aside perilipin (see [Fig. 17-2](#)). Finally, epinephrine stimulates glucagon secretion and inhibits insulin secretion, reinforcing its effect of mobilizing fuels and inhibiting fuel storage.

Cortisol Signals Stress, Including Low Blood Glucose

A variety of stressors (anxiety, fear, pain, hemorrhage, infection, low blood glucose, starvation) stimulate release of the glucocorticoid **cortisol** from the adrenal cortex (see [Fig. 23-6](#)).

 Cortisol acts on muscle, liver, and adipose tissue to supply the organism with fuel to withstand the stress. Cortisol is a relatively slow-acting hormone that alters metabolism by changing the kinds and amounts of certain enzymes synthesized in its target cells, rather than by regulating the activity of existing enzyme molecules.

In adipose tissue, cortisol leads to an increased release of fatty acids from stored TAGs. The exported fatty acids serve as fuel for other tissues, and the glycerol is used for gluconeogenesis in the liver. Cortisol stimulates the breakdown of nonessential muscle proteins and the export of amino acids to the liver, where they serve as precursors for gluconeogenesis. In the liver, cortisol promotes gluconeogenesis by stimulating synthesis of PEP carboxykinase; glucagon has the same effect, whereas insulin has the opposite effect. Glucose produced in this way is stored in the liver as glycogen or exported immediately to tissues that need glucose for fuel. The net effect of these metabolic changes is to restore blood glucose to its normal level and to increase glycogen stores, ready to support the fight-or-flight response commonly associated with stress. The effects of cortisol therefore counterbalance those of insulin. During extended periods of stress, the continued release of cortisol loses its positive adaptive value and begins to cause damage to muscle and bone and to impair endocrine and immune function.



Cushing disease is a medical condition in which a tumor on the pituitary gland causes the adrenal glands to overproduce cortisol. It is treated by surgery to remove the tumor, followed by chemotherapy to kill remaining tumor cells. Addison disease results from underproduction of cortisol, and is treated by administering hydrocortisone (the pharmaceutical name for cortisol). ■

SUMMARY 23.3 *Hormonal Regulation of Fuel Metabolism*

■ Fluctuations in blood glucose (normally 70 to 100 mg/100 mL, or about 4.5 mM) due to dietary intake or vigorous exercise are counterbalanced by a variety of hormonally triggered changes in metabolism in several organs.

■ High blood glucose elicits the release of insulin, which speeds the uptake of glucose by tissues and favors the storage of fuels as glycogen and triacylglycerols while inhibiting fatty acid mobilization in adipose tissue.

■ Low blood glucose triggers release of glucagon, which stimulates glucose release from liver glycogen and shifts fuel metabolism in liver and muscle to fatty acid oxidation, sparing glucose for use by the brain. In prolonged fasting, TAGs become the principal fuel; the liver converts the fatty acids to ketone bodies for export to other tissues, including the brain.

■ Epinephrine prepares the body for increased activity by mobilizing glucose from glycogen and other precursors, releasing the glucose into the blood.

■ Cortisol, released in response to a variety of stressors (including low blood glucose), stimulates gluconeogenesis from amino acids and glycerol in the liver, thus raising blood glucose and counterbalancing the effects of insulin.

23.4 Obesity and the Regulation of Body Mass



In the U.S. population, more than 40% of adults are obese, including more than 10% who are severely obese, as defined in terms of body mass index (BMI), calculated as $(\text{weight in kg})/(\text{height in m})^2$. A BMI below 25 is considered normal; an individual with a BMI of 25 to 30 is overweight; a BMI greater than 30 indicates **obesity**; a BMI greater than 40 indicates severe obesity. Obesity is life-threatening. It significantly increases the likelihood of developing type 2 diabetes, as well as heart attack, stroke, and cancers of the colon, breast, prostate, and endometrium. Consequently, there is great interest in understanding how body mass and the storage of fats in adipose tissue are regulated. ■

To a first approximation, obesity is the result of taking in more calories in the diet than are expended by the body's fuel-consuming activities. The body can deal with an excess of dietary calories in three ways: (1) convert excess fuel to fat and store it in adipose tissue, (2) burn excess fuel by extra exercise, and (3) “waste” fuel by diverting it to heat production (thermogenesis) by uncoupled mitochondria. 

In mammals, a complex set of hormonal and neuronal signals acts to keep fuel intake and energy expenditure in balance so as to hold the amount of adipose tissue at a suitable level. Dealing effectively with obesity requires understanding how the various checks and balances

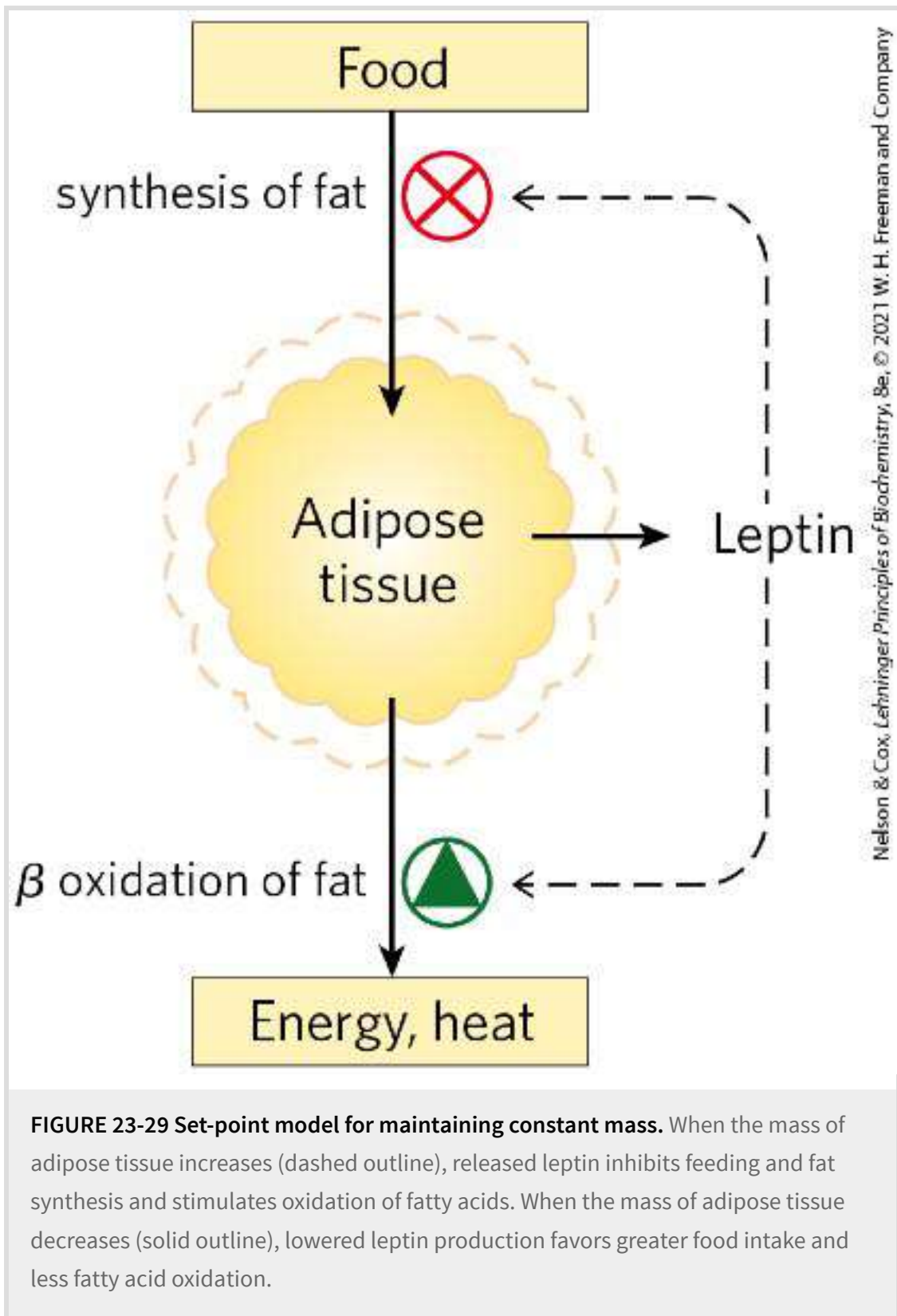
work under normal conditions and how these homeostatic mechanisms can fail.

Adipose Tissue Has Important Endocrine Functions

One early hypothesis to explain body-mass homeostasis, the “adiposity negative-feedback” model, postulated a mechanism that inhibits eating behavior and increases energy expenditure whenever body weight exceeds a certain value, called the set point; the inhibition is relieved when body weight drops below the set point ([Fig. 23-29](#)). This model predicts that a feedback signal originating in adipose tissue influences the brain centers that control eating behavior and metabolic and motor activity. The first such signal to be discovered was leptin, in 1994.

Subsequent research revealed that  adipose tissue is an important endocrine organ that produces peptide hormones, known as **adipokines**. Adipokines may act locally (autocrine and paracrine action) or systemically (endocrine action), carrying information about the adequacy of the energy reserves (TAGs) stored in adipose tissue to other tissues and to the brain.

Normally, adipokines produce changes in fuel metabolism and feeding behavior that reestablish adequate fuel reserves and maintain body mass. When adipokines are over- or underproduced, the resulting dysregulation may result in life-threatening disease.



Leptin (Greek *leptos*, “thin”) is an adipokine (167 amino acid residues) that, on reaching the brain, acts on receptors in the

hypothalamus to curtail appetite. Leptin was first identified as the product of a gene designated *OB* (obese) in laboratory mice. Mice with two defective copies of this gene (*ob/ob* genotype; lowercase letters signify a mutant form of the gene) show the behavior and physiology of animals in a constant state of starvation: their plasma cortisol levels are elevated; they exhibit unrestrained appetite, are unable to stay warm, grow abnormally large, and do not reproduce. As a consequence of unrestrained appetite, they become severely obese, weighing as much as three times more than normal mice ([Fig. 23-30](#)). They also have metabolic disturbances similar to those seen in diabetes, and they are insulin-resistant. When leptin is injected into *ob/ob* mice, they eat less, lose weight, and increase their locomotor activity and thermogenesis.



The mouse on the left got no leptin and consequently ate more food and was less active; it weighs 67 g.

A second mouse gene, designated *DB* (diabetic), also has a role in appetite regulation. Mice with two defective copies (*db/db*) are obese and diabetic. The *DB* gene encodes the **leptin receptor**. When the receptor is defective, the signaling function of leptin is lost.

The leptin receptor is expressed primarily in regions of the brain known to regulate feeding behavior — neurons of the [arcuate nucleus](#) of the hypothalamus ([Fig. 23-31a](#)). Leptin carries the message that fat reserves are sufficient, and it promotes reduction of fuel intake and increase in expenditure of energy. Leptin-receptor interaction in the hypothalamus alters the release of neuronal signals to the region of the brain that affects appetite. Leptin also stimulates the sympathetic nervous system, increasing blood pressure, heart rate, and thermogenesis by uncoupling the mitochondria of brown adipocytes ([Fig. 23-31b](#)). Recall that the uncoupling protein UCP1 forms a channel in the inner mitochondrial membrane that allows protons to reenter the mitochondrial matrix without passing through the ATP synthase complex. This permits constant oxidation of fuel (fatty acids in a brown or beige adipocyte) without ATP synthesis, dissipating energy as heat and consuming dietary calories or stored fats in potentially large amounts.

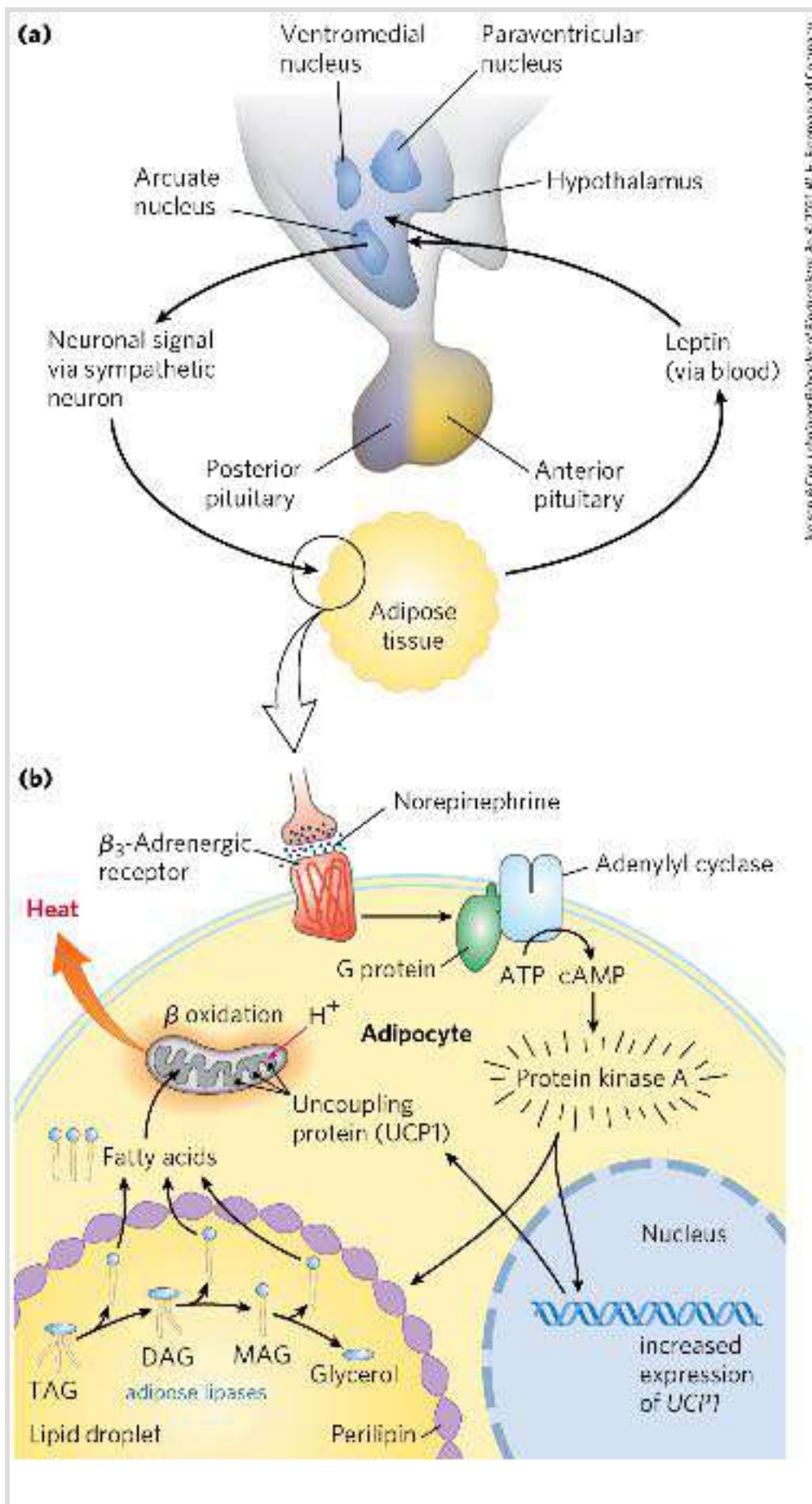


FIGURE 23-31 Hypothalamic regulation of food intake and energy expenditure. (a) Role of the hypothalamus in its interaction with adipose tissue. The hypothalamus receives input (leptin) from adipose tissue and responds with neuronal signals to adipocytes. (b) This signal (norepinephrine) activates protein kinase A, which triggers mobilization of fatty acids from TAG and their uncoupled oxidation in mitochondria, generating heat but not ATP. DAG, diacylglycerol; MAG, monoacylglycerol.

Leptin Stimulates Production of Anorexigenic Peptide Hormones

Two types of neurons in the arcuate nucleus control fuel intake and metabolism ([Fig. 23-32](#)). The **orexigenic** (appetite-stimulating) neurons stimulate eating by producing and releasing **neuropeptide Y (NPY)**, which causes the next neuron in the circuit to send the signal to the brain: Eat! The blood level of NPY rises during starvation and is elevated in mice with either two defective copies of the leptin gene (*ob/ob*) or two defective copies of the leptin receptor gene (*db/db*). The high NPY concentration presumably contributes to the obesity of these mice, which eat voraciously.

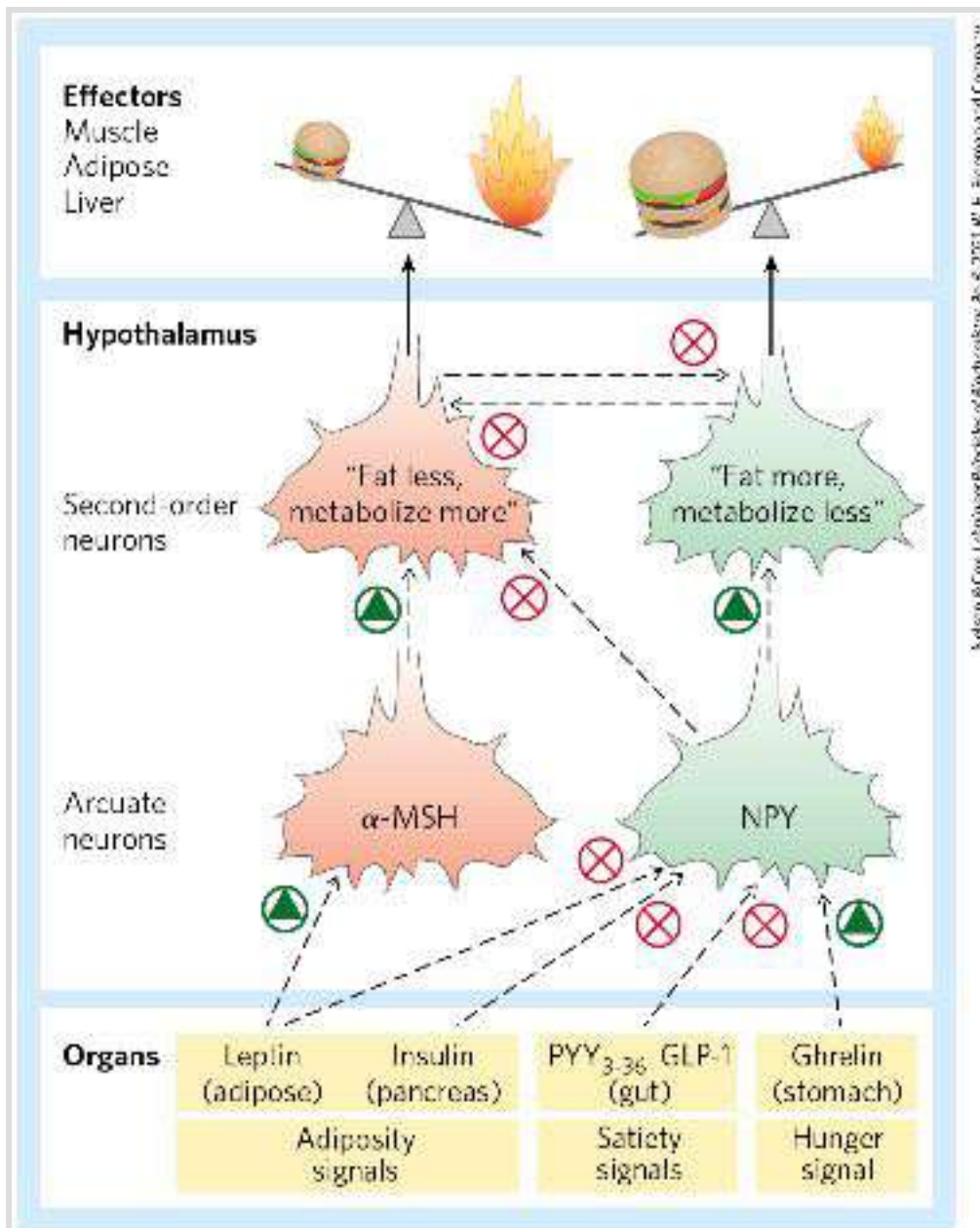


FIGURE 23-32 Hormones that control eating. In the arcuate nucleus of the hypothalamus, two sets of neurosecretory cells receive hormonal input and relay neuronal signals to the cells of muscle, adipose tissue, and liver. Leptin and insulin are released from adipose tissue and the pancreas, respectively, in proportion to the mass of body fat. The two hormones act on anorexigenic neurosecretory cells to trigger release of α -MSH (melanocyte-stimulating hormone); α -MSH carries the signal to second-order neurons in the hypothalamus, which puts out the signals to eat less and metabolize more fuel. Leptin and insulin also act on orexigenic neurosecretory cells to inhibit the release of NPY, reducing the “eat more” signal sent to the tissues. As described later in the text, the gastric hormone

ghrelin *stimulates* appetite by activating the NPY-expressing cells; PYY₃₋₃₆, released from the colon, *inhibits* these neurons and decreases appetite. Each of the two types of neurosecretory cells inhibits hormone production by the other, so any stimulus that activates orexigenic cells inactivates anorexigenic cells, and vice versa. This strengthens the effect of stimulatory inputs.

The **anorexigenic** (appetite-suppressing) neurons in the arcuate nucleus produce α -**melanocyte-stimulating hormone** (α -MSH; also known as melanocortin), formed from its polypeptide precursor pro-opiomelanocortin (POMC; [Fig. 23-5](#)). Release of α -MSH causes the next neuron in the circuit to send the signal to the brain: Stop eating!

The amount of leptin released by adipose tissue depends on both the number and the size of adipocytes. When weight loss decreases the mass of lipid tissue, leptin levels in the blood decrease, the production of NPY increases, and the processes in adipose tissue shown in [Figure 23-31](#) are reversed. Uncoupling is diminished, slowing thermogenesis and saving fuel, and fat mobilization slows in response to reduced signaling by cAMP. Consumption of more food, combined with more efficient utilization of fuel, results in replenishment of the fat reserve in adipose tissue, bringing the system back into balance.

Leptin Triggers a Signaling Cascade That Regulates Gene Expression

The leptin signal is transduced by a mechanism also used by receptors for some growth factors. The leptin receptor has a single transmembrane segment. When leptin binds to the extracellular domains of two monomers, they dimerize and undergo phosphorylation on several Tyr residues. This initiates a chain of events that ends in the nucleus with the increased synthesis of target genes including the gene for POMC, from which α -MSH is produced.

The increased catabolism and thermogenesis triggered by leptin are due in part to increased synthesis of the mitochondria in brown and beige adipocytes. Leptin stimulates UCP1 synthesis by altering synaptic transmissions from neurons of the arcuate nucleus to adipose and other tissues via the sympathetic nervous system. The consequent increased release of norepinephrine in these tissues acts through β_3 -adrenergic receptors to stimulate transcription of the *UCP1* gene. The resulting uncoupling of electron transfer from oxidative phosphorylation consumes fat and is thermogenic ([Fig. 23-31](#)).

Might human obesity be the result of insufficient leptin production and therefore be treatable by the injection of leptin? Blood levels of leptin are, in fact, usually much *higher* in obese animals (including humans) than in animals of normal body mass (except, of course, in *ob/ob* mutants, which cannot make leptin). Some downstream element in the leptin response system must be defective in obese individuals, and the elevation in leptin is the result of an (unsuccessful) attempt to overcome the leptin

resistance. In those very rare humans with extreme obesity who have a defective leptin gene (*OB*), leptin injection does result in dramatic weight loss. In the vast majority of obese individuals, however, the *OB* gene is intact. In clinical trials, the injection of leptin did not have the weight-reducing effect observed in obese *ob/ob* mice. Clearly, most cases of human obesity involve one or more factors in addition to leptin.

Adiponectin Acts through AMPK to Increase Insulin Sensitivity

Adiponectin is a peptide hormone produced almost exclusively in adipose tissue, an adipokine that sensitizes other organs to the effects of insulin. Adiponectin circulates in the blood and powerfully affects the metabolism of fatty acids and carbohydrates in liver and muscle. It increases the uptake of fatty acids from the blood by myocytes and the rate at which fatty acids undergo β oxidation in muscle. It also blocks fatty acid synthesis and gluconeogenesis in hepatocytes, and stimulates glucose uptake and catabolism in muscle and liver.

These effects of adiponectin are indirect and not fully understood, but the **AMP-activated protein kinase (AMPK)** mediates many of them. Acting through its GPCR, adiponectin triggers phosphorylation and activation of AMPK. Recall that AMPK is activated by factors that signal the need to shift metabolism toward energy generation and away from energy-

requiring biosynthesis ([Fig. 23-33](#); see also [p. 503](#)). When activated, AMPK profoundly affects the metabolism of individual cells and, through its actions in the brain, the metabolism of the whole animal.

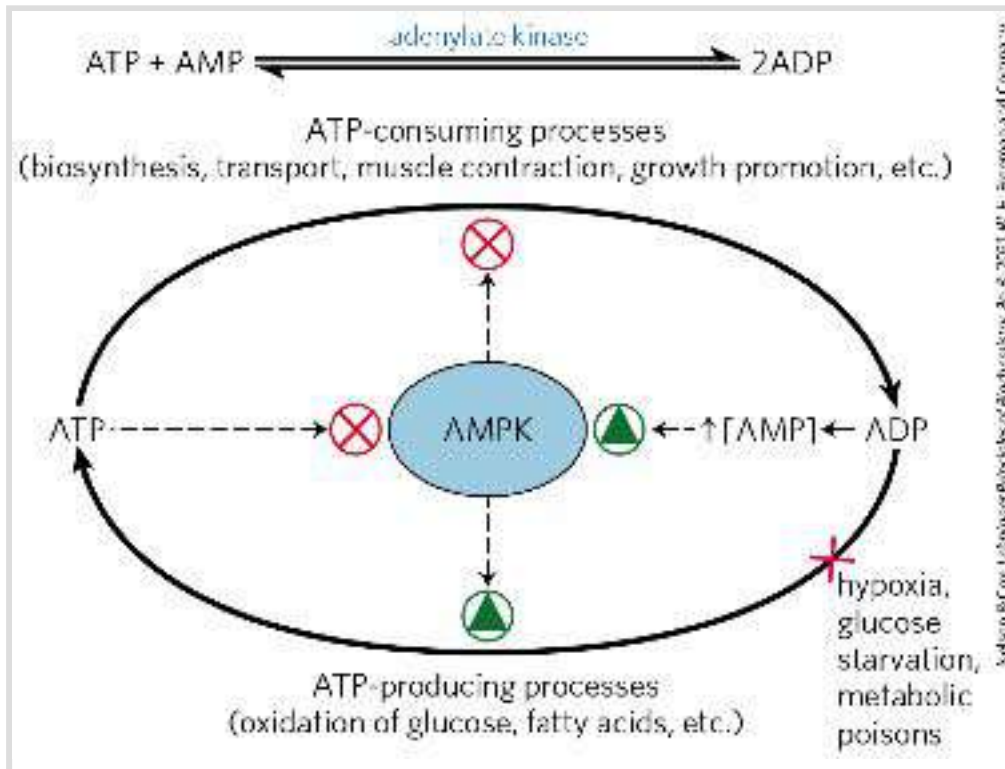


FIGURE 23-33 The role of AMP-activated protein kinase (AMPK) in maintaining energy homeostasis. ADP produced by synthetic reactions is converted to ATP and AMP by adenylate kinase. AMP activates AMPK, which reciprocally regulates ATP-consuming and ATP-generating pathways by phosphorylating key enzymes (see [Fig. 23-34](#)). Conditions or agents that inhibit ATP production by catabolic reactions (such as hypoxia, lack of glucose, or metabolic poisons) raise [AMP], activate AMPK, and stimulate catabolism. Cellular or organismal activities that consume ATP (muscle contraction, growth) increase [AMP] and stimulate catabolic reactions to replenish ATP. When [ATP] is high, ATP prevents AMP binding to AMPK, thus lowering AMPK activity and slowing catabolism.

AMPK Coordinates Catabolism and Anabolism in Response to Metabolic Stress

AMPK has emerged as a central player in the coordination of metabolic pathways, organism activity, and feeding behavior ([Fig. 23-34](#)). This heterotrimeric, AMP-activated protein kinase monitors the energy and nutrient status *in individual cells* and shifts metabolism toward energy generation when necessary to maintain metabolic homeostasis. Furthermore, by responding to a variety of hormone signals, AMPK in the hypothalamus acts to keep the *whole organism* in energetic balance ([Fig. 23-8](#)).

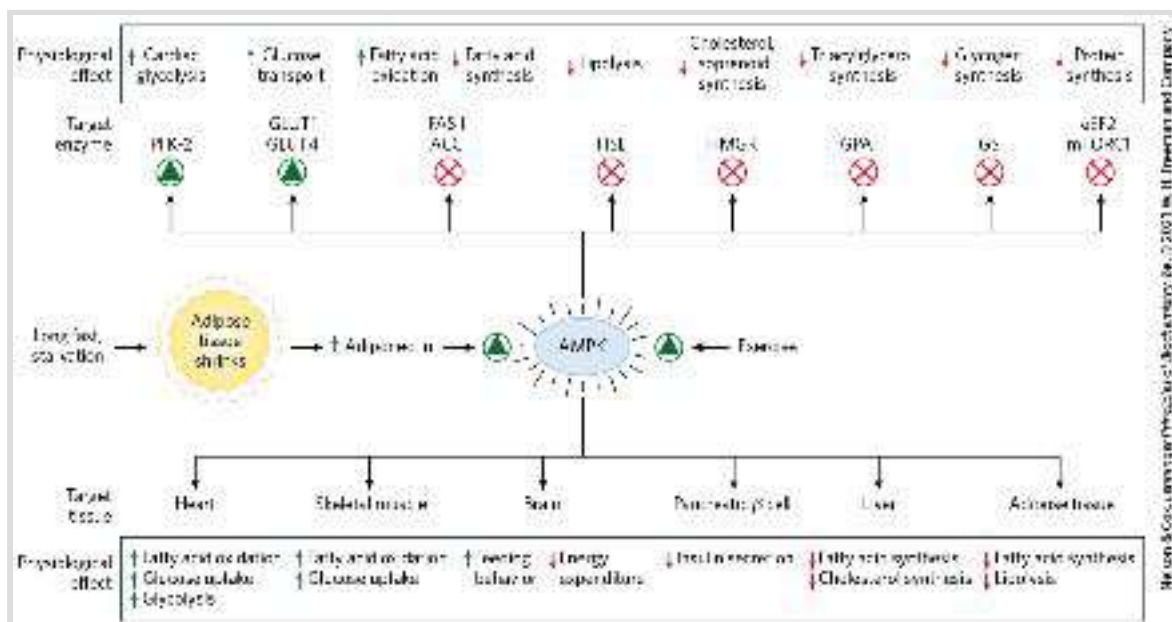


FIGURE 23-34 Formation of adiponectin and its actions through AMPK. Extended fasting or starvation decreases triacylglycerol reserves in adipose tissue, which triggers adiponectin production and release from adipocytes. Adiponectin acts through its plasma membrane receptors in various cell types and organs to inhibit energy-consuming processes and stimulate energy-producing processes. It acts in the brain to stimulate feeding behavior and inhibit energy-consuming physical activity, and in brown

fat to inhibit thermogenesis. Adiponectin exerts some of its metabolic effects by activating AMPK, which regulates (by phosphorylation) specific enzymes in key metabolic processes. PFK-2, phosphofructokinase-2; GLUT1 and GLUT4, glucose transporters; FAS I, fatty acid synthase I; ACC, acetyl-CoA carboxylase; HSL, hormone-sensitive lipase; HMGR, HMG-CoA reductase; GPAT, an acyl transferase; GS, glycogen synthase; eEF2, eukaryotic elongation factor 2 (required for protein synthesis; see [Chapter 27](#)); and mTORC1, mammalian target of rapamycin complex 1 (a protein kinase complex that regulates protein synthesis on the basis of nutrient availability). Exercise, through conversion of ATP to ADP and AMP, also stimulates AMPK.

AMPK monitors the energy status of a cell through its response to increased $[AMP]/[ATP]$. Many of the energy-consuming reactions in cells convert ATP to ADP or AMP. Adenylate kinase catalyzes the reaction $2ADP \rightarrow AMP + ATP$, so $[AMP]$ is a sensitive measure of the cell's energy status. AMPK is allosterically activated by AMP binding, and ATP prevents AMP binding, so the enzyme is activated when the cell is energetically depleted (high $[AMP]$) and inactivated when energy is plentiful (high $[ATP]$, and high $[ATP]/[AMP]$). AMPK responds to the energetic needs of the whole organism through a second mode of regulation. The enzyme is activated 100-fold by phosphorylation of Thr¹⁷² by liver kinase B1 (LKB1), which is itself subject to regulation by upstream components, including adiponectin. When activated by phosphorylation and AMP binding, AMPK phosphorylates specific enzymes in metabolic pathways that are crucial to energy homeostasis ([Fig. 23-34](#)).

When AMPK senses depletion of ATP in an individual cell, lipid synthesis is inhibited and use of lipid as fuel is stimulated. One enzyme regulated by AMPK in the liver and in white adipose

tissue is acetyl-CoA carboxylase, which produces malonyl-CoA, the first intermediate committed to fatty acid synthesis. Malonyl-CoA is a powerful inhibitor of the enzyme carnitine acyltransferase 1, which starts the process of β oxidation by transporting fatty acids into the mitochondrion (see [Fig. 17-6](#)). By phosphorylating and inactivating acetyl-CoA carboxylase, AMPK inhibits fatty acid synthesis while relieving inhibition (by malonyl-CoA) of β oxidation (see [Fig. 17-13](#)). Cholesterol synthesis, a heavy energy consumer, is also inhibited by AMPK, which phosphorylates and inactivates HMG-CoA reductase, an enzyme central to sterol synthesis (see [Fig. 21-34](#)). Similarly, AMPK inhibits fatty acid synthase and acyl transferase, effectively blocking the synthesis of TAGs. In addition to its effects on lipid metabolism, AMPK inhibits the synthesis of glycogen and of protein (see [Fig. 23-35](#)). Inadequate supplies of oxygen (hypoxia) or blood glucose (hypoglycemia) are among the stressors that trigger AMPK activation.

In the hypothalamus, AMPK is positioned to receive a variety of signals from throughout the body ([Fig. 23-8](#)). Ghrelin and adiponectin signal “empty stomach” and “fat tissue depleted,” and, like low blood glucose, they elicit hypothalamic signals, mediated by AMPK, that stimulate feeding and inhibit energy-requiring biosynthetic processes. Leptin, as we have seen, brings to the brain the signal “adipose tissue is full,” slowing catabolic processes and favoring growth and biosynthesis.

At the cytoplasmic surface of lysosomes, AMPK interacts with a second central regulator of cellular activity, the protein kinase mTOR. This enzyme, at the center of an enormous complex, mTORC1, gauges whether sufficient nutrients and low molecular weight substrates are available to support cell growth and proliferation. Together, the two protein kinases, AMPK and mTOR, control major aspects of a cell's activity and fate.

The mTORC1 Pathway Coordinates Cell Growth with the Supply of Nutrients and Energy

The highly conserved Ser/Thr kinase **mTOR** forms a complex, **mTORC1**, with a scaffold protein, raptor, and other regulatory proteins. mTORC1 is recruited to the cytosolic surface of the lysosome through raptor by another complex, Ragulator, and a number of associated Rag G proteins that sense amino acid sufficiency in the cell. The Ragulator-Rag complex is tethered to the lysosomal membrane by covalently linked lipids. When mTORC1 is docked to the Ragulator-Rag complex, it is in contact with another G protein, **Rheb** ([Fig. 23-35](#)). This massive complex integrates signals from inside and outside of the lysosome about the energy status of the cell, the availability of critical amino acids needed for protein synthesis, and the presence of growth factors. When these signals indicate that the cell has what it needs to grow, GTP-bound Rheb activates the protein kinase activity of mTOR, which then phosphorylates many different

proteins required for transcription, increased ribosome synthesis, and expression of genes encoding the enzymes of lipid synthesis and mitochondrial proliferation ([Fig. 23-36](#)). Because the lysosome is where the cell recycles defective or unneeded components, or extracellular substances brought into the cell through phagocytosis, it is, in effect, the warehouse of parts for cellular construction. The mTORC1-Ragulator-Rag complex is the supply chain manager determining whether the assembly line can operate.

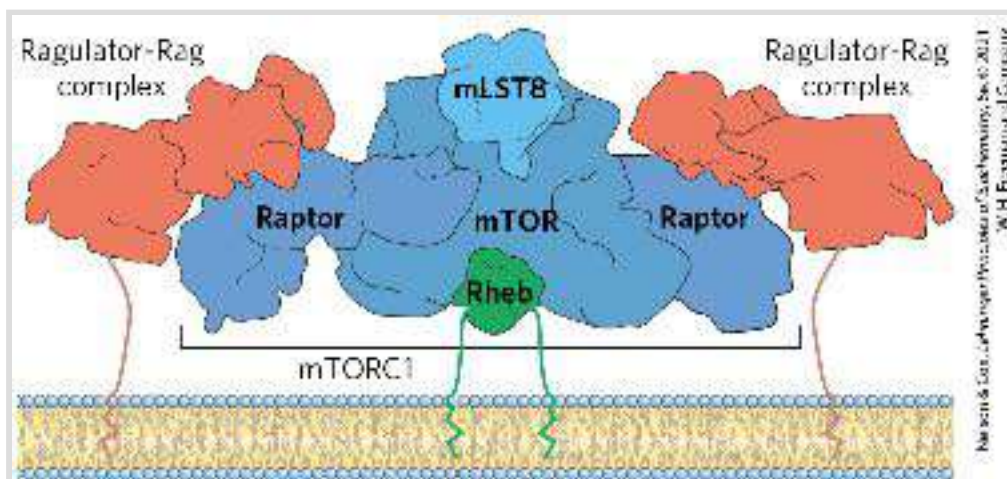
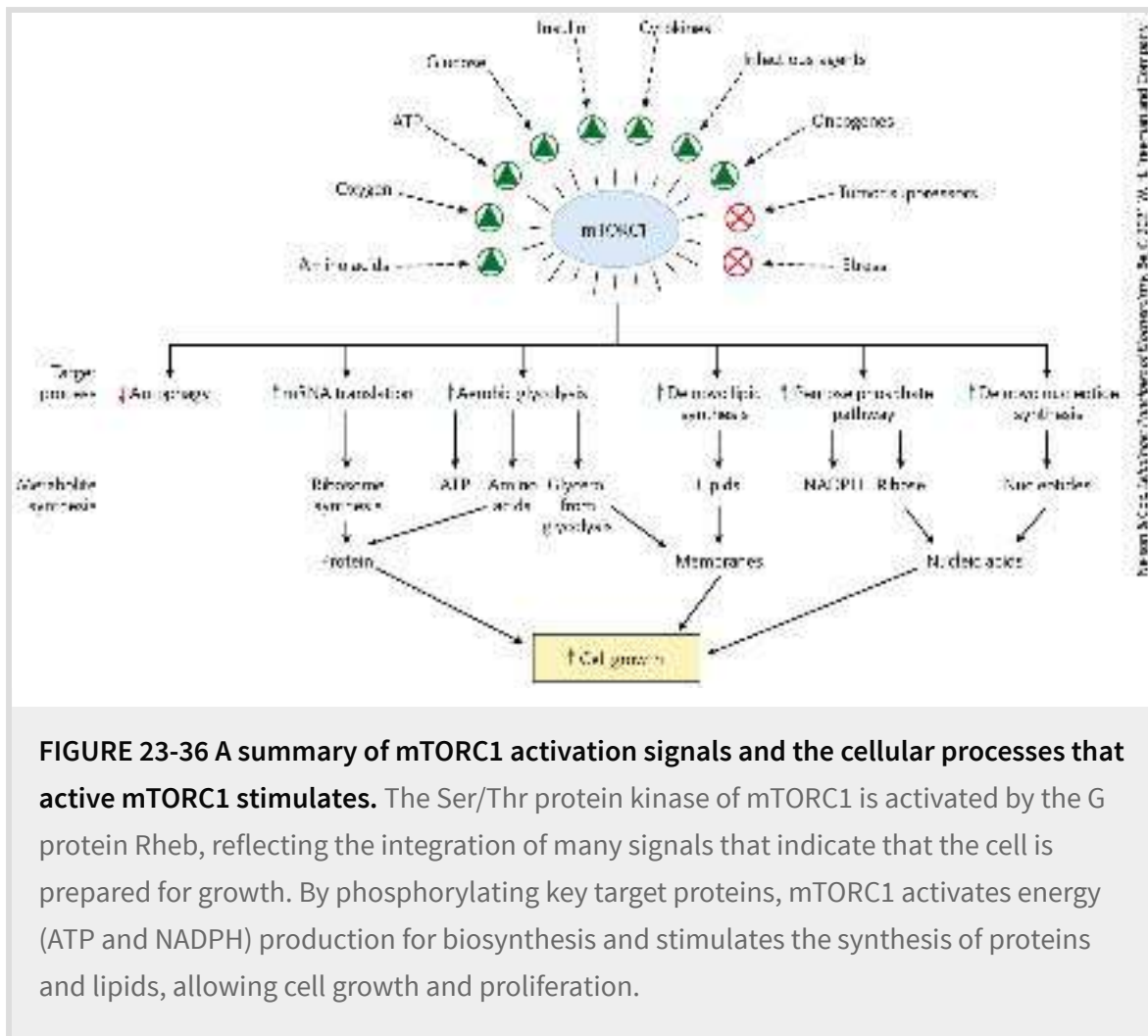


FIGURE 23-35 The mTORC1-Ragulator-Rag complex on the lysosomal surface. This model integrates the structures of mTORC1 and Ragulator-Rag, separately determined by cryo-EM. Raptor acts as a scaffold protein organizing the assembly, which puts the mTOR protein kinase in contact with the G protein Rheb for activation. [Information from J. H. Park et al., *Trends Biochem. Sci.* 45:367, 2019, Fig. 1; K. B. Rogala et al., *Science* 366:468, 2019, Fig. 5A.]



Fasting results in inactivation of mTORC1 by AMPK, leading to increased breakdown of protein and glycogen in the liver and muscle and mobilization of TAGs in adipose tissue. Chronic activation of mTORC1 by overeating results in excess deposition of TAGs in adipose tissue, as well as in liver and muscle, which may contribute to insulin insensitivity and type 2 diabetes. Mutations that produce constantly activated mTORC1 are commonly associated with human cancers.

Diet Regulates the Expression of Genes Central to Maintaining Body Mass

Proteins in a family of ligand-activated transcription factors known as **peroxisome proliferator-activated receptors (PPARs)** respond to changes in dietary lipid by altering the expression of genes involved in fat and carbohydrate metabolism. (These transcription factors were first recognized for their roles in peroxisome synthesis — thus their names.) Their normal ligands are fatty acids or fatty acid derivatives. PPAR γ , PPAR α , and PPAR δ act in the nucleus by forming heterodimers with another nuclear receptor, RXR, then binding to regulatory regions of DNA near the genes under their control and changing the rate of transcription of those genes ([Fig. 23-37](#)).

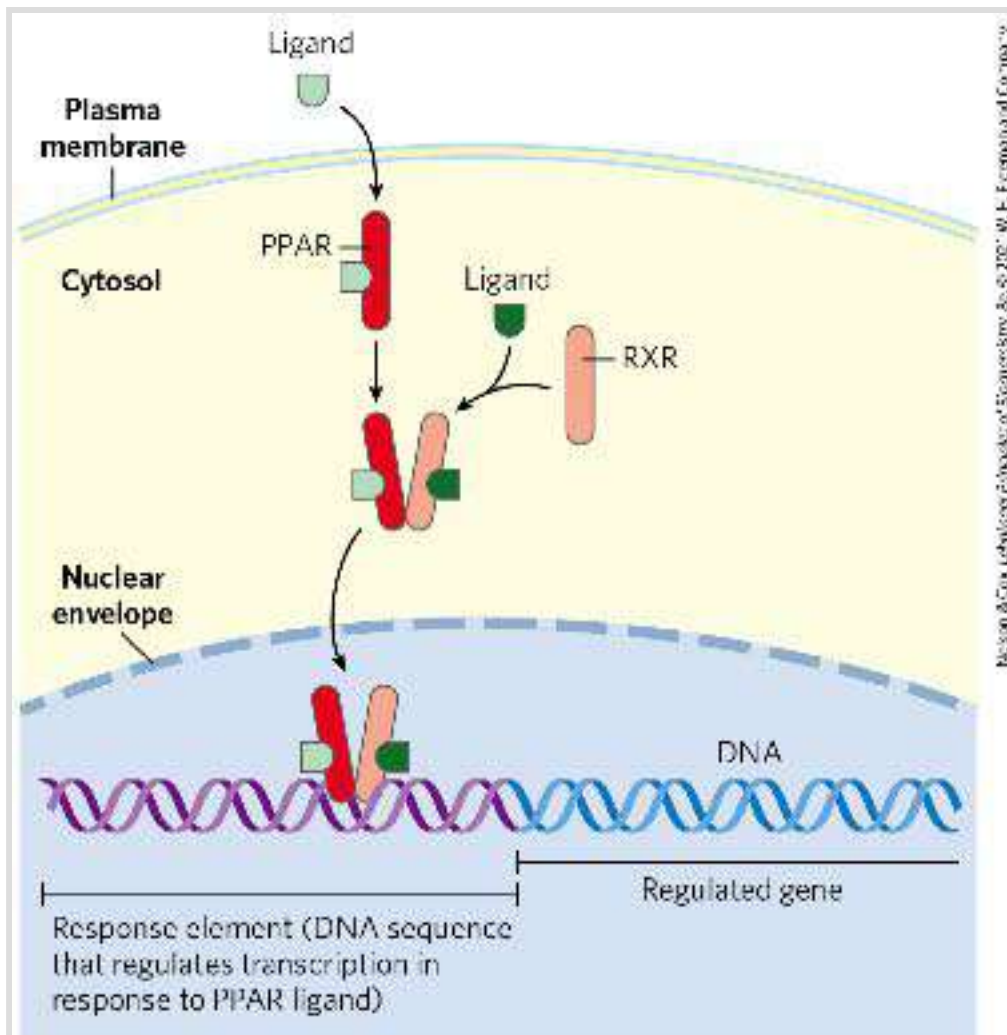


FIGURE 23-37 Mode of action of PPARs. PPARs are transcription factors that, when bound to their cognate ligand, form heterodimers with the nuclear receptor RXR. The dimer binds specific regions of DNA known as response elements, stimulating transcription of genes in those regions. [Information from R. M. Evans et al., *Nat. Med.* 10:355, 2004, Fig. 3.]

PPAR γ turns on genes that act in the differentiation of fibroblasts into adipocytes and genes that encode proteins required for lipid synthesis and storage in adipocytes ([Fig. 23-38](#)). PPAR γ is activated by the thiazolidinedione drugs that are used to treat type 2 diabetes (discussed below).

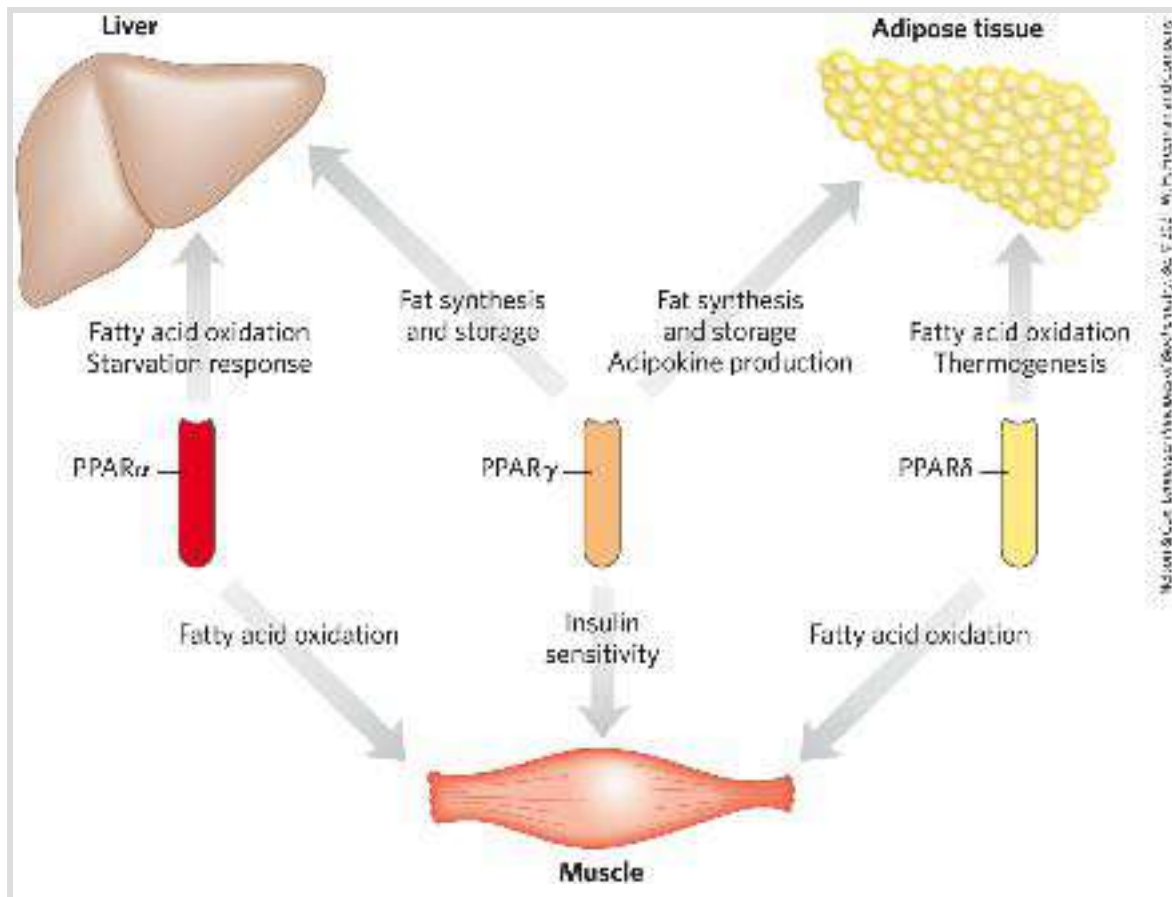


FIGURE 23-38 Metabolic integration by PPARs. The three PPAR isoforms regulate lipid and glucose homeostasis through their coordinated effects on gene expression in liver, muscle, and adipose tissue. PPAR α and PPAR δ regulate lipid utilization; PPAR γ regulates lipid storage and the insulin sensitivity of various tissues.

PPAR α is expressed in liver, kidney, heart, skeletal muscle, and brown adipose tissue. The ligands that activate this transcription factor include eicosanoids and free fatty acids. In hepatocytes, PPAR α turns on the genes necessary for the uptake and β oxidation of fatty acids and for the formation of ketone bodies during fasting.

PPAR δ is a key regulator of fat oxidation, which responds to changes in dietary lipid. PPAR δ acts in liver and muscle, stimulating the transcription of at least nine genes encoding

proteins for β oxidation and for energy dissipation through uncoupling of mitochondria. By stimulating fatty acid breakdown in uncoupled mitochondria, PPAR δ causes fat depletion, weight loss, and thermogenesis.

Short-Term Eating Behavior Is Influenced by Ghrelin, PPY_{3–36}, and Cannabinoids

The peptide hormone **ghrelin** is produced in cells lining the stomach. It is a powerful appetite stimulant that works on a shorter time scale (between meals) than leptin and insulin. Ghrelin receptors are located in the hypothalamus, affecting appetite, as well as in heart muscle and adipose tissue. Ghrelin acts through a GPCR to generate the second messenger IP₃, which mediates the hormone's action. The concentration of ghrelin in the blood fluctuates strikingly throughout the day, peaking just before a meal and dropping sharply just after a meal ([Fig. 23-39](#)). Injection of ghrelin into humans produces immediate sensations of intense hunger. Individuals with Prader-Willi syndrome, whose blood levels of ghrelin are exceptionally high, have an uncontrollable appetite, leading to extreme obesity that often results in death before the age of 30.

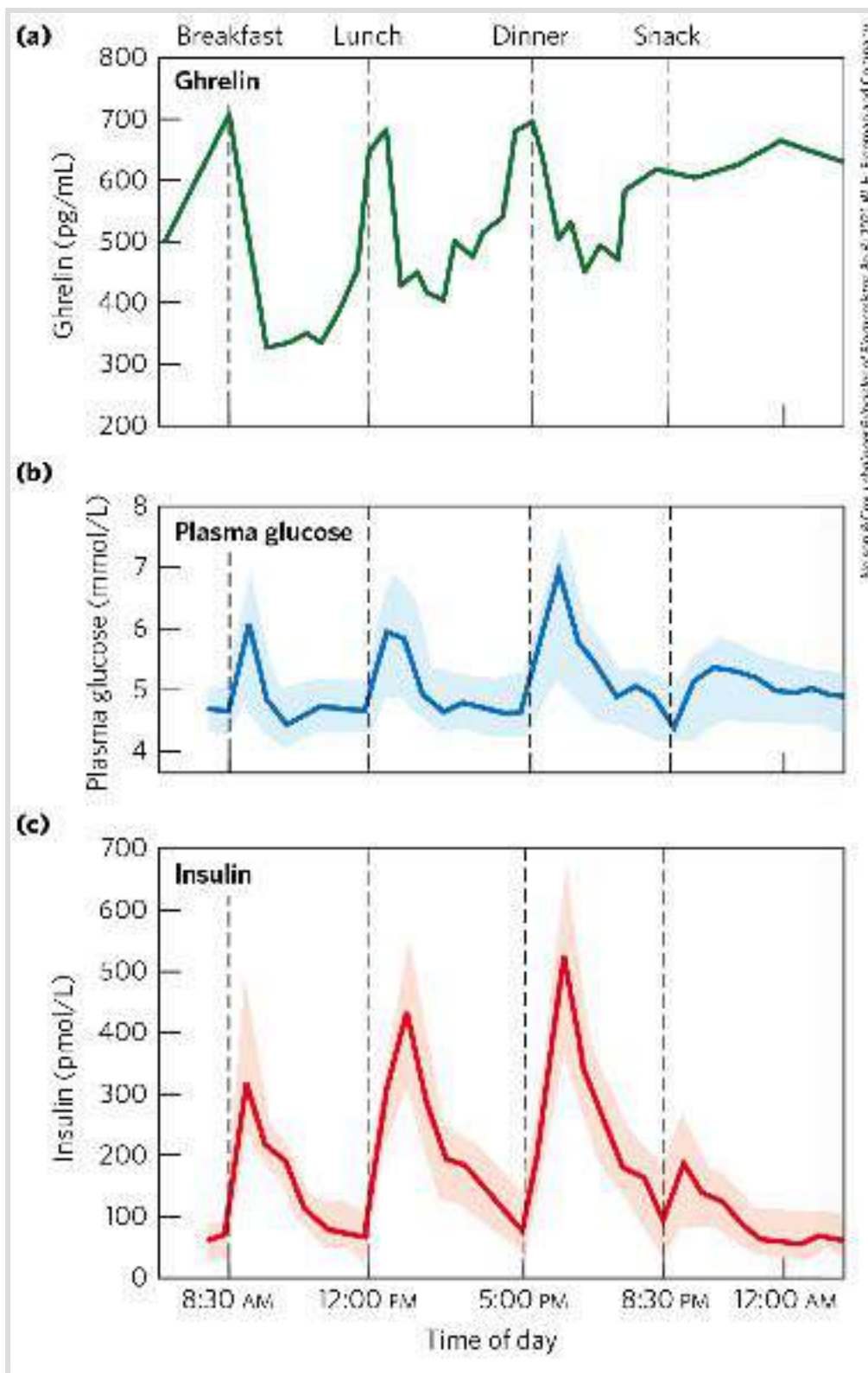


FIGURE 23-39 Variations in blood concentrations of glucose, ghrelin, and insulin relative to meal times. (a) Plasma levels of ghrelin rise sharply just *before* the usual time for meals (7 AM breakfast, 12 noon lunch, 5:30 PM dinner) and drop precipitously just after meals, paralleling subjective feelings of hunger. (b) Plasma glucose rises sharply *after* a meal, (c)

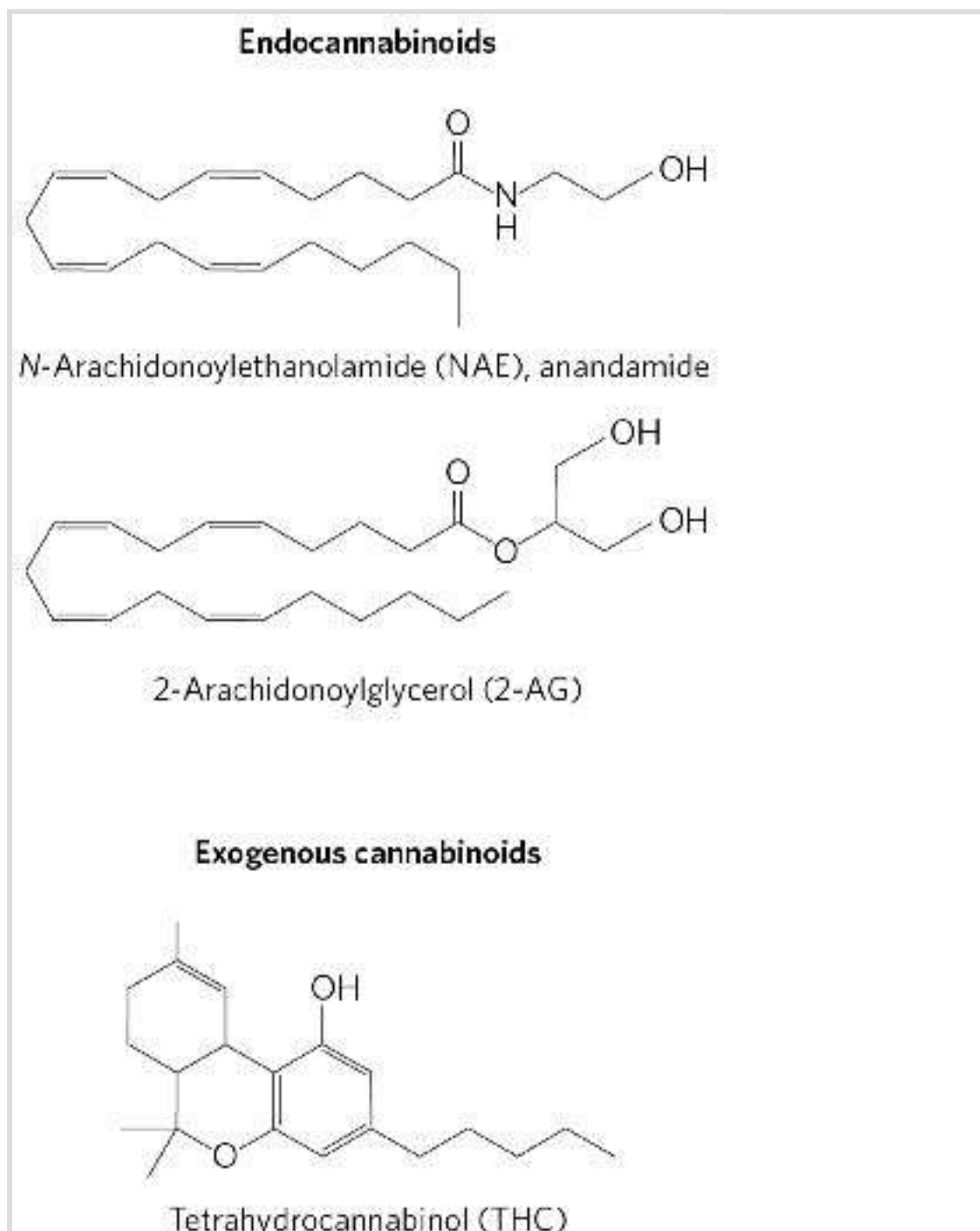
followed immediately by a rise in insulin level in response to the increased blood glucose. [(a, c) Data from D. E. Cummings et al., *Diabetes* 50:1714, 2001, Fig. 1. (b) Data from M. D. Feher and C. J. Bailey, *Br. J. Diabet. Vasc. Dis.* 4:39, 2004.]

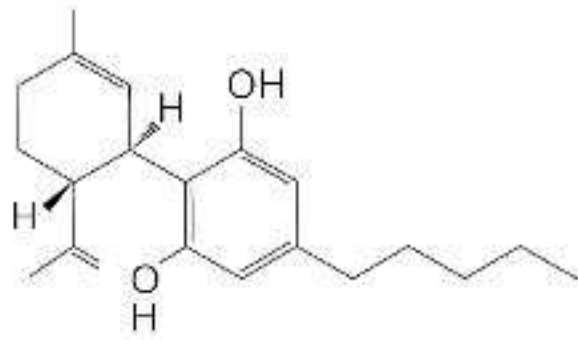
PYY₃₋₃₆ is a peptide hormone (34 amino acid residues) secreted by endocrine cells in the lining of the small intestine and colon in response to food entering from the stomach. The level of PYY₃₋₃₆ in the blood rises after a meal and remains high for some hours. The hormone is carried in the blood to the arcuate nucleus, where it acts on orexigenic neurons, inhibiting NPY release and reducing hunger ([Fig. 23-32](#)). Humans injected with PYY₃₋₃₆ feel little hunger and eat less than normal amounts for about 12 hours.

Endocannabinoids ([Fig. 23-40](#)) are eicosanoid lipid messengers that act through specific GPCRs in the brain and peripheral nervous system to increase appetite, heighten the sensory response to food (especially sweet and fatty foods), and elevate mood. When food enters the mouth, neuronal signals travel to the brain, and from there through the vagus nerve to the intestine, which then produces and releases endocannabinoids. The receptors for endocannabinoids control ion channels of sensory neurons, changing their membrane potentials and sending signals to the brain. Palatable food sensed in this way motivates further consumption of that food. The taste of fats (which are particularly high in caloric content) causes cannabinoid release, which effectively triggers further consumption. Well-conserved across vertebrate species, this system probably evolved to

maximize the intake of food and to guard against starvation. In mammals, cannabinoid action stimulates an increase in fat mass and inhibits energy loss by motor activity or thermogenesis.

Cannabinoid receptors also mediate the psychoactive effects of Δ^9 -tetrahydrocannabinol ([Fig. 23-40](#)), the main active ingredient in marijuana, long known for its stimulating effect on appetite.





Cannabidiol (CBD)

FIGURE 23-40 Cannabinoids. Two endocannabinoids produced by animals, and two products from the cannabis plant. Endocannabinoids carry retrograde signals: secreted by a postsynaptic cell, they diffuse across the synaptic cleft and activate GPCR receptors in the presynaptic cell. The receptors are of two types: CB₁ in the central nervous system, and CB₂ in the peripheral nervous system. Δ^9 -Tetrahydrocannabinol (THC), the psychoactive compound in cannabis, is an agonist of both types of cannabinoid receptors in animals; cannabidiol (CBD) does not bind either.

Microbial Symbionts in the Gut Influence Energy Metabolism and Adipogenesis

An adult human is host to about 10^{14} microbial cells that inhabit the gut. These microbes function as a major endocrine organ, producing a variety of metabolites with profound effects on host metabolism, feeding behavior, and body mass. Lean and obese individuals harbor different combinations of microbial symbionts in the gut. Investigation of this observation led to the discovery that gut microbes release fermentation products — the short-chain fatty acids acetate, propionate, butyrate, and lactate — that

enter the bloodstream and trigger metabolic changes in adipose tissue ([Fig. 23-41](#)). Propionate, for example, drives the expansion of white adipose tissue by acting on GPCRs in the plasma membranes of several cell types, including adipocytes. These receptors trigger differentiation of precursor cells (preadipocytes) into adipocytes and inhibit lipolysis in existing adipocytes, leading to an increase in white adipose tissue mass — that is, obesity. Gut microbes also convert primary bile acids, synthesized in the liver, into the secondary bile acids deoxycholate and lithocholate, which enter the bloodstream and act through GPCRs and steroid receptors to activate beige adipocytes to produce UCP1 and increase energy expenditure.

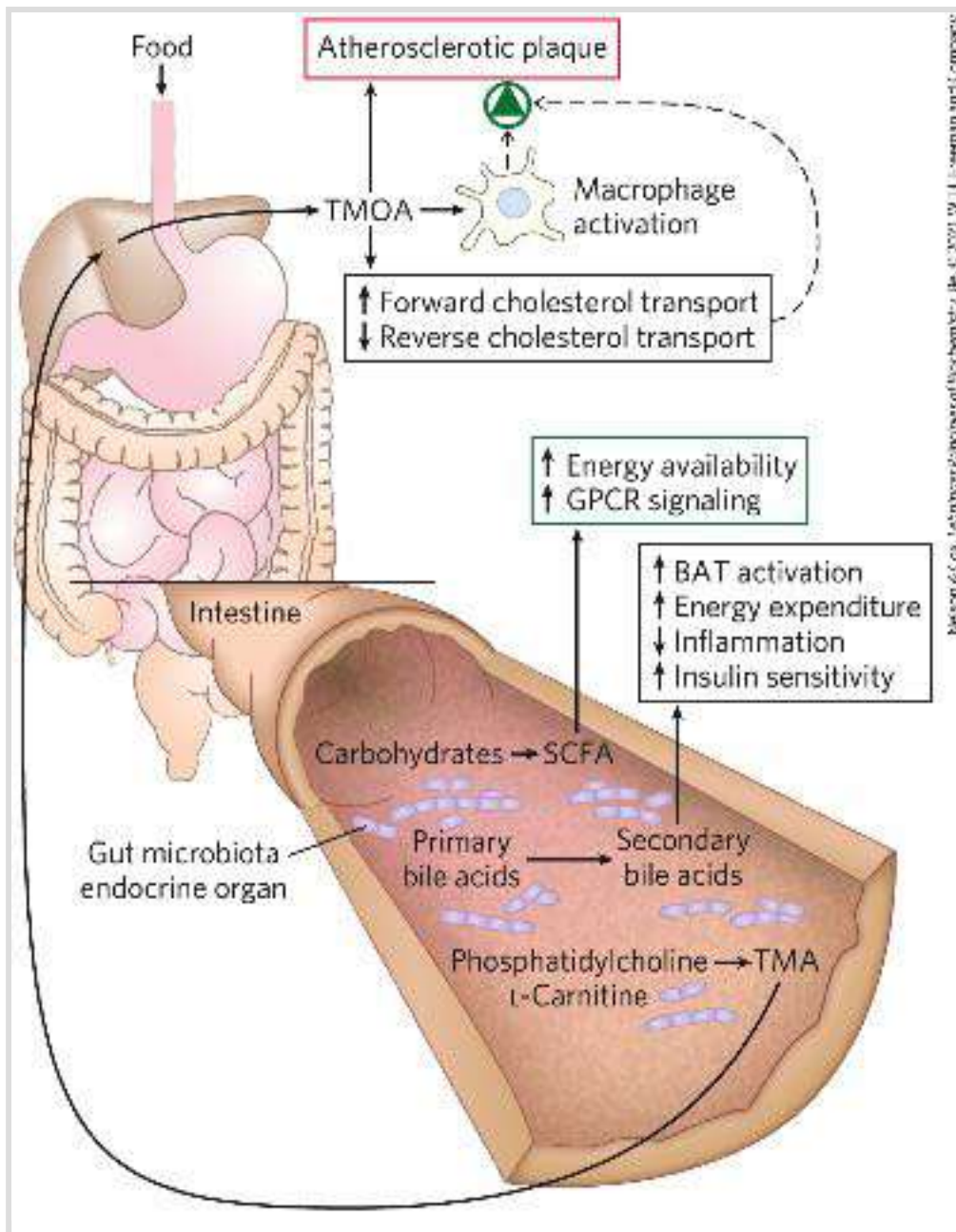


FIGURE 23-41 Effects of gut microbe metabolism on health. The enormous number and diversity of microorganisms (the microbiota) in the colon generate metabolic products that may have significant effects on health, both positive and negative. For example, the metabolism of primary bile acids by microbiota produces secondary products that act through nuclear receptors to stimulate thermogenesis in host brown adipose tissue (BAT) and increase both energy consumption and insulin sensitivity, while reducing inflammation. Metabolism of undigested carbohydrates by microbiota produces short-chain fatty acids (SCFAs) that signal expansion of the host's white adipose tissue (WAT), promoting obesity. SCFAs produced by microbiota are also a readily metabolizable energy source for

the host. Microbial production of the SCFA propionate prevents lipogenesis in the liver and lowers blood cholesterol, both favorable to health. On the other hand, metabolic conversion of phosphatidylcholine and L-carnitine to trimethylamine (TMA), and its further conversion in liver to trimethylamine-*N*-oxide (TMAO), results in receptor-mediated changes in cholesterol transport and macrophage activity. The combination of altered sterol transport and increased macrophage activity can lead to formation of atherosclerotic plaque (see [Fig. 21-46](#)).

These findings raise the possibility of preventing obesity by altering the makeup of the microbial community in the gut. Weight reduction might be accomplished either by adding, directly to the gut, microbial species ([probiotics](#)) that disfavor adipogenesis, or by adding to the diet nutrients ([prebiotics](#)) that favor the dominance of probiotic microbes. For example, experiments in mice have shown that fructans, polymers of fructose that are indigestible by animals, favor a specific microbial community. When this combination of microorganisms is present, fat storage in white adipose tissue and liver decreases, and there is none of the decrease in insulin sensitivity that is associated with obesity and lipid deposition in the liver (see below). Investigators have transplanted fecal material from lean mice to fat ones, and found that a new collection of microbes became established in the gut of the recipient animals, and the animals lost weight.



Endocrine cells in the lining of the intestinal tract secrete peptides — the anorexigenic PYY₃₋₃₆ and GLP-1 and the orexigenic ghrelin — that modulate food intake and energy expenditure. Interaction with specific microbes in the gut, or with

their fermentation products, may trigger release of these peptides. Understanding how diet and microbial symbionts interact to affect energy metabolism and adipogenesis is an important key to understanding the development of obesity, metabolic syndrome, and type 2 diabetes and is a major challenge for the future.

The exquisitely interconnected system of neuroendocrine controls of food intake and metabolism presumably evolved to protect against starvation and to eliminate counterproductive accumulation of fat (extreme obesity). The difficulty most people face in trying to lose weight testifies to the remarkable effectiveness of these controls.

SUMMARY 23.4 *Obesity and the Regulation of Body Mass*

■ Adipose tissue produces leptin, a hormone that regulates feeding behavior and energy expenditure so as to maintain adequate reserves of fat. Leptin production and release increase with the number and size of adipocytes.

■ Leptin acts on receptors in the brain, causing the release of anorexigenic (appetite-suppressing) peptides, including α -MSH, that act in the brain to inhibit eating.

■ Adiponectin stimulates fatty acid uptake and oxidation and inhibits fatty acid synthesis. It also sensitizes muscle and liver to insulin. At least some of the actions of adiponectin are mediated by AMPK, which is also activated by metabolic stress (low [AMP]), and by exercise.

- The protein kinase complex mTORC1 gauges the supply of essential amino acids and other metabolites, triggering cell growth if all required nutrients are available. It thus complements AMPK in determining the energetic status of a cell.
- Expression of the enzymes of lipid synthesis is under tight and complex regulation. PPARs are transcription factors that determine the rate of synthesis of many enzymes involved in lipid metabolism and adipocyte differentiation.
- Ghrelin, a hormone produced in the stomach, acts on orexigenic (appetite-stimulating) neurons in the arcuate nucleus to produce hunger before a meal. PYY₃₋₃₆, a peptide hormone of the intestine, acts at the same site to lessen hunger after a meal. Endocannabinoids signal the availability of sweet or fatty food and stimulate its consumption.
- Microbial symbionts in the gut produce fermentation products and secondary bile acids, which influence release of gut hormones that regulate body mass.

23.5 Diabetes Mellitus



Diabetes mellitus is a relatively common disease: about 9% of the U.S. population, and nearly 25% of the U.S. population over the age of 65, show some degree of abnormality in glucose metabolism that is indicative of diabetes or a tendency toward the condition. There are two major clinical classes of diabetes mellitus: **type 1 diabetes**, sometimes referred to as insulin-dependent diabetes mellitus (IDDM), and **type 2 diabetes**, or non-insulin-dependent diabetes mellitus (NIDDM), also called insulin-resistant diabetes. The discovery of insulin and its role in diabetes led to its development as a pharmaceutical, saving millions of lives ([Box 23-2](#)). ■

BOX 23-2 MEDICINE

The Arduous Path to Purified Insulin

Millions of people with type 1 diabetes mellitus inject themselves daily with pure insulin to compensate for the lack of production of this critical hormone by their own pancreatic β cells. Insulin injection is not a cure for diabetes, but it allows people who otherwise would have died young to lead long and productive lives. The discovery of insulin, which began with an accidental observation, illustrates the combination of serendipity and careful experimentation that led to the discovery of many hormones.

In 1889, Oskar Minkowski, a young assistant at the Medical College of Strasbourg, and Josef von Mering, at the Hoppe-Seyler Institute in Strasbourg, had a friendly disagreement about whether the pancreas, known to contain lipases, was important in fat digestion in dogs. To resolve the issue, they began an experiment on the digestion of fats. They surgically removed the pancreas from a dog, but before their experiment got any farther, Minkowski noticed that

the dog was now producing far more urine than normal (a common symptom of untreated diabetes). Also, the dog's urine had glucose levels far above normal (another symptom of diabetes). These findings suggested that lack of some pancreatic product caused diabetes.

Minkowski tried unsuccessfully to prepare an extract of dog pancreas that would reverse the effect of removing the pancreas — that is, would lower the urinary or blood glucose levels. Despite considerable effort, no significant progress was made in the isolation or characterization of the “antidiabetic factor” until the summer of 1921, when Frederick G. Banting, a young scientist working in the laboratory of J. J. R. MacLeod at the University of Toronto, and a student assistant, Charles Best, took up the problem. By that time, several lines of evidence pointed to a group of specialized cells in the pancreas (the islets of Langerhans; see [Fig. 23-23](#)) as the source of the antidiabetic factor, which came to be called insulin (from Latin *insula*, “island”).

Taking precautions to prevent proteolysis by the pancreatic proteases trypsin and chymotrypsin, Banting and Best (later aided by biochemist J. B. Collip) succeeded in December 1921 in preparing a purified pancreatic extract that cured the symptoms of experimentally induced diabetes in dogs. On January 25, 1922 (just one month later!), their insulin preparation was injected into Leonard Thompson, a 14-year-old boy severely ill with diabetes mellitus. Within days, the levels of ketone bodies and glucose in Thompson's urine dropped dramatically; the extract saved his life and the lives of other seriously ill children who also received these early preparations ([Fig. 1](#)). In 1923, Banting and MacLeod won the Nobel Prize for their isolation of insulin. Banting immediately announced that he would share his prize with Best; MacLeod shared his with Collip.

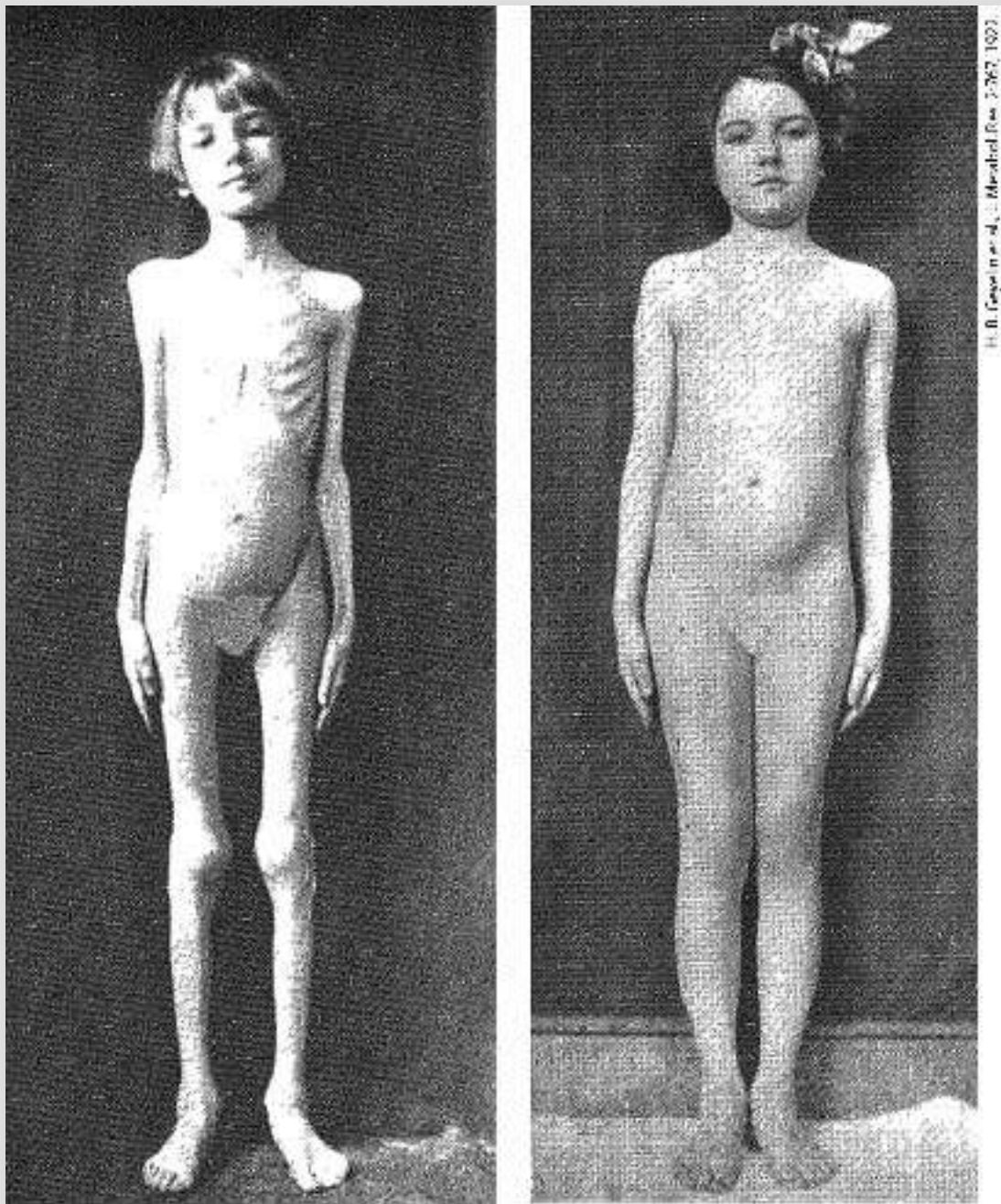


FIGURE 1 A child with type 1 diabetes, before (left) and after (right) three months of treatment with an early preparation of insulin.

By 1923, pharmaceutical companies were supplying thousands of patients throughout the world with insulin extracted from porcine pancreas. With the development of genetic engineering techniques in the 1980s, it became possible to produce unlimited quantities of human insulin by inserting the cloned human gene for insulin into a microorganism, which was then cultured on an industrial scale. Many people with diabetes are now fitted with

implanted insulin pumps, which release adjustable amounts of insulin on demand to meet changing needs at meal times and during exercise. There is a reasonable prospect that, in the future, transplanted pancreatic tissue will provide a source of insulin that responds as well as a normal pancreas, releasing insulin into the bloodstream only when blood glucose rises.

Diabetes Mellitus Arises from Defects in Insulin Production or Action



Type 1 diabetes usually begins early in life, and symptoms quickly become severe. This disease responds to insulin injection, because the metabolic defect stems from an autoimmune destruction of pancreatic β cells and a consequent inability to produce sufficient insulin. Type 1 diabetes requires both insulin therapy and careful, lifelong control of the balance between dietary intake, activity, and insulin dose. Characteristic symptoms of untreated type 1 (and type 2) diabetes are excessive thirst and frequent urination, leading to the intake of large volumes of water. These symptoms are due to excretion of large amounts of glucose in the urine. (“Diabetes mellitus” means “excessive excretion of sweet urine.”)

Type 2 diabetes is slower to develop (typically but not always in obese adults), and the symptoms are milder and often go unrecognized at first. This is really a group of diseases in which the regulatory activity of insulin is disordered: insulin is produced, but some feature of the insulin-response system is defective. Many individuals with this disorder are insulin-

resistant. The connection between type 2 diabetes and obesity (discussed below) is an active area of research.

The pathology of diabetes includes cardiovascular disease, renal failure, blindness, and neuropathy. In 2019, the global mortality from diabetes was estimated at 4.2 million and rising. It is essential to understand diabetes and its relationship to obesity and to find countermeasures that prevent or reverse the damage done by this disease.

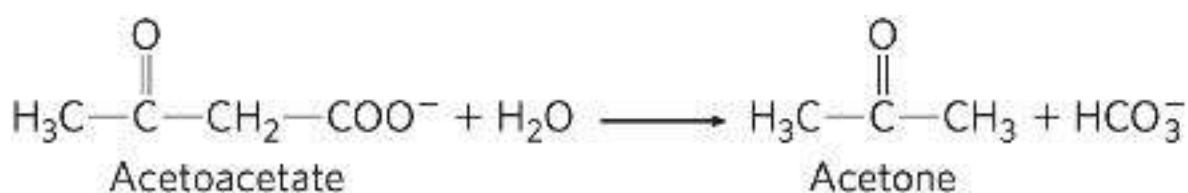
 Individuals with either type of diabetes are unable to take up glucose efficiently from the blood. Recall that insulin triggers the movement of GLUT4 glucose transporters to the plasma membrane in muscle and adipose tissue (see [Fig. 12-23](#) and [Box 11-1](#)). Biochemical measurements on blood and urine samples are essential in the diagnosis and treatment of diabetes. A sensitive diagnostic criterion is the level of HbA1c, a glucose derivative of hemoglobin, which forms in the blood and reflects the average blood glucose level (see [Box 7-2](#)). Another measurement to confirm the diagnosis of diabetes is the **glucose-tolerance test**. The individual fasts overnight, then drinks a test dose of 100 g of glucose dissolved in a glass of water. The blood glucose concentration is measured before the test dose and at 30 min intervals for several hours thereafter. A healthy individual assimilates the glucose readily, the blood glucose rising to no more than about 9 or 10 mM; little or no glucose appears in the urine. In diabetes, individuals assimilate the test dose of glucose poorly; their blood glucose level rises dramatically and returns to

the fasting level very slowly. Because the blood glucose levels exceed the kidney threshold (about 10 mM), glucose also appears in the urine. ■

Carboxylic Acids (Ketone Bodies) Accumulate in the Blood of Those with Untreated Diabetes



With glucose unavailable to cells, fatty acids become the principal fuel, which leads to another characteristic metabolic change in diabetes: excessive but incomplete oxidation of fatty acids in the liver. The acetyl-CoA produced by β oxidation cannot be completely oxidized by the citric acid cycle, because the high $[\text{NADH}]/[\text{NAD}^+]$ ratio produced by β oxidation inhibits the cycle (recall that three steps of the cycle convert NAD^+ to NADH). Accumulation of acetyl-CoA leads to overproduction of the ketone bodies β -hydroxybutyrate and acetoacetate, which cannot be used by extrahepatic tissues as fast as they are made in the liver (see [Figs. 17-15](#), [17-16](#)). In addition to β -hydroxybutyrate and acetoacetate, the blood of individuals with diabetes contains small amounts of acetone, which results from the spontaneous decarboxylation of acetoacetate:



The overproduction of ketone bodies, called **ketosis**, results in greatly increased concentrations of ketone bodies in the blood (ketonemia) and urine (ketonuria). The ketone bodies are carboxylic acids, which ionize, releasing protons. In uncontrolled diabetes, this acid production can overwhelm the capacity of the blood's bicarbonate buffering system and produce a lowering of blood pH called **acidosis** or, in combination with ketosis, **ketoacidosis**, a potentially life-threatening condition. ■

In Type 2 Diabetes the Tissues Become Insensitive to Insulin



In the industrialized world, where the food supply is more than adequate, there is a growing epidemic of obesity and the type 2 diabetes associated with it.  The hallmark of **type 2 diabetes** is the development of insulin resistance, a state in which more insulin is needed to bring about the biological effects produced by a lower concentration of insulin in the normal, healthy state. In the early stages of the disease, pancreatic β cells secrete enough insulin to overcome the lower insulin sensitivity of muscle and liver. But the β cells eventually fail, and the lack of insulin becomes apparent in the body's inability to regulate blood glucose. The intermediate stage, preceding type 2 diabetes mellitus, is sometimes called [metabolic syndrome](#). This is typified by obesity, especially in the abdomen; hypertension (high blood pressure); abnormal blood lipids (high TAG and LDL, low HDL); slightly high fasting blood glucose; and a reduced ability to

clear glucose in the glucose-tolerance test. Individuals with metabolic syndrome often also show changes in blood proteins associated with abnormal clotting (high fibrinogen concentration) or inflammation (high concentration of the C-reactive peptide, not to be confused with the C peptide generated during the proteolytic maturation of insulin). About 30% of the adult population in the United States has these indicators of metabolic syndrome.

What predisposes individuals with metabolic syndrome to develop type 2 diabetes? According to the “lipid toxicity” hypothesis ([Fig. 23-42](#)), the action of PPAR δ on adipocytes normally keeps the cells ready to synthesize and store triacylglycerols — the adipocytes are insulin-sensitive and produce leptin, which leads to their continued intracellular deposition of TAGs. However, excess caloric intake in obese individuals causes adipocytes to become filled with TAGs, leaving adipose tissue unable to meet any further demand for TAG storage. Lipid-filled adipose tissue releases protein factors, including MCP-1 (monocyte chemotaxis protein-1), that attract macrophages, which infiltrate the tissue and may eventually represent as much as 50% of the adipose tissue by mass. Macrophages trigger the inflammatory response mediated by TNF α release, which impairs TAG deposition in adipocytes and favors release of free fatty acids into the blood. These excess fatty acids enter liver and muscle cells, where they are converted to TAGs that accumulate as lipid droplets. This ectopic (Greek *ektopos*, “out of place”) deposition of TAGs inhibits GLUT4

transporter movement to the plasma membrane, which leads to insulin insensitivity in liver and muscle, the hallmark of type 2 diabetes.

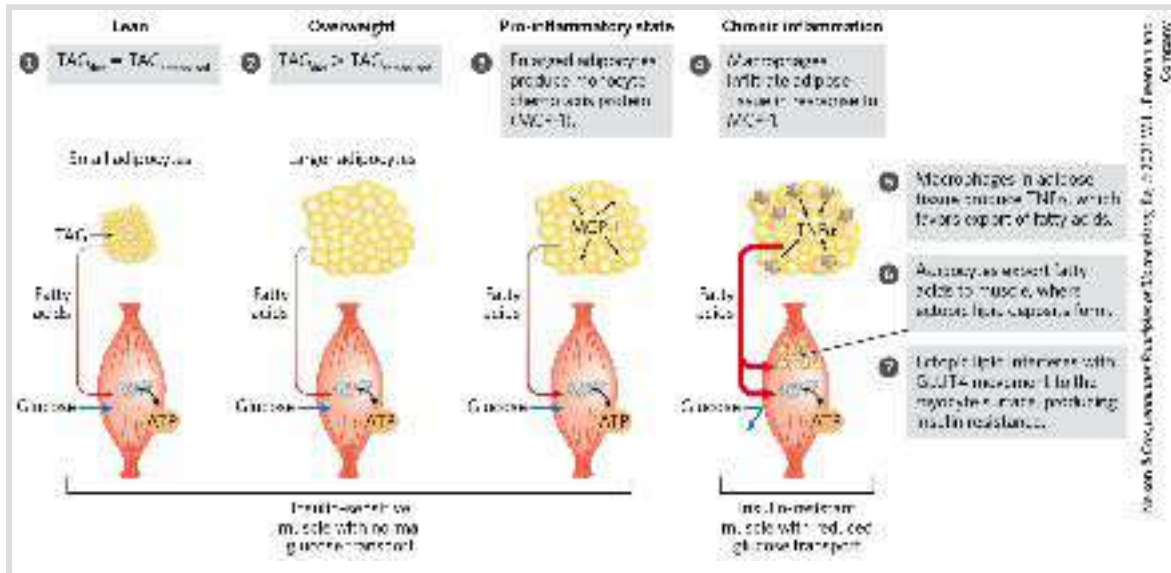


FIGURE 23-42 Overloading adipocytes with triacylglycerols triggers inflammation in fat tissue and ectopic lipid deposition and insulin resistance in muscle. In an individual of healthy body mass, dietary TAG uptake equals TAG oxidation for energy. In overweight individuals, excess caloric intake results in enlarged adipocytes, engorged with TAG and unable to store more. Enlarged adipocytes secrete MCP-1 (monocyte chemotaxis protein-1), attracting macrophages. Macrophages infiltrate the adipose tissue and produce $TNF\ \alpha$ (tumor necrosis factor α), which triggers lipid breakdown and release of fatty acids into the blood. The fatty acids enter myocytes, where they accumulate in small lipid droplets. This ectopic lipid storage in muscle somehow causes insulin resistance, perhaps by triggering lipid-activated protein kinases that inactivate some element in the insulin-signaling pathway. GLUT4 glucose transporters leave the myocyte surface, preventing glucose entry into muscle; the myocyte has now become insulin-resistant. It cannot use blood glucose for its fuel, so fatty acids are mobilized from adipose tissue and become the primary fuel. The increased influx of fatty acids into muscle leads to further deposition of ectopic lipids. In some individuals, insulin resistance develops into type 2 diabetes. [Information from A. Guilherme et al., *Mol. Cell Biol.* 9:367, 2008, Fig. 1.]

According to this hypothesis, excess stored fatty acids and TAGs are toxic to liver and muscle. Some individuals are less well equipped genetically to handle this burden of ectopic lipids and are more susceptible to the cellular damage that leads to development of type 2 diabetes. Insulin resistance probably involves impairment of several of the mechanisms by which insulin acts on metabolism, which include changes in protein levels and changes in the activities of signaling enzymes and transcription factors. For example, both adiponectin synthesis in adipocytes and adiponectin levels in the blood decrease with obesity and increase with weight loss.

There are genetic factors that predispose individuals to type 2 diabetes. Although 80% of people with type 2 diabetes are obese, most obese individuals do not develop type 2 diabetes. Given the complexity of the regulatory mechanisms we have discussed in this chapter, it is not surprising that the genetics of diabetes is complex, involving interactions among variant genes and environmental factors, including diet and lifestyle. A growing number of genetic loci have been reliably linked to type 2 diabetes; variation in any of these “diabetogenes” alone would cause a relatively small increase in the likelihood of developing type 2 diabetes. ■

Type 2 Diabetes Is Managed with Diet, Exercise, Medication, and Surgery



Studies show that at least four factors improve the health of individuals with type 2 diabetes: dietary restriction, regular exercise, drugs that increase insulin sensitivity or insulin production, and surgery that reroutes food passage through the gastrointestinal tract. Dietary restriction (and accompanying weight loss) reduces the overall burden of handling fatty acids. The lipid composition of the diet influences, through PPARs and other transcription factors, the expression of genes that encode proteins involved in burning fat.

Exercise contributes to weight loss directly by consuming calories. Exercise also increases release of **irisin** from muscle into the blood. Irisin increases the expression of *UCP1* genes in white adipose tissue and also stimulates the development of beige adipocytes, so that even after the exercise ends, energy continues to be used in thermogenesis. Exercise activates AMPK, as does adiponectin; AMPK shifts metabolism toward fat oxidation and inhibits fat synthesis.

Several classes of drugs are used in the management of type 2 diabetes ([Table 23-7](#)). Their targets include AMPK, the K^+ channels in β cells, PPARs, and GLP receptors.

TABLE 23-7 Treatments for Type 2 Diabetes Mellitus

Intervention/treatment	Direct target	Effect of treatment
Weight loss	Adipose tissue;	Reduces lipid burden; increases capacity for lipid storage in adipose tissue;

	reduction in TAG content	restores insulin sensitivity
Exercise	AMPK, activated by increasing [AMP]/[ATP]	Aids weight loss (see Fig. 23-34)
Bariatric surgery	Unknown	Leads to weight loss, better control of blood glucose
Sulfonylureas: glipizide (Glucotrol), glyburide (Micronase), glimepiride (Amaryl)	Pancreatic β cells; K^+ channels blocked	Stimulates insulin secretion by pancreas (see Fig. 23-24)
Biguanides: metformin (Glucophage)	AMPK, activated	Increases glucose uptake by muscle; decreases glucose production in liver
Thiazolidinediones: rosiglitazone (Avandia), pioglitazone (Actos)	PPAR γ	Stimulates expression of genes potentiating the action of insulin in liver, muscle, adipose tissue; increases glucose uptake; decreases glucose synthesis in liver
GLP-1 modulators: exenatide (Byetta), sitagliptin (Januvia), dulaglutide (Trulicity)	Glucagon-like peptide-1, dipeptide protease IV	Enhances insulin secretion by pancreas

In cases of extreme obesity, dramatic weight loss can be achieved by bariatric surgery, which reroutes food movement through the

stomach and small intestine. In many cases, this procedure also moderates or even reverses type 2 diabetes. In Roux-en-Y gastric bypass (RYGBP, named for César Roux, the Swiss surgeon who developed the procedure), the stomach is reduced to a small pouch attached to the esophagus, and the middle region of the small intestine (the jejunum) is attached directly to the pouch. Food bypasses most of the stomach and the duodenum, and goes primarily to the “Roux limb” of the intestine. Stomach acid and digestive enzymes travel through bypassed portions of the gut to join the food in the common channel. People who undergo RYGBP surgery not only experience dramatic weight loss but also are less hungry. Remarkably, this surgery also reverses type 2 diabetes in many cases. The explanations for these effects are likely to lie in altered communication among the gut, the brain, and other organs. This may result from changes in the kind and amount of peptide hormones (such as GLP-1 and PYY₃₋₃₆) secreted in the intestine that signal satiety and inhibit feeding behavior. The last word has not been written on this issue. ■

SUMMARY 23.5 *Diabetes Mellitus*

■ Metabolic syndrome, which includes obesity, hypertension, elevated blood lipids, and insulin resistance, is often the prelude to type 2 diabetes. Uncontrolled diabetes is characterized by high glucose levels in the blood and urine and the production and excretion of ketone bodies.

■ In diabetes, insulin is either not produced or not recognized by the tissues, and the uptake of blood glucose is defective. Lacking access to glucose, cells rely on fatty acid oxidation, which results in ketone body formation, producing ketoacidosis.

■ The insulin resistance that characterizes type 2 diabetes may be a consequence of abnormal lipid storage in muscle and liver, in response to a lipid intake that cannot be accommodated by adipose tissue.

■ Effective treatments for type 2 diabetes include exercise, appropriate diet, and drugs that increase insulin sensitivity or insulin production. Surgical alteration of the digestive tract leads to weight loss and often reverses type 2 diabetes.

Chapter Review

KEY TERMS

Terms in bold are defined in the glossary.

hormone

neuroendocrine system

metabotropic

ionotropic

endocrine

paracrine

autocrine

hypothalamus

hepatocyte

white adipose tissue (WAT)

adipocyte

brown adipose tissue (BAT)

uncoupling protein 1 (UCP1)

thermogenesis

beige adipose tissue

myocyte

plasma proteins

ATP-gated K^+ channels

sulfonylurea drugs

glucagon

cortisol

body mass index (BMI)

adipokines

leptin

arcuate nucleus

orexigenic

anorexigenic

α -melanocyte-stimulating hormone (α -MSH)

adiponectin

AMP-activated protein kinase (AMPK)

mTORC1 (mechanistic target of rapamycin complex 1)

PPAR (peroxisome proliferator-activated receptor)

ghrelin

endocannabinoids

probiotics

prebiotics

diabetes mellitus

type 1 diabetes

type 2 diabetes

glucose-tolerance test

ketosis

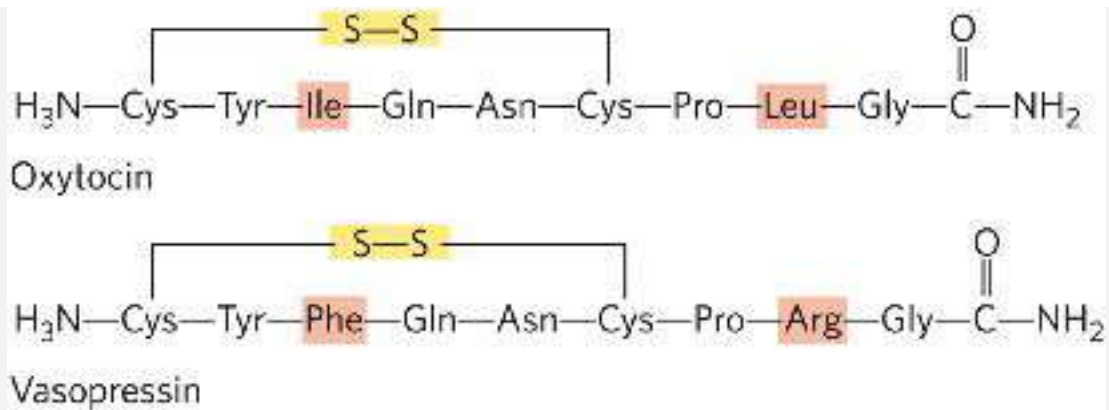
acidosis

ketoacidosis

metabolic syndrome

PROBLEMS

1. Peptide Hormone Activity Explain how two peptide hormones as structurally similar as oxytocin and vasopressin can have such different effects (see [Fig. 23-8](#)).



2. Metabolism of Glutamate in the Brain Brain tissue takes up glutamate from the blood, transforms it into glutamine, and releases the glutamine into the blood. What does this metabolic conversion accomplish? How does this conversion take place? The amount of glutamine produced in the brain can exceed the amount of glutamate entering from the blood. How does this extra glutamine arise? (Hint: You may want to review amino acid catabolism in [Chapter 18](#); recall that NH_4^+ is very toxic to the brain.)

3. Proteins as Fuel during Fasting When muscle proteins undergo catabolism in skeletal muscle during a fast, what are the fates of the amino acids?

4. Absence of Glycerol Kinase in Adipose Tissue The biosynthesis of triacylglycerols requires glycerol 3-phosphate. Adipocytes, specialized for the synthesis and degradation of TAGs, cannot use glycerol directly because they lack glycerol kinase, which catalyzes the reaction



How does adipose tissue obtain the glycerol 3-phosphate necessary for TAG synthesis?

5. Oxygen Consumption during Exercise A sedentary adult consumes about 0.05 L of O₂ in 10 seconds. A sprinter running a 100 m race consumes about 1 L of O₂ in 10 seconds. After finishing the race, the sprinter continues to breathe at an elevated (but declining) rate for some minutes, consuming an extra 4 L of O₂ above the amount consumed by the sedentary individual.

- a. Why does the need for O₂ increase dramatically during the sprint?
- b. Why does the demand for O₂ remain high after the sprinter finishes the race?

6. Thiamine Deficiency and Brain Function Individuals with thiamine deficiency show some characteristic neurological signs and symptoms, including loss of reflexes, anxiety, and mental confusion. Why might thiamine deficiency manifest as changes in brain function?

7. Potency of Hormones Under normal conditions, the human adrenal medulla secretes epinephrine (C₉H₁₃NO₃) at a rate sufficient to maintain a concentration of 10⁻¹⁰ M in circulating blood. To appreciate what that concentration means, calculate the volume of water that you would need to dissolve 1.0 g (about 1 teaspoon) of epinephrine to a concentration equal to that in blood.

8. Regulation of Hormone Levels in the Blood The half-life of most hormones in the blood is relatively short. For example,

when researchers inject radioactively labeled insulin into an animal, half of the labeled hormone disappears from the blood within 30 min.

- a. What is the importance of the relatively rapid inactivation of circulating hormones?
- b. In what ways can the organism make rapid changes in the level of a circulating peptide hormone?

9. Water-Soluble versus Lipid-Soluble Hormones On the basis of their physical properties, hormones fall into one of two categories: those that are very soluble in water but relatively insoluble in lipids (e.g., epinephrine) and those that are relatively insoluble in water but highly soluble in lipids (e.g., steroid hormones). In their role as regulators of cellular activity, most water-soluble hormones do not enter their target cells. The lipid-soluble hormones, by contrast, do enter their target cells and ultimately act in the nucleus. What is the relationship between solubility, the location of receptors, and the mode of action of these two classes of hormones?

10. Metabolic Differences between Muscle and Liver in a “Fight-or-Flight” Situation When an animal confronts a “fight-or-flight” situation, the release of epinephrine promotes glycogen breakdown in the liver and skeletal muscle. The end product of glycogen breakdown in the liver is glucose; the end product in skeletal muscle is pyruvate.

- a. What is the reason for the different products of glycogen breakdown in the two tissues?
- b. What is the advantage of these specific glycogen breakdown routes to an animal that must fight or flee?

11. **Excessive Amounts of Insulin Secretion:**

Hyperinsulinism Certain malignant tumors of the pancreas cause excessive production of insulin by β cells. Affected individuals exhibit shaking and trembling, weakness and fatigue, sweating, and hunger.

- a. What is the effect of hyperinsulinism on the metabolism of carbohydrates, amino acids, and lipids by the liver?
- b. What are the causes of the observed symptoms? Suggest why this condition, if prolonged, leads to brain damage.

12. **Thermogenesis Caused by Thyroid Hormones** Thyroid hormones are intimately involved in regulating the basal metabolic rate. Liver tissue of animals given excess thyroxine shows an increased rate of O_2 consumption and increased heat output (thermogenesis), but the ATP concentration in the tissue is normal. Different explanations have been offered for the thermogenic effect of thyroxine. One is that excess thyroxine causes uncoupling of oxidative phosphorylation in mitochondria. How could such an effect account for the observations? Another explanation suggests that thermogenesis is due to an increased rate of ATP utilization by the thyroxine-stimulated tissue. Is this a reasonable explanation? Why or why not?

13. **Function of Prohormones** What are the possible advantages of synthesizing hormones as prohormones?

14. **Sources of Glucose during Starvation** The typical human adult uses about 160 g of glucose per day. Of this, the brain

alone uses 120 g. The body's available reserve of glucose (~20 g of circulating glucose and ~190 g of glycogen) is adequate for about one day. After the glucose reserve has been depleted during starvation, how does the body obtain more glucose?

15. Parabiotic *ob/ob* Mice By careful surgery, researchers can connect the circulatory systems of two mice so that the same blood circulates through both animals. In these **parabiotic** mice, products released into the blood by one animal reach the other animal via the shared circulation. Both animals are free to eat independently. Suppose a researcher parabiotically joins a mutant *ob/ob* mouse (both copies of the *OB* gene are defective) and a normal *OB/OB* mouse (both copies of the *OB* gene are functional). What would happen to the weight of each mouse in the parabiotic pair?

16. Calculation of Body Mass Index A biochemistry professor weighs 260 lb (118 kg) and is 5 feet 8 inches (173 cm) tall. What is his body mass index (BMI)? How much weight would he have to lose to bring his BMI down to 25 (normal)?

17. Insulin Secretion Predict the effects on insulin secretion by pancreatic β cells of exposure to the potassium ionophore valinomycin. Explain your prediction.

18. Effects of a Deleted Insulin Receptor A strain of mice specifically lacking the insulin receptor of liver is found to have mild fasting hyperglycemia (blood glucose = 132 mg/dL, vs. 101 mg/dL in controls) and a more striking hyperglycemia in the fed state (glucose = 363 mg/dL, vs. 135 mg/dL in

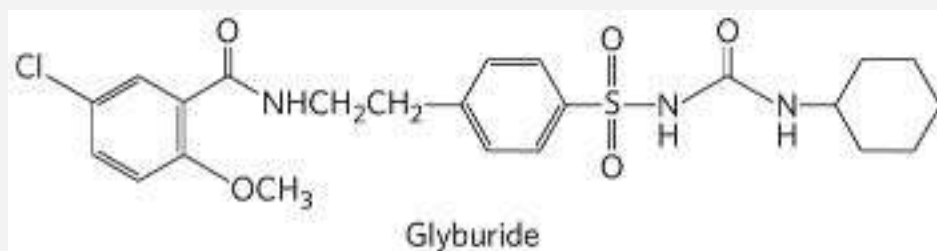
controls). The mice have higher than normal levels of glucose 6-phosphatase in the liver and elevated levels of insulin in the blood. Explain these observations.

19.  **Decisions on Drug Safety** The drug rosiglitazone (Avandia) is effective in lowering blood glucose levels in patients with type 2 diabetes, but a few years after rosiglitazone came into widespread use, it seemed that using the drug came with an increased risk of heart attack. In response, the U.S. Food and Drug Administration (FDA) severely restricted the conditions under which it could be prescribed. Two years later, after additional studies had been completed, the FDA lifted the restrictions, and today rosiglitazone is available by prescription in the United States, with no special limitations. Many other countries ban it completely. If it were your responsibility to decide whether this drug should remain on the market (labeled with suitable warnings about its side effects) or should be withdrawn from the market altogether, what factors would you weigh in making your decision?

20.  **Type 2 Diabetes Medication** The drugs acarbose (Precose) and miglitol (Glyset), used in the treatment of type 2 diabetes mellitus, inhibit α -glucosidases in the brush border of the small intestine. These enzymes degrade oligosaccharides derived from glycogen or starch to monosaccharides. Suggest a possible mechanism for the salutary effect of these drugs for individuals with diabetes. What side effects, if any, would you expect from these drugs? Why? (Hint: Review lactose intolerance, [p. 523](#).)

DATA ANALYSIS PROBLEM

21. Cloning the Sulfonylurea Receptor of the Pancreatic β Cell Glyburide, a member of the sulfonylurea family of drugs, is used to treat type 2 diabetes. It binds to and closes the ATP-gated K^+ channel shown in [Figures 23-26](#) and [23-27](#).

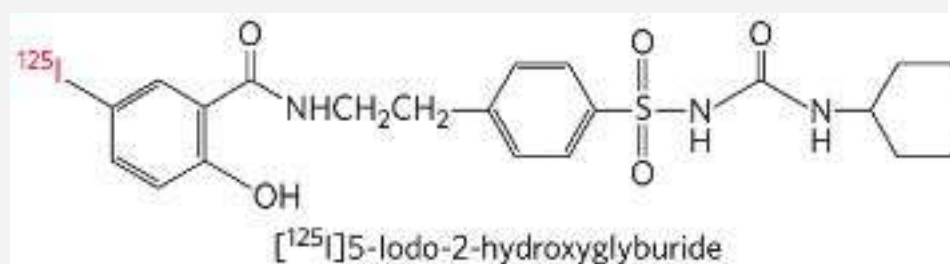


- Given the mechanism shown in [Figure 23-27](#), would treatment with glyburide result in increased or decreased insulin secretion by pancreatic β cells? Explain your reasoning.
- How does treatment with glyburide help reduce the symptoms of type 2 diabetes?
- Would you expect glyburide to be useful for treating type 1 diabetes? Explain your answer.

[Aguilar-Bryan and coauthors \(1995\)](#) cloned the gene for the sulfonylurea receptor (SUR) portion of the ATP-gated K^+ channel from hamsters. The research team went to great lengths to ensure that the gene they cloned was, in fact, the SUR-encoding gene. Here we explore how it is possible for researchers to

demonstrate that they have cloned the gene of interest rather than another gene.

The first step was to obtain pure SUR protein. As was already known, drugs such as glyburide bind SUR with very high affinity ($K_d < 10$ nM), and SUR has a molecular weight of 140 to 170 kDa. Aguilar-Bryan and coworkers made use of the high-affinity glyburide binding to tag the SUR protein with a radioactive label that would serve as a marker to purify the protein from a cell extract. First, they made a radiolabeled derivative of glyburide, using radioactive iodine (^{125}I):



- d. In preliminary studies, the ^{125}I -labeled glyburide derivative (hereafter, $[^{125}\text{I}]$ glyburide) was shown to have the same K_d and binding characteristics as unaltered glyburide. Why was it necessary to demonstrate this? (What alternative possibilities did it rule out?)

Even though $[^{125}\text{I}]$ glyburide bound to SUR with high affinity, a significant amount of the labeled drug would probably dissociate from the SUR protein during purification. To prevent this, $[^{125}\text{I}]$ glyburide had to be covalently cross-linked to SUR. There are many methods for covalent cross-linking; Aguilar-Bryan and

coworkers used UV light. When aromatic molecules are exposed to short-wave UV, they enter an excited state and readily form covalent bonds with nearby molecules. By cross-linking the radiolabeled glyburide to the SUR protein, the researchers could simply track the ^{125}I radioactivity to follow SUR through the purification procedure.

The research team treated hamster HIT cells (which express SUR) with [^{125}I]glyburide and UV light, purified the ^{125}I -labeled 140 kDa protein, and sequenced its 25 residue amino-terminal segment; they found the sequence PLAFCGTENHSAAYRVDQGVLNNGC. The investigators then generated antibodies that bound to two short peptides in this sequence, one binding to PLAFCGTE and the other to HSAAYRVDQGV, and showed that these antibodies bound the purified ^{125}I -labeled 140 kDa protein.

- e. Why was it necessary to include this antibody-binding step?

Next, the researchers designed PCR primers based on the sequences above, and then cloned a gene from a hamster cDNA library that encoded a protein with these sequences (see [Chapter 9](#) on biotechnology methods). The cloned putative *SUR* cDNA hybridized to an mRNA of the appropriate length that was present in cells known to contain SUR. The putative *SUR* cDNA did not hybridize to any mRNA fraction of the mRNAs isolated from hepatocytes, which do not express SUR.

- f. Why was it necessary to include this putative *SUR* cDNA–mRNA hybridization step?

Finally, the cloned gene was inserted into and expressed in COS cells, which do not normally express the *SUR* gene. The investigators mixed these cells with [^{125}I]glyburide, with or without a large excess of unlabeled glyburide, exposed the cells to UV light, and measured the radioactivity of the 140 kDa protein produced. Their results are shown in the table.

Experiment	Cell type	Added putative <i>SUR</i> cDNA?	Added excess unlabeled glyburide?	^{125}I label in 140 kDa protein
1	HIT	No	No	+++
2	HIT	No	Yes	–
3	COS	No	No	–
4	COS	Yes	No	+++
5	COS	Yes	Yes	–

- g. Why was no ^{125}I -labeled 140 kDa protein found in experiment 2?
- h. How would you use the information in the table to argue that the cDNA encoded *SUR*?
- i. What other information would you want to collect to be more confident that you had cloned the *SUR* gene?

Reference

Aguilar-Bryan, L., C.G. Nichols, S.W. Wechsler, J.P. Clement, IV, A.E. Boyd, III, G. González, H. Herrera-Sosa, K. Nguy, J. Bryan, and D.A. Nelson. 1995. Cloning of the β cell high-affinity sulfonylurea receptor: a regulator of insulin secretion. *Science* 268:423–426.